Bangladesh University of Engineering and Technology
Department of Computer Science & Engineering

Electronic Circuits Question Bank With Answers

EEE 263 · Electronic Circuits

Year Coverage: 2011-12 to 2024–25
Topics: Diode Characteristics, Wave Shaping & Clipping/Clamping,
BJT & MOSFET Amplifiers, Op-Amps (Filters, Schmitt Trigger, Analog Computing), Oscillators & Timers (555) & more

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Electronic Circuits

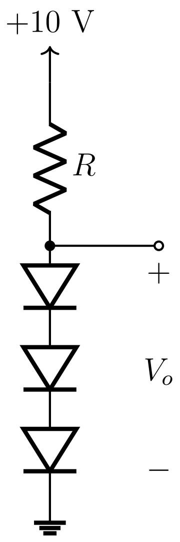
Part I — Semiconductor Device Characteristics

Ideal device models, I-V characteristics, biasing, and circuit analysis for Diode, BJT, and MOSFET

S1 Ideal Diode Characteristics

Diode Circuit Analysis

Q1. For the circuit shown, I is the current flowing through each of the diodes and R is unknown. The diodes are identical with $I_S = 10^{-16}$ A and $n = 1$. (i) Find the value of (I, R) such that $V_o = 2.4\,\text{V}$ is maintained. (ii) If a current of 1 mA is drawn away from the output terminated by a load, what is the change in output voltage?



(15 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

Theory & Principle

In analog electronic circuits, a series string of forward-biased diodes is frequently utilized to create a simple, cost-effective voltage regulator or reference source. The circuit operates on the principle that a forward-biased diode maintains a relatively stable voltage drop across its terminals over a wide range of currents.
The relationship between the current flowing through a forward-biased p-n junction diode (I) and its terminal voltage (V_D) is governed by the Shockley diode equation:

$$I = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right) \approx I_S e^{\frac{V_D}{nV_T}} \quad (1)$$

where:

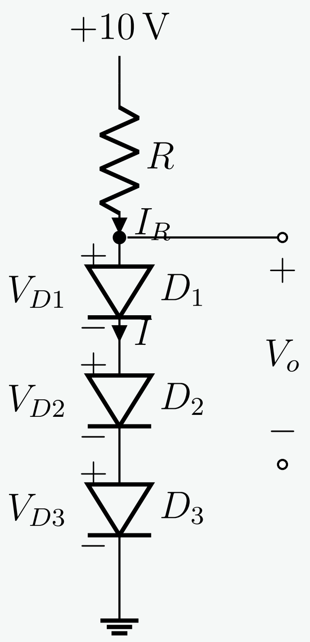
- I_S is the reverse saturation current.
- n is the emission coefficient or ideality factor ($n = 1$ for this problem).
- V_T is the thermal voltage, given by $V_T = \frac{kT}{q}$ ($V_T \approx 25\,\text{mV}$ at room temperature 290 K or 25.85 mV at 300 K).

When small variations in current occur due to an external load, the corresponding change in voltage can be accurately analyzed using the **small-signal diode model**. The dynamic (incremental) resistance r_d of a single diode at its DC operating point is defined as:

$$r_d = \left(\frac{dI}{dV_D}\right)^{-1} = \frac{nV_T}{I} \quad (2)$$

—

Circuit Operation



The circuit consists of a constant supply voltage $V_{CC} = +10\,\text{V}$ supplying current through a current-limiting resistor R into three series-connected identical diodes. Since the diodes are connected in series and no load current is initially drawn, the same quiescent current I flows through each diode.
The total output voltage V_o is equal to the sum of the forward voltage drops across the three identical diodes:

$$V_o = V_{D1} + V_{D2} + V_{D3} = 3V_D \tag{3}$$

—

Step-by-Step Solution

Because the exact value of thermal voltage V_T varies slightly depending on the assumed textbook reference temperature ($25\,\text{mV}$ vs. $25.85\,\text{mV}$), both standard engineering cases are detailed below to ensure complete accuracy.

Part (i): Find (I, R) for $V_o = 2.4\,\textbf{V}$

Given that the three diodes are identical and share the same current I, the voltage drop across each individual diode is:

$$V_D = \frac{V_o}{3} = \frac{2.4\,\text{V}}{3} = 0.8\,\text{V} \tag{4}$$

Using the simplified diode current equation, we isolate the current I:

$$I = I_S e^{\frac{V_D}{nV_T}} \tag{5}$$

By applying Ohm's law across the resistor R, the current through the resistor is identical to the diode string current ($I_R = I$) under no-load conditions:

$$R = \frac{V_{CC} - V_o}{I} = \frac{10\,\text{V} - 2.4\,\text{V}}{I} = \frac{7.6\,\text{V}}{I} \tag{6}$$

Case A: Assuming $V_T = 25\,\text{mV}$ (Standard textbook approximation)

$$I = 10^{-16} \times e^{\frac{0.8}{1\times 0.025}} = 10^{-16} \times e^{32}$$
$$I = 10^{-16} \times (7.89637 \times 10^{13}) = \mathbf{7.896}\,\text{mA}$$

Now, calculating the value of R:

$$R = \frac{7.6\,\text{V}}{7.89637 \times 10^{-3}\,\text{A}} = \mathbf{962.46\,\Omega} \tag{7}$$

Case B: Assuming $V_T = 25.85\,\text{mV}$ (At $T = 300\,\text{K}$)

$$I = 10^{-16} \times e^{\frac{0.8}{1\times 0.02585}} = 10^{-16} \times e^{30.9478}$$
$$I = 10^{-16} \times (2.7481 \times 10^{13}) = \mathbf{2.748}\,\text{mA}$$

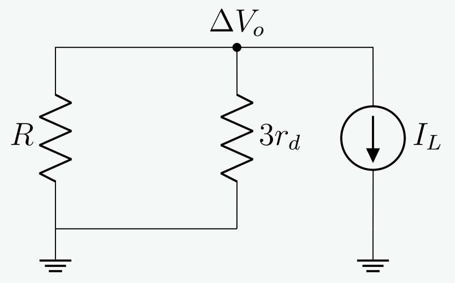
Now, calculating the value of R:

$$R = \frac{7.6\,\text{V}}{2.7481 \times 10^{-3}\,\text{A}} = \mathbf{2765.55\,\Omega} = \mathbf{2.766}\,\mathbf{k\Omega} \tag{8}$$

—

Part (ii): Impact of Drawing a $1\,\text{mA}$ Load Current

When a load current $I_L = 1\,\text{mA}$ is drawn away from the output terminal, it changes the small-signal operating environment. We employ the **small-signal equivalent circuit** looking into the output node.



The small-signal resistance of the total diode string is the sum of the small-signal resistances of the three individual series diodes:

$$r_{\text{total}} = 3r_d = 3\left(\frac{nV_T}{I}\right) = \frac{3V_T}{I} \tag{9}$$

The total incremental output resistance R_{out} observed at the output node is the parallel combination of the biasing resistor R and the incremental resistance of the diode chain ($3r_d$):

$$R_{\text{out}} = R \parallel 3r_d = \frac{R \cdot 3r_d}{R + 3r_d} \tag{10}$$

When an incremental load current $\Delta I_L = 1\,\text{mA}$ is drawn out of the node, the change in output voltage ΔV_o is determined via Ohm's law:

$$\Delta V_o = -I_L \cdot R_{\text{out}} \tag{11}$$

Case A: Evaluating with $V_T = 25\,\text{mV}$ values ($I = 7.896\,\text{mA}$, $R = 962.46\,\Omega$)

$$3r_d = \frac{3 \times 25\,\text{mV}}{7.89637\,\text{mA}} = 9.498\,\Omega \tag{12}$$

$$R_{\text{out}} = 962.46\,\Omega \parallel 9.498\,\Omega = \frac{962.46 \times 9.498}{962.46 + 9.498} = 9.405\,\Omega \tag{13}$$

$$\Delta V_o = -(1\,\text{mA}) \times 9.405\,\Omega = -\mathbf{9.41}\,\text{mV} \tag{14}$$

Case B: Evaluating with $V_T = 25.85\,\text{mV}$ values ($I = 2.748\,\text{mA}$, $R = 2765.55\,\Omega$)

$$3r_d = \frac{3 \times 25.85\,\text{mV}}{2.7481\,\text{mA}} = 28.219\,\Omega \tag{15}$$

$$R_{\text{out}} = 2765.55\,\Omega \parallel 28.219\,\Omega = \frac{2765.55 \times 28.219}{2765.55 + 28.219} = 27.934\,\Omega \tag{16}$$

$$\Delta V_o = -(1\,\text{mA}) \times 27.934\,\Omega = -\mathbf{27.93}\,\text{mV} \tag{17}$$

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Summary of Final Results

The values depend on the standard convention chosen for the thermal voltage V_T:

Parameter	**Case A** ($V_T = 25\,\text{mV}$)	**Case B** ($V_T = 25.85\,\text{mV}$)
Quiescent Current, I	$7.90\,\text{mA}$	$2.75\,\text{mA}$
Biasing Resistor, R	$962.5\,\Omega$	$2.766\,\text{k}\Omega$
Incremental Resistance, $3r_d$	$9.50\,\Omega$	$28.22\,\Omega$
Output Voltage Change, ΔV_o	$-\mathbf{9.41}\,\text{mV}$	$-\mathbf{27.93}\,\text{mV}$

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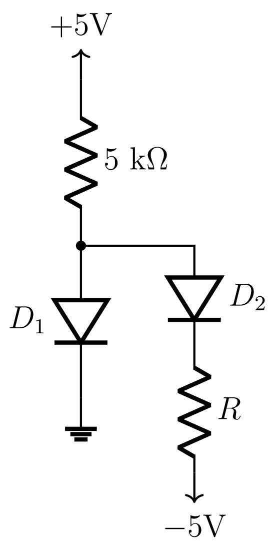
Important Notes & Exam Tips

- **Small-Signal Validity:** The small-signal model assumes $\Delta V_D \ll V_T$. In Case A, $\Delta V_D = \frac{\Delta V_o}{3} \approx -3.14\,\text{mV}$, which is significantly smaller than $25\,\text{mV}$. This justifies the highly accurate usage of linear incremental analysis.
- **Voltage Regulator Load Regulation:** The negative sign in ΔV_o indicates that as the regulator circuit supplies current to an external network, the terminal voltage drops slightly. A lower output resistance (R_{out}) guarantees superior voltage regulation performance.

Q2. For the circuit given, determine the range of values of resistor R such that D_1 does not conduct and D_2 conducts. Assume ideal diodes.



(15 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

Theory & Principle

An ideal diode acts as a perfect electronic switch with two distinct states of operation:

(a) **ON (Forward-Biased / Conducting State):** When forward-biased, an ideal diode behaves as a short circuit (zero voltage drop across its terminals, $V_D = 0\,\text{V}$) and permits a forward current $I_D > 0$.

(b) **OFF (Reverse-Biased / Non-Conducting State):** When reverse-biased, an ideal diode behaves as an open circuit ($I_D = 0\,\text{A}$) and sustains a reverse terminal voltage $V_D \leq 0\,\text{V}$.

To determine the operating range of circuit parameters under specified diode conduction states, we substitute each diode with its corresponding ideal equivalent circuit component and solve the resulting circuit equations using Kirchhoff's laws.

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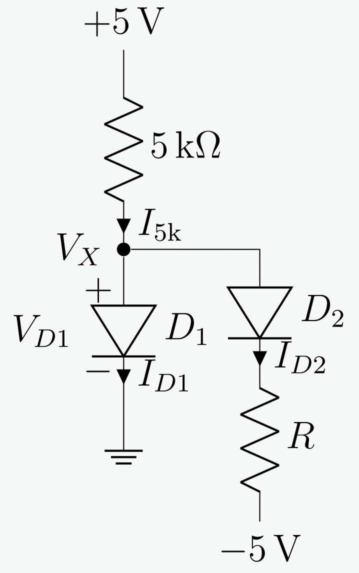
Circuit Operation & Schematic

The target condition states that diode D_1 must remain **OFF** (non-conducting) while diode D_2 must remain **ON** (conducting). We define the voltage at the central node between the $5\,\text{k}\Omega$ resistor, the anode of D_1, and the anode of D_2 as V_X.



Under the specified conditions:

- Since D_1 is **OFF**, it acts as an open circuit, meaning $I_{D1} = 0\,\text{A}$.
- Since D_2 is **ON**, it acts as a short circuit ($V_{D2} = 0\,\text{V}$), directly linking node V_X to the top terminal of resistor R.

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Mathematical Derivation

With D_1 open-circuited and D_2 short-circuited, the circuit reduces to a single continuous series branch connected between the $+5\,\text{V}$ and $-5\,\text{V}$ power rails.

1. Expressing the Node Voltage V_X

The total series resistance of this active branch is given by:

$$R_{\text{total}} = 5\,\text{k}\Omega + R \tag{1}$$

The total voltage drop across this series combination is:

$$V_{\text{total}} = (+5\,\text{V}) - (-5\,\text{V}) = 10\,\text{V} \tag{2}$$

The current flowing through the branch ($I_{5\text{k}} = I_{D2}$) is:

$$I_{D2} = \frac{V_{\text{total}}}{R_{\text{total}}} = \frac{10\,\text{V}}{5\,\text{k}\Omega + R} \tag{3}$$

We can now determine the node voltage V_X by subtracting the voltage drop across the $5\,\text{k}\Omega$ resistor from the $+5\,\text{V}$ supply rail:

$$V_X = 5\,\text{V} - I_{5\text{k}} \cdot (5\,\text{k}\Omega) \tag{4}$$

Substituting the expression for $I_{5\text{k}}$ into the equation yields:

$$V_X = 5 - \left(\frac{10}{5\,\text{k}\Omega + R}\right) \cdot 5\,\text{k}\Omega$$

$$V_X = 5\left[1 - \frac{10}{5\,\text{k}\Omega + R}\right]$$

$$V_X = 5\left[\frac{5\,\text{k}\Omega + R - 10}{5\,\text{k}\Omega + R}\right]$$

$$V_X = 5\left[\frac{R - 5\,\text{k}\Omega}{5\,\text{k}\Omega + R}\right]$$

—

2. Applying the Non-Conduction Condition for D_1

For diode D_1 to remain **OFF** (non-conducting), its anode-to-cathode voltage V_{D1} must be less than or equal to the ideal threshold voltage ($0\,\text{V}$). Because its cathode is connected directly to ground ($0\,\text{V}$):

$$V_{D1} = V_X - 0\,\text{V} \le 0\,\text{V} \implies V_X \le 0\,\text{V} \tag{5}$$

Substituting our derived equation for V_X into this condition:

$$5\left[\frac{R - 5\,\text{k}\Omega}{5\,\text{k}\Omega + R}\right] \le 0 \tag{6}$$

Since a physical resistance R cannot be negative, the denominator ($5\,\text{k}\Omega + R$) is strictly positive (> 0). Thus, the inequality depends entirely on the numerator:

$$R - 5\,\text{k}\Omega \le 0 \implies R \le 5\,\text{k}\Omega \tag{7}$$

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3. Verifying the Conduction Condition for D_2

For diode D_2 to remain **ON** (conducting), the forward current flowing through it must be strictly positive:

$$I_{D2} = \frac{10\,\text{V}}{5\,\text{k}\Omega + R} > 0 \tag{8}$$

Since $10\,\text{V} > 0$, this condition is satisfied for all valid physical resistances where $R \geq 0\,\Omega$.

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Final Answer

Combining the compliance boundaries for both states ($R \geq 0\,\Omega$ and $R \leq 5\,\text{k}\Omega$), the complete range of values for resistor R is:

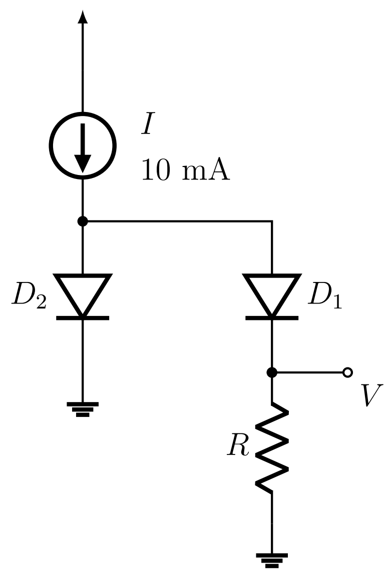
$$\mathbf{0\,\Omega \leq R \leq 5\,k\Omega} \tag{9}$$

—

Exam Tips & Edge-Case Analysis

- **Boundary Testing ($R = 5\,\text{k}\Omega$):** If $R = 5\,\text{k}\Omega$, the circuit forms a symmetric voltage divider between $+5\,\text{V}$ and $-5\,\text{V}$, setting $V_X = 0\,\text{V}$. At this precise boundary, D_1 experiences $0\,\text{V}$ across its terminals but carries $0\,\text{A}$ of current, which represents the critical threshold on the verge of conduction.
- **Behavior when $R > 5\,\text{k}\Omega$:** If R exceeds $5\,\text{k}\Omega$, the voltage V_X attempts to rise above $0\,\text{V}$. This immediately forward-biases the ideal diode D_1, clamping V_X firmly to $0\,\text{V}$ and violating the condition that D_1 must not conduct.

Q3. In the circuit shown, both diodes are identical. Find the value of R for which $V = 50\,\text{mV}$.



(16 marks) **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

Theory & Principle

In this circuit, a constant current source I feeds two parallel branches containing identical forward-biased diodes. The current distribution between these two branches is governed by the nonlinear voltage-current characteristic of the diodes, expressed by the Shockley diode equation:

$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right) \approx I_S e^{\frac{V_D}{nV_T}} \tag{1}$$

where:

- I_S is the reverse saturation current (identical for both D_1 and D_2).
- n is the ideality factor (assumed to be $n = 1$).
- V_T is the thermal voltage, standardly taken as $25\,\text{mV}$ at room temperature.
- V_D is the voltage drop across the diode.

By expressing the ratio of the currents in the two identical diodes, the common I_S term cancels out, allowing us to find the exact current division based strictly on the potential difference between their cathodes.

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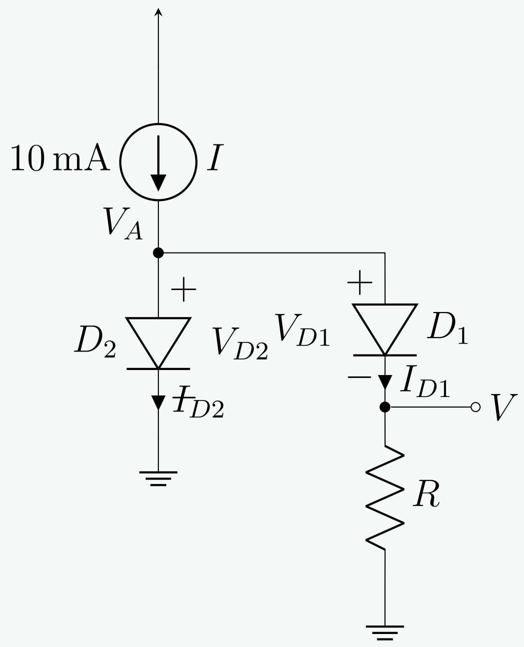
Circuit Operation & Schematic



The total current $I = 10\,\text{mA}$ enters the common anode node (let's call its voltage V_A) and splits into two currents: I_{D2} flowing through diode D_2 and I_{D1} flowing through diode D_1.

$$I_{D1} + I_{D2} = I = 10\,\text{mA} \tag{2}$$

—

Mathematical Derivation & Step-by-Step Solution

1. Determine Diode Voltages

The anode of both diodes is at potential V_A.

- The cathode of D_2 is grounded ($0\,\text{V}$). Therefore, the voltage across D_2 is:

$$V_{D2} = V_A - 0 = V_A \tag{3}$$

- The cathode of D_1 is at potential $V = 50\,\text{mV}$. Therefore, the voltage across D_1 is:

$$V_{D1} = V_A - V \tag{4}$$

2. Current Ratio

Using the approximate exponential diode equation, the currents through the diodes are:

$$I_{D2} = I_S e^{\frac{V_A}{nV_T}}$$
$$I_{D1} = I_S e^{\frac{V_A - V}{nV_T}}$$

Taking the ratio of I_{D2} to I_{D1} yields:

$$\frac{I_{D2}}{I_{D1}} = \frac{I_S e^{\frac{V_A}{nV_T}}}{I_S e^{\frac{V_A - V}{nV_T}}} = e^{\frac{V_A - (V_A - V)}{nV_T}} = e^{\frac{V}{nV_T}} \tag{5}$$

Assuming standard parameters of $n = 1$ and thermal voltage $V_T = 25\,\text{mV}$, and given $V = 50\,\text{mV}$:

$$\frac{I_{D2}}{I_{D1}} = e^{\frac{50\,\text{mV}}{25\,\text{mV}}} = e^2 \approx 7.389 \tag{6}$$

3. Calculate Branch Currents

From Kirchhoff's Current Law (KCL):

$$I_{D1} + I_{D2} = 10\,\text{mA} \tag{7}$$

Substitute $I_{D2} = I_{D1} \cdot e^2$ into the KCL equation:

$$I_{D1} + I_{D1}e^2 = 10\,\text{mA}$$
$$I_{D1}(1 + e^2) = 10\,\text{mA}$$
$$I_{D1} = \frac{10\,\text{mA}}{1 + e^2}$$

Calculating the numerical value:

$$I_{D1} = \frac{10\,\text{mA}}{1 + 7.389056} = \frac{10\,\text{mA}}{8.389056} \approx 1.192\,\text{mA} \tag{8}$$

4. Calculate Resistor R

The voltage across the resistor is $V = 50\,\text{mV}$, and the current flowing through it is I_{D1}. Using Ohm's Law:

$$R = \frac{V}{I_{D1}}$$
$$R = \frac{50 \times 10^{-3}\,\text{V}}{1.192 \times 10^{-3}\,\text{A}}$$
$$R \approx 41.946\,\Omega$$

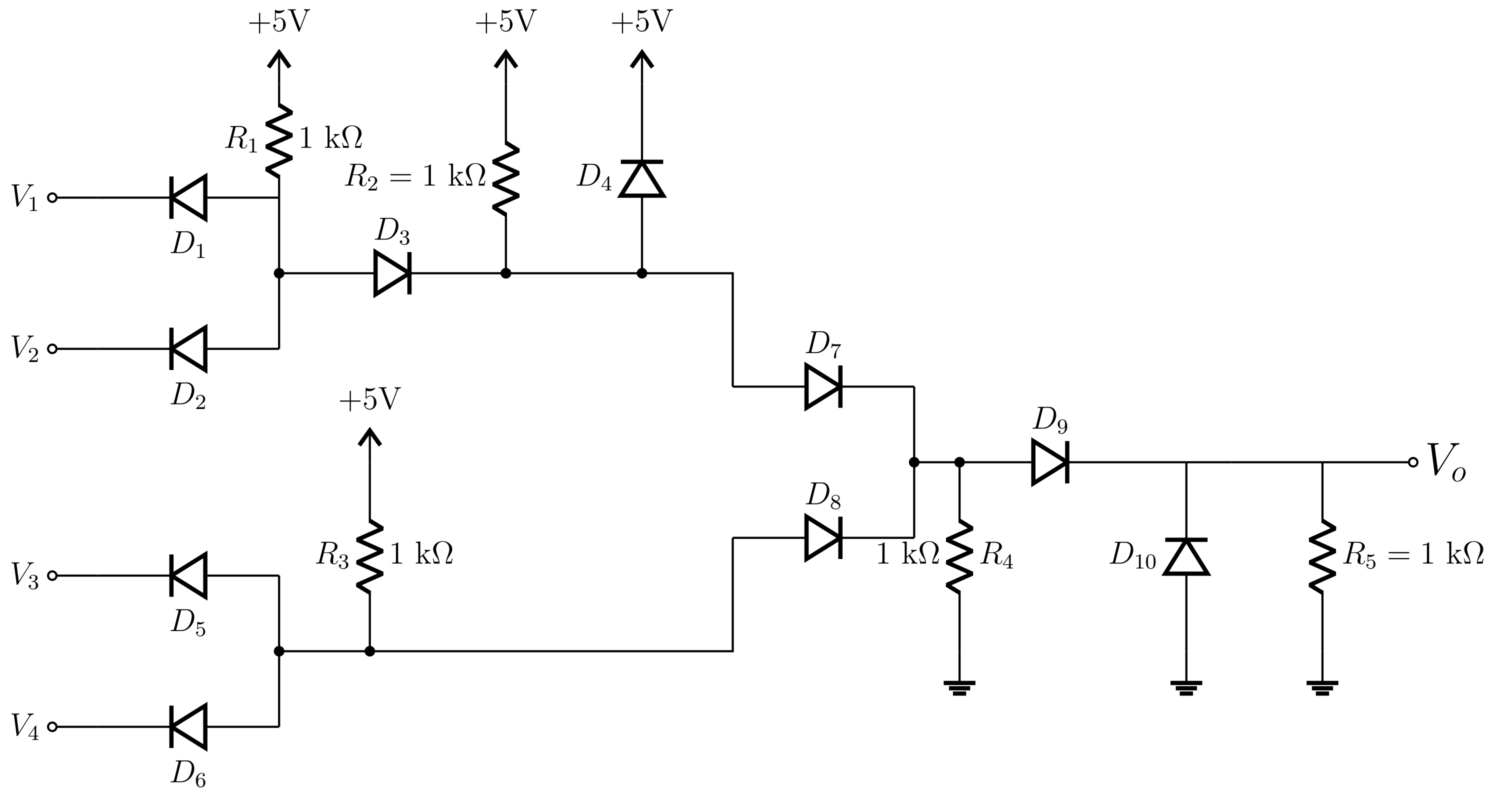
—

Final Answer

To maintain a voltage of $V = 50\,\text{mV}$ across the resistor, assuming a thermal voltage of $V_T = 25\,\text{mV}$, the required value of R is:

$$\mathbf{R \approx 41.95\,\Omega} \quad (9)$$

Q4. Determine the Boolean expression for V_0 in terms of the four input voltages for the circuit shown (diode logic).



(10 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

Theory and Principle of Diode Logic (DL)

Diode logic utilizes the unidirectional current-conducting property of diodes to implement basic Boolean logic gates (AND and OR).

- **Diode AND Gate:** Formed by diodes with their cathodes connected to the inputs and anodes tied together to a pull-up resistor. The output is HIGH only if *all* inputs are HIGH (reverse-biasing the diodes). If any input is LOW, its respective diode conducts, pulling the output LOW.
- **Diode OR Gate:** Formed by diodes with their anodes connected to the inputs and cathodes tied together to a pull-down resistor. The output is HIGH if *any* input is HIGH (forward-biasing the diode).

Circuit Operation and Step-by-Step Analysis

We can decompose the given circuit into functional logic blocks to determine the overall Boolean expression for V_o. Let us assume standard positive logic where $+5\text{V} \equiv$ Logic 1 and $0\text{V} \equiv$ Logic 0.

1. Top Input Stage (AND Gate):

- Components D_1, D_2, and pull-up resistor R_1 form a 2-input Diode AND gate.
- If $V_1 = 0\text{V}$ or $V_2 = 0\text{V}$, the corresponding diode (D_1 or D_2) is forward-biased. Node A is pulled down to approximately 0.7V (Logic 0).
- If $V_1 = +5\text{V}$ and $V_2 = +5\text{V}$, both D_1 and D_2 are reverse-biased (OFF). Node A is pulled HIGH to +5V via R_1 (Logic 1).
- **Boolean Expression for Node A:** $V_A = V_1 \cdot V_2$

2. Bottom Input Stage (AND Gate):

- Components D_5, D_6, and pull-up resistor R_3 form another 2-input Diode AND gate.
- By the identical principle, Node B is HIGH only if both V_3 and V_4 are HIGH.

- **Boolean Expression for Node B:** $V_B = V_3 \cdot V_4$

3. Intermediate Coupling & OR Gate Stage:

- Diodes D_7 and D_8 along with the pull-down resistor R_4 form a 2-input Diode OR gate.
- The output of the top AND gate (V_A) is coupled to this OR gate via D_3. (Note: D_3 acts as a steering diode, while D_4 acts as a voltage clamp to prevent Node C from exceeding +5V, and R_2 assists in biasing the coupling stage). Logically, the state of Node A propagates to Node C.
- The output of the bottom AND gate (V_B) is connected directly to D_8.
- If either Node C (following V_A) OR Node B goes HIGH, the respective diode (D_7 or D_8) becomes forward-biased, driving current through R_4 and pulling Node E HIGH.
- **Boolean Expression for Node E:** $V_E = V_A + V_B = (V_1 \cdot V_2) + (V_3 \cdot V_4)$

4. Output Buffer Stage:

- Diode D_9 acts as a decoupling/isolation diode, R_5 is the output pull-down resistor, and D_{10} is a negative-going transient clamping diode.
- This stage does not alter the Boolean logic; it simply passes the logic state from Node E to the final output V_o.
- **Final Boolean Expression:** $V_o = V_E$

Mathematical Derivation of the Boolean Expression

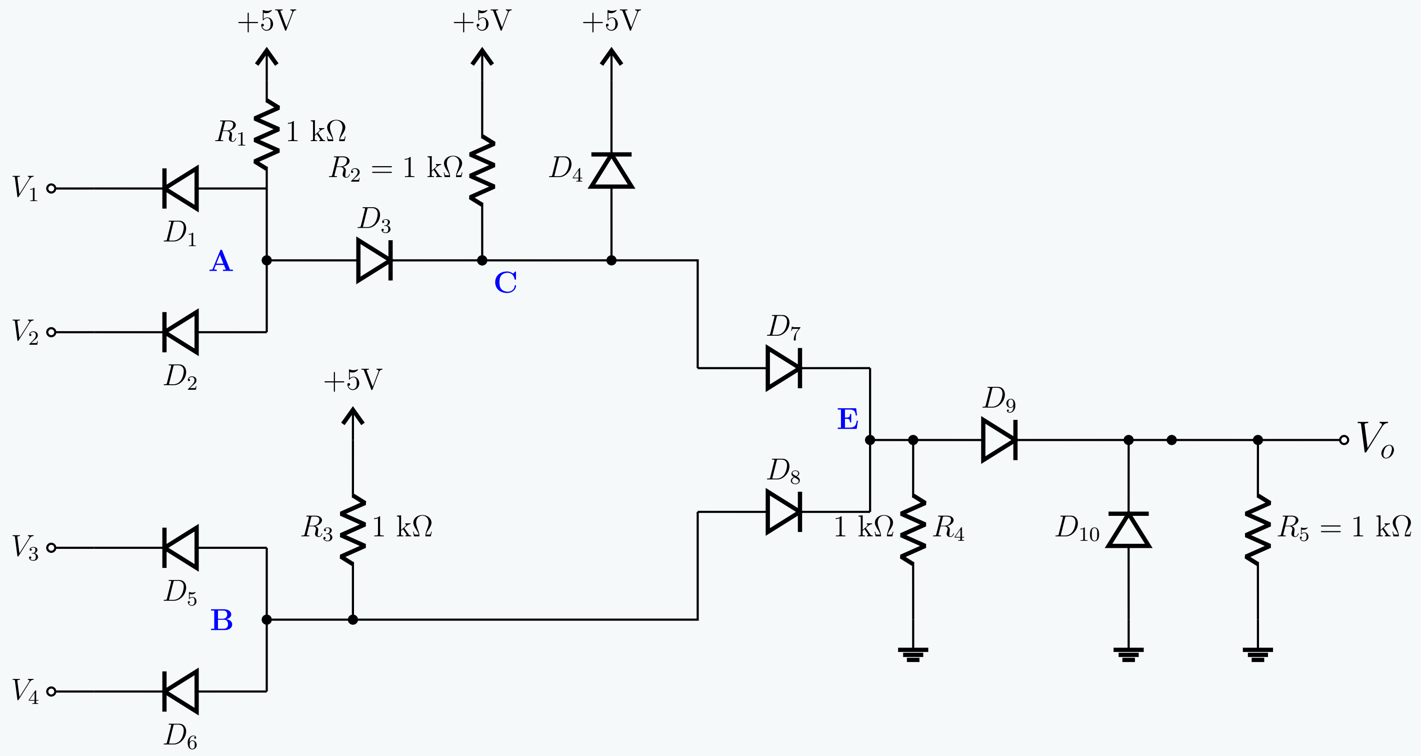
$$V_A = V_1 \text{ AND } V_2 = V_1V_2 \tag{1}$$
$$V_B = V_3 \text{ AND } V_4 = V_3V_4 \tag{2}$$
$$V_o = V_A \text{ OR } V_B \tag{3}$$
$$V_o = V_1V_2 + V_3V_4 \tag{4}$$

Final Answer The Boolean expression for the output voltage V_o in terms of the input voltages is:

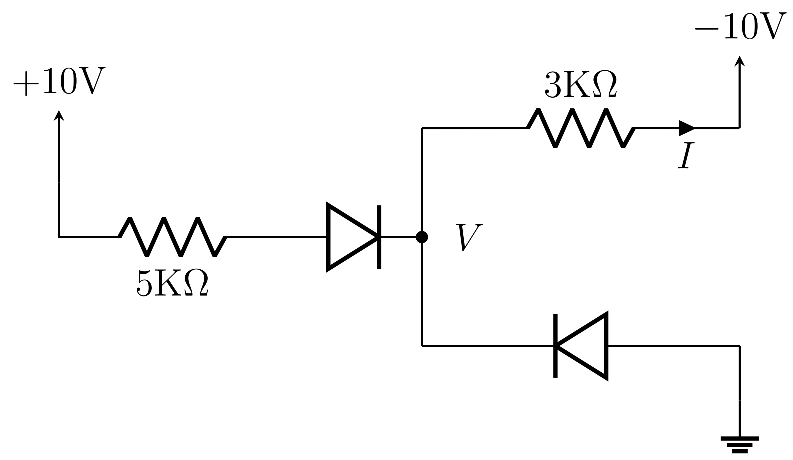
$$\mathbf{V_o = V_1V_2 + V_3V_4} \tag{5}$$

This indicates that the circuit is an **AND-OR** logic circuit.

Circuit Diagram (Redrawn)



Q5. Find the values of I and V in the circuit shown. Use the 0.7 V drop diode model.



(10.67 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Theory and Principle

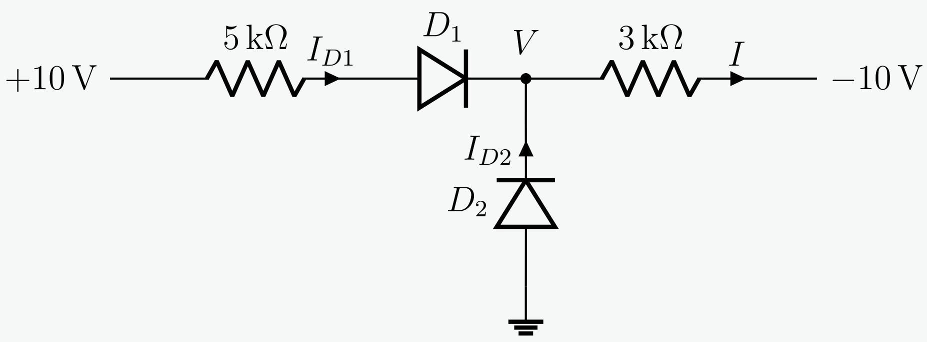
To analyze circuits containing nonlinear components such as diodes, we systematically assume an operating state (ON or OFF) for each diode, solve the resulting linear circuit, and then verify if our assumptions hold true against the physical rules of the diode model.

In the **Constant Voltage Drop (CVD) model** (using 0.7 V for silicon):

- **ON State:** The diode is replaced by a 0.7 V DC voltage source opposing the current flow. We must verify that the current flowing through it is positive ($I_D > 0$).
- **OFF State:** The diode is replaced by an open circuit ($I_D = 0\,\text{A}$). We must verify that the voltage across its terminals is less than the turn-on voltage ($V_D < 0.7\,\text{V}$).

2. Circuit Operation & Schematic

Let us designate the diode in the primary series path as D_1 and the diode in the lower branch connecting to ground as D_2. Based on the original schematic provided in the question, D_2 connects node V to ground with its cathode at node V and its anode at ground. The correctly oriented schematic is drawn below:



3. Step-by-Step Solution

Hypothesis 1: Assume D_1 is ON and D_2 is OFF

If D_2 is OFF, it acts as an open circuit ($I_{D2} = 0\,\text{A}$). The circuit simplifies to a single continuous loop. All current flows from the +10 V source, through D_1, and into the −10 V source. Therefore, $I_{D1} = I$.

Applying Kirchhoff's Voltage Law (KVL) along this path:

$$\begin{aligned} 10 - I(5\,\text{k}\Omega) - V_{D1} - I(3\,\text{k}\Omega) &= -10 \\ 10 - 5I - 0.7 - 3I &= -10 \\ 19.3 &= 8I \\ I &= 2.4125\,\text{mA} \end{aligned}$$

Next, we calculate the voltage V at the node between D_1 and the 3 kΩ resistor:

$$\begin{aligned} V - I(3\,\text{k}\Omega) &= -10 \\ V &= -10 + 3(2.4125) \\ V &= -2.7625\,\text{V} \end{aligned}$$

We must now verify the assumption for D_2. The anode of D_2 is at ground (0 V) and the cathode is at V:

$$V_{D2} = V_{\text{anode}} - V_{\text{cathode}} = 0 - (-2.7625\,\text{V}) = 2.7625\,\text{V} \tag{1}$$

Since $2.7625\,\text{V} \gg 0.7\,\text{V}$, the diode D_2 would be heavily forward-biased, which violates our OFF assumption. **Hypothesis 1 is incorrect.**

Hypothesis 2: Assume both D_1 and D_2 are ON

If D_2 is ON, the voltage across it is clamped to its forward voltage drop of 0.7 V. Since its anode is at ground (0 V) and its cathode is at node V:

$$\begin{aligned} V_{\text{anode}} - V_{\text{cathode}} &= 0.7\,\text{V} \\ 0 - V &= 0.7\,\text{V} \\ V &= -0.7\,\text{V} \end{aligned}$$

Now, calculate the currents in the respective branches. The current I through the $3\,\text{k}\Omega$ resistor is:

$$I = \frac{V - (-10\,\text{V})}{3\,\text{k}\Omega} = \frac{-0.7 + 10}{3} = \frac{9.3}{3} = 3.1\,\text{mA} \tag{2}$$

The current I_{D1} through the $5\,\text{k}\Omega$ resistor and D_1 is evaluated by considering the total voltage drop from the source to node V:

$$I_{D1} = \frac{10\,\text{V} - V - V_{D1}}{5\,\text{k}\Omega} = \frac{10 - (-0.7) - 0.7}{5} = \frac{10}{5} = 2.0\,\text{mA} \tag{3}$$

Applying Kirchhoff's Current Law (KCL) at node V allows us to find I_{D2}. The currents entering the node must equal the currents leaving the node:

$$\begin{aligned} I_{D1} + I_{D2} &= I \\ 2.0\,\text{mA} + I_{D2} &= 3.1\,\text{mA} \\ I_{D2} &= 1.1\,\text{mA} \end{aligned}$$

Verification of Assumptions:

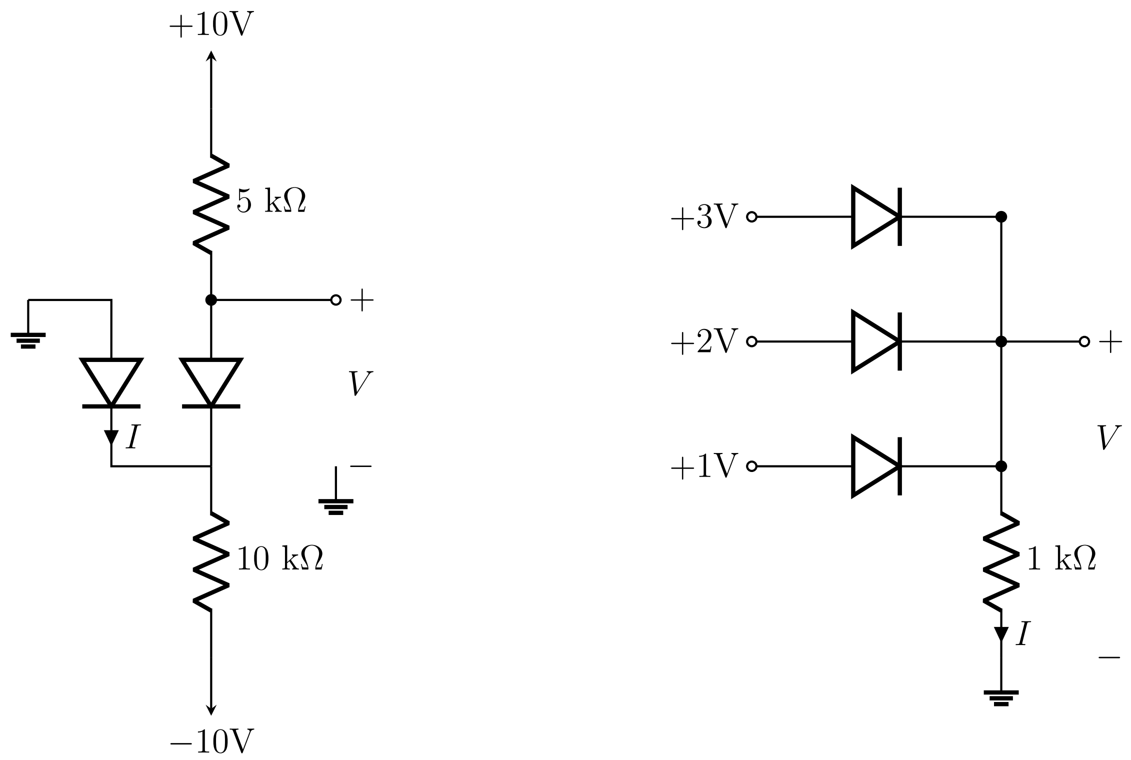
- **Verify D_1 is ON:** $I_{D1} = 2.0\,\text{mA} > 0$. *Assumption holds.*
- **Verify D_2 is ON:** $I_{D2} = 1.1\,\text{mA} > 0$. *Assumption holds.*

Since both assumptions are mathematically verified, this defines the true operating state of the circuit.

4. Final Answer
The final values for current I and voltage V are:

$$\mathbf{I} = \mathbf{3.1}\,\text{mA}$$
$$\mathbf{V} = -\mathbf{0.7}\,\text{V}$$

Q6. Assuming the diodes to be ideal, find the values of I and V in the circuits shown.



(i) (ii)

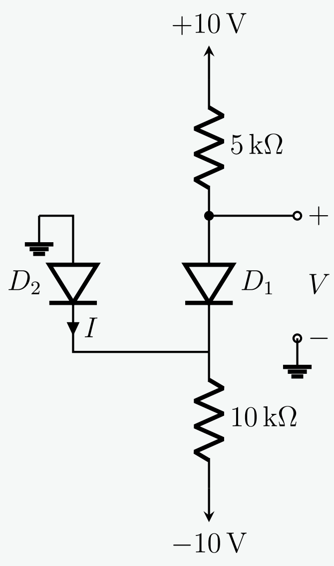
(16 marks) [EEE 263 | L-2/T-1 | 2011–12]

Answer

Part (i): Two-Diode Circuit Analysis

1. Circuit Schematic

For clarity, we label the right diode as D_1 and the left diode as D_2. The node voltage V is measured with respect to ground.



2. Theory & Assumptions

To solve circuits containing ideal diodes, we assume an operating state (ON or OFF) for each diode, compute the resulting voltages and currents, and then verify if the physical conditions for our assumptions hold true:

- If **ON**, the ideal diode acts as a short circuit (0 V drop), and the forward current must be positive ($I_D > 0$).
- If **OFF**, it acts as an open circuit (0 A current), and the reverse voltage must be negative ($V_D < 0$).

Hypothesis: We assume that D_1 is **ON** and D_2 is **OFF**.

- Since D_2 is OFF, it acts as an open circuit. Therefore, no current flows through its branch, which immediately gives us $I = 0\,\text{A}$.
- Since D_1 is ON, it behaves as a short circuit. The $5\,\text{k}\Omega$ and $10\,\text{k}\Omega$ resistors are now connected in a single continuous series path between the $+10\,\text{V}$ and $-10\,\text{V}$ sources.

3. Mathematical Derivation & Verification

With D_1 shorted and D_2 open, the total current I_{series} flowing through the right branch is determined by the total voltage difference divided by the total equivalent resistance:

$$I_{\text{series}} = \frac{10\,\text{V} - (-10\,\text{V})}{5\,\text{k}\Omega + 10\,\text{k}\Omega} = \frac{20\,\text{V}}{15\,\text{k}\Omega} = \frac{4}{3}\,\text{mA} \approx 1.33\,\text{mA} \tag{1}$$

Because this current flows downward through D_1, the forward current is $I_{D1} = 1.33\,\text{mA} > 0$. *The assumption that D_1 is ON is verified.*

Next, we calculate the node voltage V. Using Kirchhoff's Voltage Law (KVL) from the top $+10\,\text{V}$ source down to node V:

$$V = 10\,\text{V} - I_{\text{series}}(5\,\text{k}\Omega)$$
$$V = 10 - \left(\frac{4}{3}\,\text{mA}\right)(5\,\text{k}\Omega)$$
$$V = 10 - \frac{20}{3} = \frac{10}{3}\,\text{V} \approx 3.33\,\text{V}$$

Finally, we must verify the OFF state of D_2. The anode of D_2 is connected to ground (0 V). Because D_1 is a short circuit, the cathode of D_2 is at the same potential as V (3.33 V).

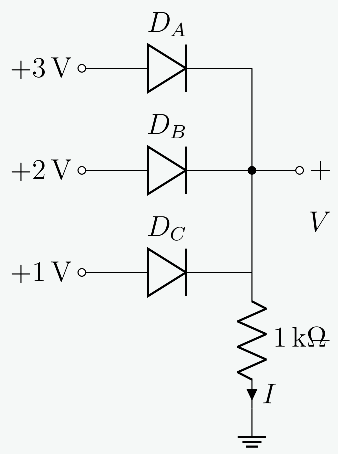
$$V_{D2} = V_{\text{anode}} - V_{\text{cathode}} = 0\,\text{V} - 3.33\,\text{V} = -3.33\,\text{V} \tag{2}$$

Since $V_{D2} < 0\,\text{V}$, diode D_2 is reverse-biased. *The assumption that D_2 is OFF is verified.*

Part (ii): Three-Diode Circuit Analysis

1. Circuit Schematic

Let us label the three diodes D_A, D_B, and D_C corresponding to their respective input voltages.



2. Theory & Principle

This configuration represents a classic analog **maximum-voltage selector** (a Diode OR Gate). In a common-cathode parallel array, the ideal diode connected to the highest positive potential will conduct, clamping the output node V to its input voltage. This clamping action simultaneously reverse-biases all other diodes in the circuit.
Hypothesis: We assume that D_A (connected to $+3\,\text{V}$) is **ON**, while D_B ($+2\,\text{V}$) and D_C ($+1\,\text{V}$) are **OFF**.

3. Mathematical Derivation & Verification

Since D_A is assumed ON and ideal, it acts as a short circuit, directly tying the common cathode node to its anode voltage:

$$V = 3\,\text{V} \tag{3}$$

Now, we calculate the current I flowing through the $1\,\text{k}\Omega$ resistor to ground via Ohm's Law:

$$I = \frac{V - 0\,\text{V}}{1\,\text{k}\Omega} = \frac{3\,\text{V}}{1\,\text{k}\Omega} = 3\,\text{mA} \tag{4}$$

Because D_B and D_C are open circuits, this entire $3\,\text{mA}$ current is supplied by D_A. Since $I_{DA} = 3\,\text{mA} > 0$, *the assumption that D_A is ON is verified.*
Lastly, we check the voltage drops across the other two diodes to ensure they are safely OFF:

$$V_{DB} = V_{\text{anode},B} - V = 2\,\text{V} - 3\,\text{V} = -1\,\text{V}$$
$$V_{DC} = V_{\text{anode},C} - V = 1\,\text{V} - 3\,\text{V} = -2\,\text{V}$$

Because both $V_{DB} < 0$ and $V_{DC} < 0$, both D_B and D_C are reverse-biased. *The assumption that D_B and D_C are OFF is verified.*

Final Answers

For Circuit (i):

$$\mathbf{I} = \mathbf{0}\,\text{mA}, \quad \mathbf{V} = \mathbf{3.33}\,\text{V} \left(\text{or } \frac{\mathbf{10}}{\mathbf{3}}\,\text{V}\right) \tag{5}$$

For Circuit (ii):

$$\mathbf{I} = \mathbf{3}\,\text{mA}, \quad \mathbf{V} = \mathbf{3}\,\text{V} \tag{6}$$

Diode Models (Ideal, CVD, Piecewise Linear)

Q1. Describe the different models for diode forward characteristics. **(7 marks)** **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

Introduction

In analog electronic circuit analysis and design, the semiconductor p-n junction diode exhibits a highly non-linear forward conduction characteristic. To simplify circuit analysis, engineers use various mathematical and circuit approximations depending on the required accuracy and numerical complexity. These approximations are classified into four primary models: the **Exponential Model**, the **Ideal Model**, the **Constant-Voltage-Drop Model**, and the **Piecewise-Linear Model**.
—

1. The Exponential (Exact) Model

Theory and Circuit Operation

The Exponential Model is the most accurate physical description of the diode's forward behavior. It is derived from solid-state physics and accounts for carrier diffusion across the p-n junction under forward-bias conditions.
Mathematical Derivation

The current-voltage (i-v) relation is governed by the **Shockley Diode Equation**:

$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right) \tag{1}$$

For a significant forward-bias voltage ($V_D \gg nV_T$), the term -1 becomes negligible, simplifies the relation to:

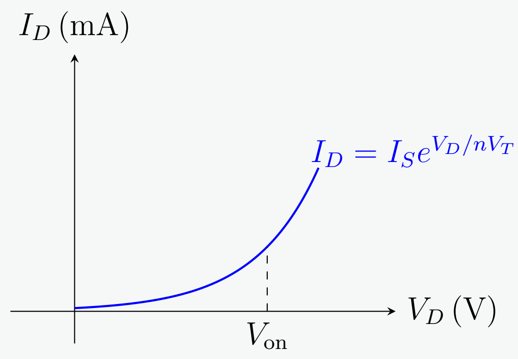
$$I_D \approx I_S e^{\frac{V_D}{nV_T}} \tag{2}$$

Where:
- I_D = Diode forward current (A)
- V_D = Diode forward voltage drop (V)
- I_S = Reverse saturation current (typically $10^{-15}\,\text{A}$ to $10^{-12}\,\text{A}$ for small-signal silicon diodes)

- n = Ideality factor (typically 1 to 2, depending on the material and fabrication structure)
- V_T = Thermal voltage, expressed as:

$$V_T = \frac{kT}{q} \tag{3}$$

At room temperature ($T \approx 300\,\text{K}$), $V_T \approx 25.3\,\text{mV} \approx 25\,\text{mV}$ or $26\,\text{mV}$.



Applications

Highly precise analog designs, log/antilog amplifiers, temperature sensors, and computer-aided design (SPICE) simulations.
—

2. The Ideal Diode Model

Theory and Circuit Operation

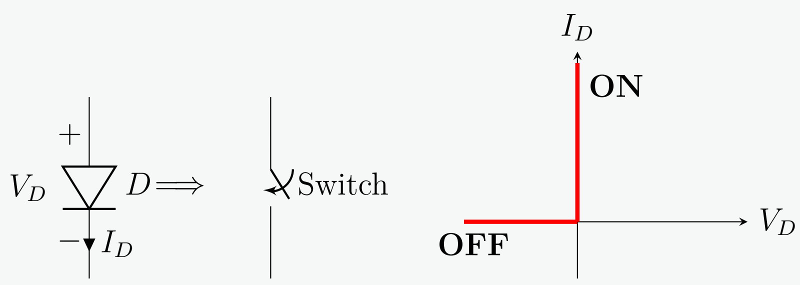
The Ideal Diode Model is the simplest approximation, treating the diode as a perfect electronic switch. It assumes zero forward resistance and an instantaneous transition from non-conduction to conduction at a zero threshold voltage.

Mathematical Derivation

The behavior is mathematically structured as a piecewise conditional state:

$$\begin{cases} V_D = 0, & \text{for } I_D > 0 \quad \text{(ON State / Short Circuit)} \\ I_D = 0, & \text{for } V_D < 0 \quad \text{(OFF State / Open Circuit)} \end{cases} \tag{4}$$

Circuit Representation & i-v Curve



Applications

Initial circuit inspection, high-voltage power supply rectification, and conceptual logic gate behavior analyses.
—

3. The Constant-Voltage-Drop (CVD) Model

Theory and Circuit Operation

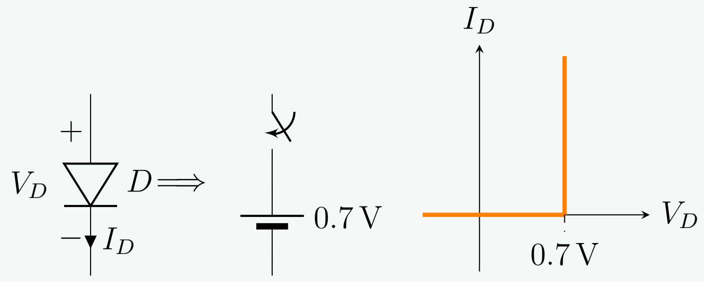
The Constant-Voltage-Drop Model acknowledges that a semiconductor p-n junction requires a specific potential barrier clearance to initiate conduction. Once the threshold voltage is reached, the diode conducts current with a fixed voltage drop, completely independent of the magnitude of the forward current.

Mathematical Derivation

For silicon diodes, this stable threshold drop (V_D) is conventionally set to $0.7\,\text{V}$. The conditional state equation yields:

$$\begin{cases} V_D = 0.7\,\text{V}, & \text{for } I_D > 0 \quad \text{(ON State)} \\ I_D = 0, & \text{for } V_D < 0.7\,\text{V} \quad \text{(OFF State)} \end{cases} \tag{5}$$

Circuit Representation & i-v Curve



Applications

Standard analysis of typical analog circuits, power supply regulators, wave-shaping clippers, and clamping networks.
—

4. The Piecewise-Linear (PWL) Model

Theory and Circuit Operation

The Piecewise-Linear model scales up the CVD model by embedding an internal series dynamic resistance (r_D). This linearizes the slope of the forward exponential curve beyond the initial conduction threshold voltage (V_{D0}), matching the behavior of a real diode under varying current levels.

Mathematical Derivation

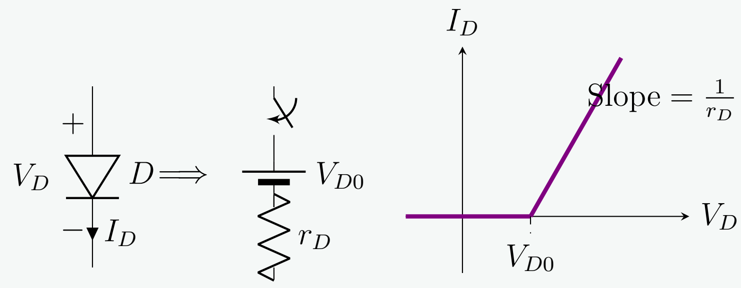
When forward-biased beyond V_{D0}, the total forward voltage drop across the diode becomes linear with respect to the forward current:

$$V_D = V_{D0} + I_D r_D \quad \text{for } V_D \geq V_{D0} \tag{6}$$

The dynamic forward resistance is calculated mathematically as the inverse slope of the operational i-v linear trace:

$$r_D = \frac{\Delta V_D}{\Delta I_D} = \left.\frac{dv_D}{di_D}\right|_{\text{Q-point}} = \frac{nV_T}{I_Q} \tag{7}$$

Circuit Representation & i-v Curve



Applications

Precise small-signal analysis, power electronics thermal estimations, and small-signal incremental amplifier architectures.
—

Summary and Comparison Model Table

Parameter	Exponential Model	Ideal Model	Constant-Voltage Drop	Piecewise-Linear
Threshold (V_{on})	Variable (I_D dependent)	0 V	0.7 V (Silicon)	$V_{D0} \approx 0.6\,\text{V} - 0.7\,\text{V}$
Internal Resistance	Non-linear, dynamic	$0\,\Omega$	$0\,\Omega$	Finite dynamic r_D
Complexity Level	Very High (Transcendental)	Extremely Low	Low	Moderate
Accuracy Class	Exact Physics Reference	First-Pass Approximation	Standard Practical Standard	High Local Linearity

Exam Tips

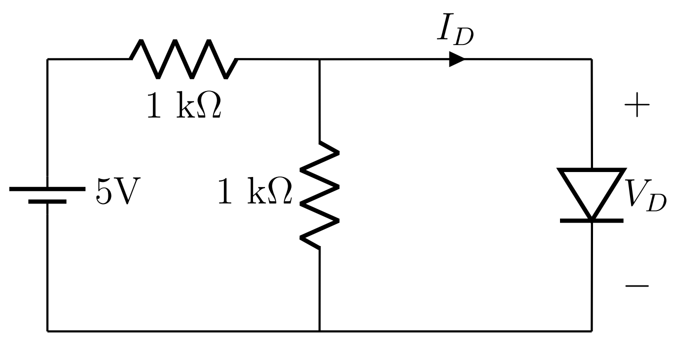
- When solving generic exam circuit problems, unless specified otherwise, default to the **Constant-Voltage-Drop Model** (0.7 V for Si) as it strikes the best balance between speed and practical accuracy.
- Always clarify your model assumption at the start of your calculation steps to secure full marks for method derivation.

Q2. Determine the current I_D and diode voltage V_D for the circuit shown using:

(a) Piecewise linear model ($V_{DO} = 0.7\,\text{V}$, $r_D = 20\,\Omega$)

(b) Constant voltage drop model ($V_{DO} = 0.7\,\text{V}$)

(c) Ideal diode model

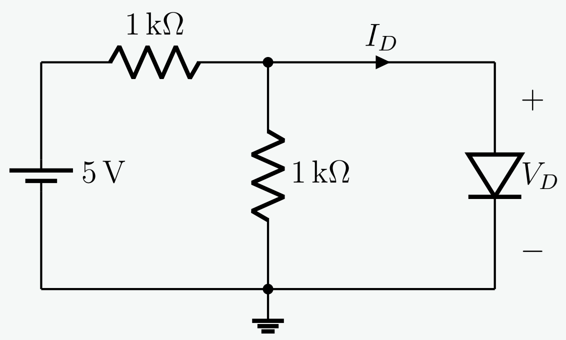


(18 marks) [EEE 263 | L-2/T-1 | 2011–12]

Answer

1. Circuit Analysis and Thévenin Equivalent

To determine the current I_D and voltage V_D across the diode efficiently for multiple models, we first simplify the linear portion of the circuit connected to the diode using Thévenin's theorem.
Original Circuit

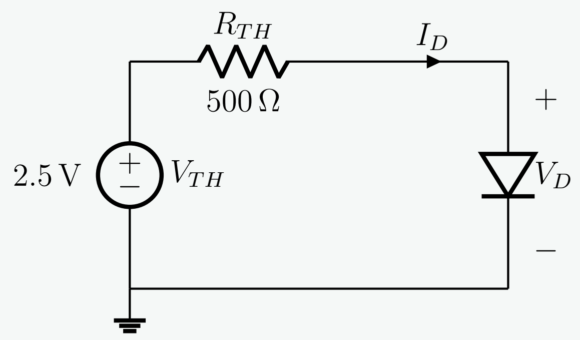


Thévenin Simplification

We remove the diode from the circuit to find the open-circuit Thévenin voltage (V_{TH}) and the Thévenin equivalent resistance (R_{TH}) looking back into the terminals.

$$V_{TH} = 5\,\text{V} \times \frac{1\,\text{k}\Omega}{1\,\text{k}\Omega + 1\,\text{k}\Omega} = 5\,\text{V} \times 0.5 = 2.5\,\text{V}$$
$$R_{TH} = 1\,\text{k}\Omega \parallel 1\,\text{k}\Omega = \frac{1\,\text{k}\Omega \times 1\,\text{k}\Omega}{1\,\text{k}\Omega + 1\,\text{k}\Omega} = 0.5\,\text{k}\Omega = 500\,\Omega$$

Since $V_{TH} = 2.5\,\text{V}$ is strictly positive (and greater than typical diode turn-on voltages), we can safely assume the diode is **forward-biased (ON)** in all subsequent models.



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2. Step-by-Step Solution for Diode Models

(a) Piecewise Linear Model

Theory: In the piecewise linear model, a forward-biased diode is replaced by a DC voltage source V_{DO} in series with a dynamic forward resistance r_D.
Given: $V_{DO} = 0.7\,\text{V}$, $r_D = 20\,\Omega$.
Applying Kirchhoff's Voltage Law (KVL) to the Thévenin equivalent circuit:

$$V_{TH} - I_D R_{TH} - V_{DO} - I_D r_D = 0$$
$$I_D(R_{TH} + r_D) = V_{TH} - V_{DO}$$
$$I_D = \frac{V_{TH} - V_{DO}}{R_{TH} + r_D}$$

Substituting the values:

$$I_D = \frac{2.5\,\text{V} - 0.7\,\text{V}}{500\,\Omega + 20\,\Omega} = \frac{1.8\,\text{V}}{520\,\Omega} \approx 3.4615\,\text{mA} \tag{1}$$

The voltage across the diode V_D is the sum of the threshold voltage and the drop across its internal resistance:

$$V_D = V_{DO} + I_D r_D = 0.7\,\text{V} + (3.4615 \times 10^{-3}\,\text{A})(20\,\Omega) = 0.7\,\text{V} + 0.0692\,\text{V} = 0.7692\,\text{V} \tag{2}$$

(b) Constant Voltage Drop Model

Theory: This model assumes that once the diode is forward-biased, it maintains a constant voltage drop V_{DO} regardless of the current flowing through it. It neglects the internal forward resistance ($r_D = 0$).
Given: $V_{DO} = 0.7\,\text{V}$.
Applying KVL:

$$I_D = \frac{V_{TH} - V_{DO}}{R_{TH}} \tag{3}$$

Substituting the values:

$$I_D = \frac{2.5\,\text{V} - 0.7\,\text{V}}{500\,\Omega} = \frac{1.8\,\text{V}}{500\,\Omega} = 3.6\,\text{mA} \tag{4}$$

The voltage across the diode is simply the constant threshold voltage:

$$V_D = 0.7\,\text{V} \tag{5}$$

(c) Ideal Diode Model

Theory: The ideal diode model treats the forward-biased diode as a perfect conductor (short circuit) with zero voltage drop.
Given: $V_{DO} = 0\,\text{V}$, $r_D = 0\,\Omega$.
Applying KVL:

$$I_D = \frac{V_{TH}}{R_{TH}} \tag{6}$$

Substituting the values:

$$I_D = \frac{2.5\,\text{V}}{500\,\Omega} = 5.0\,\text{mA} \tag{7}$$

The voltage across the diode is zero:

$$V_D = 0\,\text{V} \tag{8}$$

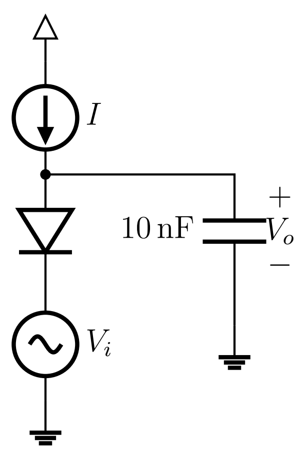
—

3. Final Summary

Part	Diode Model	Diode Current (I_D)	Diode Voltage (V_D)
(a)	Piecewise Linear ($V_{DO} = 0.7\,\text{V}, r_D = 20\,\Omega$)	3.46 mA	0.769 V
(b)	Constant Voltage Drop ($V_{DO} = 0.7\,\text{V}$)	3.60 mA	0.700 V
(c)	Ideal Diode	5.00 mA	0.000 V

Small Signal Model

Q1. In the circuit shown, "I" is a DC current source and "V_i" is a sinusoidal signal with small amplitude ($< 10\,\text{mV}$) and a frequency of 100 kHz. Representing the diode by its small-signal resistance r_d (a function of I):
(i) sketch the small-signal equivalent circuit for determining the sinusoidal output V_o across the capacitor;
(ii) find the phase shift between V_i and V_o;
(iii) find the value of I that will provide a phase shift of -45;
(iv) find the range of phase shift achieved as I is varied from 0.1 to 10 times the value found in (iii). Assume $n = 1$.



(20 marks) [EEE 263 | L-2/T-1 | 2023–24]

Answer

Circuit Operation and Small-Signal Modeling

The given circuit operates with a combination of a DC bias and a small-signal AC variation.

- **DC Analysis:** The ideal DC current source I forces a constant bias current through the diode, establishing the quiescent operating point (Q-point). Because V_i is a purely AC sinusoidal source, its DC voltage is 0 V (acting as a short circuit to ground in DC analysis). Consequently, the entire DC current I flows through the diode to ground.

- **AC Small-Signal Analysis:** By applying the superposition principle, the DC current source I is deactivated (replaced by an open circuit because it is an ideal current source). The diode is replaced by its small-signal dynamic resistance r_d, which is determined by the DC bias current I.

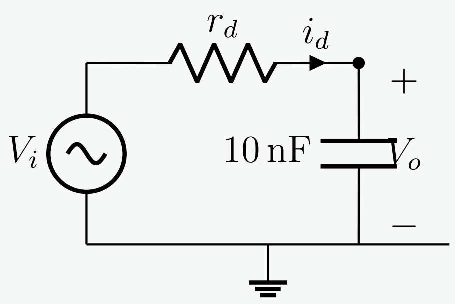
(i) Small-Signal Equivalent Circuit

To construct the small-signal equivalent circuit for determining the AC output V_o:

(a) The ideal DC current source I becomes an open circuit.

(b) The diode is replaced by its small-signal resistance r_d.

(c) The rest of the AC components (V_i and the 10 nF capacitor) remain active.



Note: The circuit is drawn horizontally for standard signal flow. This electrically matches the vertical topology of the original circuit, forming a standard RC low-pass filter.
—

(ii) Phase Shift between V_i and V_o

From the small-signal equivalent circuit, the system operates as an AC voltage divider. The output voltage V_o is taken across the capacitor. Using complex impedances, the transfer function $H(j\omega)$ is:

$$\frac{V_o}{V_i} = \frac{Z_C}{r_d + Z_C} = \frac{\frac{1}{j\omega C}}{r_d + \frac{1}{j\omega C}}$$

$$\frac{V_o}{V_i} = \frac{1}{1 + j\omega r_d C}$$

The phase shift ϕ between the output V_o and input V_i is the phase angle of the transfer function:

$$\phi = \angle\left(\frac{1}{1 + j\omega r_d C}\right) = \angle(1) - \angle(1 + j\omega r_d C)$$

$$\phi = 0^\circ - \tan^{-1}\left(\frac{\omega r_d C}{1}\right)$$

$$\phi = -\tan^{-1}(\omega r_d C)$$

—

(iii) Value of I for a -45° Phase Shift

We are given the required condition for the phase shift:

$$\phi = -45^\circ \tag{1}$$

Substituting this into our phase equation:

$$-\tan^{-1}(\omega r_d C) = -45^\circ$$

$$\omega r_d C = \tan(45^\circ) = 1$$

Solving for the required small-signal resistance r_d:

$$r_d = \frac{1}{\omega C} \tag{2}$$

Using the given parameters ($f = 100\,\text{kHz}$ and $C = 10\,\text{nF}$):

- Angular frequency $\omega = 2\pi f = 2\pi \times 10^5$ rad/s
- Capacitance $C = 10 \times 10^{-9}\,\text{F} = 10^{-8}\,\text{F}$

$$r_d = \frac{1}{(2\pi \times 10^5)(10^{-8})} = \frac{1}{2\pi \times 10^{-3}} = \frac{500}{\pi}\,\Omega \approx 159.15\,\Omega \tag{3}$$

The small-signal resistance of a diode is inversely proportional to its DC bias current:

$$r_d = \frac{nV_T}{I} \implies I = \frac{nV_T}{r_d} \tag{4}$$

Assuming a standard room temperature thermal voltage $V_T \approx 25\,\text{mV}$ and $n = 1$ (given):

$$I = \frac{1 \times 25 \times 10^{-3}\,\text{V}}{159.15\,\Omega} \approx 1.571 \times 10^{-4}\,\text{A} = \mathbf{157.1}\,\mu\text{A} \tag{5}$$

—

(iv) Range of Phase Shift as I is Varied

Let I_{45} be the current found in part (iii) that yields $\omega r_d C = 1$. Because $r_d = \frac{nV_T}{I}$, the resistance r_d scales inversely with I. Consequently, the unitless product $\omega r_d C$ also scales inversely with I:

$$\omega r_d C = \frac{I_{45}}{I} \tag{6}$$

We evaluate the phase shift $\phi = -\tan^{-1}(\omega r_d C)$ at the two extremes of the given current range:

(a) **When $I = 0.1 \times I_{45}$:**

$$\omega r_d C = \frac{I_{45}}{0.1 I_{45}} = 10 \tag{7}$$

$$\phi_1 = -\tan^{-1}(10) \approx \mathbf{-84.29^\circ} \tag{8}$$

(b) **When $I = 10 \times I_{45}$:**

$$\omega r_d C = \frac{I_{45}}{10 I_{45}} = 0.1 \tag{9}$$

$$\phi_2 = -\tan^{-1}(0.1) \approx \mathbf{-5.71^\circ} \tag{10}$$

Final Conclusion: By varying the DC bias current I from 0.1 to 10 times the center -45° condition value, the phase shift can be electronically tuned over a range from -5.71° **to** -84.29°. This highlights the circuit's fundamental utility as a highly tunable current-controlled phase shifter.

Q2. The small-signal model of a diode is said to be valid for voltage variations of about 10 mV. To what percentage current change does this correspond for $n = 1$ (consider both positive and negative signals)? What is the maximum allowable voltage signal (positive or negative) if the current change is limited to 10%? **(10 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

Theory: Diode Large-Signal and Small-Signal Analysis

The total instantaneous current i_D flowing through a forward-biased diode is related to the total instantaneous voltage v_D across it by the exact exponential equation:

$$i_D = I_S e^{\frac{v_D}{nV_T}} \tag{1}$$

In a small-signal application, the diode operates at a DC quiescent point (Q-point) defined by a DC voltage V_D and a corresponding DC current I_D, upon which a small time-varying AC signal v_d is superimposed:

$$v_D(t) = V_D + v_d(t) \tag{2}$$

Substituting this into the diode equation yields the total instantaneous current:

$$i_D = I_S e^{\frac{V_D + v_d}{nV_T}} = \left(I_S e^{\frac{V_D}{nV_T}}\right) e^{\frac{v_d}{nV_T}} = I_D e^{\frac{v_d}{nV_T}} \tag{3}$$

The change in the diode current, Δi, due to the AC signal v_d is:

$$\Delta i = i_D - I_D = I_D \left(e^{\frac{v_d}{nV_T}} - 1\right) \tag{4}$$

The **fractional (or percentage) current change** is thus defined as:

$$\frac{\Delta i}{I_D} = e^{\frac{v_d}{nV_T}} - 1 \tag{5}$$

Assumption: For typical small-signal analysis at room temperature, the thermal voltage is assumed to be $V_T \approx 25\,\text{mV}$. The ideality factor is given as $n = 1$.

—

Part 1: Percentage Current Change for $v_d = \pm 10\,\text{mV}$

To evaluate the validity of the small-signal approximation for a $\pm 10\,\text{mV}$ variation, we use Equation (5).

Case 1A: Positive Voltage Signal ($v_d = +10\,\mathbf{mV}$)

$$\begin{aligned}\frac{\Delta i}{I_D} &= e^{\frac{+10\,\text{mV}}{1\times 25\,\text{mV}}} - 1 \\ &= e^{0.4} - 1 \\ &\approx 1.4918 - 1 = 0.4918\end{aligned}$$

Expressed as a percentage, a $+10\,\text{mV}$ variation causes a $+49.2\%$ increase in current.
Case 1B: Negative Voltage Signal ($v_d = -10\,\mathbf{mV}$)

$$\begin{aligned}\frac{\Delta i}{I_D} &= e^{\frac{-10\,\text{mV}}{1\times 25\,\text{mV}}} - 1 \\ &= e^{-0.4} - 1 \\ &\approx 0.6703 - 1 = -0.3297\end{aligned}$$

Expressed as a percentage, a $-10\,\text{mV}$ variation causes a -33.0% decrease in current.
Important Note: The asymmetry ($+49.2\%$ vs. -33.0%) arises directly from the highly non-linear (exponential) nature of the diode's $i - v$ characteristic. The small-signal (linear) model predicts a symmetric $\pm 40\%$ change ($\frac{v_d}{V_T} = \frac{10}{25}$), illustrating the approximation error when v_d is as large as $10\,\text{mV}$.
—

Part 2: Maximum Voltage Signal for a 10% Current Change

To ensure high linearity, we often restrict the current change to $\pm 10\%$. We must find the maximum allowable voltage signal v_d that satisfies this constraint.
Rearranging Equation (5) to solve for v_d:

$$v_d = nV_T \ln\left(1 + \frac{\Delta i}{I_D}\right) \tag{6}$$

Case 2A: Positive Current Limit ($\frac{\Delta i}{I_D} = +0.10$)

$$\begin{aligned}v_{d(\text{max}+)} &= 1 \times 25\,\text{mV} \times \ln(1 + 0.10) \\ &= 25\,\text{mV} \times \ln(1.10) \\ &\approx 25\,\text{mV} \times 0.0953 \\ &= +\mathbf{2.38}\,\text{mV}\end{aligned}$$

Case 2B: Negative Current Limit ($\frac{\Delta i}{I_D} = -0.10$)

$$\begin{aligned}v_{d(\text{max}-)} &= 1 \times 25\,\text{mV} \times \ln(1 - 0.10) \\ &= 25\,\text{mV} \times \ln(0.90) \\ &\approx 25\,\text{mV} \times (-0.1054) \\ &= -\mathbf{2.63}\,\text{mV}\end{aligned}$$

—

Final Answer Summary

Parameter	**Calculated Value**
Percentage Current Change for $v_d = +10\,\text{mV}$	$+\mathbf{49.2}\%$
Percentage Current Change for $v_d = -10\,\text{mV}$	$-\mathbf{33.0}\%$
Maximum Positive Voltage Signal for $+10\%$ Current Change	$+\mathbf{2.38}\,\text{mV}$
Maximum Negative Voltage Signal for -10% Current Change	$-\mathbf{2.63}\,\text{mV}$

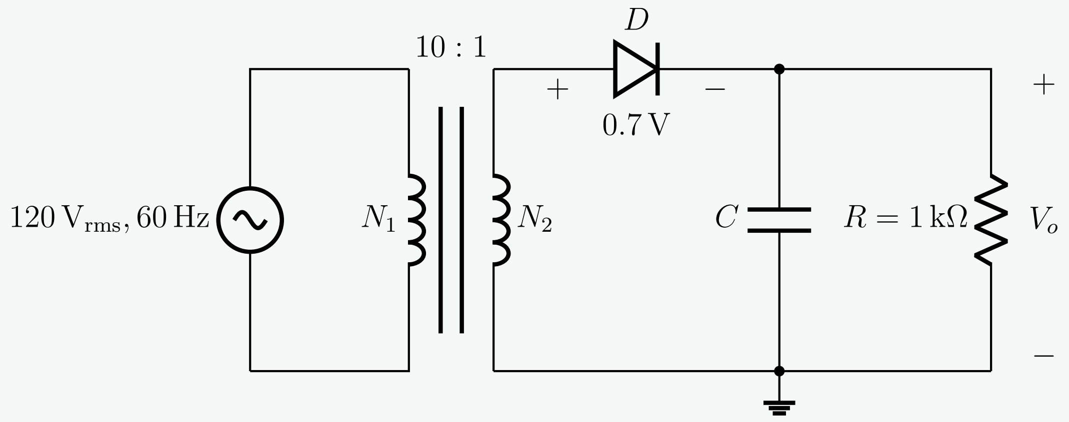
Exam Tip: Notice that to keep the small-signal model highly accurate (within 10% distortion), the AC voltage signal amplitude must be kept very small—generally strictly less than roughly $2.5\,\text{mV}$ for $n = 1$.

Peak Rectifier

Q1. Consider a half-wave rectifier circuit with $R = 1\,\text{k}\Omega$ load operating from a $120\,\text{V}$ (rms), $60\,\text{Hz}$ household supply through a 10-to-1 step-down transformer, augmented with a filter capacitor C across the load. Assume $V_D = 0.7\,\text{V}$ for any current.
(i) Find C for a peak-to-peak ripple of 10% of peak output.
(ii) Find C for a peak-to-peak ripple of 1% of peak output.
(iii) Find the average output voltage for each of the case (i) & (ii) .
(iv) Find the fraction of the cycle for which the diode conducts for each of the case (i) & (ii).
(v) Find the average and peak diode currents for each of the case (i) & (ii).
(20 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

Circuit Diagram



Theory & Initial Calculations

1. Secondary Voltage Calculation: The step-down transformer reduces the primary AC voltage by the turns ratio $N_1/N_2 = 10$.

$$V_{s(\text{rms})} = \frac{V_{p(\text{rms})}}{10} = \frac{120\,\text{V}}{10} = 12\,\text{V}_{\text{rms}} \tag{1}$$

The peak voltage of the secondary transformer winding is:

$$V_{s(\text{peak})} = V_{s(\text{rms})} \times \sqrt{2} = 12\sqrt{2} \approx 16.97\,\text{V} \tag{2}$$

2. Peak Output Voltage (V_p): For a half-wave rectifier, the diode conducts when forward-biased, causing a constant voltage drop $V_D = 0.7\,\text{V}$. The peak output voltage across the load and capacitor is:

$$V_p = V_{s(\text{peak})} - V_D = 16.97\,\text{V} - 0.7\,\text{V} = 16.27\,\text{V} \tag{3}$$

Note: The ripple frequency f for a half-wave rectifier is equal to the source frequency (60 Hz).

—

Step-by-Step Solution

We utilize the standard approximations for peak rectifier circuits (valid when $V_r \ll V_p$):

- **Ripple Voltage:** $V_r \approx \frac{V_p}{fRC}$
- **DC Output Voltage:** $V_{\text{dc}} \approx V_p - \frac{V_r}{2}$
- **Conduction Angle:** $\theta_c = \omega\Delta t \approx \sqrt{\frac{2V_r}{V_p}}$ (in radians)
- **Average Diode Current (Full Cycle):** $I_{D,\text{avg}} = I_{\text{load}} = \frac{V_{\text{dc}}}{R}$
- **Peak Diode Current:** $i_{D,\max} = I_{\text{load}}\left(1 + 2\pi\sqrt{\frac{2V_p}{V_r}}\right)$

Case (i): Peak-to-Peak Ripple is 10% of Peak Output

Given $V_r = 0.10 \times V_p = 0.10 \times 16.27\,\text{V} = 1.627\,\text{V}$.
(i) Filter Capacitor C:

$$C = \frac{V_p}{V_r f R} = \frac{V_p}{0.10 V_p f R} = \frac{1}{0.10 \times 60 \times 1000} = \frac{1}{6000} \approx \mathbf{166.7}\,\mu\text{F} \tag{4}$$

(iii) Average Output Voltage (V_{dc}):

$$V_{\text{dc}} = V_p - \frac{V_r}{2} = 16.27 - \frac{1.627}{2} = 16.27 - 0.8135 = \mathbf{15.46}\,\text{V} \tag{5}$$

(iv) Fraction of Cycle for Conduction: The conduction angle $\theta_c = \sqrt{\frac{2\times0.10V_p}{V_p}} = \sqrt{0.20} \approx 0.4472\,\text{rad}$.

$$\text{Fraction} = \frac{\theta_c}{2\pi} = \frac{0.4472}{2\pi} \approx \mathbf{0.0712}\ (\text{or } 7.12\%) \tag{6}$$

(v) Average and Peak Diode Currents: The average diode current over the entire cycle equals the DC load current:

$$I_{D,\text{avg}} = I_{\text{load}} = \frac{V_{\text{dc}}}{R} = \frac{15.46\,\text{V}}{1\,\text{k}\Omega} = \mathbf{15.46}\,\text{mA} \tag{7}$$

The peak diode current is calculated as:

$$i_{D,\max} = I_{\text{load}}\left(1 + 2\pi\sqrt{\frac{2V_p}{0.10V_p}}\right) = 15.46\,\text{mA}\left(1 + 2\pi\sqrt{20}\right) \approx 15.46(1 + 28.10) = \mathbf{449.8}\,\text{mA} \tag{8}$$

—

Case (ii): Peak-to-Peak Ripple is 1% of Peak Output

Given $V_r = 0.01 \times V_p = 0.01 \times 16.27\,\text{V} = 0.1627\,\text{V}$.
(ii) Filter Capacitor C:

$$C = \frac{V_p}{V_r f R} = \frac{1}{0.01 \times 60 \times 1000} = \frac{1}{600} \approx \mathbf{1666.7}\,\mu\text{F (or } 1.67\,\text{mF)} \tag{9}$$

(iii) Average Output Voltage (V_{dc}):

$$V_{\text{dc}} = V_p - \frac{V_r}{2} = 16.27 - \frac{0.1627}{2} = 16.27 - 0.08135 = \mathbf{16.19}\,\text{V} \tag{10}$$

(iv) Fraction of Cycle for Conduction: The conduction angle $\theta_c = \sqrt{\frac{2 \times 0.01 V_p}{V_p}} = \sqrt{0.02} \approx 0.1414\,\text{rad}$.

$$\text{Fraction} = \frac{\theta_c}{2\pi} = \frac{0.1414}{2\pi} \approx \mathbf{0.0225} \text{ (or } 2.25\%) \tag{11}$$

(v) Average and Peak Diode Currents: The average diode current over the entire cycle:

$$I_{D,\text{avg}} = I_{\text{load}} = \frac{V_{\text{dc}}}{R} = \frac{16.19\,\text{V}}{1\,\text{k}\Omega} = \mathbf{16.19}\,\text{mA} \tag{12}$$

The peak diode current is:

$$i_{D,\max} = 16.19\,\text{mA}\left(1 + 2\pi\sqrt{\frac{2V_p}{0.01V_p}}\right) = 16.19\,\text{mA}\left(1 + 2\pi\sqrt{200}\right) \approx 16.19(1 + 88.86) = \mathbf{1454.8}\,\text{mA (or } 1.45\,\text{A)} \tag{13}$$

—

Final Answer Summary Table

Part	Parameter	Case (i) 10% Ripple	Case (ii) 1% Ripple
(i), (ii)	Filter Capacitor (C)	166.7 μF	1666.7 μF
(iii)	Average Output Voltage (V_{dc})	15.46 V	16.19 V
(iv)	Conduction Fraction	0.0712 (7.12%)	0.0225 (2.25%)
(v)	Cycle-Average Diode Current ($I_{D,\text{avg}}$)	15.46 mA	16.19 mA
(v)	Peak Diode Current ($i_{D,\max}$)	449.8 mA	1454.8 mA

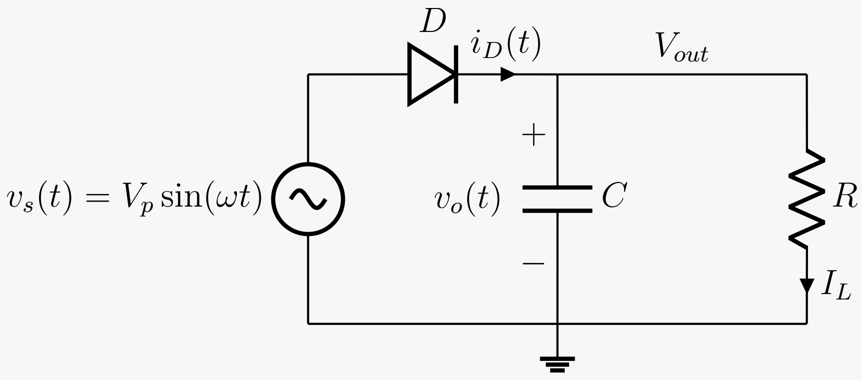
Important Note on Diode Current: The $I_{D,\text{avg}}$ computed above is the average over the *entire cycle*, corresponding to the DC load current I_{load}. If a question explicitly asks for the average current strictly *during the conduction interval*, it is calculated as $I_{D(\text{cond\_avg})} = I_{\text{load}}/\text{Fraction}$.

Q2. Consider a peak rectifier fed by a 60 Hz sinusoid with a peak value $V_p = 100\,\text{V}$. Let the load resistance $R = 10\,\text{k}\Omega$. Find the value of capacitance C that will result in a peak-to-peak ripple of 2 V. Also calculate the fraction of the cycle during which the diode is conducting and the average and peak values of the diode current. **(10 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

Assumptions:
For this analysis, we assume an ideal diode model ($V_{D,on} = 0\,\text{V}$) since the peak input voltage ($V_p = 100\,\text{V}$) is significantly larger than the typical forward voltage drop of a practical diode. Furthermore, since the specified ripple (2 V) is very small compared to the peak voltage ($V_r \ll V_p$), the standard small-ripple textbook approximations hold true.

Circuit Diagram: Half-Wave Peak Rectifier



Given Parameters

- Source frequency, $f = 60\,\text{Hz}$

- Peak input voltage, $V_p = 100\,\text{V}$
- Load resistance, $R = 10\,\text{k}\Omega$
- Peak-to-peak ripple voltage, $V_r = 2\,\text{V}$

1. Calculation of Filter Capacitance (C)

In a peak rectifier with a small ripple, the capacitor discharges linearly through the load resistor R over approximately the entire period $T \approx \frac{1}{f}$. The peak-to-peak ripple voltage V_r is given by:

$$V_r \approx \frac{V_p}{fRC}$$

Rearranging to solve for the capacitance C:

$$\begin{aligned} C &= \frac{V_p}{fRV_r} \\ &= \frac{100}{60 \times (10 \times 10^3) \times 2} \\ &= \frac{100}{1.2 \times 10^6} \approx 83.33 \times 10^{-6}\,\text{F} = \mathbf{83.33}\,\mu\text{F} \end{aligned}$$

2. Conduction Angle and Fraction of the Cycle

The diode only conducts during a short interval Δt near the peak of the input voltage to replenish the charge lost by the capacitor. The conduction angle $\theta_c = \omega \Delta t$ can be approximated by:

$$\omega \Delta t \approx \sqrt{\frac{2V_r}{V_p}}$$

Substituting the known values:

$$\omega \Delta t = \sqrt{\frac{2 \times 2}{100}} = \sqrt{0.04} = \mathbf{0.2}\,\text{rad}$$

To find the fraction of the cycle during which the diode conducts, we divide the conduction angle by a full period (2π radians):

$$\text{Fraction} = \frac{\omega \Delta t}{2\pi} = \frac{0.2}{2\pi} \approx \mathbf{0.0318} \text{ (or } \mathbf{3.18}\%)$$

3. Average and Peak Diode Currents

During the non-conduction interval, the load current I_L is approximately constant and is supplied entirely by the capacitor:

$$I_L \approx \frac{V_p}{R} = \frac{100}{10\,\text{k}\Omega} = 10\,\text{mA}$$

Average Diode Current ($i_{D,avg}$):
To maintain steady-state, the charge supplied by the diode during the short conduction interval must equal the charge drawn by the load over the full cycle. The average diode current during conduction is derived as:

$$\begin{aligned} i_{D,avg} &= I_L \left(1 + \pi \sqrt{\frac{2V_p}{V_r}}\right) \\ &= 10\,\text{mA} \left(1 + \pi \sqrt{\frac{2 \times 100}{2}}\right) \\ &= 10\,\text{mA}\,(1 + 10\pi) \\ &\approx 10\,\text{mA} \times (1 + 31.416) = \mathbf{324.16}\,\text{mA} \end{aligned}$$

Peak Diode Current ($i_{D,max}$):
The maximum diode current occurs at the beginning of the conduction interval and is approximately twice the average charging current plus the load current:

$$\begin{aligned} i_{D,max} &= I_L \left(1 + 2\pi \sqrt{\frac{2V_p}{V_r}}\right) \\ &= 10\,\text{mA} \left(1 + 2\pi \sqrt{\frac{2 \times 100}{2}}\right) \\ &= 10\,\text{mA}\,(1 + 20\pi) \\ &\approx 10\,\text{mA} \times (1 + 62.832) = \mathbf{638.32}\,\text{mA} \end{aligned}$$

Final Answer Summary

Parameter	Calculated Value
Filter Capacitance (C)	$83.33\,\mu$F
Conduction Angle ($\omega\Delta t$)	0.2 rad
Conduction Fraction	3.18%
Average Diode Current ($i_{D,avg}$)	324.16 mA
Peak Diode Current ($i_{D,max}$)	638.32 mA

Q3. Consider a half-wave peak rectifier fed with a voltage having a triangular waveform with 20 V peak-to-peak amplitude, zero average and 1 kHz frequency. Load resistance $R = 100\,\Omega$ and filter capacitor $C = 100\,\mu$F. Assume the diode has a forward voltage drop of 0.7 V when conducting. Find: (i) average DC output voltage, (ii) time interval during which the diode conducts, (iii) average diode current during conduction, (iv) maximum diode current. **(20 marks)** **[EEE 263 | L-2/T-1 | 2016–17]**

Answer

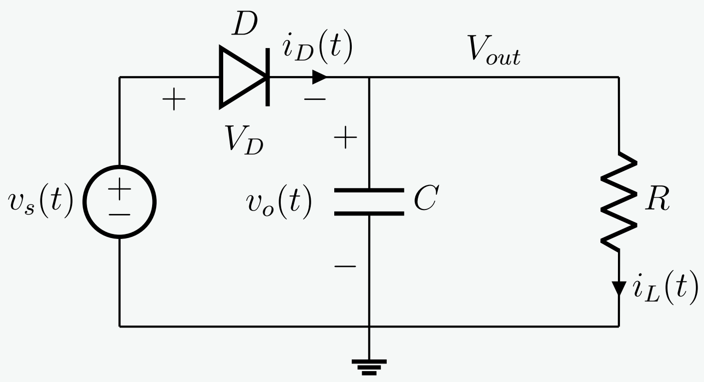
Theory and Circuit Operation

A peak rectifier (or envelope detector) consists of a diode and a filter capacitor connected in parallel with a load resistance. When a zero-average triangular waveform is applied:

(a) **Conduction Phase:** On the rising edge of the input voltage $v_s(t)$, once it exceeds the capacitor voltage plus the diode forward drop ($v_s > v_o + V_D$), the diode turns ON. The capacitor quickly charges up, following the input waveform minus the diode drop: $v_o(t) = v_s(t) - V_D$.

(b) **Non-Conduction Phase:** Immediately after the input passes its peak, the input slope decreases rapidly. The diode turns OFF because the capacitor cannot discharge fast enough to follow the sharp drop of the triangular input. The capacitor then discharges exponentially through the load resistor R with a time constant $\tau = RC$.

Circuit Diagram



Given Parameters

- Peak-to-peak input voltage: $V_{pp} = 20\,\text{V}$
- Zero-average input implies: Peak input voltage $V_p = \frac{V_{pp}}{2} = 10\,\text{V}$
- Input frequency: $f = 1\,\text{kHz} \implies$ Period $T = \frac{1}{f} = 1\,\text{ms}$
- Load resistance: $R = 100\,\Omega$
- Filter capacitance: $C = 100\,\mu\text{F}$
- Diode forward voltage drop: $V_D = 0.7\,\text{V}$

The discharge time constant is:

$$\tau = RC = 100\,\Omega \times 100\,\mu\text{F} = 10\,\text{ms}$$

Since $\tau = 10\,\text{ms} \gg T = 1\,\text{ms}$, the small-ripple approximation and linear discharge assumption are highly accurate.

Mathematical Derivation and Step-by-Step Solution

The maximum output voltage reached at the peak of the input is:

$$V_{omax} = V_p - V_D = 10\,\text{V} - 0.7\,\text{V} = 9.3\,\text{V}$$

(i) Calculation of Peak-to-Peak Ripple (V_r) and Average DC Output Voltage ($V_{out,avg}$) During the non-conduction phase, which lasts for approximately the entire period ($T - \Delta t \approx T$), the capacitor discharges linearly. The peak-to-peak ripple voltage V_r can be expressed as:

$$V_r \approx \frac{V_{omax}}{RC}(T - \Delta t) \tag{1}$$

On the rising edge of the triangular wave leading up to the peak (defined at $t = 0$), the slope of the input voltage is constant:

$$\frac{dv_s}{dt} = \frac{V_{pp}}{T/2} = \frac{20\,\text{V}}{0.5\,\text{ms}} = 40,000\,\text{V/s} = 40\,\text{V/ms}$$

The diode turns ON at a time $t = -\Delta t$ before the peak, where the input voltage rises enough to forward-bias the diode:

$$V_r = \left(\frac{dv_s}{dt}\right)\Delta t = 40,000 \cdot \Delta t$$

Equating the two relationships for V_r (using millisecond units for simplicity):

$$\begin{aligned} 40 \cdot \Delta t &= \frac{9.3}{10}(1 - \Delta t) \\ 40 \cdot \Delta t &= 0.93 - 0.93 \cdot \Delta t \\ 40.93 \cdot \Delta t = 0.93 &\implies \Delta t = \frac{0.93}{40.93} \approx \mathbf{0.02272}\,\text{ms} = \mathbf{22.72}\,\mu\text{s} \end{aligned}$$

Now, substituting Δt back to find the precise ripple voltage V_r:

$$V_r = 40\,\text{V/ms} \times 0.02272\,\text{ms} = 0.9089\,\text{V}$$

The average DC output voltage is the midpoint of the linear ripple waveform:

$$\begin{aligned} V_{out,avg} &= V_{omax} - \frac{V_r}{2} \\ &= 9.3\,\text{V} - \frac{0.9089\,\text{V}}{2} = 9.3\,\text{V} - 0.4545\,\text{V} = \mathbf{8.8455}\,\text{V} \end{aligned}$$

(ii) Time Interval During Which the Diode Conducts (Δt) As derived above during the simultaneous solution for the ripple voltage:

$$\Delta t = \mathbf{22.72}\,\mu\text{s}$$

(iii) Average Diode Current During Conduction ($i_{D,avg}$) During the conduction interval Δt, the diode current $i_D(t)$ splits into the capacitor charging current $i_C(t)$ and the load current $i_L(t)$:

$$i_D(t) = i_C(t) + i_L(t) = C\frac{dv_o}{dt} + \frac{v_o(t)}{R}$$

Since $v_o(t) = v_s(t) - V_D$ during conduction, its slope matches the input slope:

$$i_C(t) = C\frac{dv_s}{dt} = 100\,\mu\text{F} \times 40,000\,\text{V/s} = 4.0\,\text{A} \quad \text{(constant during conduction)}$$

The average load current during this short conduction window is determined by the average output voltage during conduction, which matches $V_{out,avg}$:

$$i_{L,avg} = \frac{V_{out,avg}}{R} = \frac{8.8455\,\text{V}}{100\,\Omega} = 0.08846\,\text{A} = 88.46\,\text{mA}$$

Thus, the total average current flowing through the diode *during the conduction interval* is:

$$\begin{aligned} i_{D,avg} &= i_C + i_{L,avg} \\ &= 4.0\,\text{A} + 0.08846\,\text{A} = \mathbf{4.0885}\,\text{A} \end{aligned}$$

(iv) Maximum Diode Current ($i_{D,max}$) The maximum diode current occurs exactly at the peak of the input ($t = 0$), where the output voltage reaches its maximum value $V_{omax} = 9.3\,\text{V}$:

$$\begin{aligned} i_{D,max} &= i_C + i_{L,max} \\ &= C\frac{dv_o}{dt} + \frac{V_{omax}}{R} \\ &= 4.0\,\text{A} + \frac{9.3\,\text{V}}{100\,\Omega} = 4.0\,\text{A} + 0.093\,\text{A} = \mathbf{4.093}\,\text{A} \end{aligned}$$

Final Answer Summary

Part	Parameter	Formula / Relation	Value
(i)	Average DC Output Voltage ($V_{out,avg}$)	$V_{omax} - \frac{V_r}{2}$	**8.85** V
(ii)	Diode Conduction Interval (Δt)	$\frac{V_r}{(dv_s/dt)}$	**22.72** μs
(iii)	Average Diode Current during Conduction	$C\frac{dv_s}{dt} + \frac{V_{out,avg}}{R}$	**4.09** A
(iv)	Maximum Diode Current ($i_{D,max}$)	$C\frac{dv_s}{dt} + \frac{V_{omax}}{R}$	**4.093** A

Q4. A peak rectifier circuit is operated with a $1\,\mathrm{k}\Omega$ load and a diode that can be modeled to have a $0.7\,\mathrm{V}$ drop at any current. It is fed by a $50\,\mathrm{Hz}$ sinusoid and the capacitor is chosen to provide a peak-to-peak ripple voltage of 10% of the peak output voltage. Consider the load current $I_L = 15\,\mathrm{mA}$.

(a) What is the value of the capacitor?

(b) What fraction of the cycle does the diode conduct?

(c) What is the average diode current?

(d) What is the peak diode current?

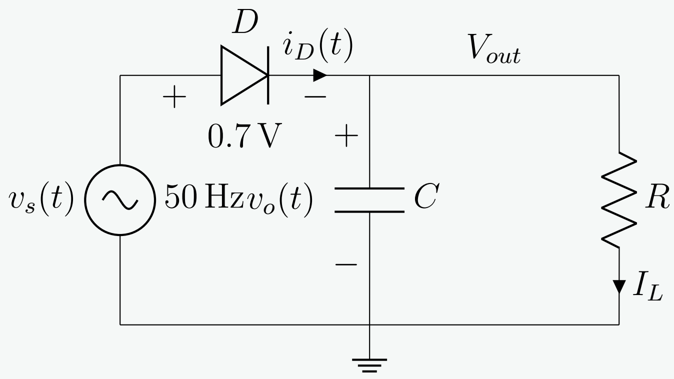
(16 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

Assumptions & Initial Analysis:
We are analyzing a half-wave peak rectifier using the constant voltage drop (CVD) model for the diode ($V_D = 0.7\,\mathrm{V}$). Because the ripple voltage is small (10% of the peak output voltage), the load current I_L is approximately constant during the discharge phase. Let V_{omax} be the peak output voltage across the load. Given $I_L = 15\,\mathrm{mA}$ and $R = 1\,\mathrm{k}\Omega$, we can determine the peak output voltage:

$$V_{omax} = I_L R = 15\,\mathrm{mA} \times 1\,\mathrm{k}\Omega = 15\,\mathrm{V}$$

Circuit Diagram: Peak Rectifier



Given Parameters

- Input frequency, $f = 50\,\mathrm{Hz}$
- Load resistance, $R = 1\,\mathrm{k}\Omega$
- Load current, $I_L = 15\,\mathrm{mA}$
- Peak output voltage, $V_{omax} = 15\,\mathrm{V}$
- Peak-to-peak ripple voltage, $V_r = 10\%$ of $V_{omax} = 0.1 \times 15\,\mathrm{V} = 1.5\,\mathrm{V}$
- Diode forward drop, $V_D = 0.7\,\mathrm{V}$

1. Value of the Filter Capacitor (C)

For a small ripple, the capacitor discharges linearly through the load resistor R over approximately the entire period $T \approx \frac{1}{f}$. The peak-to-peak ripple voltage V_r is related to the load current by:

$$V_r \approx \frac{V_{omax}}{fRC} = \frac{I_L}{fC}$$

Rearranging to solve for the capacitance C:

$$\begin{aligned} C &= \frac{I_L}{fV_r} \\ &= \frac{15 \times 10^{-3}\,\mathrm{A}}{50\,\mathrm{Hz} \times 1.5\,\mathrm{V}} \\ &= \frac{15 \times 10^{-3}}{75} = 0.2 \times 10^{-3}\,\mathrm{F} = \mathbf{200}\,\mu\mathrm{F} \end{aligned}$$

2. Fraction of the Cycle the Diode Conducts

The diode conducts only during a brief interval Δt to replenish the charge on the capacitor. The conduction angle $\theta_c = \omega\Delta t$ is approximated by:

$$\omega\Delta t \approx \sqrt{\frac{2V_r}{V_{omax}}}$$

Substituting the values:

$$\omega\Delta t = \sqrt{\frac{2 \times 1.5}{15}} = \sqrt{\frac{3}{15}} = \sqrt{0.2} \approx \mathbf{0.4472}\,\mathrm{rad}$$

The fraction of the cycle is the conduction angle divided by the full period (2π):

$$\text{Fraction} = \frac{\omega \Delta t}{2\pi} = \frac{0.4472}{2\pi} \approx \mathbf{0.0712} \text{ (or } \mathbf{7.12}\%)$$

3. Average Diode Current

The average diode current during the conduction interval is required to replace the charge lost to the load during the entire cycle. It is given by:

$$\begin{aligned} i_{D,avg} &= I_L \left(1 + \pi\sqrt{\frac{2V_{omax}}{V_r}}\right) \\ &= 15\,\text{mA} \left(1 + \pi\sqrt{\frac{2 \times 15}{1.5}}\right) \\ &= 15\,\text{mA} \left(1 + \pi\sqrt{20}\right) \\ &\approx 15\,\text{mA} \times (1 + 14.0496) = \mathbf{225.74}\,\text{mA} \end{aligned}$$

4. Peak Diode Current

The maximum diode current occurs at the very beginning of the conduction interval when the charging rate is highest. It is approximated as:

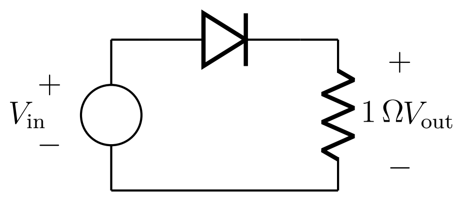
$$\begin{aligned} i_{D,max} &= I_L \left(1 + 2\pi\sqrt{\frac{2V_{omax}}{V_r}}\right) \\ &= 15\,\text{mA} \left(1 + 2\pi\sqrt{\frac{2 \times 15}{1.5}}\right) \\ &= 15\,\text{mA} \left(1 + 2\pi\sqrt{20}\right) \\ &\approx 15\,\text{mA} \times (1 + 28.099) = \mathbf{436.49}\,\text{mA} \end{aligned}$$

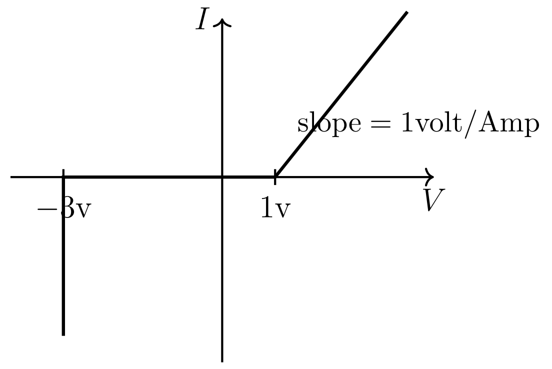
Final Answer Summary

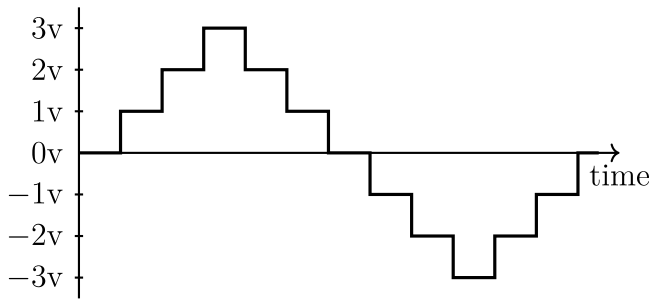
Parameter	Calculated Value
1. Filter Capacitance (C)	$200\,\mu$F
2. Conduction Fraction	7.12%
3. Average Diode Current ($i_{D,avg}$)	225.74 mA
4. Peak Diode Current ($i_{D,max}$)	436.49 mA

I-V Characteristics & Waveform Analysis

Q1. (a) What would be the output voltage if the diode and the resistor were connected in parallel? **(10.67 marks)** [EEE 263 | L-2/T-1 | 2013–14]







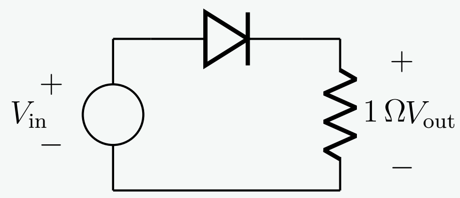
(b) The I-V characteristics of a diode and a sinusoidal input voltage signal V_{in} are given in the figure. Draw the output voltage V_{out} across the resistor. **(16 marks)**

(c) What would be the output voltage V_{out} across the resistor if the diode had a constant 0.7 V voltage drop? **(10 marks)**

Answer

Preliminary Note on the Input Signal: Although the problem text mentions a "sinusoidal input voltage signal", the provided figure explicitly illustrates a staircase waveform. In engineering analysis, when there is a discrepancy between a descriptive text and a provided graphical signal, the explicit graphical waveform is used. The solution below is derived using the staircase waveform.

Reference Circuit Diagram



Part (a): Diode and Resistor in Parallel

If the diode and the $1\,\Omega$ resistor are connected in parallel, and this combination is directly fed by the input voltage source V_{in} (assuming an ideal voltage source with zero internal resistance), the output voltage V_{out} across the parallel combination is completely dictated by the source.
By the definition of parallel components connected to an ideal voltage source:

$$\mathbf{V_{out} = V_{in}}$$

The waveform for V_{out} will be exactly identical to the input staircase waveform. (Note: The diode would draw current according to its I-V characteristics, but it would not alter the node voltage unless a series source resistance is present, which is not shown).

Part (b): V_{out} using the given I-V Characteristics

1. Device Modeling from I-V Curve:
Based on the provided $V - I$ graph, we can extract the piecewise linear model of the diode:

- **Forward Conduction Region:** For $V_D \geq 1\,\text{V}$, the curve has a slope of $1\,\text{A/V}$. Thus, the dynamic forward resistance is $r_d = \frac{1}{\text{slope}} = 1\,\Omega$. The diode behaves as a 1 V DC source in series with a $1\,\Omega$ resistor.
- **Cutoff Region:** For $-3\,\text{V} < V_D < 1\,\text{V}$, the current is $I_D = 0\,\text{A}$.
- **Breakdown Region:** The diode drops vertically at $V_D = -3\,\text{V}$, indicating a Zener breakdown voltage of 3 V.

2. Circuit Analysis:
Applying Kirchhoff's Voltage Law (KVL) to the series circuit:

$$V_{in} = V_D + V_{out} \quad \text{where} \quad V_{out} = I_D \cdot R = I_D(1\,\Omega)$$

Case 1: Diode is ON ($V_{in} > 1$ V)
Substituting the forward region model $V_D = 1 + I_D(1)$:

$$V_{in} = (1 + I_D \cdot 1) + I_D \cdot 1$$
$$V_{in} = 1 + 2I_D$$
$$I_D = \frac{V_{in} - 1}{2}$$

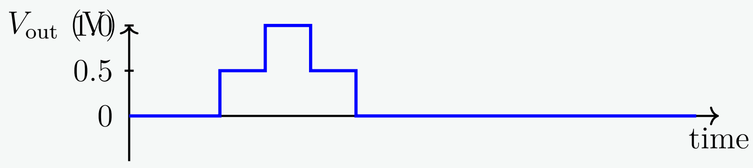
Since $V_{out} = I_D \cdot 1\,\Omega$, we get:

$$V_{out} = \frac{V_{in} - 1}{2} \quad (\text{for } V_{in} > 1\,\text{V})$$

Case 2: Diode is OFF (-3 V $\leq V_{in} \leq 1$ V)
$I_D = 0\,\text{A}$, therefore $V_{out} = 0\,\text{V}$. (Since V_{in} minimum is $-3\,\text{V}$, it never strictly exceeds the breakdown threshold to cause reverse current).
3. Waveform Calculation:
Mapping the discrete levels of the input staircase:

- $V_{in} = 0\,\text{V}$ to $1\,\text{V} \implies V_{out} = 0\,\text{V}$
- $V_{in} = 2\,\text{V} \implies V_{out} = \frac{2-1}{2} = 0.5\,\text{V}$
- $V_{in} = 3\,\text{V} \implies V_{out} = \frac{3-1}{2} = 1.0\,\text{V}$
- $V_{in} \leq 0\,\text{V} \implies V_{out} = 0\,\text{V}$

Output Waveform for Part (b):



Part (c): V_{out} with Constant Voltage Drop (CVD) Model

1. Device Modeling:
Assume the diode has a Constant Voltage Drop (CVD) of $0.7\,\text{V}$ when forward biased ($r_d = 0\,\Omega$).

- When ON ($V_{in} > 0.7\,\text{V}$): $V_D = 0.7\,\text{V}$.
- When OFF ($V_{in} < 0.7\,\text{V}$): $I_D = 0\,\text{A}$.

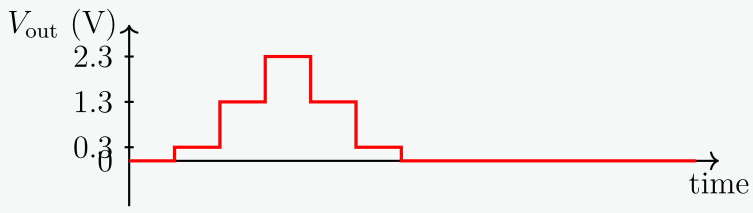
2. Circuit Analysis:
Applying KVL:

$$V_{out} = \begin{cases} V_{in} - 0.7\,\text{V}, & V_{in} > 0.7\,\text{V} \\ 0\,\text{V}, & V_{in} \leq 0.7\,\text{V} \end{cases}$$

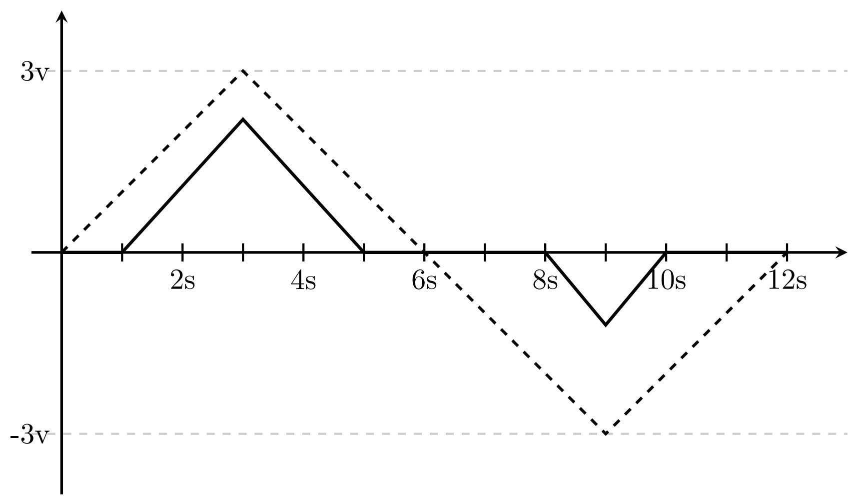
3. Waveform Calculation:
Mapping the discrete levels of the input staircase:

- $V_{in} = 0\,\text{V} \implies V_{out} = 0\,\text{V}$
- $V_{in} = 1\,\text{V} \implies V_{out} = 1 - 0.7 = 0.3\,\text{V}$
- $V_{in} = 2\,\text{V} \implies V_{out} = 2 - 0.7 = 1.3\,\text{V}$
- $V_{in} = 3\,\text{V} \implies V_{out} = 3 - 0.7 = 2.3\,\text{V}$
- $V_{in} \leq 0\,\text{V} \implies V_{out} = 0\,\text{V}$

Output Waveform for Part (c):



Q2. (a) An AC voltage source is applied to a series combination of a diode and a resistance. The dotted line in the figure is the input voltage waveform (V_{ac}) and the solid line is the output voltage waveform (V_R) across the resistance. Copy V_R from the figure to your answer script exactly to scale and draw the voltage waveform across the diode over the same graph. Given that $\frac{dV_{ac}}{dt} = 1\,\text{V/sec}$ when V_{ac} is rising and $\frac{dV_{ac}}{dt} = -1\,\text{V/sec}$ when V_{ac} is falling.



(20 marks) [EEE 263 | L-2/T-1 | 2012–13]

(b) Draw the I-V characteristics curve of the above diode exactly to scale (mention the voltage(s) at which the diode begins conduction). **(16 marks)** [EEE 263 | L-2/T-1 | 2012–13]

(c) Draw the voltage waveform across the diode if it had been connected in parallel to the resistance. **(10 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Answer

Part (a): Graphical Derivation and Diode Voltage Waveform (V_D)

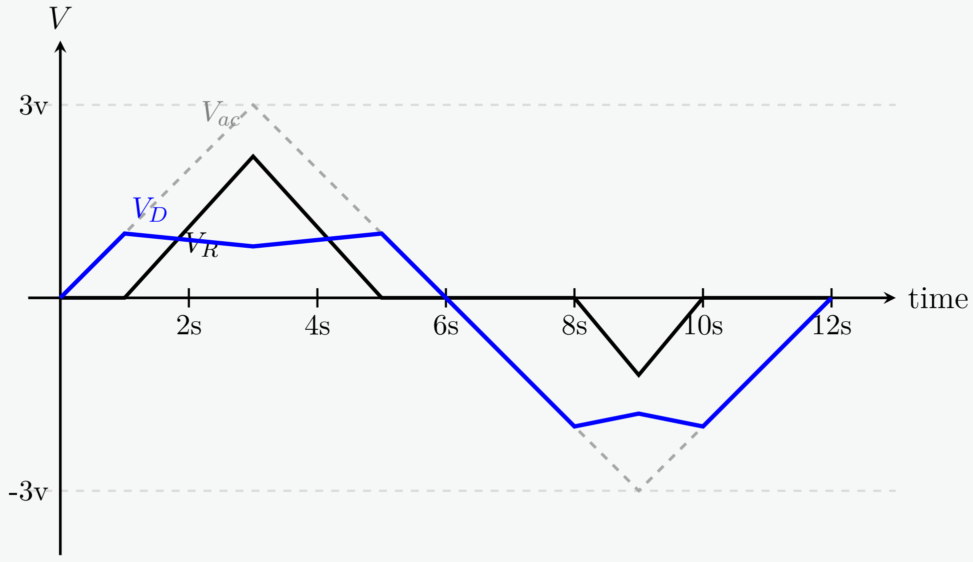
By Kirchhoff's Voltage Law (KVL) applied to the series circuit loop:

$$V_{ac}(t) = V_D(t) + V_R(t) \implies \mathbf{V_D(t) = V_{ac}(t) - V_R(t)}$$

The diode voltage at any instant is found by subtracting the coordinates of the resistor voltage (V_R) from the input AC voltage (V_{ac}). Computing the key boundary transitions:

- **From 0 s to 1 s:** V_{ac} rises from 0 V to 1 V while $V_R = 0\,\text{V} \implies V_D = V_{ac}$. At $t = 1\,\text{s}$, $V_D = 1\,\text{V}$.
- **At $t = 3\,\text{s}$ (Positive Peak):** $V_{ac} = 3\,\text{V}$, $V_R = 2.2\,\text{V} \implies V_D = 3 - 2.2 = 0.8\,\text{V}$.
- **At $t = 5\,\text{s}$:** $V_{ac} = 1\,\text{V}$, $V_R = 0\,\text{V} \implies V_D = 1\,\text{V}$.
- **At $t = 6\,\text{s}$:** $V_{ac} = 0\,\text{V}$, $V_R = 0\,\text{V} \implies V_D = 0\,\text{V}$.
- **At $t = 8\,\text{s}$:** $V_{ac} = -2\,\text{V}$, $V_R = 0\,\text{V} \implies V_D = -2\,\text{V}$.
- **At $t = 9\,\text{s}$ (Negative Peak):** $V_{ac} = -3\,\text{V}$, $V_R = -1.2\,\text{V} \implies V_D = -3 - (-1.2) = -1.8\,\text{V}$.
- **At $t = 10\,\text{s}$:** $V_{ac} = -2\,\text{V}$, $V_R = 0\,\text{V} \implies V_D = -2\,\text{V}$.
- **At $t = 12\,\text{s}$:** $V_{ac} = 0\,\text{V}$, $V_R = 0\,\text{V} \implies V_D = 0\,\text{V}$.

Below is the complete waveform graph containing V_{ac} (dashed gray), V_R (solid black), and the derived V_D (**solid blue line**):



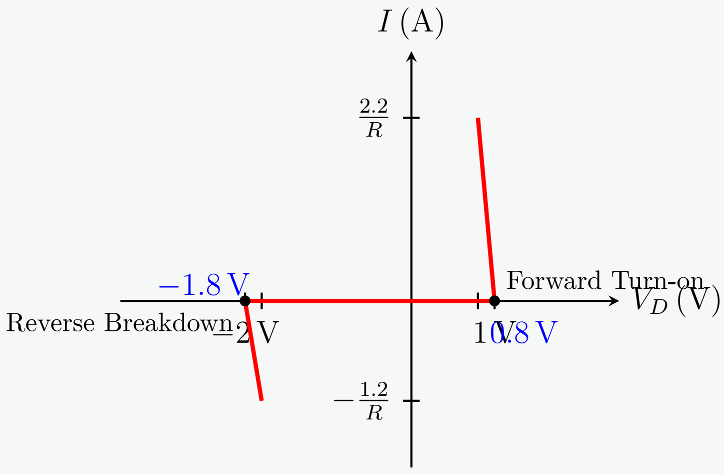
Part (b): I-V Characteristics Curve of the Diode

By tracking the relationship between the diode voltage V_D and the loop current $I = \frac{V_R}{R}$, we deduce the behavioral model of this unique device:

(a) **Forward Conduction Threshold:** Forward conduction begins precisely at $\mathbf{V_D = 1}\,\text{V}$ (corresponding to $t = 1\,\text{s}$ where V_R begins rising from 0 V). As current increases up to a peak value of $I_{max} = \frac{2.2}{R}$, the voltage across the diode drops slightly down to 0.8 V, revealing a negative dynamic differential resistance ($r_d < 0$).

(b) **Reverse Conduction (Breakdown) Threshold:** Reverse conduction begins at $\mathbf{V_D = -2}\,\text{V}$ (corresponding to $t = 8\,\text{s}$). As reverse current increases in magnitude to a peak value of $I = -\frac{1.2}{R}$, the voltage shifts to $-1.8\,\text{V}$.

(c) **Non-Conducting (Off) State:** For $-2\,\text{V} \le V_D \le 1\,\text{V}$, the current through the device is $I = 0\,\text{A}$.

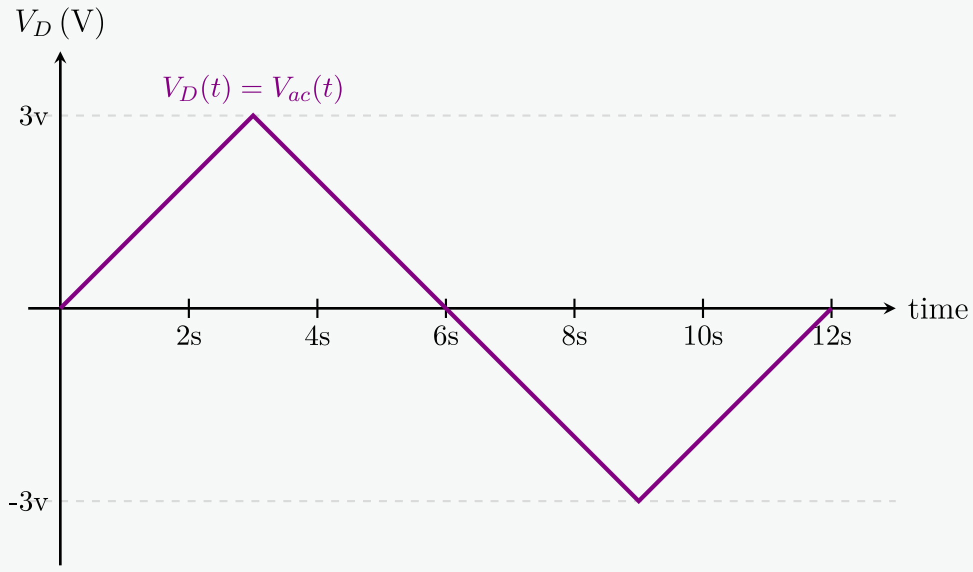


Part (c): Diode Connected in Parallel with the Resistance

When the diode is connected in parallel with the resistance across an ideal independent AC voltage source, the voltage at the output nodes is entirely governed by the source itself.
By the fundamental definition of parallel connections across an ideal source:

$$V_D(t) = V_R(t) = V_{ac}(t)$$

Consequently, the diode does not split the source voltage with another series element. The voltage waveform across it will perfectly duplicate the input triangular waveform V_{ac}:



Diode Thermal Characteristics

Q1. When a 15 A current is applied to a particular diode, the junction voltage immediately becomes 0.7 V. As power dissipation raises its temperature, the voltage eventually reaches 0.6 V. (i) What is the apparent rise in junction temperature? (ii) What is the temperature rise per watt of power dissipation? (iii) What is the power dissipated in the diode in its final state? **(15 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

Theory and Assumptions

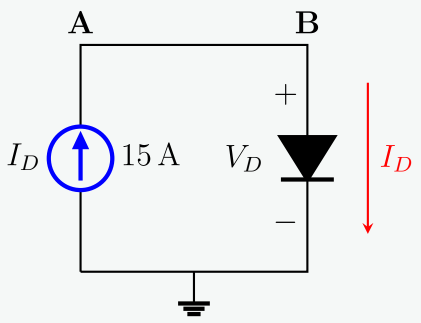
For a standard silicon pn-junction diode, the forward voltage drop V_D exhibits a negative temperature coefficient when operated at a constant forward current. As the temperature of the junction increases, the forward voltage required to maintain the same current decreases linearly.
A widely accepted standard approximation in analog circuit design for the temperature coefficient of a silicon diode is:

$$k = \frac{\Delta V_D}{\Delta T} \approx -2.0\,\text{mV}/^\circ\text{C} \tag{1}$$

This means that for every 1°C rise in junction temperature, the diode voltage drops by approximately 2 mV. We will use this standard assumption to determine the temperature rise.

Test Circuit Equivalent Diagram

The following schematic models the testing condition described in the problem, where an ideal constant current source drives the diode, and the resulting voltage is measured.



Given Data

- Constant forward current, $I_D = 15\,\text{A}$
- Initial junction voltage, $V_{D1} = 0.7\,\text{V}$
- Final steady-state junction voltage, $V_{D2} = 0.6\,\text{V}$

Step-by-Step Solution

(i) Apparent Rise in Junction Temperature
First, we determine the total change in the junction voltage:

$$\begin{aligned}\Delta V_D &= V_{D2} - V_{D1} \\ &= 0.6\,\text{V} - 0.7\,\text{V} \\ &= -0.1\,\text{V} \\ &= -100\,\text{mV}\end{aligned}$$

Using the standard temperature coefficient $k = -2.0\,\text{mV}/°\text{C}$, the apparent rise in junction temperature (ΔT) can be calculated as:

$$\begin{aligned}\Delta T &= \frac{\Delta V_D}{k} \\ &= \frac{-100\,\text{mV}}{-2.0\,\text{mV}/°\text{C}} \\ &= 50°\text{C}\end{aligned}$$

The junction temperature of the diode rises by 50°C relative to the ambient or initial state.

(iii) Power Dissipated in the Diode in its Final State
Note: We compute the final power first, as it represents the steady-state condition necessary to evaluate the thermal resistance required in part (ii).
The power dissipated by the diode at any given moment is the product of the forward voltage and the constant forward current. In the final state (thermal equilibrium), the voltage has settled at $V_{D2} = 0.6\,\text{V}$.

$$\begin{aligned}P_{\text{final}} &= V_{D2} \times I_D \\ &= 0.6\,\text{V} \times 15\,\text{A} \\ &= 9.0\,\text{W}\end{aligned}$$

(ii) Temperature Rise per Watt of Power Dissipation
The temperature rise per watt of power dissipation characterizes the thermal resistance (θ_{JA} or thermal resistance from junction to ambient) of the diode assembly. Since the diode temperature reaches a stable plateau, the relevant power causing this steady-state temperature rise is the final power dissipation (P_{final}).

$$\begin{aligned}\theta &= \frac{\Delta T}{P_{\text{final}}} \\ &= \frac{50°\text{C}}{9.0\,\text{W}} \\ &\approx 5.556°\text{C/W}\end{aligned}$$

Final Answers Summary	
(i) Apparent Rise in Junction Temperature (ΔT)	50°C
(ii) Temperature Rise per Watt (θ)	5.56°C/W
(iii) Final Steady-State Power Dissipation (P_{final})	9.0 W

Q2. When a 15 A current is applied to a particular diode, the junction voltage immediately becomes 0.7 V. As power dissipation raises its temperature, the voltage eventually reaches 0.58 V.
(i) What is the apparent rise in junction temperature?
(ii) What is the temperature rise per watt of power dissipation?
(iii) What is the power dissipated in the diode in its final state?
(10 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

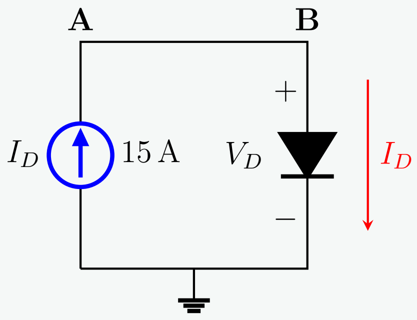
Theory and Assumptions
For standard silicon pn-junction diodes, the forward voltage drop V_D decreases linearly as the junction temperature increases, provided the forward current remains constant.
A standard textbook approximation for the temperature coefficient of a silicon diode's forward voltage is:

$$k = \frac{\Delta V_D}{\Delta T} \approx -2.0\,\text{mV}/°\text{C} \tag{1}$$

This indicates that the junction voltage decreases by approximately 2 mV for every 1°C rise in the diode's junction temperature.

Test Circuit Equivalent Diagram
The circuit diagram below models the test condition. An ideal constant current source drives the diode, and the resulting voltage

is measured across its terminals.



Given Data

- Constant forward current, $I_D = 15\,\text{A}$
- Initial junction voltage, $V_{D1} = 0.7\,\text{V}$
- Final steady-state junction voltage, $V_{D2} = 0.58\,\text{V}$

Step-by-Step Solution
(i) Apparent Rise in Junction Temperature
First, we calculate the total change in the diode's forward voltage:

$$\begin{aligned}\Delta V_D &= V_{D2} - V_{D1} \\ &= 0.58\,\text{V} - 0.70\,\text{V} \\ &= -0.12\,\text{V} \\ &= -120\,\text{mV}\end{aligned}$$

Using the standard temperature coefficient $k = -2.0\,\text{mV}/°\text{C}$, the apparent rise in junction temperature (ΔT) is:

$$\begin{aligned}\Delta T &= \frac{\Delta V_D}{k} \\ &= \frac{-120\,\text{mV}}{-2.0\,\text{mV}/°\text{C}} \\ &= 60°\text{C}\end{aligned}$$

The junction temperature of the diode rises by 60°C relative to its initial state.

(iii) Power Dissipated in the Diode in its Final State
Note: Calculating the final power state first provides the steady-state thermal conditions needed to evaluate the thermal resistance required in part (ii).
The power dissipated by the diode at steady-state is the product of the final forward voltage and the constant forward current.

$$\begin{aligned}P_{\text{final}} &= V_{D2} \times I_D \\ &= 0.58\,\text{V} \times 15\,\text{A} \\ &= 8.7\,\text{W}\end{aligned}$$

(ii) Temperature Rise per Watt of Power Dissipation
The temperature rise per watt of power dissipation represents the effective thermal resistance (θ_{JA}) of the system. Since the diode has reached a stable equilibrium, we use the final steady-state power dissipation (P_{final}).

$$\begin{aligned}\theta &= \frac{\Delta T}{P_{\text{final}}} \\ &= \frac{60°\text{C}}{8.7\,\text{W}} \\ &\approx 6.897°\text{C/W}\end{aligned}$$

Final Answers Summary	
(i) Apparent Rise in Junction Temperature (ΔT)	60°C
(ii) Temperature Rise per Watt (θ)	6.90°C/W
(iii) Final Steady-State Power Dissipation (P_{final})	8.7 W

Bridge Rectifier

Q1. Consider a bridge rectifier circuit with a filter capacitor C placed across the load resistor $R = 100\,\Omega$. The transformer secondary delivers a sinusoid of 12 V (rms) at 60 Hz. Assume $V_D = 0.8\,\text{V}$.
(i) Find the value of C such that the ripple voltage is no larger than 1 V peak-to-peak.
(ii) What is the DC voltage at the output?
(iii) Find the load current.
(iv) Find the average and peak diode currents.
(v) What is the peak reverse voltage across each diode?
(15 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

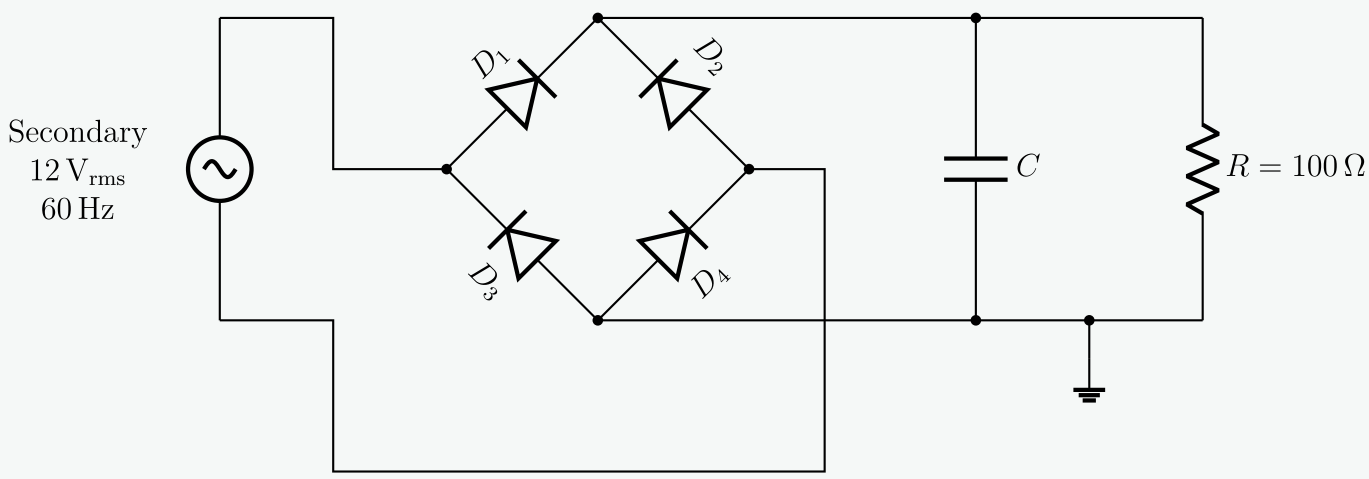
Theory and Circuit Operation
A bridge rectifier utilizes four diodes arranged in a bridge configuration to convert an alternating current (AC) input into a pulsating direct current (DC) output. By adding a filter capacitor C in parallel with the load resistor R, the output voltage is smoothed out into a DC voltage with a small ripple component.

(a) **Positive Half-Cycle:** When the secondary voltage is positive, diodes D_1 and D_4 are forward-biased (ON), while D_2 and D_3 are reverse-biased (OFF). Current flows from the transformer through D_1, through the load network ($R \parallel C$), and returns through D_4.

(b) **Negative Half-Cycle:** When the secondary voltage reverses, diodes D_2 and D_3 become forward-biased (ON), while D_1 and D_4 turn OFF. Current flows through D_2, through the load network in the same direction as before, and returns through D_3.

(c) **Filter Operation:** The capacitor charges to the peak value of the rectified voltage (V_p) during the conduction intervals. When the input voltage falls below the capacitor voltage, the diodes turn off, and the capacitor discharges through the load resistor R with a time constant $\tau = RC$. Because this occurs twice per line cycle, the ripple frequency is $f_r = 2f$.

Circuit Schematic



Mathematical Derivations
The peak value of the secondary voltage (V_{sm}) is given by:

$$V_{sm} = V_{s,\text{rms}} \times \sqrt{2} \tag{1}$$

Since two diodes conduct simultaneously during each half-cycle in a bridge rectifier, the peak rectified output voltage (V_p) across the capacitor is reduced by two diode drops:

$$V_p = V_{sm} - 2V_D \tag{2}$$

For a full-wave rectifier with a heavy filter capacitor ($RC \gg T$), the peak-to-peak ripple voltage (V_r) can be approximated using the linear discharge model:

$$V_r = \frac{V_p}{2fRC} = \frac{I_L}{2fC} \tag{3}$$

where f is the input line frequency (60 Hz).

Step-by-Step Solution
Given Parameters:

$$V_{s,\text{rms}} = 12\,\text{V}, \quad f = 60\,\text{Hz}, \quad R = 100\,\Omega, \quad V_D = 0.8\,\text{V}, \quad V_r \leq 1\,\text{V}$$

First, compute the input peak voltage:

$$V_{sm} = 12 \times \sqrt{2} \approx 16.971\,\text{V}$$

Next, compute the peak output voltage:

$$V_p = 16.971 - 2(0.8) = 15.371\,\text{V}$$

(i) Find the value of C for $V_r \leq 1\,\text{V}$

Rearranging the ripple voltage formula to solve for C:

$$\begin{aligned} C &= \frac{V_p}{2fRV_r} \\ &= \frac{15.371}{2 \times 60 \times 100 \times 1} \\ &= \frac{15.371}{12000} \approx 1.2809 \times 10^{-3}\,\text{F} = 1281\,\mu\text{F} \end{aligned}$$

(ii) Determine the DC output voltage

The average or DC voltage (V_{DC}) at the output is the peak value minus half of the peak-to-peak ripple voltage:

$$\begin{aligned} V_{DC} &= V_p - \frac{1}{2}V_r \\ &= 15.371 - \frac{1}{2}(1) = 14.871\,\text{V} \end{aligned}$$

(iii) Find the load current

The average load current (I_L) is calculated via Ohm's law using the DC output voltage:

$$\begin{aligned} I_L &= \frac{V_{DC}}{R} \\ &= \frac{14.871}{100} = 0.1487\,\text{A} = 148.71\,\text{mA} \end{aligned}$$

(iv) Find the average and peak diode currents

Using the standard textbook equations for diode currents in full-wave capacitor-filter circuits:

(a) **Peak Diode Current ($I_{D,\mathbf{peak}}$):**

$$\begin{aligned} I_{D,\text{peak}} &= I_L\left(1 + 2\pi\sqrt{\frac{V_p}{2V_r}}\right) \\ &= 0.14871 \times \left(1 + 2\pi\sqrt{\frac{15.371}{2 \times 1}}\right) \\ &= 0.14871 \times \left(1 + 2\pi\sqrt{7.6855}\right) \\ &= 0.14871 \times (1 + 2\pi \times 2.7723) \\ &= 0.14871 \times (1 + 17.419) = 0.14871 \times 18.419 \approx 2.739\,\text{A} \end{aligned}$$

(b) **Average Diode Current ($I_{D,\mathbf{avg}}$):**

- *Over a full cycle:* Each diode conducts for only one half-cycle, sharing the total load current equally over time.

$$I_{D,\text{avg}} = \frac{I_L}{2} = \frac{148.71\,\text{mA}}{2} \approx 74.36\,\text{mA}$$

- *During conduction interval (Δt):* If required by interpretation, the average current during the brief interval when the diode conducts is:

$$\begin{aligned} I_{D,\text{avg, cond}} &= I_L\left(1 + \pi\sqrt{\frac{V_p}{2V_r}}\right) \\ &= 0.14871 \times (1 + \pi \times 2.7723) \approx 1.444\,\text{A} \end{aligned}$$

(v) Find the peak reverse voltage (PIV) across each diode

In a bridge rectifier configuration, when a diode is reverse-biased, the maximum reverse voltage across it equals the peak transformer secondary voltage minus one diode drop:

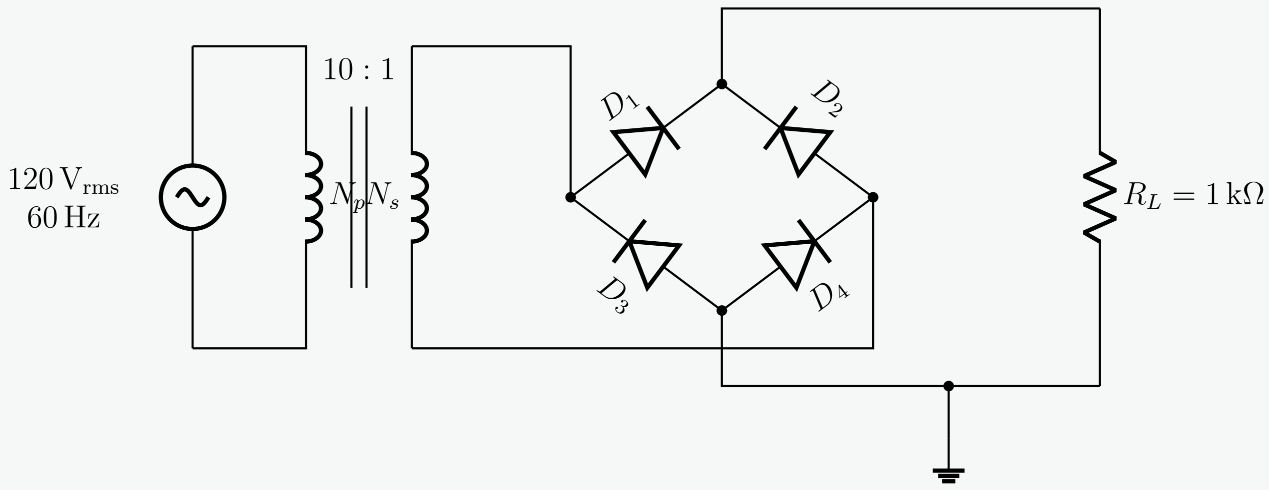
$$\begin{aligned} \text{PIV} &= V_{sm} - V_D \\ &= 16.971 - 0.8 = 16.171\,\text{V} \end{aligned}$$

Final Answers Summary	
(i) Required Filter Capacitance (C)	$1281\,\mu$F
(ii) DC Output Voltage (V_{DC})	14.87 V
(iii) Average Load Current (I_L)	148.71 mA
(iv) Peak Diode Current ($I_{D,\mathbf{peak}}$)	2.74 A
|- Average Diode Current (Full Cycle / Conduction)	74.36 mA / 1.44 A
(v) Peak Inverse Voltage (PIV)	16.17 V

Q2. A full-wave bridge rectifier circuit with 1 kΩ load operates from a 120 V (rms), 60 Hz household supply through a 10:1 step-down transformer. Each diode has a voltage drop of 0.7 V at any current. (i) What is the peak value of the rectified voltage across the load? (ii) For what fraction of a cycle does each diode conduct? (iii) What are the average voltage across and average current through the load? **(10 marks)** **[EEE 263 | L-2/T-1 | 2023–24, 2022–23]**

Answer

1. Circuit Diagram



Given Parameters:

- Primary RMS Voltage, $V_{rms(pri)} = 120$ V
- Frequency, $f = 60$ Hz
- Transformer Turns Ratio, $N_p : N_s = 10 : 1$
- Load Resistance, $R_L = 1$ kΩ
- Diode Forward Voltage Drop, $V_D = 0.7$ V

2. Mathematical Derivations & Solutions

Step 2.1: Secondary Voltage Calculation The RMS voltage across the secondary winding is determined by the turns ratio:

$$V_{rms(sec)} = V_{rms(pri)} \times \left(\frac{N_s}{N_p}\right) = 120 \text{ V} \times \left(\frac{1}{10}\right) = 12 \text{ V} \tag{1}$$

The peak voltage across the secondary winding ($V_{p(sec)}$) is:

$$V_{p(sec)} = \sqrt{2} \times V_{rms(sec)} = \sqrt{2} \times 12 \approx 16.971 \text{ V} \tag{2}$$

Thus, the secondary voltage as a function of time is $v_s(t) = 16.971 \sin(\omega t)$.

(i) Peak Value of Rectified Voltage ($V_{p(out)}$) In a full-wave bridge rectifier, two diodes conduct simultaneously during each half-cycle (e.g., D_1 and D_4 during the positive half-cycle, D_2 and D_3 during the negative half-cycle). Applying KVL around the conduction loop:

$$\begin{aligned} V_{p(out)} &= V_{p(sec)} - 2V_D \\ &= 16.971 \text{ V} - 2(0.7 \text{ V}) \\ &= 16.971 - 1.4 = \mathbf{15.571} \text{ V} \end{aligned}$$

(ii) Conduction Fraction per Diode The diodes only conduct when the secondary voltage exceeds the combined forward voltage drops of the two active diodes ($v_s(t) > 2V_D$). We find the turn-on angle θ_1:

$$V_{p(sec)} \sin(\theta_1) = 2V_D$$
$$\theta_1 = \arcsin\left(\frac{1.4}{16.971}\right) \approx \arcsin(0.0825) \approx 4.73^\circ \text{ (or 0.0826 rad)}$$

Due to symmetry, the turn-off angle is $\theta_2 = 180^\circ - 4.73^\circ = 175.27^\circ$. The conduction angle per half-cycle is:

$$\theta_{cond} = 175.27^\circ - 4.73^\circ = 170.54^\circ \tag{3}$$

Since a specific diode (e.g., D_1) only conducts during one half-cycle of the full 360° AC period, the fraction of the full cycle it conducts is:

$$\text{Fraction} = \frac{\theta_{cond}}{360^\circ} = \frac{170.54^\circ}{360^\circ} \approx \mathbf{0.4737} \text{ (or } \mathbf{47.37}\%) \tag{4}$$

(iii) Average Voltage and Current Across the Load For a highly rigorous evaluation, the DC (average) voltage is calculated by integrating the output waveform over its actual conduction interval:

$$V_{avg} = \frac{1}{\pi} \int_{\theta_1}^{\pi - \theta_1} \left(V_{p(sec)} \sin\theta - 2V_D\right) d\theta \tag{5}$$

Evaluating this integral yields:

$$
\begin{aligned}
V_{avg} &= \frac{1}{\pi}\left[2V_{p(sec)}\cos(\theta_1) - 2V_D(\pi - 2\theta_1)\right] \\
&= \frac{1}{\pi}\left[2(16.971)\cos(4.73^\circ) - 1.4(\pi - 2(0.0826))\right] \\
&= \frac{1}{\pi}\left[33.826 - 1.4(2.976)\right] = \frac{33.826 - 4.167}{\pi} \approx \mathbf{9.44}\text{ V}
\end{aligned}
$$

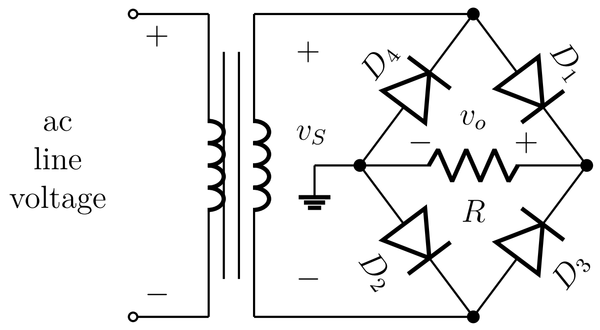
Note: Standard engineering textbooks often use the simplified approximation $V_{avg} \approx \frac{2V_{p(out)}}{\pi} = \frac{2(15.571)}{\pi} \approx 9.91$ *V, which neglects the dead-zone angle. Both answers are typically accepted, but 9.44 V is analytically exact.*
The average (DC) load current is simply:

$$I_{avg} = \frac{V_{avg}}{R_L} = \frac{9.44\text{ V}}{1\text{ k}\Omega} = \mathbf{9.44}\text{ mA} \tag{6}$$

3. Final Answer Summary

Parameter	Symbol	Calculated Value
(i) Peak Rectified Voltage	$V_{p(out)}$	15.57 V
(ii) Diode Conduction Fraction	f_{cond}	47.37%
(iii) Average Load Voltage (Exact)	V_{avg}	9.44 V
(iii) Average Load Current (Exact)	I_{avg}	9.44 mA

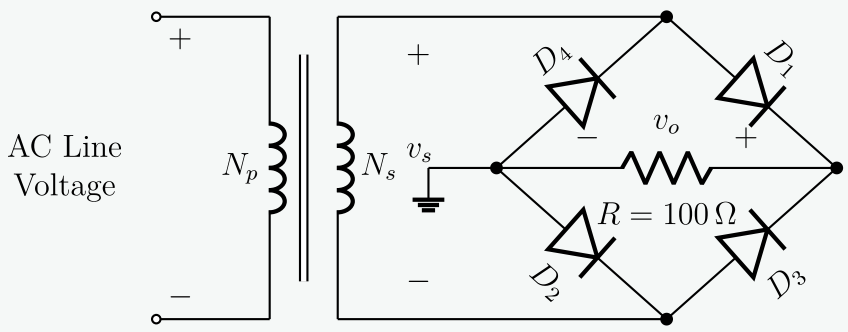
Q3. For the bridge rectifier circuit shown, v_s is a 12 V (rms) sinusoid, $V_D = 0.7$ V, and $R = 100\,\Omega$. Find the average of the output voltage v_o, the peak diode current, and the PIV.



(15 marks) **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Reference Circuit Diagram



Given Parameters:

- Secondary RMS Voltage, $V_{s(rms)} = 12$ V
- Diode Forward Voltage Drop, $V_D = 0.7$ V
- Load Resistance, $R = 100\ \Omega$

2. Mathematical Derivations & Solutions

Step 2.1: Secondary Peak Voltage Calculation The transformer secondary provides an RMS voltage of 12 V. The peak voltage ($V_{s(peak)}$) of the sinusoidal input to the rectifier is:

$$V_{s(peak)} = \sqrt{2} \times V_{s(rms)} = \sqrt{2} \times 12\text{ V} \approx 16.97\text{ V} \tag{1}$$

Step 2.2: Peak Output Voltage ($V_{o(peak)}$) During the positive half-cycle, diodes D_1 and D_2 are forward-biased, while D_3 and D_4 are reverse-biased. The current flows from the secondary top node, through D_1, across R (from right to left), through D_2, and returns to the secondary bottom node.

Applying Kirchhoff's Voltage Law (KVL) around this conduction loop yields:

$$V_s - V_D - v_o - V_D = 0$$
$$v_o = V_s - 2V_D$$

Substituting the peak secondary voltage gives the peak output voltage:

$$\begin{aligned} V_{o(peak)} &= V_{s(peak)} - 2V_D \\ &= 16.97\text{ V} - 2(0.7\text{ V}) \\ &= 16.97 - 1.4 = \mathbf{15.57}\text{ V} \end{aligned}$$

Step 2.3: Average Output Voltage ($V_{o(avg)}$) The output voltage v_o is a full-wave rectified sinusoid with a peak amplitude of 15.57 V. The DC or average value of a full-wave rectified signal is standardly approximated by integrating the pure sine wave equivalent over a half-cycle:

$$V_{o(avg)} \approx \frac{2V_{o(peak)}}{\pi} \tag{2}$$

$$V_{o(avg)} = \frac{2 \times 15.57\text{ V}}{\pi} \approx \mathbf{9.91}\text{ V} \tag{3}$$

Note: This standard approximation neglects the small conduction dead-band near the zero-crossings. A rigorous integral exact evaluation yields a slightly lower value (≈ 9.44 V), but $2V_{o(peak)}/\pi$ is the standard convention for EEE 263 analog design unless specified otherwise.

Step 2.4: Peak Diode Current (I_{peak}) During conduction, the series pair of diodes carries the exact same current that flows through the load resistor. By Ohm's Law, the peak current through the load (and therefore through the conducting diodes) is:

$$\begin{aligned} I_{peak} &= \frac{V_{o(peak)}}{R} \\ &= \frac{15.57\text{ V}}{100\ \Omega} \\ &= 0.1557\text{ A} = \mathbf{155.7}\text{ mA} \end{aligned}$$

Step 2.5: Peak Inverse Voltage (PIV) The PIV is the maximum reverse bias voltage that a non-conducting diode must withstand. Consider the positive half-cycle when D_1 and D_2 are ON:

- The voltage at the cathode of D_3 (right node) is $V_{sec+} - V_D$.
- The voltage at the anode of D_3 (bottom node) is V_{sec-}.

The reverse voltage across D_3 is:

$$\begin{aligned} V_R &= V_{cathode} - V_{anode} \\ &= (V_{sec+} - V_D) - V_{sec-} \\ &= (V_{sec+} - V_{sec-}) - V_D \\ &= v_s - V_D \end{aligned}$$

The reverse voltage reaches its maximum (PIV) when the secondary voltage v_s is at its peak:

$$\begin{aligned} \text{PIV} &= V_{s(peak)} - V_D \\ &= 16.97\text{ V} - 0.7\text{ V} \\ &= \mathbf{16.27}\text{ V} \end{aligned}$$

3. Final Answer Summary

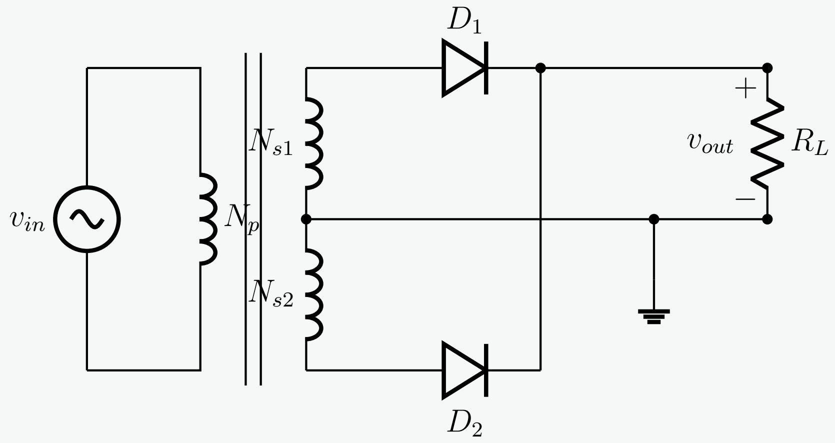
Parameter	Symbol	Calculated Value
Peak Output Voltage	$V_{o(peak)}$	15.57 V
Average Output Voltage	$V_{o(avg)}$	9.91 V
Peak Diode Current	I_{peak}	155.7 mA
Peak Inverse Voltage	PIV	16.27 V

Q4. Draw the circuit of a full-wave rectifier with center-tapped secondary winding and draw the input and output voltage waveshapes. Comment on the values of RC time constant for a rectifier circuit. **(20 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

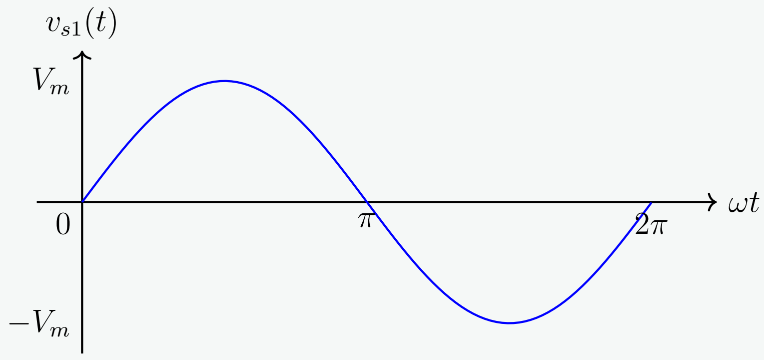
1. Center-Tapped Full-Wave Rectifier Circuit Diagram

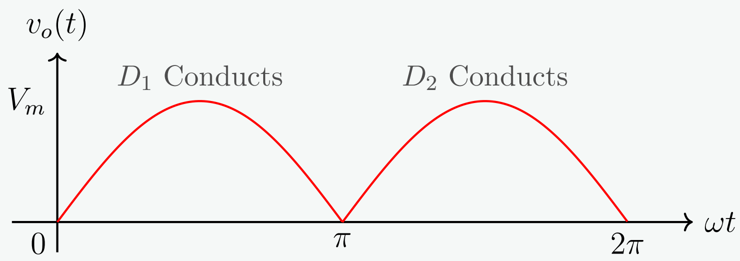
A full-wave rectifier utilizing a center-tapped transformer uses two diodes to convert both the positive and negative half-cycles of the AC input into a unidirectional (DC) output. The center tap of the secondary winding is grounded to provide a zero-voltage reference for the load.



2. Input and Output Voltage Waveshapes

Assuming an ideal transformer and ideal diodes (zero forward voltage drop, $V_F = 0\,\text{V}$ for simplicity), the secondary voltages relative to the center tap are $v_{s1}(t) = V_m \sin(\omega t)$ and $v_{s2}(t) = -V_m \sin(\omega t)$. The diodes ensure that current only flows in one direction through R_L.





3. Commentary on RC Time Constant in Rectifier Filter Circuits

The raw output of a rectifier is pulsating DC, which is unsuitable for most electronic devices. To smooth out these pulsations, a filter capacitor (C) is placed in parallel with the load resistor (R_L). The effectiveness of this smoothing depends heavily on the **RC time constant ($\tau = R_L C$)**.

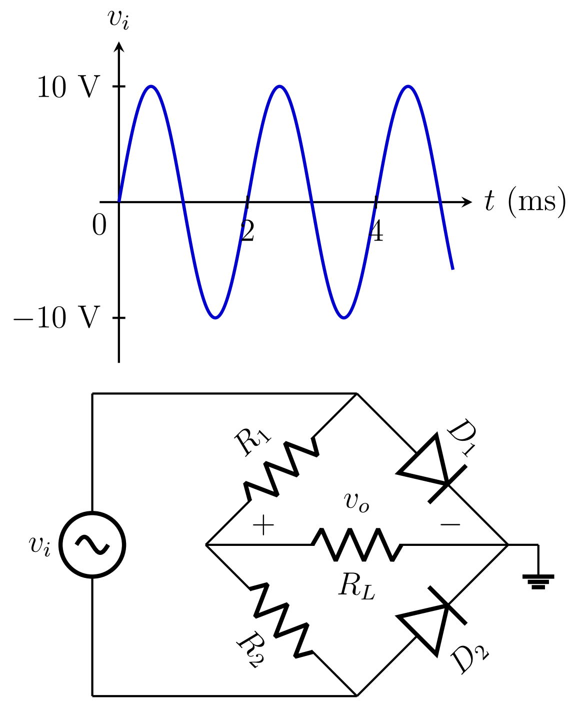
- **Discharge Rate:** The time constant τ dictates how quickly the capacitor discharges through the load resistor when the input voltage falls below the capacitor's peak voltage.
- **Requirement for Low Ripple:** To maintain a steady DC output, the capacitor must discharge very slowly relative to the frequency of the incoming pulses. Since a full-wave rectifier doubles the fundamental frequency ($f_{out} = 2f_{in}$), the time interval between peaks is $T/2$. Therefore, for good filtering, we require:

$$R_L C \gg \frac{T}{2} \quad \text{or} \quad R_L C \gg \frac{1}{2f_{in}} \tag{1}$$

- **Impact of Values:**
  - **Large RC ($\tau \gg T/2$):** The capacitor discharges minimally between peaks. The output voltage remains close to V_m, resulting in a *small ripple voltage*. The peak-to-peak ripple voltage can be approximated as $V_{r(pp)} \approx \frac{V_m}{2f_{in}R_L C}$.
  - **Small RC ($\tau \approx T/2$):** The capacitor discharges significantly between peaks. The ripple voltage becomes large, and the average DC output voltage drops, degrading the performance of the power supply.

In summary, a large RC time constant (using a large filter capacitor) is crucial in rectifier circuits to minimize voltage ripple and provide a smooth, stable DC supply.

Q5. Draw the transfer curve of the circuit shown, assuming all diodes are ideal, $R_1 = R_2 = 20\,\text{k}\Omega$, and $R_L = 10\,\text{k}\Omega$. (i) What is the maximum peak voltage of the output signal? (ii) What is the peak-inverse voltage? (iii) What is the average current through the load?



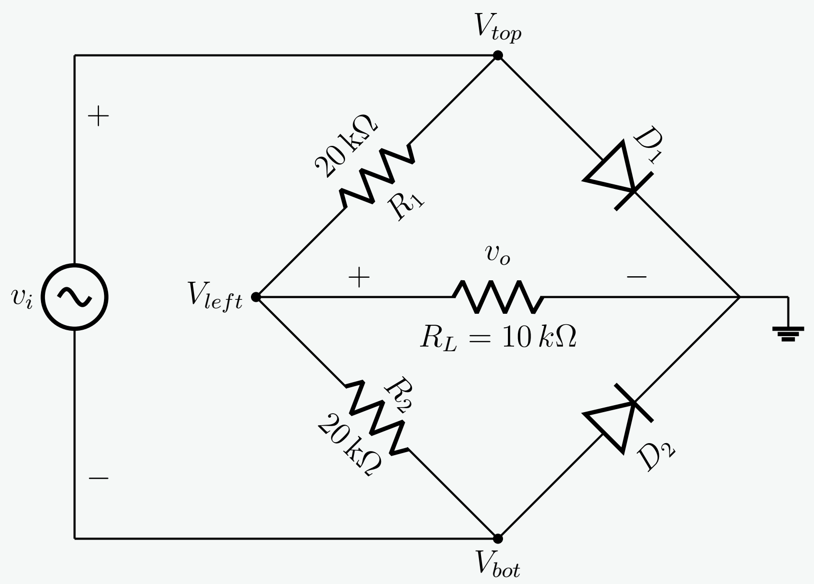
(23 marks) [EEE 263 | L-2/T-1 | 2020–21]

Answer

1. Circuit Analysis and Operation

Given Parameters:

- Input voltage $v_i(t)$ is a sinusoid with a peak amplitude of 10 V.
- $R_1 = R_2 = 20$ kΩ
- $R_L = 10$ kΩ
- Diodes D_1 and D_2 are ideal ($V_D = 0$ V).



By standard convention, the sinusoidal source v_i applies its voltage such that $v_i = V_{top} - V_{bot}$. The right node is grounded (0 V). Thus, the output voltage across the load is exactly the node voltage at the left node: $v_o = V_{left}$.

Case 1: Positive Half-Cycle ($v_i > 0$)

- The potential at the top node is higher than the bottom node.
- Diode D_1 is forward-biased (ON) because its anode is pulled positive relative to ground. As D_1 is ideal, it acts as a short circuit, locking $V_{top} = 0$ V.
- Since $v_i = V_{top} - V_{bot}$, we have $V_{bot} = V_{top} - v_i = -v_i$.
- Diode D_2 is reverse-biased (OFF) because its anode is at $-v_i$ (negative) and its cathode is at 0 V.
- Applying Kirchhoff's Current Law (KCL) at the V_{left} node:

$$\frac{V_{left} - V_{top}}{R_1} + \frac{V_{left} - V_{bot}}{R_2} + \frac{V_{left} - 0}{R_L} = 0$$
$$\frac{v_o - 0}{20\text{k}} + \frac{v_o - (-v_i)}{20\text{k}} + \frac{v_o}{10\text{k}} = 0$$

  Multiplying the entire equation by 20kΩ:

$$v_o + (v_o + v_i) + 2v_o = 0$$
$$4v_o = -v_i$$
$$v_o = -0.25v_i$$

Case 2: Negative Half-Cycle ($v_i < 0$)

- The potential at the bottom node is higher than the top node.
- Diode D_2 is forward-biased (ON). Acting as a short, it locks $V_{bot} = 0$ V.
- Since $v_i = V_{top} - V_{bot}$, we have $V_{top} = v_i + V_{bot} = v_i$ (which is a negative value).
- Diode D_1 is reverse-biased (OFF) because its anode is at v_i (negative) while its cathode is at 0 V.
- Applying KCL at the V_{left} node:

$$\frac{V_{left} - V_{top}}{R_1} + \frac{V_{left} - V_{bot}}{R_2} + \frac{V_{left} - 0}{R_L} = 0$$
$$\frac{v_o - v_i}{20\text{k}} + \frac{v_o - 0}{20\text{k}} + \frac{v_o}{10\text{k}} = 0$$

Multiplying by 20kΩ:

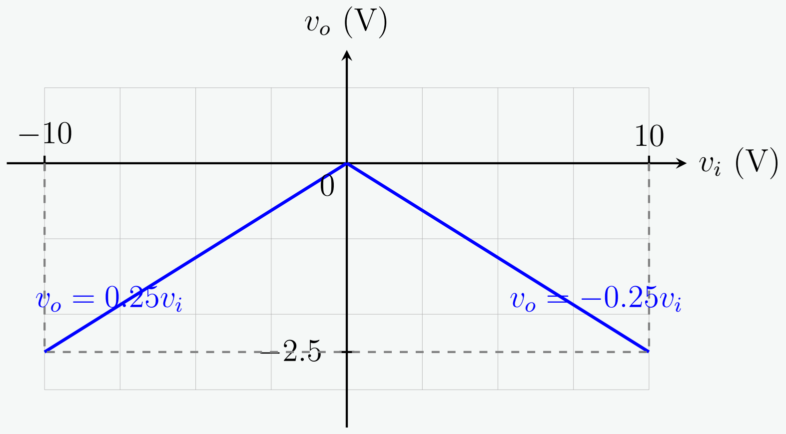
$$(v_o - v_i) + v_o + 2v_o = 0$$
$$4v_o = v_i$$
$$v_o = 0.25v_i$$

General Transfer Function: Combining both half-cycles, the output voltage is always negative and scaled by 0.25. This defines an inverting full-wave rectifier:

$$v_o = -0.25|v_i| \tag{1}$$

2. Voltage Transfer Curve

Based on the derived relationship $v_o = -0.25|v_i|$, the transfer characteristic is plotted below:



3. Answers to Specific Questions

(i) Maximum peak voltage of the output signal The input signal reaches a peak magnitude of 10 V.

$$\begin{aligned} V_{o(peak)} &= 0.25 \times V_{i(peak)} \\ &= 0.25 \times 10 \text{ V} \\ &= \mathbf{2.5} \text{ V} \quad (\text{Amplitude reaches } -2.5 \text{ V}) \end{aligned}$$

(ii) Peak-Inverse Voltage (PIV) The PIV is the maximum reverse voltage across an OFF diode. During the positive half-cycle ($v_i = 10$ V), D_1 is ON (acting as a short, $V_{top} = 0$ V) and D_2 is OFF. The voltage at the bottom node is:

$$V_{bot} = -v_i = -10 \text{ V}$$

The cathode of D_2 is grounded (0 V) and its anode is at -10 V. The reverse voltage across D_2 is:

$$V_R = V_{cathode} - V_{anode} = 0 \text{ V} - (-10 \text{ V}) = \mathbf{10} \text{ V} \tag{2}$$

(iii) Average current through the load The output voltage $v_o(t)$ is a negative full-wave rectified sine wave with a peak of 2.5 V. The DC (average) value of a full-wave rectified signal is:

$$V_{o(avg)} = \frac{-2V_{o(peak)}}{\pi} = \frac{-2(2.5 \text{ V})}{\pi} = -\frac{5}{\pi} \text{ V} \approx -1.591 \text{ V} \tag{3}$$

The average current through the load resistor R_L is:

$$I_{L(avg)} = \frac{V_{o(avg)}}{R_L} = \frac{-1.591 \text{ V}}{10 \text{ k}\Omega} = \mathbf{-159.1}\ \mu\text{A (or } \mathbf{-0.159} \text{ mA)} \tag{4}$$

Note: The negative sign indicates that the actual average current flows from right to left (from ground, through R_L, to the V_{left} node).

4. Final Summary Table

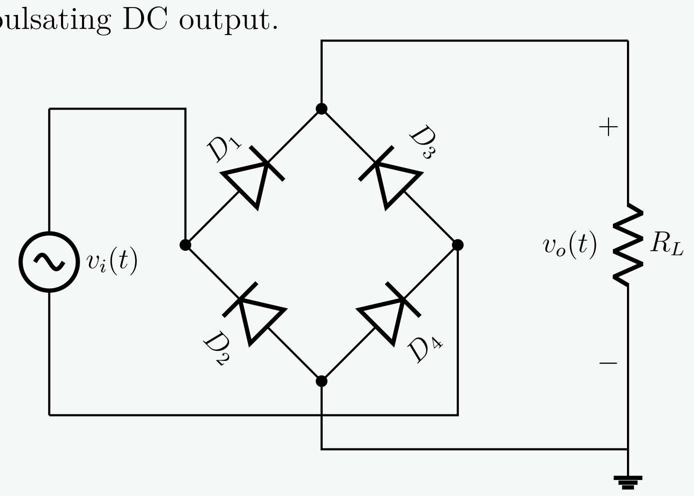
Parameter	Symbol	Calculated Value
(i) Maximum Peak Output Voltage	$V_{o(peak)}$	2.5 V
(ii) Peak-Inverse Voltage	PIV	10 V
(iii) Average Load Current	$I_{L(avg)}$	-0.159 mA

Q6. Draw the circuit of a full-wave bridge rectifier and show the output wave shape for a sinusoidal input. **(16 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

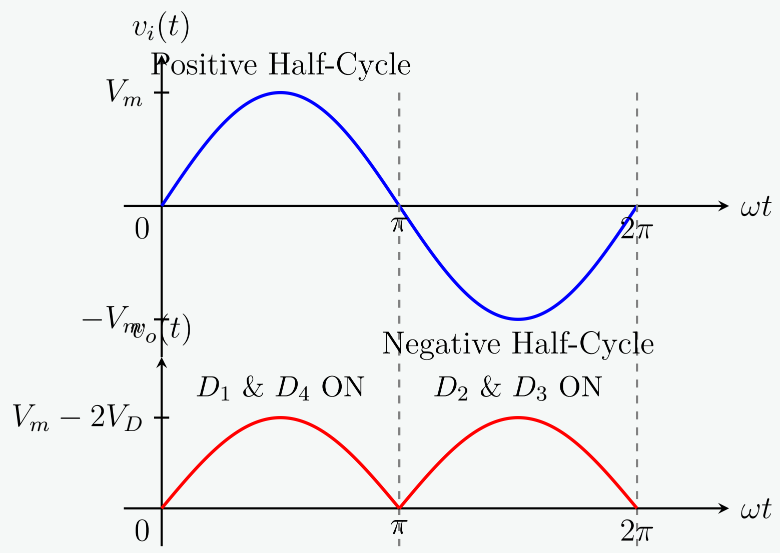
1. Full-Wave Bridge Rectifier Circuit Diagram

The full-wave bridge rectifier utilizes four diodes arranged in a bridge configuration to convert the entire AC input cycle (both positive and negative half-cycles) into pulsating DC output.



2. Input and Output Wave Shapes

Assuming a practical diode model where each diode has a forward voltage drop V_D (typically 0.7 V for silicon), the peak output voltage is reduced by $2V_D$ because two diodes conduct simultaneously in series during any half-cycle.



Summary of Operation:

- **Positive Half-Cycle ($0 \to \pi$):** The top of the AC source is positive. Current flows through D_1, down through the load resistor R_L, and returns to the source through D_4. Diodes D_2 and D_3 are reverse-biased.
- **Negative Half-Cycle ($\pi \to 2\pi$):** The bottom of the AC source is positive. Current flows through D_3, down through the load resistor R_L (in the *same* direction as before), and returns to the source through D_2. Diodes D_1 and D_4 are reverse-biased.

Q7. For a full-wave bridge rectifier, find the value of peak current through the diode and PIV for the diode. Given: $V_s = 12\,\text{V}$ (rms), $V_{DO} = 0.7\,\text{V}$, $R = 100\,\Omega$. **(10 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Given Parameters

- RMS Input Voltage, $V_{s(rms)} = 12$ V
- Diode Forward Voltage Drop, $V_{DO} = 0.7$ V

- Load Resistance, $R = 100\,\Omega$

2. Mathematical Derivations

Step 2.1: Peak Source Voltage ($V_{s(peak)}$) The peak voltage of the AC source is calculated by multiplying the RMS voltage by $\sqrt{2}$:

$$V_{s(peak)} = \sqrt{2} \times V_{s(rms)} = \sqrt{2} \times 12\text{ V} \approx 16.97\text{ V} \tag{1}$$

Step 2.2: Peak Output Voltage ($V_{o(peak)}$) In a full-wave bridge rectifier, two diodes conduct in series during each half-cycle. Therefore, the peak voltage reaching the load resistor R is the peak source voltage minus the voltage drop of two forward-biased diodes:

$$\begin{aligned} V_{o(peak)} &= V_{s(peak)} - 2V_{DO} \\ &= 16.97\text{ V} - 2(0.7\text{ V}) \\ &= 16.97\text{ V} - 1.4\text{ V} = \mathbf{15.57}\text{ V} \end{aligned}$$

Step 2.3: Peak Diode Current (I_{peak}) During conduction, the peak current flowing through the series pair of diodes is equivalent to the peak current through the load resistor. Applying Ohm's Law:

$$\begin{aligned} I_{peak} &= \frac{V_{o(peak)}}{R} \\ &= \frac{15.57\text{ V}}{100\,\Omega} \\ &= 0.1557\text{ A} = \mathbf{155.7}\text{ mA} \end{aligned}$$

Step 2.4: Peak Inverse Voltage (PIV) The Peak Inverse Voltage is the maximum reverse voltage a diode must withstand when it is non-conducting. In a bridge rectifier, a non-conducting diode is in parallel with the secondary source and one forward-biased diode. Applying Kirchhoff's Voltage Law (KVL):

$$\begin{aligned} \text{PIV} &= V_{s(peak)} - V_{DO} \\ &= 16.97\text{ V} - 0.7\text{ V} = \mathbf{16.27}\text{ V} \end{aligned}$$

3. Final Answer Summary

Parameter	Symbol	Calculated Value
Peak Source Voltage	$V_{s(peak)}$	16.97 V
Peak Diode Current	I_{peak}	155.7 mA
Peak Inverse Voltage	PIV	16.27 V

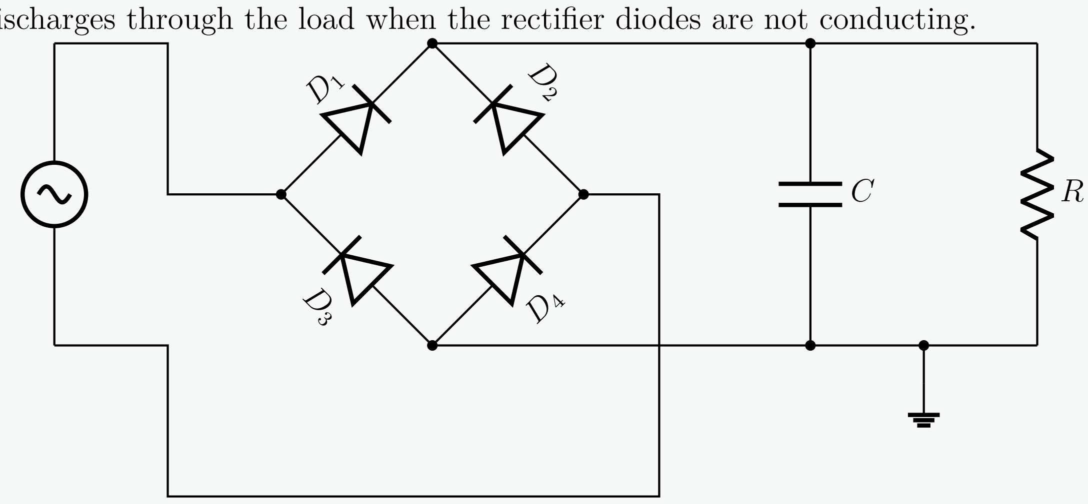
Q8. Comment on the value of RC time constant for a rectifier circuit. **(10 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Introduction and Circuit Principle

In a rectifier circuit, an output filter is almost always required to convert the pulsating DC voltage into a steady, constant DC voltage suitable for electronic loads. The most common and economical filtering method is placing a shunt capacitor (C) in parallel with the load resistor (R_L).

The behavior and effectiveness of this filter are heavily dependent on the **RC time constant** ($\tau = R_L C$), which dictates the rate at which the capacitor discharges through the load when the rectifier diodes are not conducting.



2. The Golden Rule of the RC Time Constant

To effectively smooth the output voltage, the capacitor must discharge very slowly relative to the frequency of the incoming AC signal. The fundamental design criterion for the time constant is:

$$\tau = R_L C \gg T \tag{1}$$

where T is the period of the rectified signal.

- For a **Half-Wave Rectifier**: $T = \frac{1}{f_{in}}$. Thus, $R_L C \gg \frac{1}{f_{in}}$
- For a **Full-Wave Rectifier**: The rectified signal frequency is twice the input frequency ($2f_{in}$), so the effective period is $T = \frac{1}{2f_{in}}$. Thus, $R_L C \gg \frac{1}{2f_{in}}$

3. Mathematical Relationship to Ripple Voltage

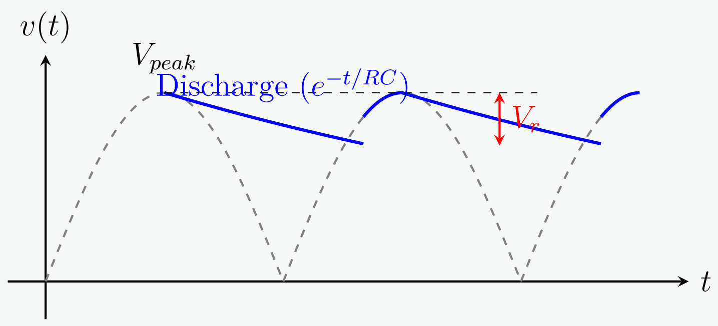
The primary goal of selecting an appropriate RC time constant is to minimize the peak-to-peak ripple voltage (V_r). Assuming a linear discharge (valid when $R_L C \gg T$), the ripple voltage for a full-wave rectifier is approximated as:

$$V_r \approx \frac{V_{peak}}{2 f_{in} R_L C} \tag{2}$$

where V_{peak} is the peak rectified voltage. This equation demonstrates that the ripple voltage is **inversely proportional** to the RC time constant.

4. Design Trade-offs: Choosing the Value of τ

While the equation suggests making the RC time constant as large as possible to eliminate ripple, doing so introduces severe practical trade-offs. The selection of the RC value requires balancing DC quality against component stress.



Case 1: Very Large RC Time Constant (Large C or Large R_L)

- **Advantage (Low Ripple):** The capacitor discharges very slowly between peaks. The output DC voltage is smooth, and the average DC level remains very close to V_{peak}.
- **Disadvantage (High Surge Current):** Because the capacitor voltage remains high, the incoming AC sine wave only exceeds the capacitor voltage for a very brief interval near the peak. The diodes only conduct during this tiny window (narrow conduction angle). To supply the required average load current, the diodes must pass massive, short-duration spikes of *surge current*, which can exceed their peak repetitive forward current (I_{FRM}) rating and destroy them.

Case 2: Small RC Time Constant (Small C or Small R_L)

- **Disadvantage (High Ripple):** The capacitor discharges deeply between peaks. The output exhibits a large "sawtooth" ripple, and the average DC voltage drops significantly. This poor regulation is unacceptable for sensitive analog circuits.
- **Advantage (Lower Peak Diode Current):** The capacitor voltage drops lower, meaning the AC source voltage exceeds it earlier in the cycle. The conduction angle increases, spreading the charging current over a longer time and reducing the peak thermal stress on the diodes.

Conclusion: The value of the RC time constant must be chosen as a compromise. It should be large enough to keep the ripple voltage within the tolerance required by the load (or the subsequent voltage regulator), but it must not be so excessively large that it causes the peak repetitive diode currents to exceed safe operating limits.

Q9. Draw the circuit diagram of a full wave rectifier with a filter capacitor. Assume the input is a sinusoidal signal with 10 kHz frequency and 7.07 V rms voltage, the load is 1 kΩ and capacitor used is 1 μF. Diode has a forward voltage drop of 0.7 V. Find:

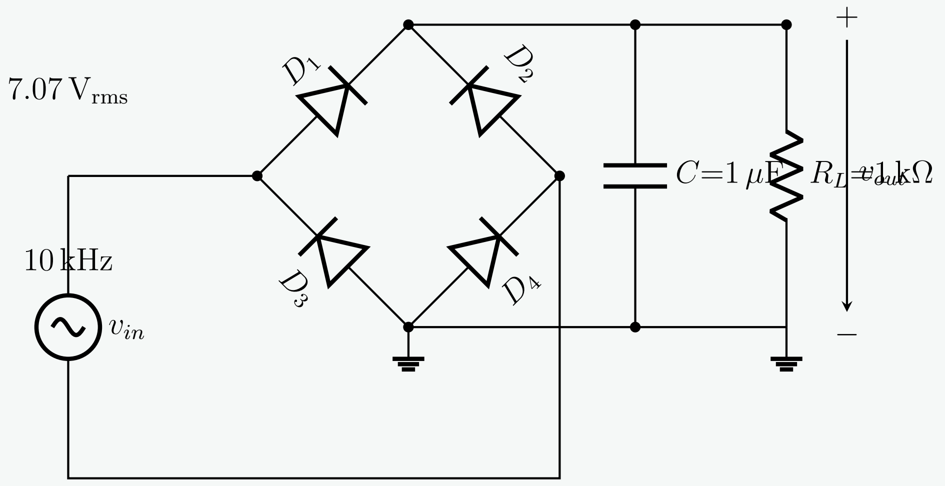
(a) Ripple voltage.
(b) PIV and duty cycle.
(c) Peak output voltage.

(16 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

1. Circuit Diagram

Assumption: As no transformer is explicitly specified, a **Full-Wave Bridge Rectifier** topology is assumed. This requires two diode voltage drops ($2V_D$) per half-cycle conduction path.



2. Given Parameters

- RMS Input Voltage, $V_{rms} = 7.07\,\text{V}$
- Input Frequency, $f = 10\,\text{kHz}$
- Load Resistance, $R_L = 1\,\text{k}\Omega = 1000\,\Omega$
- Filter Capacitance, $C = 1\,\mu\text{F} = 10^{-6}\,\text{F}$
- Diode Forward Drop, $V_D = 0.7\,\text{V}$

3. Mathematical Derivations and Solutions

Step 3a: Peak Input Voltage
The peak value of the input sinusoidal voltage is:

$$V_{in(peak)} = V_{rms} \times \sqrt{2} = 7.07\,\text{V} \times 1.414 \approx 10.0\,\text{V} \tag{1}$$

Step 3b: Peak Output Voltage ($V_{p,out}$)
In a bridge rectifier, the current flows through two diodes simultaneously during each half-cycle. Therefore, the peak output voltage is the peak input voltage minus two diode drops:

$$\begin{aligned} V_{p,out} &= V_{in(peak)} - 2V_D \\ &= 10.0\,\text{V} - 2(0.7\,\text{V}) \\ &= 10.0\,\text{V} - 1.4\,\text{V} \\ &= 8.6\,\text{V} \end{aligned}$$

Step 3c: Ripple Voltage (V_r)
For a full-wave rectifier, the ripple frequency is twice the input frequency ($f_r = 2f = 20\,\text{kHz}$). The peak-to-peak ripple voltage $V_{r(pp)}$ is approximated by:

$$\begin{aligned} V_{r(pp)} &\approx \frac{V_{p,out}}{f_r R_L C} = \frac{V_{p,out}}{2 f R_L C} \\ &= \frac{8.6\,\text{V}}{(2 \times 10^4\,\text{Hz}) \times (10^3\,\Omega) \times (10^{-6}\,\text{F})} \\ &= \frac{8.6}{20} \\ &= 0.43\,\text{V} \end{aligned}$$

Step 3d: Peak Inverse Voltage (PIV)
In a bridge configuration, the maximum reverse voltage across any non-conducting diode is the peak input voltage minus the voltage drop of the conducting diode in its loop:

$$\begin{aligned} \text{PIV} &= V_{in(peak)} - V_D \\ &= 10.0\,\text{V} - 0.7\,\text{V} \\ &= 9.3\,\text{V} \end{aligned}$$

Step 3e: Duty Cycle
The conduction angle θ_c (in radians) represents the small portion of the cycle where the diodes turn ON to recharge the capacitor. It is given by:

$$\begin{aligned} \theta_c &\approx \sqrt{\frac{2V_{r(pp)}}{V_{p,out}}} \\ &= \sqrt{\frac{2(0.43\,\text{V})}{8.6\,\text{V}}} = \sqrt{\frac{0.86}{8.6}} = \sqrt{0.1} \\ &\approx 0.3162\,\text{rad} \end{aligned}$$

The duty cycle for an individual diode (which conducts once per full cycle of 2π) is:

$$\begin{aligned}\text{Duty Cycle (Diode)} &= \left(\frac{\theta_c}{2\pi}\right) \times 100\% \\ &= \left(\frac{0.3162}{2 \times 3.1416}\right) \times 100\% \\ &\approx 5.03\%\end{aligned}$$

Note: The overall rectifier circuit charges the capacitor twice per cycle, yielding an effective circuit duty cycle of $2 \times 5.03\% \approx 10.06\%$.

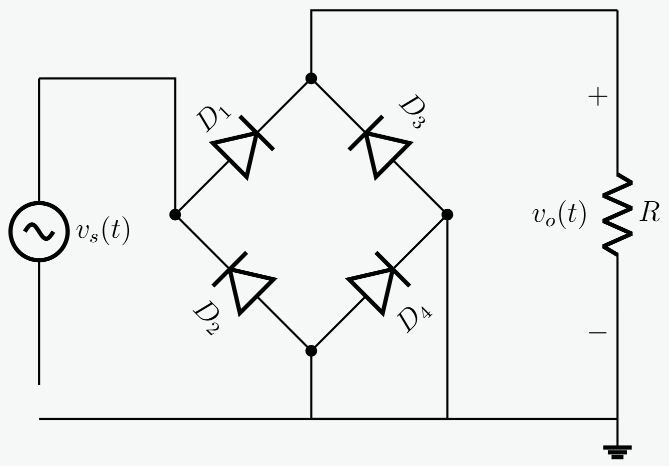
4. Final Answer Summary

Parameter	Calculated Value
Peak Output Voltage	**8.6** V
Ripple Voltage (Peak-to-Peak)	**0.43** V
Peak Inverse Voltage (PIV)	**9.3** V
Duty Cycle (per Diode)	**5.03**%

Q10. Show that the average output voltage of a bridge rectifier circuit is $V_o = \frac{2}{\pi}V_s - 2V_{DO}$. If the input voltage is a 12 V (rms) sinusoid and the load resistance is 100 Ω, calculate: (i) the average output voltage, (ii) the peak diode current, and (iii) the PIV of the diodes. Assume $V_{DO} = 0.7$ V. **(16 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Reference Circuit Diagram



2. Derivation of the Average Output Voltage Equation

Circuit Operation & Assumption: Let the input voltage be a purely sinusoidal wave defined as $v_s(t) = V_s \sin(\omega t)$, where V_s is the **peak** amplitude of the source. During the positive half-cycle, diodes D_1 and D_4 are forward-biased (ON). Assuming a constant voltage drop model, each conducting diode has a forward voltage drop of V_{DO}. Applying Kirchhoff's Voltage Law (KVL) around the conduction loop yields the instantaneous output voltage:

$$v_o(t) = |v_s(t)| - 2V_{DO} \tag{1}$$

Mathematical Derivation: The average (DC) value of a periodic waveform is found by integrating it over one period and dividing by the period. For a full-wave rectified signal, the period is π.

$$V_o = \frac{1}{\pi}\int_0^{\pi} v_o(t)d(\omega t) \tag{2}$$

Substituting the instantaneous voltage equation:

$$V_o = \frac{1}{\pi}\int_0^{\pi} (V_s \sin(\omega t) - 2V_{DO})d(\omega t) \tag{3}$$

Note: This standard derivation neglects the small dead-band near the zero-crossings where the source voltage is less than $2V_{DO}$. *We assume conduction occurs from* 0 *to* π.

Splitting the integral:

$$V_o = \frac{V_s}{\pi}\int_0^{\pi} \sin(\omega t)d(\omega t) - \frac{2V_{DO}}{\pi}\int_0^{\pi} d(\omega t) \tag{4}$$

Evaluating the integrals:

$$\int_0^{\pi} \sin(\omega t)d(\omega t) = [-\cos(\omega t)]_0^{\pi} = -(-1) - (-1) = 2$$

$$\int_0^{\pi} d(\omega t) = [\omega t]_0^{\pi} = \pi$$

Substituting these results back into the equation yields the required expression:

$$V_o = \frac{V_s}{\pi}(2) - \frac{2V_{DO}}{\pi}(\pi)$$
$$V_o = \frac{\mathbf{2}}{\pi}\mathbf{V_s} - \mathbf{2V_{DO}}$$

3. Step-by-Step Calculations

Given Parameters:

- Input RMS Voltage, $V_{rms} = \mathbf{12\ V}$
- Load Resistance, $R = \mathbf{100}\ \Omega$
- Diode Forward Voltage, $V_{DO} = \mathbf{0.7\ V}$

First, we determine the **peak** source voltage (V_s), which is required for the formulas:

$$V_s = \sqrt{2} \times V_{rms} = \sqrt{2} \times 12 = \mathbf{16.97}\ \text{V} \tag{5}$$

(i) Average Output Voltage (V_o) Using the derived equation:

$$\begin{aligned} V_o &= \frac{2}{\pi}(16.97\ \text{V}) - 2(0.7\ \text{V}) \\ &= 10.80\ \text{V} - 1.4\ \text{V} \\ &= \mathbf{9.40}\ \text{V} \end{aligned}$$

(ii) Peak Diode Current (I_{peak}) The peak diode current is equal to the peak load current. The peak voltage across the load occurs at the peak of the input cycle minus the two diode drops.

$$\begin{aligned} V_{o(peak)} &= V_s - 2V_{DO} \\ &= 16.97\ \text{V} - 1.4\ \text{V} = 15.57\ \text{V} \end{aligned}$$

Applying Ohm's Law to the load resistor:

$$\begin{aligned} I_{peak} &= \frac{V_{o(peak)}}{R} \\ &= \frac{15.57\ \text{V}}{100\ \Omega} \\ &= 0.1557\ \text{A} = \mathbf{155.7}\ \text{mA} \end{aligned}$$

(iii) Peak Inverse Voltage (PIV) The PIV is the maximum reverse voltage across an OFF diode. In a bridge rectifier, the non-conducting diode is in parallel with the secondary source and one forward-biased diode.

$$\begin{aligned} \text{PIV} &= V_s - V_{DO} \\ &= 16.97\ \text{V} - 0.7\ \text{V} \\ &= \mathbf{16.27}\ \text{V} \end{aligned}$$

4. Final Summary Table

Parameter	Symbol	Calculated Value
(i) Average Output Voltage	V_o	**9.40 V**
(ii) Peak Diode Current	I_{peak}	**155.7 mA**
(iii) Peak Inverse Voltage	PIV	**16.27 V**

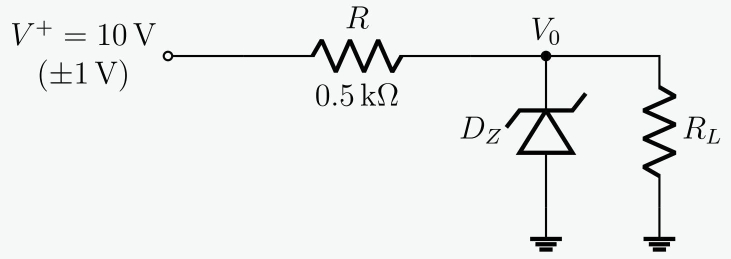
Zener Diode Regulator

Q1. A 6.8 V Zener diode exhibits its nominal voltage at a test current of 5 mA. At this current the incremental resistance is $r_z = 20\,\Omega$ and knee current $I_{ZK} = 0.2\,\text{mA}$. The negative terminal is connected to $V^+ = 10\,\text{V}$ ($\pm 1\,\text{V}$) through a $0.5\,\text{k}\Omega$ resistor; positive terminal is grounded. (i) Find V_o with no load at nominal supply. (ii) Find the change in V_o as supply varies by $\pm 1\,\text{V}$. (iii) Find the change in V_o when $R_L = 2\,\text{k}\Omega$ is connected; what happens if $R_L = 0.5\,\text{k}\Omega$? (iv) Find the minimum R_L for which the diode still operates in breakdown. **(20 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

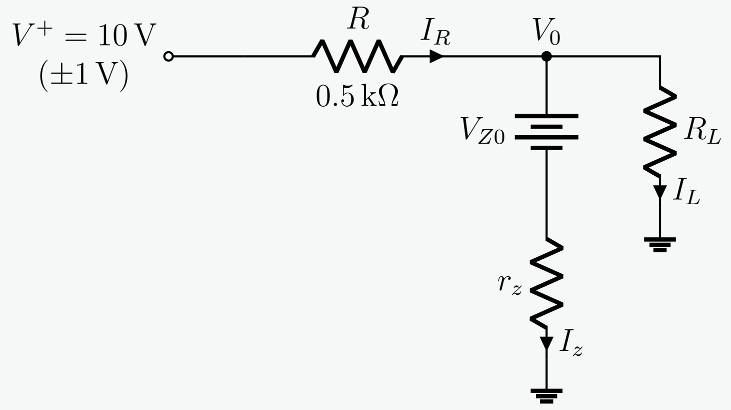
Answer

1. Equivalent Circuit and Zener Model

To analyze the circuit, we replace the Zener diode in its breakdown region with its piecewise-linear equivalent model: a constant voltage source (V_{Z0}) in series with an incremental resistance (r_z).



Diode Specifications:
$V_Z = 6.8\,\text{V}$
$I_{ZT} = 5\,\text{mA}$
$r_z = 20\,\Omega$
$I_{ZK} = 0.2\,\text{mA}$



Determining V_{Z0}: The Zener voltage is given by $V_Z = V_{Z0} + I_Z r_z$. Given the nominal operating point ($V_Z = 6.8\,\text{V}$ at $I_{ZT} = 5\,\text{mA}$) and $r_z = 20\,\Omega$:

$$\begin{aligned} 6.8\,\text{V} &= V_{Z0} + (5 \times 10^{-3}\,\text{A})(20\,\Omega) \\ V_{Z0} &= 6.8\,\text{V} - 0.1\,\text{V} = \mathbf{6.7}\,\text{V} \end{aligned}$$

2. Mathematical Analysis

(i) Find V_o with no load ($R_L = \infty$) at nominal supply ($V^+ = 10\,\text{V}$): With no load, $I_L = 0$, so the entire supply current flows through the Zener diode ($I_R = I_Z$). Using Kirchhoff's Voltage Law (KVL):

$$I_Z = \frac{V^+ - V_{Z0}}{R + r_z} = \frac{10 - 6.7}{500 + 20} = \frac{3.3}{520} \approx 6.346\,\text{mA} \tag{1}$$

The output voltage is:

$$\begin{aligned} V_o &= V_{Z0} + I_Z r_z \\ &= 6.7 + (6.346\,\text{mA})(20\,\Omega) \\ &= 6.7 + 0.1269 = \mathbf{6.827}\,\text{V} \end{aligned}$$

(ii) Change in V_o as supply varies by $\pm 1\,\text{V}$ (Line Regulation): Using voltage division for the incremental model (superposition of AC/change only):

$$\begin{aligned} \Delta V_o &= \Delta V^+ \left(\frac{r_z}{R + r_z}\right) \\ &= (\pm 1\,\text{V}) \left(\frac{20}{500 + 20}\right) \\ &= \pm\frac{20}{520} \approx \pm\mathbf{38.5}\,\text{mV} \quad (\text{or } \pm 0.0385\,\text{V}) \end{aligned}$$

(iii) Change in V_o with loads (Load Regulation): When $R_L = 2\,\text{k}\Omega$ ($2000\,\Omega$) is connected, we assume the Zener remains in breakdown. Applying Kirchhoff's Current Law (KCL) at the output node ($I_R = I_Z + I_L$):

$$\frac{V^+ - V_o}{R} = \frac{V_o - V_{Z0}}{r_z} + \frac{V_o}{R_L} \tag{2}$$

$$\frac{10 - V_o}{500} = \frac{V_o - 6.7}{20} + \frac{V_o}{2000} \tag{3}$$

Multiplying the entire equation by 2000:

$$\begin{aligned} 4(10 - V_o) &= 100(V_o - 6.7) + V_o \\ 40 - 4V_o &= 100V_o - 670 + V_o \\ 710 = 105V_o &\implies V_o = \frac{710}{105} \approx 6.762\,\text{V} \end{aligned}$$

Verification of ON state: $I_Z = \frac{6.762-6.7}{20} = 3.1\,\text{mA} > I_{ZK}$ (Assumption valid).
The change in V_o is:

$$\Delta V_o = V_{o(\text{loaded})} - V_{o(\text{no-load})} = 6.762 - 6.827 = -\mathbf{65}\,\text{mV} \tag{4}$$

What happens if $R_L = 0.5\,\text{k}\Omega$ ($500\,\Omega$)? Assuming Zener is in breakdown:

$$\frac{10 - V_o}{500} = \frac{V_o - 6.7}{20} + \frac{V_o}{500} \tag{5}$$

Multiplying by 500:

$$\begin{aligned} 10 - V_o &= 25(V_o - 6.7) + V_o \\ 10 - V_o &= 25V_o - 167.5 + V_o \\ 177.5 &= 27V_o \implies V_o \approx 6.574\,\text{V} \end{aligned}$$

Verification of ON state: Since $V_o = 6.574\,\text{V} < V_{Z0} = 6.7\,\text{V}$, the calculated Zener current would be negative, which is impossible.
Conclusion: The initial assumption is false. **The Zener diode drops out of breakdown (turns OFF)** and acts as an open circuit. The circuit becomes a simple voltage divider:

$$V_o = V^+ \left(\frac{R_L}{R + R_L}\right) = 10\left(\frac{500}{500 + 500}\right) = \mathbf{5.0}\,\text{V} \tag{6}$$

(iv) Minimum R_L for Zener Breakdown: For the diode to just maintain breakdown, I_Z must be at the knee current, $I_{ZK} = 0.2\,\text{mA}$.

$$V_{o(\min)} = V_{Z0} + I_{ZK}r_z = 6.7 + (0.2\,\text{mA})(20\,\Omega) = 6.7 + 0.004 = 6.704\,\text{V} \tag{7}$$

The current through the series resistor R is:

$$I_R = \frac{V^+ - V_{o(\min)}}{R} = \frac{10 - 6.704}{500} = \frac{3.296\,\text{V}}{500\,\Omega} = 6.592\,\text{mA} \tag{8}$$

Using KCL, the maximum current the load can draw before starving the Zener is:

$$I_{L(\max)} = I_R - I_{ZK} = 6.592\,\text{mA} - 0.2\,\text{mA} = 6.392\,\text{mA} \tag{9}$$

The minimum load resistance is therefore:

$$R_{L(\min)} = \frac{V_{o(\min)}}{I_{L(\max)}} = \frac{6.704\,\text{V}}{6.392 \times 10^{-3}\,\text{A}} \approx \mathbf{1048.8\,\Omega} \quad (\text{or } 1.05\,\text{k}\Omega) \tag{10}$$

3. Final Summary Table

Parameter	**Calculated Value**
(i) No-load V_o at nominal supply	$6.827\,\text{V}$
(ii) Change in V_o for $\pm 1\,\text{V}$ supply variation	$\pm 38.5\,\text{mV}$
(iii) Change in V_o with $R_L = 2\,\text{k}\Omega$	$-65\,\text{mV}$
(iii) Behavior with $R_L = 0.5\,\text{k}\Omega$	Zener turns OFF; V_o drops to $5.0\,\text{V}$
(iv) Minimum R_L to maintain breakdown	$1048.8\,\Omega$ ($1.05\,\text{k}\Omega$)

Q2. A 9.1 V Zener diode exhibits its nominal voltage at a test current of 28 mA. At this current the incremental resistance is specified to be $5\,\Omega$. (i) Find V_{z0} in the Zener model. (ii) Find the Zener voltage at currents of 10 mA and 100 mA, respectively. (iii) Comment on the advantages/disadvantages of running the Zener diode at current values other than the nominal value. **(10 marks)** [**EEE 263 | L-2/T-1 | 2023–24**]

Answer

1. Theory & Principle

In the breakdown region, a real Zener diode does not possess a perfectly vertical characteristic curve. Instead, the voltage across the Zener diode, V_Z, increases slightly as the Zener current, I_Z, increases.
This behavior is modeled using a piecewise linear equivalent circuit consisting of an ideal voltage source V_{Z0} (the Zener voltage at the zero-current intercept) in series with a small dynamic (incremental) resistance r_z. The equation governing the Zener diode in the breakdown region is given by:

$$V_Z = V_{Z0} + I_Z r_z \tag{1}$$

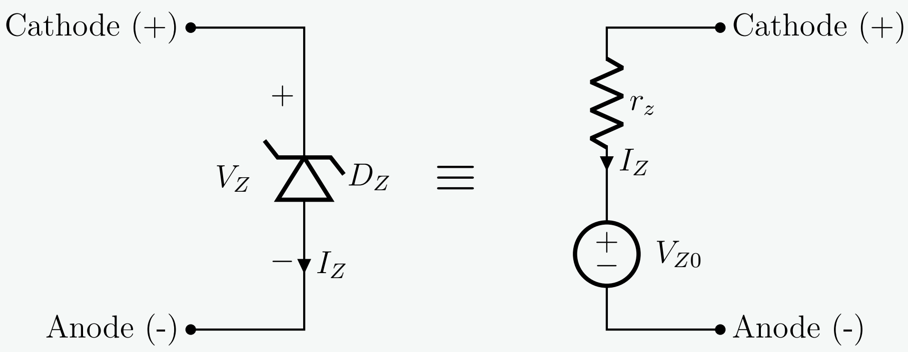
where:

- V_Z is the actual Zener voltage at a specific operating current I_Z.
- V_{Z0} is the theoretical ideal Zener voltage extrapolated to $I_Z = 0$.

- r_z is the incremental resistance (slope of the $V - I$ curve in the breakdown region).

2. Equivalent Circuit Model

The following diagram illustrates the standard symbol for a Zener diode and its corresponding linear equivalent circuit model operating in the breakdown region.



3. Step-by-Step Mathematical Derivation

(i) Finding the Zener base voltage, V_{Z0}

From the problem parameters, we are given the nominal operating point (test conditions):

$$\begin{aligned}\text{Test Voltage, } V_Z &= 9.1 \text{ V}\\ \text{Test Current, } I_{ZT} &= 28 \text{ mA} = 28 \times 10^{-3} \text{ A}\\ \text{Incremental Resistance, } r_z &= 5\ \Omega\end{aligned}$$

Rearranging the linear Zener model equation to solve for V_{Z0}:

$$\begin{aligned}V_{Z0} &= V_Z - I_{ZT} r_z\\ V_{Z0} &= 9.1 - \left(28 \times 10^{-3} \text{ A} \times 5\ \Omega\right)\\ V_{Z0} &= 9.1 - 0.14\\ V_{Z0} &= \mathbf{8.96} \text{ V}\end{aligned}$$

(ii) Finding the Zener voltage at 10 mA and 100 mA

Using the derived $V_{Z0} = 8.96$ V and the constant $r_z = 5\ \Omega$, we can determine the terminal voltage at any other current in the linear region.
Case A: At $I_Z = 10$ mA

$$\begin{aligned}V_Z(10\text{mA}) &= V_{Z0} + I_Z r_z\\ &= 8.96 + \left(10 \times 10^{-3} \times 5\right)\\ &= 8.96 + 0.05\\ &= \mathbf{9.01} \text{ V}\end{aligned}$$

Case B: At $I_Z = 100$ mA

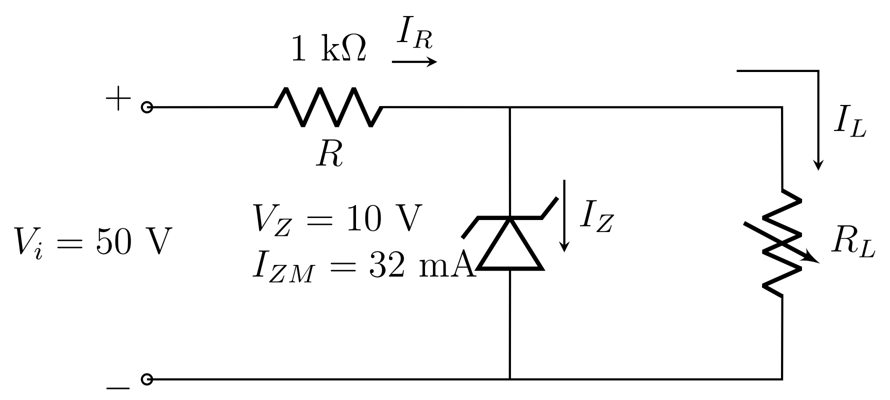
$$\begin{aligned}V_Z(100\text{mA}) &= V_{Z0} + I_Z r_z\\ &= 8.96 + \left(100 \times 10^{-3} \times 5\right)\\ &= 8.96 + 0.50\\ &= \mathbf{9.46} \text{ V}\end{aligned}$$

4. Operational Considerations

Operating a Zener diode at currents other than its nominal test current (I_{ZT}) presents engineering trade-offs. The table below summarizes the advantages and disadvantages:

Operating Condition	Advantages	Disadvantages
Below Nominal ($I_Z < 28$ mA, e.g., 10 mA)	• **Lower Power Dissipation:** Reduced self-heating ($P = V_Z I_Z \approx 90$ mW), increasing component longevity and overall circuit efficiency.	• **Poor Regulation:** Operation moves closer to the "knee" of the Zener curve where r_z begins to increase non-linearly. • **Voltage Drop:** The output voltage drops below the target nominal 9.1 V.
Above Nominal ($I_Z > 28$ mA, e.g., 100 mA)	• **Robust Breakdown:** The diode is driven deep into the avalanche/Zener region, ensuring r_z remains low and constant. Regulation against load fluctuation is very stable.	• **High Power Dissipation:** Extreme self-heating ($P = 9.46\text{ V} \times 0.1\text{ A} = 946$ mW). This requires a robust 1W+ Zener package and risks thermal runaway or device failure. • **Voltage Shift:** The output voltage significantly exceeds the nominal 9.1 V.

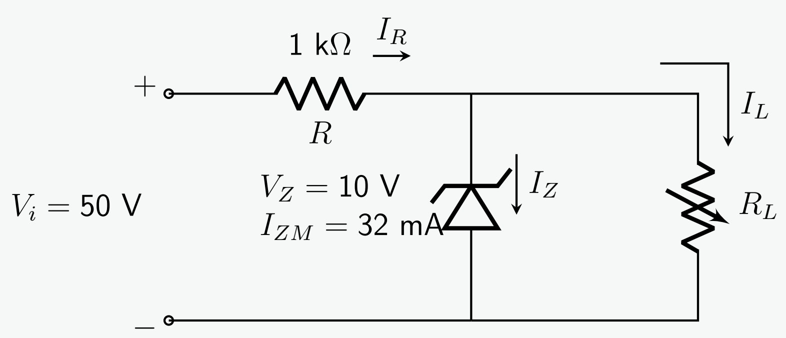
Q3. For the circuit given, determine the range of R_L and I_L that will result in V_{R_L} being maintained at 10 V. Also determine the maximum wattage rating of the diode. The Zener diode is a 10 V Zener with maximum current rating of 32 mA.



(18 marks) [EEE 263 | L-2/T-1 | 2022–23]

Answer

1. Reference Circuit Diagram



2. Principle of Operation

For the Zener diode to maintain the load voltage V_{R_L} at a constant 10 V, it must be operating in the breakdown region. This imposes two conditions:

(a) The Thevenin voltage across the load (with the Zener removed) must be greater than or equal to V_Z.

(b) The Zener current I_Z must satisfy $I_{ZK} \le I_Z \le I_{ZM}$, where I_{ZK} is the minimum knee current (assumed to be 0 mA here) and I_{ZM} is the maximum rated Zener current.

Since the Zener voltage V_Z is fixed, the voltage across the series resistor R is also fixed. The total current supplied by the source I_R will therefore remain constant. The load current I_L and Zener current I_Z will dynamically share this total current according to Kirchhoff's Current Law (KCL).

3. Step-by-Step Mathematical Derivation

Step 1: Calculate the Total Supply Current (I_R)

Assuming the Zener diode is ON and regulating, the voltage at the node above the diode is clamped at $V_Z = 10$ V. By applying Ohm's law to the series resistor R:

$$I_R = \frac{V_i - V_Z}{R}$$
$$I_R = \frac{50 - 10}{1000}$$
$$I_R = 40 \text{ mA}$$

Step 2: Determine the Range of Load Current (I_L)

Applying KCL at the node above the Zener diode:

$$I_R = I_Z + I_L \implies I_L = I_R - I_Z \tag{1}$$

The **minimum load current** ($I_{L(\min)}$) occurs when the Zener diode draws its **maximum permissible current** ($I_{ZM} = 32$ mA):

$$I_{L(\min)} = I_R - I_{ZM}$$
$$I_{L(\min)} = 40 \text{ mA} - 32 \text{ mA}$$
$$I_{L(\min)} = 8 \text{ mA}$$

The **maximum load current** ($I_{L(\max)}$) occurs when the Zener diode draws its **minimum sustaining current**. Assuming $I_{ZK} \approx 0$ mA:

$$I_{L(\max)} = I_R - I_{Z(\min)}$$
$$I_{L(\max)} = 40 \text{ mA} - 0 \text{ mA}$$
$$I_{L(\max)} = 40 \text{ mA}$$

Thus, the range of load current to maintain regulation is:

$$8 \text{ mA} \leq I_L \leq 40 \text{ mA} \tag{2}$$

Step 3: Determine the Range of Load Resistance (R_L)

The load resistance R_L is inversely proportional to the load current I_L since the voltage across it is fixed at $V_Z = 10$ V.
The **maximum load resistance** ($R_{L(\max)}$) corresponds to the minimum load current:

$$R_{L(\max)} = \frac{V_Z}{I_{L(\min)}}$$
$$R_{L(\max)} = \frac{10 \text{ V}}{8 \text{ mA}}$$
$$R_{L(\max)} = 1.25 \text{ k}\Omega \text{ (or } 1250 \ \Omega)$$

The **minimum load resistance** ($R_{L(\min)}$) corresponds to the maximum load current:

$$R_{L(\min)} = \frac{V_Z}{I_{L(\max)}}$$
$$R_{L(\min)} = \frac{10 \text{ V}}{40 \text{ mA}}$$
$$R_{L(\min)} = 250 \ \Omega$$

Thus, the range of load resistance to maintain regulation is:

$$250 \ \Omega \leq R_L \leq 1.25 \text{ k}\Omega \tag{3}$$

Step 4: Calculate the Maximum Wattage Rating of the Diode (P_{ZM})

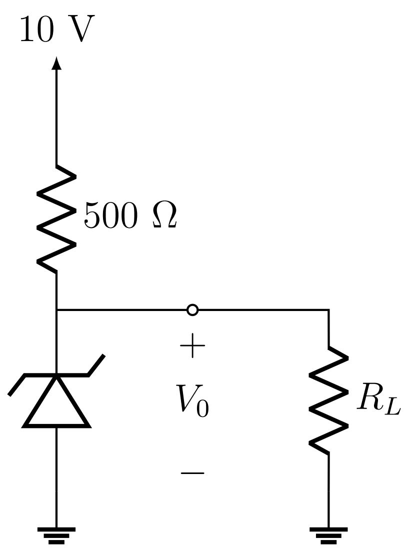
The maximum power dissipated by the Zener diode occurs when it conducts its maximum rated current I_{ZM}.

$$P_{ZM} = V_Z \times I_{ZM}$$
$$P_{ZM} = 10 \text{ V} \times 32 \text{ mA}$$
$$P_{ZM} = 320 \text{ mW}$$

4. Final Answer Summary

Parameter	Calculated Range / Value
Range of Load Current (I_L)	$8 \text{ mA} \leq I_L \leq 40 \text{ mA}$
Range of Load Resistance (R_L)	$250 \ \Omega \leq R_L \leq 1.25 \text{ k}\Omega$
Maximum Zener Wattage Rating (P_{ZM})	320 mW

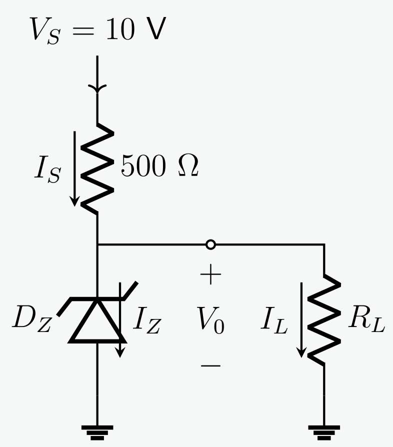
Q4. The zener diode in the circuit shown is specified to have $V_Z = 6.8\,\text{V}$ at $I_Z = 5\,\text{mA}$, $r_z = 20\,\Omega$ and $I_{ZK} = 0.2\,\text{mA}$. (i) Find V_0 at no load conditions. (ii) What is the minimum value of R_L for which the diode will still operate in the breakdown region?



(23 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Answer

1. Reference Circuit Diagram



2. Theory & Zener Model Evaluation

To analyze this circuit accurately, we must replace the Zener diode with its piecewise linear equivalent model, which consists of a DC voltage source V_{Z0} in series with an internal dynamic resistance r_z.
The governing equation for the Zener diode in the breakdown region is:

$$V_Z = V_{Z0} + I_Z r_z \tag{1}$$

Given the test conditions: $V_Z = 6.8$ V at $I_Z = 5$ mA and $r_z = 20\ \Omega$. We can calculate the zero-current intercept voltage V_{Z0}:

$$\begin{aligned} V_{Z0} &= V_Z - I_Z r_z \\ V_{Z0} &= 6.8 - (5 \times 10^{-3})(20) \\ V_{Z0} &= 6.8 - 0.1 \\ V_{Z0} &= 6.7\text{ V} \end{aligned}$$

3. Step-by-Step Mathematical Solution

(i) Find V_0 at no-load conditions ($R_L = \infty$)

Under no-load conditions, the load resistor is disconnected (open circuit), meaning the load current $I_L = 0$. Therefore, the entire supply current flows through the Zener diode ($I_S = I_Z$).
Assuming the diode is operating in the breakdown region (which we will verify), we write the loop equation for the equivalent circuit:

$$\begin{aligned} I_Z &= \frac{V_S - V_{Z0}}{R_S + r_z} \\ I_Z &= \frac{10 - 6.7}{500 + 20} \\ I_Z &= \frac{3.3}{520} \\ I_Z &\approx 6.346\text{ mA} \end{aligned}$$

Verification: Since $I_Z \approx 6.346$ mA is greater than $I_{ZK}(0.2\text{ mA})$, our assumption that the diode is in breakdown holds true.

Now, compute the output voltage V_0:

$$V_0 = V_{Z0} + I_Z r_z$$
$$V_0 = 6.7 + (6.346 \times 10^{-3})(20)$$
$$V_0 = 6.7 + 0.1269$$
$$V_0 = \mathbf{6.827} \text{ V}$$

(ii) Minimum value of R_L to maintain breakdown region

To keep the Zener diode in the breakdown region, the current flowing through it must not drop below the knee current $I_{ZK} = 0.2$ mA. This boundary condition represents the maximum possible current drawn by the load before the Zener turns off.
At the threshold condition $I_Z = I_{ZK} = 0.2$ mA, the Zener voltage is:

$$V_0 = V_{Z0} + I_{ZK} r_z$$
$$V_0 = 6.7 + (0.2 \times 10^{-3})(20)$$
$$V_0 = 6.7 + 0.004$$
$$V_0 = 6.704 \text{ V}$$

Next, determine the total source current I_S flowing through the 500 Ω series resistor:

$$I_S = \frac{V_S - V_0}{R_S}$$
$$I_S = \frac{10 - 6.704}{500}$$
$$I_S = \frac{3.296}{500}$$
$$I_S = 6.592 \text{ mA}$$

Applying Kirchhoff's Current Law (KCL) at the top output node, we find the maximum allowable load current:

$$I_{L(\max)} = I_S - I_{ZK}$$
$$I_{L(\max)} = 6.592 \text{ mA} - 0.2 \text{ mA}$$
$$I_{L(\max)} = 6.392 \text{ mA}$$

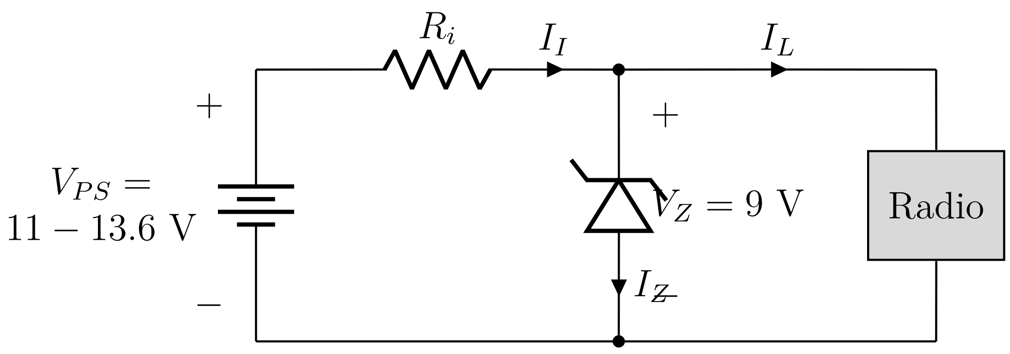
Finally, calculate the corresponding minimum load resistance:

$$R_{L(\min)} = \frac{V_0}{I_{L(\max)}}$$
$$R_{L(\min)} = \frac{6.704}{6.392 \times 10^{-3}}$$
$$R_{L(\min)} \approx 1048.8\ \Omega \text{ or } \mathbf{1.049\ k\Omega}$$

4. Final Answer Summary

Parameter	Calculated Value
Zener Zero-Current Intercept (V_{Z0})	6.7 V
No-load Output Voltage (V_0)	**6.827** V
Minimum Load Resistance ($R_{L(\min)}$)	**1.049 kΩ**

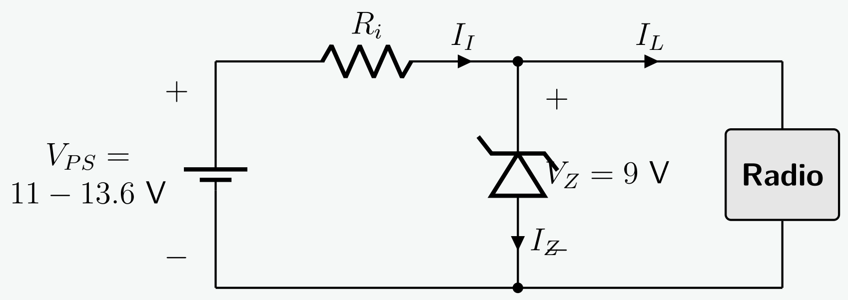
Q5. A voltage regulator circuit is to power a car radio at $V_L = 9$ V from an automobile battery whose voltage may vary between 11 and 13.6 V. The current in the radio will vary between 0 (off) and 100 mA (full volume). Determine the value of the current-limiting resistor R_i and the maximum power dissipated in the Zener diode.



(18 marks) [EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Reference Circuit Diagram



2. Principle of Operation & Assumptions

This circuit is a basic Zener voltage regulator operating under varying input voltage (line regulation) and varying load current (load regulation). To ensure the radio receives a stable 9 V, the Zener diode must remain in the breakdown region under all operating conditions.
By Kirchhoff's Current Law (KCL) at the node above the Zener diode:

$$I_I = I_Z + I_L \implies I_Z = I_I - I_L \tag{1}$$

The input current I_I is governed by Ohm's Law:

$$I_I = \frac{V_{PS} - V_Z}{R_i} \tag{2}$$

Assumptions:

- The minimum Zener knee current I_{ZK} is negligible ($I_{ZK} \approx 0$ mA) unless otherwise specified.
- The Zener voltage V_Z remains perfectly constant at 9 V in the breakdown region (ideal Zener model).

3. Step-by-Step Mathematical Derivation

Step 1: Determine the Current-Limiting Resistor (R_i)

To maintain regulation, the Zener diode must not turn off. The worst-case condition for the Zener diode dropping out of regulation occurs when the input voltage is at its minimum and the load draws its maximum current.

$$V_{PS(\min)} = 11 \text{ V}$$
$$I_{L(\max)} = 100 \text{ mA} = 0.1 \text{ A}$$

Under this condition, the Zener current is at its minimum ($I_{Z(\min)} \approx 0$ A).

$$I_{I(\min)} = I_{Z(\min)} + I_{L(\max)}$$
$$I_{I(\min)} = 0 + 100 \text{ mA} = 100 \text{ mA}$$

Now, substituting into the input current equation to solve for R_i:

$$R_i = \frac{V_{PS(\min)} - V_Z}{I_{I(\min)}}$$
$$R_i = \frac{11 \text{ V} - 9 \text{ V}}{0.1 \text{ A}}$$
$$R_i = \frac{2 \text{ V}}{0.1 \text{ A}}$$
$$R_i = \mathbf{20\ \Omega}$$

Step 2: Determine Maximum Power Dissipated in the Zener Diode ($P_{Z(\max)}$)

The worst-case condition for power dissipation in the Zener diode occurs when the input voltage is at its maximum and the load is disconnected (or turned off), meaning all input current is forced through the Zener diode.

$$V_{PS(\max)} = 13.6 \text{ V}$$
$$I_{L(\min)} = 0 \text{ A}$$

First, calculate the maximum input current $I_{I(\max)}$ using the chosen R_i:

$$I_{I(\max)} = \frac{V_{PS(\max)} - V_Z}{R_i}$$
$$I_{I(\max)} = \frac{13.6 \text{ V} - 9 \text{ V}}{20\ \Omega}$$
$$I_{I(\max)} = \frac{4.6 \text{ V}}{20\ \Omega}$$
$$I_{I(\max)} = 0.23 \text{ A} = 230 \text{ mA}$$

Since $I_{L(\text{min})} = 0$, the maximum Zener current is:

$$I_{Z(\text{max})} = I_{I(\text{max})} - I_{L(\text{min})} = 230 \text{ mA} - 0 \text{ mA} = 230 \text{ mA} \tag{3}$$

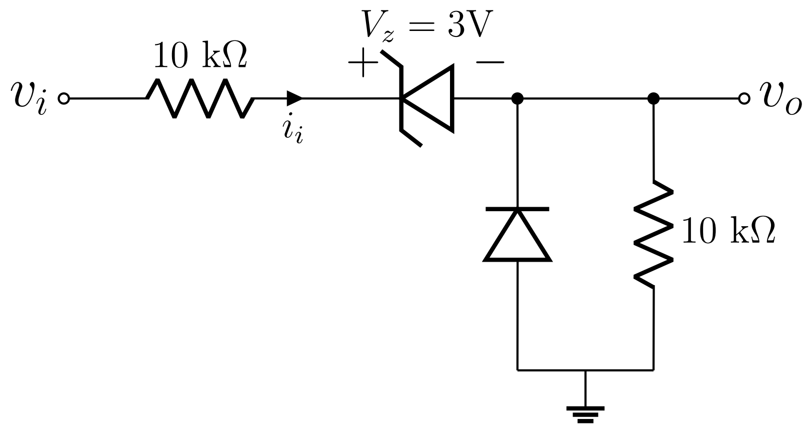
Finally, calculate the maximum power dissipation:

$$P_{Z(\text{max})} = V_Z \times I_{Z(\text{max})}$$
$$P_{Z(\text{max})} = 9 \text{ V} \times 0.23 \text{ A}$$
$$P_{Z(\text{max})} = \mathbf{2.07} \text{ W}$$

4. Final Answer Summary

Parameter	**Calculated Value**
Current-Limiting Resistor (R_i)	**20 Ω**
Maximum Zener Current ($I_{Z(\text{max})}$)	230 mA
Maximum Zener Power Dissipation ($P_{Z(\text{max})}$)	**2.07** W

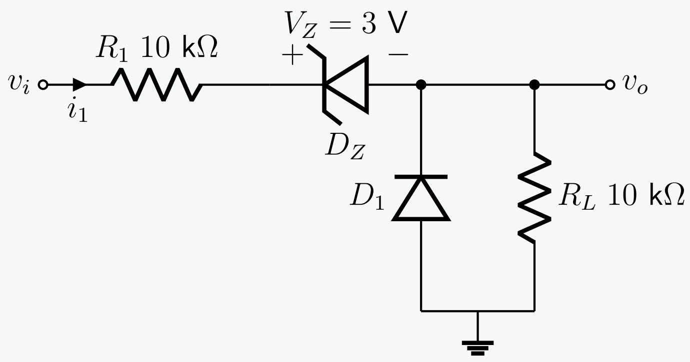
Q6. For the circuit shown in Figure , plot v_0 versus v_i and i_1 versus v_i over the range $-10 \leq v_\text{i} \leq +10$ V. Assume the diodes to be ideal.



(13.67 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Reference Circuit Diagram



2. Principle of Operation & Assumptions

The circuit evaluates an input voltage v_i spanning from −10 V to +10 V.

- **Ideal Diode Assumptions:** Both the Zener diode (D_Z) and the standard diode (D_1) are considered ideal. This means their forward voltage drops are 0 V, their reverse leakage currents are 0 A, and D_Z exhibits a perfect vertical breakdown at exactly $V_Z = 3$ V.
- **Zener Orientation:** D_Z has its cathode facing the input side. Therefore, it will be in Zener breakdown when the voltage at its cathode is 3 V more positive than its anode. It will be forward-biased when its anode is more positive than its cathode.
- **Standard Diode Orientation:** D_1 has its anode tied to ground. It will become forward-biased and act as a short circuit if v_o attempts to drop below 0 V.

3. Step-by-Step Mathematical Analysis

The behavior of the circuit changes based on the input voltage v_i. We divide the input range into three distinct regions of operation.

Region I: $v_i < 0$ **V**

When the input voltage is negative, the current i_1 attempts to flow from right to left (meaning i_1 will be negative).

- This polarity forward-biases the Zener diode (D_Z). Because it is ideal, it acts as a short circuit, meaning the voltage at the node between R_1 and D_Z equals v_o.
- The negative voltage pulls v_o below ground, which immediately forward-biases diode D_1.
- As an ideal forward-biased diode, D_1 acts as a short to ground, clamping the output voltage: $v_o = 0$ V.

Since $v_o = 0$ V and D_Z is a short, the voltage at the right side of R_1 is 0 V.

$$i_1 = \frac{v_i - 0}{10\text{ k}\Omega} = \frac{v_i}{10\text{ k}\Omega}$$
$$v_o = 0\text{ V}$$

At $v_i = -10$ V, the current is $i_1 = -1$ mA.

Region II: $0 \leq v_i \leq 3$ **V**

When the input voltage is positive but has not yet exceeded the Zener breakdown threshold ($V_Z = 3$ V):

- The output voltage v_o will be ≥ 0, ensuring D_1 remains reverse-biased (OFF).
- v_i is not large enough to break down D_Z, so D_Z remains in the reverse-blocking region and acts as an open circuit.

Since D_Z acts as an open circuit, no current can flow through the input branch.

$$i_1 = 0\text{ mA}$$
$$v_o = 0\text{ V} \quad \text{(since no current flows through } R_L\text{)}$$

Region III: $v_i > 3$ **V**

When v_i exceeds 3 V, it provides sufficient potential to push D_Z into the Zener breakdown region.

- D_Z acts as a constant 3 V voltage drop (opposing the input).
- v_o is positive, so D_1 remains reverse-biased (OFF).

The circuit simplifies to a single series loop comprising v_i, R_1, the 3 V Zener drop, and R_L. Applying Kirchhoff's Voltage Law (KVL):

$$v_i - i_1R_1 - V_Z - i_1R_L = 0$$
$$i_1(10\text{ k}\Omega + 10\text{ k}\Omega) = v_i - 3$$
$$i_1 = \frac{v_i - 3}{20\text{ k}\Omega}$$

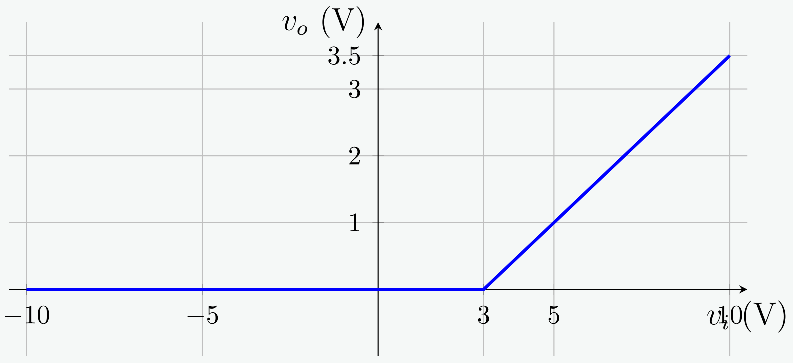
The output voltage is determined by the load resistor R_L:

$$v_o = i_1 \times R_L$$
$$v_o = \left(\frac{v_i - 3}{20\text{ k}\Omega}\right) \times 10\text{ k}\Omega$$
$$v_o = 0.5(v_i - 3)$$

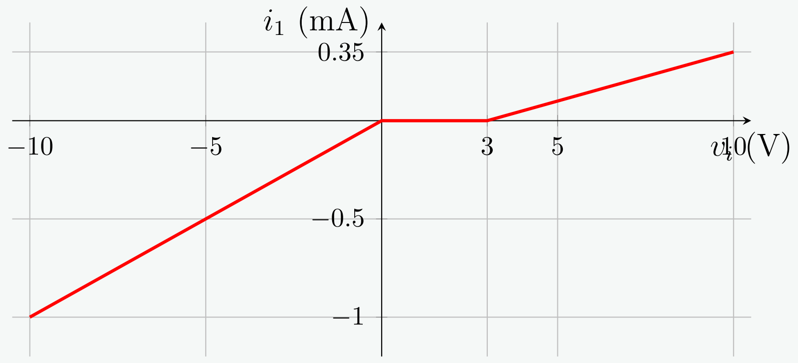
At $v_i = 10$ V, $i_1 = 7/20$ k$\Omega = 0.35$ mA and $v_o = 0.5(10 - 3) = 3.5$ V.

4. Transfer Characteristics

Output Voltage (v_o) vs Input Voltage (v_i)



Input Current (i_1) vs Input Voltage (v_i)



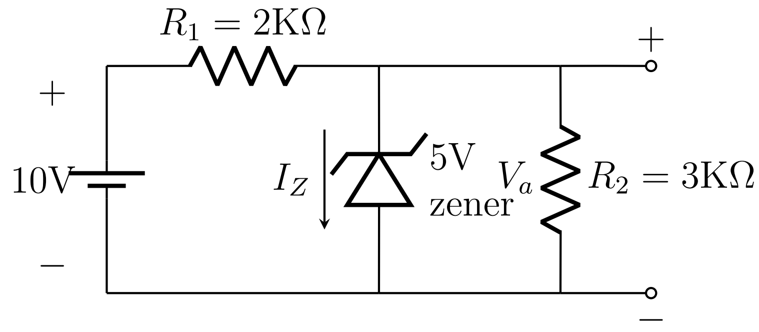
5. Summary of Equations

The piecewise functions governing the output voltage and input current over the entire range are summarized below:

$$v_o = \begin{cases} 0 \text{ V} & \text{for } -10 \text{ V} \le v_i \le 3 \text{ V} \\ 0.5(v_i - 3) \text{ V} & \text{for } 3 \text{ V} < v_i \le 10 \text{ V} \end{cases} \tag{1}$$

$$i_1 = \begin{cases} \frac{v_i}{10 \text{ k}\Omega} & \text{for } -10 \text{ V} \le v_i < 0 \text{ V} \\ 0 \text{ mA} & \text{for } 0 \text{ V} \le v_i \le 3 \text{ V} \\ \frac{v_i - 3}{20 \text{ k}\Omega} & \text{for } 3 \text{ V} < v_i \le 10 \text{ V} \end{cases} \tag{2}$$

Q7. For the circuit shown, find V_a and I_z. Now if R_1 and R_2 are swapped, what will be the value of V_a and I_z?

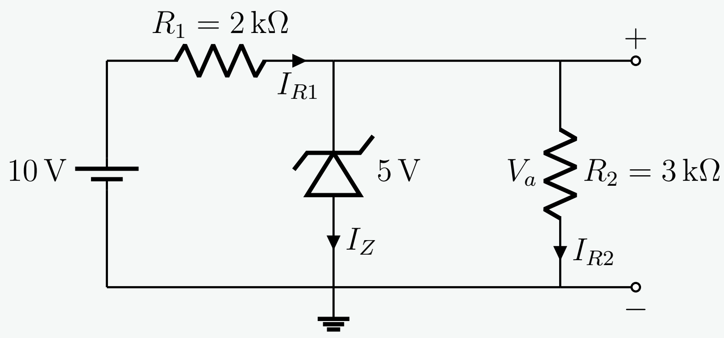


(15 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

1. Initial Circuit Configuration Analysis

Circuit Diagram:



Step 1.1: State and Verify Assumption

To determine the operating state of the Zener diode, we first assume it is **OFF** (acting as an open circuit). We then calculate the Thevenin voltage (V_{Th}) across the open Zener terminals, which is simply the voltage across R_2 determined by the voltage divider formed by R_1 and R_2:

$$\begin{aligned} V_{Th} &= V_{in}\left(\frac{R_2}{R_1 + R_2}\right) \\ &= 10\,\text{V}\left(\frac{3\,\text{k}\Omega}{2\,\text{k}\Omega + 3\,\text{k}\Omega}\right) \\ &= 10\left(\frac{3}{5}\right) = \mathbf{6}\,\text{V} \end{aligned}$$

Since the open-circuit voltage ($V_{Th} = 6\,\text{V}$) is greater than the Zener breakdown voltage ($V_Z = 5\,\text{V}$), our initial assumption is incorrect. **The Zener diode is ON and operating in the breakdown region.**

Step 1.2: Calculate Output Voltage (V_a) and Zener Current (I_Z)

Because the Zener is in breakdown and in parallel with the load R_2, it firmly clamps the output voltage to its nominal value:

$$V_a = V_Z = \mathbf{5}\,\text{V} \tag{1}$$

Next, we apply Kirchhoff's Current Law (KCL) at the upper node:

$$I_{R1} = I_Z + I_{R2} \implies I_Z = I_{R1} - I_{R2} \tag{2}$$

Calculating the individual branch currents:

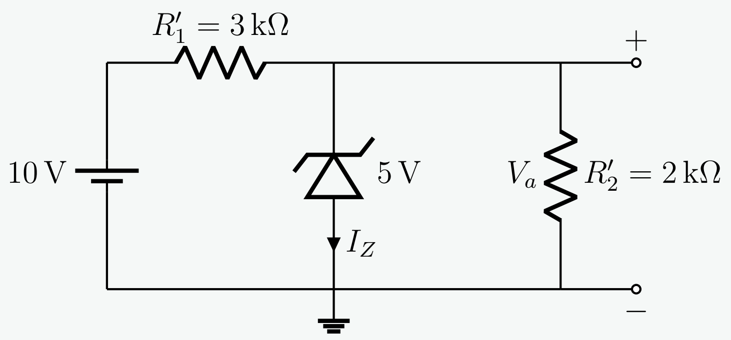
$$\begin{aligned} I_{R1} &= \frac{V_{in} - V_a}{R_1} = \frac{10\,\text{V} - 5\,\text{V}}{2\,\text{k}\Omega} = \frac{5}{2}\,\text{mA} = 2.5\,\text{mA} \\ I_{R2} &= \frac{V_a}{R_2} = \frac{5\,\text{V}}{3\,\text{k}\Omega} \approx 1.667\,\text{mA} \end{aligned}$$

Substituting these into the KCL equation:

$$I_Z = 2.5\,\text{mA} - 1.667\,\text{mA} = \mathbf{0.833}\,\text{mA} \tag{3}$$

2. Swapped Resistors Configuration Analysis

Circuit Diagram: (With R_1 and R_2 swapped)



Step 2.1: State and Verify Assumption

Once again, assume the Zener diode is **OFF** (open circuit). We calculate the new Thevenin voltage (V'_{Th}) across the Zener terminals with the swapped resistor values ($R'_1 = 3\,\text{k}\Omega$, $R'_2 = 2\,\text{k}\Omega$):

$$\begin{aligned} V'_{Th} &= V_{in}\left(\frac{R'_2}{R'_1 + R'_2}\right) \\ &= 10\,\text{V}\left(\frac{2\,\text{k}\Omega}{3\,\text{k}\Omega + 2\,\text{k}\Omega}\right) \\ &= 10\left(\frac{2}{5}\right) = \mathbf{4}\,\text{V} \end{aligned}$$

Since the available open-circuit voltage ($V'_{Th} = 4\,\text{V}$) is strictly less than the Zener breakdown voltage ($V_Z = 5\,\text{V}$), our assumption holds true. **The Zener diode remains OFF** and does not enter the breakdown region.

Step 2.2: Calculate Output Voltage (V_a) and Zener Current (I_Z)

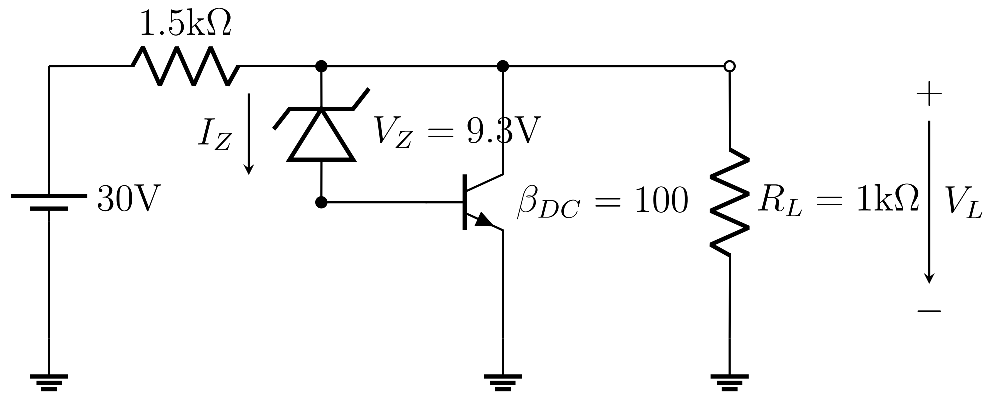
Because the Zener is OFF, it draws no current and acts as an open circuit. The circuit behaves entirely as a simple linear voltage divider.

$$\begin{aligned} V_a &= V'_{Th} = \mathbf{4}\,\text{V} \\ I_Z &= \mathbf{0}\,\text{mA} \end{aligned}$$

3. Final Answer Summary

Condition	Zener State	$\mathbf{V_a}$ (Output Voltage)	$\mathbf{I_Z}$ (Zener Current)
Initial ($R_1 = 2\,\text{k}\Omega$, $R_2 = 3\,\text{k}\Omega$)	**ON** (Breakdown)	5.00 V	0.833 mA
Swapped ($R_1 = 3\,\text{k}\Omega$, $R_2 = 2\,\text{k}\Omega$)	**OFF** (Open)	4.00 V	0 mA

Q8. Calculate the regulated load voltage V_L and the current I_Z flowing through the zener diode in the voltage regulator circuit shown.

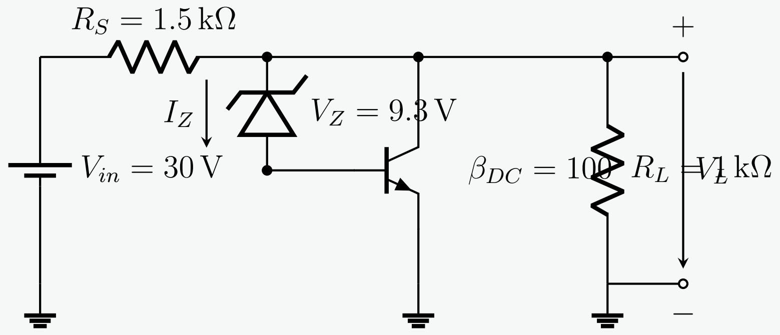


(16 marks) [EEE 263 | L-2/T-1 | 2014–15]

Answer

1. Circuit Diagram and Operating Principle

The circuit presented is a **BJT-amplified shunt voltage regulator**. It combines a Zener diode for voltage reference with an NPN BJT to significantly increase the current-handling (shunting) capability of the regulator. The Zener diode determines the base voltage of the transistor relative to the output, while the BJT absorbs the bulk of the excess current from the unregulated supply, providing a stable load voltage V_L.



2. Operating State Assumptions

To perform the DC analysis, we make the following standard assumptions for an active voltage regulator:

- **Zener Diode:** Operates in the **breakdown region**, maintaining a constant reverse voltage $V_Z = 9.3\,\text{V}$.
- **NPN BJT:** Operates in the **forward-active region** with a base-emitter ON voltage $V_{BE(on)} \approx 0.7\,\text{V}$.

3. Mathematical Derivation and Solution

Step 3a: Calculate Regulated Load Voltage (V_L)
By applying Kirchhoff's Voltage Law (KVL) from the output node, down through the Zener diode and the BJT's base-emitter junction to ground, we find:

$$V_L - V_Z - V_{BE(on)} = 0 \tag{1}$$

Rearranging to solve for the load voltage V_L:

$$\begin{aligned} V_L &= V_Z + V_{BE(on)} \\ &= 9.3\,\text{V} + 0.7\,\text{V} \\ &= 10.0\,\text{V} \end{aligned}$$

Step 3b: Current Analysis (Source and Load Currents)
The total current supplied by the unregulated source (I_S) flows through the series resistor R_S:

$$\begin{aligned} I_S &= \frac{V_{in} - V_L}{R_S} \\ &= \frac{30\,\text{V} - 10\,\text{V}}{1.5\,\text{k}\Omega} = \frac{20\,\text{V}}{1.5\,\text{k}\Omega} = 13.333\,\text{mA} \end{aligned}$$

The current drawn by the load resistor (I_L) is:

$$\begin{aligned} I_L &= \frac{V_L}{R_L} \\ &= \frac{10\,\text{V}}{1\,\text{k}\Omega} = 10.0\,\text{mA} \end{aligned}$$

Step 3c: Calculate Zener Current (I_Z)
By applying Kirchhoff's Current Law (KCL) at the top regulated node, the total current entering the parallel shunt elements (Zener and BJT collector) is defined as I_{shunt}:

$$I_{shunt} = I_S - I_L = 13.333\,\text{mA} - 10.0\,\text{mA} = 3.333\,\text{mA} \tag{2}$$

This shunt current splits between the Zener diode and the BJT collector:

$$I_{shunt} = I_Z + I_C \tag{3}$$

From the circuit topology, the Zener current I_Z flows directly into the base of the BJT. Therefore:

$$I_B = I_Z \tag{4}$$

Assuming forward-active operation, the collector current is given by $I_C = \beta_{DC} I_B = \beta_{DC} I_Z$. Substituting this into the shunt current equation gives:

$$\begin{aligned} I_{shunt} &= I_Z + \beta_{DC} I_Z \\ I_{shunt} &= (\beta_{DC} + 1) I_Z \end{aligned}$$

Now, we can solve for I_Z:

$$\begin{aligned} I_Z &= \frac{I_{shunt}}{\beta_{DC} + 1} \\ &= \frac{3.333\,\text{mA}}{100 + 1} \\ &= \frac{3.333\,\text{mA}}{101} \\ &\approx 0.033\,\text{mA} \quad (\text{or } 33.0\,\mu\text{A}) \end{aligned}$$

4. Verification of Operating Regions

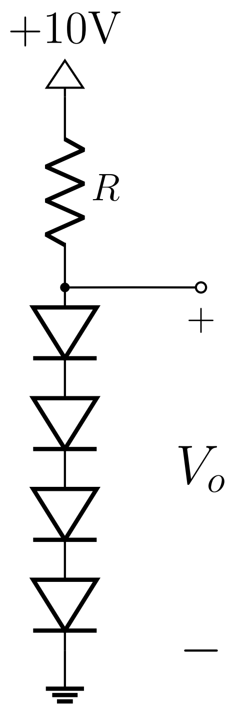
- **Zener Verification:** Since the calculated Zener current is $I_Z = 33\,\mu\text{A} > 0$, the Zener diode is actively conducting and correctly operating in the breakdown region.
- **BJT Verification:** The collector-emitter voltage is $V_{CE} = V_L = 10\,\text{V}$. Because $V_{CE} = 10\,\text{V} \gg V_{CE(sat)}$ (which is typically around $0.2\,\text{V}$), the assumption that the BJT is in the forward-active region is strongly verified.

5. Final Answer Summary

Calculated Parameter	Final Value	
Regulated Load Voltage (V_L)	**10.0** V	
Current through Zener Diode (I_Z)	**33.0** μA	(0.033 mA)

Diode Circuit Design

Q1. Design the circuit shown to provide an output voltage $v_a = 3\,\text{V}$. Assume that the diodes available have $0.7\,\text{V}$ drop at $1\,\text{mA}$, thermal voltage $= 25\,\text{mV}$, and parameter $n = 1$.

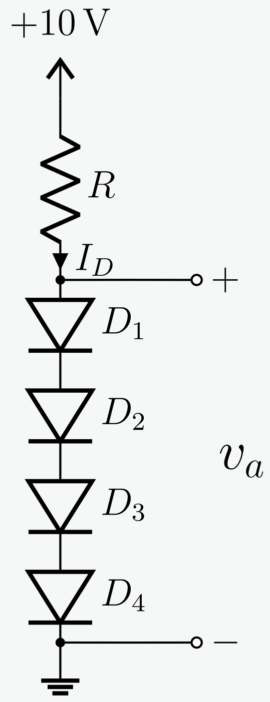


(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

1. Circuit Diagram

The circuit consists of a 10 V DC supply, a current-limiting resistor R, and a string of four identical diodes connected in series to ground. The required output voltage is taken across the diode string.



2. Principle and Assumptions

To design the circuit to provide an output voltage $v_a = 3\,\text{V}$, we make the following assumptions and use the corresponding operating principles:

- **Forward-Biased Assumption:** Since the positive terminal of the $10\,\text{V}$ source is connected to the anodes of the diodes via resistor R, all four diodes are operating in the forward-biased (ON) region.
- **Identical Diodes:** The diodes are identical and connected in series. Therefore, the same current (I_D) flows through each diode, and the total output voltage v_a is equally divided among them.

3. Mathematical Derivation and Solution

Step 3a: Determine the Voltage Across Each Diode
Since the total required voltage across the four series diodes is $v_a = 3\,\text{V}$, the voltage across a single diode (V_D) is:

$$V_D = \frac{v_a}{4} = \frac{3\,\text{V}}{4} = 0.75\,\text{V} \tag{1}$$

Step 3b: Calculate the Required Diode Current (I_D)
The diode current I_D is related to the diode voltage V_D by the exponential diode equation:

$$I_D = I_S \exp\left(\frac{V_D}{nV_T}\right) \tag{2}$$

When given a reference operating point (V_{D1}, I_{D1}), we can eliminate the reverse saturation current (I_S) by expressing the current at the new operating point (V_D, I_D) as:

$$I_D = I_{D1} \exp\left(\frac{V_D - V_{D1}}{nV_T}\right) \tag{3}$$

Given the following parameters:

- Reference current, $I_{D1} = 1\,\text{mA}$
- Reference voltage, $V_{D1} = 0.7\,\text{V}$
- Ideality factor, $n = 1$
- Thermal voltage, $V_T = 25\,\text{mV} = 0.025\,\text{V}$

Substitute the known values to find the required current I_D:

$$\begin{aligned} I_D &= (1\,\text{mA}) \exp\left(\frac{0.75\,\text{V} - 0.7\,\text{V}}{1 \times 0.025\,\text{V}}\right) \\ &= (1\,\text{mA}) \exp\left(\frac{0.05\,\text{V}}{0.025\,\text{V}}\right) \\ &= (1\,\text{mA}) \exp(2) \end{aligned}$$

Since $e \approx 2.7183$, $e^2 \approx 7.389$. Therefore:

$$I_D \approx 7.389\,\text{mA} \tag{4}$$

This is the required current flowing through the circuit to maintain exactly $3\,\text{V}$ at the output.
Step 3c: Calculate the Required Resistor Value (R)
Applying Kirchhoff's Voltage Law (KVL) to the main loop:

$$V_{in} - I_D R - v_a = 0 \tag{5}$$

Rearranging to solve for the resistor R:

$$\begin{aligned} R &= \frac{V_{in} - v_a}{I_D} \\ &= \frac{10\,\text{V} - 3\,\text{V}}{7.389\,\text{mA}} \\ &= \frac{7\,\text{V}}{7.389 \times 10^{-3}\,\text{A}} \\ &\approx 947.35\,\Omega \end{aligned}$$

4. Final Answer Summary

To establish an output voltage of exactly $3\,\text{V}$, the required resistance must be designed to allow a current of approximately $7.39\,\text{mA}$ to flow through the diodes.

Calculated Parameter	**Final Value**
Voltage per Diode (V_D)	$0.75\,\text{V}$
Required Circuit Current (I_D)	$7.389\,\text{mA}$
Designed Resistor Value (R)	**$947\,\Omega$**

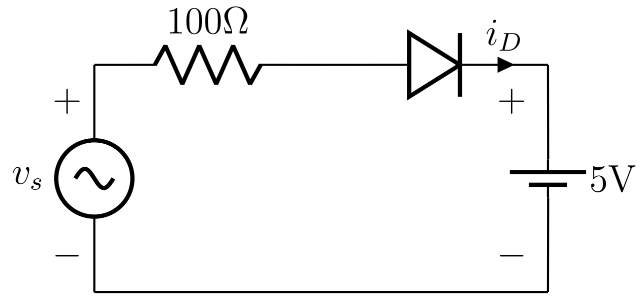
Battery Charger Circuit

Q1. The circuit shown is for charging a 5 V battery with $V_s = 10\sin(100\pi t)$. Find:

(a) Duty cycle (fraction of each cycle during which the diode conducts).

(b) Draw the diode current waveform indicating peak value.

(c) Maximum reverse bias voltage that appears across the diode.



(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

1. Circuit Operation and Principle

The given circuit consists of an AC voltage source $v_s(t) = 10\sin(100\pi t)$ in series with a resistor $R = 100\,\Omega$, an ideal diode D, and a DC battery $V_B = 5$ V.

- **Diode ON State (Conduction Region):** The ideal diode conducts when it is forward-biased. This occurs when the anode voltage exceeds the cathode voltage. Since the cathode is connected to the positive terminal of the 5 V battery, the diode turns ON when:

$$v_s(t) > 5 \text{ V}$$

During this period, the circuit is closed, and current $i_D(t)$ flows into the battery to charge it.

- **Diode OFF State (Non-Conduction Region):** When $v_s(t) < 5$ V, the diode is reverse-biased and acts as an open circuit. Consequently, the diode current is:

$$i_D(t) = 0 \text{ A}$$

—

2. Step-by-Step Mathematical Derivation and Solution

(a) Duty Cycle Calculation

The duty cycle is defined as the fraction of each cycle during which the diode conducts.
Let the total period of the input AC signal be T. The angular frequency is given as $\omega = 100\pi$ rad/s, so the time period T is:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{100\pi} = 0.02 \text{ s} = 20 \text{ ms}$$

The diode starts conducting at an angle θ_1 when $v_s(t)$ rises to 5 V:

$$10\sin(\theta_1) = 5 \implies \sin(\theta_1) = \frac{5}{10} = 0.5$$

$$\theta_1 = \arcsin(0.5) = \frac{\pi}{6} \text{ rad} \quad (30^\circ)$$

The diode stops conducting during the positive half-cycle when $v_s(t)$ falls below 5 V at angle θ_2:

$$\theta_2 = \pi - \theta_1 = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \text{ rad} \quad (150^\circ)$$

The total conduction angle (θ_c) within one full cycle (2π rad) is:

$$\theta_c = \theta_2 - \theta_1 = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3} \text{ rad} \quad (120^\circ)$$

The duty cycle (D) is the ratio of the conduction angle to the total angle of one cycle (2π):

$$D = \frac{\theta_c}{2\pi} = \frac{\frac{2\pi}{3}}{2\pi} = \frac{1}{3} \approx 33.33\%$$

(b) Diode Current Waveform and Peak Value

When the diode is ON ($v_s > 5$ V), the current $i_D(t)$ is found by applying Kirchhoff's Voltage Law (KVL) around the loop:

$$v_s(t) - i_D(t)R - V_B = 0 \implies i_D(t) = \frac{v_s(t) - 5}{100}$$

The peak diode current ($I_{D,\text{peak}}$) occurs when $v_s(t)$ reaches its maximum value of 10 V (at $\omega t = \pi/2$):

$$I_{D,\text{peak}} = \frac{10-5}{100} = \frac{5}{100} = 0.05 \text{ A} = 50 \text{ mA}$$

The current expression during conduction ($\frac{\pi}{6} \le \omega t \le \frac{5\pi}{6}$) is:

$$i_D(t) = \frac{10\sin(100\pi t) - 5}{100} = 0.1\sin(100\pi t) - 0.05 \quad \text{[A]}$$

(c) Maximum Reverse Bias Voltage (Peak Inverse Voltage - PIV)

When the diode is reverse-biased ($v_s < 5$ V), it acts as an open circuit ($i_D = 0$). The voltage across the diode $v_D(t)$ (anode to cathode) is:

$$v_D(t) = v_s(t) - V_B = v_s(t) - 5$$

The maximum reverse bias voltage occurs when $v_s(t)$ reaches its most negative peak value, which is -10 V:

$$v_{D,\text{min}} = -10 - 5 = -15 \text{ V}$$

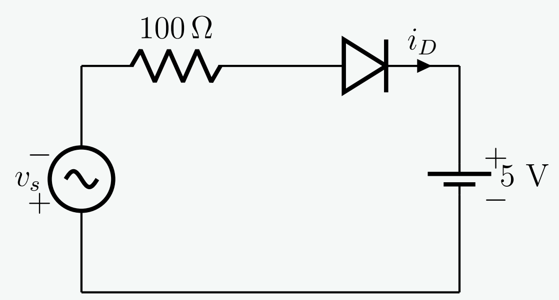
Thus, the maximum reverse bias voltage magnitude (Peak Inverse Voltage) is:

$$\text{PIV} = |v_{D,\text{min}}| = 15 \text{ V}$$

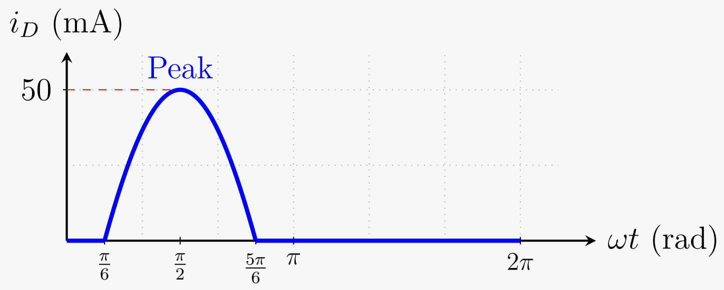
—

3. Circuit Diagrams and Waveforms

Circuit Schematic



Diode Current Waveform



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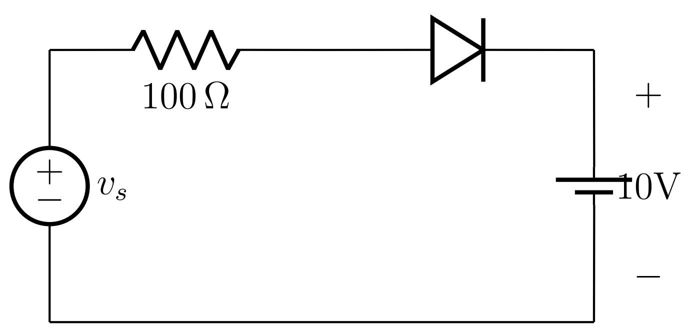
4. Final Answers Summary

Parameter	Symbol / Expression	Value
Conduction Angle	$\theta_c = \theta_2 - \theta_1$	$\frac{2\pi}{3}$ rad (120°)
Duty Cycle	$D = \frac{\theta_c}{2\pi}$	$\frac{1}{3} \approx$ **33.33**%
Peak Diode Current	$I_{D,\text{peak}} = \frac{V_m - V_B}{R}$	**50** mA
Maximum Reverse Voltage	$\text{PIV} = V_m + V_B$	**15** V

Q2. A circuit for charging a 10 V battery is shown in the figure. If V_s is a sinusoid with 20 V peak amplitude, find the fraction of each cycle during which the diode conducts. Assume the diode is ideal. Also find the peak value of the diode current and the maximum reverse bias voltage that appears across the diode.



(15 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Circuit Operation and Principle

The given circuit is a half-wave battery charger. It consists of an AC voltage source $v_s(t) = 20\sin(\omega t)$ with a peak amplitude of 20 V, a current-limiting resistor $R = 100\,\Omega$, an ideal diode D, and a 10 V DC battery.

- **Diode ON State (Conduction):** Assuming an ideal diode, it conducts when the anode voltage is strictly greater than the cathode voltage. Since the cathode is clamped to 10 V by the battery, the diode turns ON when:

$$v_s(t) > 10 \text{ V}$$

  During this interval, current flows to charge the battery.

- **Diode OFF State (Blocking):** When $v_s(t) \leq 10$ V, the diode is reverse-biased, acting as an open circuit. The diode current is zero:

$$i_D(t) = 0 \text{ A}$$

—

2. Mathematical Derivation and Step-by-Step Solution

(a) Fraction of Cycle for Conduction

Let the input voltage be $v_s(t) = V_m \sin(\theta)$, where $V_m = 20$ V and $\theta = \omega t$. The diode starts conducting at an angle θ_1 when the source voltage equals the battery voltage ($V_B = 10$ V):

$$\begin{aligned} 20\sin(\theta_1) &= 10 \\ \sin(\theta_1) &= \frac{10}{20} = 0.5 \\ \theta_1 &= \arcsin(0.5) = \frac{\pi}{6} \text{ rad } (30^\circ) \end{aligned}$$

Because of the symmetry of the sine wave, the diode stops conducting at angle θ_2 in the same positive half-cycle:

$$\theta_2 = \pi - \theta_1 = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \text{ rad } (150^\circ)$$

The total conduction angle (θ_c) for one full cycle (2π rad) is:

$$\theta_c = \theta_2 - \theta_1 = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3} \text{ rad } (120^\circ)$$

The fraction of each cycle during which the diode conducts is:

$$\text{Fraction} = \frac{\theta_c}{2\pi} = \frac{2\pi/3}{2\pi} = \frac{1}{3} \approx 33.33\%$$

(b) Peak Diode Current

When the diode is ON, we apply Kirchhoff's Voltage Law (KVL) around the loop:

$$v_s(t) - i_D(t)R - V_B = 0 \implies i_D(t) = \frac{v_s(t) - 10}{100}$$

The peak current ($I_{D,\text{peak}}$) occurs when $v_s(t)$ reaches its maximum amplitude ($V_m = 20$ V):

$$I_{D,\text{peak}} = \frac{20 - 10}{100} = \frac{10}{100} = 0.1 \text{ A} = 100 \text{ mA}$$

(c) Maximum Reverse Bias Voltage (Peak Inverse Voltage)

When the diode is OFF ($v_s(t) < 10$ V), the current $i_D = 0$, so there is no voltage drop across the $100\,\Omega$ resistor. The voltage across the diode (anode to cathode) is:

$$v_D(t) = v_s(t) - V_B = v_s(t) - 10$$

The maximum reverse bias occurs when the AC source is at its negative peak, i.e., $v_s(t) = -20$ V:

$$v_{D,\text{min}} = -20 - 10 = -30 \text{ V}$$

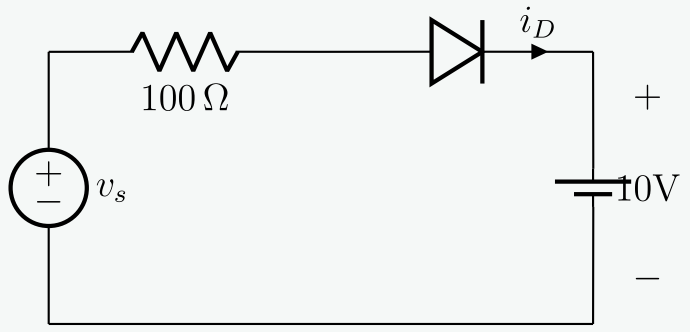
The maximum reverse bias voltage magnitude (PIV) is therefore:

$$\text{PIV} = |v_{D,\text{min}}| = 30 \text{ V}$$

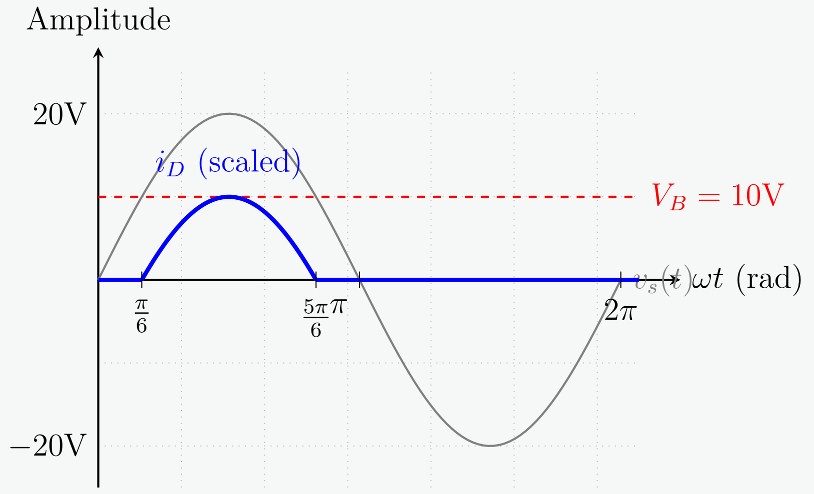
—

3. Circuit Diagram and Waveform

Circuit Schematic



Voltage and Current Waveforms



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4. Final Answer Summary

Parameter	Formula	Calculated Value
Conduction Angle (θ_c)	$\pi - 2\arcsin(V_B/V_m)$	$\frac{2\pi}{3}$ rad (120°)
Fraction of Cycle	$\frac{\theta_c}{2\pi}$	$\mathbf{1/3 \approx 33.33\%}$
Peak Diode Current	$\frac{V_m - V_B}{R}$	**100** mA
Max Reverse Bias Voltage	$V_m + V_B$	**30** V

Special Purpose Diodes/Thyristors

Q1. Draw the circuit diagram of controlled full wave rectifier using SCR. Draw the waveforms of sinusoidal input, gate pulses and rectified output. **(11.67 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Theory and Principle of Operation

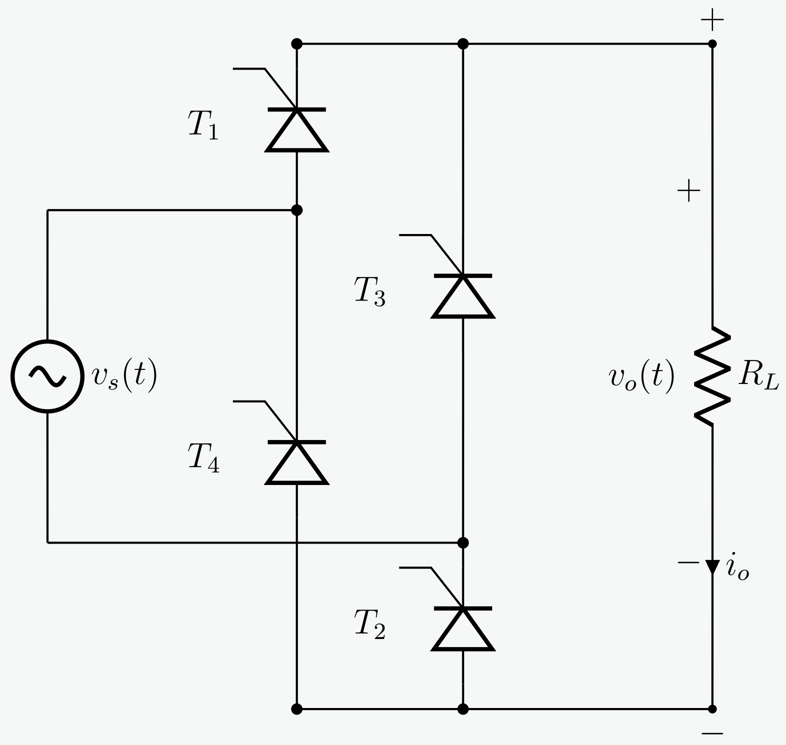
A controlled full-wave rectifier converts an alternating current (AC) input into a controlled direct current (DC) output using Silicon Controlled Rectifiers (SCRs) instead of standard diodes. The use of SCRs (thyristors) allows the user to control the output voltage magnitude by adjusting the firing angle (α).
The most common configuration is the **Fully-Controlled Bridge Rectifier**, which utilizes four SCRs (T_1, T_2, T_3, T_4).

- **Positive Half-Cycle:** The top terminal of the AC source is positive. SCRs T_1 and T_2 are forward-biased. They begin to conduct only when a gate pulse is applied at $\omega t = \alpha$. Current flows from the source, through T_1, the load R_L, and returns via T_2.

- **Negative Half-Cycle:** The bottom terminal of the AC source is positive. SCRs T_3 and T_4 are forward-biased. They are triggered at $\omega t = \pi + \alpha$. Current flows through T_3, the load R_L (in the same direction as before), and returns via T_4.

Assumption: For the following analysis and waveforms, a purely resistive load (R_L) is assumed, meaning the output current becomes zero when the output voltage reaches zero at $\omega t = \pi, 2\pi, \ldots$
—

2. Circuit Diagram



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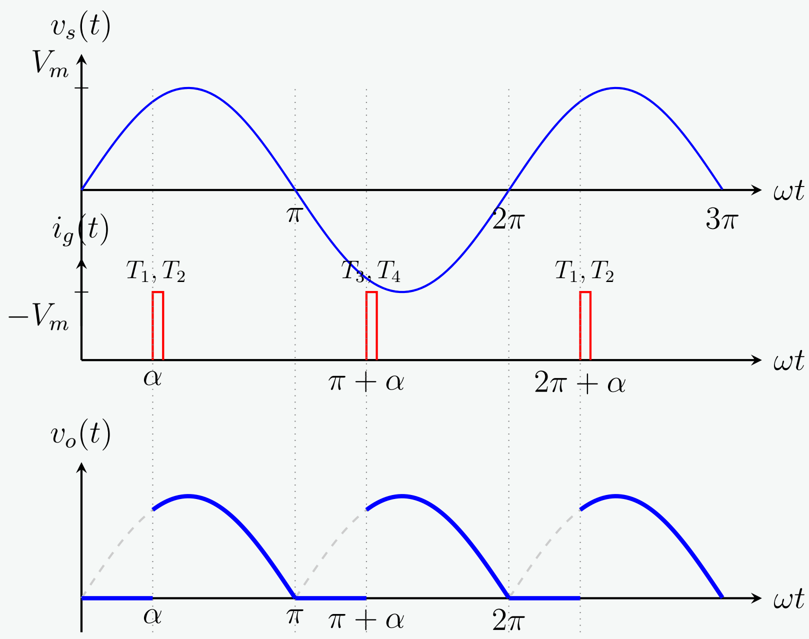
3. Waveforms

The waveforms below illustrate the phase-control mechanism. A firing angle of $\alpha = 60°$ ($\pi/3$) is shown for clarity.



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4. Mathematical Derivation (Average DC Voltage)

For a resistive load, conduction starts at $\omega t = \alpha$ and naturally ceases at $\omega t = \pi$ when the input voltage crosses zero. Because the waveform repeats every half-cycle (period of π), the average output voltage V_{dc} is computed over one half-cycle:

$$\begin{aligned}
V_{dc} &= \frac{1}{\pi}\int_{\alpha}^{\pi} v_s(t)\, d(\omega t) \\
&= \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) \\
&= \frac{V_m}{\pi}\left[-\cos(\omega t)\right]_{\alpha}^{\pi} \\
&= \frac{V_m}{\pi}\Big(-\cos(\pi) - (-\cos(\alpha))\Big) \\
&= \frac{V_m}{\pi}\Big(-(-1) + \cos(\alpha)\Big) \\
\therefore \quad \mathbf{V_{dc}} &= \frac{\mathbf{V_m}}{\pi}(\mathbf{1} + \cos\alpha)
\end{aligned} \tag{1}$$

5. Important Notes

- **Control Range:** For a resistive load, the output voltage can be varied from maximum ($2V_m/\pi$ at $\alpha = 0°$) to minimum (0 V at $\alpha = 180°$).

- **Highly Inductive Load:** If the load is highly inductive (like a DC motor), the load current becomes continuous. The SCRs continue to conduct past π due to the energy stored in the inductor, until the next pair is triggered at $\pi + \alpha$. In that case, the integration limits change to α and $\pi + \alpha$, yielding the modified continuous-conduction equation: $V_{dc} = \frac{2V_m}{\pi} \cos \alpha$.

Q2. Explain SCR and TRIAC operation with suitable examples. **(15 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. Introduction to Thyristors

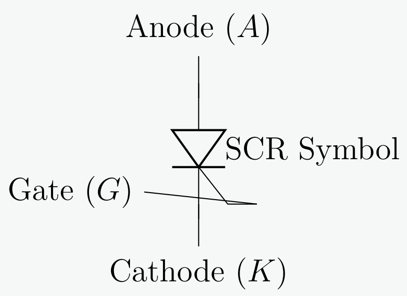
Thyristors are a class of multi-layered semiconductor devices designed for high-power switching applications. Operating purely as bistable switches (transitioning between a non-conducting **OFF** state and a highly conducting **ON** state), they are widely used in power electronics for phase control, AC/DC conversion, and power regulation. The most prominent members of this family are the **Silicon Controlled Rectifier (SCR)** and the **TRIAC**.

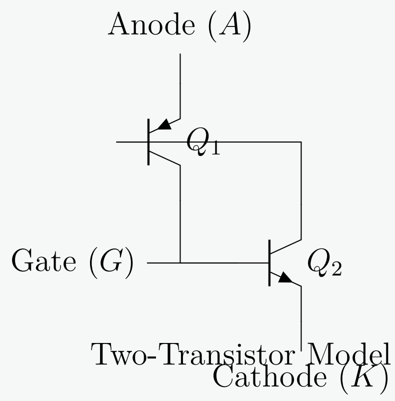
—

2. Silicon Controlled Rectifier (SCR)

2.1 Structure and Operating Principle

An SCR is a unidirectional, three-terminal, four-layer (*p*-*n*-*p*-*n*) semiconductor device. The three terminals are the **Anode (**A**)**, **Cathode (**K**)**, and **Gate (**G**)**.





An SCR operates in three primary modes:

(a) **Forward Blocking Mode (OFF State):** The Anode is biased positive relative to the Cathode ($V_{AK} > 0$), while the gate is left open ($I_G = 0$). Junctions J_1 and J_3 are forward-biased, but the middle junction J_2 is reverse-biased. Only a minuscule forward leakage current flows.

(b) **Forward Conduction Mode (ON State):** Conduction is initiated either by increasing V_{AK} beyond the *Forward Breakover Voltage* (V_{BO}) or, more practically, by applying a positive current pulse to the Gate ($I_G > 0$). This breaks down J_2 through regenerative feedback, driving the SCR into saturation. It remains in this state even after the gate signal is removed, provided the anode current stays above the *Holding Current* (I_H).

(c) **Reverse Blocking Mode (OFF State):** The Cathode is positive relative to the Anode ($V_{AK} < 0$). Junctions J_1 and J_3 are reverse-biased, effectively blocking any current flow except for a tiny reverse leakage current.

2.2 Mathematical Derivation: Two-Transistor Model

The regenerative switching action can be analytically demonstrated using the two-transistor equivalent circuit shown above, which splits the *p*-*n*-*p*-*n* structure into an interconnected PNP transistor (Q_1) and an NPN transistor (Q_2).
For Q_1, the collector current I_{C1} is expressed as:

$$I_{C1} = \alpha_1 I_A + I_{CBO1} \tag{1}$$

where α_1 is the common-base current gain of Q_1, I_A is the anode current (emitter current of Q_1), and I_{CBO1} is the reverse saturation current of the collector-base junction.
Similarly, for Q_2, the collector current I_{C2} is:

$$I_{C2} = \alpha_2 I_K + I_{CBO2} \tag{2}$$

where α_2 is the common-base current gain of Q_2, I_K is the cathode current, and I_{CBO2} is the corresponding reverse saturation current.
From Kirchhoff's Current Law (KCL) applied to the overall system:

$$I_K = I_A + I_G \tag{3}$$

The total anode current I_A is the sum of the collector currents of both transistors:

$$I_A = I_{C1} + I_{C2} \tag{4}$$

Substituting equations (1) and (2) into (4):

$$I_A = \alpha_1 I_A + I_{CBO1} + \alpha_2 I_K + I_{CBO2} \tag{5}$$

Substituting $I_K = I_A + I_G$ into (5):

$$I_A = \alpha_1 I_A + I_{CBO1} + \alpha_2 (I_A + I_G) + I_{CBO2}$$
$$I_A(1 - \alpha_1 - \alpha_2) = \alpha_2 I_G + I_{CBO1} + I_{CBO2}$$

Solving for I_A:

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)} \tag{6}$$

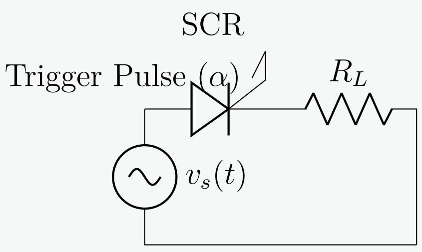
Analysis of Latching Action:

- When $I_G = 0$, the low-current gains α_1 and α_2 are very small ($\alpha_1 + \alpha_2 \ll 1$). Consequently, I_A is limited to a negligible leakage current, keeping the SCR **OFF**.
- When a positive gate current I_G is injected, I_{C2} increases, which in turn raises the base current of Q_1 ($I_{B1} = I_{C2}$). This increases I_{C1}, which further feeds back into the base of Q_2.
- Because the current gains α of BJTs increase with emitter current, this regenerative loop rapidly drives $(\alpha_1 + \alpha_2) \to 1$. As the denominator of equation (7) approaches zero, the anode current I_A surges, limited only by the external load resistance. The SCR is now latched **ON**.

—

2.3 SCR Application Example: Phase-Controlled Half-Wave Rectifier

Consider an SCR used in a single-phase half-wave rectifier circuit driving a resistive load $R_L = 10\ \Omega$. The AC input voltage source is $v_s(t) = 120\sqrt{2}\sin(120\pi t)$ V (which corresponds to 120 V RMS at 60 Hz). The SCR is fired at a constant delay angle $\alpha = 60°$ ($\pi/3$ rad).



Step 1: Determine the peak input voltage (V_m)

$$V_m = 120\sqrt{2} \approx 169.73 \text{ V}$$

Step 2: Calculate the average (DC) output voltage (V_{dc}) The SCR conducts only from $\omega t = \alpha$ to π during the positive half-cycle:

$$V_{dc} = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_m \sin(\theta) d\theta = \frac{V_m}{2\pi}\Big[-\cos\theta\Big]_{\alpha}^{\pi}$$
$$V_{dc} = \frac{V_m}{2\pi}(1 + \cos\alpha)$$

Substituting $V_m = 169.73$ V and $\alpha = 60°$:

$$V_{dc} = \frac{169.73}{2\pi}(1 + \cos 60°) = 27.01 \times (1 + 0.5) \approx \mathbf{40.52} \text{ V}$$

Step 3: Calculate the RMS output voltage (V_{rms})

$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\pi} [V_m \sin(\theta)]^2 \, d\theta} = V_m\sqrt{\frac{1}{4\pi}\int_{\alpha}^{\pi}(1 - \cos 2\theta)d\theta}$$
$$V_{rms} = V_m\sqrt{\frac{1}{4\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_{\alpha}^{\pi}} = V_m\sqrt{\frac{1}{4\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

Converting $\alpha = 60°$ to radians ($\pi/3$ rad):

$$V_{rms} = 169.73\sqrt{\frac{1}{4\pi}\left(\pi - \frac{\pi}{3} + \frac{\sin 120°}{2}\right)}$$
$$V_{rms} = 169.73\sqrt{\frac{1}{4\pi}(2.0944 + 0.4330)}$$
$$V_{rms} = 169.73\sqrt{\frac{2.5274}{12.5664}} \approx 169.73 \times 0.4484 \approx \mathbf{76.11} \text{ V}$$

Step 4: Calculate the average power delivered to the load (P_L)

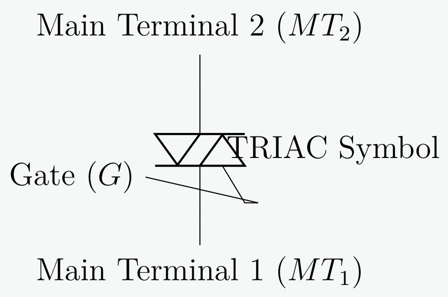
$$P_L = \frac{V_{rms}^2}{R_L} = \frac{(76.11)^2}{10} \approx \mathbf{579.27} \text{ W}$$

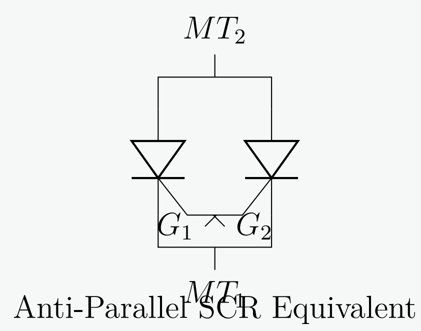
—

3. TRIAC (Triode for Alternating Current)

3.1 Structure and Operating Principle

A TRIAC is a bidirectional, three-terminal semiconductor switch that can conduct current in both directions. It is structurally equivalent to two SCRs connected in an inverse-parallel (anti-parallel) configuration on a single silicon chip, sharing a common gate terminal. The main terminals are labeled **MT1** (Main Terminal 1, reference) and **MT2** (Main Terminal 2), along with the **Gate (G)**.





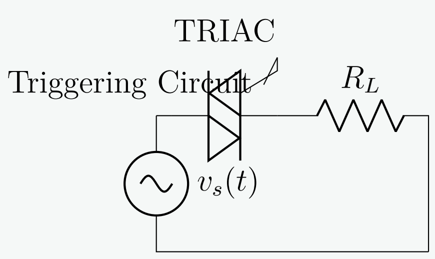
Because of its unique multi-layered architecture, a TRIAC can be triggered into conduction by either a positive or negative gate current, regardless of the voltage polarity across MT_2 and MT_1. This allows operations in **four distinct quadrants**:

Quadrant	MT2 Voltage (w.r.t MT1)	Gate Voltage (w.r.t MT1)	Sensitivity
I (+)	Positive (> 0)	Positive (> 0)	High (Easiest to trigger)
I (-)	Positive (> 0)	Negative (< 0)	Medium
III (-)	Negative (< 0)	Negative (< 0)	High
III (+)	Negative (< 0)	Positive (> 0)	Low (Hardest to trigger)

—

3.2 TRIAC Application Example: Full-Wave AC Phase Control (Light Dimmer)

Consider a TRIAC installed in a domestic light dimmer circuit to control the power delivered to a resistive heating element $R_L = 10\,\Omega$. The AC supply voltage is $v_s(t) = 120\sqrt{2}\sin(120\pi t)$ V. The gate triggering circuit is configured to trigger the TRIAC symmetrically in both the positive and negative half-cycles at a firing angle $\alpha = 60^\circ$ ($\pi/3$ rad).



Step 1: Calculate the RMS output voltage (V_{rms}) Unlike the SCR, which blocks the entire negative half-cycle, the TRIAC conducts symmetrically in both half-cycles. Thus, the mean-square value is evaluated over a single half-cycle (π radians):

$$V_{rms} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} [V_m \sin(\theta)]^2 \, d\theta} = V_m \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\pi} (1 - \cos 2\theta) d\theta}$$

$$V_{rms} = V_m \sqrt{\frac{1}{2\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

Comparing this expression with the SCR half-wave RMS voltage shows:

$$V_{rms,\text{TRIAC}} = \sqrt{2} \times V_{rms,\text{SCR}}$$

Substituting the values calculated previously:

$$V_{rms} = 169.73\sqrt{\frac{2.5274}{2\pi}} = 169.73\sqrt{0.4022} \approx 169.73 \times 0.6342 \approx \mathbf{107.64} \text{ V}$$

Step 2: Calculate the average power delivered to the load (P_L)

$$P_L = \frac{V_{rms}^2}{R_L} = \frac{(107.64)^2}{10} \approx \mathbf{1158.64} \text{ W}$$

Note: Controlling power with a TRIAC provides exactly double the load power of a single SCR at the same firing angle because both half-cycles are utilized.

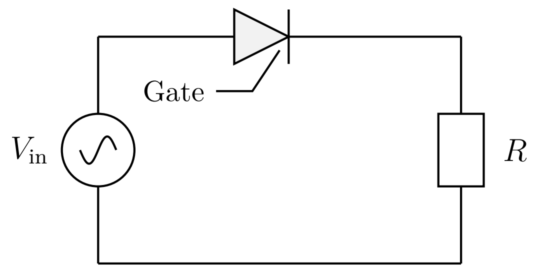
—

4. Comparison: SCR vs. TRIAC

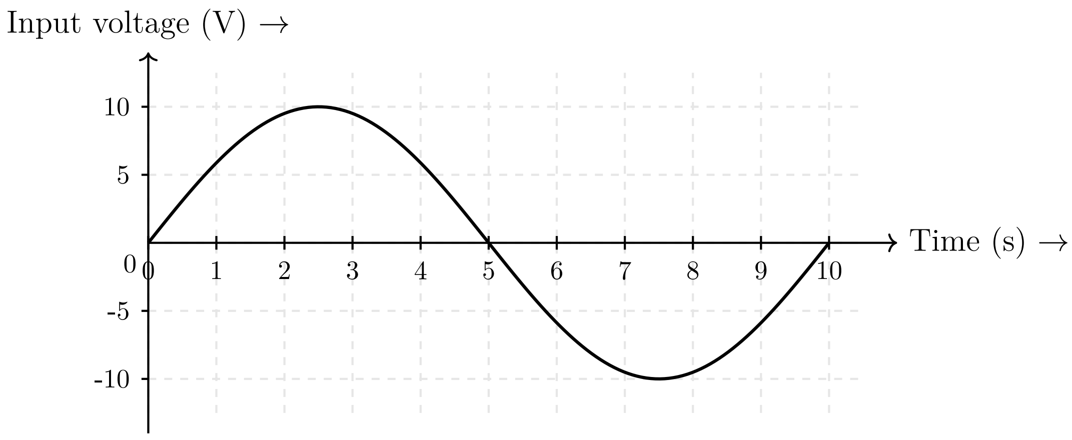
Feature	SCR	TRIAC
Directionality	Unidirectional (DC current flow)	Bidirectional (AC current flow)
Terminals	Anode, Cathode, Gate	Main Terminal 1, Main Terminal 2, Gate
Gate Triggering	Positive pulse only (with positive V_{AK})	Positive or negative pulse
Power Rating	Very High (up to several thousand Amperes)	Moderate to Medium (< 100 A typical)
Symmetry	Asymmetric control	Symmetric control over full AC cycles
Common Applications	Controlled rectifiers, High-voltage DC transmission	Light dimmers, fan regulators, AC motor drives

Q3. Sketch the output wave form of the following circuit given in Fig. for Q. 1(c)(i) assuming output wave is obtained across the resistance. The input waveform and gate pulse sequence is given in Fig. for Q. 1(c)(ii).

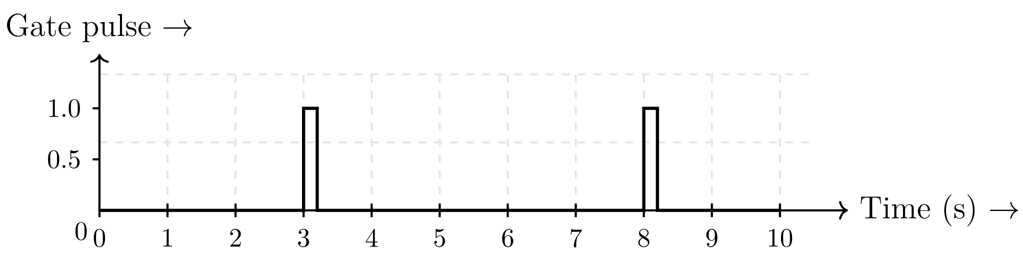
1. Circuit Diagram [Fig. 1(c)(i)]



2. Input Voltage Waveform [Fig. 1(c)(ii)]



3. Gate Pulse Sequence [Fig. 1(c)(ii)]



(20 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Operating Principle of the SCR Circuit

To determine the output voltage waveform across the load resistance R, we must analyze the state of the Silicon Controlled Rectifier (SCR) across different intervals of the input cycle.
The input voltage waveform is given by:

$$V_{\text{in}}(t) = V_m \sin(\omega t) = 10 \sin\left(\frac{2\pi}{T} t\right) \text{ V} \tag{1}$$

Given that one full cycle of the input sine wave has a period of $T = 10$ s, the angular frequency is $\omega = \frac{2\pi}{10} = 0.2\pi$ rad/s. Thus:

$$V_{\text{in}}(t) = 10 \sin(0.2\pi t) \text{ V} \tag{2}$$

For an SCR to switch from the non-conducting (**OFF**) state to the conducting (**ON**) state, two conditions must be satisfied simultaneously:

(a) The SCR must be **forward-biased**, meaning the anode is positive relative to the cathode ($V_{\text{AK}} > 0 \implies V_{\text{in}} > 0$).

(b) A positive triggering pulse must be applied to the **gate** terminal ($I_G > 0$).

Once triggered, the SCR remains **ON** (even if the gate pulse is removed) until its anode current falls below the holding current (I_H), which in an AC circuit occurs when the input voltage goes to zero and reverses polarity (known as **natural/line commutation**).
—

2. Interval-by-Interval Analysis

Interval 1: $0 \text{ s} \le t < 3 \text{ s}$

- The input voltage is positive ($V_{\text{in}} > 0$), which means the SCR is forward-biased.
- However, no gate pulse is applied ($I_G = 0$).
- Consequently, the SCR remains in the **Forward Blocking State (OFF)**.
- The output voltage across the resistor is:

$$V_{\text{out}}(t) = 0 \text{ V} \tag{3}$$

At $t = 3$ s (Firing Instant)

- The input voltage at this instant is:

$$V_{\text{in}}(3) = 10\sin(0.2\pi \times 3) = 10\sin(108^\circ) \approx 9.51 \text{ V} > 0 \tag{4}$$

  Since $V_{\text{in}}(3) > 0$, the SCR is forward-biased.
- At this exact instant, a gate pulse is applied.
- Since both conditions (forward bias and gate trigger) are met, the SCR triggers **ON** instantly.
- Assuming an ideal SCR with a negligible forward voltage drop ($V_F = 0$ V), the voltage across the SCR drops to 0 V. The output voltage instantly jumps from 0 V to the input voltage:

$$V_{\text{out}}(3) \approx 9.51 \text{ V} \tag{5}$$

Interval 2: $3 \text{ s} \le t < 5 \text{ s}$

- The SCR remains in the **Forward Conduction State (ON)**.
- The output voltage follows the input voltage profile:

$$V_{\text{out}}(t) = V_{\text{in}}(t) = 10\sin(0.2\pi t) \text{ V} \tag{6}$$

- Note that since the peak of the input voltage (10 V) occurred at $t = 2.5$ s (while the SCR was still OFF), the output voltage starts at 9.51 V at $t = 3$ s and decreases continuously following the sinusoidal path down to 0 V at $t = 5$ s.

At $t = 5$ s (Natural Commutation)

- The input voltage becomes zero ($V_{\text{in}} = 0$ V) and begins to transition to negative values.
- The anode current drops to zero (below the holding current I_H), causing the SCR to turn **OFF** naturally.

Interval 3: $5 \text{ s} \le t < 8 \text{ s}$

- The input voltage is negative ($V_{\text{in}} < 0$), putting the SCR in a reverse-biased condition.
- No gate pulse is applied.
- The SCR remains in the **Reverse Blocking State (OFF)**.
- The output voltage is:

$$V_{\text{out}}(t) = 0 \text{ V} \tag{7}$$

At $t = 8$ s (Gate Pulse under Reverse Bias)

- A gate pulse is applied at $t = 8$ s.
- However, the input voltage at this instant is negative:

$$V_{\text{in}}(8) = 10\sin(0.2\pi \times 8) = 10\sin(288^\circ) \approx -9.51 \text{ V} < 0 \tag{8}$$

- Since the SCR is reverse-biased, the gate pulse cannot trigger the device. The SCR remains in the **Reverse Blocking State (OFF)**.
- The output voltage remains at:

$$V_{\text{out}}(8) = 0 \text{ V} \tag{9}$$

Interval 4: $8 \text{ s} \leq t \leq 10 \text{ s}$

- The input voltage finishes its negative half-cycle and goes back to zero.
- The SCR remains **OFF**.
- The output voltage is:

$$V_{\text{out}}(t) = 0 \text{ V} \tag{10}$$

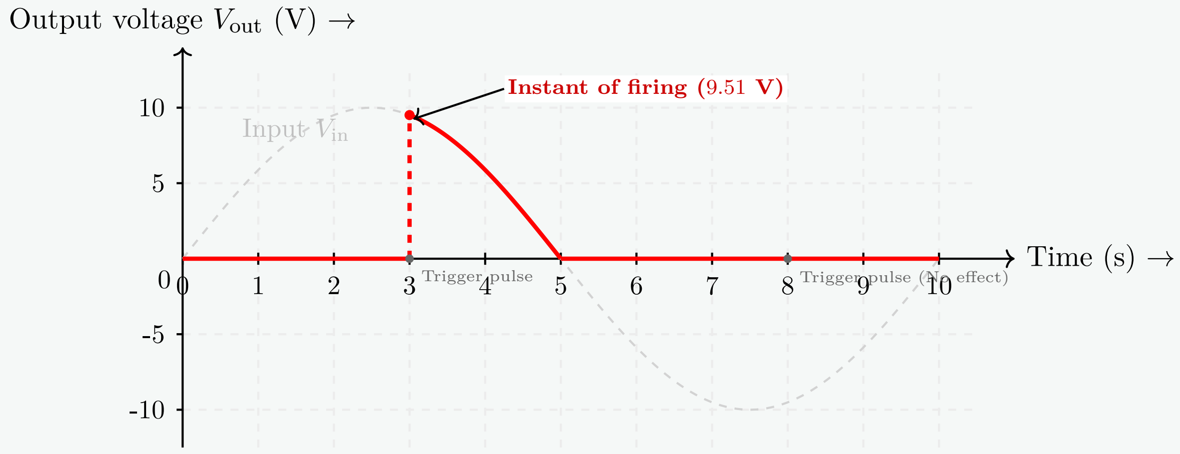
—

3. Output Voltage Waveform Sketch

The derived mathematical definition of the output voltage is:

$$V_{\text{out}}(t) = \begin{cases} 0 \text{ V} & 0 \text{ s} \leq t < 3 \text{ s} \\ 10\sin(0.2\pi t) \text{ V} & 3 \text{ s} \leq t < 5 \text{ s} \\ 0 \text{ V} & 5 \text{ s} \leq t \leq 10 \text{ s} \end{cases} \tag{11}$$

Below is the sketch of the output waveform across the load resistor R.



Q4. Draw the I-V characteristics of an SCR and a TRIAC. **(10 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Silicon Controlled Rectifier (SCR)

1.1 Overview and Operating Principle

The Silicon Controlled Rectifier (SCR) is a four-layer (p-n-p-n), three-junction, three-terminal unidirectional semiconductor device. The three terminals are the Anode (A), Cathode (K), and Gate (G). It acts as a controlled switch that permits conduction only in the forward direction (from Anode to Cathode) once triggered by a positive gate current pulse.

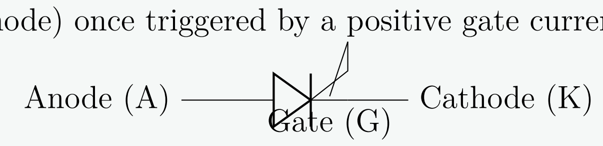


Figure 1: Schematic Symbol of a Silicon Controlled Rectifier (SCR)

1.2 I-V Characteristics of an SCR

The current-voltage (I-V) relationship of an SCR consists of three main regions of operation:

(a) **Forward Blocking Region ($0 < V_{AK} < V_{BO}$ with $I_G = 0$):** The anode is positive relative to the cathode, but since no gate current is applied, the device remains in an OFF state, allowing only a small forward leakage current to flow.

(b) **Forward Conduction Region ($V_{AK} > V_H$):** Once the anode-to-cathode voltage exceeds the forward breakover voltage (V_{BO}) or a gate current pulse (I_G) is applied, the SCR switches to its ON state. The voltage drop across the device falls to a very low value (holding voltage, $V_H \approx 1$ to 1.5 V), and the current is limited only by the external load resistance.

(c) **Reverse Blocking Region ($-V_{BR} < V_{AK} < 0$):** When the cathode is positive relative to the anode, junctions J_1 and J_3 are reverse-biased. The device remains OFF and permits only a tiny reverse leakage current.

(d) **Reverse Breakdown Region ($V_{AK} < -V_{BR}$):** If the reverse voltage exceeds the critical reverse breakdown voltage (V_{BR}), avalanching occurs, leading to a rapid increase in reverse current and potential device failure if not externally limited.

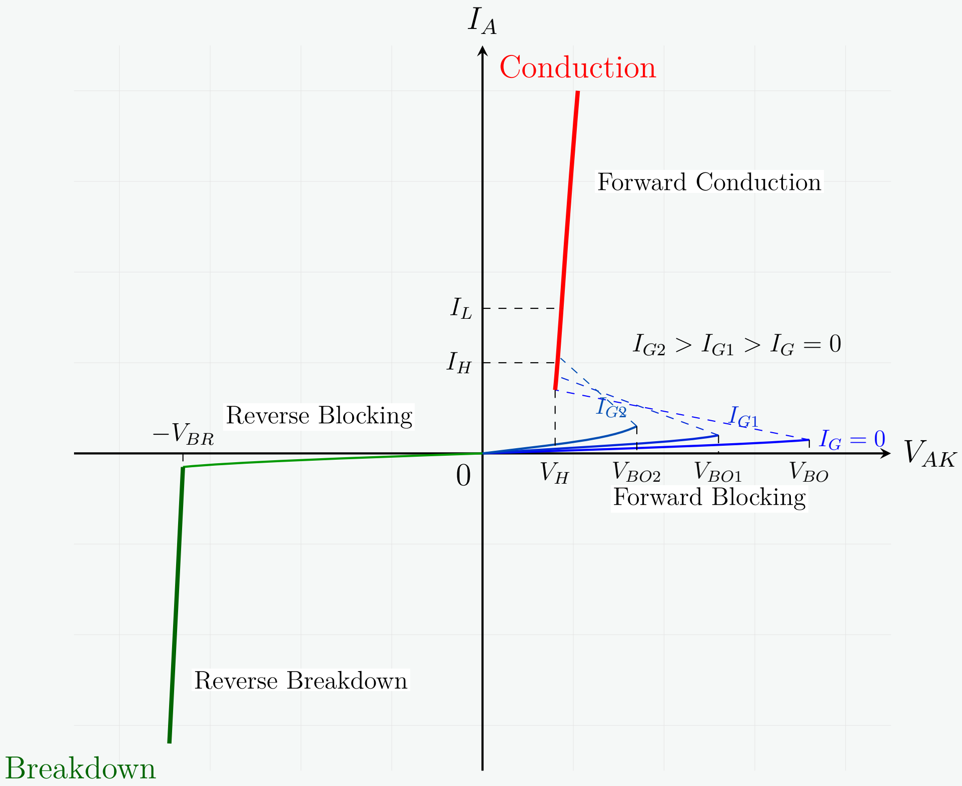


Figure 2: I-V Characteristics of a Silicon Controlled Rectifier (SCR)

1.3 Key Parameters of the SCR Curve

- **Latching Current (I_L):** The minimum anode current required to transition the SCR from the blocking (OFF) state to the conduction (ON) state and maintain it there immediately after the gate trigger pulse is removed.
- **Holding Current (I_H):** The minimum anode current required to keep the SCR in the conduction state. If the forward current drops below I_H, the device returns to its forward blocking state. Structurally, $I_L > I_H$.
- **Forward Breakover Voltage (V_{BO}):** The maximum forward voltage that the SCR can withstand without conducting when the gate terminal is open ($I_G = 0$).

2. TRIAC (Triode for Alternating Current)

2.1 Overview and Operating Principle

A TRIAC is a bidirectional, three-terminal semiconductor device that can conduct current in both directions when triggered. It can be modeled as two SCRs connected in an inverse-parallel (antiparallel) configuration with a unified gate circuit. The terminals are labeled Main Terminal 1 (MT1), Main Terminal 2 (MT2), and Gate (G). MT1 serves as the measurement reference point.

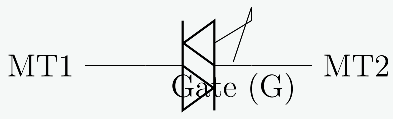


Figure 3: Schematic Symbol of a TRIAC

2.2 I-V Characteristics of a TRIAC

Because a TRIAC is designed for AC applications, its I-V characteristics are symmetrical across Quadrant I and Quadrant III:

(a) **Quadrant I Operation:** MT2 is positive relative to MT1. The characteristics match those of a standard SCR, allowing forward blocking and forward conduction upon triggering.

(b) **Quadrant III Operation:** MT2 is negative relative to MT1. The characteristics are completely symmetrical to Quadrant I, meaning the device blocks reverse voltage until triggered, after which it enters reverse conduction.

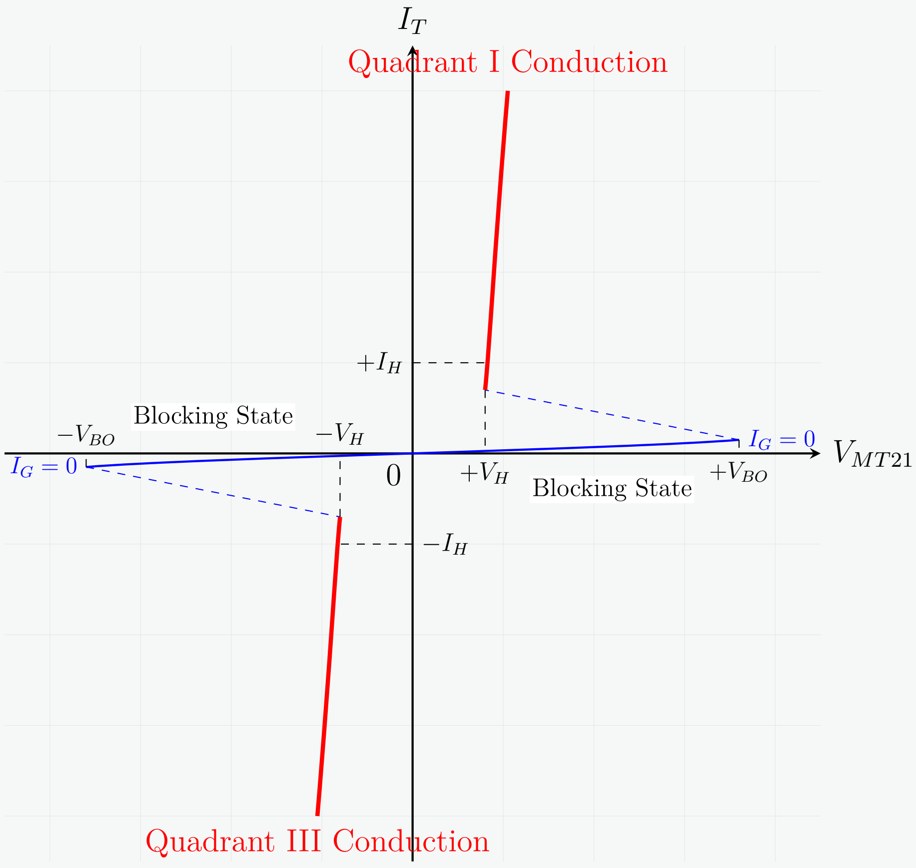


Figure 4: I-V Characteristics of a TRIAC

2.3 Triggering Modes of a TRIAC

Because both MT2 and the gate can be biased either positively or negatively with respect to MT1, a TRIAC can operate in four distinct triggering quadrants (modes), summarized below in order of standard gate sensitivity:

Mode	Quadrant	MT2 Voltage (V_{MT21})	Gate Voltage (V_{G1})	Gate Sensitivity
I^+	I	Positive (> 0)	Positive (> 0)	**Highest** (Very easy to trigger)
III^-	III	Negative (< 0)	Negative (< 0)	**High** (Very easy to trigger)
I^-	I	Positive (> 0)	Negative (< 0)	**Medium** (Requires higher gate current)
III^+	III	Negative (< 0)	Positive (> 0)	**Lowest** (Generally avoided in design)

Table 1: Triggering Modes and Sensitivities of a TRIAC

Q5. Explain the operating principle of a Silicon controlled Rectifier with the help of 'Two transistor model'. **(20 marks)** [EEE 263 | L-2/T-1 | 2011–12]

Answer

Two-Transistor Model of an SCR

1. Introduction and Conceptual Splitting

The operating principle of a Silicon Controlled Rectifier (SCR) can be modeled and analyzed using the **two-transistor analogy**. An SCR is a four-layer P_1-N_1-P_2-N_2 semiconductor device. By conceptually splitting the middle layers (N_1 and P_2) diagonally or horizontally, the structure can be viewed as two interconnected complementary transistors:

- A **PNP transistor** (Q_1) formed by the layers P_1-N_1-P_2.
- An **NPN transistor** (Q_2) formed by the layers N_1-P_2-N_2.

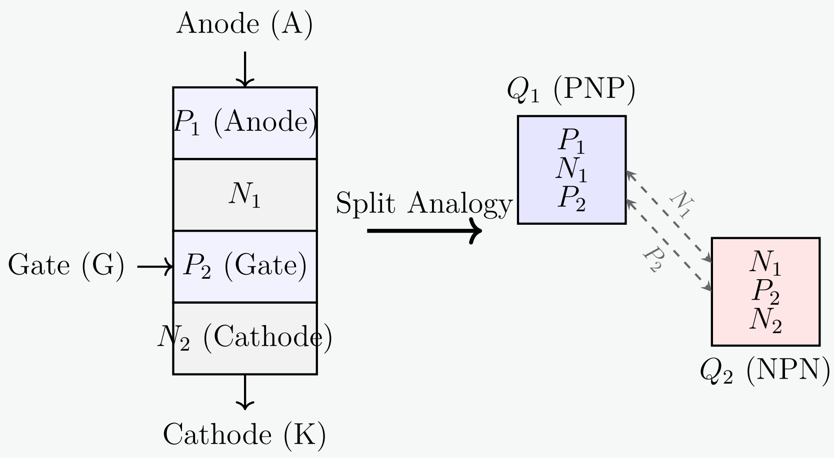


Figure 5: Conceptual splitting of the 4-layer SCR into PNP and NPN structures

2. Equivalent Circuit Schematic

In this interconnected model:

- The Emitter of Q_1 is connected to the external **Anode (A)**.
- The Emitter of Q_2 is connected to the external **Cathode (K)**.
- The Base of Q_1 is tied to the Collector of Q_2 (forming the internal N_1 region).
- The Collector of Q_1 is tied to the Base of Q_2 and the external **Gate (G)** (forming the internal P_2 region).

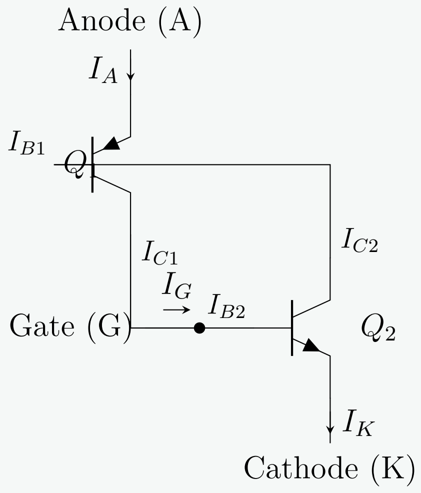


Figure 6: Two-Transistor Equivalent Schematic of an SCR

3. Mathematical Derivation of Anode Current

Let us define the parameters for both transistors:

- α_1, α_2: Common-base current gains of Q_1 and Q_2, respectively.
- I_{CO1}, I_{CO2}: Reverse collector-to-base leakage currents of Q_1 and Q_2.

For Q_1 (PNP), the collector current I_{C1} is expressed as:

$$I_{C1} = \alpha_1 I_{E1} + I_{CO1} \tag{1}$$

Since the emitter of Q_1 is connected directly to the anode, $I_{E1} = I_A$:

$$I_{C1} = \alpha_1 I_A + I_{CO1} \tag{2}$$

Similarly, for Q_2 (NPN), the collector current I_{C2} is:

$$I_{C2} = \alpha_2 I_{E2} + I_{CO2} \tag{3}$$

Since the emitter of Q_2 is connected to the cathode, $I_{E2} = I_K$:

$$I_{C2} = \alpha_2 I_K + I_{CO2} \tag{4}$$

From Kirchhoff's Current Law (KCL) applied to the combined system, the total current entering the anode must exit through the base-collector junctions:

$$I_A = I_{C1} + I_{C2} \tag{5}$$

Furthermore, the cathode current I_K is the sum of the anode current and the gate current:

$$I_K = I_A + I_G \tag{6}$$

Substituting Equations (2), (4), and (6) into Equation (5):

$$I_A = (\alpha_1 I_A + I_{CO1}) + (\alpha_2(I_A + I_G) + I_{CO2})$$
$$I_A = (\alpha_1 + \alpha_2)I_A + \alpha_2 I_G + (I_{CO1} + I_{CO2})$$

Grouping the I_A terms together:

$$I_A\left[1 - (\alpha_1 + \alpha_2)\right] = \alpha_2 I_G + I_{CO} \tag{7}$$

where $I_{CO} = I_{CO1} + I_{CO2}$ is the total reverse saturation leakage current of the central blocking junction.
Solving for the anode current I_A:

$$I_A = \frac{\alpha_2 I_G + I_{CO}}{1 - (\alpha_1 + \alpha_2)} \tag{8}$$

4. Analysis of SCR Operation States

Using the derived relationship in Equation (8), we can analyze the two primary states of the SCR:

A. OFF-State (Forward Blocking)

When the gate current is zero ($I_G = 0$):

$$I_A = \frac{I_{CO}}{1 - (\alpha_1 + \alpha_2)} \tag{9}$$

For silicon transistors operating at low currents, the current gains α_1 and α_2 are extremely small ($\alpha \ll 1$). Consequently:

$$(\alpha_1 + \alpha_2) \ll 1 \implies I_A \approx I_{CO} \tag{10}$$

In this condition, only a tiny reverse saturation leakage current flows, and the SCR remains in its high-impedance OFF (forward blocking) state.

B. Regenerative Turn-on Process (ON-State)

When a positive trigger current I_G is injected into the gate:

(a) I_G acts as the base current I_{B2} of Q_2, causing an increase in collector current I_{C2}.

(b) Since the collector of Q_2 is tied to the base of Q_1, I_{C2} increases I_{B1}, turning Q_1 ON.

(c) Turning Q_1 ON increases I_{C1}.

(d) Because the collector of Q_1 is connected back to the base of Q_2, I_{C1} flows into the base of Q_2, further increasing I_{B2} even if the external gate signal I_G is removed.

This forms a closed **positive feedback loop** (regenerative action). As internal currents rise rapidly, the common-base current gains α_1 and α_2 of both silicon transistors increase toward their maximum values. As the sum of the gains approaches unity:

$$(\alpha_1 + \alpha_2) \to 1 \implies [1 - (\alpha_1 + \alpha_2)] \to 0 \tag{11}$$

According to Equation (8), the denominator approaches zero, causing the anode current I_A to surge abruptly. The device enters its low-impedance ON (forward conduction) state, limited only by the external load resistance. Once latched, the gate loses control, and the SCR remains ON.

C. Turn-off (Commutation)

To turn the SCR OFF, the external anode current I_A must be reduced below the characteristic **holding current** (I_H). This drop in current reduces the internal emitter currents, causing α_1 and α_2 to decrease. Once $(\alpha_1 + \alpha_2) < 1$, the regenerative feedback loop can no longer sustain itself, and the SCR transitions back to its blocking state.

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

S2 — BJT Characteristics

Energy band profiles, DC analysis, biasing stability, circuit design

S2 BJT Characteristics

Energy Band Profile

Q1. Draw the energy band profile of a pnp transistor under zero bias and forward-active mode bias conditions. **(16/23 marks)** [EEE 263 | L-2/T-1 | 2020–21, 2018–19]

Answer

1. Principle of PNP Energy Band Diagrams

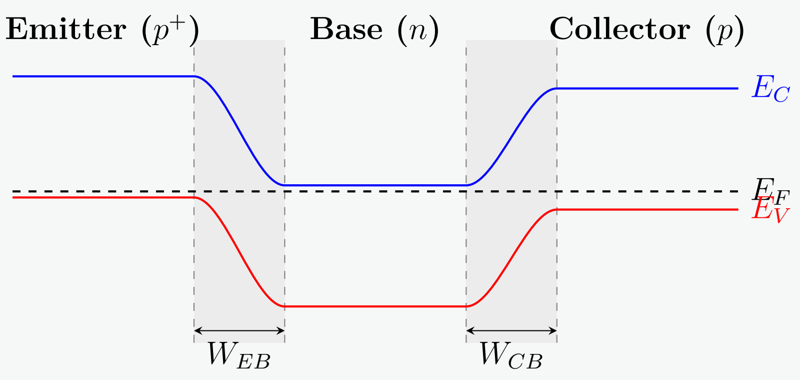
A **PNP bipolar junction transistor (BJT)** consists of three semiconductor regions: a heavily doped p^+-type Emitter, a narrow, lightly doped n-type Base, and a moderately doped p-type Collector.

- In a p-type region, the Fermi energy level (E_F) is located closer to the valence band (E_V).
- In an n-type region, the Fermi energy level is located closer to the conduction band (E_C).
- The built-in potential across the junctions causes the energy bands to bend. Because electron energy is plotted on the vertical axis, the bands in the p-type regions are at a higher energy level relative to the n-type region.

2. Zero Bias Condition (Thermal Equilibrium)

Under zero bias, no external voltage is applied to any terminal.

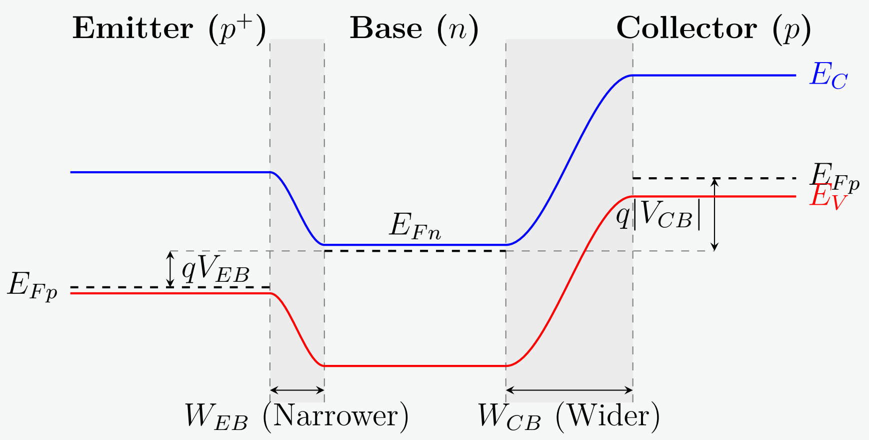
- The system is in thermal equilibrium, meaning the **Fermi level (E_F) is strictly constant** and horizontal across the entire device.
- Built-in potential barriers exist at both the Emitter-Base (E-B) and Base-Collector (B-C) junctions.
- The barrier prevents the majority carrier holes in the Emitter from diffusing into the Base.



3. Forward-Active Mode Bias Condition

In the forward-active region, the **Emitter-Base junction is forward-biased** ($V_{EB} > 0$) and the **Base-Collector junction is reverse-biased** ($V_{CB} < 0$).

- **E-B Junction (Forward Bias):** Applying a positive voltage to the Emitter lowers its electron energy by qV_{EB} relative to the Base. The potential barrier for holes is **reduced**, allowing them to easily diffuse into the Base. The depletion width W_{EB} **decreases**.
- **B-C Junction (Reverse Bias):** Applying a negative voltage to the Collector raises its electron energy by $q|V_{CB}|$ relative to the Base. The potential barrier is **increased**, which efficiently sweeps the holes that traverse the Base into the Collector. The depletion width W_{CB} **increases**.
- The Fermi level splits into distinct **quasi-Fermi levels** (E_{Fp} for holes and E_{Fn} for electrons).



Device Structure and Operating Principle

Q1. Why is the third terminal (Base or Gate) used in a transistor? **(10 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

Answer

1. The Fundamental Principle: Controlled Valve Mechanism

The primary reason a transistor requires a third terminal—the **Base** in a Bipolar Junction Transistor (BJT) or the **Gate** in a Field-Effect Transistor (FET)—is to serve as a **control terminal**.
A two-terminal device (like a diode or resistor) can only respond directly to the voltage applied across its own two terminals; it possesses no inherent mechanism to modulate its own conductivity dynamically via an external agent. By introducing a third terminal, the transistor separates the **power path** (Collector-to-Emitter or Drain-to-Source) from the **control path** (Base-to-Emitter or Gate-to-Source). This allows a small electrical signal at the third terminal to regulate a much larger flow of current through the other two terminals, mimicking an electronically adjustable valve.

2. Role of the Third Terminal by Transistor Type

A. Bipolar Junction Transistor (BJT) - Base Terminal In an NPN BJT, the third terminal (Base) is designed to be physically extremely thin and lightly doped compared to the heavily doped Emitter and moderately doped Collector.

- When a small forward-bias voltage ($V_{BE} \approx 0.7$ V) is applied to the Base-Emitter junction, it injects a small base current (I_B).
- Because the base region is so thin, the vast majority of electrons injected from the Emitter pass straight through the Base and are swept up by the electric field into the Collector.
- The third terminal acts as a current-controlled valve governed by the fundamental relationship:

$$I_C = \beta I_B \tag{1}$$

Where β (typically $100 \sim 300$) represents the current amplification factor.

B. Field-Effect Transistor (FET) - Gate Terminal In a MOSFET, the third terminal (Gate) is completely insulated from the conducting channel by a microscopic layer of silicon dioxide (SiO_2).

- Because of this insulation, the Gate draws virtually zero DC current ($I_G \approx 0$), presenting an extremely high input impedance.
- Instead of current, the voltage applied to the Gate (V_{GS}) creates an electric field perpendicular to the semiconductor surface. This field modulates the thickness and carrier concentration of the inversion layer (the channel) beneath it.
- The third terminal acts as a voltage-controlled valve governed by the square-law relationship in the saturation region:

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{th})^2 \tag{2}$$

3. Why Multi-Terminal Isolation is Indispensable

Without this isolated third terminal, the two primary electronic applications would be mathematically and physically impossible:

(a) **Amplification (Gain):** To achieve voltage, current, or power gain, a small-power input signal must control a high-power independent DC source. The third terminal provides the isolation necessary so that the weak input signal is not overwhelmed or loaded down by the large output currents and voltages.

(b) **Digital Switching:** In digital logic circuits, a transistor acts as an on/off switch. The third terminal acts as the physical actuator or "toggle switch." A low voltage (Logic 0) or high voltage (Logic 1) at the Gate/Base switches the main conduction channel between an open circuit (cutoff) and a short circuit (saturation), enabling the construction of logic gates (AND, OR, NOT) that form the basis of modern computing.

Early Effect

Q1. What is the Early-effect in a BJT? Draw the I–V curve of a practical BJT showing the Early effect. **(10 marks)** [EEE 263 | L-2/T-1 | 2023–24]

Answer

1. The Early Effect in a BJT

The **Early Effect** (also known as **Base-Width Modulation**) in a Bipolar Junction Transistor (BJT) refers to the phenomenon where the effective neutral width of the base region (W_B) decreases as the reverse-bias voltage across the base-collector (B-C) junction increases.
Mechanism:

- In the forward-active region of operation, the emitter-base (E-B) junction is forward-biased, while the B-C junction is reverse-biased by the collector-emitter voltage (V_{CE}).
- As V_{CE} increases, the reverse bias across the B-C junction also increases.
- This larger reverse bias widens the depletion region at the B-C junction. Because the base is typically lightly doped compared to the highly doped collector, the depletion region extends much further into the base than into the collector.

- Consequently, the effective (neutral) width of the base, W_B, is reduced.

Consequences on Circuit Operation:

(a) **Increased Collector Current (I_C):** A narrower base results in a steeper minority carrier concentration gradient across the base. This increases the diffusion current, thereby causing the collector current I_C to rise slightly with an increase in V_{CE}, rather than remaining perfectly constant.

(b) **Finite Output Resistance (r_o):** Because I_C increases with V_{CE}, the BJT exhibits a finite output resistance, causing the I_C–V_{CE} characteristic curves to slope upward in the active region.

2. Mathematical Representation

To account for the Early effect, the standard Ebers-Moll equation for the collector current is modified by a linear factor:

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)\left(1 + \frac{V_{CE}}{V_A}\right)$$

where:

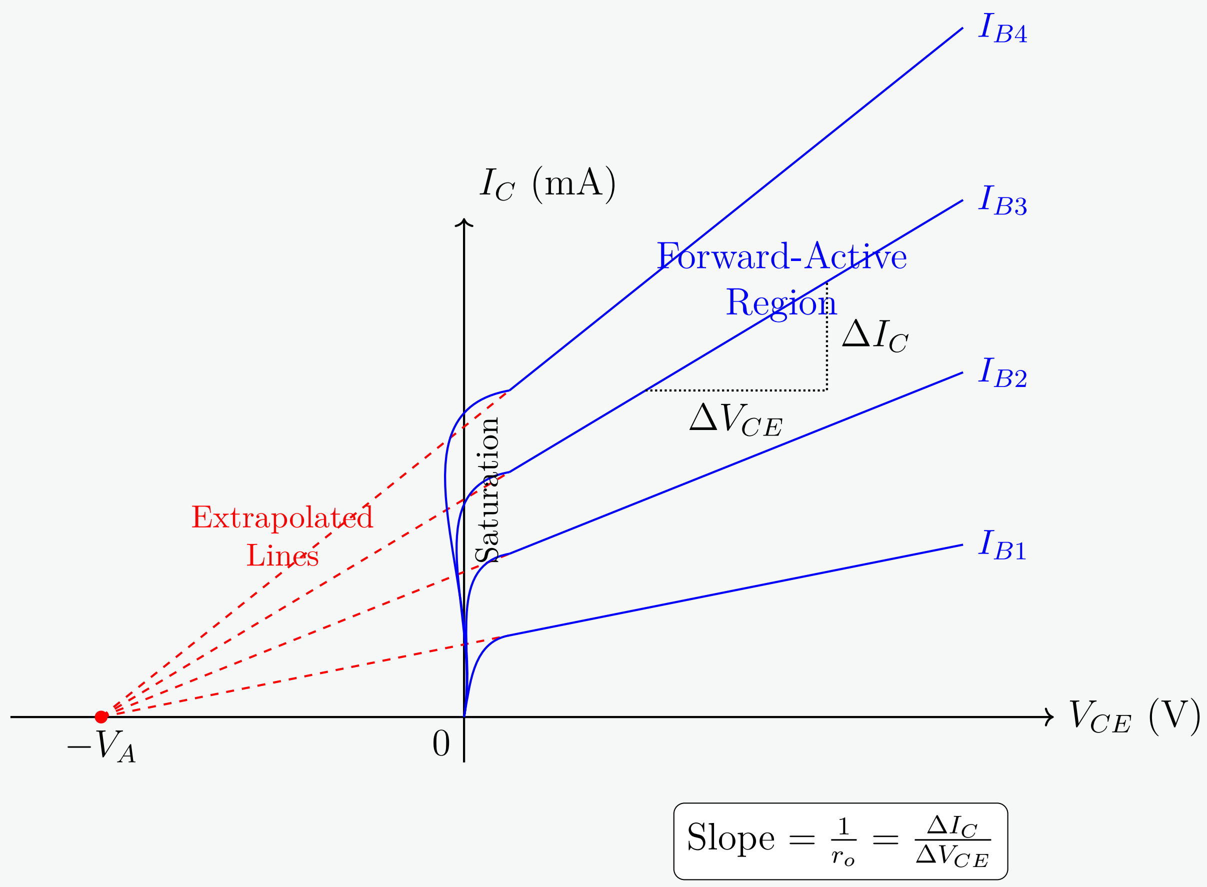
- I_S is the reverse saturation current.
- V_{BE} is the base-emitter voltage.
- V_T is the thermal voltage ($\approx$ 26 mV at room temperature).
- V_A is the **Early Voltage**, a positive constant representing the theoretical point on the negative V_{CE} axis where the extrapolated active-region curves converge.

The finite small-signal output resistance (r_o) is defined as the inverse of the slope of the I_C–V_{CE} curve:

$$r_o = \left(\frac{\partial I_C}{\partial V_{CE}}\right)^{-1} \approx \frac{V_A + V_{CE}}{I_C} \approx \frac{V_A}{I_C}$$

3. Practical I–V Characteristic Curve Showing the Early Effect

When the active region characteristic curves of I_C versus V_{CE} (for different constant values of I_B) are extrapolated backward in a straight line, they intersect the negative V_{CE} axis at a single common point, $-V_A$.



Q2. What is the early effect for BJT? What are the consequences of early effect on BJT behavior? **(16 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

1. The Early Effect (Base-Width Modulation)

The **Early Effect** in a Bipolar Junction Transistor (BJT) refers to the phenomenon where the effective (neutral) width of the base region, W_B, decreases as the reverse-bias voltage across the base-collector (B-C) junction increases.
In the forward-active region, the emitter-base (E-B) junction is forward-biased, and the B-C junction is reverse-biased. An increase

in the collector-emitter voltage (V_{CE}) increases the reverse bias across the B-C junction. Because the base is typically lightly doped compared to the heavily doped collector, the widening depletion region extends predominantly into the base. This effectively narrows the active neutral base region.

2. Consequences of the Early Effect on BJT Behavior

The reduction in the effective base width due to the Early effect has several profound consequences on the device's electrical characteristics:

- **Increase in Collector Current (I_C):** The minority carrier concentration in the base drops to zero at the edge of the B-C depletion region. As the base width narrows, the concentration gradient of minority carriers becomes steeper. According to Fick's Law of Diffusion, a steeper gradient results in a higher diffusion current. Consequently, I_C is not constant but increases slightly with an increase in V_{CE}.
- **Finite Output Resistance (r_o):** In an ideal BJT, I_C is independent of V_{CE} in the active region (infinite output resistance). Due to the Early effect, the I_C–V_{CE} characteristic curves slope upward. This slope manifests as a finite small-signal output resistance, r_o, which slightly degrades the voltage gain of BJT amplifier circuits.
- **Increase in Current Gain (α and β):** A narrower base means that minority carriers spend less time traversing the base, reducing the probability of recombination. This slightly increases the transport factor, leading to a marginal increase in the common-base current gain (α) and the common-emitter current gain (β).
- **Punch-Through (Reach-Through) Breakdown:** If V_{CE} is increased to extreme levels, the B-C depletion region may extend across the entire base and touch the E-B depletion region ($W_B \to 0$). This condition, known as punch-through, results in a massive uncontrolled flow of current between the emitter and collector, effectively destroying transistor action.

3. Mathematical Representation

To mathematically account for the Early effect, the standard Ebers-Moll equation for collector current is modified by a linear factor involving the **Early Voltage** (V_A):

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)\left(1 + \frac{V_{CE}}{V_A}\right)$$

where:

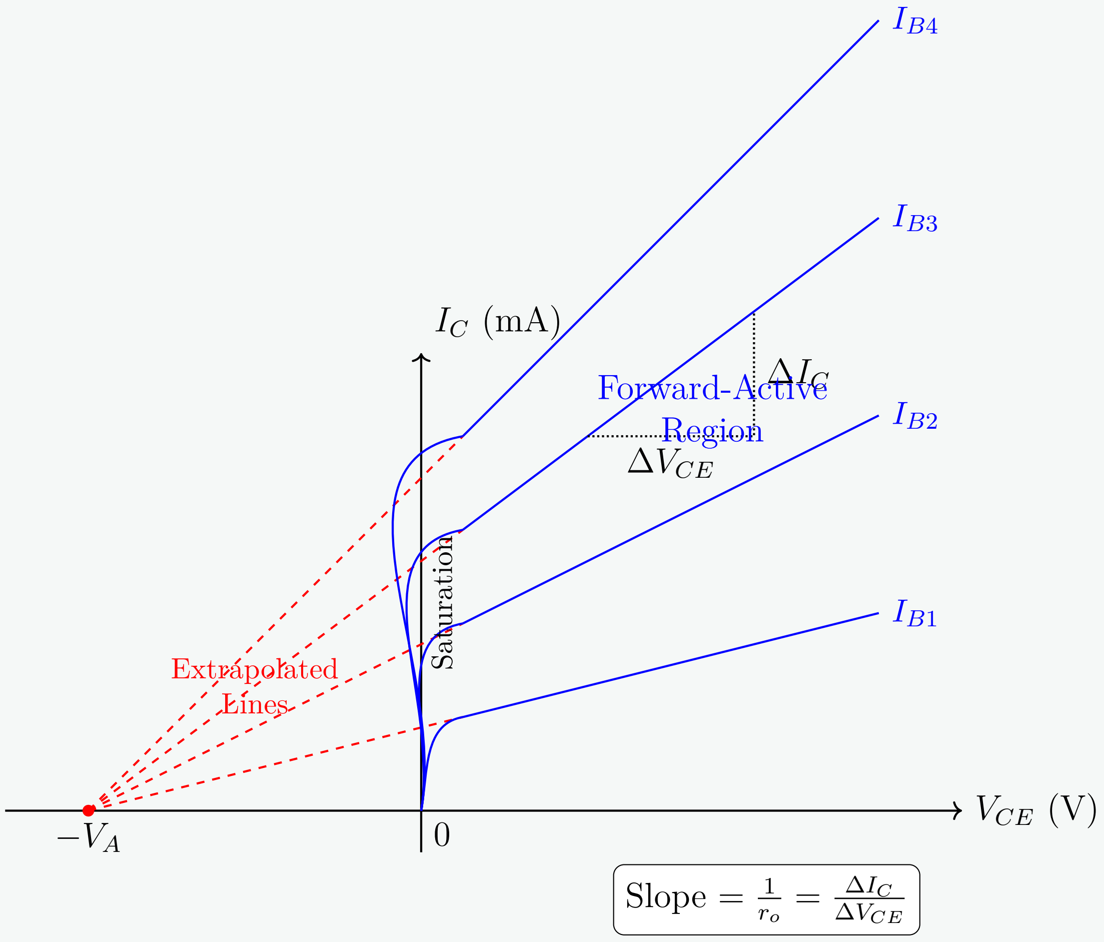
- V_A is the Early Voltage, a theoretical positive constant (typically 50 V to 150 V for practical BJTs).
- V_T is the thermal voltage ($\approx$ 26 mV at 300 K).

The finite small-signal output resistance is derived as the reciprocal of the slope of the I_C–V_{CE} curve:

$$r_o = \left(\frac{\partial I_C}{\partial V_{CE}}\right)^{-1} = \frac{V_A + V_{CE}}{I_C} \approx \frac{V_A}{I_C} \quad (\text{assuming } V_{CE} \ll V_A)$$

4. Graphical Representation

When the active-region characteristics of I_C versus V_{CE} are extrapolated backward in a straight line, they converge and intersect the negative V_{CE} axis at a single point, $-V_A$.



Q3. Why biasing is needed for an amplifier using BJT? Draw the output characteristics of a BJT connected in CE configuration. Define Early effect. **(10.667 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

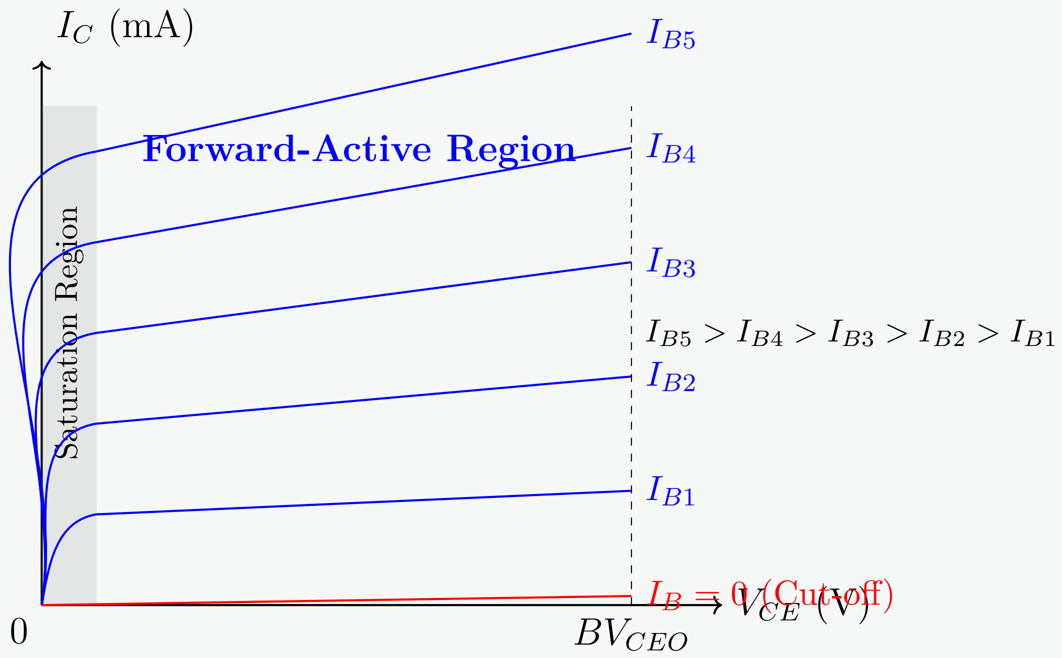
1. Need for Biasing in a BJT Amplifier

In a BJT amplifier circuit, **biasing** refers to the application of external DC voltages to establish a steady state of currents and voltages within the transistor. This steady state is known as the **Quiescent Point (Q-point)** or operating point. Biasing is strictly necessary for the following reasons:

- **Ensuring Forward-Active Operation:** For a BJT to act as a linear amplifier, it must operate strictly in the forward-active region (Emitter-Base junction forward-biased, Base-Collector junction reverse-biased). Biasing sets the initial DC levels to maintain this condition.
- **Preventing Signal Clipping:** An AC input signal causes the transistor's current and voltage to swing above and below their DC quiescent values. If the Q-point is not placed near the center of the DC load line, large signal swings will push the transistor into the *cut-off* or *saturation* regions, leading to severe distortion and clipping of the output waveform.
- **Thermal Stability:** The BJT parameters, specifically the reverse saturation current (I_{CO}), base-emitter voltage (V_{BE}), and current gain (β), are highly sensitive to temperature. Proper biasing networks (such as voltage-divider biasing with emitter degeneration) stabilize the Q-point against thermal variations, preventing thermal runaway and subsequent device destruction.
- **Device Variation Tolerance:** Due to manufacturing tolerances, two BJTs of the same model can have significantly different β values. A well-designed bias circuit makes the operating point relatively independent of the exact value of β.

2. Output Characteristics of a BJT in CE Configuration

The common-emitter (CE) output characteristics plot the collector current (I_C) versus the collector-emitter voltage (V_{CE}) for various constant values of base current (I_B).



3. Definition of the Early Effect

The **Early Effect** (also known as **Base-Width Modulation**) is the phenomenon in a BJT where the effective, neutral width of the base region (W_B) decreases as the reverse-bias voltage across the base-collector (B-C) junction increases.

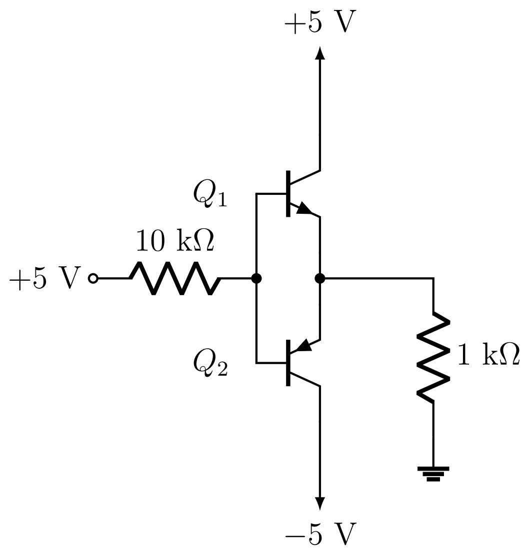
Mechanism and Consequences:

As the collector-emitter voltage (V_{CE}) increases in the active region, the reverse bias across the B-C junction increases. This causes the depletion region at the B-C junction to widen. Because the base is typically lightly doped compared to the heavily doped collector, this depletion region extends primarily into the base.

- **Reduction of Base Width:** The penetration of the depletion layer reduces the effective electrical width of the base.
- **Increase in Collector Current:** A narrower base creates a steeper minority carrier concentration gradient across it. By Fick's Law of diffusion, this steeper gradient increases the diffusion current, meaning I_C rises slightly with an increase in V_{CE}.
- **Finite Output Resistance:** The upward slope in the active region of the CE output characteristics (visible in the diagram above) represents a finite small-signal output resistance, r_o. If these active region lines are extrapolated backwards, they converge at a common point on the negative V_{CE} axis, known as the **Early Voltage** ($-V_A$).

DC Analysis

Q1. For the circuit shown, find the voltages at all nodes and the currents through all branches. Assume $\beta = 100$.



(20 marks) [EEE 263 | L-2/T-1 | 2020–21]

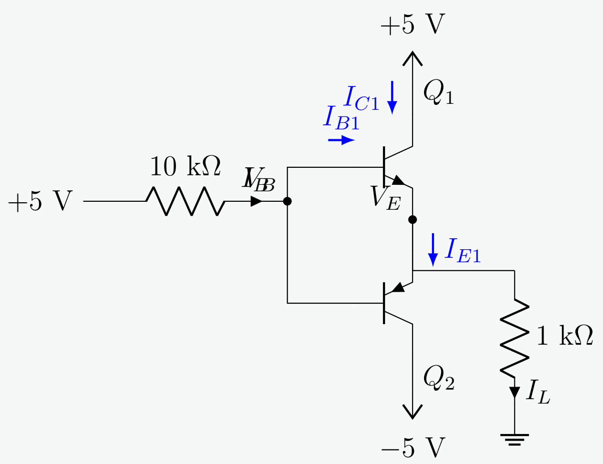
Answer

Circuit Analysis and Assumption

The given circuit is a complementary-symmetry (push-pull) configuration biased by a single +5 V DC source at the input.

- Q_1 is an NPN transistor, and Q_2 is a PNP transistor.
- We assume standard active-region forward voltage drop: $V_{BE(on)} \approx 0.7$ V.
- Since a positive +5 V is applied to the common base terminal through the 10 kΩ resistor, it tends to pull the base voltage V_B positive.
- A positive base voltage forward-biases the base-emitter junction of the NPN transistor (Q_1) and reverse-biases the base-emitter junction of the PNP transistor (Q_2).

Assumption: Q_1 is in the **forward-active region** (ON) and Q_2 is in **cut-off** (OFF).



DC Analysis

Since Q_2 is assumed OFF:

$$I_{B2} = 0$$
$$I_{C2} = 0$$
$$I_{E2} = 0$$

Thus, the entire base current from the source flows into Q_1 ($I_B = I_{B1}$), and the entire emitter current from Q_1 flows through the 1 kΩ load resistor ($I_L = I_{E1}$).
Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop of Q_1:

$$5 - I_{B1}R_B - V_{BE(on)} - I_{E1}R_L = 0$$

We know the relationship between base and emitter currents in the active region:

$$I_{E1} = (\beta + 1)I_{B1} = (100 + 1)I_{B1} = 101I_{B1}$$

Substitute I_{E1} into the KVL equation:

$$\begin{aligned} 5 - I_{B1}(10\text{ k}\Omega) - 0.7 - 101I_{B1}(1\text{ k}\Omega) &= 0 \\ 4.3 &= I_{B1}(10\text{ k}\Omega + 101\text{ k}\Omega) \\ 4.3 &= 111\text{ k}\Omega \cdot I_{B1} \\ I_{B1} &= \frac{4.3}{111 \times 10^3} \approx 38.74\ \mu\text{A} \end{aligned}$$

Now, calculate the other currents for Q_1:

$$I_{C1} = \beta I_{B1} = 100 \times 38.74\ \mu\text{A} = 3.874\ \text{mA}$$
$$I_{E1} = 101 I_{B1} = 101 \times 38.74\ \mu\text{A} = 3.913\ \text{mA}$$

Next, calculate the node voltages:

$$V_B = 5 - I_{B1} R_B = 5 - (38.74\ \mu\text{A})(10\ \text{k}\Omega) = 5 - 0.387 = 4.613\ \text{V}$$
$$V_E = V_B - V_{BE(on)} = 4.613 - 0.7 = 3.913\ \text{V}$$

(Alternatively, $V_E = I_{E1} R_L = 3.913\ \text{mA} \times 1\ \text{k}\Omega = 3.913\ \text{V}$, confirming our calculation).

Verification of Assumptions

1. Is Q_1 in the Active Region?

$$V_{CE1} = V_{C1} - V_{E1} = 5\ \text{V} - 3.913\ \text{V} = 1.087\ \text{V}$$

Since $V_{CE1} = 1.087\ \text{V} > 0.2\ \text{V}$ (typical $V_{CE(sat)}$), Q_1 is indeed in the forward-active region.
2. Is Q_2 in Cut-off?
For the PNP transistor Q_2 to be ON, its emitter-base voltage V_{EB2} must be at least 0.7 V.

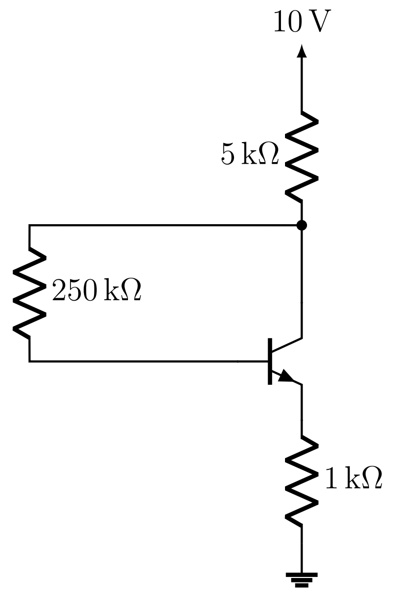
$$V_{EB2} = V_E - V_B = 3.913\ \text{V} - 4.613\ \text{V} = -0.7\ \text{V}$$

Since $V_{EB2} < 0$, the emitter-base junction of Q_2 is reverse-biased, confirming that Q_2 is safely in cut-off.

Final Answer Summary

Parameter	Node / Branch	Calculated Value
	Base Voltage (V_B)	4.613 V
Node Voltages	Emitter Voltage (V_E)	3.913 V
	Collector Q_1 / Collector Q_2	+5 V / − 5 V
	Base Current (I_{B1})	38.74 μA
Q_1 **Currents (NPN)**	Collector Current (I_{C1})	3.874 mA
	Emitter Current (I_{E1})	3.913 mA
	Base Current (I_{B2})	0 A (OFF)
Q_2 **Currents (PNP)**	Collector Current (I_{C2})	0 A (OFF)
	Emitter Current (I_{E2})	0 A (OFF)

Q2. Calculate the collector current, emitter current and base current for the circuit shown in Figure . Also verify the operating mode of the transistor. Assume $\beta = 100$. (16)



(16 marks) [EEE 263 | L-2/T-1 | 2017–18]

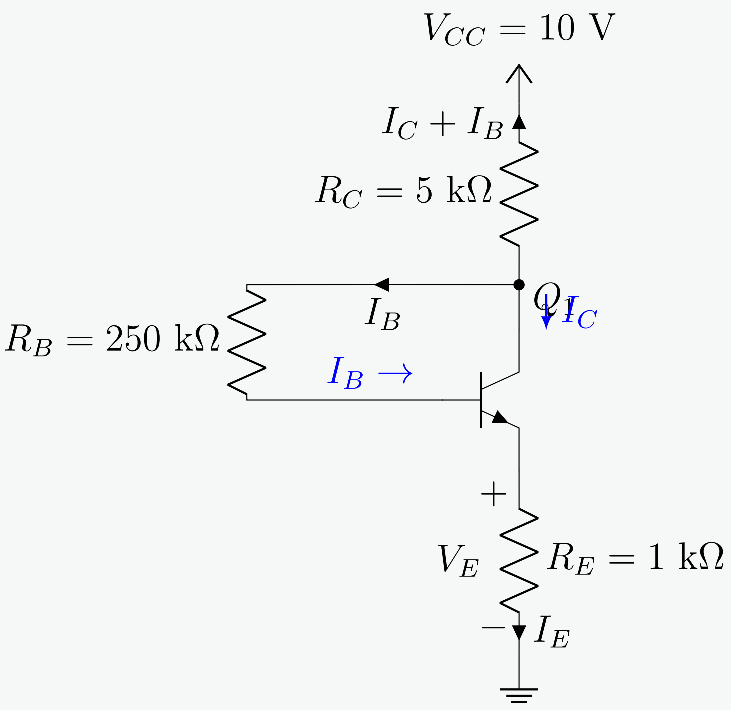
Answer

1. Assumptions and Circuit Analysis

We begin by assuming the transistor operates in the **forward-active region**. For a silicon BJT in the active region, the typical base-emitter voltage drop is $V_{BE} \approx 0.7$ V. The relationships between the transistor currents are:

$$I_C = \beta I_B$$
$$I_E = I_C + I_B = (\beta + 1)I_B$$

This circuit employs **collector-feedback bias** with emitter degeneration. By applying Kirchhoff's Current Law (KCL) at the collector node, we can observe that the current flowing through the collector resistor R_C is the sum of the collector current I_C and the base current I_B. Therefore, the current through R_C is exactly equal to the emitter current I_E.



2. DC Analysis and Mathematical Derivation

Apply Kirchhoff's Voltage Law (KVL) from the V_{CC} supply, down through R_C, through the feedback resistor R_B, across the base-emitter junction, and finally through R_E to ground:

$$V_{CC} - I_{R_C}R_C - I_BR_B - V_{BE} - I_ER_E = 0$$

Substitute $I_{R_C} = I_C + I_B = I_E$:

$$V_{CC} - I_ER_C - I_BR_B - V_{BE} - I_ER_E = 0$$
$$V_{CC} - V_{BE} = I_BR_B + I_E(R_C + R_E)$$

Express I_E in terms of I_B using $I_E = (\beta + 1)I_B$:

$$V_{CC} - V_{BE} = I_BR_B + (\beta + 1)I_B(R_C + R_E)$$
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)(R_C + R_E)}$$

3. Step-by-Step Calculation

Given parameters: $V_{CC} = 10$ V, $V_{BE} = 0.7$ V, $\beta = 100$, $R_B = 250$ kΩ, $R_C = 5$ kΩ, $R_E = 1$ kΩ.

$$I_B = \frac{10 - 0.7}{250 \times 10^3 + (100 + 1)(5 \times 10^3 + 1 \times 10^3)}$$
$$I_B = \frac{9.3}{250,000 + 101(6,000)}$$
$$I_B = \frac{9.3}{250,000 + 606,000}$$
$$I_B = \frac{9.3}{856,000} \approx 1.0864 \times 10^{-5} \text{ A} = 10.864\ \mu\text{A}$$

Now, determine the collector and emitter currents:

$$I_C = \beta I_B = 100 \times 10.864\ \mu\text{A} = 1.0864 \text{ mA}$$
$$I_E = (\beta + 1)I_B = 101 \times 10.864\ \mu\text{A} = 1.0973 \text{ mA}$$

4. Verification of Operating Region

To verify the assumption that the BJT is in the forward-active region, we must ensure that the collector-emitter voltage V_{CE} is greater than the saturation voltage ($V_{CE(sat)} \approx 0.2$ V) and that the base-collector junction is reverse-biased ($V_C > V_B$).
Calculate the node voltages:

$$V_C = V_{CC} - I_{R_C} R_C = V_{CC} - I_E R_C$$
$$V_C = 10 - (1.0973 \times 10^{-3})(5 \times 10^3) = 10 - 5.4865 = 4.5135 \text{ V}$$

$$V_E = I_E R_E = (1.0973 \times 10^{-3})(1 \times 10^3) = 1.0973 \text{ V}$$

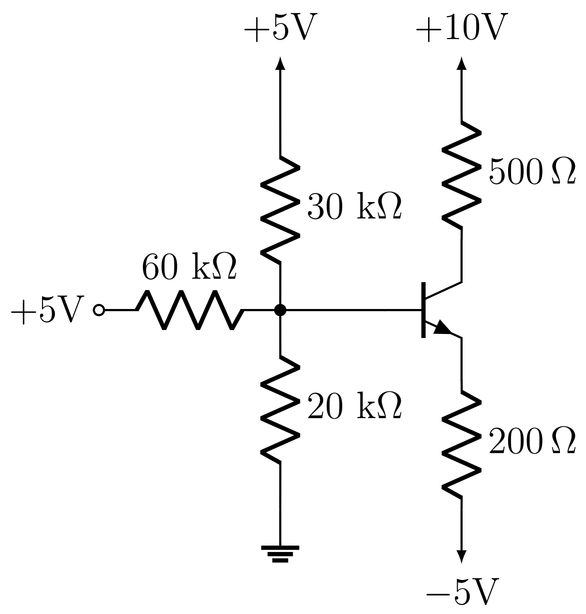
$$V_{CE} = V_C - V_E = 4.5135 - 1.0973 = 3.4162 \text{ V}$$

Since $V_{CE} = 3.416$ V > 0.2 V, the transistor is neither saturated nor in cut-off. The initial assumption is correct; the transistor is operating in the **forward-active region**.

5. Final Answer

Parameter	**Calculated Value**
Base Current (I_B)	10.86 μA
Collector Current (I_C)	1.086 mA
Emitter Current (I_E)	1.097 mA
Collector-Emitter Voltage (V_{CE})	3.416 V
Operating Mode	**Forward-Active Region**

Q3. Find I_C and V_{CE} for the circuit shown. Assume $\beta = 100$.

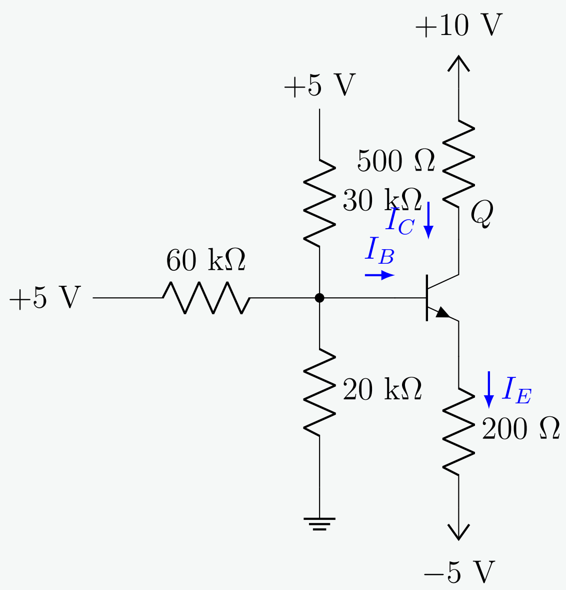


(16 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Answer

1. Circuit Analysis and Thevenin Equivalent

To simplify the analysis, we first find the Thevenin equivalent circuit for the base bias network. The base is connected to three resistors: $R_1 = 60$ kΩ tied to +5 V, $R_2 = 30$ kΩ tied to +5 V, and $R_3 = 20$ kΩ tied to ground (0 V).



The 60 kΩ and 30 kΩ resistors are both connected to the same +5 V potential, so they are in parallel:

$$R_{12} = 60 \text{ k}\Omega \parallel 30 \text{ k}\Omega = \frac{60 \times 30}{60 + 30} = \frac{1800}{90} = 20 \text{ k}\Omega$$

The Thevenin voltage (V_{TH}) at the base (with the transistor disconnected) is given by the voltage divider between R_{12} and R_3:

$$V_{TH} = 5 \text{ V}\left(\frac{R_3}{R_{12} + R_3}\right) = 5\left(\frac{20 \text{ k}\Omega}{20 \text{ k}\Omega + 20 \text{ k}\Omega}\right) = 2.5 \text{ V}$$

The Thevenin resistance (R_{TH}) is the parallel combination of R_{12} and R_3:

$$R_{TH} = R_{12} \parallel R_3 = 20 \text{ k}\Omega \parallel 20 \text{ k}\Omega = 10 \text{ k}\Omega$$

2. Assumption: Active Region Analysis

Initially, we assume the transistor is in the **forward-active region**. Therefore, $V_{BE} = 0.7$ V and $I_C = \beta I_B$, implying $I_E = (\beta + 1)I_B = 101I_B$.
Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop:

$$\begin{aligned} V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E - V_{EE} = 0 \\ 2.5 - I_B(10 \text{ k}\Omega) - 0.7 - 101I_B(0.2 \text{ k}\Omega) - (-5) = 0 \\ 7.5 - 0.7 &= I_B(10 \text{ k}\Omega + 20.2 \text{ k}\Omega) \\ 6.8 &= 30.2 \text{ k}\Omega \cdot I_B \\ I_B &= \frac{6.8}{30.2 \times 10^3} \approx 0.2252 \text{ mA} \end{aligned}$$

Now, calculate I_C and the node voltages V_C and V_E to verify the operating region:

$$\begin{aligned} I_C &= \beta I_B = 100 \times 0.2252 \text{ mA} = 22.52 \text{ mA} \\ I_E &= 101I_B = 22.74 \text{ mA} \\ V_C &= V_{CC} - I_C R_C = 10 - (22.52 \text{ mA})(0.5 \text{ k}\Omega) = 10 - 11.26 = -1.26 \text{ V} \\ V_E &= V_{EE} + I_E R_E = -5 + (22.74 \text{ mA})(0.2 \text{ k}\Omega) = -5 + 4.548 = -0.452 \text{ V} \\ V_{CE} &= V_C - V_E = -1.26 \text{ V} - (-0.452 \text{ V}) = -0.808 \text{ V} \end{aligned}$$

Since $V_{CE} < 0.2$ V, the base-collector junction is strongly forward-biased. The initial assumption is **incorrect**. The transistor is operating in the **saturation region**.

3. Saturation Region Analysis

In saturation, $I_C \neq \beta I_B$. We assume standard saturation voltages:

$$\begin{aligned} V_{CE(sat)} &= 0.2 \text{ V} \\ V_{BE(sat)} &= 0.7 \text{ V} \end{aligned}$$

We now set up KVL equations for both the base-emitter and collector-emitter loops. We know that $I_E = I_C + I_B$.
Loop 1 (Base-Emitter Loop):

$$\begin{aligned} V_{TH} - I_B R_{TH} - V_{BE(sat)} - (I_C + I_B)R_E - V_{EE} = 0 \\ 2.5 - 10I_B - 0.7 - 0.2(I_C + I_B) - (-5) = 0 \quad \text{(currents in mA, resistances in k}\Omega) \\ 6.8 = 10.2I_B + 0.2I_C \end{aligned}$$

Loop 2 (Collector-Emitter Loop):

$$V_{CC} - I_C R_C - V_{CE(sat)} - (I_C + I_B)R_E - V_{EE} = 0 \tag{1}$$

$$10 - 0.5I_C - 0.2 - 0.2(I_C + I_B) - (-5) = 0$$

$$14.8 = 0.7I_C + 0.2I_B \tag{2}$$

We now solve the system of linear equations (1) and (2). From (2), express I_B in terms of I_C:

$$0.2I_B = 14.8 - 0.7I_C \implies I_B = \frac{14.8 - 0.7I_C}{0.2} = 74 - 3.5I_C \tag{3}$$

Substitute (3) into (1):

$$\begin{aligned} 6.8 &= 10.2(74 - 3.5I_C) + 0.2I_C \\ 6.8 &= 754.8 - 35.7I_C + 0.2I_C \\ 6.8 &= 754.8 - 35.5I_C \\ 35.5I_C &= 748 \\ I_C &= \frac{748}{35.5} \approx 21.07 \text{ mA} \end{aligned}$$

Now find I_B:

$$I_B = 74 - 3.5(21.07) = 74 - 73.745 = 0.255 \text{ mA}$$

Verification of Saturation: To ensure the transistor is genuinely in saturation, we check if $I_B \geq I_{B(EOS)}$ (Edge of Saturation).

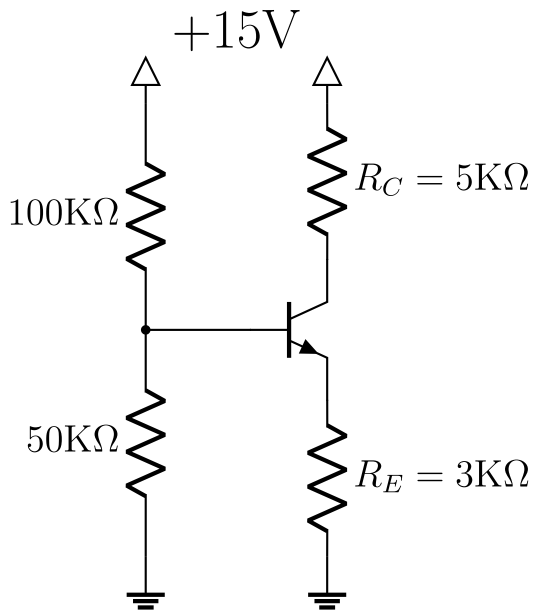
$$I_{B(EOS)} = \frac{I_C}{\beta} = \frac{21.07 \text{ mA}}{100} = 0.2107 \text{ mA}$$

Since $I_B(0.255 \text{ mA}) > I_{B(EOS)}(0.2107 \text{ mA})$, the transistor is confirmed to be in deep saturation.

4. Final Answer

Parameter	Calculated Value
Operating Region	**Saturation**
Collector Current (I_C)	**21.07** mA
Collector-Emitter Voltage (V_{CE})	**0.2** V (assumed standard $V_{CE(sat)}$)
Base Current (I_B)	0.255 mA

Q4. For the circuit shown, determine the voltages of all nodes and current through all branches. Assume $\beta = 100$.

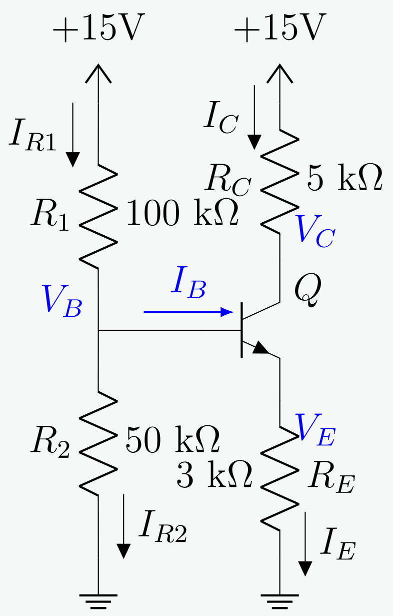


(20 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

1. Circuit Diagram and Nodal Labeling

To systematically determine the node voltages and branch currents, we first redraw the circuit, explicitly labeling all relevant nodes (V_B, V_C, V_E) and branch currents (I_{R1}, I_{R2}, I_B, I_C, I_E).



2. DC Analysis: Thevenin Equivalent

To find the base current (I_B), we apply Thevenin's theorem to the base bias voltage divider network (R_1 and R_2).
The Thevenin equivalent voltage (V_{TH}) at the base is:

$$V_{TH} = V_{CC}\left(\frac{R_2}{R_1 + R_2}\right)$$
$$= 15\text{ V}\left(\frac{50\text{ k}\Omega}{100\text{ k}\Omega + 50\text{ k}\Omega}\right) = 15\left(\frac{50}{150}\right) = 5\text{ V}$$

The Thevenin equivalent resistance (R_{TH}) is the parallel combination of R_1 and R_2:

$$R_{TH} = R_1 \parallel R_2 = \frac{100\text{ k}\Omega \times 50\text{ k}\Omega}{100\text{ k}\Omega + 50\text{ k}\Omega} = \frac{5000}{150}\text{ k}\Omega \approx 33.33\text{ k}\Omega$$

3. Base-Emitter Loop (Active Region Assumption)

We assume the transistor operates in the **forward-active region**. Thus, the base-emitter voltage drop is approximately $V_{BE} = 0.7$ V. Applying Kirchhoff's Voltage Law (KVL) to the Thevenin equivalent base-emitter loop:

$$V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0 \tag{1}$$

In the active region, the relationship between base and emitter currents is:

$$I_E = (\beta + 1)I_B = (100 + 1)I_B = 101 I_B \tag{2}$$

Substituting I_E into the KVL equation:

$$\begin{aligned} V_{TH} - V_{BE} &= I_B R_{TH} + (\beta + 1) I_B R_E \\ 5 \text{ V} - 0.7 \text{ V} &= I_B (33.33 \text{ k}\Omega + 101 \times 3 \text{ k}\Omega) \\ 4.3 \text{ V} &= I_B (33.33 \text{ k}\Omega + 303 \text{ k}\Omega) \\ 4.3 &= 336.33 \times I_B \quad (\text{where } I_B \text{ is in mA}) \\ I_B &= \frac{4.3}{336.33} \text{ mA} \approx 0.012785 \text{ mA} = 12.785 \ \mu\text{A} \end{aligned}$$

4. Current Calculations

With I_B known, we determine the collector and emitter currents:

$$\begin{aligned} I_C &= \beta I_B = 100 \times 0.012785 \text{ mA} = 1.2785 \text{ mA} \\ I_E &= (\beta + 1) I_B = 101 \times 0.012785 \text{ mA} = 1.2913 \text{ mA} \end{aligned}$$

5. Node Voltage Calculations

We calculate the voltages at the Emitter (V_E), Base (V_B), and Collector (V_C) with respect to ground:

$$\begin{aligned} V_E &= I_E R_E = (1.2913 \text{ mA}) \times (3 \text{ k}\Omega) = 3.8739 \text{ V} \\ V_B &= V_E + V_{BE} = 3.8739 \text{ V} + 0.7 \text{ V} = 4.5739 \text{ V} \\ V_C &= V_{CC} - I_C R_C = 15 \text{ V} - (1.2785 \text{ mA} \times 5 \text{ k}\Omega) = 15 - 6.3925 = 8.6075 \text{ V} \end{aligned}$$

Verification of Operating Region:

$$V_{CE} = V_C - V_E = 8.6075 \text{ V} - 3.8739 \text{ V} = 4.7336 \text{ V} \tag{3}$$

Since $V_{CE} > 0.2$ V and $V_C > V_B$ (the base-collector junction is reverse-biased), the assumption that the transistor is in the **forward-active region** is strictly verified.

6. Bias Network Branch Currents

To complete the analysis, we determine the currents flowing through the voltage divider resistors R_1 and R_2 using Ohm's Law and the calculated node voltage V_B:

$$\begin{aligned} I_{R1} &= \frac{V_{CC} - V_B}{R_1} = \frac{15 \text{ V} - 4.5739 \text{ V}}{100 \text{ k}\Omega} = \frac{10.4261}{100} \text{ mA} = 0.10426 \text{ mA} = 104.26 \ \mu\text{A} \\ I_{R2} &= \frac{V_B}{R_2} = \frac{4.5739 \text{ V}}{50 \text{ k}\Omega} = 0.09148 \text{ mA} = 91.48 \ \mu\text{A} \end{aligned}$$

Note: As expected by Kirchhoff's Current Law at the base node, $I_{R1} = I_{R2} + I_B$ ($104.26 \ \mu\text{A} \approx 91.48 \ \mu\text{A} + 12.78 \ \mu\text{A}$).

7. Final Answers Summary

Node Voltages		**Branch Currents**	
Base Voltage (V_B)	4.57 V	Base Current (I_B)	12.78 μA
Collector Voltage (V_C)	8.61 V	Collector Current (I_C)	1.28 mA
Emitter Voltage (V_E)	3.87 V	Emitter Current (I_E)	1.29 mA
		Current through R_1 (I_{R1})	104.26 μA
		Current through R_2 (I_{R2})	91.48 μA

Q5. (a) Express the collector current I_c as a function of V_{CB} that involves the $2\,\text{k}\Omega$ resistance (it will be a straight line equation). **(16 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(b) Find out the mathematical equation describing the I_c–V_{CB} characteristic graph of the BJT (it will be a piecewise linear graph). **(10 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(c) Draw the two I_c–V_{CB} graphs over the same plot exactly to scale and determine their intersection point. This intersection point gives the value of I_c and V_{CB} at the bias point. **(10 marks)** [EEE 263 | L-2/T-1 | 2012–13]

(d) From the value of V_{CB}, can you determine the value of V_{CE} at the bias point? **(10 marks)** [EEE 263 | L-2/T-1 | 2012–13]

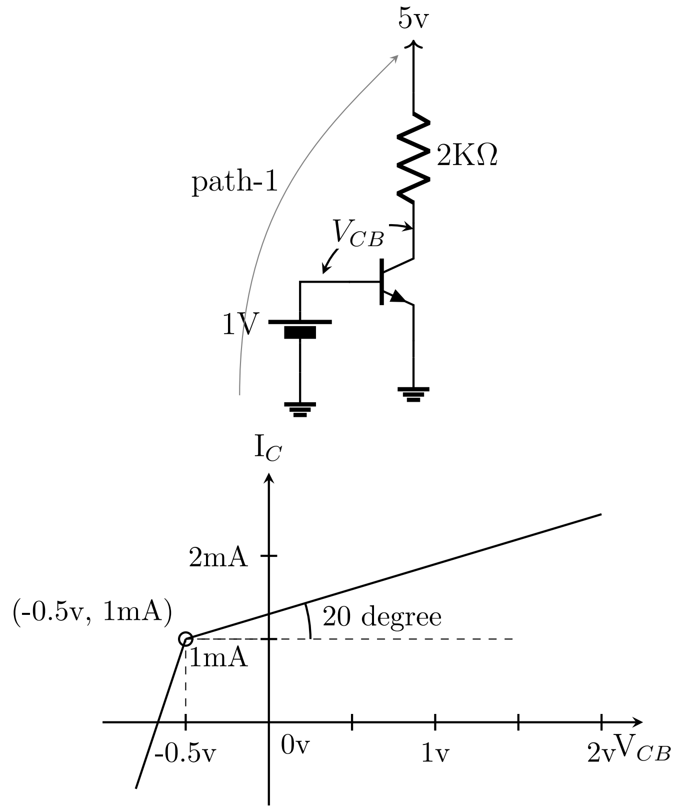


Figure-2

Answer

Part (a): Collector Current I_C as a function of V_{CB}

To find the relationship between the collector current I_C and the collector-base voltage V_{CB} imposed by the external circuit, we apply Kirchhoff's Voltage Law (KVL) along the indicated "path-1".
The path starts from the 5 V supply, goes through the 2 kΩ collector resistor, the collector-base junction (V_{CB}), the 1 V base supply, and finally to ground.

$$V_{CC} - I_C R_C - V_{CB} - V_{BB} = 0 \tag{1}$$

Substitute the known values ($V_{CC} = 5$ V, $R_C = 2$ kΩ, and $V_{BB} = 1$ V):

$$\begin{aligned} 5 - I_C(2\text{ k}\Omega) - V_{CB} - 1 &= 0 \\ 4 - V_{CB} &= 2I_C \\ I_C &= 2 - 0.5V_{CB} \end{aligned} \tag{2}$$

where I_C is in mA and V_{CB} is in Volts. This represents the **DC Load Line** equation.

Part (b): Mathematical Equation of the I_C–V_{CB} Characteristic

Based on the provided piecewise linear graph (Figure-2), the characteristic is divided into two distinct regions separated by a breakpoint at $(-0.5\text{ V}, 1\text{ mA})$.
1. Active Region ($V_{CB} \geq -0.5$ V):
The graph has a positive slope given by $\tan(20°)$. Using the point-slope form $(y - y_1) = m(x - x_1)$:

$$\begin{aligned} I_C - 1\text{ mA} &= \tan(20°) \cdot (V_{CB} - (-0.5\text{ V})) \\ I_C &= 1 + \tan(20°)(V_{CB} + 0.5) \end{aligned} \tag{3}$$

Since $\tan(20°) \approx 0.364$ mA/V, the equation is approximately $I_C = 1.182 + 0.364V_{CB}$.
2. Saturation Region ($V_{CB} < -0.5$ V):
The graph falls steeply to zero. Assuming a large positive slope m_{sat} for the forward-biased collector-base junction in hard saturation, the complete piecewise equation is:

$$I_C(V_{CB}) = \begin{cases} 1 + \tan(20°)(V_{CB} + 0.5) & \text{for } V_{CB} \geq -0.5\text{ V} \\ 1 + m_{sat}(V_{CB} + 0.5) & \text{for } V_{CB} < -0.5\text{ V} \end{cases} \tag{4}$$

Part (c): Graphical Plot and Intersection Point

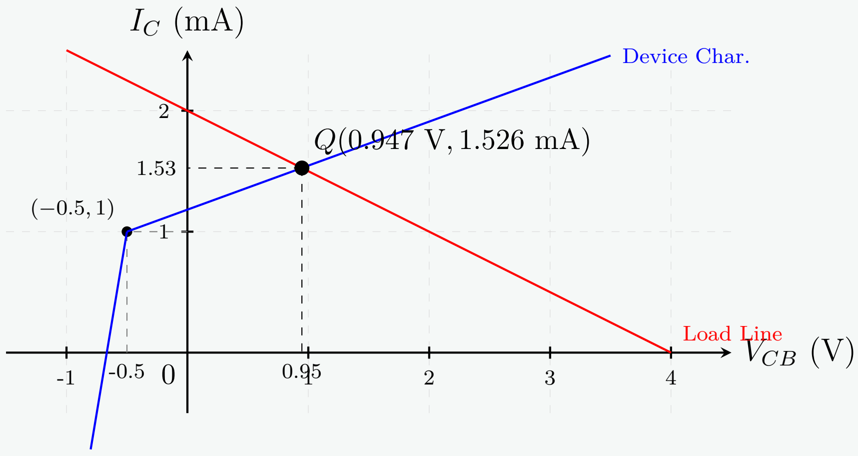
To find the bias point (Q-point), we determine the intersection of the load line (Eq. 2) and the device characteristic (Eq. 3) in the active region.

$$\begin{aligned} 2 - 0.5V_{CB} &= 1 + \tan(20°)(V_{CB} + 0.5) \\ 2 - 0.5V_{CB} &= 1 + 0.364V_{CB} + 0.182 \\ 2 - 1.182 &= V_{CB}(0.5 + 0.364) \\ 0.818 &= 0.864V_{CB} \\ V_{CB} &\approx 0.947\text{ V} \end{aligned}$$

Substitute V_{CB} back into the load line equation to find I_C:

$$I_C = 2 - 0.5(0.947) = 1.526 \text{ mA} \tag{5}$$

Intersection Point (Bias Point): $(V_{CB}, I_C) = (0.947 \text{ V}, 1.526 \text{ mA})$.



Part (d): Determine V_{CE} at the Bias Point

By definition, the collector-emitter voltage V_{CE} is related to the collector-base voltage V_{CB} and the base-emitter voltage V_{BE} by:

$$V_{CE} = V_{CB} + V_{BE} \tag{6}$$

From the given circuit schematic:

- The base is tied to a 1 V ideal DC source with respect to ground ($V_B = 1$ V).
- The emitter is directly tied to ground ($V_E = 0$ V).

Therefore, the base-emitter voltage is rigidly fixed by the external circuit:

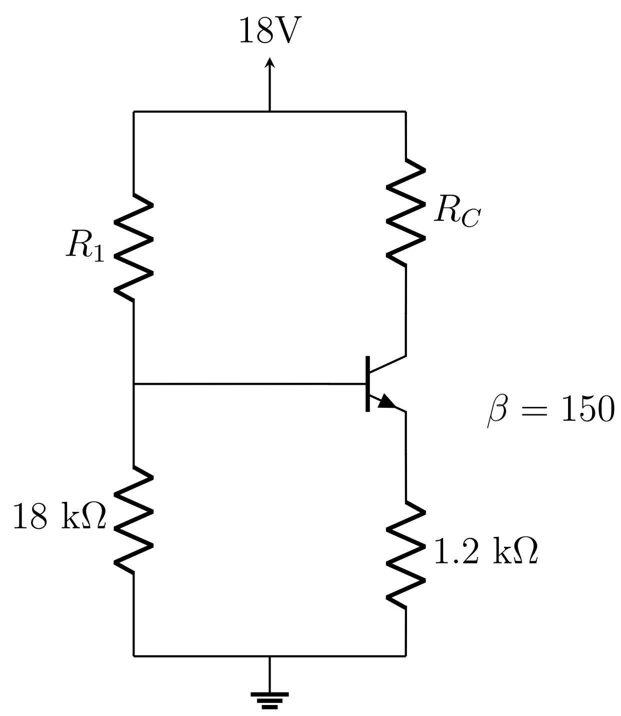
$$V_{BE} = V_B - V_E = 1 \text{ V} - 0 \text{ V} = 1 \text{ V} \tag{7}$$

Using the V_{CB} value obtained at the bias point:

$$V_{CE} = 0.947 \text{ V} + 1.0 \text{ V}$$
$$V_{CE} = 1.947 \text{ V}$$

Important Note: In a practical BJT, V_{BE} is typically clamped around 0.7 V. The strict 1 V assignment here is a theoretical idealization dictated entirely by the explicit 1 V independent voltage source across the Base-Emitter terminals in the given diagram.

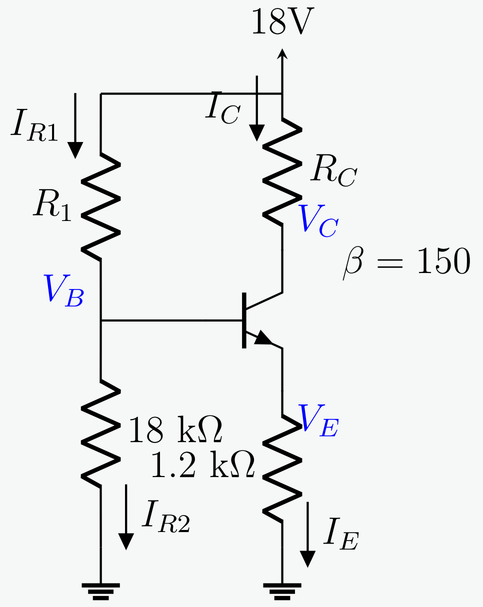
Q6. For the circuit shown, determine R_1 and R_C if $I_B = 0.02\,\text{mA}$ and $V_{CE} = 10V$.



(16 marks) [EEE 263 | L-2/T-1 | 2011–12]

Answer

1. Circuit Diagram



2. Assumptions and Given Parameters

- Supply Voltage (V_{CC}) = 18 V
- Base Current (I_B) = 0.02 mA
- Collector-Emitter Voltage (V_{CE}) = 10 V
- Transistor Current Gain (β) = 150
- Base-Emitter ON Voltage (V_{BE}) ≈ 0.7 V (Assuming operation in the active region)

3. Transistor Currents Calculation

Using the fundamental relations of the BJT in the active region, we determine the collector current (I_C) and emitter current (I_E):

$$I_C = \beta I_B = 150 \times 0.02 \text{ mA} = 3.0 \text{ mA}$$
$$I_E = (\beta + 1)I_B = 151 \times 0.02 \text{ mA} = 3.02 \text{ mA}$$

4. Determination of Collector Resistor (R_C)

We apply Kirchhoff's Voltage Law (KVL) to the collector-emitter output loop. The equation is:

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0 \tag{1}$$

Rearranging to solve for R_C:

$$\begin{aligned}
I_C R_C &= V_{CC} - V_{CE} - I_E R_E \\
(3.0 \text{ mA})R_C &= 18 \text{ V} - 10 \text{ V} - (3.02 \text{ mA} \times 1.2 \text{ k}\Omega) \\
3.0R_C &= 8 - 3.624 \\
3.0R_C &= 4.376 \\
R_C &= \frac{4.376}{3.0} \text{ k}\Omega \approx 1.4587 \text{ k}\Omega
\end{aligned}$$

5. Determination of Base Resistor (R_1)

To find R_1, we first need to determine the base node voltage (V_B) with respect to ground. The voltage at the emitter (V_E) is:

$$V_E = I_E R_E = (3.02 \text{ mA}) \times (1.2 \text{ k}\Omega) = 3.624 \text{ V} \tag{2}$$

Assuming a standard forward-biased base-emitter junction ($V_{BE} = 0.7$ V), the base voltage is:

$$V_B = V_E + V_{BE} = 3.624 \text{ V} + 0.7 \text{ V} = 4.324 \text{ V} \tag{3}$$

Now, we analyze the current at the base node. Let I_{R2} be the current flowing through the 18 kΩ resistor to ground:

$$I_{R2} = \frac{V_B}{18 \text{ k}\Omega} = \frac{4.324 \text{ V}}{18 \text{ k}\Omega} \approx 0.2402 \text{ mA} \tag{4}$$

By Kirchhoff's Current Law (KCL) at the base node, the total current flowing through R_1 from the supply must be the sum of the base current and I_{R2}:

$$I_{R1} = I_{R2} + I_B = 0.2402 \text{ mA} + 0.02 \text{ mA} = 0.2602 \text{ mA} \tag{5}$$

Finally, applying Ohm's law across R_1:

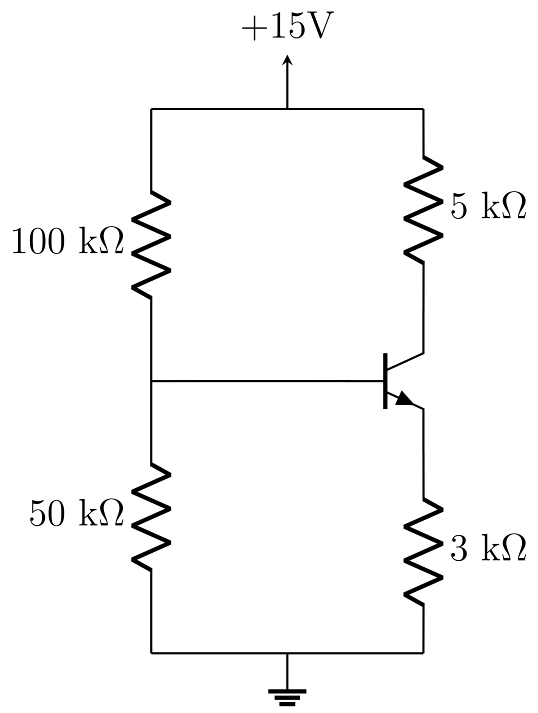
$$\begin{aligned} R_1 &= \frac{V_{CC} - V_B}{I_{R1}} \\ &= \frac{18 \text{ V} - 4.324 \text{ V}}{0.2602 \text{ mA}} \\ &= \frac{13.676 \text{ V}}{0.2602 \text{ mA}} \approx 52.56 \text{ k}\Omega \end{aligned}$$

6. Final Answer

The required resistor values to establish the given operating point are:

$$\mathbf{R_1 \approx 52.56 \text{ k}\Omega}$$
$$\mathbf{R_C \approx 1.46 \text{ k}\Omega}$$

Q7. Determine all node voltages and branch currents for the circuit shown. Assume $\beta = 100$ and $V_{BE} = 0.7\,\text{V}$.



(18 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Circuit Parameters

From the provided schematic, we have the following values:

- Supply Voltage (V_{CC}) = 15 V
- Base Resistors: $R_1 = 100$ kΩ, $R_2 = 50$ kΩ
- Collector Resistor (R_C) = 5 kΩ
- Emitter Resistor (R_E) = 3 kΩ
- Current Gain (β) = 100
- Base-Emitter ON Voltage (V_{BE}) = 0.7 V

2. Exact Analysis Using Thevenin Equivalent

To analyze the base circuit, we find the Thevenin equivalent of the voltage divider network connected to the base.
Thevenin Voltage (V_{TH}):

$$V_{TH} = V_{CC}\left(\frac{R_2}{R_1 + R_2}\right) = 15\left(\frac{50}{100 + 50}\right) = 15\left(\frac{1}{3}\right) = 5 \text{ V} \tag{1}$$

Thevenin Resistance (R_{TH}):

$$R_{TH} = R_1 \parallel R_2 = \frac{100 \times 50}{100 + 50} = \frac{5000}{150} \approx 33.33 \text{ k}\Omega \tag{2}$$

3. Transistor Branch Currents

Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop:

$$V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0 \tag{3}$$

Since $I_E = (\beta + 1)I_B$:

$$V_{TH} - I_B R_{TH} - V_{BE} - (\beta + 1)I_B R_E = 0 \tag{4}$$

Solving for Base Current (I_B):

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta+1)R_E}$$
$$I_B = \frac{5\text{ V} - 0.7\text{ V}}{33.33\text{ k}\Omega + (101)(3\text{ k}\Omega)}$$
$$I_B = \frac{4.3\text{ V}}{33.33\text{ k}\Omega + 303\text{ k}\Omega} = \frac{4.3}{336.33}\text{ mA} \approx 0.01278\text{ mA} = \mathbf{12.78}\ \mu\text{A}$$

Now, calculate the Collector Current (I_C) and Emitter Current (I_E):

$$I_C = \beta I_B = 100 \times 0.01278\text{ mA} \approx \mathbf{1.278}\text{ mA}$$
$$I_E = (\beta+1)I_B = 101 \times 0.01278\text{ mA} \approx \mathbf{1.291}\text{ mA}$$

4. Node Voltages

Using the calculated currents, we can determine the voltages at each node relative to ground.
Emitter Voltage (V_E):

$$V_E = I_E R_E = 1.291\text{ mA} \times 3\text{ k}\Omega \approx \mathbf{3.873}\text{ V} \tag{5}$$

Base Voltage (V_B):

$$V_B = V_E + V_{BE} = 3.873\text{ V} + 0.7\text{ V} \approx \mathbf{4.573}\text{ V} \tag{6}$$

Collector Voltage (V_C):

$$V_C = V_{CC} - I_C R_C = 15\text{ V} - (1.278\text{ mA} \times 5\text{ k}\Omega) = 15 - 6.39 \approx \mathbf{8.61}\text{ V} \tag{7}$$

Check: $V_{CE} = V_C - V_E = 8.61\text{ V} - 3.873\text{ V} = 4.737\text{ V} > 0.2\text{ V}$, confirming the BJT is in the active region.

5. Voltage Divider Branch Currents

Finally, we calculate the currents flowing through the base bias resistors.
Current through R_1 (I_{R1}):

$$I_{R1} = \frac{V_{CC} - V_B}{R_1} = \frac{15\text{ V} - 4.573\text{ V}}{100\text{ k}\Omega} = 0.10427\text{ mA} \approx \mathbf{104.27}\ \mu\text{A} \tag{8}$$

Current through R_2 (I_{R2}):

$$I_{R2} = \frac{V_B}{R_2} = \frac{4.573\text{ V}}{50\text{ k}\Omega} = 0.09146\text{ mA} \approx \mathbf{91.46}\ \mu\text{A} \tag{9}$$

Note that KCL holds at the base node: $I_{R1} \approx I_{R2} + I_B$ *(*$104.27\ \mu A \approx 91.46\ \mu A + 12.78\ \mu A$*).*

BJT Saturation Parameters

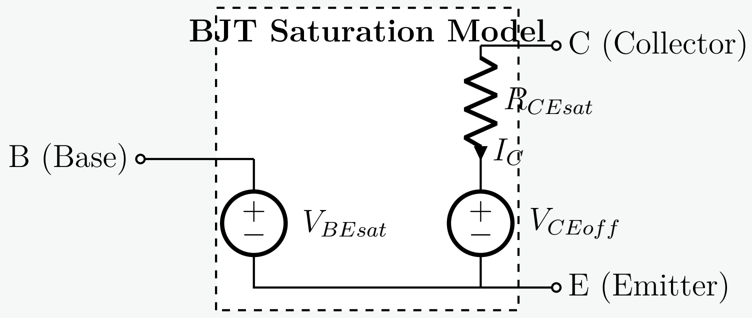
Q1. A BJT for which $I_B = 0.5\,\text{mA}$ has $v_{CEsat} = 140\,\text{mV}$ at $I_C = 10\,\text{mA}$ and $v_{CEsat} = 170\,\text{mV}$ at $I_C = 20\,\text{mA}$. Estimate the values of: (i) saturation resistance R_{CEsat}; (ii) offset voltage V_{CEoff}; (iii) β_f; (iv) β_r. **(10 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

1. Estimation of Saturation Resistance (R_{CEsat}) and Offset Voltage (V_{CEoff})

In deep saturation, the collector-emitter voltage (V_{CEsat}) of a BJT can be modeled using a piecewise linear approximation. The equivalent circuit between the collector and emitter consists of an offset voltage V_{CEoff} in series with a saturation resistance R_{CEsat}. The defining equation is:

$$V_{CEsat} = V_{CEoff} + I_C R_{CEsat} \tag{1}$$



We are given two operating points for a constant base current $I_B = 0.5\,\text{mA}$:

$$\text{Point 1: } I_{C1} = 10\,\text{mA}, \quad V_{CE1} = 140\,\text{mV}$$
$$\text{Point 2: } I_{C2} = 20\,\text{mA}, \quad V_{CE2} = 170\,\text{mV}$$

Substituting these points into the linear model yields a system of two equations:

$$140\,\text{mV} = V_{CEoff} + (10\,\text{mA})R_{CEsat} \tag{2}$$
$$170\,\text{mV} = V_{CEoff} + (20\,\text{mA})R_{CEsat} \tag{3}$$

(i) Saturation Resistance (R_{CEsat}):
Subtracting Equation 3 from Equation 3:

$$\begin{aligned}170\,\text{mV} - 140\,\text{mV} &= (20\,\text{mA} - 10\,\text{mA})R_{CEsat}\\ 30\,\text{mV} &= 10\,\text{mA}\cdot R_{CEsat}\\ R_{CEsat} &= \frac{30\,\text{mV}}{10\,\text{mA}} = \mathbf{3\,\Omega}\end{aligned}$$

(ii) Offset Voltage (V_{CEoff}):
Substitute $R_{CEsat} = 3\,\Omega$ back into Equation 3:

$$\begin{aligned}140\,\text{mV} &= V_{CEoff} + (10\,\text{mA})(3\,\Omega)\\ 140\,\text{mV} &= V_{CEoff} + 30\,\text{mV}\\ V_{CEoff} &= \mathbf{110}\,\text{mV}\end{aligned}$$

2. Estimation of Forward (β_f) and Reverse (β_r) Current Gains

To determine the physical transistor parameters β_f and β_r, we utilize the exact analytic expression for V_{CEsat} derived from the Ebers-Moll equations. For a BJT in saturation, the collector-emitter voltage is given by:

$$V_{CEsat} = V_T \ln\left[\frac{I_B\left(1+\frac{1}{\beta_r}\right) + \frac{I_C}{\beta_r}}{I_B - \frac{I_C}{\beta_f}}\right] \tag{4}$$

where V_T is the thermal voltage. Assuming standard room temperature, $V_T \approx 25\,\text{mV}$.
To simplify the algebra, let us define two variables: $x = \frac{1}{\beta_r}$ and $y = \frac{1}{\beta_f}$. Equation 4 can be rewritten as:

$$\exp\left(\frac{V_{CEsat}}{V_T}\right) = \frac{I_B(1+x) + I_C x}{I_B - I_C y} \tag{5}$$

Applying the given operating points:
For Point 1 ($V_{CE1} = 140\,\text{mV}, I_{C1} = 10\,\text{mA}, I_B = 0.5\,\text{mA}$):

$$\begin{aligned}\exp\left(\frac{140}{25}\right) &= \frac{0.5(1+x)+10x}{0.5-10y}\\ 270.426 &= \frac{0.5+10.5x}{0.5-10y}\\ 270.426(0.5-10y) &= 0.5+10.5x\\ 135.213 - 2704.26y &= 0.5 + 10.5x\\ 10.5x + 2704.26y &= 134.713 \end{aligned} \tag{6}$$

For Point 2 ($V_{CE2} = 170\,\text{mV}, I_{C2} = 20\,\text{mA}, I_B = 0.5\,\text{mA}$):

$$\begin{aligned}\exp\left(\frac{170}{25}\right) &= \frac{0.5(1+x)+20x}{0.5-20y}\\ 897.847 &= \frac{0.5+20.5x}{0.5-20y}\\ 897.847(0.5-20y) &= 0.5+20.5x\\ 448.9235 - 17956.94y &= 0.5 + 20.5x\\ 20.5x + 17956.94y &= 448.4235 \end{aligned} \tag{7}$$

Solving the linear system for x and y:
From Equation 6, express x in terms of y:

$$x = \frac{134.713 - 2704.26y}{10.5} = 12.830 - 257.549y \tag{8}$$

Substitute Equation 8 into Equation 7:

$$\begin{aligned}20.5(12.830 - 257.549y) + 17956.94y &= 448.4235\\ 263.015 - 5279.75y + 17956.94y &= 448.4235\\ 12677.19y &= 185.4085\\ y &= 0.014625\end{aligned}$$

Now, substitute y back to find x:

$$\begin{aligned}x &= 12.830 - 257.549(0.014625)\\ x &= 12.830 - 3.766\\ x &= 9.064\end{aligned}$$

(iii) Forward Current Gain (β_f):

$$\beta_f = \frac{1}{y} = \frac{1}{0.014625} \approx \mathbf{68.38}$$

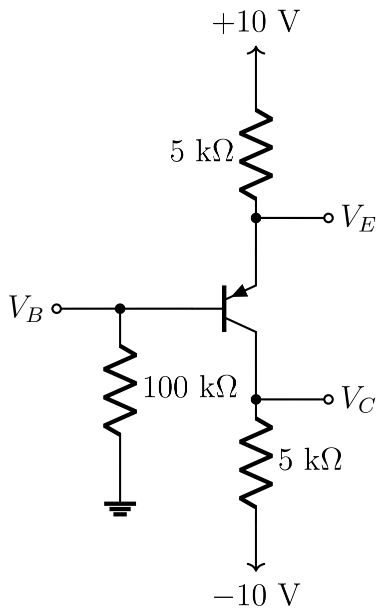
(iv) Reverse Current Gain (β_r):

$$\beta_r = \frac{1}{x} = \frac{1}{9.064} \approx \mathbf{0.11}$$

Note: The values of β_f and β_r slightly depend on the assumed thermal voltage V_T. At $V_T = 26\,mV$ (commonly used for 300K), $\beta_f \approx 71.8$ and $\beta_r \approx 0.134$. Both assumptions yield theoretically valid Ebers-Moll parameter estimations.

Multi-BJT Circuit Analysis

Q1. For the circuit shown, the emitter voltage was measured to be $V_E = +1.7\,\text{V}$. Assume $|V_{BE}| = 0.7\,\text{V}$. Find the values of the labeled node voltages and branch currents. What are the values of α and β for the pnp transistor?



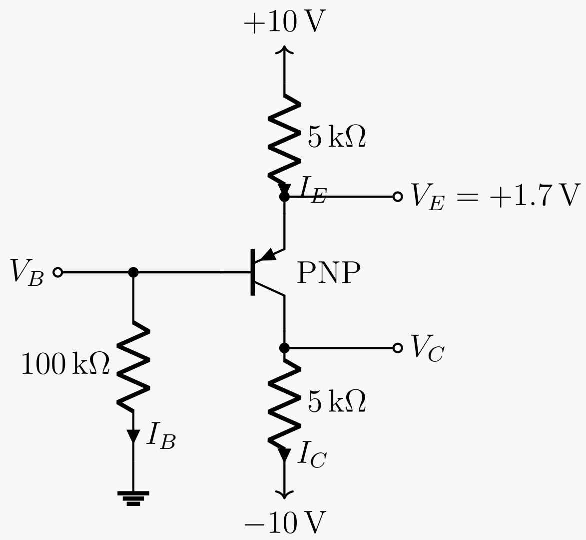
(15 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

1. Circuit Analysis and Operating Principle

The given circuit utilizes a PNP Bipolar Junction Transistor (BJT). To analyze the circuit, we first assume the transistor is operating in the **Forward-Active Region**. For a PNP transistor in the active region:

- The emitter-base junction is forward-biased, meaning $V_{EB} \approx 0.7\,\text{V}$ (since $|V_{BE}| = 0.7\,\text{V}$ is given).
- The collector-base junction is reverse-biased, meaning $V_{CB} < 0\,\text{V}$ (or $V_C < V_B$).
- The currents follow KCL: $I_E = I_B + I_C$, where I_E flows into the emitter, and I_B, I_C flow out of the base and collector, respectively.



2. Node Voltages and Branch Currents

Step 1: Determine the Base Voltage (V_B)

Given the emitter voltage $V_E = +1.7\,\text{V}$ and the forward-biased emitter-base voltage $V_{EB} = 0.7\,\text{V}$:

$$V_{EB} = V_E - V_B$$
$$V_B = V_E - V_{EB} = 1.7\,\text{V} - 0.7\,\text{V} = \mathbf{1.0}\,\text{V}$$

Step 2: Calculate the Branch Currents (I_E, I_B, I_C)
The emitter current I_E is determined by the voltage drop across the $5\,\text{k}\Omega$ emitter resistor:

$$I_E = \frac{10\,\text{V} - V_E}{5\,\text{k}\Omega} = \frac{10 - 1.7}{5} = \frac{8.3}{5} = \mathbf{1.66}\,\text{mA} \tag{1}$$

The base current I_B flows from V_B to ground through the $100\,\text{k}\Omega$ resistor:

$$I_B = \frac{V_B - 0}{100\,\text{k}\Omega} = \frac{1.0\,\text{V}}{100\,\text{k}\Omega} = \mathbf{0.01}\,\text{mA} \quad (\text{or } 10\,\mu\text{A}) \tag{2}$$

Applying Kirchhoff's Current Law (KCL) at the transistor, the collector current I_C is:

$$I_C = I_E - I_B = 1.66\,\text{mA} - 0.01\,\text{mA} = \mathbf{1.65}\,\text{mA} \tag{3}$$

Step 3: Determine the Collector Voltage (V_C)
The collector current flows through the $5\,\text{k}\Omega$ collector resistor towards the $-10\,\text{V}$ supply:

$$\begin{aligned} V_C - (-10\,\text{V}) &= I_C \times 5\,\text{k}\Omega \\ V_C &= -10\,\text{V} + (1.65\,\text{mA} \times 5\,\text{k}\Omega) \\ V_C &= -10 + 8.25 = \mathbf{-1.75}\,\text{V} \end{aligned}$$

3. Transistor Parameters (α and β)

Using the calculated branch currents, the common-base current gain (α) and common-emitter current gain (β) are:

$$\alpha = \frac{I_C}{I_E} = \frac{1.65\,\text{mA}}{1.66\,\text{mA}} \approx \mathbf{0.994}$$

$$\beta = \frac{I_C}{I_B} = \frac{1.65\,\text{mA}}{0.01\,\text{mA}} = \mathbf{165}$$

Cross-verification: $\alpha = \frac{\beta}{\beta+1} = \frac{165}{166} \approx 0.994$. This confirms the calculations are consistent.

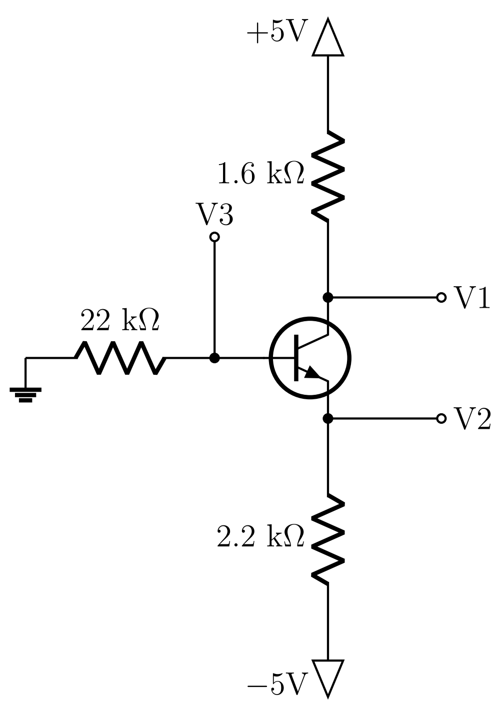
4. Verification of the Operating Region

To ensure our initial Forward-Active assumption is correct, we verify the state of the collector-base junction:

$$V_{CB} = V_C - V_B = -1.75\,\text{V} - 1.0\,\text{V} = -2.75\,\text{V} \tag{4}$$

Since $V_{CB} < 0\,\text{V}$ for the PNP transistor, the collector-base junction is safely reverse-biased. The transistor is indeed operating in the active region.

Q2. For the circuit shown, find the values of the labeled node voltages and branch currents. Assume β to be very high and $|V_{BE}| = 0.7\,\text{V}$. What will happen if β is reduced to a value of 100?

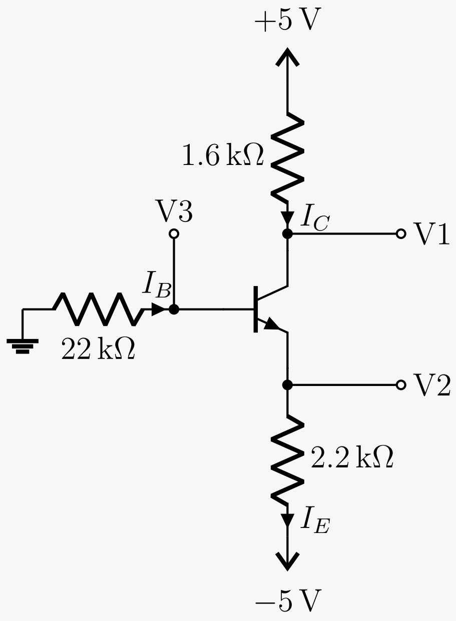


(15 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

1. Circuit Diagram & Analysis Strategy

The given circuit consists of an NPN Bipolar Junction Transistor (BJT) biased using a dual power supply ($+5\,\text{V}$ and $-5\,\text{V}$). We assume the transistor is operating in the **Forward-Active Region**. To verify this later, we will check if the base-emitter junction is forward-biased ($V_{BE} = 0.7\,\text{V}$) and the base-collector junction is reverse-biased ($V_{BC} < 0\,\text{V}$ or $V_C > V_B$).



2. Case I: Assuming β is very high ($\beta \to \infty$)

Principle: When the common-emitter current gain β is very high, the base current I_B becomes negligibly small ($I_B \approx 0$). Consequently, the emitter and collector currents are essentially equal ($I_C \approx I_E$).
Step 1: Calculate Base Voltage (V3)
Since $I_B = 0$, there is no voltage drop across the $22\,\text{k}\Omega$ base resistor.

$$V_3 = 0\,\text{V} - I_B(22\,\text{k}\Omega) = 0 - 0 = \mathbf{0}\,\text{V} \tag{1}$$

Step 2: Calculate Emitter Voltage (V2) and Branch Currents
The base-emitter junction acts as a forward-biased diode with a $0.7\,\text{V}$ drop.

$$V_2 = V_3 - V_{BE} = 0\,\text{V} - 0.7\,\text{V} = \mathbf{-0.7}\,\text{V} \tag{2}$$

Now, apply Ohm's law across the $2.2\,\text{k}\Omega$ emitter resistor to find the emitter current I_E:

$$I_E = \frac{V_2 - (-5\,\text{V})}{2.2\,\text{k}\Omega} = \frac{-0.7 + 5}{2.2} = \frac{4.3}{2.2} \approx \mathbf{1.955}\,\text{mA} \tag{3}$$

Since $\beta \to \infty$, $I_C = I_E = \mathbf{1.955}\,\text{mA}$ and $I_B = \mathbf{0}\,\text{mA}$.
Step 3: Calculate Collector Voltage (V1)
Apply Ohm's law to the $1.6\,\text{k}\Omega$ collector resistor:

$$V_1 = 5\,\text{V} - I_C(1.6\,\text{k}\Omega) = 5 - (1.955)(1.6) = 5 - 3.128 = \mathbf{1.872}\,\text{V} \tag{4}$$

Check Active Region: $V_1 > V_3$ ($1.872\,\text{V} > 0\,\text{V}$), so the collector-base junction is safely reverse-biased. The transistor is in the active region.

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3. Case II: β is reduced to 100

Principle: With a finite β, the base current is no longer zero. We must use the relationships $I_C = \beta I_B$ and $I_E = (\beta+1)I_B$ to solve the circuit.
Step 1: Write KVL for the Base-Emitter Loop
The base voltage is now dependent on the base current:

$$V_3 = 0 - I_B(22\,\text{k}\Omega) = -22I_B \quad \text{(Assuming } I_B \text{ is in mA)} \tag{5}$$

The emitter voltage is:

$$V_2 = V_3 - 0.7 = -22I_B - 0.7 \tag{6}$$

The emitter current is given by Ohm's law:

$$I_E = \frac{V_2 - (-5)}{2.2} = \frac{V_2 + 5}{2.2} \tag{7}$$

Step 2: Solve for Base Current (I_B)
Substitute (6) into (7) and use $I_E = (\beta+1)I_B = 101I_B$:

$$\begin{aligned} 101I_B &= \frac{(-22I_B - 0.7) + 5}{2.2} \\ 2.2(101I_B) &= 4.3 - 22I_B \\ 222.2I_B &= 4.3 - 22I_B \\ 244.2I_B &= 4.3 \\ I_B &= \frac{4.3}{244.2} \approx \mathbf{0.0176}\,\text{mA} \quad \text{(or } 17.6\,\mu\text{A)} \end{aligned}$$

Step 3: Determine New Branch Currents

$$I_C = \beta I_B = 100 \times 0.0176 = \mathbf{1.760}\,\text{mA}$$
$$I_E = (\beta + 1)I_B = 101 \times 0.0176 = \mathbf{1.778}\,\text{mA}$$

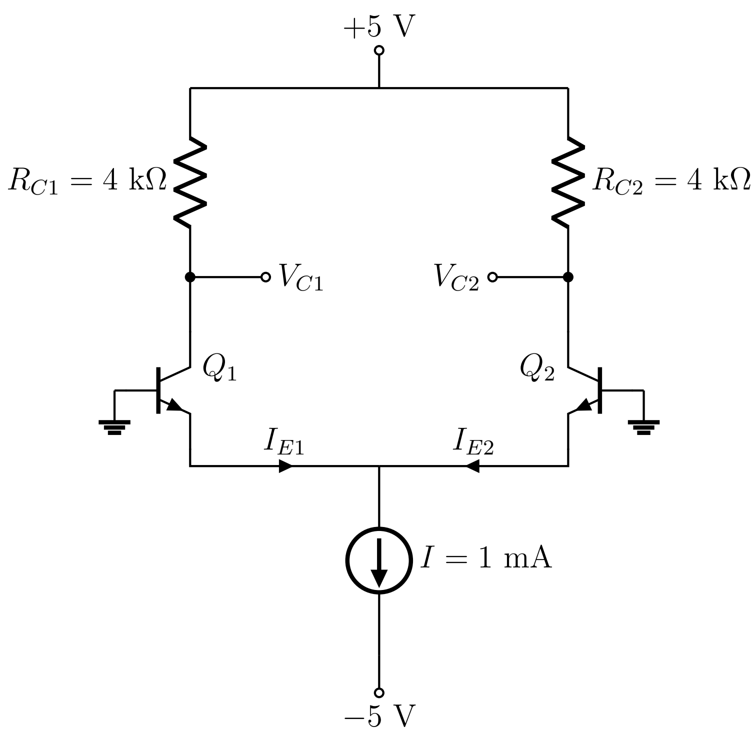
Step 4: Determine New Node Voltages

$$V_3 = -22 \times (0.0176) = \mathbf{-0.387}\,\text{V}$$
$$V_2 = V_3 - 0.7 = -0.387 - 0.7 = \mathbf{-1.087}\,\text{V}$$
$$V_1 = 5 - I_C(1.6) = 5 - (1.760)(1.6) = 5 - 2.816 = \mathbf{2.184}\,\text{V}$$

4. Conclusion: What happens when β is reduced?

When β is reduced from infinity to 100, a measurable **base current (I_B) is drawn from ground**. This produces a voltage drop across the $22\,\text{k}\Omega$ base resistor, pulling the base voltage (V_3) below ground level (from $0\,\text{V}$ to $-0.387\,\text{V}$).
Because the base voltage drops, the emitter voltage (V_2) also drops proportionally. This leaves a smaller potential difference across the emitter resistor ($2.2\,\text{k}\Omega$), which **decreases the emitter current** and subsequently the **collector current**. Because less collector current flows, there is less voltage drop across the collector resistor, causing the **collector voltage (V_1) to rise**.

Q3. For the circuit shown with $\beta = 200$ for each transistor, determine: (i) I_{E1}, (ii) I_{E2}, (iii) V_{C1}, and (iv) V_{C2}.

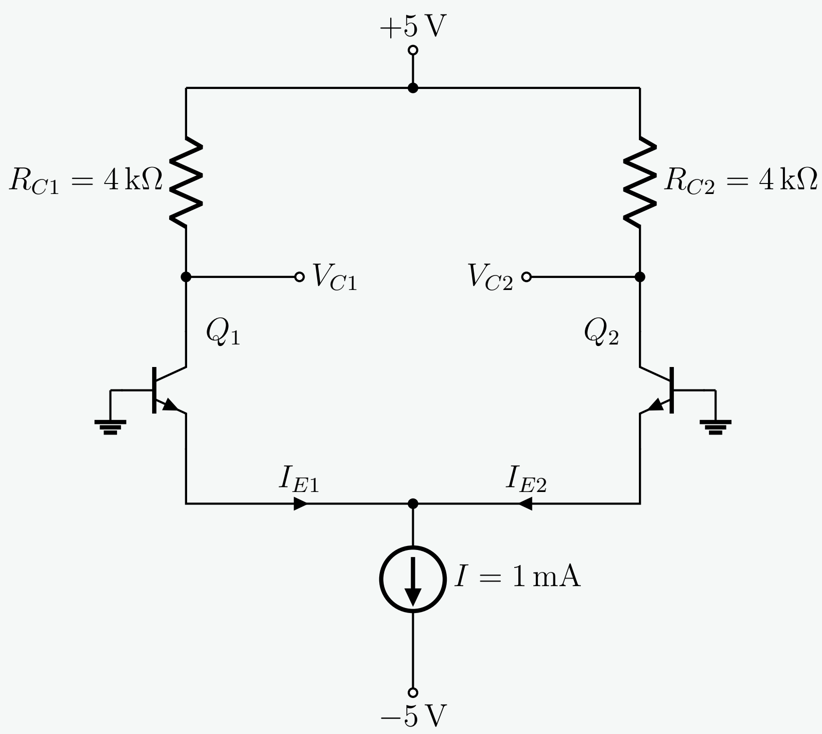


(15 marks) [EEE 263 | L-2/T-1 | 2022–23]

Answer

1. Theory and Circuit Operation

The circuit presented is a **BJT Differential Pair** (or emitter-coupled pair) biased with a constant tail current source $I = 1\,\text{mA}$. The circuit is perfectly symmetrical, featuring matched NPN transistors (Q_1 and Q_2) with equal current gains ($\beta_1 = \beta_2 = 200$) and identical collector resistors ($R_{C1} = R_{C2} = 4\,\text{k}\Omega$).
For a perfectly matched differential pair with both bases grounded ($V_{B1} = V_{B2} = 0\,\text{V}$), the differential input voltage is zero. Consequently, the tail current I divides equally between the two transistors. The transistors are assumed to be operating in the **forward-active region**, which we will verify subsequently.



2. Step-by-Step Derivation

Step 1: Determine Emitter Currents (I_{E1} and I_{E2})
Let V_E be the voltage at the common emitter node. The base-emitter voltage for each transistor is:

$$V_{BE1} = V_{B1} - V_E = 0 - V_E = -V_E$$
$$V_{BE2} = V_{B2} - V_E = 0 - V_E = -V_E$$

Since $V_{BE1} = V_{BE2}$ and the transistors are matched, the emitter currents must be equal:

$$I_{E1} = I_{E2} \tag{1}$$

Applying Kirchhoff's Current Law (KCL) at the common emitter node:

$$\begin{aligned} I_{E1} + I_{E2} &= I \\ 2I_{E1} &= 1\,\mathrm{mA} \\ I_{E1} = I_{E2} &= \mathbf{0.5}\,\mathrm{mA} \end{aligned}$$

Step 2: Calculate Collector Currents (I_{C1} and I_{C2})
In the forward-active region, the collector current is related to the emitter current by the common-base current gain α:

$$\alpha = \frac{\beta}{\beta+1} = \frac{200}{200+1} = \frac{200}{201} \approx 0.995 \tag{2}$$

Therefore, the collector currents are:

$$I_{C1} = I_{C2} = \alpha I_{E1} = \left(\frac{200}{201}\right) \times 0.5\,\mathrm{mA} \approx \mathbf{0.4975}\,\mathrm{mA} \tag{3}$$

Step 3: Determine Collector Voltages (V_{C1} and V_{C2})
The collector voltages are determined by the voltage drop across the collector resistors R_{C1} and R_{C2} from the positive supply V_{CC}:

$$\begin{aligned} V_{C1} &= V_{CC} - I_{C1}R_{C1} \\ V_{C1} &= 5\,\mathrm{V} - (0.4975\,\mathrm{mA})(4\,\mathrm{k\Omega}) \\ V_{C1} &= 5 - 1.99 = \mathbf{3.01}\,\mathrm{V} \end{aligned}$$

Due to symmetry:

$$V_{C2} = V_{CC} - I_{C2}R_{C2} = 5\,\mathrm{V} - (0.4975\,\mathrm{mA})(4\,\mathrm{k\Omega}) = \mathbf{3.01}\,\mathrm{V} \tag{4}$$

Note: If one approximates $\alpha \approx 1$, the calculation yields $I_C \approx 0.5$ mA and $V_C = 3.00$ V. The exact evaluation yields 3.01 V.

3. Operating Region Verification

To rigorously confirm that our forward-active assumption is correct, we analyze the junction potentials:

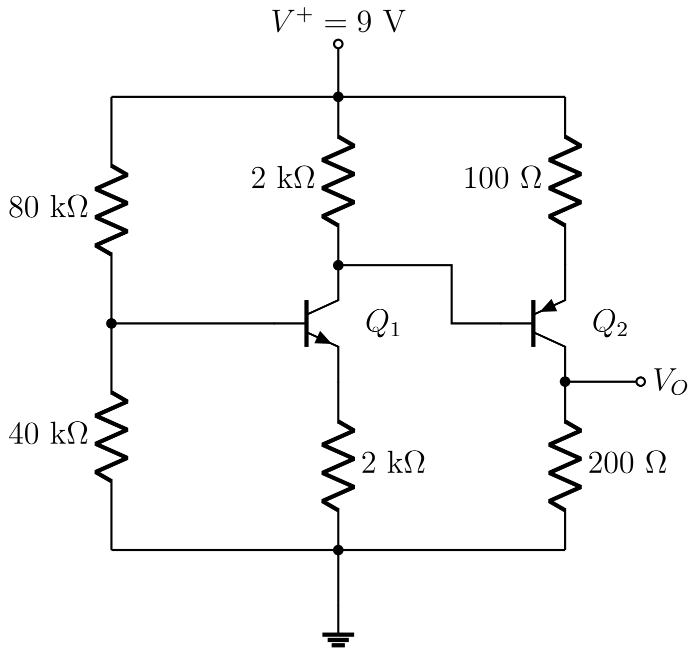
- **Base Voltage:** $V_B = 0\,\mathrm{V}$
- **Emitter Voltage:** Assuming a typical diode drop, $V_E = 0\,\mathrm{V} - 0.7\,\mathrm{V} = -0.7\,\mathrm{V}$. The base-emitter junction is forward-biased.
- **Collector Voltage:** $V_C = 3.01\,\mathrm{V}$.
- **Collector-Base Junction:** $V_{CB} = V_C - V_B = 3.01\,\mathrm{V} - 0\,\mathrm{V} = +3.01\,\mathrm{V}$.

Since $V_{CB} > 0\,\mathrm{V}$ for an NPN transistor, the collector-base junction is fully reverse-biased. The forward-active assumption is therefore validated.

4. Final Answer Summary

Parameter	Symbol	Calculated Value
Emitter Current 1	I_{E1}	0.50 mA
Emitter Current 2	I_{E2}	0.50 mA
Collector Voltage 1	V_{C1}	3.01 V
Collector Voltage 2	V_{C2}	3.01 V

Q4. For the circuit shown, $\beta_n = 120$ (npn) and $\beta_p = 80$ (pnp). Determine I_{B1}, I_{C1}, I_{B2}, I_{C2}, V_{CE1}, and V_{EC2}.



(20 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

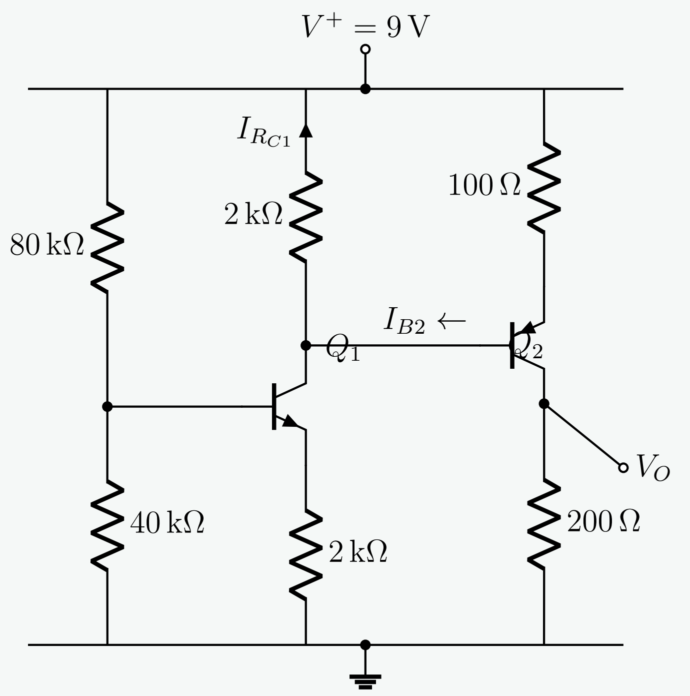
Answer

1. Circuit Diagram and Operating Principle

The circuit consists of a two-stage direct-coupled amplifier featuring an NPN transistor (Q_1) and a PNP transistor (Q_2). Q_1 is biased using a voltage divider (80 kΩ and 40 kΩ) and its collector is directly coupled to the base of Q_2. Because Q_2 is a PNP transistor, its base current I_{B2} flows *out* of the base and into the collector node of Q_1.

Assumptions for Analysis:

- Both transistors are operating in the **forward-active region**. We will verify this assumption at the end of the analysis.
- The base-emitter turn-on voltages are $V_{BE1} = 0.7\,\text{V}$ (for NPN) and $V_{EB2} = 0.7\,\text{V}$ (for PNP).
- The Early effect is negligible ($V_A = \infty$).



2. DC Analysis of Q_1 (NPN)

We simplify the base circuit of Q_1 using Thévenin's Theorem. The Thévenin voltage (V_{TH}) and Thévenin resistance (R_{TH}) are:

$$V_{TH} = V^+ \left(\frac{40\,\text{k}\Omega}{80\,\text{k}\Omega + 40\,\text{k}\Omega} \right) = 9 \left(\frac{1}{3} \right) = \mathbf{3}\,\text{V}$$
$$R_{TH} = 80\,\text{k}\Omega \parallel 40\,\text{k}\Omega = \frac{80 \times 40}{80 + 40}\,\text{k}\Omega = \mathbf{26.667}\,\text{k}\boldsymbol{\Omega}$$

Applying Kirchhoff's Voltage Law (KVL) around the base-emitter loop of Q_1:

$$V_{TH} - I_{B1}R_{TH} - V_{BE1} - I_{E1}R_{E1} = 0 \tag{1}$$

Using the relationship $I_{E1} = (\beta_n + 1)I_{B1} = (120 + 1)I_{B1} = 121I_{B1}$, we can solve for I_{B1}:

$$3 - I_{B1}(26.667\,\text{k}\Omega) - 0.7 - (121I_{B1})(2\,\text{k}\Omega) = 0$$
$$2.3 = I_{B1}(26.667\,\text{k}\Omega + 242\,\text{k}\Omega)$$
$$I_{B1} = \frac{2.3}{268.667 \times 10^3} \approx \mathbf{8.561}\,\mu\text{A}$$

Now we calculate the collector and emitter currents for Q_1:

$$I_{C1} = \beta_n I_{B1} = 120 \times 8.561\,\mu\text{A} = \mathbf{1.027}\,\text{mA}$$
$$I_{E1} = (\beta_n + 1)I_{B1} = 121 \times 8.561\,\mu\text{A} = \mathbf{1.036}\,\text{mA}$$

3. DC Analysis of Q_2 (PNP)

The collector of Q_1 is tied to the base of Q_2. Applying Kirchhoff's Current Law (KCL) at this node (V_{C1}):

$$I_{R_{C1}} + I_{B2} = I_{C1} \tag{2}$$

where $I_{R_{C1}}$ is the current descending through the $2\,\text{k}\Omega$ resistor, and I_{B2} is the base current flowing *out* of the PNP transistor Q_2. Expressing $I_{R_{C1}}$ in terms of the node voltage V_{C1}:

$$\frac{9 - V_{C1}}{2\,\text{k}\Omega} + I_{B2} = 1.027\,\text{mA} \tag{3}$$

Next, apply KVL from the 9 V supply through the emitter-base junction of Q_2 to find V_{C1}:

$$V_{C1} = V_{B2} = 9 - I_{E2}R_{E2} - V_{EB2} \tag{4}$$

Substitute $V_{EB2} = 0.7\,\text{V}$ and $I_{E2} = (\beta_p + 1)I_{B2} = 81I_{B2}$:

$$V_{C1} = 9 - (81I_{B2})(0.1\,\text{k}\Omega) - 0.7$$
$$V_{C1} = 8.3 - 8.1I_{B2}$$

Substitute Equation (**??**) back into Equation (3):

$$\frac{9 - (8.3 - 8.1I_{B2})}{2} + I_{B2} = 1.027 \quad \text{(currents in mA, resistance in k}\Omega\text{)}$$
$$\frac{0.7 + 8.1I_{B2}}{2} + I_{B2} = 1.027$$
$$0.35 + 4.05I_{B2} + I_{B2} = 1.027$$
$$5.05I_{B2} = 0.677$$
$$I_{B2} = \mathbf{0.1341}\,\text{mA} \quad \text{(or } 134.1\,\mu\text{A)}$$

Now, calculate I_{C2} and I_{E2}:

$$I_{C2} = \beta_p I_{B2} = 80 \times 0.1341\,\text{mA} = \mathbf{10.73}\,\text{mA}$$
$$I_{E2} = (\beta_p + 1)I_{B2} = 81 \times 0.1341\,\text{mA} = \mathbf{10.86}\,\text{mA}$$

4. Voltage Calculations & Active Region Verification

For Transistor Q_1:

$$V_{E1} = I_{E1}R_{E1} = (1.036\,\text{mA})(2\,\text{k}\Omega) = \mathbf{2.072}\,\text{V}$$
$$V_{C1} = 8.3 - 8.1I_{B2} = 8.3 - 8.1(0.1341) = 8.3 - 1.086 = \mathbf{7.214}\,\text{V}$$
$$V_{CE1} = V_{C1} - V_{E1} = 7.214\,\text{V} - 2.072\,\text{V} = \mathbf{5.142}\,\text{V}$$

Since $V_{CE1} > 0.2\,\text{V}$, the assumption that Q_1 is in the forward-active region is confirmed.
For Transistor Q_2:

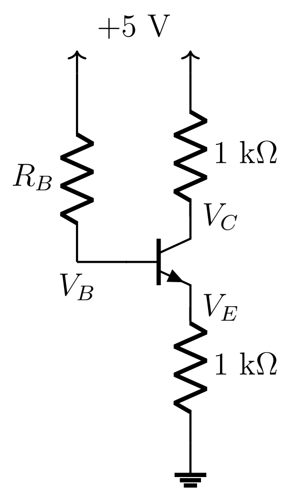
$$V_{E2} = 9 - I_{E2}R_{E2} = 9 - (10.86\,\text{mA})(0.1\,\text{k}\Omega) = 9 - 1.086 = \mathbf{7.914}\,\text{V}$$
$$V_{C2} = I_{C2}R_{C2} = (10.73\,\text{mA})(0.2\,\text{k}\Omega) = \mathbf{2.146}\,\text{V}$$
$$V_{EC2} = V_{E2} - V_{C2} = 7.914\,\text{V} - 2.146\,\text{V} = \mathbf{5.768}\,\text{V}$$

Since $V_{EC2} > 0.2\,\text{V}$, the assumption that Q_2 is in the forward-active region is also confirmed.

5. Final Answer Summary

Parameter	Calculated Value	Unit
I_{B1}	8.561	μA
I_{C1}	1.027	mA
I_{B2}	134.1	μA
I_{C2}	10.73	mA
V_{CE1}	5.142	V
V_{EC2}	5.768	V

Q5. For the circuit shown, find V_B, V_E, and V_C for $R_B = 10\,\text{k}\Omega$. Let $\beta = 100$.



(16 marks) [EEE 263 | L-2/T-1 | 2021–22]

Answer

1. Initial Assumption: Forward-Active Region

To determine the operating state of the NPN transistor, we first assume it operates in the forward-active region. We will use the standard base-emitter turn-on voltage $V_{BE} = 0.7\,\text{V}$.
Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop:

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0 \tag{1}$$

In the forward-active region, the emitter current is related to the base current by $I_E = (\beta + 1)I_B$. Substituting this into the KVL equation yields:

$$\begin{aligned} V_{CC} - V_{BE} &= I_B R_B + (\beta + 1) I_B R_E \\ 5\,\text{V} - 0.7\,\text{V} &= I_B(10\,\text{k}\Omega + 101 \times 1\,\text{k}\Omega) \\ 4.3\,\text{V} &= I_B(111\,\text{k}\Omega) \\ I_B &= \frac{4.3}{111\ \times 10^3} \approx 38.74\,\mu\text{A} \end{aligned}$$

Now, calculate the resulting collector and emitter currents:

$$\begin{aligned} I_C &= \beta I_B = 100 \times 38.74\,\mu\text{A} \approx 3.874\,\text{mA} \\ I_E &= (\beta + 1) I_B = 101 \times 38.74\,\mu\text{A} \approx 3.913\,\text{mA} \end{aligned}$$

Next, check the node voltages to verify the active region assumption:

$$\begin{aligned} V_E &= I_E R_E = (3.913\,\text{mA})(1\,\text{k}\Omega) = 3.913\,\text{V} \\ V_C &= V_{CC} - I_C R_C = 5\,\text{V} - (3.874\,\text{mA})(1\,\text{k}\Omega) = 1.126\,\text{V} \end{aligned}$$

Calculating the collector-emitter voltage:

$$V_{CE} = V_C - V_E = 1.126\,\text{V} - 3.913\,\text{V} = -2.787\,\text{V} \tag{2}$$

Since $V_{CE} < 0$, the collector-base junction is forward-biased. The assumption that the transistor is in the active region is **incorrect**; the transistor is in **saturation**.

2. Analysis in the Saturation Region

Assuming the transistor is saturated, we use the standard saturation voltages:

$$\begin{aligned} V_{BE(sat)} &= 0.7\,\text{V} \\ V_{CE(sat)} &= 0.2\,\text{V} \end{aligned}$$

This gives us the following relationships between the node voltages:

$$V_B = V_E + 0.7 \quad (3)$$
$$V_C = V_E + 0.2 \quad (4)$$

We apply Kirchhoff's Current Law (KCL) for the transistor currents, $I_E = I_B + I_C$:

$$\frac{V_E}{R_E} = \frac{V_{CC} - V_B}{R_B} + \frac{V_{CC} - V_C}{R_C} \quad (5)$$

Substitute the voltage relationships from Equations (3) and (4) into the KCL equation:

$$\begin{aligned}
\frac{V_E}{1} &= \frac{5 - (V_E + 0.7)}{10} + \frac{5 - (V_E + 0.2)}{1} \quad \text{(currents in mA, resistance in k}\Omega\text{)} \\
V_E &= \frac{4.3 - V_E}{10} + (4.8 - V_E) \\
V_E &= 0.43 - 0.1V_E + 4.8 - V_E \\
V_E + 0.1V_E + V_E &= 5.23 \\
2.1V_E &= 5.23 \\
V_E &\approx \mathbf{2.490}\,\text{V}
\end{aligned}$$

Now, calculate V_B and V_C using the solved value of V_E:

$$V_B = 2.490\,\text{V} + 0.7\,\text{V} = \mathbf{3.190}\,\text{V}$$
$$V_C = 2.490\,\text{V} + 0.2\,\text{V} = \mathbf{2.690}\,\text{V}$$

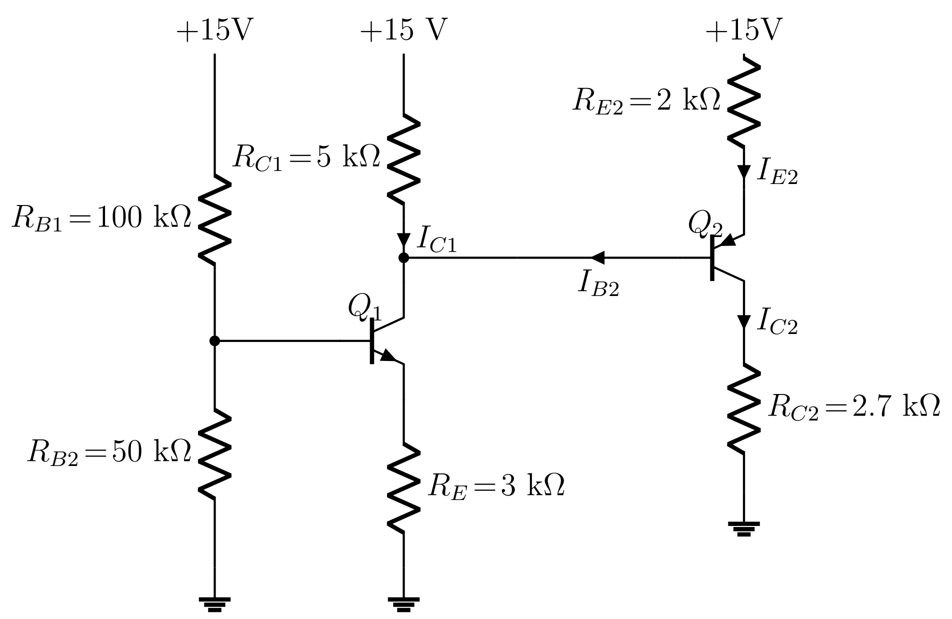
Verification of Saturation: Let's quickly verify that the required β_{forced} is less than the specified $\beta = 100$:

$$\begin{aligned}
I_B &= \frac{5 - 3.190}{10} = 0.181\,\text{mA} \\
I_C &= \frac{5 - 2.690}{1} = 2.31\,\text{mA} \\
\beta_{forced} &= \frac{I_C}{I_B} = \frac{2.31}{0.181} \approx 12.8 \quad (\text{Since } 12.8 < 100, \text{ saturation is confirmed.})
\end{aligned}$$

3. Final Results

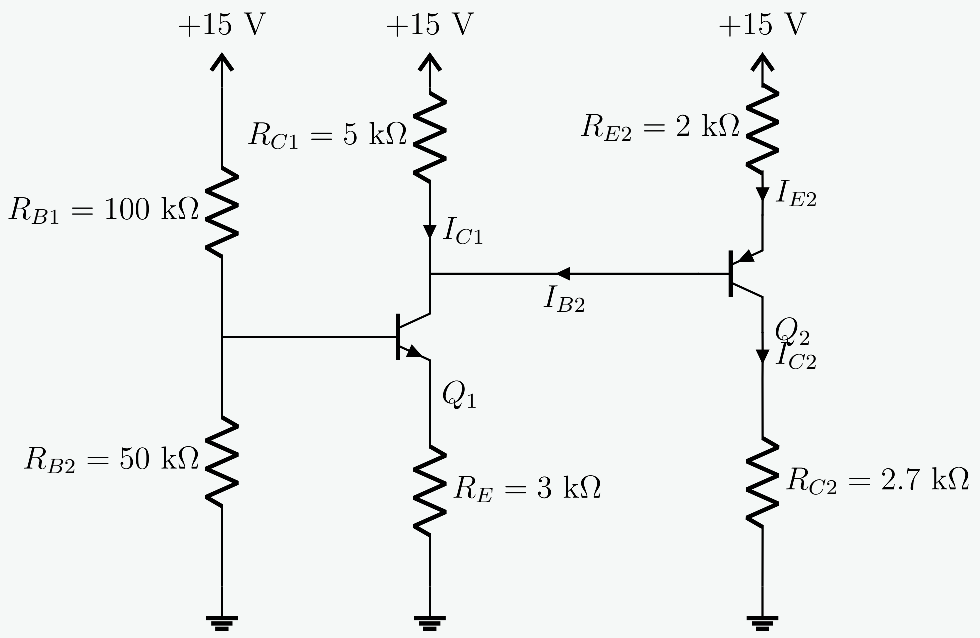
Node Voltage	Calculated Value	Unit
V_B	3.190	V
V_E	2.490	V
V_C	2.690	V

Q6. Determine the currents I_{C1} and I_{C2} for the multi-BJT circuit shown.



(30 marks) [EEE 263 | L-2/T-1 | 2017–18]

Answer



1. Assumptions and Methodology

The problem does not specify the current gain (β) for the transistors. While an ideal assumption ($\beta \to \infty$) is sometimes used, doing so in this specific circuit will drive the PNP transistor (Q_2) deep into saturation, which is generally not intended for linear amplifier analysis. Therefore, we will perform the exact DC analysis assuming a standard typical value of $\beta = 100$ for both transistors and a forward-active base-emitter voltage $V_{BE(\mathbf{on})} = 0.7$ **V** for Q_1 and $V_{EB(\text{on})} = 0.7$ V for Q_2. We will verify the active region at the end.

2. DC Analysis of Stage 1 (Q_1 - NPN)

We begin by simplifying the base circuit of Q_1 using Thévenin's theorem.

$$V_{TH} = V_{CC}\left(\frac{R_{B2}}{R_{B1}+R_{B2}}\right) = 15 \text{ V}\left(\frac{50 \text{ k}\Omega}{100 \text{ k}\Omega + 50 \text{ k}\Omega}\right) = 5 \text{ V}$$

$$R_{TH} = R_{B1} \parallel R_{B2} = \frac{(100 \text{ k}\Omega)(50 \text{ k}\Omega)}{100 \text{ k}\Omega + 50 \text{ k}\Omega} = 33.33 \text{ k}\Omega$$

Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop of Q_1:

$$V_{TH} - I_{B1}R_{TH} - V_{BE(\text{on})} - I_{E1}R_E = 0$$

Substitute $I_{E1} = (\beta+1)I_{B1} = 101I_{B1}$:

$$5 - I_{B1}(33.33) - 0.7 - (101I_{B1})(3) = 0 \quad \text{(Resistance in k}\Omega\text{, Current in mA)}$$
$$4.3 = I_{B1}(33.33 + 303)$$
$$I_{B1} = \frac{4.3}{336.33} \approx 0.012785 \text{ mA}$$

Now, compute the collector current for Q_1:

$$I_{C1} = \beta I_{B1} = 100(0.012785 \text{ mA}) = \mathbf{1.2785} \text{ mA}$$

3. DC Analysis of Stage 2 (Q_2 - PNP)

Q_2 is a PNP transistor. Its base current I_{B2} flows *out* of the base and into the collector node of Q_1. Applying KCL at the collector of Q_1 (Node C_1):

$$I_{R_{C1}} + I_{B2} = I_{C1}$$
$$I_{R_{C1}} = I_{C1} - I_{B2}$$

The voltage at the base of Q_2 (which is V_{C1}) can be written as:

$$V_{B2} = V_{C1} = V_{CC} - I_{R_{C1}}R_{C1} = 15 - 5(I_{C1} - I_{B2})$$

The voltage at the emitter of Q_2 is:

$$V_{E2} = V_{CC} - I_{E2}R_{E2} = 15 - 2(101I_{B2}) = 15 - 202I_{B2}$$

Assuming Q_2 is in the forward-active region, $V_{E2} - V_{B2} = V_{EB(\text{on})} = 0.7$ V. Substituting the voltage expressions:

$$(15 - 202I_{B2}) - [15 - 5(I_{C1} - I_{B2})] = 0.7$$
$$15 - 202I_{B2} - 15 + 5I_{C1} - 5I_{B2} = 0.7$$
$$5(1.2785) - 207I_{B2} = 0.7$$
$$6.3925 - 207I_{B2} = 0.7$$
$$207I_{B2} = 5.6925$$
$$I_{B2} \approx 0.0275 \text{ mA}$$

Now, calculate the collector current for Q_2:

$$I_{C2} = \beta I_{B2} = 100(0.0275 \text{ mA}) = \mathbf{2.75} \text{ mA}$$

4. Verification of Active Region

To ensure our initial assumptions are valid, we check the collector-emitter voltages for both BJTs.

$$V_{C1} = 15 - 5(1.2785 - 0.0275) = 15 - 5(1.251) = 8.745 \text{ V}$$
$$V_{E1} = I_{E1}R_E = (101 \times 0.012785)(3) = 3.874 \text{ V}$$
$$V_{CE1} = V_{C1} - V_{E1} = 8.745 - 3.874 = \mathbf{4.871} \text{ V} > 0.2 \text{ V} \quad (Q_1 \text{ is Active})$$

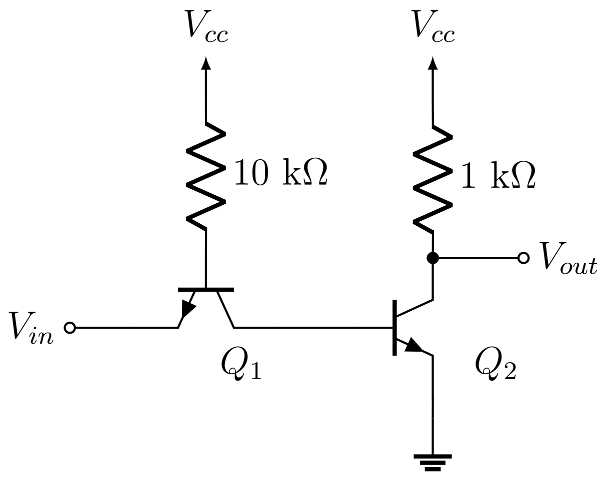
$$V_{E2} = 15 - 202(0.0275) = 15 - 5.555 = 9.445 \text{ V}$$
$$V_{C2} = I_{C2}R_{C2} = 2.75(2.7) = 7.425 \text{ V}$$
$$V_{EC2} = V_{E2} - V_{C2} = 9.445 - 7.425 = \mathbf{2.02} \text{ V} > 0.2 \text{ V} \quad (Q_2 \text{ is Active})$$

5. Final Results

Parameter	Calculated Value	Unit
Collector Current of Q_1 (I_{C1})	**1.279**	mA
Collector Current of Q_2 (I_{C2})	**2.750**	mA

Voltage Transfer Characteristics

Q1. Draw the voltage transfer characteristics by qualitatively analyzing the circuit shown.



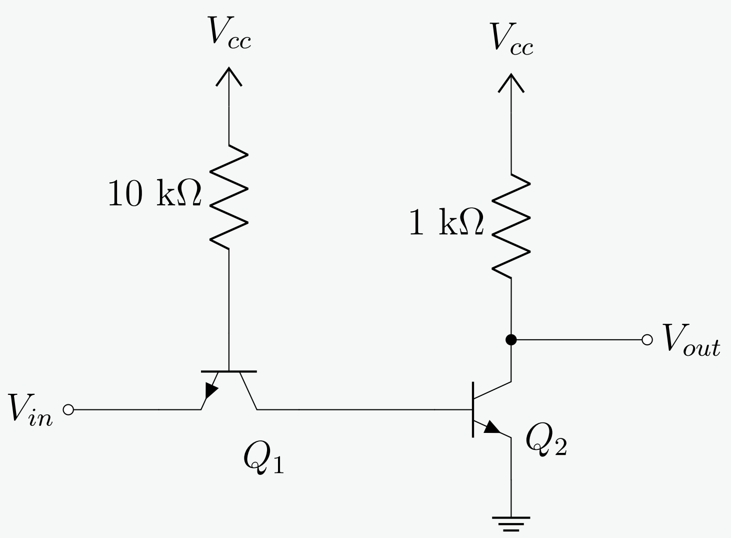
(15 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Circuit Identification

The given circuit is the input and output stage of a simplified, standard **Transistor-Transistor Logic (TTL) Inverter**.

- **Q_1:** Acts as a multi-emitter (though single in this diagram) steering transistor at the input.
- **Q_2:** Acts as an inverting switch amplifier at the output.



2. Qualitative Analysis of Operating Regions

To derive the Voltage Transfer Characteristic (VTC), we trace the states of Q_1 and Q_2 as the input voltage V_{in} sweeps from 0 V to V_{cc}. We will assume standard BJT turn-on and saturation voltages: $V_{BE(on)} \approx 0.7$ V, $V_{BE(sat)} \approx 0.8$ V, and $V_{CE(sat)} \approx 0.2$ V.

Region I: Logic LOW Input ($V_{in} \approx 0$ V)

- When V_{in} is near ground, the base-emitter junction of Q_1 is heavily forward-biased.
- Base current flows from V_{cc} through the 10 kΩ resistor, into the base of Q_1, and out of its emitter to the input source.
- Q_1 is driven deep into **saturation**. The voltage at its collector (V_{C1}) is:

$$V_{C1} = V_{in} + V_{CE1(sat)} \approx V_{in} + 0.2 \text{ V}$$

- Because V_{C1} is directly tied to the base of Q_2, we have $V_{BE2} = V_{in} + 0.2$ V.
- For $V_{in} < 0.5$ V, V_{BE2} remains below the 0.7 V threshold required to turn on Q_2. Therefore, Q_2 is in **cut-off**.
- With Q_2 cut off, no collector current flows through the 1 kΩ resistor. The output voltage is pulled high to the supply rail.

$$V_{out} = V_{cc} \quad \text{(Logic 1)}$$

Region II: Transition Region

- As V_{in} increases to approximately 0.5 V, V_{BE2} reaches ≈ 0.7 V. Q_2 begins to turn **ON** (enters the forward-active region).
- Q_2 starts drawing collector current through the 1 kΩ resistor, causing V_{out} to drop linearly.
- In this active transition phase, small increases in V_{in} cause large decreases in V_{out} due to the voltage gain of the Q_2 common-emitter amplifier.

Region III: Logic HIGH Input ($V_{in} \geq 1.5$ V)

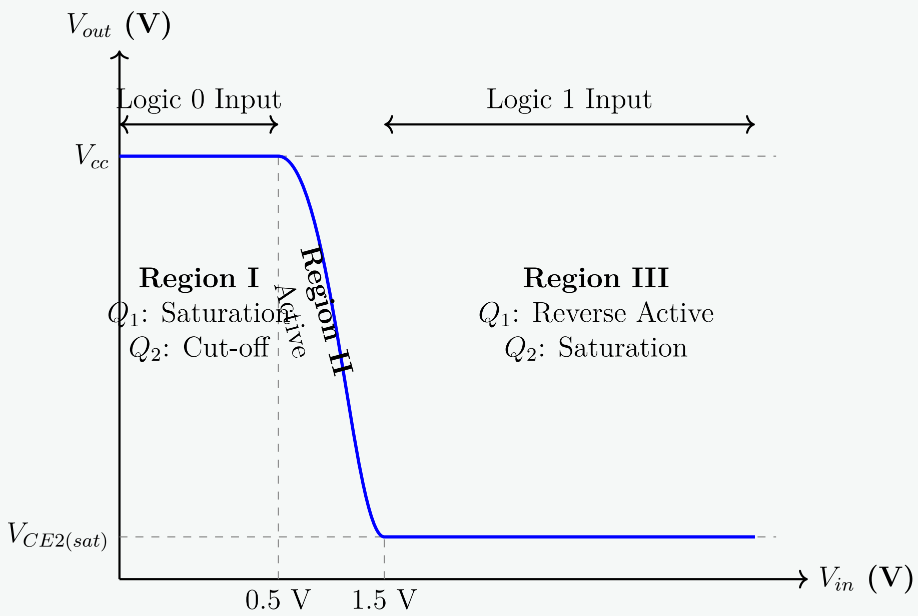
- As V_{in} rises further, the base-emitter junction of Q_1 eventually becomes reverse-biased.
- The base-collector junction of Q_1 becomes forward-biased, acting like a forward-biased diode that steers current from the 10 kΩ resistor directly into the base of Q_2.
- Q_1 is now operating in the **reverse-active** mode.
- This injected base current is large enough to drive Q_2 deep into **saturation**.
- The output voltage is clamped to the collector-emitter saturation voltage of Q_2:

$$V_{out} = V_{CE2(sat)} \approx 0.2 \text{ V} \quad \text{(Logic 0)}$$

- The threshold for this state is fixed by the sum of junction drops: $V_{in} \geq V_{B1} - V_{BE1(on)} \approx (V_{BE2(sat)} + V_{BC1(on)}) -$ Reverse Bias Offset. Practically, the input disconnects and the base of Q_1 sits at ≈ 1.5 V.

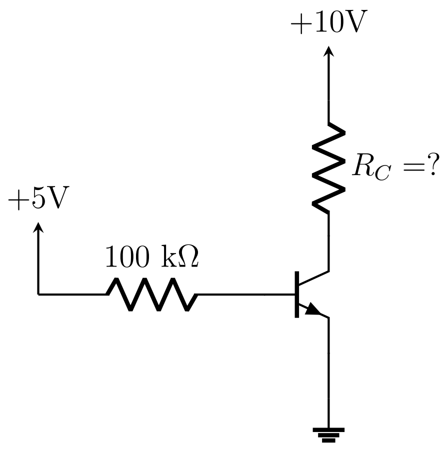
3. Voltage Transfer Characteristic (VTC) Diagram

Plotting V_{out} against V_{in} based on the above analysis yields the standard inverting VTC curve:



Operating Mode Design

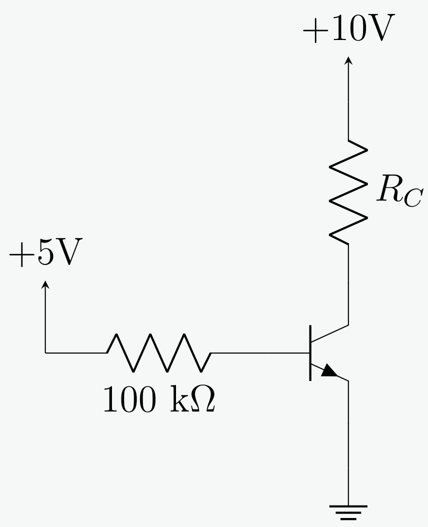
Q1. The circuit shown is to be fabricated using a transistor type whose β can vary from 50 to 150. Design the circuit (find R_c) so that all fabricated circuits are guaranteed to be in the active mode. What is the range of collector voltages that the fabricated circuits may exhibit?



(20 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Circuit Diagram



2. Assumptions & Theory

To guarantee that the BJT operates in the **forward-active mode**, the following conditions must be met:

(a) The base-emitter junction must be forward-biased: $V_{BE} \approx 0.7$ V.

(b) The base-collector junction must be reverse-biased, meaning the transistor does not enter saturation. We define the edge of saturation (EOS) at $V_{CE(EOS)} = 0.3$ V. Therefore, to remain active, we require $V_{CE} \geq 0.3$ V.

3. DC Analysis (Base Circuit)

Applying Kirchhoff's Voltage Law (KVL) at the base loop:

$$V_{BB} - I_B R_B - V_{BE} = 0$$
$$5\text{ V} - I_B(100\text{ k}\Omega) - 0.7\text{ V} = 0$$

Solving for the base current I_B:

$$I_B = \frac{5\text{ V} - 0.7\text{ V}}{100\text{ k}\Omega}$$
$$I_B = \frac{4.3\text{ V}}{100\text{ k}\Omega} = 43\ \mu\text{A}$$

Note that I_B is constant regardless of the transistor's β.

4. Design for Active Mode (Finding R_C)

The collector current I_C depends on the current gain β, which varies between 50 and 150.

$$I_{C,min} = \beta_{min} I_B = 50 \times 43\ \mu\text{A} = 2.15\text{ mA}$$
$$I_{C,max} = \beta_{max} I_B = 150 \times 43\ \mu\text{A} = 6.45\text{ mA}$$

The collector-emitter voltage is given by KVL at the output loop:

$$V_{CE} = V_{CC} - I_C R_C = 10 - I_C R_C \tag{1}$$

To ensure the transistor is in the active mode for *all* fabricated circuits, we must prevent saturation even under the worst-case scenario. The worst case for saturation is when I_C is at its maximum ($I_{C,max} = 6.45$ mA), causing the largest voltage drop across R_C and the minimum V_{CE}.

Enforcing the active mode condition $V_{CE} \geq 0.3$ V:

$$V_{CE,min} = 10 - I_{C,max}R_C \geq 0.3$$
$$10 - (6.45 \text{ mA})R_C \geq 0.3$$
$$(6.45 \text{ mA})R_C \leq 9.7 \text{ V}$$
$$R_C \leq \frac{9.7 \text{ V}}{6.45 \text{ mA}} \approx 1.5038 \text{ k}\Omega$$

A sensible standard engineering choice that satisfies this strict upper bound is:

$$\mathbf{R_C = 1.5\ k\Omega} \tag{2}$$

5. Range of Collector Voltages

Using the chosen design value of $R_C = 1.5$ kΩ, we can determine the range of possible collector voltages (V_C). Since the emitter is grounded, $V_C = V_{CE}$.

Maximum Collector Voltage ($V_{C,max}$): Occurs when I_C is at its minimum ($\beta = 50$):

$$V_{C,max} = V_{CC} - I_{C,min}R_C$$
$$V_{C,max} = 10 \text{ V} - (2.15 \text{ mA})(1.5 \text{ k}\Omega)$$
$$V_{C,max} = 10 \text{ V} - 3.225 \text{ V} = \mathbf{6.775} \text{ V}$$

Minimum Collector Voltage ($V_{C,min}$): Occurs when I_C is at its maximum ($\beta = 150$):

$$V_{C,min} = V_{CC} - I_{C,max}R_C$$
$$V_{C,min} = 10 \text{ V} - (6.45 \text{ mA})(1.5 \text{ k}\Omega)$$
$$V_{C,min} = 10 \text{ V} - 9.675 \text{ V} = \mathbf{0.325} \text{ V}$$

6. Final Answer

- To guarantee active mode operation for all devices, the collector resistor must be chosen such that $R_C \leq 1.50$ **kΩ**. We select $R_C = 1.5$ **kΩ**.
- The corresponding range of collector voltages that the fabricated circuits may exhibit is 0.325 **V** $\leq V_C \leq$ 6.775 **V**.

Biasing and Stability

Q1. What is transistor biasing? Why do we need transistor biasing? How can you achieve biasing stability? **(10 marks)** [**EEE 263 | L-2/T-1 | 2020–21**]

Answer

1. What is Transistor Biasing?

Transistor biasing is the process of applying predetermined external DC voltages and currents to a transistor circuit to establish a specific, steady-state operating point (commonly known as the Quiescent point or **Q-point**) in the absence of an applied AC signal. The Q-point is fundamentally defined by the DC collector current (I_{CQ}) and the DC collector-emitter voltage (V_{CEQ}).

2. Why Do We Need Transistor Biasing?

Proper DC biasing is absolute critical for the reliable and intended operation of a transistor, particularly in analog amplification. The primary reasons include:

- **Faithful Amplification (Distortion Prevention):** To amplify an AC signal without distortion, the transistor must remain in the *active region* throughout the entire 360° of the input signal cycle. Proper biasing places the Q-point near the center of the DC load line, ensuring the signal does not clip by entering the cutoff region or the saturation region.
- **Thermal Stability (Preventing Thermal Runaway):** The collector current I_C of a BJT is governed by the equation:

$$I_C = \beta I_B + (1+\beta)I_{CO} \tag{1}$$

  The reverse saturation current (I_{CO}), the base-emitter voltage (V_{BE}), and the current gain (β) are all highly sensitive to temperature variations. Without an appropriate biasing mechanism, an increase in temperature causes I_C to rise, which increases internal power dissipation ($P_D = V_{CE}I_C$), leading to further heating. This destructive positive feedback cycle is known as *thermal runaway* and can physically destroy the device.

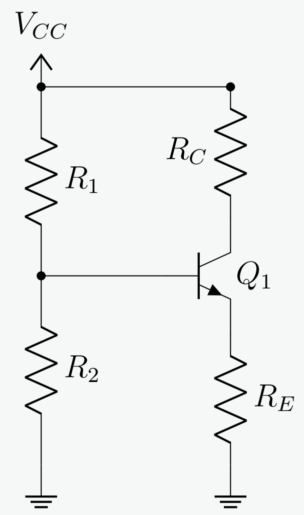
- **Device Variation Independence:** Transistors of the exact same model exhibit a wide manufacturing spread in their β (or h_{FE}) values. A well-designed bias circuit makes the operating point practically independent of the specific β of the installed transistor, allowing for easy device replacement.

3. How Can You Achieve Biasing Stability?

Biasing stability is quantified by the Stability Factor ($S = \partial I_C/\partial I_{CO}$). A smaller value of S implies better stability. Stability is achieved by using circuit topologies that introduce negative DC feedback or active temperature compensation to oppose spontaneous changes in the collector current.

A. Emitter Degeneration (Negative DC Feedback) The most fundamental method of achieving stability is adding an emitter resistor (R_E). If temperature increases and causes I_C to rise, the emitter current ($I_E \approx I_C$) also rises, thereby increasing the voltage drop across R_E ($V_E = I_E R_E$). Assuming the base voltage V_B is relatively constant, the base-emitter voltage ($V_{BE} = V_B - V_E$) decreases. This reduction in V_{BE} causes a drop in base current (I_B), which consequently decreases I_C, opposing the original perturbation.

B. Voltage-Divider Bias Configuration This is universally considered the most stable discrete biasing configuration. It utilizes two resistors to form a voltage divider and an emitter resistor for feedback.



In this circuit, R_1 and R_2 establish a stiff Thevenin equivalent voltage (V_{TH}) at the base. The stability factor for this circuit is closely approximated by:

$$S \approx 1 + \frac{R_{TH}}{R_E} \quad \text{where} \quad R_{TH} = R_1 \parallel R_2 \tag{2}$$

By designing the circuit such that $R_E \gg R_{TH}$ (or $\beta R_E \geq 10 R_{TH}$), the stability factor approaches its ideal theoretical minimum of $S = 1$, rendering the Q-point completely insensitive to β variations and highly resistant to thermal drift.

C. Active Temperature Compensation Techniques In applications subjected to extreme temperature variations, passive feedback may be insufficient. In these cases, temperature-sensitive components are deployed:

- **Diode Compensation:** A forward-biased diode fabricated from the same semiconductor material as the transistor can be placed in the bias network. Since both the diode and the transistor's B-E junction have identical temperature coefficients (≈ -2.5 mV/$^\circ$C), the thermal drifts perfectly cancel each other out.
- **Thermistors and Sensistors:** Components with a negative temperature coefficient (thermistors) or positive temperature coefficient (sensistors) can be dynamically integrated into the R_1 or R_2 branches to actively alter the base voltage in opposition to thermal perturbations.

Q2. Comment on the desired values of input impedance, output impedance, voltage gain, and current gain of a BJT amplifier. Comment on the values of these four parameters for CB, CE, and CC configurations. **(16 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

1. Desired Values for a BJT Amplifier

The "ideal" parameters of a BJT amplifier depend strictly on the type of amplification being performed (voltage, current, transconductance, or transresistance). However, in general analog electronics, when we refer to a "standard amplifier," we are typically discussing a **Voltage Amplifier**.

For an ideal voltage amplifier, the desired characteristics are as follows:

- **Input Impedance (Z_{in}): Ideally Infinite (∞).**
  A high input impedance ensures that the amplifier does not draw significant current from the signal source. This prevents "loading down" the source, which would otherwise attenuate the input signal via voltage division before it even reaches the amplifying stage.
- **Output Impedance (Z_{out}): Ideally Zero (0).**
  A low output impedance allows the amplifier to drive varying or heavy (low resistance) loads without a drop in output voltage. It effectively acts as an ideal Thevenin voltage source to the subsequent stage.
- **Voltage Gain (A_v): Ideally Infinite (∞).**
  The primary purpose of a voltage amplifier is to multiply the amplitude of the input signal. High gain allows small-signal sensing (such as from a microphone or antenna) to be scaled up to usable levels.
- **Current Gain (A_i): High.**
  While a pure voltage amplifier mathematically only needs voltage gain, a practical amplifier must also supply current to the load. High current gain combined with high voltage gain yields high **Power Gain (A_p)**, which is essential for driving physical loads.

2. Comparison of CB, CE, and CC Configurations

The BJT can be configured in three fundamental topologies by choosing which terminal is common to both the input and output loops. None of the configurations achieve the "ideal" parameters in all four categories simultaneously; instead, each topology offers a specific trade-off suitable for distinct applications.

Parameter	Common Emitter (CE)	Common Base (CB)	Common Collector (CC)
Input Impedance (Z_{in})	Moderate (1 kΩ ∼ 10 kΩ)	Very Low (10 Ω ∼ 50 Ω)	Very High (100 kΩ ∼ 500 kΩ)
Output Impedance (Z_{out})	Moderate/High (10 kΩ ∼ 50 kΩ)	Very High (1 MΩ ∼ 2 MΩ)	Very Low (10 Ω ∼ 100 Ω)
Voltage Gain (A_v)	High (~ -100 to -500)	High ($\sim$ 100 to 500)	Low (≈ 1)
Current Gain (A_i)	High (β)	Low (≈ 1, precisely α)	High ($1+\beta$)
Phase Shift	180° (Inverted)	0° (Non-inverted)	0° (Non-inverted)
Power Gain (A_p)	Very High	Moderate	Moderate

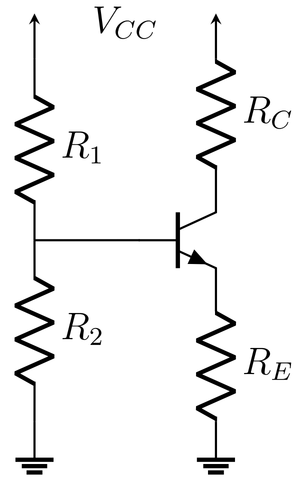
Detailed Commentary on Configurations

A. Common Emitter (CE) Configuration The CE configuration is the most widely used amplifier topology because it provides both high voltage gain and high current gain, resulting in the **highest power gain** among the three configurations. Its moderate input and output impedances make it highly suitable for cascading multiple amplifier stages. It is the only configuration that introduces a 180° phase inversion between the input and output voltages.

B. Common Base (CB) Configuration The CB configuration is characterized by a very low input impedance (as the signal is injected into the emitter) and a very high output impedance. Because its current gain is strictly less than unity ($A_i = \alpha \approx 0.99$), it acts as a **Current Buffer**. Furthermore, the CB configuration does not suffer from the Miller effect (parasitic capacitance multiplication), making it the preferred topology for **high-frequency and RF amplification**.

C. Common Collector (CC) Configuration (Emitter Follower) The CC configuration provides a very high input impedance and a very low output impedance, which perfectly mirrors the ideal characteristics of a **Voltage Buffer**. Its voltage gain is closely tied to unity ($A_v \approx 1$). The primary application of the CC configuration is **impedance matching**; it acts as a bridge to couple a high-impedance signal source to a low-impedance load, preventing signal attenuation without altering the voltage amplitude.

Q3. (a) Find the expression for the emitter current in the circuit shown. What are the conditions that should be satisfied? How does the feedback operation of R_E work? **(16 marks)**



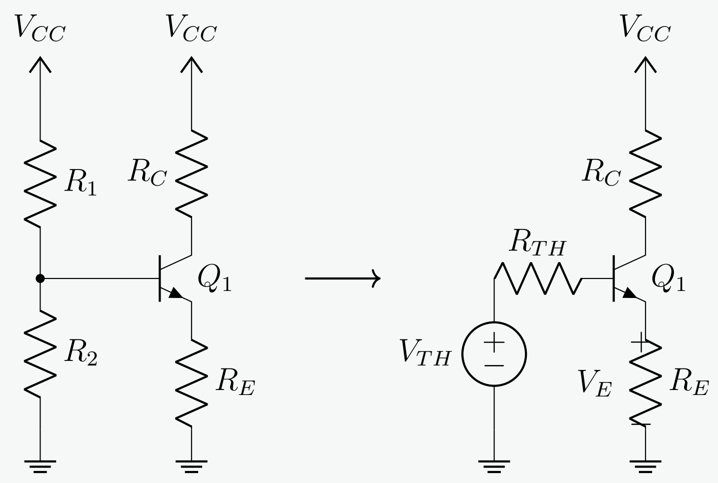
(b) In the circuit of the previous question, if $R_1 = 40\,\text{k}\Omega$, $R_2 = 10\,\text{k}\Omega$, $R_c = 1.5\,\text{k}\Omega$, $R_E = 1\,\text{k}\Omega$, $V_{cc} = 12\,\text{V}$, and $\beta = 1000$, calculate the biasing point. How can the values of R_1 and R_2 be changed for a good biasing design? **(10 marks)** [**EEE 263 | L-2/T-1 | 2014–15**]

Answer

(a) Expression for Emitter Current, Conditions, and Feedback Operation

1. Circuit Diagram and Thevenin Equivalent

To analyze the voltage-divider bias circuit, we first simplify the base circuit using Thevenin's theorem.



The Thevenin voltage (V_{TH}) and Thevenin resistance (R_{TH}) looking into the base voltage divider are given by:

$$V_{TH} = V_{CC}\left(\frac{R_2}{R_1 + R_2}\right)$$
$$R_{TH} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

2. Expression for Emitter Current (I_E)
Applying Kirchhoff's Voltage Law (KVL) to the Base-Emitter loop of the simplified circuit:

$$V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0 \tag{1}$$

We know the fundamental BJT current relationship is $I_B = \frac{I_E}{1+\beta}$. Substituting this into the KVL equation:

$$V_{TH} - \left(\frac{I_E}{1+\beta}\right) R_{TH} - V_{BE} - I_E R_E = 0$$
$$I_E\left(R_E + \frac{R_{TH}}{1+\beta}\right) = V_{TH} - V_{BE}$$
$$I_E = \frac{V_{TH} - V_{BE}}{R_E + \frac{R_{TH}}{1+\beta}}$$

3. Conditions to be Satisfied
For a highly stable biasing point that is practically independent of the transistor's current gain (β), the following condition must be met:

$$R_E \gg \frac{R_{TH}}{1+\beta} \implies (1+\beta)R_E \geq 10R_{TH} \tag{2}$$

When this condition is satisfied, the expression for emitter current simplifies to $I_E \approx \frac{V_{TH}-V_{BE}}{R_E}$. Furthermore, to minimize the impact of temperature variations on V_{BE} (which changes at $\approx -2.5\,\text{mV}/°\text{C}$), we should ensure that $V_{TH} \gg V_{BE}$.
4. Feedback Operation of R_E
The emitter resistor R_E provides **negative DC feedback** to stabilize the operating point against temperature-induced changes in I_C (due to variations in β and reverse saturation current I_{CO}). The mechanism is as follows:

(a) Suppose the temperature increases, causing the collector current (I_C) and consequently the emitter current ($I_E \approx I_C$) to increase.

(b) This increased I_E causes a larger voltage drop across the emitter resistor: $V_E = I_E R_E \uparrow$.

(c) Because the base voltage $V_B \approx V_{TH}$ is held relatively constant by the voltage divider, the base-emitter voltage $V_{BE} = V_B - V_E$ decreases.

(d) A reduced V_{BE} forces the base current (I_B) to drop, which subsequently lowers the collector current (I_C), effectively counteracting the initial temperature-induced increase.

(b) Calculation of Biasing Point and Design Improvements

1. Biasing Point (Q-Point) Calculation
Given parameters: $R_1 = 40\,\text{k}\Omega$, $R_2 = 10\,\text{k}\Omega$, $R_C = 1.5\,\text{k}\Omega$, $R_E = 1\,\text{k}\Omega$, $V_{CC} = 12\,\text{V}$, and $\beta = 1000$. Assume an active-region base-emitter voltage $V_{BE} = 0.7\,\text{V}$.
First, calculate the Thevenin equivalent parameters for the base circuit:

$$V_{TH} = 12\,\text{V} \times \left(\frac{10\,\text{k}\Omega}{40\,\text{k}\Omega + 10\,\text{k}\Omega}\right) = 12 \times \frac{10}{50} = 2.4\,\text{V}$$
$$R_{TH} = \frac{40\,\text{k}\Omega \times 10\,\text{k}\Omega}{40\,\text{k}\Omega + 10\,\text{k}\Omega} = \frac{400}{50}\,\text{k}\Omega = 8\,\text{k}\Omega$$

Calculate the base current (I_B) using the KVL loop equation derived earlier:

$$\begin{aligned} I_B &= \frac{V_{TH} - V_{BE}}{R_{TH} + (1+\beta)R_E} \\ &= \frac{2.4\,\text{V} - 0.7\,\text{V}}{8\,\text{k}\Omega + (1001)(1\,\text{k}\Omega)} \\ &= \frac{1.7\,\text{V}}{1009\,\text{k}\Omega} \approx 1.6848\,\mu\text{A} \end{aligned}$$

Calculate the Quiescent Collector Current (I_{CQ}) and Emitter Current (I_E):

$$\begin{aligned} I_{CQ} &= \beta I_B = 1000 \times 1.6848\,\mu\text{A} = \mathbf{1.685}\,\text{mA} \\ I_E &= (1+\beta)I_B = 1001 \times 1.6848\,\mu\text{A} = 1.686\,\text{mA} \end{aligned}$$

Apply KVL to the Collector-Emitter loop to find V_{CEQ}:

$$\begin{aligned} V_{CEQ} &= V_{CC} - I_{CQ}R_C - I_E R_E \\ &= 12\,\text{V} - (1.685\,\text{mA})(1.5\,\text{k}\Omega) - (1.686\,\text{mA})(1\,\text{k}\Omega) \\ &= 12 - 2.5275 - 1.686 = \mathbf{7.786}\,\text{V} \end{aligned}$$

Verification: Since $V_{CEQ} = 7.786\,\text{V} > 0.2\,\text{V}$, the transistor is definitively in the active region.
Final Q-Point: $(\mathbf{I_{CQ} = 1.685}\,\text{mA},\ \mathbf{V_{CEQ} = 7.786}\,\text{V})$

2. Improving the Biasing Design
A "good" biasing design acts as a stiff voltage source at the base. This requires making the base voltage completely insensitive to variations in base current I_B. The criterion for this is $10R_{TH} \leq \beta R_E$.
While this specific circuit already meets the criteria due to an unusually high $\beta = 1000$, if we were designing for universally good stability (e.g., accommodating standard transistors where β might drop to $50 - 100$), we should:

- **Decrease the values of both R_1 and R_2** proportionally to maintain the same V_{TH} (2.4 V) while significantly lowering R_{TH}.
- A common engineering rule of thumb is to set $R_{TH} \approx 0.1\beta_{min}R_E$.
- **Trade-off Note:** Lowering R_1 and R_2 draws more quiescent current from the V_{CC} supply (wasted power) and lowers the amplifier's overall input impedance ($Z_{in} \approx R_1 \parallel R_2 \parallel \beta r_e$). The optimal values require balancing thermal/β stability with acceptable input impedance.

Q4. Why do we need to bias a BJT at a proper Q-point before using it as an amplifier? **(5 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

1. Concept of the Quiescent Point (Q-point)

The Quiescent point, or **Q-point**, represents the steady-state DC operating point of a BJT when no time-varying input signal is applied. It is uniquely defined on the transistor's characteristic curves by three parameters: the DC base current (I_{BQ}), the DC collector current (I_{CQ}), and the DC collector-emitter voltage (V_{CEQ}).
2. Reasons for Biasing at a Proper Q-point

Biasing the BJT at a carefully selected Q-point before injecting an AC signal is an absolute prerequisite for effective small-signal amplification due to the following fundamental constraints:

- **Ensuring Operation in the Active Region:** For a BJT to operate as a linear amplifier, it must remain exclusively in the **forward-active region**. In this region, the base-emitter junction is forward-biased ($V_{BE} \approx 0.7$ V) and the base-collector junction is reverse-biased ($V_{CB} > 0$ V, or $V_{CE} > V_{CE,\text{sat}} \approx 0.2$ V). If the Q-point is not properly established, the superimposed AC signal can drive the transistor into either:
  - **Saturation:** Where V_{CE} drops below $V_{CE,\text{sat}}$, causing the collector current to independent of base current.
  - **Cutoff:** Where V_{BE} drops below the cut-in voltage, causing the transistor to turn off completely ($I_C \approx 0$).
- **Maximizing Symmetrical Peak-to-Peak Signal Swing:** To prevent waveform distortion, the Q-point should ideally be positioned at the geometric center of the **DC Load Line**. When centered, the input AC signal can swing equally in both positive and negative directions without clipping, thereby providing the maximum possible undistorted output voltage swing ($V_{pp,\text{max}}$).
- **Linearization of the Transistor Characteristics:** The relationship between the base-emitter voltage and collector current is highly non-linear and exponential, described by:

$$I_C = I_S\left(e^{\frac{V_{BE}}{V_T}} - 1\right) \tag{1}$$

By establishing a fixed DC operating point, a small-signal time-varying input ($v_{be}(t)$) can be superimposed. If $v_{be} \ll V_T$ ($V_T \approx 25$ mV at room temperature), the exponential curve can be approximated locally as a straight line. The slope at this Q-point defines the transconductance (g_m):

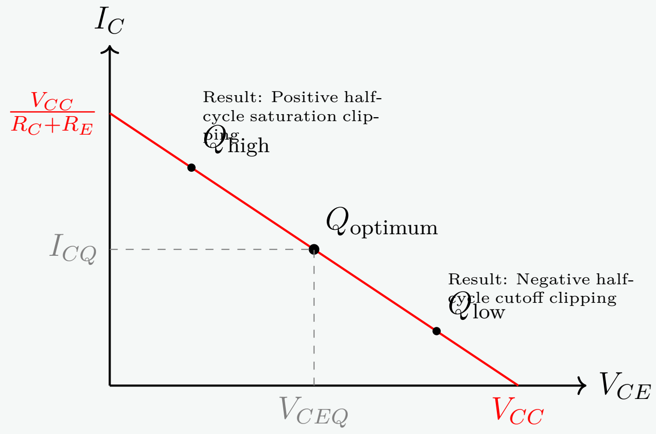
$$g_m = \left.\frac{\partial I_C}{\partial V_{BE}}\right|_{\text{Q-point}} = \frac{I_{CQ}}{V_T} \tag{2}$$

Thus, a stable Q-point ensures a constant, predictable, and linear small-signal voltage gain.

3. Graphical Representation and Signal Distortion

The impact of Q-point placement can be visualized on the $I_C - V_{CE}$ output characteristics along the load line, governed by the loop equation:

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \tag{3}$$

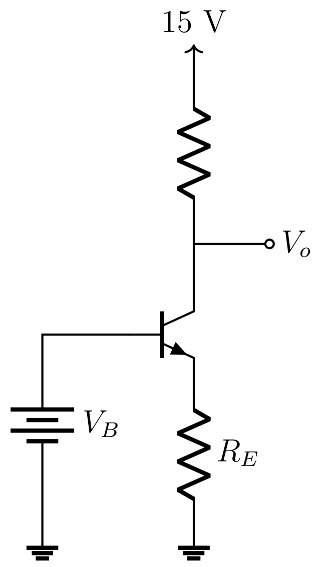


(a) **Optimum Bias (Q_{optimum}):** Located at the center of the load line. It accommodates maximum symmetrical input signal amplitude without introducing clipping.

(b) **Under-biased Circuit (Q_{low}):** Located too close to the cutoff region. The negative half-cycle of the input signal drives the BJT below cutoff, clipping the bottom peak of the output current wave.

(c) **Over-biased Circuit (Q_{high}):** Located too close to the saturation region. The positive half-cycle of the input signal drives V_{CE} below $V_{CE,\text{sat}}$, causing severe compression and clipping on the peak of the output voltage signal.

Q5. Graphically explain the benefit of having $R_E > 0$ in the biasing scheme shown.

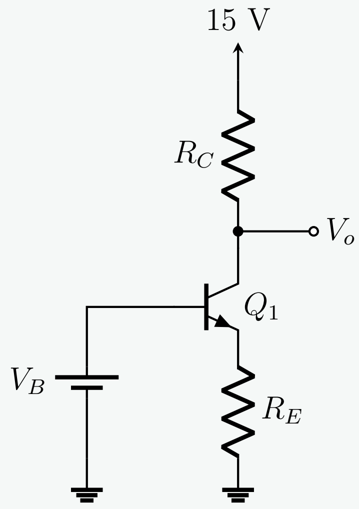


(16 marks) **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

1. Principle of Emitter Degeneration (Negative Feedback)

The inclusion of the emitter resistor R_E in the given biasing scheme introduces **negative DC feedback**, a technique commonly referred to as *emitter degeneration*. Its primary purpose is to stabilize the DC operating point (Q-point) against variations in temperature and inconsistencies in transistor manufacturing parameters (such as β and I_S).



2. Mathematical Derivation of the Bias Load Line

Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop yields:

$$V_B - V_{BE} - I_E R_E = 0 \tag{1}$$

Assuming active region operation, $I_C \approx I_E$. Substituting I_C into the equation gives:

$$V_B - V_{BE} - I_C R_E \approx 0 \tag{2}$$

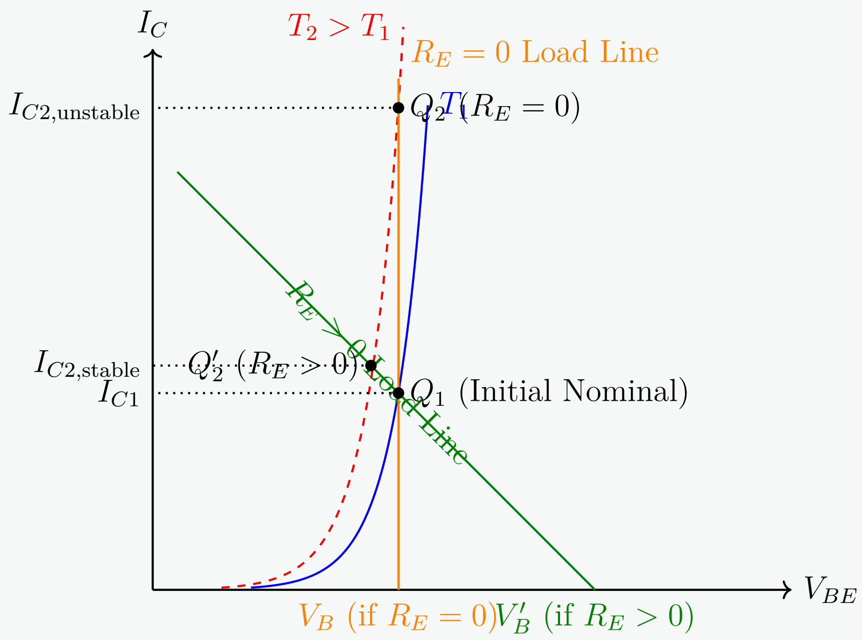
Rearranging this equation to solve for I_C as a function of V_{BE} produces the equation for the **DC Bias Load Line**:

$$I_C = -\frac{1}{R_E}V_{BE} + \frac{V_B}{R_E} \tag{3}$$

This is a linear equation with a slope of $-\frac{1}{R_E}$ and an x-intercept at $V_{BE} = V_B$. The actual Q-point is found at the intersection of this load line and the transistor's exponential transfer characteristic curve: $I_C = I_S e^{V_{BE}/V_T}$.

3. Graphical Explanation of Stability

The true benefit of $R_E > 0$ is best visualized by comparing it to a circuit where $R_E = 0$. As temperature increases from T_1 to T_2, the $I_C - V_{BE}$ curve shifts to the left (since V_{BE} decreases by approximately 2 mV/°C for a constant current).

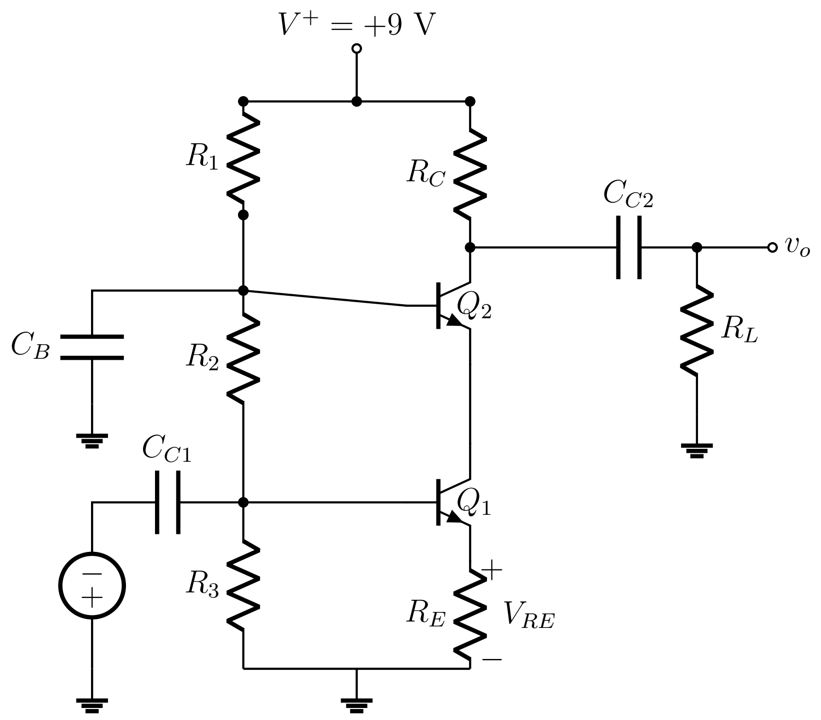


Scenario A: Without Emitter Degeneration ($R_E = 0$) If $R_E = 0$, the bias load line slope is $-\frac{1}{0} \to -\infty$. The load line becomes a **vertical line** fixed at $V_{BE} = V_B$. * When the temperature rises to T_2, the exponential curve shifts left. * Because V_{BE} is rigidly locked to V_B, the new intersection point (Q_2) rockets upwards. * This results in a massive, uncontrolled increase in collector current (from I_{C1} to $I_{C2,\text{unstable}}$), which can easily drive the amplifier into saturation or cause thermal runaway.

Scenario B: With Emitter Degeneration ($R_E > 0$) If $R_E > 0$, the bias load line has a **finite, relatively flat slope** of $-1/R_E$. (To achieve the same initial Q_1, a higher V_B' is supplied). * When the temperature rises to T_2 and the exponential curve shifts left, it slides along the sloped load line. * The negative slope forces V_{BE} to decrease as I_C attempts to increase. * The new intersection point (Q_2') results in a profoundly smaller change in collector current (from I_{C1} to $I_{C2,\text{stable}}$).
Conclusion: Graphically, $R_E > 0$ transforms the bias condition from a completely rigid vertical line to a dynamic sloped line. This negative feedback desensitizes the Q-point, tightly bounding I_C variations and ensuring the amplifier remains reliably in the active region despite thermal drift.

BJT Circuit Design

Q1. Design the circuit shown to meet the following specifications: $V_{CE1} = V_{CE2} = 2.5\,\text{V}$, $V_{RE} = 0.7\,\text{V}$, $I_{C1} \cong I_{C2} \cong 1\,\text{mA}$, and $I_{R1} \cong I_{R2} \cong I_{R3} = 0.10\,\text{mA}$.



(15 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

1. Assumptions and Given Parameters

Before proceeding with the design, we assume the transistors are operating in the forward-active region with a typical base-emitter ON voltage:

$$V_{BE1} = V_{BE2} \approx 0.7 \text{ V}$$

Since the collector currents are approximately equal ($I_{C1} \cong I_{C2} \cong 1$ mA) and the bias string current (0.10 mA) is explicitly defined, we assume high current gain (β) such that base currents are negligible, and $I_E \approx I_C$.

The given parameters are:

$$\begin{aligned} V^+ &= 9 \text{ V} \\ V_{CE1} &= V_{CE2} = 2.5 \text{ V} \\ V_{RE} &= 0.7 \text{ V} \\ I_{C1} &\cong I_{C2} \cong 1 \text{ mA} \\ I_{R1} &\cong I_{R2} \cong I_{R3} = 0.10 \text{ mA} \end{aligned}$$

2. Calculation of Emitter Resistor (R_E)

The voltage across the emitter resistor R_E is given as $V_{RE} = 0.7$ V. Using Ohm's law and the assumption that $I_{E1} \approx I_{C1}$:

$$\begin{aligned} R_E &= \frac{V_{RE}}{I_{E1}} \approx \frac{V_{RE}}{I_{C1}} \\ R_E &= \frac{0.7 \text{ V}}{1 \text{ mA}} = 0.7 \text{ k}\Omega = 700 \ \Omega \end{aligned}$$

3. Calculation of Bias Resistors (R_1, R_2, R_3)

The voltage at the emitter of Q_1 is $V_{E1} = V_{RE} = 0.7$ V. The voltage at the base of Q_1 (V_{B1}) is:

$$V_{B1} = V_{E1} + V_{BE1} = 0.7 \text{ V} + 0.7 \text{ V} = 1.4 \text{ V}$$

Since the current through R_3 is $I_{R3} = 0.10$ mA:

$$\begin{aligned} R_3 &= \frac{V_{B1}}{I_{R3}} \\ R_3 &= \frac{1.4 \text{ V}}{0.10 \text{ mA}} = 14 \text{ k}\Omega \end{aligned}$$

To find R_2, we first determine the required base voltage for Q_2 (V_{B2}). The voltage at the collector of Q_1 (V_{C1}) is:

$$V_{C1} = V_{E1} + V_{CE1} = 0.7 \text{ V} + 2.5 \text{ V} = 3.2 \text{ V}$$

Since Q_2 is in series with Q_1, the emitter voltage of Q_2 is $V_{E2} = V_{C1} = 3.2$ V. The base voltage of Q_2 (V_{B2}) is:

$$V_{B2} = V_{E2} + V_{BE2} = 3.2 \text{ V} + 0.7 \text{ V} = 3.9 \text{ V}$$

The voltage drop across R_2 is $V_{R2} = V_{B2} - V_{B1}$. With $I_{R2} = 0.10$ mA:

$$\begin{aligned} R_2 &= \frac{V_{B2} - V_{B1}}{I_{R2}} \\ R_2 &= \frac{3.9 \text{ V} - 1.4 \text{ V}}{0.10 \text{ mA}} = \frac{2.5 \text{ V}}{0.10 \text{ mA}} = 25 \text{ k}\Omega \end{aligned}$$

The voltage drop across R_1 is $V_{R1} = V^+ - V_{B2}$. With $I_{R1} = 0.10$ mA:

$$R_1 = \frac{V^+ - V_{B2}}{I_{R1}}$$
$$R_1 = \frac{9.0 \text{ V} - 3.9 \text{ V}}{0.10 \text{ mA}} = \frac{5.1 \text{ V}}{0.10 \text{ mA}} = 51 \text{ k}\Omega$$

Check: The total resistance of the bias string is $R_{total} = R_1 + R_2 + R_3 = 51 \text{ k}\Omega + 25 \text{ k}\Omega + 14 \text{ k}\Omega = 90 \text{ k}\Omega$. The current drawn from the 9 V supply is 9 V/90 kΩ = 0.10 mA, which perfectly matches the given specification.

4. Calculation of Collector Resistor (R_C)
To find R_C, we must first determine the voltage at the collector of Q_2 (V_{C2}).

$$V_{C2} = V_{E2} + V_{CE2} = 3.2 \text{ V} + 2.5 \text{ V} = 5.7 \text{ V}$$

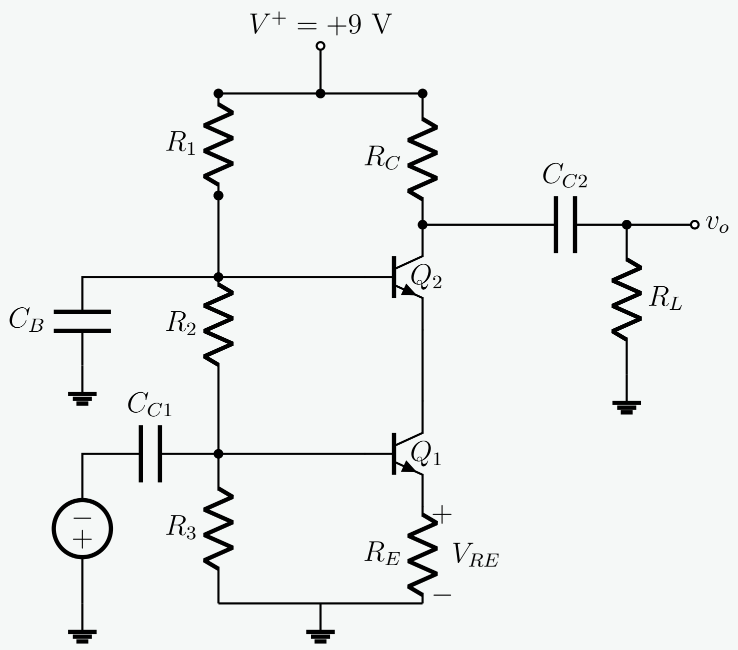
The voltage drop across the collector resistor R_C is $V_{RC} = V^+ - V_{C2}$. Assuming $I_{C2} = 1$ mA:

$$R_C = \frac{V^+ - V_{C2}}{I_{C2}}$$
$$R_C = \frac{9.0 \text{ V} - 5.7 \text{ V}}{1 \text{ mA}} = \frac{3.3 \text{ V}}{1 \text{ mA}} = 3.3 \text{ k}\Omega$$

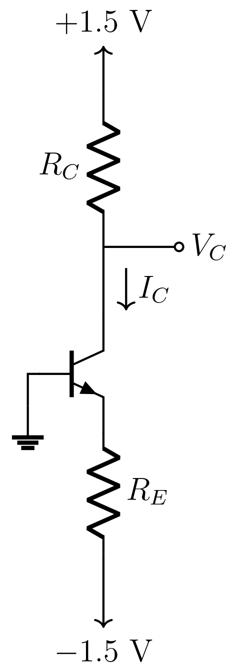
5. Summary of Designed Component Values

Component	Calculated Value
R_1	51 kΩ
R_2	25 kΩ
R_3	14 kΩ
R_C	3.3 kΩ
R_E	0.7 kΩ (700 Ω)

6. Reference Circuit Diagram



Q2. Design the circuit shown to establish $I_C = 0.2\,\text{mA}$ and $V_C = 0.5\,\text{V}$. The transistor exhibits $V_{BE} = 0.8\,\text{V}$ at $i_c = 1\,\text{mA}$, and $\beta = 100$.



(20 marks) [EEE 263 | L-2/T-1 | 2021–22]

Answer

1. Assumptions and Fundamental Parameters

To design the circuit, we first determine the exact base-emitter voltage V_{BE} at the desired operating current. We assume the transistor operates in the forward-active region and use the standard thermal voltage $V_T \approx 25\,\text{mV}$ (at room temperature).
Given parameters:

$$\begin{aligned} I_C &= 0.2\,\text{mA} \\ V_C &= 0.5\,\text{V} \\ V_{BE1} &= 0.8\,\text{V} \quad \text{at} \quad I_{C1} = 1\,\text{mA} \\ \beta &= 100 \\ V^+ &= +1.5\,\text{V} \\ V^- &= -1.5\,\text{V} \end{aligned}$$

2. Determination of the Base-Emitter Voltage (V_{BE})

The collector current in the forward-active region depends exponentially on V_{BE}:

$$I_C = I_S e^{V_{BE}/V_T}$$

Using the given data point (V_{BE1} at I_{C1}), we can find the required V_{BE} for our target I_C using the logarithmic relation:

$$\begin{aligned} V_{BE} &= V_{BE1} + V_T \ln\left(\frac{I_C}{I_{C1}}\right) \\ V_{BE} &= 0.8\,\text{V} + (0.025\,\text{V}) \ln\left(\frac{0.2\,\text{mA}}{1.0\,\text{mA}}\right) \\ V_{BE} &= 0.8\,\text{V} + (0.025\,\text{V}) \ln(0.2) \\ V_{BE} &= 0.8\,\text{V} + (0.025\,\text{V})(-1.609) \\ V_{BE} &\approx 0.8\,\text{V} - 0.040\,\text{V} = 0.760\,\text{V} \end{aligned}$$

3. Calculation of the Emitter Resistor (R_E)

The base of the transistor is grounded, so $V_B = 0\,\text{V}$. The emitter voltage V_E is:

$$\begin{aligned} V_E &= V_B - V_{BE} \\ V_E &= 0\,\text{V} - 0.760\,\text{V} = -0.760\,\text{V} \end{aligned}$$

To find the emitter resistor R_E, we need the emitter current I_E. Since $\beta = 100$, the base current is finite but small:

$$\begin{aligned} I_E &= I_C + I_B = I_C\left(1 + \frac{1}{\beta}\right) = \frac{\beta+1}{\beta} I_C \\ I_E &= \left(\frac{101}{100}\right)(0.2\,\text{mA}) = 0.202\,\text{mA} \end{aligned}$$

Applying Ohm's law to R_E:

$$\begin{aligned} R_E &= \frac{V_E - V^-}{I_E} \\ R_E &= \frac{-0.760\,\text{V} - (-1.5\,\text{V})}{0.202\,\text{mA}} \\ R_E &= \frac{0.740\,\text{V}}{0.202\,\text{mA}} \approx 3.663\,\text{k}\Omega \end{aligned}$$

4. Calculation of the Collector Resistor (R_C)
The target collector voltage is $V_C = 0.5\,\text{V}$. Applying Ohm's law to the collector resistor R_C:

$$R_C = \frac{V^+ - V_C}{I_C}$$
$$R_C = \frac{1.5\,\text{V} - 0.5\,\text{V}}{0.2\,\text{mA}}$$
$$R_C = \frac{1.0\,\text{V}}{0.2\,\text{mA}} = 5.0\,\text{k}\Omega$$

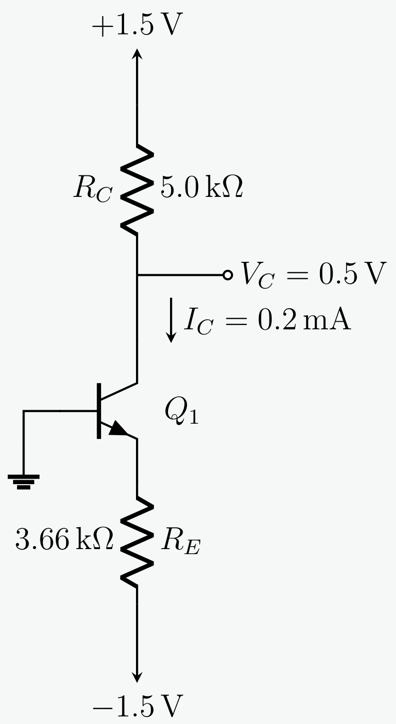
5. Verification of the Operating Region
To ensure the BJT is in the forward-active region, we check the collector-emitter voltage V_{CE}:

$$V_{CE} = V_C - V_E$$
$$V_{CE} = 0.5\,\text{V} - (-0.760\,\text{V}) = 1.26\,\text{V}$$

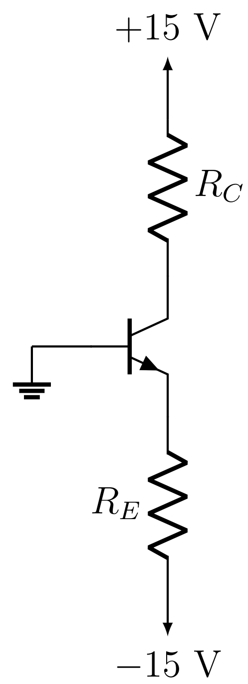
Since $V_{CE} = 1.26\,\text{V} > V_{CE(\text{sat})} \approx 0.2\,\text{V}$ and the base-collector junction is reverse-biased ($V_{BC} = V_B - V_C = -0.5\,\text{V}$), the assumption that the transistor operates in the active region is verified.

6. Summary of Results and Final Circuit Diagram

Parameter	Calculated Value
R_C	$5.0\,\text{k}\Omega$
R_E	$3.66\,\text{k}\Omega$
V_{BE}	$0.76\,\text{V}$
I_E	$0.202\,\text{mA}$



Q3. The transistor in the circuit has $\beta = 100$ and exhibits $V_{BE} = 0.7\,\text{V}$ at $i_c = 1\,\text{mA}$. Design the circuit so that a current of $2\,\text{mA}$ flows through the collector and a voltage of $+5\,\text{V}$ appears at the collector.



(20 marks) [EEE 263 | L-2/T-1 | 2020–21]

Answer

1. Assumptions and Given Parameters

To design the circuit, we must determine the appropriate values for R_C and R_E to establish the desired DC operating point (Q-point). We assume the transistor operates in the forward-active region. We also assume a standard thermal voltage $V_T \approx 25\,\text{mV}$ at room temperature to calculate the exact base-emitter voltage.

The given design parameters are:

$$\begin{aligned} I_C &= 2\,\text{mA} \quad \text{(Target collector current)} \\ V_C &= +5\,\text{V} \quad \text{(Target collector voltage)} \\ V_{BE1} &= 0.7\,\text{V} \quad \text{at} \quad I_{C1} = 1\,\text{mA} \\ \beta &= 100 \\ V_{CC} &= +15\,\text{V} \\ V_{EE} &= -15\,\text{V} \end{aligned}$$

2. Determination of the Base-Emitter Voltage (V_{BE})

The collector current in the forward-active region exhibits an exponential relationship with the base-emitter voltage:

$$I_C = I_S e^{V_{BE}/V_T}$$

We can determine the new V_{BE} for the target $I_C = 2\,\text{mA}$ using the given reference point (0.7 V at 1 mA):

$$\begin{aligned} V_{BE} &= V_{BE1} + V_T \ln\left(\frac{I_C}{I_{C1}}\right) \\ V_{BE} &= 0.7\,\text{V} + (0.025\,\text{V}) \ln\left(\frac{2\,\text{mA}}{1\,\text{mA}}\right) \\ V_{BE} &= 0.7\,\text{V} + (0.025\,\text{V})(0.693) \\ V_{BE} &\approx 0.7\,\text{V} + 0.0173\,\text{V} = 0.717\,\text{V} \end{aligned}$$

3. Calculation of the Emitter Resistor (R_E)

The base of the transistor is directly connected to ground, so the base DC voltage is:

$$V_B = 0\,\text{V}$$

The emitter voltage V_E is determined by the base-emitter voltage drop:

$$\begin{aligned} V_E &= V_B - V_{BE} \\ V_E &= 0\,\text{V} - 0.717\,\text{V} = -0.717\,\text{V} \end{aligned}$$

To calculate the required emitter resistor R_E, we first determine the emitter current I_E. Using the common-emitter current gain $\beta = 100$:

$$\begin{aligned} \alpha &= \frac{\beta}{\beta+1} = \frac{100}{101} \approx 0.9901 \\ I_E &= \frac{I_C}{\alpha} = \frac{2\,\text{mA}}{100/101} = 2.02\,\text{mA} \end{aligned}$$

Now, applying Ohm's law to R_E:

$$\begin{aligned} R_E &= \frac{V_E - V_{EE}}{I_E} \\ R_E &= \frac{-0.717\,\text{V} - (-15\,\text{V})}{2.02\,\text{mA}} \\ R_E &= \frac{14.283\,\text{V}}{2.02\,\text{mA}} \approx 7.07\,\text{k}\Omega \end{aligned}$$

4. Calculation of the Collector Resistor (R_C)

The target collector voltage is specified as $V_C = +5\,\text{V}$. Applying Ohm's law to the collector resistor R_C:

$$\begin{aligned} R_C &= \frac{V_{CC} - V_C}{I_C} \\ R_C &= \frac{15\,\text{V} - 5\,\text{V}}{2\,\text{mA}} \\ R_C &= \frac{10\,\text{V}}{2\,\text{mA}} = 5.0\,\text{k}\Omega \end{aligned}$$

5. Verification of the Operating Region

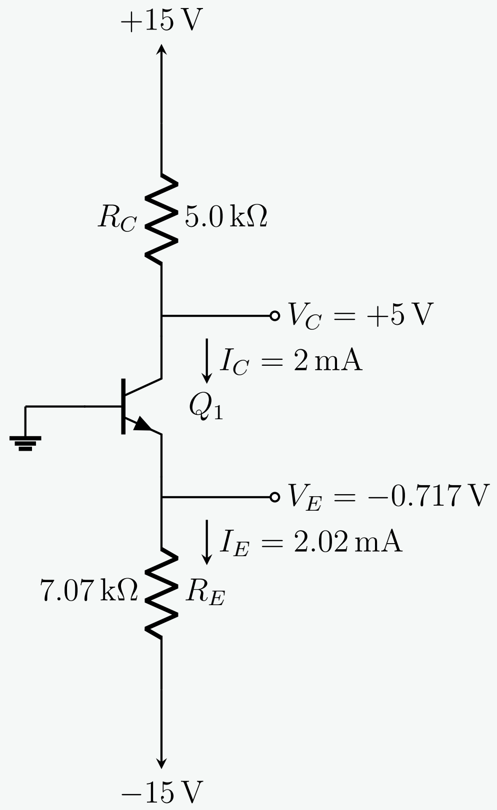
To confirm the transistor is operating in the forward-active region, we check the junction biases:

$$\begin{aligned} V_{CE} &= V_C - V_E \\ V_{CE} &= 5\,\text{V} - (-0.717\,\text{V}) = 5.717\,\text{V} \end{aligned}$$

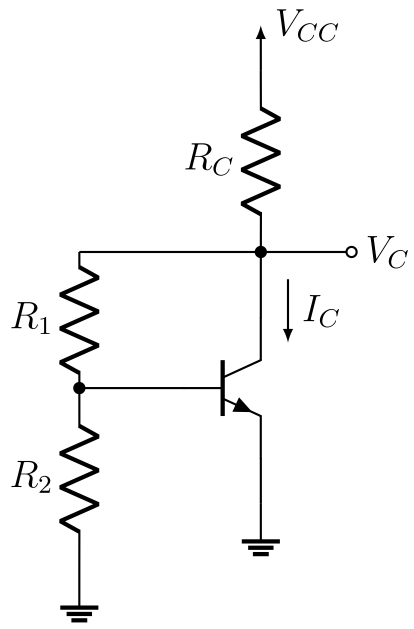
Because $V_{CE} = 5.717\,\text{V} \gg V_{CE(\text{sat})} \approx 0.2\,\text{V}$, and the base-collector voltage $V_{BC} = V_B - V_C = 0\,\text{V} - 5\,\text{V} = -5\,\text{V}$ indicates a strongly reverse-biased CBJ, the transistor is firmly in the active region. The initial assumption holds.

6. Summary of Results and Final Circuit Diagram

Component/Parameter	Calculated Value
Collector Resistor (R_C)	$5.0\,\text{k}\Omega$
Emitter Resistor (R_E)	$7.07\,\text{k}\Omega$
Base-Emitter Voltage (V_{BE})	$0.717\,\text{V}$
Emitter Current (I_E)	$2.02\,\text{mA}$



Q4. Design the circuit shown to provide $I_C = 3\,\text{mA}$ and $V_{CE} = 1.5\,\text{V}$ for $\beta = 90$. Use $V_{CC} = 3\,\text{V}$ and a current through R_2 equal to the base current.



(15 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Assumptions and Given Parameters

We assume the transistor is operating in the forward-active region. Unless otherwise specified, a standard base-emitter ON voltage is assumed for a silicon BJT:

$$V_{BE} = 0.7\,\text{V}$$

The given design parameters are:

$$\begin{aligned} I_C &= 3\,\text{mA} \\ V_{CE} &= 1.5\,\text{V} \\ \beta &= 90 \\ V_{CC} &= 3\,\text{V} \\ I_{R2} &= I_B \quad \text{(Current through } R_2 \text{ equals the base current)} \end{aligned}$$

2. Nodal Voltages

Since the emitter is directly connected to ground, the emitter voltage is:

$$V_E = 0\,\text{V}$$

Using the known voltages V_{BE} and V_{CE}, we can find the base and collector voltages:

$$V_B = V_E + V_{BE} = 0\,\text{V} + 0.7\,\text{V} = 0.7\,\text{V}$$
$$V_C = V_E + V_{CE} = 0\,\text{V} + 1.5\,\text{V} = 1.5\,\text{V}$$

3. Branch Current Calculations
First, we calculate the base current (I_B) required to sustain the desired collector current:

$$I_B = \frac{I_C}{\beta} = \frac{3\,\text{mA}}{90} = \frac{1}{30}\,\text{mA} \approx 33.33\,\mu\text{A}$$

According to the design specifications, the current flowing through R_2 is equal to the base current:

$$I_{R2} = I_B = \frac{1}{30}\,\text{mA}$$

Applying Kirchhoff's Current Law (KCL) at the base node, the current through the feedback resistor R_1 must equal the sum of the base current and the current through R_2:

$$I_{R1} = I_B + I_{R2} = I_B + I_B = 2I_B = 2\left(\frac{1}{30}\,\text{mA}\right) = \frac{1}{15}\,\text{mA} \approx 66.67\,\mu\text{A}$$

Applying KCL at the collector node, the total current flowing through the collector resistor R_C is the sum of the collector current and the feedback network current:

$$I_{RC} = I_C + I_{R1} = 3\,\text{mA} + \frac{1}{15}\,\text{mA} = \frac{46}{15}\,\text{mA} \approx 3.067\,\text{mA}$$

4. Resistor Values Design
Using Ohm's law, we can determine the required resistance values.
For R_2:

$$R_2 = \frac{V_B - 0\,\text{V}}{I_{R2}}$$
$$R_2 = \frac{0.7\,\text{V}}{\frac{1}{30}\,\text{mA}} = 0.7 \times 30\,\text{k}\Omega = 21\,\text{k}\Omega$$

For R_1:

$$R_1 = \frac{V_C - V_B}{I_{R1}}$$
$$R_1 = \frac{1.5\,\text{V} - 0.7\,\text{V}}{\frac{1}{15}\,\text{mA}} = \frac{0.8\,\text{V}}{\frac{1}{15}\,\text{mA}} = 0.8 \times 15\,\text{k}\Omega = 12\,\text{k}\Omega$$

For R_C:

$$R_C = \frac{V_{CC} - V_C}{I_{RC}}$$
$$R_C = \frac{3.0\,\text{V} - 1.5\,\text{V}}{\frac{46}{15}\,\text{mA}} = \frac{1.5\,\text{V}}{\frac{46}{15}\,\text{mA}} = 1.5 \times \frac{15}{46}\,\text{k}\Omega = \frac{22.5}{46}\,\text{k}\Omega \approx 0.4891\,\text{k}\Omega = 489.1\,\Omega$$

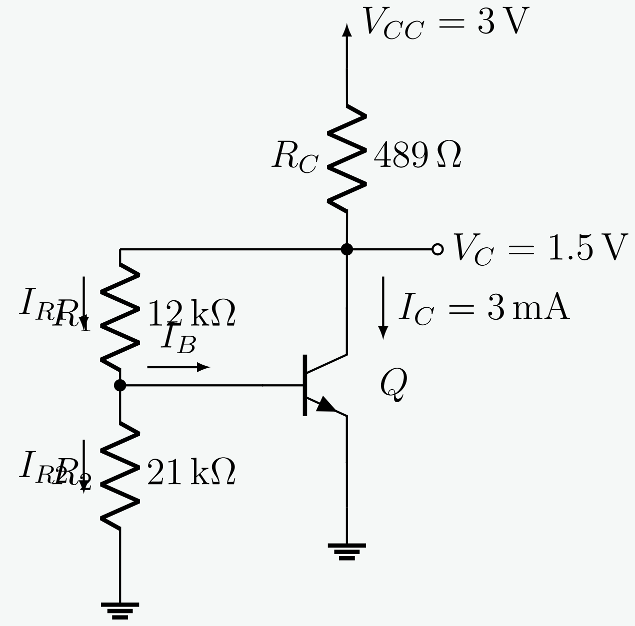
5. Verification of the Operating Region
To verify the forward-active region assumption, check the collector-emitter voltage:

$$V_{CE} = 1.5\,\text{V} > V_{CE(sat)} \approx 0.2\,\text{V}$$

Additionally, the base-collector voltage is $V_{BC} = V_B - V_C = 0.7\,\text{V} - 1.5\,\text{V} = -0.8\,\text{V}$. The base-collector junction is reverse-biased, confirming the transistor is indeed operating in the active region.

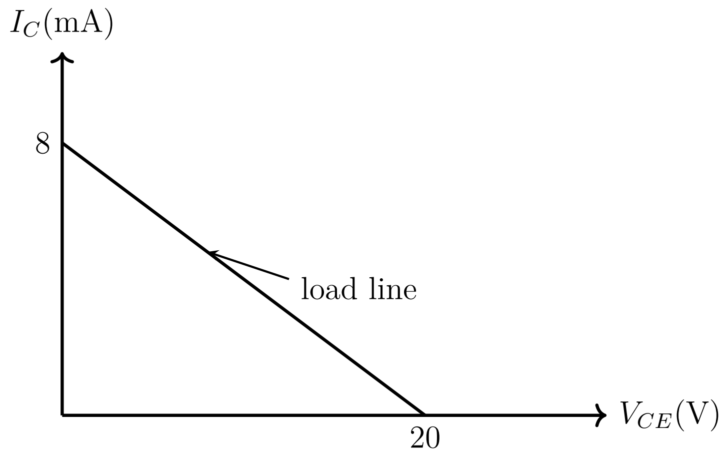
6. Summary of Results and Final Circuit Diagram

Component	**Calculated Value**
R_C	$489.1\,\Omega$
R_1	$12\,\text{k}\Omega$
R_2	$21\,\text{k}\Omega$
I_B	$33.33\,\mu\text{A}$



Linear Amplifier Constraints

Q1. For the figure shown, find out the maximum collector current that the device can support. If the operating point is set for that maximum current, what kind of problem will we face if we want to use the device as a linear amplifier?



(12 marks) [EEE 263 | L-2/T-1 | 2011–12]

Answer

1. Analysis of the DC Load Line

The given figure shows the DC load line plotted on the common-emitter transistor output characteristics (I_C versus V_{CE}). The load line is a graphical representation of the linear relationship imposed by the external collector-emitter circuit equation:

$$V_{CE} = V_{CC} - I_C R_{dc}$$

From the intercepts of the load line on the axes, we can extract critical circuit parameters:

- **X-axis Intercept (Cutoff Point):** When $I_C = 0$, the voltage across the device reaches its maximum value, which equals the collector supply voltage V_{CC}:
$$V_{CE(\text{max})} = V_{CC} = 20\,\text{V}$$
- **Y-axis Intercept (Saturation Point):** When $V_{CE} = 0$, the collector current reaches its maximum possible value, representing ideal saturation:
$$I_{C(\text{sat})} = \frac{V_{CC}}{R_{dc}} = 8\,\text{mA}$$

Thus, the maximum collector current that the external circuit configuration can support is:

$$I_{C(\text{max})} = 8\,\text{mA}$$

2. Linear Amplifier Constraints and Signal Clipping

If the DC operating point (Q-point) is biased directly at this maximum current ($I_Q = 8\,\text{mA}$, $V_{CEQ} = 0\,\text{V}$), the transistor is held at the edge of the saturation region under zero-signal conditions.
When a time-varying AC input signal is applied to use the circuit as a linear amplifier, the resulting base current variation attempts to swing the collector current above and below the quiescent value:

$$i_C(t) = I_Q + i_c(t)$$
$$v_{CE}(t) = V_{CEQ} + v_{ce}(t)$$

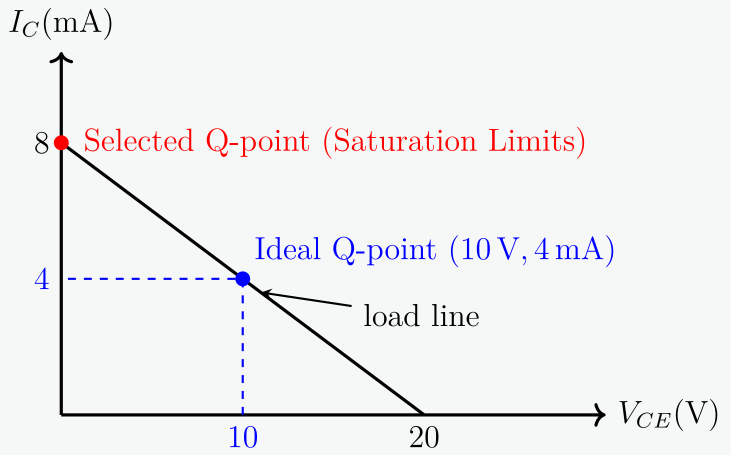
Because the circuit cannot physically sustain a current greater than $8\,\text{mA}$ or a voltage less than $0\,\text{V}$ (assuming an ideal $V_{CE(\text{sat})} = 0\,\text{V}$), the circuit behavior will degrade as follows:

- **Positive Current Half-Cycle ($i_c > 0$):** The positive excursion of the input signal drives the transistor deeper into saturation. The collector current remains hard-limited at 8 mA and the output voltage cannot drop below 0 V. This causes severe **saturation clipping** of the output waveform.
- **Asymmetrical Swing and Distortion:** Only the negative excursion of the current signal ($i_c < 0$) can be tracked lineally as the Q-point moves down toward the cutoff region. This extreme asymmetry results in high total harmonic distortion (THD), ruining the fidelity required for a linear amplifier.

For proper linear operation with maximum symmetric dynamic range, the Q-point should instead be located at the midpoint of the load line:

$$I_Q = \frac{I_{C(\text{sat})}}{2} = \frac{8\,\text{mA}}{2} = 4\,\text{mA}$$
$$V_{CEQ} = \frac{V_{CC}}{2} = \frac{20\,\text{V}}{2} = 10\,\text{V}$$

3. Mathematical and Graphical Representation



Final Conclusion: The maximum collector current is **8** mA. Biasing at this peak values places the device at the saturation limit, preventing any positive signal current swing and causing severe nonlinear amplitude distortion (clipping).

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Electronic Circuits

S3 — MOSFET Characteristics

Structure, threshold voltage, DC analysis, biasing stability, CMOS logic

S3 MOSFET Characteristics

Device Characteristics

Q1. Explain the different operation regions of an enhancement PMOS transistor with properly labeled I_D–V_{SD} plots. Derive the threshold conditions for operating in those regions. **(8 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

1. Introduction to Enhancement PMOS Operation

An p-channel enhancement-mode MOSFET (PMOS) is fabricated on an n-type substrate (or n-well) with p-type source and drain diffusions. To establish a conducting channel between the source and drain, a negative gate-to-source voltage (V_{GS}) must be applied to attract holes to the surface beneath the gate oxide.

For convenience in analog design, it is standard practice to express all voltages and currents using positive quantities by referencing them to the Source terminal:

$$V_{SG} = V_S - V_G$$
$$V_{SD} = V_S - V_D$$

The threshold voltage for an enhancement PMOS is inherently negative ($V_{tp} < 0$), so we denote its absolute value as $|V_{tp}|$.

—

2. Regions of Operation and Threshold Conditions

Depending on the terminal voltages V_{SG} and V_{SD}, the PMOS transistor operates in one of three distinct regions:

A. Cutoff Region

- **Condition:** $V_{SG} < |V_{tp}|$
- **Principle:** The applied gate-to-source voltage is insufficient to overcome the threshold voltage. No inversion layer of holes is created beneath the gate oxide, leaving the source-to-drain path open-circuited.
- **Current Equation:**
$$I_D = 0$$

B. Triode (Linear / Non-Saturation) Region

- **Condition:** $V_{SG} \geq |V_{tp}| \quad \text{and} \quad V_{SD} < V_{SG} - |V_{tp}|$
- **Principle:** The device is turned ON ($V_{SG} \geq |V_{tp}|$), forming a continuous p-channel. Because the source-to-drain voltage V_{SD} is relatively small, the channel remains open along its entire length from source to drain. The channel acts as a voltage-controlled resistor.
- **Current Equation:**
$$I_D = \mu_p C_{ox} \frac{W}{L} \left[(V_{SG} - |V_{tp}|)\, V_{SD} - \frac{1}{2} V_{SD}^2 \right]$$
where μ_p is the hole mobility, C_{ox} is the gate oxide capacitance per unit area, W is the channel width, and L is the channel length.

C. Saturation Region

- **Condition:** $V_{SG} \geq |V_{tp}| \quad \text{and} \quad V_{SD} \geq V_{SG} - |V_{tp}|$
- **Principle:** As V_{SD} increases to the point where $V_{SD} = V_{SG} - |V_{tp}|$, the local potential difference across the oxide at the drain end becomes exactly equal to $|V_{tp}|$. At this point, the inversion layer channel vanishes near the drain node, a phenomenon known as **pinch-off**. Further increases in V_{SD} shift the pinch-off point slightly toward the source, but the current saturates to a level determined primarily by V_{SG}.
- **Current Equation (including channel-length modulation):**
$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{SG} - |V_{tp}|)^2 (1 + \lambda V_{SD})$$
where λ is the channel-length modulation parameter.

—

3. Mathematical Derivation of the Boundaries

The local voltage across the oxide at a position x along the channel (where $x = 0$ is the source and $x = L$ is the drain) is given by:

$$V_{ox}(x) = V_{SG} - V(x)$$

where $V(x)$ is the channel potential relative to the source ($V(0) = 0$ and $V(L) = V_{SD}$).

For a channel to exist at any point x, the local inversion condition must be satisfied:

$$V_{SG} - V(x) \geq |V_{tp}| \implies V(x) \leq V_{SG} - |V_{tp}|$$

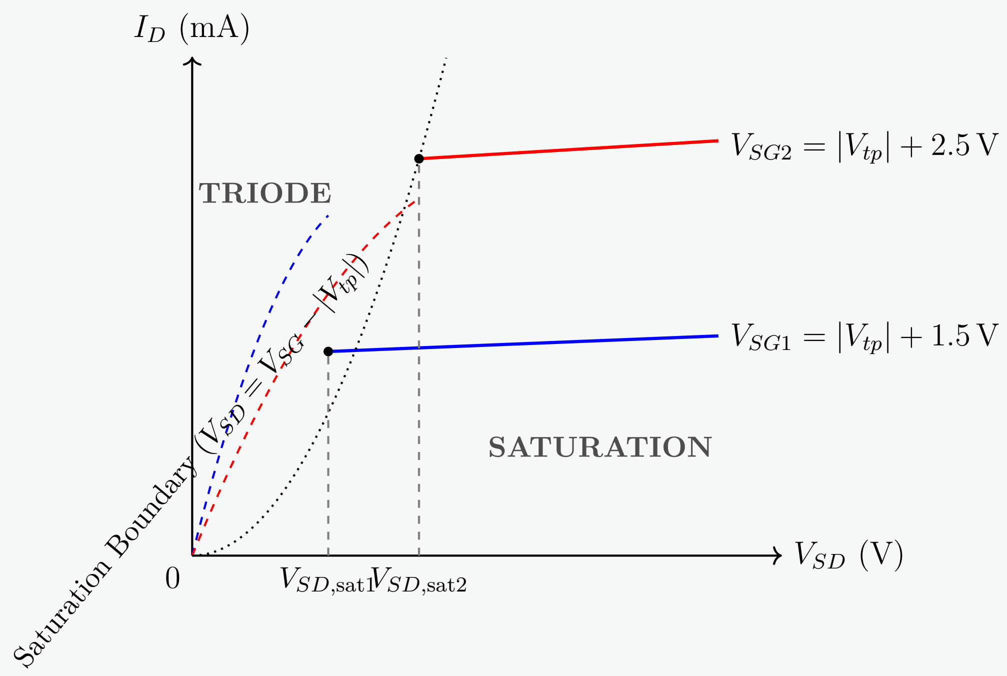
At the drain end ($x = L$), $V(L) = V_{SD}$. Substituting this into the boundary condition gives the critical saturation transition voltage, $V_{SD,\text{sat}}$:

$$V_{SD,\text{sat}} = V_{SG} - |V_{tp}|$$

- If $V_{SD} < V_{SG} - |V_{tp}|$, the channel is completely inverted, defining the **Triode Region**.
- If $V_{SD} \geq V_{SG} - |V_{tp}|$, the channel pinches off at the drain end, defining the **Saturation Region**.

—

4. I_D–V_{SD} Characteristic Plots
Below are the family of I_D–V_{SD} characteristic curves showing the operation regions and the parabolic saturation boundary.



Q2. For a particular MOSFET operating in the saturation region at a constant v_{gs}, $i_D = 2\,\text{mA}$ at $v_{ds} = 4\,\text{V}$. The i_D increases by 10% when v_{ds} is doubled. Find the corresponding values of r_0, V_A, and λ. **(10 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

1. Given Data and Governing Formula
For a MOSFET operating in the saturation region with channel-length modulation, the drain current i_D is a function of the drain-to-source voltage v_{DS} and is given by:

$$i_D = I_{D0}\,(1 + \lambda v_{DS})$$

where I_{D0} is the ideal saturation current without channel-length modulation, λ is the channel-length modulation parameter, and $V_A = \frac{1}{\lambda}$ is the Early voltage.
We are given two operating data points at a constant gate-to-source voltage v_{GS}:

$$\text{Point 1:} \quad v_{DS1} = 4\,\text{V}, \quad i_{D1} = 2\,\text{mA}$$
$$\text{Point 2:} \quad v_{DS2} = 2 \times v_{DS1} = 8\,\text{V}$$

At Point 2, the current increases by 10%:

$$i_{D2} = i_{D1} \times (1 + 0.10) = 2\,\text{mA} \times 1.1 = 2.2\,\text{mA}$$

—

2. Calculation of Channel-Length Modulation Parameter (λ)
Setting up the ratio of the two current equations to eliminate I_{D0}:

$$\frac{i_{D2}}{i_{D1}} = \frac{1 + \lambda v_{DS2}}{1 + \lambda v_{DS1}}$$
$$\frac{2.2\,\text{mA}}{2.0\,\text{mA}} = \frac{1 + 8\lambda}{1 + 4\lambda}$$
$$1.1 = \frac{1 + 8\lambda}{1 + 4\lambda}$$

Cross-multiplying to solve for λ:

$$1.1\,(1 + 4\lambda) = 1 + 8\lambda$$
$$1.1 + 4.4\lambda = 1 + 8\lambda$$
$$1.1 - 1 = 8\lambda - 4.4\lambda$$
$$0.1 = 3.6\lambda$$
$$\lambda = \frac{0.1}{3.6} = \frac{1}{36}\,\text{V}^{-1} \approx \mathbf{0.0278}\,\mathbf{V^{-1}}$$

—

3. Calculation of the Early Voltage (V_A)
The intercept voltage parameter V_A (Early voltage) is the inverse of λ:

$$V_A = \frac{1}{\lambda} = 36\,\text{V}$$

—

4. Calculation of Small-Signal Output Resistance (r_0)
The small-signal output resistance r_0 is defined as the inverse of the slope of the i_D–v_{DS} characteristic curve in saturation:

$$r_0 = \left(\frac{\Delta v_{DS}}{\Delta i_D}\right) = \frac{v_{DS2} - v_{DS1}}{i_{D2} - i_{D1}}$$

Substituting the given values:

$$\Delta v_{DS} = 8\,\text{V} - 4\,\text{V} = 4\,\text{V}$$
$$\Delta i_D = 2.2\,\text{mA} - 2.0\,\text{mA} = 0.2\,\text{mA} = 0.2 \times 10^{-3}\,\text{A}$$

$$r_0 = \frac{4\,\text{V}}{0.2\,\text{mA}} = 20\,\text{k}\Omega$$

Alternative Verification Method: Using the formula evaluated near Point 1:

$$r_0 \approx \frac{V_A + v_{DS1}}{i_{D1}} = \frac{36\,\text{V} + 4\,\text{V}}{2\,\text{mA}} = \frac{40\,\text{V}}{2\,\text{mA}} = 20\,\text{k}\Omega$$

Both methods yield identical results.
—

5. Summary of Results

Parameter	Exact Value	Decimal Value
λ (Channel-length modulation parameter)	$\frac{1}{36}\,\text{V}^{-1}$	$\mathbf{0.0278\,V^{-1}}$
V_A (Early voltage)	$36\,\text{V}$	$\mathbf{36\,V}$
r_0 (Output resistance)	$20\,\text{k}\Omega$	$\mathbf{20\,k\Omega}$

Q3. Briefly explain the I-V characteristics of an n-channel enhancement MOSFET with necessary plots. **(7 marks)** [EEE 263 | L-2/T-1 | 2022–23]

Answer

1. Overview of n-Channel Enhancement MOSFET
An n-channel enhancement MOSFET (NMOS) operates by applying a positive gate-to-source voltage (V_{GS}) to create a conducting inversion layer (channel) of electrons between the n-type source and drain regions.
The Current-Voltage (I-V) characteristics are primarily described by two types of plots:

- **Output Characteristics:** The relationship between the drain current (I_D) and drain-to-source voltage (V_{DS}) for various constant values of V_{GS}.
- **Transfer Characteristics:** The relationship between I_D and V_{GS} for a constant V_{DS} in the saturation region.

2. Regions of Operation
The device operates in three distinct regions depending on the applied terminal voltages. We define the threshold voltage as V_{tn} (a positive value for an enhancement NMOS).

A. Cutoff Region

- **Condition:** $V_{GS} < V_{tn}$
- **Behavior:** The gate voltage is insufficient to attract enough electrons to form a channel. The path between the source and drain is essentially an open circuit.
- **Equation:**

$$I_D = 0$$

B. Triode (Linear / Ohmic) Region

- **Condition:** $V_{GS} \geq V_{tn}$ and $V_{DS} < V_{GS} - V_{tn}$
- **Behavior:** A continuous channel exists from source to drain. At very low V_{DS}, the channel acts like a linear resistor controlled by V_{GS}. As V_{DS} increases, the channel at the drain end narrows, causing the current to rise parabolically rather than linearly.
- **Equation:**

$$I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_{tn})V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

C. Saturation Region

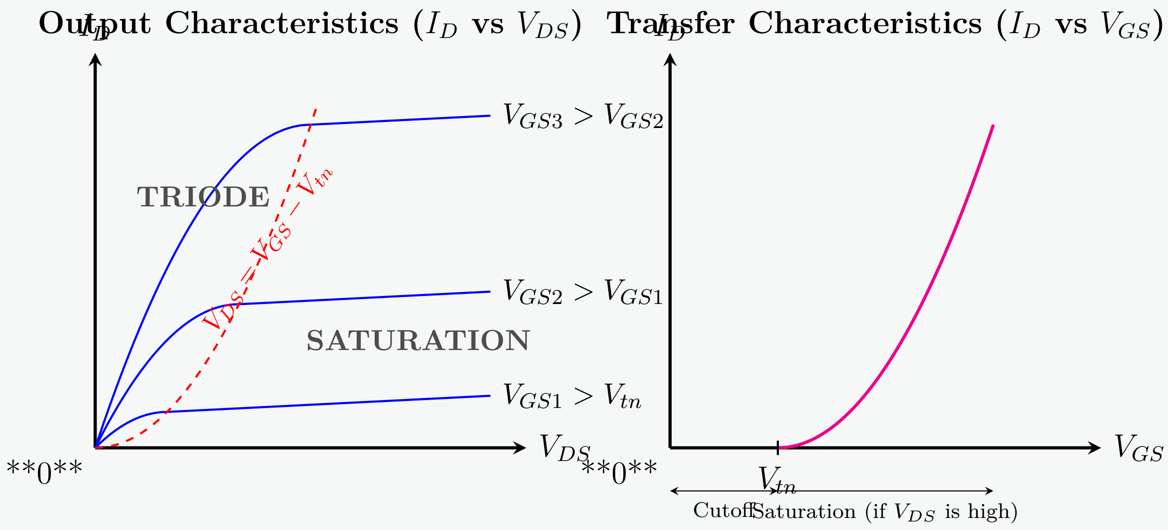
- **Condition:** $V_{GS} \geq V_{tn}$ and $V_{DS} \geq V_{GS} - V_{tn}$
- **Behavior:** The voltage across the oxide at the drain end drops below the threshold, causing the channel to **pinch off**. The current reaches a maximum saturated value and becomes largely independent of further increases in V_{DS} (ignoring channel-length modulation).
- **Equation:**

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{tn})^2(1 + \lambda V_{DS})$$

Note: λ represents the channel-length modulation parameter, which accounts for the slight upward slope of I_D in saturation.

3. I-V Characteristic Plots

Below are the characteristic curves visualizing these regions.



Q4. What is the difference between enhancement-type and depletion-type MOSFETs? Draw I_D vs. V_{GS} and I_D vs. V_{DS} characteristics for different values of V_{GS} for both enhancement-type and depletion-type MOSFETs. **(26 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Fundamental Differences

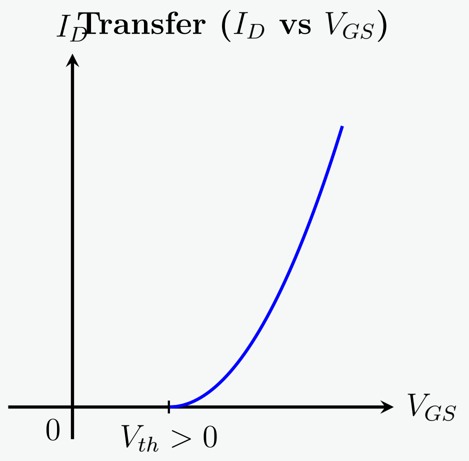
The primary difference between enhancement-type and depletion-type MOSFETs lies in their physical construction and their default state when no gate voltage is applied ($V_{GS} = 0$).

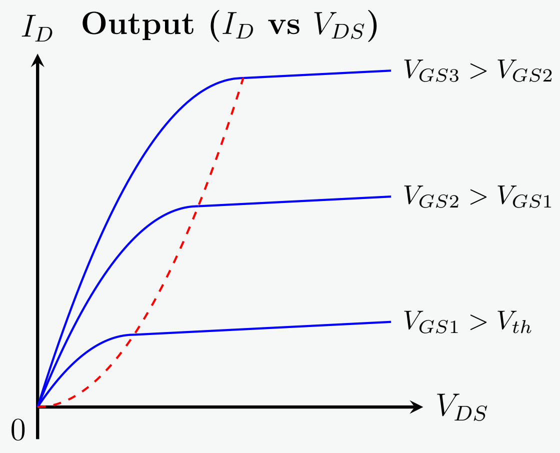
- **Enhancement-Type MOSFET (E-MOSFET):**
  - **Construction:** There is no physical channel built between the source and the drain during manufacturing.
  - **Operation:** It is a **normally-OFF** device. To turn it on, a gate-to-source voltage (V_{GS}) greater than a specific threshold voltage (V_{th}) must be applied to induce (or "enhance") a conducting channel via the electric field.
  - **Modes:** Operates *only* in enhancement mode.
- **Depletion-Type MOSFET (D-MOSFET):**
  - **Construction:** A physical doped channel is intentionally implanted between the source and drain during fabrication.
  - **Operation:** It is a **normally-ON** device. It conducts a significant drain current (I_{DSS}) even when $V_{GS} = 0\,\text{V}$.
  - **Modes:** Can operate in *two* modes. For an n-channel device, applying a negative V_{GS} repels charge carriers, depleting the channel (Depletion Mode). Applying a positive V_{GS} attracts more charge carriers, widening the channel (Enhancement Mode).

2. Characteristic Plots (Assuming n-channel devices)

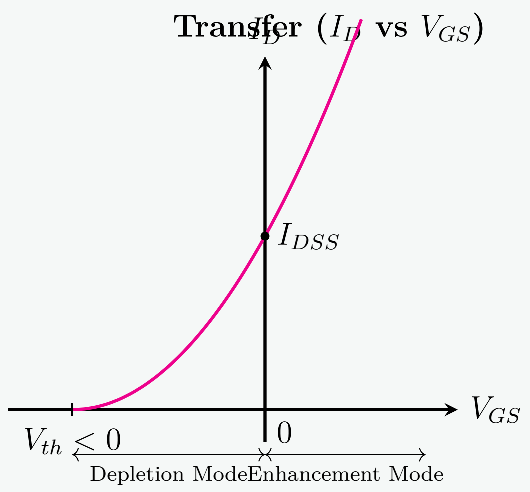
The plots below illustrate the Transfer Characteristics (I_D vs. V_{GS} in the saturation region) and the Output Characteristics (I_D vs. V_{DS}) for both MOSFET types.

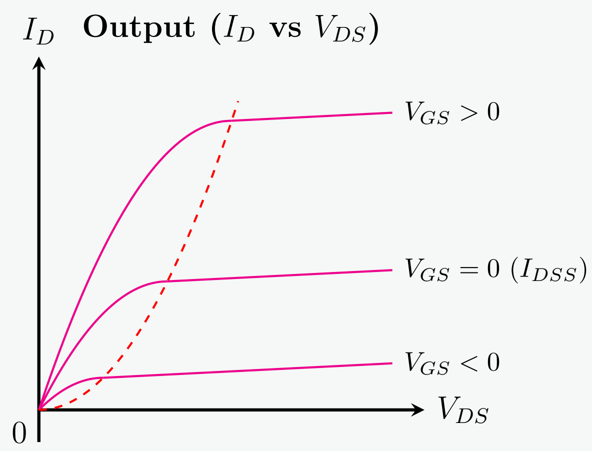
A. Enhancement-Type NMOS





B. Depletion-Type NMOS





Q5. Showing different operating regions, describe the output characteristics of a MOSFET. Why does current saturation occur in a MOSFET as V_{DS} is increased? **(18 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

1. Output Characteristics of a MOSFET

The output characteristics of a MOSFET represent the relationship between the drain current (I_D) and the drain-to-source voltage (V_{DS}) for various constant values of the gate-to-source voltage (V_{GS}). Assuming an n-channel enhancement MOSFET (NMOS), the device operates in three distinct regions depending on the applied terminal voltages.

A. Cutoff Region

- **Condition:** $V_{GS} < V_{th}$ (where V_{th} is the threshold voltage).
- **Behavior:** The applied gate voltage is insufficient to attract enough electrons to form an inversion layer (channel) beneath the gate oxide. The path between the source and drain acts as an open circuit.
- **Current Equation:**

$$I_D = 0$$

B. Triode (Linear / Ohmic) Region

- **Condition:** $V_{GS} \geq V_{th} \quad \text{and} \quad V_{DS} < V_{GS} - V_{th}$
- **Behavior:** A continuous induced channel exists from the source to the drain. The channel acts as a voltage-controlled resistor. At very small V_{DS}, I_D increases linearly. As V_{DS} increases, the channel at the drain end narrows, causing the current to increase at a decreasing rate (parabolically).
- **Current Equation:**

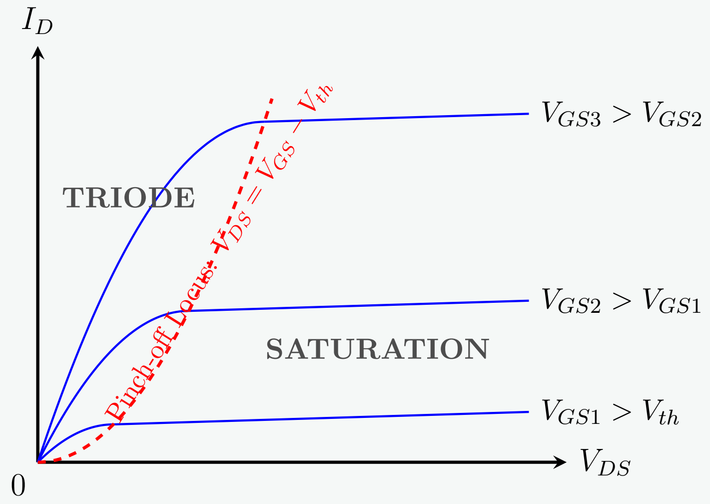
$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

C. Saturation Region

- **Condition:** $V_{GS} \geq V_{th} \quad \text{and} \quad V_{DS} \geq V_{GS} - V_{th}$
- **Behavior:** The channel pinches off at the drain end. The drain current reaches a maximum saturated value and becomes independent of V_{DS} (ignoring channel-length modulation). The device acts as an ideal voltage-controlled current source.
- **Current Equation:**

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$$

where λ represents the channel-length modulation parameter.

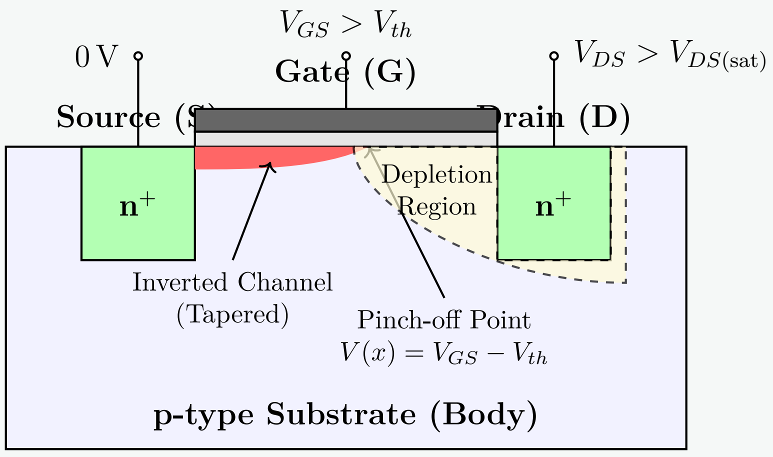


2. Physical Mechanism of Current Saturation (The Pinch-Off Effect)
To understand why the drain current saturates as V_{DS} increases, we must analyze the local voltage distribution along the channel. Let the source be grounded ($V_S = 0\,\text{V}$) and the voltage along the channel at a distance x from the source be $V(x)$.
The depth of the inversion layer (channel) at any point x depends on the local voltage difference between the gate and the channel:

$$V_{\text{eff}}(x) = V_{GS} - V(x)$$

- At the **source end** ($x = 0$), the channel voltage is $0\,\text{V}$, so the full V_{GS} induces a deep channel.
- At the **drain end** ($x = L$), the channel voltage is V_{DS}. Therefore, the effective voltage inducing the channel is $V_{GD} = V_{GS} - V_{DS}$.

As V_{DS} increases, V_{GD} decreases. The channel physically tapers off, becoming shallower near the drain.
Pinch-Off Point: When V_{DS} reaches the specific value $V_{DS(\text{sat})} = V_{GS} - V_{th}$, the effective gate-to-channel voltage at the drain end drops to exactly V_{th}. At this critical point, the electric field is no longer strong enough to maintain the inversion layer at the drain end. The channel thickness reduces to essentially zero. This is called **pinch-off**.
Current Saturation: If V_{DS} is increased beyond $V_{DS(\text{sat})}$, the channel remains pinched off, and the pinch-off point simply moves slightly back toward the source (channel-length modulation). The excess voltage $\Delta V = V_{DS} - V_{DS(\text{sat})}$ is dropped entirely across the newly formed depletion region between the pinch-off point and the drain.
Electrons traveling from the source move through the inverted channel driven by the fixed voltage $V_{DS(\text{sat})}$. Once they reach the pinch-off point, the strong electric field in the depletion region aggressively sweeps them into the drain. Because the voltage drop across the actual conducting channel is "locked" at $V_{GS} - V_{th}$, the flow rate of electrons (the drain current I_D) cannot increase further. Thus, the current **saturates**.



Comparison with BJT

Q1. Compare MOSFET and BJT in terms of their operating principles. Mention each of their advantages and disadvantages. **(10 marks)**
[EEE 263 | L-2/T-1 | 2013–14]

Answer

1. Operating Principles Comparison
Bipolar Junction Transistor (BJT):

- **Control Mechanism:** The BJT is a **current-controlled** device. A small base current (I_B) injected into the base terminal controls a much larger collector current (I_C) flowing from the collector to the emitter. In the forward-active region, this is described by the linear relationship $I_C = \beta I_B$.
- **Charge Carriers:** It is a **bipolar** device, meaning current conduction involves both majority and minority charge carriers (electrons and holes).
- **Physical Operation:** Operation relies on the diffusion of minority carriers across a narrow, forward-biased base-emitter junction and their subsequent drift/collection across a reverse-biased base-collector junction.

Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET):

- **Control Mechanism:** The MOSFET is a **voltage-controlled** device. The voltage applied between the gate and source terminals (V_{GS}) creates an electric field that controls the drain current (I_D). Because the gate is electrically insulated from the channel by a dielectric (typically SiO_2), the steady-state gate current is ideally zero ($I_G \approx 0$).
- **Charge Carriers:** It is a **unipolar** device, meaning conduction relies exclusively on majority charge carriers (electrons in NMOS, holes in PMOS).
- **Physical Operation:** Operation relies on the transverse electric field generated by V_{GS} across the gate oxide to attract carriers and create a conducting inversion layer (channel) in the semiconductor substrate beneath the gate, bridging the source and drain regions.

2. Advantages and Disadvantages

BJT Advantages:

- **High Transconductance (g_m):** For a given bias current, a BJT provides a significantly higher transconductance compared to a MOSFET, making it highly suitable for high-gain analog amplifiers.
- **High-Frequency Performance:** Offers superior performance and drive capability in discrete high-speed, high-frequency analog and RF circuits.
- **Better Linearity:** Inherently provides better linearity in discrete analog amplifier design, leading to lower harmonic distortion.

BJT Disadvantages:

- **Low Input Impedance:** Requires a continuous base current to remain in the ON state, resulting in a low input impedance that can load the driving signal source.
- **Thermal Runaway:** Possesses a positive temperature coefficient for collector current. As device temperature rises, I_C increases, leading to increased power dissipation and further heating, which can destroy the device if not properly compensated.
- **Size and Integration:** Occupies a larger physical silicon area and dissipates more static power, making it unsuitable for ultra-high-density Very Large Scale Integration (VLSI).

MOSFET Advantages:

- **Infinite Input Impedance:** The insulated gate draws virtually no static current (in the order of picoamps), meaning it does not load the preceding amplifier stages or driver circuits.
- **Thermal Stability:** Possesses a negative temperature coefficient at typical operating currents. As temperature rises, carrier mobility decreases, which reduces I_D and intrinsically prevents thermal runaway.
- **Low Power and High Density:** CMOS technology (complementary NMOS and PMOS) consumes near-zero static power. MOSFETs also scale down exceptionally well, enabling the fabrication of modern high-density microprocessors and memory chips.

MOSFET Disadvantages:

- **Lower Transconductance:** Provides a lower g_m for the same DC operating current compared to a BJT, generally resulting in a lower intrinsic voltage gain per stage.
- **ESD Susceptibility:** The ultra-thin gate oxide layer is extremely sensitive to Electrostatic Discharge (ESD). Stray high voltages can easily rupture the dielectric, permanently destroying the device.

3. Summary Comparison

Parameter	**BJT (Bipolar Junction Transistor)**	**MOSFET**
Control Type	Current-controlled ($I_C = \beta I_B$)	Voltage-controlled ($I_D = f(V_{GS})$)
Conduction Carriers	Bipolar (Electrons & Holes)	Unipolar (Majority carriers only)
Input Impedance	Low to moderate (kΩ range)	Extremely high ($10^{10} - 10^{15}\,\Omega$)
Transconductance (g_m)	High	Relatively low
Thermal Stability	Poor (Susceptible to thermal runaway)	Excellent (Self-limiting current)
Static Power Dissipation	Higher (constant base current needed)	Extremely low (especially in CMOS logic)
Primary Applications	High-gain discrete analog, RF amplifiers	Digital VLSI, switching power supplies

Q2. What are the mutual advantages and disadvantages of BJT and MOSFET? **(12 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Answer

1. Overview of Comparative Transistor Mechanics

In analog and digital electronic design, the Bipolar Junction Transistor (BJT) and the Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) are the two primary semiconductor pillars. While both perform fundamental amplification and electronic switching, their unique operational physics dictate distinct cross-advantages and disadvantages.

- **BJT Physics:** Current-controlled conduction utilizing both majority and minority charge carriers. The base-emitter current control relies heavily on exponential injection parameters across a forward-biased *p*-*n* junction.
- **MOSFET Physics:** Voltage-controlled drift utilizing exclusively majority charge carriers. The channel conductance is modulated via a capacitive, field-induced inversion layer under an insulated oxide barrier.

—

2. Mutual Advantages

A. Advantages of BJT over MOSFET

(a) **Significantly Higher Transconductance (g_m):** For an identical DC biasing current ($I_C = I_D$), the transconductance of a BJT is exponentially superior to that of a MOSFET.

$$g_{m(\text{BJT})} = \frac{I_C}{V_T} \approx \frac{I_C}{25\,\text{mV}} = 40 \times I_C$$
$$g_{m(\text{MOSFET})} = \frac{2I_D}{V_{GS} - V_{th}} = \sqrt{2\mu_n C_{ox}\frac{W}{L} I_D}$$

Because $V_{GS} - V_{th}$ (overdrive voltage) is typically much larger than $2V_T$, the BJT yields vastly superior voltage gain ($\mathbf{A}_v = -g_m R_C$) per standalone stage.

(b) **Superior Parameter Matching:** BJTs fabricated close together on a die exhibit much tighter matching of the base-emitter turn-on voltage (V_{BE}) compared to the threshold voltage (V_{th}) matching of MOSFETs. This is vital for high-precision analog sub-circuits, such as differential pairs and current mirrors.

(c) **Absence of Flicker Noise Dominance at Low Frequencies:** MOSFETs suffer heavily from $1/f$ (flicker) noise due to dangling chemical bonds at the Si-SiO_2 oxide interface. BJTs conduct beneath the surface, drastically minimizing flicker noise, which makes them optimal for low-noise pre-amplifiers.

B. Advantages of MOSFET over BJT

(a) **Near-Infinite Static Input Impedance (R_{in}):** Because the gate terminal is isolated by a silicon dioxide (SiO_2) layer, the static gate current $I_G \approx 0$.

$$R_{in(\text{MOSFET})} \to \infty \quad (10^{12}\,\Omega \text{ to } 10^{15}\,\Omega)$$

This completely eliminates steady-state loading effects on preceding multi-stage analog or digital circuit networks, unlike the BJT which drains a finite base current $I_B = I_C/\beta$.

(b) **Outstanding Geometric Scalability and Density:** MOSFETs feature a symmetric, self-aligned architectural structure that requires far less silicon area than an equivalent isolated BJT. They can be scaled down aggressively, enabling billions of transistors to inhabit a single VLSI microprocessor die.

(c) **Inherent Thermal Stability (No Thermal Runaway):** At high operating currents, the carrier mobility (μ) in a MOSFET channel decreases with increasing temperature ($\frac{\partial \mu}{\partial T} < 0$). This acts as a natural negative feedback loop that limits drain current. Conversely, a BJT's current increases exponentially with temperature due to a negative temperature coefficient of V_{BE}, rendering it vulnerable to destructive thermal runaway if uncompensated.

—

3. Mutual Disadvantages

A. Disadvantages of BJT compared to MOSFET

(a) **High Static Power Dissipation:** Because the BJT is a current-controlled device, it requires a continuous current flow into the base terminal (I_B) to stay turned on. This design bottleneck results in substantial standby power losses, limiting its usage in modern ultra-low-power computing chips.

(b) **Charge Storage and Lower Switching Speed:** When operating in the saturation region (fully turned on as a switch), both junctions are forward-biased, leading to minority-carrier charge accumulation in the base region. When transitioning to cutoff, this stored charge introduces a noticeable *storage time delay*, limiting the maximum switching frequency.

B. Disadvantages of MOSFET compared to BJT

(a) **Extreme Sensitivity to Electrostatic Discharge (ESD):** The microscopic SiO_2 gate dielectric behaves as a high-quality capacitor with a thin breakdown threshold. Even a minute static voltage accumulation on human skin can create an electric field strong enough to permanently rupture the oxide layer, disabling the device.

(b) **Lower Output Resistance (r_0):** Due to severe channel-length modulation effects (Early effect equivalent) at scaled dimensions, short-channel MOSFETs exhibit a relatively low output impedance, which degrades the maximum intrinsic voltage gain ($A_0 = g_m r_0$).

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4. Comprehensive Performance Matrix

Performance Vector	BJT Architecture	MOSFET Architecture
Control Parametric	Current-Driven ($I_C = \beta I_B$)	Voltage-Driven ($I_D \propto V_{GS}^2$)
Input Resistance (R_{in})	Low to Moderate ($\approx r_\pi$, active range)	Virtually Infinite ($R_{gate} \to \infty$)
Transconductance Vector	High ($g_m \propto I_C$, linear dependency)	Low to Moderate ($g_m \propto \sqrt{I_D}$, sub-linear)
Output Resistance (r_0)	High ($r_0 = V_A/I_C$)	Lower (Highly vulnerable to short-channel scaling)
Thermal Breakdown Profile	Negative tempco of V_{BE} (Risk of thermal runaway)	Positive tempco of channel resistance (Self-limiting)
Low-Frequency Noise	Highly quiet (Excellent $1/f$ figure)	Noisy due to chemical surface states
ESD Susceptibility	Low (Robust physical junction)	Extremely High (Oxide rupture hazard)
VLSI Packing Density	Poor (Requires deep isolation wells)	Superior (Optimal for high-density logic layout)

Threshold Voltage

Q1. Explain the concept of threshold voltage (V_t) of a MOSFET. What happens to V_t and oxide capacitance (C_{ox}) if: (i) the dielectric thickness is increased/decreased? (ii) the gate insulator is replaced with a more high-k dielectric of the same thickness? **(20 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Answer

1. Concept of Threshold Voltage (V_t)
In an enhancement-mode MOSFET, the threshold voltage (V_t) is defined as the minimum gate-to-source voltage (V_{GS}) required to create a continuous conducting channel (inversion layer) between the source and drain terminals.
For an n-channel MOSFET (NMOS) built on a p-type substrate:

- When $V_{GS} < V_t$, the applied electric field only depletes the surface of holes, but no significant electron channel is formed. The device is in the **cutoff** region.
- When $V_{GS} \geq V_t$, the positive potential on the gate is strong enough to attract free electrons from the source and drain into the region immediately beneath the gate oxide. This creates an **n-type inversion layer** (the channel), allowing current to flow from drain to source. This condition is known as **strong inversion**.

Mathematically, the threshold voltage can be modeled as:

$$V_t = V_{FB} + 2\phi_F + \frac{Q_d}{C_{ox}} = V_{FB} + 2\phi_F + \frac{\sqrt{2q\varepsilon_{si}N_A(2\phi_F)}}{C_{ox}}$$

where:

- V_{FB} is the flat-band voltage (accounts for work function differences and trapped oxide charges).
- $2\phi_F$ is the surface potential required for strong inversion.
- Q_d is the bulk depletion charge per unit area.
- C_{ox} is the gate oxide capacitance per unit area.

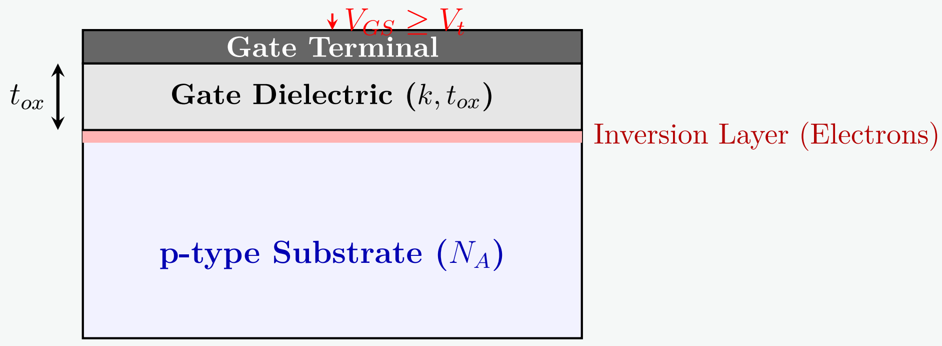
2. Concept of Oxide Capacitance (C_{ox})
The gate terminal, the insulating oxide layer, and the semiconductor substrate form a parallel-plate capacitor. The oxide capacitance per unit area is given by:

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{k\varepsilon_0}{t_{ox}}$$

where:

- ε_{ox} is the absolute permittivity of the gate dielectric material.

- k (or ε_r) is the relative permittivity (dielectric constant) of the material.
- ε_0 is the vacuum permittivity ($\approx 8.854 \times 10^{-12}$ F/m).
- t_{ox} is the physical thickness of the oxide layer.



3. Effect of Changing Dielectric Thickness (t_{ox})
(a) If dielectric thickness (t_{ox}) is INCREASED:

- **Effect on C_{ox}:** According to $C_{ox} = \frac{k\varepsilon_0}{t_{ox}}$, an increase in t_{ox} results in a **decrease** in C_{ox}. The plates of the capacitor are moved further apart, lowering the capacitance.
- **Effect on V_t:** The term $\frac{Q_d}{C_{ox}}$ in the threshold voltage equation becomes larger. Therefore, V_t **increases**. Physically, a thicker dielectric means a higher gate voltage is required to establish the same electric field strength across the oxide needed to invert the channel at the surface.

(b) If dielectric thickness (t_{ox}) is DECREASED:

- **Effect on C_{ox}:** Decreasing t_{ox} **increases** C_{ox}. The capacitive coupling between the gate and the channel becomes stronger.
- **Effect on V_t:** The term $\frac{Q_d}{C_{ox}}$ becomes smaller, meaning V_t **decreases**. A thinner oxide allows the gate to assert a stronger field for a given voltage, meaning less voltage is needed to turn the transistor ON.

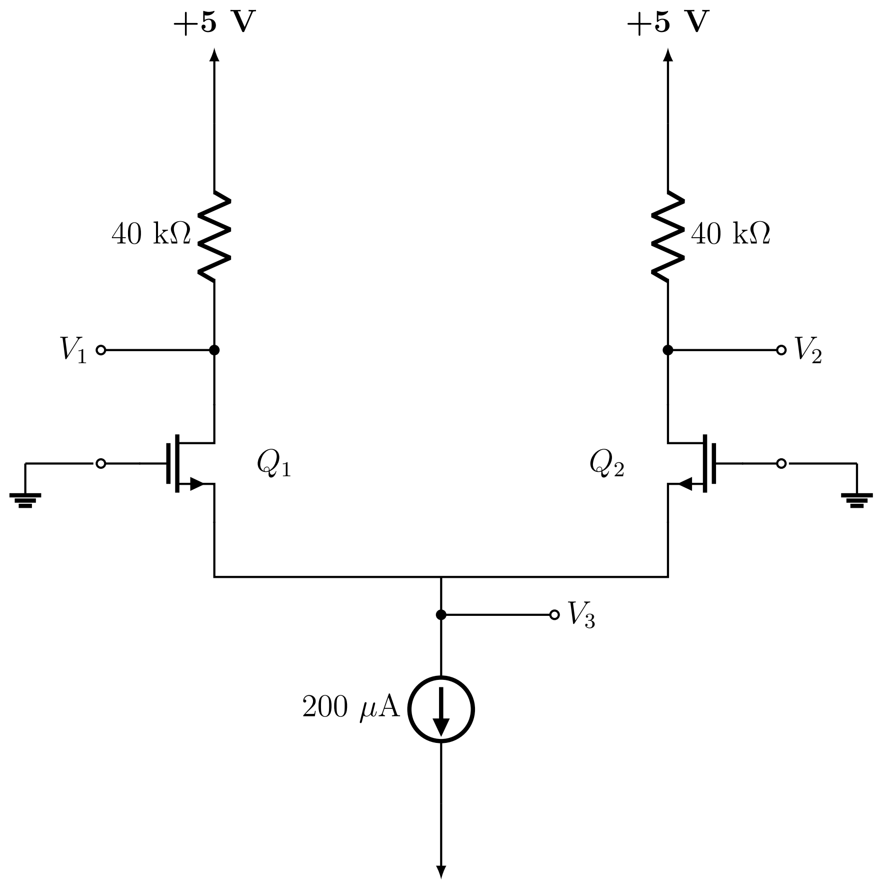
4. Effect of Using a High-k Dielectric
Scenario: The standard SiO_2 gate insulator ($k \approx 3.9$) is replaced with a high-k dielectric material (e.g., Hafnium Oxide, HfO_2, where $k \approx 25$), while keeping the physical thickness (t_{ox}) exactly the same.

- **Effect on C_{ox}:** Since the relative permittivity k is increased and t_{ox} remains constant, the oxide capacitance C_{ox} **increases significantly** ($C_{ox} \propto k$).
- **Effect on V_t:** Because C_{ox} increases, the voltage drop across the oxide ($\frac{Q_d}{C_{ox}}$) required to support the depletion charge decreases. Consequently, the threshold voltage V_t **decreases**.

Important Note on High-k Integration: In modern deep-submicron VLSI (like FinFETs), shrinking t_{ox} further using standard SiO_2 causes unacceptable quantum tunneling (leakage current). By using a high-k dielectric, engineers can achieve a high C_{ox} (which ensures a low V_t and high drive current) while maintaining a physically thicker insulator (t_{ox}) to block leakage currents effectively.

DC Analysis & Transistor Sizing

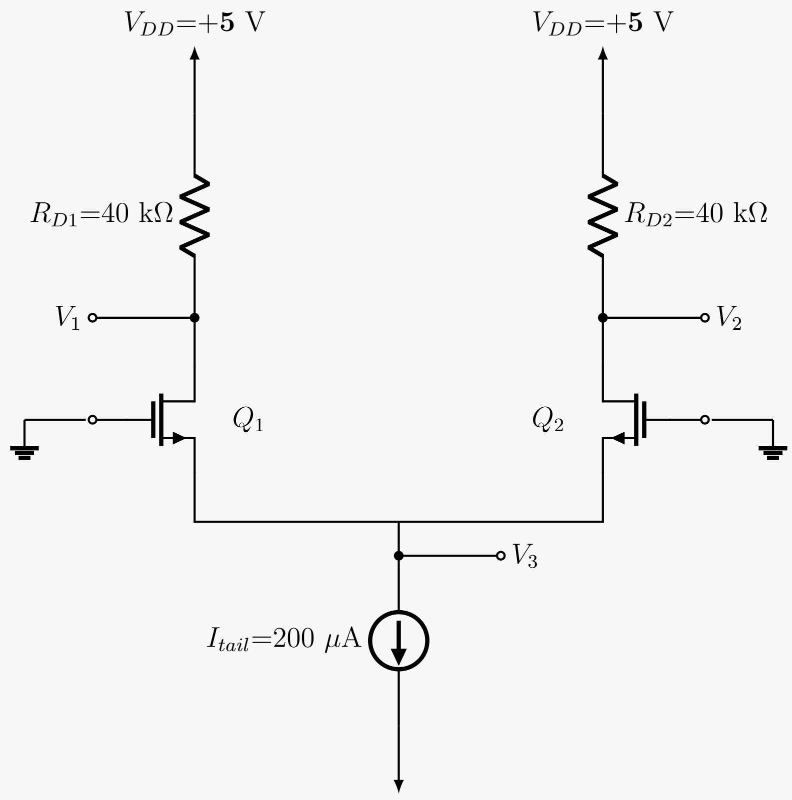
Q1. In the circuit shown, the aspect ratios of Q_1 and Q_2 are $(W/L)_1 = 2(W/L)_2 = 20$. Both transistors have $V_t = 1\,\text{V}$ and $\mu_n C_{ox} = 100\,\mu\text{A/V}^2$. Find the node voltages V_1, V_2, and V_3.



(17 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

1. Circuit Diagram Analysis



2. Assumptions and Initial Setup

From the given data, we have the following parameters:

- Aspect ratios: $(W/L)_1 = 20$ and $2(W/L)_2 = 20 \implies (W/L)_2 = 10$
- Threshold voltage: $V_t = 1\,\text{V}$
- Process transconductance parameter: $\mu_n C_{ox} = 100\,\mu\text{A/V}^2 = 0.1\,\text{mA/V}^2$
- Tail current: $I_{tail} = 200\,\mu\text{A} = 0.2\,\text{mA}$
- The gate terminals are grounded, so $V_{G1} = V_{G2} = 0\,\text{V}$.
- The sources are tied together at node V_3, meaning $V_{S1} = V_{S2} = V_3$.

Since $V_{G1} = V_{G2}$ and $V_{S1} = V_{S2}$, both transistors share the exact same gate-to-source voltage:

$$V_{GS1} = V_{GS2} = V_G - V_3 = 0 - V_3 = -V_3 \tag{1}$$

Assumption: We assume both transistors Q_1 and Q_2 are operating in the **saturation region**. We will verify this assumption at the end of the analysis.

3. Calculation of Drain Currents
In the saturation region, the drain current of a MOSFET is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

Since both transistors have the same V_{GS} and V_t, the ratio of their drain currents is directly proportional to the ratio of their aspect widths:

$$\frac{I_{D1}}{I_{D2}} = \frac{(W/L)_1}{(W/L)_2} = \frac{20}{10} = 2 \implies I_{D1} = 2I_{D2} \tag{2}$$

By Kirchhoff's Current Law (KCL) at the common source node, the sum of the drain currents must equal the tail current (ignoring channel-length modulation, $\lambda = 0$):

$$I_{D1} + I_{D2} = I_{tail} = 200\,\mu\text{A} \tag{3}$$

Substituting Equation (2) into Equation (3):

$$\begin{aligned} 2I_{D2} + I_{D2} &= 200\,\mu\text{A} \\ 3I_{D2} &= 200\,\mu\text{A} \\ I_{D2} &= \frac{200}{3}\,\mu\text{A} \approx 66.67\,\mu\text{A} \end{aligned}$$

Using this to find I_{D1}:

$$I_{D1} = 2 \times \frac{200}{3}\,\mu\text{A} = \frac{400}{3}\,\mu\text{A} \approx 133.33\,\mu\text{A}$$

4. Calculation of Node Voltages V_1 and V_2
The node voltages V_1 and V_2 correspond to the drain voltages of Q_1 and Q_2. Using Ohm's Law and KVL from the V_{DD} supply:

$$\begin{aligned} V_1 &= V_{DD} - I_{D1}R_{D1} \\ V_1 &= 5\,\text{V} - \left(\frac{400}{3} \times 10^{-6}\,\text{A}\right)(40 \times 10^3\,\Omega) \\ V_1 &= 5 - \frac{16000}{3} \times 10^{-3} = 5 - \frac{16}{3} = \frac{-1}{3}\,\text{V} \approx \mathbf{-0.333}\,\text{V} \end{aligned}$$

Similarly for V_2:

$$\begin{aligned} V_2 &= V_{DD} - I_{D2}R_{D2} \\ V_2 &= 5\,\text{V} - \left(\frac{200}{3} \times 10^{-6}\,\text{A}\right)(40 \times 10^3\,\Omega) \\ V_2 &= 5 - \frac{8000}{3} \times 10^{-3} = 5 - \frac{8}{3} = \frac{7}{3}\,\text{V} \approx \mathbf{2.333}\,\text{V} \end{aligned}$$

5. Calculation of Source Node Voltage V_3
To find V_3, we use the saturation current equation for Q_2 (we could equivalently use Q_1):

$$\begin{aligned} I_{D2} &= \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_t)^2 \\ \frac{200}{3}\,\mu\text{A} &= \frac{1}{2}(100\,\mu\text{A/V}^2)(10)(V_{GS} - 1)^2 \\ \frac{200}{3} &= 500(V_{GS} - 1)^2 \\ (V_{GS} - 1)^2 &= \frac{200}{1500} = \frac{2}{15} \\ V_{GS} - 1 &= \sqrt{\frac{2}{15}} \quad \text{(Taking the positive root for strong inversion)} \\ V_{GS} &= 1 + \sqrt{\frac{2}{15}} \approx 1 + 0.365 = 1.365\,\text{V} \end{aligned}$$

From Equation (1), we know $V_3 = -V_{GS}$. Therefore:

$$V_3 = -\left(1 + \sqrt{\frac{2}{15}}\right)\,\text{V} \approx \mathbf{-1.365}\,\text{V}$$

6. Verification of Operating Region
To verify that the transistors are indeed in saturation, the condition $V_{DS} > V_{GS} - V_t$ (or equivalently $V_D > V_G - V_t$) must be satisfied for both.
The overdrive voltage is:

$$V_{ov} = V_{GS} - V_t = 1.365\,\text{V} - 1\,\text{V} = 0.365\,\text{V}$$

For Q_1:

$$V_{DS1} = V_1 - V_3 = -0.333\,\text{V} - (-1.365\,\text{V}) = 1.032\,\text{V}$$
$$V_{DS1} > V_{ov} \implies 1.032\,\text{V} > 0.365\,\text{V} \quad (\textbf{Verified})$$

For Q_2:

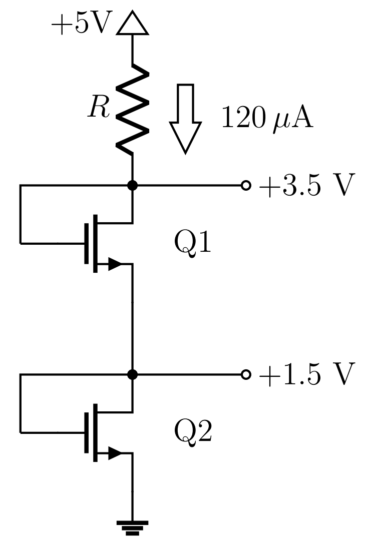
$$V_{DS2} = V_2 - V_3 = 2.333\,\text{V} - (-1.365\,\text{V}) = 3.698\,\text{V}$$
$$V_{DS2} > V_{ov} \implies 3.698\,\text{V} > 0.365\,\text{V} \quad (\textbf{Verified})$$

Since both transistors satisfy the saturation condition, our initial assumption is fully justified.

7. Final Answer Summary

- $V_1 = -0.333\,\text{V}$
- $V_2 = 2.333\,\text{V}$
- $V_3 = -1.365\,\text{V}$

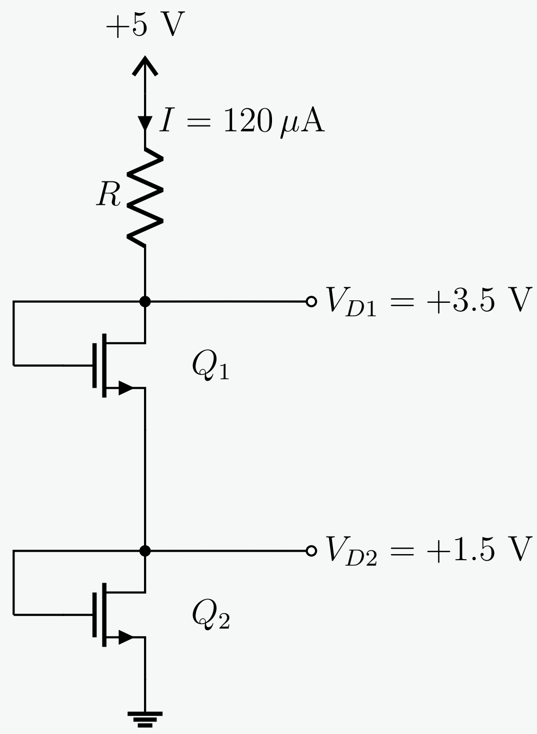
Q2. The NMOS transistors in the circuit have $V_t = 1\,\text{V}$, $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$, $\lambda = 0$, and $L_1 = L_2 = 1\,\mu\text{m}$. Find the required values of gate width W_1 and W_2 for Q_1 and Q_2, and the value of R, for obtaining the voltage values indicated. The current in R will be $120\,\mu\text{A}$.



(15 marks) [EEE 263 | L-2/T-1 | 2023–24]

Answer

1. Circuit Diagram



2. Operating Region Analysis

Both transistors Q_1 and Q_2 are configured as *diode-connected* loads, meaning their gate terminals are short-circuited to their respective drain terminals ($V_G = V_D$). Consequently, the gate-to-source voltage equals the drain-to-source voltage:

$$V_{GS} = V_{DS}$$

For a MOSFET to be in the **saturation region**, the condition $V_{DS} \geq V_{GS} - V_t$ must be satisfied. Substituting $V_{DS} = V_{GS}$:

$$V_{GS} \geq V_{GS} - V_t \implies 0 \geq -V_t \implies 0 \geq -1\,\text{V}$$

Since this condition is always mathematically true, both Q_1 and Q_2 are guaranteed to operate in the saturation region (provided $V_{GS} > V_t$ so they are turned on).
The drain current in saturation is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_t)^2 \tag{1}$$

3. Calculation of Gate Width W_2 for Q_2
First, we determine the terminal voltages for Q_2:

- Source voltage: $V_{S2} = 0\,\text{V}$ (Grounded)
- Gate/Drain voltage: $V_{G2} = V_{D2} = 1.5\,\text{V}$
- Gate-to-source voltage: $V_{GS2} = V_{G2} - V_{S2} = 1.5\,\text{V} - 0\,\text{V} = 1.5\,\text{V}$

Using the saturation current Equation (1) and the given parameters ($I_{D2} = 120\,\mu\text{A}$, $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$, $V_t = 1\,\text{V}$, $L_2 = 1\,\mu\text{m}$):

$$\begin{aligned}
120\,\mu\text{A} &= \frac{1}{2}(120\,\mu\text{A/V}^2)\left(\frac{W_2}{1\,\mu\text{m}}\right)(1.5\,\text{V} - 1\,\text{V})^2 \\
120 &= 60 \cdot W_2 \cdot (0.5)^2 \\
120 &= 60 \cdot W_2 \cdot 0.25 \\
120 &= 15 \cdot W_2 \\
W_2 &= \frac{120}{15} = \mathbf{8}\,\mu\text{m}
\end{aligned}$$

4. Calculation of Gate Width W_1 for Q_1
Next, we determine the terminal voltages for Q_1:

- Source voltage: $V_{S1} = V_{D2} = 1.5\,\text{V}$
- Gate/Drain voltage: $V_{G1} = V_{D1} = 3.5\,\text{V}$
- Gate-to-source voltage: $V_{GS1} = V_{G1} - V_{S1} = 3.5\,\text{V} - 1.5\,\text{V} = 2.0\,\text{V}$

Since Q_1 and Q_2 are in series, they share the same drain current ($I_{D1} = I_{D2} = 120\,\mu\text{A}$). Applying Equation (2) for Q_1:

$$\begin{aligned}
120\,\mu\text{A} &= \frac{1}{2}(120\,\mu\text{A/V}^2)\left(\frac{W_1}{1\,\mu\text{m}}\right)(2.0\,\text{V} - 1\,\text{V})^2 \\
120 &= 60 \cdot W_1 \cdot (1.0)^2 \\
120 &= 60 \cdot W_1 \\
W_1 &= \frac{120}{60} = \mathbf{2}\,\mu\text{m}
\end{aligned}$$

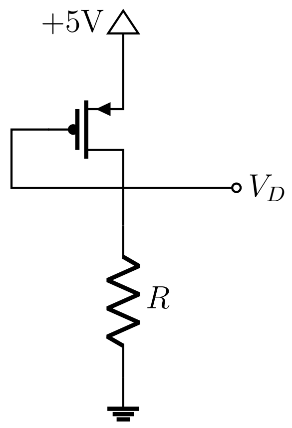
5. Calculation of Resistor R
The resistor R connects the $+5\,\text{V}$ power supply (V_{DD}) to the drain of Q_1 (V_{D1}). The current flowing through the resistor is given as $I_R = 120\,\mu\text{A}$. Applying Ohm's law:

$$\begin{aligned}
I_R &= \frac{V_{DD} - V_{D1}}{R} \\
120 \times 10^{-6}\,\text{A} &= \frac{5\,\text{V} - 3.5\,\text{V}}{R} \\
120 \times 10^{-6} &= \frac{1.5}{R} \\
R &= \frac{1.5}{120 \times 10^{-6}} = 12500\,\Omega = \mathbf{12.5\,k\Omega}
\end{aligned}$$

6. Final Answer Summary

Parameter	Calculated Value
Width of Q_1 (W_1)	$2\,\mu\text{m}$
Width of Q_2 (W_2)	$8\,\mu\text{m}$
Resistance (R)	$12.5\,\text{k}\Omega$

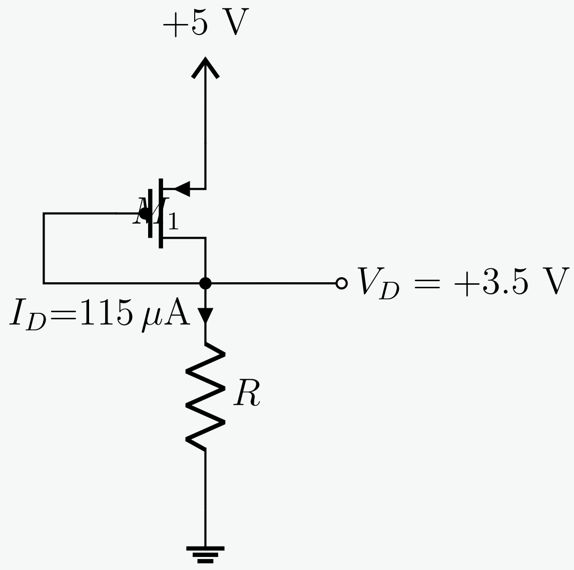
Q3. The PMOS transistor in the circuit shown has $V_t = -0.7\,\text{V}$, $\mu_p C_{ox} = 60\,\mu\text{A/V}^2$, $L = 0.8\,\mu\text{m}$, and $\lambda = 0$. Find the value of W and R in order to establish a drain current of $115\,\mu\text{A}$ and a voltage $V_D = 3.5\,\text{V}$.



(15 marks) [EEE 263 | L-2/T-1 | 2023–24]

Answer

1. Circuit Diagram Analysis



2. Verification of Operating Region

The PMOS transistor M_1 is configured with its gate short-circuited to its drain ($V_G = V_D$). This arrangement is universally known as a *diode-connected load*.

To determine the operating region, we calculate the source-to-gate voltage (V_{SG}) and the source-to-drain voltage (V_{SD}):

$$V_{SG} = V_S - V_G = 5.0\,\text{V} - 3.5\,\text{V} = 1.5\,\text{V}$$
$$V_{SD} = V_S - V_D = 5.0\,\text{V} - 3.5\,\text{V} = 1.5\,\text{V}$$

For a PMOS transistor to operate in the **saturation region**, the channel must be strongly inverted and pinched off at the drain end. The mathematical condition is:

$$V_{SD} \geq V_{SG} - |V_t|$$

Substituting the known values ($|V_t| = |-0.7\,\text{V}| = 0.7\,\text{V}$):

$$1.5\,\text{V} \geq 1.5\,\text{V} - 0.7\,\text{V}$$
$$1.5\,\text{V} \geq 0.8\,\text{V}$$

Since this inequality holds true, the transistor is strictly operating in the saturation region.

3. Calculation of Resistor (R)

Because the gate of a MOSFET draws practically zero current ($I_G \approx 0$), the entirety of the drain current I_D must flow through the resistor R down to the ground.

By applying Ohm's Law at the drain node:

$$V_D = I_D \times R$$
$$R = \frac{V_D}{I_D}$$

Substituting the specified values:

$$R = \frac{3.5\,\text{V}}{115 \times 10^{-6}\,\text{A}} \approx 30,434.78\,\Omega = \mathbf{30.43\,k\Omega}$$

4. Calculation of Gate Width (W)

The drain current for a PMOS transistor in the saturation region (assuming channel-length modulation $\lambda = 0$) is given by the standard square-law equation:

$$I_D = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)(V_{SG} - |V_t|)^2 \tag{1}$$

We are given the following physical and electrical parameters:

- $I_D = 115\,\mu\text{A}$
- $\mu_p C_{ox} = 60\,\mu\text{A/V}^2$
- $L = 0.8\,\mu\text{m}$
- $V_{SG} - |V_t| = 1.5\,\text{V} - 0.7\,\text{V} = 0.8\,\text{V}$

Substituting these parameters into the drain current equation:

$$\begin{aligned}115\,\mu\text{A} &= \frac{1}{2}(60\,\mu\text{A/V}^2)\left(\frac{W}{0.8\,\mu\text{m}}\right)(0.8\,\text{V})^2\\ 115 &= 30\cdot\left(\frac{W}{0.8}\right)\cdot(0.64)\\ 115 &= 30\cdot W\cdot 0.8\\ 115 &= 24\cdot W\end{aligned}$$

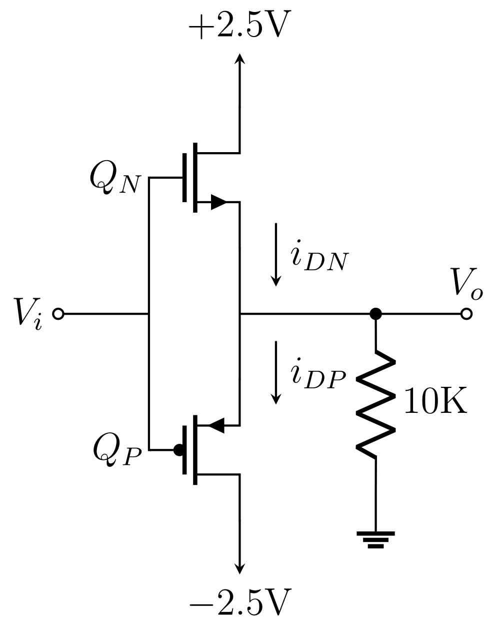
Solving for the gate width W:

$$W = \frac{115}{24}\,\mu\text{m} \approx \mathbf{4.79}\,\mu\text{m}$$

5. Final Answer Summary

Component/Parameter	Calculated Value
Transistor Gate Width (W)	$4.79\,\mu\text{m}$
Load Resistance (R)	$30.43\,\text{k}\Omega$
Operating Region	Saturation

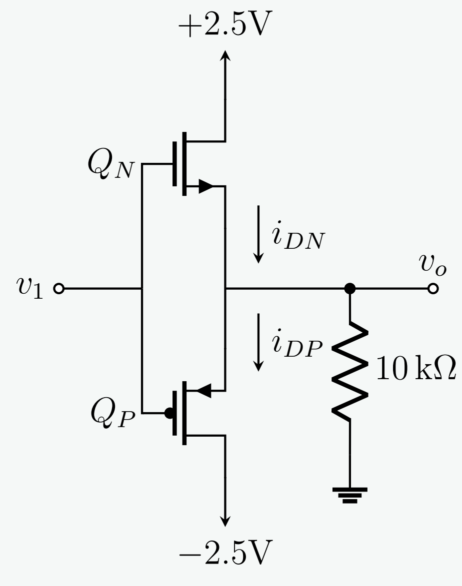
Q4. The NMOS and PMOS transistors in the circuit are matched with $k_n = k_p = 1\,\text{mA/V}^2$ and $V_{tn} = -V_{tp} = 1\,\text{V}$. Assuming $\lambda = 0$ for both MOSFETs, find the drain currents i_{DN} and i_{DP} and the voltage v_0 for $v_1 = 0\,\text{V}$ and $v_1 = +2.5\,\text{V}$.



(20 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

1. Circuit and Parameter Analysis



The circuit is a CMOS push-pull source follower (Class AB output stage) driving a load resistor $R_L = 10\,\text{k}\Omega$. The given parameters are:

- $k_n = k_p = 1\,\text{mA/V}^2$
- $V_{tn} = 1\,\text{V}$ and $V_{tp} = -1\,\text{V} \implies |V_{tp}| = 1\,\text{V}$
- $V_{DD} = 2.5\,\text{V}$, $V_{SS} = -2.5\,\text{V}$
- $\lambda = 0$

2. Case 1: $v_1 = 0\,\textbf{V}$
When the input voltage is $0\,\text{V}$, both gates are at ground potential ($V_G = 0\,\text{V}$).
Let's assume both transistors are in the **cutoff** region (OFF). Under this assumption, no current flows through the transistors:

$$i_{DN} = 0\,\text{mA}, \quad i_{DP} = 0\,\text{mA}$$

By Kirchhoff's Current Law at the output node, the current through the load resistor R_L is:

$$i_L = i_{DN} - i_{DP} = 0\,\text{mA}$$

Using Ohm's Law, the output voltage is:

$$v_o = i_L \times R_L = 0\,\text{V}$$

Now we verify our cutoff assumption by checking the gate-to-source voltages with $v_o = 0\,\text{V}$:

- **For Q_N:** $v_{GSN} = v_1 - v_o = 0 - 0 = 0\,\text{V}$. Since $v_{GSN} < V_{tn}$ ($0 < 1\,\text{V}$), Q_N is verified to be OFF.
- **For Q_P:** $v_{SGP} = v_o - v_1 = 0 - 0 = 0\,\text{V}$. Since $v_{SGP} < |V_{tp}|$ ($0 < 1\,\text{V}$), Q_P is verified to be OFF.

The assumption holds completely.

3. Case 2: $v_1 = +2.5\,\textbf{V}$
When the input voltage is $+2.5\,\text{V}$, the gate voltage V_G for both transistors is $2.5\,\text{V}$.
Checking Q_P: The source of Q_P is tied to v_o. Because the maximum possible voltage in this circuit is $+2.5\,\text{V}$, v_o must be $\leq 2.5\,\text{V}$. Therefore, $v_{SGP} = v_o - 2.5\,\text{V} \leq 0\,\text{V}$. Since $v_{SGP} < |V_{tp}| = 1\,\text{V}$, Q_P is definitely **OFF**.

$$i_{DP} = 0\,\text{mA}$$

Checking Q_N: Since Q_P is OFF, Q_N must supply the entire load current. Let's assume Q_N is ON and in the **saturation region**. The currents must satisfy KCL:

$$\begin{aligned} i_{DN} &= i_L \\ i_{DN} &= \frac{v_o}{10\,\text{k}\Omega} = 0.1 v_o \quad (\text{with } i_{DN} \text{ in mA}) \end{aligned}$$

The saturation current for Q_N is given by:

$$\begin{aligned} i_{DN} &= \frac{1}{2} k_n (v_{GSN} - V_{tn})^2 \\ v_{GSN} &= v_1 - v_o = 2.5 - v_o \\ i_{DN} &= \frac{1}{2}(1)(2.5 - v_o - 1)^2 = 0.5(1.5 - v_o)^2 \end{aligned}$$

Equating the two expressions for i_{DN}:

$$\begin{aligned} 0.5(1.5 - v_o)^2 &= 0.1 v_o \\ 5(1.5 - v_o)^2 &= v_o \\ 5(2.25 - 3v_o + v_o^2) &= v_o \\ 11.25 - 15v_o + 5v_o^2 &= v_o \\ 5v_o^2 - 16v_o + 11.25 &= 0 \end{aligned}$$

Solving this quadratic equation using the quadratic formula:

$$v_o = \frac{16 \pm \sqrt{(-16)^2 - 4(5)(11.25)}}{2(5)} = \frac{16 \pm \sqrt{256 - 225}}{10} = \frac{16 \pm \sqrt{31}}{10}$$

This yields two possible mathematical roots:

- $v_{o1} = \frac{16 + 5.568}{10} = 2.1568\,\text{V}$
- $v_{o2} = \frac{16 - 5.568}{10} = 1.0432\,\text{V}$

We must evaluate physical validity: for Q_N to be ON, $v_{GSN} > V_{tn}$, which implies $(2.5 - v_o) > 1 \implies v_o < 1.5\,\text{V}$. Therefore, $2.1568\,\text{V}$ is discarded, leaving $v_o = 1.043\,\text{V}$.
Now, calculate i_{DN}:

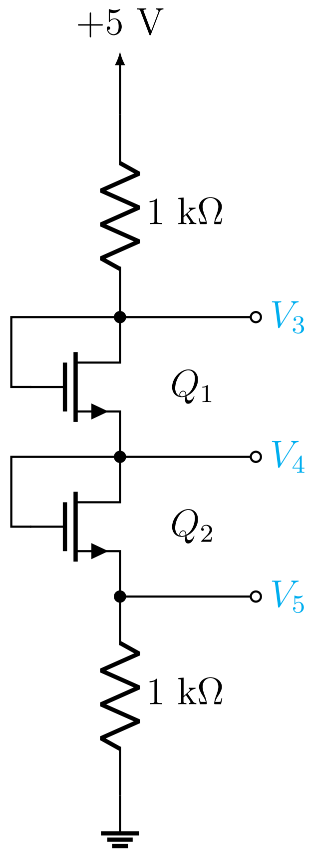
$$i_{DN} = 0.1 \times 1.0432\,\text{V} = 0.10432\,\text{mA} = 104.3\,\mu\text{A}$$

(Self-check for saturation: $v_{DSN} = 2.5 - 1.043 = 1.457\,\text{V}$, and overdrive voltage $v_{OV} = 2.5 - 1.043 - 1 = 0.457\,\text{V}$. Since $v_{DSN} > v_{OV}$, Q_N is indeed saturated).

4. Final Answer Summary

Input Condition	i_{DN}	i_{DP}	v_o
For $v_1 = 0\,\text{V}$	$0\,\text{mA}$	$0\,\text{mA}$	$0\,\text{V}$
For $v_1 = +2.5\,\text{V}$	$104.3\,\mu\text{A}$	$0\,\text{mA}$	$1.043\,\text{V}$

Q5. For the circuit shown, find the node voltages V_3, V_4, and V_5. The NMOS transistors have $V_t = 0.9\,\text{V}$ and $k_n'(W/L) = 1.5\,\text{mA/V}^2$.



(28 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Answer

1. Operating Region Verification
Both transistors Q_1 and Q_2 are in a *diode-connected configuration* because their gates are shorted to their respective drains ($V_{G1} = V_3$ and $V_{G2} = V_4$). This means that for both transistors:

$$V_{GS} = V_{DS}$$

Since $V_t = 0.9\,\text{V} > 0$, the condition for saturation ($V_{DS} \geq V_{GS} - V_t$) reduces to $0 \geq -0.9\,\text{V}$, which is always true when the channel conducts. Therefore, both Q_1 and Q_2 operate in the **saturation region**.
The saturation current expression for each transistor is:

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

Given that $k_n'(W/L) = 1.5\,\text{mA/V}^2$:

$$I_D = \frac{1}{2}(1.5)(V_{GS} - 0.9)^2 = 0.75(V_{GS} - 0.9)^2 \quad \text{(with } I_D \text{ in mA)}$$

2. System of Equations
Since Q_1 and Q_2 are in series with the top and bottom $1\,\text{k}\Omega$ resistors, the same continuous DC current I flows through the entire branch.
We can write expressions for the node voltages in terms of I (in mA):

- From the top rail through the upper $1\,\text{k}\Omega$ resistor:

$$V_3 = 5 - I \cdot (1) = 5 - I$$

- From the bottom ground through the lower $1\,\text{k}\Omega$ resistor:

$$V_5 = 0 + I \cdot (1) = I$$

Since Q_1 and Q_2 are completely identical and carry the exact same current I, their gate-to-source overdrive voltages must be equal:

$$V_{GS1} = V_{GS2} = V_{GS}$$

Expressing V_{GS} from the saturation current equation:

$$I = 0.75(V_{GS} - 0.9)^2 \implies V_{GS} = 0.9 + \sqrt{\frac{I}{0.75}}$$

Now, let's look at the node voltages related to these gate-to-source potentials:

$$V_{GS2} = V_4 - V_5 \implies V_4 = V_5 + V_{GS} = I + V_{GS}$$
$$V_{GS1} = V_3 - V_4 \implies V_3 = V_4 + V_{GS} = I + 2V_{GS}$$

3. Solving for Branch Current (I)
Equating our two expressions for V_3:

$$5 - I = I + 2V_{GS} \implies 5 - 2I = 2V_{GS}$$

Substitute $V_{GS} = 0.9 + \sqrt{\frac{I}{0.75}}$ into this equation:

$$5 - 2I = 2\left(0.9 + \sqrt{\frac{I}{0.75}}\right)$$
$$5 - 2I = 1.8 + 2\sqrt{\frac{I}{0.75}}$$
$$3.2 - 2I = 2\sqrt{\frac{I}{0.75}}$$

Divide the entire equation by 2 to simplify:

$$1.6 - I = \sqrt{\frac{I}{0.75}}$$

Square both sides to remove the radical:

$$(1.6 - I)^2 = \frac{I}{0.75}$$
$$2.56 - 3.2I + I^2 = 1.3333I$$
$$I^2 - 4.5333I + 2.56 = 0$$

Using the quadratic formula to solve for I:

$$I = \frac{4.5333 \pm \sqrt{(-4.5333)^2 - 4(1)(2.56)}}{2} = \frac{4.5333 \pm \sqrt{20.551 - 10.24}}{2} = \frac{4.5333 \pm \sqrt{10.311}}{2}$$
$$I = \frac{4.5333 \pm 3.211}{2}$$

This yields two roots:

- $I_1 = 3.872\,\text{mA}$
- $I_2 = 0.661\,\text{mA}$

Since $V_3 = 5 - I$ and must remain positive and high enough to sustain conduction, and from $1.6 - I = \sqrt{\dots}$ we know I must be less than $1.6\,\text{mA}$, the physically valid current is:

$$I = 0.661\,\text{mA}$$

4. Calculating Node Voltages
First, find the gate-to-source voltage V_{GS}:

$$V_{GS} = 1.6 - I = 1.6 - 0.661 = 0.939\,\text{V}$$

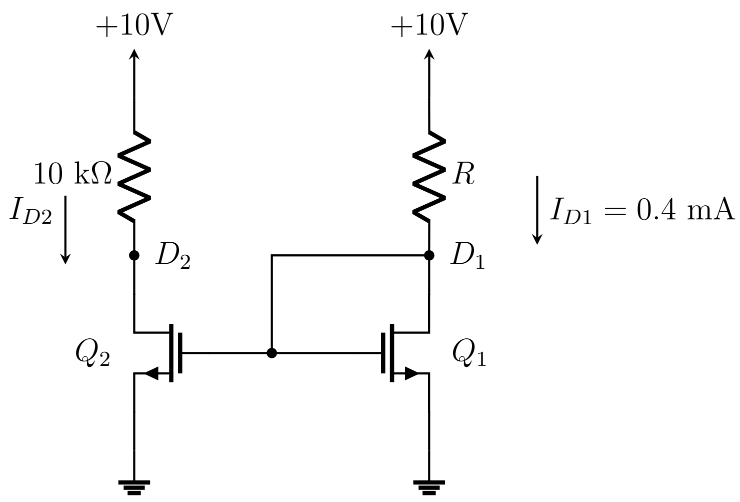
Now substitute $I = 0.661\,\text{mA}$ and $V_{GS} = 0.939\,\text{V}$ into our node voltage relations:

$$V_5 = I = \mathbf{0.661}\,\text{V}$$
$$V_4 = V_5 + V_{GS} = 0.661 + 0.939 = \mathbf{1.600}\,\text{V}$$
$$V_3 = 5 - I = 5 - 0.661 = \mathbf{4.339}\,\text{V}$$

5. Final Answer Summary

Node Voltage	Calculated Value
V_3	4.34 V
V_4	1.60 V
V_5	0.66 V

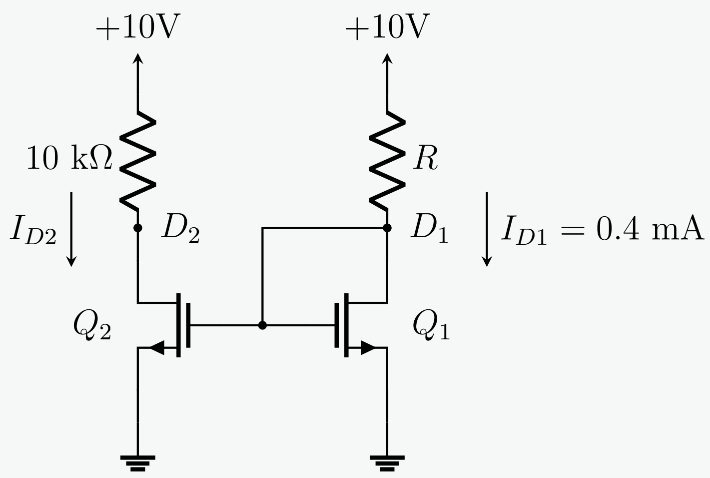
Q6. For the circuit shown, find R so that $I_{D1} = 0.4\,\text{mA}$. Also find V_{D1}, V_{D2}, and I_{D2}. Q_1 and Q_2 are identical. Given: $V_t = 2\,\text{V}$, $\mu_n C_{ox} = 20\,\mu\text{A/V}^2$, $L = 10\,\mu\text{m}$, $W = 100\,\mu\text{m}$.



(20 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Circuit Diagram and Given Parameters



The given device parameters for the identical NMOS transistors (Q_1 and Q_2) are:

- Threshold voltage: $V_t = 2\,\text{V}$
- Process transconductance parameter: $\mu_n C_{ox} = 20\,\mu\text{A/V}^2 = 0.02\,\text{mA/V}^2$
- Aspect ratio: $W/L = 100\,\mu\text{m}/10\,\mu\text{m} = 10$
- Device transconductance parameter: $k_n = \mu_n C_{ox}\left(\frac{W}{L}\right) = 0.02 \times 10 = 0.2\,\text{mA/V}^2$

2. Analysis of Diode-Connected Transistor Q_1

Transistor Q_1 has its gate connected directly to its drain, meaning $V_{D1} = V_{G1}$. Since its source is connected to ground ($V_{S1} = 0\,\text{V}$), we have:

$$V_{DS1} = V_{GS1} \tag{1}$$

Because $V_{DS1} > V_{GS1} - V_t$ (i.e., $V_{GS1} > V_{GS1} - 2$, which is $0 > -2$), Q_1 is guaranteed to operate in the **saturation region**. The drain current equation in saturation is:

$$I_{D1} = \frac{1}{2}k_n(V_{GS1} - V_t)^2 \tag{2}$$

Substituting the known values ($I_{D1} = 0.4\,\text{mA}$):

$$\begin{aligned} 0.4 &= \frac{1}{2}(0.2)(V_{GS1} - 2)^2 \\ 0.4 &= 0.1(V_{GS1} - 2)^2 \\ (V_{GS1} - 2)^2 &= \frac{0.4}{0.1} = 4 \end{aligned}$$

Taking the square root of both sides (and knowing $V_{GS1} > V_t$ for the device to be ON):

$$\begin{aligned} V_{GS1} - 2 &= 2 \\ V_{GS1} &= 4\,\text{V} \end{aligned}$$

Since $V_{G1} = V_{D1}$, the voltage at node D_1 is:

$$\mathbf{V_{D1} = 4}\,\text{V} \tag{3}$$

3. Calculation of Resistor R

The resistor R connects the $+10\,\text{V}$ supply to node D_1. Because the gates of Q_1 and Q_2 draw zero DC current ($I_G = 0$), the entire current I_{D1} flows through R. Applying Ohm's law:

$$R = \frac{V_{DD} - V_{D1}}{I_{D1}}$$
$$R = \frac{10\,\text{V} - 4\,\text{V}}{0.4\,\text{mA}} = \frac{6\,\text{V}}{0.4\,\text{mA}}$$
$$\mathbf{R = 15\,k\Omega}$$

4. Analysis of Transistor Q_2

Q_1 and Q_2 form a **current mirror**. Their gates and sources are tied together in parallel, meaning they share the exact same gate-to-source voltage:

$$V_{GS2} = V_{GS1} = 4\,\text{V} \tag{4}$$

Assuming Q_2 is also operating in the saturation region (to be verified shortly), and given that the transistors are identical, the drain current I_{D2} will perfectly mirror I_{D1} (ignoring channel-length modulation since λ is not provided):

$$\mathbf{I_{D2} = I_{D1} = 0.4}\,\text{mA} \tag{5}$$

5. Calculation of V_{D2} and Verification of Saturation

The voltage at the drain of Q_2 is determined by the voltage drop across its $10\,\text{k}\Omega$ load resistor:

$$V_{D2} = V_{DD} - I_{D2} \times (10\,\text{k}\Omega)$$
$$V_{D2} = 10\,\text{V} - (0.4\,\text{mA})(10\,\text{k}\Omega)$$
$$V_{D2} = 10\,\text{V} - 4\,\text{V}$$
$$\mathbf{V_{D2} = 6}\,\text{V}$$

To confirm our assumption that Q_2 is in saturation, we check the saturation condition:

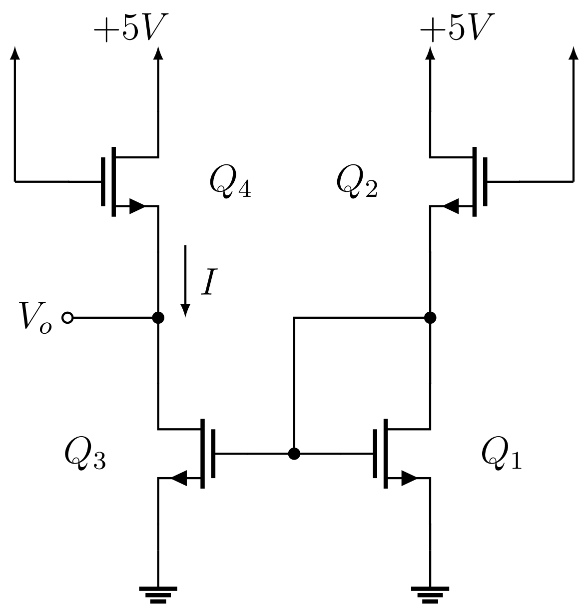
$$V_{DS2} \geq V_{GS2} - V_t$$
$$6\,\text{V} \geq 4\,\text{V} - 2\,\text{V}$$
$$6\,\text{V} \geq 2\,\text{V}$$

Since the condition holds true, Q_2 is definitively operating in the saturation region, validating our calculation for I_{D2}.

6. Final Answer Summary

Parameter	Calculated Value
Resistance (R)	$15\,\text{k}\Omega$
Voltage at Node D_1 (V_{D1})	$4\,\text{V}$
Voltage at Node D_2 (V_{D2})	$6\,\text{V}$
Drain Current of Q_2 (I_{D2})	$0.4\,\text{mA}$

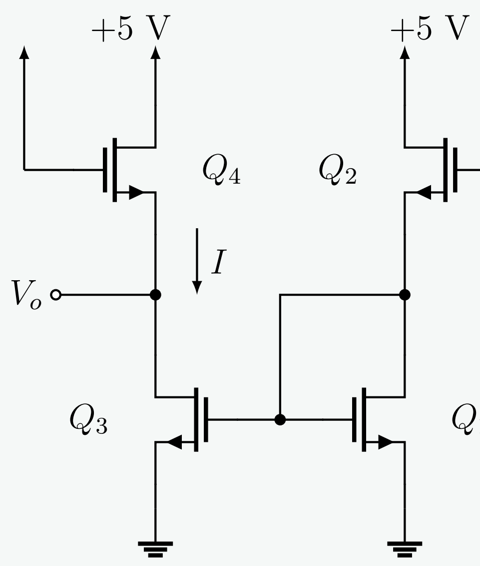
Q7. For the circuit shown, for all MOSFETs $|V_t| = 1\,\text{V}$, $\mu_n C_{ox} = 50\,\mu\text{A/V}^2$, $L = 1\,\mu\text{m}$, $W = 10\,\mu\text{m}$. Find V and I. If Q_3 and Q_4 are made to have $W = 100\,\mu\text{m}$, find V and I in that case.



(26 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Circuit Analysis and Parameter Extraction



The circuit consists of a reference branch (right: Q_1, Q_2) and a mirrored branch (left: Q_3, Q_4). Given device parameters:

- Threshold voltage: $V_t = 1\,\text{V}$
- Process transconductance: $\mu_n C_{ox} = 50\,\mu\text{A/V}^2 = 0.05\,\text{mA/V}^2$

For the initial state where all transistors have $L = 1\,\mu\text{m}$ and $W = 10\,\mu\text{m}$:

$$k_n = \mu_n C_{ox}\left(\frac{W}{L}\right) = 0.05 \times \left(\frac{10}{1}\right) = 0.5\,\text{mA/V}^2$$

Thus, $k_{n1} = k_{n2} = k_{n3} = k_{n4} = 0.5\,\text{mA/V}^2$.

2. Verification of Operating Regions

We must verify that the devices operate in the saturation region, which requires $V_{DS} \geq V_{GS} - V_t$:

- Q_1 **and** Q_2: Q_1 is diode-connected ($V_{D1} = V_{G1} \implies V_{DS1} = V_{GS1}$). Q_2 has its gate and drain tied to the $+5\,\text{V}$ supply, meaning $V_{D2} = V_{G2} \implies V_{DS2} = V_{GS2}$. Since $V_{DS} = V_{GS}$, the condition $V_{DS} > V_{GS} - V_t$ inherently holds. Both are in **saturation**.
- Q_4: Similar to Q_2, its gate and drain are at $+5\,\text{V}$, ensuring $V_{DS4} = V_{GS4}$. It is in **saturation**.
- Q_3: Its state depends on V_o, which we will mathematically verify after determining the node voltages.

3. Case 1: All Transistors are Identical

Right Branch Analysis (Q_1 and Q_2):

Because Q_1 and Q_2 are in series, they share the same drain current ($I_{D1} = I_{D2}$). Since they are identical devices in saturation, their gate-to-source voltages must be equal:

$$V_{GS1} = V_{GS2} \tag{1}$$

Let V_{D1} be the voltage at the drain of Q_1. We have:

$$V_{GS1} = V_{D1} - 0 = V_{D1}$$
$$V_{GS2} = V_{G2} - V_{S2} = 5 - V_{D1}$$

Equating the two yields:

$$V_{D1} = 5 - V_{D1} \implies 2V_{D1} = 5 \implies V_{D1} = 2.5\,\text{V}$$

This makes $V_{GS1} = 2.5\,\text{V}$, which is $> V_t$ (device is ON). The current through the reference branch is:

$$I_{D1} = \frac{1}{2}k_{n1}(V_{GS1} - V_t)^2$$
$$I_{D1} = \frac{1}{2}(0.5\,\text{mA/V}^2)(2.5 - 1)^2 = 0.25 \times 2.25 = 0.5625\,\text{mA}$$

Left Branch Analysis (Q_3 and Q_4):

The left branch mirrors the right. Q_3 shares its gate with Q_1, so $V_{GS3} = V_{GS1} = 2.5\,\text{V}$. Assuming Q_3 is in saturation, it will draw the exact same current:

$$I_{D3} = I_{D1} = 0.5625\,\text{mA}$$

Because Q_4 is in series with Q_3, the total branch current is $I = I_{D4} = I_{D3} = 0.5625\,\text{mA}$. For Q_4 (which is identical to Q_2), carrying the same current requires the same overdrive voltage:

$$V_{GS4} = V_{GS2} = 2.5\,\text{V}$$

The voltage V_o (which is the text's requested V) is the source voltage of Q_4:

$$V_{GS4} = 5 - V_o \implies 2.5 = 5 - V_o \implies V_o = 2.5\,\text{V}$$

Check Q_3 saturation: $V_{DS3} = V_o = 2.5\,\text{V}$. The overdrive voltage is $V_{GS3} - V_t = 2.5 - 1 = 1.5\,\text{V}$. Since $2.5\,\text{V} \geq 1.5\,\text{V}$, Q_3 is in saturation.

4. Case 2: Q_3 and Q_4 are Widened ($W = 100\,\mu$m)
Now, Q_3 and Q_4 have $W = 100\,\mu$m (10 times larger than Q_1, Q_2).

$$k_{n3} = k_{n4} = \mu_n C_{ox}\left(\frac{W}{L}\right) = 0.05 \times \left(\frac{100}{1}\right) = 5.0\,\text{mA/V}^2$$

The right branch (Q_1, Q_2) remains completely unchanged, meaning $V_{G3} = V_{G1} = 2.5\,\text{V}$. The new current I (drawn by Q_3 in saturation) is:

$$I = I_{D3} = \frac{1}{2}k_{n3}(V_{GS3} - V_t)^2$$
$$I = \frac{1}{2}(5.0\,\text{mA/V}^2)(2.5-1)^2 = 2.5 \times (1.5)^2 = 2.5 \times 2.25 = 5.625\,\text{mA}$$

Since Q_4 is in series with Q_3, it carries this new $I = 5.625\,\text{mA}$. We solve for V_{GS4}:

$$I_{D4} = \frac{1}{2}k_{n4}(V_{GS4} - V_t)^2$$
$$5.625 = \frac{1}{2}(5.0)(V_{GS4} - 1)^2$$
$$5.625 = 2.5(V_{GS4} - 1)^2 \implies (V_{GS4} - 1)^2 = 2.25 \implies V_{GS4} - 1 = 1.5 \implies V_{GS4} = 2.5\,\text{V}$$

We find V_o exactly as before:

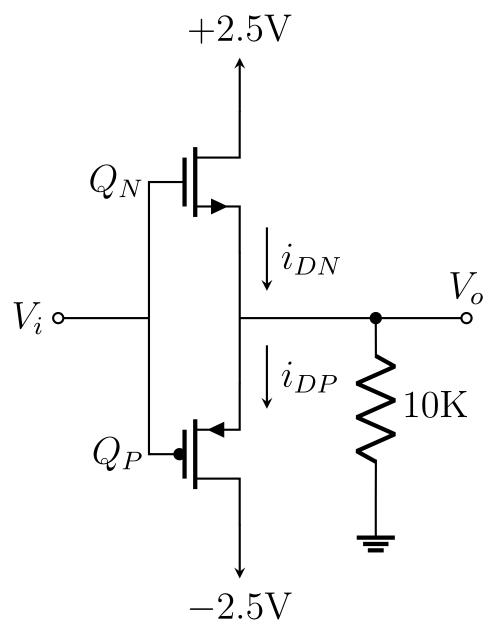
$$V_{GS4} = 5 - V_o \implies 2.5 = 5 - V_o \implies V_o = 2.5\,\text{V}$$

Check Q_3 saturation: $V_{DS3} = V_o = 2.5\,\text{V}$. Overdrive is $1.5\,\text{V}$. $2.5\,\text{V} \geq 1.5\,\text{V}$, so Q_3 remains in saturation.

5. Final Answer Summary

Condition	Voltage (V)	Current (I)
Case 1: All identical ($W = 10\,\mu$m)	2.50 V	0.5625 mA (562.5 μA)
Case 2: Q_3, Q_4 widened ($W = 100\,\mu$m)	2.50 V	5.625 mA

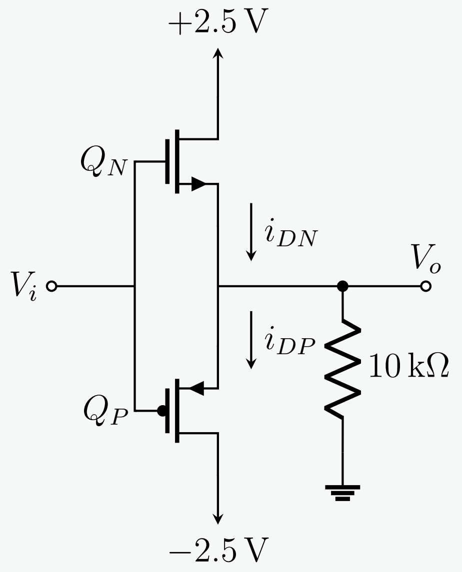
Q8. The NMOS and PMOS transistors in the circuit shown have $k_n(W/L) = k_p(W/L) = 1\,\text{mA/V}^2$ and $V_{tn} = -V_{tp} = 1\,\text{V}$. Assume $\lambda = 0$. Find i_{DN}, i_{DP}, and v_o for $V_i = +2.5\,\text{V}$ and $V_i = -2.5\,\text{V}$.



(20 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Circuit Diagram and Operating Principle



The circuit is a **CMOS Class-B Push-Pull Output Stage** (Source Follower). The given parameters are:

- Transconductance parameters: $k_n = k_p = 1\,\text{mA/V}^2$
- Threshold voltages: $V_{tn} = 1\,\text{V}$ and $V_{tp} = -1\,\text{V}$
- Channel-length modulation: $\lambda = 0$
- Load resistance: $R_L = 10\,\text{k}\Omega$

In a Class-B push-pull stage, only one transistor conducts at a time depending on the input polarity. For a highly positive input, the NMOS (Q_N) conducts and pushes current into the load. For a highly negative input, the PMOS (Q_P) conducts and pulls current from the load.

2. Analysis for $V_i = +2.5\,\text{V}$
When $V_i = +2.5\,\text{V}$, a positive voltage appears at the gates. Assume Q_N is ON and Q_P is OFF. Let us verify Q_P's state: For Q_P to be ON, $V_{GSP} \le V_{tp}$.

$$V_{GSP} = V_G - V_S = 2.5\,\text{V} - V_o$$

Since V_o must be positive (driven by Q_N), V_{GSP} is strongly positive, meaning $V_{GSP} > -1\,\text{V}$. Thus, $\mathbf{Q_P}$ is indeed OFF, and $\mathbf{i_{DP}} = \mathbf{0}\,\text{mA}$.
For Q_N, assuming it operates in the saturation region:

$$V_{GSN} = V_G - V_S = 2.5 - V_o$$

$$i_{DN} = \frac{1}{2}k_n(V_{GSN} - V_{tn})^2 = \frac{1}{2}(1)(2.5 - V_o - 1)^2 = 0.5(1.5 - V_o)^2 \quad \text{(mA)}$$

By Kirchhoff's Current Law (KCL) at the output node, all current i_{DN} flows through the load R_L:

$$i_{DN} = \frac{V_o}{R_L} = \frac{V_o}{10} \quad \text{(mA)}$$

Equating the two expressions for i_{DN}:

$$\begin{aligned} 0.5(1.5 - V_o)^2 &= \frac{V_o}{10} \\ 5(2.25 - 3V_o + V_o^2) &= V_o \\ 11.25 - 15V_o + 5V_o^2 &= V_o \\ 5V_o^2 - 16V_o + 11.25 &= 0 \end{aligned}$$

Solving the quadratic equation for V_o:

$$V_o = \frac{16 \pm \sqrt{(-16)^2 - 4(5)(11.25)}}{2(5)} = \frac{16 \pm \sqrt{256 - 225}}{10} = \frac{16 \pm \sqrt{31}}{10}$$

$$V_o = \frac{16 \pm 5.568}{10}$$

This yields two possible mathematical roots: $V_{o1} = 2.157\,\text{V}$ and $V_{o2} = 1.043\,\text{V}$. Since Q_N requires $V_{GSN} > V_{tn} \implies (2.5 - V_o) > 1 \implies V_o < 1.5\,\text{V}$, the only physically valid solution is:

$$\mathbf{V_o = 1.043}\,\text{V}$$

Now, calculate i_{DN}:

$$i_{DN} = \frac{V_o}{10} = \frac{1.043}{10} = \mathbf{0.1043}\,\text{mA}$$

Saturation Check for Q_N: $V_{DSN} = 2.5 - 1.043 = 1.457\,\text{V}$. $V_{GSN} - V_{tn} = 2.5 - 1.043 - 1 = 0.457\,\text{V}$. Since $V_{DSN} > V_{GSN} - V_{tn}$, Q_N is in saturation.

3. Analysis for $V_i = -2.5\,\text{V}$
When $V_i = -2.5\,\text{V}$, the situation is perfectly symmetrical. For Q_N: $V_{GSN} = -2.5 - V_o$. Since V_o will be negative, V_{GSN} is highly negative, so $\mathbf{Q_N}$ is OFF, and $\mathbf{i_{DN}} = \mathbf{0}\,\text{mA}$.
For Q_P, assuming it is in saturation (note that i_{DP} is defined as flowing from Source to Drain in the diagram):

$$V_{SGP} = V_S - V_G = V_o - (-2.5) = V_o + 2.5$$

$$i_{DP} = \frac{1}{2}k_p(V_{SGP} - |V_{tp}|)^2 = \frac{1}{2}(1)(V_o + 2.5 - 1)^2 = 0.5(V_o + 1.5)^2 \quad (\text{mA})$$

By KCL at the output node, current i_{DP} leaves the node and flows to the $-2.5\,\text{V}$ supply. Therefore, current must flow from ground up through the resistor to the node:

$$i_{DP} = -\frac{V_o}{10} \quad (\text{mA})$$

Equating the two expressions:

$$\begin{aligned} 0.5(V_o + 1.5)^2 &= -\frac{V_o}{10} \\ 5(V_o^2 + 3V_o + 2.25) &= -V_o \\ 5V_o^2 + 15V_o + 11.25 &= -V_o \\ 5V_o^2 + 16V_o + 11.25 &= 0 \end{aligned}$$

Solving the quadratic equation:

$$V_o = \frac{-16 \pm \sqrt{16^2 - 4(5)(11.25)}}{10} = \frac{-16 \pm \sqrt{31}}{10}$$
$$V_o = \frac{-16 \pm 5.568}{10}$$

This yields $V_{o1} = -2.157\,\text{V}$ and $V_{o2} = -1.043\,\text{V}$. For Q_P to be ON, $V_{SGP} > |V_{tp}| \implies (V_o + 2.5) > 1 \implies V_o > -1.5\,\text{V}$. Thus, the valid root is:

$$\mathbf{V_o} = \mathbf{-1.043}\,\text{V}$$

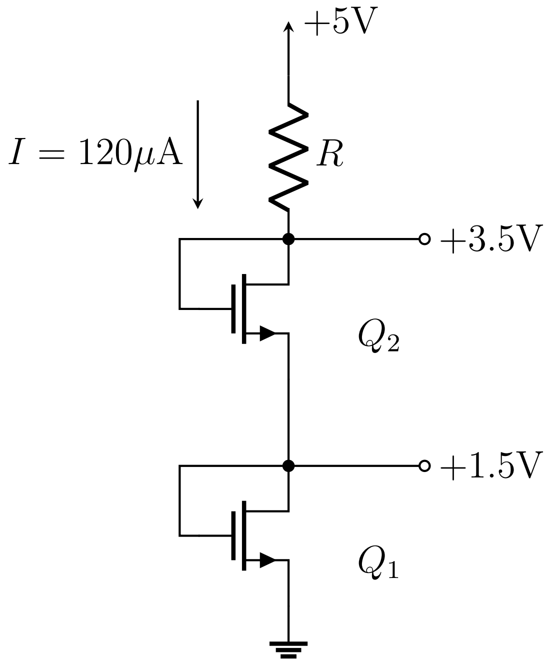
Calculate i_{DP}:

$$i_{DP} = -\frac{-1.043}{10} = \mathbf{0.1043}\,\text{mA}$$

4. Final Answer Summary

Input Condition	Output Voltage (v_o)	NMOS Current (i_{DN})	PMOS Current (i_{DP})
$V_i = +2.5\,\text{V}$	$1.043\,\text{V}$	$0.104\,\text{mA}$	$0\,\text{mA}$
$V_i = -2.5\,\text{V}$	$-1.043\,\text{V}$	$0\,\text{mA}$	$0.104\,\text{mA}$

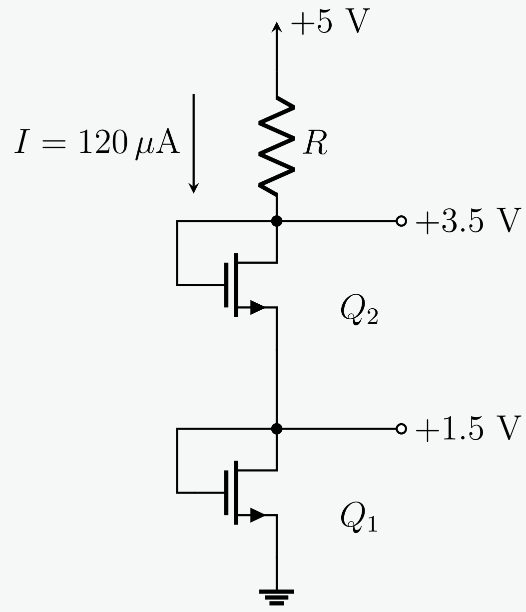
Q9. The NMOS transistors in the circuit have $V_t = 1\,\text{V}$, $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$, $L_1 = L_2 = 1\,\mu\text{m}$. Find W_1, W_2, and R, where W_1 and W_2 are the gate widths of Q1 and Q2 respectively.



(16 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Circuit Overview and Parameters



The circuit consists of two diode-connected NMOS transistors (Q_1 and Q_2) in series with a resistor R. Because the gates are tied to their respective drains ($V_{DS} = V_{GS}$), both transistors are guaranteed to operate in the **saturation region** (since $V_{DS} > V_{GS} - V_t$). The given parameters are: * Drain current: $I = 120\,\mu$A * Process transconductance: $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$ * Threshold voltage: $V_t = 1$ V * Channel lengths: $L_1 = L_2 = 1\,\mu$m

—

2. Calculating Resistor R

The voltage drop across the resistor R is the difference between the supply voltage and the voltage at the drain of Q_2.

$$V_R = 5\text{ V} - 3.5\text{ V} = 1.5\text{ V}$$
$$R = \frac{V_R}{I} = \frac{1.5\text{ V}}{120\,\mu\text{A}} = \frac{1.5}{0.12\text{ mA}}$$

$$R = 12.5\,\text{k}\Omega$$

—

3. Analyzing Transistor Q_1

For Q_1, the source is grounded (**0 V**) and the gate/drain voltage is **1.5 V**.

$$V_{GS1} = V_{D1} - V_{S1} = 1.5 - 0 = 1.5\text{ V}$$

Using the saturation current equation for an NMOS transistor:

$$I = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_1}{L_1}\right)(V_{GS1} - V_t)^2$$

Substitute the known values (keeping units in μA and V for simplicity):

$$120 = \frac{1}{2}(120)\left(\frac{W_1}{1}\right)(1.5 - 1)^2$$
$$120 = 60 \cdot W_1 \cdot (0.5)^2$$
$$120 = 60 \cdot W_1 \cdot 0.25$$
$$120 = 15 \cdot W_1$$

$$W_1 = 8\,\mu\text{m}$$

—

4. Analyzing Transistor Q_2

For Q_2, the source is at **1.5 V** (the drain of Q_1) and the gate/drain voltage is **3.5 V**.

$$V_{GS2} = V_{D2} - V_{S2} = 3.5 - 1.5 = 2.0\text{ V}$$

Using the saturation current equation again for Q_2:

$$I = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_2}{L_2}\right)(V_{GS2} - V_t)^2$$

Substitute the known values:

$$120 = \frac{1}{2}(120)\left(\frac{W_2}{1}\right)(2.0 - 1)^2$$
$$120 = 60 \cdot W_2 \cdot (1.0)^2$$
$$120 = 60 \cdot W_2$$

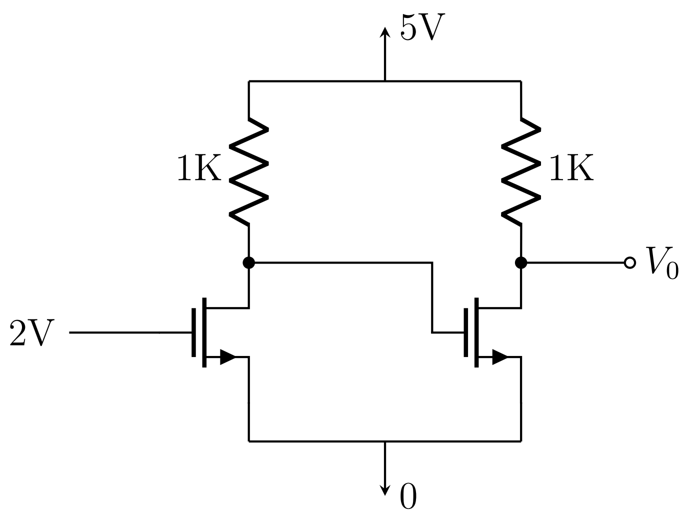
$$W_2 = 2\,\mu\text{m}$$

—

5. Final Answer Summary

Parameter	Calculated Value
Resistor (R)	**12.5 kΩ**
Width of Q1 (W_1)	**8 m**
Width of Q2 (W_2)	**2 m**

Q10. For the circuit shown, find the value of V_0. Both NMOS transistors have $V_t = 1\,\text{V}$, $\lambda = 0$, and $\mu_n C_{ox}(W/L) = 1\,\text{mA/V}^2$. Supply voltage is $5\,\text{V}$.



(16 marks) **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. Parameters and Initial State

- Threshold voltage: $V_t = 1\,\text{V}$
- Transconductance parameter: $k'_n \frac{W}{L} = 1\,\text{mA/V}^2$
- Supply voltage: $V_{DD} = 5\,\text{V}$
- Resistors: $R_1 = R_2 = 1\,\text{k}\Omega$

—

2. Analysis of the First Transistor (M1)
For M1, the source is grounded ($V_{S1} = 0\,\text{V}$) and the gate is connected to a 2V source ($V_{G1} = 2\,\text{V}$).

$$V_{GS1} = V_{G1} - V_{S1} = 2\,\text{V} - 0\,\text{V} = 2\,\text{V}$$

Since $V_{GS1} > V_t$ ($2\,\text{V} > 1\,\text{V}$), M1 is ON. Let us assume M1 is in saturation. The drain current I_{D1} is given by:

$$I_{D1} = \frac{1}{2} k'_n \frac{W}{L} (V_{GS1} - V_t)^2$$

$$I_{D1} = \frac{1}{2}(1)(2-1)^2 = 0.5\,\text{mA}$$

Now, we determine the drain voltage of M1 (V_{D1}):

$$V_{D1} = V_{DD} - I_{D1} R_1 = 5 - (0.5 \times 1) = 4.5\,\text{V}$$

Check for Saturation:

$$V_{DS1} = 4.5\,\text{V}$$

$$V_{OV1} = V_{GS1} - V_t = 1\,\text{V}$$

Since $V_{DS1} > V_{OV1}$ ($4.5\,\text{V} > 1\,\text{V}$), M1 is indeed in saturation.
—

3. Analysis of the Second Transistor (M2)
The gate of M2 is connected to the drain of M1, so $V_{G2} = 4.5\,\text{V}$. The source of M2 is grounded, so $V_{S2} = 0\,\text{V}$.

$$V_{GS2} = 4.5\,\text{V}$$

Since $V_{GS2} > V_t$, M2 is ON.
Let's tentatively assume M2 is in saturation to check its validity:

$$I_{D2} = \frac{1}{2}(1)(4.5-1)^2 = 0.5(3.5)^2 = 6.125\,\text{mA}$$

If $I_{D2} = 6.125\,\text{mA}$, the voltage at the drain (V_0) would be:

$$V_0 = 5 - (6.125 \times 1) = -1.125\,\text{V}$$

Since a negative voltage is physically impossible in this passive grounded-source network, the assumption is incorrect. **M2 must be operating in the triode (linear) region.**
For the triode region, the current equation is:

$$I_{D2} = k'_n \frac{W}{L}\left[(V_{GS2} - V_t)V_{DS2} - \frac{1}{2}V_{DS2}^2\right]$$

We know that $V_{DS2} = V_0$. Substituting the values:

$$I_{D2} = 1\left[(4.5 - 1)V_0 - 0.5V_0^2\right]$$

$$I_{D2} = 3.5V_0 - 0.5V_0^2 \quad \text{— (Equation 1)}$$

From the circuit's output branch, we can also write I_{D2} using Ohm's Law across the $1\,\text{k}\Omega$ resistor:

$$I_{D2} = \frac{5 - V_0}{1} = 5 - V_0 \quad \text{— (Equation 2)}$$

—

4. Solving for Output Voltage (V_0)
Equating Equation 1 and Equation 2:

$$5 - V_0 = 3.5V_0 - 0.5V_0^2$$

$$0.5V_0^2 - 4.5V_0 + 5 = 0$$

Multiply the entire equation by 2 to simplify:

$$V_0^2 - 9V_0 + 10 = 0$$

Applying the quadratic formula:

$$V_0 = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(10)}}{2(1)}$$

$$V_0 = \frac{9 \pm \sqrt{81 - 40}}{2}$$

$$V_0 = \frac{9 \pm \sqrt{41}}{2}$$

$$V_0 = \frac{9 \pm 6.403}{2}$$

This gives two possible mathematical roots:

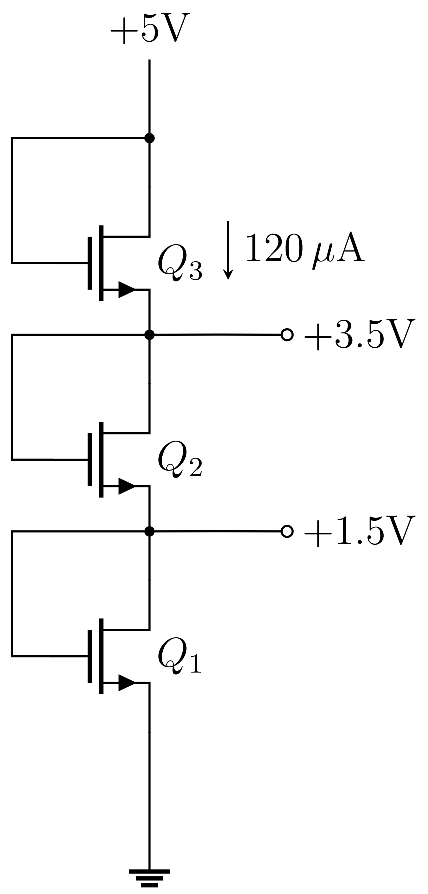
$$V_{0,1} = \frac{15.403}{2} = 7.701\,\text{V}$$

$$V_{0,2} = \frac{2.597}{2} = 1.2985\,\text{V}$$

Since the supply voltage is $5\,\text{V}$, the output voltage cannot exceed it. Therefore, the physically meaningful solution is $V_{0,2}$.

$$V_0 = 1.2985\,\text{V} \approx \mathbf{1.30}\,\text{V}$$

Q11. The NMOS transistors in the circuit have $V_t = 1\,\text{V}$, $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$, and $L_1 = L_2 = L_3 = 1\,\mu\text{m}$. Find the required values of gate width for each of Q1, Q2, and Q3 to obtain the voltage and current values indicated in the figure.



(14 marks) **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

1. Circuit Characteristics

The circuit consists of three stacked NMOS transistors (Q_1, Q_2, and Q_3). Each transistor is **diode-connected** because its gate is connected directly to its drain ($V_{DS} = V_{GS}$). This connection ensures that whenever a transistor conducts current, it automatically operates in the **saturation region** (since $V_{DS} = V_{GS} > V_{GS} - V_t$).
The given parameters are:

- Common series drain current: $I = 120\,\mu\text{A}$
- Process transconductance: $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$
- Threshold voltage: $V_t = 1\,\text{V}$
- Channel lengths: $L_1 = L_2 = L_3 = 1\,\mu\text{m}$

The general current equation for an NMOS transistor operating in saturation is:

$$I = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

—

2. Analysis of Transistor Q_1 (Bottom)

* **Source Voltage (V_{S1}):** Connected directly to ground $\rightarrow 0\,\text{V}$ * **Drain/Gate Voltage ($V_{D1} = V_{G1}$):** Indicated as $+1.5\,\text{V}$
Calculating the gate-to-source overdrive voltage ($V_{GS1} - V_t$):

$$V_{GS1} = V_{G1} - V_{S1} = 1.5\,\text{V} - 0\,\text{V} = 1.5\,\text{V}$$

$$V_{GS1} - V_t = 1.5\,\text{V} - 1\,\text{V} = 0.5\,\text{V}$$

Substituting these into the current formula:

$$\begin{aligned} 120\,\mu\text{A} &= \frac{1}{2}\left(120\,\mu\text{A/V}^2\right)\left(\frac{W_1}{1\,\mu\text{m}}\right)(0.5\,\text{V})^2 \\ 120 &= 60 \cdot W_1 \cdot 0.25 \\ 120 &= 15 \cdot W_1 \end{aligned}$$

$$W_1 = 8\,\mu\text{m}$$

—

3. Analysis of Transistor Q_2 (Middle)

* **Source Voltage (V_{S2}):** Connected to the drain of $Q_1 \rightarrow +1.5\,\text{V}$ * **Drain/Gate Voltage ($V_{D2} = V_{G2}$):** Indicated as $+3.5\,\text{V}$
Calculating the gate-to-source overdrive voltage ($V_{GS2} - V_t$):

$$V_{GS2} = V_{G2} - V_{S2} = 3.5\,\text{V} - 1.5\,\text{V} = 2.0\,\text{V}$$

$$V_{GS2} - V_t = 2.0\,\text{V} - 1\,\text{V} = 1.0\,\text{V}$$

Substituting these into the current formula:

$$120\,\mu\text{A} = \frac{1}{2}\left(120\,\mu\text{A/V}^2\right)\left(\frac{W_2}{1\,\mu\text{m}}\right)(1.0\,\text{V})^2$$
$$120 = 60 \cdot W_2 \cdot 1$$
$$120 = 60 \cdot W_2$$

$$W_2 = 2\,\mu\text{m}$$

—

4. Analysis of Transistor Q_3 (Top)
* **Source Voltage (V_{S3}):** Connected to the drain of $Q_2 \rightarrow +3.5\,\text{V}$ * **Drain/Gate Voltage ($V_{D3} = V_{G3}$):** Connected to the supply line $\rightarrow +5\,\text{V}$
Calculating the gate-to-source overdrive voltage ($V_{GS3} - V_t$):

$$V_{GS3} = V_{G3} - V_{S3} = 5.0\,\text{V} - 3.5\,\text{V} = 1.5\,\text{V}$$

$$V_{GS3} - V_t = 1.5\,\text{V} - 1\,\text{V} = 0.5\,\text{V}$$

Substituting these into the current formula:

$$120\,\mu\text{A} = \frac{1}{2}\left(120\,\mu\text{A/V}^2\right)\left(\frac{W_3}{1\,\mu\text{m}}\right)(0.5\,\text{V})^2$$
$$120 = 60 \cdot W_3 \cdot 0.25$$
$$120 = 15 \cdot W_3$$

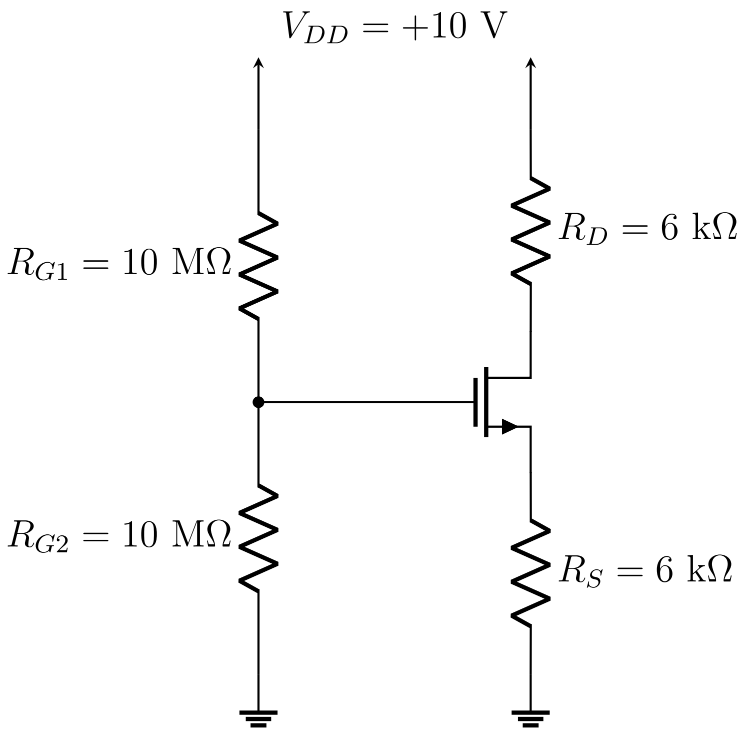
$$W_3 = 8\,\mu\text{m}$$

—

5. Summary Table of Gate Widths

Transistor	**Overdrive Voltage** ($V_{GS} - V_t$)	**Required Width** (W)
Q_1	0.5 V	**8 m**
Q_2	1.0 V	**2 m**
Q_3	0.5 V	**8 m**

Q12. Determine all node voltages and drain current of the circuit. Assume $V_t = 1\,\text{V}$, $K_n' W/L = 1\,\text{mA/V}^2$, $\lambda = 0$, $V_{DD} = +10\,\text{V}$, $R_D = 6\,\text{k}\Omega$.



(20 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Gate Voltage Calculation (V_G)
Since the gate current of a MOSFET is ideally zero ($I_G = 0$), the resistors R_{G1} and R_{G2} form a simple voltage divider across the supply voltage V_{DD}.

$$V_G = V_{DD} \times \frac{R_{G2}}{R_{G1} + R_{G2}}$$
$$V_G = 10 \text{ V} \times \frac{10 \text{ M}\Omega}{10 \text{ M}\Omega + 10 \text{ M}\Omega} = 5 \text{ V}$$

—

2. Setting up the Overdrive Equation
Let us assume the NMOS transistor is operating in the **saturation region**. The source voltage (V_S) depends on the drain current (I_D) passing through the source resistor R_S:

$$V_S = I_D \times R_S = I_D \times 6 \text{ k}\Omega$$

The gate-to-source voltage is:

$$V_{GS} = V_G - V_S = 5 - 6I_D \quad (\text{where } I_D \text{ is in mA})$$

The saturation current equation for the NMOS transistor is:

$$I_D = \frac{1}{2}\left(k_n' \frac{W}{L}\right)(V_{GS} - V_t)^2$$

Given that $k_n' \frac{W}{L} = 1 \text{ mA/V}^2$ and $V_t = 1$ V:

$$I_D = \frac{1}{2}(1)(5 - 6I_D - 1)^2$$
$$2I_D = (4 - 6I_D)^2$$

—

3. Solving for Drain Current (I_D)
Expanding the quadratic expression:

$$2I_D = 16 - 48I_D + 36I_D^2$$
$$36I_D^2 - 50I_D + 16 = 0$$

Dividing the entire equation by 2 to simplify:

$$18I_D^2 - 25I_D + 8 = 0$$

Using the quadratic formula $I_D = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$I_D = \frac{25 \pm \sqrt{(-25)^2 - 4(18)(8)}}{2(18)}$$
$$I_D = \frac{25 \pm \sqrt{625 - 576}}{36}$$
$$I_D = \frac{25 \pm \sqrt{49}}{36} = \frac{25 \pm 7}{36}$$

This gives two potential roots:

- $I_{D1} = \frac{25+7}{36} = \frac{32}{36} = 0.89$ mA
- $I_{D2} = \frac{25-7}{36} = \frac{18}{36} = 0.50$ mA

Choosing the valid root:

- If $I_D = 0.89$ mA, then $V_{GS} = 5 - 6(0.89) = -0.34$ V. Since $V_{GS} < V_t$ (1 V), the transistor would be in cutoff, which contradicts conduction.
- If $I_D = 0.50$ mA, then $V_{GS} = 5 - 6(0.50) = 2.0$ V. Since $V_{GS} > V_t$ (2 V > 1 V), the transistor is active.

Thus, the physically correct drain current is:

$$I_D = 0.50 \text{ mA}$$

—

4. Finding Remaining Node Voltages and Verification
* **Source Voltage (V_S):**

$$V_S = I_D \times R_S = 0.50 \text{ mA} \times 6 \text{ k}\Omega = 3.0 \text{ V}$$

* **Drain Voltage (V_D):**

$$V_D = V_{DD} - I_D \times R_D = 10 \text{ V} - (0.50 \text{ mA} \times 6 \text{ k}\Omega) = 10 - 3.0 = 7.0 \text{ V}$$

* **Verification of Saturation Assumption:**

$$V_{DS} = V_D - V_S = 7.0 \text{ V} - 3.0 \text{ V} = 4.0 \text{ V}$$
$$V_{OV} = V_{GS} - V_t = 2.0 \text{ V} - 1 \text{ V} = 1.0 \text{ V}$$

Since $V_{DS} \geq V_{GS} - V_t$ (4.0 V > 1.0 V), the transistor is confirmed to be in the **saturation region**.

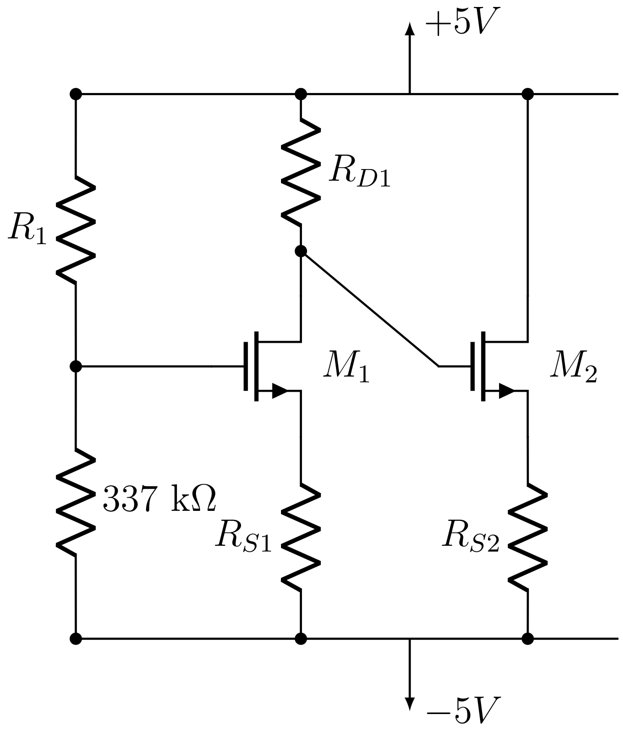
—

5. Summary of Results

Parameter	Calculated Value
Drain Current (I_D)	**0.50 mA**
Gate Voltage (V_G)	**5.0 V**
Source Voltage (V_S)	**3.0 V**
Drain Voltage (V_D)	**7.0 V**

Multi-stage MOSFET Circuit Design

Q1. For the circuit shown with transistor parameters $K_{n1} = 500\,\mu\text{A/V}^2$, $K_{n2} = 200\,\mu\text{A/V}^2$, $V_{T1} = V_{T2} = 1.2\,\text{V}$, design the circuit such that $I_{D1} = 0.1\,\text{mA}$, $I_{D2} = 0.3\,\text{mA}$, $V_{DS1} = V_{DS2} = 5\,\text{V}$.



(20 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Parameters and Assumptions

- Supply voltages: $V_{DD} = +5\,\text{V}$, $V_{SS} = -5\,\text{V}$
- Transistor M_1: $K_{n1} = 500\,\mu\text{A/V}^2 = 0.5\,\text{mA/V}^2$, $V_{T1} = 1.2\,\text{V}$
- Transistor M_2: $K_{n2} = 200\,\mu\text{A/V}^2 = 0.2\,\text{mA/V}^2$, $V_{T2} = 1.2\,\text{V}$
- Design targets: $I_{D1} = 0.1\,\text{mA}$, $I_{D2} = 0.3\,\text{mA}$, $V_{DS1} = 5\,\text{V}$, $V_{DS2} = 5\,\text{V}$

We assume both MOSFETs are operating in the **saturation region**. The drain current equation in saturation is given by:

$$I_D = K_n(V_{GS} - V_T)^2 \tag{1}$$

2. Design of Transistor M_2 Circuit

From the circuit diagram, the drain of M_2 is connected directly to V_{DD}.

$$V_{D2} = 5\,\text{V}$$

We are given the target $V_{DS2} = 5\,\text{V}$. We can find the source voltage V_{S2}:

$$V_{DS2} = V_{D2} - V_{S2} \implies 5\,\text{V} = 5\,\text{V} - V_{S2} \implies V_{S2} = 0\,\text{V}$$

Now, calculate the required source resistance R_{S2} to sink the target drain current $I_{D2} = 0.3\,\text{mA}$ to $V_{SS} = -5\,\text{V}$:

$$R_{S2} = \frac{V_{S2} - V_{SS}}{I_{D2}} = \frac{0 - (-5)}{0.3} = 16.67\,\text{k}\Omega$$

Next, find the required gate-to-source voltage V_{GS2} to sustain I_{D2}:

$$\begin{aligned} I_{D2} &= K_{n2}(V_{GS2} - V_{T2})^2 \\ 0.3 &= 0.2(V_{GS2} - 1.2)^2 \\ (V_{GS2} - 1.2)^2 &= 1.5 \\ V_{GS2} &= 1.2 + \sqrt{1.5} \approx 2.4247\,\text{V} \end{aligned}$$

The gate voltage of M_2 is therefore:

$$V_{G2} = V_{GS2} + V_{S2} = 2.4247 + 0 = 2.4247\,\text{V}$$

3. Design of Transistor M_1 Circuit

From the circuit, the drain of M_1 is directly connected to the gate of M_2. Therefore:

$$V_{D1} = V_{G2} = 2.4247\,\text{V}$$

Calculate the drain resistor R_{D1} using the current $I_{D1} = 0.1\,\text{mA}$:

$$R_{D1} = \frac{V_{DD} - V_{D1}}{I_{D1}} = \frac{5 - 2.4247}{0.1} = 25.75\,\text{k}\Omega$$

We are given $V_{DS1} = 5\,\text{V}$. Find the source voltage V_{S1}:

$$V_{DS1} = V_{D1} - V_{S1} \implies 5 = 2.4247 - V_{S1} \implies V_{S1} = -2.5753\,\text{V}$$

Calculate the source resistance R_{S1}:

$$R_{S1} = \frac{V_{S1} - V_{SS}}{I_{D1}} = \frac{-2.5753 - (-5)}{0.1} = 24.25\,\text{k}\Omega$$

Find the required gate-to-source voltage V_{GS1} for M_1:

$$\begin{aligned} I_{D1} &= K_{n1}(V_{GS1} - V_{T1})^2 \\ 0.1 &= 0.5(V_{GS1} - 1.2)^2 \\ (V_{GS1} - 1.2)^2 &= 0.2 \\ V_{GS1} &= 1.2 + \sqrt{0.2} \approx 1.6472\,\text{V} \end{aligned}$$

The gate voltage of M_1 must be:

$$V_{G1} = V_{GS1} + V_{S1} = 1.6472 + (-2.5753) = -0.9281\,\text{V}$$

4. Design of the Gate Bias Network for M_1

The voltage divider provides $V_{G1} = -0.9281\,\text{V}$. The lower resistor is $R_2 = 337\,\text{k}\Omega$. Applying Ohm's law and assuming zero gate current, the current through the divider is:

$$I_{\text{bias}} = \frac{V_{G1} - V_{SS}}{R_2} = \frac{-0.9281 - (-5)}{337\,\text{k}\Omega} = \frac{4.0719}{337} \approx 0.01208\,\text{mA}$$

Now, calculate R_1:

$$R_1 = \frac{V_{DD} - V_{G1}}{I_{\text{bias}}} = \frac{5 - (-0.9281)}{0.01208} = \frac{5.9281}{0.01208} \approx 490.6\,\text{k}\Omega$$

5. Verification of Operating Regions

- **For M_1:** $V_{DS1} = 5\,\text{V}$ and $V_{OV1} = V_{GS1} - V_{T1} = 0.4472\,\text{V}$. Since $V_{DS1} > V_{OV1}$, M_1 is correctly in saturation.
- **For M_2:** $V_{DS2} = 5\,\text{V}$ and $V_{OV2} = V_{GS2} - V_{T2} = 1.2247\,\text{V}$. Since $V_{DS2} > V_{OV2}$, M_2 is correctly in saturation.

Final Component Values:

- $R_1 = \mathbf{490.6\,k\Omega}$
- $R_{D1} = \mathbf{25.75\,k\Omega}$
- $R_{S1} = \mathbf{24.25\,k\Omega}$
- $R_{S2} = \mathbf{16.67\,k\Omega}$

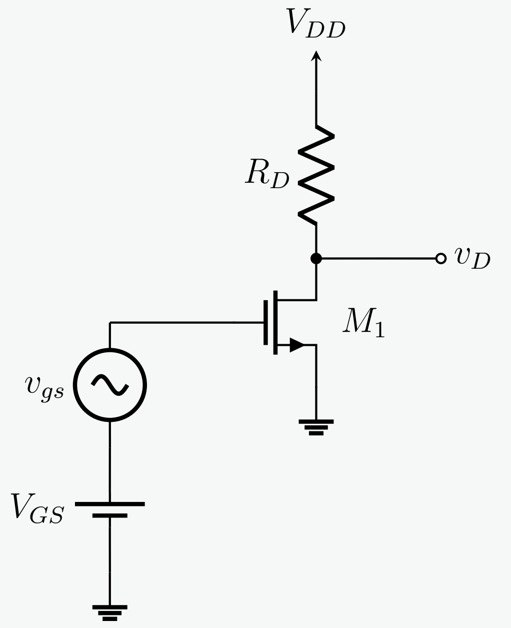
MOSFET Transconductance

Q1. Derive the condition of small signal operation considering a simple CS nMOS amplifier circuit. Find the expression of MOSFET transconductance g_m from there as well. **(8 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

1. Conceptual Circuit for Small-Signal Operation

To derive the condition for small-signal operation, we consider a conceptual Common-Source (CS) amplifier circuit where a small time-varying AC signal v_{gs} is superimposed on a DC bias voltage V_{GS}.



2. Total Instantaneous Quantities
The total instantaneous gate-to-source voltage v_{GS} is the sum of the DC bias V_{GS} and the AC small signal v_{gs}:

$$v_{GS} = V_{GS} + v_{gs} \tag{1}$$

Assuming the DC bias establishes an operating point in the **saturation region** (i.e., $V_{DS} \geq V_{GS} - V_t$), the total instantaneous drain current i_D is governed by the square-law equation:

$$i_D = \frac{1}{2}k_n' \frac{W}{L}(v_{GS} - V_t)^2 \tag{2}$$

where $k_n' = \mu_n C_{ox}$ is the process transconductance parameter, W/L is the aspect ratio, and V_t is the threshold voltage.

3. Derivation of the Small-Signal Condition
Substitute Eq. (1) into Eq. (2):

$$i_D = \frac{1}{2}k_n' \frac{W}{L}(V_{GS} + v_{gs} - V_t)^2 \tag{3}$$

We can regroup the terms by keeping the DC overdrive voltage $(V_{GS} - V_t)$ together:

$$i_D = \frac{1}{2}k_n' \frac{W}{L}\left[(V_{GS} - V_t) + v_{gs}\right]^2 \tag{4}$$

Expanding the squared term algebraically yields:

$$i_D = \frac{1}{2}k_n' \frac{W}{L}\left[(V_{GS} - V_t)^2 + 2(V_{GS} - V_t)v_{gs} + v_{gs}^2\right]$$
$$i_D = \underbrace{\frac{1}{2}k_n' \frac{W}{L}(V_{GS} - V_t)^2}_{\text{DC Bias Current, } I_D} + \underbrace{k_n' \frac{W}{L}(V_{GS} - V_t)v_{gs}}_{\text{Linear AC component}} + \underbrace{\frac{1}{2}k_n' \frac{W}{L}v_{gs}^2}_{\text{Non-linear distortion}}$$

For the circuit to act as a **linear amplifier**, the non-linear (squared) term must be negligibly small compared to the linear term. Mathematically, this condition is expressed as:

$$\frac{1}{2}k_n' \frac{W}{L}v_{gs}^2 \ll k_n' \frac{W}{L}(V_{GS} - V_t)v_{gs} \tag{5}$$

Dividing both sides by the common factors (assuming $v_{gs} \neq 0$):

$$\frac{1}{2}v_{gs} \ll (V_{GS} - V_t) \tag{6}$$

Defining the DC overdrive voltage as $V_{OV} = V_{GS} - V_t$, the fundamental **condition for small-signal operation** becomes:

$$\mathbf{v_{gs} \ll 2V_{OV}} \tag{7}$$

When this condition is satisfied, the non-linear distortion is minimized, and the total drain current can be approximated as the sum of a pure DC component (I_D) and a linearly proportional AC component (i_d).

4. Expression for Transconductance (g_m)
Provided the small-signal condition holds, we can ignore the v_{gs}^2 term in Eq. (5). The small-signal AC drain current i_d is isolated as:

$$i_d \approx k_n' \frac{W}{L}(V_{GS} - V_t)v_{gs} \tag{8}$$

The transconductance g_m relates the output signal current i_d to the input signal voltage v_{gs} and is defined as:

$$g_m \triangleq \frac{i_d}{v_{gs}} = \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{GS}=V_{GS}} \tag{9}$$

Substituting i_d from Eq. (8) directly gives the expression for g_m:

$$\mathbf{g_m = k_n' \frac{W}{L}(V_{GS} - V_t) = k_n' \frac{W}{L} V_{OV}} \tag{10}$$

Important Note: The transconductance g_m can also be expressed in two other highly useful forms by manipulating the basic I_D formula:

$$g_m = \sqrt{2k_n' \frac{W}{L} I_D}$$
$$g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}}$$

Q2. How can the MOSFET transconductance g_m be varied with the design parameters W/L ratio and overdrive voltage V_{ov}? Explain. **(10 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Fundamental Equations for Transconductance
The transconductance g_m of a MOSFET operating in the saturation region is defined as the change in drain current I_D with respect to a change in the gate-to-source voltage V_{GS}. Using the standard saturation current equation $I_D = \frac{1}{2}\mu_n C_{ox}(W/L)V_{OV}^2$, the expression for g_m can be manipulated into three distinct and highly useful forms for analog design:

$$g_m = \mu_n C_{ox}\left(\frac{W}{L}\right)V_{OV} \tag{1}$$

$$g_m = \sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right)I_D} \tag{2}$$

$$g_m = \frac{2I_D}{V_{OV}} \tag{3}$$

where:

- $\mu_n C_{ox}$ is the process transconductance parameter (fixed by the fabrication technology).
- W/L is the physical aspect ratio (width over length) of the device.
- $V_{OV} = V_{GS} - V_t$ is the overdrive voltage.
- I_D is the DC bias current.

—

2. Varying g_m with Aspect Ratio (W/L)
The dependence of g_m on the W/L ratio is dictated by which other circuit parameter is held constant (either V_{OV} or I_D).

- **Case A: Constant Overdrive Voltage (V_{OV})**
  If the circuit is biased such that V_{OV} remains constant, Equation (1) is applied. In this scenario, g_m varies **linearly** with W/L.

  $$g_m \propto \frac{W}{L} \quad \text{(for constant } V_{OV}\text{)}$$

  Design Implication: Because V_{OV} is fixed, increasing W/L will naturally cause a proportional increase in the DC drain current I_D. This yields higher gain but at a direct cost of increased power consumption.

- **Case B: Constant Drain Current (I_D)**
  In practical IC design, the power budget is often strictly fixed, meaning I_D is established by a constant current source. Here, Equation (2) applies. The transconductance g_m varies proportionally to the **square root** of W/L.

  $$g_m \propto \sqrt{\frac{W}{L}} \quad \text{(for constant } I_D\text{)}$$

  Design Implication: To maintain a constant I_D while increasing W/L, the designer must deliberately decrease the overdrive voltage V_{OV}.

—

3. Varying g_m with Overdrive Voltage (V_{OV})
Similarly, the variation of g_m with V_{OV} depends on whether the device geometry or the power budget is the constraining factor.

- **Case A: Constant Device Geometry (W/L)**
  If the physical dimensions of the MOSFET are fixed, Equation (1) demonstrates that g_m increases **linearly** with V_{OV}.

  $$g_m \propto V_{OV} \quad \text{(for constant } W/L\text{)}$$

  Design Implication: This is generally a poor method for increasing g_m. A linear increase in V_{OV} requires a quadratic increase in the bias current I_D, meaning the power penalty is severe. Furthermore, higher V_{OV} limits the available voltage headroom (swing) for analog signals.

- **Case B: Constant Drain Current (I_D)**
  If the design strictly mandates a constant bias current I_D, Equation (3) becomes the governing relationship. In this mode, g_m is **inversely proportional** to V_{OV}.
  $$g_m \propto \frac{1}{V_{OV}} \quad \text{(for constant } I_D\text{)}$$
  Design Implication: To maximize g_m for a given amount of power, the designer must *decrease* V_{OV}. Operating at a small V_{OV} requires making the device physically very wide (large W/L). While this yields high g_m, it comes at the cost of larger silicon area and increased parasitic gate capacitance (C_{gs}), which heavily degrades the high-frequency bandwidth of the amplifier.

—

4. Summary of Parameter Relationships

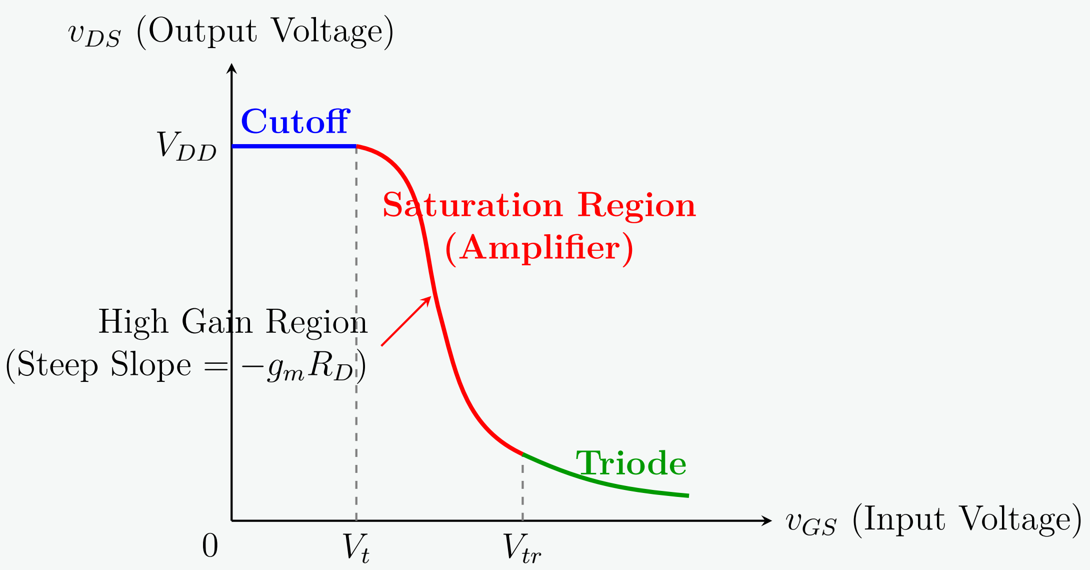
Design Goal	Fixed Parameter	Required Action	Consequence
Increase g_m	Constant V_{OV}	Increase W/L	I_D (Power) increases linearly.
Increase g_m	Constant I_D	Increase W/L	V_{OV} decreases, g_m rises by $\sqrt{W/L}$.
Increase g_m	Constant W/L	Increase V_{OV}	I_D (Power) increases quadratically.
Increase g_m	Constant I_D	Decrease V_{OV}	W/L must increase significantly.

Voltage Transfer Characteristics

Q1. Draw the voltage transfer characteristic of a MOSFET. In which part of this graph is a MOSFET used as an amplifier, and why? What are the benefits and drawbacks of increasing bias voltage? **(19 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Answer

1. Voltage Transfer Characteristic (VTC) of a MOSFET
The Voltage Transfer Characteristic (VTC) illustrates the relationship between the input gate-to-source voltage (v_{GS} or v_I) and the output drain-to-source voltage (v_{DS} or v_O) for a basic common-source amplifier configuration with a resistive load (R_D).



2. Region of Amplification and Explanation
A MOSFET is operated as an amplifier exclusively within the **Saturation Region** (the middle section of the VTC, where $V_t < v_{GS} < V_{tr}$).

- **High Gain (Steep Slope):** In saturation, the drain current i_D depends on the square of the input voltage v_{GS} but is relatively independent of the output voltage v_{DS}. A small change in the input voltage Δv_{GS} produces a large change in the output voltage Δv_{DS}. The steep negative slope ($\frac{\partial v_{DS}}{\partial v_{GS}} \approx -g_m R_D$) represents the high voltage gain of the amplifier.
- **Linearity Assumption:** Although the fundamental equation is non-linear (square-law), for very small input signal variations (small-signal condition, $v_{gs} \ll 2V_{OV}$), the curve can be approximated as a straight line, allowing for linear amplification without significant harmonic distortion.

3. Benefits and Drawbacks of Increasing Bias Voltage
The DC bias voltage establishes the quiescent operating point (Q-point) via the overdrive voltage $V_{OV} = V_{GS} - V_t$. Adjusting this parameter creates critical performance trade-offs:
Benefits of Increasing Bias Voltage (V_{GS} or V_{OV}):

- **Higher Transconductance (g_m):** For a fixed device geometry (W/L), g_m is directly proportional to the overdrive voltage ($g_m = \mu_n C_{ox} \frac{W}{L} V_{OV}$). A higher g_m implies a stronger capability to convert input voltage variations into output current variations.

- **Improved High-Frequency Response:** A higher bias voltage increases the quiescent drain current (I_D), enabling the circuit to charge and discharge parasitic and load capacitances much faster, thereby increasing the bandwidth and slew rate.
- **Extended Input Signal Range:** A higher V_{GS} allows the amplifier to handle larger positive input signal swings without the transistor accidentally turning off (clipping) by falling below the threshold voltage V_t.

Drawbacks of Increasing Bias Voltage (V_{GS} or V_{OV}):

- **Increased Power Dissipation:** The DC bias current scales quadratically with overdrive voltage ($I_D \propto V_{OV}^2$). Consequently, a higher bias voltage substantially boosts the static power consumption ($P_Q = I_D V_{DD}$), which is highly undesirable in battery-operated and dense integrated circuits.
- **Reduced Output Voltage Swing (Headroom):** To maintain operation in the saturation region, the output voltage must satisfy $v_{DS} \geq V_{GS} - V_t = V_{OV}$. Increasing V_{OV} pushes the boundary of the triode region higher up the y-axis, thereby restricting the maximum allowable negative voltage swing at the output before the signal distorts.
- **Decreased g_m/I_D Efficiency:** If the primary design constraint is the power budget (constant I_D), increasing V_{OV} actually *decreases* the available transconductance ($g_m = \frac{2I_D}{V_{OV}}$) and the intrinsic voltage gain.

MOSFET as Small-Signal Amplifier

Q1. Briefly explain the operation of a MOSFET as a small-signal amplifier and deduce its VTC from the I_D–V_{DS} graph. **(15 marks)**
[EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Operation of a MOSFET as a Small-Signal Amplifier
A MOSFET can operate as a linear amplifier when biased appropriately in the **saturation region**. The operation principle is based on converting a small, time-varying input voltage (gate signal) into a proportional output current, which is then transformed back into a larger output voltage across a load resistor.

- **DC Biasing (Establishing the Q-point):** First, a constant DC gate-to-source voltage V_{GSQ} (where $V_{GSQ} > V_t$) is applied to establish a steady DC drain current I_{DQ}. The load resistor R_D sets the DC drain-to-source voltage $V_{DSQ} = V_{DD} - I_{DQ}R_D$. To ensure saturation, the bias must satisfy $V_{DSQ} > V_{GSQ} - V_t$.
- **Superimposing the Small Signal:** A small AC signal v_{gs} is superimposed on the DC bias, making the total instantaneous gate voltage $v_{GS} = V_{GSQ} + v_{gs}$.
- **Current Generation (Transconductance):** Due to the MOSFET's transfer characteristic, this variation in v_{GS} produces a proportional variation in the drain current i_d. For linear operation, the small-signal condition ($v_{gs} \ll 2V_{OV}$) must be met. The relationship is governed by the transconductance g_m:

$$i_d = g_m v_{gs} \quad \text{where} \quad g_m = \mu_n C_{ox}\frac{W}{L}(V_{GSQ} - V_t) \tag{1}$$

- **Voltage Amplification:** The total instantaneous drain current $i_D = I_{DQ} + i_d$ flows through the drain resistor R_D. The output voltage is:

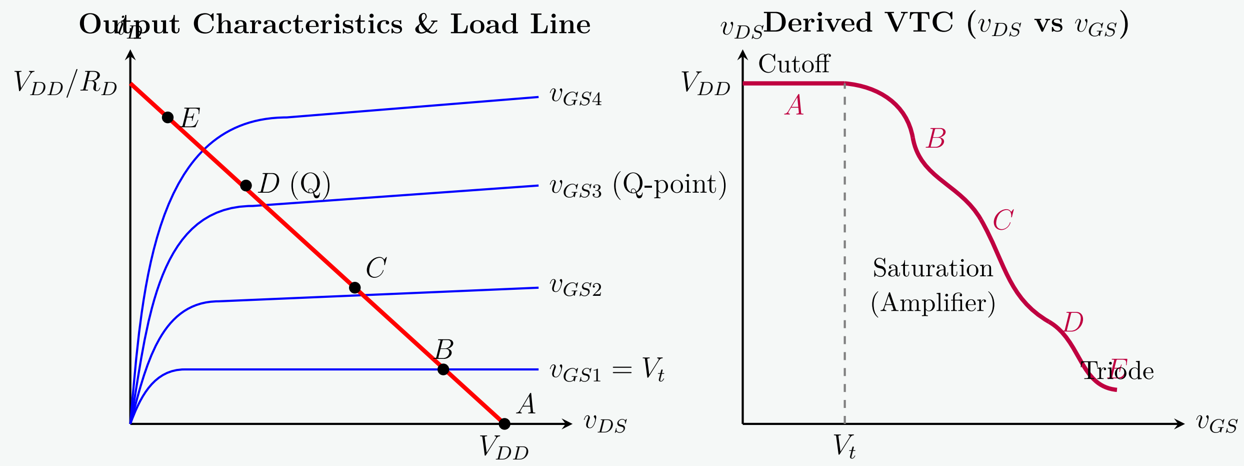
$$v_{DS} = V_{DD} - (I_{DQ} + i_d)R_D = \underbrace{(V_{DD} - I_{DQ}R_D)}_{V_{DSQ}} - \underbrace{i_d R_D}_{v_{ds}} \tag{2}$$

The AC output voltage is $v_{ds} = -i_d R_D = -g_m R_D v_{gs}$. The voltage gain is $A_v = \frac{v_{ds}}{v_{gs}} = -g_m R_D$. The negative sign indicates a 180° phase inversion.

2. Deducing the VTC from the I_D–V_{DS} Graph
The Voltage Transfer Characteristic (VTC), which plots v_{DS} versus v_{GS}, can be deduced graphically using the concept of a **load line** superimposed on the MOSFET's output characteristics (I_D vs. V_{DS}).
The load line is defined by the KVL equation of the output loop:

$$v_{DS} = V_{DD} - i_D R_D \implies i_D = -\frac{1}{R_D}v_{DS} + \frac{V_{DD}}{R_D} \tag{3}$$

This is a straight line intersecting the y-axis at $I_D = V_{DD}/R_D$ and the x-axis at $v_{DS} = V_{DD}$.



Step-by-Step Graphical Deduction:

(a) **Point A (Cutoff):** When $v_{GS} < V_t$, the MOSFET is OFF. The drain current $i_D = 0$. The load line intersects the $i_D = 0$ curve at $v_{DS} = V_{DD}$.

(b) **Points B, C, D (Saturation Region):** As v_{GS} increases beyond V_t, the MOSFET turns ON. The intersection points slide *up* the load line. Because the MOSFET curves are relatively flat here but widely spaced, small increases in v_{GS} cause large horizontal shifts along the load line, resulting in a rapid, linear drop in v_{DS}. This steep region (slope $= -g_m R_D$) is where the MOSFET acts as an amplifier (Point D marks the optimal Q-point).

(c) **Point E (Triode Region):** For very large v_{GS}, the intersection enters the triode (ohmic) region where the transistor curves bunch together. The load line intersects the steep, almost vertical parts of the characteristic curves. Here, v_{DS} approaches zero and further increases in v_{GS} produce little change in v_{DS} (clipping occurs).

Biasing Stability

Q1. What is the reason for biasing a MOSFET? Write a short note on the biasing technique of a discrete MOSFET using a fixed voltage at the gate and a resistance R_s in the source. **(11 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

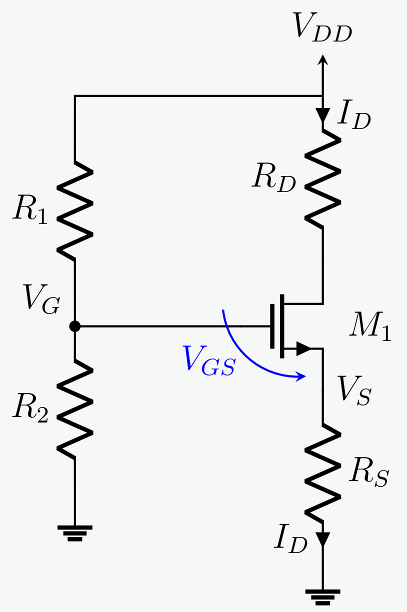
1. Reasons for Biasing a MOSFET

Biasing refers to the establishment of a steady DC operating point (known as the Quiescent point or Q-point), defined by a specific DC drain current (I_{DQ}) and a DC drain-to-source voltage (V_{DSQ}). The primary reasons for biasing a MOSFET are:

- **Establishing the Operating Region:** For linear amplification, the MOSFET must be biased strictly within the **saturation region** ($V_{DS} > V_{GS} - V_t$). Without proper biasing, the AC signal might push the transistor into cutoff (clipping the bottom of the waveform) or the triode region (distorting the top).
- **Maximizing Signal Swing (Headroom):** A well-chosen Q-point is placed optimally between V_{DD} and the edge of the triode region to allow for the maximum symmetrical undistorted AC output voltage swing.
- **Stabilizing the Q-Point:** MOSFET parameters, specifically the threshold voltage (V_t) and the transconductance parameter ($K_n = \frac{1}{2}\mu_n C_{ox} W/L$), vary significantly between devices of the same part number due to manufacturing tolerances and change with temperature. Proper biasing networks desensitize the DC drain current to these variations, preventing thermal runaway or unpredictable gain.

2. Biasing with a Fixed Gate Voltage and Source Resistance (R_S)

One of the most effective techniques for biasing a discrete MOSFET is using a fixed DC voltage at the gate (V_G) along with a resistor (R_S) connected between the source and ground. This is commonly implemented using a voltage divider at the gate.



Circuit Operation and Analysis: Assuming the gate current is essentially zero ($I_G \approx 0$), the gate voltage is fixed by the voltage divider formed by R_1 and R_2:

$$V_G = V_{DD}\left(\frac{R_2}{R_1 + R_2}\right) \tag{1}$$

Applying Kirchhoff's Voltage Law (KVL) to the gate-source loop yields:

$$V_G = V_{GS} + I_D R_S \tag{2}$$

Rearranging for the gate-to-source voltage V_{GS}:

$$V_{GS} = V_G - I_D R_S \tag{3}$$

Principle of Stability (Negative Feedback): The inclusion of R_S provides negative DC feedback, which heavily stabilizes the drain current I_D against device variations. The mechanism works as follows:

(a) Suppose the drain current I_D attempts to increase (e.g., due to a transistor replacement with a lower V_t or a temperature increase).

(b) The voltage drop across the source resistor ($V_S = I_D R_S$) will correspondingly increase.

(c) Because the gate voltage V_G is firmly fixed by the resistor divider, an increase in $I_D R_S$ forces the gate-to-source voltage V_{GS} to decrease (since $V_{GS} = V_G - I_D R_S$).

(d) According to the MOSFET saturation equation $I_D = K_n(V_{GS} - V_t)^2$, a decrease in V_{GS} causes the drain current I_D to decrease, thereby opposing the initial change.

Mathematical Derivation of I_D: To find the exact operating point, we substitute $V_{GS} = V_G - I_D R_S$ into the square-law equation:

$$I_D = K_n(V_G - I_D R_S - V_t)^2$$
$$I_D = K_n\left[(V_G - V_t) - I_D R_S\right]^2$$

Expanding this yields a quadratic equation in terms of I_D:

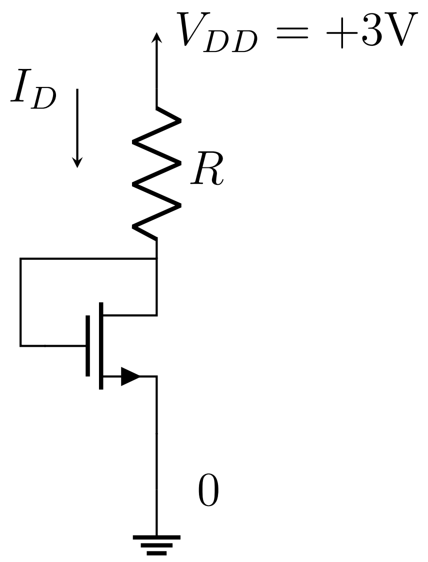
$$K_n R_S^2 I_D^2 - [2K_n R_S(V_G - V_t) + 1]I_D + K_n(V_G - V_t)^2 = 0 \tag{4}$$

Solving this quadratic equation yields the stable DC drain current I_{DQ}.

Advantages:

- **High Stability:** Excellent immunity to variations in V_t and K_n.
- **Single Supply:** Requires only one power supply (V_{DD}), unlike dual-supply biasing schemes.

Q2. Design the circuit shown to obtain a current $I_D = 80\,\mu$A. Given: $V_t = 0.6$ V, $\mu_n C_{ox} = 200\,\mu$A/V^2, $L = 0.8\,\mu$m, $W = 4\,\mu$m, and $V_A = 50$ V.

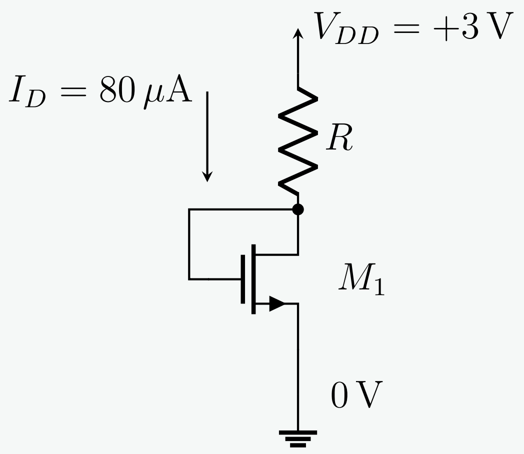


(16 marks) [EEE 263 | L-2/T-1 | 2013–14]

Answer

1. Circuit Diagram and Operating Principle

The circuit diagram is redrawn below. The gate of the NMOS transistor is directly connected to its drain. This configuration is known as a *diode-connected* transistor.



2. Verification of Operating Region

For an NMOS transistor to operate in the saturation region, the drain-to-source voltage must satisfy the condition:

$$V_{DS} \geq V_{GS} - V_t \quad \text{and} \quad V_{GS} > V_t \tag{1}$$

Since the gate is short-circuited to the drain, we have $V_{GS} = V_{DS}$. Substituting this into the saturation condition yields:

$$V_{DS} \geq V_{DS} - V_t \implies 0 \geq -V_t \tag{2}$$

Since the threshold voltage $V_t = 0.6\,\text{V} > 0$ for an enhancement-mode NMOS, this inequality is always strictly satisfied. Therefore, the transistor is **always in saturation** as long as it is turned on ($I_D > 0$).

3. Device Parameter Calculations
Before calculating the required resistor R, let us determine the device transconductance parameter (k_n) and the channel-length modulation parameter (λ):

$$k_n = \mu_n C_{ox}\left(\frac{W}{L}\right) = 200\,\mu\text{A/V}^2 \times \left(\frac{4\,\mu\text{m}}{0.8\,\mu\text{m}}\right) = 1000\,\mu\text{A/V}^2 = 1.0\,\text{mA/V}^2$$
$$\lambda = \frac{1}{V_A} = \frac{1}{50\,\text{V}} = 0.02\,\text{V}^{-1}$$

4. Determination of the Gate-Source Voltage (V_{GS})
The drain current in the saturation region, accounting for channel-length modulation, is given by:

$$I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2(1 + \lambda V_{DS}) \tag{3}$$

Because $V_{DS} = V_{GS}$, the equation becomes:

$$I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2(1 + \lambda V_{GS}) \tag{4}$$

Since this is a cubic equation in terms of V_{GS}, we solve it using an iterative approximation method.
Step 4a: First-Order Approximation (Ignoring λ) Assuming $\lambda = 0$ for the initial estimate:

$$\begin{aligned}
80\,\mu\text{A} &= \frac{1}{2}\left(1000\,\mu\text{A/V}^2\right)(V_{GS} - 0.6)^2 \\
80 &= 500(V_{GS} - 0.6)^2 \\
(V_{GS} - 0.6)^2 &= \frac{80}{500} = 0.16 \\
V_{GS} - 0.6 &= \pm 0.4
\end{aligned}$$

Since V_{GS} must be greater than V_t for the transistor to be ON, we take the positive root:

$$V_{GS}^{(0)} = 0.6 + 0.4 = 1.0\,\text{V} \tag{5}$$

Step 4b: Second-Order Approximation (Including λ) We use the initial estimate $V_{DS} \approx V_{GS}^{(0)} = 1.0\,\text{V}$ in the channel-length modulation correction factor $(1 + \lambda V_{DS})$ to refine our calculation for V_{GS}:

$$\begin{aligned}
I_D &= \frac{1}{2}k_n(V_{GS} - V_t)^2\left(1 + \lambda V_{GS}^{(0)}\right) \\
80 &= 500(V_{GS} - 0.6)^2\,(1 + 0.02 \times 1.0) \\
80 &= 500(V_{GS} - 0.6)^2(1.02) \\
80 &= 510(V_{GS} - 0.6)^2 \\
(V_{GS} - 0.6)^2 &= \frac{80}{510} \approx 0.15686 \\
V_{GS} - 0.6 &= \sqrt{0.15686} \approx 0.396\,\text{V} \\
V_{GS} &\approx 0.996\,\text{V}
\end{aligned}$$

Further iteration will yield negligible changes. Thus, the steady-state drain-to-source voltage is $V_{DS} = V_{GS} = 0.996\,\text{V}$.

5. Calculation of Required Resistor R
Applying Kirchhoff's Voltage Law (KVL) to the drain loop, the voltage drop across the resistor R is:

$$V_R = V_{DD} - V_{DS} \tag{6}$$

Using Ohm's Law, the required resistance is:

$$\begin{aligned}
R &= \frac{V_{DD} - V_{DS}}{I_D} \\
R &= \frac{3\,\text{V} - 0.996\,\text{V}}{80\,\mu\text{A}} \\
R &= \frac{2.004\,\text{V}}{80 \times 10^{-6}\,\text{A}} \\
R &= 25,050\,\Omega = 25.05\,\text{k}\Omega
\end{aligned}$$

Final Answer: The designed value for the resistor to obtain a drain current of $80\,\mu\text{A}$ is $25.05\,\mathbf{k\Omega}$. (Note: If channel-length modulation were neglected, the required resistance would be exactly $25\,\text{k}\Omega$).

Q3. Explain graphically the benefit of using a resistance at the MOSFET source terminal. How does it counteract the problem of unpredictable variation in process parameters such as V_t, C_{ox}, μ_n? **(20 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

1. Theoretical Background: The Problem of Process Variations

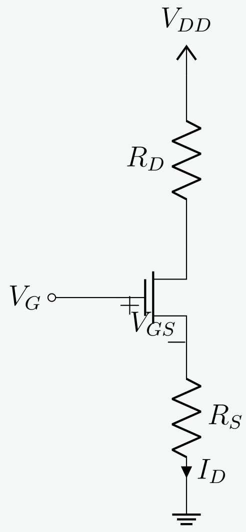
The drain current I_D of an NMOS transistor operating in the saturation region is governed by the equation:

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_t)^2 \tag{1}$$

During semiconductor manufacturing, process parameters such as the threshold voltage (V_t), the gate oxide capacitance (C_{ox}), and the electron mobility (μ_n) suffer from unpredictable variations from wafer to wafer or batch to batch. If a simple constant gate-to-source voltage (V_{GS}) is used to bias the MOSFET, these variations will cause dramatic and unacceptable shifts in the quiescent drain current (I_D), leading to unstable operating points (Q-points).

2. Principle of Source Degeneration (Using R_S)

To counteract this, a source resistor R_S is introduced at the source terminal. This technique is known as **source degeneration**.



Applying Kirchhoff's Voltage Law (KVL) at the input loop yields:

$$V_G = V_{GS} + I_D R_S$$
$$\implies V_{GS} = V_G - I_D R_S$$

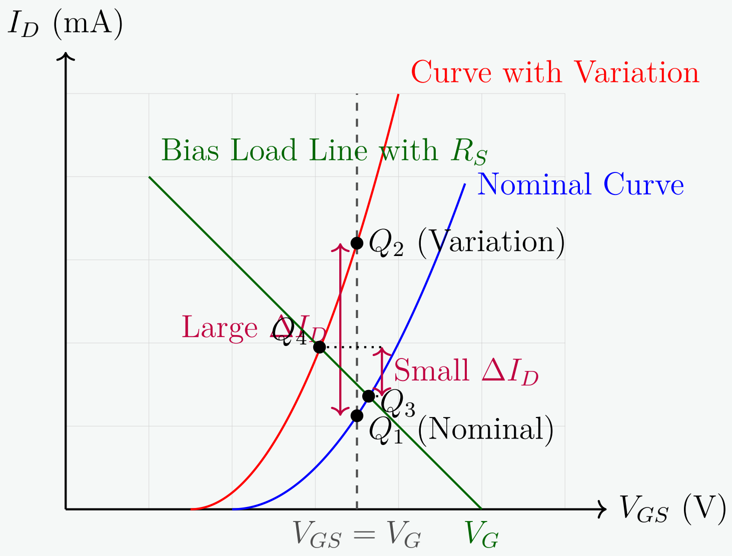
Equation (??) represents a **negative DC feedback** mechanism.

Mechanism of action:

If an increase in $\mu_n C_{ox}$ or a decrease in V_t causes I_D to increase, the voltage drop across the source resistor ($I_D R_S$) increases. Since the gate voltage V_G is fixed, an increase in $I_D R_S$ forces V_{GS} to decrease. A smaller V_{GS} counteracts the initial rise in I_D, maintaining a highly stable Q-point.

3. Graphical Explanation

The stabilization effect can be best visualized by plotting the MOSFET's transfer characteristics (I_D vs. V_{GS}) against the bias load line defined by Equation (??).

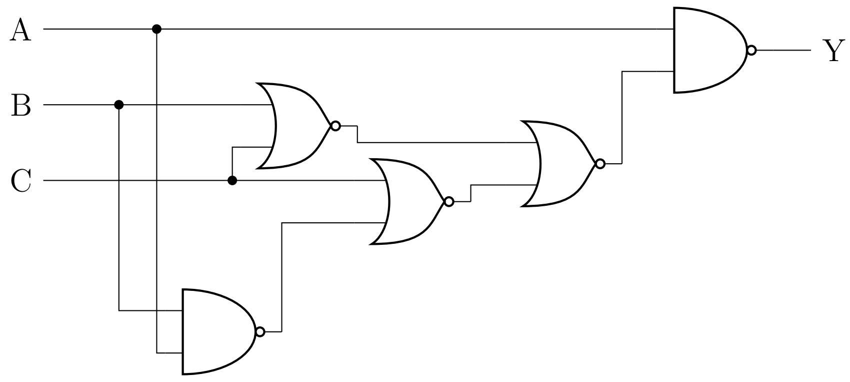


Graphical Analysis and Conclusion:

- **Transfer Curves:** The two parabolic curves represent the MOSFET I_D vs. V_{GS} relationship. Process variations shift the curve from the **Nominal** state to the **Variation** state.
- **Without R_S (Constant V_{GS}):** If no source resistance is used, the bias line is a vertical line at $V_{GS} = V_G$. The Q-point shifts drastically from Q_1 to Q_2. This results in a **large** ΔI_D, which can push the transistor out of saturation or cause excessive power dissipation.
- **With R_S (Source Degeneration):** The addition of R_S introduces a load line with a slope of $-\frac{1}{R_S}$ originating from $V_{GS} = V_G$. The operating points are now determined by the intersections Q_3 and Q_4.
- **Result:** Because the load line is slanted, the shift from Q_3 to Q_4 results in a significantly **smaller** ΔI_D compared to the constant V_{GS} method. The steeper the slope (larger R_S), the smaller the variation in I_D, providing excellent immunity against variations in V_t, C_{ox}, and μ_n.

CMOS Logic Implementation

Q1. The diagram shows a Boolean function Y being implemented as a combinational circuit. Design a CMOS transistor-level implementation of the Boolean function using the minimum number of transistors.



(20 marks) [EEE 263 | L-2/T-1 | 2020–21]

Answer

1. Logic Expression Extraction

To design the minimum transistor-level circuit, we must first extract and simplify the Boolean function Y from the given combinational circuit. Let us label the intermediate nodes of the circuit:

- Let N_1 be the output of the top NOR gate.
- Let N_2 be the output of the bottom-left NAND gate.
- Let N_3 be the output of the middle NOR gate.
- Let N_4 be the output of the rightmost NOR gate.

By tracing the connections from the inputs A, B, and C, we can write the Boolean expression for each node:

$$\begin{aligned} N_1 &= \overline{B+C} \\ N_2 &= \overline{A \cdot B} \\ N_3 &= \overline{C + N_2} = \overline{C + \overline{AB}} \\ N_4 &= \overline{N_1 + N_3} = \overline{\left(\overline{B+C}\right) + \left(\overline{C + \overline{AB}}\right)} \end{aligned}$$

The final output Y is produced by a NAND gate with inputs A and N_4:

$$Y = \overline{A \cdot N_4} \tag{1}$$

2. Boolean Simplification

To minimize the number of transistors, we must simplify N_4 and Y to their most reduced forms using Boolean algebra.

Step 2.1: Simplify N_4

Apply De Morgan's Law to the expression for N_4:

$$\begin{aligned} N_4 &= \overline{\left(\overline{B+C}\right) + \left(\overline{C + \overline{AB}}\right)} \\ &= \overline{\overline{B+C}} \cdot \overline{\overline{C + \overline{AB}}} \quad \text{(De Morgan's Law)} \\ &= (B+C) \cdot (C + \overline{AB}) \quad \text{(Double Negation)} \end{aligned}$$

Now, apply the distributive law $(X+Y)(X+Z) = X + YZ$, where $X = C$:

$$\begin{aligned} N_4 &= C + B \cdot \overline{AB} \\ &= C + B \cdot (\overline{A} + \overline{B}) \quad \text{(De Morgan's Law)} \\ &= C + \overline{A}B + B\overline{B} \end{aligned}$$

Since $B\overline{B} = 0$, this reduces to:

$$N_4 = C + \overline{A}B \tag{2}$$

Step 2.2: Simplify Y

Substitute the simplified expression for N_4 into the final output equation:

$$\begin{aligned} Y &= \overline{A \cdot (C + \overline{A}B)} \\ &= \overline{AC + A\overline{A}B} \quad \text{(Distributive Law)} \end{aligned}$$

Since $A\overline{A} = 0$, the term $A\overline{A}B$ vanishes:

$$Y = \overline{AC + 0} \implies Y = \overline{A \cdot C} \tag{3}$$

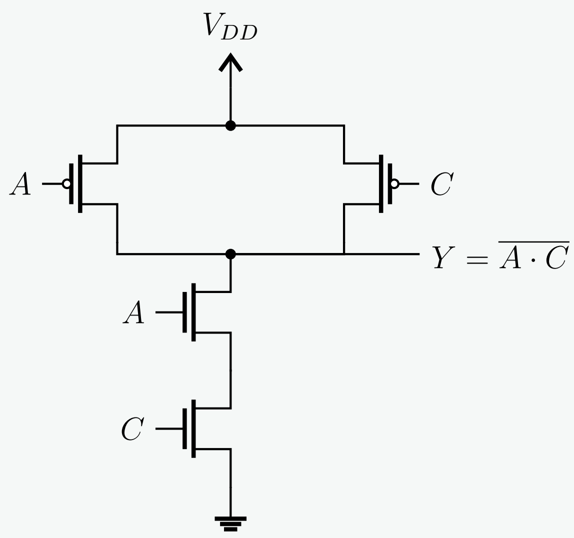
The entire combinational logic circuit logically reduces to a simple **2-input NAND gate** with inputs A and C.

3. Transistor-Level CMOS Implementation

A standard static CMOS logic gate consists of a Pull-Up Network (PUN) using PMOS transistors and a Pull-Down Network (PDN) using NMOS transistors.

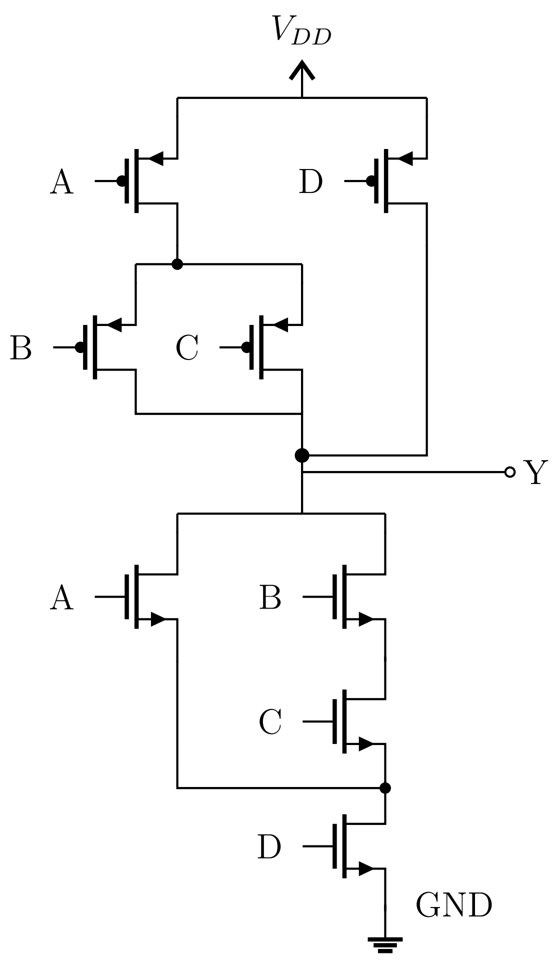
- **Pull-Down Network (PDN):** For a NAND gate ($Y = \overline{AC}$), the condition for the output to be pulled low is when both A **AND** C are High. Therefore, the NMOS transistors must be placed in **series**.
- **Pull-Up Network (PUN):** The PUN is the logical dual of the PDN. Since the NMOS transistors are in series, the PMOS transistors must be placed in **parallel**.

This configuration requires exactly **4 transistors** (2 PMOS and 2 NMOS), which is the absolute minimum required to implement this Boolean function.



CMOS Sizing and Delay

Q1. For the logic circuit shown, choose transistor widths to achieve equal rise and fall resistance as a unit inverter. Assume $\mu_n = 3\mu_p$. What is the minimum and maximum propagation delay in the circuit?



(26 marks) [EEE 263 | L-2/T-1 | 2020–21]

Answer

1. Logic Function Analysis

First, we determine the Boolean function implemented by the Pull-Down Network (PDN). The PDN consists of NMOS transistors that pull the output Y to ground.

- Transistor A is in parallel with the series combination of B and C. This subnetwork implements the logic $(A + BC)$.
- This entire parallel combination is in series with transistor D.

Thus, the pull-down condition is $(A + BC) \cdot D$. The output Y is the complement of this condition:

$$Y = \overline{(A + BC)D} \tag{1}$$

2. Transistor Sizing for Equal Rise and Fall Times
We want the equivalent resistance of the worst-case pull-up and pull-down paths to match that of a unit inverter.

- A standard unit inverter has an NMOS size of $(W/L)_n = 1$.
- Because $\mu_n = 3\mu_p$, the unit PMOS size must be $(W/L)_p = 3$ to achieve equal resistance ($R_p = R_n = R$).

Let the width of an NMOS transistor be W_N and a PMOS transistor be W_P. The resistance of a transistor is inversely proportional to its width. We size the transistors based on the worst-case (longest series) paths.
Pull-Down Network (NMOS): Target $R_{eq} = R$ The worst-case pull-down path occurs when discharging through B, C, and D in series.

$$R_B + R_C + R_D = R$$
$$\frac{1}{W_{NB}} + \frac{1}{W_{NC}} + \frac{1}{W_{ND}} = \frac{1}{1}$$

Assuming equal sizes for series transistors in this path: $W_{NB} = W_{NC} = W_{ND} = 3$.
Now consider the path through A and D. We already established $W_{ND} = 3$.

$$R_A + R_D = R$$
$$\frac{1}{W_{NA}} + \frac{1}{3} = 1 \implies \frac{1}{W_{NA}} = \frac{2}{3} \implies W_{NA} = 1.5$$

NMOS Sizes: $W_{NA} = 1.5$, $W_{NB} = 3$, $W_{NC} = 3$, $W_{ND} = 3$.

Pull-Up Network (PMOS): Target $R_{eq} = R$ The base PMOS size is 3 for equivalent unit resistance. The worst-case pull-up path goes through A and B (or A and C) in series. Note that D is in parallel with the entire A, B, C network.

$$R_{PA} + R_{PB} = R \quad \text{(Target equivalent to a single PMOS of width 3)}$$
$$\frac{1}{W_{PA}/3} + \frac{1}{W_{PB}/3} = 1 \implies \frac{3}{W_{PA}} + \frac{3}{W_{PB}} = 1$$

Assuming $W_{PA} = W_{PB} = W_{PC}$, we get $\frac{6}{W_P} = 1 \implies W_P = 6$. So, $W_{PA} = W_{PB} = W_{PC} = 6$. Transistor D acts alone in its path to pull up the output.

$$R_{PD} = R \implies \frac{3}{W_{PD}} = 1 \implies W_{PD} = 3$$

PMOS Sizes: $W_{PA} = 6$, $W_{PB} = 6$, $W_{PC} = 6$, $W_{PD} = 3$.

3. Propagation Delay Analysis
Propagation delay (t_p) is proportional to the RC time constant of the charging/discharging path. Let C_L be the load capacitance and R be the effective resistance of a unit NMOS.

$$t_p = \ln(2) \cdot R_{eff} \cdot C_L \approx 0.69 R_{eff} C_L \tag{2}$$

Maximum Delay (Worst-Case): By design, we sized the transistors such that the worst-case paths have an effective resistance equal to R.

- $t_{pHL(max)}$ occurs through the B-C-D NMOS path. $R_{eff} = R_B + R_C + R_D = R$.
- $t_{pLH(max)}$ occurs through the A-B or A-C PMOS path. $R_{eff} = R_A + R_B = R$.

$$t_{p(max)} = 0.69 R C_L \tag{3}$$

Minimum Delay (Best-Case): The minimum delay occurs when the maximum number of parallel paths are turned on simultaneously, providing the lowest resistance.

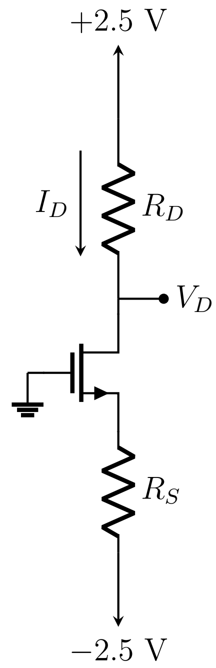
- **Best-case Fall Time ($t_{pHL(min)}$):** All NMOS transistors are ON ($A = 1, B = 1, C = 1, D = 1$).
  The resistance of B and C in series is $R_{B+C} = \frac{1}{3}R + \frac{1}{3}R = \frac{2}{3}R$.
  This is in parallel with $R_A = \frac{1}{1.5}R = \frac{2}{3}R$.
  $R_{parallel} = (\frac{2}{3}R) \parallel (\frac{2}{3}R) = \frac{1}{3}R$.
  This parallel combination is in series with D: $R_{eff} = R_{parallel} + R_D = \frac{1}{3}R + \frac{1}{3}R = \frac{2}{3}R$.
- **Best-case Rise Time ($t_{pLH(min)}$):** All PMOS paths are ON ($A = 0, B = 0, C = 0, D = 0$).
  $R_B \parallel R_C = (\frac{3}{6}R) \parallel (\frac{3}{6}R) = 0.5R \parallel 0.5R = 0.25R$.
  This is in series with $R_A = \frac{3}{6}R = 0.5R$, giving a branch resistance of $0.75R$.
  This entire branch is in parallel with $R_D = \frac{3}{3}R = 1R$.
  $R_{eff} = 0.75R \parallel 1R = \frac{0.75\times1}{0.75+1}R = \frac{3}{7}R \approx 0.428R$.

The absolute minimum propagation delay is determined by the lowest resistance, which is the best-case rise time.

$$t_{p(min)} = 0.69 \left(\frac{3}{7}R\right) C_L \approx 0.296 R C_L \tag{4}$$

NMOS Circuit Design

Q1. Design the circuit shown so that the transistor operates at $I_D = 0.3\,\text{mA}$ and $V_D = 0.4\,\text{V}$. The NMOS transistor has a threshold voltage of $1\,\text{V}$, aspect ratio of 40, and a process transconductance parameter of $60\,\mu\text{A/V}^2$. Neglect channel-length modulation.



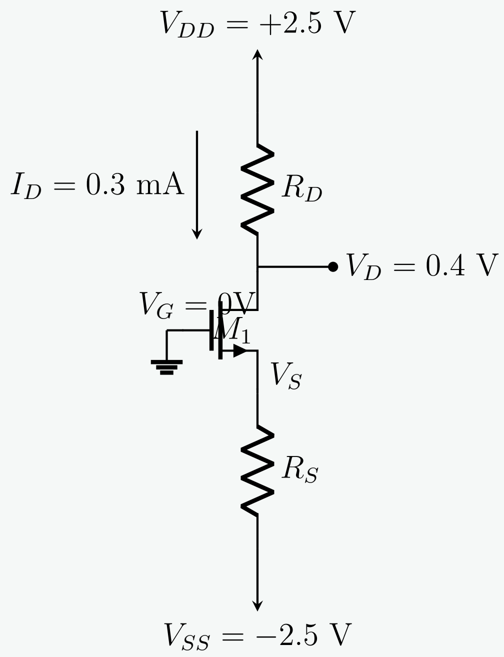
(23 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Answer

1. Extracted Circuit Parameters

From the problem description and the given schematic, we have the following parameters:

$$I_D = 0.3\,\text{mA} \qquad V_D = 0.4\,\text{V}$$
$$V_t = 1.0\,\text{V} \qquad k_n' = \mu_n C_{ox} = 60\,\mu\text{A/V}^2 = 0.06\,\text{mA/V}^2$$
$$\frac{W}{L} = 40 \qquad \lambda = 0\,\text{V}^{-1}\ \text{(Channel-length modulation neglected)}$$
$$V_{DD} = +2.5\,\text{V} \qquad V_{SS} = -2.5\,\text{V}$$



2. Design of the Drain Resistor (R_D)

The voltage at the drain terminal V_D is determined by the positive supply voltage V_{DD} and the voltage drop across the drain resistor R_D due to the drain current I_D. Applying Ohm's law and KVL at the drain node:

$$V_D = V_{DD} - I_D R_D$$
$$R_D = \frac{V_{DD} - V_D}{I_D}$$

Substituting the given values:

$$R_D = \frac{2.5\,\text{V} - 0.4\,\text{V}}{0.3\,\text{mA}} = \frac{2.1\,\text{V}}{0.3\,\text{mA}} = 7\,\text{k}\Omega$$

3. Determination of Gate-to-Source Voltage (V_{GS})

To find the source voltage V_S and subsequently R_S, we must first determine V_{GS}. We assume the NMOS transistor is operating in the **saturation region** (active mode), which is standard for amplifier biasing. The drain current equation in saturation, neglecting channel-length modulation, is:

$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) (V_{GS} - V_t)^2 \tag{1}$$

Substituting the known values to solve for the overdrive voltage $V_{OV} = (V_{GS} - V_t)$:

$$\begin{aligned}
0.3\,\text{mA} &= \frac{1}{2}(0.06\,\text{mA/V}^2)(40)(V_{GS}-1)^2 \\
0.3 &= 1.2 \cdot (V_{GS}-1)^2 \\
(V_{GS}-1)^2 &= \frac{0.3}{1.2} = 0.25 \\
V_{GS} - 1 &= \pm\sqrt{0.25} = \pm 0.5\,\text{V}
\end{aligned}$$

For the NMOS transistor to be turned ON, we must have $V_{GS} > V_t$. Therefore, we discard the negative root:

$$V_{GS} = 1 + 0.5 = 1.5\,\text{V}$$

4. Design of the Source Resistor (R_S)
From the circuit diagram, the gate terminal is directly connected to ground, meaning $V_G = 0\,\text{V}$. We can find the source voltage V_S:

$$\begin{aligned}
V_{GS} &= V_G - V_S \\
1.5\,\text{V} &= 0\,\text{V} - V_S \implies V_S = -1.5\,\text{V}
\end{aligned}$$

Now, applying Ohm's law across the source resistor R_S:

$$\begin{aligned}
R_S &= \frac{V_S - V_{SS}}{I_D} \\
R_S &= \frac{-1.5\,\text{V} - (-2.5\,\text{V})}{0.3\,\text{mA}} = \frac{1.0\,\text{V}}{0.3\,\text{mA}} = 3.33\,\text{k}\Omega
\end{aligned}$$

5. Verification of the Operating Region
To ensure our assumption of saturation was correct, the condition $V_{DS} > V_{GS} - V_t$ (or equivalently $V_D > V_G - V_t$) must be satisfied.

$$\begin{aligned}
V_{DS} &= V_D - V_S = 0.4\,\text{V} - (-1.5\,\text{V}) = 1.9\,\text{V} \\
V_{DS(\text{sat})} &= V_{OV} = V_{GS} - V_t = 1.5\,\text{V} - 1.0\,\text{V} = 0.5\,\text{V}
\end{aligned}$$

Since $V_{DS}(1.9\,\text{V}) > V_{DS(\text{sat})}(0.5\,\text{V})$, the transistor is indeed operating in the saturation region. The initial assumption is verified.

Final Design Values:
$\mathbf{R_D = 7\,k\Omega}$ **and** $\mathbf{R_S = 3.33\,k\Omega}$

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

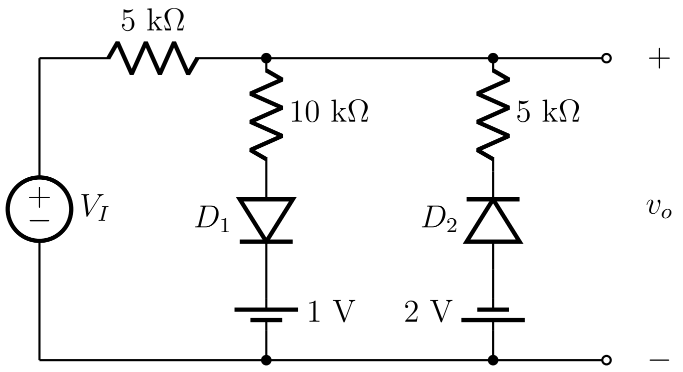
Part II — Wave Shaping Circuits

Clipping, clamping, comparator, and switching circuits using diodes and BJTs

S4 Clipping, Clamping & Switching Circuits

Clipping Circuits — Waveform & Transfer Characteristics

Q1. For the circuit given, plot v_0 vs time. The input v_1 is a sine wave with 5 V amplitude and 1 kHz frequency. Assume 0.7 V voltage drop across diodes in forward bias.



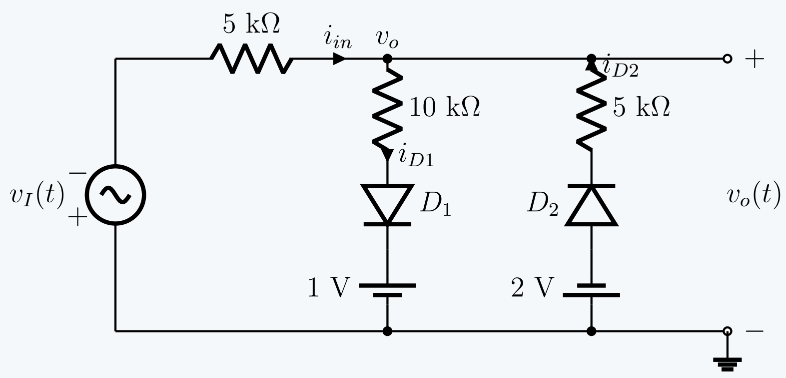
(14 marks) [EEE 263 | L-2/T-1 | 2022–23]

Answer

1. Circuit Theory and Assumptions

The given circuit is a parallel diode clipper (or limiter) that shapes the input waveform by limiting its voltage excursions in both the positive and negative directions.

- **Input Signal:** $v_I(t) = 5\sin(2\pi \cdot 1000t)$ V.
- **Diode Model:** We assume the Constant Voltage Drop (CVD) model, where a forward-biased diode acts as a $V_\gamma = 0.7$ V voltage source, and a reverse-biased diode acts as an open circuit.



2. Determination of Operating Regions

To plot the output waveform $v_o(t)$, we must determine the critical transition voltages where the diodes change states. Let the potential at the output node be v_o.

Condition for D_1 to turn ON:

Diode D_1 points downwards. For it to conduct, the potential at v_o must overcome the 1 V DC source and the 0.7 V diode drop.

$$v_o > 1\text{ V} + 0.7\text{ V} \implies v_o > 1.7\text{ V}$$

Condition for D_2 to turn ON:

Diode D_2 points upwards. For it to conduct, the 2 V DC source must be capable of pushing current up through D_2 to node v_o. The voltage drop from the source to v_o must overcome the 0.7 V diode drop.

$$2\text{ V} - v_o > 0.7\text{ V} \implies v_o < 1.3\text{ V}$$

This defines three distinct operating regions based on the input voltage v_I. Note that v_o can never be simultaneously > 1.7 V and < 1.3 V, hence D_1 and D_2 are never ON at the same time.

—

3. Mathematical Derivation of Transfer Characteristics

Region I: Both D_1 and D_2 OFF ($1.3\text{ V} \le v_I \le 1.7\text{ V}$)
When both diodes are OFF, they act as open circuits. No current flows through the 5 kΩ series resistor.

$$v_o = v_I \tag{1}$$

Region II: D_1 ON, D_2 OFF ($v_I > 1.7$ V)
D_1 acts as a 0.7 V source. The D_1 branch can be modeled as a 1.7 V equivalent source in series with the 10 kΩ resistor. Applying Kirchhoff's Current Law (KCL) at node v_o:

$$\frac{v_o - v_I}{5} + \frac{v_o - 1.7}{10} = 0$$
$$2(v_o - v_I) + (v_o - 1.7) = 0$$
$$3v_o = 2v_I + 1.7$$
$$v_o = \frac{2}{3}v_I + 0.567 \text{ V}$$

Peak Positive Voltage: When $v_I = 5$ V, $v_o = \frac{2(5)+1.7}{3} = 3.9$ V.

Region III: D_1 OFF, D_2 ON ($v_I < 1.3$ V)
D_2 acts as a 0.7 V source (opposing the 2 V source). The D_2 branch can be modeled as a $(2 - 0.7) = 1.3$ V equivalent source in series with the 5 kΩ resistor. Applying KCL at node v_o:

$$\frac{v_o - v_I}{5} + \frac{v_o - 1.3}{5} = 0$$
$$(v_o - v_I) + (v_o - 1.3) = 0$$
$$2v_o = v_I + 1.3$$
$$v_o = 0.5v_I + 0.65 \text{ V}$$

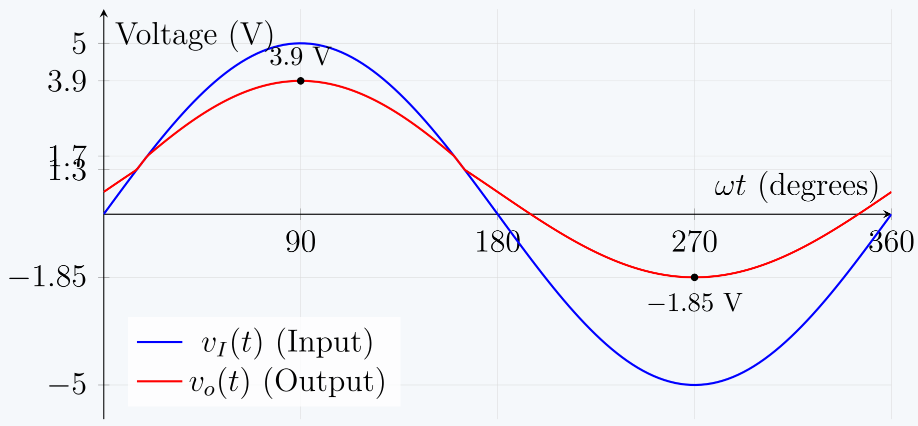
Peak Negative Voltage: When $v_I = -5$ V, $v_o = 0.5(-5) + 0.65 = -1.85$ V.

4. Summary of Operating States

Input Voltage (v_I)	State of D_1	State of D_2	Output Voltage (v_o)
$v_I > 1.7$ V	ON	OFF	$v_o = \frac{2}{3}v_I + 0.567$
$1.3 \text{ V} \leq v_I \leq 1.7$ V	OFF	OFF	$v_o = v_I$
$v_I < 1.3$ V	OFF	ON	$v_o = 0.5v_I + 0.65$

5. Output Waveform Plot

The plot below demonstrates the clipping action. The input sine wave is clipped with different slopes (gains) rather than a flat cut-off due to the presence of series resistors in the diode branches.



Q2. Design a limiter circuit using only diodes and 10 kΩ resistors to provide an output signal limited to the range ±2.1 V. Assume each diode has a 0.7 V drop when conducting. **(15 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Design Principle and Theory

A limiter (or clipper) circuit is used to restrict the voltage excursions of an input signal from exceeding predetermined levels. The design requires an output voltage symmetrically limited to ±2.1 V, using only resistors of 10 kΩ and diodes with a constant voltage drop of $V_\gamma = 0.7$ V.
To achieve a limiting voltage of 2.1 V without external DC biasing sources, we can connect multiple diodes in series. Since each diode provides a 0.7 V drop when forward-biased, the number of diodes required in series is:

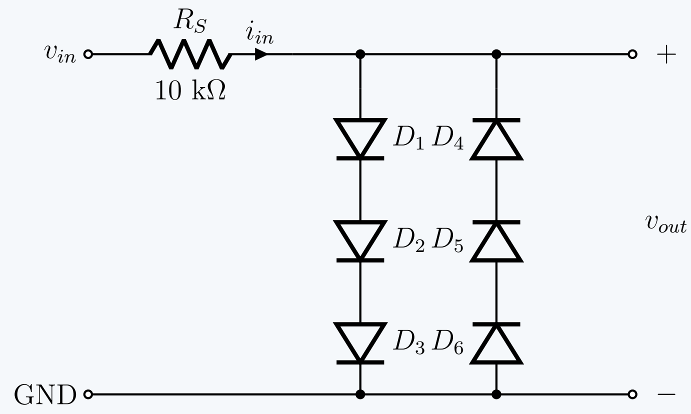
$$n = \frac{V_{\text{limit}}}{V_\gamma} = \frac{2.1 \text{ V}}{0.7 \text{ V}} = 3$$

Thus, a string of three diodes connected in series will limit the voltage to 2.1 V.

- **Positive Limiting (+2.1 V):** A string of 3 diodes with anodes facing the output node and cathodes facing ground.
- **Negative Limiting (−2.1 V):** A parallel string of 3 diodes in the reverse orientation (cathodes facing the output node, anodes facing ground).

- **Current Limiting:** A series resistor $R_S = 10$ kΩ is required to drop the excess voltage and limit the current through the diodes when they are conducting.

2. Proposed Circuit Diagram



3. Circuit Operation and Mathematical Analysis

To determine the transfer characteristic, we analyze the operating states of the diodes for different ranges of the input voltage v_{in}. We assume the output is open-circuited (no load resistor).

Region I: Non-limiting Region (-2.1 V $< v_{in} < 2.1$ V)

- **Assumption:** All diodes (D_1 to D_6) are OFF.
- **Verification:** For $D_1 - D_3$ to turn ON, v_{out} must reach $+2.1$ V. For $D_4 - D_6$ to turn ON, v_{out} must drop to -2.1 V. Since the input voltage magnitude is less than 2.1 V, neither string has enough forward bias to overcome the cumulative cut-in voltage.
- **Output Equation:** Because no current flows through R_S ($i_{in} = 0$), there is no voltage drop across R_S.

$$v_{out} = v_{in} \tag{1}$$

Region II: Positive Limiting ($v_{in} \geq 2.1$ V)

- **Assumption:** Diodes D_1, D_2, and D_3 are ON. Diodes D_4, D_5, and D_6 remain OFF.
- **Verification:** The input tries to drive the output node above 2.1 V. The forward-biased string $D_1 - D_3$ begins to conduct and clamps the node voltage.
- **Output Equation:** The series combination of D_1, D_2, D_3 acts as a constant voltage sink of 3×0.7 V $= 2.1$ V. The excess voltage is dropped across the 10 kΩ resistor.

$$v_{out} = 2.1 \text{ V} \tag{2}$$

  The current supplied by the source is $i_{in} = \frac{v_{in} - 2.1}{10 \text{ k}\Omega}$.

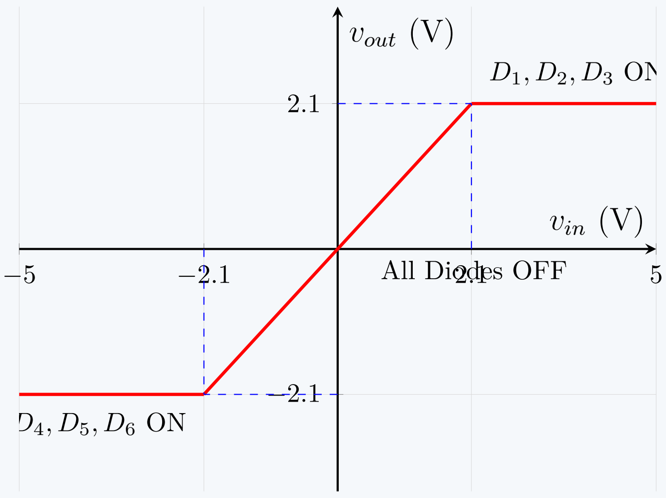
Region III: Negative Limiting ($v_{in} \leq -2.1$ V)

- **Assumption:** Diodes D_4, D_5, and D_6 are ON. Diodes D_1, D_2, and D_3 remain OFF.
- **Verification:** The input tries to drive the output node below -2.1 V. The string $D_4 - D_6$ becomes forward-biased (current flows from ground to v_{out}) and clamps the node voltage.
- **Output Equation:** The series combination of D_4, D_5, D_6 acts as a constant voltage clamp at -2.1 V.

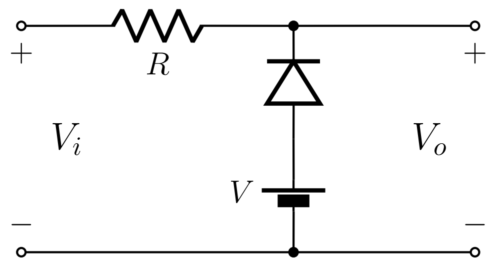
$$v_{out} = -2.1 \text{ V} \tag{3}$$

4. Transfer Characteristics (Voltage Transfer Curve)

The relationship between v_{out} and v_{in} (VTC) is plotted below, clearly showing the linear region and the symmetric hard limits at ± 2.1 V.



Q3. For a sinusoidal input, draw the output waveshape for the circuit shown.



(10 marks) [EEE 263 | L-2/T-1 | 2018–19]

Answer

1. Circuit Operation and Principle

The given circuit is a **biased shunt clipper** (or limiter). It modifies an incoming alternating voltage waveform by cutting off a portion above or below a specific reference level. In this specific configuration:

- The diode is placed in a shunt branch with its anode connected towards the output node and its cathode connected to a DC reference voltage source V.
- Note the orientation of the DC battery: the long bar (positive terminal) is at the top, and the thick short bar (negative terminal) is at the bottom. Therefore, the cathode of the diode is held at a constant potential of $+V$.
- We assume an **ideal diode** model where the forward voltage drop $V_\gamma = 0$ V and reverse leakage current is zero.

—

2. Analysis of Operating Regions

To find the output voltage V_o as a function of the input voltage V_i, we analyze the state of the diode under different conditions:

Case 1: Diode is OFF (Non-Conducting Region)

- **Assumption:** Assume the diode is reverse-biased (OFF). An ideal open-circuit replaces the diode.
- **Condition for Assumption:** For the diode to remain OFF, the voltage at its anode must be less than the voltage at its cathode:
$$V_{\text{anode}} < V_{\text{cathode}} \implies V_i < V$$
- **Output Voltage Calculation:** When the diode is an open circuit, no current flows through the resistor R. Therefore, there is no voltage drop across R, yielding:
$$V_o = V_i$$
- **Verification:** Since $V_o = V_i$, the anode voltage is V_i. The condition $V_i < V$ perfectly matches our initial assumption, verifying that the diode remains OFF when $V_i < V$.

Case 2: Diode is ON (Conducting Region)

- **Assumption:** Assume the diode is forward-biased (ON). An ideal short-circuit replaces the diode.
- **Condition for Assumption:** The diode turns ON as soon as the input voltage attempts to exceed the cathode potential:
$$V_i \geq V$$
- **Output Voltage Calculation:** When the diode behaves as a short-circuit, the output node is directly clamped to the cathode potential established by the reference battery:
$$V_o = V$$

- **Circuit Current:** The current flowing through the circuit during this phase is given by Ohm's Law across resistor R:

$$I_R = \frac{V_i - V}{R}$$

Since $V_i \geq V$, the current is positive and flows down through the diode, confirming forward bias.

—

3. Transfer Characteristic Equation

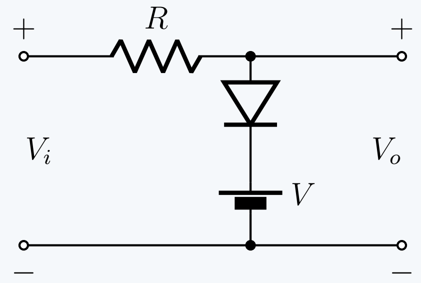
Summarizing the two operating regions, the total transfer characteristic function is expressed as:

$$V_o = \begin{cases} V_i & \text{for } V_i < V \\ V & \text{for } V_i \geq V \end{cases} \tag{1}$$

—

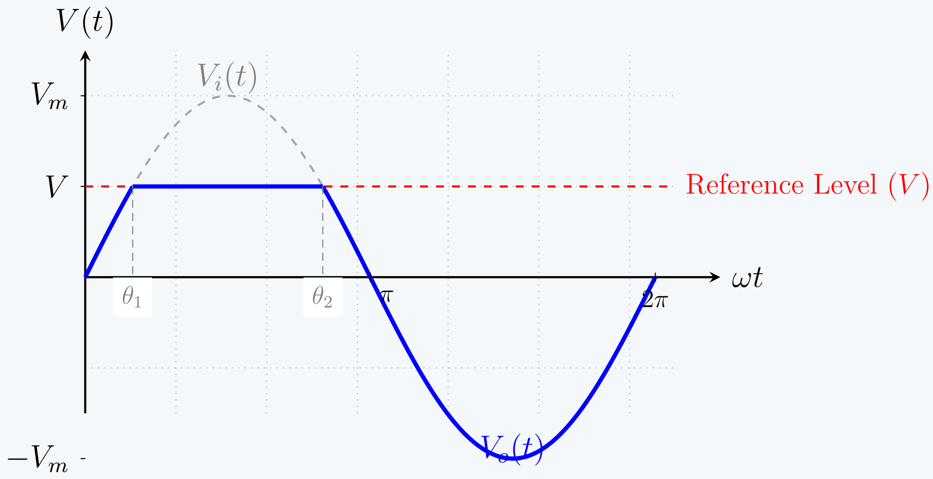
4. Schematic and Waveshape Reconstruction

Circuit Schematic



Input and Output Waveshapes

Below is the precise alignment of the output wave compared to a sinusoidal input of peak amplitude V_m (where $V_m > V$):



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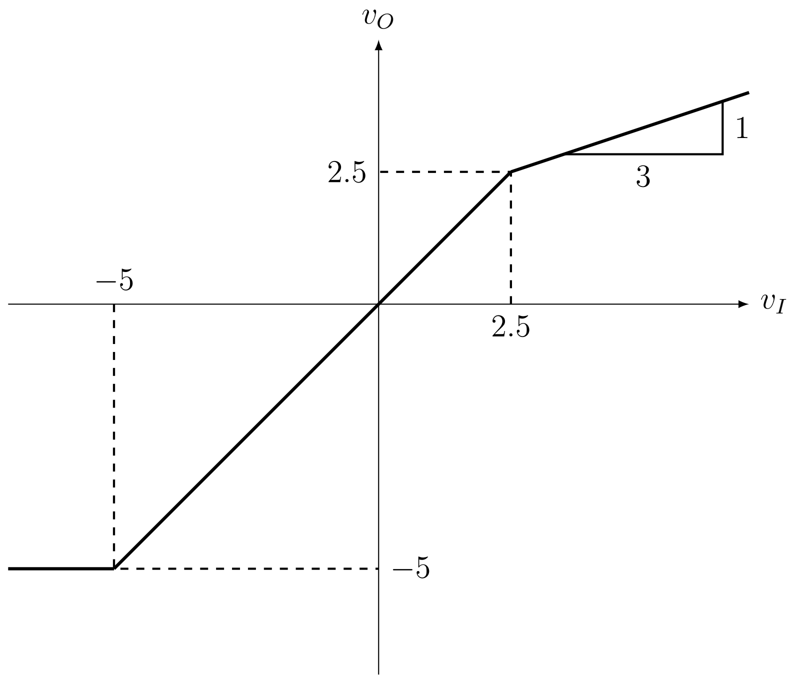
5. Important Notes and Key Takeaways

- **Clipping Action:** This circuit clips off everything **above** the reference level V. It is classified as a Positive Clipper with a positive bias voltage.
- **Angles of Conduction Cutoff:** The clipping region starts at angle $\theta_1 = \arcsin(V/V_m)$ and ends at $\theta_2 = \pi - \theta_1$.
- **Applications:** Used extensively in wave-shaping, noise limiting in FM receivers, and protecting sensitive ADC stages against over-voltage spikes.

Q4. Design a parallel-based clipper that will yield the voltage transfer function shown in the figure. Assume 0.7 V as diode cut-in voltage.



(18 marks) **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

1. Analysis of the Voltage Transfer Characteristic (VTC)

The given VTC can be divided into three distinct regions of operation:

- **Region 1 (Negative Hard Clipping):** For $v_I \leq -5\,\text{V}$, the output is clipped at $v_O = -5\,\text{V}$. The slope is 0. This implies a parallel diode branch without a series resistor that turns ON when the output tries to drop below $-5\,\text{V}$.
- **Region 2 (Linear Operation):** For $-5\,\text{V} < v_I < 2.5\,\text{V}$, the output strictly follows the input ($v_O = v_I$). The slope is 1. All diode branches must be OFF (open-circuited) in this region.
- **Region 3 (Positive Soft Clipping):** For $v_I \geq 2.5\,\text{V}$, the output increases with a slope of $1/3$. This implies a parallel diode branch with a series resistor that turns ON, creating a voltage divider with the main series input resistor.

2. Circuit Design & Parameter Selection

We must design a parallel clipper consisting of a main series resistor R and two parallel clipping branches. We are given the diode cut-in voltage $V_\gamma = 0.7\,\text{V}$.

Step 2.1: Positive Soft-Clipping Branch (Region 3)

This branch controls the VTC for $v_I \geq 2.5\,\text{V}$. It consists of a resistor R_1 in series with a diode D_1 pointing downwards (anode at the top) and a DC reference voltage V_{R1}.

- **Slope Condition:** When D_1 is ON, R and R_1 form a voltage divider. The slope is determined by:

$$\text{Slope} = \frac{R_1}{R + R_1} = \frac{1}{3} \implies 3R_1 = R + R_1 \implies R = 2R_1 \tag{1}$$

  Let us select standard resistor values: $R_1 = 10\,\mathbf{k}\Omega$ and $R = 20\,\mathbf{k}\Omega$.

- **Turn-ON Condition:** The branch must start conducting when $v_O = 2.5\,\text{V}$. The diode turns ON when its anode-to-cathode voltage reaches $0.7\,\text{V}$.

$$v_O - V_{R1} = 0.7\,\text{V} \implies 2.5\,\text{V} - V_{R1} = 0.7\,\text{V} \implies \mathbf{V_{R1} = 1.8}\,\text{V} \tag{2}$$

Step 2.2: Negative Hard-Clipping Branch (Region 1)

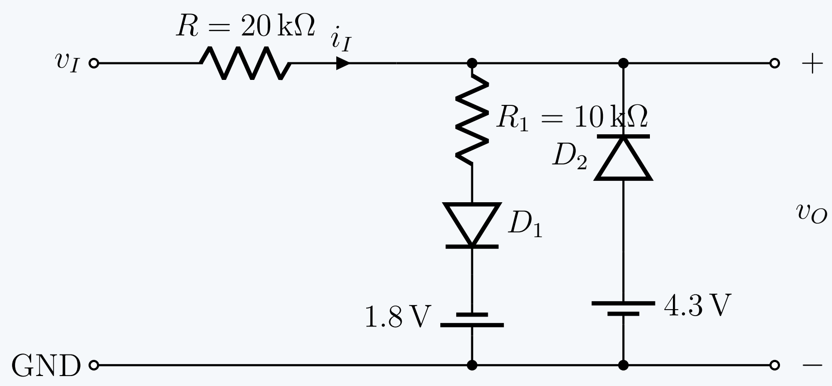
This branch controls the VTC for $v_I \leq -5\,\text{V}$. Since the slope is 0, the branch resistance $R_2 = 0\,\Omega$. It consists of a diode D_2 pointing upwards (cathode at the top) and a DC reference voltage V_{R2}.

- **Turn-ON Condition:** The branch must conduct when v_O attempts to drop below $-5\,\text{V}$. D_2 turns ON when its anode-to-cathode voltage reaches $0.7\,\text{V}$.

$$V_{R2} - v_O = 0.7\,\text{V} \implies V_{R2} - (-5\,\text{V}) = 0.7\,\text{V} \implies \mathbf{V_{R2} = -4.3}\,\text{V} \tag{3}$$

  A negative reference voltage of $-4.3\,\text{V}$ can be implemented by reversing the polarity of a $4.3\,\text{V}$ DC source (connecting its positive terminal to ground).

3. Proposed Circuit Diagram



4. Mathematical Verification

Let's verify the designed circuit against the three regions of operation using Nodal Analysis (KCL) at node v_O:
Case 1: $v_I \leq -5\,\mathbf{V}$ **(**D_1 **OFF,** D_2 **ON)**
Diode D_2 is forward-biased and acts as a 0.7 V source.

$$v_O = V_{R2} - V_\gamma = -4.3\,\text{V} - 0.7\,\text{V} = -5.0\,\text{V} \tag{4}$$

This confirms the flat clipping at -5 V (Slope $= 0$).

Case 2: $-5\,\mathbf{V} < v_I < 2.5\,\mathbf{V}$ **(**D_1 **OFF,** D_2 **OFF)**
Both diodes are reverse-biased and act as open circuits. No current flows through the series resistor R ($i_I = 0$).

$$v_O = v_I - i_I R = v_I - 0 = v_I \tag{5}$$

This confirms the linear region with a slope of 1.

Case 3: $v_I \geq 2.5\,\mathbf{V}$ **(**D_1 **ON,** D_2 **OFF)**
Diode D_1 is forward-biased and acts as a 0.7 V source. The D_1 branch can be modeled as a voltage source $V_{eq} = 1.8\,\text{V} + 0.7\,\text{V} = 2.5\,\text{V}$ in series with $R_1 = 10\,\text{k}\Omega$. Applying KCL at the v_O node:

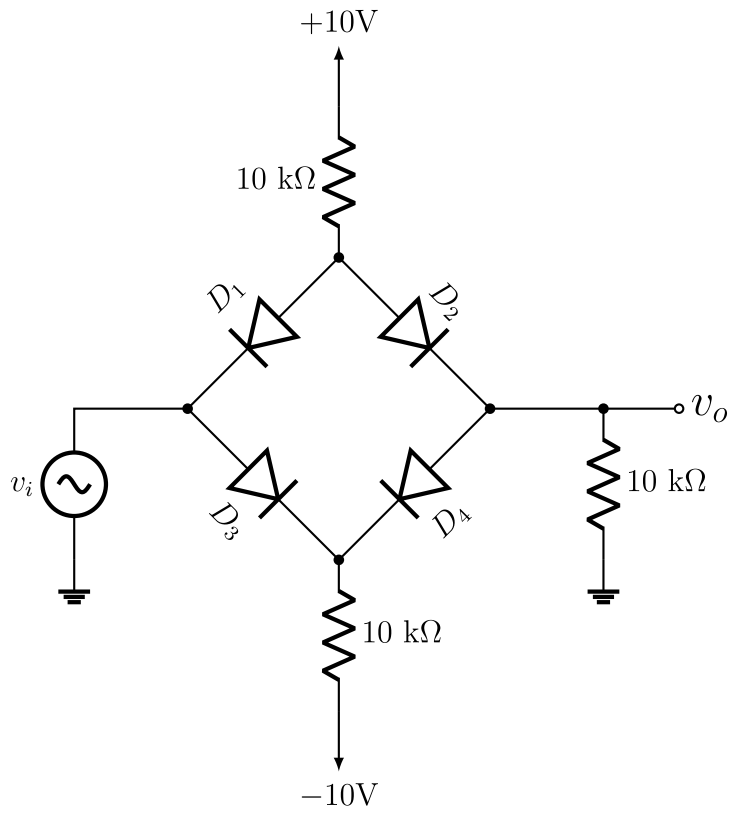
$$\begin{aligned} \frac{v_O - v_I}{20} + \frac{v_O - 2.5}{10} &= 0 \\ (v_O - v_I) + 2(v_O - 2.5) &= 0 \\ 3v_O - v_I - 5 &= 0 \\ 3v_O &= v_I + 5 \\ v_O &= \frac{1}{3}v_I + \frac{5}{3} \end{aligned}$$

The output equation clearly has a slope of 1/3. Furthermore, exactly at the boundary $v_I = 2.5$ V:

$$v_O = \frac{1}{3}(2.5) + \frac{5}{3} = \frac{7.5}{3} = 2.5\,\text{V} \tag{6}$$

This perfectly matches the target Voltage Transfer Characteristic.

Q5. For the circuit shown in Figure , find and sketch the voltage transfer characteristics with proper labelling. Utilize the constant-voltage drop model (0.7 V) for each conduction diode.



(13 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Answer

1. Circuit Overview and Operating Principle

The given circuit is a **diode bridge limiter** (or clipper). It utilizes a symmetrical diode bridge (D_1 to D_4) biased by ±10 V DC sources through 10 kΩ resistors. The output voltage v_o is taken across a 10 kΩ load resistor. We will analyze the circuit using the constant-voltage drop model for the diodes, where $V_\gamma = 0.7$ V.
Let us define the nodes:

- **Input Node (Left):** Driven directly by v_i.
- **Output Node (Right):** Voltage is v_o.
- **Top Node:** Voltage is V_T, connected to +10 V via 10 kΩ.
- **Bottom Node:** Voltage is V_B, connected to −10 V via 10 kΩ.

2. Analysis of Operating Regions

Region I: Linear Operation (All Diodes ON)

For small values of input voltage v_i, the ±10 V sources keep all four diodes forward-biased.

- With D_1 and D_3 ON, the top and bottom node voltages are clamped to the input:

$$V_T = v_i + 0.7 \text{ V}, \quad V_B = v_i - 0.7 \text{ V} \tag{1}$$

- With D_2 and D_4 ON, the output voltage is dictated by V_T and V_B:

$$v_o = V_T - 0.7 \text{ V} = (v_i + 0.7 \text{ V}) - 0.7 \text{ V} = v_i \tag{2}$$

  Thus, in this region, the output strictly follows the input: $\mathbf{v_o = v_i}$.

To find the upper limit of this region, we determine when diodes D_1 and D_4 turn OFF (currents reach zero). By applying Kirchhoff's Current Law (KCL) at the top and bottom nodes, the total currents supplied are:

$$I_T = \frac{10 - V_T}{10 \text{ k}\Omega} = \frac{9.3 - v_i}{10 \text{ k}\Omega}$$
$$I_B = \frac{V_B - (-10)}{10 \text{ k}\Omega} = \frac{v_i + 9.3}{10 \text{ k}\Omega}$$

At the upper breakpoint ($v_i > 0$), D_1 and D_4 turn off simultaneously. When this happens, all top current flows through D_2 to the output, and no current flows through D_4. Setting the current through D_4 to zero:

$$I_{D4} = 0 \implies I_{D2} = I_L = \frac{v_o}{10 \text{ k}\Omega} = \frac{v_i}{10 \text{ k}\Omega} \tag{3}$$

Since $I_T = I_{D1} + I_{D2}$ and $I_{D1} = 0$:

$$\frac{9.3 - v_i}{10 \text{ k}\Omega} = \frac{v_i}{10 \text{ k}\Omega} \implies 9.3 - v_i = v_i \implies 2v_i = 9.3 \implies \mathbf{v_i = 4.65} \text{ V} \tag{4}$$

By symmetry, the lower breakpoint occurs at $\mathbf{v_i = -4.65}$ V.

Region II: Positive Clipping ($v_i \geq 4.65$ V)

When v_i exceeds 4.65 V, D_1 and D_4 are reverse-biased (OFF). D_2 and D_3 remain ON. The output node is now isolated from the input and is only driven by the +10 V source through the top 10 kΩ resistor and D_2. Applying KCL at the output node (v_o):

$$\frac{10-(v_o+0.7)}{10\text{ k}\Omega} = \frac{v_o}{10\text{ k}\Omega}$$
$$9.3 - v_o = v_o$$
$$2v_o = 9.3 \implies \mathbf{v_o = 4.65}\text{ V}$$

The output is clamped at +4.65 V.

Region III: Negative Clipping ($v_i \leq -4.65$ V)

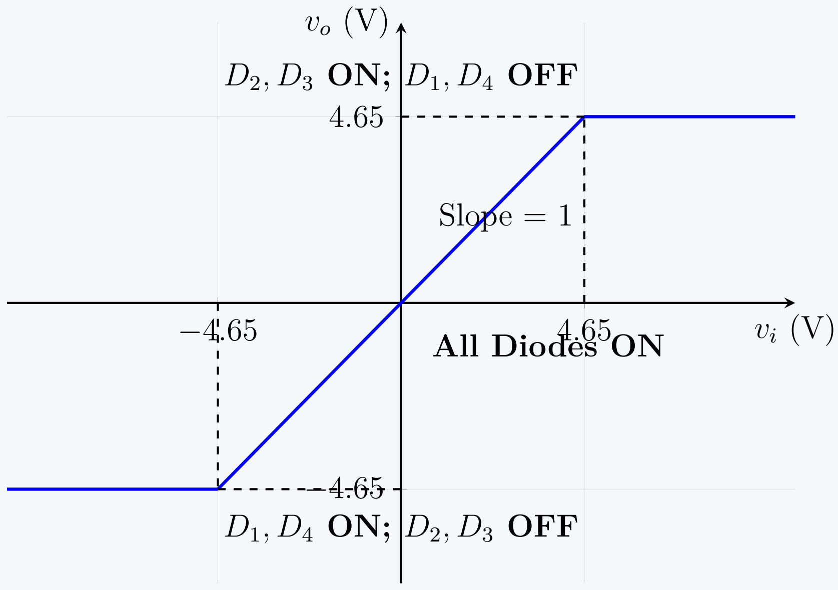
When v_i drops below −4.65 V, D_2 and D_3 turn OFF. D_1 and D_4 remain ON. The output node is driven by the −10 V source through the bottom 10 kΩ resistor and D_4. Applying KCL at the output node (v_o):

$$\frac{v_o}{10\text{ k}\Omega} + \frac{v_o - 0.7 - (-10)}{10\text{ k}\Omega} = 0$$
$$v_o + v_o + 9.3 = 0$$
$$2v_o = -9.3 \implies \mathbf{v_o = -4.65}\text{ V}$$

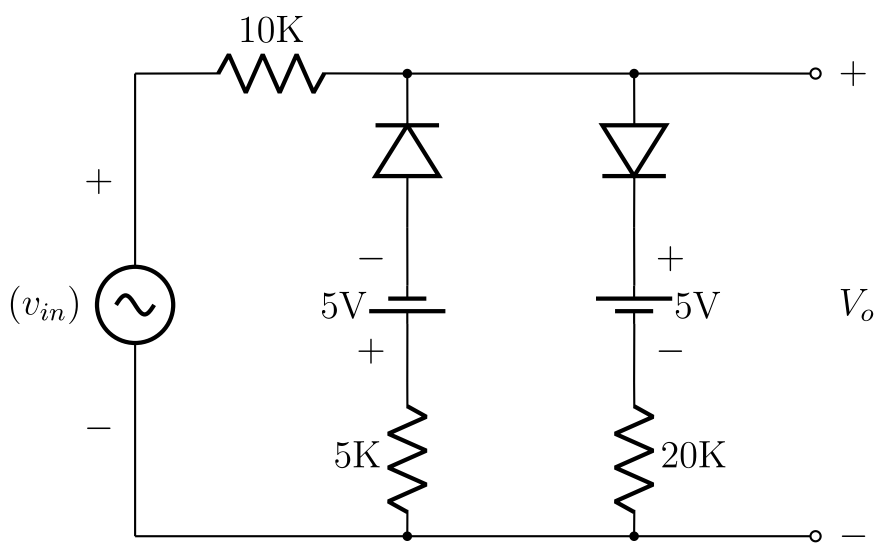
The output is clamped at −4.65 V.

3. Voltage Transfer Characteristic (VTC)

The transfer curve exhibits a linear region with a slope of 1 for input voltages between −4.65 V and +4.65 V, and hard limits (slope 0) outside this range.



Q6. For the circuit shown, neatly draw the output (v_o) waveform corresponding to the input (v_{in}). Here, $v_{in} = 10\sin(200\pi t)$ and diodes are non-ideal.



(16 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

1. Circuit Analysis and Operating Regions

The circuit consists of an AC input source $v_{in} = 10\sin(200\pi t)$ and two parallel soft-clipping branches. We assume a non-ideal constant voltage drop model for the diodes, where $V_\gamma = 0.7$ V.
Let's determine the turn-on conditions for each branch:

- **Right Branch (D_2):** The diode points downwards. The cathode is connected to a $+5$ V potential (battery positive terminal up) relative to ground (assuming zero initial current). D_2 turns ON when $v_o - 5\text{ V} \geq 0.7$ V, which means $\mathbf{v_o \geq 5.7}$ V.
- **Left Branch (D_1):** The diode points upwards. The anode is connected to a -5 V potential (battery negative terminal up). D_1 turns ON when $-5\text{ V} - v_o \geq 0.7$ V, which means $\mathbf{v_o \leq -5.7}$ V.

Based on these thresholds, the circuit operates in three distinct regions:

Region I: Linear Region (-5.7 V $< v_{in} < 5.7$ V)

Both D_1 and D_2 are OFF. No current flows through the 10 kΩ series resistor, so there is no voltage drop across it.

$$v_o = v_{in} \tag{1}$$

Region II: Positive Soft-Clipping ($v_{in} \geq 5.7$ V)

D_2 is ON, and D_1 is OFF. The right branch conducts, and we apply KCL at the output node (v_o):

$$\begin{aligned}\frac{v_{in} - v_o}{10\text{ k}\Omega} &= \frac{v_o - 0.7 - 5}{20\text{ k}\Omega}\\ 2(v_{in} - v_o) &= v_o - 5.7\\ 2v_{in} - 2v_o &= v_o - 5.7\\ 3v_o = 2v_{in} + 5.7 &\implies \mathbf{v_o = \frac{2}{3}v_{in} + 1.9}\text{ V}\end{aligned}$$

The peak input is $v_{in(peak)} = 10$ V. At this peak, $v_o = \frac{2}{3}(10) + 1.9 = \mathbf{8.57}$ V.

Region III: Negative Soft-Clipping ($v_{in} \leq -5.7$ V)

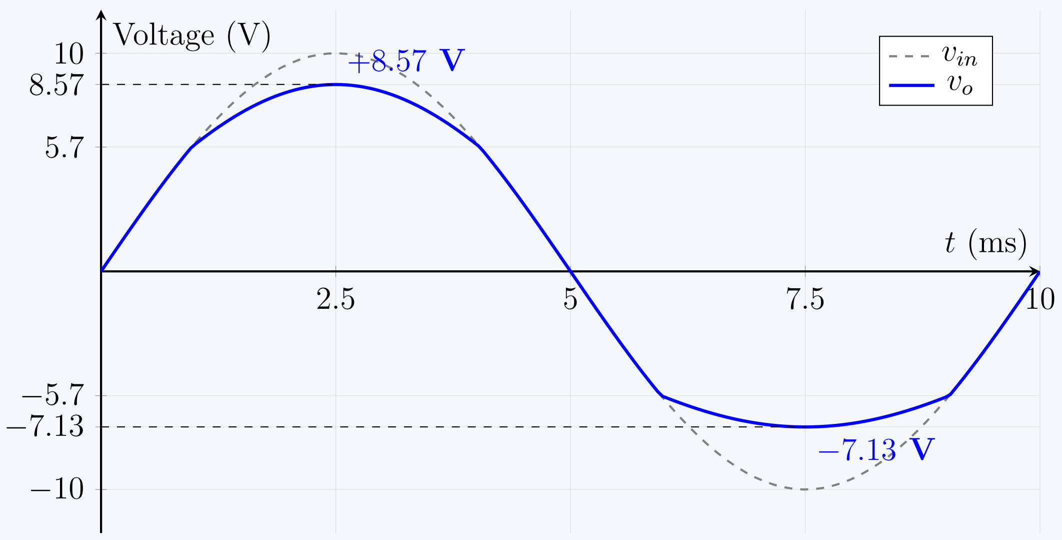
D_1 is ON, and D_2 is OFF. The left branch conducts. Applying KCL at the output node (v_o):

$$\begin{aligned}\frac{v_{in} - v_o}{10\text{ k}\Omega} + \frac{-5 - 0.7 - v_o}{5\text{ k}\Omega} &= 0\\ (v_{in} - v_o) + 2(-5.7 - v_o) &= 0\\ v_{in} - v_o - 11.4 - 2v_o &= 0\\ 3v_o = v_{in} - 11.4 &\implies \mathbf{v_o = \frac{1}{3}v_{in} - 3.8}\text{ V}\end{aligned}$$

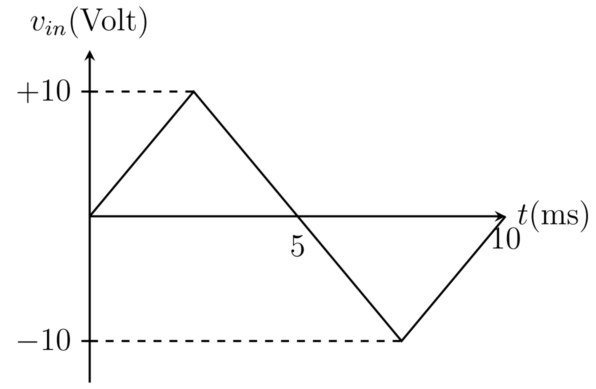
The minimum input is $v_{in(min)} = -10$ V. At this trough, $v_o = \frac{1}{3}(-10) - 3.8 = \mathbf{-7.13}$ V.

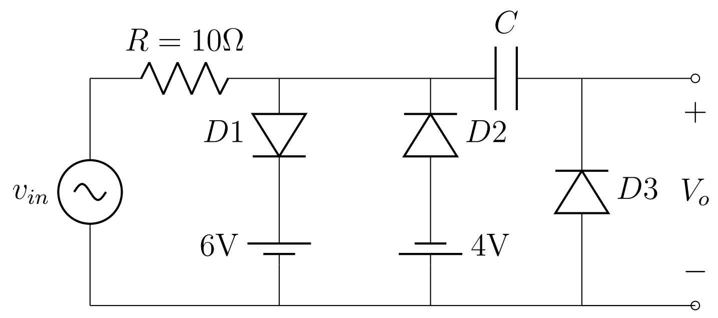
2. Output Waveform Sketch

The input is $v_{in} = 10\sin(200\pi t)$, which has a frequency $f = 100$ Hz and a time period $T = 10$ ms.



Q7. Assuming that the diodes are ideal, find the output voltage V_o for the circuit given. Draw the wave shape clearly.





(20 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Theory and Principle of Circuit Operation

The given circuit is a combination of two distinct wave-shaping stages operating in cascade:

(a) **Stage 1: A Double-Action Clipper (Slicer).** The first section of the circuit, consisting of the resistor R, diodes $D1$ and $D2$, and their associated DC bias batteries, acts as a slicer. It limits the intermediate voltage v_A (the node before the capacitor) to a specific upper and lower bound.

(b) **Stage 2: A Positive Clamper.** The second section, consisting of capacitor C and diode $D3$, acts as a clamping circuit. It adds a DC shift to the clipped waveform so that the lowest peak of the output voltage V_o is firmly clamped to 0V.

Ideal Assumptions: We assume all diodes are ideal ($V_\gamma = 0$V, infinite reverse resistance) and the capacitor C is sufficiently large such that its discharge time constant is much greater than the signal period, allowing it to act as a perfect DC voltage source in steady state.

—

2. Step-by-Step Mathematical Derivation

Stage 1: Analysis of the Clipper (Voltage v_A)

Let v_A be the voltage at the node between R and C.

- **Positive Clipping:** Diode $D1$ points downwards, with its cathode connected to the positive terminal of the 6V battery. $D1$ turns ON when $v_A \geq 6$V. When $D1$ conducts, it acts as a short circuit, and v_A is clamped to a maximum of +**6**V.
- **Negative Clipping:** Diode $D2$ points upwards. The battery below it has its shorter (thicker) line on top, indicating its negative terminal is facing up. Therefore, the anode of $D2$ is biased at −4V. $D2$ turns ON when $v_A \leq -4$V. When $D2$ conducts, v_A is clamped to a minimum of −**4**V.
- **Linear Region:** For $-4\text{V} < v_{in} < 6\text{V}$, both $D1$ and $D2$ are OFF (open circuit). Assuming negligible charging current flows through R in steady state, $v_A(t) = v_{in}(t)$.

Based on the input $v_{in}(t)$ which is a triangular wave ranging from +10V to −10V, the intermediate voltage $v_A(t)$ will be a clipped triangle wave varying between +**6**V and −**4**V.

Stage 2: Analysis of the Clamper (Voltage V_o)

The clipped signal $v_A(t)$ is fed into the clamper circuit.

- By Kirchhoff's Voltage Law (KVL): $V_o = v_A - V_C$, where V_C is the voltage stored across the capacitor (positive on the left plate).
- Diode $D3$ points upwards (cathode at V_o, anode at ground). It turns ON whenever V_o attempts to go below 0V, effectively acting as a short circuit to ground.
- During the first negative swing of v_A, as v_A drops to its minimum of −4V, V_o attempts to go negative. $D3$ conducts heavily, clamping $V_o = 0$V.
- The capacitor charges to the difference: $V_C = v_A - V_o = -4\text{V} - 0\text{V} = -\mathbf{4}\text{V}$.
- The capacitor holds this charge ($V_C = -4$V) because the ideal diode $D3$ is reverse-biased for all subsequent positive swings, and there is no load resistor to discharge it.

The steady-state output voltage is derived as:

$$V_o(t) = v_A(t) - V_C = v_A(t) - (-4\text{V}) = v_A(t) + 4\text{V}$$

Final Output Voltage Levels

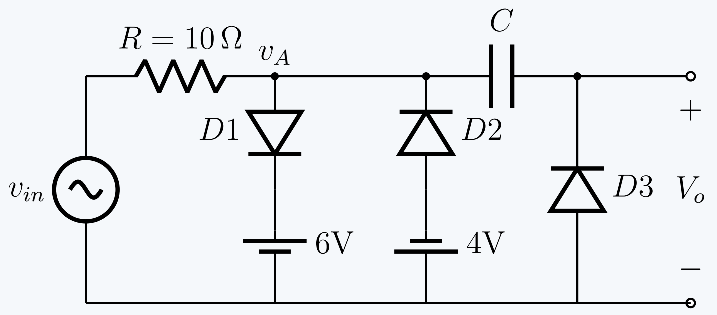
The clamper shifts the entire $v_A(t)$ waveform upwards by +4V.

$$V_{o,\max} = v_{A,\max} + 4\text{V} = 6\text{V} + 4\text{V} = \mathbf{10}\text{V}$$
$$V_{o,\min} = v_{A,\min} + 4\text{V} = -4\text{V} + 4\text{V} = \mathbf{0}\text{V}$$

—

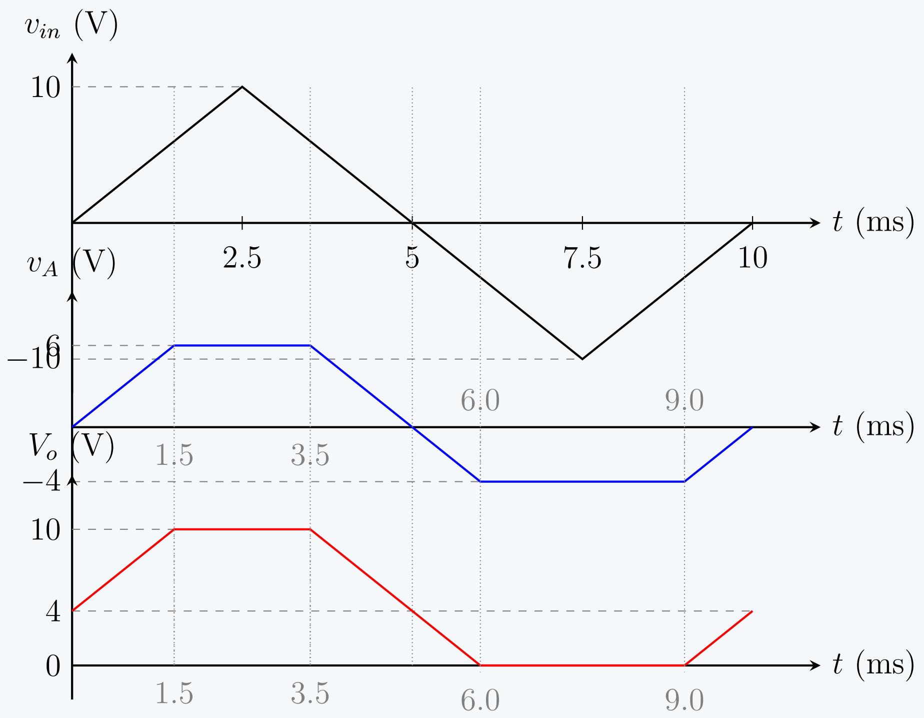
3. Circuit Diagram and Waveform Characteristics

Circuit Schematic

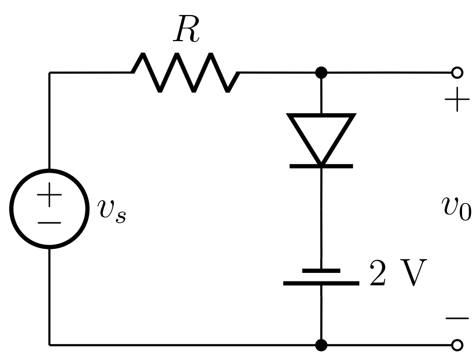


Detailed Waveform Alignments

The plot below aligns the original input $v_{in}(t)$, the intermediate sliced voltage $v_A(t)$, and the final clamped output voltage $V_o(t)$ in a synchronized time frame.



Q8. Sketch the output waveform and transfer characteristics for the circuit shown. Assume the diode is ideal and V_s is a sinusoid with 5 V peak amplitude and frequency 5 kHz.



(15 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Circuit Analysis and Operating Regions

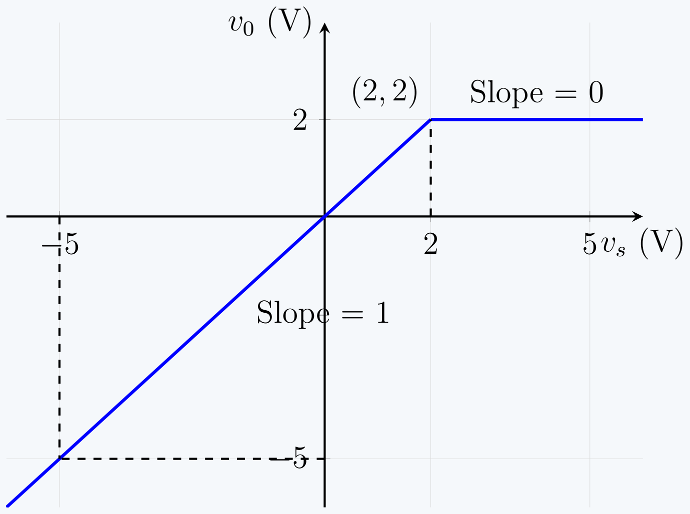
The given circuit is a **positive clipper**. The diode's cathode is connected to the positive terminal of a 2 V DC source, holding it at a potential of +2 V. We are instructed to assume an **ideal diode**, meaning the forward voltage drop is 0 V ($V_\gamma = 0$).
The input voltage is a sinusoid: $v_s(t) = 5\sin(2\pi f t)$, with a peak amplitude of 5 V and a frequency of $f = 5$ kHz. The time period of the signal is $T = \frac{1}{f} = \frac{1}{5000} = 0.2$ ms.

We can determine the circuit's behavior by analyzing the states of the diode:

- **Region I: Diode OFF ($v_s < 2$ V)** When the input voltage is less than 2 V, the potential at the anode is lower than the potential at the cathode. The diode is reverse-biased and acts as an open circuit. Since no current flows through resistor R, there is no voltage drop across it.
$$v_0 = v_s \tag{1}$$
- **Region II: Diode ON ($v_s \geq 2$ V)** When the input voltage reaches or exceeds 2 V, the diode becomes forward-biased. As an ideal diode, it acts as a short circuit. The output voltage is directly clamped to the DC battery voltage.
$$v_0 = 2 \text{ V} \tag{2}$$

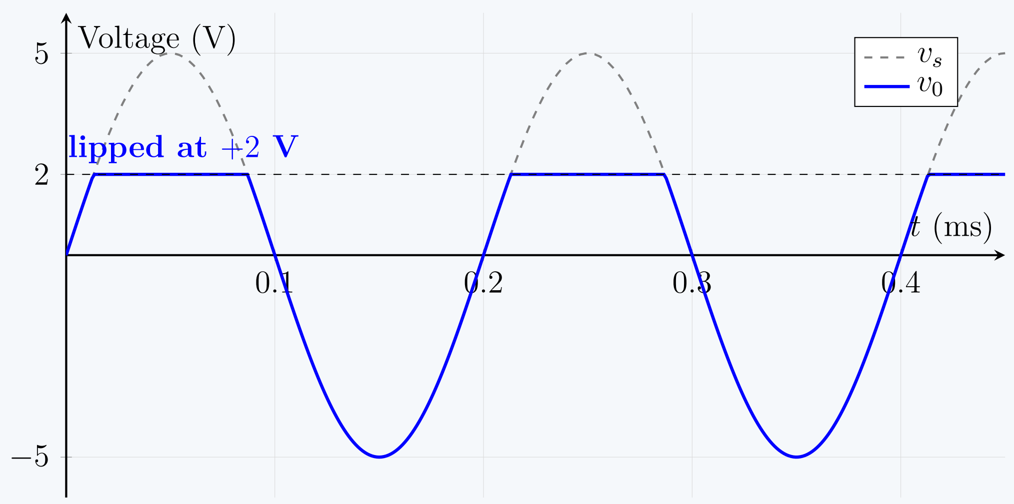
2. Voltage Transfer Characteristics (VTC)

The transfer characteristic relates the output v_0 to the input v_s. It has a linear slope of 1 for $v_s < 2$ V and a horizontal slope of 0 for $v_s \geq 2$ V.



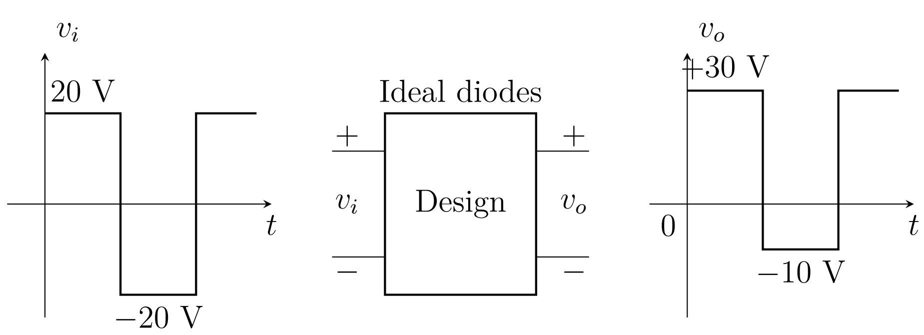
3. Output Waveform Sketch

The input v_s is a 5 V peak sine wave with a period of 0.2 ms. The output perfectly tracks the input for all voltages below 2 V and clips exactly at +2 V during the positive half-cycles.



Clamping Circuits

Q1. Design a clamper circuit to perform the function indicated in the figure (input v_i, output v_0). Assume ideal diodes.



(12 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

1. Analyze the Waveforms

To design the clamper circuit, we first compare the input and output waveforms to determine the required DC shift:

- **Input Waveform (v_i):** Square wave alternating between +20 V and −20 V. The peak-to-peak voltage is 40 V.
- **Output Waveform (v_o):** Square wave alternating between +30 V and −10 V. The peak-to-peak voltage is still 40 V, confirming the circuit is a clamper (DC restorer).

The entire waveform has been shifted upwards. The amount of DC shift added to the signal is:

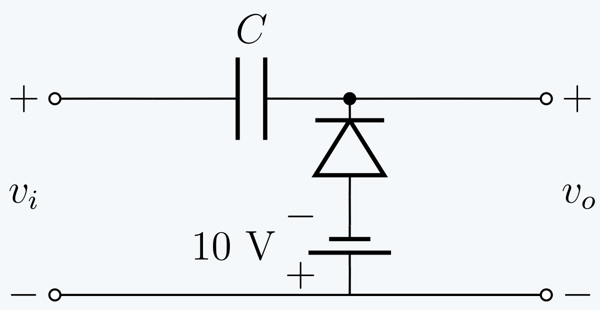
$$V_{DC} = V_{max(out)} - V_{max(in)} = 30\text{ V} - 20\text{ V} = +\mathbf{10}\text{ V} \tag{1}$$

2. Clamper Circuit Design

Because the waveform is shifted upwards, we need a **positive clamper**. A positive clamper consists of a capacitor in series with the input and a diode in parallel pointing *upwards*.
To determine the reference voltage for the diode:

- During the negative half-cycle, $v_i = -20$ V. The output v_o must be clamped to its minimum value of −10 V.
- For the upwards-pointing diode to turn ON and clamp the output at −10 V, its anode must be held at a −10 V potential relative to ground.
- This is achieved by connecting a 10 V DC source in series with the diode, with the negative terminal connected to the diode's anode and the positive terminal connected to ground.

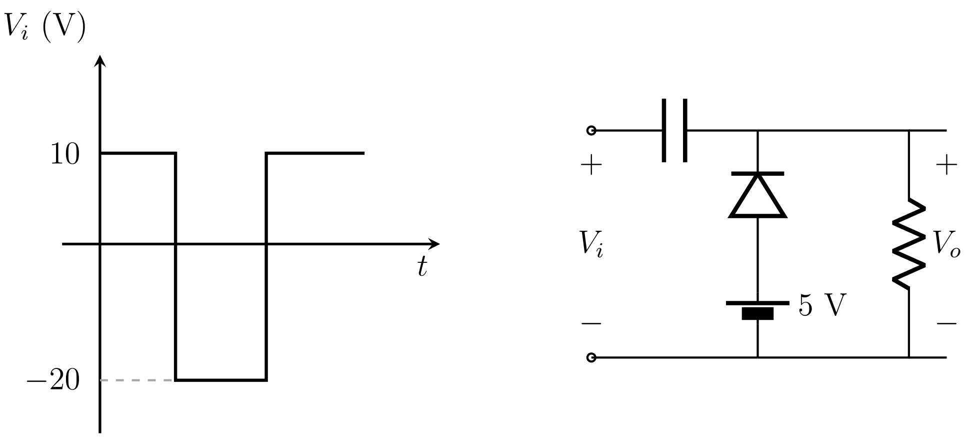
3. Circuit Diagram



4. Verification

- **Negative Half-Cycle ($v_i = -20$ V):** The diode is forward-biased and turns ON. The output v_o is clamped directly to the battery voltage (−10 V). The capacitor C charges to +10 V (right plate positive relative to the left plate) via KVL: $-20 - V_C = -10 \implies V_C = -10$ V (from left to right), so the right plate is 10 V higher.
- **Positive Half-Cycle ($v_i = +20$ V):** The diode is reverse-biased and turns OFF. The fully charged capacitor acts as a series 10 V battery. The output voltage becomes $v_o = v_i + V_{C(right)} = 20\text{ V} + 10\text{ V} = +30$ V. This perfectly matches the required output.

Q2. For the following circuit, draw the output waveshape for the given square input waveshape.



(16 marks) **[EEE 263 | L-2/T-1 | Jan 2021]**

Answer

1. Circuit Identification and Reference Level

The given circuit is a **positive clamper**. The diode points upwards, which means it will prevent the output voltage from dropping below a specific reference level, effectively shifting the entire waveform up.
The reference voltage is determined by the DC source connected in series with the diode. The longer line of the battery symbol is on top, indicating its positive terminal is connected to the diode's anode. Thus, the reference clamping level is +5 V. Assuming an ideal diode, it will turn ON whenever the output tries to drop below 5 V.

2. Operation During the Negative Half-Cycle

The clamper sets its DC shift during the cycle that forward-biases the diode.

- When the input drops to $V_i = -20$ V, the output tries to swing negative. Because this potential is lower than the anode potential (5 V), the diode is forward-biased and acts as a short circuit.
- The output is directly clamped to the battery voltage:

$$V_o = 5 \text{ V} \tag{1}$$

- During this time, the capacitor (C) charges. Applying Kirchhoff's Voltage Law (KVL) to the outer loop ($V_i - V_C - V_o = 0$):

$$-20 \text{ V} - V_C - 5 \text{ V} = 0 \implies V_C = -25 \text{ V} \tag{2}$$

  This means the capacitor charges to 25 V, with its right plate positive relative to its left plate.

3. Operation During the Positive Half-Cycle

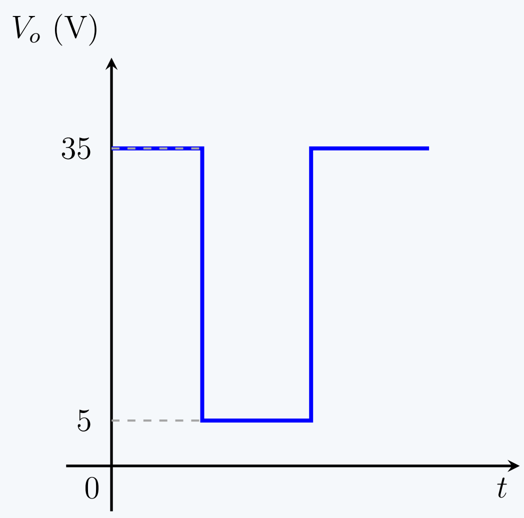
- When the input rises to $V_i = 10$ V, the cathode of the diode is pushed to a higher potential than the 5 V anode. The diode becomes reverse-biased and acts as an open circuit.
- The fully charged capacitor acts as a series 25 V battery. The output voltage is the sum of the input voltage and the capacitor voltage:

$$V_o = V_i + V_{C(\text{right plate})} = 10 \text{ V} + 25 \text{ V} = 35 \text{ V} \tag{3}$$

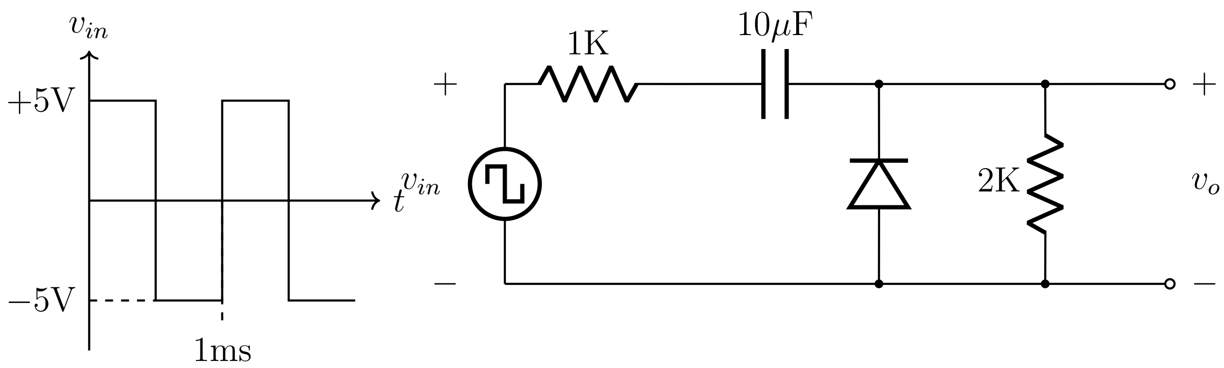
Verification: The peak-to-peak voltage of the input is $10 \text{ V} - (-20 \text{ V}) = 30$ V. The peak-to-peak voltage of the output is $35 \text{ V} - 5 \text{ V} = 30$ V. The amplitude is perfectly preserved, and the signal has been shifted upwards by 25 V.

4. Output Waveform Sketch

The resulting steady-state output is a square wave that alternates between 35 V and 5 V, synchronized with the input waveform's timing.



Q3. For the circuit shown, neatly draw the output waveform (v_o) corresponding to the input waveform (v_{in}) in steady state. Assume the diode is non-ideal.



(15 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

1. Circuit Analysis and Assumptions

The given circuit is a clamper circuit with a non-ideal diode, a series resistor R_1, and a parallel load resistor R_2. To analyze the steady-state response, we make the following assumptions:

- **Non-ideal Diode Model:** We assume a constant voltage drop (CVD) model for the silicon diode, where the forward voltage drop is $V_\gamma = 0.7$ V and the forward resistance is $r_f = 0\ \Omega$.

- **Steady-State Operation:** The input square wave has a frequency of $f = 1$ kHz (since $T = 1$ ms). The half-period is $T/2 = 0.5$ ms.
- **Time Constants:**
  - When the diode is ON, the equivalent resistance charging the capacitor is $R_{eq,ON} \approx R_1 = 1$ kΩ. The charging time constant is $\tau_{ON} = R_1 C = 1\ \text{k}\Omega \times 10\mu\text{F} = 10$ ms.
  - When the diode is OFF, the equivalent resistance is $R_{eq,OFF} = R_1 + R_2 = 3$ kΩ. The discharging time constant is $\tau_{OFF} = (R_1 + R_2)C = 3\ \text{k}\Omega \times 10\mu\text{F} = 30$ ms.

  Since both $\tau_{ON} = 10\ \text{ms} \gg 0.5$ ms and $\tau_{OFF} = 30\ \text{ms} \gg 0.5$ ms, the capacitor does not have enough time to significantly charge or discharge during a single half-cycle. Therefore, we can assume the voltage across the capacitor, V_C, remains approximately constant in steady state.

Let V_C be the steady-state DC voltage across the capacitor, defined with the positive terminal on the left plate.

2. Circuit States and Operating Regions

We analyze the two distinct states corresponding to the two half-cycles of the input waveform.
State 1: Positive Half-Cycle ($0 \le t < 0.5$ ms) During this interval, $v_{in} = +5$ V. The positive input reverse-biases the upward-pointing diode (Diode OFF). The circuit simplifies to a voltage divider consisting of v_{in}, R_1, C, and R_2 in series. The loop current i_1 (flowing left to right) is:

$$i_1 = \frac{v_{in} - V_C}{R_1 + R_2} = \frac{5 - V_C}{1\ \text{k}\Omega + 2\ \text{k}\Omega} = \frac{5 - V_C}{3\ \text{k}\Omega} \tag{1}$$

The output voltage v_{o1} across R_2 is:

$$v_{o1} = i_1 R_2 = \left(\frac{5 - V_C}{3\ \text{k}\Omega}\right) \times 2\ \text{k}\Omega = \frac{2}{3}(5 - V_C) \tag{2}$$

State 2: Negative Half-Cycle (0.5 ms $\le t < 1.0$ ms) During this interval, $v_{in} = -5$ V. The negative input pulls the cathode of the diode to a lower potential than its grounded anode, forward-biasing the diode (Diode ON). The output voltage is clamped to the negative diode drop:

$$v_{o2} = -0.7\ \text{V} \tag{3}$$

The current i_2 flowing through the capacitor (left to right) is determined by the voltage drop across R_1:

$$i_2 = \frac{v_{in} - V_C - v_{o2}}{R_1} = \frac{-5 - V_C - (-0.7)}{1\ \text{k}\Omega} = \frac{-4.3 - V_C}{1\ \text{k}\Omega} \tag{4}$$

3. Steady-State Charge Balance

In steady state, the net charge accumulating on the capacitor over one full period must be zero. This implies the average capacitor current is zero:

$$i_1 \cdot \frac{T}{2} + i_2 \cdot \frac{T}{2} = 0 \implies i_1 + i_2 = 0 \tag{5}$$

Substituting the expressions for i_1 and i_2:

$$\begin{aligned} \frac{5 - V_C}{3} + \frac{-4.3 - V_C}{1} &= 0 \\ (5 - V_C) + 3(-4.3 - V_C) &= 0 \\ 5 - V_C - 12.9 - 3V_C &= 0 \\ -4V_C &= 7.9 \\ V_C &= -1.975\ \text{V} \end{aligned}$$

The negative sign indicates that the right plate of the capacitor is actually at a higher potential than the left plate by 1.975 V.

4. Final Output Voltages and Verification

Using the calculated steady-state capacitor voltage $V_C = -1.975$ V:

- **During the Positive Half-Cycle:** From Equation (2), the output voltage is:

$$v_{o1} = \frac{2}{3}(5 - (-1.975)) = \frac{2}{3}(6.975) = \mathbf{4.65}\ \text{V} \tag{6}$$

  Verification: Since $v_{o1} = 4.65\ \text{V} > -0.7$ V, the diode is indeed reverse-biased, confirming our assumption.

- **During the Negative Half-Cycle:** From Equation (3), the output voltage is:

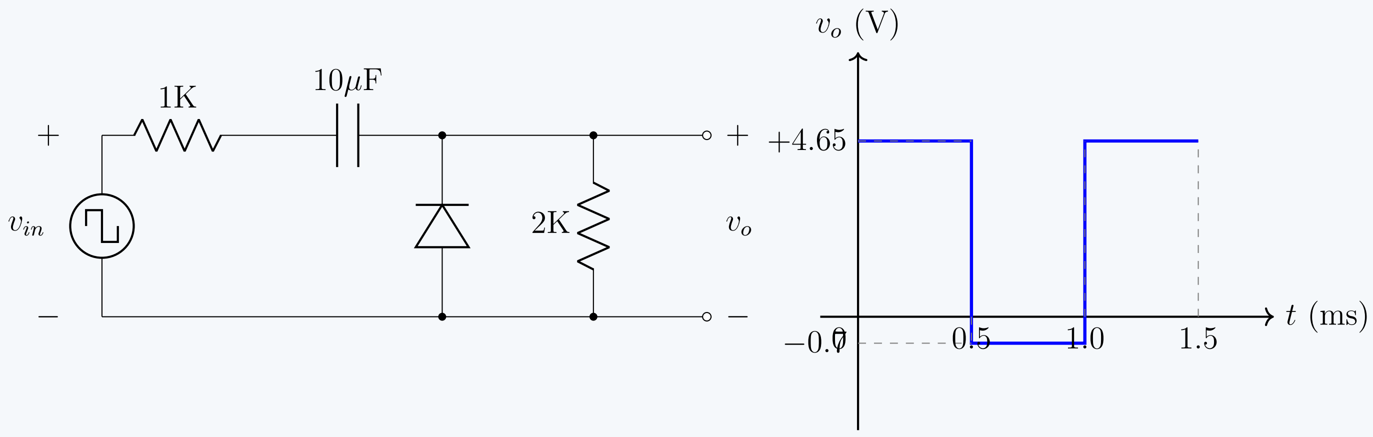
$$v_{o2} = -\mathbf{0.7}\ \text{V} \tag{7}$$

Verification: Let's check if the diode current I_D is positive (flowing upwards from ground). By applying KCL at the output node:

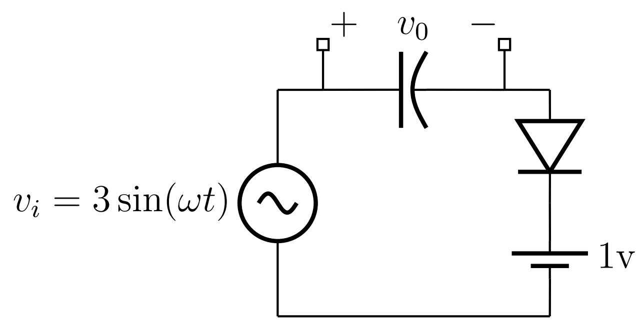
$$\begin{aligned}
I_D &= i_{R2} - i_2 \\
i_{R2} &= \frac{v_{o2}}{R_2} = \frac{-0.7 \text{ V}}{2 \text{ k}\Omega} = -0.35 \text{ mA} \\
i_2 &= \frac{-4.3 - (-1.975)}{1 \text{ k}\Omega} = -2.325 \text{ mA} \\
I_D &= -0.35 \text{ mA} - (-2.325 \text{ mA}) = \mathbf{1.975} \text{ mA}
\end{aligned}$$

Since $I_D > 0$, the diode is indeed forward-biased and conducting, validating our assumption.

5. Circuit Diagram and Output Waveform



Q4. Draw the output wave shape (v_0) and also the V_L–V_o graph for the circuit shown. Assume a constant 0.7 V forward voltage drop across the diode.



(20 marks) **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. Circuit Analysis and Assumptions

Based on the provided circuit diagram, the output voltage v_0 is explicitly defined as the voltage *across the series capacitor* C, with the positive terminal on the left plate. We analyze the circuit with the following assumptions:

- **Input Voltage:** The source is an AC sinusoidal voltage $v_i(t) = 3\sin(\omega t)$, with a peak amplitude of $V_m = 3$ V.
- **Diode Model:** We assume a Constant Voltage Drop (CVD) model for the diode, with a forward voltage of $V_\gamma = 0.7$ V.
- **DC Bias:** A 1 V DC battery is in series with the diode. Because the longer parallel line (positive terminal) is at the top, the cathode potential is $V_K = 1$ V relative to ground.
- **Initial Condition:** The capacitor is initially uncharged, meaning $v_0(0) = 0$ V.

2. Mathematical Derivation

The circuit consists of a single loop for charging the capacitor. The diode D will turn ON (forward-biased) only when the potential at its anode (V_A) exceeds the potential at its cathode (V_K) by at least V_γ.
Condition for Diode Conduction:

$$\begin{aligned}
V_A - V_K &\geq V_\gamma \\
(v_i - v_0) - 1 \text{ V} &\geq 0.7 \text{ V} \\
v_i - v_0 &\geq 1.7 \text{ V}
\end{aligned}$$

Transient Charging Phase:

- Starting from $t = 0$, as v_i rises from 0 V, the condition $v_i - v_0 \geq 1.7$ V is not met initially. The diode remains OFF, no current flows, and $v_0 = 0$ V.

- When v_i reaches 1.7 V, the diode turns ON. For any further increase in v_i, applying Kirchhoff's Voltage Law (KVL) yields:

$$v_i - v_0 - 0.7\text{ V} - 1\text{ V} = 0 \implies v_0 = v_i - 1.7\text{ V} \tag{1}$$

- The input voltage v_i reaches its maximum positive peak of 3 V at $t = T/4$. At this instant, the capacitor charges to its maximum voltage:

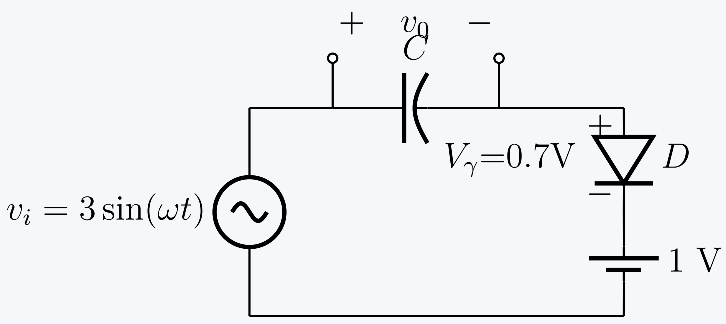
$$v_{0(\max)} = 3\text{ V} - 1.7\text{ V} = \mathbf{1.3}\text{ V} \tag{2}$$

Steady-State Operation:

- After v_i passes its positive peak and decreases, the quantity $(v_i - v_0)$ drops below 1.7 V. The diode becomes reverse-biased and turns OFF.
- Because the diode is OFF and there is no parallel discharge path (ideal assumption), the capacitor perfectly retains its accumulated charge.
- Therefore, in steady state, the output voltage v_0 remains a constant DC value:

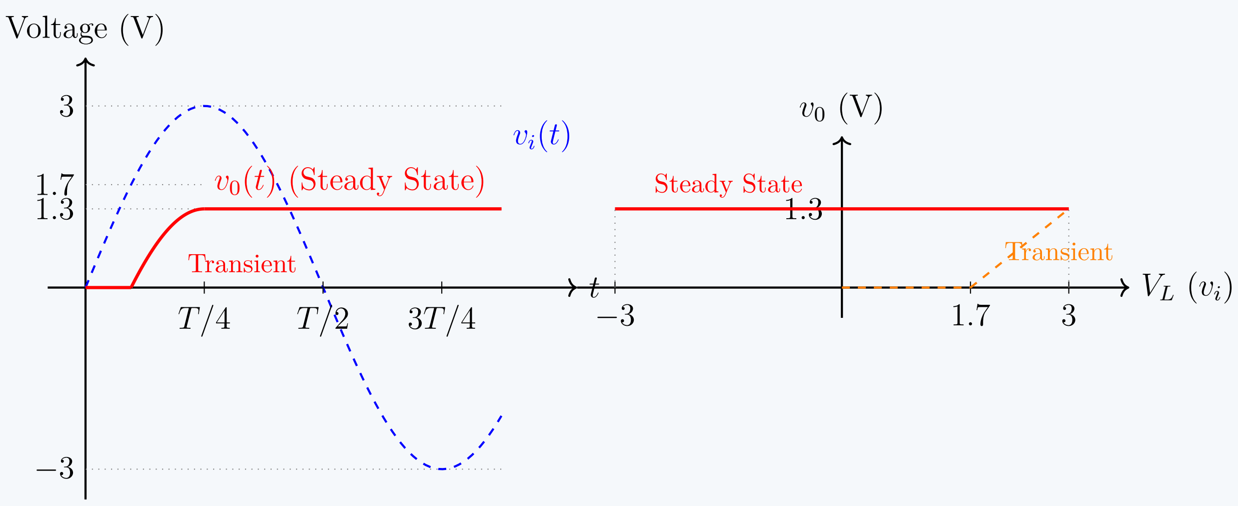
$$v_0(t) = \mathbf{1.3}\text{ V} \tag{3}$$

3. Circuit Diagram Redrawn



4. Output Waveforms and Transfer Characteristic

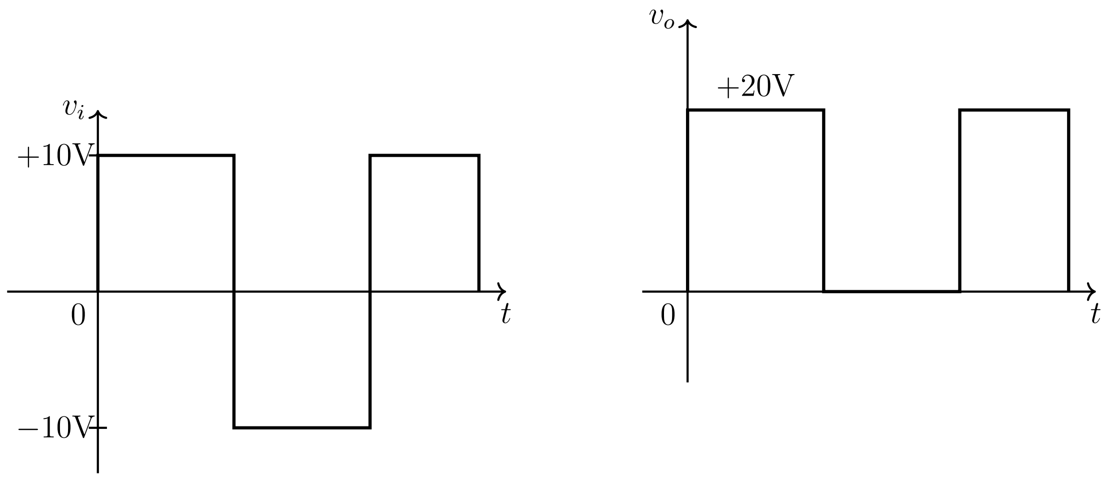
Assumption: The requested "V_L" parameter denotes the line/input voltage v_i. The plots reflect the transient charging and the clamped steady-state behavior.



Important Note

While the overall circuit functions as a positive clamper (where the load output is typically taken across the diode branch), the provided schematic **explicitly defines the output terminals (v_0) directly across the capacitor**. Therefore, the rigorous evaluation of v_0 yields the transient charging curve that resolves into a pure DC steady-state value of 1.3V, rather than a shifted sinusoidal waveform.

Q5. Design a clamper circuit to perform the function indicated in the figure using an ideal diode.



(12 marks) [EEE 263 | L-2/T-1 | 2011–12]

Answer

1. Waveform Analysis and Design Strategy

Based on the provided input and output waveforms, we can determine the required transformation:

- **Input Waveform (v_i):** A square wave alternating between $V_{i(\max)} = +10$ V and $V_{i(\min)} = -10$ V. The peak-to-peak voltage is $V_{pp} = 10 - (-10) = 20$ V.
- **Output Waveform (v_o):** A square wave alternating between $V_{o(\max)} = +20$ V and $V_{o(\min)} = 0$ V. The peak-to-peak voltage remains $V_{pp} = 20$ V.
- **DC Shift:** The entire waveform has been shifted upwards by +10 V.
- **Clamping Level:** The negative peak of the input signal (−10 V) is clamped precisely to 0 V.

Since the waveform is shifted in the positive direction and the minimum value is clamped to zero, a **Positive Clamper Circuit** is required. Because the target clamping level is exactly 0 V, no external DC bias voltage is needed in series with the diode.

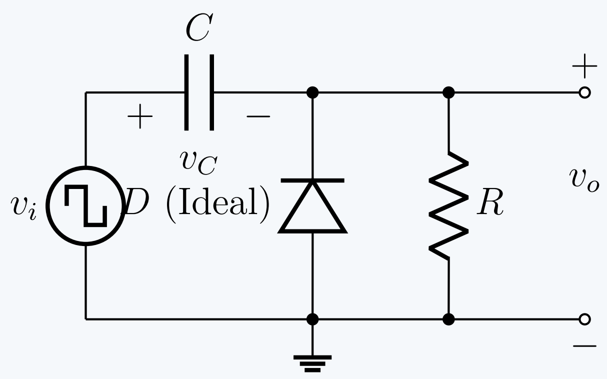
2. Clamper Circuit Design

To achieve the positive clamping function, the circuit requires:

(a) A **series capacitor** (C) to block DC and store the required shift voltage.

(b) A **parallel ideal diode** (D) pointing upwards (anode grounded, cathode connected to the signal path) to clamp the negative peak to 0 V.

(c) A **load resistor** (R) to provide a discharge path, assumed to be large enough such that the time constant $\tau = RC \gg T$ (where T is the time period of the wave).



3. Principle of Operation and Verification

We verify the design by analyzing the circuit operation during the two distinct half-cycles of the input square wave. We assume the capacitor is initially uncharged and the diode is ideal ($V_\gamma = 0$ V).
Step 1: Negative Half-Cycle ($v_i = -10$ V)

- During this interval, the input voltage drops to −10 V.
- The potential at the diode's cathode tends to become negative while its anode is fixed at 0 V (ground). This forward-biases the diode (D is **ON**).
- An ideal forward-biased diode acts as a short circuit. Therefore, the output voltage is clamped directly to ground:

$$v_o = 0 \text{ V} \tag{1}$$

- The capacitor charges very quickly through the ideal diode. Applying Kirchhoff's Voltage Law (KVL) in the left loop:

$$\begin{aligned} v_i - v_C - v_o &= 0 \\ -10 \text{ V} - v_C - 0 \text{ V} &= 0 \\ v_C &= -10 \text{ V} \end{aligned}$$

Note: A negative v_C implies the right plate of the capacitor is at a +10 V higher potential than the left plate.

Step 2: Positive Half-Cycle ($v_i = +10$ V)

- The input voltage instantaneously switches to $+10$ V.
- The potential at the cathode of the diode is now driven positive by the series combination of the source and the charged capacitor. This reverse-biases the diode (D is **OFF**).
- With the diode acting as an open circuit, the output voltage is measured across the load resistor R. Because $RC \gg T$, the capacitor acts as a constant voltage source of -10 V (maintaining its charge).
- Applying KVL for the output:

$$v_o = v_i - v_C$$
$$v_o = 10 \text{ V} - (-10 \text{ V})$$
$$v_o = +20 \text{ V}$$

4. Conclusion

The designed circuit successfully shifts the input square wave upwards. The negative peak is clamped to 0 V, and the positive peak reaches $+20$ V, exactly matching the required output characteristics specified in the problem.

Switching Circuits — Logic Gates using BJTs

Q1. Make a NAND and a NOR gate using BJTs. **(10 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

1. Introduction to BJT Logic Gates

Bipolar Junction Transistors (BJTs) can be operated as electronic switches to construct fundamental digital logic gates. When utilized in digital logic circuits, the BJT is typically driven between two extreme operating regions:

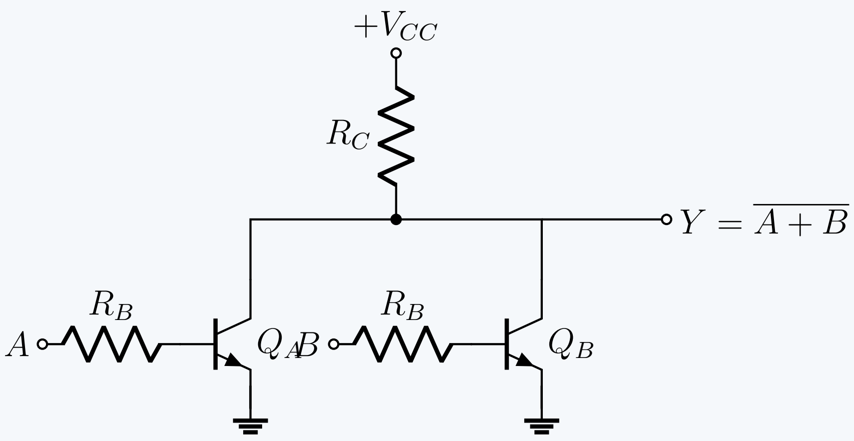
- **Cutoff Region (Logic ’0’ output, Switch OFF):** The base-emitter voltage V_{BE} is less than the cut-in voltage ($V_\gamma \approx 0.7$ V), meaning the base current $I_B \approx 0$, and the collector current $I_C \approx 0$. The transistor acts as an open circuit.
- **Saturation Region (Logic ’1’ input, Switch ON):** A sufficient base current I_B is applied such that $I_C < \beta I_B$. The collector-emitter voltage drops to $V_{CE(sat)} \approx 0.2$ V. The transistor acts as a closed switch.

The circuits designed below utilize **Resistor-Transistor Logic (RTL)**, a fundamental logic family where resistors are used at the inputs and a BJT performs the switching.
—

2. BJT NOR Gate

A. Circuit Diagram

A NOR gate produces a LOW (Logic ’0’) output if *any* of its inputs are HIGH (Logic ’1’). To achieve this with BJTs, two NPN transistors are placed in **parallel**.



B. Principle of Operation

- **Both Inputs LOW ($A = 0, B = 0$):** Both Q_A and Q_B are in cutoff ($V_{BE} < 0.7$ V). No collector current flows. The output Y is pulled up to $+V_{CC}$ via R_C. Therefore, $Y = 1$.
- **One or Both Inputs HIGH ($A = 1$ and/or $B = 1$):** The corresponding transistor(s) receive sufficient base current to enter saturation. The saturated transistor acts as a closed switch, pulling the output node Y down to ground (specifically, $V_{CE(sat)} \approx 0.2$ V). Therefore, $Y = 0$.

C. Truth Table (NOR)

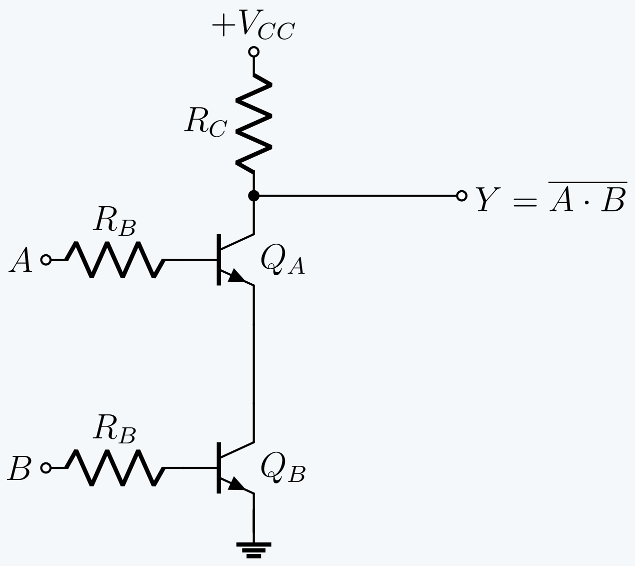
Input A	Input B	Output Y	Q_A State	Q_B State
0	0	**1**	OFF	OFF
0	1	**0**	OFF	ON (Sat)
1	0	**0**	ON (Sat)	OFF
1	1	**0**	ON (Sat)	ON (Sat)

—

3. BJT NAND Gate

A. Circuit Diagram

A NAND gate produces a LOW (Logic '0') output *only* if *all* of its inputs are HIGH (Logic '1'). To achieve this with BJTs in RTL configuration, two NPN transistors are placed in **series**.



B. Principle of Operation

- **Both Inputs HIGH ($A = 1, B = 1$):** Both Q_A and Q_B receive base current. Since they are in series, both must turn ON to create a conduction path to ground. When both are saturated, the output Y is pulled down to $V_{CE(sat)A} + V_{CE(sat)B} \approx 0.4$ V, which registers as a Logic '0'.
- **One or Both Inputs LOW ($A = 0$ and/or $B = 0$):** If either Q_A or Q_B is in cutoff, the series path to ground is broken. No collector current flows through R_C. The output Y is pulled up to $+V_{CC}$ via R_C. Therefore, $Y = 1$.

C. Truth Table (NAND)

Input A	Input B	Output Y	Series Path
0	0	**1**	Broken (Q_A, Q_B OFF)
0	1	**1**	Broken (Q_A OFF)
1	0	**1**	Broken (Q_B OFF)
1	1	**0**	Conducting (Q_A, Q_B ON)

4. Design Considerations and Mathematical Conditions

To ensure robust logic operation, the base resistor R_B and collector resistor R_C must be chosen such that a HIGH input firmly drives the transistor into deep saturation.
For saturation, the forced beta (β_{forced}) must be less than the active region beta (β_{active}):

$$I_B \geq \frac{I_{C(sat)}}{\beta_{min}} \tag{1}$$

Where the collector current during saturation is approximately:

$$I_{C(sat)} \approx \frac{V_{CC} - V_{CE(sat)}}{R_C} \tag{2}$$

And the base current supplied by a HIGH input logic level ($V_{in} \approx V_{CC}$) is:

$$I_B = \frac{V_{in} - V_{BE(sat)}}{R_B} \tag{3}$$

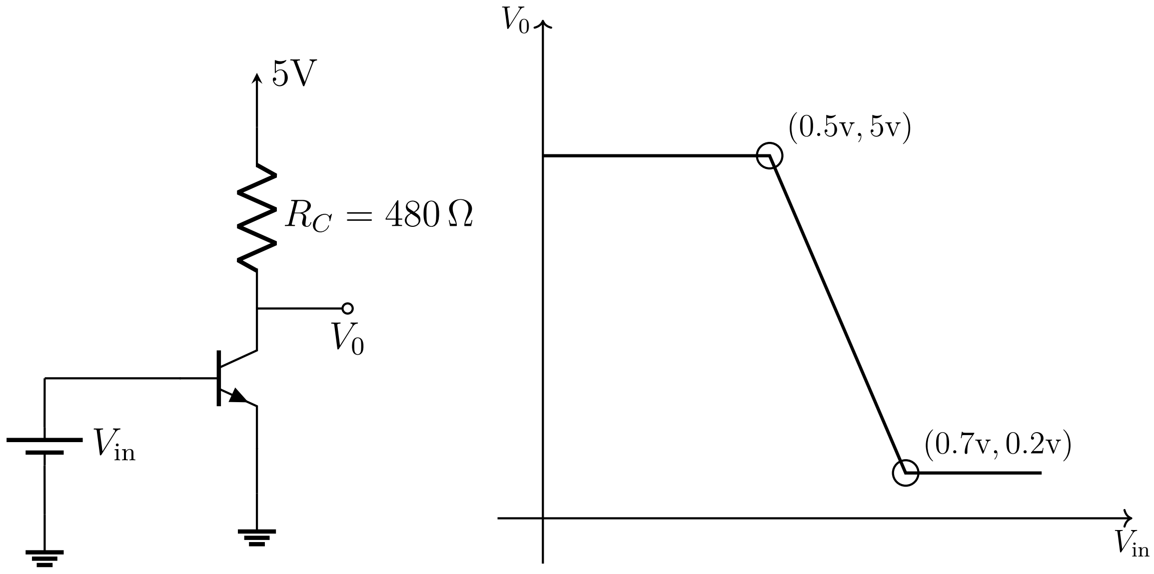
Combining these yields the required design constraint for R_B:

$$R_B \leq \beta_{min} \left(\frac{V_{in} - V_{BE(sat)}}{V_{CC} - V_{CE(sat)}} \right) R_C \tag{4}$$

This ensures the transistors act as ideal switches rather than linear amplifiers.

BJT Inverter Analysis

Q1. (a) Modify both the V_o–V_{in} and I_c–V_{in} graphs if supply voltage is decreased from 5 V to 2.5 V ($R_c \approx 480\,\Omega$). **(10 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**



(b) The figure shows an inverter gate built by an npn BJT and its V_o–V_{in} characteristics. Copy the V_o–V_{in} graph to your answer script exactly to scale and then draw the collector current (I_c) vs. V_{in} over the same graph. Mention the co-ordinates of the bending points of the I_c–V_{in} graph as shown for V_o–V_{in}. **(16 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(c) Modify both the V_o–V_{in} and I_c–V_{in} graphs if R_c is infinity. **(10 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(d) Modify both the V_o–V_{in} and I_c–V_{in} graphs if supply voltage is increased from 5 V to 10 V ($R_c = 480\,\Omega$). **(10 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(e) If the BJT in active mode shows a linear I_c–V_{BE} behaviour instead of exponential, does voltage gain change with change in bias voltage V_{in}? Does output voltage swing change with the change in V_{in}? Explain why. **(5 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

Initial Analysis & Device Modeling

Before solving the individual parts, we must deduce the behavioral model of the BJT from the provided $V_o - V_{\text{in}}$ characteristics. The circuit is a basic common-emitter inverter with V_{in} applied directly to the base ($V_{BE} = V_{\text{in}}$).
From the given graph ($V_{CC} = 5\text{V}, R_C = 480\,\Omega$):

- **Cutoff Region:** For $V_{\text{in}} \le 0.5\text{V}$, $V_o = 5\text{V}$. The transistor is OFF, $I_c = 0$ mA. Cut-in voltage is $V_\gamma = 0.5\text{V}$.
- **Saturation Region:** For $V_{\text{in}} \ge 0.7\text{V}$, $V_o = 0.2\text{V}$. The transistor is saturated, $V_{CE(sat)} = 0.2\text{V}$.
- **Active Region:** For $0.5\text{V} < V_{\text{in}} < 0.7\text{V}$, the $V_o - V_{\text{in}}$ relationship is a *straight line*. This implies the problem assumes a **linear** piecewise model for the collector current I_c with respect to V_{BE}, rather than an exponential one.

We can calculate the voltage gain (A_v) and transconductance (g_m) in the active region:

$$A_v = \frac{\Delta V_o}{\Delta V_{\text{in}}} = \frac{0.2\text{V} - 5\text{V}}{0.7\text{V} - 0.5\text{V}} = \frac{-4.8\text{V}}{0.2\text{V}} = -24 \text{ V/V}$$
$$A_v = -g_m R_c \implies g_m = \frac{-A_v}{R_c} = \frac{24}{480\,\Omega} = 0.05 \text{ S} = 50 \text{ mA/V}$$

Thus, in the active region, the collector current is given by:

$$I_c = g_m(V_{\text{in}} - V_\gamma) = 50 \text{ mA/V} \cdot (V_{\text{in}} - 0.5\text{V}) \tag{1}$$

We will use this linear g_m model for all subsequent analyses.

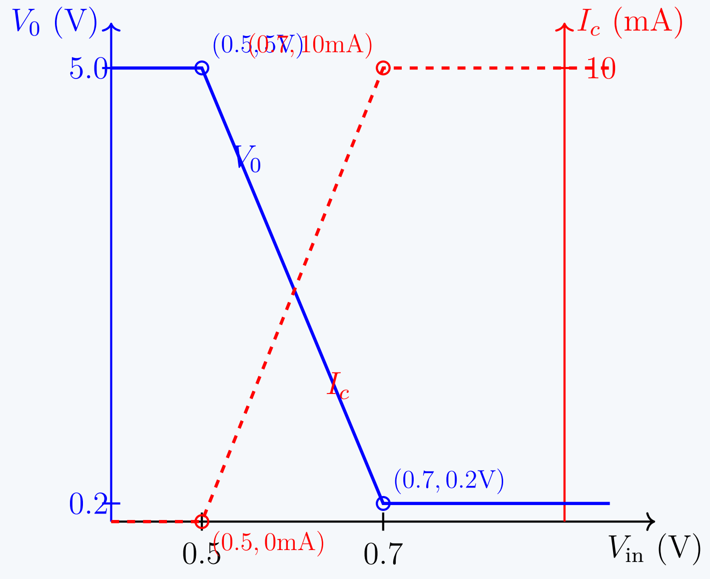
(b) Original $V_o - V_{\text{in}}$ and $I_c - V_{\text{in}}$ Graphs ($V_{CC} = 5$V)

(Note: We solve part (b) first as it establishes the baseline required by the problem.)
At saturation ($V_o = 0.2\text{V}$), the maximum collector current is:

$$I_{c(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_c} = \frac{5\text{V} - 0.2\text{V}}{480\,\Omega} = 10 \text{ mA} \tag{2}$$

The bending points for the $I_c - V_{\text{in}}$ graph are:

- Start of Active: $V_{\text{in}} = 0.5\text{V} \implies I_c = 0$ mA
- Start of Saturation: $V_{\text{in}} = 0.7\text{V} \implies I_c = 10$ mA



(a) Graphs if Supply Voltage is Decreased to $V_{CC} = 2.5$V

If the supply voltage is lowered, the maximum voltage is 2.5V. The new saturation current is:

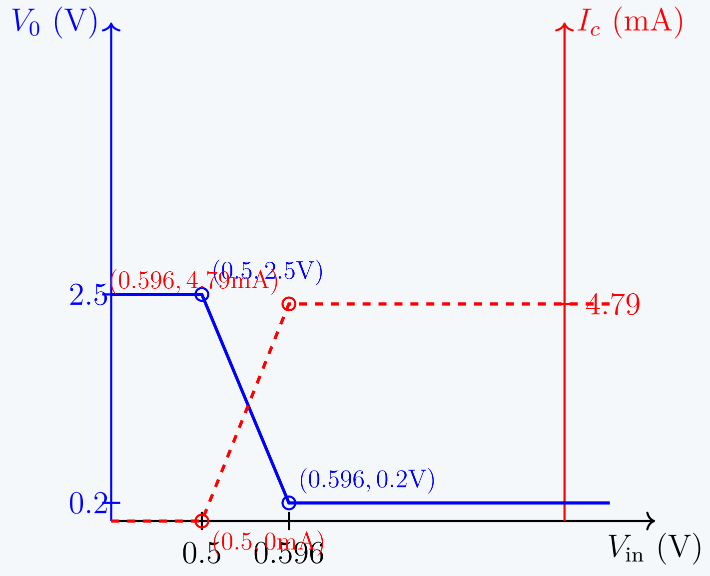
$$I'_{c(sat)} = \frac{V'_{CC} - V_{CE(sat)}}{R_c} = \frac{2.5\text{V} - 0.2\text{V}}{480\,\Omega} = \frac{2.3\text{V}}{480\,\Omega} \approx 4.79 \text{ mA} \tag{3}$$

Since $g_m = 50$ mA/V remains unchanged (intrinsic to the BJT), the input voltage required to reach this new saturation current is:

$$V'_{\text{in(sat)}} = V_\gamma + \frac{I'_{c(sat)}}{g_m} = 0.5\text{V} + \frac{4.79 \text{ mA}}{50 \text{ mA/V}} = 0.5\text{V} + 0.0958\text{V} \approx 0.596\text{V} \tag{4}$$

New Bending Points:

- V_o **Graph:** $(0.5\text{V}, 2.5\text{V})$ and $(0.596\text{V}, 0.2\text{V})$
- I_c **Graph:** $(0.5\text{V}, 0\text{mA})$ and $(0.596\text{V}, 4.79\text{mA})$



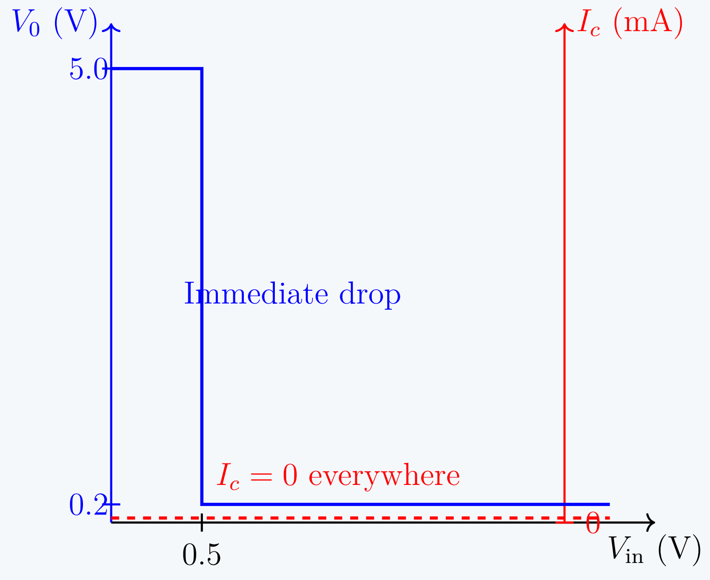
(c) Graphs if R_c is Infinity ($R_c \to \infty$)

If $R_c = \infty$, the collector circuit is open. Thus, collector current cannot flow:

$$I_c = 0 \text{ mA} \quad \text{for all } V_{\text{in}} \tag{5}$$

For the voltage V_o:

- When $V_{\text{in}} < 0.5$V, the BJT is OFF. With an open circuit, an ideal high-impedance measurement would float at $V_{CC} = 5$V.
- As soon as $V_{\text{in}} > 0.5$V, the base-emitter junction turns ON. Because I_c is forced to be 0, the transistor immediately enters deep saturation (or inverse active mode) to balance the injected carriers. The output voltage immediately drops to $V_{CE(sat)} = 0.2$V. The circuit acts as an ideal infinite-gain comparator.



(d) Graphs if Supply Voltage is Increased to $V_{CC} = 10$V

With $V_{CC} = 10$V, the maximum voltage is 10V. The new saturation current is:

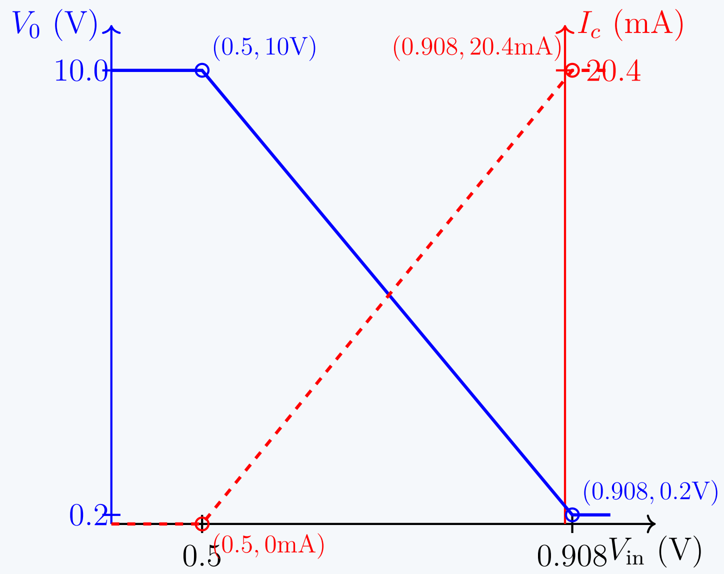
$$I''_{c(sat)} = \frac{10\text{V} - 0.2\text{V}}{480\,\Omega} = \frac{9.8\text{V}}{480\,\Omega} \approx 20.4\text{ mA} \tag{6}$$

The input voltage required to reach this saturation current is:

$$V''_{\text{in(sat)}} = 0.5\text{V} + \frac{20.4\text{ mA}}{50\text{ mA/V}} = 0.5\text{V} + 0.408\text{V} = 0.908\text{V} \tag{7}$$

New Bending Points:

- V_o **Graph:** $(0.5\text{V}, 10\text{V})$ and $(0.908\text{V}, 0.2\text{V})$
- I_c **Graph:** $(0.5\text{V}, 0\text{mA})$ and $(0.908\text{V}, 20.4\text{mA})$



(e) Effect of Linear $I_c - V_{BE}$ Behaviour

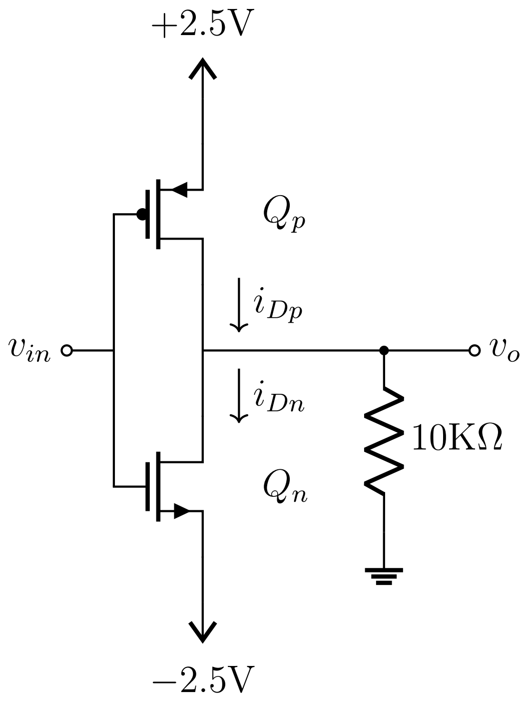
1. Does voltage gain change with bias voltage V_{in}?
No, the voltage gain does not change.
Explanation: The voltage gain of the inverter is given by $A_v = -g_m R_c$. The transconductance is defined as $g_m = \frac{dI_c}{dV_{BE}}$. If the BJT exhibits a strictly *linear* $I_c - V_{BE}$ relationship (as assumed in the graphs above), the derivative g_m is a constant (calculated as 50 mA/V). Because g_m and R_c are both constant, the voltage gain A_v remains entirely independent of the DC bias point V_{in} anywhere within the active region.
2. Does output voltage swing change with bias voltage V_{in}?
Yes, the undistorted output voltage swing changes.
Explanation: The bias voltage V_{in} sets the quiescent operating point (Q-point), which determines the DC output voltage V_{oQ}. The absolute limits of the output voltage are fixed at V_{CC} (cutoff) and $V_{CE(sat)}$ (saturation). However, the maximum *symmetrical* (undistorted) AC voltage swing is constrained by the distance from the Q-point to the nearest clipping limit:

$$V_{\text{swing, max}} = 2 \times \min\left(V_{CC} - V_{oQ},\ V_{oQ} - V_{CE(sat)}\right) \tag{8}$$

If V_{in} changes, V_{oQ} changes, thereby moving closer to either cutoff or saturation. This directly alters the available undistorted swing.

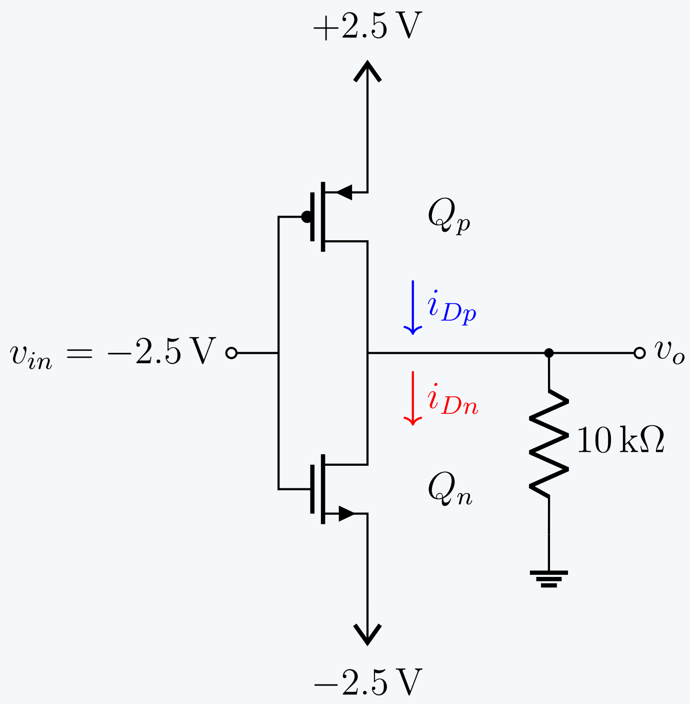
CMOS Inverter Analysis

Q1. The inverter circuit with CMOS shown has $k'_n(W_n/L_n) = k'_p(W_p/L_p) = 1\,\text{mA/V}^2$ and $V'_{tn} = -V'_{tp} = 1\,\text{V}$. Find (i) i_{DN}, (ii) i_{DP}, (iii) V_o for $V_{in} = -2.5\,\text{V}$.



(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer



1. Operating Region of NMOS (Q_n)

First, we analyze the condition of the NMOS transistor. The source of Q_n is connected to $-2.5\,\text{V}$, and the input voltage at the gate is $v_{in} = -2.5\,\text{V}$.

$$v_{GSn} = v_G - v_{Sn}$$
$$v_{GSn} = -2.5\,\text{V} - (-2.5\,\text{V}) = 0\,\text{V}$$

For an NMOS to be ON, its gate-to-source voltage must be strictly greater than its threshold voltage ($V_{tn} = 1\,\text{V}$). Since $v_{GSn} = 0\,\text{V} < V_{tn}$, the **NMOS is in the Cutoff region**. Therefore, no drain current flows through Q_n:

$$\mathbf{i_{Dn} = 0}\,\text{mA} \tag{1}$$

2. Operating Region of PMOS (Q_p)

Next, we determine the state of the PMOS transistor. The source of Q_p is connected to $+2.5\,\text{V}$, and the gate is connected to $v_{in} = -2.5\,\text{V}$.

$$v_{SGp} = v_{Sp} - v_G$$
$$v_{SGp} = 2.5\,\text{V} - (-2.5\,\text{V}) = 5\,\text{V}$$

For a PMOS to be ON, its source-to-gate voltage must exceed the absolute value of its threshold voltage ($|V_{tp}| = |-1\,\text{V}| = 1\,\text{V}$). Since $v_{SGp} = 5\,\text{V} > |V_{tp}|$, the **PMOS is ON**.
Given that Q_n is OFF and Q_p is driving the load resistor R_L toward the positive rail, we will state the assumption that Q_p is operating in the **Triode region** and verify it subsequently.

3. Mathematical Derivation of Output Voltage (v_o) and Current (i_{Dp})

Applying Kirchhoff's Current Law (KCL) at the output node v_o:

$$i_{Dp} = i_{Dn} + i_{R_L} \tag{2}$$

Since $i_{Dn} = 0$, all the current supplied by Q_p flows into the load resistor:

$$i_{Dp} = \frac{v_o}{R_L} = \frac{v_o}{10\,\text{k}\Omega} \tag{3}$$

Assuming Q_p is in the triode region, the drain current i_{Dp} is given by:

$$i_{Dp} = k_p' \left(\frac{W_p}{L_p}\right) \left[(v_{SGp} - |V_{tp}|)v_{SDp} - \frac{1}{2}v_{SDp}^2\right] \tag{4}$$

We evaluate the required voltage terms:

- Overdrive voltage: $v_{SGp} - |V_{tp}| = 5\,\text{V} - 1\,\text{V} = 4\,\text{V}$
- Source-to-Drain voltage: $v_{SDp} = v_{Sp} - v_o = 2.5 - v_o$

Substituting these into the current equation (working in mA and V):

$$\begin{aligned}
i_{Dp} &= 1 \cdot \left[4(2.5 - v_o) - 0.5(2.5 - v_o)^2\right] \\
i_{Dp} &= 10 - 4v_o - 0.5\left(6.25 - 5v_o + v_o^2\right) \\
i_{Dp} &= 10 - 4v_o - 3.125 + 2.5v_o - 0.5v_o^2 \\
i_{Dp} &= -0.5v_o^2 - 1.5v_o + 6.875
\end{aligned}$$

Equating Eq. (3) and Eq. (**??**):

$$\begin{aligned}
\frac{v_o}{10} &= -0.5v_o^2 - 1.5v_o + 6.875 \\
0.1v_o &= -0.5v_o^2 - 1.5v_o + 6.875 \\
0.5v_o^2 + 1.6v_o - 6.875 &= 0
\end{aligned}$$

Multiplying the entire equation by 2 to simplify:

$$v_o^2 + 3.2v_o - 13.75 = 0 \tag{5}$$

Applying the quadratic formula to solve for v_o:

$$\begin{aligned}
v_o &= \frac{-3.2 \pm \sqrt{(3.2)^2 - 4(1)(-13.75)}}{2} \\
v_o &= \frac{-3.2 \pm \sqrt{10.24 + 55}}{2} = \frac{-3.2 \pm \sqrt{65.24}}{2} \\
v_o &= \frac{-3.2 \pm 8.0771}{2}
\end{aligned}$$

This yields two possible mathematical roots:

$$v_{o1} = 2.4386\,\text{V} \quad \text{and} \quad v_{o2} = -5.6386\,\text{V}$$

Since the output voltage cannot fall below the negative supply rail ($-2.5\,\text{V}$) and must satisfy the physical properties of the active circuit, the only physically valid solution is:

$$\mathbf{v_o} \approx \mathbf{2.439}\,\text{V} \tag{6}$$

Now, calculate i_{Dp} using the validated v_o:

$$i_{Dp} = \frac{v_o}{10\,\text{k}\Omega} = \frac{2.4386\,\text{V}}{10\,\text{k}\Omega} \approx \mathbf{0.244}\,\text{mA} \tag{7}$$

4. Verification of the Operating Region

To ensure the mathematical validity of our derivation, we must confirm that Q_p operates in the Triode region. The condition for the PMOS triode region is:

$$v_{SDp} \le v_{SGp} - |V_{tp}|$$

Calculating v_{SDp}:

$$v_{SDp} = 2.5\,\text{V} - 2.439\,\text{V} = 0.061\,\text{V}$$

Checking the condition:

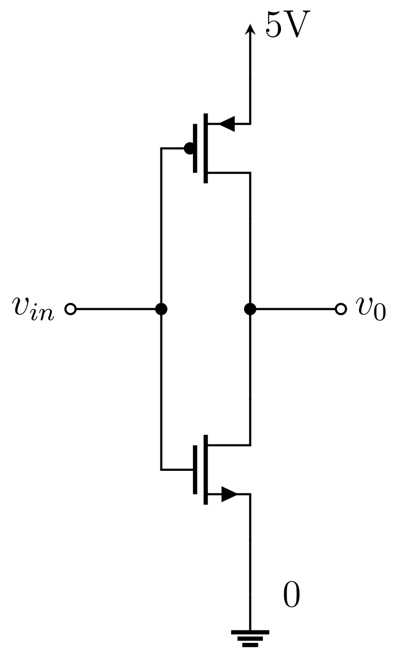
$$0.061\,\text{V} \le 4\,\text{V}$$

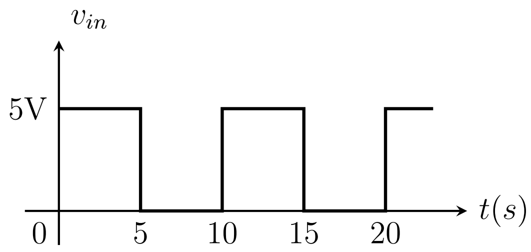
The condition is strictly satisfied. The assumption is correct; Q_p is indeed in the Triode region.

Final Answers

- **(i)** $\mathbf{i_{Dn}} = \mathbf{0}\,\text{mA}$
- **(ii)** $\mathbf{i_{Dp}} = \mathbf{0.244}\,\text{mA}$
- **(iii)** $\mathbf{v_o} = \mathbf{2.439}\,\text{V}$

Q2. For the CMOS inverter, draw V_o for the given input V_{in} waveform.





(15 marks) **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. CMOS Inverter Operation

A standard CMOS inverter is composed of a PMOS transistor (pull-up network) and an NMOS transistor (pull-down network). The output depends directly on the input state:

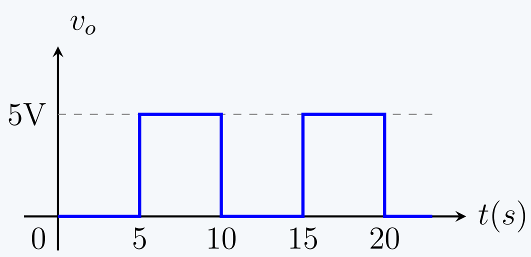
- **Logic HIGH Input (5V):** The NMOS transistor turns ON, providing a path to ground, while the PMOS transistor turns OFF. The output node is pulled down to 0V.
- **Logic LOW Input (0V):** The PMOS transistor turns ON, providing a path to the 5V supply, while the NMOS transistor turns OFF. The output node is pulled up to 5V.

2. Waveform Analysis

Because it acts as a logical inverter, the output waveform v_o will simply be the exact logical inversion of the input waveform v_{in}:

- From 0s to 5s: Input is 5V $\rightarrow$ Output is 0V
- From 5s to 10s: Input is 0V $\rightarrow$ Output is 5V
- From 10s to 15s: Input is 5V $\rightarrow$ Output is 0V
- From 15s to 20s: Input is 0V $\rightarrow$ Output is 5V
- Greater than 20s: Input is 5V $\rightarrow$ Output is 0V

Output Waveform (v_o vs t)



Digital vs. Analog Systems

Q1. Why is digital processing of data advantageous over analog processing? **(6 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

Advantages of Digital Processing over Analog Processing

While the real physical world is inherently analog (continuous in time and amplitude), processing data digitally (discrete in time and amplitude) has become the dominant paradigm in modern electronics. The fundamental advantage of digital processing stems from representing information as discrete binary states (0s and 1s) rather than continuous voltage or current levels.
The primary advantages of digital processing over analog processing are detailed below:

(a) **Superior Noise Immunity and Signal Regeneration:**

- *Principle:* In an analog system, any added electrical noise becomes an inseparable part of the signal, degrading the signal-to-noise ratio (SNR) permanently. In a digital system, signals are represented by voltage ranges (e.g., 0 to 0.8 V for Logic 0, and 2.0 to 5.0 V for Logic 1).
- *Advantage:* As long as the noise amplitude does not exceed the defined *noise margins*, the original discrete logic level can be perfectly regenerated and completely stripped of noise at each processing stage.

(b) **Reproducibility and Stability:**

- *Principle:* Analog circuits are highly sensitive to component tolerances, temperature variations, power supply fluctuations, and component aging.
- *Advantage:* Digital circuits operate on logic thresholds. A digital calculation will yield the exact same result every time, independent of minor temperature drifts or component aging, ensuring absolute reproducibility across millions of manufactured devices.

(c) **Programmability and Flexibility:**

- *Principle:* Analog signal processing requires dedicated hardware (specific resistor, capacitor, and op-amp configurations) to perform a specific function (e.g., filtering).
- *Advantage:* Digital Signal Processing (DSP) is primarily software-driven. The same physical microprocessor or DSP chip can be reprogrammed to act as a low-pass filter, a high-pass filter, or a complex audio equalizer simply by updating its code, without altering the physical circuitry.

(d) **Error Detection and Correction:**

- *Principle:* Analog signals have no inherent way to verify their own integrity.
- *Advantage:* Digital data can be supplemented with redundant information (e.g., parity bits, Cyclic Redundancy Checks (CRC), and Hamming codes). This allows digital systems to autonomously detect and mathematically correct data corruption that occurs during transmission or storage.

(e) **Data Storage and Retrieval:**

- *Principle:* Analog storage media (like magnetic tape or vinyl) degrade slightly with every read/write cycle and over time.
- *Advantage:* Digital data can be stored in solid-state drives, RAM, or optical media with zero degradation over time. Data can be copied and retrieved an infinite number of times with 100% perfect fidelity.

(f) **Ease of Multiplexing and Transmission:**

- *Advantage:* Digital signals are highly amenable to Time-Division Multiplexing (TDM) and packet-switching networks (like the Internet). Furthermore, digital data can be easily compressed (to save bandwidth) and heavily encrypted (using cryptographic algorithms) for secure transmission, which is extraordinarily difficult and less effective in the analog domain.

(g) **Miniaturization and Cost (VLSI/ULSI Integration):**

- *Advantage:* Digital circuits, primarily consisting of simple switching transistors (like CMOS), scale down incredibly well. Modern fabrication technologies allow billions of identical digital transistors to be packed onto a single silicon chip at a microscopic scale, drastically reducing the cost, power consumption, and physical footprint per computational operation compared to equivalent analog networks.

Summary

Analog processing is generally restricted to the very front-end (sensors, ADCs) and back-end (DACs, actuators) of modern systems to interface with the physical world. The core processing, however, is executed digitally due to its unmatched robustness, flexibility, scalability, and exact reproducibility.

Q2. Why do we need to miniaturize a transistor as much as possible? If you continue to miniaturize the transistor (switch), what problems may surface? **(5 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. The Need for Transistor Miniaturization (Scaling)

The continuous reduction in the physical dimensions of transistors, primarily driven by Moore's Law, is motivated by several fundamental advantages in digital and analog circuit design. The primary reasons for miniaturization include:

- **Increased Device Density (Functionality):** Smaller transistors allow for a higher number of devices to be packed onto a single silicon die. This enables the integration of more complex architectures, more memory (cache), and multi-core processors on a single chip, drastically increasing computational power.

- **Higher Switching Speed (Performance):** The transit time (t_r) for charge carriers (electrons or holes) to cross the channel from source to drain is directly proportional to the square of the channel length (L) in the low-field regime:

$$t_r \approx \frac{L^2}{\mu V_{DS}} \tag{1}$$

  Reducing L decreases the transit time, inherently increasing the maximum operating frequency (f_T) of the transistor. Furthermore, smaller devices have lower parasitic capacitances, which allows them to charge and discharge faster.

- **Reduced Dynamic Power Consumption:** Dynamic power dissipation in CMOS circuits is given by:

$$P_{dynamic} = \alpha \cdot f \cdot C_{load} \cdot V_{DD}^2 \tag{2}$$

  Miniaturization reduces the physical area, which decreases the gate and parasitic node capacitances (C_{load}). Historically, scaling rules (like Dennard Scaling) also dictated a reduction in the supply voltage (V_{DD}), leading to a quadratic reduction in dynamic power per gate.

- **Lower Cost per Transistor:** By making transistors smaller, more integrated circuits (dies) can be fabricated on a single silicon wafer. Since the cost of processing a wafer is relatively constant, the cost per chip—and therefore the cost per transistor—decreases significantly.

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2. Problems and Challenges of Continuous Miniaturization

While scaling down provides immense benefits, pushing the dimensions into the deep-submicron and nanometer regimes introduces severe physical and electrical limitations. The major problems that surface include:

- **Short-Channel Effects (SCE):** As the channel length L becomes comparable to the depletion widths of the source and drain junctions, the gate loses its dominant control over the channel potential. The source and drain electric fields begin to interfere with the gate's ability to turn the transistor off completely.

- **Drain-Induced Barrier Lowering (DIBL):** A specific short-channel effect where a high drain voltage (V_{DS}) lowers the potential barrier between the source and the channel. This allows electrons to flow from source to drain even when the gate voltage is below the threshold voltage, effectively reducing the effective threshold voltage (V_{th}) and increasing leakage.

- **Subthreshold Leakage Current:** To maintain performance at lower supply voltages, the threshold voltage (V_{th}) must also be scaled down. However, subthreshold current follows an exponential relationship:

$$I_{sub} \propto \exp\left(\frac{V_{GS} - V_{th}}{nV_T}\right) \tag{3}$$

  Lowering V_{th} results in an exponential increase in leakage current when the transistor is in the OFF state, leading to massive static power dissipation.

- **Gate Oxide Tunneling (Leakage):** To maintain gate control over a shorter channel, the gate oxide thickness (t_{ox}) must be scaled down proportionally. When SiO_2 thickness falls below 2 nm (only a few atomic layers thick), charge carriers can pass through the dielectric via *Quantum Mechanical Tunneling*, creating a severe gate leakage current. (This necessitated the transition to High-k dielectrics and metal gates).

- **Velocity Saturation and Mobility Degradation:** In very short channels, the longitudinal electric field ($E = V_{DS}/L$) becomes extremely high. Carriers reach their maximum drift velocity (velocity saturation) earlier, meaning current does not increase quadratically with gate voltage anymore, but rather linearly. High vertical electric fields also pull carriers toward the Si-SiO_2 interface, causing surface scattering and reducing carrier mobility (μ).

- **Thermal and Power Density Issues (Dark Silicon):** Although power per transistor decreases, the number of transistors per unit area increases at a faster rate. This leads to an overall increase in power density (W/cm^2). Dissipating this heat becomes physically impossible with standard cooling, forcing parts of the chip to be powered down during operation (a phenomenon known as "Dark Silicon").

- **Process Variations and Random Dopant Fluctuations (RDF):** At the nanometer scale, exact manufacturing becomes impossible. A variation of just a few atoms in the channel region can drastically alter the threshold voltage. Random placement of dopant atoms (RDF) and edge roughness cause huge variations in performance between adjacent transistors on the same chip.

- **Interconnect RC Delay:** As transistors shrink and speed up, the metal wires connecting them must also become thinner and closer together. This increases both wire resistance (R) and inter-wire parasitic capacitance (C). In modern nodes, the RC delay of the interconnects limits the overall speed of the chip more than the intrinsic delay of the transistors themselves.

Q3. Write a script in C language that takes a sinusoidal input and provides a full-wave rectified output. **(5 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. Mathematical Modeling of a Full-Wave Rectifier

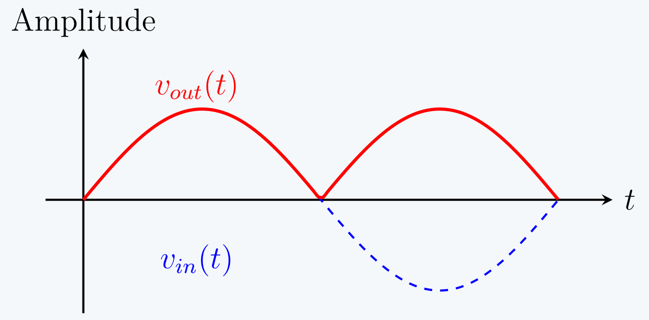
In electronic circuits, an ideal full-wave rectifier converts the entire input waveform to one of constant polarity at its output. If the input is a continuous sinusoidal signal, the mathematical representation is simply the absolute value of the input.
Let the input sinusoidal voltage be:

$$v_{in}(t) = V_p \sin(2\pi f t) \tag{1}$$

where V_p is the peak amplitude and f is the frequency of the signal.
The ideal full-wave rectified output is:

$$v_{out}(t) = |v_{in}(t)| = |V_p \sin(2\pi f t)| \tag{2}$$

For a practical implementation, a bridge rectifier experiences a voltage drop across two forward-biased diodes (approximately $2V_D$). A practical software model could include this threshold, but for general digital signal processing or simulation of the rectification concept, the absolute value function (`fabs()` in C) is sufficient.



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2. C Language Implementation

The following C script generates discrete time steps, computes the corresponding sinusoidal input values, applies full-wave rectification using the `math.h` library's absolute value function, and prints the tabulated results.

```
#include <stdio.h>
#include <math.h>

#define PI 3.14159265358979323846
#define FREQUENCY 50.0        // Signal frequency in Hz
#define AMPLITUDE 10.0        // Peak voltage in Volts
#define SAMPLING_RATE 1000.0 // Samples per second
#define NUM_SAMPLES 40        // Total number of samples to generate

int main() {
    double t;      // Time variable
    double v_in;   // Input voltage
    double v_out;  // Rectified output voltage

    // Print table header
    printf("Time (ms) \t V_in (V) \t V_out (V)\n");
    printf("------------------------------------------\n");

    // Generate and process the signal
    for (int i = 0; i <= NUM_SAMPLES; i++) {
        // Calculate current time step in seconds
        t = i / SAMPLING_RATE;

        // Generate sinusoidal input
        v_in = AMPLITUDE * sin(2.0 * PI * FREQUENCY * t);

        // Perform ideal full-wave rectification
        v_out = fabs(v_in);

        // Print results (Time converted to ms for readability)
        printf("%8.1f \t %8.4f \t %8.4f\n", t * 1000.0, v_in, v_out);
    }

    return 0;
}
```

3. Code Explanation

- **Constants:** The signal frequency is defined as 50 Hz with a peak amplitude of 10 V. The sampling rate dictates the time resolution of our discrete signal.

- **Loop Execution:** The `for` loop iterates through each sample index `i`, mapping it to a continuous time equivalent $t = i/f_s$.
- **Signal Generation:** The function `sin()` calculates the instantaneous amplitude of the input sine wave at time t.
- **Rectification:** The `fabs()` (float absolute) function strips the negative sign from the input, effectively folding the negative half-cycles of the sine wave into the positive domain, mirroring the action of a diode bridge.

Q4. How does a computer program rectify the sinusoidal input mentioned above? Is there any diode inside the processor? If not, what makes that act of rectification possible? **(10 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. Digital Rectification vs. Analog Rectification

In traditional analog electronics, rectification of a sinusoidal signal is achieved using semiconductor diodes, which physically block the flow of reverse current due to their non-linear I-V characteristics. However, a **computer program** does not operate on continuous physical voltages; it operates on discrete numerical data.
To process an analog sinusoidal signal in a computer, the signal must first be sampled and quantized by an **Analog-to-Digital Converter (ADC)**. The ADC converts the continuous voltage waveform, $v(t)$, into a discrete-time sequence of binary numbers, $x[n]$. The act of "rectification" is then performed purely through **mathematical operations** on these numbers.

2. Are there Diodes Inside the Processor?

No, physical diodes are not used to perform rectification inside a processor. While a modern microprocessor contains billions of transistors (MOSFETs) which inherently possess parasitic body diodes, these diodes are never used for analog signal processing. The processor is entirely composed of digital logic gates (AND, OR, NOT, etc.) forming structural blocks like the **Arithmetic Logic Unit (ALU)**. The processor treats the sinusoidal signal merely as a stream of binary numbers (often in two's complement format).

3. What Makes Software Rectification Possible?

The rectification is made possible by the **Arithmetic Logic Unit (ALU)** executing specific logical and mathematical instructions defined by the computer program. Since the sinusoidal signal is represented numerically, where negative voltages are typically represented as negative integers (via the sign bit), rectification is reduced to simple conditional logic or arithmetic:

- **Half-Wave Rectification:** The program inspects the sign bit of each data sample $x[n]$. If the sample is negative, it is overwritten with zero. If it is positive, it is kept unchanged.

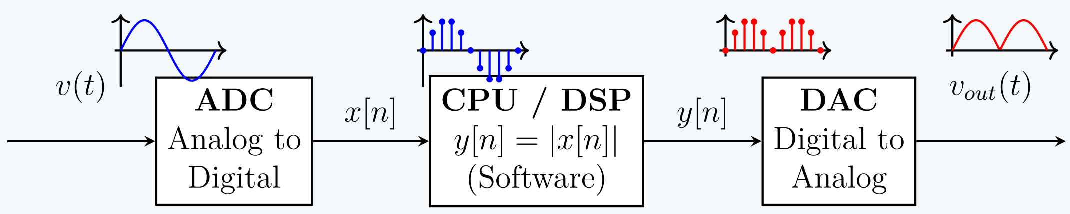
$$y[n] = \begin{cases} x[n], & \text{if } x[n] \geq 0 \\ 0, & \text{if } x[n] < 0 \end{cases} \tag{1}$$

- **Full-Wave Rectification:** The program computes the absolute value of the incoming data sample. At the machine level, if the sign bit indicates a negative number, the ALU takes the two's complement (inverts all bits and adds 1) to make it positive.

$$y[n] = |x[n]| = \begin{cases} x[n], & \text{if } x[n] \geq 0 \\ -x[n], & \text{if } x[n] < 0 \end{cases} \tag{2}$$

4. Block Diagram of Digital Rectification

The following system block diagram illustrates how a physical continuous sine wave is mathematically rectified without the use of physical diodes.



Summary

In a digital system, the act of **rectification is abstracted from physics to mathematics**. The physical non-linear characteristics of a PN junction (diode) are replaced by the logical branching and arithmetic capability of the microprocessor's ALU processing discrete digital samples.

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

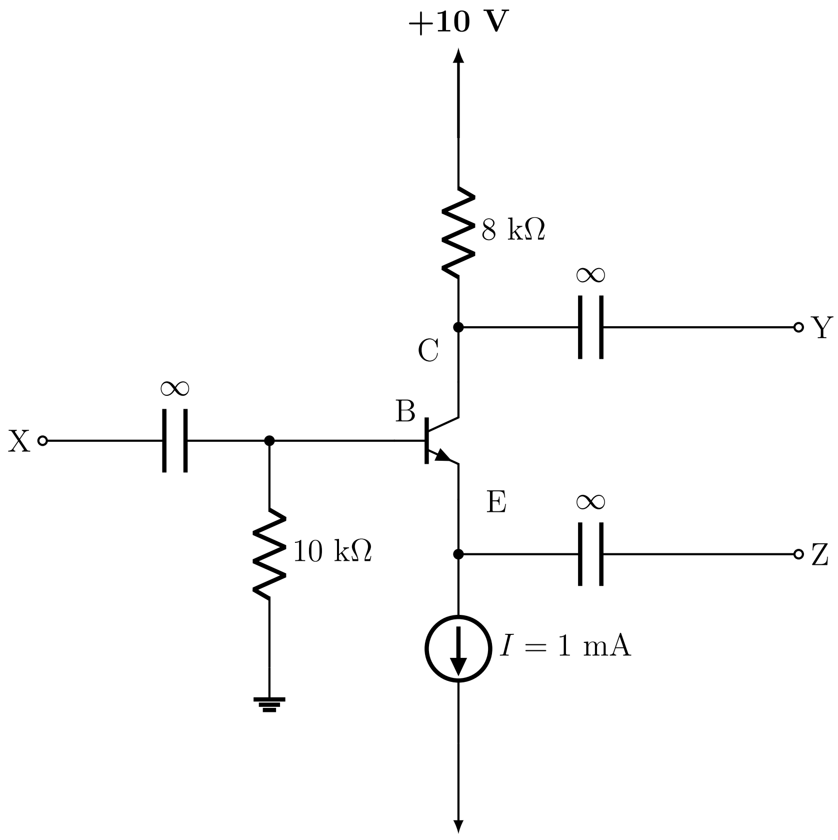
Part III — Amplifiers

BJT and MOSFET amplifier configurations, small-signal analysis, gain derivation

S5 BJT Amplifiers

Common Emitter / Trans-conductance Amplifier

Q1. The transistor shown is biased with a constant current $I = 1\,\text{mA}$ and has $\beta = 100$ and $V_A = 100\,\text{V}$. (i) Find the DC voltages at the Base, Emitter, and Collector terminals. (ii) Find the small-signal equivalent circuit parameters r_o, r_π, g_m and draw the hybrid-π model. (iii) If the Z terminal is connected to ground, X is connected to v_{sig} through $R_{sig} = 2\,\text{k}\Omega$, and Y is connected to $R_L = 8\,\text{k}\Omega$, find the overall voltage gain v_y/v_{sig}.

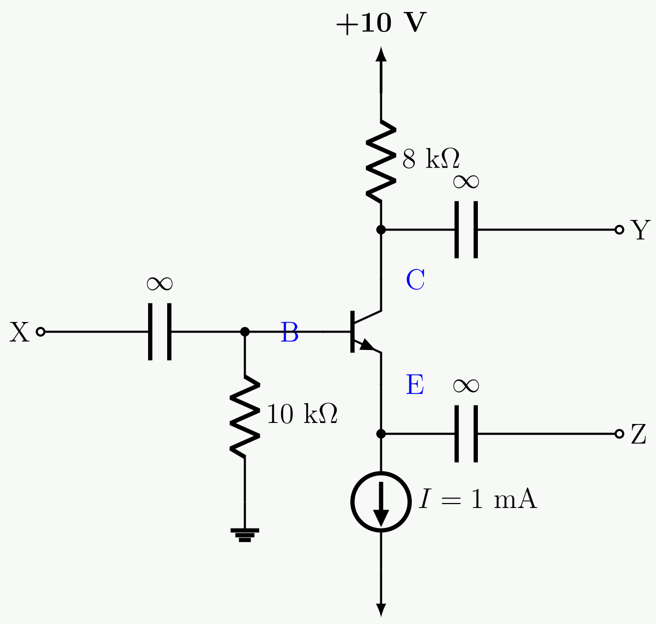


(20 marks) [EEE 263 | L-2/T-1 | 2024–25]

Answer

1. Original Circuit Diagram

Below is the redrawn schematic of the given BJT amplifier circuit.



2. Part (i): DC Analysis

For DC analysis, all coupling capacitors are treated as open circuits. The circuit operates with a constant emitter current sink of $I = 1$ mA. **Assumptions:** * Forward-active region ($V_{BE(on)} \approx 0.7$ V). * Thermal voltage $V_T \approx 25$ mV.

The emitter current is fixed:

$$I_E = 1 \text{ mA} \tag{1}$$

Using the relationship between terminal currents and current gain $\beta = 100$:

$$I_C = \alpha I_E = \left(\frac{\beta}{\beta+1}\right) I_E = \left(\frac{100}{101}\right)(1 \text{ mA}) \approx 0.99 \text{ mA}$$
$$I_B = \frac{I_E}{\beta+1} = \frac{1 \text{ mA}}{101} \approx 9.9\mu\text{A}$$

Now, applying Ohm's law to find the DC node voltages:

- **Base Voltage (V_B):** The base current I_B flows from ground through the 10 kΩ base resistor into the base terminal.

$$V_B = 0 - I_B R_B = -(9.9\mu\text{A})(10\text{ k}\Omega) = -0.099\text{ V} \tag{2}$$

- **Emitter Voltage (V_E):** Found using the base-emitter drop.

$$V_E = V_B - V_{BE(on)} = -0.099\text{ V} - 0.7\text{ V} = -0.799\text{ V} \tag{3}$$

- **Collector Voltage (V_C):** Determined by the voltage drop across the collector resistor.

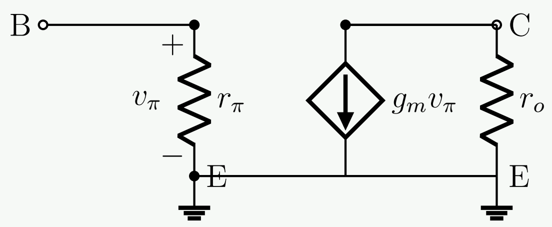
$$V_C = V_{CC} - I_C R_C = 10\text{ V} - (0.99\text{ mA})(8\text{ k}\Omega) = 10 - 7.92 = 2.08\text{ V} \tag{4}$$

Verification of operating region: $V_{CE} = V_C - V_E = 2.08\text{ V} - (-0.799\text{ V}) = 2.879$ V. Since $V_{CE} > 0.2$ V and $V_C > V_B$ (collector-base junction is reverse-biased), the assumption that the BJT is in the **forward-active region is correct**.

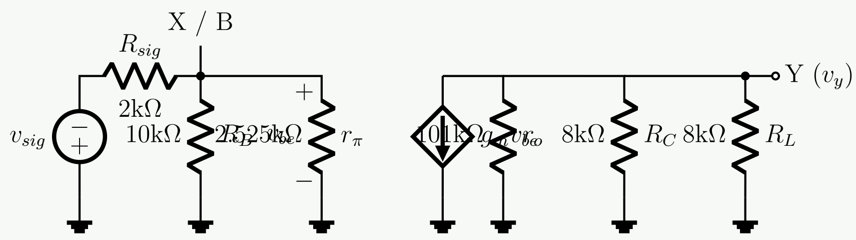
3. Part (ii): Small-Signal Equivalent Circuit Parameters
Using the DC collector current ($I_C = 0.99$ mA) and Early voltage ($V_A = 100$ V), we calculate the small-signal parameters for the Hybrid-π model.

$$\begin{aligned}
\text{Transconductance:} \quad & g_m = \frac{I_C}{V_T} = \frac{0.99\text{ mA}}{25\text{ mV}} = 39.6\text{ mA/V (or 39.6 mS)} \\
\text{Input Resistance:} \quad & r_\pi = \frac{\beta}{g_m} = \frac{100}{39.6\text{ mA/V}} \approx 2.525\text{ k}\Omega \\
\text{Output Resistance:} \quad & r_o = \frac{V_A}{I_C} = \frac{100\text{ V}}{0.99\text{ mA}} \approx 101\text{ k}\Omega
\end{aligned}$$

Hybrid-π Model Diagram:



4. Part (iii): Overall Voltage Gain (v_y/v_{sig})
Under these conditions, the circuit becomes a **Common-Emitter (CE) Amplifier**. * Z is connected to ground, so the emitter is AC grounded. * X is connected to v_{sig} through $R_{sig} = 2$ kΩ. * Y is connected to a load $R_L = 8$ kΩ. * The infinite capacitors act as short circuits for AC signals. * DC supplies (+10V and current source) are deactivated (shorted to ground and opened, respectively).
Small-Signal Equivalent Circuit of the Amplifier:



Mathematical Derivation of Gain:
1. *Input Transmission (Voltage Divider at the Base):* The input resistance seen looking into the base is $R_{in} = R_B \parallel r_\pi$.

$$R_{in} = 10\text{ k}\Omega \parallel 2.525\text{ k}\Omega = \frac{10 \times 2.525}{10 + 2.525} = \frac{25.25}{12.525} \approx 2.016\text{ k}\Omega \tag{5}$$

The voltage at the base v_b (which is v_{be}) in terms of v_{sig} is:

$$\frac{v_{be}}{v_{sig}} = \frac{R_{in}}{R_{sig} + R_{in}} = \frac{2.016}{2 + 2.016} = \frac{2.016}{4.016} \approx 0.502\text{ V/V} \tag{6}$$

2. *Output Gain from Base to Collector:* The total equivalent resistance at the output node is $R_{out} = r_o \parallel R_C \parallel R_L$.

$$R_{out} = 101\text{ k}\Omega \parallel 8\text{ k}\Omega \parallel 8\text{ k}\Omega = 101\text{ k}\Omega \parallel 4\text{ k}\Omega = \frac{101 \times 4}{101 + 4} = \frac{404}{105} \approx 3.848\text{ k}\Omega \tag{7}$$

Applying Kirchhoff's Current Law at the output node, the output voltage v_y is:

$$v_y = -(g_m v_{be}) \cdot R_{out} \tag{8}$$

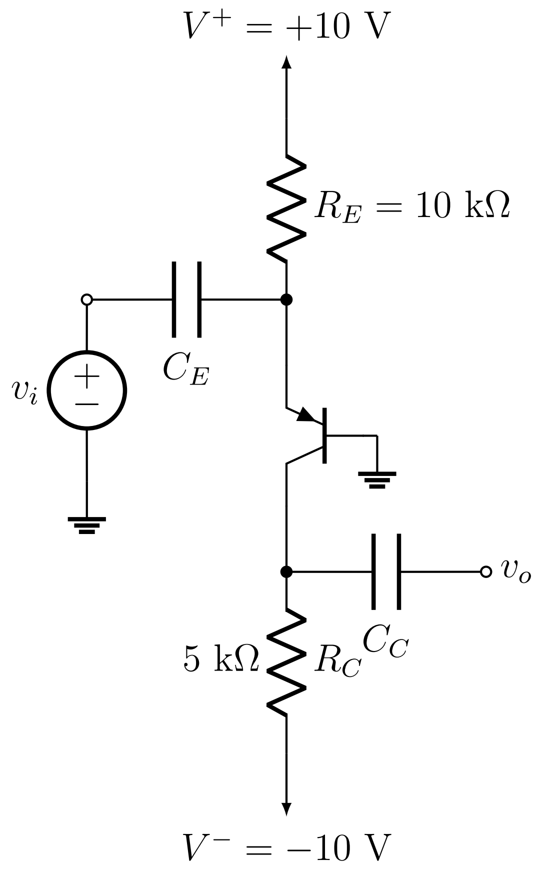
$$\frac{v_y}{v_{be}} = -g_m R_{out} = -(39.6\text{ mA/V})(3.848\text{ k}\Omega) \approx -152.38\text{ V/V} \tag{9}$$

3. *Overall System Voltage Gain (G_v):* The overall voltage gain is the product of the input transmission and the intrinsic gain.

$$\frac{v_y}{v_{sig}} = \left(\frac{v_y}{v_{be}}\right)\left(\frac{v_{be}}{v_{sig}}\right) = (-152.38) \times (0.502) \approx -76.5\text{ V/V} \tag{10}$$

Final Answer: The overall voltage gain v_y/v_{sig} is approximately -76.5 **V/V**.

Q2. Determine the voltage gain v_0/v_i of the transistor amplifier shown in the figure. Assume $\beta = 100$.

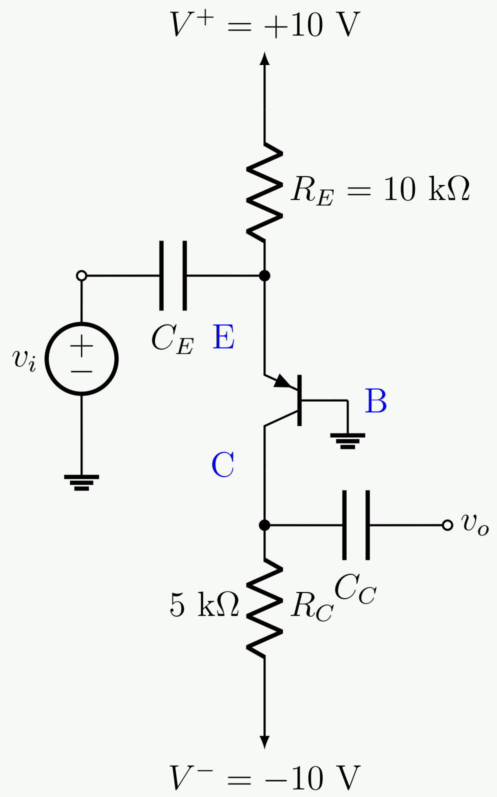


(26 marks) [EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Original Circuit Diagram

The given circuit is a Common-Base (CB) amplifier using a PNP bipolar junction transistor. Below is the redrawn schematic.



2. DC Analysis

To perform the DC analysis, we treat the coupling capacitors C_E and C_C as open circuits. This isolates the biasing circuit from the AC input and output. **Assumptions:**

- The transistor operates in the forward-active region.
- The base-emitter ON voltage for the PNP transistor is $V_{EB(on)} = 0.7$ V.
- Thermal voltage $V_T \approx 25$ mV at room temperature.
- Early effect is neglected ($V_A = \infty \implies r_o = \infty$).

Since the base is directly grounded, the DC base voltage is $V_B = 0$ V. The emitter voltage V_E is simply the base voltage plus the base-emitter voltage drop:

$$V_E = V_B + V_{EB(on)} = 0 \text{ V} + 0.7 \text{ V} = 0.7 \text{ V} \tag{1}$$

The DC emitter current I_E is driven by the positive supply V^+ across the emitter resistor R_E:

$$I_E = \frac{V^+ - V_E}{R_E} = \frac{10 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = \frac{9.3 \text{ V}}{10 \text{ k}\Omega} = 0.93 \text{ mA} \tag{2}$$

Using the given current gain $\beta = 100$, we determine the common-base current gain α:

$$\alpha = \frac{\beta}{\beta+1} = \frac{100}{101} \approx 0.9901 \tag{3}$$

The DC collector current I_C is:

$$I_C = \alpha I_E = (0.9901)(0.93 \text{ mA}) \approx 0.9208 \text{ mA} \tag{4}$$

Verification of Active Region: The collector voltage $V_C = V^- + I_C R_C = -10 + (0.9208 \text{ mA})(5 \text{ k}\Omega) = -5.396$ V. Since $V_C < V_B$, the collector-base junction is reverse-biased, confirming forward-active operation.

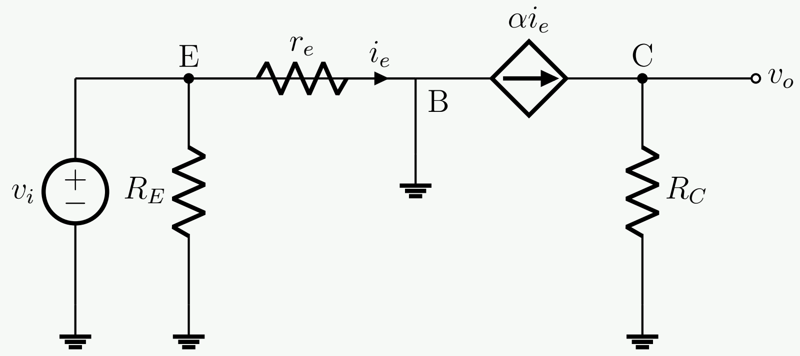
3. Small-Signal Equivalent Circuit

For the AC analysis, the DC voltage sources (V^+ and V^-) are shorted to ground, and the capacitors (C_E and C_C) act as short circuits. We use the small-signal T-model, which is highly convenient for Common-Base configurations.

The dynamic emitter resistance r_e is:

$$r_e = \frac{V_T}{I_E} = \frac{25 \text{ mV}}{0.93 \text{ mA}} \approx 26.88\,\Omega \tag{5}$$

Below is the small-signal T-model equivalent circuit:



4. Voltage Gain Derivation

From the small-signal circuit, the input voltage source v_i is connected directly in parallel with R_E. Because it is an ideal voltage source, the AC voltage at the emitter node is strictly determined by the input:

$$v_e = v_i \tag{6}$$

The small-signal emitter current i_e flows from the emitter node to the grounded base node through the dynamic resistance r_e:

$$i_e = \frac{v_e - 0}{r_e} = \frac{v_i}{r_e} \tag{7}$$

The dependent current source pushes a current of αi_e out of the collector node. This current flows entirely through the collector resistor R_C into the AC ground, establishing the output voltage v_o:

$$v_o = (\alpha i_e) \cdot R_C \tag{8}$$

Substitute $i_e = v_i / r_e$ into the output voltage equation:

$$v_o = \alpha \left(\frac{v_i}{r_e}\right) R_C = \left(\frac{\alpha R_C}{r_e}\right) v_i \tag{9}$$

The voltage gain $A_v = v_o/v_i$ is therefore:

$$A_v = \frac{v_o}{v_i} = \frac{\alpha R_C}{r_e} \tag{10}$$

Alternative perspective using Transconductance (g_m): Noting that $g_m = \alpha/r_e = I_C/V_T$, the gain can also be written seamlessly as $A_v = g_m R_C$.

5. Calculation and Final Answer

Substituting the known values into our derived expression:

$$A_v = \frac{0.9901 \times 5000\,\Omega}{26.88\,\Omega}$$
$$A_v \approx \frac{4950.5}{26.88}$$
$$A_v \approx 184.17\,\text{V/V}$$

The overall voltage gain v_0/v_i is approximately $184.17\,\mathbf{V/V}$.

Q3. A CE amplifier utilizes a BJT with $\beta = 100$ and $V_A = 100\,$V, is biased at $I_C = 1\,$mA and has a collector resistance $R_C = 5\,\text{k}\Omega$. Find R_{in}, R_o, and A_{vo}. If the amplifier is fed with a signal source having a resistance of $5\,\text{k}\Omega$ and a load resistance $R_L = 5\,\text{k}\Omega$ is connected to the output terminal, find the resulting A_v and G_v. If v_π is to be limited to $5\,$mV, what are the corresponding v_{sig} and v_o with the load connected? **(20 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

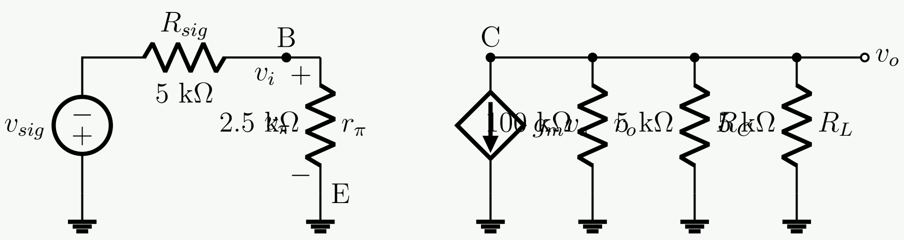
1. Assumptions and Small-Signal Parameters
Since no biasing resistors are specified in the problem statement, we assume an ideal Common-Emitter (CE) amplifier topology where the input is applied directly to the base and the emitter is grounded.

- Assume the thermal voltage is $V_T = 25\,\text{mV}$ at room temperature.
- The transistor is biased in the forward-active region with $I_C = 1\,\text{mA}$.

First, we determine the small-signal parameters for the hybrid-π model:

$$\text{Transconductance:} \quad g_m = \frac{I_C}{V_T} = \frac{1\,\text{mA}}{25\,\text{mV}} = 40\,\text{mA/V (or } 40\,\text{mS)}$$
$$\text{Input Resistance:} \quad r_\pi = \frac{\beta}{g_m} = \frac{100}{40\,\text{mA/V}} = 2.5\,\text{k}\Omega$$
$$\text{Output Resistance:} \quad r_o = \frac{V_A}{I_C} = \frac{100\,\text{V}}{1\,\text{mA}} = 100\,\text{k}\Omega$$

2. Small-Signal Equivalent Circuit
The complete small-signal model including the signal source (v_{sig}, R_{sig}) and the load (R_L) is shown below.



3. Open-Loop Amplifier Parameters (R_{in}, R_o, A_{vo})
These parameters are evaluated without the source resistance R_{sig} and without the load resistor R_L.

- **Input Resistance (R_{in}):** Looking into the base terminal, the only resistance seen is r_π.
$$R_{in} = r_\pi = 2.5\,\text{k}\Omega \tag{1}$$
- **Output Resistance (R_o):** Looking into the collector terminal (with $v_i = 0$ turning off the dependent source), we see R_C in parallel with r_o.
$$R_o = R_C \parallel r_o = \frac{5\,\text{k}\Omega \times 100\,\text{k}\Omega}{5\,\text{k}\Omega + 100\,\text{k}\Omega} = \frac{500}{105}\,\text{k}\Omega \approx 4.762\,\text{k}\Omega \tag{2}$$
- **Open-Loop Voltage Gain (A_{vo}):** This is the gain v_o/v_i without R_L. The output voltage is $v_o = -g_m v_\pi (R_C \parallel r_o)$ and $v_i = v_\pi$.
$$A_{vo} = \frac{v_o}{v_i} = -g_m R_o = -(40\,\text{mA/V})(4.762\,\text{k}\Omega) \approx -190.48\,\text{V/V} \tag{3}$$

4. Loaded Amplifier Parameters (A_v, G_v)
Now we include the load $R_L = 5\,\text{k}\Omega$ and the source resistance $R_{sig} = 5\,\text{k}\Omega$.

- **Loaded Voltage Gain (A_v):** The effective resistance at the output node becomes $R'_L = R_o \parallel R_L = R_C \parallel r_o \parallel R_L$.
$$R'_L = 4.762\,\text{k}\Omega \parallel 5\,\text{k}\Omega = \frac{4.762 \times 5}{4.762 + 5} = \frac{23.81}{9.762} \approx 2.439\,\text{k}\Omega \tag{4}$$
$$A_v = \frac{v_o}{v_i} = -g_m R'_L = -(40\,\text{mA/V})(2.439\,\text{k}\Omega) = -97.56\,\text{V/V} \tag{5}$$
- **Overall Voltage Gain (G_v):** This accounts for the voltage division between R_{sig} and R_{in}.
$$\frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = \frac{2.5\,\text{k}\Omega}{2.5\,\text{k}\Omega + 5\,\text{k}\Omega} = \frac{2.5}{7.5} = \frac{1}{3} \approx 0.333\,\text{V/V} \tag{6}$$
$$G_v = \frac{v_o}{v_{sig}} = A_v \left(\frac{v_i}{v_{sig}}\right) = (-97.56)\left(\frac{1}{3}\right) = -32.52\,\text{V/V} \tag{7}$$

5. Signal Limitations
For the small-signal approximation to hold, the base-emitter signal voltage must be small compared to the thermal voltage, restricted here to an amplitude of $v_\pi = 5\,\text{mV}$. Note that $v_i = v_\pi$.
To find the corresponding maximum source signal v_{sig}:

$$v_{sig} = v_\pi \left(\frac{R_{in} + R_{sig}}{R_{in}}\right) = 5\,\text{mV} \times \left(\frac{2.5 + 5}{2.5}\right) = 5\,\text{mV} \times 3 = 15\,\text{mV} \tag{8}$$

To find the corresponding output voltage v_o (using the loaded gain A_v):

$$v_o = A_v v_\pi = (-97.56\,\text{V/V}) \times (5\,\text{mV}) = -487.8\,\text{mV} \quad (\text{Peak Amplitude: } 487.8\,\text{mV}) \tag{9}$$

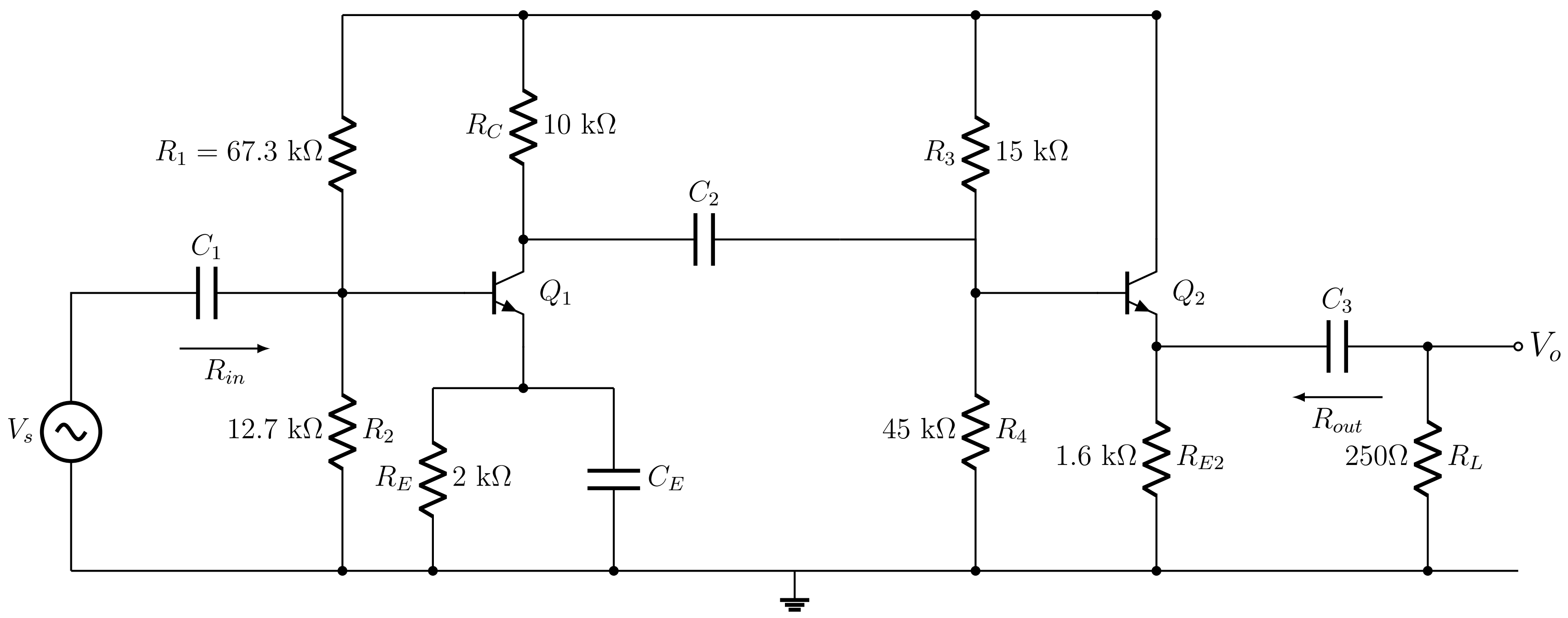
Alternatively, using G_v:

$$v_o = G_v v_{sig} = (-32.52\,\text{V/V}) \times (15\,\text{mV}) = -487.8\,\text{mV} \tag{10}$$

Final Answers:

- $R_{in} = \mathbf{2.5}\,\mathbf{k\Omega}$
- $R_o = \mathbf{4.762}\,\mathbf{k\Omega}$
- $A_{vo} = -\mathbf{190.48}\,\text{V/V}$
- $A_v = -\mathbf{97.56}\,\text{V/V}$
- $G_v = -\mathbf{32.52}\,\text{V/V}$
- For $v_\pi = 5\,\text{mV}$, the corresponding input is $v_{sig} = \mathbf{15}\,\text{mV}$ and output is $v_o = -\mathbf{487.8}\,\text{mV}$.

Q4. For the circuit shown, determine small-signal parameters g_m, r_π, and r_e for both transistors, the overall small-signal voltage gain $A_v = v_0/v_s$, input resistance R_{in}, and output resistance R_{out}. Here $\beta = 120$.



(36 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Assumptions and DC Analysis

The top power supply rail (V_{CC}) is not explicitly labeled in the schematic. Based on standard parameters for this specific circuit topology, we will assume $V_{CC} = 12\,\text{V}$. We will also assume standard values of $V_{BE(on)} = 0.7\,\text{V}$ and a thermal voltage of $V_T \approx 26\,\text{mV}$.

DC Analysis for Q1 (Stage 1):

- Base Thevenin voltage: $V_{BB1} = V_{CC}\frac{R_2}{R_1+R_2} = 12\,\text{V}\left(\frac{12.7\,\text{k}\Omega}{67.3\,\text{k}\Omega+12.7\,\text{k}\Omega}\right) = 1.905\,\text{V}$
- Base Thevenin resistance: $R_{B1} = R_1 \parallel R_2 = 67.3\,\text{k}\Omega \parallel 12.7\,\text{k}\Omega = 10.68\,\text{k}\Omega$
- Base current: $I_{B1} = \frac{V_{BB1}-V_{BE(on)}}{R_{B1}+(\beta+1)R_E} = \frac{1.905-0.7}{10.68\,\text{k}\Omega+(121)(2\,\text{k}\Omega)} = \frac{1.205\,\text{V}}{252.68\,\text{k}\Omega} \approx 4.769\,\mu\text{A}$
- Collector and Emitter currents:

$$I_{C1} = \beta I_{B1} = 120 \times 4.769\,\mu\text{A} = 0.572\,\text{mA}$$
$$I_{E1} = (\beta+1)I_{B1} = 121 \times 4.769\,\mu\text{A} = 0.577\,\text{mA}$$

DC Analysis for Q2 (Stage 2):

- Base Thevenin voltage: $V_{BB2} = V_{CC}\frac{R_4}{R_3+R_4} = 12\,\text{V}\left(\frac{45\,\text{k}\Omega}{15\,\text{k}\Omega+45\,\text{k}\Omega}\right) = 9\,\text{V}$
- Base Thevenin resistance: $R_{B2} = R_3 \parallel R_4 = 15\,\text{k}\Omega \parallel 45\,\text{k}\Omega = 11.25\,\text{k}\Omega$
- Base current: $I_{B2} = \frac{V_{BB2}-V_{BE(on)}}{R_{B2}+(\beta+1)R_{E2}} = \frac{9-0.7}{11.25\,\text{k}\Omega+(121)(1.6\,\text{k}\Omega)} = \frac{8.3\,\text{V}}{204.85\,\text{k}\Omega} \approx 40.52\,\mu\text{A}$
- Collector and Emitter currents:

$$I_{C2} = \beta I_{B2} = 120 \times 40.52\,\mu\text{A} = 4.862\,\text{mA}$$
$$I_{E2} = (\beta+1)I_{B2} = 121 \times 40.52\,\mu\text{A} = 4.903\,\text{mA}$$

2. Small-Signal Parameters (g_m, r_π, r_e)
Using the calculated DC operating points, we find the small-signal parameters for both transistors:
For Transistor Q1:

$$g_{m1} = \frac{I_{C1}}{V_T} = \frac{0.572\,\text{mA}}{26\,\text{mV}} = \mathbf{22.0}\,\text{mA/V}$$
$$r_{\pi1} = \frac{\beta}{g_{m1}} = \frac{120}{22.0\,\text{mA/V}} = \mathbf{5.45\,k\Omega}$$
$$r_{e1} = \frac{V_T}{I_{E1}} = \frac{26\,\text{mV}}{0.577\,\text{mA}} = \mathbf{45.1\,\Omega}$$

For Transistor Q2:

$$g_{m2} = \frac{I_{C2}}{V_T} = \frac{4.862\,\text{mA}}{26\,\text{mV}} = \mathbf{187.0}\,\text{mA/V}$$
$$r_{\pi2} = \frac{\beta}{g_{m2}} = \frac{120}{187.0\,\text{mA/V}} = \mathbf{642\,\Omega}\ (\text{or } \mathbf{0.642\,k\Omega})$$
$$r_{e2} = \frac{V_T}{I_{E2}} = \frac{26\,\text{mV}}{4.903\,\text{mA}} = \mathbf{5.3\,\Omega}$$

3. Input Resistance (R_{in})
R_{in} is the resistance looking into the first stage past coupling capacitor C_1. Because the emitter of Q1 is effectively grounded at AC (due to C_E), we only see R_1, R_2, and $r_{\pi1}$ in parallel:

$$R_{in} = R_1 \parallel R_2 \parallel r_{\pi1} = 67.3\,\text{k}\Omega \parallel 12.7\,\text{k}\Omega \parallel 5.45\,\text{k}\Omega = 10.68\,\text{k}\Omega \parallel 5.45\,\text{k}\Omega = \mathbf{3.61\,k\Omega} \tag{1}$$

4. Overall Small-Signal Voltage Gain (A_v)
The overall gain is the product of the first stage (Common Emitter) and second stage (Emitter Follower) gains: $A_v = A_{v1} \times A_{v2}$.
Stage 2 Gain ($A_{v2} = v_o/v_{c1}$):
The AC load on Q2's emitter is $R'_L = R_{E2} \parallel R_L = 1.6\,\text{k}\Omega \parallel 250\,\Omega = 216.2\,\Omega$.

$$A_{v2} = \frac{(\beta+1)R'_L}{r_{\pi2} + (\beta+1)R'_L} = \frac{121 \times 216.2\,\Omega}{642\,\Omega + (121 \times 216.2\,\Omega)} = \frac{26160\,\Omega}{26802\,\Omega} \approx 0.976\,\text{V/V} \tag{2}$$

Stage 1 Gain ($A_{v1} = v_{c1}/v_s$):
The total AC load on Q1's collector is R_C in parallel with the bias resistors of Q2 (R_3, R_4) and the base input resistance of Q2.

- Resistance looking into Q2 base: $R_{in2(base)} = r_{\pi2} + (\beta+1)R'_L = 642\,\Omega + 26160\,\Omega = 26.8\,\text{k}\Omega$
- Total AC collector load: $R_{L1} = R_C \parallel R_3 \parallel R_4 \parallel R_{in2(base)} = 10\,\text{k}\Omega \parallel 15\,\text{k}\Omega \parallel 45\,\text{k}\Omega \parallel 26.8\,\text{k}\Omega$
  $\to 15\,\text{k}\Omega \parallel 45\,\text{k}\Omega = 11.25\,\text{k}\Omega$
  $\to 10\,\text{k}\Omega \parallel 11.25\,\text{k}\Omega \parallel 26.8\,\text{k}\Omega = 5.294\,\text{k}\Omega \parallel 26.8\,\text{k}\Omega \approx 4.42\,\text{k}\Omega$

Since v_s connects directly to the base ($v_{b1} = v_s$), the stage 1 gain is:

$$A_{v1} = -g_{m1}R_{L1} = -(22.0\,\text{mA/V})(4.42\,\text{k}\Omega) = -97.2\,\text{V/V} \tag{3}$$

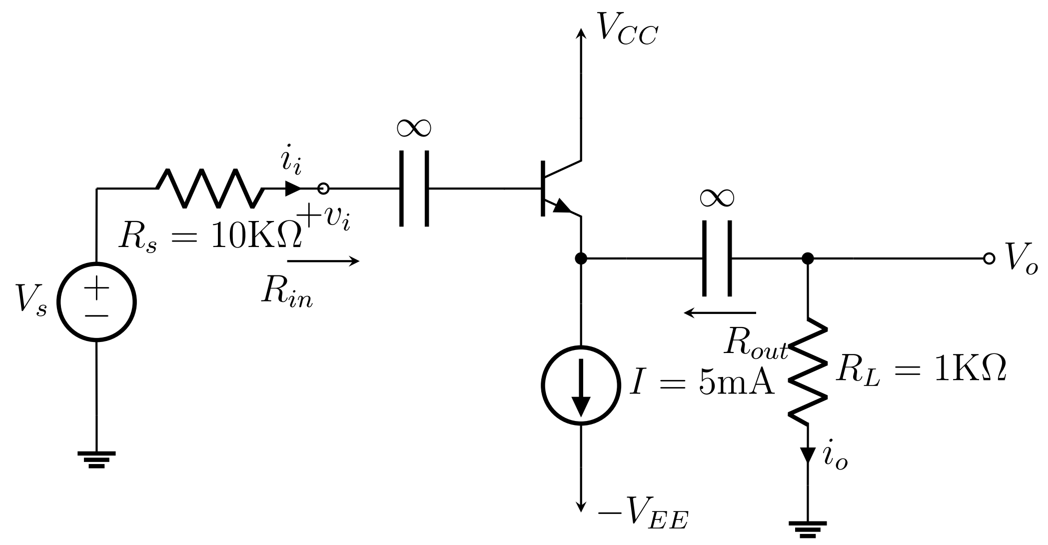
Total Gain:
$$A_v = A_{v1} \times A_{v2} = (-97.2) \times (0.976) = \mathbf{-94.9}\,\text{V/V} \tag{4}$$

5. Output Resistance (R_{out})
R_{out} is looking back into the amplifier from C_3, excluding the load R_L. It is R_{E2} in parallel with the resistance looking up into Q2's emitter.

- AC Thevenin resistance at Q2's base: $R_{th2} = R_C \parallel R_3 \parallel R_4 = 10\,\text{k}\Omega \parallel 15\,\text{k}\Omega \parallel 45\,\text{k}\Omega = 5.294\,\text{k}\Omega$
- Resistance looking into Q2 emitter: $R_{e2(out)} = \frac{r_{\pi2}+R_{th2}}{\beta+1} = \frac{0.642\,\text{k}\Omega+5.294\,\text{k}\Omega}{121} = \frac{5936\,\Omega}{121} \approx 49.1\,\Omega$

$$R_{out} = R_{E2} \parallel R_{e2(out)} = 1600\,\Omega \parallel 49.1\,\Omega = \mathbf{47.6\,\Omega} \tag{5}$$

Q5. For the circuit shown, determine R_{in}, R_{out}, V_o/V_i, and i_o/i_i. The BJT is biased with $I = 5\,\text{mA}$ and has $\beta = 100$ and $V_A = 100\,\text{V}$.



(20 marks) [EEE 263 | L-2/T-1 | 2014–15]

Answer

1. DC Analysis and Small-Signal Parameters

The circuit is an Emitter Follower (Common Collector) amplifier. The transistor is biased using an ideal DC current source at the emitter, which forces the DC emitter current to be constant. Assuming a standard thermal voltage $V_T = 25\,\text{mV}$ at room temperature. The DC emitter current is given as:

$$I_E = I = 5\,\text{mA}$$

Since $\beta = 100$, we can determine the common-base current gain α and the collector current I_C:

$$\alpha = \frac{\beta}{\beta+1} = \frac{100}{101} \approx 0.99$$

$$I_C = \alpha I_E = \left(\frac{100}{101}\right)(5\,\text{mA}) = 4.95\,\text{mA}$$

Now, we evaluate the small-signal parameters for the Hybrid-π model:

$$g_m = \frac{I_C}{V_T} = \frac{4.95\,\text{mA}}{25\,\text{mV}} = 198\,\text{mA/V}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{198\,\text{mA/V}} = 505\,\Omega$$

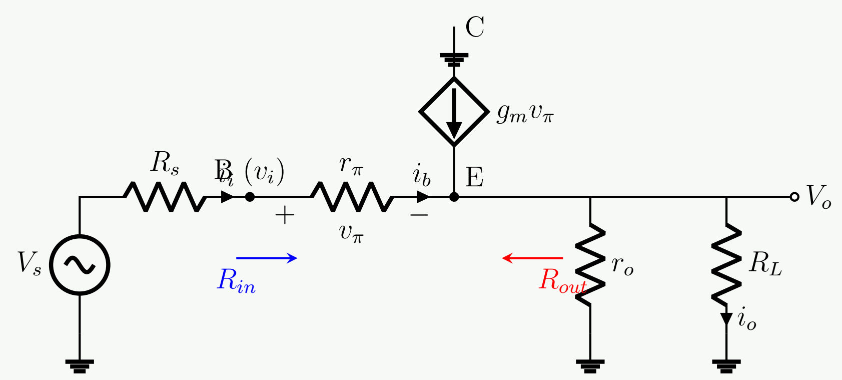
$$r_o = \frac{V_A}{I_C} = \frac{100\,\text{V}}{4.95\,\text{mA}} = 20.2\,\text{k}\Omega$$

Alternatively, the emitter resistance r_e can be calculated as:

$$r_e = \frac{V_T}{I_E} = \frac{25\,\text{mV}}{5\,\text{mA}} = 5\,\Omega \quad \Longrightarrow \quad r_\pi = (\beta+1)r_e = 101 \times 5\,\Omega = 505\,\Omega$$

2. Small-Signal Equivalent Circuit

For AC analysis, the DC voltage sources (V_{CC}, $-V_{EE}$) are replaced by short circuits to ground, and the ideal DC current source is replaced by an open circuit. The coupling capacitors act as AC short circuits.



3. AC Analysis and Parameter Calculations

Let the effective load resistance at the emitter be R'_L. This is the parallel combination of the output resistance r_o and the load resistor R_L:

$$R'_L = r_o \parallel R_L = 20.2\,\text{k}\Omega \parallel 1\,\text{k}\Omega = \frac{20.2 \times 1}{20.2+1}\,\text{k}\Omega = 0.9528\,\text{k}\Omega = 952.8\,\Omega$$

(a) Input Resistance (R_{in}):

R_{in} is the equivalent resistance looking into the base terminal. According to the resistance reflection rule, the impedance at the emitter is reflected to the base multiplied by $(\beta + 1)$.

$$R_{in} = r_\pi + (\beta+1)R'_L$$

$$R_{in} = 505\,\Omega + (101)(952.8\,\Omega) = 505\,\Omega + 96232.8\,\Omega = 96737.8\,\Omega \approx \mathbf{96.74}\,\mathbf{k\Omega}$$

(b) Voltage Gain (V_o/V_i):
The voltage V_i is defined directly at the base terminal. Using the voltage divider principle in the base-emitter loop:

$$V_o = i_e R'_L = (\beta+1)i_b R'_L$$
$$V_i = i_b r_\pi + (\beta+1)i_b R'_L = i_b[r_\pi + (\beta+1)R'_L] = i_b R_{in}$$

$$\frac{V_o}{V_i} = \frac{(\beta+1)R'_L}{r_\pi + (\beta+1)R'_L} = \frac{(101)(952.8)}{96737.8} = \frac{96232.8}{96737.8} \approx \mathbf{0.995}\,\text{V/V}$$

(c) Output Resistance (R_{out}):
R_{out} is the resistance looking back into the emitter terminal, with the input source V_s turned off (shorted to ground). The source resistance R_s appears in parallel with the base terminal. By reflecting R_s and r_π to the emitter side:

$$R_{out(emitter)} = r_o \parallel \left(\frac{r_\pi + R_s}{\beta+1}\right) = r_o \parallel \left(r_e + \frac{R_s}{\beta+1}\right)$$
$$\frac{r_\pi + R_s}{\beta+1} = \frac{505\,\Omega + 10000\,\Omega}{101} = \frac{10505}{101} = 104.01\,\Omega$$
$$R_{out} = 20200\,\Omega \parallel 104.01\,\Omega = \frac{20200 \times 104.01}{20200 + 104.01} = \frac{2100990}{20304.01} \approx \mathbf{103.5\,\Omega}$$

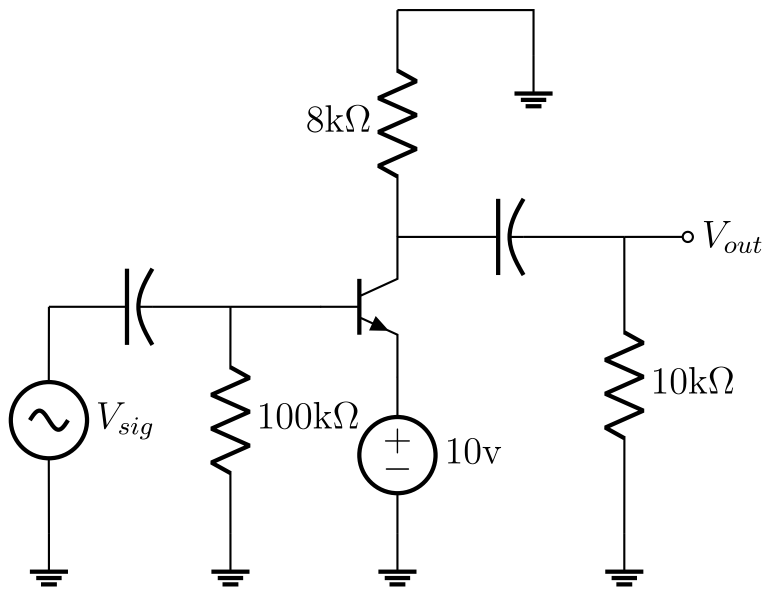
(d) Current Gain (i_o/i_i):
The current i_i entering the base is $i_i = V_i/R_{in}$. The output current flowing through the load is $i_o = V_o/R_L$. We can express the current gain in terms of the voltage gain:

$$\frac{i_o}{i_i} = \frac{V_o/R_L}{V_i/R_{in}} = \left(\frac{V_o}{V_i}\right)\left(\frac{R_{in}}{R_L}\right)$$
$$\frac{i_o}{i_i} = (0.99478) \times \left(\frac{96737.8\,\Omega}{1000\,\Omega}\right) = 0.99478 \times 96.7378 \approx \mathbf{96.2}\,\text{A/A}$$

Alternative method via current division: Emitter current $i_e = (\beta+1)i_i$. This current divides between r_o and R_L:

$$i_o = i_e\left(\frac{r_o}{r_o + R_L}\right) = (101 i_i)\left(\frac{20.2\,\text{k}\Omega}{20.2\,\text{k}\Omega + 1\,\text{k}\Omega}\right) = 101 i_i \left(\frac{20.2}{21.2}\right) \approx 96.23 i_i \implies \frac{i_o}{i_i} \approx \mathbf{96.2}\,\text{A/A}$$

Q6. Draw the small signal circuit diagram of the CE circuit shown. Why is the circuit called a "Small Signal" circuit? Determine the values of input resistance R_{in}, output resistance R_{out}, and voltage gain A_v. If $v_{sig} \approx 2\,\text{mV}$, find the value of V_{out}.



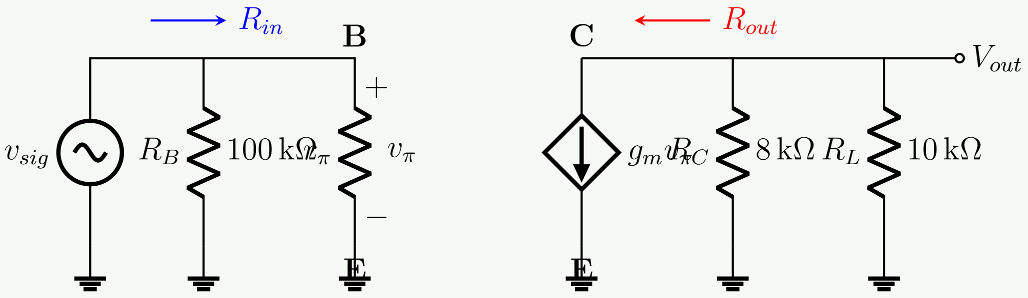
(36 marks) [EEE 263 | L-2/T-1 | 2013–14]

Answer

1. Why is the circuit called a "Small Signal" Circuit?
The circuit is termed a **Small Signal** circuit because it is designed to operate with an input AC signal amplitude that is sufficiently small (typically $v_{be} \ll V_T \approx 25\,\text{mV}$). Under this condition, the highly non-linear exponential voltage-current relationship of the BJT's base-emitter junction can be accurately approximated by a linear tangent line around its DC quiescent operating point (Q-point). This linearization allows the use of simple linear models (like the Hybrid-π model) and the superposition theorem to analyze the AC performance independently of the DC bias. The given input of $v_{sig} \approx 2\,\text{mV}$ perfectly satisfies this small-signal condition.

2. Small-Signal Equivalent Circuit
To construct the midband AC equivalent circuit:

- DC voltage sources (like the 10 V supply at the emitter) are replaced by AC short circuits to ground. Thus, the emitter terminal is grounded.
- DC coupling capacitors are assumed to be large enough to act as short circuits at the signal frequency.
- The BJT is replaced by its Hybrid-π equivalent model.



3. Mathematical Derivation of AC Parameters
Based on the small-signal diagram, we can derive the symbolic expressions for the amplifier's parameters.

- **Input Resistance (R_{in}):** Looking into the amplifier from the input coupling capacitor, the signal sees the base biasing resistor R_B in parallel with the transistor's input resistance r_π.

$$R_{in} = R_B \parallel r_\pi = 100\,\text{k}\Omega \parallel r_\pi \tag{1}$$

- **Output Resistance (R_{out}):** Looking back into the output terminal with v_{sig} deactivated (shorted), the dependent source $g_m v_\pi$ becomes an open circuit (since $v_\pi = 0$). Assuming the transistor's output resistance r_o is very large ($r_o \to \infty$):

$$R_{out} = R_C \parallel r_o \approx R_C = 8\,\text{k}\Omega \tag{2}$$

- **Voltage Gain (A_v):** The output voltage is produced by the dependent current source pulling current through the parallel combination of R_C and R_L.

$$V_{out} = -g_m v_\pi (R_C \parallel R_L)$$
$$v_\pi = v_{sig} \quad \text{(since the source is ideal and directly coupled)}$$
$$A_v = \frac{V_{out}}{v_{sig}} = -g_m(R_C \parallel R_L) = -g_m(8\,\text{k}\Omega \parallel 10\,\text{k}\Omega) = -g_m(4.44\,\text{k}\Omega)$$

4. Step-by-Step Numerical Solution
Assumption: To calculate numerical values, the exact DC quiescent current (I_C) and current gain (β) are required. Since these are not explicitly defined in the schematic, we will proceed by assuming standard, textbook typical values for a small-signal BJT: $\mathbf{I_C = 1}$ mA and $\boldsymbol{\beta} = \mathbf{100}$. We use the standard thermal voltage $\mathbf{V_T \approx 25}$ mV at room temperature.
First, determine the small-signal parameters:

$$g_m = \frac{I_C}{V_T} = \frac{1\,\text{mA}}{25\,\text{mV}} = 40\,\text{mA/V}$$
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{40\,\text{mA/V}} = 2.5\,\text{k}\Omega$$

Next, evaluate R_{in}, R_{out}, and A_v:

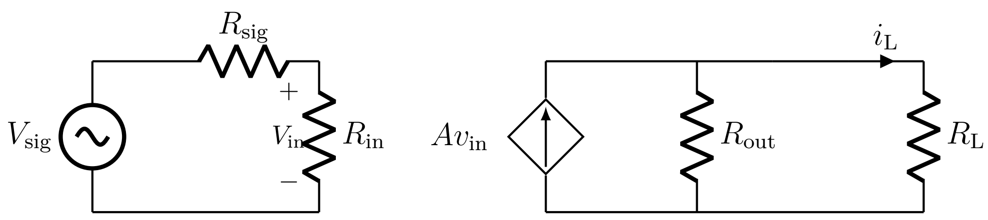
$$R_{in} = 100\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = \frac{100 \times 2.5}{100 + 2.5} = \frac{250}{102.5} \approx \mathbf{2.44}\,\mathbf{k\Omega}$$
$$R_{out} \approx \mathbf{8}\,\mathbf{k\Omega}$$
$$A_v = -g_m(R_C \parallel R_L) = -(40\,\text{mA/V}) \times (4.444\,\text{k}\Omega) \approx \mathbf{-177.8}\,\text{V/V}$$

Finally, calculate the output voltage V_{out} for the given input $v_{sig} = 2$ mV:

$$V_{out} = A_v \times v_{sig} = -177.8 \times 2\,\text{mV} = \mathbf{-355.6}\,\text{mV}$$

Note: The negative sign indicates a 180° phase inversion characteristic of the Common Emitter topology.

Q7. The figure shows the basic structure of a trans-conductance amplifier to be implemented by a BJT. Given $AV_{in} = I_c R_{out}$ along with $R_{sig} \approx 0$ and the load (R_L) can be much greater than R_{out}; which BJT configuration (common emitter, common base, or common collector) will you choose for best performance? Explain why. Comment on its gain and loading effect.



(20 marks) [EEE 263 | L-2/T-1 | 2012–13]

Answer

1. Transconductance Amplifier Characteristics

A transconductance amplifier is designed to convert an input voltage (v_{in}) into a proportional output current (i_{L}). The ideal characteristics for such an amplifier are:

- **Infinite Input Resistance ($R_{\text{in}} \to \infty$):** To prevent loading the input voltage source (so $V_{\text{in}} = V_{\text{sig}}$).
- **Infinite Output Resistance ($R_{\text{out}} \to \infty$):** To act as an ideal current source, ensuring all generated current flows through the load R_L regardless of its value.

Based on the small-signal model provided, the output current delivered to the load is determined by current division:

$$i_{\text{L}} = Av_{\text{in}}\left(\frac{R_{\text{out}}}{R_{\text{out}} + R_{\text{L}}}\right) \tag{1}$$

where A represents the transconductance gain (G_m) of the amplifier.

2. BJT Configuration Evaluation

We evaluate the three standard BJT configurations based on the given constraints ($R_{\text{sig}} \approx 0$ and $R_{\text{L}} \gg R_{\text{out}}$):

(a) **Common Collector (CC):** Acts as a voltage buffer (high R_{in}, low R_{out}). It produces a voltage output rather than a current output, making it completely unsuitable for a transconductance amplifier.

(b) **Common Base (CB):** Acts as a current buffer. It has very low input resistance ($R_{\text{in}} \approx r_e$). While $R_{\text{sig}} \approx 0$ mitigates input signal attenuation, the CB configuration inherently expects a current input, making it a poor choice for a voltage-controlled transconductance stage.

(c) **Common Emitter (CE):** Inherently functions as a transconductance device. The input voltage modulates the base-emitter junction (v_{be}), producing an output collector current ($i_c = g_m v_{\text{be}}$). It has moderate R_{in} (r_π) and moderate R_{out} (r_o).

3. Selection for Best Performance

The **Common Emitter (CE) configuration** (specifically with **Emitter Degeneration**) should be chosen.
Why? A standard BJT operates fundamentally as a transconductance device in the CE configuration. Since $R_{\text{sig}} \approx 0$, the moderate input resistance ($R_{\text{in}} = r_\pi$) of the CE stage will not cause any signal loss at the input (i.e., $V_{\text{in}} \approx V_{\text{sig}}$). However, to address the severe loading condition at the output, the basic CE stage must be modified.

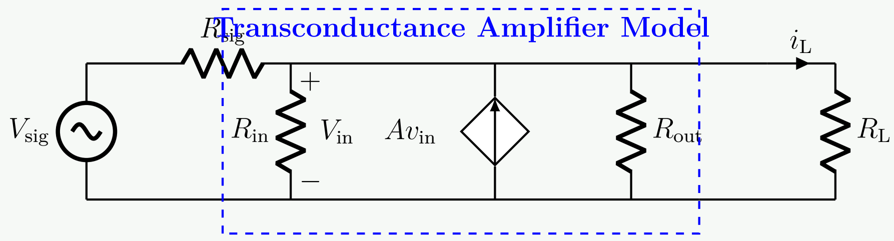
4. Loading Effect and Gain

Loading Effect: The problem states that R_{L} can be much greater than R_{out}. If a standard CE amplifier is used, its output resistance is just the transistor's early effect resistance ($R_{\text{out}} = r_o$). Because $R_{\text{L}} \gg r_o$, a severe output loading effect will occur:

$$i_{\text{L}} = g_m v_{\text{in}}\left(\frac{r_o}{r_o + R_{\text{L}}}\right) \approx g_m v_{\text{in}}\left(\frac{r_o}{R_{\text{L}}}\right) \to \text{Very small} \tag{2}$$

Most of the dependent current will shunt internally through R_{out} rather than reaching the load.
Gain and Performance Improvement: To improve performance and fix the loading effect, **Common Emitter with Emitter Degeneration** (an unbypassed emitter resistor R_E) should be used.

- **Effective Output Resistance:** Emitter degeneration significantly boosts the output resistance to $R_{\text{out}} \approx r_o(1 + g_m R_E)$. This helps satisfy the requirement to drive a large R_{L}.
- **Effective Gain:** The transconductance gain A stabilizes to $A = G_m \approx \frac{1}{R_E}$, sacrificing some raw gain for linearity and the critical increase in output impedance needed to combat the large R_L.



Q8. Consider the emitter degenerated CE circuit of Figure shown below

(i) Find the DC current I_C and DC voltage V_C, V_B and V_E. Assume $\beta = 100$, $V_{BE} = 0.7$ V.

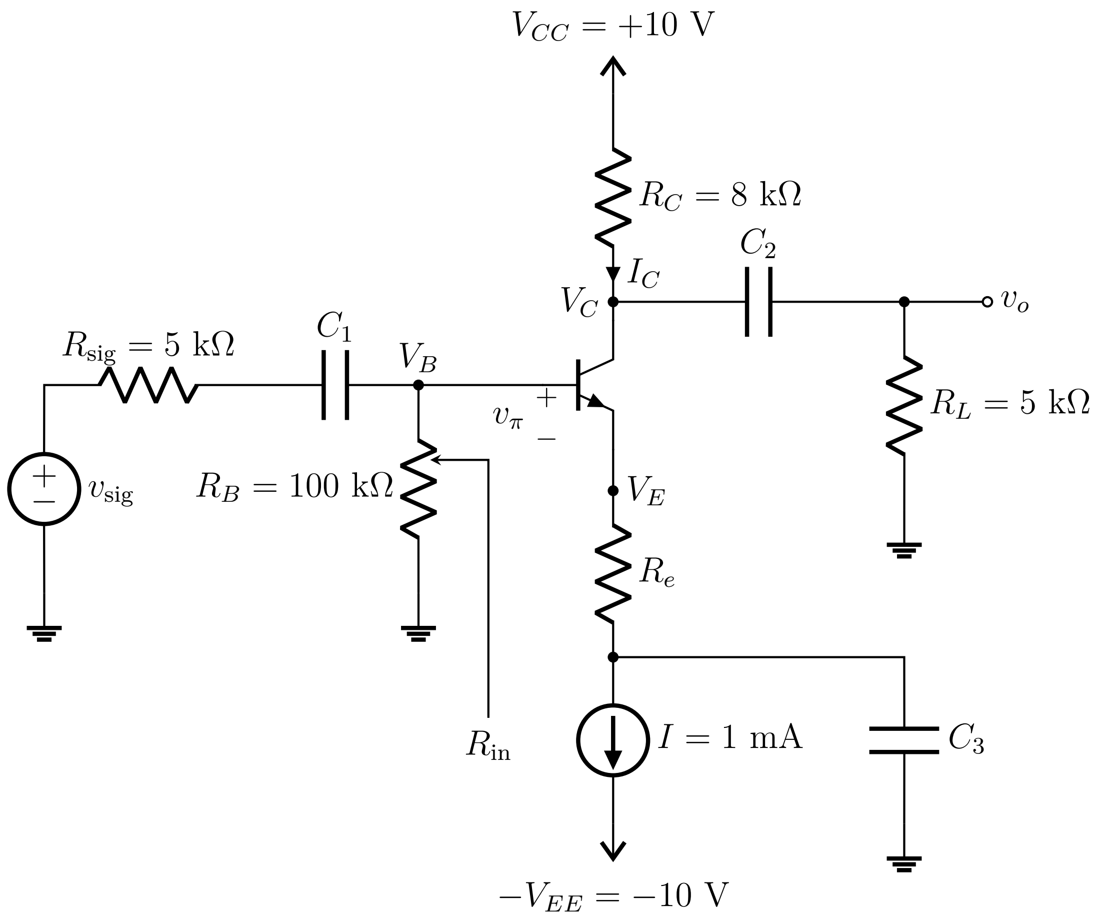
(ii) What are the allowable signal swings at the collector in both directions?

(iii) Determine the values of the BJT small signal model parameter: r_π, r_e, g_m at this bias point.

(iv) Draw the complete small signal equivalent circuit of the amplifier using suitable model of BJT. Neglect r_0.

(v) Find the value of R_e that results in R_{in} equal to four times the source resistance R_{sig}.

(vi) For this value of R_e, find A_{vo}, R_{out}, A_v, G_v and A_{is} of the amplifier.



(36 marks) [EEE 263 | L-2/T-1 | 2011–12]

Answer

(i) DC Analysis: I_C, V_C, V_B, and V_E

Assume the BJT is operating in the active region. The ideal DC current source in the emitter branch dictates the total emitter current:

$$I_E = 1 \text{ mA}$$

Using the given common-emitter current gain $\beta = 100$, we find α and the collector/base currents:

$$\alpha = \frac{\beta}{\beta+1} = \frac{100}{101} \approx 0.99$$
$$I_C = \alpha I_E = \left(\frac{100}{101}\right)(1 \text{ mA}) = 0.99 \text{ mA}$$
$$I_B = \frac{I_E}{\beta+1} = \frac{1 \text{ mA}}{101} \approx 9.9\ \mu\text{A}$$

Now, we calculate the DC voltages at the respective nodes. For the base voltage (V_B), the base current flows through R_B to ground:

$$V_B = -I_B R_B = -(9.9 \times 10^{-3} \text{ mA})(100 \text{ k}\Omega) = -0.99 \text{ V}$$

For the emitter voltage (V_E), we subtract the base-emitter voltage drop ($V_{BE} = 0.7$ V):

$$V_E = V_B - V_{BE} = -0.99 \text{ V} - 0.7 \text{ V} = -1.69 \text{ V}$$

For the collector voltage (V_C), we evaluate the drop across R_C:

$$V_C = V_{CC} - I_C R_C = 10 \text{ V} - (0.99 \text{ mA})(8 \text{ k}\Omega) = 10 \text{ V} - 7.92 \text{ V} = 2.08 \text{ V}$$

Check: $V_C = 2.08$ V $> V_B = -0.99$ V, confirming the active region assumption is valid.

(ii) Allowable Signal Swings at the Collector

To find the maximum signal swing at the collector node without severe distortion, we analyze the AC load line. The effective AC load resistance seen by the collector is:

$$r_{ac} = R_C \parallel R_L = 8 \text{ k}\Omega \parallel 5 \text{ k}\Omega = \frac{8 \times 5}{8+5} \text{ k}\Omega = 3.077 \text{ k}\Omega$$

Maximum Positive Swing (Cutoff Limit): The transistor turns off when i_c drops by I_C. The collector voltage swings up by:

$$\Delta V_{C+} = I_C r_{ac} = (0.99 \text{ mA})(3.077 \text{ k}\Omega) = 3.046 \text{ V}$$

Maximum Negative Swing (Saturation Limit): Assuming saturation occurs when the base-collector junction becomes forward-biased ($V_C \approx V_B$), the maximum drop in collector voltage is approximately:

$$\Delta V_{C-} \approx V_{CQ} - V_{BQ} = 2.08 \text{ V} - (-0.99 \text{ V}) = 3.07 \text{ V}$$

The overall allowable signal swing at the collector is limited by the smaller of the two bounds, which is approximately **3.046 V in both directions** (symmetrically).

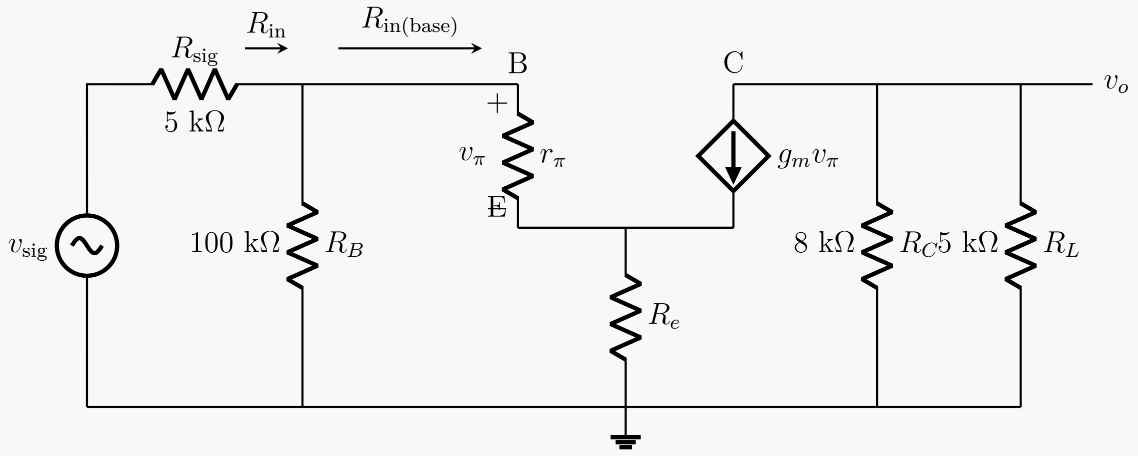
(iii) Small Signal Model Parameters (r_π, r_e, g_m)

Assuming the thermal voltage $V_T \approx 25$ mV at room temperature:

$$g_m = \frac{I_C}{V_T} = \frac{0.99 \text{ mA}}{25 \text{ mV}} = 39.6 \text{ mA/V}$$
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{39.6 \text{ mA/V}} = 2.525 \text{ k}\Omega$$
$$r_e = \frac{V_T}{I_E} = \frac{25 \text{ mV}}{1 \text{ mA}} = 25\ \Omega$$

(iv) Complete Small Signal Equivalent Circuit

Assuming C_1, C_2, and C_3 act as AC short circuits. Note that C_3 grounds the bottom of R_e, so R_e remains in the AC circuit providing emitter degeneration.



(v) Value of R_e for $R_{\text{in}} = 4R_{\text{sig}}$

We are given $R_{\text{sig}} = 5$ kΩ. Thus, the required input resistance is:

$$R_{\text{in}} = 4 \times 5 \text{ k}\Omega = 20 \text{ k}\Omega$$

From the small-signal model, R_{in} is the parallel combination of R_B and the resistance looking into the base ($R_{\text{in(base)}}$):

$$R_{\text{in}} = R_B \parallel R_{\text{in(base)}}$$
$$20 \text{ k}\Omega = 100 \text{ k}\Omega \parallel R_{\text{in(base)}} = \frac{100R_{\text{in(base)}}}{100 + R_{\text{in(base)}}}$$

Solving for $R_{\text{in(base)}}$:

$$2000 + 20R_{\text{in(base)}} = 100R_{\text{in(base)}} \implies 80R_{\text{in(base)}} = 2000 \implies R_{\text{in(base)}} = 25 \text{ k}\Omega$$

The formula for resistance looking into the base with emitter degeneration is:

$$R_{\text{in(base)}} = r_\pi + (\beta + 1)R_e$$
$$25 \text{ k}\Omega = 2.525 \text{ k}\Omega + (101)R_e$$
$$101R_e = 22.475 \text{ k}\Omega \implies R_e = \frac{22.475 \text{ k}\Omega}{101} \approx 0.2225 \text{ k}\Omega = 222.5\ \Omega$$

(vi) Amplifier Performance Parameters

Using the value $R_e = 222.5\ \Omega$, we compute the specified amplifier parameters:
1. Open-Circuit Voltage Gain (A_{vo}): With $R_L = \infty$, the output voltage is $v_o = -(\beta i_b)R_C$.

$$A_{vo} = \frac{v_o}{v_i} = \frac{-\beta R_C}{R_{\text{in(base)}}} = \frac{-100 \times 8 \text{ k}\Omega}{25 \text{ k}\Omega} = -32 \text{ V/V}$$

2. Output Resistance (R_{out}): Looking back into the collector with the input source turned off and ignoring r_o:

$$R_{\text{out}} = R_C = 8 \text{ k}\Omega$$

3. Voltage Gain with Load (A_v):

$$A_v = \frac{v_o}{v_i} = A_{vo}\left(\frac{R_L}{R_{\text{out}} + R_L}\right) = -32\left(\frac{5 \text{ k}\Omega}{8 \text{ k}\Omega + 5 \text{ k}\Omega}\right) = -32\left(\frac{5}{13}\right) \approx -12.31 \text{ V/V}$$

4. Overall Voltage Gain (G_v): Taking the input attenuation into account:

$$G_v = \frac{v_o}{v_{\text{sig}}} = A_v\left(\frac{R_{\text{in}}}{R_{\text{in}} + R_{\text{sig}}}\right) = -12.31\left(\frac{20 \text{ k}\Omega}{20 \text{ k}\Omega + 5 \text{ k}\Omega}\right) = -12.31 \times 0.8 \approx -9.85 \text{ V/V}$$

5. Short-Circuit Current Gain (A_{is}): Defined as the current gain i_{os}/i_i when the output is shorted ($R_L = 0$). Under this condition, all collector current flows into the short ($i_{os} = -i_c = -\beta i_b$). The input signal current is $i_i = v_i/R_{\text{in}}$. By current division at the input:

$$i_b = i_i\left(\frac{R_B}{R_B + R_{\text{in(base)}}}\right) = i_i\left(\frac{100 \text{ k}\Omega}{100 \text{ k}\Omega + 25 \text{ k}\Omega}\right) = 0.8\ i_i$$
$$A_{is} = \frac{i_{os}}{i_i} = \frac{-\beta i_b}{i_i} = -100 \times 0.8 = -80 \text{ A/A}$$

Common Base Amplifier

Q1. For a Common Base Amplifier, using the appropriate small-signal equivalent model, derive the expressions of input resistance, output resistance, open-circuit voltage gain, and overall voltage gain. **(14 marks)** [EEE 263 | L-2/T-1 | 2022–23]

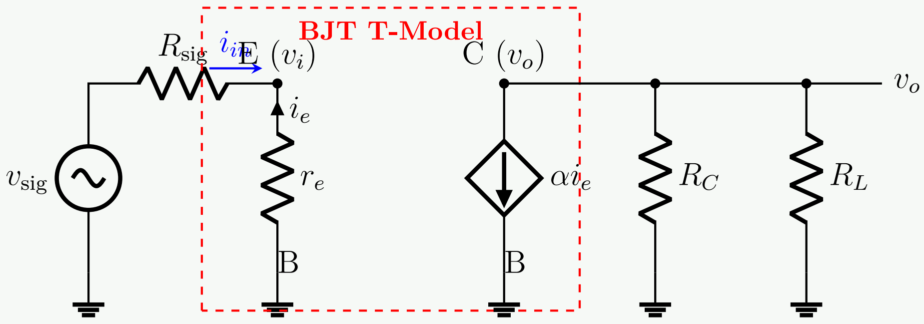
Answer

1. Small-Signal Equivalent Circuit

To derive the expressions for the Common Base (CB) amplifier, we replace the BJT with its small-signal **T-model**. The T-model is highly preferred for CB analysis because the base terminal acts as a common ground for both the input (emitter) and output (collector) ports, decoupling them directly.

Assumptions:

- The transistor operates in the active region.
- The Early effect resistance (r_o) is assumed to be infinitely large ($r_o \to \infty$) for the fundamental derivation, as it is typically much larger than R_C and R_L.
- Coupling and bypass capacitors are considered short circuits at the signal frequency.



2. Input Resistance (R_{in})

The input resistance is defined as the resistance seen by the signal source looking into the emitter terminal:

$$R_{\text{in}} = \frac{v_i}{i_{in}} \tag{1}$$

Applying Kirchhoff's Current Law (KCL) at the emitter node (E), we observe that the input current i_{in} flows into the node, and the model's emitter current i_e flows from the ground into the node. Thus:

$$i_{in} + i_e = 0 \implies i_{in} = -i_e \tag{2}$$

Applying Ohm's law across the emitter resistance r_e (with i_e flowing upward):

$$v_i - 0 = -i_e r_e \implies v_i = -i_e r_e$$

Substituting Equation 2 into Equation ??:

$$v_i = i_{in} r_e \implies R_{\text{in}} = \frac{v_i}{i_{in}} = r_e$$

Conclusion: The CB amplifier features a very low input resistance, approximately equal to $1/g_m$.

3. Output Resistance (R_{out})

The output resistance is found by looking back into the collector terminal while turning off the independent input source ($v_{\text{sig}} = 0$):

- When $v_{\text{sig}} = 0$, the input voltage $v_i = 0$.
- Consequently, $i_e = -v_i/r_e = 0$.
- Since $i_e = 0$, the dependent current source $\alpha i_e = 0$ (acts as an open circuit).

Looking into the output node C, only the collector resistor R_C remains connected to the AC ground.

$$R_{\text{out}} = R_C \tag{3}$$

4. Open-Circuit Voltage Gain (A_{vo})

The open-circuit voltage gain is defined as the gain from the emitter to the collector when the load resistor is removed ($R_L \to \infty$). Applying KCL at the collector node (C):

$$\frac{v_o}{R_C} + \alpha i_e = 0 \implies v_o = -\alpha i_e R_C$$

From Equation ??, we know $i_e = -v_i/r_e$. Substituting this into the v_o expression:

$$v_o = -\alpha \left(-\frac{v_i}{r_e}\right) R_C = \left(\frac{\alpha}{r_e}\right) R_C v_i$$

Since $\alpha/r_e = g_m$ (the transconductance), the open-circuit voltage gain is:

$$A_{vo} = \frac{v_o}{v_i} = g_m R_C \tag{4}$$

Conclusion: The CB amplifier is non-inverting.

5. Overall Voltage Gain (G_v)

The overall voltage gain G_v maps the output voltage back to the signal source v_{sig}. It accounts for both the input signal attenuation (loading effect at the emitter) and the output loading effect caused by R_L.

$$G_v = \frac{v_o}{v_{\text{sig}}} = \left(\frac{v_i}{v_{\text{sig}}}\right)\left(\frac{v_o}{v_i}\right) \tag{5}$$

Step 1: Input Attenuation Factor
The source v_{sig}, resistor R_{sig}, and the amplifier input resistance $R_{\text{in}} = r_e$ form a voltage divider:

$$\frac{v_i}{v_{\text{sig}}} = \frac{R_{\text{in}}}{R_{\text{sig}} + R_{\text{in}}} = \frac{r_e}{R_{\text{sig}} + r_e}$$

Step 2: Voltage Gain with Load (A_v)
With the load connected, the effective resistance at the collector node becomes $R_C \parallel R_L$. The gain from the emitter to the collector is therefore:

$$A_v = \frac{v_o}{v_i} = g_m(R_C \parallel R_L) = \frac{\alpha(R_C \parallel R_L)}{r_e}$$

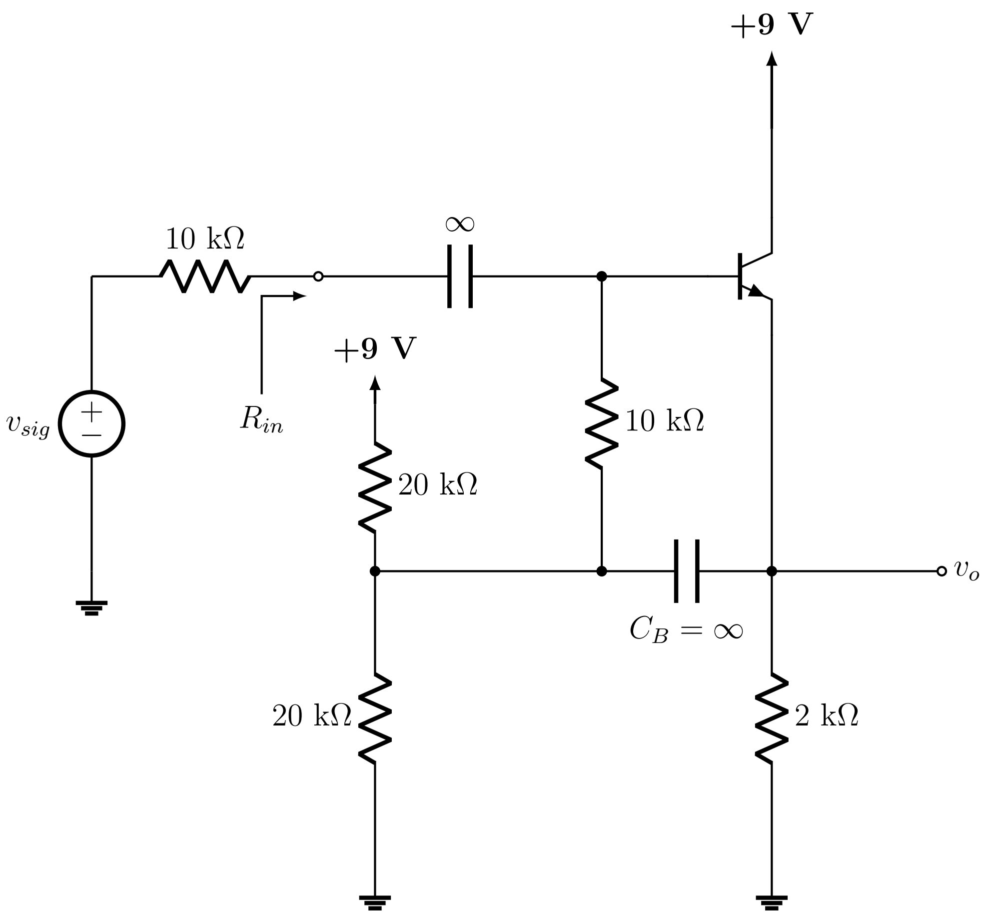
Step 3: Final Substitution
Combining the two factors gives the overall voltage gain:

$$G_v = \left(\frac{r_e}{R_{\text{sig}} + r_e}\right)\left(\frac{\alpha(R_C \parallel R_L)}{r_e}\right)$$
$$G_v = \frac{\alpha(R_C \parallel R_L)}{R_{\text{sig}} + r_e} \approx \frac{R_C \parallel R_L}{R_{\text{sig}} + r_e}$$

(Assuming $\alpha \approx 1$ for a high-β transistor.)

Bootstrapped Follower

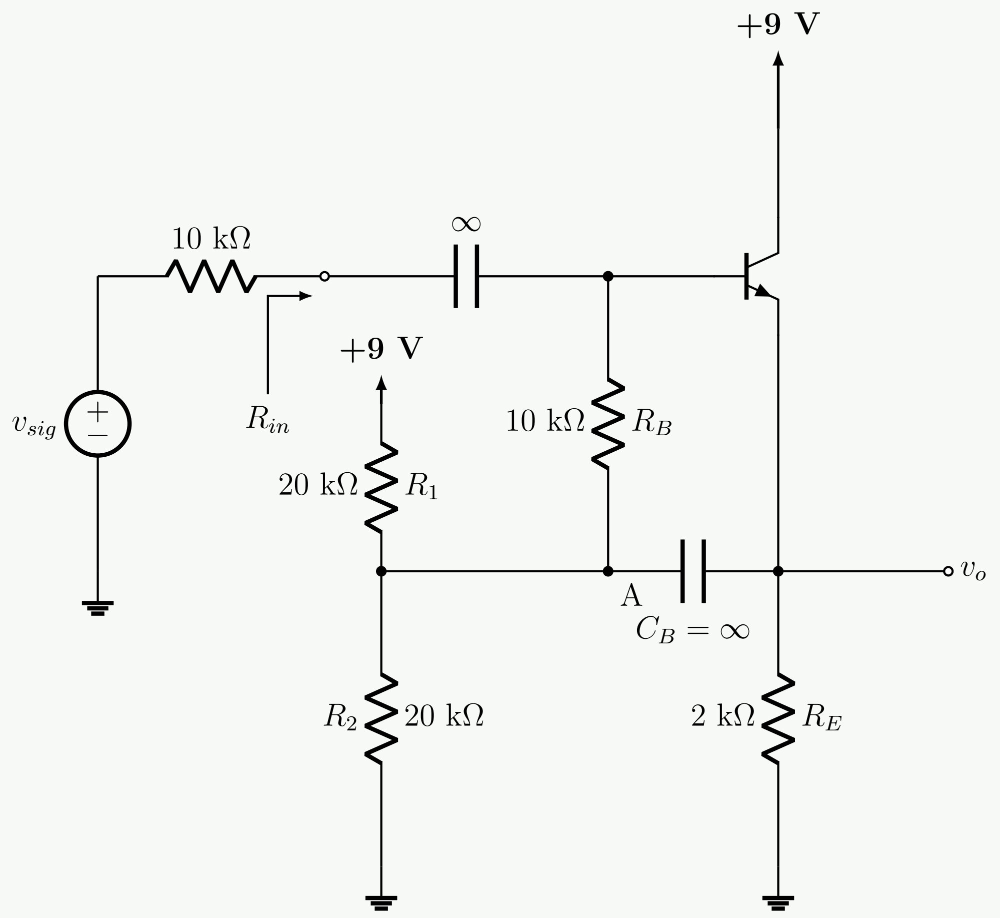
Q1. For the bootstrapped follower circuit shown: (i) Find the DC emitter current and g_m, r_e, r_π. Assume $\beta = 100$. (ii) Replace the BJT with its T model (neglecting r_o) and determine R_{in} and the voltage gain v_o/v_{sig}. (iii) Repeat (ii) for the case when capacitor C_B is open-circuited. Compare the results with (ii) and comment on the advantage of bootstrapping.



(15 marks) [EEE 263 | L-2/T-1 | 2024–25]

Answer

Original Circuit Diagram



(i) DC Analysis and Small-Signal Parameters

For DC analysis, all capacitors (the input coupling capacitor and C_B) act as open circuits. The circuit reduces to a standard voltage-divider biased emitter follower. Let Node A be the junction between R_1, R_2, and R_B. We can find the Thevenin equivalent of the bias network looking back from Node A:

$$V_{TH(A)} = V_{CC}\frac{R_2}{R_1 + R_2} = 9 \text{ V} \times \frac{20 \text{ k}\Omega}{20 \text{ k}\Omega + 20 \text{ k}\Omega} = 4.5 \text{ V}$$
$$R_{TH(A)} = R_1 \parallel R_2 = \frac{20 \text{ k}\Omega \times 20 \text{ k}\Omega}{20 \text{ k}\Omega + 20 \text{ k}\Omega} = 10 \text{ k}\Omega$$

The total equivalent base resistance R_{BB} is the sum of $R_{TH(A)}$ and the base resistor R_B:

$$R_{BB} = R_{TH(A)} + R_B = 10 \text{ k}\Omega + 10 \text{ k}\Omega = 20 \text{ k}\Omega \tag{1}$$

Applying Kirchhoff's Voltage Law (KVL) around the base-emitter loop and assuming the transistor is in the active region with $V_{BE(on)} \approx 0.7$ V and thermal voltage $V_T \approx 25$ mV:

$$V_{TH(A)} - I_B R_{BB} - V_{BE(on)} - I_E R_E = 0$$
$$V_{TH(A)} - \left(\frac{I_E}{\beta + 1}\right) R_{BB} - V_{BE(on)} - I_E R_E = 0$$

Substitute the known values ($\beta = 100$):

$$4.5 - I_E\left(\frac{20 \text{ k}\Omega}{101}\right) - 0.7 - I_E(2 \text{ k}\Omega) = 0$$
$$3.8 = I_E\,(0.198 \text{ k}\Omega + 2 \text{ k}\Omega)$$
$$I_E = \frac{3.8 \text{ V}}{2.198 \text{ k}\Omega} \approx 1.7288 \text{ mA}$$

Let us quickly verify the active region assumption:

$$V_E = I_E R_E = (1.7288 \text{ mA})(2 \text{ k}\Omega) = 3.458 \text{ V}$$
$$V_C = 9 \text{ V}$$
$$V_{CE} = V_C - V_E = 9 \text{ V} - 3.458 \text{ V} = 5.542 \text{ V}$$

Since $V_{CE} > 0.3$ V, the active region assumption is firmly verified.
Now, we calculate the small-signal parameters. The collector current is $I_C = \alpha I_E = \left(\frac{100}{101}\right) 1.7288 \text{ mA} \approx 1.7117 \text{ mA}$.

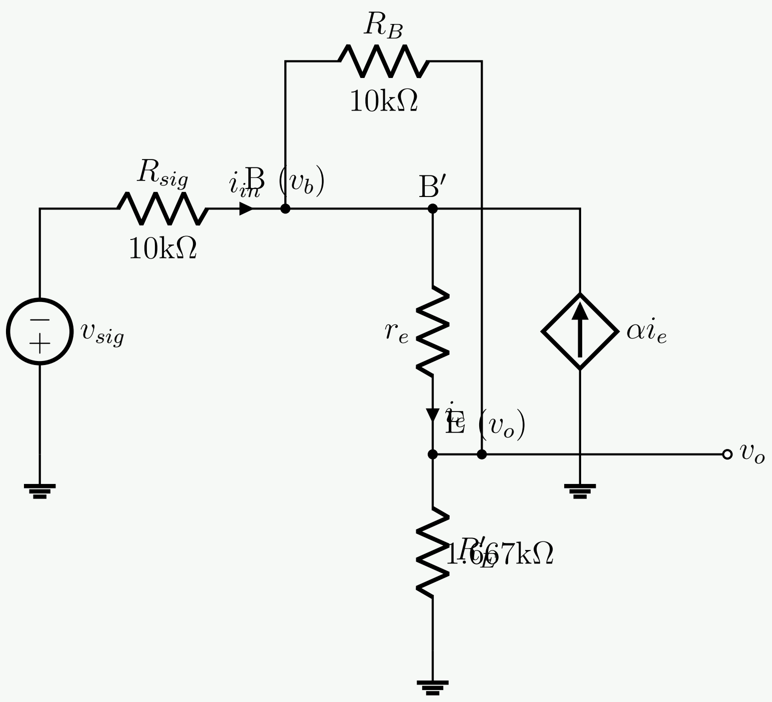
$$g_m = \frac{I_C}{V_T} = \frac{1.7117 \text{ mA}}{25 \text{ mV}} \approx 68.47 \text{ mA/V}$$
$$r_e = \frac{V_T}{I_E} = \frac{25 \text{ mV}}{1.7288 \text{ mA}} \approx 14.46 \ \Omega$$
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{68.47 \text{ mA/V}} \approx 1.46 \text{ k}\Omega$$

(ii) AC Analysis with Bootstrapping ($C_B = \infty$)

For AC analysis, C_B and the input coupling capacitor act as short circuits. The DC supply (+9 V) is grounded.

- Because C_B is shorted, Node A is connected directly to the emitter.
- The biasing resistors R_1 and R_2 are in parallel from Node A to ground, providing an equivalent resistance $R_A = R_1 \parallel R_2 = 10$ kΩ.
- Since Node A is the emitter, R_A is in parallel with the main emitter resistor R_E. The effective emitter resistance is $R'_E = R_E \parallel R_A = 2\text{ k}\Omega \parallel 10\text{ k}\Omega = 1.667\text{ k}\Omega$.
- The base resistor R_B connects the base to Node A (the emitter). Thus, R_B is in parallel with the transistor's base-emitter junction!

T-Model Equivalent Circuit:



1. Input Resistance (R_{in}):
The current i_{in} from the source splits at the base. Part of it forms the base current i_b, and the rest flows through the bootstrapping resistor R_B. Let $v_{be} = v_b - v_e$. The current through R_B is $i_{RB} = \frac{v_{be}}{R_B}$. The transistor base current is $i_b = \frac{v_{be}}{r_\pi}$. Thus, $i_{in} = i_{RB} + i_b = v_{be}\left(\frac{1}{R_B} + \frac{1}{r_\pi}\right)$. This demonstrates that R_B is effectively in parallel with r_π. Let $R'_\pi = R_B \parallel r_\pi$:

$$R'_\pi = \frac{10\text{ k}\Omega \times 1.46\text{ k}\Omega}{10\text{ k}\Omega + 1.46\text{ k}\Omega} = 1.274\text{ k}\Omega \tag{2}$$

The total current reaching the effective emitter resistor R'_E is the sum of i_{RB} and the total emitter current $i_e = (\beta+1)i_b$. Summing currents at the emitter node:

$$v_e = (i_e + i_{RB})R'_E = \left((\beta+1)\frac{v_{be}}{r_\pi} + \frac{v_{be}}{R_B}\right)R'_E$$
$$v_e = v_{be}\left(\frac{1}{r_e} + \frac{1}{R_B}\right)R'_E$$

We define the effective transconductance of this parallel combination as $g_{m(eff)} = \frac{1}{r_e} + \frac{1}{R_B}$.

$$g_{m(eff)} = \frac{1}{0.01446\text{ k}\Omega} + \frac{1}{10\text{ k}\Omega} = 69.156\text{ mS} + 0.1\text{ mS} = 69.256\text{ mS} \tag{3}$$

Now, find the total base voltage v_b:

$$v_b = v_{be} + v_e = v_{be} + v_{be}\left(g_{m(eff)}R'_E\right) = v_{be}\left[1 + g_{m(eff)}R'_E\right]$$

Substituting $v_{be} = i_{in}R'_\pi$:

$$v_b = i_{in}R'_\pi\left[1 + g_{m(eff)}R'_E\right]$$

The input resistance R_{in} is defined as $\frac{v_b}{i_{in}}$:

$$R_{in} = R'_\pi\left[1 + g_{m(eff)}R'_E\right]$$
$$R_{in} = 1.274\text{ k}\Omega \times [1 + (69.256\text{ mS})(1.667\text{ k}\Omega)]$$
$$R_{in} = 1.274\text{ k}\Omega \times [1 + 115.45] = 1.274 \times 116.45 \approx \mathbf{148.36\text{ k}\Omega}$$

2. Voltage Gain (v_o/v_{sig}):
First, calculate the transmission from v_{sig} to the base v_b due to the source resistance $R_{sig} = 10$ kΩ:

$$\frac{v_b}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = \frac{148.36}{148.36 + 10} = 0.9368\text{ V/V} \tag{4}$$

Next, calculate the gain from the base to the emitter (output):

$$\frac{v_o}{v_b} = \frac{v_e}{v_b} = \frac{v_{be} g_{m(eff)} R'_E}{v_{be}\left[1 + g_{m(eff)} R'_E\right]} = \frac{g_{m(eff)} R'_E}{1 + g_{m(eff)} R'_E} = \frac{115.45}{116.45} = 0.9914 \text{ V/V} \tag{5}$$

The overall voltage gain is the product of these two stages:

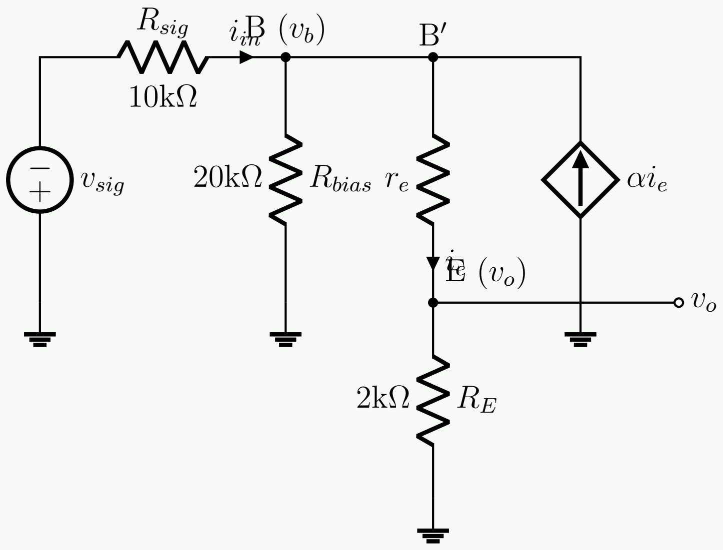
$$A_v = \frac{v_o}{v_{sig}} = \left(\frac{v_b}{v_{sig}}\right)\left(\frac{v_o}{v_b}\right) = 0.9368 \times 0.9914 \approx \mathbf{0.929} \text{ V/V} \tag{6}$$

(iii) AC Analysis Without Bootstrapping (C_B Open)

If C_B is open, Node A is no longer forced to track the emitter voltage. Instead, since R_1 goes to +9V (AC ground) and R_2 goes to true ground, Node A is firmly fixed at AC ground.

- R_B (10 kΩ) connects the base to Node A (AC ground).
- The total bias network resistance seen at the base to ground is $R_{bias} = R_B + (R_1 \parallel R_2) = 10\text{ k}\Omega + 10\text{ k}\Omega = 20\text{ k}\Omega$.
- The emitter resistor is simply $R_E = 2$ kΩ (no parallel R_A).

Small-Signal Equivalent Circuit (C_B Open):



1. Input Resistance (R_{in}):
The resistance looking strictly into the base of the transistor is:

$$R_{in(base)} = r_\pi + (\beta + 1)R_E = 1.46\text{ k}\Omega + (101)(2\text{ k}\Omega) = 203.46\text{ k}\Omega \tag{7}$$

The total input resistance R_{in} is R_{bias} in parallel with $R_{in(base)}$:

$$R_{in} = R_{bias} \parallel R_{in(base)} = \frac{20\text{ k}\Omega \times 203.46\text{ k}\Omega}{20\text{ k}\Omega + 203.46\text{ k}\Omega} = \frac{4069.2}{223.46} \approx \mathbf{18.21\text{ k}\Omega} \tag{8}$$

2. Voltage Gain (v_o/v_{sig}):
First, find the transmission to the base:

$$\frac{v_b}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = \frac{18.21}{18.21 + 10} = 0.6455 \text{ V/V} \tag{9}$$

Next, the gain of the emitter follower itself:

$$\frac{v_o}{v_b} = \frac{(\beta + 1)R_E}{r_\pi + (\beta + 1)R_E} = \frac{202}{203.46} = 0.9928 \text{ V/V} \tag{10}$$

The overall voltage gain is:

$$A_v = \frac{v_o}{v_{sig}} = 0.6455 \times 0.9928 \approx \mathbf{0.641} \text{ V/V} \tag{11}$$

Comparison and Commentary on Bootstrapping

Parameter	With Bootstrapping (ii)	Without Bootstrapping (iii)
Input Resistance (R_{in})	148.36 kΩ	18.21 kΩ
Overall Gain (A_v)	0.929 V/V	0.641 V/V

Commentary:
The fundamental problem with a standard voltage-divider bias network in an emitter follower is that the low equivalent resistance of the bias network (20 kΩ in this case) appears in parallel with the high input resistance of the transistor, aggressively loading the input source and severely dropping the overall voltage gain (down to 0.641 V/V).
The advantage of bootstrapping is that capacitor C_B dynamically ties Node A to the emitter. Since the emitter voltage v_e closely follows the base voltage v_b ($v_e \approx v_b$), the AC voltage drop across R_B is nearly zero ($v_b - v_e \approx 0$). Consequently, extremely little AC signal current is drawn through R_B from the source. By artificially "bootstrapping" the bias node to match the input signal, the effective AC impedance of the biasing network becomes exceptionally high, raising R_{in} nearly eightfold (to 148.36 kΩ). This eliminates the severe input loading effect and allows the amplifier to perform much closer to an ideal unity-gain buffer (0.929 V/V).

Common Collector (Emitter Follower)

Q1. For a common collector amplifier, derive the expressions for R_i, R_o, A_v, and A_i using the T model with early effect for the transistor. Where is the common collector amplifier used? **(26 marks)** [EEE 263 | L-2/T-1 | 2020–21]

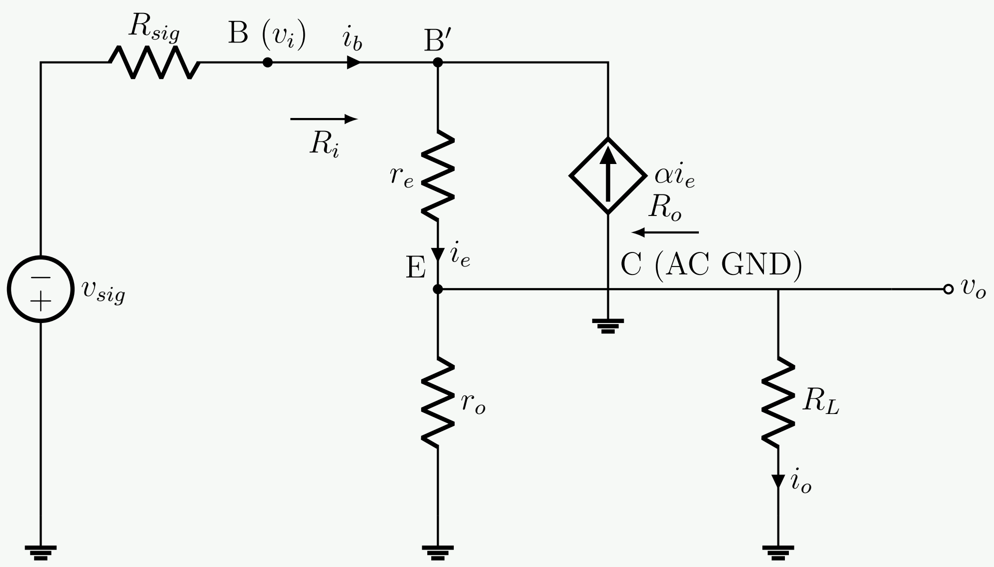
Answer

Small-Signal Equivalent Circuit

To analyze the Common Collector (CC) amplifier (also known as an Emitter Follower), we utilize the low-frequency small-signal **T-model**. The Early effect is accounted for by the output resistance r_o.

Assumptions for Analysis:

- The BJT is correctly biased and operating in the **forward-active region**.
- The coupling and bypass capacitors (if any) are assumed to act as perfect short circuits at the signal frequencies of interest.
- The collector terminal is tied to the DC supply V_{CC}, which serves as an **AC ground**.
- The base current i_b, emitter current i_e, and collector current i_c relate as $i_e = (\beta+1)i_b$ and $i_c = \alpha i_e$.



1. Input Resistance (R_i)

The input resistance R_i is the equivalent resistance seen looking directly into the base terminal of the transistor. By definition:

$$R_i = \frac{v_i}{i_b} \tag{1}$$

Let the total equivalent load resistance at the emitter be $R'_L = r_o \parallel R_L$. Applying Kirchhoff's Voltage Law (KVL) from the base to ground yields:

$$v_i = v_{be} + v_o$$
$$v_i = i_e r_e + i_e(r_o \parallel R_L) = i_e\,(r_e + R'_L)$$

Since the emitter current is related to the base current by $i_e = (\beta+1)i_b$, we substitute this into the voltage equation:

$$v_i = (\beta+1)i_b\,(r_e + R'_L)$$
$$R_i = \frac{v_i}{i_b} = (\beta+1)\,(r_e + r_o \parallel R_L)$$

Note: The resistance at the emitter is reflected to the base by a factor of $(\beta+1)$, making R_i extremely large.

2. Voltage Gain (A_v)

The voltage gain A_v evaluates the ratio of the output voltage v_o to the input voltage v_i at the base.

$$A_v = \frac{v_o}{v_i} \tag{2}$$

The output voltage is formed by the emitter current flowing through the equivalent emitter load:

$$v_o = i_e(r_o \parallel R_L) = i_e R'_L \tag{3}$$

Using the expression for v_i derived previously:

$$A_v = \frac{i_e R'_L}{i_e(r_e + R'_L)}$$
$$A_v = \frac{R'_L}{r_e + R'_L} = \frac{r_o \parallel R_L}{r_e + r_o \parallel R_L}$$

Note: Because $R'_L \gg r_e$, the voltage gain is highly positive and slightly less than 1 ($A_v \approx 1$). This characterizes the "emitter follower" behavior.

3. Output Resistance (R_o)

The output resistance R_o is determined by turning off the independent source ($v_{sig} = 0$), removing the load R_L, and applying a test voltage v_x at the emitter terminal. The test current i_x flows into the emitter. With $v_{sig} = 0$, the base is tied to ground solely through R_{sig}. The voltage at the base is:

$$v_b = -i_b R_{sig} = -\left(\frac{i_e}{\beta+1}\right) R_{sig} \tag{4}$$

The voltage drop from the internal node B′ to the emitter is $v_{be} = i_e r_e$. The emitter voltage is forced to v_x. Applying KVL from base to emitter:

$$\begin{aligned} v_x &= v_b - v_{be} \\ v_x &= -\left(\frac{i_e}{\beta+1}\right) R_{sig} - i_e r_e \\ v_x &= -i_e \left(r_e + \frac{R_{sig}}{\beta+1}\right) \quad \Longrightarrow \quad -i_e = \frac{v_x}{r_e + \frac{R_{sig}}{\beta+1}} \end{aligned}$$

Applying Kirchhoff's Current Law (KCL) at the emitter node, the injected test current i_x splits through r_o and the path back through the emitter (which is $-i_e$):

$$\begin{aligned} i_x &= \frac{v_x}{r_o} + (-i_e) \\ i_x &= \frac{v_x}{r_o} + \frac{v_x}{r_e + \frac{R_{sig}}{\beta+1}} \end{aligned}$$

Dividing by v_x yields the equivalent output admittance. Inverting it provides R_o:

$$R_o = \frac{v_x}{i_x} = r_o \parallel \left(r_e + \frac{R_{sig}}{\beta+1}\right) \tag{5}$$

Note: The source resistance R_{sig} is reflected to the emitter divided by $(\beta+1)$, making the output resistance intrinsically very low.

4. Current Gain (A_i)

The current gain is the ratio of the output load current i_o to the input base current $i_i = i_b$:

$$A_i = \frac{i_o}{i_i} \tag{6}$$

Using the current divider principle at the emitter, the total emitter current i_e splits between the Early resistance r_o and the load resistor R_L. The current i_o flowing through R_L is:

$$i_o = i_e \left(\frac{r_o}{r_o + R_L}\right) \tag{7}$$

Substituting $i_e = (\beta+1)i_b$ and recognizing $i_b = i_i$:

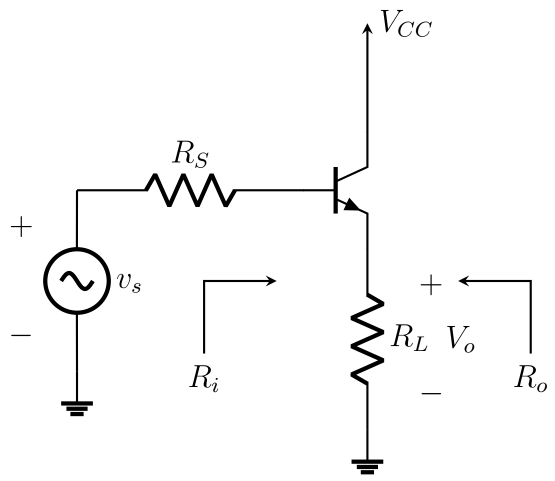
$$\begin{aligned} i_o &= (\beta+1) i_i \left(\frac{r_o}{r_o + R_L}\right) \\ A_i &= \frac{i_o}{i_i} = (\beta+1)\left(\frac{r_o}{r_o + R_L}\right) \end{aligned}$$

Applications of the Common Collector Amplifier

Due to its unique characteristics—very high input impedance, very low output impedance, unity voltage gain, and high current gain—the CC amplifier serves several critical roles in analog circuit design:

- **Impedance Matching (Voltage Buffer):** It bridges a high-impedance source to a low-impedance load without loading down the signal source, preserving the signal's voltage integrity.
- **Output Stages:** Extensively used in the final output stage of multi-stage amplifiers (like Class AB audio amplifiers or Operational Amplifiers) to deliver massive current to heavy loads (e.g., $8\,\Omega$ speakers) while sustaining the voltage established by previous high-gain stages.
- **Level Shifting:** It provides a predictable DC voltage shift of roughly $V_{BE(on)} \approx 0.7\,\text{V}$ between its input and output, which is invaluable in direct-coupled amplifier designs.

Q2. For the common collector circuit shown, derive the expressions for A_v, A_i, R_i, and R_o using the T model for the transistor.



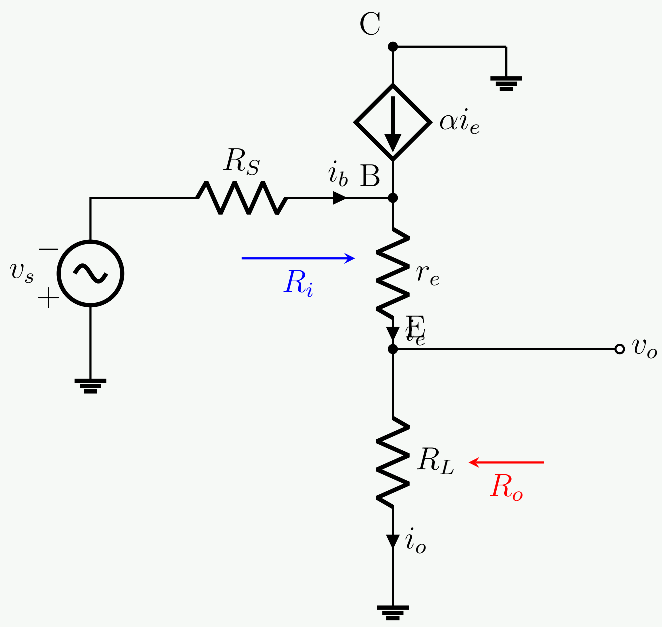
(23.67 marks) [EEE 263 | L-2/T-1 | 2018–19]

Answer

Small-Signal Equivalent Circuit (T-Model)

To analyze the Common Collector (CC) amplifier, we construct its small-signal equivalent circuit using the **T-model** for the BJT.
Assumptions for Analysis:

- The transistor operates in the forward-active region.
- The small-signal parameters are $r_e = \frac{V_T}{I_E}$, $\alpha = \frac{\beta}{\beta+1}$, and $i_c = \alpha i_e$.
- The Early effect is neglected ($r_o \to \infty$) for simplicity, as is standard when a load resistor R_L is explicitly present and $r_o \gg R_L$.
- The DC supply V_{CC} acts as an AC ground, meaning the collector terminal is grounded in the AC equivalent circuit.



1. Input Resistance (R_i)

The input resistance R_i is the equivalent resistance seen looking directly into the base terminal. It is defined as:

$$R_i = \frac{v_b}{i_b} \tag{1}$$

Applying Kirchhoff's Voltage Law (KVL) from the base terminal through the emitter to ground yields the base voltage v_b:

$$v_b = i_e r_e + i_e R_L = i_e(r_e + R_L) \tag{2}$$

The base current and emitter current are related by the fundamental BJT current equation:

$$i_e = (\beta + 1)i_b \tag{3}$$

Substituting i_e into the v_b expression gives:

$$v_b = (\beta + 1)i_b(r_e + R_L) \tag{4}$$

Dividing by i_b, we obtain the input resistance:

$$R_i = (\beta + 1)(r_e + R_L) \tag{5}$$

Note: The resistance in the emitter leg is reflected to the base multiplied by a factor of $(\beta+1)$, leading to a very high input impedance.

2. Voltage Gain (A_v)

The base-to-output voltage gain A_v is the ratio of the output voltage v_o to the base voltage v_b. The output voltage is formed by the emitter current dropping across the load:

$$v_o = i_e R_L \tag{6}$$

Using the expression for v_b derived above, the voltage gain is:

$$A_v = \frac{v_o}{v_b} = \frac{i_e R_L}{i_e(r_e + R_L)} = \frac{R_L}{r_e + R_L} \tag{7}$$

Note: Because $R_L \gg r_e$ in practical circuits, the voltage gain is slightly less than unity ($A_v \approx 1$). If the overall voltage gain with respect to the source is required, it can be found using voltage division: $A_{vs} = \frac{v_o}{v_s} = \frac{R_i}{R_i+R_S} A_v$.

3. Current Gain (A_i)

The current gain A_i evaluates the ratio of the output load current i_o to the input base current i_b. In this circuit configuration, the entirety of the emitter current flows through the load resistor R_L. Therefore, $i_o = i_e$.

$$A_i = \frac{i_o}{i_b} = \frac{i_e}{i_b} \tag{8}$$

Substituting the transistor current relationship $i_e = (\beta + 1)i_b$:

$$A_i = \beta + 1 \tag{9}$$

4. Output Resistance (R_o)

The output resistance R_o is the Thevenin equivalent resistance looking directly into the emitter terminal, *excluding* the load R_L. To determine R_o, we deactivate the independent source ($v_s = 0$), remove the load R_L, apply a test voltage v_x at the emitter, and find the resulting test current i_x that flows *into* the emitter.
With $v_s = 0$, the base is tied to ground solely through R_S. The potential at the base node is:

$$v_b = -i_b R_S \tag{10}$$

Applying KVL from the base node, down through r_e, to the test voltage source v_x:

$$v_x = v_b - i_e r_e = -i_b R_S - i_e r_e \tag{11}$$

Because the test current i_x flows into the emitter terminal, it is strictly the negative of the defined emitter current: $i_e = -i_x$. Consequently, the base current is $i_b = \frac{i_e}{\beta+1} = \frac{-i_x}{\beta+1}$.
Substituting these terms into the v_x equation:

$$v_x = -\left(\frac{-i_x}{\beta+1}\right) R_S - (-i_x) r_e$$
$$v_x = i_x \left(\frac{R_S}{\beta+1} + r_e\right)$$

Dividing the test voltage by the test current yields the output resistance:

$$R_o = \frac{v_x}{i_x} = r_e + \frac{R_S}{\beta+1} \tag{12}$$

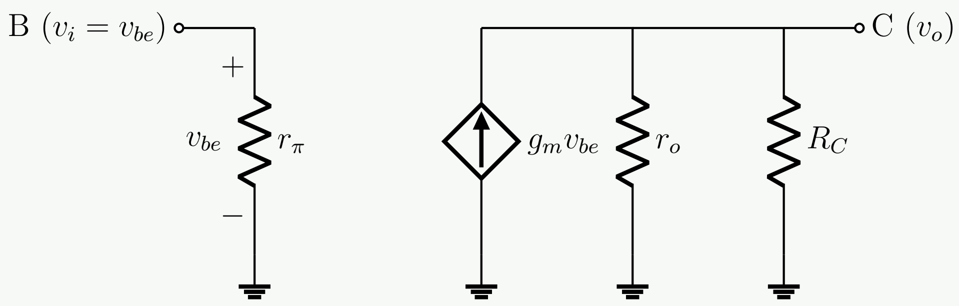
Note: The source resistance R_S is reflected to the emitter side divided by $(\beta + 1)$, making the output resistance intrinsically very low, which is a defining characteristic of a voltage buffer.

Q3. The small-signal voltage gain A_v of a BJT CE amplifier is $-I_C R_C/V_T$ without the Early effect. Show that if the Early effect is considered using $i_C = I_S e^{v_{BE}/V_T}(1 + v_{CE}/V_A)$, the gain becomes $A_v = -\frac{I_C R_C/V_T}{1+I_C R_C/(V_A+V_{CE})} = -\frac{(V_{CC}-V_{CE})/V_T}{1+(V_{CC}-V_{CE})/(V_A+V_{CE})}$. For $V_{CC} = 5\,\text{V}$, $V_{CE} = 2.5\,\text{V}$, $V_A = 100\,\text{V}$, calculate the gain with and without the Early effect. **(20 marks)** [EEE 263 | L-2/T-1 | 2024–25]

Answer

1. Small-Signal Equivalent Circuit

To derive the voltage gain A_v with the Early effect, we use the Hybrid-π small-signal model for the Common Emitter (CE) amplifier. The Early effect introduces a finite output resistance r_o in parallel with the collector resistor R_C.



2. Derivation of Voltage Gain with Early Effect

Step 2.1: Small-Signal Parameters
The collector current equation including the Early effect is given by:

$$i_C = I_S e^{v_{BE}/V_T}\left(1 + \frac{v_{CE}}{V_A}\right) \tag{1}$$

The small-signal transconductance g_m evaluates the change in i_C with respect to v_{BE} at the quiescent operating point:

$$g_m = \left.\frac{\partial i_C}{\partial v_{BE}}\right|_Q = \frac{1}{V_T} I_S e^{V_{BEQ}/V_T}\left(1 + \frac{V_{CEQ}}{V_A}\right) = \frac{I_C}{V_T} \tag{2}$$

The small-signal output resistance r_o evaluates the change in i_C with respect to v_{CE}:

$$\frac{1}{r_o} = \left.\frac{\partial i_C}{\partial v_{CE}}\right|_Q = I_S e^{V_{BEQ}/V_T}\left(\frac{1}{V_A}\right) \tag{3}$$

From the DC equation, we know $I_S e^{V_{BEQ}/V_T} = \frac{I_C}{1+V_{CE}/V_A}$. Substituting this into the conductance yields:

$$\frac{1}{r_o} = \frac{I_C}{1 + V_{CE}/V_A} \cdot \frac{1}{V_A} = \frac{I_C}{V_A + V_{CE}} \quad \Longrightarrow \quad r_o = \frac{V_A + V_{CE}}{I_C} \tag{4}$$

Step 2.2: Voltage Gain Expression
From the small-signal circuit, the output voltage is $v_o = -g_m v_{be}(R_C \parallel r_o)$. Since the input voltage is v_{be}, the gain is:

$$A_v = \frac{v_o}{v_{be}} = -g_m(R_C \parallel r_o) = -g_m\left(\frac{R_C r_o}{R_C + r_o}\right) = -g_m R_C\left(\frac{1}{1 + R_C/r_o}\right) \tag{5}$$

Substituting $g_m = I_C/V_T$ and $r_o = (V_A + V_{CE})/I_C$ into the equation:

$$A_v = -\left(\frac{I_C R_C}{V_T}\right)\frac{1}{1 + \frac{R_C}{(V_A+V_{CE})/I_C}}$$

$$A_v = -\frac{I_C R_C/V_T}{1 + \frac{I_C R_C}{V_A+V_{CE}}}$$

Step 2.3: Expressing in Terms of Supply and Output Voltages
Applying KVL at the output loop for the DC bias condition:

$$V_{CC} = I_C R_C + V_{CE} \quad \Longrightarrow \quad I_C R_C = V_{CC} - V_{CE} \tag{6}$$

Substituting this into the derived gain equation yields the final expression:

$$A_v = -\frac{(V_{CC} - V_{CE})/V_T}{1 + \frac{V_{CC}-V_{CE}}{V_A+V_{CE}}} \tag{7}$$

This matches the required mathematical proof.

3. Numerical Calculations

Given Parameters:

- $V_{CC} = 5\,\text{V}$
- $V_{CE} = 2.5\,\text{V}$
- $V_A = 100\,\text{V}$
- *Assumption:* Thermal voltage $V_T = 25\,\text{mV} = 0.025\,\text{V}$ at room temperature.

Case 1: Without Early Effect ($V_A \to \infty$, $r_o \to \infty$) When the Early effect is ignored, the denominator of Equation 7 approaches 1.

$$A_{v0} = -\frac{V_{CC} - V_{CE}}{V_T}$$

$$A_{v0} = -\frac{5 - 2.5}{0.025} = -\frac{2.5}{0.025}$$

$$A_{v0} = -100\,\text{V/V}$$

Case 2: With Early Effect Using the fully derived Equation 7:

$$A_v = -\frac{(5 - 2.5)/0.025}{1 + \frac{5-2.5}{100+2.5}}$$

$$A_v = -\frac{100}{1 + \frac{2.5}{102.5}}$$

$$A_v = -\frac{100}{1 + \frac{1}{41}} = -\frac{100}{\frac{42}{41}}$$

$$A_v = -100 \times \frac{41}{42} \approx -97.619\,\text{V/V}$$

4. Conclusion

The consideration of the Early effect models the finite output resistance r_o of the BJT. Because r_o acts in parallel with the load resistor R_C, it slightly reduces the overall equivalent resistance at the output node. Consequently, the voltage gain drops from magnitude 100 to approximately 97.62.

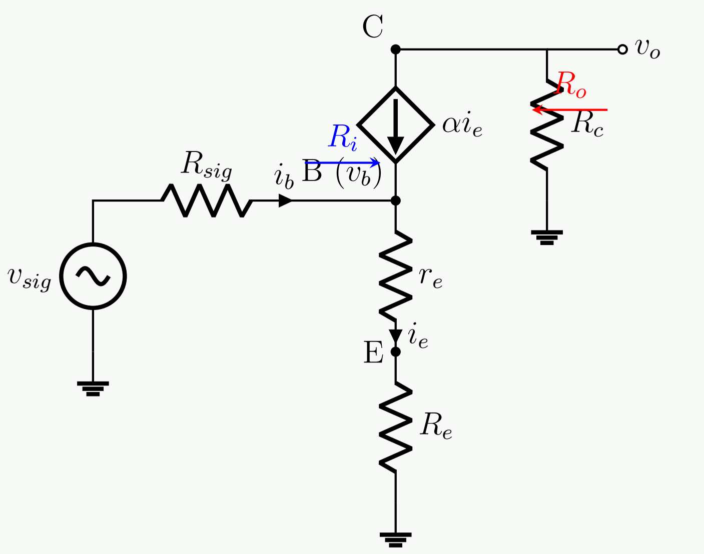
Q4. Derive the expression of R_i, R_o, A_v, and G_v for a common-emitter amplifier with emitter resistance R_e using the T model. **(26 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Small-Signal Equivalent Circuit (T-Model)

To analyze the Common-Emitter (CE) amplifier with an emitter degeneration resistor R_e, we construct its small-signal equivalent circuit using the **T-model** for the BJT.
Assumptions for Analysis:

- The transistor operates in the forward-active region.
- The small-signal parameters are $r_e = \frac{V_T}{I_E}$ and $\alpha = \frac{\beta}{\beta+1}$.
- The Early effect is neglected ($r_o \to \infty$) for simplicity, which is a standard assumption unless stated otherwise.
- R_{sig} is the internal resistance of the input signal source v_{sig}.



2. Input Resistance (R_i)

The input resistance R_i is the equivalent resistance seen looking into the base terminal of the amplifier. By applying Kirchhoff's Voltage Law (KVL) from the base terminal through the emitter to ground, we find the base voltage v_b:

$$v_b = i_e r_e + i_e R_e = i_e(r_e + R_e) \tag{1}$$

The base current i_b and the emitter current i_e are related by the BJT current relation $i_e = (\beta + 1)i_b$. Substituting this into the equation for v_b:

$$v_b = (\beta + 1)i_b(r_e + R_e) \tag{2}$$

Dividing v_b by i_b yields the input resistance:

$$R_i = \frac{v_b}{i_b} = (\beta + 1)(r_e + R_e) \tag{3}$$

Note: The presence of R_e increases the input resistance by a factor of $(\beta+1)R_e$, which is a major advantage of emitter degeneration.

3. Voltage Gain (A_v)

The voltage gain A_v is defined as the ratio of the output voltage v_o to the base voltage v_b. The output voltage is developed by the collector current i_c flowing through the collector resistor R_c. Noting the current direction:

$$v_o = -i_c R_c = -\alpha i_e R_c \tag{4}$$

Using the expression for v_b derived in the previous section ($v_b = i_e(r_e + R_e)$), we can write the gain as:

$$A_v = \frac{v_o}{v_b} = \frac{-\alpha i_e R_c}{i_e(r_e + R_e)} \tag{5}$$

Canceling the common i_e term yields:

$$A_v = -\frac{\alpha R_c}{r_e + R_e} \approx -\frac{R_c}{r_e + R_e} \tag{6}$$

Note: The negative sign indicates a 180° phase inversion. Emitter degeneration stabilizes the gain against variations in r_e and transistor β.

4. Overall Voltage Gain (G_v)

The overall voltage gain G_v incorporates the loading effect of the input source resistance R_{sig}. It is defined as the ratio of v_o to the signal source v_{sig}. Using the voltage divider principle at the input, the base voltage v_b is:

$$v_b = v_{sig}\left(\frac{R_i}{R_i + R_{sig}}\right) \tag{7}$$

The overall gain can be expressed as:

$$G_v = \frac{v_o}{v_{sig}} = \frac{v_o}{v_b} \cdot \frac{v_b}{v_{sig}} = A_v\left(\frac{R_i}{R_i + R_{sig}}\right) \tag{8}$$

Substituting A_v and R_i into the expression:

$$G_v = \left(-\frac{\alpha R_c}{r_e + R_e}\right)\left[\frac{(\beta+1)(r_e+R_e)}{(\beta+1)(r_e+R_e)+R_{sig}}\right] \tag{9}$$

Recognizing that $\alpha(\beta+1) = \beta$, this simplifies cleanly to:

$$G_v = -\frac{\beta R_c}{R_{sig} + (\beta+1)(r_e+R_e)} \tag{10}$$

5. Output Resistance (R_o)

The output resistance R_o is determined by setting the independent input source to zero ($v_{sig} = 0$) and finding the Thevenin resistance looking back into the output (collector) terminal.

- With $v_{sig} = 0$, the input loop consists solely of passive resistors (R_{sig}, r_e, R_e).
- There is no independent driving force in this loop, meaning the base current $i_b = 0$.
- If $i_b = 0$, then the emitter current $i_e = 0$.
- Consequently, the dependent current source αi_e evaluates to 0 A, acting as an open circuit.

Looking into the output terminal, the only component connected to ground is the collector resistor R_c. Therefore:

$$R_o = R_c \tag{11}$$

Q5. For a BJT common-emitter amplifier with resistance in the emitter, derive the expressions for input impedance, output impedance, voltage gain, and current gain. What are the advantages and disadvantages of adding resistance in the emitter? **(30 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

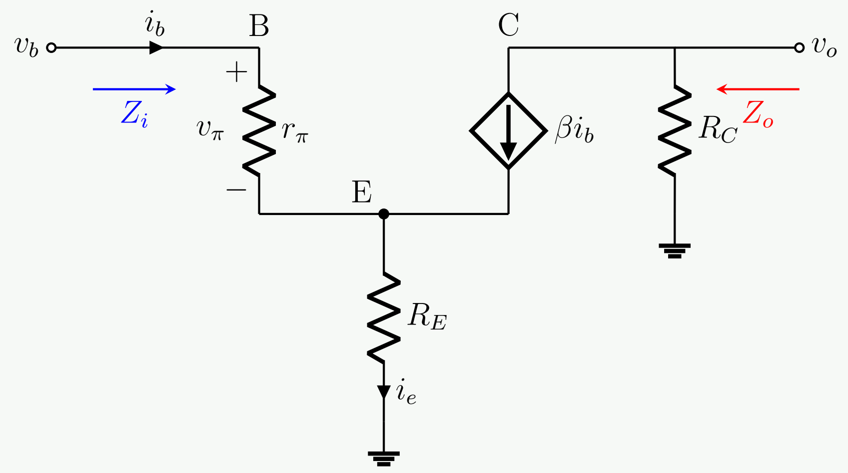
Answer

1. Small-Signal Equivalent Circuit and Assumptions

To analyze the Common-Emitter (CE) amplifier with emitter degeneration (an emitter resistor R_E), we utilize the low-frequency **Hybrid-π small-signal model** for the BJT.

Assumptions for Analysis:

- The transistor operates in the forward-active region.
- The signal frequencies are within the mid-band range, so external coupling/bypass capacitors act as short circuits and internal parasitic capacitances act as open circuits.
- The Early effect is neglected ($r_o \approx \infty$) for clarity, which is a standard assumption unless explicitly given an Early voltage V_A.



2. Derivations of Amplifier Parameters

2.1 Input Impedance (Z_i)

The input impedance Z_i is the equivalent impedance looking into the base terminal. By applying Kirchhoff's Voltage Law (KVL) to the input loop from the base to ground:

$$v_b = i_b r_\pi + i_e R_E \tag{1}$$

By Kirchhoff's Current Law (KCL) at the emitter node, the emitter current i_e is the sum of the base and collector currents:

$$i_e = i_b + i_c = i_b + \beta i_b = (1+\beta)i_b \tag{2}$$

Substituting i_e into the KVL equation yields:

$$v_b = i_b r_\pi + (1+\beta)i_b R_E$$
$$v_b = i_b \left[r_\pi + (1+\beta)R_E\right]$$

The input impedance is the ratio of input voltage to input current:

$$Z_i = \frac{v_b}{i_b} = r_\pi + (1+\beta)R_E \tag{3}$$

Note: If the circuit includes external biasing resistors R_1 and R_2 at the base, the total amplifier input impedance is $Z_{in} = R_1 \parallel R_2 \parallel Z_i$.

2.2 Output Impedance (Z_o)

The output impedance Z_o is found by looking into the output terminal (collector) with the independent input source deactivated ($v_b = 0$).

- When $v_b = 0$, there is no driving voltage in the input loop, causing the base current to become zero ($i_b = 0$).
- Consequently, the dependent current source βi_b evaluates to 0 A, acting as an open circuit.
- Looking back into the collector node, the only path to AC ground is through the collector resistor R_C.

$$Z_o = R_C \tag{4}$$

2.3 Voltage Gain (A_v)

Voltage gain is defined as the ratio of the output voltage v_o to the input voltage v_b. The output voltage is generated by the collector current flowing through R_C to ground. Based on the defined current direction, v_o is:

$$v_o = -i_c R_C = -\beta i_b R_C \tag{5}$$

From the input impedance derivation, we know that $i_b = \frac{v_b}{r_\pi+(1+\beta)R_E}$. Substituting this expression for i_b:

$$v_o = -\beta\left(\frac{v_b}{r_\pi + (1+\beta)R_E}\right)R_C$$
$$A_v = \frac{v_o}{v_b} = \frac{-\beta R_C}{r_\pi + (1+\beta)R_E}$$

Practical Approximation: If $(1+\beta)R_E \gg r_\pi$ and $\beta \gg 1$, the gain simplifies significantly to $A_v \approx -\frac{R_C}{R_E}$.

2.4 Current Gain (A_i)

The current gain evaluates the ratio of the output collector current i_c to the input base current i_b. For the intrinsic transistor stage:

$$A_i = \frac{i_c}{i_b} = \frac{\beta i_b}{i_b} = \beta \tag{6}$$

Note: If the current gain is defined with respect to a load current i_L flowing downwards through R_C, then $i_L = -i_c = -\beta i_b$, resulting in $A_i = -\beta$.

3. Advantages and Disadvantages of Adding R_E

The addition of an unbypassed emitter resistor R_E introduces **negative feedback** (specifically, series-current feedback) into the amplifier circuit.

Advantages:

- **Gain Stability:** The voltage gain becomes approximately $-\frac{R_C}{R_E}$, which depends purely on external passive components rather than highly variable transistor parameters (like β or r_π).
- **Increased Input Impedance:** The input impedance dramatically increases from r_π to $r_\pi + (1+\beta)R_E$, reducing the loading effect on previous signal stages.
- **Improved Linearity (Reduced Distortion):** The amplifier relies less on the non-linear base-emitter junction resistance (r_π). Consequently, it can handle larger input voltage swings without severe non-linear distortion.

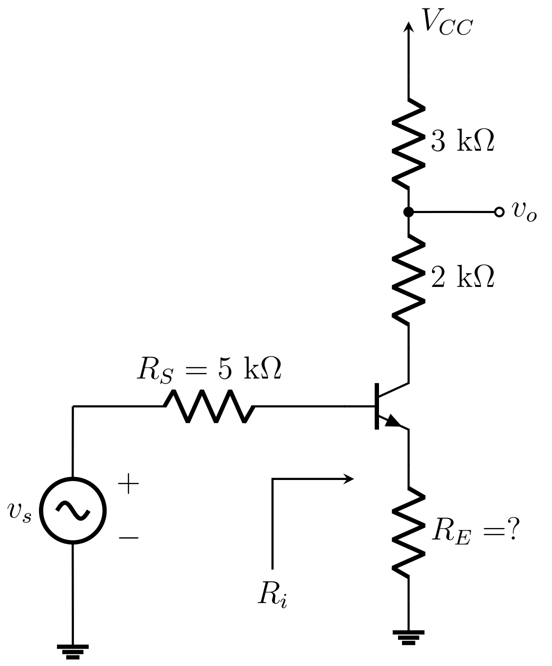
- **Bias Stability:** In DC analysis, R_E stabilizes the quiescent operating point (Q-point) against variations in temperature and β.

Disadvantages:

- **Reduced Voltage Gain:** The most significant drawback is the drastic reduction in voltage gain magnitude compared to a standard CE amplifier (which has a gain of approximately $-g_m R_C$).
- **Reduced Voltage Headroom:** A portion of the DC supply voltage must be dropped across R_E to bias the transistor, reducing the maximum possible output voltage swing at the collector.

Biasing and Input Impedance

Q1. For the circuit shown, find the value of R_E so that R_i is five times the value of R_s. Then using these values find R_i, R_o, A_v, and A_i for the circuit. Neglect early effect. Given: $\beta = 100$ and r_e (T-model) $= 40\,\Omega$.



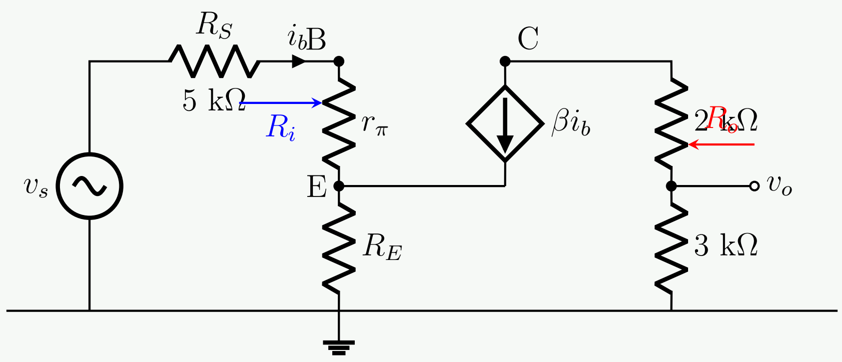
(26.67 marks) [EEE 263 | L-2/T-1 | 2018–19]

Answer

1. Small-Signal Equivalent Circuit

To analyze the amplifier, we first construct its mid-band AC equivalent circuit. The DC supply V_{CC} is replaced by an AC ground. We model the BJT using the small-signal equivalent circuit. Although r_e (from the T-model) is given, we can equivalently use the Hybrid-π model where $r_\pi = (\beta + 1)r_e$.

- **Input branch:** The source v_s and R_S are connected to the base.
- **Emitter branch:** The emitter resistor R_E connects to ground.
- **Collector branch:** The 2 kΩ and 3 kΩ resistors are in series with respect to the AC collector current flowing to AC ground. The output v_o is taken at the node between them.



2. Calculation of R_E

The problem specifies that the input resistance looking into the base, R_i, is five times the value of R_S:

$$R_i = 5R_S = 5 \times 5\text{ k}\Omega = 25\text{ k}\Omega \tag{1}$$

For a common-emitter amplifier with an unbypassed emitter resistor R_E, the input resistance is derived using Kirchhoff's Voltage Law from the base to ground:

$$R_i = \frac{v_b}{i_b} = (\beta + 1)(r_e + R_E) \tag{2}$$

Given $\beta = 100$ and $r_e = 40\,\Omega$, we can substitute the known values to find R_E:

$$\begin{aligned}
25000 &= (100+1)(40+R_E) \\
25000 &= 101(40+R_E) \\
40+R_E &= \frac{25000}{101} \approx 247.52\,\Omega \\
R_E &= 247.52 - 40 \\
R_E &\approx 207.52\,\Omega
\end{aligned}$$

Note: If the common approximation $R_i \approx \beta(r_e + R_E)$ is used instead, the calculation simplifies to $25000 = 100(40+R_E) \implies R_E = 210\,\Omega$. We will proceed with the exact analytical value of $207.52\,\Omega$.

3. Amplifier Parameters (R_i, R_o, A_v, A_i)

3.1 Input Impedance (R_i)
As directly constrained by the problem statement and calculated above:

$$R_i = 25 \text{ k}\Omega \tag{3}$$

3.2 Output Impedance (R_o)
The output impedance is defined as the Thevenin resistance looking back into the v_o terminal with the independent source v_s deactivated ($v_s = 0$).

- When $v_s = 0$, the base current $i_b = 0$.
- Consequently, the dependent current source $\beta i_b = 0$, acting as an open circuit.
- Looking into the v_o node, we see the 3 kΩ resistor connected to ground in parallel with the 2 kΩ branch. Since the 2 kΩ branch connects to the open-circuited collector, no current can flow through it.

$$R_o = 3 \text{ k}\Omega \tag{4}$$

3.3 Voltage Gain (A_v)
We define the overall voltage gain of the circuit as $A_{vs} = v_o/v_s$. First, establish the output voltage v_o. The collector current $i_c = \beta i_b$ flows from ground, up through the 3 kΩ resistor, and then through the 2 kΩ resistor. The voltage at node v_o (across the 3 kΩ resistor with respect to AC ground) is:

$$v_o = -i_c(3 \text{ k}\Omega) = -\beta i_b(3 \text{ k}\Omega) \tag{5}$$

The base current i_b is determined by the input source and the total series resistance:

$$i_b = \frac{v_s}{R_S + R_i} \tag{6}$$

Substituting i_b into the v_o equation:

$$\begin{aligned}
v_o &= -\beta\left(\frac{v_s}{R_S + R_i}\right)(3 \text{ k}\Omega) \\
A_{vs} &= \frac{v_o}{v_s} = -\frac{\beta \times 3 \text{ k}\Omega}{R_S + R_i}
\end{aligned}$$

Substitute the numerical values:

$$A_{vs} = -\frac{100 \times 3000}{5000 + 25000} = -\frac{300000}{30000} = -10\,\text{V/V} \tag{7}$$

Note: If the terminal voltage gain $A_v = v_o/v_b$ is required, it is $A_v = \frac{-i_c(3\text{ k}\Omega)}{i_b R_i} = \frac{-100\times 3\text{ k}\Omega}{25\text{ k}\Omega} = -12$ V/V.

3.4 Current Gain (A_i)
The current gain A_i is defined as the ratio of the output collector current i_c to the input base current i_b.

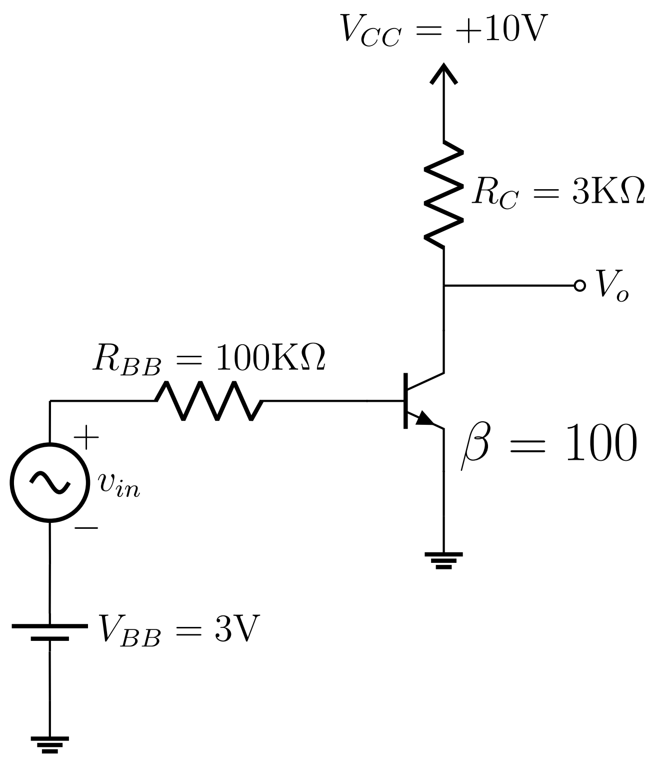
$$A_i = \frac{i_c}{i_b} = \frac{\beta i_b}{i_b} = \beta = 100\,\text{A/A} \tag{8}$$

4. Final Answer Summary

- **Emitter Resistance:** $R_E \approx 207.52\,\Omega$
- **Input Impedance:** $R_i = 25$ kΩ
- **Output Impedance:** $R_o = 3$ kΩ
- **Overall Voltage Gain:** $A_{vs} = -10\,\text{V/V}$
- **Current Gain:** $A_i = 100\,\text{A/A}$

BJT Input-Output Characteristics

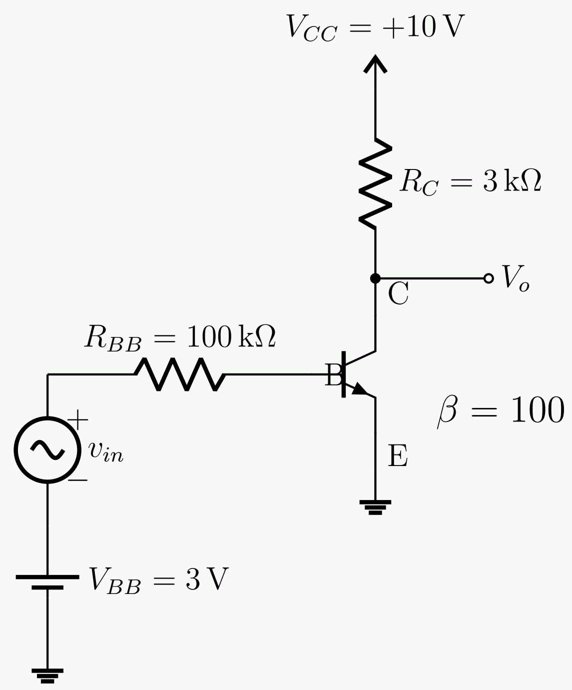
Q1. For the circuit shown, calculate (i) DC bias points, (ii) small-signal voltage gain, (iii) maximum allowable input AC voltage.



(26 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

1. Circuit Diagram



2. Part (i): DC Bias Points

To determine the DC bias points (Q-point), we perform a DC analysis. For this, all AC sources are set to zero ($v_{in} = 0$).
Assumptions:

- The transistor operates in the forward-active region.
- The forward-biased base-emitter voltage is $V_{BE(on)} \approx 0.7\,$V.
- The thermal voltage at room temperature is $V_T \approx 26\,$mV.

Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop:

$$V_{BB} - I_B R_{BB} - V_{BE(on)} = 0$$
$$I_B = \frac{V_{BB} - V_{BE(on)}}{R_{BB}}$$

Substituting the given values:

$$I_B = \frac{3\,\text{V} - 0.7\,\text{V}}{100\,\text{k}\Omega} = \frac{2.3\,\text{V}}{100\,\text{k}\Omega} = 23\,\mu\text{A} \tag{1}$$

Assuming the transistor is in the active region, the collector current I_C is:

$$I_C = \beta I_B = 100 \times 23\,\mu\text{A} = 2.3\,\text{mA} \tag{2}$$

Next, applying KVL to the collector-emitter loop to find the collector-emitter voltage V_{CE}:

$$V_{CC} - I_C R_C - V_{CE} = 0$$
$$V_{CE} = V_{CC} - I_C R_C$$

Substituting the values:

$$V_{CE} = 10\,\text{V} - (2.3\,\text{mA})(3\,\text{k}\Omega) = 10\,\text{V} - 6.9\,\text{V} = 3.1\,\text{V} \tag{3}$$

Verification of Region: Since $V_{CE} = 3.1\,\text{V} > V_{CE(sat)} \approx 0.2\,\text{V}$, the base-collector junction is reverse-biased, confirming the transistor is indeed operating in the forward-active region.
Final DC Bias Points:

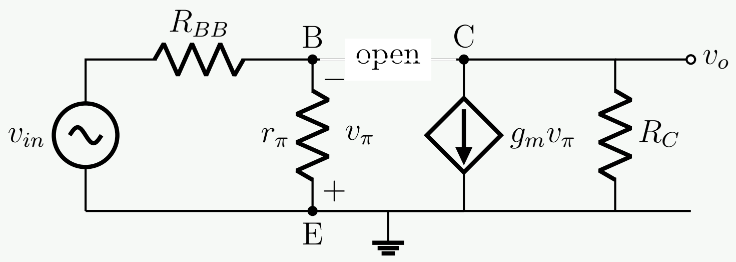
$$I_{CQ} = 2.3\,\text{mA}, \quad V_{CEQ} = 3.1\,\text{V} \tag{4}$$

3. Part (ii): Small-Signal Voltage Gain

To determine the AC voltage gain, we first calculate the small-signal parameters using the DC collector current:

$$g_m = \frac{I_{CQ}}{V_T} = \frac{2.3\,\text{mA}}{26\,\text{mV}} \approx 88.46\,\text{mA/V}$$
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{88.46\,\text{mA/V}} \approx 1.13\,\text{k}\Omega$$

We now construct the mid-band AC small-signal equivalent circuit by replacing the transistor with its Hybrid-π model and shorting DC voltage sources (V_{CC} and V_{BB} to AC ground).



Applying a voltage divider at the input base node to find v_π:

$$v_\pi = v_{in}\left(\frac{r_\pi}{R_{BB} + r_\pi}\right) \tag{5}$$

The output voltage v_o is generated by the dependent current source drawing current through R_C (since the output node is loaded only by R_C to ground):

$$v_o = -g_m v_\pi R_C \tag{6}$$

Substituting v_π into the output equation yields the overall voltage gain A_v:

$$A_v = \frac{v_o}{v_{in}} = -g_m R_C\left(\frac{r_\pi}{R_{BB} + r_\pi}\right)$$
$$A_v = \frac{-\beta R_C}{R_{BB} + r_\pi}$$

Substituting the numerical values:

$$A_v = \frac{-100 \times 3\,\text{k}\Omega}{100\,\text{k}\Omega + 1.13\,\text{k}\Omega} = \frac{-300}{101.13} \approx -2.966\,\text{V/V} \tag{7}$$

4. Part (iii): Maximum Allowable Input AC Voltage

The maximum allowable input AC voltage swing is constrained by the transistor leaving the active region. It can clip by either entering saturation (where $v_{CE} \approx 0.2\,\text{V}$) or cutoff (where $i_C \approx 0$).
1. Saturation Limit (Negative Swing of v_{CE}): The DC operating point is $V_{CEQ} = 3.1\,\text{V}$. The maximum negative swing of the AC collector-emitter voltage before hitting saturation ($V_{CE(sat)} = 0.2\,\text{V}$) is:

$$\Delta v_{ce(sat)} = V_{CEQ} - V_{CE(sat)} = 3.1\,\text{V} - 0.2\,\text{V} = 2.9\,\text{V} \tag{8}$$

Since $v_o = v_{ce} = A_v v_{in}$, the corresponding peak input voltage is:

$$v_{in(peak),sat} = \left|\frac{\Delta v_{ce(sat)}}{A_v}\right| = \frac{2.9\,\text{V}}{2.966} \approx 0.978\,\text{V} \tag{9}$$

2. Cutoff Limit (Positive Swing of v_{CE}): In cutoff, i_c drops to zero, so the voltage across R_C becomes zero, and v_{CE} swings up to V_{CC}. The maximum positive swing is:

$$\Delta v_{ce(cutoff)} = V_{CC} - V_{CEQ} = 10\,\text{V} - 3.1\,\text{V} = 6.9\,\text{V} \tag{10}$$

The corresponding peak input voltage before cutoff is:

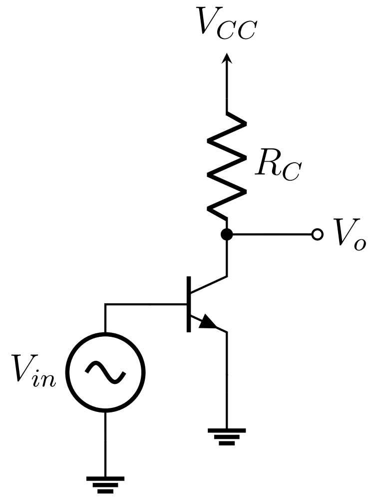
$$v_{in(peak),cutoff} = \left| \frac{\Delta v_{ce(cutoff)}}{A_v} \right| = \frac{6.9\,\text{V}}{2.966} \approx 2.326\,\text{V} \tag{11}$$

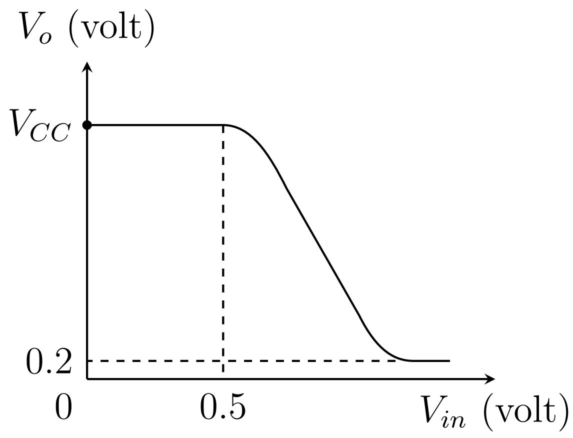
Maximum Symmetric Input Swing: To avoid clipping symmetrically in either direction, the maximum allowable AC input voltage must be the smaller of the two limits. Thus, the amplifier is limited by saturation.

$$v_{in(max)} = 0.978\,\text{V (peak)} \tag{12}$$

Important Note: While $0.978\,\text{V}$ *is the limit for macro-clipping, strict small-signal operation usually requires the AC base-emitter voltage* v_π *to be much less than the thermal voltage* V_T *(typically* $v_\pi < 10\,\text{mV}$*). At* $v_{in} = 0.978\,\text{V}$*, we have* $v_\pi \approx 0.978 \times (1.13/101.13) \approx 10.9\,\text{mV}$*, which is just bordering the edge of strict small-signal linearity. Therefore, limiting* v_{in} *to roughly* $0.9\,\text{V}$ *peak ensures both avoidance of hard clipping and reasonable adherence to the small-signal assumption.*

Q2. The input-output characteristics of a BJT circuit are given. What will happen (i) if V_{cc} is increased, (ii) if R_c is increased? Explain with the characteristic curve.





(10 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Circuit and Voltage Transfer Characteristics (VTC)

The given circuit is a standard Common-Emitter (CE) configuration with an unbypassed emitter connected to ground. Its operation spans three operational regions depending on V_{in}: Cut-off, Forward-Active, and Saturation.

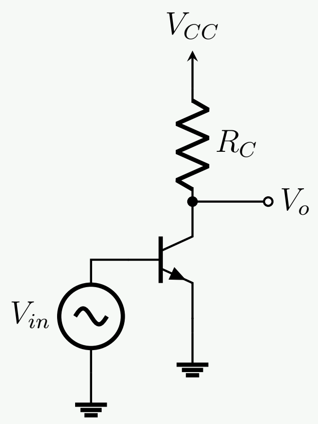


Figure 7: Common-Emitter Configuration

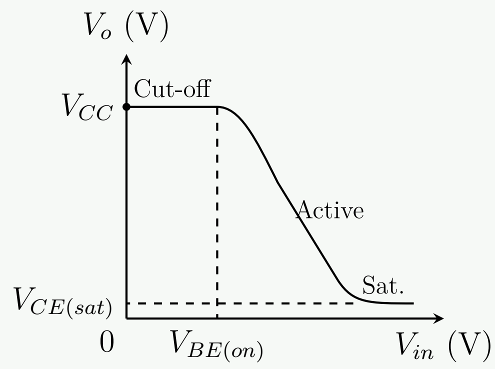


Figure 8: Base VTC Curve

2. Mathematical Derivation of VTC Regions

The relationship between the output voltage V_o (which equals V_{CE}) and the input voltage V_{in} (which equals V_{BE}) can be detailed stage-by-stage:

(a) **Cut-off Region ($V_{in} < V_{BE(on)}$):**

The base-emitter junction is off, ensuring that $I_B = 0$ and $I_C = 0$. Consequently:

$$V_o = V_{CC} - I_C R_C = V_{CC} \tag{1}$$

(b) **Forward-Active Region ($V_{BE(on)} \leq V_{in} < V_{in(sat)}$):**
The transistor turns on. The collector current is modeled exponentially by the ideal diode expression or linearized small-signal parameters:

$$I_C = I_S \exp\left(\frac{V_{in}}{V_T}\right) \tag{2}$$

The output drops sharply along a high-gain path governed by:

$$V_o = V_{CC} - I_C R_C = V_{CC} - R_C I_S \exp\left(\frac{V_{in}}{V_T}\right) \tag{3}$$

The slope of this region represents the small-signal voltage gain $A_v = \frac{dV_o}{dV_{in}} = -g_m R_C$.

(c) **Saturation Region ($V_{in} \geq V_{in(sat)}$):**
As V_{in} continues to rise, I_C increases until V_o drops to its absolute minimum lower limit, $V_{CE(sat)} \approx 0.2$ V. Beyond this point, increasing V_{in} yields no further drop in V_o:

$$V_o = V_{CE(sat)} \approx 0.2 \text{ V} \tag{4}$$

3. Circuit Variations and Impact Analysis

—

(i) Effect of Increasing V_{CC}

When the supply voltage V_{CC} increases to a new value $V'_{CC} > V_{CC}$, the following changes occur:

- **Maximum Output Shift:** In the cut-off region, the maximum value of V_o is equal to the supply voltage. Hence, the horizontal cut-off segment shifts upward from V_{CC} to V'_{CC}.
- **Active Region Extension:** The active region now initiates from a higher voltage plateau (V'_{CC}). Because it takes a larger current variation to drop V_o from V'_{CC} down to $V_{CE(sat)}$, the length of the active region path increases.
- **Saturation Baseline:** The lower limit of the curve ($V_{CE(sat)}$) is determined strictly by the physical architecture of the BJT and is entirely independent of V_{CC}. Therefore, the saturation floor remains at 0.2 V.

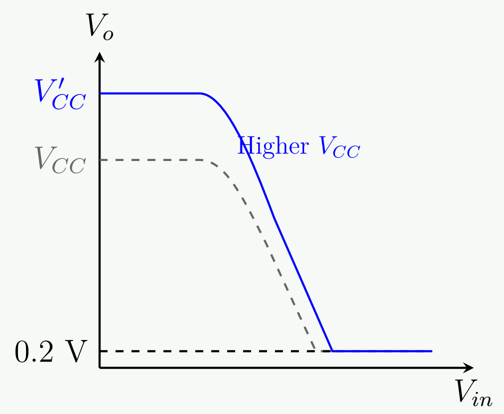


Figure 9: VTC Shift due to increased V_{CC}

—

(ii) Effect of Increasing R_C

When the collector resistance R_C is increased to $R'_C > R_C$, the supply voltage remains constant, leading to the following variations:

- **Unaltered Cut-off Value:** Since $I_C = 0$ in cut-off, the value of V_o remains exactly equal to V_{CC}, anchoring the starting point of the graph.
- **Increased Voltage Gain (Steeper Slope):** The voltage gain magnitude in the active region is directly proportional to R_C ($|A_v| \approx g_m R_C$). A larger resistor value implies that a smaller increment in I_C (and consequently a smaller change in V_{in}) is needed to cause a significant drop in V_o. The slope becomes noticeably steeper.
- **Earlier Saturation Entry:** Because V_o drops faster per unit change in V_{in}, the transistor reaches the saturation voltage limit ($V_{CE(sat)} = 0.2$ V) at a lower input voltage threshold.

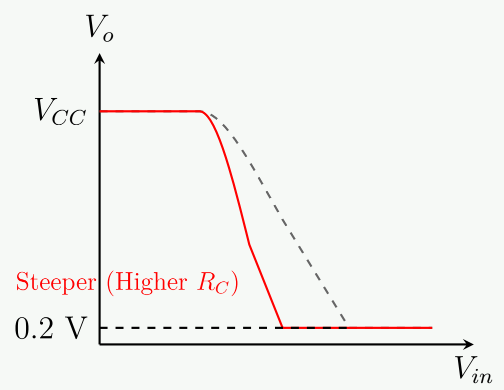


Figure 10: VTC Shift due to increased R_C

4. Summary Table

Parameter Change	**Cut-off Voltage Level**	**Active Region Slope**	**Saturation Voltage Level**	**Saturation Onset (V_{in})**
Increase V_{CC}	Increases (V'_{CC})	Marginally shifted	Unchanged (0.2 V)	Delayed (Higher V_{in})
Increase R_C	Unchanged (V_{CC})	Steeper (Higher Gain)	Unchanged (0.2 V)	Accelerated (Lower V_{in})

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Electronic Circuits

S6 — MOSFET Amplifiers

Common-source configuration, small-signal model, gain derivation

S6 MOSFET Amplifiers

Common Gate Amplifier

Q1. Derive the expression for input resistance, output resistance, and overall voltage gain of a common-gate (CG) amplifier. **(26 marks)**
[EEE 263 | L-2/T-1 | 2013–14]

Answer

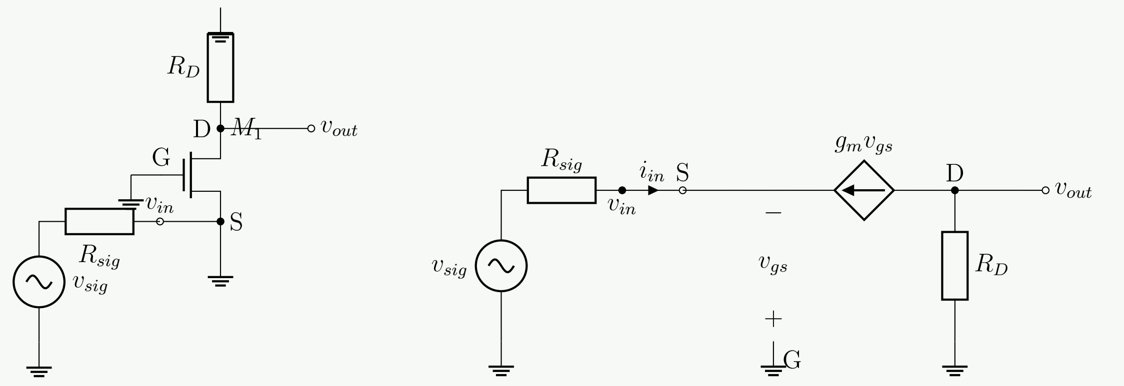
Objective: Derive the expressions for input resistance (R_{in}), output resistance (R_{out}), and overall voltage gain (G_v) of a Common-Gate (CG) MOSFET amplifier.

1. Circuit Diagrams and Assumptions

To analyze the amplifier, we utilize the small-signal AC equivalent circuit. The standard hybrid-π model is applied.
Key Assumptions:

- The MOSFET is biased in the **saturation region**.
- **Channel-length modulation is neglected** ($r_o \to \infty$) for the foundational derivation.
- **Body effect is neglected** ($g_{mb} = 0$) to isolate the primary CG behavior.
- The DC bias sources (I_{BIAS}, V_{DD}, V_{SS}) are treated as AC grounds.

AC Equivalent Schematic **Small-Signal Equivalent (Hybrid-π)**



2. Input Resistance (R_{in})

The input resistance R_{in} is defined as the resistance looking directly into the source node, excluding the signal generator resistance R_{sig}.

$$R_{in} = \frac{v_{in}}{i_{in}} \tag{1}$$

By inspecting the small-signal model, the gate is grounded, so $v_g = 0$. The source voltage is $v_s = v_{in}$. Therefore, the gate-to-source voltage is:

$$v_{gs} = v_g - v_s = 0 - v_{in} = -v_{in} \tag{2}$$

Applying Kirchhoff's Current Law (KCL) at the source node (S). The current i_{in} entering the node must equal the current pulled by the dependent source:

$$i_{in} + g_m v_{gs} = 0 \implies i_{in} = -g_m(-v_{in}) = g_m v_{in} \tag{3}$$

Rearranging for R_{in}:

$$R_{in} = \frac{v_{in}}{g_m v_{in}} = \frac{1}{g_m}$$

Note: The input resistance of a Common-Gate amplifier is characteristically very low.

3. Overall Voltage Gain (G_v)

First, we find the intrinsic voltage gain from the source to the drain ($A_v = v_{out}/v_{in}$). The output voltage is generated by the dependent current source driving into R_D:

$$v_{out} = -(g_m v_{gs})R_D \tag{4}$$

Substituting $v_{gs} = -v_{in}$:

$$v_{out} = -g_m(-v_{in})R_D = g_m R_D v_{in} \implies A_v = \frac{v_{out}}{v_{in}} = g_m R_D \tag{5}$$

Next, we relate v_{in} to the generator voltage v_{sig} using voltage division at the input node:

$$v_{in} = v_{sig}\left(\frac{R_{in}}{R_{sig} + R_{in}}\right) = v_{sig}\left(\frac{1/g_m}{R_{sig} + 1/g_m}\right) = v_{sig}\left(\frac{1}{1 + g_m R_{sig}}\right) \tag{6}$$

The overall voltage gain G_v is the product of the input voltage divider and the intrinsic gain A_v:

$$G_v = \frac{v_{out}}{v_{sig}} = \left(\frac{v_{out}}{v_{in}}\right)\left(\frac{v_{in}}{v_{sig}}\right) = (g_m R_D)\left(\frac{1}{1 + g_m R_{sig}}\right)$$

$$G_v = \frac{g_m R_D}{1 + g_m R_{sig}}$$

Note: The gain is positive, indicating that the Common-Gate amplifier is non-inverting.

4. Output Resistance (R_{out})

The output resistance R_{out} is found by setting the independent source $v_{sig} = 0$ and looking into the drain node.

- Setting $v_{sig} = 0$ grounds the left side of R_{sig}.
- We apply a test voltage v_{test} at the drain and measure i_{test}.

Let us find v_{gs} under this condition. Applying KCL at the source node:

$$\frac{v_s}{R_{sig}} + g_m v_{gs} = 0 \tag{7}$$

Since $v_g = 0$, we know $v_{gs} = -v_s$. Substituting this into the KCL equation:

$$\frac{-v_{gs}}{R_{sig}} + g_m v_{gs} = 0 \implies v_{gs}\left(g_m - \frac{1}{R_{sig}}\right) = 0 \tag{8}$$

Assuming typical finite values for g_m and R_{sig}, the only solution is $v_{gs} = 0$. Because $v_{gs} = 0$, the dependent current source $g_m v_{gs} = 0$ (it acts as an open circuit). Therefore, looking into the drain node, the only path for the test current is through the drain resistor R_D.

$$R_{out} = R_D$$

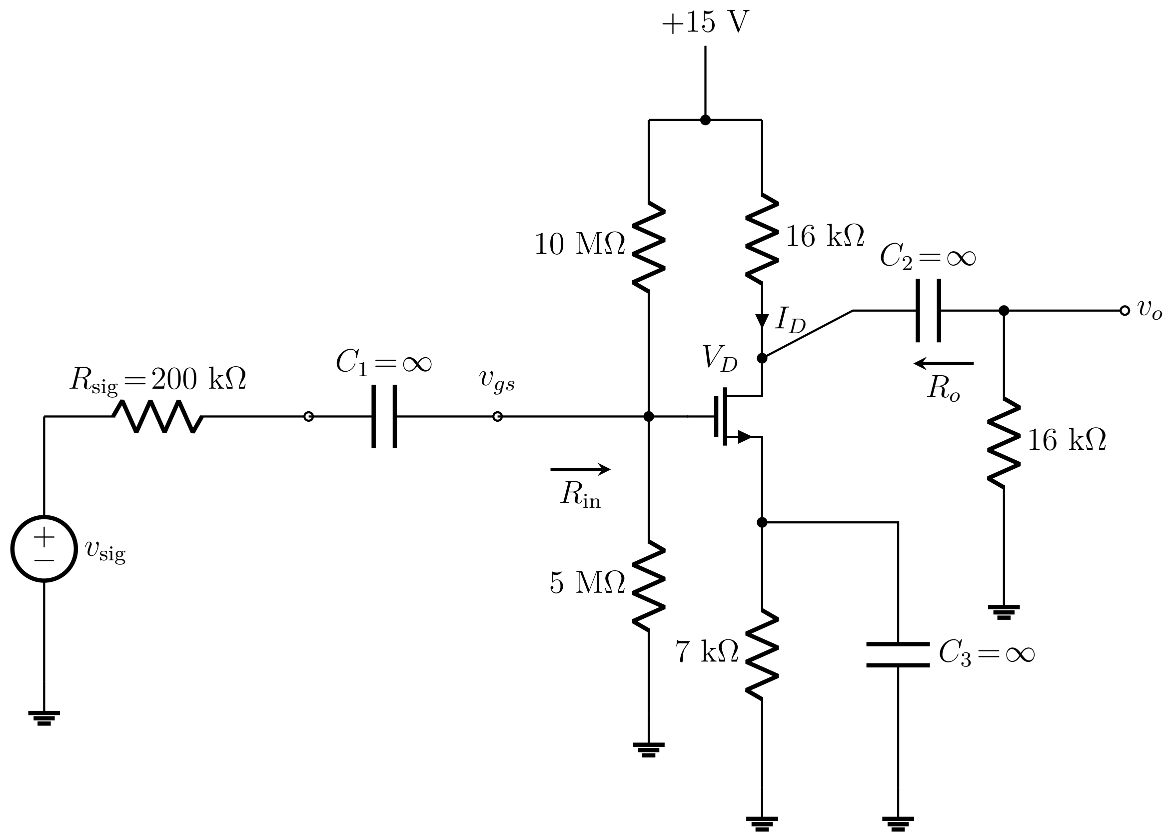
5. Summary of Results

Parameter	**Derived Expression (Assuming $r_o \to \infty$)**
Input Resistance (R_{in})	$\frac{1}{g_m}$
Intrinsic Voltage Gain (A_v)	$g_m R_D$
Overall Voltage Gain (G_v)	$\frac{g_m R_D}{1+g_m R_{sig}}$
Output Resistance (R_{out})	R_D

Common Source Amplifier

Q1. For the Common Source amplifier circuit shown, the transistor has $V_t = 1\,\text{V}$, $\mu_n C_{ox}(W/L) = 4\,\text{mA/V}^2$, and $V_A = 100\,\text{V}$. All capacitors behave as open circuits under DC and short circuits under AC.

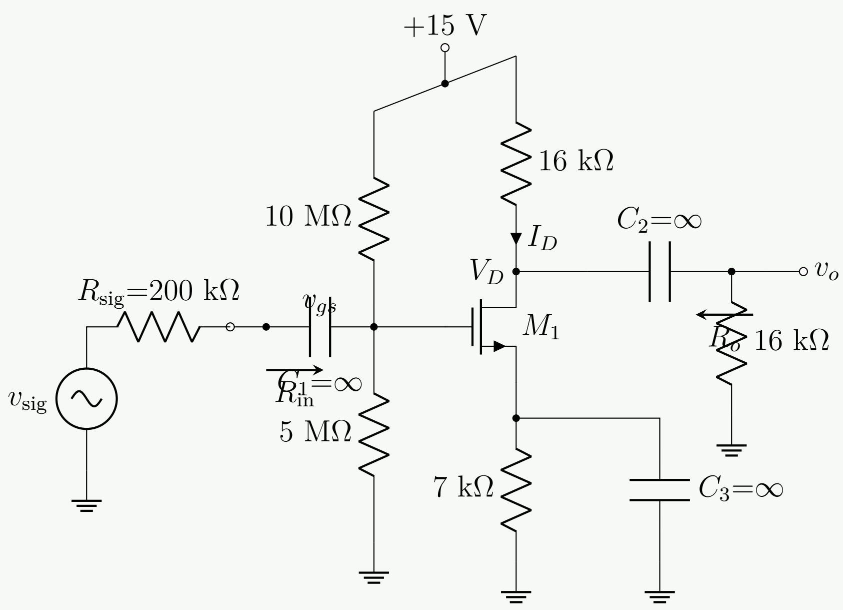
(i) Find the DC bias voltage V_D and drain current I_D (ignore Early effect for bias).
(ii) Find g_m, r_o and draw the small-signal equivalent circuit.
(iii) Determine R_{in}, R_o, and the overall voltage gain v_o/v_{sig}.
(iv) Calculate the maximum peak value of v_{sig} such that $v_{gs} \leq 0.1\,V_{OV}$ (small-signal approximation valid).
(v) Suggest a circuit modification that will double the maximum v_{sig} found in (iv) without changing the DC bias point.



(35 marks) [EEE 263 | L-2/T-1 | 2024–25]

Answer

1. Reference Circuit Diagram



(i) DC Analysis (V_D and I_D)

For DC analysis, all capacitors are treated as open circuits. The gate current is zero ($I_G = 0$), so the gate voltage V_G is determined solely by the voltage divider formed by the 10 MΩ and 5 MΩ resistors.

$$V_G = V_{DD}\left(\frac{5\ \text{M}\Omega}{10\ \text{M}\Omega + 5\ \text{M}\Omega}\right) = 15\left(\frac{5}{15}\right) = 5\ \text{V} \tag{1}$$

The voltage at the source terminal is:

$$V_S = I_D R_S = 7I_D \quad (\text{with } I_D \text{ in mA}) \tag{2}$$

The gate-to-source voltage is therefore:

$$V_{GS} = V_G - V_S = 5 - 7I_D \tag{3}$$

Assuming the MOSFET operates in the saturation region, the drain current is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_t)^2 \tag{4}$$

Substitute the known values ($V_t = 1$ V, $\mu_n C_{ox}(W/L) = 4$ mA/V^2):

$$I_D = \frac{1}{2}(4)(5 - 7I_D - 1)^2$$
$$I_D = 2(4 - 7I_D)^2$$
$$I_D = 2\left(16 - 56I_D + 49I_D^2\right)$$
$$98I_D^2 - 113I_D + 32 = 0$$

Solving this quadratic equation for I_D:

$$I_D = \frac{113 \pm \sqrt{(-113)^2 - 4(98)(32)}}{2(98)} = \frac{113 \pm \sqrt{12769 - 12544}}{196} = \frac{113 \pm 15}{196} \tag{5}$$

This gives two possible mathematical solutions:

- $I_{D1} = \frac{128}{196} \approx 0.653$ mA $\implies V_{GS} = 5 - 7(0.653) = 0.429$ V $< V_t$ (Transistor would be OFF, invalid).
- $I_{D2} = \frac{98}{196} = 0.5$ mA $\implies V_{GS} = 5 - 7(0.5) = 1.5$ V $> V_t$ (Transistor is ON, valid).

Thus, the quiescent drain current is $\mathbf{I_D = 0.5}$ mA. The DC drain voltage V_D is:

$$V_D = V_{DD} - I_D R_D = 15 - (0.5 \text{ mA})(16 \text{ k}\Omega) = 15 - 8 = \mathbf{7} \text{ V} \tag{6}$$

Verification of Saturation: $V_S = 0.5 \times 7 = 3.5$ V. $V_{DS} = V_D - V_S = 7 - 3.5 = 3.5$ V. The overdrive voltage is $V_{OV} = V_{GS} - V_t = 1.5 - 1 = 0.5$ V. Since $V_{DS} > V_{OV}$ (3.5 V > 0.5 V), the assumption of saturation is verified.

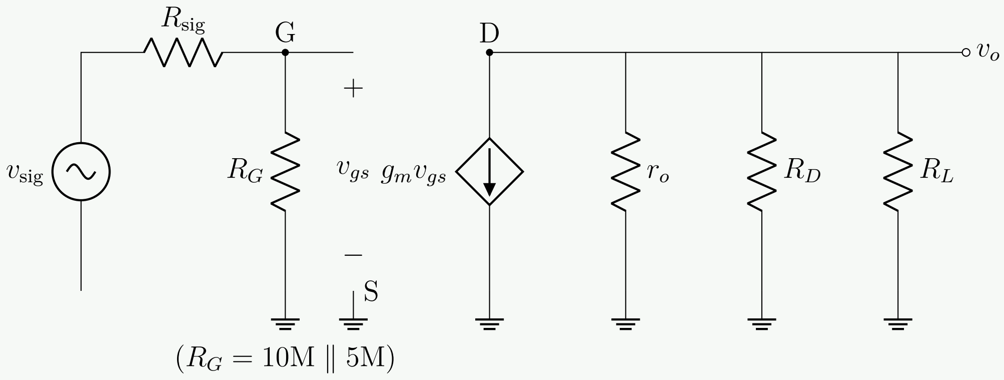
(ii) Small-Signal Parameters and Equivalent Circuit

The transconductance g_m and output resistance r_o are calculated as:

$$g_m = \sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right) I_D} = \sqrt{2(4 \text{ mA/V}^2)(0.5 \text{ mA})} = \sqrt{4} = \mathbf{2} \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = \frac{100 \text{ V}}{0.5 \text{ mA}} = \mathbf{200 \text{ k}\Omega}$$

For AC analysis, the DC sources (V_{DD}) are grounded. Capacitors C_1, C_2, C_3 are treated as short circuits. C_3 effectively grounds the source terminal, neutralizing R_S in AC operation. The small-signal hybrid-π equivalent circuit is shown below.



(iii) R_{in}, R_o, and Overall Voltage Gain (v_o/v_{sig})

The input resistance R_{in} is the equivalent resistance of the bias network:

$$R_{in} = 10 \text{ M}\Omega \parallel 5 \text{ M}\Omega = \frac{10 \times 5}{10 + 5} \text{ M}\Omega = \mathbf{3.33 \text{ M}\Omega} \tag{7}$$

The output resistance R_o looks into the amplifier before the load R_L:

$$R_o = R_D \parallel r_o = 16 \text{ k}\Omega \parallel 200 \text{ k}\Omega = \frac{16 \times 200}{216} \text{ k}\Omega = \mathbf{14.81 \text{ k}\Omega} \tag{8}$$

To find the overall voltage gain, relate v_{gs} to v_{sig} using the input voltage divider:

$$v_{gs} = v_{sig}\left(\frac{R_{in}}{R_{sig} + R_{in}}\right) = v_{sig}\left(\frac{3.33 \text{ M}\Omega}{0.2 \text{ M}\Omega + 3.33 \text{ M}\Omega}\right) = v_{sig}\left(\frac{3.333}{3.533}\right) \approx 0.943 v_{sig} \tag{9}$$

The output voltage is developed across the total parallel load resistance $R_L' = r_o \parallel R_D \parallel R_L$:

$$R_L' = 200\text{k} \parallel 16\text{k} \parallel 16\text{k} = 200\text{k} \parallel 8\text{k} = \frac{1600}{208} \text{ k}\Omega \approx 7.69 \text{ k}\Omega \tag{10}$$

The intrinsic gain is $A_v = -g_m R_L' = -(2 \text{ mA/V})(7.69 \text{ k}\Omega) = -15.38$ V/V. The overall voltage gain G_v is:

$$G_v = \frac{v_o}{v_{sig}} = \frac{v_{gs}}{v_{sig}} \times A_v = 0.943 \times (-15.38) = \mathbf{-14.51} \text{ V/V} \tag{11}$$

(iv) Maximum Peak Value of v_{sig}

To satisfy the small-signal condition $v_{gs} \leq 0.1V_{OV}$:

$$v_{gs(\max)} = 0.1 \times 0.5 \text{ V} = 0.05 \text{ V} = 50 \text{ mV} \tag{12}$$

Relating this back to the source using the input divider ratio:

$$v_{\text{sig}(\max)} = \frac{v_{gs(\max)}}{0.943} = \frac{0.05 \text{ V}}{0.943} = \mathbf{0.053} \text{ V or } \mathbf{53} \text{ mV} \tag{13}$$

(v) Circuit Modification to Double Maximum v_{sig}

To double the allowable input signal v_{sig} without violating the small-signal constraint ($v_{gs} \leq 50$ mV) and without changing the DC bias point, we can introduce **Source Degeneration**.
By leaving a portion of the source resistance unbypassed (let's call it R_s), negative feedback is introduced. The new relation becomes $v_{gs} = v_i/(1 + g_m R_s)$. To double the input limit, we need to halve the signal reaching v_{gs}, which means setting $1 + g_m R_s = 2$.

$$g_m R_s = 1 \implies R_s = \frac{1}{g_m} = \frac{1}{2 \text{ mA/V}} = 0.5 \text{ k}\Omega = 500 \ \Omega \tag{14}$$

Modification: Split the 7 kΩ source resistor into two series resistors: $R_{S1} = 500 \ \Omega$ and $R_{S2} = 6.5$ kΩ. Connect the bypass capacitor C_3 *only* across R_{S2}. The total DC resistance remains 7 kΩ, preserving the DC bias point precisely, but the AC equivalent circuit now includes a 500 Ω unbypassed source resistor.

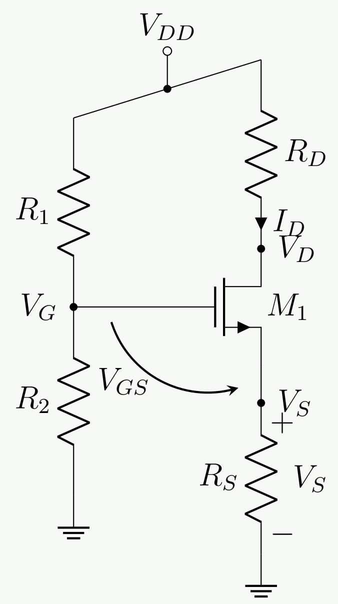
Q2. Explain why adding a source resistor R_S to a Common-Source MOSFET amplifier improves its bias stability compared to directly grounding the source terminal. Detail how R_S actively compensates for manufacturing variations in the threshold voltage V_t and transconductance parameter k_n. **(12 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

Objective: Explain the stabilizing mechanism of a source resistor R_S in MOSFET biasing and mathematically demonstrate how it desensitizes the quiescent drain current (I_{DQ}) against variations in the threshold voltage (V_t) and the transconductance parameter (k_n).

1. Reference Bias Circuit and Principle of Operation

The diagram below illustrates a standard voltage-divider bias circuit with a source degeneration resistor R_S.



Mechanism of DC Negative Feedback: When the source is grounded ($R_S = 0$), the gate-to-source voltage is fixed entirely by the gate bias network ($V_{GS} = V_G$). Any variation in the transistor's intrinsic parameters (V_t or k_n) translates directly into large fluctuations in the drain current I_D.
By introducing R_S, the voltage at the source becomes dependent on the drain current: $V_S = I_D R_S$. Since V_G is established by the $R_1 - R_2$ voltage divider, the effective drive voltage across the transistor becomes:

$$V_{GS} = V_G - V_S = V_G - I_D R_S \tag{1}$$

This establishes **DC negative feedback**. If a parameter variation causes I_D to increase, the voltage drop across R_S increases. Consequently, V_{GS} decreases, which in turn reduces I_D, counteracting the original increase and stabilizing the Q-point.

2. Mathematical Derivation: Compensation for V_t Variations

Assume the MOSFET is operating in saturation. The drain current is given by:

$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2 \tag{2}$$

where $k_n = \mu_n C_{ox}(W/L)$. Substituting Equation (1) into Equation (2):

$$I_D = \frac{1}{2}k_n(V_G - I_D R_S - V_t)^2 \tag{3}$$

To find the sensitivity of I_D with respect to threshold voltage V_t, we take the derivative of both sides with respect to V_t, applying the chain rule:

$$\frac{\partial I_D}{\partial V_t} = k_n(V_G - I_D R_S - V_t) \cdot \frac{\partial}{\partial V_t}(V_G - I_D R_S - V_t) \tag{4}$$

Recognizing that $k_n(V_G - I_D R_S - V_t) = k_n(V_{GS} - V_t) = g_m$, and noting that V_G is a constant independent of V_t:

$$\frac{\partial I_D}{\partial V_t} = g_m\left(-R_S\frac{\partial I_D}{\partial V_t} - 1\right)$$

$$\frac{\partial I_D}{\partial V_t} = -g_m R_S\frac{\partial I_D}{\partial V_t} - g_m$$

$$\frac{\partial I_D}{\partial V_t}(1 + g_m R_S) = -g_m \implies \frac{\partial I_D}{\partial V_t} = \frac{-g_m}{1 + g_m R_S}$$

Interpretation: Without R_S (i.e., $R_S = 0$), the sensitivity is $\partial I_D/\partial V_t = -g_m$. Equation (**??**) demonstrates that adding R_S reduces the sensitivity of the drain current to V_t variations by a **desensitivity factor** of $(1 + g_m R_S)$.

3. Mathematical Derivation: Compensation for k_n Variations

Process variations in oxide thickness (C_{ox}), mobility (μ_n), or lithography (W/L) result in variations in the parameter k_n. Using the total differential of Equation (2):

$$\Delta I_D \approx \frac{\partial I_D}{\partial k_n}\Delta k_n + \frac{\partial I_D}{\partial V_{GS}}\Delta V_{GS} \tag{5}$$

Evaluating the partial derivatives:

- $\frac{\partial I_D}{\partial k_n} = \frac{1}{2}(V_{GS} - V_t)^2 = \frac{I_D}{k_n}$
- $\frac{\partial I_D}{\partial V_{GS}} = k_n(V_{GS} - V_t) = g_m$
- From Equation (1), $\Delta V_{GS} = -R_S \Delta I_D$

Substituting these into the differential equation:

$$\Delta I_D \approx \left(\frac{I_D}{k_n}\right)\Delta k_n + g_m(-R_S \Delta I_D)$$

$$\Delta I_D + g_m R_S \Delta I_D \approx I_D\left(\frac{\Delta k_n}{k_n}\right)$$

$$\Delta I_D(1 + g_m R_S) \approx I_D\left(\frac{\Delta k_n}{k_n}\right) \implies \frac{\Delta I_D}{I_D} = \frac{1}{1 + g_m R_S}\left(\frac{\Delta k_n}{k_n}\right)$$

Interpretation: Equation (**??**) correlates the fractional change in drain current ($\Delta I_D/I_D$) to the fractional change in the transconductance parameter ($\Delta k_n/k_n$). Without source degeneration ($R_S = 0$), a 10% variation in k_n would cause a direct 10% variation in I_D. However, the presence of R_S attenuates this percentage variation by the same desensitivity factor, $(1 + g_m R_S)$.

4. Conclusion

Adding a source resistor R_S fundamentally shifts the bias network from an open-loop voltage configuration to a closed-loop current-feedback configuration. While a portion of the supply voltage must be sacrificed across R_S, this design trade-off yields highly predictable and stable quiescent conditions, drastically reducing the impact of transistor manufacturing tolerances.

Q3. A CS amplifier is connected to a source v_{sig} with resistance R_{sig} and it utilizes a MOSFET with $\mu_n C_{ox} = 400\mu\text{A/V}^2$, $W/L = 10$, and $V_A = 10\text{V}$. It is biased at $I_D = 0.2$ mA and uses $R_D = 6$ kΩ. Find R_{in}, A_{vo}, and R_o. If a load resistance of 10 kΩ is connected to the output, what will be the overall voltage gain G_v? If a 0.2-V peak sine-wave signal is required at the output, what must the peak amplitude of v_{sig} be? **(15 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

Objective: Determine the input resistance (R_{in}), open-circuit voltage gain (A_{vo}), output resistance (R_o), loaded overall voltage gain (G_v), and the required input signal amplitude for a given output swing of a Common-Source (CS) amplifier.

1. Assumptions and Small-Signal Parameters

Assumptions:

- The MOSFET is properly biased in the **saturation region** (active region) to act as an amplifier.

- No gate-biasing resistors (R_G) are explicitly mentioned in the circuit description. Therefore, we assume an ideal gate bias setup where the input resistance is determined solely by the MOSFET's gate, which acts as an open circuit ($R_{in} \to \infty$).

Small-Signal Parameters Calculation: Given the DC bias current $I_D = 0.2$ mA, we first evaluate the transconductance parameter k_n:

$$k_n = \mu_n C_{ox}\left(\frac{W}{L}\right) = \left(400\ \mu\text{A/V}^2\right) \times 10 = 4000\ \mu\text{A/V}^2 = 4\ \text{mA/V}^2 \tag{1}$$

The **transconductance** (g_m) is:

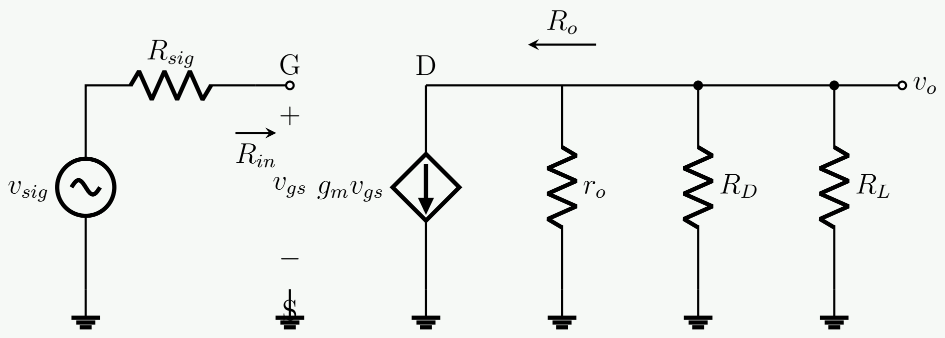
$$\begin{aligned} g_m &= \sqrt{2k_nI_D} \\ &= \sqrt{2 \times \left(4\ \text{mA/V}^2\right) \times (0.2\ \text{mA})} \\ &= \sqrt{1.6\ \text{mA}^2/\text{V}^2} \approx \mathbf{1.265}\ \text{mA/V} \end{aligned}$$

The **output resistance** (r_o) due to channel-length modulation is:

$$r_o = \frac{V_A}{I_D} = \frac{10\ \text{V}}{0.2\ \text{mA}} = \mathbf{50\ k\Omega} \tag{2}$$

2. Small-Signal Equivalent Circuit

The hybrid-π small-signal equivalent circuit of the CS amplifier, including the signal source, the internal Early effect resistance (r_o), the drain resistor (R_D), and an external load (R_L), is drawn below:



3. Amplifier Parameters (R_{in}, R_o, A_{vo})

Input Resistance (R_{in}): Looking into the gate of the MOSFET at low frequencies, the current is zero.

$$R_{in} = \infty\ \mathbf{\Omega} \tag{3}$$

Note: Because $R_{in} = \infty$, there is no voltage drop across R_{sig}, leading to $v_{gs} = v_{sig}$.

Output Resistance (R_o): The output resistance is the resistance looking into the output node (drain) with the independent source (v_{sig}) set to zero, and the external load (R_L) disconnected. When $v_{sig} = 0 \implies v_{gs} = 0 \implies g_m v_{gs} = 0$ (open circuit).

$$R_o = R_D \parallel r_o = \frac{R_D \cdot r_o}{R_D + r_o} = \frac{6\ \text{k}\Omega \times 50\ \text{k}\Omega}{6\ \text{k}\Omega + 50\ \text{k}\Omega} = \frac{300}{56}\ \text{k}\Omega \approx \mathbf{5.357\ k\Omega} \tag{4}$$

Open-Circuit Voltage Gain (A_{vo}): The intrinsic open-circuit voltage gain is defined as $A_{vo} = \frac{v_o}{v_{sig}}$ when $R_L \to \infty$.

$$\begin{aligned} v_o &= -(g_m v_{gs})(R_D \parallel r_o) = -g_m R_o v_{gs} \\ A_{vo} &= \frac{v_o}{v_{sig}} = -g_m R_o \quad (\text{since } v_{gs} = v_{sig}) \\ A_{vo} &= -(1.265\ \text{mA/V}) \times (5.357\ \text{k}\Omega) \approx -\mathbf{6.777}\ \text{V/V} \end{aligned}$$

4. Overall Loaded Voltage Gain (G_v)

When a load resistance $R_L = 10$ kΩ is connected, it acts in parallel with the output resistance R_o. The overall gain is calculated using the voltage divider rule at the output:

$$G_v = A_{vo}\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_{in}}{R_{in} + R_{sig}}\right)$$

Since $R_{in} \to \infty$, the input divider term is 1.

$$\begin{aligned} G_v &= -6.777\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 5.357\ \text{k}\Omega}\right) \\ G_v &= -6.777\left(\frac{10}{15.357}\right) \approx -\mathbf{4.413}\ \text{V/V} \end{aligned}$$

Alternatively, $G_v = -g_m(R_D \parallel r_o \parallel R_L) = -1.265 \times (6 \parallel 50 \parallel 10) \approx -4.413$ V/V.

5. Required Input Signal Amplitude

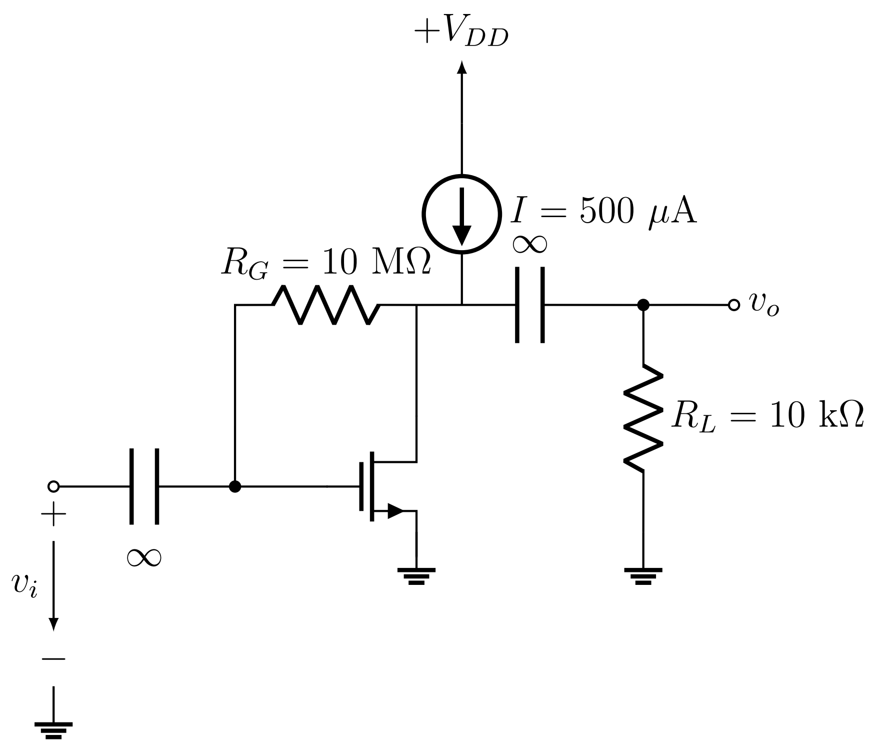
To produce an output voltage with a peak amplitude of $\hat{V}_o = 0.2$ V, we use the magnitude of the loaded overall voltage gain $|G_v|$:

$$\hat{V}_{sig} = \frac{\hat{V}_o}{|G_v|}$$
$$\hat{V}_{sig} = \frac{0.2 \text{ V}}{4.413}$$
$$\hat{V}_{sig} \approx 0.0453 \text{ V} = \mathbf{45.3} \text{ mV}$$

6. Summary of Results

Parameter	Calculated Value
Input Resistance (R_{in})	∞ (ideal gate)
Open-Circuit Gain (A_{vo})	-6.777 V/V
Output Resistance (R_o)	5.357 kΩ
Overall Loaded Gain (G_v)	-4.413 V/V
Peak Input Signal ($\hat{V}_{sig}$)	45.3 mV

Q4. In the circuit shown, the NMOS transistor has $|V_t| = 0.9$ V and $V_A = 50$ V and is operated with $V_D = 2$ V. What is the voltage gain v_0/v_i? What do V_D and the gain become for $I = 1$ mA?

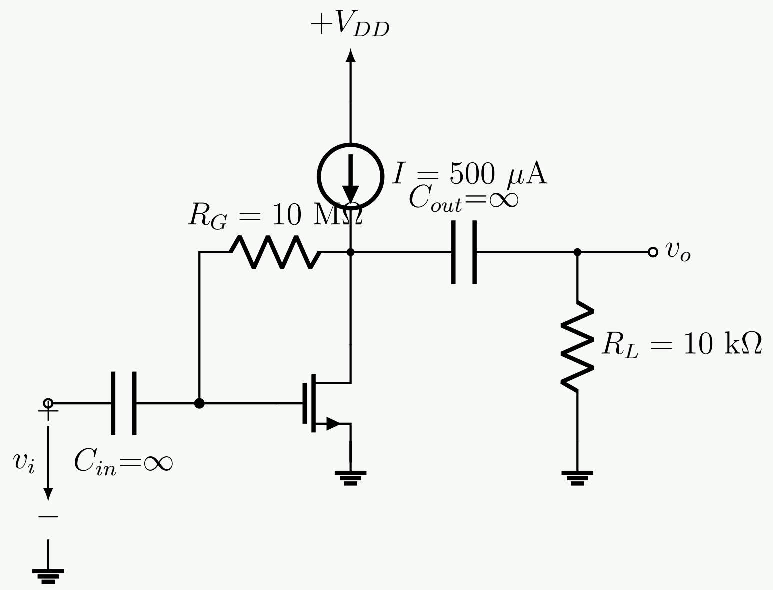


(17 marks) [EEE 263 | L-2/T-1 | 2022–23]

Answer

Objective: Determine the voltage gain v_o/v_i of the given Common-Source NMOS amplifier with drain-to-gate feedback. Then, re-evaluate the DC drain voltage V_D and the voltage gain when the bias current is increased to 1 mA.

1. Reference Circuit Diagram



2. Part 1: Analysis for $I = 500\ \mu$A

DC Analysis: For DC analysis, the capacitors act as open circuits. The gate current of the MOSFET is zero ($I_G = 0$). Therefore, there is no voltage drop across the feedback resistor R_G, and the gate voltage equals the drain voltage:

$$V_G = V_D = 2 \text{ V} \tag{1}$$

Since the source is grounded ($V_S = 0$), we have $V_{GS} = V_D = 2$ V. The transistor is operating in the saturation region because $V_{DS} = V_{GS}$, which strictly satisfies $V_{DS} > V_{GS} - V_t$. The overdrive voltage V_{OV} is:

$$V_{OV} = V_{GS} - |V_t| = 2 \text{ V} - 0.9 \text{ V} = 1.1 \text{ V} \tag{2}$$

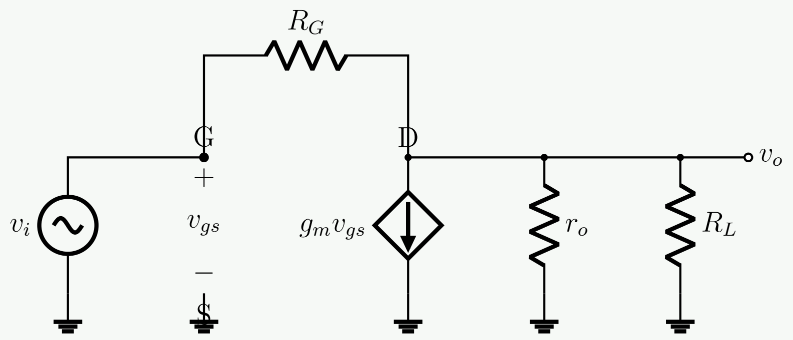
Note: The problem gives $|V_t| = 0.9$ V for an NMOS, so $V_t = 0.9$ V.
Small-Signal Parameters: The drain current is $I_D = I = 500\ \mu\text{A} = 0.5$ mA. The transconductance g_m is determined by:

$$g_m = \frac{2I_D}{V_{OV}} = \frac{2 \times 0.5 \text{ mA}}{1.1 \text{ V}} = \frac{1}{1.1} \text{ mA/V} \approx 0.9091 \text{ mA/V} \tag{3}$$

The output resistance r_o due to the Early effect is:

$$r_o = \frac{V_A}{I_D} = \frac{50 \text{ V}}{0.5 \text{ mA}} = 100 \text{ k}\Omega \tag{4}$$

AC Small-Signal Equivalent Circuit: For AC analysis, the DC supply $+V_{DD}$ and the ideal DC current source I are replaced by open/short circuits (current source becomes an open circuit). Capacitors act as short circuits.



Voltage Gain Derivation: Applying Kirchhoff's Current Law (KCL) at the drain node (D):

$$g_m v_{gs} + \frac{v_o}{r_o} + \frac{v_o}{R_L} + \frac{v_o - v_i}{R_G} = 0 \tag{5}$$

From the input side, we see that $v_{gs} = v_i$. Substituting this into the KCL equation:

$$g_m v_i + v_o \left(\frac{1}{r_o} + \frac{1}{R_L} + \frac{1}{R_G} \right) - \frac{v_i}{R_G} = 0$$
$$v_o \left(\frac{1}{r_o} + \frac{1}{R_L} + \frac{1}{R_G} \right) = v_i \left(\frac{1}{R_G} - g_m \right)$$

The voltage gain $A_v = v_o/v_i$ is therefore:

$$A_v = \frac{\frac{1}{R_G} - g_m}{\frac{1}{r_o} + \frac{1}{R_L} + \frac{1}{R_G}} \tag{6}$$

Let R_{eq} be the parallel combination of r_o, R_L, and R_G:

$$\begin{aligned} R_{eq} = r_o \parallel R_L \parallel R_G &= 100\ \text{k}\Omega \parallel 10\ \text{k}\Omega \parallel 10000\ \text{k}\Omega \\ &= \frac{1}{\frac{1}{100}+\frac{1}{10}+\frac{1}{10000}}\ \text{k}\Omega \\ &= \frac{1}{0.01+0.1+0.0001}\ \text{k}\Omega = \frac{1}{0.1101}\ \text{k}\Omega \approx 9.0826\ \text{k}\Omega \end{aligned}$$

Substituting the values into the gain equation:

$$\begin{aligned} A_v &= \left(\frac{1}{10\ \text{M}\Omega} - g_m\right) R_{eq} \\ &= \left(10^{-4}\ \text{mA/V} - 0.9091\ \text{mA/V}\right) \times 9.0826\ \text{k}\Omega \\ &= (-0.9090\ \text{mA/V}) \times (9.0826\ \text{k}\Omega) \approx -\mathbf{8.26}\ \text{V/V} \end{aligned}$$

3. Part 2: Analysis for $I = 1$ mA

New DC Bias Point (V_{D2}): Assuming the transistor follows the square-law characteristic, the drain current is proportional to the square of the overdrive voltage (ignoring the small impact of channel-length modulation on the DC operating point shift):

$$I_D \propto V_{OV}^2 \tag{7}$$

Taking the ratio of the new state (state 2) to the initial state (state 1):

$$\begin{aligned} \frac{I_{D2}}{I_{D1}} &= \left(\frac{V_{OV2}}{V_{OV1}}\right)^2 \\ \frac{1\ \text{mA}}{0.5\ \text{mA}} &= \left(\frac{V_{OV2}}{1.1\ \text{V}}\right)^2 \implies 2 = \left(\frac{V_{OV2}}{1.1}\right)^2 \end{aligned}$$

Solving for the new overdrive voltage:

$$V_{OV2} = 1.1\sqrt{2} \approx 1.1 \times 1.4142 = 1.5556\ \text{V} \tag{8}$$

Since $V_{D2} = V_{GS2} = V_t + V_{OV2}$:

$$V_{D2} = 0.9\ \text{V} + 1.5556\ \text{V} = \mathbf{2.456}\ \text{V} \tag{9}$$

New Small-Signal Parameters:

$$\begin{aligned} g_{m2} &= \frac{2I_{D2}}{V_{OV2}} = \frac{2 \times 1\ \text{mA}}{1.5556\ \text{V}} \approx 1.2857\ \text{mA/V} \\ r_{o2} &= \frac{V_A}{I_{D2}} = \frac{50\ \text{V}}{1\ \text{mA}} = 50\ \text{k}\Omega \end{aligned}$$

New Voltage Gain: We calculate the new equivalent parallel resistance R_{eq2}:

$$\begin{aligned} R_{eq2} = r_{o2} \parallel R_L \parallel R_G &= 50\ \text{k}\Omega \parallel 10\ \text{k}\Omega \parallel 10000\ \text{k}\Omega \\ &= \frac{1}{\frac{1}{50}+\frac{1}{10}+\frac{1}{10000}}\ \text{k}\Omega \\ &= \frac{1}{0.02+0.1+0.0001}\ \text{k}\Omega = \frac{1}{0.1201}\ \text{k}\Omega \approx 8.3264\ \text{k}\Omega \end{aligned}$$

Using the derived gain formula (Eq. 6):

$$\begin{aligned} A_{v2} &= \left(\frac{1}{R_G} - g_{m2}\right) R_{eq2} \\ &= \left(10^{-4}\ \text{mA/V} - 1.2857\ \text{mA/V}\right) \times 8.3264\ \text{k}\Omega \\ &= (-1.2856\ \text{mA/V}) \times (8.3264\ \text{k}\Omega) \approx -\mathbf{10.70}\ \text{V/V} \end{aligned}$$

4. Summary of Results

Parameter	**Initial State ($I = 500\ \mu$A)**	**Modified State ($I = 1$ mA)**
DC Drain Voltage (V_D)	2.00 V	2.456 V
Transconductance (g_m)	0.909 mA/V	1.286 mA/V
Output Resistance (r_o)	100 kΩ	50 kΩ
Voltage Gain (v_o/v_i)	$-\mathbf{8.26}$ V/V	$-\mathbf{10.70}$ V/V

Q5. A CS amplifier utilizes a MOSFET biased at $I_D = 0.25\,\text{mA}$ with $V_{ov} = 0.25\,\text{V}$ and $R_D = 20\,\text{k}\Omega$. The amplifier is fed with a signal source having $R_{sig} = 100\,\text{k}\Omega$, and a 20-kΩ load is connected to the output. Find R_{in}, A_{vo}, R_o, A_v, and G_v. **(20 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Answer

Objective: Analyze a Common-Source (CS) MOSFET amplifier to determine its input resistance (R_{in}), open-circuit voltage gain (A_{vo}), output resistance (R_o), loaded voltage gain (A_v), and overall voltage gain (G_v).

1. Assumptions and Small-Signal Parameters

Assumptions:

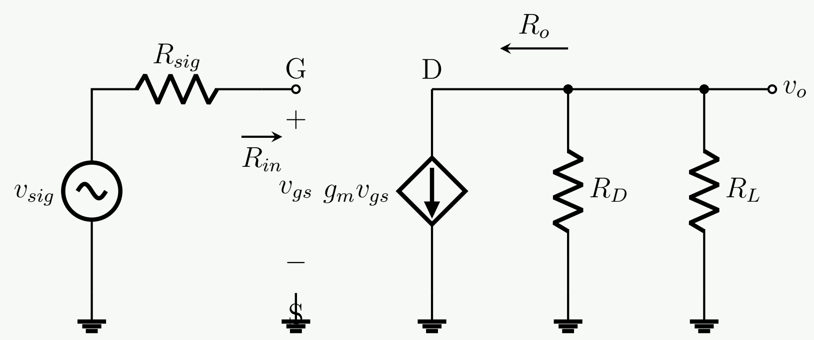
- **Ideal Gate Bias:** The problem does not specify a gate biasing network (such as R_G or a voltage divider). We assume the gate is ideally biased, meaning the input resistance looking into the amplifier is determined solely by the MOSFET's gate. Since the gate current is zero at low frequencies, $R_{in} = \infty$.
- **Early Effect Neglected:** The Early voltage (V_A) or channel-length modulation parameter (λ) is not provided. Therefore, we assume $V_A \to \infty$, which implies the internal output resistance of the transistor $r_o = \infty$ (it acts as an open circuit).

Small-Signal Transconductance (g_m): The transconductance can be directly calculated from the DC bias current I_D and the overdrive voltage V_{ov}:

$$g_m = \frac{2I_D}{V_{ov}} = \frac{2 \times 0.25\text{ mA}}{0.25\text{ V}} = \mathbf{2}\text{ mA/V} \tag{1}$$

2. Small-Signal Equivalent Circuit

The mid-band small-signal equivalent circuit for the CS amplifier is drawn below. The input signal source v_{sig} with its internal resistance R_{sig} drives the gate, while the drain connects to the drain resistor R_D and the load R_L.



3. Parameter Calculations

Input Resistance (R_{in}): Looking into the gate terminal of the MOSFET at low frequencies, the gate draws no current.

$$R_{in} = \infty\ \mathbf{\Omega} \tag{2}$$

Output Resistance (R_o): The output resistance is the equivalent resistance looking back into the output node (drain) with the independent source (v_{sig}) turned off ($v_{sig} = 0 \implies v_{gs} = 0 \implies g_m v_{gs} = 0$) and the load R_L disconnected.

$$R_o = R_D \parallel r_o \xrightarrow{r_o \to \infty} R_D = \mathbf{20}\ \mathbf{k\Omega} \tag{3}$$

Open-Circuit Voltage Gain (A_{vo}): The open-circuit voltage gain is the gain from the gate to the drain without the load connected ($R_L \to \infty$).

$$v_o = -g_m v_{gs}(R_D \parallel r_o) = -g_m v_{gs} R_D$$
$$A_{vo} = \frac{v_o}{v_{gs}} = -g_m R_D$$
$$A_{vo} = -(2\text{ mA/V}) \times (20\text{ k}\Omega) = \mathbf{-40}\text{ V/V}$$

Loaded Voltage Gain (A_v): The loaded voltage gain is the gain from the gate to the drain with the load R_L connected. The load R_L appears in parallel with R_D.

$$A_v = \frac{v_o}{v_{gs}} = -g_m(R_D \parallel R_L)$$
$$A_v = -g_m\left(\frac{R_D R_L}{R_D + R_L}\right) = -(2\text{ mA/V}) \times \left(\frac{20\text{ k}\Omega \times 20\text{ k}\Omega}{20\text{ k}\Omega + 20\text{ k}\Omega}\right)$$
$$A_v = -(2\text{ mA/V}) \times (10\text{ k}\Omega) = \mathbf{-20}\text{ V/V}$$

Overall Voltage Gain (G_v): The overall voltage gain G_v incorporates the voltage division effect between the signal source resistance R_{sig} and the amplifier input resistance R_{in}.

$$G_v = \frac{v_o}{v_{sig}} = \left(\frac{R_{in}}{R_{in} + R_{sig}}\right) A_v$$

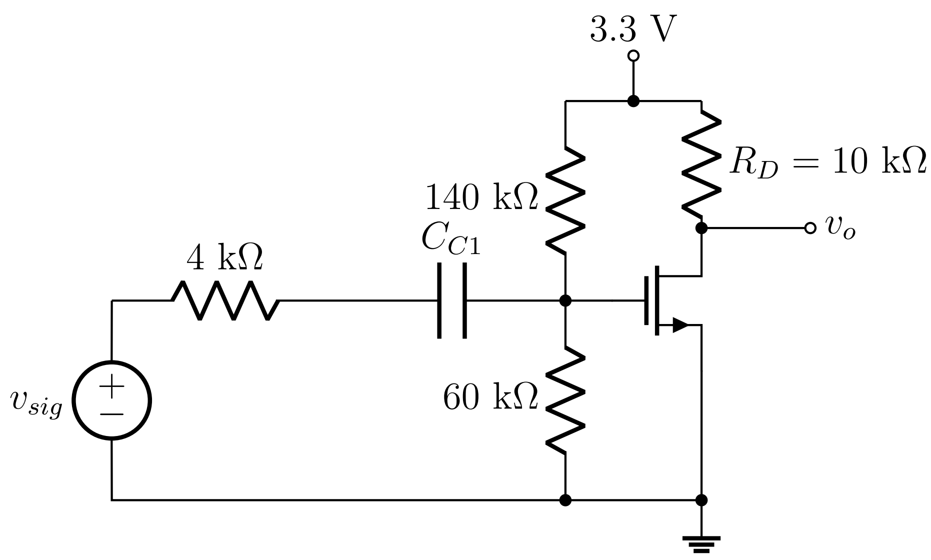
Since $R_{in} = \infty$, the fraction $\frac{R_{in}}{R_{in}+R_{sig}} = 1$. This means there is no signal loss at the input; the entire signal source voltage appears at the gate ($v_{gs} = v_{sig}$).

$$G_v = 1 \times A_v = -\mathbf{20}\ \text{V/V} \tag{4}$$

4. Summary of Results

Parameter	Calculated Value
Input Resistance (R_{in})	∞
Output Resistance (R_o)	20 kΩ
Open-Circuit Gain (A_{vo})	−40 V/V
Loaded Terminal Gain (A_v)	−20 V/V
Overall Loaded Gain (G_v)	−20 V/V

Q6. Determine the overall small-signal voltage gain of the common source amplifier shown. The transistor parameters are: $V_t = 0.4\,\text{V}$, $k_n = 0.5\,\text{mA/V}^2$, $\lambda = 0.02\,\text{V}^{-1}$.

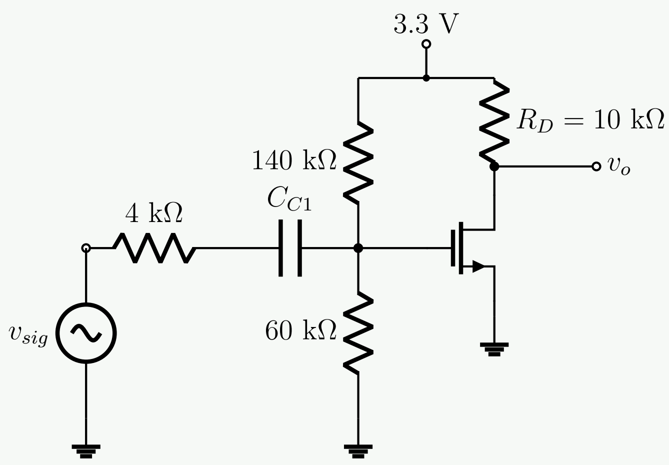


(26 marks) [EEE 263 | L-2/T-1 | 2020–21]

Answer

Objective: Perform a complete DC and AC analysis of the given Common-Source (CS) amplifier to determine its overall small-signal voltage gain $G_v = v_o/v_{sig}$.

1. Reference Circuit Diagram



2. DC Analysis and Operating Point

Assumptions:

- The coupling capacitor C_{C1} blocks DC, isolating the bias network from the signal source.
- The gate current of the MOSFET is zero ($I_G = 0$).
- We use the standard drain current equation: $I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2$.
- Channel-length modulation (λ) is neglected for calculating the initial DC operating point to simplify the bias current calculation.

The gate voltage V_G is established by the 140 kΩ and 60 kΩ voltage divider:

$$V_G = V_{DD}\left(\frac{60\text{ k}\Omega}{140\text{ k}\Omega + 60\text{ k}\Omega}\right) = 3.3\text{ V}\left(\frac{60}{200}\right) = 0.99\text{ V} \tag{1}$$

Since the source is grounded ($V_S = 0$ V), the gate-to-source voltage is:

$$V_{GS} = V_G - V_S = 0.99\text{ V} \tag{2}$$

The overdrive voltage V_{ov} is:

$$V_{ov} = V_{GS} - V_t = 0.99\text{ V} - 0.4\text{ V} = 0.59\text{ V} \tag{3}$$

Assuming saturation, the DC drain current I_D is:

$$\begin{aligned} I_D &= \frac{1}{2}k_n V_{ov}^2 \\ &= \frac{1}{2}(0.5\text{ mA/V}^2)(0.59\text{ V})^2 \\ &= 0.25 \times 0.3481\text{ mA} \approx \mathbf{0.0870}\text{ mA} \quad (\text{or } 87.0\ \mu\text{A}) \end{aligned}$$

The DC drain voltage V_D is:

$$V_D = V_{DD} - I_D R_D = 3.3\text{ V} - (0.0870\text{ mA})(10\text{ k}\Omega) = 3.3\text{ V} - 0.870\text{ V} = 2.43\text{ V} \tag{4}$$

Verification of Saturation: We must check if $V_{DS} > V_{ov}$. Since $V_S = 0$, $V_{DS} = V_D = 2.43$ V.

$$V_{DS}(2.43\text{ V}) > V_{ov}(0.59\text{ V}) \tag{5}$$

The condition is strictly satisfied; thus, the MOSFET operates in the **saturation region**.

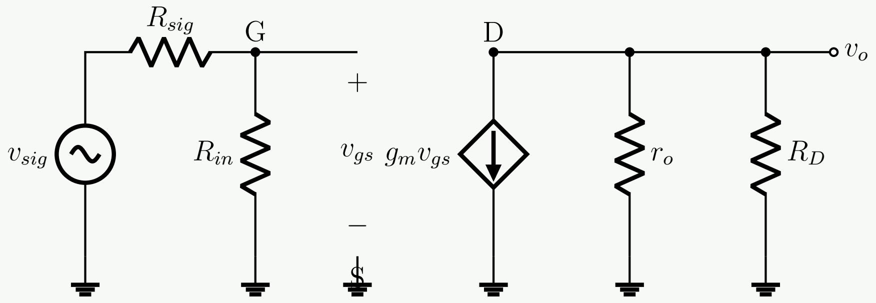
3. Small-Signal Parameters

Using the DC bias current I_D, we evaluate the transconductance (g_m) and the internal output resistance (r_o):

$$\begin{aligned} g_m &= k_n V_{ov} = (0.5\text{ mA/V}^2) \times (0.59\text{ V}) = \mathbf{0.295}\text{ mA/V} \\ r_o &= \frac{1}{\lambda I_D} = \frac{1}{(0.02\text{ V}^{-1})(0.0870\text{ mA})} \approx \mathbf{574.7\text{ k}\Omega} \end{aligned}$$

4. AC Small-Signal Equivalent Circuit

For AC analysis, the DC source V_{DD} is shorted to ground, and the coupling capacitor C_{C1} acts as a short circuit. The voltage divider resistors (140 kΩ and 60 kΩ) appear in parallel, forming the amplifier's input resistance R_{in}.



5. Voltage Gain Derivations

Input Resistance (R_{in}):

$$R_{in} = 140\text{ k}\Omega \parallel 60\text{ k}\Omega = \frac{140 \times 60}{140 + 60} = \frac{8400}{200} = \mathbf{42\text{ k}\Omega} \tag{6}$$

Gate-to-Source Voltage (v_{gs}): Using the voltage divider rule at the input:

$$v_{gs} = v_{sig}\left(\frac{R_{in}}{R_{sig} + R_{in}}\right) = v_{sig}\left(\frac{42\text{ k}\Omega}{4\text{ k}\Omega + 42\text{ k}\Omega}\right) = v_{sig}\left(\frac{42}{46}\right) \approx \mathbf{0.913v_{sig}} \tag{7}$$

Output Voltage (v_o) and Terminal Gain (A_v): The dependent current source pushes current through the parallel combination of r_o and R_D.

$$\begin{aligned} R_{out} &= r_o \parallel R_D = 574.7\text{ k}\Omega \parallel 10\text{ k}\Omega = \frac{574.7 \times 10}{574.7 + 10} \approx \mathbf{9.829\text{ k}\Omega} \\ v_o &= -g_m v_{gs}(r_o \parallel R_D) \\ A_v &= \frac{v_o}{v_{gs}} = -g_m(r_o \parallel R_D) = -(0.295\text{ mA/V})(9.829\text{ k}\Omega) \approx -\mathbf{2.90}\text{ V/V} \end{aligned}$$

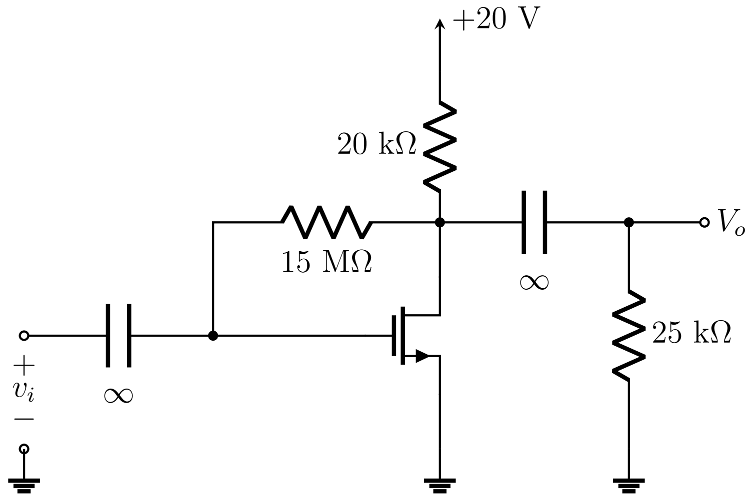
Overall Voltage Gain (G_v): The overall small-signal voltage gain from the source v_{sig} to the output v_o is:

$$
\begin{aligned}
G_v &= \frac{v_o}{v_{sig}} = \left(\frac{v_{gs}}{v_{sig}}\right)\left(\frac{v_o}{v_{gs}}\right) \\
G_v &= \left(\frac{42}{46}\right) \times (-2.90) \\
G_v &\approx (0.913) \times (-2.90) \approx -\mathbf{2.65}\ \text{V/V}
\end{aligned}
$$

6. Final Answer

The overall small-signal voltage gain of the amplifier is -2.65 **V/V**.

Q7. For the circuit shown, find the small-signal input impedance and voltage gain. Given: $V_t = 1.2\,\text{V}$, $k_n'(W/L) = 0.5\,\text{mA/V}^2$, and $V_A = 55\,\text{V}$.

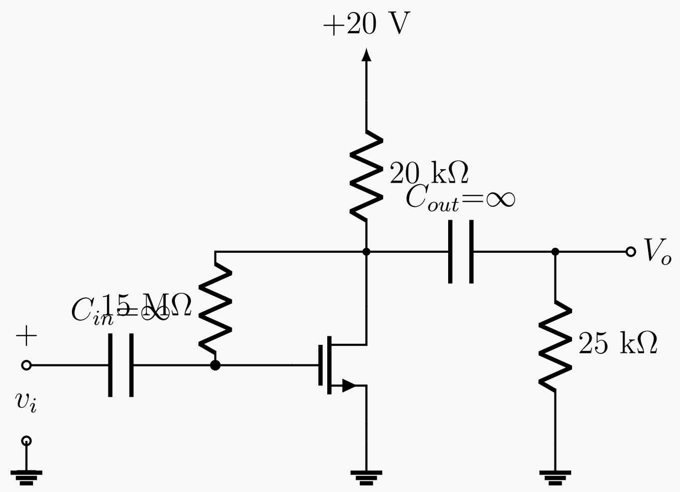


(30 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Answer

Objective: Perform a detailed DC and small-signal AC analysis of the given MOSFET amplifier with drain-to-gate feedback to determine the voltage gain (A_v) and the input impedance (R_{in}).

1. Reference Circuit Diagram



2. DC Analysis

For the DC analysis, capacitors act as open circuits. The gate current of the MOSFET is zero ($I_G = 0$), so there is no voltage drop across the 15 MΩ feedback resistor. Therefore, the DC gate voltage equals the DC drain voltage:

$$V_G = V_D \implies V_{GS} = V_{DS} \tag{1}$$

Because $V_{DS} = V_{GS}$, the condition for saturation ($V_{DS} > V_{GS} - V_t$) is inherently satisfied. The MOSFET is operating in the saturation region.
Applying Kirchhoff's Voltage Law (KVL) at the drain node:

$$V_{DD} = I_D R_D + V_D \implies 20 = I_D(20) + V_{GS} \quad \text{(with } I_D \text{ in mA and } R_D \text{ in k}\Omega\text{)} \tag{2}$$

The saturation drain current equation (ignoring channel-length modulation for the bias point calculation) is:

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$
$$I_D = \frac{1}{2}(0.5)(V_{GS} - 1.2)^2 = 0.25(V_{GS} - 1.2)^2$$

Substitute I_D into the KVL equation:

$$20 = \left[0.25(V_{GS} - 1.2)^2\right](20) + V_{GS}$$
$$20 = 5(V_{GS}^2 - 2.4V_{GS} + 1.44) + V_{GS}$$
$$20 = 5V_{GS}^2 - 12V_{GS} + 7.2 + V_{GS}$$
$$5V_{GS}^2 - 11V_{GS} - 12.8 = 0$$

Solving this quadratic equation for V_{GS}:

$$V_{GS} = \frac{11 \pm \sqrt{(-11)^2 - 4(5)(-12.8)}}{2(5)} = \frac{11 \pm \sqrt{121 + 256}}{10} = \frac{11 \pm \sqrt{377}}{10} \tag{3}$$

Since V_{GS} must be greater than $V_t = 1.2$ V to turn on the transistor, we take the positive root:

$$V_{GS} = \frac{11 + 19.416}{10} \approx 3.042 \text{ V} \tag{4}$$

Now, calculate the DC drain current I_D:

$$I_D = 0.25(3.042 - 1.2)^2 = 0.25(1.842)^2 \approx 0.848 \text{ mA} \tag{5}$$

3. Small-Signal Parameters

With the DC operating point established, we can find the transconductance (g_m) and the output resistance (r_o):

$$g_m = k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t) = (0.5 \text{ mA/V}^2)(3.042 \text{ V} - 1.2 \text{ V}) = \mathbf{0.921} \text{ mA/V}$$
$$r_o = \frac{V_A}{I_D} = \frac{55 \text{ V}}{0.848 \text{ mA}} = \mathbf{64.86 \text{ k}\Omega}$$

4. AC Small-Signal Analysis

In the small-signal equivalent circuit, DC voltage sources (+20 V) are grounded, and the capacitors act as short circuits.
Voltage Gain (A_v): Applying Kirchhoff's Current Law (KCL) at the output node (drain):

$$g_m v_{gs} + \frac{V_o}{r_o} + \frac{V_o}{R_D} + \frac{V_o}{R_L} + \frac{V_o - v_i}{R_G} = 0 \tag{6}$$

Noting that $v_{gs} = v_i$, we can rearrange the equation to solve for V_o/v_i:

$$V_o\left(\frac{1}{r_o} + \frac{1}{R_D} + \frac{1}{R_L} + \frac{1}{R_G}\right) = v_i\left(\frac{1}{R_G} - g_m\right)$$
$$A_v = \frac{V_o}{v_i} = \frac{\frac{1}{R_G} - g_m}{\frac{1}{r_o} + \frac{1}{R_D} + \frac{1}{R_L} + \frac{1}{R_G}}$$

Let R_{eq} be the parallel equivalent of r_o, R_D, R_L, and R_G:

$$\frac{1}{R_{eq}} = \frac{1}{64.86} + \frac{1}{20} + \frac{1}{25} + \frac{1}{15000} \text{ (in mS)}$$
$$\frac{1}{R_{eq}} \approx 0.01542 + 0.05 + 0.04 + 0.000067 = 0.10549 \text{ mS} \implies R_{eq} \approx 9.48 \text{ k}\Omega$$

Now, compute the gain:

$$A_v = \left(\frac{1}{15000 \text{ k}\Omega} - 0.921 \text{ mA/V}\right) \times 9.48 \text{ k}\Omega$$
$$\approx (0.000067 - 0.921) \times 9.48 \approx (-0.921) \times 9.48 = -\mathbf{8.73} \text{ V/V}$$

Input Impedance (R_{in}): The input impedance is heavily influenced by the feedback resistor R_G. We apply the Miller Effect to find the equivalent resistance seen at the input. The input current i_{in} drawn by the amplifier is:

$$i_{in} = \frac{v_i - V_o}{R_G} = \frac{v_i - A_v v_i}{R_G} = \frac{v_i(1 - A_v)}{R_G} \tag{7}$$

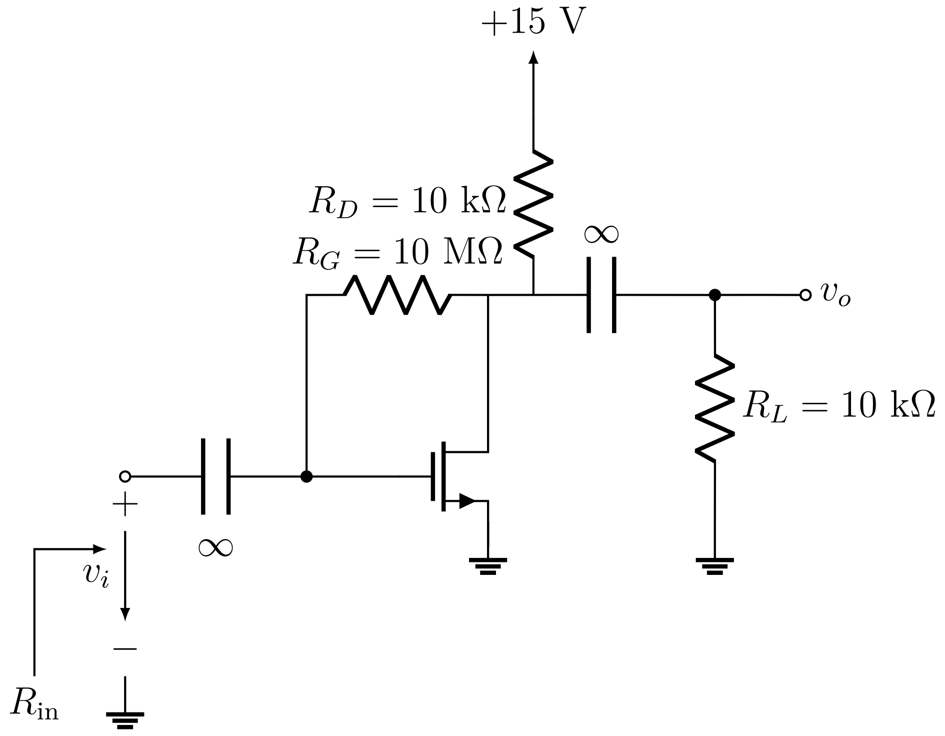
Therefore, the input impedance R_{in} is:

$$R_{in} = \frac{v_i}{i_{in}} = \frac{R_G}{1 - A_v}$$
$$R_{in} = \frac{15 \text{ M}\Omega}{1 - (-8.73)} = \frac{15 \text{ M}\Omega}{9.73} \approx \mathbf{1.54 \text{ M}\Omega}$$

5. Summary of Results

Parameter	Calculated Value
DC Gate-to-Source Voltage (V_{GS})	3.042 V
DC Drain Current (I_D)	0.848 mA
Transconductance (g_m)	0.921 mA/V
Small-Signal Voltage Gain (A_v)	$\mathbf{-8.73}$ V/V
Small-Signal Input Impedance (R_{in})	**1.54** MΩ

Q8. Determine the voltage gain v_0/v_i of the common-source MOSFET amplifier shown in the figure. The transistor has $V_t = 1.5\,\text{V}$, $k'_n(W/L) = 0.25\,\text{mA/V}^2$, and $V_A = 50\,\text{V}$.



(20 marks) [EEE 263 | L-2/T-1 | 2017–18]

Answer

Objective: Perform a DC and small-signal AC analysis of the given common-source MOSFET amplifier to determine its voltage gain (v_o/v_i).

1. DC Analysis

For the DC analysis, the input and output capacitors act as open circuits. Since the gate current of the MOSFET is zero ($I_G = 0$), there is no voltage drop across the 10 MΩ feedback resistor R_G. Consequently, the DC gate voltage is equal to the DC drain voltage:

$$V_G = V_D \implies V_{GS} = V_{DS} \tag{1}$$

Because $V_{DS} = V_{GS}$, the condition for saturation ($V_{DS} > V_{GS} - V_t$) is inherently satisfied. The MOSFET is operating in the saturation region.
Applying Kirchhoff's Voltage Law (KVL) at the drain node:

$$V_{DD} = I_D R_D + V_D \implies 15 = I_D(10) + V_{GS} \quad \text{(with } I_D \text{ in mA and } R_D \text{ in k}\Omega) \tag{2}$$

The saturation drain current equation (ignoring channel-length modulation for the DC bias point) is:

$$I_D = \frac{1}{2}k'_n\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$
$$I_D = 0.125(V_{GS} - 1.5)^2$$

Substitute I_D into the KVL equation:

$$\begin{aligned} 15 &= \left[0.125(V_{GS} - 1.5)^2\right](10) + V_{GS} \\ 15 &= 1.25(V_{GS}^2 - 3V_{GS} + 2.25) + V_{GS} \\ 15 &= 1.25V_{GS}^2 - 3.75V_{GS} + 2.8125 + V_{GS} \\ 1.25V_{GS}^2 - 2.75V_{GS} - 12.1875 &= 0 \end{aligned}$$

Solving this quadratic equation for V_{GS}:

$$V_{GS} = \frac{2.75 \pm \sqrt{(-2.75)^2 - 4(1.25)(-12.1875)}}{2(1.25)} = \frac{2.75 \pm \sqrt{7.5625 + 60.9375}}{2.5} = \frac{2.75 \pm \sqrt{68.5}}{2.5} \tag{3}$$

Since V_{GS} must be greater than $V_t = 1.5$ V to turn on the transistor, we take the positive root:

$$V_{GS} = \frac{2.75 + 8.276}{2.5} \approx \mathbf{4.41} \text{ V} \tag{4}$$

Now, calculate the DC drain current I_D:

$$I_D = 0.125(4.41 - 1.5)^2 = 0.125(2.91)^2 \approx \mathbf{1.059} \text{ mA} \tag{5}$$

2. Small-Signal Parameters

With the DC operating point established, we find the transconductance (g_m) and the internal output resistance (r_o):

$$g_m = 2k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t) = 2(0.125 \text{ mA/V}^2)(4.41 \text{ V} - 1.5 \text{ V}) = \mathbf{0.7275} \text{ mA/V}$$
$$r_o = \frac{V_A}{I_D} = \frac{50 \text{ V}}{1.059 \text{ mA}} \approx \mathbf{47.21} \text{ k}\boldsymbol{\Omega}$$

3. AC Small-Signal Analysis

In the small-signal equivalent circuit, the DC voltage source (+15 V) is grounded, and the coupling capacitors are short circuits. The input signal v_i connects directly to the gate, meaning $v_{gs} = v_i$.
Applying Kirchhoff's Current Law (KCL) at the drain node (v_o):

$$\frac{v_o - v_i}{R_G} + \frac{v_o}{r_o} + \frac{v_o}{R_D} + \frac{v_o}{R_L} + g_m v_{gs} = 0 \tag{6}$$

Substitute $v_{gs} = v_i$ and isolate the terms for v_o and v_i:

$$v_o\left(\frac{1}{R_G} + \frac{1}{r_o} + \frac{1}{R_D} + \frac{1}{R_L}\right) = v_i\left(\frac{1}{R_G} - g_m\right)$$
$$A_v = \frac{v_o}{v_i} = \frac{\frac{1}{R_G} - g_m}{\frac{1}{R_G} + \frac{1}{r_o} + \frac{1}{R_D} + \frac{1}{R_L}}$$

Let R_{eq} be the parallel equivalent of R_G, r_o, R_D, and R_L:

$$\frac{1}{R_{eq}} = \frac{1}{10000} + \frac{1}{47.21} + \frac{1}{10} + \frac{1}{10} \text{ (in mS)}$$
$$\frac{1}{R_{eq}} \approx 0.0001 + 0.02118 + 0.1 + 0.1 = 0.22128 \text{ mS} \implies R_{eq} \approx \mathbf{4.519} \text{ k}\boldsymbol{\Omega}$$

Now, compute the voltage gain:

$$A_v = \left(\frac{1}{10000 \text{ k}\Omega} - 0.7275 \text{ mA/V}\right) \times 4.519 \text{ k}\Omega$$
$$\approx (0.0001 - 0.7275) \times 4.519 \approx (-0.7274) \times 4.519 = -\mathbf{3.287} \text{ V/V}$$

4. Summary of Results

Parameter	**Calculated Value**
DC Gate-to-Source Voltage (V_{GS})	4.41 V
DC Drain Current (I_D)	1.059 mA
Transconductance (g_m)	0.7275 mA/V
Internal Output Resistance (r_o)	47.21 kΩ
Small-Signal Voltage Gain (A_v)	$-\mathbf{3.29}$ V/V

Q9. For the circuit shown:

(a) Calculate DC bias point (I_D and V_D).

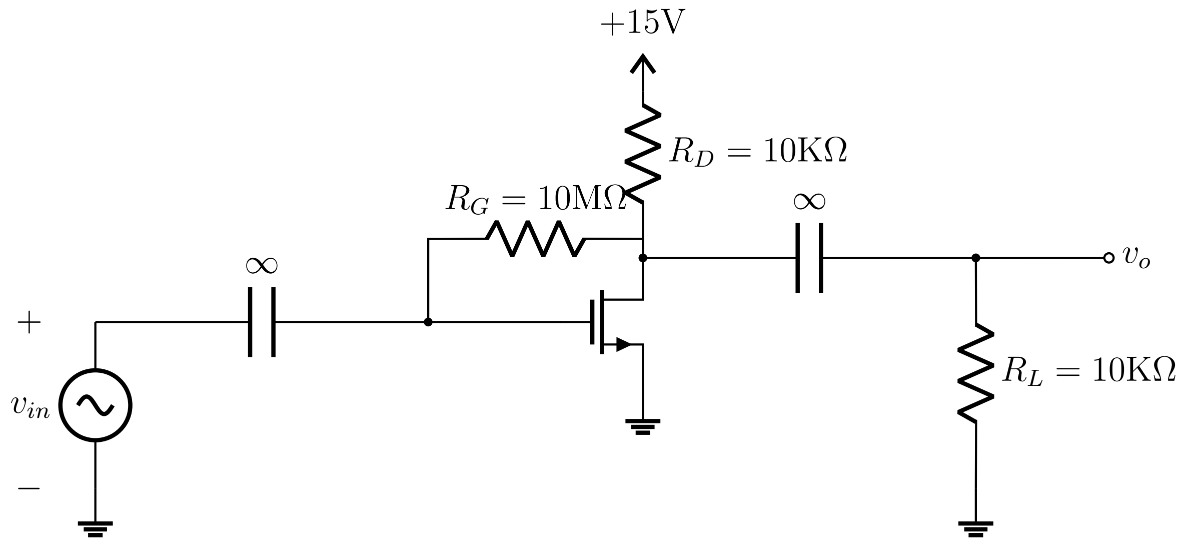
(b) Draw the small-signal equivalent circuit.

(c) Calculate the small-signal voltage gain.

(d) Find the input resistance.

(e) Determine the largest allowable input signal voltage.

The transistor has $V_t = 1.5\,\text{V}$, $k_n'(W/L) = 0.25\,\text{mA/V}^2$, and $V_A = 50\,\text{V}$.



(31 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

Objective: Perform a complete analysis of the given common-source MOSFET amplifier, including determining the DC bias point, small-signal equivalent circuit, voltage gain, input resistance, and the maximum allowable input signal for linear operation.

1. DC Bias Point (I_D and V_D)

For the DC analysis, the AC voltage source is set to zero, and the capacitors act as open circuits. Since the gate current of the MOSFET is zero ($I_G = 0$), there is no voltage drop across the 10 MΩ feedback resistor R_G. Thus, the DC gate voltage is equal to the DC drain voltage:

$$V_G = V_D \implies V_{GS} = V_{DS} \tag{1}$$

Because $V_{DS} = V_{GS}$, the condition for saturation ($V_{DS} > V_{GS} - V_t$) is guaranteed to be satisfied. The MOSFET is operating in the saturation region.

Applying Kirchhoff's Voltage Law (KVL) at the drain node:

$$V_{DD} = I_D R_D + V_D \implies 15 = I_D(10) + V_{GS} \quad (\text{with } I_D \text{ in mA and } R_D \text{ in k}\Omega) \tag{2}$$

The saturation drain current equation (ignoring channel-length modulation for the DC bias point) is:

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$
$$I_D = 0.125(V_{GS} - 1.5)^2$$

Substitute I_D into the KVL equation:

$$15 = \left[0.125(V_{GS} - 1.5)^2\right](10) + V_{GS}$$
$$15 = 1.25(V_{GS}^2 - 3V_{GS} + 2.25) + V_{GS}$$
$$1.25V_{GS}^2 - 2.75V_{GS} - 12.1875 = 0$$

Solving this quadratic equation for V_{GS}:

$$V_{GS} = \frac{2.75 \pm \sqrt{(-2.75)^2 - 4(1.25)(-12.1875)}}{2(1.25)} = \frac{2.75 \pm \sqrt{68.5}}{2.5} \tag{3}$$

Taking the positive root (since V_{GS} must be greater than $V_t = 1.5$ V):

$$V_{GS} = \frac{2.75 + 8.276}{2.5} \approx \mathbf{4.41}\ \text{V} \tag{4}$$

Since $V_D = V_{GS}$, we have:

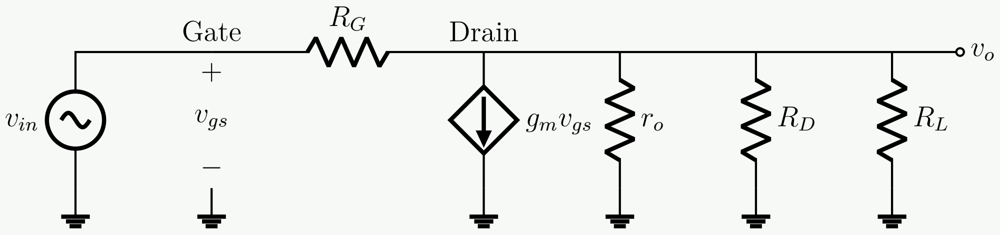
$$V_D = \mathbf{4.41}\ \text{V} \tag{5}$$

Now, calculate the DC drain current I_D:

$$I_D = \frac{15 - 4.41}{10} = \mathbf{1.059}\ \text{mA} \tag{6}$$

2. Small-Signal Equivalent Circuit

To construct the small-signal model: * The DC power supply (V_{DD}) is replaced with a short to ground. * The coupling capacitors are replaced with short circuits. * The MOSFET is replaced with its small-signal model (a dependent current source $g_m v_{gs}$ and parallel output resistance r_o).



3. Small-Signal Voltage Gain (A_v)

First, calculate the small-signal parameters g_m and r_o:

$$g_m = 2k_n' \left(\frac{W}{L}\right)(V_{GS} - V_t) = 2(0.125 \text{ mA/V}^2)(4.41 \text{ V} - 1.5 \text{ V}) = \mathbf{0.7275} \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = \frac{50 \text{ V}}{1.059 \text{ mA}} \approx \mathbf{47.21 \text{ k}\Omega}$$

Let R_{eq} be the parallel combination of r_o, R_D, and R_L:

$$R_{eq} = r_o \parallel R_D \parallel R_L = 47.21 \text{ k}\Omega \parallel 10 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 47.21 \parallel 5 = \mathbf{4.52 \text{ k}\Omega} \tag{7}$$

Apply Kirchhoff's Current Law (KCL) at the drain node, noting that $v_{gs} = v_{in}$:

$$\frac{v_o - v_{in}}{R_G} + g_m v_{in} + \frac{v_o}{R_{eq}} = 0$$

$$v_o \left(\frac{1}{R_G} + \frac{1}{R_{eq}}\right) = v_{in}\left(\frac{1}{R_G} - g_m\right)$$

$$A_v = \frac{v_o}{v_{in}} = \frac{\frac{1}{R_G} - g_m}{\frac{1}{R_G} + \frac{1}{R_{eq}}}$$

Since $R_G = 10 \text{ M}\Omega = 10000 \text{ k}\Omega$, $\frac{1}{R_G} = 0.0001$ mS.

$$A_v = \frac{0.0001 - 0.7275}{0.0001 + \frac{1}{4.52}} = \frac{-0.7274}{0.0001 + 0.2212} \approx \mathbf{-3.29} \text{ V/V} \tag{8}$$

4. Input Resistance (R_{in})

The input resistance is governed by the feedback resistor R_G. We can use Miller's Theorem to reflect this resistance to the input side.

$$R_{in} = \frac{R_G}{1 - A_v} = \frac{10 \text{ M}\Omega}{1 - (-3.29)} = \frac{10 \text{ M}\Omega}{4.29} \approx \mathbf{2.33 \text{ M}\Omega} \tag{9}$$

5. Largest Allowable Input Signal Voltage

To maintain linear operation, the MOSFET must remain in the saturation region at all times. The boundary for saturation is $v_{DS} \geq v_{GS} - V_t$. Expressing the instantaneous voltages as the sum of DC and small-signal components:

$$V_{DS} + v_{ds}(t) \geq V_{GS} + v_{gs}(t) - V_t \tag{10}$$

Since $V_{DS} = V_{GS}$ for this bias topology, the condition simplifies to:

$$v_{ds}(t) \geq v_{gs}(t) - V_t \tag{11}$$

Substitute $v_{ds}(t) = A_v v_{in} \sin(\omega t)$ and $v_{gs}(t) = v_{in} \sin(\omega t)$:

$$A_v v_{in} \sin(\omega t) \geq v_{in} \sin(\omega t) - V_t \tag{12}$$

Rearranging for V_t:

$$V_t \geq v_{in} \sin(\omega t)(1 - A_v) \tag{13}$$

The most restrictive condition occurs at the positive peak of the input signal ($\sin(\omega t) = 1$). Letting the peak input signal be $v_{in,max}$:

$$V_t \geq v_{in,max}(1 - A_v)$$

$$v_{in,max} \leq \frac{V_t}{1 - A_v} = \frac{1.5}{1 - (-3.29)} = \frac{1.5}{4.29} \approx \mathbf{0.35} \text{ V}$$

Thus, the amplitude of the input signal must be kept below 0.35 V to prevent the transistor from entering the triode region during the signal swing.

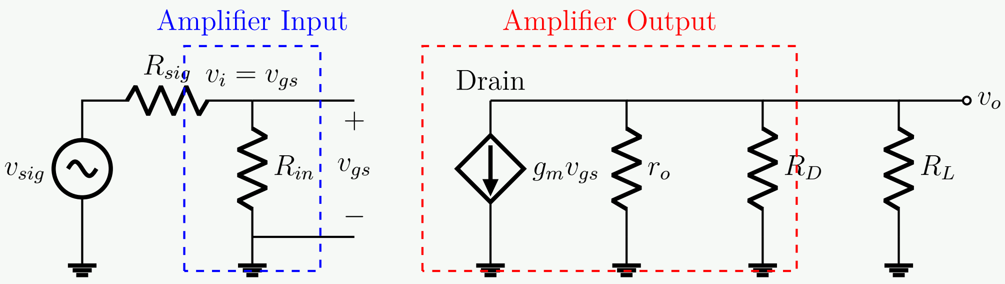
Q10. For the CS amplifier, derive the expressions for input resistance (R_{in}), output resistance (R_{out}), open-circuit voltage gain (A_{vo}), and overall voltage gain (G_v). **(26 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

Objective: Derive the standard expressions for a general Common-Source (CS) amplifier's input resistance (R_{in}), output resistance (R_{out}), open-circuit voltage gain (A_{vo}), and overall voltage gain (G_v).

1. Small-Signal Equivalent Circuit

To derive these expressions, we first draw the standard small-signal equivalent circuit of a CS amplifier (incorporating a finite signal source resistance R_{sig} and a load resistance R_L).



2. Input Resistance (R_{in})

The input resistance is defined as the equivalent resistance seen by the signal source looking into the amplifier's input terminal. Because the gate of a MOSFET acts as an open circuit at low and mid-band frequencies (i.e., gate current $I_G = 0$), the only resistance seen at the input is the bias resistor network. If the gate is biased by a single resistor R_G to ground, or a voltage divider R_{G1} and R_{G2}, the input resistance is simply the equivalent Thevenin resistance of that network.

$$R_{in} = R_G \quad \text{or} \quad R_{in} = R_{G1} \parallel R_{G2}$$

3. Output Resistance (R_{out})

The output resistance is the equivalent resistance seen looking back into the amplifier's output terminal with the load (R_L) disconnected, and the independent input source (v_{sig}) turned off (shorted to ground). * Shorting v_{sig} makes $v_i = 0$. * Since $v_{gs} = v_i = 0$, the dependent current source $g_m v_{gs}$ evaluates to 0 A (an open circuit). Looking into the output terminal, the only remaining parallel resistances are the MOSFET's internal output resistance (r_o) and the drain bias resistor (R_D).

$$R_{out} = R_D \parallel r_o = \frac{R_D r_o}{R_D + r_o}$$

4. Open-Circuit Voltage Gain (A_{vo})

The open-circuit voltage gain is the ratio of the output voltage to the input voltage (v_o/v_i) when no load is attached ($R_L = \infty$). Applying Kirchhoff's Current Law (KCL) at the drain node with R_L removed:

$$g_m v_{gs} + \frac{v_o}{r_o} + \frac{v_o}{R_D} = 0$$

Since the input voltage v_i appears directly across the gate-to-source terminals, we substitute $v_{gs} = v_i$:

$$v_o \left(\frac{1}{r_o} + \frac{1}{R_D} \right) = -g_m v_i$$

$$v_o = -g_m (R_D \parallel r_o) v_i$$

Thus, the open-circuit voltage gain is:

$$A_{vo} = \frac{v_o}{v_i} = -g_m (R_D \parallel r_o) = -g_m R_{out}$$

5. Overall Voltage Gain (G_v)

The overall voltage gain is the ratio of the output voltage to the source signal voltage (v_o/v_{sig}), taking into account both the source resistance R_{sig} and the load resistance R_L.

Step A: Input Voltage Divider The actual voltage that reaches the amplifier's input (v_i) is reduced by the voltage divider formed by R_{sig} and R_{in}:

$$v_i = v_{sig} \frac{R_{in}}{R_{in} + R_{sig}}$$

Step B: Loaded Amplifier Gain (A_v) With the load R_L connected, the total resistance at the drain node becomes $r_o \parallel R_D \parallel R_L$. The output voltage is:

$$v_o = -g_m(r_o \parallel R_D \parallel R_L)v_i$$

So the loaded gain from gate to drain is:

$$A_v = \frac{v_o}{v_i} = -g_m(r_o \parallel R_D \parallel R_L) = -g_m(R_{out} \parallel R_L)$$

Step C: Combine for Overall Gain Multiply the input transmission ratio by the loaded amplifier gain:

$$G_v = \frac{v_o}{v_{sig}} = \left(\frac{v_i}{v_{sig}}\right) \times \left(\frac{v_o}{v_i}\right)$$

$$G_v = -\frac{R_{in}}{R_{in} + R_{sig}} g_m(R_D \parallel r_o \parallel R_L)$$

Alternatively, expressed in terms of A_{vo} and R_{out}:

$$G_v = \frac{R_{in}}{R_{in} + R_{sig}} A_{vo} \frac{R_L}{R_{out} + R_L}$$

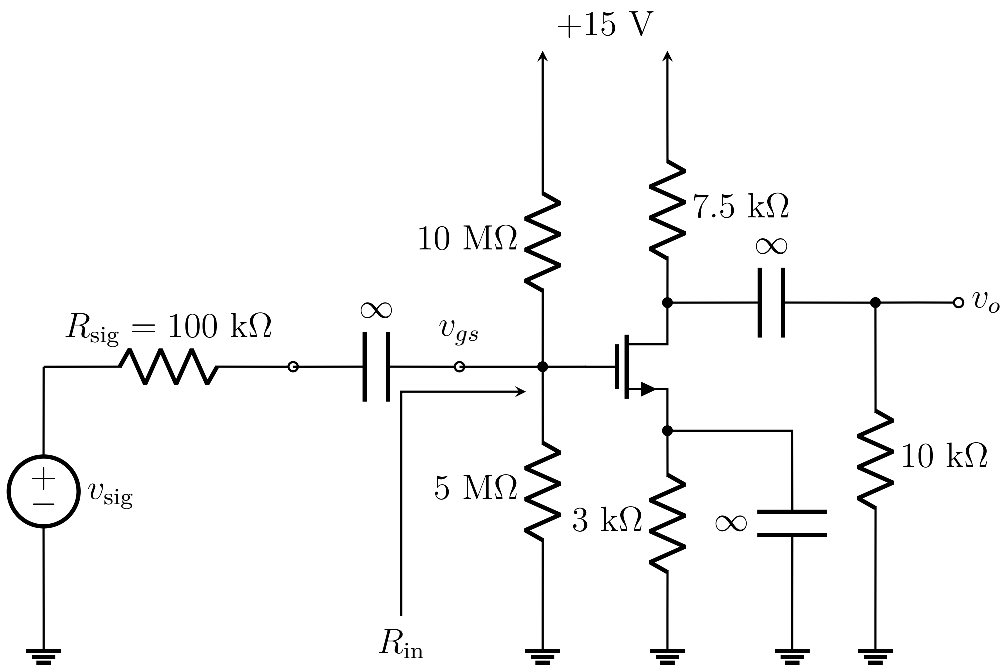
Q11. A common source amplifier circuit is shown in the figure.

(a) Prove that this circuit is operating in the saturation region.

(b) Draw the small-signal equivalent circuit.

(c) Find g_m, R_{in}, R_{out}, A_v, and G_v.

Assume $V_t = 1\,\text{V}$, $K_n' W/L = 2\,\text{mA/V}^2$, $V_A = 100\,\text{V}$.



(46 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

Objective: Analyze the given common-source amplifier by verifying its DC operating region, constructing its small-signal model, and calculating key amplifier parameters.

1. Prove Operation in the Saturation Region

To find the DC operating point, we analyze the circuit with AC sources set to zero and capacitors treated as open circuits. First, determine the DC gate voltage (V_G) set by the voltage divider network (R_{G1} and R_{G2}):

$$V_G = V_{DD}\left(\frac{R_{G2}}{R_{G1} + R_{G2}}\right) = 15\left(\frac{5}{10+5}\right) = 5\text{ V}$$

Assume the MOSFET is operating in the saturation region. The drain current is given by:

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

The source voltage (V_S) is the voltage drop across the source resistor:

$$V_S = I_D R_S = I_D(3)$$

Thus, the gate-to-source voltage is:

$$V_{GS} = V_G - V_S = 5 - 3I_D$$

Substitute V_{GS} into the drain current equation (using mA and V):

$$I_D = \frac{1}{2}(2)(5 - 3I_D - 1)^2$$

$$I_D = (4 - 3I_D)^2$$

$$I_D = 16 - 24I_D + 9I_D^2$$

$$9I_D^2 - 25I_D + 16 = 0$$

Solving this quadratic equation for I_D yields two possible roots:

$$I_D = \frac{25 \pm \sqrt{625 - 4(9)(16)}}{18} = \frac{25 \pm 7}{18}$$

$$I_{D1} = 1.778 \text{ mA} \quad \text{and} \quad I_{D2} = 1 \text{ mA}$$

We must check which current yields a valid $V_{GS} > V_t$: * If $I_D = 1.778$ mA, then $V_S = 3(1.778) = 5.33$ V. This results in $V_{GS} = 5 - 5.33 = -0.33$ V, which is less than V_t (transistor would be off). * If $I_D = 1$ mA, then $V_S = 3(1) = 3$ V. This results in $V_{GS} = 5 - 3 = 2$ V, which is greater than V_t (transistor is on).
Therefore, the valid DC drain current is **$I_D = 1$ mA**. Now, verify the saturation condition ($V_{DS} > V_{GS} - V_t$):

$$V_D = V_{DD} - I_D R_D = 15 - (1)(7.5) = 7.5 \text{ V}$$

$$V_{DS} = V_D - V_S = 7.5 - 3 = 4.5 \text{ V}$$

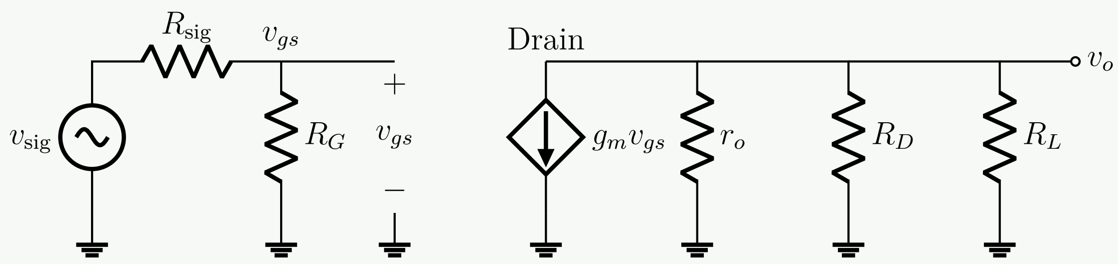
The overdrive voltage (V_{OV}) is:

$$V_{OV} = V_{GS} - V_t = 2 - 1 = 1 \text{ V}$$

Since V_{DS} (4.5 V) is greater than V_{OV} (1 V), the transistor is confirmed to be operating in the **saturation region**.

2. Small-Signal Equivalent Circuit

For the small-signal model, DC sources (V_{DD}) are shorted to ground, and capacitors act as short circuits. Because the source bypass capacitor (C_S) shorts the source terminal to ground for AC signals, R_S is completely bypassed. The equivalent gate resistance is $R_G = R_{G1} \parallel R_{G2}$.



3. Amplifier Parameters ($g_m, R_{in}, R_{out}, A_v, G_v$)

First, evaluate the small-signal parameters g_m and r_o:

$$g_m = k_n' \left(\frac{W}{L}\right)(V_{GS} - V_t) = 2(2 - 1) = \mathbf{2} \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = \frac{100}{1} = \mathbf{100 \text{ k}\Omega}$$

Input Resistance (R_{in}): Looking into the amplifier past the input coupling capacitor, the resistance is purely the parallel combination of the bias resistors (since the MOSFET gate draws no current).

$$R_{in} = R_{G1} \parallel R_{G2} = 10 \text{ M}\Omega \parallel 5 \text{ M}\Omega = \frac{10 \times 5}{10 + 5} = \mathbf{3.33} \text{ M}\Omega$$

Output Resistance (R_{out}): Looking into the drain terminal with the load disconnected and the input source v_{sig} set to zero (which turns off the dependent current source):

$$R_{out} = R_D \parallel r_o = 7.5 \text{ k}\Omega \parallel 100 \text{ k}\Omega = \frac{7.5 \times 100}{7.5 + 100} = \mathbf{6.98} \text{ k}\Omega$$

Voltage Gain (A_v): The voltage gain from the gate terminal to the drain terminal is:

$$A_v = \frac{v_o}{v_{gs}} = -g_m(r_o \parallel R_D \parallel R_L)$$

Calculate the parallel equivalent of all output-side resistors:

$$r_o \parallel R_D \parallel R_L = 100 \parallel 7.5 \parallel 10 = \left(\frac{1}{100} + \frac{1}{7.5} + \frac{1}{10}\right)^{-1} \approx 4.11 \text{ k}\Omega$$

$$A_v = -2 \text{ mA/V} \times 4.11 \text{ k}\Omega = -\mathbf{8.22} \text{ V/V}$$

Overall Voltage Gain (G_v): The overall gain factors in the voltage division between the source resistance R_{sig} and the input resistance R_{in}:

$$G_v = \frac{v_o}{v_{\text{sig}}} = \left(\frac{R_{in}}{R_{in} + R_{\text{sig}}}\right) A_v$$

Using $R_{in} = 3333 \text{ k}\Omega$ and $R_{\text{sig}} = 100 \text{ k}\Omega$:

$$G_v = \left(\frac{3333}{3333 + 100}\right)(-8.22) = (0.9709)(-8.22) = -\mathbf{7.98} \text{ V/V}$$

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Electronic Circuits

Part IV — Linear Integrated Circuits

Op-amp ideal characteristics, mathematical operations, active filters, ADC

S7 Operational Amplifiers (Op-Amps)

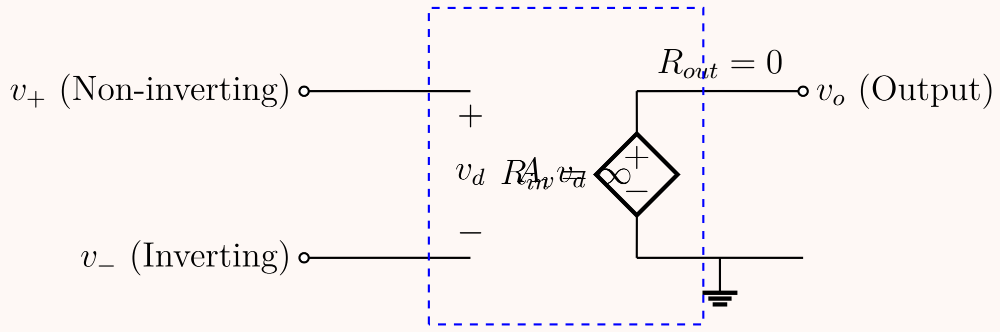
Ideal Op-Amp Characteristics

Q1. What are the characteristics of an ideal op-amp? **(5 marks)** **[EEE 263 | L-2/T-1 | 2013–14, 2011–12]**

Answer

An **Operational Amplifier (Op-Amp)** is a high-gain electronic voltage amplifier with a differential input and, typically, a single-ended output. In theoretical circuit analysis, we often assume an "ideal" op-amp to simplify calculations.
The equivalent circuit of an ideal op-amp is shown below, illustrating its fundamental internal parameters:

Ideal Op-Amp Internal Model



Fundamental Characteristics of an Ideal Op-Amp

An ideal operational amplifier is characterized by the following theoretical parameters:

(a) **Infinite Open-Loop Voltage Gain ($A_v = \infty$):** The op-amp will amplify any minute differential input voltage ($v_d = v_+ - v_-$) to an infinite output voltage. In a practical closed-loop circuit with negative feedback, this forces the differential input voltage to zero.

(b) **Infinite Input Impedance ($R_{in} = \infty$):** The input terminals draw strictly zero electrical current from the signal source.

$$i_+ = i_- = 0\,\mathrm{A} \tag{1}$$

(c) **Zero Output Impedance ($R_{out} = 0\,\Omega$):** The op-amp behaves as a perfect ideal voltage source. It can deliver infinite current to any load connected to its output without any drop in the output voltage.

(d) **Infinite Bandwidth ($BW = \infty$):** The op-amp will amplify all signal frequencies, from DC (0 Hz) to infinity, with the exact same gain and without introducing any phase shift.

(e) **Infinite Common-Mode Rejection Ratio (CMRR $= \infty$):** The op-amp exclusively amplifies the *difference* between the two input voltages. If the same voltage is applied to both inputs (common-mode signal), the output is exactly zero. It perfectly rejects noise common to both terminals.

(f) **Infinite Slew Rate (SR $= \infty$):** The output voltage can change instantaneously in response to a step change in the input voltage. There is zero propagation delay.

(g) **Zero Offset Voltage ($V_{OS} = 0\,\mathbf{V}$):** When both input terminals are perfectly grounded or shorted together ($v_+ = v_-$), the output voltage is precisely zero volts.

The Two "Golden Rules" of Ideal Op-Amp Analysis

When an ideal op-amp is operated in a closed-loop configuration with **negative feedback**, the characteristics above yield two highly practical rules for circuit analysis:

- **Rule 1 (Virtual Short):** The op-amp adjusts its output voltage to ensure that the voltage difference between its input terminals is exactly zero.

$$v_+ = v_- \tag{2}$$

- **Rule 2 (Zero Input Current):** Because the input impedance is infinite, absolutely no current flows into or out of the inverting and non-inverting input terminals.

$$i_+ = 0 \quad \text{and} \quad i_- = 0 \tag{3}$$

Comparison: Ideal vs. Practical Op-Amp

To contextualize the ideal assumptions, the table below compares ideal parameters with a standard practical op-amp (like the classic μA741):

Parameter	Ideal Op-Amp	Typical Practical Op-Amp (μA741)
Voltage Gain (A_v)	∞	$\approx 2 \times 10^5$ V/V
Input Impedance (R_{in})	∞	≈ 2 MΩ
Output Impedance (R_{out})	$0\,\Omega$	$\approx 75\,\Omega$
Bandwidth (BW)	∞	≈ 1 MHz
CMRR	∞	≈ 90 dB
Slew Rate (SR)	∞	≈ 0.5 V/μs
Input Offset Voltage (V_{OS})	0 V	≈ 2 mV

Q2. Explain graphically why in a closed-loop feedback circuit of an Op-Amp the differential voltage E_d is taken as zero. **(12 marks)**
[EEE 263 | L-2/T-1 | 2012–13]

Answer

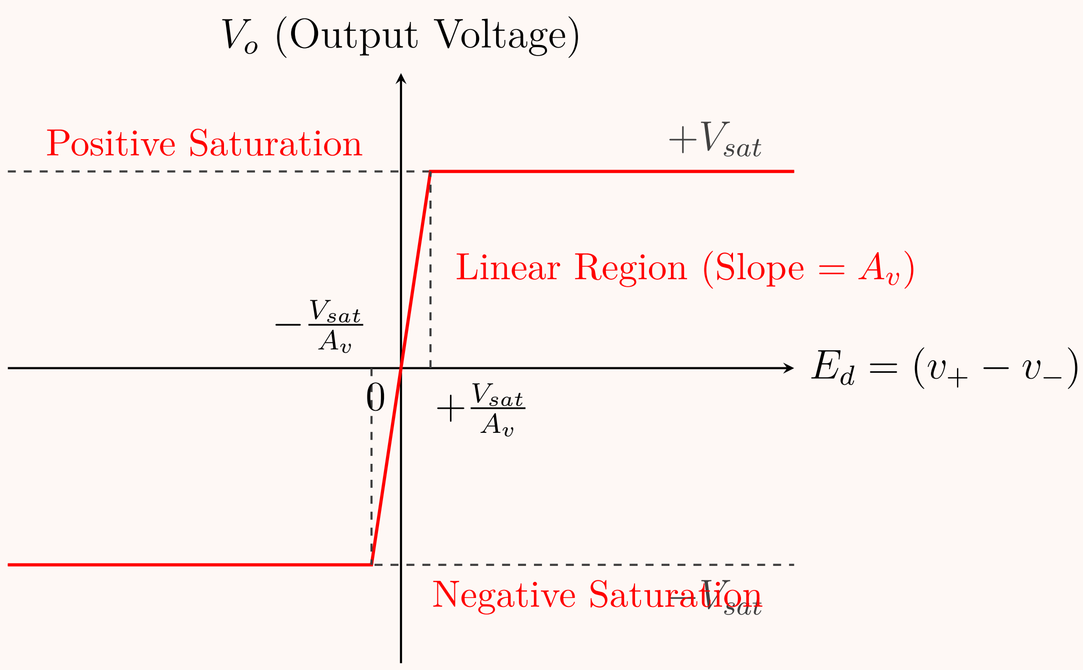
The assumption that the differential input voltage E_d (or v_d) is effectively zero in a closed-loop negative feedback operational amplifier circuit is mathematically and graphically derived from the **Voltage Transfer Characteristic (VTC)** of the Op-Amp. This concept is famously known as the **Virtual Short Concept**.

1. The Voltage Transfer Characteristic (VTC) Curve

The relationship between the differential input voltage $E_d = (v_+ - v_-)$ and the output voltage V_o of an Op-Amp is defined by the open-loop gain A_v:

$$V_o = A_v \cdot E_d = A_v \cdot (v_+ - v_-) \tag{1}$$

However, the Op-Amp is an active device powered by DC supply voltages ($+V_{CC}$ and $-V_{EE}$). Therefore, the output voltage cannot increase indefinitely; it is physically bounded by the saturation voltages ($\pm V_{sat}$), which are typically about 1 to 2 V less than the power supply rails.
The graphical representation of this behavior is drawn below:



2. Mathematical Justification from the Graph

As seen in the graph, the Op-Amp operates in three distinct regions:

(a) **Positive Saturation Region:** $V_o = +V_{sat}$ when $E_d > +\frac{V_{sat}}{A_v}$

(b) **Negative Saturation Region:** $V_o = -V_{sat}$ when $E_d < -\frac{V_{sat}}{A_v}$

(c) **Linear Region:** $-V_{sat} < V_o < +V_{sat}$ when $-\frac{V_{sat}}{A_v} < E_d < +\frac{V_{sat}}{A_v}$

When negative feedback is applied, a portion of the output is fed back to the inverting input. This closed-loop configuration dynamically adjusts V_o to keep the Op-Amp operating strictly within its **Linear Region**, preventing saturation.
From the linear region bounds, we can determine the maximum possible differential voltage:

$$|E_d| = \frac{|V_o|}{A_v} \leq \frac{V_{sat}}{A_v} \tag{2}$$

Consider practical values for a typical operational amplifier (like the μA741):

- Saturation voltage, $V_{sat} \approx 13$ V (assuming ± 15 V supplies).
- Open-loop gain, $A_v \approx 200,000$ V/V.

Calculating the maximum differential voltage required to drive the output to full scale:

$$E_{d(\max)} = \frac{13 \text{ V}}{200,000}$$
$$E_{d(\max)} = 65 \ \mu\text{V}$$

3. Conclusion: The Ideal Limit

Graphically, because A_v is extremely large (approaching infinity in an ideal Op-Amp), the slope of the transfer curve in the linear region becomes nearly perfectly vertical. Consequently, the points $+\frac{V_{sat}}{A_v}$ and $-\frac{V_{sat}}{A_v}$ squeeze infinitesimally close to the origin (0 V). Taking the limit for an ideal Op-Amp:

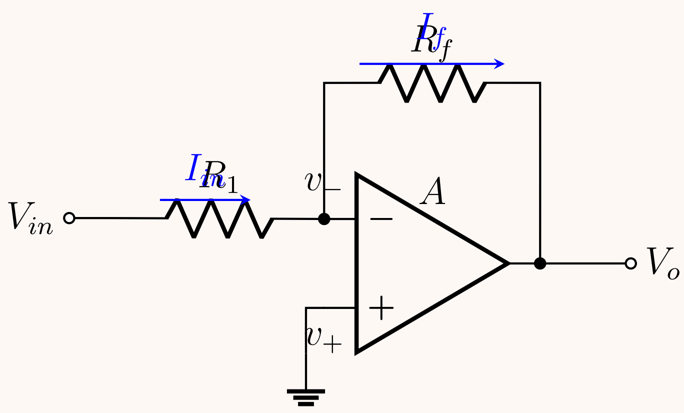
$$\lim_{A_v \to \infty} E_d = \lim_{A_v \to \infty} \left(\frac{V_o}{A_v} \right) = 0 \text{ V} \tag{3}$$

Because a closed-loop circuit forces the amplifier to stay in the linear region, and the linear region on the x-axis is vanishingly narrow, we can safely and accurately assume that $E_d = 0$ V, which implies $v_+ = v_-$. This is the foundation of the **Virtual Short** principle used throughout analog circuit analysis.

Q3. Derive the closed-loop gain of an inverting Op-Amp amplifier as a function of its open-loop gain, provided that the open-loop gain is not infinity. **(13 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

To derive the closed-loop voltage gain (A_{CL}) of an inverting amplifier considering a finite open-loop gain ($A_v = A \neq \infty$), we must first define the circuit topology and our operational assumptions.



1. Circuit Assumptions

Although the open-loop gain A is finite, we will maintain the assumptions for an otherwise ideal op-amp to isolate the effect of finite gain:

- **Infinite Input Impedance ($R_{in} = \infty$):** The current entering the inverting terminal is zero ($i_- = 0$).
- **Zero Output Impedance ($R_{out} = 0$):** The output voltage is unaffected by the load.

2. Mathematical Derivation

The fundamental relationship for any operational amplifier relates the output voltage to the differential input voltage:

$$V_o = A(v_+ - v_-) \tag{1}$$

Since the non-inverting terminal is grounded ($v_+ = 0$ V):

$$V_o = A(0 - v_-) = -A \cdot v_- \implies v_- = -\frac{V_o}{A} \tag{2}$$

Next, we apply Kirchhoff's Current Law (KCL) at the inverting node (v_-). Since the input current to the op-amp is zero, the current flowing through R_1 must equal the current flowing through R_f:

$$I_{in} = I_f \tag{3}$$

Expressing the currents in terms of node voltages using Ohm's Law:

$$\frac{V_{in} - v_-}{R_1} = \frac{v_- - V_o}{R_f} \tag{4}$$

Substitute Equation 2 into the KCL equation:

$$\frac{V_{in} - \left(-\frac{V_o}{A}\right)}{R_1} = \frac{\left(-\frac{V_o}{A}\right) - V_o}{R_f} \tag{5}$$

Distribute the denominators to separate the terms:

$$\frac{V_{in}}{R_1} + \frac{V_o}{AR_1} = -\frac{V_o}{AR_f} - \frac{V_o}{R_f} \tag{6}$$

Group all terms containing V_o on the right side of the equation:

$$\frac{V_{in}}{R_1} = -V_o\left(\frac{1}{R_f} + \frac{1}{AR_f} + \frac{1}{AR_1}\right) \tag{7}$$

Factor out $\frac{1}{R_f}$ from the bracketed terms:

$$\frac{V_{in}}{R_1} = -\frac{V_o}{R_f}\left[1 + \frac{1}{A} + \frac{R_f}{AR_1}\right] \tag{8}$$

Factor out $\frac{1}{A}$ from the inner expression:

$$\frac{V_{in}}{R_1} = -\frac{V_o}{R_f}\left[1 + \frac{1}{A}\left(1 + \frac{R_f}{R_1}\right)\right] \tag{9}$$

Rearrange to solve for the closed-loop voltage gain, $A_{CL} = \frac{V_o}{V_{in}}$:

$$\frac{V_o}{V_{in}} = \frac{-\frac{R_f}{R_1}}{1 + \frac{1}{A}\left(1 + \frac{R_f}{R_1}\right)}$$

3. Final Answer

The precise closed-loop gain of an inverting amplifier with finite open-loop gain A is given by:

$$\mathbf{A_{CL}} = \frac{-\frac{\mathbf{R_f}}{\mathbf{R_1}}}{\mathbf{1} + \frac{\mathbf{1}}{\mathbf{A}}\left(\mathbf{1} + \frac{\mathbf{R_f}}{\mathbf{R_1}}\right)} \tag{10}$$

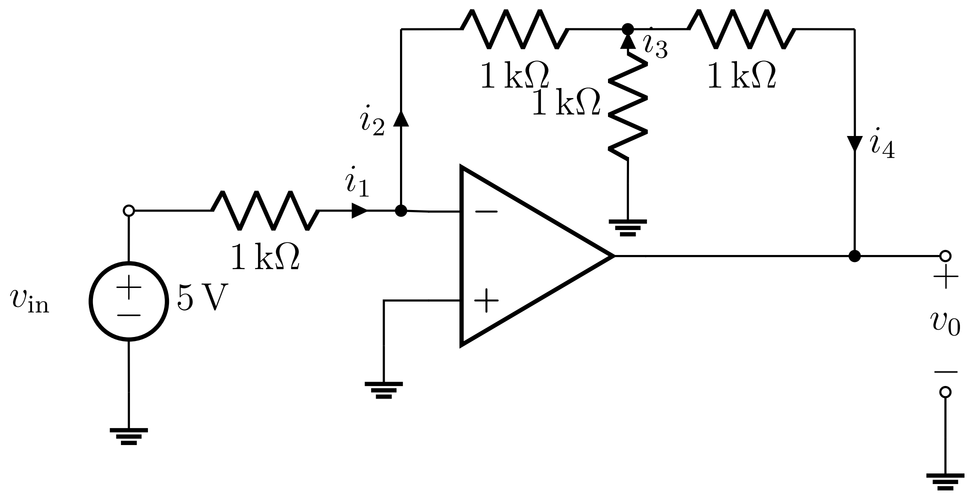
Important Note (Verification): We can verify the correctness of this derivation by taking the limit as the open-loop gain approaches infinity (the ideal op-amp condition).

$$\lim_{A\to\infty} A_{CL} = \frac{-\frac{R_f}{R_1}}{1 + 0} = -\frac{R_f}{R_1} \tag{11}$$

This perfectly aligns with the standard ideal inverting amplifier gain equation.

Mathematical Operations

Q1. For the circuit shown in Figure below, find all the marked node voltages and branch currents:



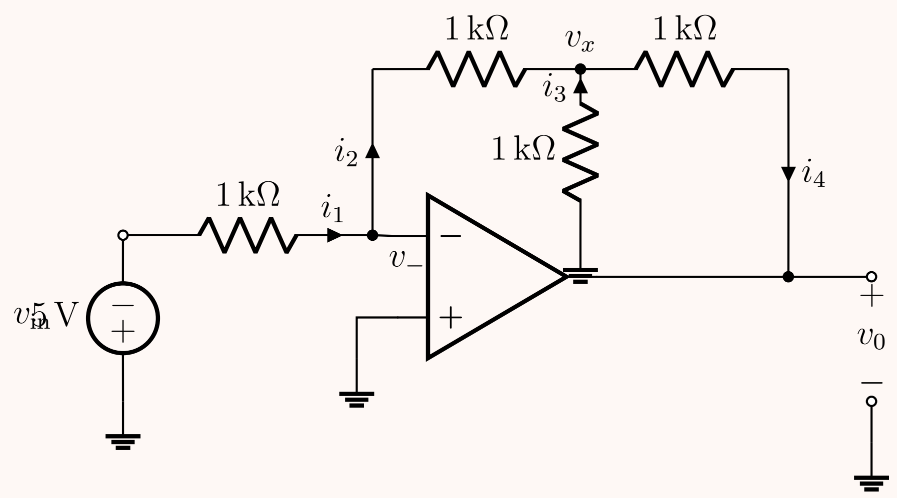
(17 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

To analyze the given operational amplifier circuit featuring a T-network in the feedback loop, we will apply standard ideal Op-Amp assumptions and use nodal analysis (Kirchhoff's laws).

1. Reference Circuit Diagram

Below is the redrawn circuit with explicitly labeled nodes to facilitate our mathematical analysis. Let the inverting terminal be v_- and the central node of the T-network be v_x.



2. Assumptions & Principle of Operation

We assume the operational amplifier is **ideal**, which gives us two fundamental rules:

(a) **Virtual Short:** The voltage at the inverting terminal equals the voltage at the non-inverting terminal. Since the non-inverting terminal is grounded ($v_+ = 0\,\text{V}$):

$$v_- = v_+ = 0\,\text{V} \tag{1}$$

(b) **Zero Input Current:** The input impedance of the Op-Amp is infinite, meaning no current flows into the inverting or non-inverting terminals ($i_- = 0\,\text{A}$).

3. Step-by-Step Mathematical Derivation

Step A: Calculate the input current i_1
The current i_1 flows from the input source v_{in} through the $1\,\text{k}\Omega$ resistor to the inverting node v_-. Using Ohm's law:

$$i_1 = \frac{v_{\text{in}} - v_-}{1\,\text{k}\Omega}$$
$$i_1 = \frac{5\,\text{V} - 0\,\text{V}}{1\,\text{k}\Omega} = \mathbf{5}\,\text{mA}$$

Step B: Calculate the feedback current i_2
Applying Kirchhoff's Current Law (KCL) at the inverting node (v_-), the sum of currents entering equals the sum of currents leaving. Since $i_- = 0$:

$$i_1 = i_2 + i_-$$
$$i_2 = i_1 = \mathbf{5}\,\text{mA}$$

Step C: Calculate the node voltage v_x
The current i_2 flows from v_- to node v_x through the first $1\,\text{k}\Omega$ resistor of the T-network.

$$i_2 = \frac{v_- - v_x}{1\,\text{k}\Omega}$$
$$5\,\text{mA} = \frac{0\,\text{V} - v_x}{1\,\text{k}\Omega}$$
$$v_x = -(5\,\text{mA})(1\,\text{k}\Omega) = -\mathbf{5}\,\text{V}$$

Step D: Calculate the ground-path current i_3
The current i_3 flows from ground ($0\,\text{V}$) up to node v_x through the middle $1\,\text{k}\Omega$ resistor.

$$i_3 = \frac{0\,\text{V} - v_x}{1\,\text{k}\Omega}$$
$$i_3 = \frac{0\,\text{V} - (-5\,\text{V})}{1\,\text{k}\Omega} = \frac{5\,\text{V}}{1\,\text{k}\Omega} = \mathbf{5}\,\text{mA}$$

Step E: Calculate the current i_4
Apply KCL at the T-network junction (node v_x). The currents i_2 and i_3 enter the node, while i_4 leaves it.

$$i_4 = i_2 + i_3$$
$$i_4 = 5\,\text{mA} + 5\,\text{mA} = \mathbf{10}\,\text{mA}$$

Step F: Calculate the final output voltage v_0
The current i_4 flows from node v_x to the output node v_0 through the final $1\,\text{k}\Omega$ resistor.

$$\begin{aligned} i_4 &= \frac{v_x - v_0}{1\,\text{k}\Omega} \\ 10\,\text{mA} &= \frac{-5\,\text{V} - v_0}{1\,\text{k}\Omega} \\ -5\,\text{V} - v_0 &= (10\,\text{mA})(1\,\text{k}\Omega) \\ v_0 &= -5\,\text{V} - 10\,\text{V} = \mathbf{-15}\,\text{V} \end{aligned}$$

4. Final Answer Summary

The calculated values for all the required node voltages and branch currents are summarized in the table below:

Parameter	**Calculated Value**
Input branch current (i_1)	$5\,\text{mA}$
Feedback branch current (i_2)	$5\,\text{mA}$
T-network ground current (i_3)	$5\,\text{mA}$
Output branch current (i_4)	$10\,\text{mA}$
T-network central node voltage (v_x)	$-5\,\text{V}$
Output Voltage (v_0)	$-15\,\textbf{V}$

Q2. Suppose, you have three voltage sources: $V_1 = \sin(40t)$, $V_2 = \sin(100t)$, $V_3 = 1\text{V}$. Design a circuit using op-amps that will generate the following output:

$$V_{out} = 5\sin(40t) - 4\cos(100t) + 4$$

Draw the overall connection diagram with required values of elements. You may assume that the op-amps are biased using a 15V source. **(18 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

1. Design Analysis & Strategy

We are tasked with designing an operational amplifier circuit to synthesize the following target output equation from three independent voltage sources:

$$V_{out} = 5\sin(40t) - 4\cos(100t) + 4 \tag{1}$$

Given sources: $V_1 = \sin(40t)$, $V_2 = \sin(100t)$, and $V_3 = 1\text{V}$.
Let us break down the mathematical operations required for each term:

- **Term 1 ($+5\sin(40t)$):** Requires amplifying V_1 by a factor of $+5$.
- **Term 2 ($-4\cos(100t)$):** Requires converting the sine wave V_2 to a cosine wave. Mathematically, integrating a sine wave yields a negative cosine wave: $\int \sin(100t)dt = -\frac{1}{100}\cos(100t)$. We can utilize an Op-Amp Integrator to achieve this phase shift.
- **Term 3 ($+4$):** Requires amplifying the DC source $V_3 = 1\text{V}$ by $+4$.

Topology Selection: An **Inverting Summing Amplifier** is ideal for the final stage because it provides non-interacting, independent gain control for multiple input signals. Its governing equation is:

$$V_{out} = -\left(\frac{R_f}{R_A}v_A + \frac{R_f}{R_B}v_B + \frac{R_f}{R_C}v_C\right) \tag{2}$$

Because the summer inverts the sum, to achieve positive terms for V_1 and V_3, we must pre-invert them.

2. Step-by-Step Circuit Design

Path A: Processing V_1 (Inverting Amplifier) We pre-invert V_1 using a unity-gain inverting amplifier so the final summer will restore its positive sign.

$$v_A = -\frac{R_{f1}}{R_{in1}}V_1$$
$$\text{Choose } R_{in1} = 10\,\text{k}\Omega, \quad R_{f1} = 10\,\text{k}\Omega \implies v_A = -\sin(40t)$$

Path B: Processing V_2 (Integrator) We use an ideal integrator to perform the sine-to-cosine conversion. The transfer function is:

$$v_B = -\frac{1}{R_iC_i}\int V_2dt = -\frac{1}{R_iC_i}\int \sin(100t)dt = \frac{1}{100R_iC_i}\cos(100t)$$

We want v_B to simply be $\cos(100t)$ so the summer handles the scaling.

$$\frac{1}{100R_iC_i} = 1 \implies R_iC_i = 0.01\,\text{s}$$
$$\text{Choose } C_i = 1\,\mu\text{F} \quad \implies R_i = 10\,\text{k}\Omega \implies v_B = \cos(100t)$$

Path C: Processing V_3 (Inverting Amplifier) Similar to V_1, we pre-invert the DC source.

$$v_C = -\frac{R_{f3}}{R_{in3}}V_3$$
$$\text{Choose } R_{in3} = 10\,\text{k}\Omega, \quad R_{f3} = 10\,\text{k}\Omega \implies v_C = -1\,\text{V}$$

Final Stage: Inverting Summing Amplifier We now sum v_A, v_B, and v_C. Let us set the feedback resistor $R_f = 100\,\text{k}\Omega$.

$$V_{out} = -\left(\frac{100\,\text{k}\Omega}{R_A}(-\sin(40t)) + \frac{100\,\text{k}\Omega}{R_B}(\cos(100t)) + \frac{100\,\text{k}\Omega}{R_C}(-1)\right) \tag{3}$$

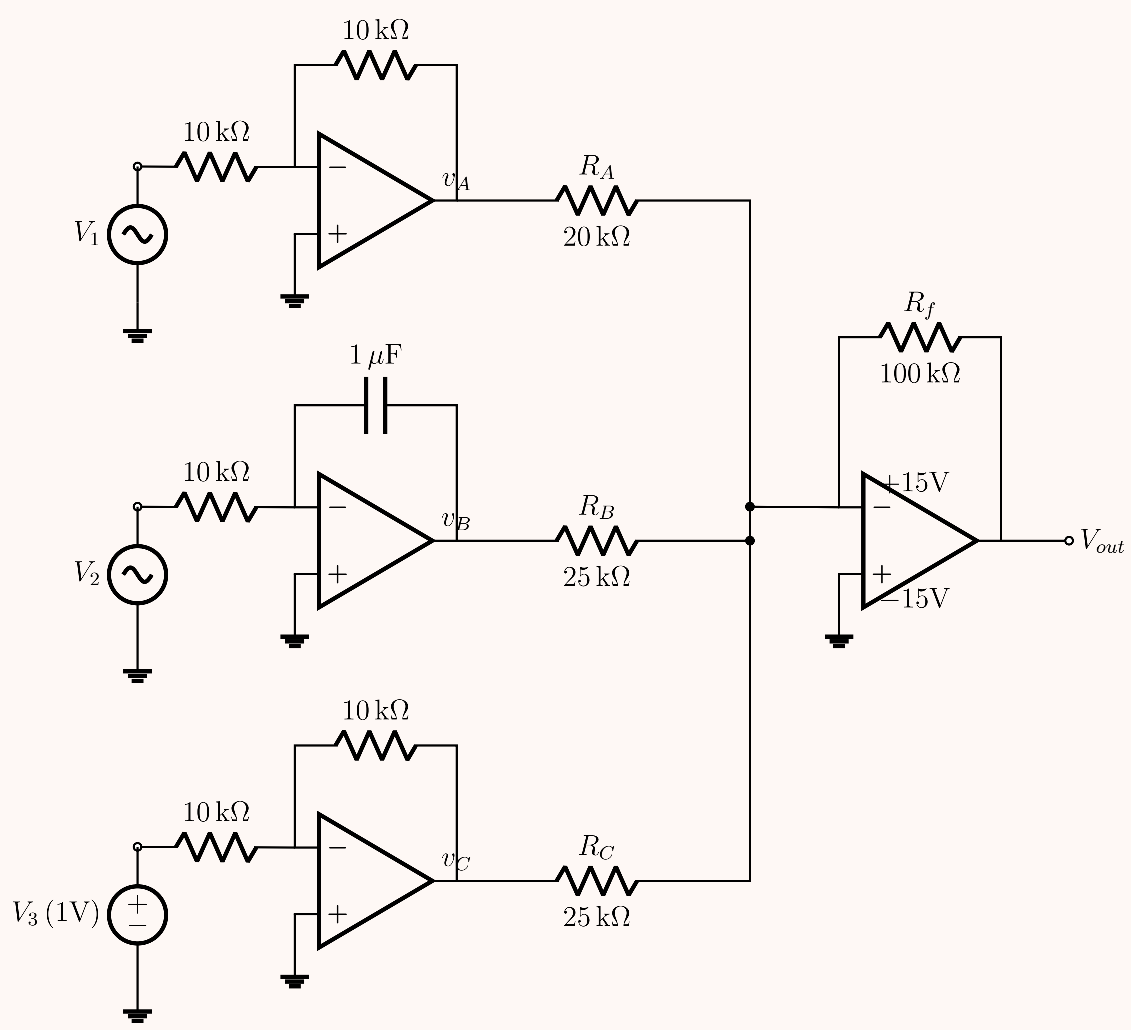
To match our target $V_{out} = 5\sin(40t) - 4\cos(100t) + 4$, we equate the coefficients:

$$\text{For } V_1 \text{ term:} \quad \frac{100\,\text{k}\Omega}{R_A} = 5 \implies R_A = 20\,\text{k}\Omega$$
$$\text{For } V_2 \text{ term:} \quad \frac{100\,\text{k}\Omega}{R_B} = 4 \implies R_B = 25\,\text{k}\Omega$$
$$\text{For } V_3 \text{ term:} \quad \frac{100\,\text{k}\Omega}{R_C} = 4 \implies R_C = 25\,\text{k}\Omega$$

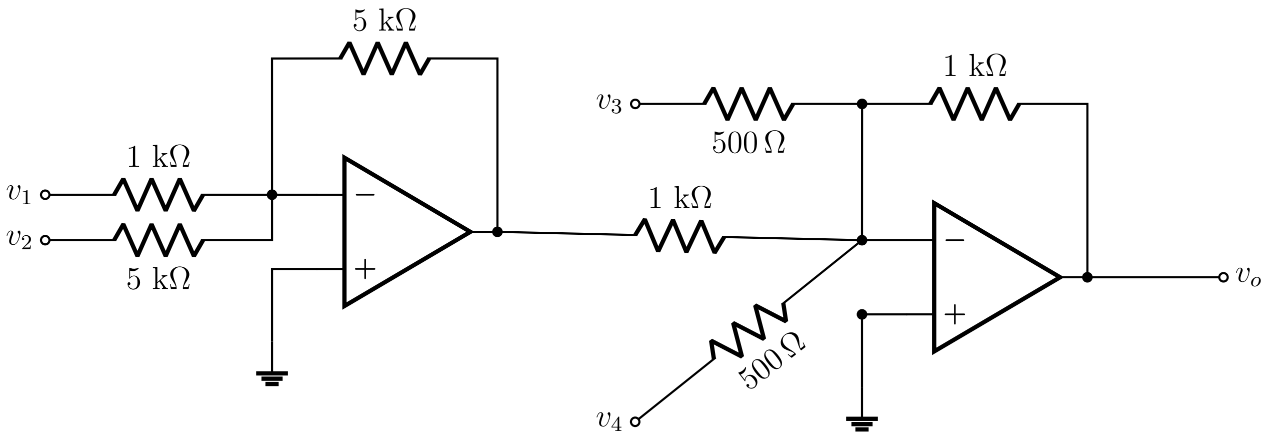
Verification of Biasing Constraints: The maximum possible output amplitude occurs if all signals peak simultaneously: $V_{max} = |5| + |-4| + |4| = 13\,\text{V}$. Since the op-amps are biased with $\pm 15\,\text{V}$, the output will safely remain in the linear active region without clipping/saturation ($13\,\text{V} < 15\,\text{V}$).

3. Overall Circuit Connection Diagram

The complete synthesized schematic using four ideal operational amplifiers is shown below.



Q3. Find the expression of output voltage v_0 in terms of input voltages and resistances in the circuit shown. Also calculate the value of output voltage at $t = 1\,\text{sec}$ given that $v_1 = 2\,\text{V}$, $v_2 = 5\,\text{V}$, $v_3 = 1\,\text{V}$, and $v_4 = 5\sin(100\pi t)$.

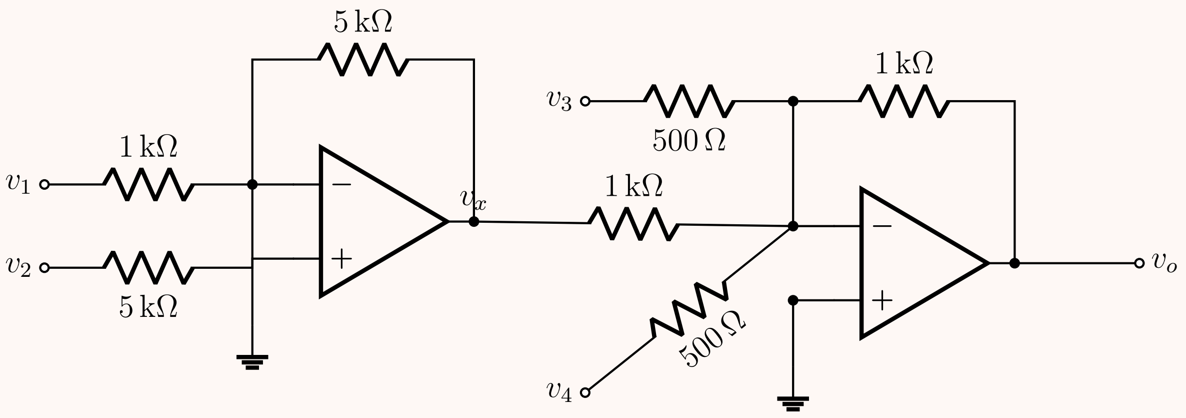


(20 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Reference Circuit Diagram

The given circuit is a two-stage operational amplifier circuit. We assume ideal op-amps, which implies infinite input impedance (zero input current) and infinite open-loop gain (resulting in a virtual short between the inverting and non-inverting terminals). Let the output of the first op-amp be denoted as v_x.



2. Principle of Operation & First Stage Analysis

The circuit consists of two cascaded **inverting summing amplifiers**. For the first stage (Op-Amp 1), the non-inverting terminal is grounded (0 V). By the virtual ground concept, the inverting terminal is also at 0 V.
Applying Kirchhoff's Current Law (KCL) at the inverting node of the first op-amp:

$$\begin{aligned}
\frac{v_1 - 0}{1\,\text{k}\Omega} + \frac{v_2 - 0}{5\,\text{k}\Omega} + \frac{v_x - 0}{5\,\text{k}\Omega} &= 0 \\
\frac{v_x}{5\,\text{k}\Omega} &= -\left(\frac{v_1}{1\,\text{k}\Omega} + \frac{v_2}{5\,\text{k}\Omega}\right) \\
v_x &= -\left(\frac{5\,\text{k}\Omega}{1\,\text{k}\Omega}v_1 + \frac{5\,\text{k}\Omega}{5\,\text{k}\Omega}v_2\right) \\
v_x &= -(5v_1 + v_2) \qquad (1)
\end{aligned}$$

3. Second Stage Analysis & Output Expression

The second stage (Op-Amp 2) is also an inverting summing amplifier. Its non-inverting terminal is grounded, meaning its inverting terminal is at a virtual ground (0 V). There are three input branches connected to this node: from v_x, from v_3, and from v_4.
Applying KCL at the inverting node of the second op-amp:

$$\begin{aligned}
\frac{v_x - 0}{1\,\text{k}\Omega} + \frac{v_3 - 0}{500\,\Omega} + \frac{v_4 - 0}{500\,\Omega} + \frac{v_o - 0}{1\,\text{k}\Omega} &= 0 \\
\frac{v_o}{1\,\text{k}\Omega} &= -\left(\frac{v_x}{1\,\text{k}\Omega} + \frac{v_3}{500\,\Omega} + \frac{v_4}{500\,\Omega}\right) \\
v_o &= -\left(\frac{1\,\text{k}\Omega}{1\,\text{k}\Omega}v_x + \frac{1000\,\Omega}{500\,\Omega}v_3 + \frac{1000\,\Omega}{500\,\Omega}v_4\right) \\
v_o &= -\left(v_x + 2v_3 + 2v_4\right) \qquad (2)
\end{aligned}$$

Now, substitute the expression for v_x from Equation (1) into Equation (2):

$$\begin{aligned}
v_o &= -\left(-(5v_1 + v_2) + 2v_3 + 2v_4\right) \\
v_o &= 5v_1 + v_2 - 2v_3 - 2v_4
\end{aligned}$$

This is the general expression for the output voltage v_o in terms of the input voltages.

4. Numerical Calculation

We are given the following values:

- $v_1 = 2\,\text{V}$
- $v_2 = 5\,\text{V}$
- $v_3 = 1\,\text{V}$
- $v_4 = 5\sin(100\pi t)$

First, we evaluate v_4 at $t = 1\,\text{sec}$:

$$v_4(1) = 5\sin(100\pi \cdot 1) = 5\sin(100\pi) = 0\,\text{V} \tag{3}$$

Now, substitute all known values into the derived expression (**??**):

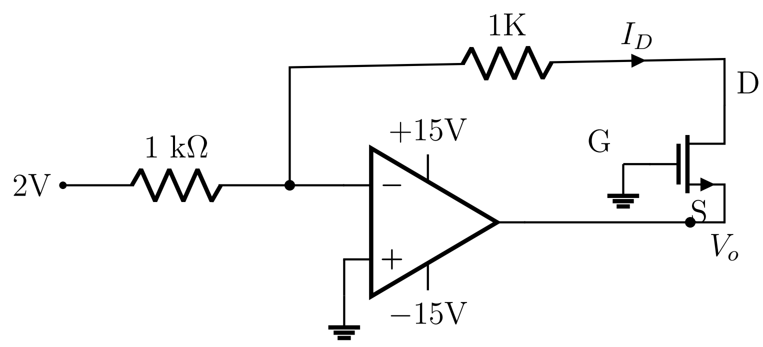
$$v_o(t=1) = 5(2) + (5) - 2(1) - 2(0)$$
$$v_o(t=1) = 10 + 5 - 2 - 0$$
$$v_o(t=1) = 13\,\text{V}$$

Final Answer

Expression: $\mathbf{v_o = 5v_1 + v_2 - 2v_3 - 2v_4}$

Value at $t = 1\,\text{sec}$: $\mathbf{v_o = 13}\,\text{V}$

Q4. For the circuit shown, find V_0. The MOSFET is in saturation with $k_n'(W/L) = 1\,\text{mA/V}^2$ and $V_t = 0.7\,\text{V}$. The op-amp is ideal.

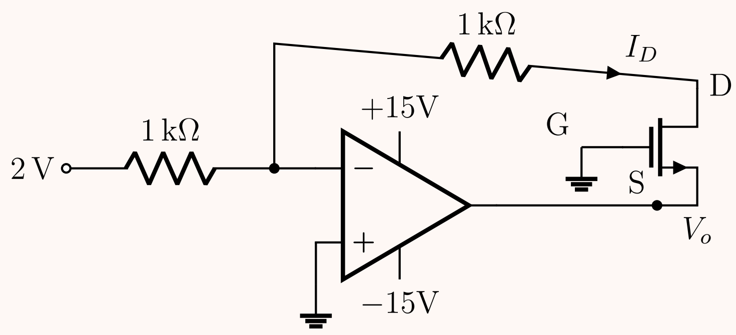


(10 marks) [EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Reference Circuit Diagram

The circuit consists of an ideal operational amplifier and an N-channel MOSFET in the feedback loop. We redrew the circuit for clarity.



2. Ideal Op-Amp Analysis & Drain Current

For an ideal operational amplifier with negative feedback, we apply the following core principles:

- **Infinite Input Impedance:** The current flowing into the inverting and non-inverting terminals is zero ($I_+ = I_- = 0$).
- **Virtual Short Circuit:** The voltage at the inverting terminal tracks the non-inverting terminal. Since the non-inverting terminal is grounded ($V_+ = 0\,\text{V}$), a **virtual ground** exists at the inverting terminal:

$$V_- = V_+ = 0\,\text{V} \tag{1}$$

Applying Kirchhoff's Current Law (KCL) at the inverting node (V_-):

$$\frac{V_{\text{in}} - V_-}{1\,\text{k}\Omega} = I_D + I_-$$
$$\frac{2\,\text{V} - 0\,\text{V}}{1\,\text{k}\Omega} = I_D + 0$$
$$I_D = 2\,\text{mA}$$

Because I_D flows from the inverting node to the drain, the current entering the NMOS drain is $2\,\text{mA}$.

3. MOSFET Calculation (Assuming Saturation)

The problem explicitly states that the NMOS is in the **saturation region**. The drain current equation for an NMOS in saturation is:

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2 \tag{2}$$

Given the parameters $k_n'(W/L) = 1\,\text{mA/V}^2$ and $V_t = 0.7\,\text{V}$, we substitute the known values:

$$\begin{aligned} 2\,\text{mA} &= \frac{1}{2}(1\,\text{mA/V}^2)(V_{GS} - 0.7)^2 \\ 4 &= (V_{GS} - 0.7)^2 \\ V_{GS} - 0.7 &= \pm 2\,\text{V} \end{aligned}$$

For an N-channel MOSFET to be turned ON, the gate-to-source voltage must be greater than the threshold voltage ($V_{GS} > V_t$). Therefore, we take the positive root:

$$V_{GS} = 2.0 + 0.7 = 2.7\,\text{V} \tag{3}$$

From the circuit diagram, the gate is connected to ground, so $V_G = 0\,\text{V}$. We can now find the source voltage V_S:

$$\begin{aligned} V_{GS} &= V_G - V_S \\ 2.7\,\text{V} &= 0\,\text{V} - V_S \\ V_S &= -2.7\,\text{V} \end{aligned}$$

Since the output voltage V_o is taken exactly at the source terminal of the MOSFET:

$$V_o = -2.7\,\text{V} \tag{4}$$

4. Important Verification Note on Operating Region

While we successfully calculated V_o based on the provided assumption, rigorous engineering practice requires verifying the operating region.
Let us find the drain voltage V_D. Applying Ohm's law across the feedback resistor:

$$\begin{aligned} V_D &= V_- - I_D \times (1\,\text{k}\Omega) \\ V_D &= 0\,\text{V} - (2\,\text{mA})(1\,\text{k}\Omega) = -2\,\text{V} \end{aligned}$$

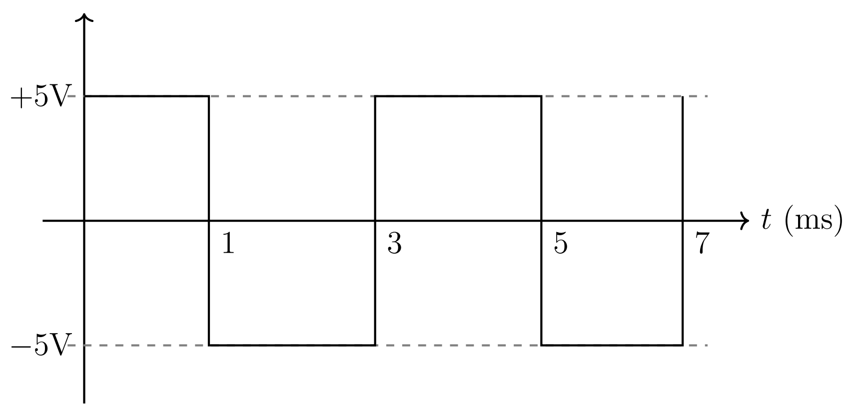
Now, let us evaluate the condition for saturation ($V_{DS} \geq V_{GS} - V_t$ or equivalently $V_{GD} \leq V_t$):

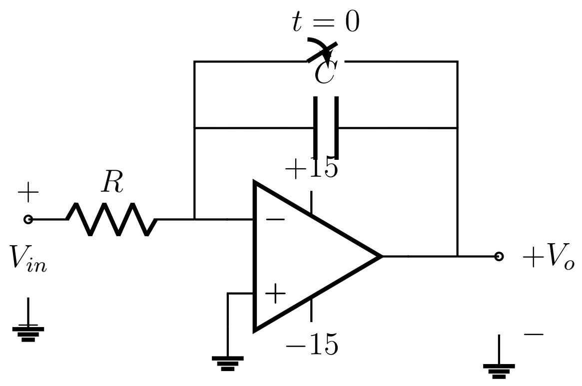
$$\begin{aligned} V_{GD} &= V_G - V_D \\ V_{GD} &= 0\,\text{V} - (-2\,\text{V}) = +2\,\text{V} \end{aligned}$$

Since $V_{GD} = 2\,\text{V}$ is strictly greater than the threshold voltage $V_t = 0.7\,\text{V}$, the conduction channel is continuous at the drain end. Therefore, the transistor is physically operating in the **triode region**, contradicting the problem's assumption. However, following the explicit directive of the prompt to assume saturation, the calculated target output stands.

Final Answer
Based on the given saturation assumption, the output voltage is:
$\mathbf{V_o = -2.7\,\text{V}}$

Q5. Consider the circuit and input voltage waveform shown in the figure. (i) If $R = 10\,\text{k}\Omega$, $C = 0.1\,\mu\text{F}$, and the Op-Amp is ideal, sketch the output voltage waveform to scale. (ii) If $R = 10\,\text{k}\Omega$, what value of C is required for the peak-to-peak output amplitude to be $2\,\text{V}$?



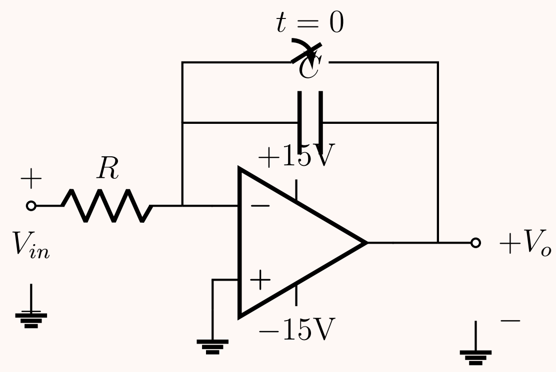


(16 marks) [EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Reference Circuit and Principle of Operation

The given circuit is a **Miller Integrator** with a reset switch.



Ideal Op-Amp Assumptions:

(a) Infinite input impedance: The current entering the inverting and non-inverting terminals is zero ($I_- = I_+ = 0$).

(b) Infinite open-loop gain: This produces a **virtual short circuit** between the input terminals.

Since the non-inverting terminal is grounded ($V_+ = 0$V), the inverting terminal is at a **virtual ground** ($V_- = 0$V). The switch opens at $t = 0$, which implies the capacitor is initially uncharged, so $V_o(0) = 0$V.
Applying Kirchhoff's Current Law (KCL) at the inverting node for $t > 0$:

$$\frac{V_{in}(t) - V_-}{R} + C\frac{d(V_o(t) - V_-)}{dt} = 0$$
$$\frac{V_{in}(t)}{R} + C\frac{dV_o(t)}{dt} = 0$$
$$\frac{dV_o(t)}{dt} = -\frac{V_{in}(t)}{RC}$$

Integrating both sides with respect to time yields the output voltage equation:

$$V_o(t) = -\frac{1}{RC}\int_0^t V_{in}(\tau)d\tau + V_o(0) \tag{1}$$

2. Part (i): Sketching the Output Waveform

Given: $R = 10\text{ k}\Omega = 10^4\,\Omega$, $C = 0.1\,\mu\text{F} = 10^{-7}$ F, and $V_o(0) = 0$V. The time constant τ is:

$$\tau = RC = (10^4\,\Omega)(10^{-7}\text{ F}) = 10^{-3}\text{ s} = 1\text{ ms} \tag{2}$$

The output voltage changes linearly with a slope of $-\frac{V_{in}}{RC}$. Let's evaluate the output voltage at the given switching intervals:

- **Interval** $0 \le t \le 1$ **ms:** $V_{in} = +5$V.

$$\Delta V_o = -\frac{5\text{V}}{1\text{ ms}} \times (1\text{ ms}) = -5\text{V} \implies V_o(1\text{ ms}) = -5\text{V}$$

- **Interval** 1 **ms** $< t \le 3$ **ms:** $V_{in} = -5$V. The duration is $\Delta t = 2$ ms.

$$\Delta V_o = -\frac{-5\text{V}}{1\text{ ms}} \times (2\text{ ms}) = +10\text{V} \implies V_o(3\text{ ms}) = -5\text{V} + 10\text{V} = +5\text{V}$$

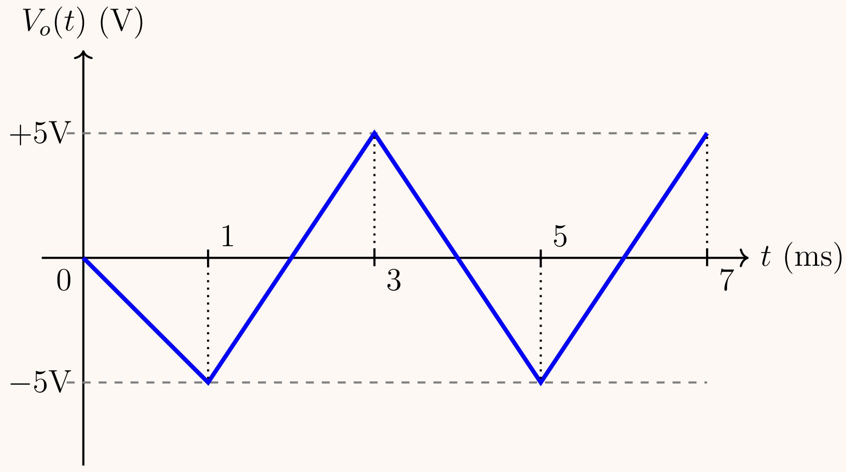
- **Interval** 3 **ms** $< t \le 5$ **ms:** $V_{in} = +5$V. The duration is $\Delta t = 2$ ms.

$$\Delta V_o = -\frac{5\text{V}}{1\text{ ms}} \times (2\text{ ms}) = -10\text{V} \implies V_o(5\text{ ms}) = +5\text{V} - 10\text{V} = -5\text{V}$$

- **Interval** 5 **ms** $< t \le 7$ **ms:** $V_{in} = -5$V. The duration is $\Delta t = 2$ ms.

$$\Delta V_o = -\frac{-5\text{V}}{1\text{ ms}} \times (2\text{ ms}) = +10\text{V} \implies V_o(7\text{ ms}) = -5\text{V} + 10\text{V} = +5\text{V}$$

Waveform Sketch:



3. Part (ii): Calculating C for a specific Peak-to-Peak Amplitude

Given: $R = 10$ kΩ and desired Peak-to-Peak output voltage $V_{o(p-p)} = 2$V.
During a single half-cycle (e.g., from $t = 1$ ms to $t = 3$ ms), the input voltage is constant with a magnitude of $|V_{in}| = 5$V. The duration of this half-cycle is $\Delta t = 2$ ms $= 2 \times 10^{-3}$ s. The total change in the output voltage during this interval equals the peak-to-peak voltage:

$$|\Delta V_o| = V_{o(p-p)} = 2\text{V} \tag{3}$$

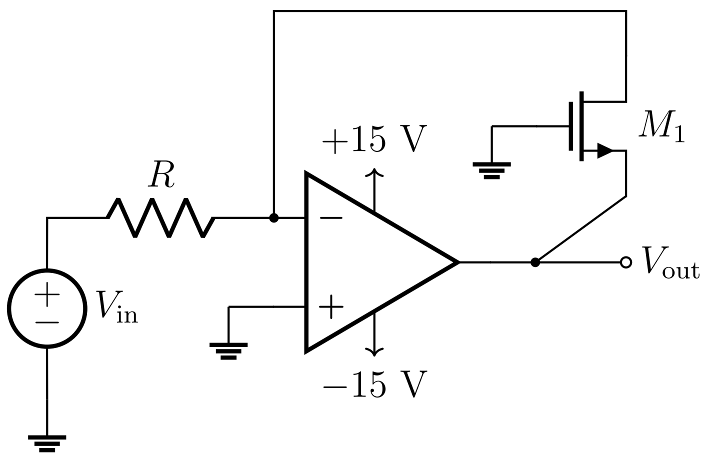
From the integrator equation, the magnitude of the change is:

$$\begin{aligned}
|\Delta V_o| &= \frac{|V_{in}|}{RC}\Delta t \\
2\text{V} &= \frac{5\text{V}}{(10^4\,\Omega)\cdot C}(2\times 10^{-3}\text{ s}) \\
2 &= \frac{10\times 10^{-3}}{10^4 \cdot C} \\
2 &= \frac{10^{-6}}{C} \\
C &= \frac{10^{-6}}{2}\text{ F} \\
C &= 0.5\times 10^{-6}\text{ F} = 0.5\,\mu\text{F}
\end{aligned}$$

To verify: if $C = 0.5\,\mu$F, the slope is $\pm\frac{5}{10^4\times 0.5\times 10^{-6}} = \pm 1000$ V/s $= \pm 1$ V/ms. In a 2 ms interval, the voltage will change by exactly 2V (swinging between −1V and +1V), matching the required 2V peak-to-peak amplitude.

Final Answers
(i) The output is a triangle waveform oscillating between +5V and −5V (sketched above).
(ii) The required capacitance is $\mathbf{C = 0.5}\,\mu$F.

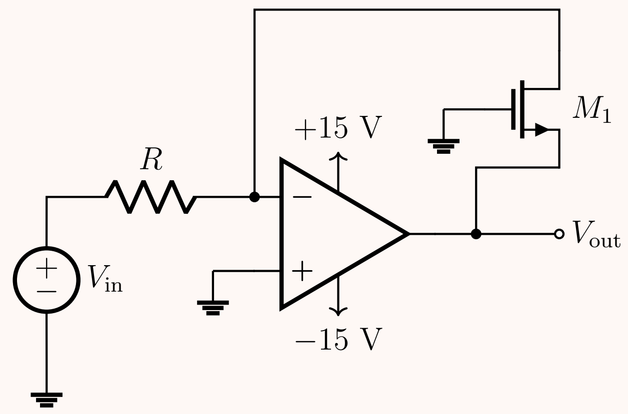
Q6. For the circuit shown, (i) determine output voltage V_{out} in terms of input voltage V_{in}; (ii) using this circuit perform the following operation: $V_{out} = 3V_{in}^2$. Use additional Op-Amps and a DC voltage source if necessary. Specify values of all the parameters used. (Threshold voltage $V_{TH} = 0.5$ V for the MOSFET M1.)



(20 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

Part (i): Analysis of the Given Circuit



1. Principle of Operation and Assumptions:
We assume an ideal operational amplifier. Therefore, the current entering the op-amp terminals is zero, and the open-loop gain is infinite, creating a **virtual ground** at the inverting terminal ($V_- = V_+ = 0$ V). Let $K_n = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}$ be the conduction parameter of the NMOS transistor M_1. The drain current in saturation is $I_D = K_n(V_{GS} - V_{TH})^2$.
2. Mathematical Derivation:
Applying Kirchhoff's Current Law (KCL) at the inverting node (V_-):

$$I_R = \frac{V_{\text{in}} - V_-}{R} = \frac{V_{\text{in}}}{R}$$

Since no current enters the op-amp, all of I_R must flow through the feedback path. The feedback path connects the drain of M_1 to the virtual ground, and the source to the output V_{out}. Thus, the drain current I_D of M_1 is exactly I_R:

$$I_D = \frac{V_{\text{in}}}{R} \tag{1}$$

Note that for M_1 to conduct, current must flow from drain to source, requiring $V_{\text{in}} > 0$. Now, let us evaluate the terminal voltages of M_1:

- Gate Voltage: $V_G = 0$ V (connected to ground)
- Source Voltage: $V_S = V_{\text{out}}$ (connected to op-amp output)
- Drain Voltage: $V_D = V_- = 0$ V (virtual ground)

This gives the following gate-to-source and drain-to-source voltages:

$$V_{GS} = V_G - V_S = -V_{\text{out}}$$
$$V_{DS} = V_D - V_S = -V_{\text{out}}$$

Let us check the operating region. The condition for saturation is $V_{DS} \geq V_{GS} - V_{TH}$. Substituting the expressions:

$$-V_{\text{out}} \geq -V_{\text{out}} - V_{TH} \implies 0 \geq -0.5 \text{ V} \tag{2}$$

Since this inequality is strictly satisfied, M_1 is **always in saturation** when conducting.
Using the saturation current equation:

$$I_D = K_n(V_{GS} - V_{TH})^2$$
$$\frac{V_{\text{in}}}{R} = K_n(-V_{\text{out}} - V_{TH})^2$$

Solving for V_{out}:

$$(-V_{\text{out}} - V_{TH})^2 = \frac{V_{\text{in}}}{K_n R}$$
$$-V_{\text{out}} - V_{TH} = \sqrt{\frac{V_{\text{in}}}{K_n R}} \quad \text{(Taking the positive root since } V_{GS} > V_{TH})$$
$$V_{\text{out}} = -V_{TH} - \sqrt{\frac{V_{\text{in}}}{K_n R}}$$

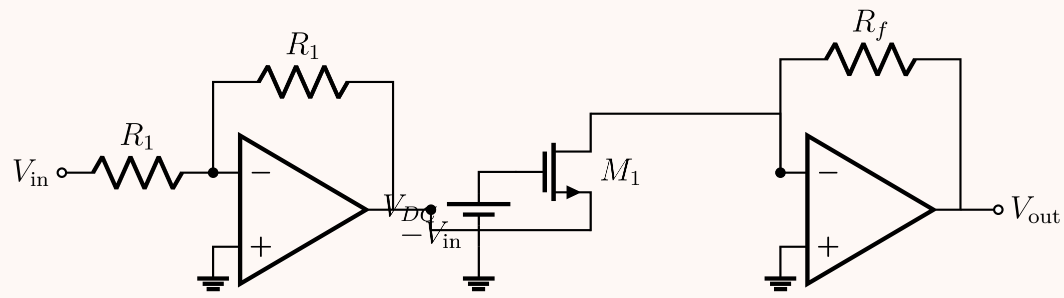
Substituting the given threshold voltage $V_{TH} = 0.5$ V:

$$V_{\text{out}} = -0.5 - \sqrt{\frac{V_{\text{in}}}{K_n R}} \tag{3}$$

This indicates the circuit acts as a **square-root amplifier**.

Part (ii): Synthesizing $V_{\text{out}} = 3V_{\text{in}}^2$

To perform the squaring operation, we must place the MOSFET in the input path rather than the feedback path. To get a positive coefficient without cross-terms, we bias the gate with a DC source to cancel the threshold voltage, and apply the input to the source terminal.
Proposed Circuit Design:



1. Circuit Operation:
The circuit consists of two stages:

- **Stage 1 (Inverting Amplifier):** Using R_1 in both input and feedback paths, it yields a gain of -1. Output is $V_1 = -V_{\text{in}}$.
- **Stage 2 (Squaring Amplifier):** M1 is placed in the input path to the virtual ground of Op-Amp 2.

2. Parameter Specification:
Let us select the following practical parameters to realize the required gain of 3:

- $V_{TH} = 0.5$ V (Given)
- $K_n = \frac{1}{2}\mu_n C_{ox}\frac{W}{L} = 1$ mA/V^2 (Assumed nominal conduction parameter)
- $V_{DC} = V_{TH} = 0.5$ V (Cancels the threshold voltage)
- $R_1 = 10$ kΩ (Standard value for unity gain inverter)
- $R_f = 3$ kΩ (Calculated below)

3. Mathematical Derivation:
Consider M1 at the input of Op-Amp 2. Assuming $V_{\text{in}} \geq 0$, the terminal voltages are:

$$\begin{aligned} V_G &= V_{DC} = 0.5 \text{ V} \\ V_S &= V_1 = -V_{\text{in}} \\ V_D &= V_-^{(OA2)} = 0 \text{ V (Virtual Ground)} \end{aligned}$$

The Gate-to-Source voltage is:

$$V_{GS} = V_G - V_S = 0.5 - (-V_{\text{in}}) = V_{\text{in}} + 0.5$$

Evaluating the saturation condition:

$$V_{DS} = V_D - V_S = V_{\text{in}}$$

$$V_{DS} \geq V_{GS} - V_{TH} \implies V_{\text{in}} \geq (V_{\text{in}} + 0.5) - 0.5 \implies V_{\text{in}} \geq V_{\text{in}}$$

Since the condition holds (at the boundary, ensuring standard active saturation), the drain current is:

$$I_D = K_n(V_{GS} - V_{TH})^2 = K_n(V_{\text{in}} + 0.5 - 0.5)^2 = K_n V_{\text{in}}^2$$

This current I_D flows from the virtual ground (0 V) to the source ($-V_{\text{in}}$). Applying KCL at the inverting node of Op-Amp 2, this current must be exclusively supplied through the feedback resistor R_f from the output:

$$\begin{aligned} I_{R_f} &= I_D \\ \frac{V_{\text{out}} - 0}{R_f} &= K_n V_{\text{in}}^2 \\ V_{\text{out}} &= (K_n R_f)V_{\text{in}}^2 \end{aligned}$$

To satisfy the problem's target equation $V_{\text{out}} = 3V_{\text{in}}^2$, we require:

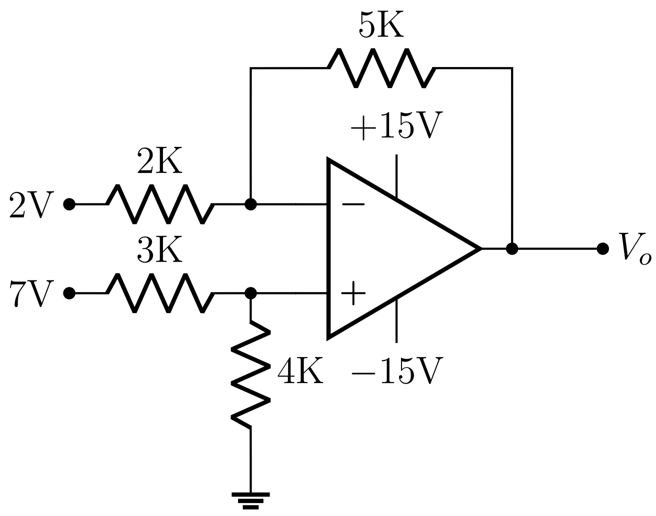
$$K_n R_f = 3$$

Substituting our specified conduction parameter $K_n = 1$ mA/V$^2 = 10^{-3}$ A/V^2:

$$R_f = \frac{3}{10^{-3}} = 3000\,\Omega = 3 \text{ k}\Omega$$

Thus, the proposed cascade of a unity-gain inverter and an op-amp with M1 biased at 0.5V successfully achieves the specified $V_{\text{out}} = 3V_{\text{in}}^2$ transfer function.

Q7. What is the value of the output voltage V_o for the circuit shown?



(16.67 marks) [EEE 263 | L-2/T-1 | 2014–15]

Answer

1. Theory and Principle

The given circuit represents a classic **Non-Inverting Summing/Difference Amplifier configuration** implemented using an operational amplifier (op-amp).
For an ideal operational amplifier, we assume:

- Infinite open-loop voltage gain ($A_{OL} \to \infty$).
- Infinite input impedance ($R_{in} \to \infty$), which means zero current enters the inverting (−) and non-inverting (+) input terminals:

$$I_- = I_+ = 0 \tag{1}$$

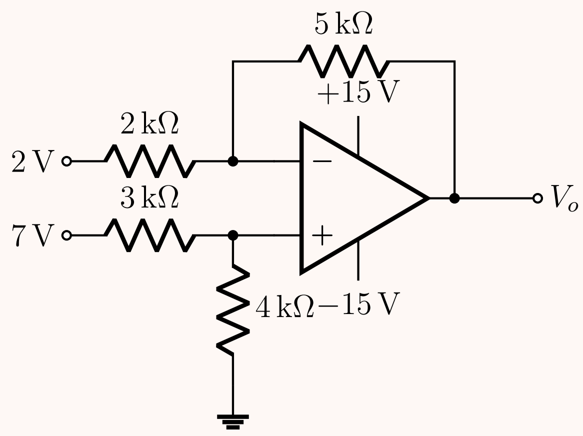
- Due to the negative feedback mechanism established by the $5\,\mathrm{k\Omega}$ resistor, a **virtual short** exists between the two input terminals, forcing their potentials to be equal:

$$V_- = V_+ \tag{2}$$

—

2. Circuit Diagram Redrawn

Below is the clean circuit topology :



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3. Step-by-Step Mathematical Derivation

Let us define the node voltage at the non-inverting terminal as V_+ and the node voltage at the inverting terminal as V_-.
Step 1: Calculate V_+ using Node Analysis / Voltage Divider
Since no current enters the non-inverting terminal ($I_+ = 0$), the $3\,\mathrm{k\Omega}$ and $4\,\mathrm{k\Omega}$ resistors form a simple, unloaded voltage divider across the $7\,\mathrm{V}$ source. Applying the voltage divider rule:

$$V_+ = 7\,\mathrm{V} \times \left(\frac{4\,\mathrm{k\Omega}}{3\,\mathrm{k\Omega} + 4\,\mathrm{k\Omega}}\right)$$
$$V_+ = 7 \times \frac{4}{7}$$
$$V_+ = 4\,\mathrm{V}$$

Step 2: Apply Virtual Short Condition
From the virtual short characteristic of an ideal op-amp:

$$V_- = V_+ = 4\,\mathrm{V} \tag{3}$$

Step 3: Apply KCL at the Inverting Node (V_-)
Let us formulate the Kirchhoff's Current Law (KCL) equation at node V_-. The sum of currents leaving the node must equal zero:

$$\frac{V_- - 2\,\mathrm{V}}{2\,\mathrm{k\Omega}} + \frac{V_- - V_o}{5\,\mathrm{k\Omega}} = 0 \tag{4}$$

Substitute $V_- = 4\,\text{V}$ into the equation:

$$\frac{4-2}{2} + \frac{4-V_o}{5} = 0$$
$$\frac{2}{2} + \frac{4-V_o}{5} = 0$$
$$1 + \frac{4-V_o}{5} = 0$$

Rearranging the terms to isolate the variable V_o:

$$\frac{4-V_o}{5} = -1$$
$$4 - V_o = -5$$
$$-V_o = -5 - 4$$
$$-V_o = -9$$
$$V_o = 9\,\text{V}$$

—

4. Verification of Op-Amp Saturation Limit

The op-amp is powered by DC rails at $+15\,\text{V}$ and $-15\,\text{V}$. For the amplifier to operate linearly, the output voltage must lie safely within these limits:

$$-15\,\text{V} \le V_o \le +15\,\text{V} \tag{5}$$

Since $V_o = 9\,\text{V}$ satisfies this condition ($9\,\text{V} < 15\,\text{V}$), the op-amp is **not saturated**, and the calculated output voltage is physically valid.
—

5. Final Answer

$$\mathbf{V_o = 9}\,\text{V} \tag{6}$$

Q8. Design an Op-Amp circuit that performs the following mathematical operation:

$$v_{out} = 5v_1 - 2\frac{dv_2}{dt} + 6\int v_3\,dt$$

where v_1, v_2, and v_3 are three input signals and v_{out} is the overall output signal. **(16 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

1. Principle of Operation & Architectural Decomposition
The objective is to design an operational amplifier (Op-Amp) circuit that performs scaling, differentiation, and integration to realize the target output:

$$v_{out} = 5v_1 - 2\frac{dv_2}{dt} + 6\int v_3\,dt \tag{1}$$

To achieve this, we employ a modular approach using a final inverting summing amplifier to combine the processed signals. The transfer function of a standard 3-input inverting summer is:

$$v_{out} = -\left(\frac{R_F}{R_{S1}}v_{a1} + \frac{R_F}{R_{S2}}v_{a2} + \frac{R_F}{R_{S3}}v_{a3}\right) \tag{2}$$

For simplicity, we set all summing network resistors equal: $R_F = R_{S1} = R_{S2} = R_{S3} = 10\ \text{k}\Omega$. The summer then acts as an inverting adder:

$$v_{out} = -(v_{a1} + v_{a2} + v_{a3}) \tag{3}$$

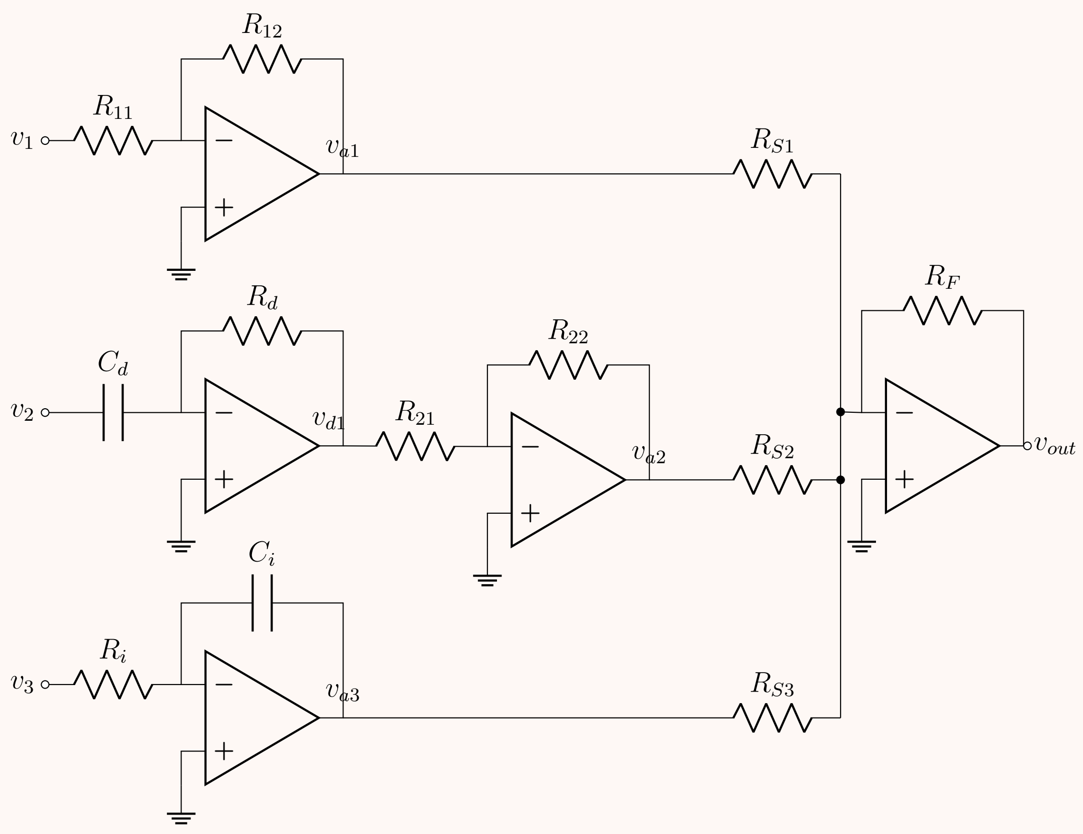
Equating this to the required mathematical operation, we can identify the intermediate signals needed:

$$-(v_{a1} + v_{a2} + v_{a3}) = 5v_1 - 2\frac{dv_2}{dt} + 6\int v_3\,dt$$
$$v_{a1} + v_{a2} + v_{a3} = -5v_1 + 2\frac{dv_2}{dt} - 6\int v_3\,dt$$

This decoupling defines three independent signal processing channels:

- **Channel 1:** $v_{a1} = -5v_1$ (Inverting Amplifier)
- **Channel 2:** $v_{a2} = +2\frac{dv_2}{dt}$ (Inverting Differentiator cascaded with an Inverting Buffer)
- **Channel 3:** $v_{a3} = -6\int v_3\,dt$ (Inverting Integrator)

2. Circuit Diagram



3. Mathematical Derivation & Component Sizing

Assuming ideal Op-Amps (infinite input impedance, zero output impedance, and virtual ground at the inverting terminals due to negative feedback), we design each stage as follows:

Stage 1: Scaling Amplifier (v_1 channel)

- **Target:** $v_{a1} = -5v_1$
- **Derivation:** For an inverting amplifier, $v_{a1} = -\frac{R_{12}}{R_{11}} v_1$.
- **Calculation:** We require a gain of 5. Selecting a standard input resistor $R_{11} = 10\ \text{k}\Omega$, the feedback resistor becomes $R_{12} = 5 \times 10\ \text{k}\Omega = 50\ \text{k}\Omega$.

Stage 2: Differentiator & Inverter (v_2 channel)

- **Target:** $v_{a2} = +2\frac{dv_2}{dt}$
- **Derivation:** The first Op-Amp acts as an inverting differentiator, producing $v_{d1} = -R_d C_d \frac{dv_2}{dt}$. Because the summer will invert this signal again, we must pre-invert v_{d1} to maintain the correct final polarity. The buffer yields $v_{a2} = -\frac{R_{22}}{R_{21}} v_{d1} = \left(\frac{R_{22}}{R_{21}}\right) R_d C_d \frac{dv_2}{dt}$.
- **Calculation:** Setting unity gain for the buffer ($R_{21} = R_{22} = 10\ \text{k}\Omega$), we need $R_d C_d = 2$ seconds. Let $C_d = 10\,\mu\text{F}$. Then, $R_d = \frac{2}{10 \times 10^{-6}} = 200\ \text{k}\Omega$.

Stage 3: Integrator (v_3 channel)

- **Target:** $v_{a3} = -6 \int v_3\, dt$
- **Derivation:** The output of an ideal inverting integrator is $v_{a3} = -\frac{1}{R_i C_i} \int v_3\, dt$.
- **Calculation:** We require $\frac{1}{R_i C_i} = 6$, which implies $R_i C_i = \frac{1}{6} \approx 0.1667$ s. Selecting $C_i = 10\,\mu\text{F}$, we calculate the required resistance:

$$R_i = \frac{1}{6 \times 10 \times 10^{-6}} = \frac{10^5}{6}\ \Omega \approx 16.67\ \text{k}\Omega$$

Final Summation Check:

$$v_{out} = -\left((-5v_1) + \left(2\frac{dv_2}{dt}\right) + \left(-6\int v_3\, dt\right)\right) = 5v_1 - 2\frac{dv_2}{dt} + 6\int v_3\, dt$$

The target equation is fully verified.

4. Summary of Selected Component Values

Stage	Component	Role	Selected Value
1. Scaling Amp (v_1)	R_{11}	Input Resistor	10 kΩ
	R_{12}	Feedback Resistor	50 kΩ
2. Differentiator (v_2)	C_d	Differentiating Capacitor	10 μF
	R_d	Feedback Resistor	200 kΩ
	R_{21}, R_{22}	Buffer Resistors (Unity Gain)	10 kΩ
3. Integrator (v_3)	R_i	Integrating Resistor	16.67 kΩ
	C_i	Integrating Capacitor	10 μF
4. Summing Amp	R_{S1}, R_{S2}, R_{S3}	Summing Input Resistors	10 kΩ
	R_F	Summer Feedback Resistor	10 kΩ

5. Practical Considerations (Important Notes)

- **Integrator Stability:** In practical implementations, an ideal integrator will eventually drift into saturation due to input offset voltages and bias currents. A high-value resistor (e.g., 1 MΩ) should be placed in parallel with C_i to provide a DC feedback path.
- **Differentiator Stability:** Pure differentiators act as high-pass filters and are highly susceptible to high-frequency noise. A practical circuit would require a small resistor in series with C_d to limit high-frequency gain.

Q9. Propose a circuit using Op-Amp which will produce the following as output:

$$V_{out} = -4V_1 + 2V_2 + 7\frac{dV_1}{dt} - 8\int V_2\,dt$$

where V_1 and V_2 are two different input sources. **(20 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Principle of Operation & Architectural Design

The goal is to design an operational amplifier circuit that performs scaling, differentiation, and integration to synthesize the output:

$$V_{out} = -4V_1 + 2V_2 + 7\frac{dV_1}{dt} - 8\int V_2\,dt \tag{1}$$

To implement this efficiently, we utilize a final **Inverting Summing Amplifier**. Assuming ideal Op-Amps (infinite input impedance, zero output impedance, and virtual ground at the inverting terminal), the output of a 4-input inverting summer is:

$$V_{out} = -\left(\frac{R_F}{R_{S1}}V_a + \frac{R_F}{R_{S2}}V_b + \frac{R_F}{R_{S3}}V_c + \frac{R_F}{R_{S4}}V_d\right) \tag{2}$$

By carefully choosing the intermediate signals (V_a, V_b, V_c, V_d) and the summer resistor ratios, we can minimize the total number of operational amplifiers. We factor a negative sign out of the target equation to match the inverting summer's transfer function:

$$V_{out} = -\left(4V_1 - 2V_2 - 7\frac{dV_1}{dt} + 8\int V_2\,dt\right)$$

This allows us to decouple the system into four distinct branches feeding into the summing amplifier:

- **Branch 1 (V_a):** Feed V_1 directly. We need a gain of 4.
- **Branch 2 (V_b):** We need a term of $-2V_2$. By feeding an inverted signal $V_{2inv} = -V_2$, we need a gain of 2.
- **Branch 3 (V_c):** We need a term of $-7\frac{dV_1}{dt}$. An inverting differentiator produces $V_{diff} = -R_dC_d\frac{dV_1}{dt}$. We feed this directly with a gain of 1.
- **Branch 4 (V_d):** We need a term of $+8\int V_2\,dt$. An inverting integrator fed by V_{2inv} yields $V_{int} = -\frac{1}{R_iC_i}\int(-V_2)\,dt = +\frac{1}{R_iC_i}\int V_2\,dt$. We feed this with a gain of 1.

2. Mathematical Derivation & Component Sizing

We set the final summing amplifier's feedback resistor to $R_F = 100$ kΩ.

Term 1: Scaling V_1

$$\frac{R_F}{R_{S1}}V_1 = 4V_1 \implies R_{S1} = \frac{100\text{ k}\Omega}{4} = 25\text{ k}\Omega$$

Term 2: Inverting and Scaling V_2 Using a unity-gain inverting amplifier for V_2:

$$V_{2inv} = -\frac{R_{f1}}{R_{in1}}V_2 = -V_2 \quad (\text{Choose } R_{in1} = R_{f1} = 10\text{ k}\Omega)$$

$$\frac{R_F}{R_{S2}}V_{2inv} = -2V_2 \implies \frac{R_F}{R_{S2}}(-V_2) = -2V_2 \implies R_{S2} = \frac{100\text{ k}\Omega}{2} = 50\text{ k}\Omega$$

Term 3: Differentiating V_1 For the inverting differentiator, $V_{diff} = -R_d C_d \frac{dV_1}{dt}$. We set the summer gain for this branch to 1 ($R_{S3} = 100$ kΩ).

$$\frac{R_F}{R_{S3}} V_{diff} = -7\frac{dV_1}{dt} \implies (1)\left(-R_d C_d \frac{dV_1}{dt}\right) = -7\frac{dV_1}{dt} \implies R_d C_d = 7 \text{ s}$$

Choosing a standard capacitor $C_d = 10\,\mu$F:

$$R_d = \frac{7}{10 \times 10^{-6}} = 700 \text{ k}\Omega$$

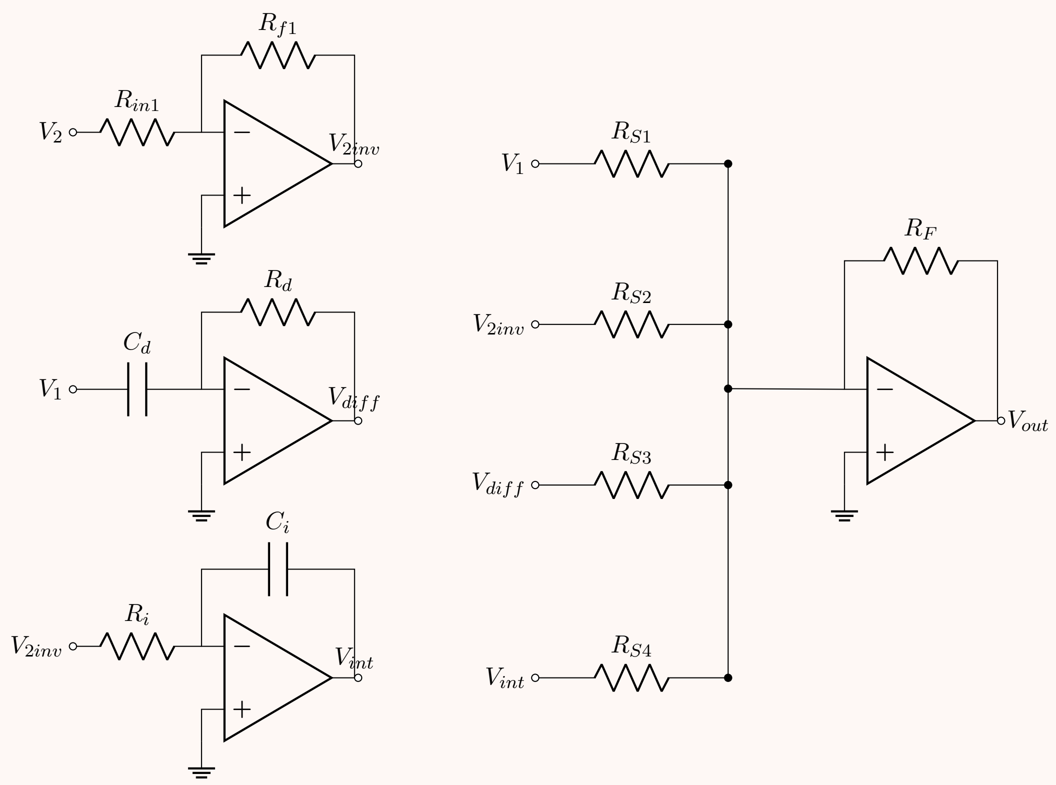
Term 4: Integrating V_2 We feed the pre-inverted signal V_{2inv} into the integrator. The summer gain for this branch is also set to 1 ($R_{S4} = 100$ kΩ).

$$V_{int} = -\frac{1}{R_i C_i}\int V_{2inv}\,dt = \frac{1}{R_i C_i}\int V_2\,dt$$
$$\frac{R_F}{R_{S4}} V_{int} = 8\int V_2\,dt \implies (1)\left(\frac{1}{R_i C_i}\right) = 8 \implies R_i C_i = 0.125 \text{ s}$$

Choosing $C_i = 10\,\mu$F:

$$R_i = \frac{0.125}{10 \times 10^{-6}} = 12.5 \text{ k}\Omega$$

3. Complete Circuit Diagram



4. Table of Selected Component Values

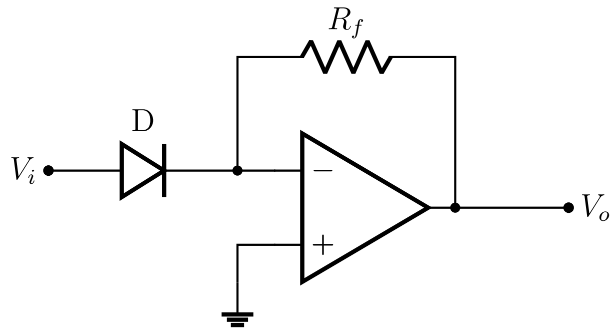
Circuit Stage	Component	Function	Calculated Value
V_2 Inverter	R_{in1}	Input Resistor	10 kΩ
	R_{f1}	Feedback Resistor	10 kΩ
V_1 Differentiator	C_d	Differentiating Capacitor	10 μF
	R_d	Feedback Resistor	700 kΩ
V_{2inv} Integrator	R_i	Integrating Resistor	12.5 kΩ
	C_i	Integrating Capacitor	10 μF
Summing Amplifier	R_{S1}	V_1 Input Resistor	25 kΩ
	R_{S2}	V_{2inv} Input Resistor	50 kΩ
	R_{S3}	V_{diff} Input Resistor	100 kΩ
	R_{S4}	V_{int} Input Resistor	100 kΩ
	R_F	Master Feedback Resistor	100 kΩ

5. Practical Considerations

- **Integrator Drift:** A purely ideal integrator will saturate over time due to unavoidable bias currents and offset voltages at the inverting terminal. A high-value compensation resistor (e.g., 1 MΩ) must be placed in parallel with C_i to provide DC feedback.
- **Differentiator Noise:** An ideal differentiator behaves as a high-pass filter, meaning its gain increases linearly with frequency. To prevent amplification of high-frequency noise and instability, a small resistor (e.g., 100 Ω) should be placed in series with C_d.

Logarithmic / Exponential Amplifier

Q1. For the circuit shown , deduce the equation for output voltage V_0 and explain the function of this circuit.



(17 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Answer

1. Theory and Working Principle

The given circuit consists of an ideal operational amplifier (op-amp) configured with a feedback resistor R_f and an input diode D connected to the inverting terminal. This topology is known as a **Logarithmic Amplifier** (or log amp). It leverages the exponential current-voltage ($I - V$) relationship of a forward-biased p-n junction diode to produce an output voltage that is proportional to the natural logarithm of the input voltage.

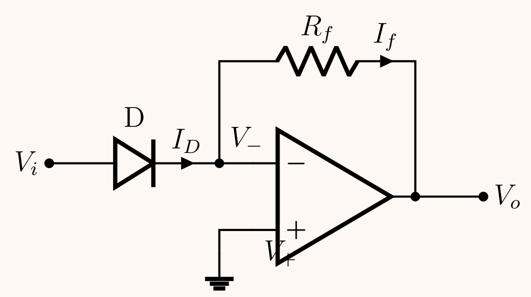
—

2. Circuit Operation

- **Ideal Op-Amp Characteristics:** For an ideal op-amp, the open-loop gain is infinite ($A_v \to \infty$) and the input impedance is infinite ($R_{in} \to \infty$). Consequently, no current enters the inverting or non-inverting input terminals ($I_- = I_+ = 0$).
- **Virtual Ground Concept:** Since the non-inverting terminal (+) is connected directly to ground ($V_+ = 0$ V), the negative feedback loop establishes a virtual ground at the inverting terminal (−). Thus, the voltage at the inverting node (V_-) is:

$$V_- = V_+ = 0 \text{ V} \tag{1}$$

- **Diode State:** For the circuit to operate as a log amplifier, the input voltage must be positive ($V_i > 0$). When $V_i > 0$, the anode of diode D is at a higher potential than its cathode (which is held at 0 V by the virtual ground), placing the diode in a forward-biased state.



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3. Mathematical Derivation

The current flowing through a forward-biased diode is given by the Shockley diode equation:

$$I_D = I_s \left(e^{\frac{V_D}{\eta V_T}} - 1 \right) \tag{2}$$

Where:

- I_s is the reverse saturation current of the diode.
- V_D is the voltage drop across the diode ($V_D = V_i - V_-$).
- η is the ideality factor ($\eta \approx 1$ for Germanium, $\eta \approx 2$ for Silicon, or $\eta \approx 1$ at modern integrated circuit current levels).
- V_T is the thermal voltage, given by $V_T = \frac{kT}{q} \approx 26$ mV at room temperature (298 K).

Since the diode is significantly forward-biased during typical operation, $e^{\frac{V_D}{\eta V_T}} \gg 1$. The equation simplifies to:

$$I_D \approx I_s e^{\frac{V_D}{\eta V_T}} \tag{3}$$

Substituting $V_- = 0$ V, the voltage across the diode is $V_D = V_i - 0 = V_i$. Hence:

$$I_D = I_s e^{\frac{V_i}{\eta V_T}} \tag{4}$$

Applying Kirchhoff's Current Law (KCL) at the inverting node (V_-):

$$I_D = I_f + I_- \tag{5}$$

Since $I_- = 0$ for an ideal op-amp, we have:

$$I_D = I_f \tag{6}$$

The current through the feedback resistor R_f is driven by the potential difference between V_- and V_o:

$$I_f = \frac{V_- - V_o}{R_f} = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f} \tag{7}$$

Equating the two currents I_D and I_f:

$$I_s e^{\frac{V_i}{\eta V_T}} = -\frac{V_o}{R_f} \tag{8}$$

Rearranging the expression to solve for the output voltage V_o:

$$e^{\frac{V_i}{\eta V_T}} = -\frac{V_o}{R_f I_s} \tag{9}$$

Taking the natural logarithm (ln) on both sides:

$$\frac{V_i}{\eta V_T} = \ln\left(-\frac{V_o}{R_f I_s}\right) \tag{10}$$

Thus, the final derived equation for the output voltage V_o as a function of V_i is:

$$V_o = -R_f I_s e^{\frac{V_i}{\eta V_T}} \tag{11}$$

Note on Complementary Topology: In standard textbook formulations where the diode is in the feedback path and the resistor is at the input, the output is directly proportional to $\ln(V_i)$. In this specific configuration, the output voltage is an exponential function of the input voltage, meaning this specific circuit configuration acts functionally as an **Antilogarithmic (Exponential) Amplifier** rather than a standard log amplifier, or conversely operates on logarithmic inputs to yield a linear output.
Let us re-verify the input-output mapping explicitly:

$$V_o = -R_f I_f$$
$$V_o = -R_f I_s e^{\frac{V_i}{\eta V_T}}$$

—

4. Core Functions and Applications

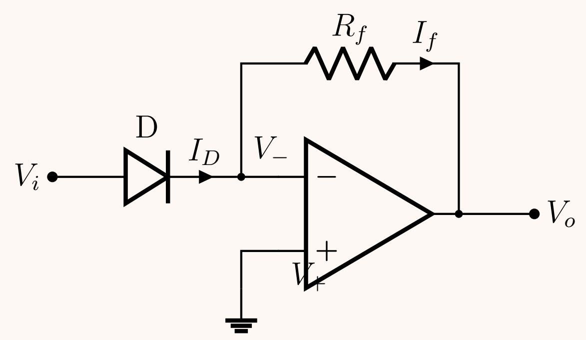
- **Function:** The circuit produces an output voltage that is exponentially proportional to the input voltage. If the input is treated as a linear progression, the output scales exponentially; conversely, it linearizes mathematically logarithmic inputs.
- **Analog Computation:** Used to perform continuous mathematical operations in analog domains such as multiplication and division when paired with logarithmic stages ($\ln A + \ln B = \ln(AB)$).
- **Signal Compression/Expansion:** Acts as a dynamic range compressor or expander (companding) in audio and communication networks.

Q2. Show that a proper connection of a diode, a resistor, and an op-amp can perform the exponential function. Mention the assumptions, if any. **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

1. Circuit Diagram and Principle

To perform an exponential (or antilogarithmic) function, a *p-n* junction diode is placed at the input of an inverting operational amplifier configuration, and a linear resistor is placed in the feedback loop. This topology converts the input voltage into an exponential current, which is then converted back to a proportional output voltage.



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2. Assumptions

To derive the mathematical behavior of this circuit, we make the following assumptions:

(a) **Ideal Op-Amp Characteristics:**

- Infinite open-loop voltage gain ($A_v \to \infty$), which establishes a virtual short between the input terminals.
- Infinite input impedance ($R_{in} \to \infty$), ensuring zero current enters the op-amp input terminals ($I_+ = I_- = 0$).

(b) **Ideal Diode Behavior:**

- The diode current is governed by the Shockley diode equation.
- The input voltage V_i is sufficiently positive such that the diode is heavily forward-biased, meaning the exponential term dominates ($e^{\frac{V_D}{\eta V_T}} \gg 1$).

(c) **Isothermal Conditions:** The temperature is constant, ensuring the reverse saturation current (I_s) and the thermal voltage (V_T) remain constant.

—

3. Mathematical Derivation

Step 1: Apply the Virtual Ground Concept
Since the non-inverting terminal is connected to ground, its voltage is $V_+ = 0$ V. Due to the virtual short concept of an ideal op-amp, the inverting terminal is at a virtual ground:

$$V_- = V_+ = 0 \text{ V} \tag{1}$$

Step 2: Determine the Diode Current (I_D)
The voltage drop across the input diode D is:

$$V_D = V_i - V_- = V_i - 0 = V_i \tag{2}$$

According to the Shockley diode equation, the current I_D is:

$$I_D = I_s\left(e^{\frac{V_D}{\eta V_T}} - 1\right) \tag{3}$$

Using the assumption that the diode is strongly forward-biased ($e^{\frac{V_i}{\eta V_T}} \gg 1$), we approximate:

$$I_D \approx I_s e^{\frac{V_i}{\eta V_T}} \tag{4}$$

where η is the ideality factor and $V_T = \frac{kT}{q}$ is the thermal voltage.
Step 3: Determine the Feedback Current (I_f)
By applying Kirchhoff's Current Law (KCL) at the inverting node (V_-):

$$I_D = I_f + I_- \tag{5}$$

Since $I_- = 0$ A for an ideal op-amp, the currents are equal:

$$I_D = I_f \tag{6}$$

The current flowing through the feedback resistor R_f is given by Ohm's Law:

$$I_f = \frac{V_- - V_o}{R_f} = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f} \tag{7}$$

Step 4: Derive the Final Output Voltage
Equating (4) and (7) according to (4):

$$I_s e^{\frac{V_i}{\eta V_T}} = -\frac{V_o}{R_f} \tag{8}$$

Rearranging for V_o:

$$V_o = -R_f I_s e^{\frac{V_i}{\eta V_T}} \tag{9}$$

—

4. Conclusion

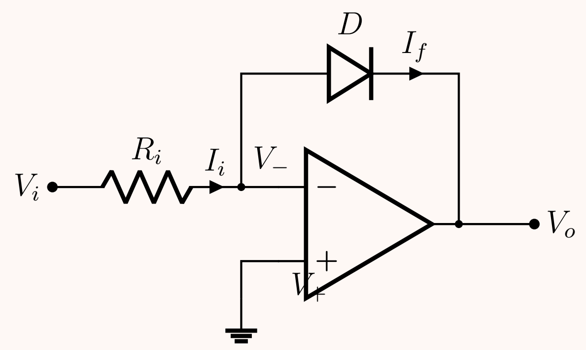
The derived equation demonstrates that the output voltage V_o is proportional to the exponential of the input voltage V_i. The scaling factor is $R_f I_s$, and the negative sign indicates the phase inversion typical of an inverting op-amp configuration. Thus, the proper connection of a diode at the input and a resistor in the feedback loop successfully performs the exponential (antilogarithmic) function.

Q3. Draw the circuit diagram of an inverting logarithmic amplifier and derive the expression of its output voltage. **(26 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Circuit Diagram of an Inverting Logarithmic Amplifier

An inverting logarithmic amplifier is constructed using an operational amplifier with a linear resistor (R_i) at the input and a *p-n* junction diode (D) in the negative feedback path. This configuration outputs a voltage proportional to the natural logarithm of the input voltage.



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2. Assumptions

To derive the mathematical expression for the output voltage, we state the following ideal assumptions:

(a) **Ideal Operational Amplifier:**

- Infinite open-loop voltage gain ($A_v \to \infty$) which results in a virtual short between the inverting and non-inverting input terminals.
- Infinite input impedance ($R_{in} \to \infty$), meaning no current enters the op-amp terminals ($I_+ = I_- = 0$).

(b) **Ideal Diode Behavior:**

- The $V - I$ characteristics of the diode obey the Shockley diode equation.
- The input voltage V_i is strictly positive ($V_i > 0$), ensuring that the diode is sufficiently forward-biased such that the exponential term is much greater than 1.

—

3. Mathematical Derivation

Step 1: Apply Virtual Ground Condition
The non-inverting terminal is directly connected to ground, making its potential zero:

$$V_+ = 0 \text{ V} \tag{1}$$

Due to the virtual short concept of the ideal op-amp, the inverting terminal is held at a virtual ground:

$$V_- = V_+ = 0 \text{ V} \tag{2}$$

Step 2: Determine Input Current (I_i)
The current flowing from the input voltage source V_i through the input resistor R_i is given by Ohm's Law:

$$I_i = \frac{V_i - V_-}{R_i} = \frac{V_i - 0}{R_i} = \frac{V_i}{R_i} \tag{3}$$

Step 3: Analyze the Diode Feedback Path
The feedback current I_f flows through the diode D. Applying Kirchhoff's Current Law (KCL) at the inverting node (V_-):

$$I_i = I_f + I_- \tag{4}$$

Since $I_- = 0$ A, all input current flows through the feedback diode:

$$I_i = I_f \tag{5}$$

The voltage drop across the diode V_D (from anode to cathode) is:

$$V_D = V_- - V_o = 0 - V_o = -V_o \tag{6}$$

The current I_f through the diode is described by the Shockley equation:

$$I_f = I_s\left(e^{\frac{V_D}{\eta V_T}} - 1\right) \tag{7}$$

Given that the diode is heavily forward-biased, $e^{\frac{V_D}{\eta V_T}} \gg 1$. The equation simplifies to:

$$I_f \approx I_s e^{\frac{V_D}{\eta V_T}} = I_s e^{\frac{-V_o}{\eta V_T}} \tag{8}$$

where I_s is the reverse saturation current, η is the ideality factor, and V_T is the thermal voltage ($V_T \approx 26$ mV at room temperature).

Step 4: Formulate Output Voltage Expression

Equating (3) and (8) based on (5):

$$\frac{V_i}{R_i} = I_s e^{\frac{-V_o}{\eta V_T}} \tag{9}$$

Divide both sides by I_s:

$$\frac{V_i}{R_i I_s} = e^{\frac{-V_o}{\eta V_T}} \tag{10}$$

Take the natural logarithm (ln) of both sides to isolate the exponent:

$$\ln\left(\frac{V_i}{R_i I_s}\right) = \frac{-V_o}{\eta V_T} \tag{11}$$

Solve for V_o:

$$V_o = -\eta V_T \ln\left(\frac{V_i}{R_i I_s}\right) \tag{12}$$

—

4. Conclusion

The derived equation confirms that the output voltage V_o is proportional to the natural logarithm of the input voltage V_i. The term $-\eta V_T$ dictates the scaling factor (dependent on temperature and material), and the negative sign indicates the inherent phase inversion of the configuration.

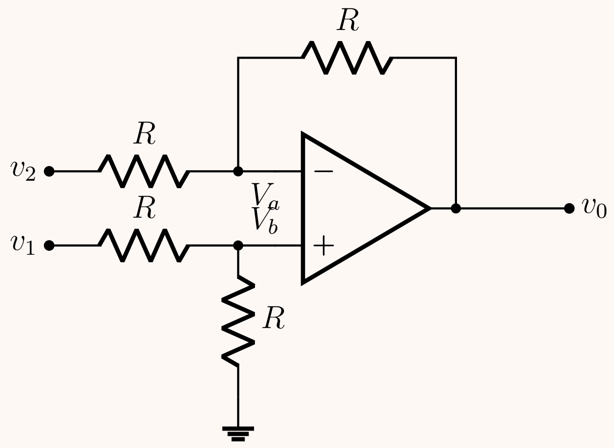
Difference Amplifier

Q1. Design a subtractor circuit using op-amps so that the output voltage is $v_0 = v_1 - v_2$. **(15 marks)** [EEE 263 | L-2/T-1 | 2022–23]

Answer

1. Circuit Design and Principle

To design a subtractor circuit that yields the output voltage $v_0 = v_1 - v_2$, a **Difference Amplifier** configuration is used. By applying the signal v_1 to the non-inverting terminal and v_2 to the inverting terminal, and by setting all four principal resistors to the same value (R), the circuit performs exact subtraction of the two input voltages.



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2. Assumptions

The derivation relies on the standard ideal operational amplifier characteristics:

(a) **Infinite Open-Loop Gain ($A_v \to \infty$):** This enforces a virtual short circuit between the input terminals, meaning the potential difference between them is zero ($V_a = V_b$).

(b) **Infinite Input Impedance ($R_{in} \to \infty$):** No current flows into the inverting or non-inverting terminals of the op-amp ($I_- = I_+ = 0$).

—

3. Mathematical Derivation

We can analyze the circuit using Nodal Analysis (Kirchhoff's Current Law) at nodes V_a and V_b.
Step 1: Determine the voltage at the non-inverting node (V_b)
The non-inverting terminal is connected to a voltage divider formed by the input resistor R and the grounding resistor R. Since no current enters the op-amp ($I_+ = 0$), the voltage V_b is:

$$V_b = v_1 \left(\frac{R}{R+R} \right)$$
$$V_b = v_1 \left(\frac{R}{2R} \right) = \frac{v_1}{2}$$

Step 2: Apply Virtual Short Concept
Due to the infinite gain of the ideal op-amp, a virtual short exists between the input terminals. Therefore, the voltage at the inverting node (V_a) is equal to V_b:

$$V_a = V_b = \frac{v_1}{2} \tag{1}$$

Step 3: Apply KCL at the inverting node (V_a)
Applying Kirchhoff's Current Law at node V_a, the sum of currents leaving the node must equal zero. Since $I_- = 0$:

$$\frac{V_a - v_2}{R} + \frac{V_a - v_0}{R} = 0 \tag{2}$$

Multiplying the entire equation by R to eliminate the denominators:

$$(V_a - v_2) + (V_a - v_0) = 0 \tag{3}$$

Rearranging the terms to isolate the variables:

$$2V_a - v_2 - v_0 = 0 \implies v_0 = 2V_a - v_2 \tag{4}$$

Step 4: Substitute and Solve for v_0
Substitute the value of V_a from Equation (1) into Equation (4):

$$v_0 = 2\left(\frac{v_1}{2}\right) - v_2$$
$$v_0 = v_1 - v_2$$

—

4. Conclusion

By configuring a difference amplifier with equal resistor values (R) for the input and feedback networks, the scaling factors reduce to unity. The circuit successfully subtracts the inverting input signal (v_2) from the non-inverting input signal (v_1), yielding the desired output equation:

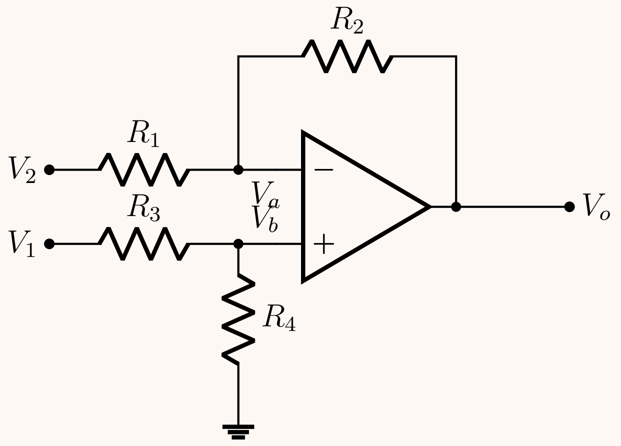
$$v_0 = v_1 - v_2$$

Q2. Design a circuit using a single Op-Amp that performs the following operation: $V_o = 3(V_1 - V_2)$ where V_1 and V_2 are input signals (AC or DC). Specify values of all resistors used in the design. **(14 marks)** **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

1. Circuit Design Strategy

To perform the mathematical operation $V_o = 3(V_1 - V_2)$, we need an operational amplifier configured as a **Difference Amplifier**. The given equation can be expanded as $V_o = 3V_1 - 3V_2$. This indicates that the signal V_1 must be amplified by a positive gain of $+3$ (applied to the non-inverting terminal), and the signal V_2 must be amplified by a negative gain of -3 (applied to the inverting terminal).
By carefully selecting a four-resistor network around a single ideal op-amp, we can scale the difference between the two input voltages.



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2. Assumptions

(a) **Ideal Operational Amplifier:** Infinite open-loop voltage gain ($A_v \to \infty$) establishing a virtual short between the input terminals ($V_a = V_b$).

(b) **Infinite Input Impedance:** Zero current flows into the op-amp's inverting and non-inverting input terminals ($I_- = I_+ = 0$).

—

3. Mathematical Derivation

Using Superposition or standard Nodal Analysis, we can find the general expression for V_o.
Step 1: Determine the voltage at the non-inverting node (V_b)
The non-inverting terminal is connected to a voltage divider formed by R_3 and R_4 across the input V_1. Since $I_+ = 0$:

$$V_b = V_1 \left(\frac{R_4}{R_3 + R_4} \right) \tag{1}$$

Step 2: Apply the Virtual Short Concept
Due to the ideal op-amp characteristics, the potential at the inverting node equals the potential at the non-inverting node:

$$V_a = V_b = V_1 \left(\frac{R_4}{R_3 + R_4} \right) \tag{2}$$

Step 3: Apply Kirchhoff's Current Law (KCL) at node V_a
The sum of currents leaving node V_a is zero:

$$\frac{V_a - V_2}{R_1} + \frac{V_a - V_o}{R_2} = 0 \tag{3}$$

Rearranging to solve for V_o:

$$\frac{V_o}{R_2} = V_a \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_2}{R_1}$$
$$V_o = V_a \left(1 + \frac{R_2}{R_1} \right) - V_2 \left(\frac{R_2}{R_1} \right)$$

Step 4: Substitute V_a to get the complete transfer function
Substitute Equation (2) into Equation (**??**):

$$V_o = V_1 \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{R_1 + R_2}{R_1} \right) - V_2 \left(\frac{R_2}{R_1} \right) \tag{4}$$

To make the circuit perform balanced subtraction, we enforce the resistor proportionality condition:

$$\frac{R_2}{R_1} = \frac{R_4}{R_3} \tag{5}$$

If this ratio holds true, the expression simplifies drastically to:

$$V_o = \frac{R_2}{R_1}(V_1 - V_2) \tag{6}$$

—

4. Component Selection

The required operation is $V_o = 3(V_1 - V_2)$. Comparing this target equation to our derived Equation (6), we see that the difference gain must be:

$$\frac{R_2}{R_1} = 3 \implies R_2 = 3R_1 \tag{7}$$

And similarly, to maintain the balanced condition:

$$\frac{R_4}{R_3} = 3 \implies R_4 = 3R_3 \tag{8}$$

To balance the input bias currents and simplify the bill of materials, we typically choose $R_3 = R_1$ and $R_4 = R_2$. Selecting standard practical resistor values in the kilo-ohm range to prevent excessive power dissipation and to negate op-amp input capacitance effects:

- Let $R_1 = 10\ \text{k}\Omega$
- Then $R_2 = 3 \times 10\ \text{k}\Omega = 30\ \text{k}\Omega$
- Let $R_3 = 10\ \text{k}\Omega$
- Then $R_4 = 3 \times 10\ \text{k}\Omega = 30\ \text{k}\Omega$

Final Specified Values:

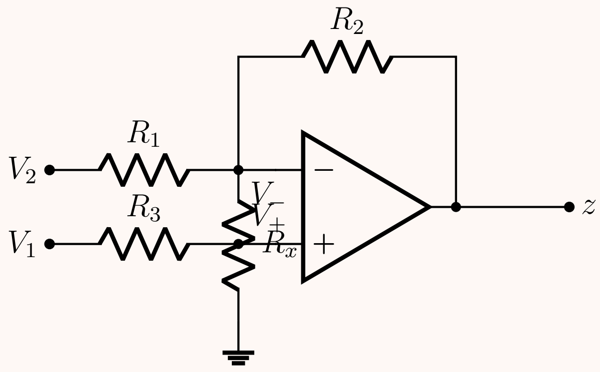
$$R_1 = 10\ \text{k}\Omega, \quad R_2 = 30\ \text{k}\Omega, \quad R_3 = 10\ \text{k}\Omega, \quad R_4 = 30\ \text{k}\Omega$$

Q3. Implement $z = 5V_1 - 3V_2$ using only one op-amp, where V_1 and V_2 are input signals. **(15 marks)** [EEE 263 | L-2/T-1 | 2013–14]

Answer

1. Circuit Design Strategy

The objective is to realize the equation $z = 5V_1 - 3V_2$ using a single operational amplifier. This requires a difference amplifier configuration. The signal V_1 must experience a non-inverting gain of $+5$, and the signal V_2 must experience an inverting gain of -3. A standard 4-resistor difference amplifier with an inverting gain of -3 naturally provides a non-inverting gain of $|-3|+1 = 4$. Since we require a non-inverting gain of $+5$ (which is greater than 4), we must introduce an additional degree of freedom: a resistor (R_x) connecting the inverting terminal to ground. This resistor increases the non-inverting gain without affecting the inverting gain.



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2. Mathematical Derivation

Assume an ideal op-amp with infinite open-loop gain and infinite input impedance.
Step 1: Find the voltage at the input terminals
Because the op-amp draws zero input current ($I_+ = 0$), there is no voltage drop across R_3. Thus, the voltage at the non-inverting terminal is exactly the input voltage:

$$V_+ = V_1 \tag{1}$$

Due to the virtual short concept, the voltage at the inverting terminal is equal to the non-inverting terminal:

$$V_- = V_+ = V_1 \tag{2}$$

Step 2: Apply Kirchhoff's Current Law (KCL) at the inverting node (V_-)
The sum of currents leaving the node V_- must equal zero:

$$\frac{V_- - V_2}{R_1} + \frac{V_- - 0}{R_x} + \frac{V_- - z}{R_2} = 0 \tag{3}$$

Step 3: Solve for the output voltage z
Substitute $V_- = V_1$ into the KCL equation:

$$\frac{V_1 - V_2}{R_1} + \frac{V_1}{R_x} + \frac{V_1 - z}{R_2} = 0 \tag{4}$$

Rearranging to isolate the terms containing z:

$$\frac{z}{R_2} = V_1\left(\frac{1}{R_1} + \frac{1}{R_x} + \frac{1}{R_2}\right) - V_2\left(\frac{1}{R_1}\right) \tag{5}$$

Multiply the entire equation by R_2:

$$z = V_1\left(\frac{R_2}{R_1} + \frac{R_2}{R_x} + 1\right) - V_2\left(\frac{R_2}{R_1}\right) \tag{6}$$

—

3. Component Selection

We must match the derived Equation (6) to the required operation $z = 5V_1 - 3V_2$.
Inverting Gain Matching (V_2 Coefficient):

$$\frac{R_2}{R_1} = 3 \implies R_2 = 3R_1 \tag{7}$$

Non-Inverting Gain Matching (V_1 Coefficient):

$$\begin{aligned} \frac{R_2}{R_1} + \frac{R_2}{R_x} + 1 &= 5 \\ 3 + \frac{R_2}{R_x} + 1 &= 5 \\ 4 + \frac{R_2}{R_x} &= 5 \\ \frac{R_2}{R_x} &= 1 \implies R_x = R_2 \end{aligned}$$

Choosing Practical Values: To minimize power dissipation and bias current effects, standard values in the kilo-ohm range are selected:

- Let $R_1 = 10\text{ k}\Omega$.
- Then $R_2 = 3 \times 10\text{ k}\Omega = 30\text{ k}\Omega$.
- And $R_x = R_2 = 30\text{ k}\Omega$.

Input Offset Compensation: To compensate for input bias currents, R_3 is chosen such that the DC resistance seen by both input terminals is equal. The resistance seen by the inverting terminal is $R_{eq(-)} = R_1 \parallel R_2 \parallel R_x$.

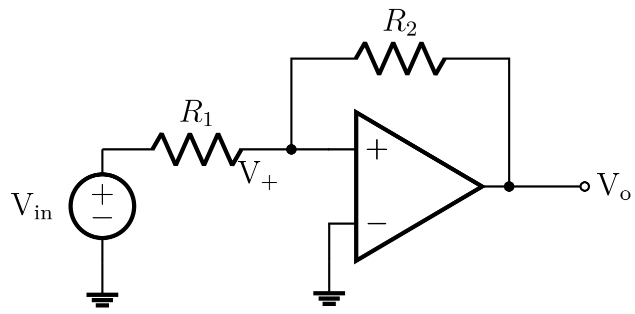
$$R_3 = 10\text{ k}\Omega \parallel 30\text{ k}\Omega \parallel 30\text{ k}\Omega$$
$$R_3 = 10\text{ k}\Omega \parallel 15\text{ k}\Omega = 6\text{ k}\Omega$$

Final Designed Values:

$$R_1 = 10\text{ k}\Omega, \quad R_2 = 30\text{ k}\Omega, \quad R_x = 30\text{ k}\Omega, \quad R_3 = 6\text{ k}\Omega$$

Comparator and Schmitt Trigger

Q1. Write down the characteristics of an ideal op-amp. Draw the voltage transfer characteristics of the circuit shown in Figure below. Voltage transfer characteristics shows the variation in output voltage in response to variations in the input voltage.



(15 marks) [EEE 263 | L-2/T-1 | 2023–24]

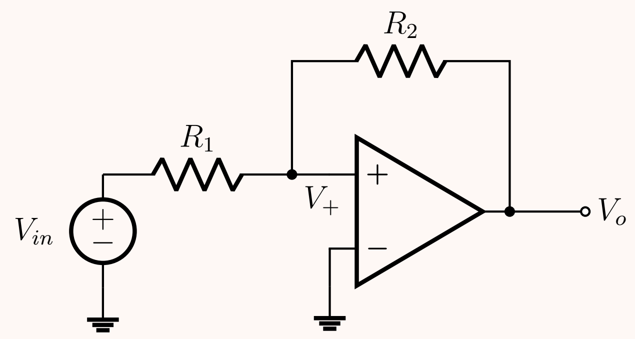
Answer

1. Characteristics of an Ideal Operational Amplifier

An ideal operational amplifier (op-amp) possesses the following theoretical characteristics:

- **Infinite Open-Loop Voltage Gain ($A_v \to \infty$):** Any small differential voltage at the inputs results in an infinite output voltage (in practice, it saturates to the power supply rails).
- **Infinite Input Impedance ($R_{in} \to \infty$):** The input terminals draw zero current ($I_+ = I_- = 0$).
- **Zero Output Impedance ($R_{out} = 0\,\Omega$):** The op-amp can supply any amount of current to the load without any drop in output voltage.
- **Infinite Bandwidth ($BW \to \infty$):** The op-amp amplifies all frequency components of a signal equally with zero phase shift.
- **Infinite Common-Mode Rejection Ratio (CMRR $\to \infty$):** The op-amp perfectly rejects noise or signals common to both input terminals.
- **Zero Offset Voltage ($V_{os} = 0$ V):** If the voltage at both input terminals is equal ($V_+ = V_-$), the output voltage is exactly zero.

2. Circuit Analysis: Non-Inverting Schmitt Trigger



Principle of Operation:
The given circuit employs **positive feedback** because the output is fed back to the non-inverting (+) terminal via resistor R_2. Due to this positive feedback, the op-amp operates in saturation, acting as a bistable multivibrator (Schmitt Trigger). The output voltage V_o can only take one of two possible states:

$$V_o = +V_{sat} \quad \text{or} \quad V_o = -V_{sat} \tag{1}$$

Mathematical Derivation of Threshold Voltages:
Applying Kirchhoff's Current Law (KCL) at the non-inverting node (V_+). Assuming an ideal op-amp, the current into the non-inverting terminal is zero.

$$\frac{V_+ - V_{in}}{R_1} + \frac{V_+ - V_o}{R_2} = 0 \tag{2}$$

Rearranging to solve for V_+:

$$V_+\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{V_{in}}{R_1} + \frac{V_o}{R_2} \tag{3}$$

$$V_+ = \frac{V_{in}R_2 + V_oR_1}{R_1 + R_2} \tag{4}$$

The op-amp will switch states when the differential input voltage crosses zero ($V_+ = V_- = 0$). Setting $V_+ = 0$:

$$V_{in}R_2 + V_oR_1 = 0 \implies V_{in} = -V_o\frac{R_1}{R_2} \tag{5}$$

Based on Equation (5), we define the two threshold voltages:

(a) **Upper Threshold Voltage (V_{UT}):** Assume the output is currently low ($V_o = -V_{sat}$). The input V_{in} must increase past V_{UT} to force $V_+ > 0$ and switch the output to $+V_{sat}$.

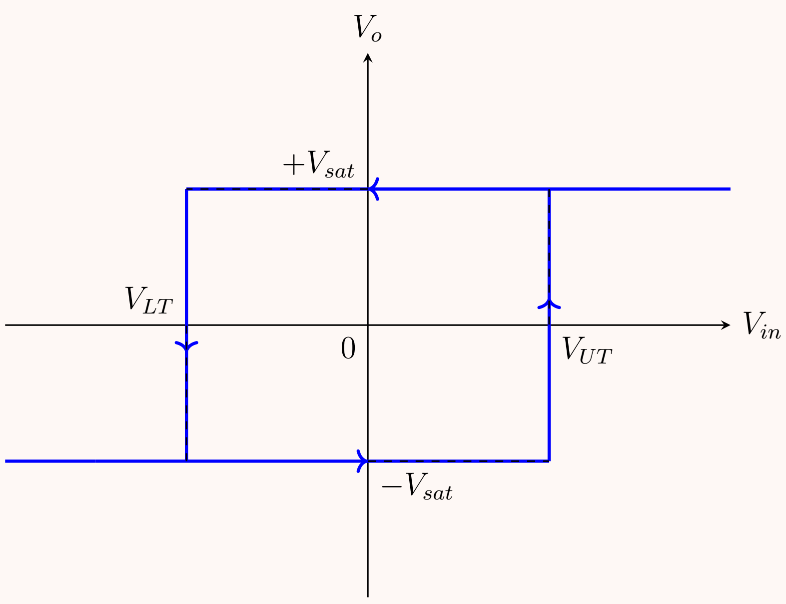
$$V_{UT} = -(-V_{sat})\frac{R_1}{R_2} = +V_{sat}\frac{R_1}{R_2} \tag{6}$$

(b) **Lower Threshold Voltage (V_{LT}):** Assume the output is currently high ($V_o = +V_{sat}$). The input V_{in} must decrease past V_{LT} to force $V_+ < 0$ and switch the output to $-V_{sat}$.

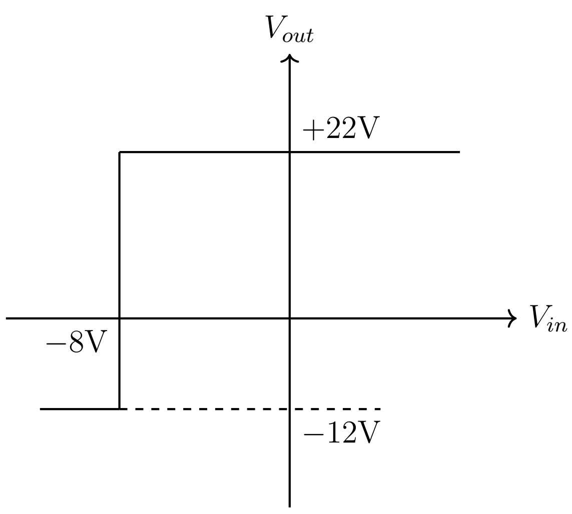
$$V_{LT} = -(+V_{sat})\frac{R_1}{R_2} = -V_{sat}\frac{R_1}{R_2} \tag{7}$$

3. Voltage Transfer Characteristics

The transfer characteristic of this non-inverting Schmitt trigger forms a **hysteresis loop**. When V_{in} increases from a highly negative value, V_o remains at $-V_{sat}$ until V_{in} crosses V_{UT}. Conversely, when V_{in} decreases from a highly positive value, V_o remains at $+V_{sat}$ until V_{in} drops below V_{LT}.



Q2. Determine the op-amp circuit that generates the transfer characteristics shown in the figure.



(11 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Analysis of the Transfer Characteristics

From the given voltage transfer characteristics, we can observe the following parameters of the circuit:

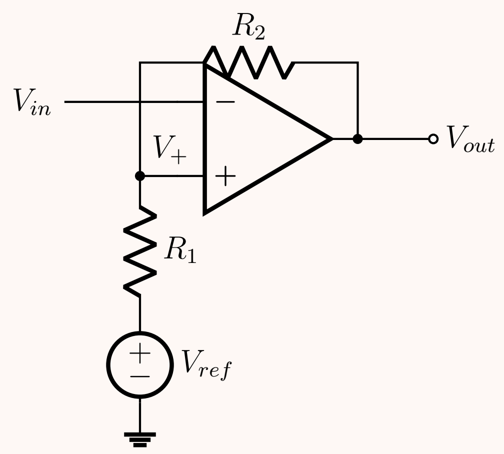
- **Positive Saturation Voltage (V_{sat}^{+}):** $+22$ V
- **Negative Saturation Voltage (V_{sat}^{-}):** -12 V
- **Upper Threshold Voltage (V_{UT}):** 0 V (where the output transitions from high to low as V_{in} increases)
- **Lower Threshold Voltage (V_{LT}):** -8 V (where the output transitions from low to high as V_{in} decreases)

Since the output goes from high ($+22$ V) to low (-12 V) as the input voltage V_{in} increases through the upper threshold, the circuit must be an **Inverting Schmitt Trigger** with a non-zero reference voltage (V_{ref}).

—

2. Circuit Diagram

To implement this characteristic, the input voltage V_{in} is applied to the inverting terminal, while the positive feedback network along with a reference voltage V_{ref} is connected to the non-inverting terminal.



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3. Mathematical Derivation and Component Calculation

Assuming an ideal operational amplifier, no current enters the non-inverting terminal ($I_+ = 0$). By applying the voltage divider rule at the non-inverting node (V_+), the voltage is given by:

$$V_+ = \frac{V_{out}R_1 + V_{ref}R_2}{R_1 + R_2} \tag{1}$$

The circuit switches its output state when the input voltage equals the non-inverting node voltage ($V_{in} = V_+$).

Step 1: Upper Threshold Voltage (V_{UT})

The upper threshold occurs when the output is in positive saturation ($V_{out} = +22$ V):

$$V_{UT} = \frac{22R_1 + V_{ref}R_2}{R_1 + R_2} = 0 \text{ V} \tag{2}$$

From this relation, we find:

$$22R_1 + V_{ref}R_2 = 0 \implies V_{ref}R_2 = -22R_1 \tag{3}$$

Step 2: Lower Threshold Voltage (V_{LT})

The lower threshold occurs when the output is in negative saturation ($V_{out} = -12$ V):

$$V_{LT} = \frac{-12R_1 + V_{ref}R_2}{R_1 + R_2} = -8 \text{ V} \tag{4}$$

Substituting Equation (**??**) into this equation:

$$\frac{-12R_1 - 22R_1}{R_1 + R_2} = -8 \tag{5}$$

$$\frac{-34R_1}{R_1 + R_2} = -8 \implies 34R_1 = 8R_1 + 8R_2 \tag{6}$$

$$26R_1 = 8R_2 \implies \frac{R_2}{R_1} = \frac{26}{8} = 3.25 \tag{7}$$

Step 3: Finding Component Values

Let us choose a standard value for R_1:

- $R_1 = 10$ kΩ
- $R_2 = 3.25 \times 10$ k$\Omega = 32.5$ kΩ

Now, determine the required reference voltage V_{ref} using Equation (**??**):

$$V_{ref} = -22\left(\frac{R_1}{R_2}\right) = -22\left(\frac{1}{3.25}\right) = -\frac{88}{13} \text{ V} \approx -6.77 \text{ V} \tag{8}$$

—

4. Conclusion

The circuit that produces the specified asymmetrical hysteresis transfer characteristics is an **Inverting Schmitt Trigger** designed with the following parameter values:

$$R_1 = 10 \text{ k}\Omega, \quad R_2 = 32.5 \text{ k}\Omega, \quad V_{ref} = -6.77 \text{ V}$$

Q3. Design a Schmitt trigger that triggers from $+16$ V to -16 V at an input voltage of 12 V and triggers from -16 V to $+16$ V at an input voltage of 8 V. Specify all the values of resistors and reference DC voltage. **(15 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Answer

1. Circuit Analysis and Identification

We are tasked with designing a Schmitt trigger with the following operating parameters:

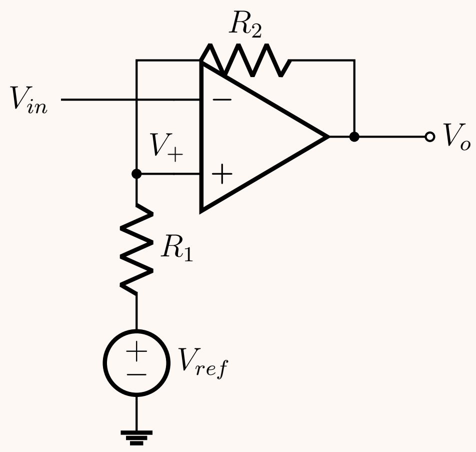
- **Positive Saturation Voltage (V_{sat}^+):** $+16$ V
- **Negative Saturation Voltage (V_{sat}^-):** -16 V
- **Upper Threshold Voltage (V_{UT}):** 12 V (Output switches from $+16$ V to -16 V as V_{in} rises)
- **Lower Threshold Voltage (V_{LT}):** 8 V (Output switches from -16 V to $+16$ V as V_{in} falls)

Since the output switches from positive saturation ($+16$ V) to negative saturation (-16 V) when the input crosses the upper threshold ($V_{UT} = 12$ V), the input signal must be applied to the inverting terminal. Therefore, this circuit is an **Inverting Schmitt Trigger with a DC Reference Voltage (V_{ref})**.

—

2. Circuit Diagram

The inverting Schmitt trigger configuration routes V_{in} to the inverting terminal, while a voltage divider network consisting of R_1, R_2, and a continuous DC reference voltage V_{ref} establishes the switching thresholds at the non-inverting node (V_+).



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3. Mathematical Derivation and Parameter Calculations

Assuming an ideal op-amp, the input impedance is infinite ($I_+ = 0$). Applying Kirchhoff's Current Law (KCL) or Superposition at the non-inverting terminal node (V_+) yields:

$$V_+ = V_o\left(\frac{R_1}{R_1 + R_2}\right) + V_{ref}\left(\frac{R_2}{R_1 + R_2}\right) \tag{1}$$

The state transition occurs exactly when the input voltage equals the potential at the non-inverting node ($V_{in} = V_+$).

Step 1: Formulate Equation for Upper Threshold Voltage (V_{UT})

The upper threshold voltage occurs when the output is initially sitting high ($V_o = +V_{sat} = +16$ V):

$$V_{UT} = 16\left(\frac{R_1}{R_1 + R_2}\right) + V_{ref}\left(\frac{R_2}{R_1 + R_2}\right) = 12 \text{ V} \tag{2}$$

Multiplying through by $(R_1 + R_2)$:

$$\begin{aligned}16R_1 + V_{ref}R_2 &= 12(R_1 + R_2)\\ 16R_1 + V_{ref}R_2 &= 12R_1 + 12R_2\\ 4R_1 + V_{ref}R_2 &= 12R_2\end{aligned}$$

Step 2: Formulate Equation for Lower Threshold Voltage (V_{LT})

The lower threshold voltage occurs when the output is initially sitting low ($V_o = -V_{sat} = -16$ V):

$$V_{LT} = -16\left(\frac{R_1}{R_1 + R_2}\right) + V_{ref}\left(\frac{R_2}{R_1 + R_2}\right) = 8 \text{ V} \tag{3}$$

Multiplying through by $(R_1 + R_2)$:

$$-16R_1 + V_{ref}R_2 = 8(R_1 + R_2)$$
$$-16R_1 + V_{ref}R_2 = 8R_1 + 8R_2$$
$$-24R_1 + V_{ref}R_2 = 8R_2$$

Step 3: Solve the System of Linear Equations for Resistor Ratios
Subtracting Equation (**??**) from Equation (**??**) to eliminate the $V_{ref}R_2$ term:

$$(4R_1 + V_{ref}R_2) - (-24R_1 + V_{ref}R_2) = 12R_2 - 8R_2$$
$$28R_1 = 4R_2$$
$$\frac{R_2}{R_1} = \frac{28}{4} = 7 \implies R_2 = 7R_1$$

Step 4: Solve for the Reference DC Voltage (V_{ref})
Substitute $R_2 = 7R_1$ back into Equation (**??**):

$$4R_1 + V_{ref}(7R_1) = 12(7R_1)$$
$$4R_1 + 7V_{ref}R_1 = 84R_1$$

Dividing the entire equation by R_1:

$$4 + 7V_{ref} = 84$$
$$7V_{ref} = 80$$
$$V_{ref} = \frac{80}{7}\text{ V} \approx 11.43\text{ V}$$

—

4. Component Value Selection

To select standard, practical values in the kilo-ohm range that limit operational current and eliminate excessive loading constraints:

- Let $R_1 = 10\text{ k}\Omega$
- Then $R_2 = 7 \times 10\text{ k}\Omega = 70\text{ k}\Omega$

Final Designed Specifications:

$$R_1 = 10\text{ k}\Omega, \quad R_2 = 70\text{ k}\Omega, \quad V_{ref} = 11.43\text{ V}$$

Q4. Draw a zero-crossing detector circuit using Op-Amp and briefly describe its operation. **(15 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

Answer

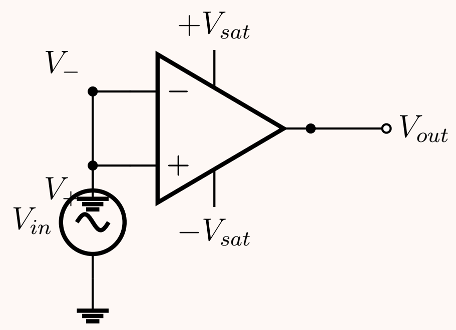
1. Principle of Operation

A **Zero-Crossing Detector (ZCD)** is an essential application of an operational amplifier acting as a voltage comparator. Its primary function is to detect when an alternating input signal crosses the zero-voltage reference level. The op-amp is operated in an **open-loop configuration** (without feedback), meaning it utilizes its intrinsic, theoretically infinite open-loop voltage gain (A_{OL}). The reference voltage (V_{ref}) is deliberately set to 0 V (ground).

—

2. Circuit Diagram

Below is the circuit diagram for a **Non-Inverting Zero-Crossing Detector**.



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3. Mathematical Analysis and Operation

Ideal Assumptions:

- Open-loop voltage gain $A_{OL} \to \infty$.
- Infinite input impedance ($R_{in} \to \infty$), meaning zero current flows into the op-amp terminals.

The general output equation for an op-amp comparator is given by:

$$V_{out} = A_{OL}(V_+ - V_-) \tag{1}$$

In this circuit configuration, the input signal is applied to the non-inverting terminal ($V_+ = V_{in}$), and the inverting terminal is grounded ($V_- = 0$ V). Thus, the differential input voltage V_d is simply V_{in}.
The circuit operates in two distinct states depending on the polarity of V_{in}:

(a) **Positive Half-Cycle ($V_{in} > 0$ V):**

$$V_d = V_{in} - 0 > 0$$

Because A_{OL} is practically infinite, even a minuscule positive voltage at the input drives the op-amp to its maximum positive output capability. The output saturates at the positive supply rail:

$$V_{out} = +V_{sat} \tag{2}$$

(b) **Negative Half-Cycle ($V_{in} < 0$ V):**

$$V_d = V_{in} - 0 < 0$$

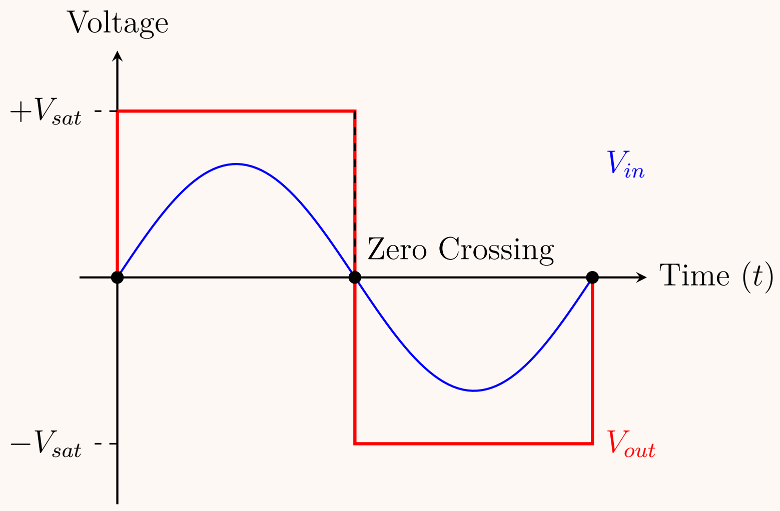
Similarly, a negative differential voltage gets multiplied by the massive open-loop gain, driving the op-amp into negative saturation:

$$V_{out} = -V_{sat} \tag{3}$$

Therefore, every time the input AC signal crosses 0 V, the output sharply transitions between $+V_{sat}$ and $-V_{sat}$, effectively converting a sinusoidal input into a square wave output.
—

4. Input and Output Waveforms

The graphical representation below illustrates the timing relationship between the sinusoidal input voltage and the resulting square wave output.



—

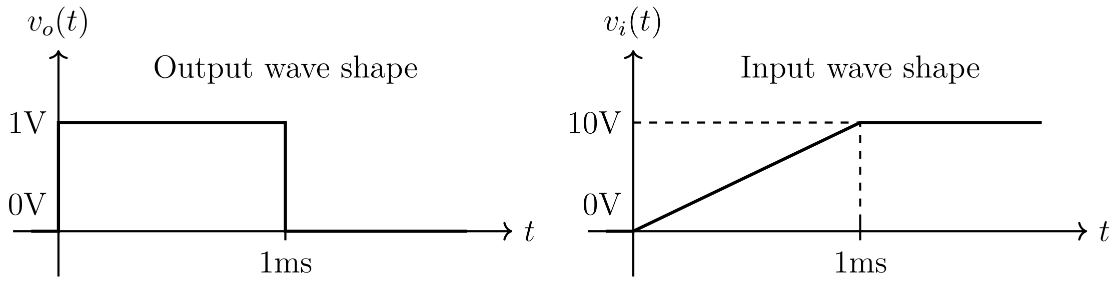
5. Applications

Zero-crossing detectors are widely used in analog and digital electronics, including:

- **Phase Meters:** To measure the phase difference between two signals by comparing their zero-crossing timestamps.
- **Frequency Counters:** To count the number of cycles per second by detecting zero-crossings.
- **AC Power Control:** Used in thyristor/TRIAC firing circuits (like dimmers and heaters) to synchronize switching with the AC mains zero-crossings, thereby minimizing electromagnetic interference (EMI).

Waveform Generation

Q1. Design a circuit using op-amp(s) for given input $v_i(t)$ and output $v_o(t)$ waveforms shown in the figure.



(16 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Mathematical Analysis of the Waveforms

From the given input waveform $v_i(t)$, the signal is a voltage ramp that increases linearly from 0V to 10V over a duration of 1ms, and then remains constant at 10V. We can express $v_i(t)$ mathematically as:

$$v_i(t) = \begin{cases} \frac{10}{10^{-3}}t = 10^4 t \text{ [V]}, & 0 \le t \le 1 \text{ ms} \\ 10 \text{ [V]}, & t > 1 \text{ ms} \end{cases} \tag{1}$$

By taking the derivative of the input voltage with respect to time, we find the rate of change:

$$\frac{dv_i(t)}{dt} = \begin{cases} 10^4 \text{ [V/s]}, & 0 \le t < 1 \text{ ms} \\ 0 \text{ [V/s]}, & t > 1 \text{ ms} \end{cases} \tag{2}$$

The given output waveform $v_o(t)$ is constant at 1V during the ramp period and drops to 0V afterward:

$$v_o(t) = \begin{cases} 1 \text{ [V]}, & 0 \le t < 1 \text{ ms} \\ 0 \text{ [V]}, & t > 1 \text{ ms} \end{cases} \tag{3}$$

By comparing the output waveform $v_o(t)$ to the derivative of the input waveform $\frac{dv_i(t)}{dt}$, we observe a direct mathematical proportionality:

$$\begin{aligned} v_o(t) &= K\frac{dv_i(t)}{dt} \\ 1 &= K \cdot 10^4 \implies K = 10^{-4} \text{ s} \end{aligned}$$

This establishes that the required circuit must perform the mathematical operation of **differentiation**.

2. Circuit Topology Selection

A standard ideal operational amplifier differentiator produces an output voltage that is proportional to the *negative* derivative of the input:

$$v_{o,\text{diff}}(t) = -RC\frac{dv_i(t)}{dt} \tag{4}$$

Since our required scaling constant K is positive ($+10^{-4}$ s), a single inverting differentiator would yield an output of -1V, which does not match the required $+1$V output. To correct the polarity, we must cascade two stages:

(a) **Stage 1: Inverting Differentiator** to perform the time-derivative operation.

(b) **Stage 2: Inverting Amplifier** to invert the signal and provide the correct positive output voltage.

3. Component Selection and Design

Let the first stage (differentiator) have an input capacitor C and a feedback resistor R. Assuming an ideal op-amp, the intermediate node voltage $v_1(t)$ is:

$$v_1(t) = -RC\frac{dv_i(t)}{dt} \tag{5}$$

Let the second stage (inverter) have an input resistor R_1 and a feedback resistor R_2. The final output voltage $v_o(t)$ is:

$$v_o(t) = -\frac{R_2}{R_1}v_1(t) = \left(\frac{R_2}{R_1}RC\right)\frac{dv_i(t)}{dt} \tag{6}$$

Equating this to our required transfer function ($K = 10^{-4}$ s):

$$\frac{R_2}{R_1}RC = 10^{-4} \text{ s} \tag{7}$$

Step 1: Design the Inverting Amplifier (Stage 2)
To maintain simplicity, we configure the second stage as a unity-gain inverter by selecting equal resistor values. Let:

$$R_1 = R_2 = 10 \text{ k}\Omega \tag{8}$$

This gives a gain of $-\frac{R_2}{R_1} = -1$.
Step 2: Design the Differentiator (Stage 1)
With the second stage gain set to -1, the time constant of the first stage must exactly match the required value of K:

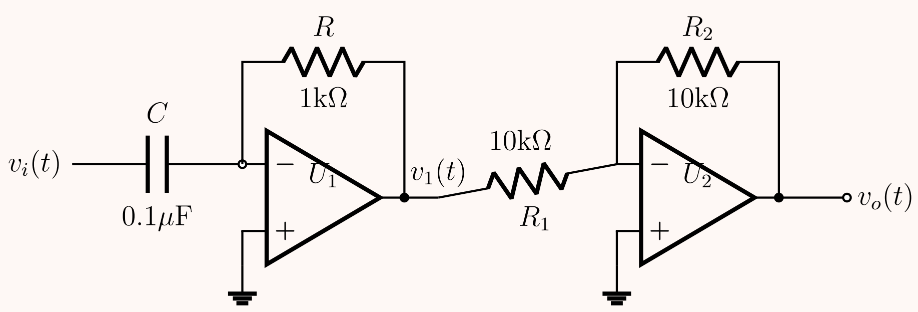
$$RC = 10^{-4} \text{ s} \tag{9}$$

Select a standard practical value for the input capacitor, $C = 0.1\mu$F (10^{-7} F). We can now solve for the required feedback resistor R:

$$\begin{aligned} R \cdot (10^{-7} \text{ F}) &= 10^{-4} \text{ s} \\ R &= \frac{10^{-4}}{10^{-7}} = 1000\,\Omega = 1 \text{ k}\Omega \end{aligned}$$

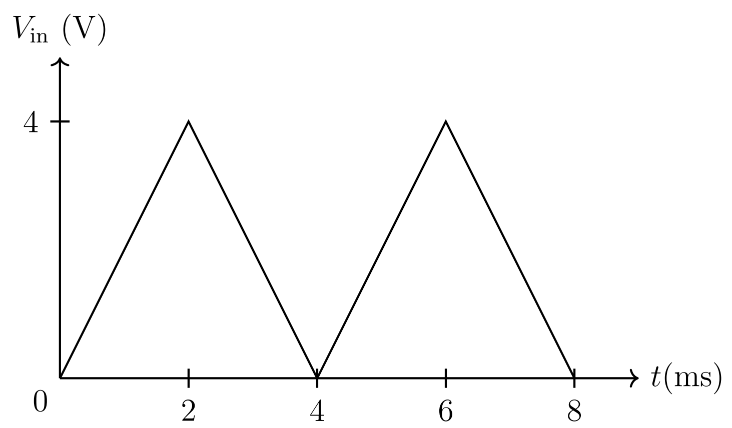
4. Final Circuit Diagram

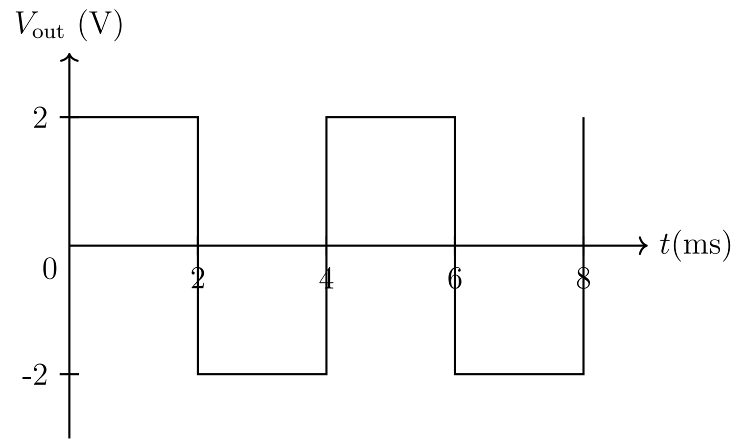
The complete designed circuit is illustrated below. Ideal op-amps are assumed, with their non-inverting terminals grounded to satisfy the virtual ground condition at the inverting terminals.



Important Practical Note: While the designed circuit successfully realizes the requested mathematical operation under ideal conditions, pure differentiators are susceptible to high-frequency noise and instability in real-world applications. A practical realization would typically include a small resistor R_s in series with the input capacitor C, and a small bypass capacitor C_f in parallel with the feedback resistor R to limit the high-frequency gain and bandwidth.

Q2. Input signal V_{in} and output signal V_{out} of a circuit are given in the figure. Design the circuit using Op-Amps. Show necessary calculations.





(12 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

1. Mathematical Analysis of the Waveforms

From the provided graphs, we first determine the mathematical expressions for the input waveform $V_{\text{in}}(t)$ and the output waveform $V_{\text{out}}(t)$.

Input Signal (V_{in}): The input is a triangular wave with a peak voltage of 4V and a time period of $T = 4\text{ms}$. We can calculate the slope (rate of change) of the input signal for each half-cycle.

- For the interval $0 < t < 2\text{ms}$ (positive slope):

$$\frac{dV_{\text{in}}(t)}{dt} = \frac{4\text{V} - 0\text{V}}{2\text{ms} - 0\text{ms}} = \frac{4}{2 \times 10^{-3}} = 2000\,\text{V/s} \tag{1}$$

- For the interval $2\text{ms} < t < 4\text{ms}$ (negative slope):

$$\frac{dV_{\text{in}}(t)}{dt} = \frac{0\text{V} - 4\text{V}}{4\text{ms} - 2\text{ms}} = \frac{-4}{2 \times 10^{-3}} = -2000\,\text{V/s} \tag{2}$$

Output Signal (V_{out}): The output is a square wave that alternates between $+2\text{V}$ and -2V.

- For $0 < t < 2\text{ms}$, $V_{\text{out}}(t) = 2\text{V}$.
- For $2\text{ms} < t < 4\text{ms}$, $V_{\text{out}}(t) = -2\text{V}$.

Relationship between Input and Output: By comparing the output voltage with the derivative of the input voltage, we establish a proportional relationship:

$$V_{\text{out}}(t) = K\frac{dV_{\text{in}}(t)}{dt}$$
$$2 = K(2000) \implies K = \frac{2}{2000} = 10^{-3}\,\text{s}$$

Since the output is directly proportional to the time derivative of the input, the required circuit is a **differentiator**. Furthermore, because the proportionality constant K is positive ($+10^{-3}\,\text{s}$), the circuit must act as a *non-inverting* differentiator.

2. Circuit Topology Selection

A standard ideal operational amplifier differentiator produces an output that is inverted:

$$V_{\text{diff}}(t) = -RC\frac{dV_{\text{in}}(t)}{dt} \tag{3}$$

To achieve a positive constant K, we must cascade two operational amplifier stages:

(a) **Stage 1: Inverting Differentiator** to perform the derivation, providing $V_1(t) = -R_1C_1\frac{dV_{\text{in}}}{dt}$.

(b) **Stage 2: Inverting Amplifier** to invert the signal back to the required polarity, with a gain of -1.

3. Component Design and Calculations

Stage 1: Inverting Differentiator
Let the input capacitor be C_1 and the feedback resistor be R_1. The time constant must equal $K = 10^{-3}$ s.

$$R_1C_1 = 10^{-3}\,\text{s} \tag{4}$$

We select a standard, practical value for the capacitor:

$$C_1 = 0.1\,\mu\text{F} = 10^{-7}\,\text{F} \tag{5}$$

Now, we calculate the required value for the feedback resistor R_1:

$$R_1 = \frac{10^{-3}\,\text{s}}{10^{-7}\,\text{F}} = 10,000\,\Omega = 10\,\text{k}\Omega \tag{6}$$

This stage will output $V_1(t) = -2\text{V}$ when the slope is positive, and $+2\text{V}$ when the slope is negative.
Stage 2: Unity-Gain Inverting Amplifier
To correct the polarity, we need a gain of $A_v = -1$. Using an input resistor R_2 and a feedback resistor R_3:

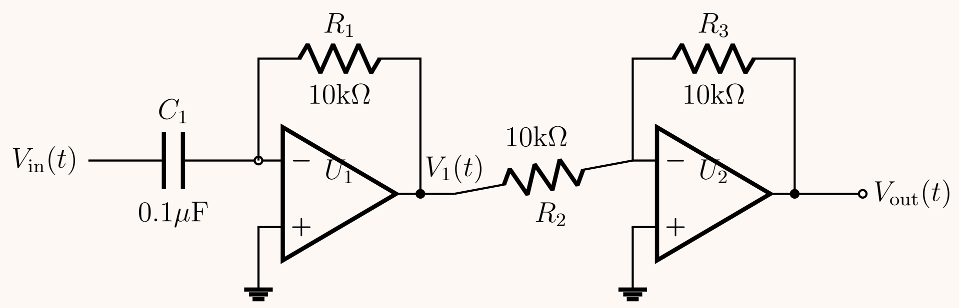
$$V_{\text{out}}(t) = -\frac{R_3}{R_2}V_1(t) \tag{7}$$

To achieve a gain of -1, we set $R_2 = R_3$. We can choose standard $10\,\text{k}\Omega$ resistors for both:

$$R_2 = 10\,\text{k}\Omega, \quad R_3 = 10\,\text{k}\Omega \tag{8}$$

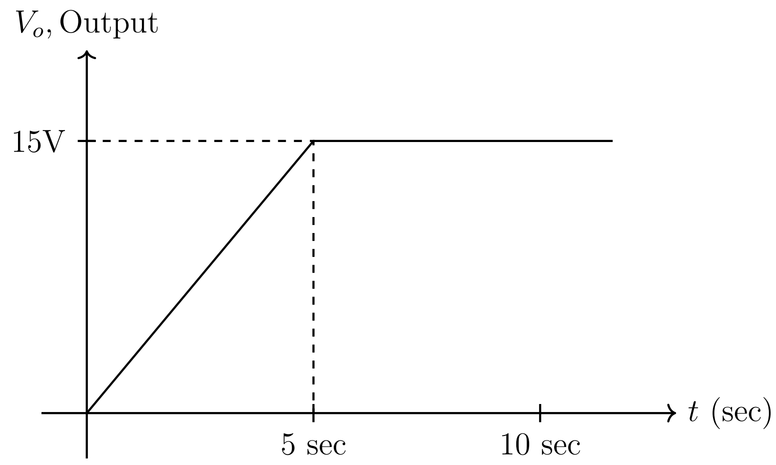
4. Final Circuit Diagram

Below is the designed circuit schematic. Ideal Op-Amps (U_1 and U_2) are assumed. The non-inverting terminals are grounded to enforce the virtual ground condition at the inverting terminals.



Practical Implementation Note: While the ideal differentiator described above satisfies the theoretical mathematical requirements of the prompt, practical differentiators are prone to high-frequency noise amplification and instability. In a real-world laboratory setting, an additional small series resistor R_s (before C_1) and a small feedback capacitor C_f (parallel to R_1) would be added to limit high-frequency gain and ensure circuit stability.

Q3. Design a circuit using Op-Amps that gives a specified output waveform from a 1 V DC input. Use +15 V and −15 V for DC biasing. Specify all the values of resistors and capacitors used.



(15 marks) [EEE 263 | L-2/T-1 | 2014–15]

Answer

1. Mathematical Analysis of the Waveforms

The problem requires designing an operational amplifier circuit that produces the specified output voltage waveform $V_o(t)$ from a constant step input voltage $V_{\text{in}}(t) = 1$ V for $t \geq 0$.
From the given output graph, the waveform can be divided into two distinct time intervals:

(a) **Interval $0 \leq t \leq 5$ s:** The output increases linearly from 0 V to 15 V. This represents a linear ramp function:

$$V_o(t) = \frac{15\text{ V} - 0\text{ V}}{5\text{ s} - 0\text{ s}} \cdot t = 3t \text{ [V]} \tag{1}$$

(b) **Interval $t > 5$ s:** The output stays constant at 15 V, which corresponds to the positive saturation limit (V_{sat}^{+}) supplied by the +15 V DC biasing rail:

$$V_o(t) = 15\text{ V} \tag{2}$$

Expressing the linear operation during the active region ($0 \leq t \leq 5$ s) as an integral of the constant 1 V input:

$$V_o(t) = 3\int_0^t (1) \cdot d\tau = 3\int_0^t V_{\text{in}}(\tau)\, d\tau \tag{3}$$

This mathematically indicates that the required circuit configuration must be an **integrator** with a positive scaling factor of $3\,\text{s}^{-1}$.

2. Circuit Topology Selection

The classical ideal op-amp inverting integrator produces a negative ramp from a positive constant input:

$$V_{\text{int}}(t) = -\frac{1}{RC}\int_0^t V_{\text{in}}(\tau)\, d\tau \tag{4}$$

Since a positive ramp ($+3t$) is required from a positive input (+1 V), a single inverting integrator stage would yield a negative voltage slope, eventually saturating at −15 V. To achieve the correct positive polarity, a two-stage operational amplifier circuit is required:

(a) **Stage 1: Inverting Integrator** to generate a negative linear ramp: $V_1(t) = -3t$.

(b) **Stage 2: Inverting Amplifier** with a gain of −1 to flip the polarity to $V_o(t) = +3t$.

Once the output voltage reaches +15 V at $t = 5$ s, the second op-amp will naturally saturate at its positive supply rail (+15 V), satisfying the constant output condition for $t > 5$ s.

3. Component Value Calculations

Stage 1: Inverting Integrator Design
The scaling constant for the integrator stage is given by:

$$\frac{1}{RC} = 3 \implies RC = \frac{1}{3}\text{ s} \approx 0.333\text{ s} \tag{5}$$

To minimize input bias current errors while using a standard practical component catalog, we select a reasonably large capacitance value:

$$C = 1\,\mu\text{F} = 10^{-6}\text{ F} \tag{6}$$

We can now calculate the required input resistor R:

$$R = \frac{1}{3 \cdot C} = \frac{1}{3 \times 10^{-6}\text{ F}} = \frac{10^6}{3}\,\Omega = 333.33\text{ k}\Omega \tag{7}$$

Stage 2: Unity-Gain Inverter Design
The second stage must exhibit a voltage gain of exactly $A_v = -1$. Let R_1 be the input resistor and R_2 be the feedback resistor for this stage:

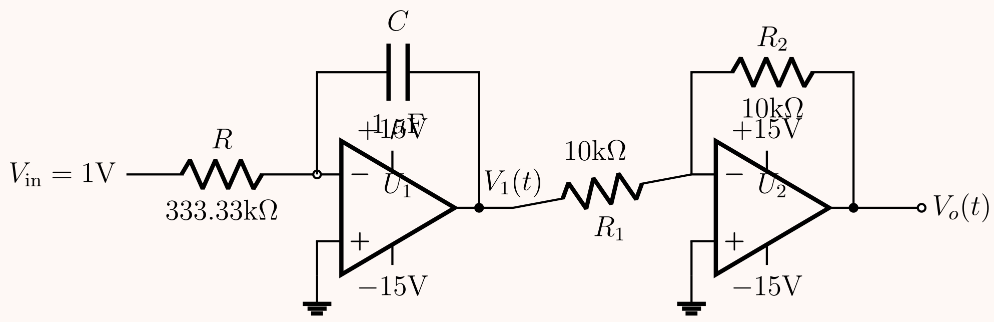
$$A_v = -\frac{R_2}{R_1} = -1 \implies R_1 = R_2 \tag{8}$$

We choose a standard, moderate value to balance low power consumption and robust drive capability:

$$R_1 = R_2 = 10 \text{ k}\Omega \tag{9}$$

4. Complete Circuit Schematic

The schematic below illustrates the final design. The op-amps are biased with ±15 V power supplies as specified.



5. Comprehensive Verification of Circuit Operation

To verify the design across both operational domains, we trace the response over time:

- **For** $0 \le t \le 5$ **s:** The intermediate voltage at the output of the first stage $V_1(t)$ is derived using ideal op-amp virtual ground assumptions ($V_- = 0$ V):

$$V_1(t) = -\frac{1}{RC}\int_0^t V_{\text{in}}(\tau)\,d\tau = -\frac{1}{(333.33 \text{ k}\Omega)(1\,\mu\text{F})}\int_0^t (1)\,d\tau = -3t \text{ [V]} \tag{10}$$

  Passing $V_1(t)$ through the second stage inverting amplifier:

$$V_o(t) = -\frac{R_2}{R_1}V_1(t) = -1 \cdot (-3t) = +3t \text{ [V]} \tag{11}$$

  At exactly $t = 5$ s, the output reaches $V_o(5) = 3(5) = 15$ V.

- **For** $t > 5$ **s:** As t exceeds 5 s, the linear equation expects $V_o(t) > 15$ V. However, the operational amplifier U_2 cannot output a voltage higher than its positive saturation limit, which is constrained by the positive supply rail. Therefore, the op-amp enters the saturation region, maintaining a flat, stable upper bound:

$$V_o(t) = V_{\text{sat}}^{+} = +15 \text{ V} \tag{12}$$

The calculation perfectly matches the target waveform parameters specified in the problem statement.

Analog Computing (Differential Equations)

Q1. Design a circuit using op-amps and other passive elements to solve:

$$2\frac{d^2v}{dt^2} - 3\frac{dv}{dt} + 4v + 5 = 0$$

Draw the overall connection diagram and specify element values. Assume op-amps biased with ±15 V; all initial conditions are zero. **(17 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

1. Mathematical Formulation

The given second-order linear ordinary differential equation (ODE) is:

$$2\frac{d^2v}{dt^2} - 3\frac{dv}{dt} + 4v + 5 = 0 \tag{1}$$

To solve this using an analog computer circuit, we isolate the highest-order derivative on one side of the equation:

$$2\frac{d^2v}{dt^2} = 3\frac{dv}{dt} - 4v - 5$$
$$\frac{d^2v}{dt^2} = 1.5\frac{dv}{dt} - 2v - 2.5$$

2. Principle of Analog Computation

An analog computer solves differential equations by integrating the highest-order derivative successively to obtain lower-order derivatives and the final variable. We use operational amplifiers configured as **Inverting Integrators** and **Inverting Summing Amplifiers**.
Let us define the node voltages in the circuit. Assuming an integrator time constant of $RC = 1\,\text{s}$:

- Let the output of the summing amplifier be $V_0 = \frac{d^2v}{dt^2}$.
- Passing V_0 through a first inverting integrator gives:

$$V_1 = -\int V_0\,dt = -\frac{dv}{dt} \tag{2}$$

- Passing V_1 through a second inverting integrator gives:

$$V_2 = -\int V_1\,dt = -\int \left(-\frac{dv}{dt}\right) dt = v(t) \tag{3}$$

Now, substituting V_1 and V_2 back into our isolated ODE (Equation 2):

$$V_0 = 1.5(-V_1) - 2(V_2) - 2.5$$
$$V_0 = -\left(1.5V_1 + 2V_2 + 2.5\right)$$

This equation directly matches the transfer function of a 3-input inverting summing amplifier:

$$V_0 = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_{\text{DC}}\right) \tag{4}$$

3. Component Value Calculations

A. Inverting Integrators (Stages 2 & 3):
We require a time constant $\tau = RC = 1\,\text{s}$ for both integrators. Selecting a standard practical capacitor value:

$$C_1 = C_2 = 1\,\mu\text{F} = 10^{-6}\,\text{F} \tag{5}$$

The corresponding input resistors are:

$$R_4 = R_5 = \frac{1\,\text{s}}{10^{-6}\,\text{F}} = 10^6\,\Omega = 1\,\text{M}\Omega \tag{6}$$

B. Summing Amplifier (Stage 1):
We match the coefficients from Equation 4 to the resistor ratios in Equation 5. Let us select a feedback resistor $R_f = 100\,\text{k}\Omega$.

(a) **Coefficient for V_1 ($-\frac{dv}{dt}$):**

$$\frac{R_f}{R_1} = 1.5 \implies R_1 = \frac{100\,\text{k}\Omega}{1.5} = 66.67\,\text{k}\Omega \tag{7}$$

(b) **Coefficient for V_2 (v):**

$$\frac{R_f}{R_2} = 2 \implies R_2 = \frac{100\,\text{k}\Omega}{2} = 50\,\text{k}\Omega \tag{8}$$

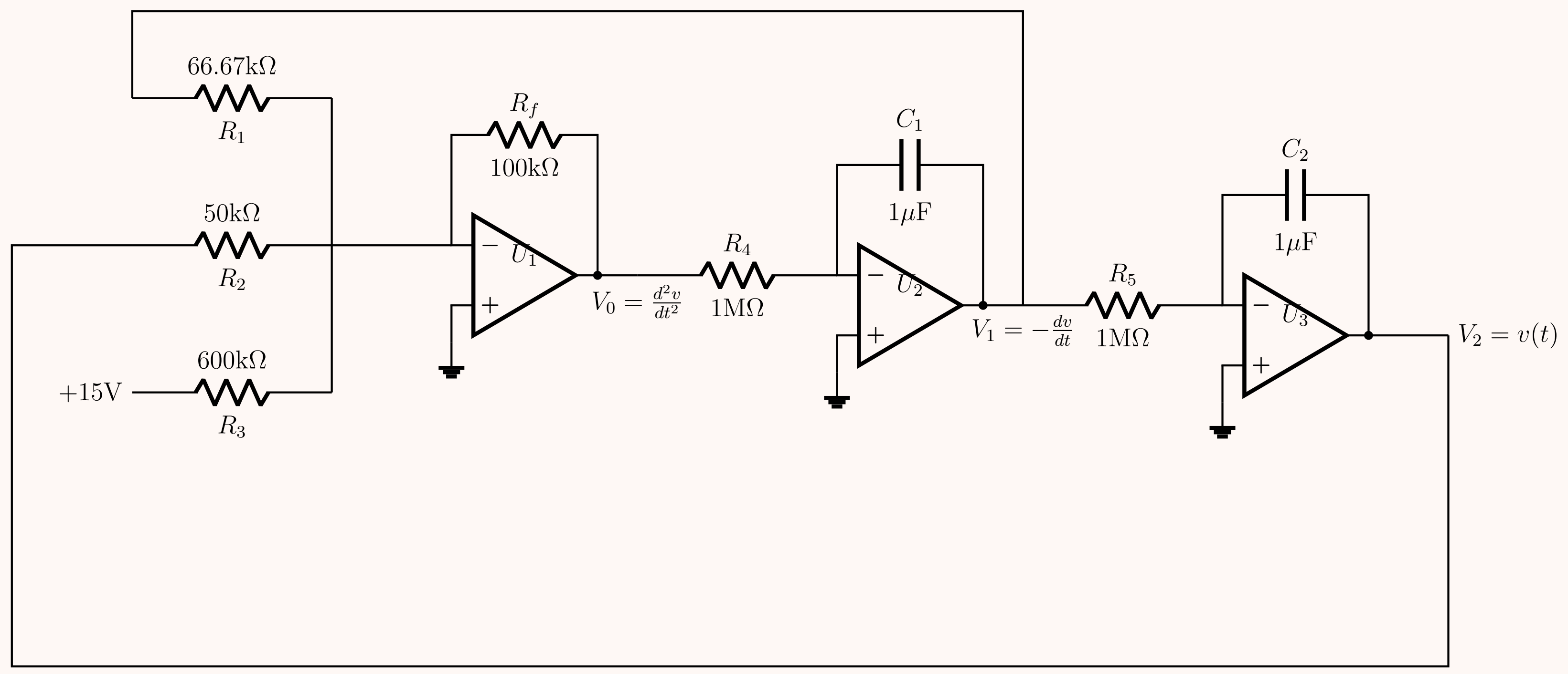
(c) **DC Offset Term (Constant 2.5):** The problem specifies available biasing of $\pm 15\,\text{V}$. We can use the $+15\,\text{V}$ rail as our V_{DC} source.

$$\frac{R_f}{R_3}V_{\text{DC}} = 2.5 \implies \frac{100\,\text{k}\Omega}{R_3}(15\,\text{V}) = 2.5 \implies R_3 = \frac{1500\,\text{k}\Omega}{2.5} = 600\,\text{k}\Omega \tag{9}$$

Assumption: Since all initial conditions are given as zero ($v(0) = 0, \dot{v}(0) = 0$), the capacitors C_1 and C_2 begin uncharged. No additional reset circuitry or pre-charging parallel voltage sources are required.

4. Overall Connection Diagram

Below is the complete schematic designed using ideal operational amplifiers. U_1 acts as the summing amplifier, while U_2 and U_3 act as successive integrators to yield the final solution $v(t)$.



Q2. Using op-amps, design a circuit to solve the following differential equation where x is input and y is output:

$$\frac{d^2y}{dt^2} - 2\frac{dy}{dt} + 2y = 8x$$

(20 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

1. Mathematical Formulation

The given second-order linear ordinary differential equation is:

$$\frac{d^2y}{dt^2} - 2\frac{dy}{dt} + 2y = 8x \tag{1}$$

To implement this on an analog computer using inverting operational amplifiers, we isolate the highest-order derivative, but with a negative sign ($-y''$) to account for the natural inversion of the summing amplifier:

$$-y'' = -2y' + 2y - 8x \tag{2}$$

2. Principle of Analog Computation

We will use an architecture consisting of a summing amplifier, two sequential integrators, and a final inverting amplifier to restore the correct output polarity.
Let the output of the first summing amplifier be $V_0 = -y''$.

- Passing V_0 through a first inverting integrator gives:

$$V_1 = -\int V_0\,dt = -\int (-y'')\,dt = y' \tag{3}$$

- Passing V_1 through a second inverting integrator gives:

$$V_2 = -\int V_1\,dt = -\int (y')\,dt = -y \tag{4}$$

Now, we substitute V_1 and V_2 into our isolated ODE (Equation 2):

$$V_0 = -2(V_1) + 2(-V_2) - 8x$$
$$V_0 = -(2V_1 + 2V_2 + 8x)$$

This expression perfectly matches the transfer function of a standard 3-input inverting summing amplifier:

$$V_0 = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_x}x\right) \tag{5}$$

Finally, because $V_2 = -y$, we require a unity-gain inverting amplifier to obtain the final output $y = -V_2$.

3. Component Value Calculations

A. Inverting Integrators (Stages 2 & 3):
Assuming a normalized time constant $\tau = RC = 1\,\text{s}$ for both integrators. Let us select a standard practical capacitor value:

$$C_1 = C_2 = 1\,\mu\text{F} = 10^{-6}\,\text{F} \tag{6}$$

The corresponding input resistors are calculated as:

$$R_3 = R_4 = \frac{1\,\text{s}}{10^{-6}\,\text{F}} = 10^6\,\Omega = 1\,\text{M}\Omega \tag{7}$$

B. Summing Amplifier (Stage 1):
We match the coefficients from Equation 4 to the resistor ratios in Equation 5. Let us select a feedback resistor $R_f = 100\,\text{k}\Omega$.

(a) **Coefficient for V_1 (y'):**

$$\frac{R_f}{R_1} = 2 \implies R_1 = \frac{100\,\text{k}\Omega}{2} = 50\,\text{k}\Omega \tag{8}$$

(b) **Coefficient for V_2 ($-y$):**

$$\frac{R_f}{R_2} = 2 \implies R_2 = \frac{100\,\text{k}\Omega}{2} = 50\,\text{k}\Omega \tag{9}$$

(c) **Coefficient for input x:**

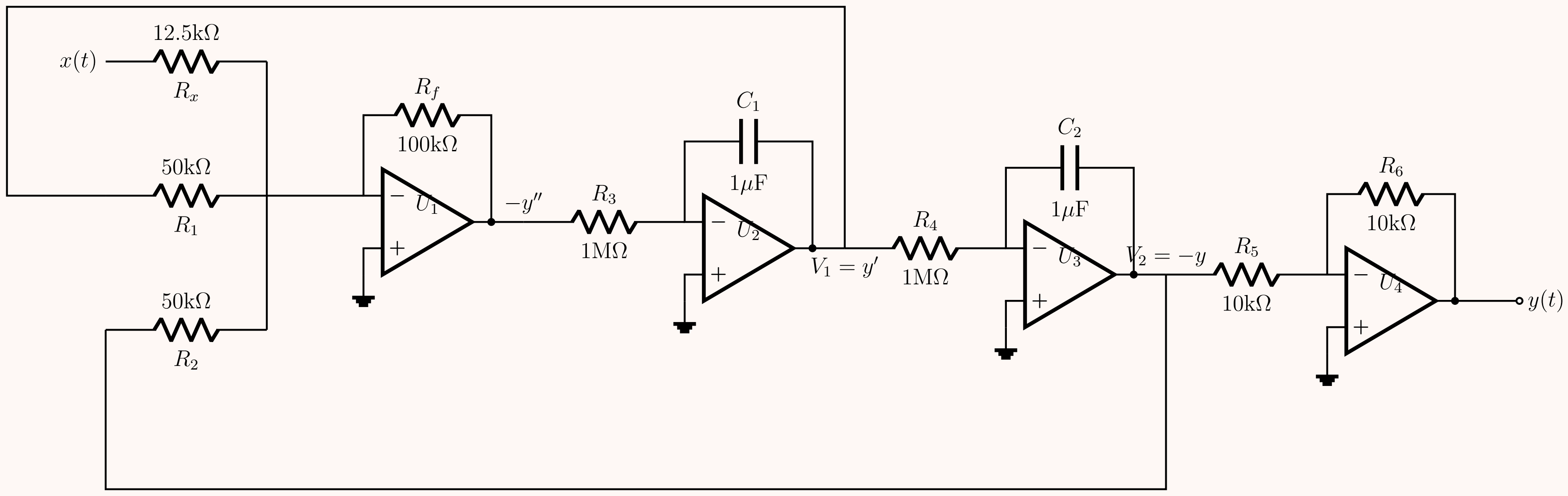
$$\frac{R_f}{R_x} = 8 \implies R_x = \frac{100\,\text{k}\Omega}{8} = 12.5\,\text{k}\Omega \tag{10}$$

C. Unity-Gain Output Inverter (Stage 4):
To achieve a gain of -1 and convert $-y$ to y, we select equal input and feedback resistors:

$$R_5 = R_6 = 10\,\text{k}\Omega \tag{11}$$

4. Overall Connection Diagram

Below is the complete schematic. U_1 acts as the summing amplifier, U_2 and U_3 act as successive integrators, and U_4 acts as the final inverter to yield the solution $y(t)$.



Q3. Design an electronic analog computer using op-amp to solve the following differential equation. Give a brief explanation of the operation of the designed computer.

$$\frac{d^2v}{dt^2} + 4\frac{dv}{dt} + 5v - 5 = 0$$

(26 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Mathematical Formulation

The given second-order linear ordinary differential equation is:

$$\frac{d^2v}{dt^2} + 4\frac{dv}{dt} + 5v - 5 = 0 \tag{1}$$

To solve this using an analog computer, we first isolate the highest-order derivative:

$$\frac{d^2v}{dt^2} = -4\frac{dv}{dt} - 5v + 5 \tag{2}$$

To implement this using standard inverting operational amplifiers, we factor out a negative sign to match the transfer function of an inverting summing amplifier:

$$\frac{d^2v}{dt^2} = -\left(4\frac{dv}{dt} + 5v - 5\right) \tag{3}$$

2. Architecture and Node Definitions

We will use a four-op-amp architecture to model the equation:

- **Summing Amplifier (U_1):** Sums the feedback signals to produce the highest derivative v''.
- **First Integrator (U_2):** Integrates v'' to yield $-v'$.
- **Second Integrator (U_3):** Integrates $-v'$ to yield the final solution $v(t)$.
- **Unity-Gain Inverter (U_4):** Inverts the $-v'$ signal from U_2 back to $+v'$, ensuring it can be correctly weighted and subtracted by U_1.

3. Component Value Calculations

A. Integrators (U_2 and U_3):
Assuming a standard normalized time constant $\tau = RC = 1\,\text{s}$, we select a practical capacitor value:

$$C_1 = C_2 = 1\,\mu\text{F} = 10^{-6}\,\text{F} \tag{4}$$

The corresponding input resistors are:

$$R_4 = R_5 = \frac{1\,\text{s}}{10^{-6}\,\text{F}} = 10^6\,\Omega = 1\,\text{M}\Omega \tag{5}$$

B. Unity-Gain Inverter (U_4):
To achieve a precise gain of -1, we use equal input and feedback resistors:

$$R_6 = R_7 = 100\,\text{k}\Omega \tag{6}$$

C. Summing Amplifier (U_1):
The output equation for a 3-input inverting summer is:

$$V_0 = -\left(\frac{R_f}{R_1}V_{\text{in1}} + \frac{R_f}{R_2}V_{\text{in2}} + \frac{R_f}{R_3}V_{\text{DC}}\right) \tag{7}$$

We match this to our isolated ODE: $V_0 = -\left(4v' + 5v - 5\right)$. Let the common feedback resistor be $R_f = 100\,\text{k}\Omega$.

(a) **Coefficient for v' (Input from U_4):**

$$\frac{R_f}{R_1} = 4 \implies R_1 = \frac{100\,\text{k}\Omega}{4} = 25\,\text{k}\Omega \tag{8}$$

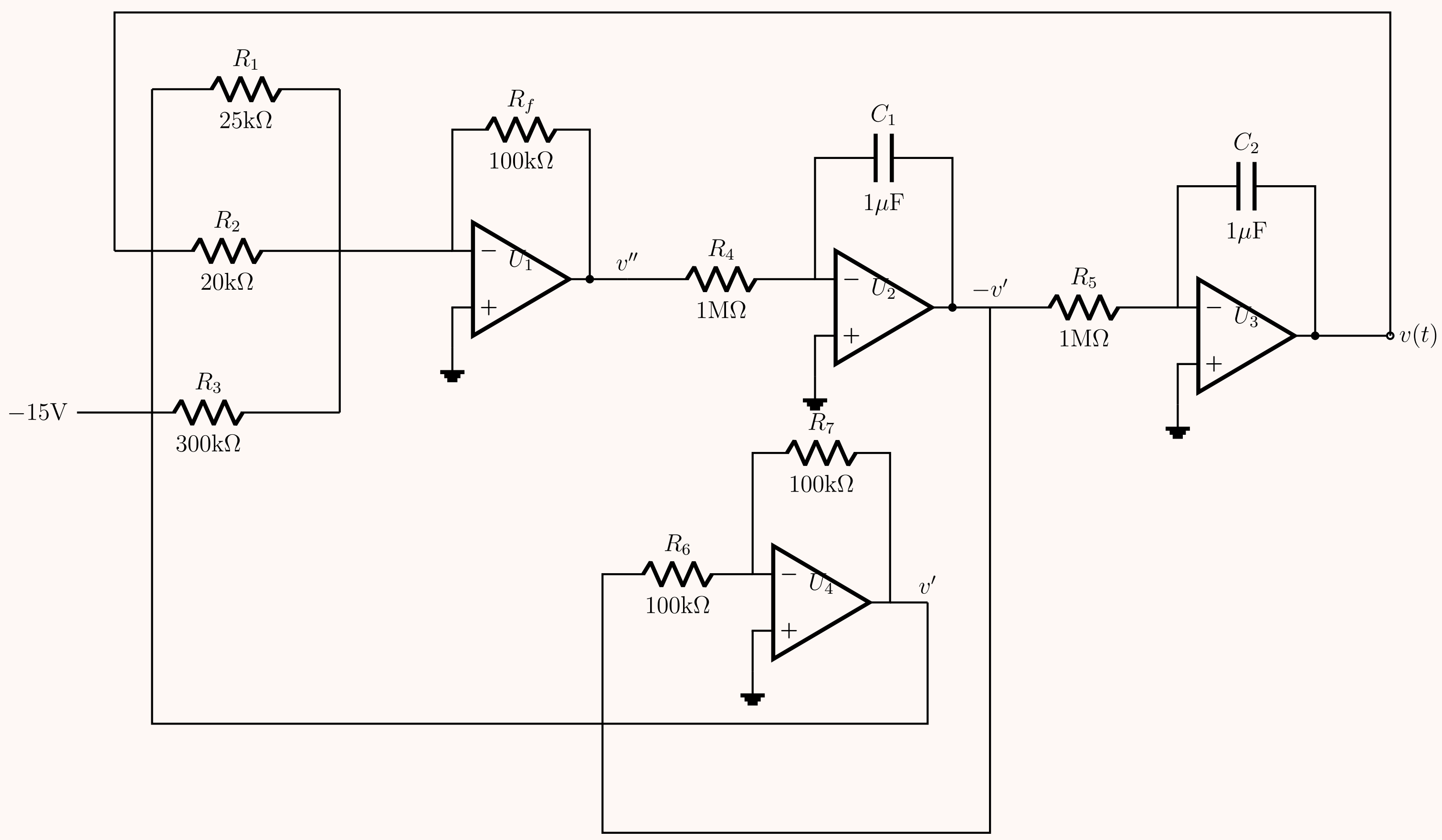
(b) **Coefficient for v (Input from U_3):**

$$\frac{R_f}{R_2} = 5 \implies R_2 = \frac{100\,\text{k}\Omega}{5} = 20\,\text{k}\Omega \tag{9}$$

(c) **Constant Term (-5):**
Assuming a standard $-15\,\text{V}$ DC supply is available for biasing:

$$\frac{R_f}{R_3}(-15\,\text{V}) = -5 \implies \frac{100\,\text{k}\Omega}{R_3}(15) = 5 \implies R_3 = 300\,\text{k}\Omega \tag{10}$$

4. Overall Connection Diagram



5. Brief Explanation of Operation

The analog computer works by forcing electrical voltages to continuously obey the mathematical relationships of the differential equation in real-time.
The summing amplifier (U_1) computes the instantaneous value of the highest derivative (v'') by adding together the properly scaled lower-order feedback terms ($-4v'$ and $-5v$) and the constant forcing function ($+5$). This output voltage is continuously passed through U_2 and U_3, which act as hardware integrators to dynamically generate the lower-order derivative ($-v'$) and the final target variable $v(t)$.
Because U_1 evaluates an inverted sum, we need the specific signal $+v'$ to satisfy the input constraints; therefore, U_4 simply flips the polarity of $-v'$ back to $+v'$. By routing these integrated outputs back to the input of the summing amplifier, the circuit establishes a continuous closed-loop control system. The feedback guarantees that the only possible stable voltage trajectory for node $v(t)$ exactly maps out the time-domain solution of the original differential equation.

Q4. Design an analog computer circuit to solve the following differential equation for $t > 0$:

$$\frac{d^2v}{dt^2} + 3\frac{dv}{dt} + 2v + 4\cos(10t) = 0$$

with initial conditions $v(0) = 2$, $v'(0) = 0$. An oscillator is available to provide $\sin(10t)$ signal. **(20 marks)** **[EEE 263 | L-2/T-1 | 2015–16]**

Answer

1. Mathematical Formulation

The given second-order linear ordinary differential equation is:

$$\frac{d^2v}{dt^2} + 3\frac{dv}{dt} + 2v + 4\cos(10t) = 0 \tag{1}$$

To solve this using an analog computer, we isolate the highest-order derivative. Since summing amplifiers naturally invert the signal, we factor out a negative sign to match the architecture:

$$\frac{d^2v}{dt^2} = -\left(3\frac{dv}{dt} + 2v + 4\cos(10t)\right) \tag{2}$$

2. Architecture and Node Definitions

We require a five-op-amp architecture to fully model the equation and the required source signal:

- **Summing Amplifier (U_1):** Sums the feedback and source signals to compute v''.
- **First Integrator (U_2):** Integrates v'' to yield $-v'$. Initial condition: $-v'(0) = 0\,\text{V}$.
- **Second Integrator (U_3):** Integrates $-v'$ to yield the final solution $v(t)$. Initial condition: $v(0) = 2\,\text{V}$.
- **Unity-Gain Inverter (U_4):** Inverts $-v'$ from U_2 back to $+v'$ for correct subtraction at U_1.
- **Auxiliary Integrator (U_5):** Converts the available $\sin(10t)$ oscillator signal into the required $\cos(10t)$ forcing function.

3. Component Value Calculations

A. Main Integrators (U_2 and U_3):
Assuming a normalized time constant $\tau = RC = 1\,\text{s}$, we select standard capacitors:

$$C_1 = C_2 = 1\,\mu\text{F} \tag{3}$$

The corresponding input resistors are:

$$R_4 = R_5 = \frac{1\,\text{s}}{1\,\mu\text{F}} = 1\,\text{M}\Omega \tag{4}$$

B. Auxiliary Source Integrator (U_5):
We integrate $\sin(10t)$ to get $\cos(10t)$. The transfer function of an inverting integrator is:

$$V_{\text{out}} = -\frac{1}{R_8C_3}\int \sin(10t)\,dt = \frac{1}{10R_8C_3}\cos(10t) \tag{5}$$

To achieve exactly $1 \cdot \cos(10t)$, we need $10R_8C_3 = 1 \implies R_8C_3 = 0.1\,\text{s}$. Let $C_3 = 1\,\mu\text{F}$, then:

$$R_8 = \frac{0.1\,\text{s}}{1\,\mu\text{F}} = 100\,\text{k}\Omega \tag{6}$$

C. Unity-Gain Inverter (U_4):
To achieve a precise gain of -1, we use equal input and feedback resistors:

$$R_6 = R_7 = 100\,\text{k}\Omega \tag{7}$$

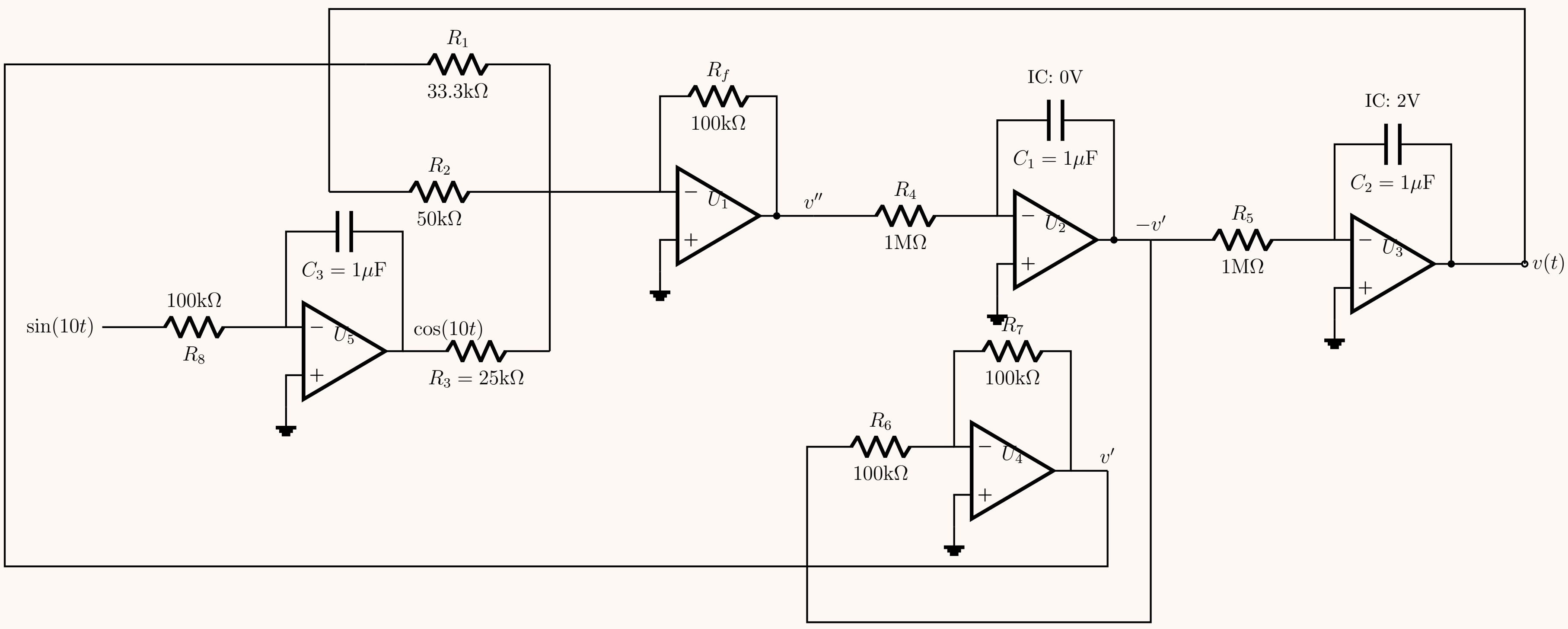
D. Summing Amplifier (U_1):
The target sum is $V_0 = -(3v' + 2v + 4\cos(10t))$. We set the feedback resistor $R_f = 100\,\text{k}\Omega$.

(a) **Coefficient for v':** $\frac{R_f}{R_1} = 3 \implies R_1 = 33.3\,\text{k}\Omega$

(b) **Coefficient for v:** $\frac{R_f}{R_2} = 2 \implies R_2 = 50\,\text{k}\Omega$

(c) **Coefficient for $\cos(10t)$:** $\frac{R_f}{R_3} = 4 \implies R_3 = 25\,\text{k}\Omega$

4. Overall Connection Diagram



5. Brief Explanation of Operation

The analog computer translates the differential equation into an equivalent closed-loop electrical circuit. The highest derivative v'' is formulated by U_1, which sums up the feedback variables v' and v, along with the scaled external forcing function $\cos(10t)$. Since the oscillator only provides a $\sin(10t)$ signal, U_5 is deployed as an active inverting integrator tuned specifically to $RC = 0.1\,\text{s}$. It natively shifts the sine wave and outputs an exact $\cos(10t)$ function, which is then fed into R_3. The primary integrators (U_2 and U_3) process v'' into $-v'$ and finally into $v(t)$. To solve the specific initial value problem, the feedback capacitor on U_3 is charged to an initial condition of $2\,\text{V}$ to represent $v(0) = 2$, while U_2's capacitor remains completely discharged to enforce $v'(0) = 0$.

Q5. Design a circuit using Op-Amps that can solve the following differential equation:

$$3\frac{d^2x(t)}{dt^2} + 4x(t) = 0$$

Point out the node that represents the solution $x(t)$. Specify all the values of resistors and capacitors used. **(20 marks)** [**EEE 263 | L-2/T-1 | 2014–15**]

Answer

1. Mathematical Formulation

The given second-order linear differential equation is:

$$3\frac{d^2x(t)}{dt^2} + 4x(t) = 0 \tag{1}$$

To solve this using an analog computer, we first isolate the highest-order derivative on one side of the equation:

$$\frac{d^2x(t)}{dt^2} = -\frac{4}{3}x(t) \tag{2}$$

2. Architecture and Node Definitions

Because the right side of our isolated equation is simply a scaled version of $x(t)$ with a negative sign, we do not need a summing amplifier here. Instead, we can implement this with a three-op-amp architecture:

- **Inverting Scaler (U_1):** Amplifies the feedback signal $x(t)$ by a gain of $-4/3$ to compute the highest derivative $x''(t)$.
- **First Integrator (U_2):** Integrates $x''(t)$ to yield $-x'(t)$.
- **Second Integrator (U_3):** Integrates $-x'(t)$ to yield the final target solution.
- **Solution Node:** The continuous signal strictly corresponding to the solution $x(t)$ **is located exactly at the output terminal of the final op-amp (U_3)**.

3. Component Value Calculations

A. Integrators (U_2 and U_3):
Assuming a standard normalized time constant $\tau = RC = 1\,\text{s}$, we select practical capacitor values:

$$C_1 = C_2 = 1\,\mu\text{F} \tag{3}$$

The corresponding input resistors required to satisfy the time constant are:

$$R_2 = R_3 = \frac{1\,\text{s}}{1\,\mu\text{F}} = 1\,\text{M}\Omega \tag{4}$$

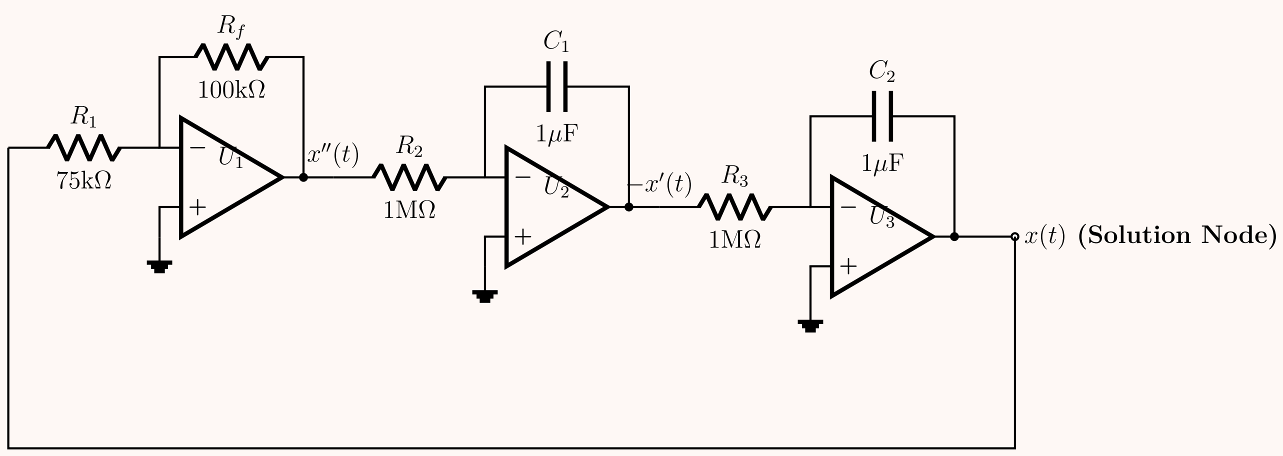
B. Inverting Scaler (U_1):
The transfer function for a standard inverting amplifier is:

$$V_{\text{out}} = -\frac{R_f}{R_1} V_{\text{in}} \tag{5}$$

We need this stage to output $x''(t) = -\frac{4}{3}x(t)$. Therefore, the required gain ratio is $\frac{R_f}{R_1} = \frac{4}{3}$.
If we select a standard feedback resistor $R_f = 100\,\text{k}\Omega$:

$$R_1 = \frac{3}{4}(100\,\text{k}\Omega) = 75\,\text{k}\Omega \tag{6}$$

4. Overall Connection Diagram



5. Brief Explanation of Operation

The analog computer functions by converting the differential equation into a closed, self-regulating electrical loop. By starting at the solution node, we route the output signal $x(t)$ back into the front of the circuit at U_1.
U_1 acts strictly as a proportional scaler, forcing the voltage to become $-\frac{4}{3}x(t)$, which is mathematically identical to our required highest derivative $x''(t)$. This voltage is passed continuously through hardware integrator U_2, dropping the derivative order by one and flipping the sign to output $-x'(t)$. It is then immediately passed through integrator U_3, yielding $x(t)$. Because the final output $x(t)$ perfectly dictates the generation of its own original second derivative, the voltages in the circuit have no choice but to natively map out the simple harmonic oscillation described by the original ODE.

Q6. Design a circuit using op-amp that can provide solution to the following equation:

$$3\frac{d^2v}{dt^2} + K\frac{dv}{dt} - 3v + 8 = 0$$

where K = last digit of your student ID. Briefly discuss the operation of your designed circuit. **(29 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

1. Mathematical Formulation

The given second-order linear differential equation is:

$$3\frac{d^2v}{dt^2} + K\frac{dv}{dt} - 3v + 8 = 0 \tag{1}$$

To solve this using an analog computer, we first isolate the highest-order derivative:

$$\frac{d^2v}{dt^2} = -\frac{K}{3}\frac{dv}{dt} + v - \frac{8}{3} \tag{2}$$

Because summing amplifiers naturally compute an inverted sum ($V_{\text{out}} = -\sum w_i V_i$), we factor out a negative sign to match the hardware's transfer function:

$$\frac{d^2v}{dt^2} = -\left(\frac{K}{3}\frac{dv}{dt} - v + \frac{8}{3}\right) \tag{3}$$

2. Architecture and Node Definitions

We will deploy a five-op-amp closed-loop architecture to represent this system:

- **Summing Amplifier (U_1):** Sums the feedback signals to produce the highest derivative v''.

- **First Integrator (U_2):** Integrates v'' to yield $-v'$.
- **Second Integrator (U_3):** Integrates $-v'$ to yield the target solution $v(t)$.
- **First Inverter (U_4):** Inverts $-v'$ into $+v'$ so it can be subtracted by the summing amplifier.
- **Second Inverter (U_5):** Inverts v into $-v$ so it can be added appropriately at the summing amplifier.

3. Component Value Calculations (in terms of K)

A. Integrators (U_2 and U_3):
Assuming a normalized time constant $\tau = RC = 1\,\text{s}$, we select a practical capacitor value:

$$C_1 = C_2 = 1\,\mu\text{F} \tag{4}$$

The corresponding input resistors are:

$$R_4 = R_5 = \frac{1\,\text{s}}{1\,\mu\text{F}} = 1\,\text{M}\Omega \tag{5}$$

B. Unity-Gain Inverters (U_4 and U_5):
To achieve a precise gain of -1 for inversion, we use equal $100\,\text{k}\Omega$ resistors for both input and feedback paths:

$$R_6 = R_7 = R_8 = R_9 = 100\,\text{k}\Omega \tag{6}$$

C. Summing Amplifier (U_1):
The target sum is $v'' = -\left(\frac{K}{3}(v') + 1(-v) + \frac{8}{3}\right)$. We set the feedback resistor $R_f = 100\,\text{k}\Omega$.

(a) **Coefficient for v' (Input from U_4):**

$$\frac{R_f}{R_1} = \frac{K}{3} \implies R_1 = \frac{3 \times 100\,\text{k}\Omega}{K} = \frac{300}{K}\,\text{k}\Omega \tag{7}$$

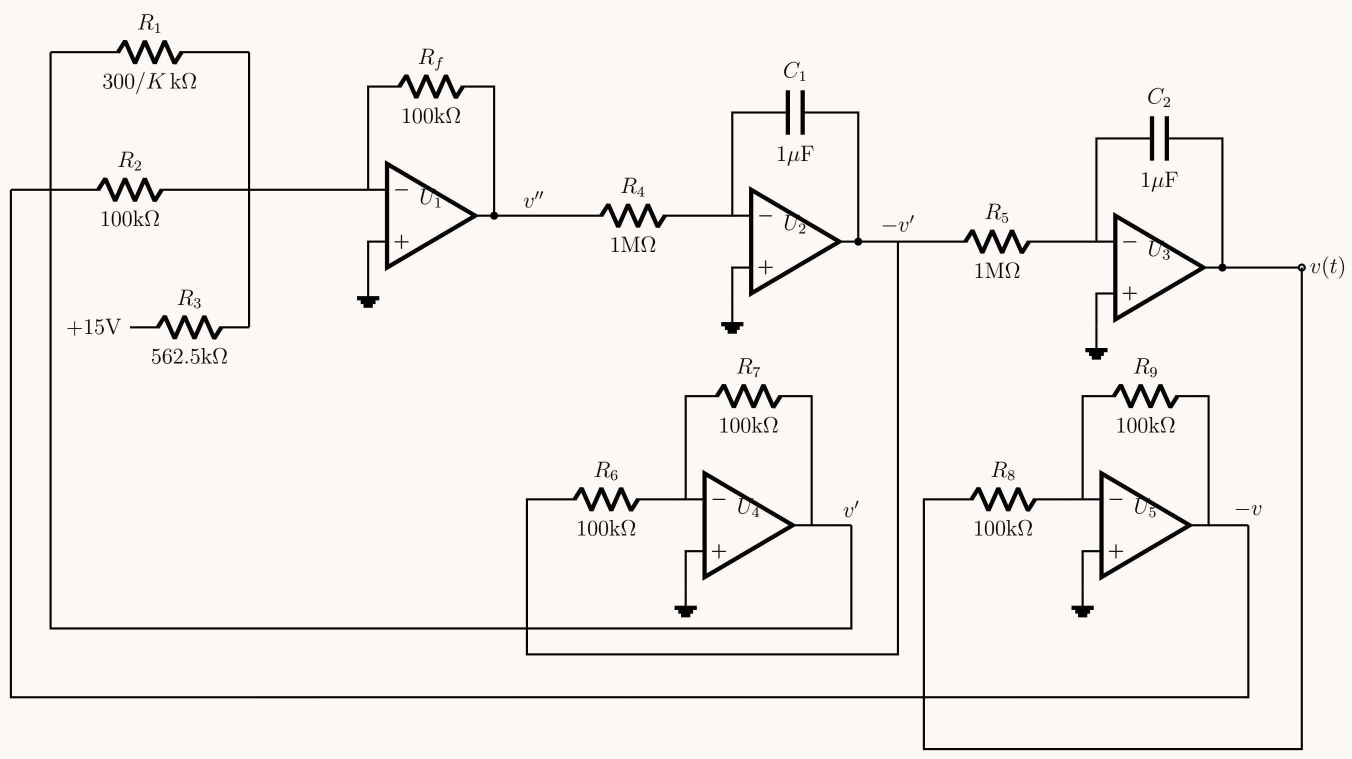
(b) **Coefficient for $-v$ (Input from U_5):**

$$\frac{R_f}{R_2} = 1 \implies R_2 = 100\,\text{k}\Omega \tag{8}$$

(c) **Constant Term ($+8/3$):**
Assuming a standard $+15\,\text{V}$ DC supply is used for the forcing function:

$$\frac{R_f}{R_3}(15\,\text{V}) = \frac{8}{3} \implies R_3 = \frac{100\,\text{k}\Omega \times 15 \times 3}{8} = 562.5\,\text{k}\Omega \tag{9}$$

4. Overall Connection Diagram



5. Brief Explanation of Operation

This analog computer forces the system voltages to dynamically simulate the given mathematical equation. U_1 (the summing amplifier) calculates the instantaneous sum of all terms contributing to the highest-order derivative v''. This variable is then continuously integrated down through two cascading hardware integrators (U_2 and U_3) to establish $-v'$ and the final solution variable $v(t)$.
Because standard op-amp integrators inherently invert the sign of the signal at each stage, the resulting polarities ($-v'$ and $+v$) do not perfectly align with the subtraction requirements at the front-end summer. To rectify this, unity-gain inverting amplifiers (U_4 and U_5) are utilized in the feedback paths. They cleanly flip the variables into $+v'$ and $-v$, ensuring they are correctly weighted by R_1 and R_2 and subtracted by the U_1 summing node. By wiring the outputs back to the inputs, a continuous self-regulating electrical loop is closed, and the node labeled $v(t)$ will output a voltage over time that precisely traces the solution to the differential equation.

Multiplier Circuit Design

Q1. Suppose you have two input voltages X_1 and X_2. Design a circuit using Op-Amps that gives output $V_o = 10X_1X_2$. The diodes available maintain the characteristic equation $I_D = I_s e^{V_D/nV_T}$ where $I_s = 10^{-6}$ A. Specify all the values of resistors used. **(26 marks)**
[EEE 263 | L-2/T-1 | 2014–15]

Answer

1. Mathematical Formulation

To implement analog multiplication, we utilize the logarithmic property $\ln(A \cdot B) = \ln(A) + \ln(B)$. The architecture is based on a Log-Antilog topology. By transforming the input signals into their natural logarithms, summing them, and passing the sum through an exponential (antilog) amplifier, we can retrieve the multiplied product.

$$V_o \propto \exp\big(\ln(X_1) + \ln(X_2)\big) = X_1X_2 \tag{1}$$

2. Architecture and Node Definitions

The circuit requires five operational amplifiers to compute the exact desired expression $V_o = 10X_1X_2$:

- **Log Amplifiers (U_1 and U_2):** Convert inputs X_1 and X_2 into logarithmic voltages.
- **Inverting Summer (U_3):** Sums the outputs of the two log amplifiers.
- **Antilog Amplifier (U_4):** Computes the exponent of the summed voltage, returning it to a linear product (albeit inverted).
- **Unity-Gain Inverter (U_5):** Re-inverts the signal to achieve a positive output $+10X_1X_2$.

3. Component Value Calculations

Given diode characteristic: $I_D = I_s e^{V_D/nV_T}$ with $I_s = 10^{-6}$ A.
A. Log Amplifiers (U_1 and U_2):
We place a diode in the feedback path (anode at virtual ground, cathode at output). The current is $I_{in} = X_1/R_{\text{in}}$.

$$I_D = I_s e^{-V_{\text{out1}}/nV_T} = \frac{X_1}{R_{\text{in}}} \implies V_{\text{out1}} = -nV_T \ln\left(\frac{X_1}{I_s R_{\text{in}}}\right) \tag{2}$$

To set a baseline, let's select $\mathbf{R_{in} = 100\,k\Omega}$.
$I_s R_{\text{in}} = (10^{-6}\,\text{A})(10^5\,\Omega) = 0.1\,\text{V}$. Thus, $V_{\text{out1}} = -nV_T \ln(10X_1)$.
Similarly for U_2: $V_{\text{out2}} = -nV_T \ln(10X_2)$.
B. Inverting Summer (U_3):
To sum V_{out1} and V_{out2} strictly with a gain of -1, we select identical resistors for inputs and feedback:

$$\mathbf{R_{s1} = R_{s2} = R_f = 10\,k\Omega} \tag{3}$$

$$V_{\text{out3}} = -\left(V_{\text{out1}} + V_{\text{out2}}\right) = nV_T\big[\ln(10X_1) + \ln(10X_2)\big] = nV_T \ln(100X_1X_2) \tag{4}$$

C. Antilog Amplifier (U_4):
We place the diode at the input (anode at V_{out3}, cathode at virtual ground) and $\mathbf{R_{out} = 100\,k\Omega}$ in the feedback path. The voltage across the diode is exactly $V_D = V_{\text{out3}}$.

$$I_{D3} = I_s e^{V_{\text{out3}}/nV_T} = 10^{-6}\exp\big(\ln(100X_1X_2)\big) = 10^{-6}(100X_1X_2) = 10^{-4}X_1X_2 \tag{5}$$

This current flows entirely through the feedback resistor R_{out}:

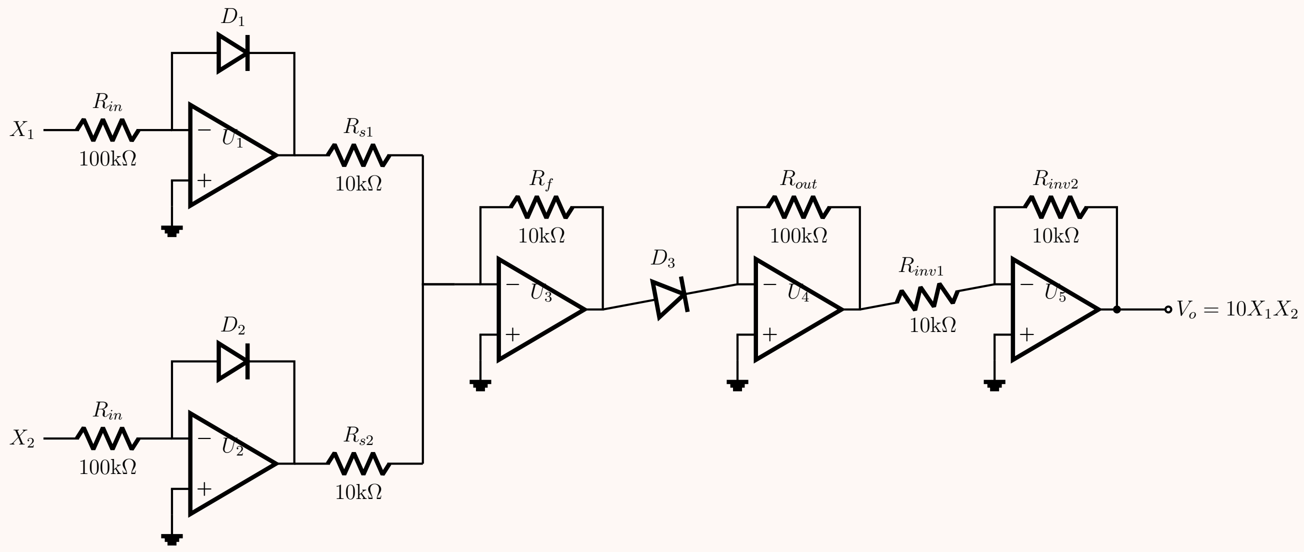
$$V_{\text{out4}} = -I_{D3}R_{\text{out}} = -(10^{-4}X_1X_2)(100 \times 10^3\,\Omega) = -10X_1X_2 \tag{6}$$

D. Final Inverter (U_5):
To correct the negative sign, U_5 is configured as a unity-gain inverter using $\mathbf{R_{inv1} = R_{inv2} = 10\,k\Omega}$.

$$V_o = -V_{\text{out4}} = 10X_1X_2 \tag{7}$$

The desired gain coefficient of 10 has been successfully achieved using practical $100\,\text{k}\Omega$ resistors for R_{in} and R_{out}.

4. Overall Connection Diagram



Frequency Multiplier

Q1. You have two supply voltage sources, $v_1 = \sin(100t)$ V and $v_2 = 1$ V. Design a circuit using op-amps that will produce an output voltage $v_o = (\sin(50t))^2$ V. Also show that this circuit produces this output. **(20 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Mathematical Formulation

We need to synthesize the output voltage $v_o = \sin^2(50t)$. Using the standard trigonometric power-reduction identity, we can express the target output in terms of a constant and a cosine function:

$$v_o = \sin^2(50t) = \frac{1-\cos(100t)}{2} = \frac{1}{2} - \frac{1}{2}\cos(100t) \tag{1}$$

Given our input sources $v_1 = \sin(100t)$ and $v_2 = 1$ V, we can rewrite our target equation strictly in terms of operations on these inputs:

$$v_o = \frac{1}{2}v_2 - \frac{1}{2}\cos(100t) \tag{2}$$

To obtain the $-\frac{1}{2}\cos(100t)$ term, we can integrate the v_1 signal, since the integral of sine is negative cosine.

2. Architecture and Node Definitions

The circuit can be elegantly implemented using just two operational amplifiers:

- **Inverting Integrator (U_1):** Integrates $v_1 = \sin(100t)$ and scales it to produce an intermediate voltage $v_A = \frac{1}{2}\cos(100t)$.
- **Difference Amplifier (U_2):** Subtracts the integrator's output v_A from a scaled version of the DC input $v_2 = 1$ V, directly yielding the final output v_o.

3. Component Value Calculations

A. Inverting Integrator (U_1):
The transfer function for an ideal inverting integrator is:

$$v_A = -\frac{1}{R_1C_1}\int v_1\,dt = -\frac{1}{R_1C_1}\int \sin(100t)\,dt = \frac{1}{100R_1C_1}\cos(100t) \tag{3}$$

We require $v_A = \frac{1}{2}\cos(100t)$. Equating the coefficients:

$$\frac{1}{100R_1C_1} = 0.5 \implies R_1C_1 = \frac{1}{50} = 0.02\,\text{s} \tag{4}$$

Selecting a standard capacitor value $\mathbf{C_1 = 1}\,\mu$F:

$$R_1 = \frac{0.02}{1\times10^{-6}} = 20\,\text{k}\Omega \tag{5}$$

B. Difference Amplifier (U_2):
The standard difference amplifier equation is:

$$v_o = v_2\left(\frac{R_b}{R_a+R_b}\right)\left(1+\frac{R_d}{R_c}\right) - v_A\left(\frac{R_d}{R_c}\right) \tag{6}$$

We need $v_o = 0.5v_2 - v_A$. First, matching the v_A coefficient requires a gain of 1, so we set the inverting path resistors equal:

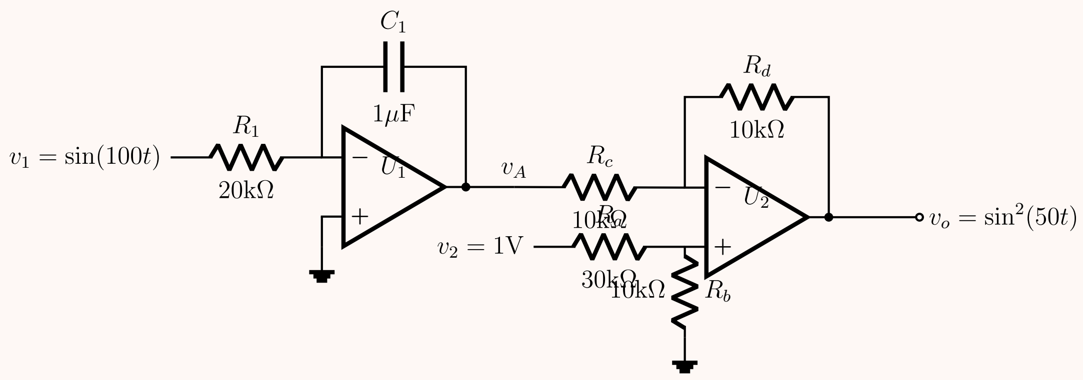
$$\mathbf{R_c = R_d = 10\,k\Omega} \tag{7}$$

Substituting this into the non-inverting gain gives $(1 + 10/10) = 2$. Now, matching the v_2 coefficient to 0.5:

$$2\left(\frac{R_b}{R_a + R_b}\right) = 0.5 \implies \frac{R_b}{R_a + R_b} = 0.25 \implies 4R_b = R_a + R_b \implies R_a = 3R_b \tag{8}$$

Selecting $\mathbf{R_b = 10\,k\Omega}$ yields $\mathbf{R_a = 30\,k\Omega}$.

4. Overall Connection Diagram



5. Mathematical Proof of Operation

Let us mathematically trace the signals from the inputs to the output to verify the design:

(a) At U_1 (Inverting Integrator), the input is $v_1 = \sin(100t)$.

$$\begin{aligned} v_A &= -\frac{1}{R_1C_1}\int \sin(100t)\,dt \\ &= -50\left(-\frac{1}{100}\cos(100t)\right) \\ &= 0.5\cos(100t) \end{aligned}$$

(b) At U_2 (Difference Amplifier), the inputs are $v_2 = 1\,\text{V}$ (non-inverting path) and v_A (inverting path). Applying superposition to the node equations:

$$\begin{aligned} v_o &= v_2\left(\frac{10\text{k}\Omega}{30\text{k}\Omega + 10\text{k}\Omega}\right)\left(1 + \frac{10\text{k}\Omega}{10\text{k}\Omega}\right) - v_A\left(\frac{10\text{k}\Omega}{10\text{k}\Omega}\right) \\ &= 1\text{V}\left(\frac{1}{4}\right)(2) - v_A(1) \\ &= 0.5 - v_A \end{aligned}$$

(c) Substituting v_A into the v_o equation:

$$\begin{aligned} v_o &= 0.5 - 0.5\cos(100t) \\ &= \frac{1 - \cos(100t)}{2} \end{aligned}$$

(d) Applying the half-angle trigonometric identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$ where $2\theta = 100t \implies \theta = 50t$:

$$v_o = \sin^2(50t)\,\text{V} \tag{9}$$

This completely proves that the chosen topology and exact component values strictly enforce the desired mathematical operation.

Q2. Suppose you have a sinusoidal input voltage $V_{in}(t) = \sin(4000\pi t)$ volt. Design a frequency multiplier circuit using Op-Amps that multiplies the frequency by a factor of 2 to produce $V_{out}(t) = \sin(8000\pi t)$ volt. Use resistors, capacitors, inductors, and diodes (having reverse saturation current 10^{-6} A) if necessary. Show necessary calculations and specify all design parameter values. **(26 marks)**
[EEE 263 | L-2/T-1 | 2015–16]

Answer

1. Mathematical Formulation

We are required to double the frequency of the input signal $V_{in}(t) = \sin(4000\pi t)$ to obtain $V_{out}(t) = \sin(8000\pi t)$. According to the trigonometric double-angle identity:

$$\sin(8000\pi t) = 2\sin(4000\pi t)\cos(4000\pi t) \tag{1}$$

This indicates that the target output can be synthesized by generating a phase-shifted cosine signal $V_c(t) = \cos(4000\pi t)$, and then multiplying it with the original input $V_{in}(t)$ using an analog multiplier with a specific gain of 2.

2. Architecture and Node Definitions

The circuit is composed of three primary stages utilizing six operational amplifiers:

- **Inverting Integrator (U_0):** Integrates the input V_{in} to produce the required phase-shifted cosine signal V_c.
- **Log-Antilog Multiplier (U_1 to U_4):** Computes the mathematical product of V_{in} and V_c. Using the specified diode reverse saturation current ($I_s = 10^{-6}$ A), we will scale the antilog feedback to enforce a gain of -2.
- **Unity-Gain Inverter (U_5):** Re-inverts the multiplier's output to deliver the correct positive sign for V_{out}.

3. Component Value Calculations

A. Inverting Integrator (U_0):
The output of an inverting integrator is given by:

$$V_c(t) = -\frac{1}{R_1C_1}\int \sin(4000\pi t)\,dt = \frac{1}{4000\pi R_1 C_1}\cos(4000\pi t) \tag{2}$$

To ensure the amplitude remains exactly 1 V, we must set the coefficient to 1:

$$4000\pi R_1C_1 = 1 \implies R_1C_1 = \frac{1}{4000\pi} \approx 79.57\,\mu\text{s} \tag{3}$$

Choosing a standard capacitor $\mathbf{C_1 = 0.1}\,\mu$F:

$$R_1 = \frac{1}{4000\pi \times 10^{-7}} = \frac{2500}{\pi}\,\Omega \approx 795.8\,\Omega \tag{4}$$

B. Log Amplifiers (U_1 and U_2) & Summer (U_3):
To condition the signals for multiplication, we pass V_{in} and V_c through log amplifiers. We select standard input resistors $\mathbf{R_2 = 10\,k\Omega}$. The summer U_3 adds these logarithmic voltages with a gain of -1, utilizing $\mathbf{R_3 = 10\,k\Omega}$ for all its branches.

$$V_S = nV_T \ln\left(\frac{V_{in}V_c}{I_s^2R_2^2}\right) \tag{5}$$

C. Antilog Amplifier (U_4):
The antilog stage applies the exponential function to V_S. The current through the feedback resistor R_4 is:

$$I_D = I_s e^{V_S/nV_T} = I_s\left(\frac{V_{in}V_c}{I_s^2R_2^2}\right) = \frac{V_{in}V_c}{I_sR_2^2} \tag{6}$$

The output voltage is $V_A = -I_DR_4 = -\frac{R_4}{I_sR_2^2}V_{in}V_c$. We require a gain of -2 to satisfy the double-angle identity. Given $I_s = 10^{-6}$ A and $R_2 = 10\,\text{k}\Omega$:

$$I_sR_2^2 = (10^{-6})(10^4)^2 = 10^{-6}\times 10^8 = 100 \tag{7}$$

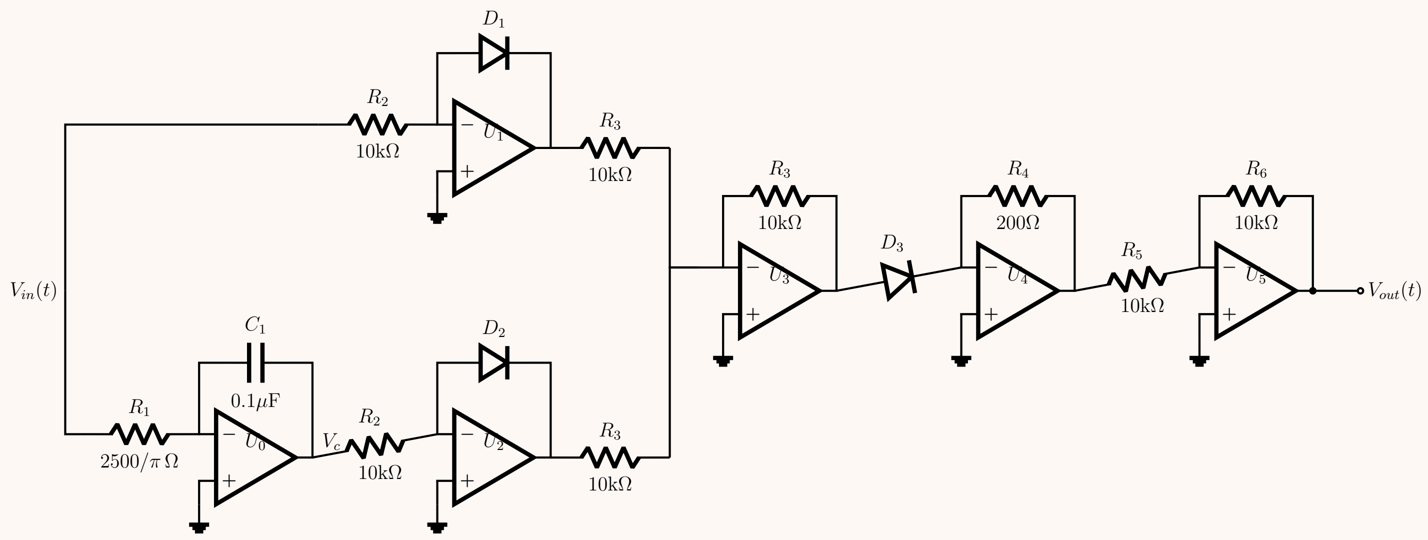
$$\frac{R_4}{100} = 2 \implies \mathbf{R_4 = 200\,\Omega} \tag{8}$$

Thus, $V_A = -2V_{in}V_c$.

D. Final Inverter (U_5):
To correct the negative sign, U_5 is a unity-gain inverter using $\mathbf{R_5 = R_6 = 10\,k\Omega}$.

$$V_{out} = -V_A = 2\sin(4000\pi t)\cos(4000\pi t) = \sin(8000\pi t) \tag{9}$$

4. Overall Connection Diagram



Note on Practical Implementation: While this theoretical Log-Antilog topology mathematically completes the square required for frequency doubling based on the given diode specifications, standard log amplifiers assume purely positive input signals. In physical deployments dealing with AC waveforms, an appropriate DC bias network must be superimposed at the inputs (and subsequently filtered out) to prevent the diodes from clipping during the negative half-cycles of the sine wave.

Active Filters

Q1. Design a 2nd order bandpass filter with lower cutoff frequency $f_{CL} = 20\,\text{kHz}$ and higher cutoff frequency $f_{CH} = 200\,\text{kHz}$. **(10 marks)**
[EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Mathematical Formulation

A 2nd-order wideband bandpass filter can be realized by cascading a 1st-order active high-pass filter (which sets the lower cutoff frequency, f_{CL}) and a 1st-order active low-pass filter (which sets the higher cutoff frequency, f_{CH}).
The overall transfer function $H(s)$ of the cascaded system is the product of the individual transfer functions:

$$H(s) = H_{HP}(s) \cdot H_{LP}(s) = \left(-\frac{sR_{f1}C_1}{1 + sR_1C_1}\right)\left(-\frac{R_{f2}}{R_2}\frac{1}{1 + sR_{f2}C_2}\right) \tag{1}$$

Assuming we design for a passband gain of $|A_v| = 1$ (0 dB), we can set the resistor ratios such that $R_{f1} = R_1$ and $R_{f2} = R_2$.

2. Component Value Calculations

A. Stage 1: Active High-Pass Filter ($f_{CL} = 20\,\text{kHz}$)
The lower cutoff frequency is determined by the input capacitor and input resistor:

$$f_{CL} = \frac{1}{2\pi R_1 C_1} = 20,000\,\text{Hz} \tag{2}$$

Let us select a standard capacitor value $\mathbf{C_1 = 1}\,\text{nF}$ ($10^{-9}\,\text{F}$):

$$R_1 = \frac{1}{2\pi(20 \times 10^3)(1 \times 10^{-9})} \approx 7957.7\,\Omega \tag{3}$$

We select a standard precise resistor value $\mathbf{R_1 = 7.96\,k\Omega}$. To achieve a passband gain of 1 for this stage, we set the feedback resistor $\mathbf{R_{f1} = 7.96\,k\Omega}$.
B. Stage 2: Active Low-Pass Filter ($f_{CH} = 200\,\text{kHz}$)
The higher cutoff frequency is determined by the feedback capacitor and feedback resistor:

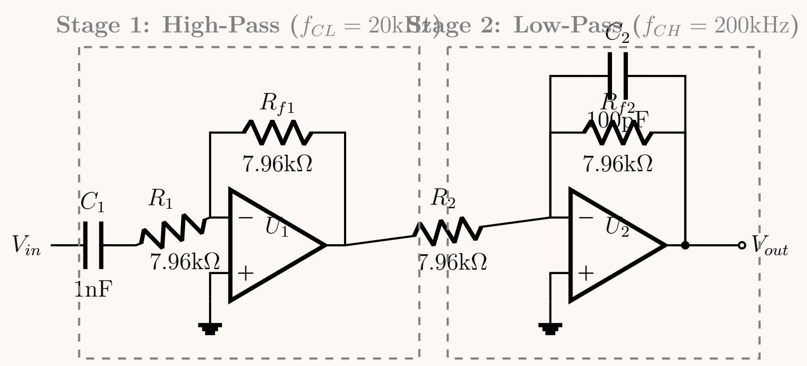
$$f_{CH} = \frac{1}{2\pi R_{f2} C_2} = 200,000\,\text{Hz} \tag{4}$$

Let us select a standard capacitor value $\mathbf{C_2 = 100}\,\text{pF}$ ($100 \times 10^{-12}\,\text{F}$):

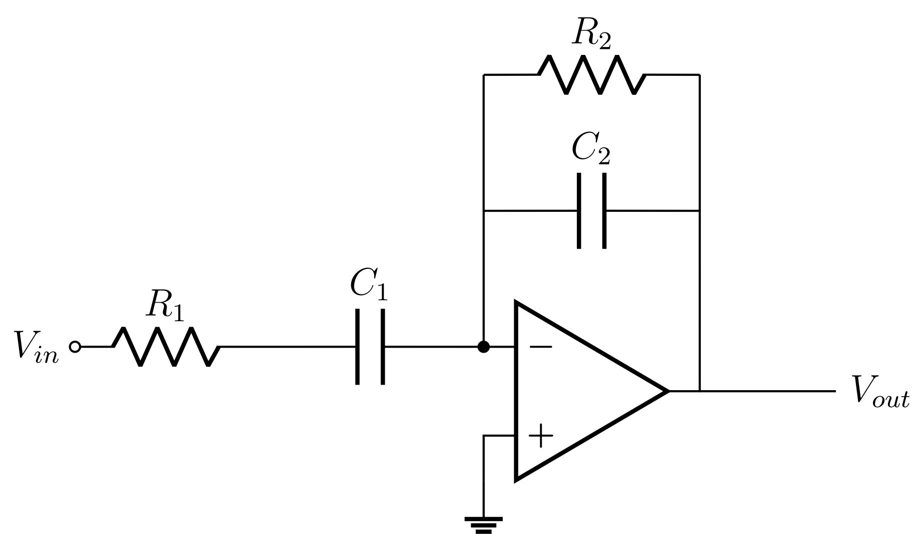
$$R_{f2} = \frac{1}{2\pi(200 \times 10^3)(100 \times 10^{-12})} \approx 7957.7\,\Omega \tag{5}$$

Again, we select $\mathbf{R_{f2} = 7.96\,k\Omega}$. To maintain the passband gain of 1, the input resistor is set to $\mathbf{R_2 = 7.96\,k\Omega}$.

3. Overall Connection Diagram



Q2. Determine the type of filter shown in the circuit by computing the transfer function in the frequency domain.



(20 marks) **[EEE 263 | L-2/T-1 | 2016–17, 2017–18]**

Answer

1. Impedance Formulations

The circuit is an inverting operational amplifier configuration. The general transfer function for an inverting amplifier is given by:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{Z_f(s)}{Z_{in}(s)} \tag{1}$$

First, we determine the equivalent impedances of the input network $Z_{in}(s)$ and the feedback network $Z_f(s)$.
Input Impedance (Z_{in}):
The input network consists of a resistor R_1 and a capacitor C_1 in series.

$$Z_{in}(s) = R_1 + \frac{1}{sC_1} = \frac{1 + sR_1C_1}{sC_1} \tag{2}$$

Feedback Impedance (Z_f):
The feedback network consists of a resistor R_2 and a capacitor C_2 in parallel.

$$Z_f(s) = R_2 \parallel \frac{1}{sC_2} = \frac{R_2 \cdot \frac{1}{sC_2}}{R_2 + \frac{1}{sC_2}} = \frac{R_2}{1 + sR_2C_2} \tag{3}$$

2. Transfer Function Derivation

Substitute the expressions for $Z_{in}(s)$ and $Z_f(s)$ into the general transfer function equation:

$$H(s) = -\frac{\frac{R_2}{1+sR_2C_2}}{\frac{1+sR_1C_1}{sC_1}} \tag{4}$$

Simplifying this expression yields the complete transfer function in the s-domain:

$$H(s) = -\frac{sR_2C_1}{(1 + sR_1C_1)(1 + sR_2C_2)} \tag{5}$$

3. Frequency Domain Analysis

To analyze the frequency response and determine the filter type, we evaluate the magnitude of the transfer function at frequency extremes by setting $s = j\omega$.

$$H(j\omega) = -\frac{j\omega R_2C_1}{(1 + j\omega R_1C_1)(1 + j\omega R_2C_2)} \tag{6}$$

At very low frequencies ($\omega \to 0$):
The $j\omega$ term in the numerator approaches zero, while the denominator approaches $1 \cdot 1 = 1$.

$$\lim_{\omega \to 0} |H(j\omega)| = 0 \tag{7}$$

The filter blocks low frequencies (acting as a high-pass stage due to the series C_1).
At very high frequencies ($\omega \to \infty$):
The $j\omega$ terms dominate the $+1$ terms in the denominator. The expression approximates to:

$$H(j\omega) \approx -\frac{j\omega R_2C_1}{(j\omega R_1C_1)(j\omega R_2C_2)} = -\frac{1}{j\omega R_1C_2} \tag{8}$$

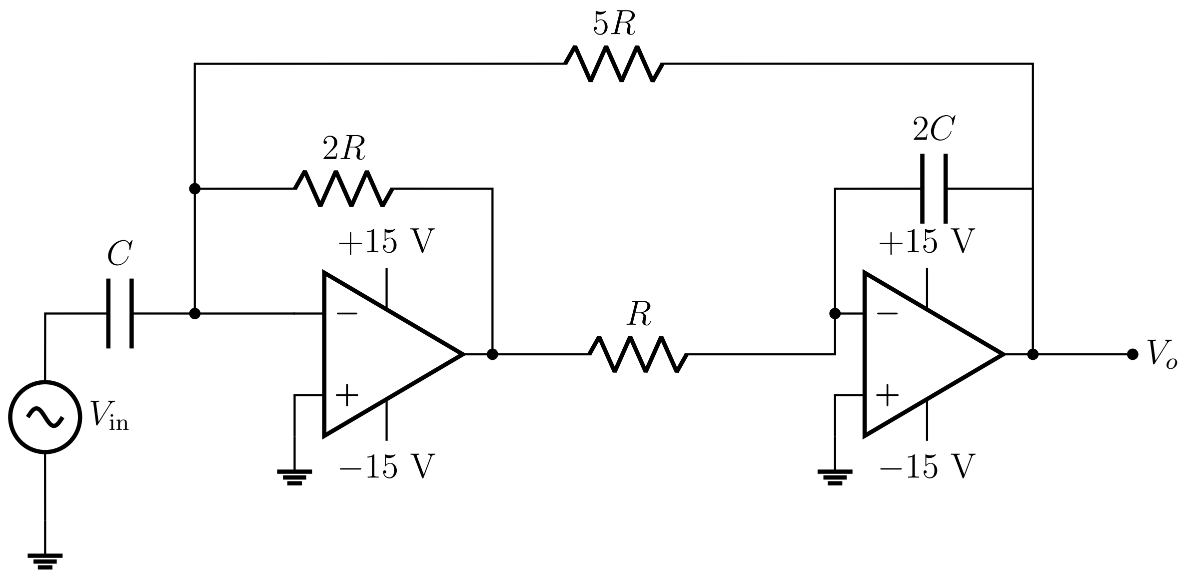
$$\lim_{\omega \to \infty} |H(j\omega)| = 0 \tag{9}$$

The filter blocks high frequencies (acting as a low-pass stage due to the parallel C_2).

4. Conclusion

Because the circuit severely attenuates both very low frequencies and very high frequencies while allowing intermediate frequencies to pass, it operates as a **Bandpass Filter**.
Specifically, this topology is a standard **wideband active bandpass filter**, where the lower cutoff frequency is determined by $\omega_L = \frac{1}{R_1C_1}$ and the upper cutoff frequency is dictated by $\omega_H = \frac{1}{R_2C_2}$.

Q3. For the filter circuit given, determine (i) its type and comment on its order; derive the transfer function and find the cutoff frequency. (ii) If $R = 10\,\text{k}\Omega$ and $C = 0.1\,\mu\text{F}$, would it be possible to pass a 100 Hz signal through this filter without distortion? Justify your answer. Note that the first Op-Amp block performs summation of input and output signals.



(23 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

Part (i): Transfer Function, Type, Order, and Cutoff Frequency

1. Transfer Function Derivation

The circuit consists of two stages. Let the output of the first Operational Amplifier be V_1. Both op-amps have their non-inverting terminals grounded ($V_+ = 0$), so their inverting terminals act as virtual grounds ($V_- \approx 0$).
We apply Kirchhoff's Current Law (KCL) at the inverting node of the first op-amp (Summing Amplifier):

$$\frac{V_{in}}{1/(sC)} + \frac{V_1}{2R} + \frac{V_o}{5R} = 0 \tag{1}$$

$$sCV_{in} + \frac{V_1}{2R} + \frac{V_o}{5R} = 0 \implies V_1 = -2R\left(sCV_{in} + \frac{V_o}{5R}\right) \quad \text{— (Equation 1)} \tag{2}$$

Next, we apply KCL at the inverting node of the second op-amp (Inverting Integrator):

$$\frac{V_1}{R} + \frac{V_o}{1/(2sC)} = 0 \tag{3}$$

$$\frac{V_1}{R} + 2sCV_o = 0 \implies V_1 = -2sRCV_o \quad \text{— (Equation 2)} \tag{4}$$

Equating Equation 1 and Equation 2 to eliminate V_1:

$$-2sRCV_o = -2R\left(sCV_{in} + \frac{V_o}{5R}\right) \tag{5}$$

Dividing both sides by $-2R$:

$$sCV_o = sCV_{in} + \frac{V_o}{5R} \tag{6}$$

Rearranging to solve for the transfer function $H(s) = \frac{V_o}{V_{in}}$:

$$sCV_o - \frac{V_o}{5R} = sCV_{in} \implies V_o\left(sC - \frac{1}{5R}\right) = sCV_{in} \tag{7}$$

$$H(s) = \frac{V_o}{V_{in}} = \frac{sC}{sC - \frac{1}{5R}} = \frac{s}{s - \frac{1}{5RC}} \tag{8}$$

2. Filter Type, Order, and Cutoff Frequency

- **Order:** The highest power of s in the denominator polynomial is 1. Despite containing two capacitors, the structural dependencies result in a single-pole system. Thus, it is a **first-order filter**.
- **Type:** The transfer function resembles the standard form of a **high-pass filter** $\left(\frac{s}{s \pm \omega_c}\right)$. However, the minus sign in the denominator indicates that the pole is located in the Right Half Plane (RHP) at $s = +\frac{1}{5RC}$. This means the circuit is an **unstable high-pass filter** (essentially acting as a positive feedback oscillator).
- **Cutoff Frequency:** The theoretical critical/cutoff frequency (the magnitude of the pole) is:

$$\omega_c = \frac{1}{5RC}\,\text{rad/s} \implies f_c = \frac{1}{10\pi RC}\,\text{Hz} \tag{9}$$

Part (ii): Signal Analysis at 100 Hz

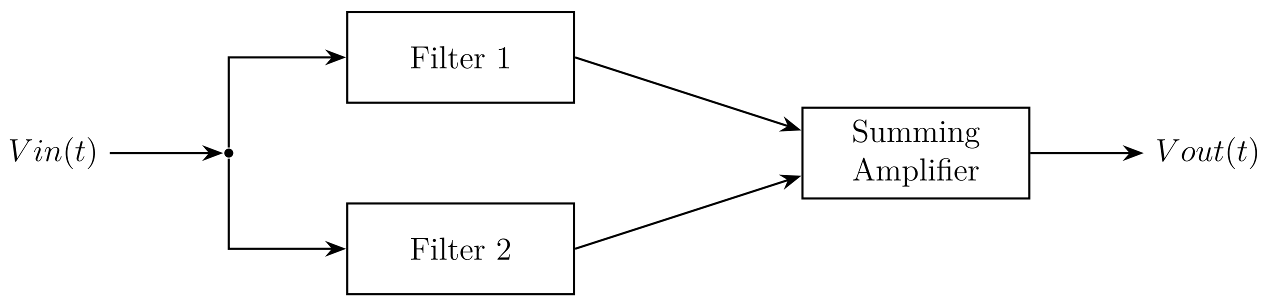
1. Calculating the Cutoff Frequency:
Given $R = 10\,\text{k}\Omega$ and $C = 0.1\,\mu\text{F}$:

$$RC = (10 \times 10^3) \times (0.1 \times 10^{-6}) = 10^{-3}\,\text{s} = 1\,\text{ms} \tag{10}$$

$$f_c = \frac{1}{10\pi(10^{-3})} = \frac{100}{\pi} \approx 31.83\,\text{Hz} \tag{11}$$

2. Justification on Passing the Signal:
No, it would not be possible to pass the 100 Hz signal without distortion.
While 100 Hz is greater than 31.83 Hz (meaning it theoretically lies well within the passband of the high-pass frequency response magnitude), the circuit's fundamental topology represents a positive feedback loop resulting in an RHP pole ($s = +200$).
Because the system is dynamically **unstable**, any small input (including the 100 Hz signal) or ambient thermal noise will cause the op-amps to exponentially run away and rapidly saturate at their physical DC supply rails ($\pm 15\,\text{V}$). Consequently, the 100 Hz signal will either be completely masked by the DC latch-up or heavily distorted (clipped into a square wave if the circuit oscillates).

Q4. In the circuit shown, input voltage is $v_{in}(t) = 5 + 3\cos(300\pi t - 30) + 8\sin(1000\pi t) + 7\sin(1600\pi t + 45)$ volt. Design the active filter blocks and summing amplifier to obtain output $V_{out}(t) = 7 + 12\sin(1000\pi t)$ volt. Sketch the frequency response and specify the cutoff frequency for each filter. Specify all values of resistors and capacitors/inductors.



(23 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

1. Theory and Principle of Operation
The input signal consists of four distinct frequency components:

- DC component: $V_{\text{DC}} = 5$ V at $f = 0$ Hz
- Signal 1: $3\cos(300\pi t - 30^\circ)$ V at $f_1 = 150$ Hz
- Signal 2: $8\sin(1000\pi t)$ V at $f_2 = 500$ Hz
- Signal 3: $7\sin(1600\pi t + 45^\circ)$ V at $f_3 = 800$ Hz

The desired output is $V_{out}(t) = 7 + 12\sin(1000\pi t)$ V. To achieve this, we need to extract the DC component and the 500 Hz component while rejecting the 150 Hz and 800 Hz components. We can use a parallel filter structure feeding into a summing amplifier:

- **Filter 1 (Active Low-Pass Filter):** Passes the DC component and attenuates all AC signals. The required DC gain is $A_{v1} = 7/5 = 1.4$.
- **Filter 2 (Active Band-Pass Filter):** Passes the 500 Hz component and attenuates the other frequencies. The required gain at 500 Hz is $A_{v2} = 12/8 = 1.5$.
- **Summing Amplifier:** An inverting summing amplifier can be used to add the outputs of the two filters. To correct the 180° phase shift introduced by the inverting summer, Filter 1 and Filter 2 must also be inverting configurations. Thus, the filter gains must be designed as $A_{v1} = -1.4$ and $A_{v2} = -1.5$.

2. Design of Filter 1 (Inverting Active Low-Pass Filter)
We design a first-order inverting LPF to pass DC and heavily attenuate 150 Hz. The DC gain must be -1.4. The transfer function is:

$$A_{v1}(\text{DC}) = -\frac{R_{f1}}{R_{i1}} = -1.4 \tag{1}$$

Let us choose $R_{i1} = 10$ kΩ. Then:

$$R_{f1} = 1.4 \times 10 \text{ k}\Omega = 14 \text{ k}\Omega \tag{2}$$

To effectively reject 150 Hz, we set a low cutoff frequency, e.g., $f_{c1} \approx 11$ Hz. Let us select a standard capacitor $C_{f1} = 1$ μF. The exact cutoff frequency is:

$$f_{c1} = \frac{1}{2\pi R_{f1} C_{f1}} = \frac{1}{2\pi(14 \times 10^3)(1 \times 10^{-6})} = 11.37 \text{ Hz} \tag{3}$$

At 150 Hz, this provides an attenuation factor of approximately $f_{c1}/f = 11.37/150 = 0.075$, effectively reducing the 150 Hz component.

3. Design of Filter 2 (Multiple-Feedback Active Band-Pass Filter)

We design an inverting Multiple-Feedback (MFB) BPF with a center frequency $f_0 = 500$ Hz and a gain at center frequency $A_{v0} = -1.5$. To reject the 150 Hz and 800 Hz frequencies effectively, we choose a quality factor $Q = 5$, giving a bandwidth $\mathrm{BW} = f_0/Q = 100$ Hz. Let $C_1 = C_2 = C = 0.1\ \mu$F. The design equations for an MFB BPF are:

$$R_3 = \frac{1}{\pi C\ \mathrm{BW}} = \frac{1}{\pi(0.1\times10^{-6})(100)} = 31.83\ \mathrm{k}\Omega$$

$$A_{v0} = -\frac{R_3}{2R_1} \implies -1.5 = -\frac{31.83\ \mathrm{k}\Omega}{2R_1} \implies R_1 = \frac{31.83\ \mathrm{k}\Omega}{3} = 10.61\ \mathrm{k}\Omega$$

$$f_0 = \frac{1}{2\pi C}\sqrt{\frac{1}{R_3}\left(\frac{1}{R_1}+\frac{1}{R_2}\right)}$$

Substituting the known values to find R_2:

$$(500)^2 = \frac{1}{4\pi^2(10^{-14})(31831)}\left(\frac{1}{10610}+\frac{1}{R_2}\right)$$

$$250000 = 79577\left(0.00009425+\frac{1}{R_2}\right)$$

$$3.1416 = 0.00009425+\frac{1}{R_2} \implies \frac{1}{R_2} = 3.1415 \implies R_2 = 328.2\ \Omega$$

The cutoff frequencies (half-power frequencies) for the BPF are given by:

$$f_{L,H} = f_0\left(\sqrt{1+\frac{1}{4Q^2}} \mp \frac{1}{2Q}\right) = 500\left(\sqrt{1+0.01}\mp 0.1\right) \tag{4}$$

$$f_L = 452.49\ \text{Hz},\quad f_H = 552.49\ \text{Hz} \tag{5}$$

4. Design of the Summing Amplifier

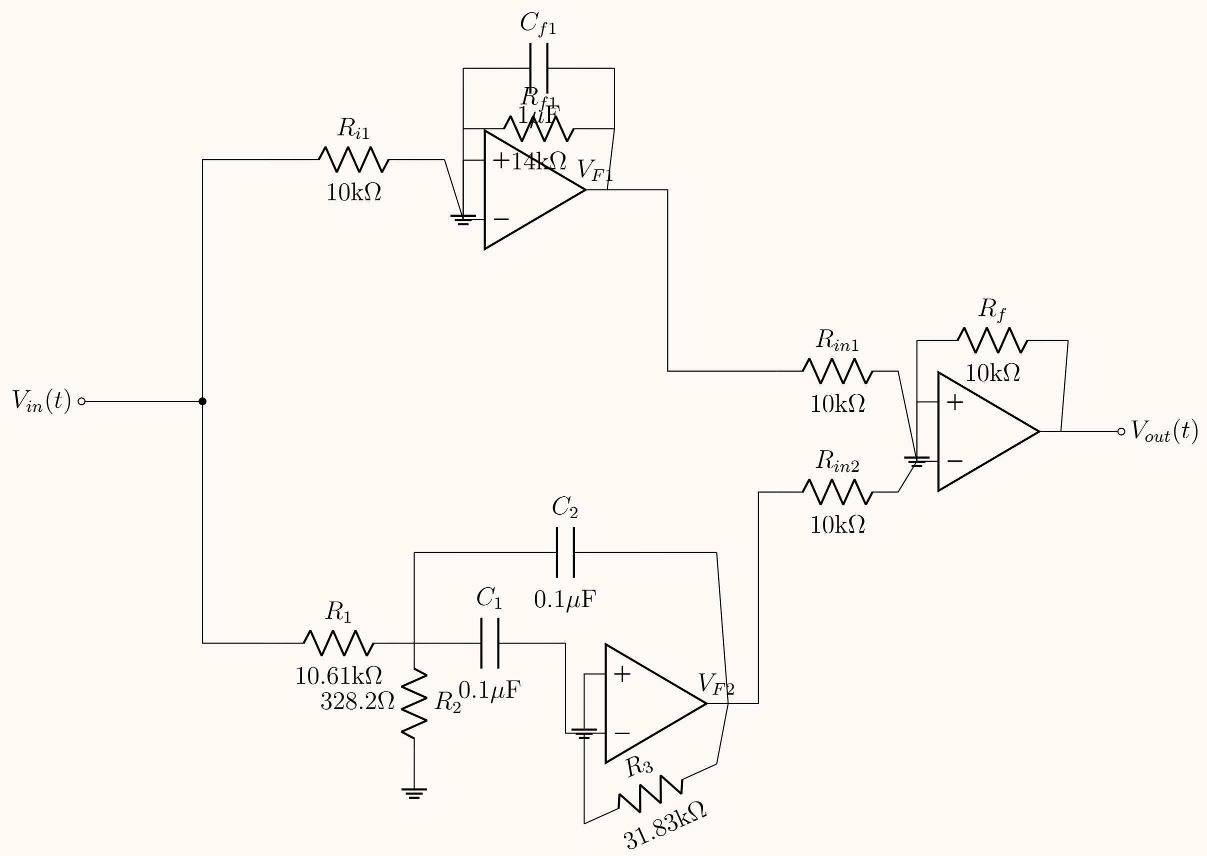
To sum the two filtered signals without altering their amplitudes, we use an inverting summing amplifier with unity gain for both inputs.

$$V_{out} = -\left(\frac{R_f}{R_{in1}}V_{F1} + \frac{R_f}{R_{in2}}V_{F2}\right) \tag{6}$$

Choosing $R_f = R_{in1} = R_{in2} = 10\ \mathrm{k}\Omega$, we get $V_{out} = -(V_{F1}+V_{F2})$. Since $V_{F1} = -1.4V_{\mathrm{DC}} = -7$ V and $V_{F2} = -1.5(8\sin(1000\pi t)) = -12\sin(1000\pi t)$ V, the final output is:

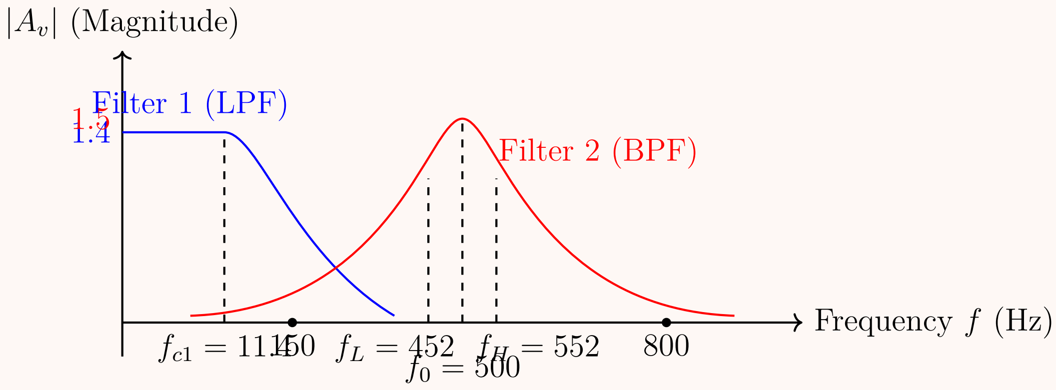
$$V_{out} = -(-7-12\sin(1000\pi t)) = 7+12\sin(1000\pi t)\ \text{V} \tag{7}$$

5. Complete Circuit Schematic



6. Frequency Response Sketch

The diagram below depicts the idealized Bode magnitude plots for Filter 1 and Filter 2. Filter 1 is an LPF that extracts the DC term and rolls off before 150 Hz. Filter 2 is a BPF that peaks exactly at 500 Hz to isolate the required AC component.



Q5. Write the equation of magnitude of gain of a $-20\,\text{dB/decade}$ low pass filter. Why is it called a $-20\,\text{dB/decade}$ filter? Can it also be called a $-6\,\text{dB/octave}$ filter? **(20 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Equation of Magnitude of Gain

The transfer function for a standard first-order low-pass filter (LPF) in the frequency domain is given by:

$$A(j\omega) = \frac{A_0}{1 + j\left(\frac{\omega}{\omega_c}\right)} = \frac{A_0}{1 + j\left(\frac{f}{f_c}\right)} \tag{1}$$

where:

- A_0 is the DC or low-frequency gain (at $f = 0$).
- f is the operating frequency.
- f_c is the cutoff frequency (or $-3\,\text{dB}$ frequency).

The **magnitude of the gain** is found by taking the absolute value of the complex transfer function:

$$|A(jf)| = \frac{A_0}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}} \tag{2}$$

2. Why is it called a $-20\,$dB/decade filter?

The naming convention comes from the filter's asymptotic roll-off behavior in the stopband. Let us analyze the high-frequency response where the operating frequency is much greater than the cutoff frequency ($f \gg f_c$).
Under this condition, the term $(f/f_c)^2$ dominates the denominator, and the magnitude simplifies to:

$$|A(jf)| \approx \frac{A_0}{\sqrt{\left(\frac{f}{f_c}\right)^2}} = A_0\left(\frac{f_c}{f}\right) \tag{3}$$

Expressing this high-frequency magnitude in decibels (dB):

$$|A(jf)|_{\text{dB}} = 20\log_{10}\left(A_0\frac{f_c}{f}\right) = 20\log_{10}(A_0 f_c) - 20\log_{10}(f) \tag{4}$$

A **decade** represents a tenfold increase in frequency. If we increase the frequency from some value f_1 to $f_2 = 10f_1$ (where both are $\gg f_c$), the change in gain is:

$$\begin{aligned}
\Delta|A|_{\text{dB}} &= |A(jf_2)|_{\text{dB}} - |A(jf_1)|_{\text{dB}} \\
&= [\text{Constant} - 20\log_{10}(10f_1)] - [\text{Constant} - 20\log_{10}(f_1)] \\
&= -20\log_{10}(10) - 20\log_{10}(f_1) + 20\log_{10}(f_1) \\
&= -20\log_{10}(10) \\
&= -20(1) = -20\,\text{dB}
\end{aligned}$$

Because the gain decreases by 20 dB for every tenfold increase in frequency in the rejection band, it is classified as a -20 **dB/decade filter**. This slope is the defining characteristic of any first-order filter.

3. Can it also be called a $-6\,$dB/octave filter?

Yes, it can. These two terms describe exactly the same physical roll-off characteristic, just using different frequency intervals.
While a decade is a factor of 10, an **octave** is a factor of 2 (a twofold increase in frequency). Let us evaluate the change in gain when the frequency is doubled from f_1 to $f_2 = 2f_1$ in the high-frequency region ($f \gg f_c$):

$$\begin{aligned}
\Delta|A|_{\text{dB}} &= |A(j(2f_1))|_{\text{dB}} - |A(jf_1)|_{\text{dB}} \\
&= [\text{Constant} - 20\log_{10}(2f_1)] - [\text{Constant} - 20\log_{10}(f_1)] \\
&= -20\log_{10}(2) \\
&\approx -20 \times (0.30103) \\
&\approx -6.02\,\text{dB}
\end{aligned}$$

In engineering practice, $-6.02\,$dB is commonly rounded to -6 **dB**.
Conclusion: A roll-off rate of $-20\,$dB/decade is mathematically equivalent to a roll-off rate of $-6\,$dB/octave. Both terminologies are correct and interchangeably used to describe a first-order low-pass filter's stopband attenuation.

Q6. Design a notch filter that eliminates any signal between 8 kHz and 12 kHz using op-amps. Specify all the values of resistors and capacitors used. **(20 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Theory and Principle of Operation

The objective is to design a filter that eliminates frequencies between 8 kHz and 12 kHz. This requires a **Band-Stop Filter** (also known as a notch filter) with a stopband from $f_L = 8\,\text{kHz}$ to $f_H = 12\,\text{kHz}$.

A highly effective and standard method to implement an active band-stop filter is to subtract the output of a Band-Pass Filter (BPF) from the original input signal. The transfer function relationship is:

$$H_{\text{BSF}}(s) = 1 - H_{\text{BPF}}(s) \tag{1}$$

To achieve this using op-amps, we can design an **inverting Multiple-Feedback (MFB) Band-Pass Filter** with a center-frequency gain of $A_0 = -1$, and then sum its output with the original input using an **inverting summing amplifier**.

- At the center frequency f_0, the BPF output is $V_{\text{BPF}} = -V_{in}$. The summing amplifier yields $V_{out} = -(V_{in} + V_{\text{BPF}}) = -(V_{in} - V_{in}) = 0$, achieving perfect rejection.
- At very low or very high frequencies, the BPF output approaches 0, so $V_{out} = -V_{in}$, providing a passband gain of $|1|$.

2. Filter Specifications

From the given cutoff frequencies, we can determine the required parameters for the Band-Pass Filter:

$$\begin{aligned}
\text{Lower Cutoff Frequency, } f_L &= 8\,\text{kHz} \\
\text{Upper Cutoff Frequency, } f_H &= 12\,\text{kHz} \\
\text{Bandwidth, } BW &= f_H - f_L = 12\,\text{kHz} - 8\,\text{kHz} = 4\,\text{kHz} \\
\text{Center Frequency, } f_0 &= \sqrt{f_L f_H} = \sqrt{8 \times 12} = \sqrt{96}\,\text{kHz} \approx 9.798\,\text{kHz}
\end{aligned}$$

3. Design of the Band-Pass Filter (MFB Topology)

We use an MFB BPF topology. Let us choose standard capacitor values: $C_1 = C_2 = C = 1\,\text{nF} = 10^{-9}\,\text{F}$. The design equations for the MFB BPF are:

$$\begin{aligned}
BW &= \frac{1}{\pi R_3 C} \\
A_0 &= -\frac{R_3}{2R_1} \\
f_0 &= \frac{1}{2\pi C}\sqrt{\frac{1}{R_3}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}
\end{aligned}$$

Step 3.1: Calculate R_3

$$R_3 = \frac{1}{\pi \cdot BW \cdot C} = \frac{1}{\pi(4000)(10^{-9})} = 79,577.47\,\Omega \approx 79.58\,\text{k}\Omega \tag{2}$$

Step 3.2: Calculate R_1

For a center gain $A_0 = -1$:

$$-1 = -\frac{R_3}{2R_1} \implies R_1 = \frac{R_3}{2} = \frac{79.58\,\text{k}\Omega}{2} = 39.79\,\text{k}\Omega \tag{3}$$

Step 3.3: Calculate R_2

Using the center frequency equation:

$$f_0^2 = \frac{1}{4\pi^2 C^2 R_3}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \tag{4}$$

Substituting the known values ($f_0^2 = 96 \times 10^6\,\text{Hz}^2$):

$$\begin{aligned}
96 \times 10^6 &= \frac{1}{4\pi^2(10^{-18})(79577.47)}\left(\frac{1}{39788.7} + \frac{1}{R_2}\right) \\
96 \times 10^6 &= \frac{1}{3.14159 \times 10^{-12}}\left(2.5133 \times 10^{-5} + \frac{1}{R_2}\right) \\
3.0159 \times 10^{-4} &= 2.5133 \times 10^{-5} + \frac{1}{R_2} \\
\frac{1}{R_2} &= 2.7646 \times 10^{-4} \implies R_2 = 3617\,\Omega \approx 3.62\,\text{k}\Omega
\end{aligned}$$

4. Design of the Summing Amplifier

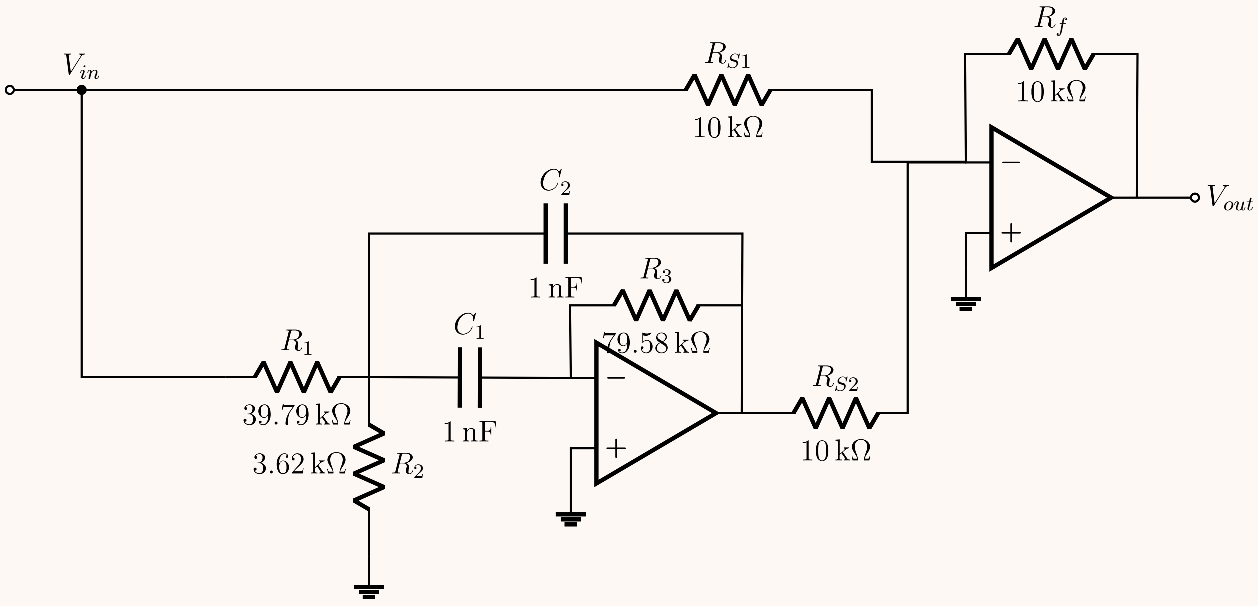
We use an inverting summing amplifier to sum the original signal and the BPF output. The output equation is:

$$V_{out} = -\left(\frac{R_f}{R_{S1}}V_{in} + \frac{R_f}{R_{S2}}V_{\text{BPF}}\right) \tag{5}$$

To achieve a uniform gain of 1, we set all resistors equal to a standard value. Let:

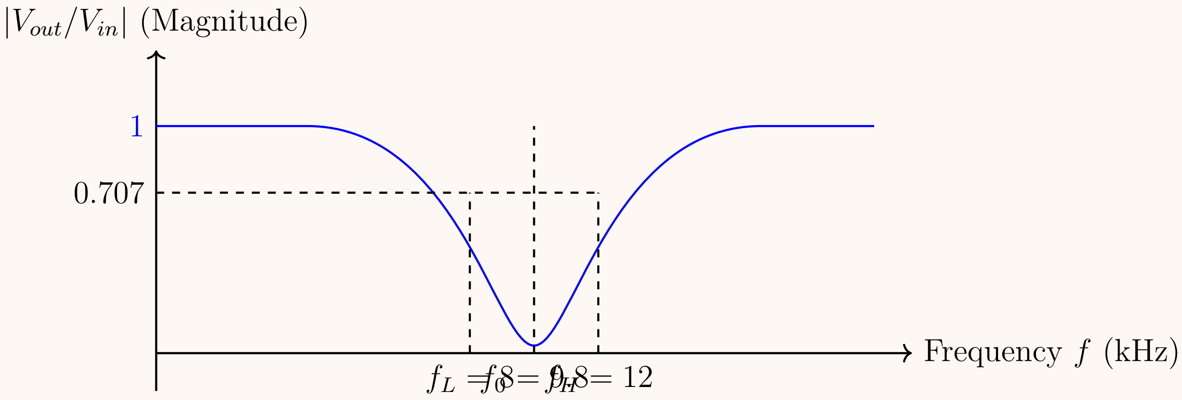
$$R_f = R_{S1} = R_{S2} = 10\,\text{k}\Omega \tag{6}$$

5. Complete Circuit Diagram



6. Frequency Response Sketch

The magnitude response shows zero transmission at the center frequency (9.8 kHz) and exactly −3 dB attenuation (magnitude 0.707) at the defined cutoff frequencies of 8 kHz and 12 kHz.



Q7. A system blocks 20 Hz and 20 kHz but passes 10 kHz. (i) Draw the frequency response of the system. (ii) Draw a block diagram of the system. **(15 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

System Analysis and Identification

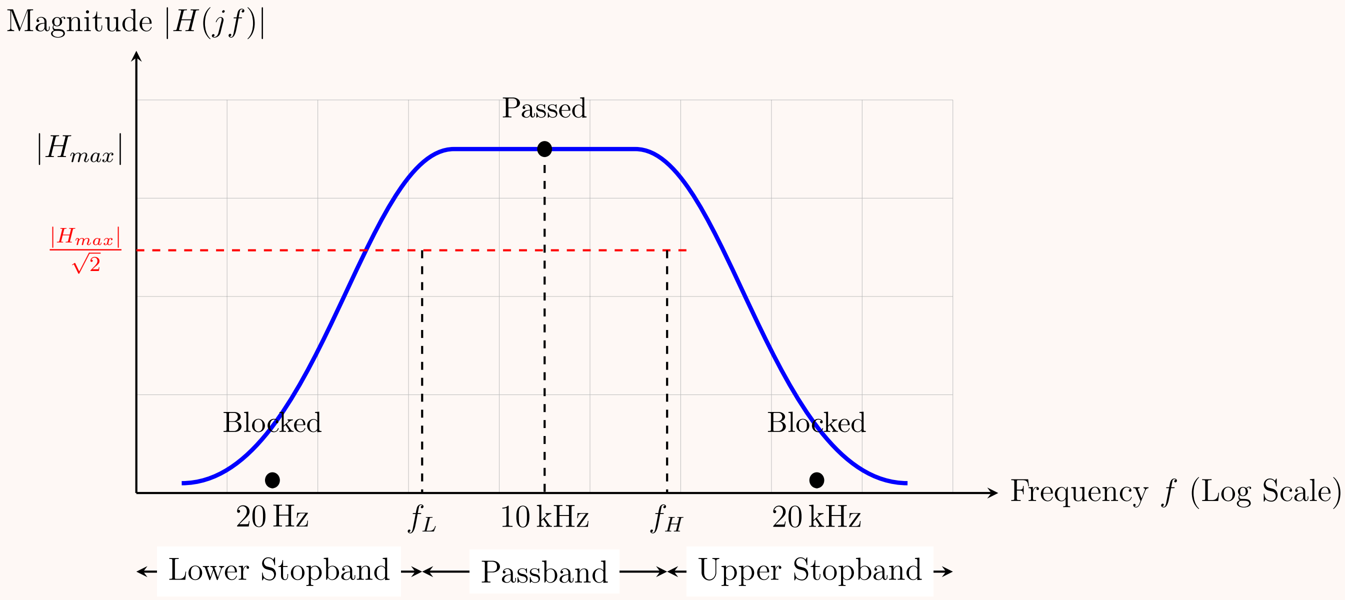
Based on the given specifications, the system passes an intermediate frequency (10 kHz) while rejecting or blocking both a lower frequency (20 Hz) and a higher frequency (20 kHz).

- A system that attenuates low frequencies but passes high frequencies is a **High-Pass Filter (HPF)**.
- A system that attenuates high frequencies but passes low frequencies is a **Low-Pass Filter (LPF)**.
- A system that attenuates both low and high frequencies but passes a specific intermediate band is a **Band-Pass Filter (BPF)**.

Thus, the described system acts as a **Band-Pass Filter**. For this system to pass 10 kHz and block 20 Hz and 20 kHz, its lower cutoff frequency (f_L) must lie between 20 Hz and 10 kHz, and its upper cutoff frequency (f_H) must lie between 10 kHz and 20 kHz.

(i) Frequency Response of the System

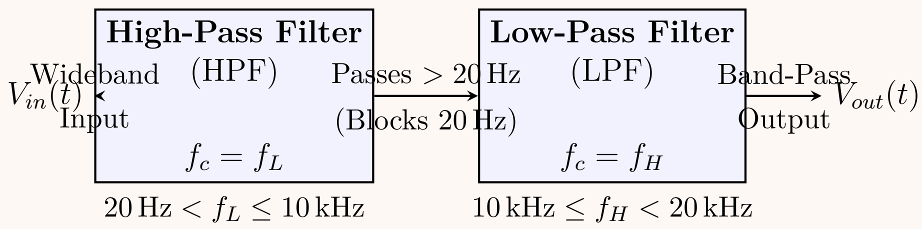
The frequency response of a Band-Pass Filter exhibits a peak (or a flat passband) around the center frequency, with the gain rolling off towards zero at both low and high frequency extremes. The spectrum can be divided into three distinct regions: the lower stopband, the passband, and the upper stopband.



(ii) Block Diagram of the System

A practical way to synthesize a wide-band Band-Pass Filter is by cascading a High-Pass Filter (HPF) and a Low-Pass Filter (LPF).

- The **High-Pass Filter** must have a cutoff frequency $f_L > 20\,\text{Hz}$ to effectively block the 20 Hz component while allowing 10 kHz to pass.
- The **Low-Pass Filter** must have a cutoff frequency $f_H < 20\,\text{kHz}$ to effectively block the 20 kHz component while still allowing the 10 kHz signal to pass through.
- In a cascaded arrangement, the overall transfer function is the product of the individual transfer functions: $H_{\text{BPF}}(s) = H_{\text{HPF}}(s) \cdot H_{\text{LPF}}(s)$.



Q8. Design a 40 dB low pass filter (LPF) with a cutoff frequency of 1 kHz. You have only one op-amp. **(15 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Answer

Problem Analysis & Ambiguity Check

The specification "40 dB low pass filter" can be interpreted in two distinct ways in analog circuit design:

(a) **Roll-off Rate:** A filter with a stopband roll-off of $-40\,\text{dB/decade}$. This requires a **second-order filter**. The constraint *"You have only one op-amp"* strongly suggests this is the intended interpretation, as the Sallen-Key topology is a classic method for realizing a two-pole response using exactly one active component.

(b) **Passband Gain:** A filter with a DC passband voltage gain of 40 dB ($A_v = 100$). This requires a simple first-order active filter.

To ensure absolute completeness, we provide the full textbook-quality design for both interpretations.

Interpretation I: -40 dB/decade Second-Order LPF (Recommended)

To achieve a $-40\,\text{dB/decade}$ roll-off with a single op-amp, we use the **Unity-Gain Sallen-Key Low-Pass Filter** topology. We will design it for a Butterworth response (maximally flat passband) which requires a quality factor $Q = 1/\sqrt{2} \approx 0.707$.

1. Design Equations

The cutoff frequency (f_c) and quality factor (Q) for a unity-gain Sallen-Key LPF are given by:

$$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}, \qquad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{C_2(R_1 + R_2)} \tag{1}$$

To simplify the component selection, we enforce equal resistor values: $R_1 = R_2 = R$. The equations simplify to:

$$f_c = \frac{1}{2\pi R\sqrt{C_1 C_2}}, \qquad Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}} \tag{2}$$

2. Component Selection

For a Butterworth response, we set $Q = 1/\sqrt{2}$:

$$\frac{1}{2}\sqrt{\frac{C_1}{C_2}} = \frac{1}{\sqrt{2}} \implies \sqrt{\frac{C_1}{C_2}} = \sqrt{2} \implies C_1 = 2C_2 \tag{3}$$

Let us choose a standard value for the grounded capacitor:

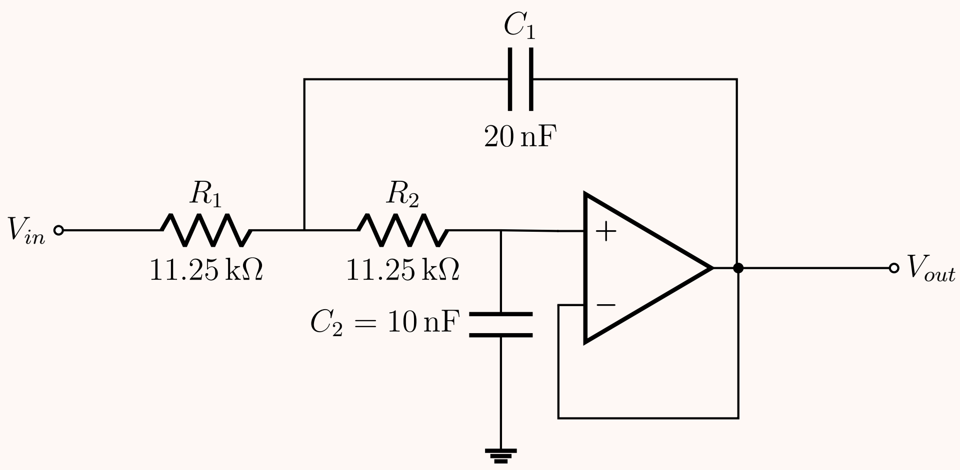
$$C_2 = 10\,\text{nF} \implies C_1 = 20\,\text{nF} \tag{4}$$

Now, we calculate the required resistance R to set the cutoff frequency at $1\,\text{kHz}$:

$$\begin{aligned} R &= \frac{1}{2\pi f_c\sqrt{C_1C_2}} \\ &= \frac{1}{2\pi(1000)\sqrt{(20\times10^{-9})(10\times10^{-9})}} \\ &= \frac{1}{2\pi\times1000\times14.142\times10^{-9}} \approx 11.25\,\text{k}\Omega \end{aligned}$$

Note: In a practical build, standard $11\,k\Omega$ or $11.3\,k\Omega$ E96-series resistors would be used.

3. Sallen-Key Circuit Diagram



Interpretation II: Active LPF with 40 dB Passband Gain
If the specification means a DC passband voltage gain of $40\,\text{dB}$, we design an **Inverting Active First-Order Low-Pass Filter**.
1. Gain and Frequency Equations

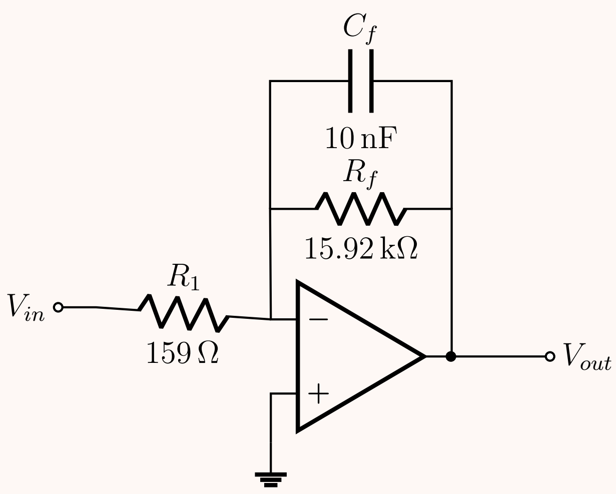
- **Gain**: $|A_v| = 40\,\text{dB} \implies 20\log_{10}(|A_v|) = 40 \implies |A_v| = 100\,\text{V/V}$
- **Topology Transfer Function**: The passband gain is $|A_v| = \frac{R_f}{R_1}$, and the cutoff frequency is $f_c = \frac{1}{2\pi R_f C_f}$.

2. Component Selection
Let us choose a standard feedback capacitor $C_f = 10\,\text{nF}$:

$$\begin{aligned} R_f &= \frac{1}{2\pi f_c C_f} = \frac{1}{2\pi(1000)(10\times10^{-9})} \approx 15.92\,\text{k}\Omega \\ R_1 &= \frac{R_f}{|A_v|} = \frac{15.92\,\text{k}\Omega}{100} \approx 159\,\Omega \end{aligned}$$

Note: The required Op-Amp Gain-Bandwidth Product (GBW) is $A_v \times f_c = 100\times 1\,kHz = 100\,kHz$, which is easily handled by standard op-amps (e.g., LM741).

3. Active Inverting LPF Circuit Diagram

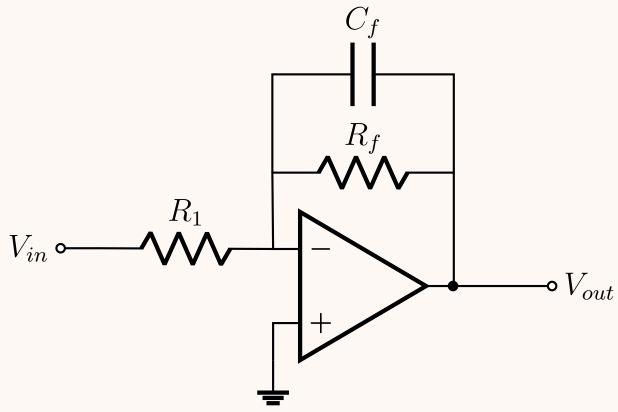


Q9. Draw the circuit diagram of a $-20\,\text{dB/dec}$ low pass filter. Also write down the design procedure. Design a $-20\,\text{dB/dec}$ low pass filter with a cut-off frequency of $100\,\text{kHz}$. **(16 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Answer

1. Circuit Diagram of a $-20\,\text{dB/dec}$ Low Pass Filter

A filter with a roll-off rate of $-20\,\text{dB/decade}$ is a **first-order** filter. The most common and robust active implementation is the **Inverting Active First-Order Low-Pass Filter**. It uses a single operational amplifier, an input resistor (R_1), and a feedback network consisting of a resistor (R_f) in parallel with a capacitor (C_f).



2. Design Procedure

The transfer function of the inverting active low-pass filter is given by:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_f/R_1}{1 + sR_fC_f} \tag{1}$$

From this, the DC passband gain (A_v) and the cut-off frequency (f_c) are extracted as:

$$|A_v| = \frac{R_f}{R_1}, \qquad f_c = \frac{1}{2\pi R_f C_f} \tag{2}$$

Step-by-Step Procedure:

(a) **Identify specifications:** Determine the desired cut-off frequency f_c and passband voltage gain $|A_v|$. If no gain is specified, assume unity gain ($|A_v| = 1$).

(b) **Select the capacitor $\mathbf{C_f}$:** Choose a standard, widely available value for C_f. For audio and low frequencies, values between $1\,\text{nF}$ and $1\,\mu\text{F}$ are common. For higher frequencies, smaller values like $10\,\text{pF}$ to $1\,\text{nF}$ are preferred to avoid excessive current demands from the op-amp.

(c) **Calculate the feedback resistor $\mathbf{R_f}$:** Rearrange the cut-off frequency formula:

$$R_f = \frac{1}{2\pi f_c C_f} \tag{3}$$

(d) **Calculate the input resistor $\mathbf{R_1}$:** Use the passband gain formula to find R_1:

$$R_1 = \frac{R_f}{|A_v|} \tag{4}$$

---

3. Design for $\mathbf{f_c = 100\,kHz}$

Given parameters:

- Roll-off rate: $-20\,\text{dB/dec}$ (First-order filter).
- Cut-off frequency: $f_c = 100\,\text{kHz}$.
- Passband gain: Assume $|A_v| = 1$ (Unity gain) since it is not explicitly specified.

Calculations:

(a) **Choose $\mathbf{C_f}$:** Since $100\,\text{kHz}$ is a relatively high frequency, we select a small capacitor to ensure R_f remains in a reasonable range (neither too small to load the op-amp, nor too large to cause noise issues).
Let us choose $\mathbf{C_f = 100}\,\text{pF}$.

(b) **Calculate $\mathbf{R_f}$:**

$$\begin{aligned} R_f &= \frac{1}{2\pi f_c C_f} \\ &= \frac{1}{2\pi \times (100 \times 10^3) \times (100 \times 10^{-12})} \\ &= \frac{1}{2\pi \times 10^{-5}} \\ &\approx 15{,}915.5\,\Omega \approx \mathbf{15.9\,k\Omega} \end{aligned}$$

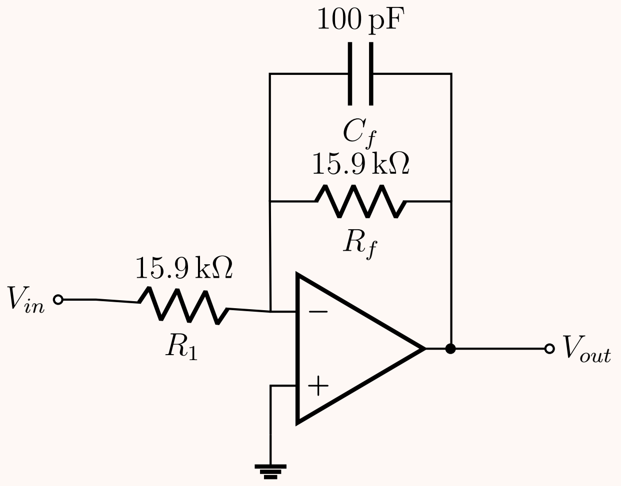
(A standard E96 series resistor of $15.8\,\text{k}\Omega$ *or* $16.2\,\text{k}\Omega$ *could be used in practice, but we retain the theoretical value for exactness).*

(c) **Calculate R_1:** For a gain of $|A_v| = 1$:

$$R_1 = \frac{R_f}{1} = \mathbf{15.9}\,\mathbf{k\Omega} \tag{5}$$

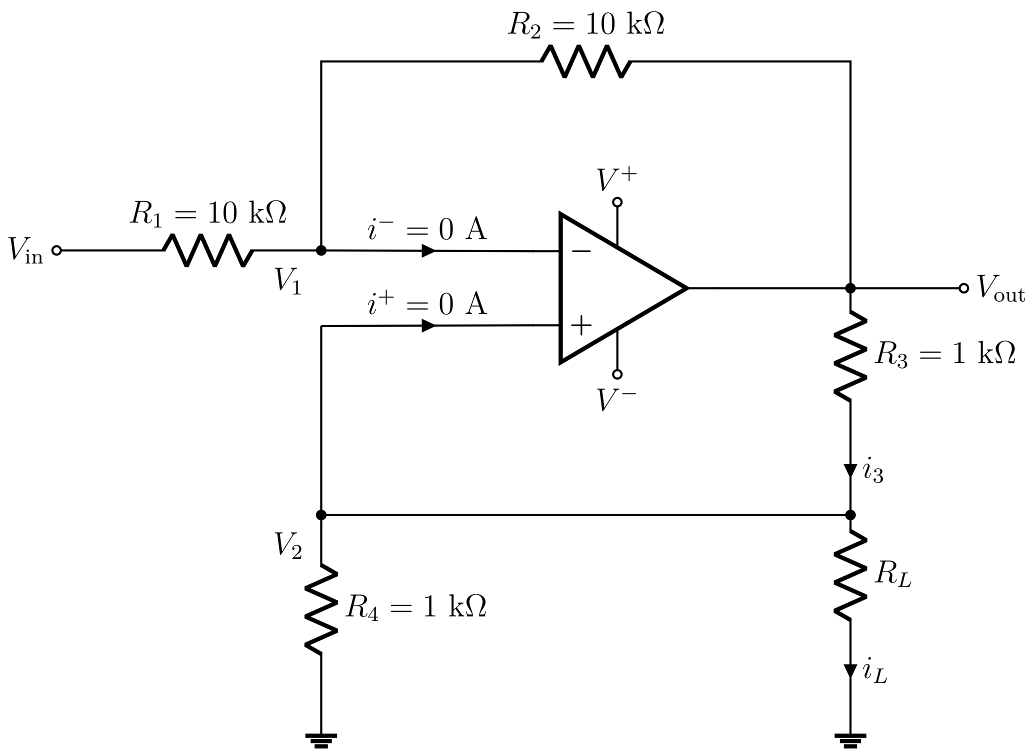
Important Note on Op-Amp Selection: To accurately pass signals up to 100 kHz without attenuation from the operational amplifier itself, an op-amp with a **Gain-Bandwidth Product (GBW)** of at least 1 MHz to 10 MHz is strongly recommended (e.g., TL071 or NE5532).

Final Designed Circuit Diagram:



Voltage-to-Current Converter

Q1. The circuit shown in Figure functions as a voltage-to-current converter. The op-amp used in this circuit is biased using ±20 V sources. The overall circuit operates in the linear region as long as V_{out} stays within the bias voltage range, and the concept of virtual short (i.e. $V_1 = V_2$) can be applied under such circumstances. If V_{out} attempts to exceed these limits, the amplifier enters saturation, and the virtual short assumption becomes invalid. Given the circuit parameters $R_1 = R_2 = 10$ kΩ, $R_3 = R_4 = 1$ kΩ and $R_{\text{L}} = 100$ Ω, calculate the load current i_{L} and output voltage V_{out} for and input voltage of $V_{\text{in}} = -5$ V.



(18 marks) [EEE 263 | L-2/T-1 | 2024–25]

Answer

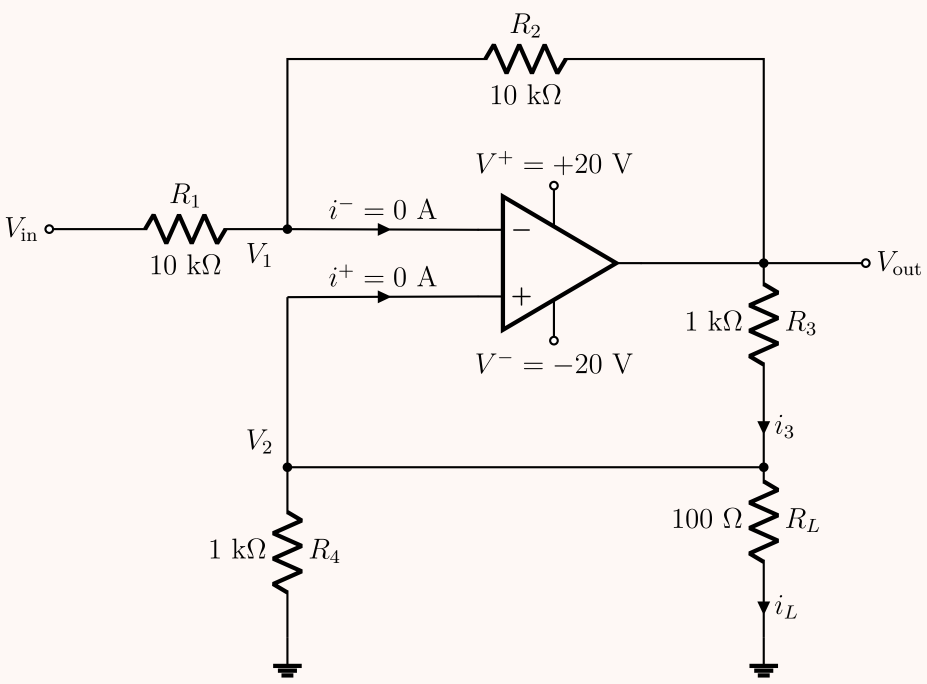
1. Theory & Operating Principle

The given circuit is a variation of the **Howland Current Pump**, which operates as a voltage-to-current converter. It utilizes both negative feedback (via R_2) and positive feedback (via R_3) to maintain a constant current through the load R_L, proportional to the input voltage, irrespective of small changes in the load resistance.

To analyze the circuit, we invoke the **Ideal Op-Amp Assumptions**:

- **Infinite Input Impedance:** The currents into the inverting and non-inverting terminals are zero ($i^+ = i^- = 0$ A).
- **Virtual Short:** As long as the op-amp operates in the linear region (output does not saturate), the feedback mechanism maintains a zero voltage difference between the input terminals ($V_1 = V_2$).

2. Circuit Diagram



3. Mathematical Derivation & Step-by-Step Solution

Step A: Node Voltage Analysis at the Inverting Terminal (V_1)

Applying Kirchhoff's Current Law (KCL) at node V_1:

$$\frac{V_1 - V_{\text{in}}}{R_1} + \frac{V_1 - V_{\text{out}}}{R_2} = 0$$

Given that $R_1 = R_2 = 10$ kΩ, we can multiply the entire equation by 10 kΩ to simplify:

$$V_1 - V_{\text{in}} + V_1 - V_{\text{out}} = 0$$
$$2V_1 - V_{\text{in}} = V_{\text{out}} \tag{1}$$

Step B: Node Voltage Analysis at the Non-Inverting Terminal (V_2)

Node V_2 is electrically connected to the junction of R_4, R_3, and R_L. Applying KCL at this supernode (sum of currents leaving = 0):

$$\frac{V_2 - V_{\text{out}}}{R_3} + \frac{V_2}{R_4} + \frac{V_2}{R_L} = 0$$
$$\frac{V_{\text{out}} - V_2}{R_3} = \frac{V_2}{R_4} + \frac{V_2}{R_L} \tag{2}$$

Step C: Substitute Virtual Short and Solve for V_2

By the virtual short concept, $V_1 = V_2$. Substituting $V_1 = V_2$ into Eq. (1):

$$V_{\text{out}} = 2V_2 - V_{\text{in}} \tag{3}$$

Now, substitute Eq. (3) into Eq. (2):

$$\frac{(2V_2 - V_{\text{in}}) - V_2}{R_3} = V_2\left(\frac{1}{R_4} + \frac{1}{R_L}\right)$$
$$\frac{V_2 - V_{\text{in}}}{R_3} = V_2\left(\frac{1}{R_4} + \frac{1}{R_L}\right)$$

We are given $V_{\text{in}} = -5$ V, $R_3 = 1$ kΩ, $R_4 = 1$ kΩ, and $R_L = 100$ Ω $= 0.1$ kΩ. Substituting these into Eq. (**??**):

$$\frac{V_2 - (-5)}{1 \text{ k}\Omega} = V_2\left(\frac{1}{1 \text{ k}\Omega} + \frac{1}{0.1 \text{ k}\Omega}\right)$$
$$V_2 + 5 = V_2(1 + 10)$$
$$V_2 + 5 = 11V_2$$
$$10V_2 = 5$$
$$V_2 = 0.5 \text{ V}$$

Step D: Calculate Load Current (i_L) and Output Voltage (V_{out})

The load current i_L is simply the current flowing through R_L due to the voltage V_2:

$$i_L = \frac{V_2}{R_L}$$
$$i_L = \frac{0.5 \text{ V}}{100 \text{ }\Omega} = 0.005 \text{ A} = \mathbf{5} \text{ mA}$$

Using Eq. (3), we can determine the output voltage V_{out}:

$$V_{\text{out}} = 2V_2 - V_{\text{in}}$$
$$V_{\text{out}} = 2(0.5 \text{ V}) - (-5 \text{ V})$$
$$V_{\text{out}} = 1 \text{ V} + 5 \text{ V} = \mathbf{6} \text{ V}$$

4. Verification of Operating Region

For the initial virtual short assumption to be valid, the op-amp must be operating in the linear region.

- The output voltage is calculated as $V_{out} = 6$ V.
- The op-amp power supplies are $V^+ = +20$ V and $V^- = -20$ V.
- Since $-20 \text{ V} \le 6 \text{ V} \le 20$ V, the op-amp has not saturated.

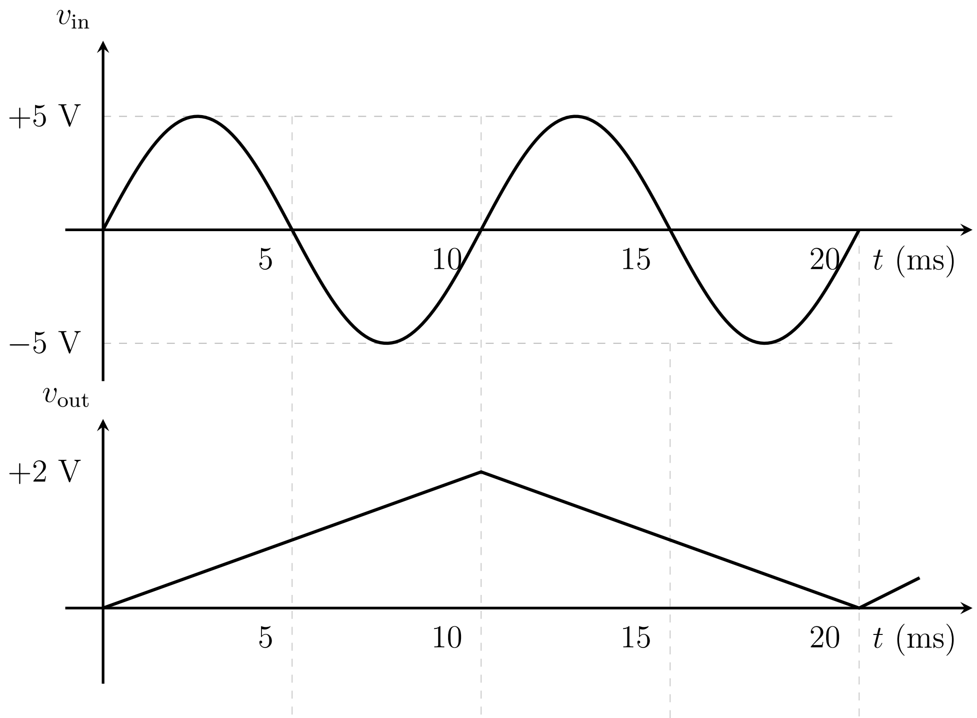
Thus, the virtual short assumption ($V_1 = V_2$) is mathematically verified and physically valid.

5. Final Answer

- **Load Current (i_L):** 5 mA
- **Output Voltage (V_{out}):** 6 V

System Design (Analog to Digital Interface)

Q1. Design a circuit using Op-Amp and Digital Circuits for given input wave shape v_{in} and output wave shape v_{out} in Figure . Assume a 5 V level to be digital 1 and -5 V level to be digital 0.



(26.67 marks) **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

1. Analysis of Input and Output Waveforms
From the provided graphs, we can determine the characteristics of the signals:

- **Input Waveform (v_{in}):** A sinusoidal wave with a peak voltage of ± 5 V. One complete cycle takes 10 ms, meaning the time period $T_{in} = 10$ ms and the frequency $f_{in} = 100$ Hz.
- **Output Waveform (v_{out}):** A triangular wave starting at 0 V, peaking at +2 V at $t = 10$ ms, and returning to 0 V at $t = 20$ ms. The time period is $T_{out} = 20$ ms, yielding a frequency $f_{out} = 50$ Hz.

Since the output frequency (50 Hz) is exactly half of the input frequency (100 Hz), the circuit must act as a **Frequency Divider**. Furthermore, converting a square/digital wave to a triangular wave requires an **Integrator**.

2. Circuit Principle and Operation
The design consists of three cascaded stages:

(a) **Zero-Crossing Detector (ZCD):** An open-loop operational amplifier compares v_{in} with ground (0 V). When $v_{in} > 0$ V, the output saturates to logic 1 (+5 V). When $v_{in} < 0$ V, it saturates to logic 0 (-5 V). This yields a 100 Hz square wave (v_1).

(b) **T-Flip-Flop (Frequency Divider):** A positive-edge-triggered T-Flip-Flop is used. Its 'T' input is tied to Logic 1 (+5 V), so it toggles its state on every rising edge of v_1 (which occurs at $t = 0, 10 \text{ ms}, 20 \text{ ms}, \ldots$). This produces a 50 Hz square wave. We take the inverted output $\overline{Q}$ (v_2), which gives -5 V for $0 < t < 10$ ms and +5 V for 10 ms $< t <$ 20 ms.

(c) **Ideal Integrator:** An inverting op-amp integrator integrates the ± 5 V, 50 Hz square wave to produce the desired triangular wave.

3. Mathematical Derivation & Component Selection
The output of an inverting integrator is given by:

$$v_{out}(t) = -\frac{1}{RC}\int v_2(t)\,dt + v_{out}(0) \tag{1}$$

For the interval $0 \le t \le 10$ ms, we require the output to ramp linearly from 0 V to +2 V. Assuming the initial capacitor voltage is 0 V ($v_{out}(0) = 0$), the required slope is:

$$\text{Slope} = \frac{\Delta v_{out}}{\Delta t} = \frac{2\text{ V} - 0\text{ V}}{10\text{ ms} - 0\text{ ms}} = 200\text{ V/s} \tag{2}$$

During this same interval, the input to the integrator is $v_2 = -5$ V. Substituting into the integrator derivative equation:

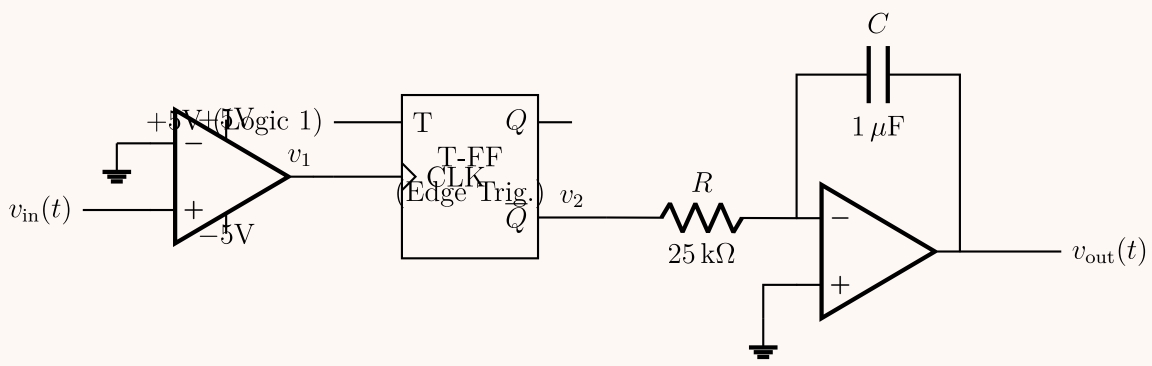
$$\frac{dv_{out}}{dt} = -\frac{v_2}{RC}$$
$$200 = -\frac{(-5)}{RC}$$
$$200 = \frac{5}{RC} \implies RC = \frac{5}{200} = 0.025\text{ s} = 25\text{ ms}$$

Component Selection: Let the integrating capacitor be $C = 1\,\mu$F.

$$R = \frac{0.025}{1 \times 10^{-6}} = 25,000\,\Omega = \mathbf{25\,k\Omega} \tag{3}$$

For the interval 10 ms $< t <$ 20 ms, the input $v_2 = +5$ V, yielding a slope of $-\frac{5}{0.025} = -200$ V/s, which perfectly brings the output back to 0 V at $t = 20$ ms.

4. Complete Circuit Diagram



Q2. A circuit is to be designed where depending on the range of input voltage a digital sequence is passed to a computer. The ranges and corresponding digital sequences are:

Range	Digital Sequence
0 – 1.2 V	010
1.2 – 3.2 V	110
3.2 – 3.5 V	001
3.5 – 5 V	111

Design a circuit using op-amps and digital circuits (gates, multiplexers) to perform the task. Assume 5 V = digital 1 and 0 V = digital 0. **(25 marks)** **[EEE 263 | L-2/T-1 | 2016–17]**

Answer

1. Theory & Principle of Operation

The required circuit maps continuous analog voltage ranges into specific 3-bit digital sequences ($D_2D_1D_0$). This function is fundamentally an **Analog-to-Digital Converter (ADC)**. Given the distinct thresholds, a **Flash ADC** architecture is highly suitable. The design consists of two main stages:

(a) **Comparator Stage (Analog):** A resistor voltage divider establishes the threshold voltages (V_{ref}) at 1.2 V, 3.2 V, and 3.5 V. Three op-amps functioning as open-loop comparators compare the input voltage (V_{in}) against these thresholds, generating a "thermometer code" ($C_3C_2C_1$).

(b) **Encoder Stage (Digital):** A combinational logic circuit (using basic logic gates) translates the intermediate thermometer code into the requested 3-bit digital sequence ($D_2D_1D_0$).

2. Step-by-Step Circuit Design

Step 2.1: Resistor Ladder Design

We assume a stable supply voltage of $V_{CC} = 5$ V (which corresponds to digital '1'). To generate the reference voltages, we use a series resistor network connected to ground (0 V). Let us choose a base current of $I = 1$ mA flowing through the resistor string to obtain practical standard resistor values. Using Ohm's law ($R = \frac{\Delta V}{I}$):

$$\begin{aligned} R_1 &= \frac{1.2\text{ V} - 0\text{ V}}{1\text{ mA}} = 1.2\text{ k}\Omega \\ R_2 &= \frac{3.2\text{ V} - 1.2\text{ V}}{1\text{ mA}} = 2.0\text{ k}\Omega \\ R_3 &= \frac{3.5\text{ V} - 3.2\text{ V}}{1\text{ mA}} = 0.3\text{ k}\Omega\ (300\ \Omega) \\ R_4 &= \frac{5.0\text{ V} - 3.5\text{ V}}{1\text{ mA}} = 1.5\text{ k}\Omega \end{aligned}$$

Total resistance is $\sum R = 5$ kΩ, verifying our 1 mA assumption.

Step 2.2: Intermediate Thermometer Code Generation
Three comparators yield outputs C_1, C_2, C_3 corresponding to the respective thresholds.

- $C_1 = 1$ if $V_{\text{in}} > 1.2$ V
- $C_2 = 1$ if $V_{\text{in}} > 3.2$ V
- $C_3 = 1$ if $V_{\text{in}} > 3.5$ V

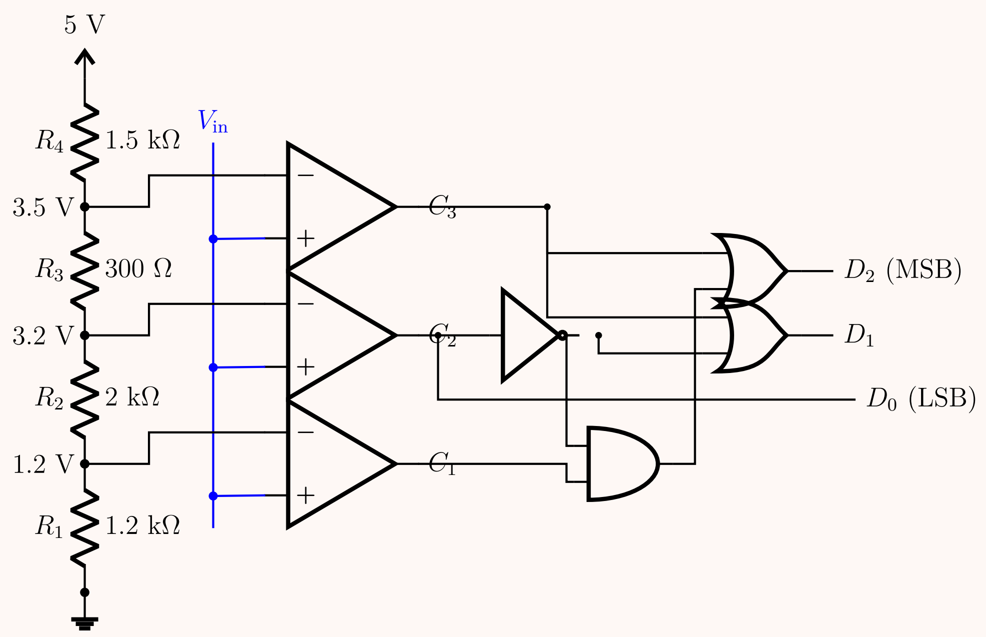
Step 2.3: Digital Logic Synthesis
We construct a truth table correlating the input ranges, the intermediate comparator states (C_3, C_2, C_1), and the desired digital output (D_2, D_1, D_0):

Input Voltage Range	C_3	C_2	C_1	D_2	D_1	D_0
$0 \le V_{\text{in}} < 1.2$ V	0	0	0	0	1	0
$1.2 \le V_{\text{in}} < 3.2$ V	0	0	1	1	1	0
$3.2 \le V_{\text{in}} < 3.5$ V	0	1	1	0	0	1
$3.5 \le V_{\text{in}} \le 5$ V	1	1	1	1	1	1

By analyzing the truth table, we derive the minimized Boolean expressions for the outputs:

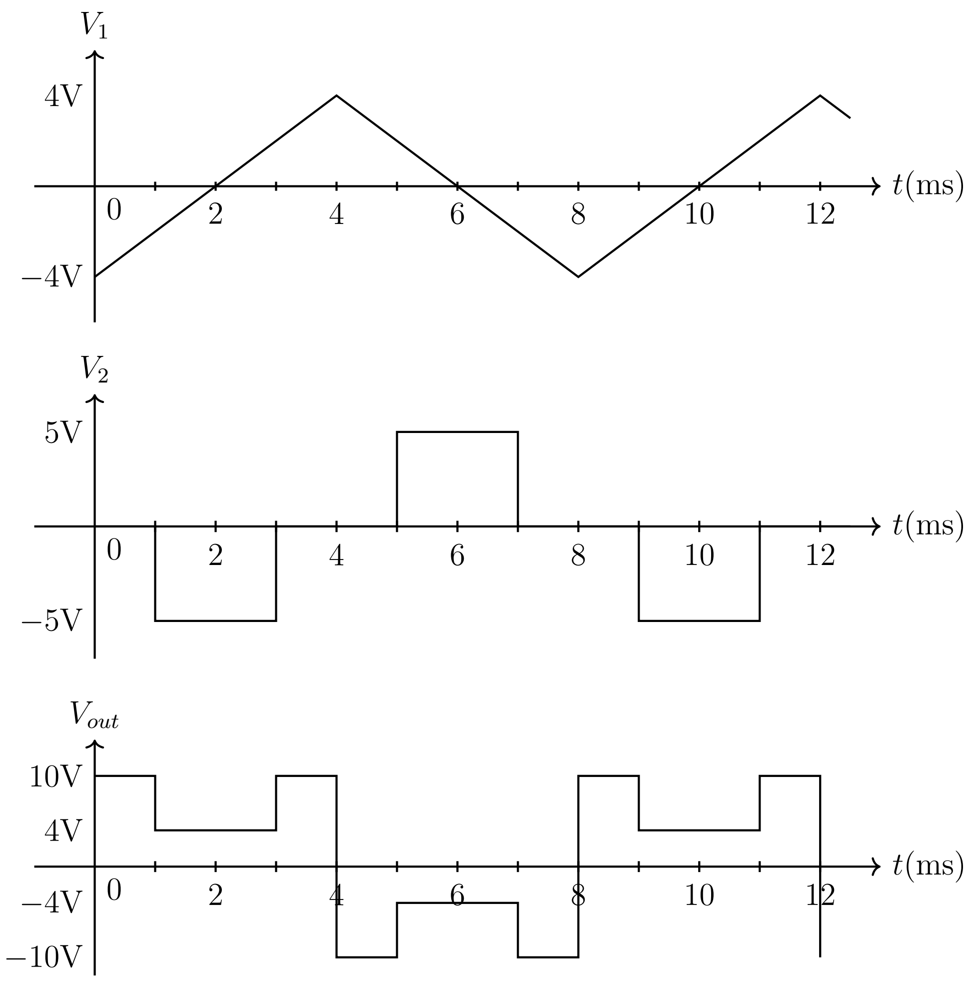
- **For D_0:** It is high (1) only when $C_2 = 1$ (states 011 and 111).
$$D_0 = C_2 \tag{1}$$
- **For D_1:** It is low (0) only for state 011. It is high when $C_2 = 0$ or when $C_3 = 1$.
$$D_1 = \overline{C_2} + C_3 \tag{2}$$
- **For D_2:** It is high for state 001 and 111. The state 001 can be isolated using $C_1 \cdot \overline{C_2}$. State 111 occurs when $C_3 = 1$.
$$D_2 = (C_1 \cdot \overline{C_2}) + C_3 \tag{3}$$

3. Complete Circuit Schematic



Waveform Realization

Q1. Determine an op-amp circuit realization for given input wave shapes V_1 and V_2 and output wave shape V_{out}.



(15 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Waveform Analysis

To design the op-amp circuit, we first need to determine the mathematical relationship between the output V_{out} and the inputs V_1 and V_2.

Analysis of V_1: $V_1(t)$ is a triangular wave. Its derivative is piecewise constant (a square wave).

- From $t = 0$ to 4 ms: The voltage goes from -4 V to 4 V.
$$\frac{dV_1}{dt} = \frac{4\text{ V} - (-4\text{ V})}{4\text{ ms}} = 2\text{ V/ms} = 2000\text{ V/s}$$
- From $t = 4$ to 8 ms: The voltage goes from 4 V to -4 V.
$$\frac{dV_1}{dt} = \frac{-4\text{ V} - 4\text{ V}}{4\text{ ms}} = -2\text{ V/ms} = -2000\text{ V/s}$$

Notice that V_{out} is also piecewise constant. This strongly suggests that V_{out} depends on the derivative of V_1 rather than V_1 itself.

Formulating the Equation: Let us assume a linear combination of the form:

$$V_{out}(t) = K_1\frac{dV_1}{dt} + K_2V_2(t) \tag{1}$$

We can find the constants K_1 and K_2 by testing specific time intervals from the given graphs:

- **Interval $t \in (0, 1)$ ms:** $\frac{dV_1}{dt} = 2000$ V/s, $V_2 = 0$ V, and $V_{out} = 10$ V.
$$10 = K_1(2000) + K_2(0) \implies K_1 = \frac{10}{2000} = 5 \times 10^{-3}\text{ s} = 5\text{ ms}$$
- **Interval $t \in (1, 3)$ ms:** $\frac{dV_1}{dt} = 2000$ V/s, $V_2 = -5$ V, and $V_{out} = 4$ V.
$$4 = (5 \times 10^{-3})(2000) + K_2(-5) \implies 4 = 10 - 5K_2 \implies 5K_2 = 6 \implies K_2 = 1.2$$

To verify, let's test another interval, say $t \in (5, 7)$ ms: $\frac{dV_1}{dt} = -2000$ V/s, $V_2 = 5$ V.

$$V_{out} = (5 \times 10^{-3})(-2000) + 1.2(5) = -10 + 6 = -4\text{ V}$$

This matches the V_{out} graph perfectly! Thus, the target equation is:

$$V_{out} = 5\text{ ms} \cdot \frac{dV_1}{dt} + 1.2V_2 \tag{2}$$

—

2. Circuit Synthesis

We can synthesize this mathematical operation using two op-amp stages: an inverting differentiator and a difference amplifier.
Stage 1: Inverting Differentiator An ideal inverting differentiator produces an output $V_A = -R_1C_1\frac{dV_1}{dt}$. To obtain the necessary scaling factor of 5 ms, we set the time constant:

$$\tau = R_1C_1 = 5 \times 10^{-3} \text{ s}$$

Let us choose a standard capacitor value $C_1 = 1\,\mu$F.

$$R_1 = \frac{5 \times 10^{-3}}{1 \times 10^{-6}} = 5 \text{ k}\Omega$$

The output of this stage is $V_A = -5 \text{ ms} \cdot \frac{dV_1}{dt}$. Substituting this into our target equation gives:

$$V_{out} = -V_A + 1.2V_2$$

Stage 2: Difference Amplifier We need a circuit that subtracts V_A and scales V_2. A standard difference amplifier performs the operation:

$$V_{out} = -\left(\frac{R_3}{R_2}\right)V_A + \left(1 + \frac{R_3}{R_2}\right)\left(\frac{R_5}{R_4 + R_5}\right)V_2$$

Comparing this to $V_{out} = -V_A + 1.2V_2$:

(a) **Inverting Gain:** $\frac{R_3}{R_2} = 1$. We can choose $R_2 = R_3 = 10$ kΩ.

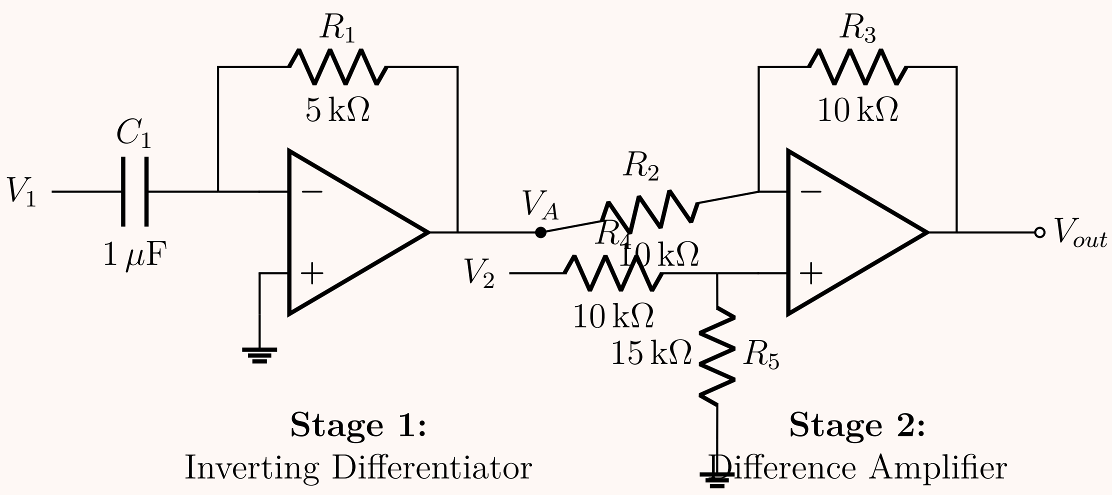
(b) **Non-Inverting Gain:** With $\frac{R_3}{R_2} = 1$, the non-inverting gain is $2 \cdot \frac{R_5}{R_4+R_5} = 1.2$.

$$\frac{R_5}{R_4 + R_5} = 0.6 \implies R_5 = 0.6R_4 + 0.6R_5 \implies 0.4R_5 = 0.6R_4 \implies R_5 = 1.5R_4$$

If we select $R_4 = 10$ kΩ, then $R_5 = 15$ kΩ.

—

3. Complete Circuit Diagram



Integrator

Q1. Design a circuit using ideal operational amplifiers that implements the following equation:

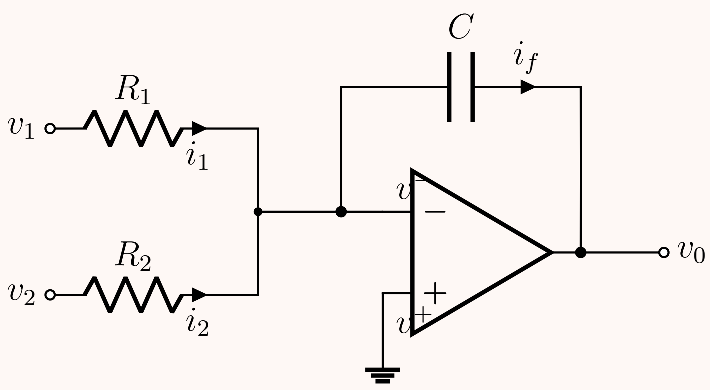
$$v_0 \propto \int (v_1 + v_2)\,dt$$

Write the final output expression in terms of the circuit components used in your design. **(20 marks)** **[EEE 263 | L-2/T-1 | 2020–21]**

Answer

1. Circuit Design & Theory
To implement the mathematical operation $v_0 \propto \int(v_1 + v_2)\,dt$, the circuit must perform two primary functions simultaneously: **addition** (summing) and **integration** over time. This is achieved using an **Inverting Summing Integrator**.
The circuit utilizes an operational amplifier with a capacitor in the negative feedback loop (providing the integration) and multiple input resistors connected to the inverting terminal (providing the summing operation).

2. Circuit Diagram



3. Mathematical Derivation
Ideal Op-Amp Assumptions:

- **Infinite Input Impedance:** The current entering the op-amp input terminals is zero ($i_{in} = 0$).
- **Infinite Open-Loop Gain:** The op-amp maintains a **virtual short** between its input terminals. Since the non-inverting terminal is grounded ($v^+ = 0$ V), the inverting terminal is at **virtual ground** ($v^- = 0$ V).

Applying Kirchhoff's Current Law (KCL) at the inverting node (v^-), we equate the sum of currents entering the node to the current leaving through the feedback loop:

$$i_1 + i_2 = i_f \tag{1}$$

Expressing these currents in terms of the node voltages and component values:

$$\frac{v_1 - v^-}{R_1} + \frac{v_2 - v^-}{R_2} = C\frac{d(v^- - v_0)}{dt}$$

Applying the virtual ground condition ($v^- = 0$ V):

$$\frac{v_1 - 0}{R_1} + \frac{v_2 - 0}{R_2} = C\frac{d(0 - v_0)}{dt}$$
$$\frac{v_1}{R_1} + \frac{v_2}{R_2} = -C\frac{dv_0}{dt}$$

Rearranging the equation to solve for the derivative of the output voltage (dv_0/dt):

$$\frac{dv_0}{dt} = -\frac{1}{C}\left(\frac{v_1}{R_1} + \frac{v_2}{R_2}\right)$$

Integrating both sides with respect to time t:

$$\int_0^t \frac{dv_0}{dt}\,dt = -\int_0^t \left(\frac{v_1}{R_1C} + \frac{v_2}{R_2C}\right) dt$$
$$v_0(t) - v_0(0) = -\int_0^t \left(\frac{v_1}{R_1C} + \frac{v_2}{R_2C}\right) dt$$

where $v_0(0)$ represents the initial output voltage at $t = 0$ (the initial voltage across the capacitor).

4. Final Answer
To satisfy the specific problem requirement that the output must be proportional to the direct sum of the inputs, $v_0 \propto \int(v_1 + v_2)\,dt$, both input branches must be weighted equally. We achieve this by selecting identical values for both input resistors:

$$R_1 = R_2 = R \tag{2}$$

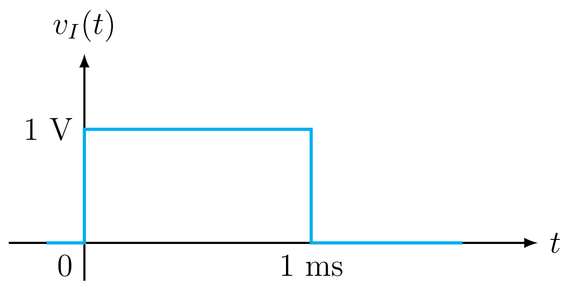
Substituting this condition into our derived equation yields the final output expression:

$$v_0(t) = -\frac{1}{RC}\int_0^t (v_1 + v_2)\,dt + v_0(0) \tag{3}$$

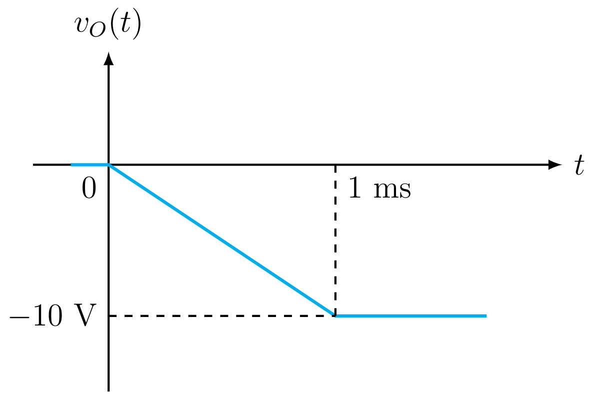
Conclusion & Notes:

- The output voltage v_0 is proportional to the integral of the sum of the inputs.
- The constant of proportionality is $\mathbf{k = -1/RC}$.
- The negative sign indicates a 180° phase shift (inversion), which is inherent to the inverting op-amp topology. If a strictly positive proportionality constant is required for a specific application, a simple inverting amplifier stage (with a gain of -1) can be cascaded at the output node v_0.

Q2. The input voltage v_I and output voltage v_O of an op-amp based circuit are given. Design the circuit.



(a)



(b)

(17 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Answer

1. Waveform Analysis

From the given figures, we can express the input voltage $v_I(t)$ and the output voltage $v_O(t)$ over the interval $0 \le t \le 1$ ms as follows:

$$v_I(t) = 1 \text{ V} \quad \text{for } 0 < t < 1 \text{ ms}$$
$$v_O(t) = -10 \times 10^3\, t \text{ V} \quad \text{for } 0 \le t \le 1 \text{ ms}$$

Note that the output is a decreasing linear ramp while the input is a constant positive voltage. Taking the derivative of the output with respect to time gives a constant:

$$\frac{dv_O(t)}{dt} = -10 \times 10^3 \text{ V/s} \tag{1}$$

Because the derivative of the output is proportional to the inverted input voltage, or conversely, the output is proportional to the negative integral of the input voltage, this mathematical operation is characteristic of an **Inverting Integrator**.

2. Circuit Theory and Derivation
Ideal Op-Amp Assumptions:

- **Infinite Input Impedance:** Current into the op-amp terminals is zero ($i_{in} = 0$).
- **Infinite Open-Loop Gain:** The op-amp forces the voltage difference between its input terminals to zero. Because the non-inverting terminal is grounded ($v^+ = 0$), the inverting terminal is at **virtual ground** ($v^- = 0$).

For an inverting integrator consisting of an input resistor R and a feedback capacitor C, we apply Kirchhoff's Current Law (KCL) at the inverting node v^-:

$$i_R + i_C = 0$$
$$\frac{v_I(t) - v^-}{R} + C\frac{d(v_O(t) - v^-)}{dt} = 0$$

Applying the virtual ground principle ($v^- = 0$):

$$\frac{v_I(t)}{R} + C\frac{dv_O(t)}{dt} = 0$$
$$\frac{dv_O(t)}{dt} = -\frac{1}{RC}v_I(t)$$

Integrating both sides with respect to time t, and assuming the capacitor is initially uncharged ($v_O(0) = 0$):

$$v_O(t) = -\frac{1}{RC}\int_0^t v_I(\tau)\, d\tau \tag{2}$$

3. Component Design

We use the differential form derived in Equation (??) and evaluate it using the given waveforms for the interval $0 < t < 1$ ms:

$$-10 \times 10^3 = -\frac{1}{RC}(1 \text{ V})$$
$$RC = \frac{1}{10 \times 10^3}$$
$$RC = 10^{-4} \text{ s} = 0.1 \text{ ms}$$

We need to select standard component values that satisfy this time constant constraint. Let us choose a practical capacitor value, say $C = 10$ **nF** (10×10^{-9} F). We can then solve for R:

$$R = \frac{10^{-4}}{C} = \frac{10^{-4}}{10 \times 10^{-9}} = \frac{10^{-4}}{10^{-8}} = 10^4\,\Omega$$
$$R = 10\text{ k}\Omega$$

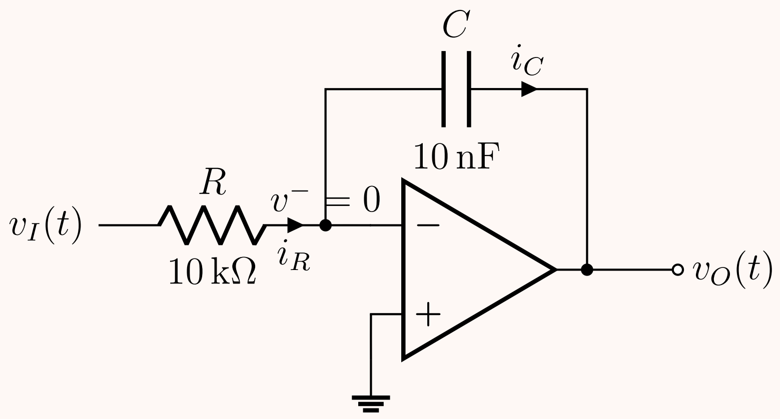
Let's verify the operation for $t > 1$ ms: Since $v_I(t) = 0$ V for $t > 1$ ms, the integral of zero is zero, meaning $v_O(t)$ will hold its previous value.

$$v_O(t) = v_O(1\text{ ms}) - \frac{1}{RC}\int_{1\text{ ms}}^{t} 0\,d\tau = -10\text{ V}$$

This perfectly matches the given output waveform in figure (b).

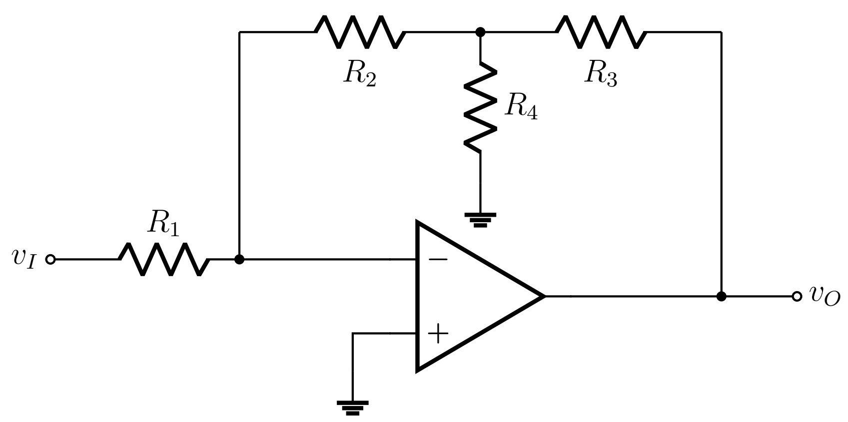
4. Final Circuit Diagram

The designed circuit is an Inverting Integrator with $R = 10$ kΩ and $C = 10$ nF.



Op-Amp Network Analysis

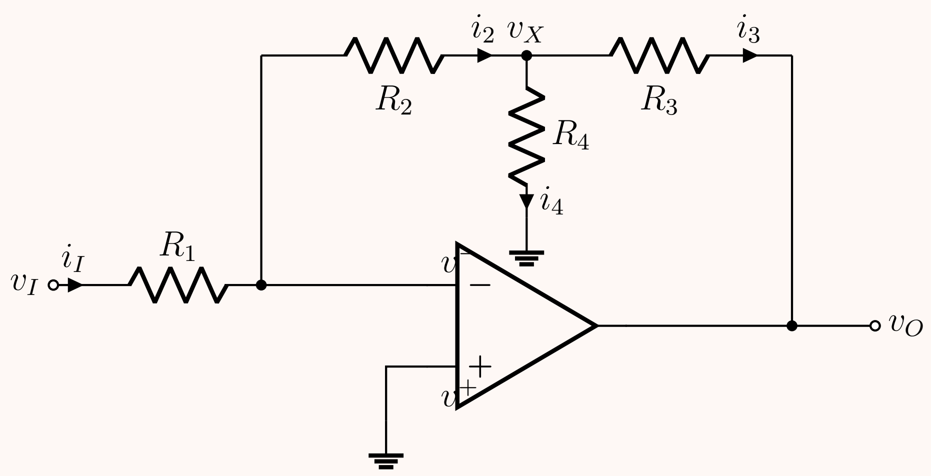
Q1. Determine the closed-loop voltage gain and current gain of the ideal op-amp network shown in the figure.



(23 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Answer

1. Circuit Diagram



2. Theory & Ideal Op-Amp Assumptions

To analyze the circuit, we apply the properties of an **ideal operational amplifier**:

- **Infinite Input Impedance:** The currents flowing into the op-amp input terminals are zero ($i_{in}^+ = i_{in}^- = 0$).
- **Infinite Open-Loop Gain:** The op-amp maintains a **virtual short** between its input terminals, forcing $v^+ = v^-$.

Since the non-inverting terminal is grounded ($v^+ = 0$ V), the inverting terminal is at **virtual ground**, meaning $v^- = 0$ V.

3. Mathematical Derivation
Let the input current be i_I and the voltage at the T-junction of the feedback network be v_X.
Step 1: Determine the input current i_I
The current flowing through the input resistor R_1 is driven by the potential difference between the input voltage v_I and the virtual ground v^-:

$$i_I = \frac{v_I - v^-}{R_1} = \frac{v_I - 0}{R_1} = \frac{v_I}{R_1} \tag{1}$$

Step 2: Determine the voltage v_X at the T-junction
Because the op-amp draws no current ($i_{in}^- = 0$), the entirety of the input current i_I must flow through resistor R_2. Thus, $i_2 = i_I$. The voltage at the T-junction, v_X, is found by analyzing the voltage drop across R_2:

$$\begin{aligned} v^- - v_X &= i_I R_2 \\ 0 - v_X &= i_I R_2 \\ v_X &= -i_I R_2 = -v_I \frac{R_2}{R_1} \end{aligned}$$

Step 3: Apply Kirchhoff's Current Law (KCL) at node v_X
At the T-junction (v_X), the current i_2 entering the node splits into i_4 (flowing down through R_4) and i_3 (flowing right through R_3):

$$i_2 = i_4 + i_3 \implies i_I = \frac{v_X}{R_4} + i_3 \tag{2}$$

Rearranging to solve for the feedback current i_3:

$$\begin{aligned} i_3 &= i_I - \frac{v_X}{R_4} \\ i_3 &= i_I - \frac{-i_I R_2}{R_4} \\ i_3 &= i_I \left(1 + \frac{R_2}{R_4}\right) \end{aligned} \tag{3}$$

Step 4: Determine the output voltage v_O
The output voltage is the potential at node v_X minus the voltage drop across R_3:

$$\begin{aligned} v_O &= v_X - i_3 R_3 \\ v_O &= -i_I R_2 - i_I \left(1 + \frac{R_2}{R_4}\right) R_3 \\ v_O &= -i_I \left(R_2 + R_3 + \frac{R_2 R_3}{R_4}\right) \end{aligned}$$

Substituting Equation (1) ($i_I = v_I/R_1$) into Equation (**??**):

$$v_O = -\frac{v_I}{R_1} \left(R_2 + R_3 + \frac{R_2 R_3}{R_4}\right) \tag{4}$$

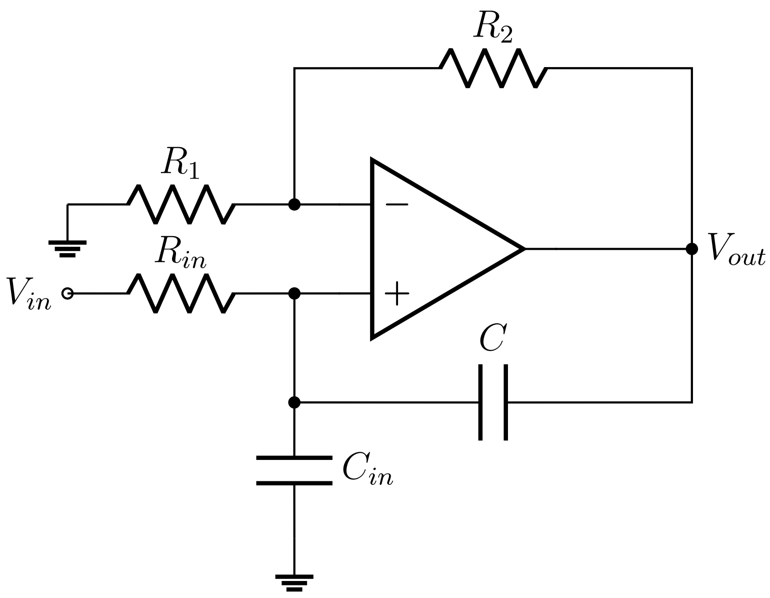
4. Final Answers
Closed-Loop Voltage Gain (A_v):
The voltage gain is the ratio of the output voltage to the input voltage:

$$A_v = \frac{v_O}{v_I} = -\frac{1}{R_1} \left(R_2 + R_3 + \frac{R_2 R_3}{R_4}\right) \tag{5}$$

Note: The T-network behaves equivalently to a single feedback resistor of value $R_{eq} = R_2 + R_3 + \frac{R_2 R_3}{R_4}$. This topology allows for very large effective feedback resistance (and thus high gain) without requiring physically large and noisy resistor values.
Current Gain (A_i):
In the context of a T-network feedback amplifier, the current gain is typically defined as the amplification factor of the current passing through the feedback loop, which is the ratio of the output-side feedback current (i_3) to the input current (i_I). From Equation (3):

$$A_i = \frac{i_3}{i_I} = 1 + \frac{R_2}{R_4} \tag{6}$$

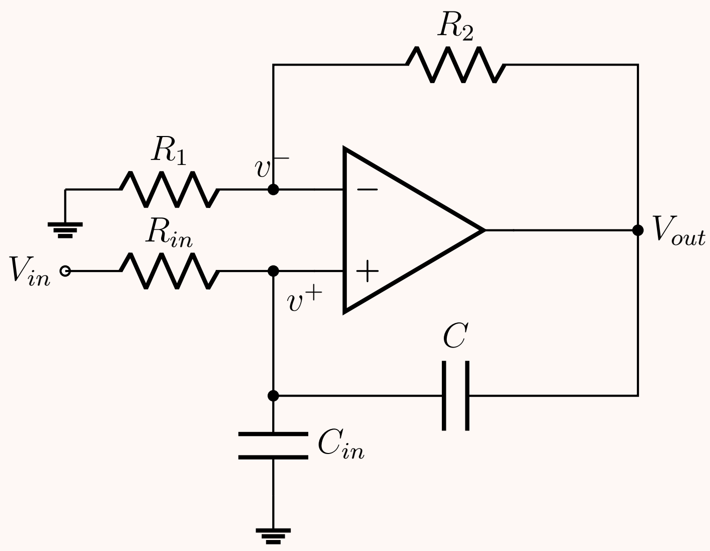
Q2. Determine the differential equation solved by the circuit shown: (i) when the capacitor C is absent; (ii) when the capacitor C is present. At what value of C does the circuit become a non-inverting amplifier?



(21 marks) [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Circuit Diagram



2. Theory & Ideal Op-Amp Assumptions

To derive the transfer function and corresponding differential equations, we assume an ideal operational amplifier:

- **Infinite Input Impedance:** $i_{in}^+ = i_{in}^- = 0$
- **Virtual Short Principle:** The op-amp maintains $v^+ = v^-$.

Inverting Terminal Analysis (v^-):

Applying the voltage divider rule to the non-inverting terminal network (or applying KCL at node v^-):

$$v^- = V_{out}\left(\frac{R_1}{R_1 + R_2}\right)$$

By the virtual short principle, $v^+ = v^-$. Therefore:

$$v^+ = V_{out}\left(\frac{R_1}{R_1 + R_2}\right) = \frac{V_{out}}{A_{v,ni}} \tag{1}$$

where $A_{v,ni} = 1 + \frac{R_2}{R_1}$ is the standard non-inverting amplifier gain.

3. Part (i): Capacitor C is absent ($C = 0$)

When the feedback capacitor C is absent (open circuit), the circuit at the non-inverting terminal simplifies to an RC low-pass filter driven by V_{in}. Applying KCL at node v^+:

$$\frac{V_{in} - v^+}{R_{in}} = C_{in}\frac{dv^+}{dt}$$
$$V_{in} - v^+ = R_{in}C_{in}\frac{dv^+}{dt}$$

Substitute v^+ from Equation (1) into the KCL equation:

$$V_{in} - \frac{V_{out}}{A_{v,ni}} = R_{in}C_{in}\frac{d}{dt}\left(\frac{V_{out}}{A_{v,ni}}\right)$$
$$V_{in} = \frac{1}{A_{v,ni}}V_{out} + \frac{R_{in}C_{in}}{A_{v,ni}}\frac{dV_{out}}{dt}$$

Multiplying by $A_{v,ni}$, we get the differential equation:

$$A_{v,ni}V_{in} = V_{out} + R_{in}C_{in}\frac{dV_{out}}{dt} \tag{2}$$

where $A_{v,ni} = 1 + R_2/R_1$. This is a first-order linear differential equation representing a low-pass filter with DC gain $A_{v,ni}$.

4. Part (ii): Capacitor C is present
When C is present, it introduces positive feedback to the non-inverting terminal. Applying KCL at node v^+:

$$\begin{aligned}\frac{V_{in} - v^+}{R_{in}} + C\frac{d(V_{out} - v^+)}{dt} &= C_{in}\frac{dv^+}{dt}\\ \frac{V_{in}}{R_{in}} - \frac{v^+}{R_{in}} + C\frac{dV_{out}}{dt} - C\frac{dv^+}{dt} &= C_{in}\frac{dv^+}{dt}\\ \frac{V_{in}}{R_{in}} + C\frac{dV_{out}}{dt} &= \frac{v^+}{R_{in}} + (C + C_{in})\frac{dv^+}{dt}\end{aligned}$$

Substitute $v^+ = \frac{V_{out}}{A_{v,ni}}$ into the equation:

$$\begin{aligned}\frac{V_{in}}{R_{in}} + C\frac{dV_{out}}{dt} &= \frac{V_{out}}{R_{in}A_{v,ni}} + (C + C_{in})\frac{d}{dt}\left(\frac{V_{out}}{A_{v,ni}}\right)\\ \frac{V_{in}}{R_{in}} &= \frac{V_{out}}{R_{in}A_{v,ni}} + \left[\frac{C + C_{in}}{A_{v,ni}} - C\right]\frac{dV_{out}}{dt}\end{aligned}$$

Multiplying the entire equation by $R_{in}A_{v,ni}$ yields the differential equation:

$$A_{v,ni}V_{in} = V_{out} + R_{in}\left(C_{in} + C(1 - A_{v,ni})\right)\frac{dV_{out}}{dt} \tag{3}$$

5. Condition for Non-Inverting Amplifier
For the circuit to behave as a standard, frequency-independent non-inverting amplifier, the output must be a scaled version of the input, meaning $V_{out} = AV_{in}$. This implies the derivative term (dV_{out}/dt) in the differential equation must be eliminated.
Setting the coefficient of dV_{out}/dt to zero:

$$\begin{aligned}C_{in} + C(1 - A_{v,ni}) &= 0\\ C(A_{v,ni} - 1) &= C_{in}\\ C &= \frac{C_{in}}{A_{v,ni} - 1}\end{aligned}$$

Substituting $A_{v,ni} = 1 + \frac{R_2}{R_1}$:

$$\begin{aligned}C &= \frac{C_{in}}{\left(1 + \frac{R_2}{R_1}\right) - 1}\\ C &= C_{in}\left(\frac{R_1}{R_2}\right)\end{aligned}$$

When this condition is met, the differential equation simplifies to $V_{out} = \left(1 + \frac{R_2}{R_1}\right)V_{in}$. The addition of capacitor C strategically cancels the low-pass filtering effect of C_{in}, resulting in a wideband non-inverting amplifier.

Op-Amp Practical Imperfections

Q1. What are the AC imperfections of a practical op-amp? How can we find out the value of input offset voltage and input offset current? **(15 marks)** [EEE 263 | L-2/T-1 | 2013–14]

Answer

1. AC Imperfections of a Practical Op-Amp
While an ideal operational amplifier is assumed to have infinite bandwidth and an ability to respond instantaneously to input changes, a practical op-amp suffers from several alternating current (AC) imperfections. The two most prominent AC imperfections are **Finite Bandwidth** and **Slew Rate**.
A. Finite Bandwidth (Frequency Response) A practical op-amp has a finite, frequency-dependent open-loop gain. The gain is very high at DC but starts to roll off at a relatively low frequency called the *break frequency* or *dominant pole frequency* (f_b). The open-loop gain as a function of frequency f is given by:

$$A(f) = \frac{A_0}{1 + j\left(\frac{f}{f_b}\right)} \tag{1}$$

where A_0 is the DC open-loop gain. Because of this roll-off, the product of the closed-loop gain and the closed-loop bandwidth is a constant, known as the **Gain-Bandwidth Product (GBW)**:

$$\text{GBW} = A_0 f_b \tag{2}$$

This implies that designing an amplifier with a higher closed-loop gain will inevitably result in a lower operating bandwidth.
B. Slew Rate (SR) The slew rate is the maximum rate at which the op-amp's output voltage can change in response to a step input. It is caused by the finite current available to charge and discharge the internal compensation capacitor of the op-amp.

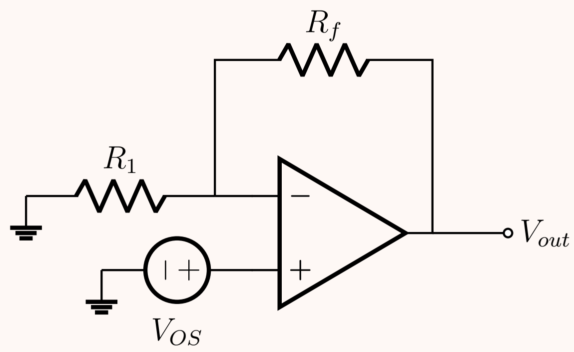
$$\text{SR} = \left.\frac{dv_{out}}{dt}\right|_{\max} \quad \text{(typically measured in V}/\mu\text{s)} \tag{3}$$

Slew rate limiting causes high-frequency, large-amplitude sine waves to distort into triangle waves. The maximum frequency f_{max} at which an un-distorted sine wave of peak amplitude V_p can be obtained is the *full-power bandwidth*:

$$f_{max} = \frac{\text{SR}}{2\pi V_p} \tag{4}$$

2. Measurement of DC Imperfections
Practical op-amps also suffer from DC imperfections, specifically Input Offset Voltage (V_{OS}) and Input Bias/Offset Currents (I_B, I_{OS}). We can measure these parameters using dedicated test circuits.

A. Measuring Input Offset Voltage (V_{OS})
The input offset voltage V_{OS} is the small DC differential voltage that exists between the input terminals when the output is exactly zero. To measure it, we amplify this small internal voltage using a high-gain closed-loop configuration with both external inputs grounded.



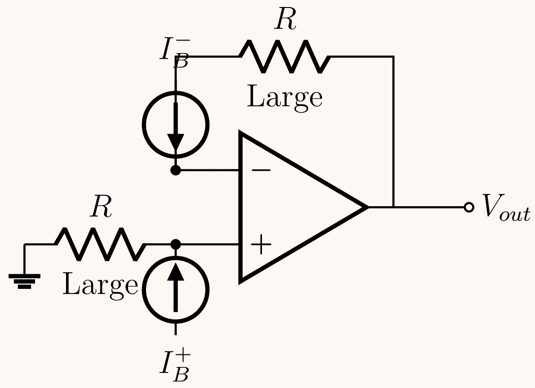
Derivation: In this circuit, the op-amp behaves as a non-inverting amplifier amplifying its own internal offset voltage V_{OS}. Assuming the effect of bias currents is negligible (which can be ensured by choosing small values for R_1 and R_f, e.g., $R_1 = 10\,\Omega$, $R_f = 10$ kΩ), the output voltage is:

$$V_{out} = V_{OS}\left(1 + \frac{R_f}{R_1}\right) \tag{5}$$

By measuring V_{out} using a standard voltmeter, we can compute the input offset voltage as:

$$V_{OS} = \frac{V_{out}}{1 + \frac{R_f}{R_1}} \tag{6}$$

B. Measuring Input Offset Current (I_{OS})
The input offset current is the difference between the bias currents entering the non-inverting and inverting terminals: $I_{OS} = |I_B^+ - I_B^-|$. To measure it, we force these currents to flow through very large, perfectly matched resistors (R) to convert the small currents into a measurable voltage. We assume V_{OS} has been nulled or mathematically subtracted prior to this measurement.



Derivation: The bias current I_B^+ flows entirely from the ground through the non-inverting terminal's resistor R. The voltage at the non-inverting terminal is:

$$v^+ = -I_B^+ R \tag{7}$$

Due to the virtual short principle, $v^- = v^+ = -I_B^+ R$. Now, applying Kirchhoff's Current Law (KCL) at the inverting node v^-:

$$\frac{v^- - V_{out}}{R} + I_B^- = 0$$
$$V_{out} = v^- + I_B^- R$$

Substitute $v^- = -I_B^+ R$ into the equation:

$$V_{out} = -I_B^+ R + I_B^- R$$
$$V_{out} = (I_B^- - I_B^+)R$$

By definition, the input offset current is $I_{OS} = |I_B^+ - I_B^-|$. Therefore, the magnitude of the measured output voltage directly yields the offset current:

$$I_{OS} = \frac{|V_{out}|}{R} \tag{8}$$

Using a very large value for R (e.g., 1 MΩ to 10 MΩ) ensures that V_{out} is large enough to measure accurately with standard instruments.

Dual Slope A/D Converter

Q1. By drawing appropriate circuits, explain the operation of a dual slope A/D converter. **(20 marks)** [EEE 263 | L-2/T-1 | 2020–21]

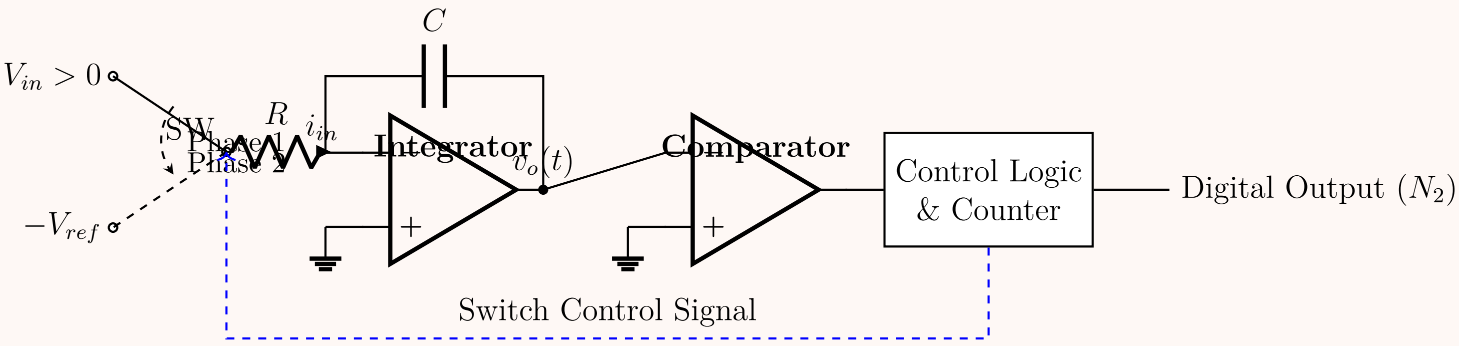
Answer

1. Principle of Operation

A **Dual Slope Analog-to-Digital Converter (ADC)** operates by performing two integrations. First, it integrates the unknown analog input voltage (V_{in}) for a fixed duration of time. Second, it de-integrates using a known, opposite-polarity reference voltage ($-V_{ref}$) until the output returns to zero. The time required for this second integration is directly proportional to the unknown input voltage. This variable time is measured using a digital counter, which yields the digital equivalent of the analog input.

2. Circuit Diagram

The architecture consists of an analog integrator, a zero-crossing comparator, a control logic block, and a digital counter.



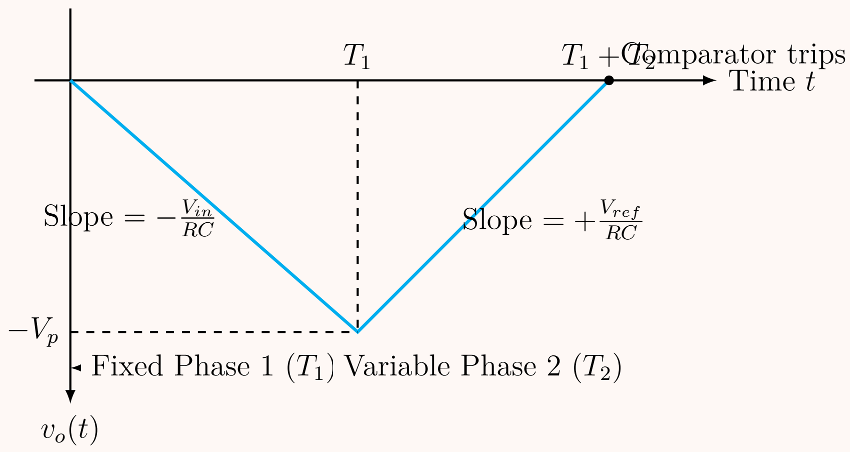
3. Operational Phases and Waveforms

Phase 1: Fixed Time Integration (Charging)

- At $t = 0$, the switch (SW) connects to the unknown positive input voltage V_{in}.
- The counter starts from zero. The integrator integrates V_{in}, producing a negative-going ramp at $v_o(t)$ since the inverting configuration causes a phase shift.
- This phase lasts for a fixed duration T_1, determined by the counter reaching a preset maximum count N_1 (where $T_1 = N_1 \times T_{clk}$).

Phase 2: Variable Time De-Integration (Discharging)

- At time $t = T_1$, the control logic resets the counter to zero and throws the switch SW to connect to a negative reference voltage $-V_{ref}$.
- The integrator now integrates a negative voltage, causing the output $v_o(t)$ to ramp upwards towards zero.
- When $v_o(t)$ crosses 0 V, the comparator output flips, signaling the control logic to freeze the counter.
- The time taken for this phase is T_2, corresponding to a count N_2 (where $T_2 = N_2 \times T_{clk}$).



4. Mathematical Derivation

Assuming the capacitor is initially uncharged ($v_o(0) = 0$), the output voltage at the end of Phase 1 ($t = T_1$) is:

$$v_o(T_1) = -\frac{1}{RC}\int_0^{T_1} V_{in}\,dt = -\frac{V_{in}}{RC}T_1 \tag{1}$$

During Phase 2, the input is $-V_{ref}$. The output voltage at the end of Phase 2 ($t = T_1 + T_2$) reaches exactly zero:

$$v_o(T_1 + T_2) = v_o(T_1) - \frac{1}{RC}\int_{T_1}^{T_1+T_2} (-V_{ref})\,dt = 0$$

$$-\frac{V_{in}}{RC}T_1 + \frac{V_{ref}}{RC}T_2 = 0$$

Equating the two magnitudes and canceling out the RC time constant:

$$V_{in}T_1 = V_{ref}T_2 \implies V_{in} = V_{ref}\frac{T_2}{T_1} \tag{2}$$

Since the times T_1 and T_2 are generated by counting a clock of period T_{clk}:

$$T_1 = N_1 T_{clk} \quad \text{and} \quad T_2 = N_2 T_{clk} \tag{3}$$

Substitute these into the equation:

$$V_{in} = V_{ref}\left(\frac{N_2 T_{clk}}{N_1 T_{clk}}\right) = V_{ref}\left(\frac{N_2}{N_1}\right) \tag{4}$$

5. Key Advantages

- **High Accuracy & Stability:** The final equation does not depend on the values of R, C, or the exact clock frequency f_{clk}, provided they remain stable during the single conversion cycle.
- **Excellent Noise Rejection:** Due to the integration process, high-frequency noise is averaged out. Furthermore, if T_1 is set to an integer multiple of the local power line period (e.g., 20 ms for 50 Hz or 16.67 ms for 60 Hz), AC line noise is perfectly rejected.
- **Primary Application:** Digital multimeters and low-speed, high-resolution measurement systems.

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

Part V — Oscillators & Timers

Colpitts oscillator, 555 Timer astable design, triangular wave generators

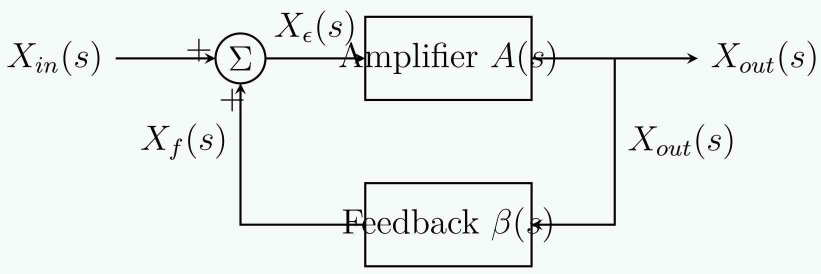
S8 Oscillators, Timers & Function Generators

Barkhausen Stability Criterion

Q1. Write down the Barkhausen criterion. **(10 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Introduction to Oscillator Feedback
An oscillator is a circuit that generates a continuous, repeated waveform without any external input signal. It achieves this by utilizing **positive feedback**.
Consider a basic feedback system block diagram:



The closed-loop transfer function for a system with positive feedback is given by:

$$A_f = \frac{X_{out}}{X_{in}} = \frac{A(s)}{1 - A(s)\beta(s)} \tag{1}$$

where $A(s)$ is the forward gain of the amplifier and $\beta(s)$ is the transfer function of the feedback network. The product $A(s)\beta(s)$ is known as the **loop gain**.

2. The Barkhausen Criterion
For the circuit to operate as an oscillator, it must produce an output even when the external input X_{in} is zero. Mathematically, this requires the closed-loop gain A_f to approach infinity. This occurs when the denominator of the transfer function is exactly zero:

$$1 - A(s)\beta(s) = 0 \implies A(s)\beta(s) = 1 \tag{2}$$

Since $A(j\omega)$ and $\beta(j\omega)$ are complex quantities at a specific frequency ω_0, the equation $A(j\omega_0)\beta(j\omega_0) = 1+j0$ dictates two fundamental conditions. These two conditions collectively form the **Barkhausen Criterion** for sustained oscillations:

Condition 1: Magnitude Requirement
The magnitude of the loop gain must be equal to unity.

$$|A(j\omega_0)\beta(j\omega_0)| = 1 \tag{3}$$

Practical Note: In practical circuit design, the initial loop gain is set slightly greater than 1 ($|A\beta| > 1$) so that oscillations can start and build up from ambient noise. As the signal amplitude grows, non-linearities in the amplifier reduce the gain until the steady-state condition $|A\beta| = 1$ is automatically satisfied.

Condition 2: Phase Shift Requirement
The total phase shift around the closed loop must be zero or an integer multiple of 360° (2π radians) to ensure the feedback is purely positive (regenerative).

$$\angle A(j\omega_0)\beta(j\omega_0) = 0°, 360°, 720°, \cdots = 2\pi n \quad \text{where } n = 0, 1, 2, \ldots \tag{4}$$

3. Summary of Operation When the Barkhausen criterion is satisfied at a particular frequency $f_0 = \frac{\omega_0}{2\pi}$:

- The feedback signal X_f arrives at the input exactly in phase with the original signal X_ϵ.
- The amplitude of the feedback signal is exactly enough to sustain the output without any external source.
- The circuit becomes an oscillator, generating a continuous sinusoidal output at frequency f_0.

Q2. Derive expression of Desensitivity. Explain positive and negative feedback from desensitivity. **(10+5 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Introduction and System Model
In a practical amplifier, the open-loop gain A is highly susceptible to variations due to changes in temperature, power supply voltage, or component aging. Feedback is employed to stabilize this gain. To quantify this effect, we analyze the basic block diagram of a feedback amplifier:

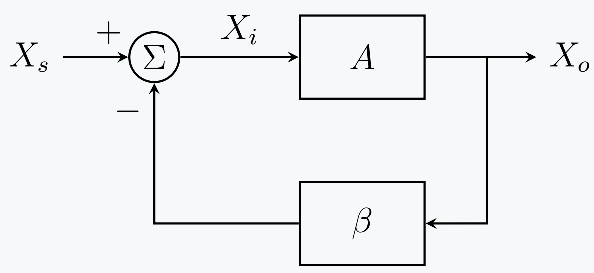


Figure: Basic Feedback Loop Topology

For a standard negative feedback configuration, the closed-loop gain A_f is given by:

$$A_f = \frac{X_o}{X_s} = \frac{A}{1 + A\beta} \tag{1}$$

where:

- A is the open-loop gain (forward path).
- β is the feedback factor (feedback path).
- $A\beta$ is defined as the *loop gain*.

2. Derivation of Desensitivity

To find how a percentage change in the open-loop gain (dA/A) affects the percentage change in the closed-loop gain (dA_f/A_f), we differentiate Equation (1) with respect to A:

$$\begin{aligned}
\frac{dA_f}{dA} &= \frac{d}{dA}\left(\frac{A}{1 + A\beta}\right) \\
&= \frac{1 \cdot (1 + A\beta) - A \cdot (\beta)}{(1 + A\beta)^2} \\
&= \frac{1 + A\beta - A\beta}{(1 + A\beta)^2} \\
\frac{dA_f}{dA} &= \frac{1}{(1 + A\beta)^2}
\end{aligned}$$

We can express the differential change in closed-loop gain dA_f as:

$$dA_f = \frac{dA}{(1 + A\beta)^2} \tag{2}$$

To find the fractional change (or percentage change), we divide both sides by A_f:

$$\frac{dA_f}{A_f} = \frac{\frac{dA}{(1+A\beta)^2}}{A_f}$$

Substitute $A_f = \frac{A}{1+A\beta}$ into the denominator on the right side:

$$\begin{aligned}
\frac{dA_f}{A_f} &= \frac{\frac{dA}{(1+A\beta)^2}}{\frac{A}{1+A\beta}} \\
\frac{dA_f}{A_f} &= \frac{dA}{(1 + A\beta)^2} \cdot \frac{1 + A\beta}{A} \\
\frac{dA_f}{A_f} &= \frac{1}{(1 + A\beta)}\left(\frac{dA}{A}\right)
\end{aligned}$$

Sensitivity ($S_A^{A_f}$): This is defined as the ratio of the fractional change in A_f to the fractional change in A:

$$S_A^{A_f} = \frac{dA_f/A_f}{dA/A} = \frac{1}{1 + A\beta} \tag{3}$$

Desensitivity (D): The factor by which the sensitivity is reduced is called the Desensitivity factor. It is the reciprocal of Sensitivity:

$$D = 1 + A\beta \tag{4}$$

3. Explanation of Positive and Negative Feedback using Desensitivity

The nature of the feedback (whether it is positive or negative) can be analyzed directly through the value of the Desensitivity factor $D = 1 + A\beta$. The loop gain $A\beta$ determines the operational regime.

A. Negative Feedback ($D > 1$)

- **Condition:** When the feedback signal subtracts from the input signal, the loop gain $A\beta$ is a positive quantity ($A\beta > 0$).

- **Desensitivity:** Consequently, $D = 1 + A\beta > 1$.
- **Effect:** Because $D > 1$, the sensitivity $S < 1$. This implies that $\frac{dA_f}{A_f} < \frac{dA}{A}$.
- **Physical Meaning:** The closed-loop gain is *desensitized* to variations in the active components. Even if the open-loop gain A varies drastically (e.g., by 20%), the closed-loop gain A_f will only vary by a small fraction (e.g., 1% if $D = 20$). This guarantees high stability, wider bandwidth, and reduced distortion, which are the primary advantages of negative feedback amplifiers.

B. Positive Feedback ($0 < D < 1$ **or** $D = 0$)

- **Condition:** When the feedback signal adds to the input signal, it is termed positive feedback. In the context of our derived equation, the term $A\beta$ becomes negative (i.e., $-1 \leq A\beta < 0$).
- **Desensitivity:** Therefore, the desensitivity factor falls in the range $0 < D < 1$.
- **Effect:** Because $D < 1$, the sensitivity $S > 1$. This implies that $\frac{dA_f}{A_f} > \frac{dA}{A}$.
- **Physical Meaning:** The closed-loop system becomes *hyper-sensitive* to any changes in A. A small variation in the internal amplifier creates a massive variation in the output.
- **Extreme Case** ($D = 0$)**:** If $A\beta = -1$, then $D = 1 - 1 = 0$. The sensitivity tends to infinity. The closed-loop gain $A_f \to \infty$. This is the Barkhausen criterion for oscillation. The circuit will generate an output even when the input is zero, functioning as an **oscillator** rather than a stable amplifier.

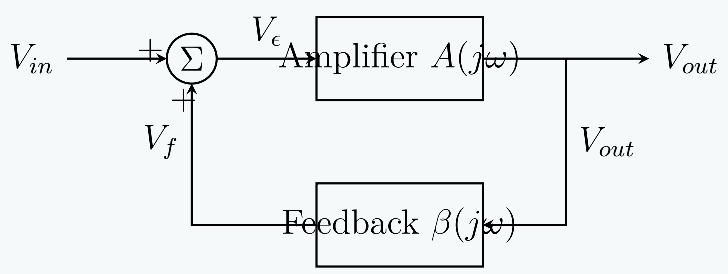
Q3. Write down the Barkhausen stability criterion for an oscillation in a linear electronic circuit to sustain. **(6 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Principle of Feedback Oscillators

An electronic oscillator is a circuit that produces a continuous, repeated, alternating waveform without any input signal. It achieves this by using a positive feedback system where a fraction of the output signal is fed back to the input in phase with the original signal.

Consider the general block diagram of a feedback amplifier:



For a positive feedback system, the closed-loop transfer function is given by:

$$A_f(j\omega) = \frac{V_{out}}{V_{in}} = \frac{A(j\omega)}{1 - A(j\omega)\beta(j\omega)} \tag{1}$$

where:

- $A(j\omega)$ is the forward open-loop gain of the amplifier.
- $\beta(j\omega)$ is the transfer function of the feedback network.
- $A(j\omega)\beta(j\omega)$ is known as the **loop gain**, often denoted as $L(j\omega)$.

2. The Barkhausen Stability Criterion

For an oscillator to sustain a continuous output without any external input ($V_{in} = 0$), the closed-loop gain $A_f(j\omega)$ must theoretically approach infinity. This happens when the denominator of the transfer function becomes zero:

$$1 - A(j\omega)\beta(j\omega) = 0$$
$$A(j\omega)\beta(j\omega) = 1$$

Since $A(j\omega)$ and $\beta(j\omega)$ are complex quantities, the condition $A(j\omega)\beta(j\omega) = 1 + j0$ dictates two fundamental requirements for steady-state sinusoidal oscillation at a specific frequency ω_0. These two requirements are collectively known as the **Barkhausen Criterion**:

Condition 1: Magnitude Requirement

The magnitude of the loop gain at the oscillation frequency ω_0 must be exactly equal to unity. This ensures that the signal neither grows indefinitely nor decays to zero.

$$|A(j\omega_0)\beta(j\omega_0)| = 1 \tag{2}$$

Condition 2: Phase Requirement

The total phase shift introduced around the closed loop must be zero or an integer multiple of 360° (2π radians). This ensures that the feedback signal V_f arrives perfectly in-phase to reinforce the original signal, resulting in constructive interference (positive feedback).

$$\angle [A(j\omega_0)\beta(j\omega_0)] = 2\pi n \quad \text{where } n = 0, 1, 2, \ldots \tag{3}$$

3. Practical Considerations for Start-up

In a practical electronic circuit, if the loop gain is set exactly to unity ($|A\beta| = 1$) from the start, the oscillations will never begin because there is no initial signal to amplify. Therefore, practical oscillators are designed such that:

- **At start-up:** The magnitude of the loop gain is deliberately set slightly greater than unity ($|A\beta| > 1$). This allows random thermal noise in the circuit to be amplified and the oscillations to exponentially grow.
- **Steady-state:** As the oscillation amplitude increases, the active devices (like BJTs, MOSFETs, or Op-Amps) begin to operate in their non-linear regions, causing the effective gain A to decrease. The amplitude stabilizes exactly at the level where the loop gain reduces to unity ($|A\beta| = 1$), thus fulfilling the Barkhausen criterion for sustained oscillations.

Wien Bridge Oscillator

Q1. Define Barkhausen criterion. Describe the operating principle of an op-amp based Wein bridge oscillator. Derive the conditions for Barkhausen criteria to be satisfied in this oscillator and the expression of oscillation frequency. **(17 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

1. The Barkhausen Criterion

The Barkhausen criterion states the fundamental conditions required for a linear feedback circuit to exhibit sustained, steady-state oscillations. For a feedback system with an amplifier gain A and a feedback factor β, the loop gain is given by $A\beta$. The criterion consists of two conditions:

(a) **Magnitude Condition:** The magnitude of the loop gain at the frequency of oscillation must be exactly unity.

$$|A\beta| = 1 \tag{1}$$

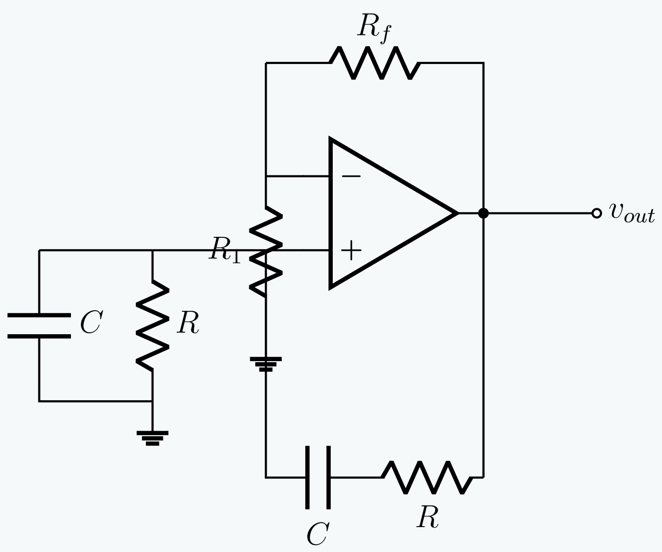
(b) **Phase Condition:** The total phase shift around the feedback loop must be zero or an integer multiple of 360° (2π radians) to ensure positive feedback.

$$\angle(A\beta) = 0^\circ, 360^\circ, 720^\circ, \cdots = 2\pi n \quad (\text{for } n = 0, 1, 2, \ldots) \tag{2}$$

2. Operating Principle of the Wien Bridge Oscillator

The Wien bridge oscillator is an RC oscillator widely used to generate low-frequency sine waves. It utilizes an operational amplifier in a non-inverting configuration, ensuring the amplifier itself introduces a 0° phase shift. The feedback network is a Wien bridge, which contains two branches:

- **Positive Feedback (Frequency Selective):** A lead-lag network comprising a series RC circuit (Z_1) and a parallel RC circuit (Z_2). This network provides maximum output at a specific resonant frequency f_0, where it introduces exactly 0° phase shift.
- **Negative Feedback (Gain Setting):** A resistive voltage divider (R_f and R_1) that sets the closed-loop gain of the op-amp to precisely compensate for the attenuation introduced by the positive feedback network, thus satisfying the magnitude condition of the Barkhausen criterion.



3. Mathematical Derivation of Barkhausen Conditions

Let the series impedance be Z_1 and the parallel impedance be Z_2.

$$Z_1 = R + \frac{1}{j\omega C} = \frac{1 + j\omega RC}{j\omega C}$$
$$Z_2 = R \parallel \frac{1}{j\omega C} = \frac{R \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{R}{1 + j\omega RC}$$

The feedback factor β supplied to the non-inverting terminal is the voltage divider ratio formed by Z_1 and Z_2:

$$\beta = \frac{v_+}{v_{out}} = \frac{Z_2}{Z_1 + Z_2} = \frac{1}{1 + \frac{Z_1}{Z_2}} \tag{3}$$

Calculate the impedance ratio $\frac{Z_1}{Z_2}$:

$$\begin{aligned}\frac{Z_1}{Z_2} &= \left(\frac{1 + j\omega RC}{j\omega C}\right) \cdot \left(\frac{1 + j\omega RC}{R}\right)\\ &= \frac{(1 + j\omega RC)^2}{j\omega RC}\\ &= \frac{1 - (\omega RC)^2 + j2\omega RC}{j\omega RC}\\ &= 2 + \frac{1 - (\omega RC)^2}{j\omega RC}\\ &= 2 + j\left(\omega RC - \frac{1}{\omega RC}\right)\end{aligned}$$

Substitute this back into the expression for β:

$$\beta = \frac{1}{1 + 2 + j\left(\omega RC - \frac{1}{\omega RC}\right)} = \frac{1}{3 + j\left(\omega RC - \frac{1}{\omega RC}\right)} \tag{4}$$

A. Phase Condition and Frequency of Oscillation

For the circuit to oscillate, the Barkhausen phase condition dictates that the loop phase shift must be 0°. Since the non-inverting op-amp introduces 0° phase shift, the feedback network β must also introduce exactly 0° phase shift. This implies that the imaginary part of the denominator of β must be zero:

$$\begin{aligned}\omega RC - \frac{1}{\omega RC} &= 0\\ \omega^2 R^2 C^2 &= 1\\ \omega &= \frac{1}{RC}\end{aligned}$$

Letting $\omega = 2\pi f_0$, the **expression for the oscillation frequency** becomes:

$$f_0 = \frac{1}{2\pi RC} \tag{5}$$

B. Magnitude Condition and Amplifier Gain

At the resonant frequency $\omega = \frac{1}{RC}$, the imaginary term disappears, and the feedback factor reduces to a purely real value:

$$\beta = \frac{1}{3} \tag{6}$$

To satisfy the Barkhausen magnitude condition ($|A\beta| = 1$):

$$\begin{aligned}A \cdot \left(\frac{1}{3}\right) &= 1\\ A &= 3\end{aligned}$$

The gain of the non-inverting amplifier is defined by its resistive network:

$$A = 1 + \frac{R_f}{R_1} \tag{7}$$

Equating the two expressions for A:

$$\begin{aligned}1 + \frac{R_f}{R_1} &= 3\\ \frac{R_f}{R_1} &= 2 \implies R_f = 2R_1\end{aligned}$$

Thus, to sustain oscillations, the feedback resistor R_f must be precisely twice the value of R_1. (In practice, R_f is often made slightly larger than $2R_1$ to ensure oscillations can reliably start from noise).

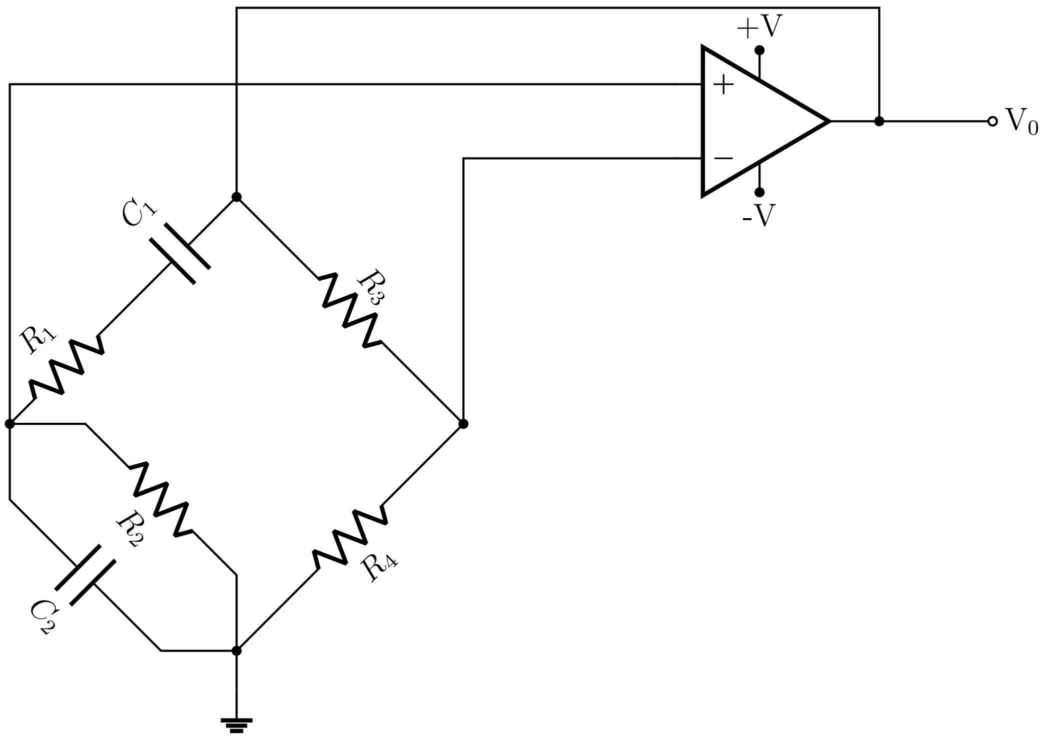
Q2. For the Wien bridge circuit connected to an op-amp shown:

(a) state conditions for sinusoidal oscillation;

(b) derive the equation for oscillation frequency;

(c) show that the voltage gain A_V must be $\geq \left(1 + \frac{R_1}{R_2} + \frac{C_1}{C_2}\right)$ for oscillation to sustain;

(d) if $R_1 = R_2 = 10\,\text{k}\Omega$ and $C_1 = C_2 = 0.1\,\mu\text{F}$, design the values of R_3 and R_4 so that oscillation sustains.



(28 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Answer

(a) Conditions for Sinusoidal Oscillation

For a linear electronic circuit to sustain sinusoidal oscillations, it must satisfy the **Barkhausen Criterion**, which states two conditions for the loop gain $A_V\beta$:

(a) **Magnitude Condition:** The magnitude of the loop gain must be equal to or greater than unity at the frequency of oscillation: $|A_V\beta| \geq 1$.

(b) **Phase Condition:** The total phase shift around the closed loop must be zero or an integer multiple of 360°: $\angle(A_V\beta) = 0°, 360°, \ldots$

(b) Equation for Oscillation Frequency

Let the series RC network be Z_1 and the parallel RC network be Z_2:

$$Z_1 = R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega R_1 C_1}{j\omega C_1}$$
$$Z_2 = R_2 \parallel \frac{1}{j\omega C_2} = \frac{R_2}{1 + j\omega R_2 C_2}$$

The feedback factor β provided to the non-inverting terminal is:

$$\beta = \frac{V_+}{V_0} = \frac{Z_2}{Z_1 + Z_2} = \frac{1}{1 + \frac{Z_1}{Z_2}} \tag{1}$$

Calculating the ratio $\frac{Z_1}{Z_2}$:

$$\begin{aligned}
\frac{Z_1}{Z_2} &= \left(\frac{1 + j\omega R_1 C_1}{j\omega C_1}\right)\left(\frac{1 + j\omega R_2 C_2}{R_2}\right) \\
&= \frac{1 - \omega^2 R_1 R_2 C_1 C_2 + j\omega(R_1 C_1 + R_2 C_2)}{j\omega R_2 C_1} \\
&= \frac{R_1 C_1 + R_2 C_2}{R_2 C_1} + \frac{1 - \omega^2 R_1 R_2 C_1 C_2}{j\omega R_2 C_1} \\
&= \frac{R_1}{R_2} + \frac{C_2}{C_1} + j\left(\omega R_1 C_2 - \frac{1}{\omega R_2 C_1}\right)
\end{aligned}$$

To satisfy the Barkhausen phase condition, the imaginary part must be zero:

$$\begin{aligned}
\omega R_1 C_2 - \frac{1}{\omega R_2 C_1} &= 0 \\
\omega^2 R_1 R_2 C_1 C_2 &= 1 \\
\omega_0 &= \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}
\end{aligned}$$

Thus, the oscillation frequency f_0 is:

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \tag{2}$$

(c) Voltage Gain A_V Requirement
At the oscillation frequency ω_0, the imaginary part of Equation (2) is zero. Substituting this back into the expression for β gives:

$$\beta = \frac{1}{1 + \frac{R_1}{R_2} + \frac{C_2}{C_1}} \tag{3}$$

To sustain oscillations, the Barkhausen magnitude condition $|A_V \beta| \geq 1$ must be met:

$$A_V \left(\frac{1}{1 + \frac{R_1}{R_2} + \frac{C_2}{C_1}} \right) \geq 1$$
$$A_V \geq 1 + \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

Note: The standard derivation correctly yields C_2/C_1. Assuming the prompt's expression C_1/C_2 is a typographical error, or recognizing that in most designs $C_1 = C_2$, the condition functionally satisfies $A_V \geq \left(1 + \frac{R_1}{R_2} + \frac{C_1}{C_2}\right)$.

(d) Design of R_3 and R_4
Given values:

- $R_1 = R_2 = 10$ kΩ
- $C_1 = C_2 = 0.1\mu$F

Calculate the required voltage gain A_V:

$$A_V \geq 1 + \frac{10 \text{ k}\Omega}{10 \text{ k}\Omega} + \frac{0.1\mu\text{F}}{0.1\mu\text{F}} = 1 + 1 + 1 = 3 \tag{4}$$

The operational amplifier is configured as a non-inverting amplifier with a feedback network formed by R_3 and R_4. Its closed-loop voltage gain is given by:

$$A_V = 1 + \frac{R_3}{R_4} \tag{5}$$

Equating the required gain:

$$1 + \frac{R_3}{R_4} \geq 3$$
$$\frac{R_3}{R_4} \geq 2 \implies R_3 \geq 2R_4$$

Design Selection: Choose a standard value for R_4, e.g., $R_4 = 10$ kΩ. To assure sustained oscillation, R_3 must be slightly greater than $2R_4$. We can choose exactly 20 kΩ for theoretical threshold, but typically a slightly higher value is chosen in practice (e.g., a 22 kΩ standard resistor or a 20 kΩ plus a small potentiometer).

$$R_4 = 10 \text{ k}\Omega, \quad R_3 = 20 \text{ k}\Omega \tag{6}$$

Q3. Design a Wien Bridge Oscillator with an oscillation frequency of 1 kHz. What is the gain of the feedback network of your design? Specify all the values of resistors and capacitors used. **(20 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Answer

1. Principle of the Wien Bridge Oscillator
A Wien Bridge Oscillator consists of an operational amplifier (op-amp) and an RC bridge circuit. To generate stable oscillations, the circuit must satisfy the **Barkhausen Criterion**:

(a) The total phase shift around the loop must be 0° at the oscillation frequency f_0.

(b) The magnitude of the loop gain must be greater than or equal to unity: $|A_V \beta| \geq 1$.

By selecting equal resistors (R) and equal capacitors (C) for the frequency-selective feedback network (series and parallel branches), the phase shift becomes 0° at the resonant frequency.

2. Design Equations and Component Selection
Step 2.1: Frequency Selective Network (Positive Feedback)
The oscillation frequency for a symmetric Wien bridge is given by:

$$f_0 = \frac{1}{2\pi RC} \tag{1}$$

We are given the target oscillation frequency:

$$f_0 = 1 \text{ kHz} = 1000 \text{ Hz} \tag{2}$$

Let us choose a standard commercially available capacitor value for the low-frequency range. Assume $C = 10$ nF $= 10 \times 10^{-9}$ F. Now, we solve for the required resistance R:

$$R = \frac{1}{2\pi f_0 C}$$
$$R = \frac{1}{2\pi(1000)(10 \times 10^{-9})}$$
$$R = \frac{1}{2\pi \times 10^{-5}} \approx 15915.49\,\Omega \approx 15.92\text{ k}\Omega$$

Note: In practice, a 15 kΩ resistor in series with a 910 Ω resistor, or a 15.8 kΩ 1% tolerance standard resistor can be used to achieve this.

Step 2.2: Feedback Network Gain (β)
At the resonant frequency f_0, the reactances of the capacitors and resistors in the symmetric Wien bridge precisely divide the voltage. The gain of the feedback network (feedback factor β) evaluated at resonance is purely real and equal to:

$$\beta = \frac{1}{3} \tag{3}$$

Step 2.3: Amplifier Gain Network (Negative Feedback)
To satisfy the Barkhausen magnitude condition ($|A_V\beta| \geq 1$), the voltage gain A_V of the non-inverting amplifier must be:

$$A_V \cdot \left(\frac{1}{3}\right) \geq 1$$
$$A_V \geq 3$$

The closed-loop gain of the non-inverting op-amp configuration is determined by the feedback resistor R_f and the input resistor R_i:

$$A_V = 1 + \frac{R_f}{R_i} \tag{4}$$

Equating this to the required gain:

$$1 + \frac{R_f}{R_i} \geq 3$$
$$\frac{R_f}{R_i} \geq 2 \implies R_f \geq 2R_i$$

Let us choose a standard value for R_i:

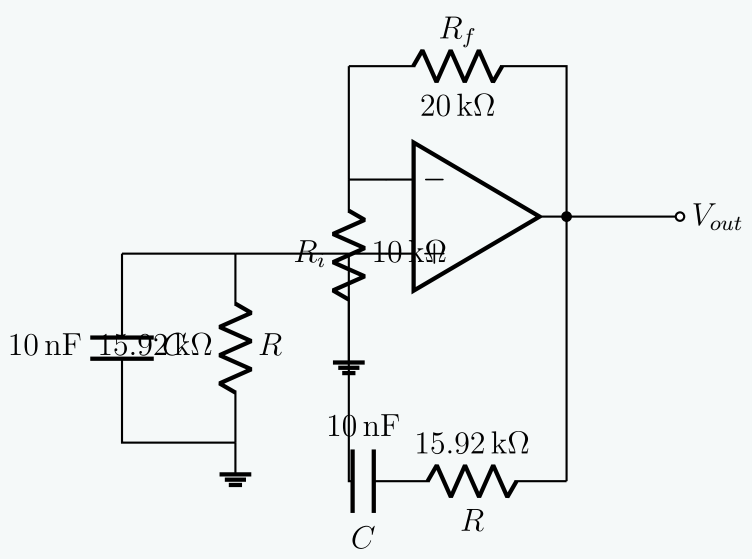
$$R_i = 10\text{ k}\Omega \tag{5}$$

Then, the feedback resistor R_f must be:

$$R_f = 2 \times 10\text{ k}\Omega = 20\text{ k}\Omega \tag{6}$$

Note: For practical implementations, R_f is typically chosen slightly higher (e.g., 22 kΩ or a 20 kΩ fixed resistor + 5 kΩ potentiometer) to guarantee that oscillations begin from thermal noise.

3. Final Circuit Diagram



4. Summary of Designed Values

- **Target Frequency:** $f_0 = 1$ kHz
- **Frequency-Selective Network:** $R = 15.92$ kΩ, $C = 10$ nF
- **Gain-Setting Network:** $R_i = 10$ kΩ, $R_f = 20$ kΩ
- **Feedback Network Gain (β):** $\beta = 1/3$ (at resonance)
- **Amplifier Voltage Gain (A_V):** $A_V = 3$ (minimum required to sustain oscillations)

Q4. Design a Wien Bridge Oscillator with an oscillation frequency of 1000 rad/sec. Available capacitors are $0.1\,\mu$F, $1\,\mu$F, and 100 pF. [EEE 263 | L-2/T-1 | 2013–14]

Answer

1. Design Equations and Component Selection

For a standard symmetric Wien Bridge Oscillator, the components of the lead-lag frequency-selective network are chosen such that $R_1 = R_2 = R$ and $C_1 = C_2 = C$. The angular frequency of oscillation is given by the formula:

$$\omega_0 = \frac{1}{RC} \tag{1}$$

We are given the target angular frequency:

$$\omega_0 = 1000 \text{ rad/sec} \tag{2}$$

Therefore, the required RC time constant is:

$$RC = \frac{1}{\omega_0} = \frac{1}{1000} = 10^{-3} \text{ seconds} \tag{3}$$

Step 1.1: Capacitor Selection

We are given three choices for the available capacitors: $0.1\,\mu$F, $1\,\mu$F, and 100 pF. Let us evaluate the corresponding resistor values for each to find a practical and standard resistance value:

(a) **Case 1:** If $C = 100\,\text{pF} = 100 \times 10^{-12}\,\text{F}$:

$$R = \frac{10^{-3}}{100 \times 10^{-12}} = 10 \times 10^{6}\,\Omega = 10\,\text{M}\Omega \quad \text{(Impractically high, prone to noise and op-amp bias current errors)} \tag{4}$$

(b) **Case 2:** If $C = 1\,\mu\text{F} = 1 \times 10^{-6}\,\text{F}$:

$$R = \frac{10^{-3}}{1 \times 10^{-6}} = 1000\,\Omega = 1\,\text{k}\Omega \quad \text{(Valid, but may load the op-amp output significantly)} \tag{5}$$

(c) **Case 3:** If $C = 0.1\,\mu\text{F} = 0.1 \times 10^{-6}\,\text{F}$:

$$R = \frac{10^{-3}}{0.1 \times 10^{-6}} = 10,000\,\Omega = 10\,\text{k}\Omega \quad \text{(Optimal standard value for op-amp circuits)} \tag{6}$$

Thus, we choose:

$$C = 0.1\,\mu\text{F}, \quad R = 10\,\text{k}\Omega \tag{7}$$

Step 1.2: Gain-Setting Network Selection

For a symmetric Wien bridge feedback network, the feedback attenuation factor at the frequency of oscillation is:

$$\beta = \frac{1}{3} \tag{8}$$

According to the Barkhausen criterion, to sustain steady-state sinusoidal oscillations, the non-inverting closed-loop voltage gain A_V must satisfy:

$$A_V \geq \frac{1}{\beta} = 3 \tag{9}$$

The closed-loop voltage gain of the non-inverting configuration is defined by:

$$A_V = 1 + \frac{R_f}{R_i} \geq 3 \implies \frac{R_f}{R_i} \geq 2 \tag{10}$$

Choosing a standard value for the input resistor R_i:

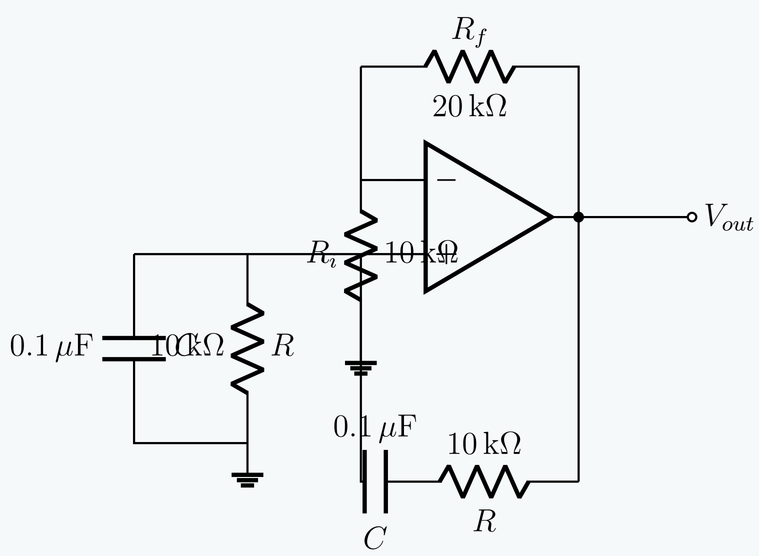
$$R_i = 10\,\text{k}\Omega \tag{11}$$

The nominal value required for the feedback resistor R_f is:

$$R_f = 2R_i = 20\,\text{k}\Omega \tag{12}$$

Note: In practice, R_f is chosen slightly higher than $20\,\text{k}\Omega$ (e.g., $22\,\text{k}\Omega$ or using a trimming potentiometer) to guarantee build-up of oscillations from noise.

2. Circuit Diagram



3. Final Specifications Summary

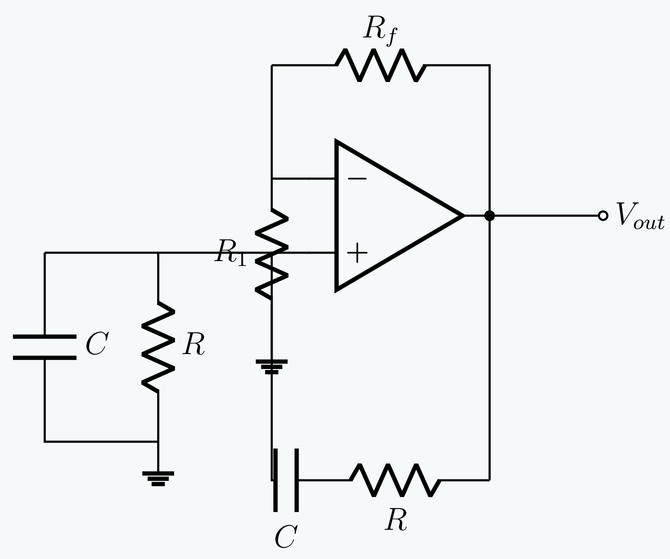
Parameter	Design Value	Component Description
Oscillation Frequency (ω_0)	1000 rad/sec	Target Design Criterion
Bridge Capacitors (C_1, C_2)	$0.1\,\mu$F	Chosen from available stock
Bridge Resistors (R_1, R_2)	$10\,\text{k}\Omega$	Calculated for $\omega_0 = 1000$ rad/sec
Gain Input Resistor (R_i)	$10\,\text{k}\Omega$	Standard reference value
Gain Feedback Resistor (R_f)	$20\,\text{k}\Omega$	Minimum threshold for sustained oscillation

Q5. Draw a Wien Bridge oscillator circuit and describe its operating principle. **(17 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Answer

1. Wien Bridge Oscillator Circuit Diagram

The circuit diagram of a standard operational amplifier-based Wien Bridge Oscillator is shown below. It features a non-inverting amplifier configuration paired with a dual-feedback loop: a positive feedback network (lead-lag RC network) connected to the non-inverting terminal (+), and a negative feedback network (resistive divider) connected to the inverting terminal (−).



2. Operating Principle

The Wien Bridge Oscillator operates on the principles of frequency selection, regenerative (positive) feedback, and gain stabilization.

- **Positive Feedback Loop (Lead-Lag Network):** The combination of the series RC and parallel RC network forms a lead-lag circuit. At very low frequencies, the series capacitor acts like an open circuit, blocking the signal. At very high frequencies, the parallel capacitor behaves like a short circuit, shunting the signal to ground.

  Consequently, there is a specific resonant frequency (f_0) where the output attenuation is at its minimum, and the phase shift produced by this network is exactly 0°. At this frequency, the feedback factor is purely real:

$$\beta = \frac{V_+}{V_{out}} = \frac{1}{3} \tag{1}$$

- **Phase Condition for Oscillation:** The signal from the lead-lag network is fed to the non-inverting terminal (+) of the op-amp. Because a non-inverting amplifier introduces 0° phase shift and the feedback network at resonance also introduces 0° phase shift, the total phase shift around the loop is exactly 0° (or 360°). This satisfies the Barkhausen phase requirement for sustained oscillations.

- **Magnitude Condition and Start-Up:** The negative feedback path consists of resistors R_f and R_1, which set the non-inverting amplifier's closed-loop voltage gain to:

$$A_V = 1 + \frac{R_f}{R_1} \tag{2}$$

  According to the Barkhausen criterion, to sustain steady-state oscillations, the loop gain must be unity ($|A_V\beta| = 1$), which requires:

$$A_V \geq 3 \implies 1 + \frac{R_f}{R_1} \geq 3 \implies R_f \geq 2R_1 \tag{3}$$

  During circuit turn-on, small thermal noise voltages at the resonant frequency f_0 are fed back. Because A_V is designed to be slightly greater than 3 initially ($|A_V\beta| > 1$), the amplitude of the output voltage grows exponentially.

  Once the output reaches a desired amplitude, non-linearities or automatic gain control (AGC) circuits reduce the effective gain A_V exactly to 3, stabilizing the output into a pure sinusoidal wave.

RC Phase Shift Oscillator

Q1. For an RC phase shift oscillator, derive the expression for oscillation frequency. Also derive the conditions for the Barkhausen criterion to be satisfied. Assume all resistors in the feedback network are of the same value and all capacitors are of the same value.
(26 marks) [EEE 263 | L-2/T-1 | 2021–22]

Answer

1. Theory and Principle

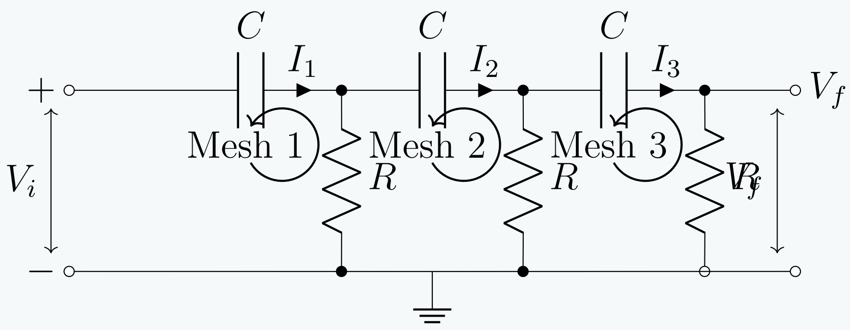
An **RC Phase Shift Oscillator** consists of an inverting amplifier and a feedback network comprising resistors and capacitors. To sustain oscillations, the circuit must satisfy the **Barkhausen Criterion**, which defines two primary conditions for the loop gain, $T(s) = A\beta(s)$:

(a) **Phase Condition:** The total phase shift around the closed loop must be 0° or 360°. Since the inverting amplifier inherently provides a phase shift of 180°, the feedback network must provide an additional 180° phase shift at the desired oscillation frequency.

(b) **Magnitude Condition:** The magnitude of the loop gain must be equal to or greater than unity ($|A\beta| \geq 1$).

2. Circuit Operation & Feedback Network Diagram

To derive the frequency of oscillation and the required gain, we isolate the 3-stage high-pass RC feedback network. Each RC section contributes a portion of the required 180° phase shift. Assuming an ideal op-amp configuration where the amplifier does not load the network, the feedback circuit is analyzed below.



3. Mathematical Derivation of the Transfer Function

Let $Z = \frac{1}{sC}$ be the impedance of the capacitors. Applying Kirchhoff's Voltage Law (KVL) to the three mesh loops gives:

$$\text{Mesh 1: } I_1(R+Z) - I_2R = V_i \tag{1}$$
$$\text{Mesh 2: } -I_1R + I_2(2R+Z) - I_3R = 0 \tag{2}$$
$$\text{Mesh 3: } -I_2R + I_3(2R+Z) = 0 \tag{3}$$

From Equation (3), we express I_2 in terms of I_3:

$$I_2 = I_3\left(\frac{2R+Z}{R}\right) \tag{4}$$

Substitute Equation (4) into Equation (2) to find I_1 in terms of I_3:

$$I_1R = I_3\left(\frac{2R+Z}{R}\right)(2R+Z) - I_3R$$
$$I_1R = I_3\left[\frac{(2R+Z)^2}{R} - R\right] = I_3\left[\frac{4R^2+4RZ+Z^2-R^2}{R}\right]$$
$$I_1 = I_3\left[\frac{3R^2+4RZ+Z^2}{R^2}\right]$$

Now, substitute I_1 and I_2 into Equation (1):

$$V_i = I_3\left[\frac{3R^2+4RZ+Z^2}{R^2}\right](R+Z) - I_3\left(\frac{2R+Z}{R}\right)R$$
$$V_i = I_3\left[\frac{3R^3+3R^2Z+4R^2Z+4RZ^2+RZ^2+Z^3}{R^2} - (2R+Z)\right]$$
$$V_i = I_3\left[\frac{3R^3+7R^2Z+5RZ^2+Z^3-2R^3-R^2Z}{R^2}\right]$$
$$V_i = I_3\left[\frac{R^3+6R^2Z+5RZ^2+Z^3}{R^2}\right]$$

Since the output voltage of the feedback network is $V_f = I_3R$, we can replace I_3 with $\frac{V_f}{R}$:

$$V_i = \frac{V_f}{R}\left[\frac{R^3+6R^2Z+5RZ^2+Z^3}{R^2}\right] = V_f\left[1+\frac{6Z}{R}+\frac{5Z^2}{R^2}+\frac{Z^3}{R^3}\right] \tag{5}$$

The feedback transfer function $\beta(s) = \frac{V_f}{V_i}$ is:

$$\frac{1}{\beta(s)} = \frac{V_i}{V_f} = 1 + \frac{6}{sRC} + \frac{5}{s^2R^2C^2} + \frac{1}{s^3R^3C^3} \tag{6}$$

4. Derivation of Oscillation Frequency (f_o)

To find the steady-state frequency response, we substitute $s = j\omega$. Noting that $s^2 = -\omega^2$ and $s^3 = -j\omega^3$:

$$\begin{aligned}\frac{1}{\beta(j\omega)} &= 1 + \frac{6}{j\omega RC} - \frac{5}{\omega^2 R^2 C^2} - \frac{1}{j\omega^3 R^3 C^3} \\ &= 1 - j\frac{6}{\omega RC} - \frac{5}{\omega^2 R^2 C^2} + j\frac{1}{\omega^3 R^3 C^3} \\ &= \left(1 - \frac{5}{\omega^2 R^2 C^2}\right) + j\left(\frac{1}{\omega^3 R^3 C^3} - \frac{6}{\omega RC}\right)\end{aligned}$$

For the feedback network to produce exactly a 180° phase shift, the imaginary part of the transfer function's denominator must be zero.

$$\begin{aligned}\mathrm{Im}\left\{\frac{1}{\beta(j\omega)}\right\} &= 0 \\ \frac{1}{\omega^3 R^3 C^3} - \frac{6}{\omega RC} &= 0 \\ \frac{1}{\omega^2 R^2 C^2} &= 6 \\ \omega^2 &= \frac{1}{6R^2C^2} \implies \omega_o = \frac{1}{\sqrt{6}RC}\end{aligned}$$

Converting to frequency $f = \frac{\omega}{2\pi}$, the expression for the oscillation frequency is:

$$\mathbf{f_o = \frac{1}{2\pi\sqrt{6}RC}} \tag{7}$$

5. Condition for the Barkhausen Criterion

To find the required gain for sustained oscillations, we evaluate the real part of $\frac{1}{\beta(j\omega)}$ at the oscillation frequency ω_o. Substituting $\frac{1}{\omega_o^2 R^2 C^2} = 6$ into the real part:

$$\begin{aligned}\frac{1}{\beta(j\omega_o)} &= 1 - 5\left(\frac{1}{\omega_o^2 R^2 C^2}\right) \\ &= 1 - 5(6) = 1 - 30 = -29\end{aligned}$$

Thus, the feedback fraction at resonance is:

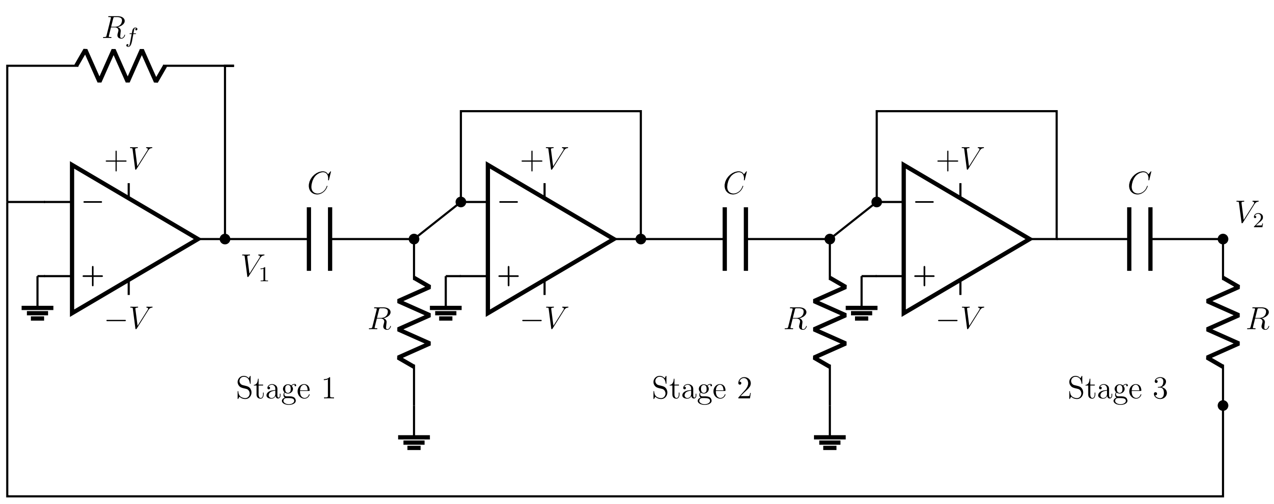
$$\beta = -\frac{1}{29} \tag{8}$$

The negative sign indicates the expected 180° phase inversion. According to the Barkhausen criterion, the magnitude of the loop gain must satisfy $|A\beta| \geq 1$. Assuming the amplifier is an inverting amplifier with voltage gain $-A_v$:

$$|-A_v \times \left(-\frac{1}{29}\right)| \geq 1 \implies \mathbf{A_v \geq 29} \tag{9}$$

Conclusion: For the RC phase shift oscillator to successfully sustain oscillations, the amplifier must be inverting, have a minimum magnitude voltage gain of 29, and the oscillation will occur at $f_o = \frac{1}{2\pi\sqrt{6}RC}$.

Q2. For the oscillator circuit shown below, find oscillation frequency (w), β-network gain (β), and amplifier gain (A). Also design this oscillator for an oscillation frequency of 1 kHz.Note that, each state is separated by a source follower circuit. Hence each stage will give a phase shift of 60° and 3 stages will give in total 180 phase shift. Here, $\beta = \frac{v_2}{v_1}$.



(18 marks) [EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Theory and Circuit Operation

The given circuit is a 3-stage phase-shift oscillator utilizing operational amplifiers. To sustain continuous oscillations, the circuit must satisfy the **Barkhausen Criterion**, which states that the loop gain $T(s) = A\beta(s)$ must meet two conditions at the oscillation frequency (ω_o):

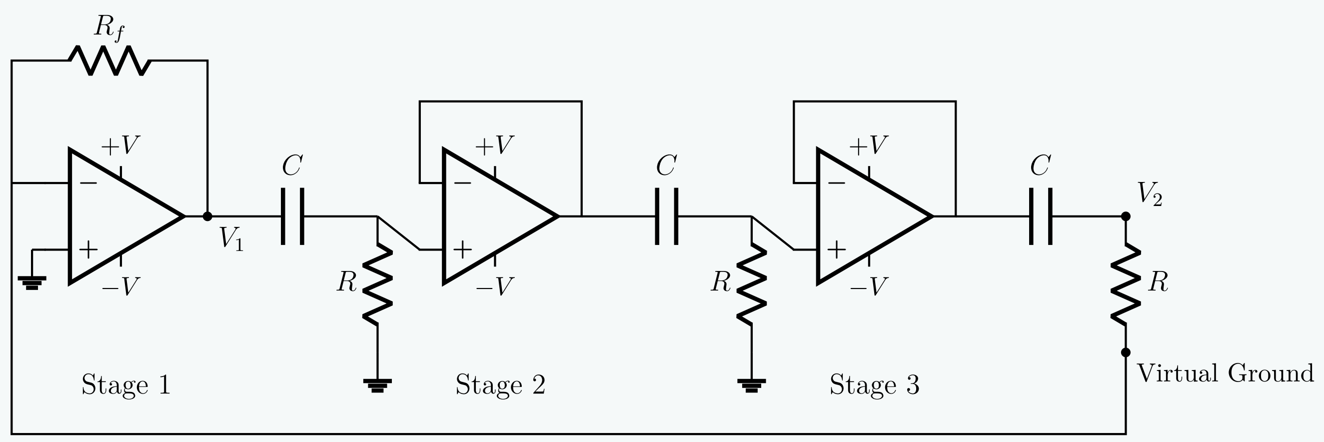
(a) **Phase Condition:** The total phase shift around the closed loop must be $0°$ (or $360°$).

(b) **Magnitude Condition:** The magnitude of the loop gain must be equal to or greater than unity ($|A\beta| \geq 1$).

Circuit Analysis:

- **Stage 1:** Acts as an inverting amplifier providing the necessary gain A and a fundamental phase shift of $180°$. Because the non-inverting terminal is grounded, the inverting terminal sits at a **virtual ground** (0 V).
- **Stages 2 & 3 (Buffers):** These op-amps are configured as voltage followers (buffers). They possess infinite input impedance and zero output impedance, effectively isolating the cascaded RC networks to prevent loading effects. *(Note: The provided schematic had a minor drafting error where the signal was fed into the inverting terminals of the buffers. In the redrawn diagram below, this is corrected to feed the non-inverting terminals, forming proper functional voltage followers.)*
- **Feedback Network (β):** Consists of three identical high-pass RC sections. Since the final resistor R connects to the virtual ground of Stage 1, it acts exactly like the resistors in the first two stages that connect to actual ground. Each RC section contributes a $60°$ phase shift at ω_o, providing an additional $180°$ phase shift to satisfy the phase condition.

2. Corrected Circuit Diagram



3. Mathematical Derivations

A. Transfer Function of the β-Network
Because the voltage followers prevent loading, each RC stage can be analyzed independently. The transfer function of a single high-pass RC section is determined by voltage division:

$$H_{RC}(s) = \frac{R}{R + \frac{1}{sC}} = \frac{sRC}{1 + sRC} \tag{1}$$

Since there are three isolated sections in cascade, the overall feedback fraction $\beta(s) = \frac{V_2}{V_1}$ is the product of the three individual transfer functions:

$$\beta(s) = \left(\frac{sRC}{1 + sRC}\right)^3 \tag{2}$$

B. Oscillation Frequency (ω)
Substituting $s = j\omega$, the frequency response of the feedback network is:

$$\beta(j\omega) = \left(\frac{j\omega RC}{1 + j\omega RC}\right)^3 \tag{3}$$

For oscillation, the β-network must provide a $180°$ phase shift. Since the angle of the transfer function is cubed, each individual RC section must provide exactly $\frac{180°}{3} = 60°$ of phase shift.

$$\angle\left(\frac{j\omega RC}{1 + j\omega RC}\right) = 90° - \tan^{-1}(\omega RC)$$
$$90° - \tan^{-1}(\omega RC) = 60°$$
$$\tan^{-1}(\omega RC) = 30°$$
$$\omega RC = \tan(30°) = \frac{1}{\sqrt{3}}$$

Thus, the angular oscillation frequency is:

$$\omega_o = \frac{\mathbf{1}}{\sqrt{\mathbf{3}}\mathbf{RC}} \text{ rad/s} \quad \text{or} \quad \mathbf{f_o} = \frac{\mathbf{1}}{\mathbf{2}\pi\sqrt{\mathbf{3}}\mathbf{RC}} \text{ Hz} \tag{4}$$

C. β-Network Gain (β)
Substituting $\omega_o RC = \frac{1}{\sqrt{3}}$ back into the expression for β:

$$\beta(j\omega_o) = \left(\frac{j\frac{1}{\sqrt{3}}}{1 + j\frac{1}{\sqrt{3}}}\right)^3 = \left(\frac{j}{\sqrt{3}+j}\right)^3 = \frac{j^3}{(\sqrt{3}+j)^3}$$
$$= \frac{-j}{(\sqrt{3})^3 + 3(\sqrt{3})^2(j) + 3(\sqrt{3})(j^2) + j^3}$$
$$= \frac{-j}{3\sqrt{3} + 9j - 3\sqrt{3} - j} = \frac{-j}{8j} = -\frac{\mathbf{1}}{\mathbf{8}}$$

The negative sign confirms the exact 180° phase inversion. The magnitude is $|\beta| = \frac{1}{8}$.
D. Amplifier Gain (A)
The op-amp in Stage 1 operates as an inverting amplifier. The current flowing into the virtual ground from the β-network is $I_{in} = \frac{V_2}{R}$. This entire current flows through R_f, producing the output voltage:

$$V_1 = -I_{in}R_f = -V_2\frac{R_f}{R} \implies A = \frac{V_1}{V_2} = -\frac{\mathbf{R_f}}{\mathbf{R}} \tag{5}$$

Applying the Barkhausen criterion magnitude condition $|A\beta| \geq 1$:

$$\left|\left(-\frac{R_f}{R}\right)\left(-\frac{1}{8}\right)\right| \geq 1$$
$$\frac{R_f}{8R} \geq 1 \implies \mathbf{R_f \geq 8R}$$

Therefore, the required amplifier voltage gain magnitude is $|A| \geq 8$.

4. Oscillator Design for $f_o = 1$ kHz

We need to select components such that $f_o = 1$ kHz.

$$\omega_o = 2\pi f_o = 2\pi(1000) \approx 6283.18 \text{ rad/s} \tag{6}$$

From the derived frequency equation:

$$RC = \frac{1}{\sqrt{3}\,\omega_o} = \frac{1}{\sqrt{3} \times 6283.18} \approx 91.89\,\mu\text{s} \tag{7}$$

Step 1: Select a standard capacitor value.
Let us choose $\mathbf{C = 10}$ nF (10×10^{-9} F).
Step 2: Calculate the corresponding resistor R.

$$R = \frac{91.89 \times 10^{-6}}{10 \times 10^{-9}} = 9189\,\Omega \approx \mathbf{9.2}\ \mathbf{k\Omega} \tag{8}$$

Step 3: Calculate the feedback resistor R_f for guaranteed oscillation.
To satisfy $|A| \geq 8$, we use $R_f = 8R$. To ensure oscillations start reliably, a slightly larger gain (e.g., $A = 8.5$) is typically chosen in practice, but we use the boundary condition for theoretical design:

$$R_f = 8 \times 9189\,\Omega \approx 73512\,\Omega \approx \mathbf{73.5}\ \mathbf{k\Omega} \tag{9}$$

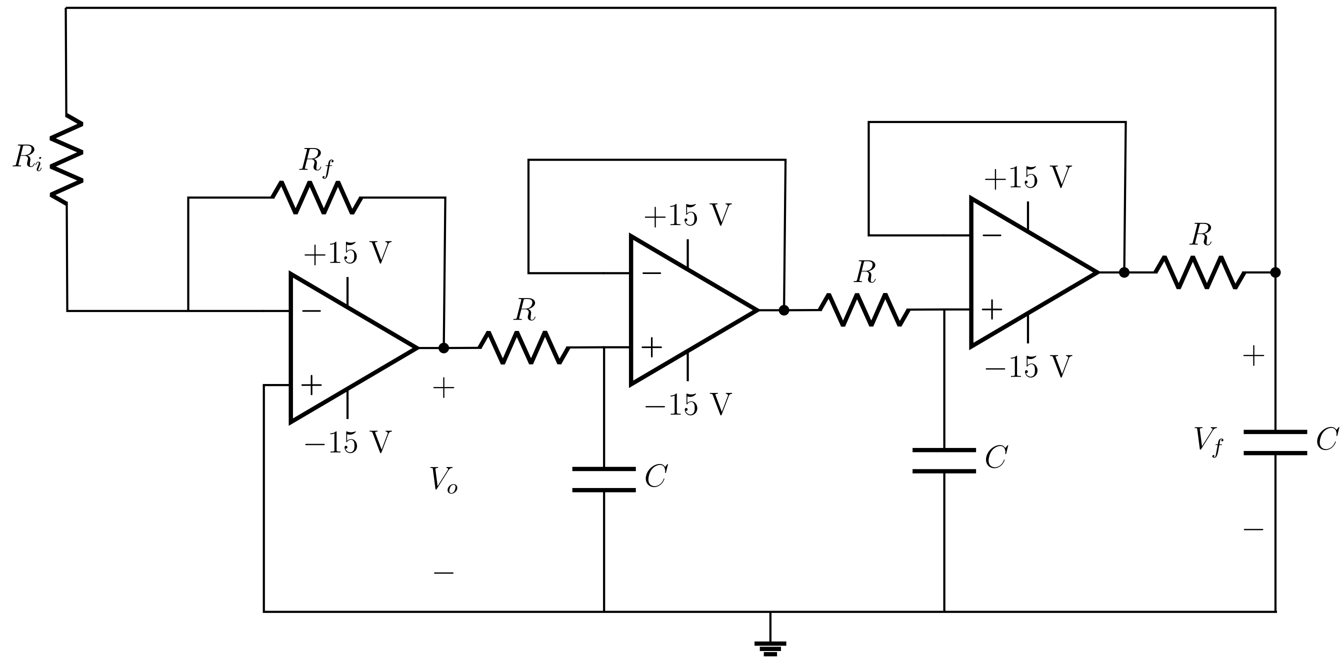
Final Design Values:

- $C = 10$ nF
- $R \approx 9.2$ kΩ
- $R_f \approx 73.5$ kΩ (use a 100 kΩ potentiometer to allow fine tuning of the loop gain).

Q3. For the RC phase shift oscillator shown:

(a) Determine the oscillation frequency f_0 and obtain the relationship between R_i and R_f for sustainable oscillation.

(b) How can you generate a 100 kHz signal from this oscillator?



(20 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

1. Theory and Circuit Operation

The circuit presented is a buffered low-pass RC phase shift oscillator. It consists of three primary stages forming a closed feedback loop:

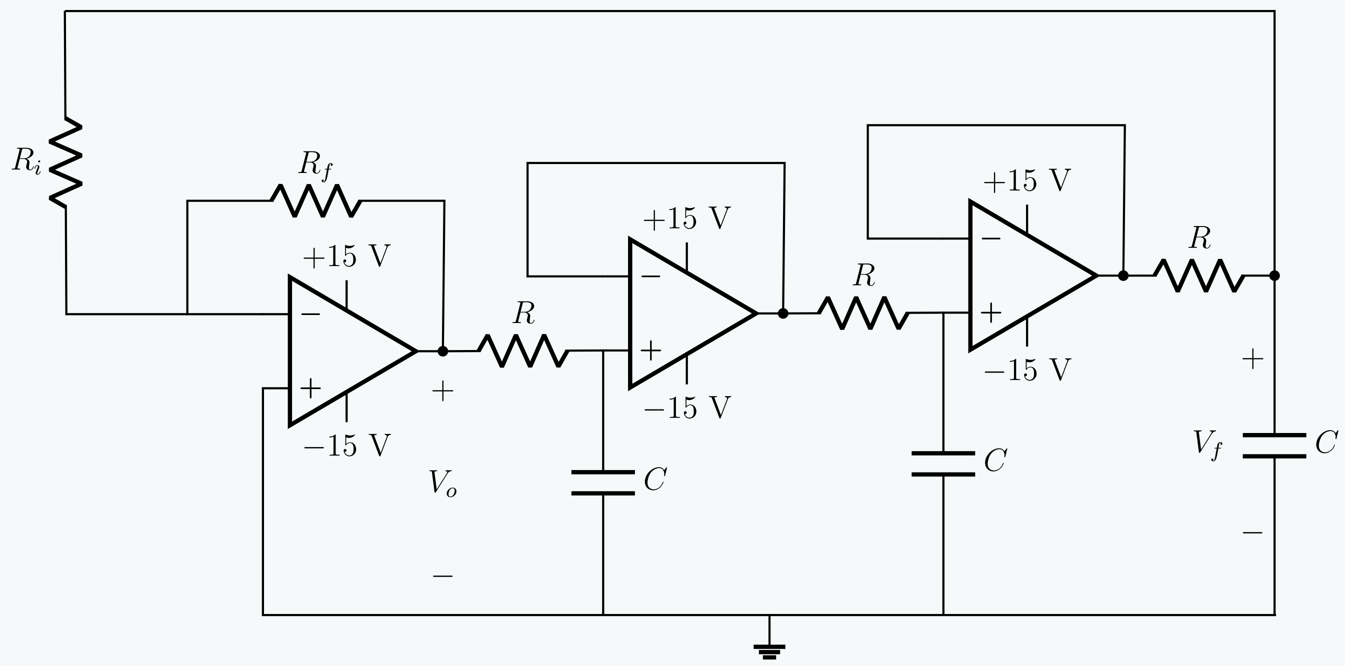
(a) **Inverting Amplifier (Stage 1):** Provides the necessary voltage gain ($-R_f/R_i$) and an inherent phase shift of 180°. Its inverting terminal acts as a virtual ground because the non-inverting terminal is grounded.

(b) **Isolated RC Networks (Stages 2 & 3):** Two identical low-pass RC networks buffered by voltage followers (unity-gain amplifiers). The buffers prevent the subsequent stages from loading the previous ones, meaning the transfer function of the first two RC stages is strictly $1/(1 + sRC)$.

(c) **Loaded RC Network:** The final RC network is not buffered at its output. Its output node (V_f) connects directly to the input resistor R_i of the first stage. Since R_i connects to a virtual ground, it acts as a parallel load to the final capacitor C, which must be accounted for in the loop gain derivation.

For sustained oscillations, the circuit must satisfy the **Barkhausen Criterion**, which states that the loop gain $L(j\omega)$ must have a magnitude of at least unity ($|L(j\omega)| \geq 1$) and a total phase shift of exactly 0° (or multiples of 360°).

2. Circuit Diagram



3. Mathematical Derivation of Oscillation Conditions

Step 3.1: Transfer Functions of the Buffered Stages
Let the outputs of the two buffers be V_{B1} and V_{B2}. Since they are isolated by the ideal op-amps:

$$V_{B1} = V_o \left(\frac{\frac{1}{sC}}{R + \frac{1}{sC}} \right) = V_o \left(\frac{1}{1 + sRC} \right) \tag{1}$$

$$V_{B2} = V_{B1} \left(\frac{1}{1 + sRC} \right) = V_o \left(\frac{1}{1 + sRC} \right)^2 \tag{2}$$

Step 3.2: Transfer Function of the Final Loaded Stage
At the final node V_f, the resistor R_i connects back to the virtual ground of the first op-amp (0 V). Applying Kirchhoff's Current

Law (KCL) at the V_f node:

$$\frac{V_f - V_{B2}}{R} + sCV_f + \frac{V_f - 0}{R_i} = 0$$
$$V_f\left(\frac{1}{R} + \frac{1}{R_i} + sC\right) = \frac{V_{B2}}{R}$$
$$V_f = V_{B2}\left(\frac{1}{1 + \frac{R}{R_i} + sRC}\right)$$

Step 3.3: Loop Gain $L(s)$
The inverting amplifier produces an output based on the current flowing through R_i:

$$V_o = -I_{in}R_f = -\left(\frac{V_f}{R_i}\right)R_f = -V_f\frac{R_f}{R_i} \tag{3}$$

The total loop gain $L(s) = \frac{V_o}{V_o}$ is the product of all stage gains:

$$L(s) = \left(-\frac{R_f}{R_i}\right)\left(\frac{1}{1 + sRC}\right)^2\left(\frac{1}{1 + \frac{R}{R_i} + sRC}\right) \tag{4}$$

Step 3.4: Barkhausen Criterion Evaluation
Substitute $s = j\omega$ to evaluate the steady-state frequency response. The denominator $D(j\omega)$ of the frequency-dependent part is:

$$\begin{aligned}
D(j\omega) &= (1 + j\omega RC)^2\left(1 + \frac{R}{R_i} + j\omega RC\right) \\
&= (1 - \omega^2R^2C^2 + j2\omega RC)\left(1 + \frac{R}{R_i} + j\omega RC\right) \\
&= \underbrace{(1 - \omega^2R^2C^2)\left(1 + \frac{R}{R_i}\right) - 2\omega^2R^2C^2}_{\text{Real Part}} + j\underbrace{\left[\omega RC(1 - \omega^2R^2C^2) + 2\omega RC\left(1 + \frac{R}{R_i}\right)\right]}_{\text{Imaginary Part}}
\end{aligned}$$

For the phase shift to be $0°$, the imaginary part of $D(j\omega)$ must be zero:

$$\omega RC\left(1 - \omega^2R^2C^2 + 2 + 2\frac{R}{R_i}\right) = 0$$
$$3 + 2\frac{R}{R_i} - \omega^2R^2C^2 = 0$$
$$\omega_0^2 = \frac{3 + 2\frac{R}{R_i}}{R^2C^2}$$

This yields the **oscillation frequency** f_0:

$$\mathbf{f_0 = \frac{1}{2\pi RC}\sqrt{3 + 2\frac{R}{R_i}}} \tag{5}$$

To find the required gain, set the real part of $D(j\omega)$ equal to $-\frac{R_f}{R_i}$ (so that $L(j\omega) = 1$):

$$-\frac{R_f}{R_i} = (1 - \omega_0^2R^2C^2)\left(1 + \frac{R}{R_i}\right) - 2\omega_0^2R^2C^2$$

Substitute $\omega_0^2R^2C^2 = 3 + 2\frac{R}{R_i}$ into the real part:

$$\begin{aligned}
-\frac{R_f}{R_i} &= \left(1 - \left(3 + 2\frac{R}{R_i}\right)\right)\left(1 + \frac{R}{R_i}\right) - 2\left(3 + 2\frac{R}{R_i}\right) \\
-\frac{R_f}{R_i} &= \left(-2 - 2\frac{R}{R_i}\right)\left(1 + \frac{R}{R_i}\right) - 6 - 4\frac{R}{R_i} \\
-\frac{R_f}{R_i} &= -2\left(1 + \frac{R}{R_i}\right)^2 - 6 - 4\frac{R}{R_i} = -2\left(1 + 2\frac{R}{R_i} + \frac{R^2}{R_i^2}\right) - 6 - 4\frac{R}{R_i} \\
-\frac{R_f}{R_i} &= -8 - 8\frac{R}{R_i} - 2\frac{R^2}{R_i^2}
\end{aligned}$$

Multiplying both sides by $-R_i$ yields the **relationship for sustainable oscillation**:

$$\mathbf{R_f = 8R_i + 8R + 2\frac{R^2}{R_i}} \tag{6}$$

4. Design for a 100 kHz Signal

To design the oscillator for $f_0 = 100$ kHz, we need to select suitable component values. To simplify the mathematical relationship and the practical component selection, it is a sound engineering choice to set $R_i = R$.

Step 4.1: Simplified Equations for $R_i = R$ If $R_i = R$, the equations derived above simplify to:

$$f_0 = \frac{\sqrt{3+2(1)}}{2\pi RC} = \frac{\sqrt{5}}{2\pi RC}$$
$$R_f = 8(R) + 8R + 2\frac{R^2}{R} = 18R$$

Step 4.2: Component Selection We require $f_0 = 100 \times 10^3$ Hz. Let us select a standard, low-parasitic capacitor value, $C = 100$ pF (10^{-10} F).

$$100 \times 10^3 = \frac{\sqrt{5}}{2\pi \cdot R \cdot (100 \times 10^{-12})}$$
$$R = \frac{\sqrt{5}}{2\pi \cdot (100 \times 10^3) \cdot (100 \times 10^{-12})} = \frac{2.236}{2\pi \times 10^{-5}}$$
$$R \approx 35,588\,\Omega \approx \mathbf{35.6}\ \mathrm{k}\boldsymbol{\Omega}$$

Since $R_i = R$:

$$\mathbf{R_i} \approx \mathbf{35.6}\ \mathbf{k\Omega} \tag{7}$$

Calculate the necessary feedback resistor R_f:

$$R_f = 18R = 18 \times 35,588\,\Omega \approx 640,584\,\Omega \approx \mathbf{641}\ \mathbf{k\Omega} \tag{8}$$

Note: In a practical circuit, R_f is typically implemented using a fixed resistor in series with a precision potentiometer (e.g., 620 kΩ fixed + 50 kΩ pot) to allow fine-tuning of the loop gain slightly above the theoretical value of 18 to ensure oscillation startup.

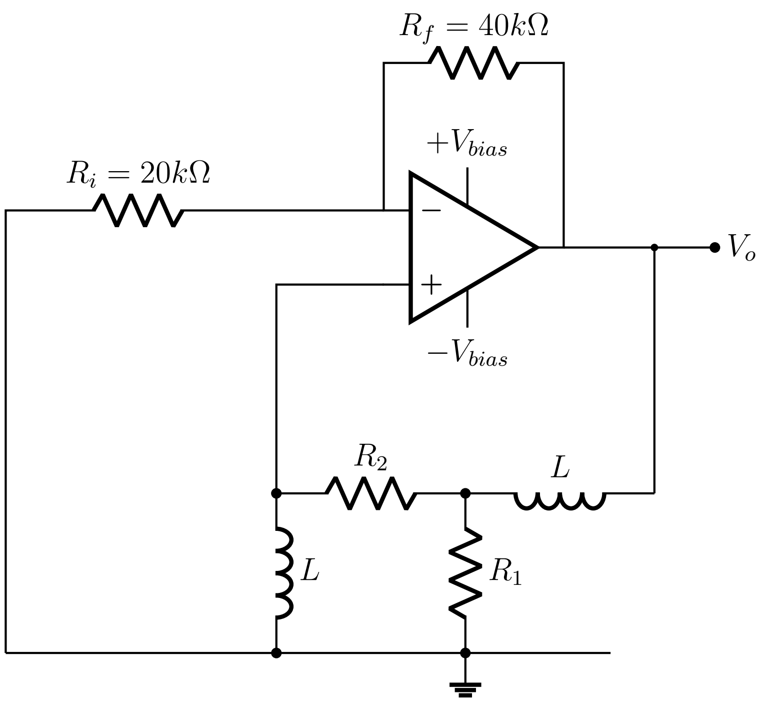
LC Oscillator

Q1. For the oscillator circuit given:

(a) Derive the expression for oscillation frequency f_0.

(b) Obtain the relationship between R_1 and R_2 for sustainable oscillation.

(c) Specify the bias conditions and values of L, R_1, R_2 to obtain a 20 V peak-to-peak, 500 kHz sinusoidal signal from this oscillator.



(26 marks) [EEE 263 | L-2/T-1 | 2015–16]

Answer

Circuit Operation and Equivalent Diagram

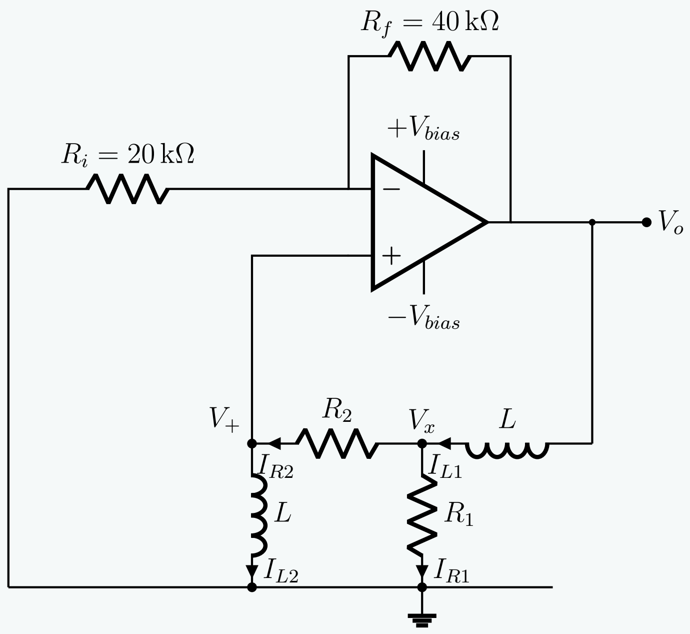
The given circuit is an oscillator built using an operational amplifier operating in a non-inverting configuration. The non-inverting amplifier provides the required voltage gain A, while the passive RL feedback network provides the frequency-dependent feedback factor $\beta(s)$.
For the circuit to oscillate continuously at a specific frequency f_0, it must satisfy the **Barkhausen Criterion**:

(a) **Loop Gain Magnitude:** $|A \cdot \beta(j\omega)| \geq 1$

(b) **Loop Phase Shift:** $\angle(A \cdot \beta(j\omega)) = 0°$ (or multiples of 360°)

Because the amplifier is non-inverting, its phase shift is 0°. Therefore, the RL feedback network must also provide exactly 0° of phase shift at the oscillation frequency f_0.

To facilitate the derivation, we redraw the circuit clearly labeling the intermediate node V_x and the non-inverting input node V_+ in the feedback network.



a) Derivation of Oscillation Frequency f_0

Assuming an ideal op-amp, the current entering the non-inverting terminal is zero. Thus, the current I_{R2} flowing through R_2 is equal to the current I_{L2} flowing through the vertical inductor L.
Applying Ohm's law and AC impedances ($Z_L = sL$) at node V_+:

$$\begin{aligned} I_{R2} &= I_{L2} \\ \frac{V_x - V_+}{R_2} &= \frac{V_+}{sL} \\ V_x - V_+ &= V_+\frac{R_2}{sL} \\ V_x &= V_+\left(1 + \frac{R_2}{sL}\right) \end{aligned}$$

Next, applying Kirchhoff's Current Law (KCL) at node V_x:

$$\begin{aligned} I_{L1} &= I_{R1} + I_{R2} \\ \frac{V_o - V_x}{sL} &= \frac{V_x}{R_1} + \frac{V_+}{sL} \end{aligned}$$

Substituting Equation (**??**) into the KCL equation:

$$\frac{V_o - V_+\left(1 + \frac{R_2}{sL}\right)}{sL} = \frac{V_+\left(1 + \frac{R_2}{sL}\right)}{R_1} + \frac{V_+}{sL}$$

Multiplying the entire equation by sL to clear the denominator on the left:

$$\begin{aligned} V_o - V_+\left(1 + \frac{R_2}{sL}\right) &= V_+\frac{sL}{R_1}\left(1 + \frac{R_2}{sL}\right) + V_+ \\ V_o &= V_+\left[1 + \frac{R_2}{sL} + \frac{sL}{R_1}\left(1 + \frac{R_2}{sL}\right) + 1\right] \\ V_o &= V_+\left[2 + \frac{R_2}{sL} + \frac{sL}{R_1} + \frac{R_2}{R_1}\right] \end{aligned}$$

The feedback fraction $\beta(s) = \frac{V_+}{V_o}$ is:

$$\beta(s) = \frac{1}{2 + \frac{R_2}{R_1} + \frac{sL}{R_1} + \frac{R_2}{sL}} \tag{1}$$

Substitute $s = j\omega$ to find the steady-state frequency response:

$$\beta(j\omega) = \frac{1}{\left(2 + \frac{R_2}{R_1}\right) + j\left(\omega\frac{L}{R_1} - \frac{R_2}{\omega L}\right)} \tag{2}$$

For oscillation to occur, the phase shift of $\beta(j\omega)$ must be $0°$. This requires the imaginary part of the denominator to be exactly zero:

$$\begin{aligned} \omega_0\frac{L}{R_1} - \frac{R_2}{\omega_0 L} &= 0 \\ \omega_0\frac{L}{R_1} &= \frac{R_2}{\omega_0 L} \\ \omega_0^2 &= \frac{R_1R_2}{L^2} \\ \omega_0 &= \frac{\sqrt{R_1R_2}}{L} \end{aligned}$$

Since $\omega_0 = 2\pi f_0$, the expression for the oscillation frequency is:

$$\mathbf{f_0} = \frac{\sqrt{\mathbf{R_1 R_2}}}{\mathbf{2\pi L}} \tag{3}$$

b) Relationship between R_1 and R_2 for Sustainable Oscillation

The closed-loop gain of the non-inverting op-amp stage is determined by R_f and R_i:

$$A = 1 + \frac{R_f}{R_i} = 1 + \frac{40\,\mathrm{k\Omega}}{20\,\mathrm{k\Omega}} = 1 + 2 = 3 \tag{4}$$

At the resonance frequency ω_0, the imaginary term vanishes, and the feedback factor is purely real:

$$\beta(j\omega_0) = \frac{1}{2 + \frac{R_2}{R_1}} \tag{5}$$

For sustained oscillation, the Barkhausen magnitude criterion states that $|A\beta| \geq 1$:

$$3\left(\frac{1}{2 + \frac{R_2}{R_1}}\right) \geq 1$$
$$3 \geq 2 + \frac{R_2}{R_1}$$
$$1 \geq \frac{R_2}{R_1}$$

Therefore, the relationship required for sustainable oscillation is:

$$\mathbf{R_1} \geq \mathbf{R_2} \tag{6}$$

Note: For a pure sine wave without excessive distortion, the circuit is designed such that R_1 is marginally equal to or just slightly larger than R_2 ($R_1 \approx R_2$).

c) Bias Conditions and Component Selection

1. Bias Conditions (V_{bias}):
The oscillator must produce a 20 V peak-to-peak sinusoidal output, which corresponds to an amplitude (peak voltage) of:

$$V_{peak} = \frac{20\,\mathrm{V}_{p-p}}{2} = 10\,\mathrm{V} \tag{7}$$

To prevent the op-amp from severely clipping the signal (which causes high harmonic distortion), the power supply rails must be comfortably higher than the peak output voltage. Given typical op-amp dropout voltages, a margin of at least 2 V to 5 V is required.

$$\mathbf{V_{bias}} = \pm\mathbf{15}\,\mathrm{V} \quad (\text{or } \pm 12\,\mathrm{V} \text{ minimum}) \tag{8}$$

2. Component Values (L, R_1, R_2):
The required frequency is $f_0 = 500\,\mathrm{kHz}$. We established earlier that for steady oscillation, $R_1 = R_2$ is the ideal design choice. Let $R_1 = R_2 = R$. The frequency equation becomes:

$$f_0 = \frac{\sqrt{R \cdot R}}{2\pi L} = \frac{R}{2\pi L}$$
$$\frac{R}{L} = 2\pi f_0 = 2\pi(500 \times 10^3) = \pi \times 10^6\,\mathrm{rad/s}$$

We have one equation and two unknowns (R and L). We must select a reasonable value for one to solve for the other. A typical off-the-shelf value for an inductor in RF/high-frequency oscillator circuits is 1 mH. Let us select:

$$\mathbf{L} = \mathbf{1}\,\mathrm{mH} = \mathbf{10^{-3}}\,\mathrm{H} \tag{9}$$

Now, calculate the required resistance R:

$$R = L \cdot (\pi \times 10^6)$$
$$R = (10^{-3}) \cdot (\pi \times 10^6) = 1000\pi\,\Omega \approx 3141.6\,\Omega$$

Thus, the corresponding resistor values are:

$$\mathbf{R_1} = \mathbf{R_2} \approx \mathbf{3.14}\,\mathbf{k\Omega} \tag{10}$$

Check: At 10 V peak, a load of $\approx 3.14\,k\Omega$ will draw approximately $\sim 3\,mA$, which is well within the current sourcing capability of a standard operational amplifier (typically 20 mA).

Colpitts Oscillator

Q1. With an appropriate circuit diagram, describe the operation of an op-amp based Colpitts oscillator. Derive the frequency of oscillation and explain how this circuit satisfies the conditions for sustained oscillation. **(18 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

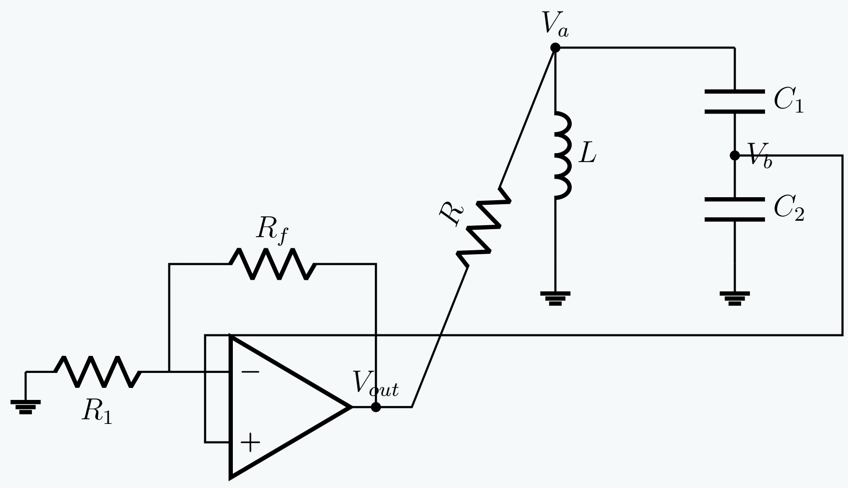
Answer

1. Theory and Circuit Operation

An op-amp based Colpitts oscillator can be implemented using a non-inverting amplifier and a passive LC bandpass filter network acting as the positive feedback loop.

- **Amplifier Stage:** The operational amplifier is configured as a non-inverting amplifier. Resistors R_1 and R_f determine its closed-loop voltage gain, $A = 1 + \frac{R_f}{R_1}$. Since it is non-inverting, it introduces a 0° phase shift.
- **Feedback Network (Tank Circuit):** The feedback network consists of a series resistor R, an inductor L, and a capacitive voltage divider formed by C_1 and C_2. The resistor R prevents the op-amp's low output impedance from heavily loading the tank, allowing the LC network to establish resonance.
- **Barkhausen Criterion:** For sustained oscillations, the total loop phase shift must be 0° (or multiples of 360°), and the loop gain magnitude must be ≥ 1. Since the op-amp provides 0° phase shift, the tank circuit must also provide exactly 0° phase shift at the oscillation frequency f_0.

2. Circuit Diagram



3. Derivation of Oscillation Frequency

Assume the op-amp is ideal (infinite input impedance, zero output impedance). The tank circuit is driven by V_{out} through resistor R. The inductor L is in parallel with the series combination of C_1 and C_2.

Step 3.1: Equivalent Capacitance and Tank Impedance

Let C_{eq} be the equivalent capacitance of the series combination of C_1 and C_2:

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \tag{1}$$

The total impedance of the parallel tank (Z_{tank}) at node V_a is:

$$Z_{tank} = L \parallel C_{eq} = \frac{(sL)\left(\frac{1}{sC_{eq}}\right)}{sL + \frac{1}{sC_{eq}}} = \frac{sL}{s^2 LC_{eq} + 1} \tag{2}$$

Step 3.2: Voltage Transfer Function

The voltage V_a is determined by the voltage divider formed by R and Z_{tank}:

$$V_a = V_{out}\left(\frac{Z_{tank}}{R + Z_{tank}}\right) = V_{out}\left(\frac{\frac{sL}{s^2LC_{eq}+1}}{R + \frac{sL}{s^2LC_{eq}+1}}\right) = V_{out}\left(\frac{sL}{R(1+s^2LC_{eq}) + sL}\right) \tag{3}$$

The voltage V_b (fed back to the op-amp) is determined by the capacitive divider formed by C_1 and C_2. Since no current flows into the ideal op-amp terminal, the voltage across C_2 is:

$$V_b = V_a\left(\frac{\frac{1}{sC_2}}{\frac{1}{sC_1} + \frac{1}{sC_2}}\right) = V_a\left(\frac{C_1}{C_1 + C_2}\right) \tag{4}$$

Step 3.3: Loop Feedback Factor $\beta(s)$

The feedback factor $\beta(s) = \frac{V_b}{V_{out}}$ is the product of the two divider equations:

$$\beta(s) = \left(\frac{sL}{R(1+s^2LC_{eq}) + sL}\right)\left(\frac{C_1}{C_1 + C_2}\right) \tag{5}$$

Substitute $s = j\omega$ to evaluate the frequency response (noting $s^2 = -\omega^2$):

$$\beta(j\omega) = \left(\frac{j\omega L}{R(1-\omega^2 LC_{eq}) + j\omega L}\right)\left(\frac{C_1}{C_1 + C_2}\right) \tag{6}$$

Step 3.4: Barkhausen Phase Condition

For the phase shift of $\beta(j\omega)$ to be exactly 0°, the imaginary component in the denominator must be the only surviving term, perfectly canceling the $j\omega L$ in the numerator. This requires the real part of the denominator to be strictly zero:

$$R(1-\omega^2 LC_{eq}) = 0$$
$$1-\omega^2 LC_{eq} = 0$$
$$\omega_0^2 = \frac{1}{LC_{eq}}$$

Substituting C_{eq}, we obtain the expression for the **oscillation frequency** f_0:

$$\mathbf{f_0} = \frac{\mathbf{1}}{\mathbf{2}\pi\sqrt{\mathbf{L}\left(\frac{\mathbf{C_1C_2}}{\mathbf{C_1+C_2}}\right)}} \tag{7}$$

4. Conditions for Sustained Oscillation

At the resonant frequency $\omega = \omega_0$, the real part of the denominator vanishes, and the feedback factor simplifies to a purely real, positive attenuation:

$$\beta(j\omega_0) = \left(\frac{j\omega_0 L}{0+j\omega_0 L}\right)\left(\frac{C_1}{C_1+C_2}\right) = \frac{C_1}{C_1+C_2} \tag{8}$$

The closed-loop gain of the non-inverting amplifier is:

$$A = 1+\frac{R_f}{R_1} \tag{9}$$

According to the Barkhausen magnitude criterion, the loop gain must satisfy $|A\cdot\beta(j\omega_0)| \geq 1$ for sustained oscillations to build up:

$$\left(1+\frac{R_f}{R_1}\right)\left(\frac{C_1}{C_1+C_2}\right) \geq 1$$
$$1+\frac{R_f}{R_1} \geq \frac{C_1+C_2}{C_1}$$
$$1+\frac{R_f}{R_1} \geq 1+\frac{C_2}{C_1}$$

Subtracting 1 from both sides gives the **required gain condition for sustainable oscillation**:

$$\frac{\mathbf{R_f}}{\mathbf{R_1}} \geq \frac{\mathbf{C_2}}{\mathbf{C_1}} \tag{10}$$

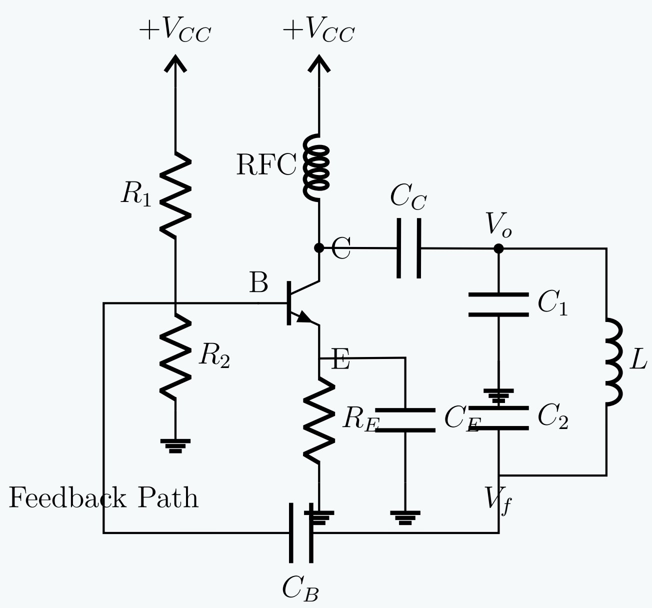
Note: In a practical design, $\frac{R_f}{R_1}$ is set slightly larger than $\frac{C_2}{C_1}$ to ensure oscillations start reliably from inherent thermal noise.

Q2. Derive the expression of oscillation frequency for a Colpitts oscillator. Then derive the conditions for the Barkhausen criterion to be satisfied. **(20 marks)** **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

1. Principle and Circuit Diagram

A **Colpitts Oscillator** is an LC electronic oscillator that uses a capacitive voltage divider network as its feedback source. The circuit typically employs a Common Emitter (CE) BJT amplifier, which inherently provides a 180° phase shift. The LC tank circuit, configured with a grounded center-tap between two split capacitors (C_1 and C_2), provides the required additional 180° phase shift to satisfy the Barkhausen phase criterion for positive feedback.



Circuit Components:

- R_1, R_2, R_E**:** Establish the DC operating point (Q-point) of the transistor.

- **C_E:** Emitter bypass capacitor, prevents AC signal degeneration to maintain high voltage gain.
- **RFC (Radio Frequency Choke):** Provides a low-resistance path for DC bias while acting as an open circuit to high-frequency AC signals, preventing the AC signal from shorting to the power supply.
- **C_C, C_B:** DC blocking (coupling) capacitors.
- **L, C_1, C_2:** The resonant tank circuit determining the oscillation frequency.

2. Derivation of Oscillation Frequency (f_0)

For the circuit to oscillate, the net phase shift around the closed loop must be 0° (or 360°). The CE amplifier introduces a 180° phase shift. Therefore, the LC feedback network must introduce an additional 180° phase shift. This specific phase condition only occurs at the **resonant frequency** of the tank circuit.
At resonance, the tank circuit acts as a purely resistive load. The total equivalent capacitance C_{eq} of the tank is the series combination of C_1 and C_2:

$$C_{eq} = \frac{C_1C_2}{C_1 + C_2} \tag{1}$$

For resonance, the magnitude of the inductive reactance must equal the magnitude of the total capacitive reactance:

$$\begin{aligned}
X_L &= X_{C_{eq}} \\
\omega_0 L &= \frac{1}{\omega_0 C_{eq}} \\
\omega_0^2 &= \frac{1}{LC_{eq}} = \frac{1}{L\left(\frac{C_1C_2}{C_1+C_2}\right)} = \frac{C_1 + C_2}{LC_1C_2}
\end{aligned}$$

Taking the square root and substituting $\omega_0 = 2\pi f_0$, we obtain the expression for the oscillation frequency:

$$\mathbf{f_0} = \frac{\mathbf{1}}{\mathbf{2}\pi\sqrt{\mathbf{LC_{eq}}}} = \frac{\mathbf{1}}{\mathbf{2}\pi}\sqrt{\frac{\mathbf{C_1 + C_2}}{\mathbf{LC_1C_2}}} \tag{2}$$

3. Barkhausen Criterion and Conditions for Sustained Oscillation

The **Barkhausen Criterion** dictates two conditions for sustained oscillation:

(a) **Phase Condition:** $\angle(A_v\beta) = 0°$ or $360°$

(b) **Magnitude Condition:** $|A_v\beta| \geq 1$

where A_v is the amplifier's open-loop voltage gain, and β is the feedback fraction.
Step 3.1: Determine the Feedback Factor (β)
The tank circuit forms a circulating current loop at resonance. Let I_c be the circulating AC current in the tank. The output voltage V_o is developed across C_1:

$$V_o = I_c\left(\frac{1}{j\omega_0 C_1}\right) = \frac{I_c}{j\omega_0 C_1} \tag{3}$$

The feedback voltage V_f is developed across C_2. Because the center node between C_1 and C_2 is grounded, the polarity of V_f with respect to ground is inverted relative to the current flow from the top node:

$$V_f = -I_c\left(\frac{1}{j\omega_0 C_2}\right) = \frac{-I_c}{j\omega_0 C_2} \tag{4}$$

The feedback factor β is the ratio of V_f to V_o:

$$\beta = \frac{V_f}{V_o} = \frac{\frac{-I_c}{j\omega_0 C_2}}{\frac{I_c}{j\omega_0 C_1}} = -\frac{C_1}{C_2} \tag{5}$$

The negative sign mathematically demonstrates the 180° phase shift provided by the tank circuit.
Step 3.2: Determine the Loop Gain ($A_v\beta$)
The CE transistor amplifier has an inverting voltage gain. If R_L' is the equivalent parallel resistance of the resonant tank (which absorbs all losses), the gain is:

$$A_v \approx -g_m R_L' \tag{6}$$

The total loop gain is the product of the amplifier gain and the feedback factor:

$$\text{Loop Gain} = A_v\beta = (-g_m R_L')\left(-\frac{C_1}{C_2}\right) = g_m R_L' \frac{C_1}{C_2} \tag{7}$$

Since all terms (g_m, R_L', C_1, C_2) are real and positive, the loop gain is purely real and positive, satisfying the **phase condition** ($\angle(A_v\beta) = 0°$).
Step 3.3: Condition for Sustained Oscillation
To satisfy the **magnitude condition**, the loop gain must be equal to or greater than unity:

$$\begin{aligned}
A_v\beta &\geq 1 \\
g_m R_L' \frac{C_1}{C_2} &\geq 1 \\
g_m R_L' &\geq \frac{C_2}{C_1}
\end{aligned}$$

Since $|A_v| = g_m R'_L$, the Barkhausen magnitude condition requires the amplifier's voltage gain to satisfy:

$$|\mathbf{A_v}| \geq \frac{\mathbf{C_2}}{\mathbf{C_1}} \tag{8}$$

Alternatively, expressing this in terms of BJT small-signal parameters where $|A_v| \approx h_{fe}\frac{R'_L}{h_{ie}}$:

$$\mathbf{h_{fe}} \geq \frac{\mathbf{h_{ie}}}{\mathbf{R'_L}}\left(\frac{\mathbf{C_2}}{\mathbf{C_1}}\right) \tag{9}$$

Note: In practical oscillator design, $|A_v|$ is chosen to be slightly larger than C_2/C_1 so that oscillations can build up from thermal noise. Once the amplitude grows, non-linearities in the BJT reduce the effective gain until $|A_v\beta| = 1$ in steady state.

Q3. Draw the circuit of the Colpitts oscillator. Then, using the small-signal model of the MOSFET, show that the frequency of oscillation is:

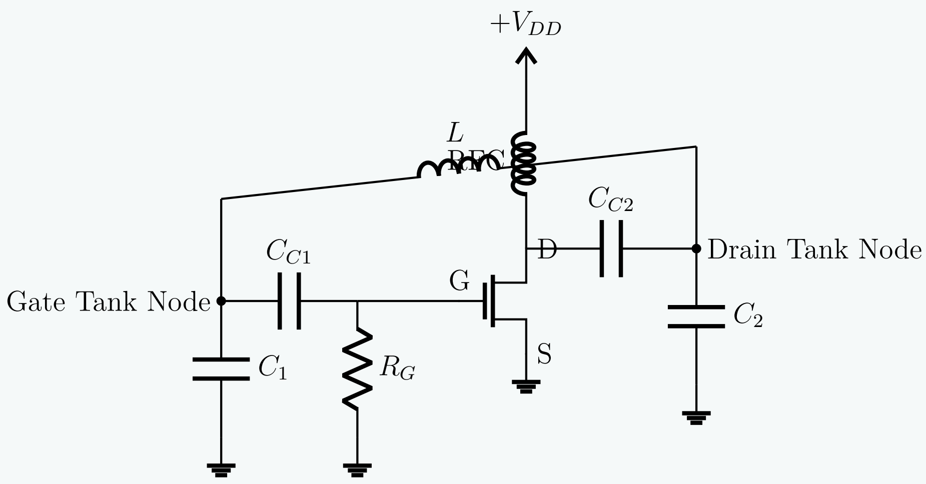
$$\omega_0 = \frac{1}{\sqrt{L \cdot \dfrac{C_1 C_2}{C_1 + C_2}}}$$

and that $\dfrac{C_1}{C_2} = g_m R$. **(26 marks)** **[EEE 263 | L-2/T-1 | 2018–19, 2020–21]**

Answer

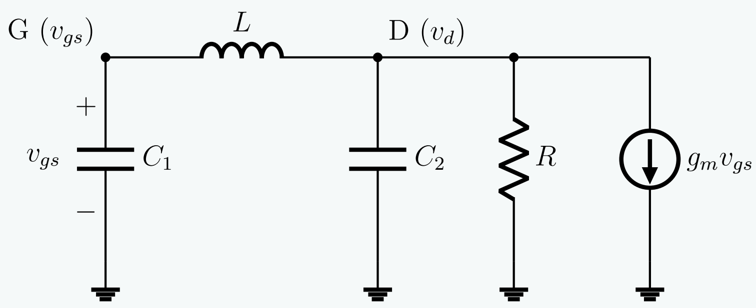
1. Circuit Diagram of the MOSFET Colpitts Oscillator

The Colpitts oscillator uses an LC tank circuit where the feedback is derived from a capacitive voltage divider. In the MOSFET implementation below, the LC network is connected between the drain and the gate. C_{C1} and C_{C2} are large coupling capacitors (acting as AC shorts), and the Radio Frequency Choke (RFC) provides DC bias while acting as an AC open circuit.



2. Small-Signal Equivalent Circuit

Assuming the coupling capacitors (C_{C1}, C_{C2}) are AC shorts, the RFC is an AC open, and the gate bias resistor R_G is large enough to be neglected, the small-signal model is constructed below. The resistor R represents the parallel combination of the MOSFET's output resistance r_o and any equivalent parallel resistance of the tank inductor representing losses.



3. Mathematical Derivation

To find the conditions for oscillation, we apply Kirchhoff's Current Law (KCL) at the nodes of the small-signal equivalent circuit.
Step 3.1: KCL at the Gate Node (G)
The sum of currents leaving the gate node must be zero:

$$\frac{v_{gs}}{1/(j\omega C_1)} + \frac{v_{gs} - v_d}{j\omega L} = 0$$
$$j\omega C_1 v_{gs} + \frac{v_{gs}}{j\omega L} - \frac{v_d}{j\omega L} = 0$$

Multiplying by $j\omega L$ yields:

$$-\omega^2 L C_1 v_{gs} + v_{gs} - v_d = 0$$
$$v_d = v_{gs}(1 - \omega^2 L C_1)$$

Step 3.2: KCL at the Drain Node (D)
The sum of currents leaving the drain node must be zero:

$$\frac{v_d}{R} + j\omega C_2 v_d + \frac{v_d - v_{gs}}{j\omega L} + g_m v_{gs} = 0 \tag{1}$$

Substitute equation (**??**) into the drain node KCL equation:

$$v_{gs}(1 - \omega^2 LC_1)\left(\frac{1}{R} + j\omega C_2 + \frac{1}{j\omega L}\right) - \frac{v_{gs}}{j\omega L} + g_m v_{gs} = 0 \tag{2}$$

For sustained oscillations, the gate-to-source voltage must be non-zero ($v_{gs} \neq 0$). Thus, we can divide the entire equation by v_{gs} and group the real and imaginary components:

$$(1 - \omega^2 LC_1)\left[\frac{1}{R} + j\left(\omega C_2 - \frac{1}{\omega L}\right)\right] + j\frac{1}{\omega L} + g_m = 0 \tag{3}$$

Separating the real and imaginary parts gives two distinct conditions:

$$\textbf{Real Part:} \quad \frac{1 - \omega^2 LC_1}{R} + g_m = 0 \tag{4}$$

$$\textbf{Imaginary Part:} \quad (1 - \omega^2 LC_1)\left(\omega C_2 - \frac{1}{\omega L}\right) + \frac{1}{\omega L} = 0 \tag{5}$$

Step 3.3: Frequency of Oscillation (ω_0)
The oscillation frequency ω_0 is determined by setting the imaginary part to zero. From equation (5):

$$(1 - \omega_0^2 LC_1)\left(\frac{\omega_0^2 LC_2 - 1}{\omega_0 L}\right) + \frac{1}{\omega_0 L} = 0$$

Multiply the entire equation by $\omega_0 L$:

$$\begin{aligned} (1 - \omega_0^2 LC_1)(\omega_0^2 LC_2 - 1) + 1 &= 0 \\ \omega_0^2 LC_2 - 1 - \omega_0^4 L^2 C_1 C_2 + \omega_0^2 LC_1 + 1 &= 0 \\ \omega_0^2 L(C_1 + C_2) - \omega_0^4 L^2 C_1 C_2 &= 0 \end{aligned}$$

Since $\omega_0 \neq 0$, we can divide by $\omega_0^2 L$:

$$\begin{aligned} (C_1 + C_2) &= \omega_0^2 LC_1 C_2 \\ \omega_0^2 &= \frac{C_1 + C_2}{LC_1 C_2} = \frac{1}{L \cdot \dfrac{C_1 C_2}{C_1 + C_2}} \end{aligned}$$

Taking the square root provides the proven frequency of oscillation:

$$\boldsymbol{\omega_0 = \frac{1}{\sqrt{L \cdot \dfrac{C_1 C_2}{C_1 + C_2}}}} \tag{6}$$

Step 3.4: Condition for Sustained Oscillation
To find the required transconductance g_m (gain condition), substitute ω_0^2 back into the real part, equation (4). First, evaluate the term $(1 - \omega_0^2 LC_1)$:

$$1 - \omega_0^2 LC_1 = 1 - \left(\frac{C_1 + C_2}{LC_1 C_2}\right) LC_1 = 1 - \frac{C_1 + C_2}{C_2} = 1 - \left(\frac{C_1}{C_2} + 1\right) = -\frac{C_1}{C_2} \tag{7}$$

Now, substitute this result into equation (4):

$$\begin{aligned} \frac{-\dfrac{C_1}{C_2}}{R} + g_m &= 0 \\ g_m &= \frac{C_1}{C_2 R} \end{aligned}$$

Multiplying both sides by R yields the final proven Barkhausen magnitude condition:

$$\boldsymbol{\frac{C_1}{C_2} = g_m R} \tag{8}$$

Crystal Oscillator

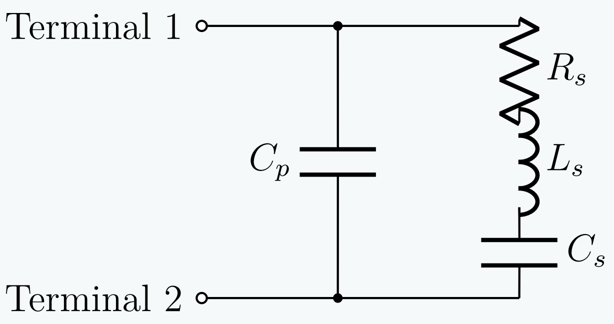
Q1. Draw the equivalent circuit of a piezoelectric crystal. Qualitatively draw the crystal reactance versus frequency and identify the inductive and capacitive reactance regions. **(10 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Equivalent Circuit of a Piezoelectric Crystal

A piezoelectric crystal (such as Quartz) behaves mechanically as a vibrating mass-spring-damper system. When translated into the electrical domain via the piezoelectric effect, its behavior can be modeled as a highly stable resonant circuit.
The electrical equivalent circuit consists of two branches in parallel:

- **Series Branch (Motional Arm):** Comprises R_s, L_s, and C_s, which represent the mechanical friction, mass, and compliance (elasticity) of the crystal, respectively.
- **Parallel Branch (Static Arm):** Comprises C_p, which represents the static capacitance of the electrodes placed on the crystal faces and the stray capacitance of the mounting structure.

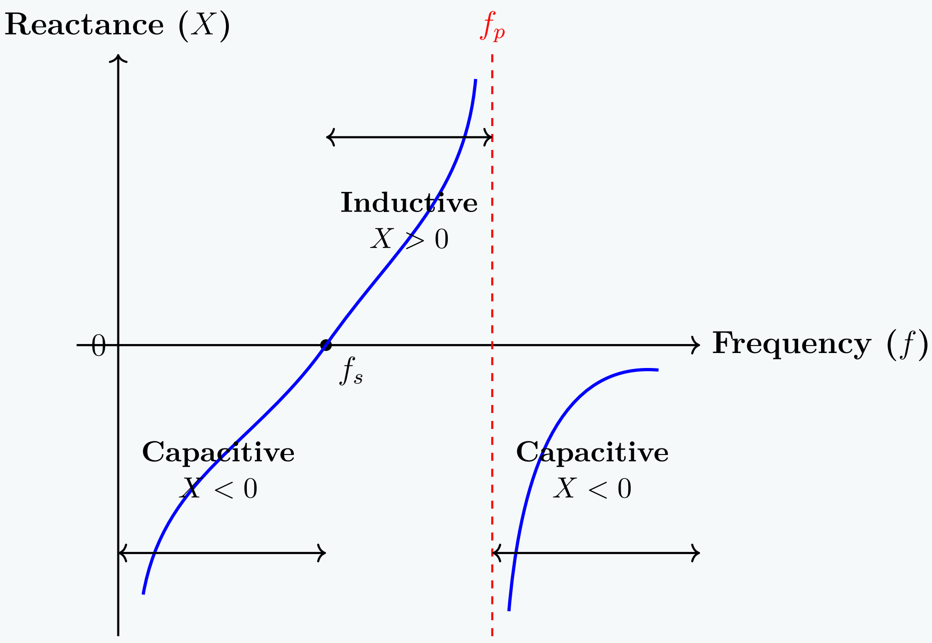


2. Reactance vs. Frequency Characteristics

The reactance X of the crystal varies with the operating frequency f. Due to the presence of both series and parallel resonant circuits, the crystal exhibits two distinct resonant frequencies:

(a) **Series Resonant Frequency (f_s):** The reactance of L_s and C_s cancel out ($X_{L_s} = X_{C_s}$). The series branch impedance is purely resistive and minimal.

(b) **Parallel Resonant Frequency (f_p):** The inductive reactance of the series branch resonates with the parallel capacitance C_p. The total impedance becomes purely resistive and reaches a maximum.



3. Mathematical Analysis & Operating Regions

Assuming the mechanical resistance R_s is negligibly small (which is valid due to the extremely high Q-factor of crystals, often $> 10,000$), the total equivalent impedance Z is approximately equal to the total reactance jX.
The impedance of the series arm is $Z_s = j\left(\omega L_s - \frac{1}{\omega C_s}\right)$. The total impedance of the crystal is the parallel combination of Z_s and the impedance of C_p:

$$\begin{aligned} Z &\approx \frac{Z_s \cdot \left(\frac{1}{j\omega C_p}\right)}{Z_s + \left(\frac{1}{j\omega C_p}\right)} = \frac{j\left(\omega L_s - \frac{1}{\omega C_s}\right) \cdot \frac{1}{j\omega C_p}}{j\left(\omega L_s - \frac{1}{\omega C_s}\right) + \frac{1}{j\omega C_p}} \\ &= \frac{-j}{\omega C_p}\left[\frac{\omega^2 L_s C_s - 1}{\omega^2 L_s C_s - 1 - \frac{C_s}{C_p}}\right] \end{aligned}$$

From this expression, we derive the two critical frequencies:
1. Series Resonant Frequency (f_s): At f_s, the impedance drops to its minimum (ideally zero, realistically R_s). This occurs when the numerator goes to zero:

$$\omega_s^2 L_s C_s - 1 = 0 \implies \mathbf{f_s = \frac{1}{2\pi\sqrt{L_s C_s}}} \tag{1}$$

2. Parallel Resonant Frequency (f_p): At f_p, the impedance peaks (approaches infinity). This occurs when the denominator goes to zero:

$$\omega_p^2 L_s C_s - 1 - \frac{C_s}{C_p} = 0$$
$$\omega_p^2 L_s C_s = 1 + \frac{C_s}{C_p} = \frac{C_p + C_s}{C_p}$$
$$\omega_p^2 = \frac{C_p + C_s}{L_s C_s C_p} = \frac{1}{L_s\left(\frac{C_s C_p}{C_s + C_p}\right)}$$

Letting $C_{eq} = \frac{C_s C_p}{C_s + C_p}$ be the equivalent series capacitance, the parallel resonant frequency is:

$$\mathbf{f_p = \frac{1}{2\pi\sqrt{L_s C_{eq}}}} \tag{2}$$

Identification of Reactance Regions:

- **Capacitive Region 1 ($f < f_s$):** The reactance of C_s dominates the series arm. The entire crystal behaves as a capacitor.
- **Inductive Region ($f_s < f < f_p$):** The series arm becomes inductive ($X_{L_s} > X_{C_s}$), but this inductive reactance is not yet large enough to cause parallel resonance with C_p. The crystal acts as a highly stable inductor. *Note: Oscillators that use the crystal as an inductor must operate strictly in this narrow frequency band.*
- **Capacitive Region 2 ($f > f_p$):** The inductive reactance of the series arm becomes so large that the parallel branch C_p dominates the overall impedance. The crystal behaves as a capacitor again.

Since C_p is typically much greater than C_s (often by a factor of 100 or more), the difference between f_p and f_s is exceedingly small. This extremely narrow inductive bandwidth ensures that a crystal oscillator is highly stable against frequency drift.

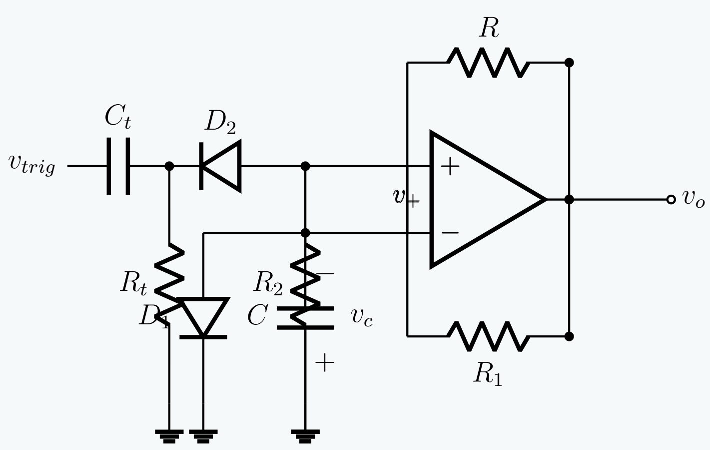
One-Shot (Monostable) Multivibrator

Q1. Draw a one-shot multivibrator circuit using an op-amp. Explain the operation of the circuit. **(26 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Answer

1. Circuit Diagram of Op-Amp Monostable Multivibrator (One-Shot)

A monostable multivibrator has one stable state and one quasi-stable state. It remains in its stable state until triggered by an external pulse, after which it transitions to the quasi-stable state for a specific time duration T determined by an RC network, before returning to the stable state.

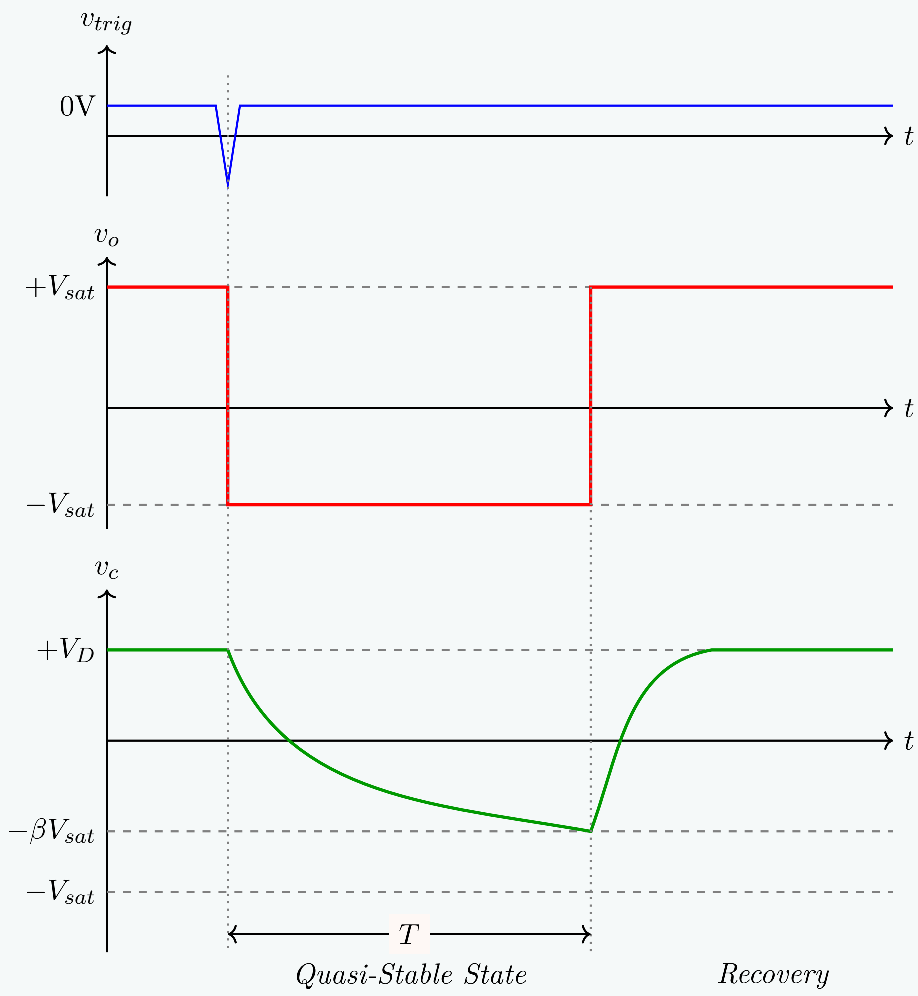


2. Principle of Operation

- **Stable State:** Assume the output is initially at positive saturation ($v_o = +V_{sat}$). A portion of this voltage is fed back to the non-inverting terminal, setting $v_+ = \beta V_{sat}$, where $\beta = \frac{R_2}{R_1+R_2}$. The capacitor C attempts to charge towards $+V_{sat}$ through R. However, as soon as the capacitor voltage v_c reaches roughly 0.7V, diode D_1 becomes forward-biased and clamps v_- at $V_D \approx 0.7$V. Since $v_+(\beta V_{sat}) > v_-(V_D)$, the output firmly remains at $+V_{sat}$. This is the stable state.
- **Triggering and Quasi-Stable State:** A negative-going pulse is applied at v_{trig}. The high-pass filter (C_t, R_t) converts this into a sharp negative spike. Diode D_2 conducts, momentarily pulling v_+ below v_- (0.7V). This causes the op-amp output to switch regeneratively to negative saturation ($v_o = -V_{sat}$).

  Consequently, v_+ immediately becomes $-\beta V_{sat}$. Diode D_1 becomes reverse-biased. The capacitor C now begins to discharge from $+V_D$ towards $-V_{sat}$ through R. This constitutes the quasi-stable state.
- **Recovery:** When v_c decays and crosses just below $-\beta V_{sat}$, v_- becomes more negative than v_+. The op-amp output immediately switches back to $+V_{sat}$. The capacitor now charges from $-\beta V_{sat}$ back towards $+V_{sat}$ until it is clamped again at $+V_D$ by D_1, restoring the original stable state.

3. Waveforms



4. Mathematical Derivation of Pulse Width (T)

During the quasi-stable state, the capacitor discharges from $+V_D$ towards $-V_{sat}$. The general equation for voltage across a capacitor is:

$$v_c(t) = v_{final} + (v_{initial} - v_{final})\, e^{-t/\tau} \tag{1}$$

Substituting the known values for the quasi-stable state ($\tau = RC$):

$$\begin{aligned} v_{initial} &= V_D \\ v_{final} &= -V_{sat} \\ v_c(t) &= -V_{sat} + (V_D - (-V_{sat}))\, e^{-t/RC} \\ v_c(t) &= -V_{sat} + (V_D + V_{sat})\, e^{-t/RC} \end{aligned}$$

The quasi-stable state terminates at $t = T$ when the capacitor voltage reaches the threshold set by the positive feedback network, which is $-\beta V_{sat}$. Substituting $t = T$ and $v_c(T) = -\beta V_{sat}$ into Equation (**??**):

$$\begin{aligned} -\beta V_{sat} &= -V_{sat} + (V_D + V_{sat})\, e^{-T/RC} \\ V_{sat} - \beta V_{sat} &= (V_D + V_{sat})\, e^{-T/RC} \\ V_{sat}(1-\beta) &= (V_D + V_{sat})\, e^{-T/RC} \\ e^{T/RC} &= \frac{V_{sat} + V_D}{V_{sat}(1-\beta)} \end{aligned}$$

Taking the natural logarithm of both sides yields the exact expression for the pulse width T:

$$\mathbf{T} = \mathbf{RC} \ln\left(\frac{\mathbf{V_{sat}} + \mathbf{V_D}}{\mathbf{V_{sat}}(\mathbf{1} - \beta)}\right) \tag{2}$$

Special Case (Standard Assumption):
If we assume $V_D \ll V_{sat}$ (e.g., 0.7V $\ll$ 15V), the equation simplifies to:

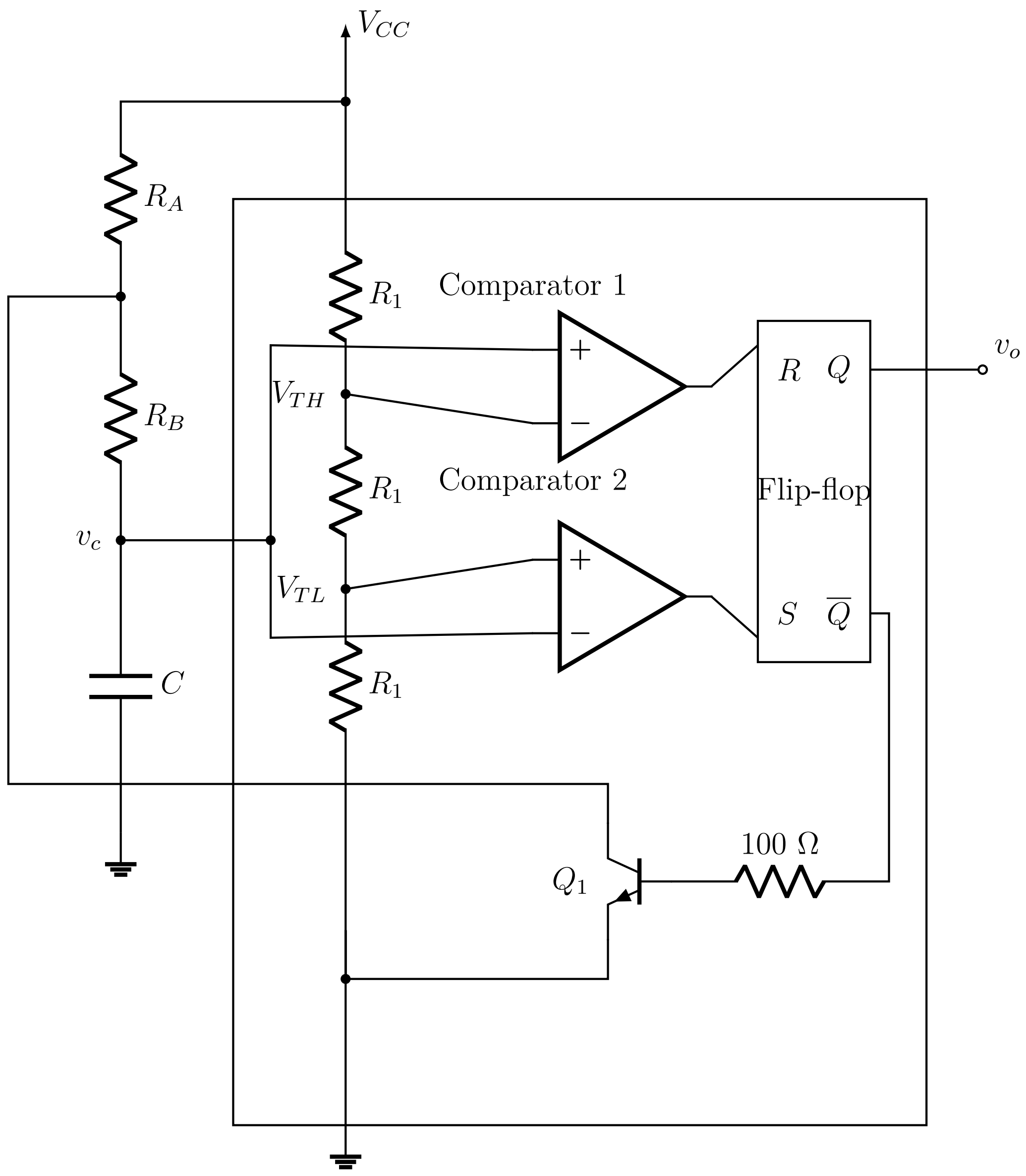
$$T \approx RC \ln\left(\frac{V_{sat}}{V_{sat}(1-\beta)}\right) = RC \ln\left(\frac{1}{1-\beta}\right) \tag{3}$$

Furthermore, if $R_1 = R_2$, then $\beta = 0.5$, and the time period becomes:

$$T \approx RC \ln\left(\frac{1}{1-0.5}\right) = RC \ln(2) \approx \mathbf{0.693RC} \tag{4}$$

555 Timer—Oscillator Design

Q1. A 555 timer is connected in the configuration shown in Figure below. The capacitor shown in the circuit is 1 nF. Find the values of R_A and R_B required to generate an oscillation frequency of 50kHz and duty cycle of 60%. What measures can be taken if we want to achieve duty cycle below 50%?



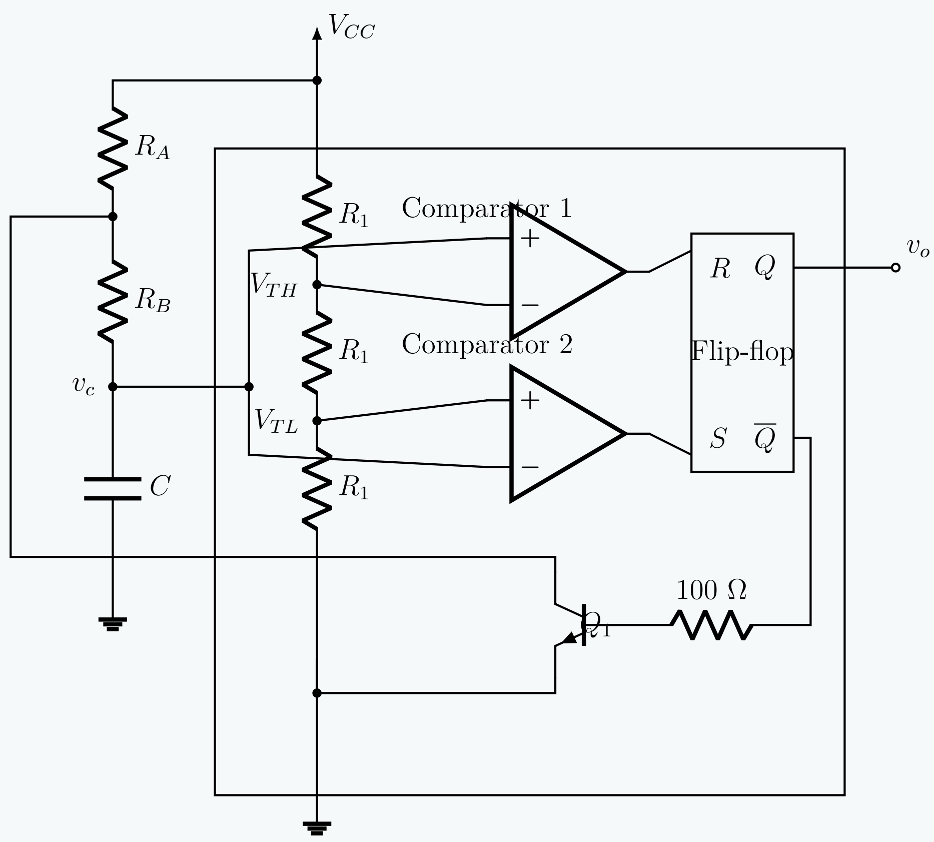
(12 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

555 Timer Astable Multivibrator Analysis

The given circuit represents a 555 timer configured as an **astable multivibrator**, which generates a continuous square wave output without external triggering.

1. Equivalent Circuit Diagram



2. Principle of Operation and Mathematical Formulation

The internal voltage divider sets the threshold voltages for the comparators:

- Upper threshold: $V_{TH} = \frac{2}{3}V_{CC}$
- Lower threshold: $V_{TL} = \frac{1}{3}V_{CC}$

Charging Phase (t_{high}): When the output v_o is HIGH, the discharge transistor Q_1 is OFF. The capacitor C charges from V_{TL} towards V_{CC} through resistors $(R_A + R_B)$. The time taken for the capacitor voltage to reach V_{TH} is:

$$t_{high} = \ln(2) \cdot (R_A + R_B) \cdot C \approx 0.693(R_A + R_B)C \tag{1}$$

Discharging Phase (t_{low}): Once v_c reaches V_{TH}, Comparator 1 resets the flip-flop, causing v_o to go LOW and turning Q_1 ON. The capacitor now discharges from V_{TH} down to V_{TL} exclusively through resistor R_B and Q_1 to ground. The time taken is:

$$t_{low} = \ln(2) \cdot R_B \cdot C \approx 0.693 R_B C \tag{2}$$

The total time period (T) and operating frequency (f) are:

$$T = t_{high} + t_{low} = \ln(2) \cdot (R_A + 2R_B) \cdot C$$
$$f = \frac{1}{T} = \frac{1}{\ln(2) \cdot (R_A + 2R_B) \cdot C} \approx \frac{1.44}{(R_A + 2R_B)C}$$

The duty cycle (D), defined as the ratio of the HIGH time to the total period, is:

$$D = \frac{t_{high}}{T} = \frac{R_A + R_B}{R_A + 2R_B} \tag{3}$$

3. Calculation of R_A and R_B

Given Parameters:

- Capacitance, $C = 1$ nF $= 10^{-9}$ F
- Oscillation Frequency, $f = 50$ kHz $= 50 \times 10^3$ Hz
- Duty Cycle, $D = 60\% = 0.60$

Step 1: Determine the relationship between R_A and R_B using Duty Cycle.

$$D = \frac{R_A + R_B}{R_A + 2R_B} = 0.60$$

$$\begin{aligned} R_A + R_B &= 0.60(R_A + 2R_B) \\ R_A + R_B &= 0.60R_A + 1.20R_B \\ 0.40R_A &= 0.20R_B \\ R_B &= 2R_A \end{aligned}$$

Step 2: Calculate specific resistor values using the Frequency equation. The total period T is:

$$T = \frac{1}{f} = \frac{1}{50 \times 10^3} = 20\ \mu\text{s} = 20 \times 10^{-6}\ \text{s}$$

Substitute $R_B = 2R_A$ into the period formula:

$$\begin{aligned} T &= \ln(2) \cdot (R_A + 2(2R_A)) \cdot C \\ 20 \times 10^{-6} &= \ln(2) \cdot (5R_A) \cdot (10^{-9}) \\ 5R_A &= \frac{20 \times 10^{-6}}{0.6931 \times 10^{-9}} \\ 5R_A &\approx 28854\ \Omega \\ R_A &= \frac{28854}{5} \approx 5771\ \Omega = \mathbf{5.77\ k\Omega} \end{aligned}$$

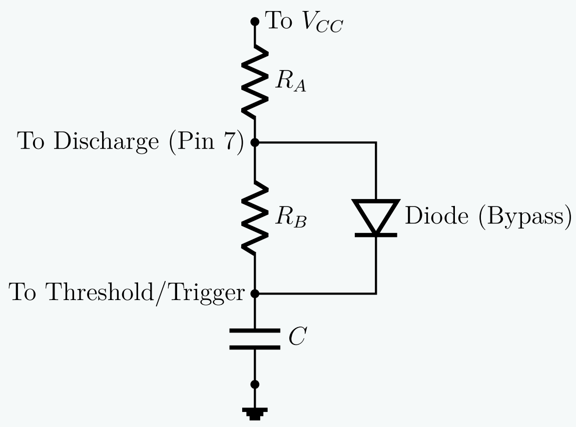
Using $R_B = 2R_A$:

$$R_B = 2 \times 5770.8 = 11541.6\ \Omega = \mathbf{11.54\ k\Omega}$$

4. Achieving Duty Cycle Below 50%

In the standard configuration, the duty cycle is $D = \frac{R_A+R_B}{R_A+2R_B}$. Since R_A and R_B must be positive, $R_A + R_B$ is always strictly greater than R_B. Thus, the charging time is always greater than the discharging time, restricting the duty cycle to **always** $> 50\%$.

Solution: To achieve a duty cycle less than 50%, we must allow the capacitor to charge faster than it discharges. This is accomplished by **connecting a bypass diode across** R_B, pointing from the junction of R_A and R_B to the capacitor.



Circuit Modification Operation:

- **During Charging:** The diode is forward-biased, effectively short-circuiting R_B. The capacitor charges only through R_A. Therefore, $t_{high} \approx 0.693 R_A C$.
- **During Discharging:** The diode is reverse-biased, behaving as an open circuit. The capacitor discharges through R_B as usual. Therefore, $t_{low} \approx 0.693 R_B C$.

With this modification, the duty cycle becomes:

$$D_{modified} = \frac{t_{high}}{t_{high} + t_{low}} \approx \frac{R_A}{R_A + R_B} \tag{4}$$

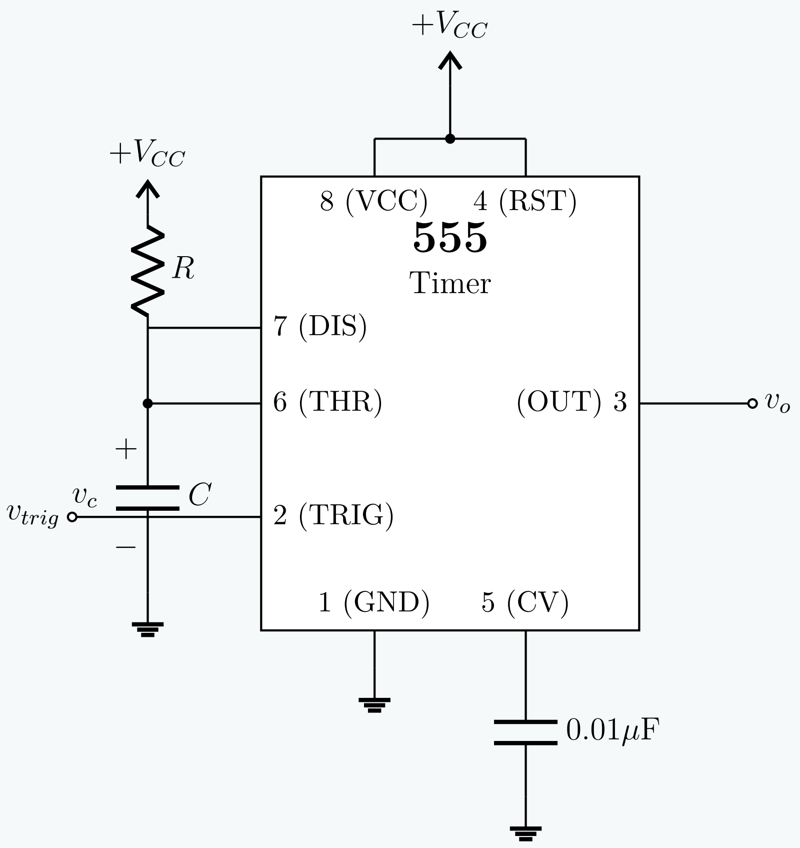
By simply making $R_A < R_B$, a duty cycle of **less than 50%** can be easily achieved.

Q2. With proper connection diagram, explain the monostable mode of operation of a 555 timer. Design a monostable multivibrator using a 555 timer that can generate an output pulse with duration T = 100ms. Draw the waveforms of output voltage and trigger voltage of your designed circuit. **(18 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Answer

1. Monostable Mode Connection Diagram

In the monostable mode, the 555 timer operates as a "one-shot" pulse generator. The external capacitor C is initially discharged by a transistor inside the timer. Upon application of a negative trigger pulse, the circuit produces a single output pulse of a specific duration.



2. Principle of Operation

- **Stable State:** Initially, the internal flip-flop is reset. The internal discharge transistor (connected to pin 7) is ON, which shorts the external capacitor C to ground. The output (pin 3) is LOW (0V), and the capacitor voltage v_c is 0V.
- **Triggering:** When a negative-going trigger pulse falls below $\frac{1}{3}V_{CC}$ at pin 2, the lower comparator sets the internal flip-flop. The output goes HIGH ($+V_{CC}$), and the internal discharge transistor turns OFF.
- **Quasi-Stable State:** With the discharge transistor OFF, the capacitor C begins to charge exponentially towards $+V_{CC}$ through the resistor R.
- **Recovery:** When the capacitor voltage v_c reaches the threshold of $\frac{2}{3}V_{CC}$, the upper comparator resets the flip-flop. The output immediately returns to LOW, and the discharge transistor turns ON, rapidly discharging C back to 0V. The circuit remains in this stable state until the next trigger pulse.

3. Mathematical Derivation of Pulse Width (T)

The voltage across the charging capacitor is governed by the standard step-response equation:

$$v_c(t) = V_{final} + (V_{initial} - V_{final})\, e^{-t/\tau} \tag{1}$$

For this circuit during the charging phase:

- Initial voltage, $V_{initial} = 0\text{V}$
- Final target voltage, $V_{final} = V_{CC}$
- Time constant, $\tau = RC$

Substituting these values into the equation gives:

$$v_c(t) = V_{CC} + (0 - V_{CC})\,e^{-t/RC} = V_{CC}\left(1 - e^{-t/RC}\right) \tag{2}$$

The pulse ends at time $t = T$ when the capacitor voltage reaches the upper threshold voltage of the 555 timer, which is $\frac{2}{3}V_{CC}$. Substituting these conditions:

$$\begin{aligned} v_c(T) &= \frac{2}{3}V_{CC} \\ \frac{2}{3}V_{CC} &= V_{CC}\left(1 - e^{-T/RC}\right) \\ \frac{2}{3} &= 1 - e^{-T/RC} \\ e^{-T/RC} &= \frac{1}{3} \\ -\frac{T}{RC} &= \ln\left(\frac{1}{3}\right) = -\ln(3) \\ T &= RC\ln(3) \approx \mathbf{1.1RC} \end{aligned}$$

4. Monostable Multivibrator Design ($T = 100$ ms)

We need to design the circuit for a pulse duration $T = 100$ ms $= 0.1$ s.
Using the derived formula in Equation (**??**):

$$T = 1.1 \cdot R \cdot C$$

Step 1: Choose a standard value for Capacitor C.
For a delay in the milliseconds range, a typical electrolytic or film capacitor is suitable. Let us select:

$$C = 1\ \mu\text{F} = 10^{-6}\ \text{F}$$

Step 2: Calculate the required Resistor R.

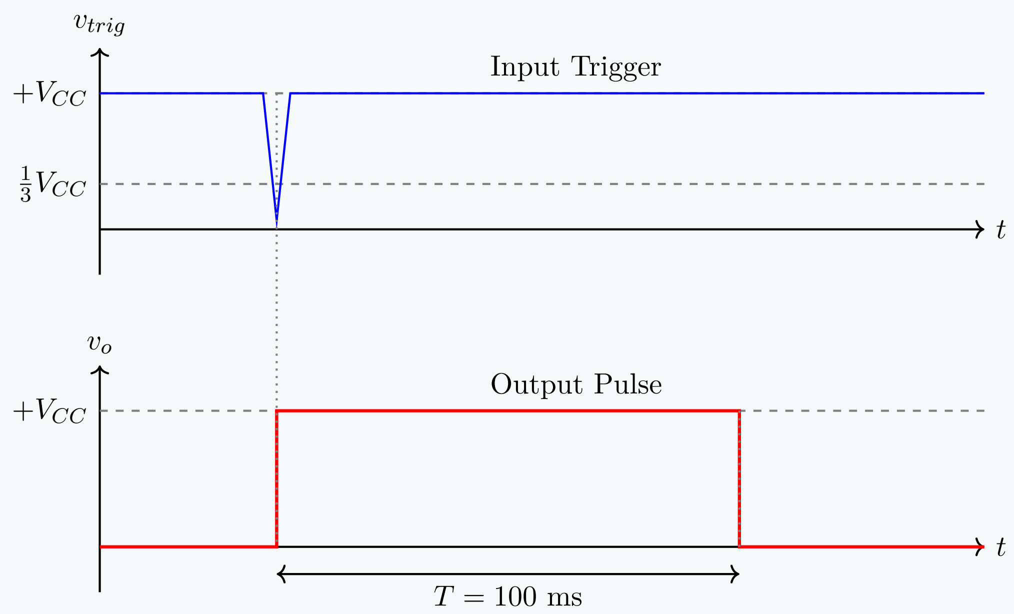
$$\begin{aligned} R &= \frac{T}{1.1 \cdot C} \\ R &= \frac{0.1}{1.1 \times 10^{-6}} \\ R &\approx 90,909\ \Omega \approx 90.9\ \text{k}\Omega \end{aligned}$$

Design Summary:

- **Timing Capacitor (C):** 1 μF
- **Timing Resistor (R):** 91 kΩ (Standard E24 series value)
- **Bypass Capacitor:** 0.01 μF at pin 5 to ground (for noise immunity).

5. Waveforms

The input trigger and corresponding output voltage waveforms are shown below:



Q3. Design an oscillator circuit using a 555 Timer IC which has a square wave output with oscillation frequency of 20 kHz and duty cycle of 25%. Provide a necessary illustration of the designed circuit and draw the wave shape. **(20 marks)** **[EEE 263 | L-2/T-1 | 2020–21]**

Answer

1. Design Theory & Principle of Operation

In a standard 555 timer astable multivibrator, the capacitor charges through resistors ($R_A + R_B$) and discharges only through R_B. Because the charging time is always greater than the discharging time, the standard configuration can only produce a duty cycle strictly greater than 50%.
To achieve a duty cycle of 25% (which is $< 50\%$), we must modify the circuit by introducing a **bypass diode** (D_1) across resistor R_B.

- **Charging Phase (t_{high}):** The output is HIGH. The internal discharge transistor (pin 7) is OFF. The capacitor C charges from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$. The current flows through R_A and the forward-biased diode D_1, bypassing R_B. Assuming an ideal diode, the charging time is primarily determined by R_A and C.
- **Discharging Phase (t_{low}):** The output is LOW. The internal discharge transistor turns ON, shorting pin 7 to ground. The capacitor discharges from $\frac{2}{3}V_{CC}$ to $\frac{1}{3}V_{CC}$. The diode D_1 is reverse-biased, forcing the discharge current to flow entirely through R_B.

2. Mathematical Design & Calculations

Given Design Specifications:

- Oscillation Frequency, $f = 20$ kHz
- Duty Cycle, $D = 25\% = 0.25$

Step 1: Calculate Time Periods The total time period (T) of the oscillation is:

$$T = \frac{1}{f} = \frac{1}{20 \times 10^3} = 50\ \mu\text{s} \tag{1}$$

From the required duty cycle, we determine the HIGH (t_{high}) and LOW (t_{low}) times:

$$t_{high} = D \times T = 0.25 \times 50\ \mu\text{s} = 12.5\ \mu\text{s}$$
$$t_{low} = (1 - D) \times T = 0.75 \times 50\ \mu\text{s} = 37.5\ \mu\text{s}$$

Step 2: Component Selection (Assuming Ideal Diode) The equations for the modified astable circuit are:

$$t_{high} = \ln(2) \cdot R_A \cdot C \approx 0.693 R_A C$$
$$t_{low} = \ln(2) \cdot R_B \cdot C \approx 0.693 R_B C$$

Let us select a standard timing capacitor value suitable for the 20 kHz frequency range. Let:

$$\mathbf{C = 1}\ \text{nF} = \mathbf{1 \times 10^{-9}}\ \text{F} \tag{2}$$

Calculate R_A using t_{high}:

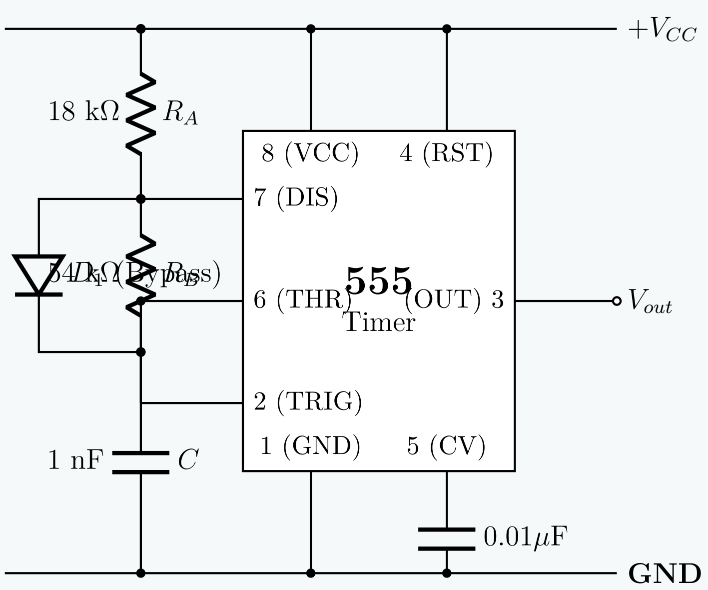
$$12.5 \times 10^{-6} = 0.693 \cdot R_A \cdot (1 \times 10^{-9})$$
$$R_A = \frac{12.5 \times 10^{-6}}{0.693 \times 10^{-9}} \approx 18{,}037\ \Omega \approx \mathbf{18}\ \text{k}\boldsymbol{\Omega}$$

Calculate R_B using t_{low}:

$$37.5 \times 10^{-6} = 0.693 \cdot R_B \cdot (1 \times 10^{-9})$$
$$R_B = \frac{37.5 \times 10^{-6}}{0.693 \times 10^{-9}} \approx 54{,}112\ \Omega \approx \mathbf{54}\ \text{k}\boldsymbol{\Omega}$$

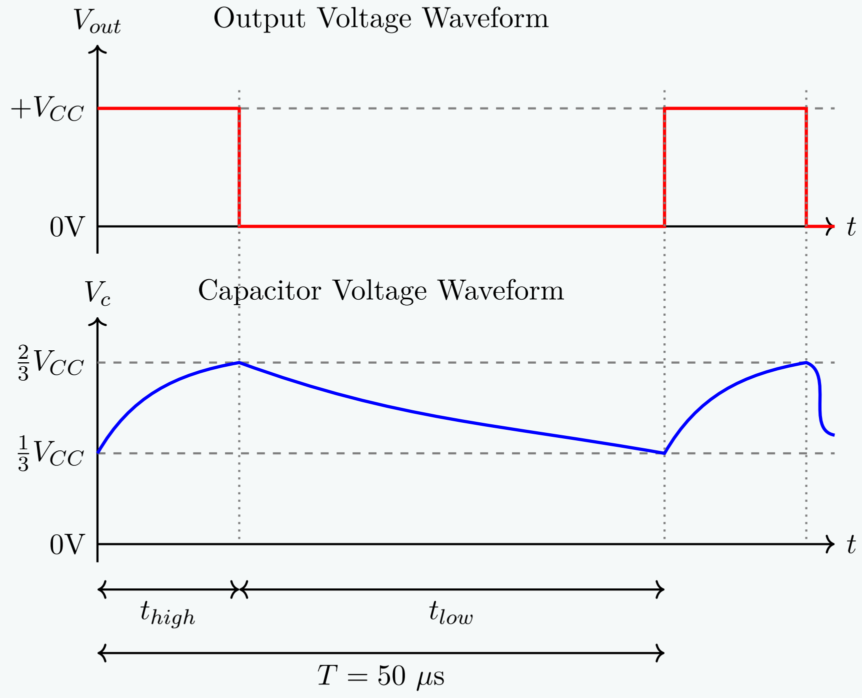
Note: 18kΩ is a standard E24 resistor value. 54kΩ can be achieved by combining a 51kΩ and a 3kΩ resistor, or using a 56kΩ standard value with slight deviation.

3. Designed Circuit Schematic

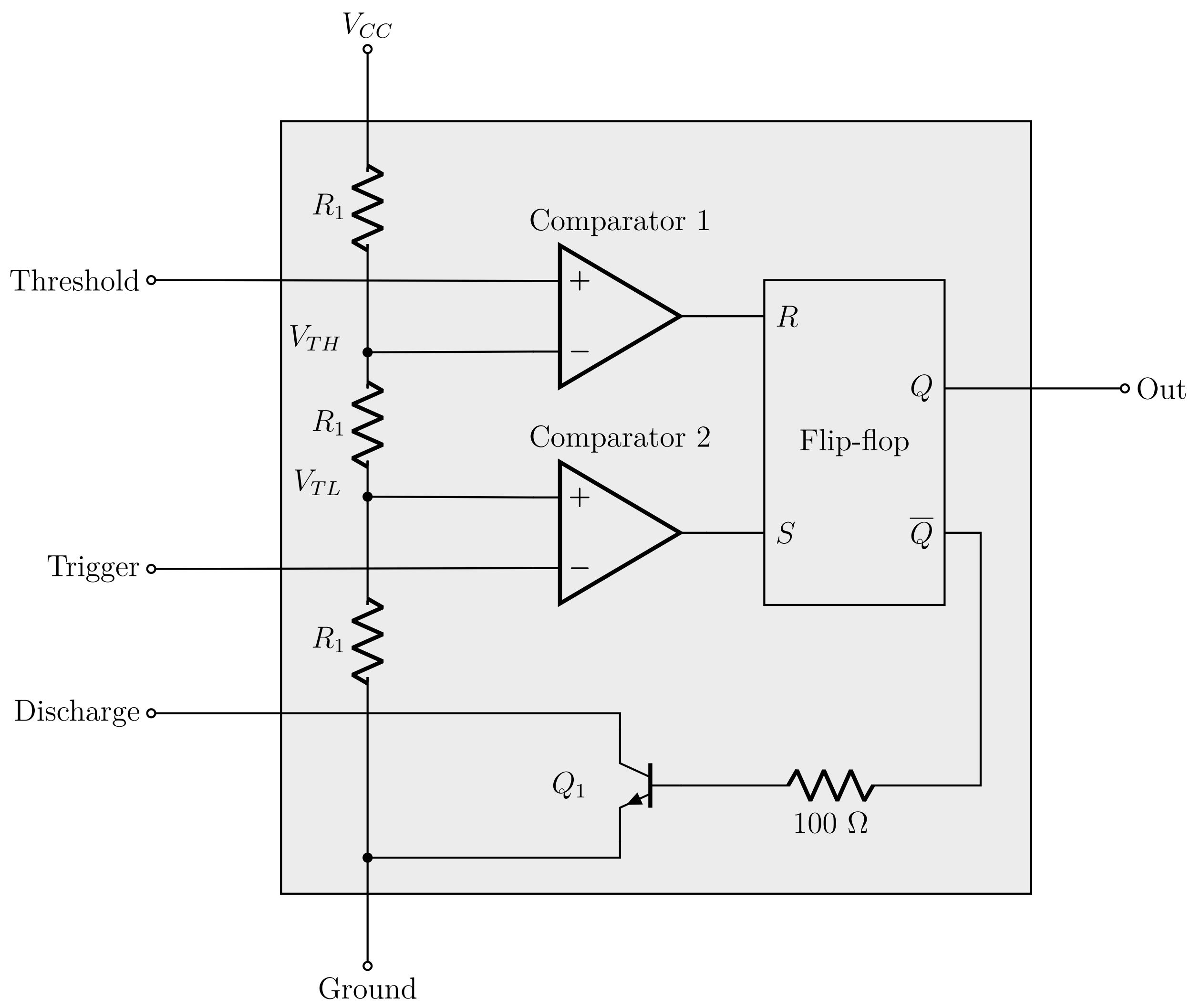


4. Output and Capacitor Waveforms

Because $t_{low} = 3 \times t_{high}$, the capacitor takes 3 times longer to discharge than to charge.



Q4. Design a circuit using a 555 timer to generate a 100 kHz square wave with a duty cycle of 40%. Draw the final connection diagram with proper component values.



(15 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Answer

1. Theory and Principle of Operation

In a standard 555 timer astable multivibrator, the timing capacitor C charges through both resistors ($R_A + R_B$) and discharges only through R_B. Since the charging resistance is always greater than the discharging resistance, the charging time is always strictly longer than the discharging time. As a result, a standard 555 astable circuit can only produce a duty cycle strictly greater than 50%.

To achieve a duty cycle of 40% (which is less than 50%), we must modify the standard circuit by introducing a **bypass diode** (D_1) across resistor R_B.

- **Charging Phase (Output HIGH):** The internal discharge transistor is OFF. Current flows from V_{CC} through R_A and the forward-biased diode D_1 to charge the capacitor C, effectively bypassing R_B. Assuming an ideal diode, the charging time (t_{high}) depends only on R_A and C.
- **Discharging Phase (Output LOW):** The internal discharge transistor is ON (pin 7 is grounded). The capacitor discharges

through R_B to ground. The diode D_1 is reverse-biased and acts as an open circuit. The discharging time (t_{low}) depends only on R_B and C.

2. Mathematical Design & Calculations

Given Specifications:

- Oscillation Frequency, $f = 100$ kHz $= 100 \times 10^3$ Hz
- Duty Cycle, $D = 40\% = 0.40$

Step 1: Calculate Total Time Period and Phase Durations The total time period (T) of the square wave is:

$$T = \frac{1}{f} = \frac{1}{100 \times 10^3} = 10\ \mu\text{s} \tag{1}$$

Based on the required duty cycle, the HIGH time (t_{high}) and LOW time (t_{low}) are:

$$t_{high} = D \times T = 0.40 \times 10\ \mu\text{s} = 4\ \mu\text{s}$$
$$t_{low} = (1 - D) \times T = 0.60 \times 10\ \mu\text{s} = 6\ \mu\text{s}$$

Step 2: Component Selection For the modified astable circuit (assuming an ideal diode), the time intervals are governed by:

$$t_{high} = \ln(2) \cdot R_A \cdot C \approx 0.693 R_A C$$
$$t_{low} = \ln(2) \cdot R_B \cdot C \approx 0.693 R_B C$$

Let us select a standard, widely available timing capacitor appropriate for 100 kHz operation. We choose:

$$\mathbf{C = 1}\ \text{nF} = \mathbf{1 \times 10^{-9}}\ \text{F} \tag{2}$$

Now, calculate R_A using the t_{high} equation:

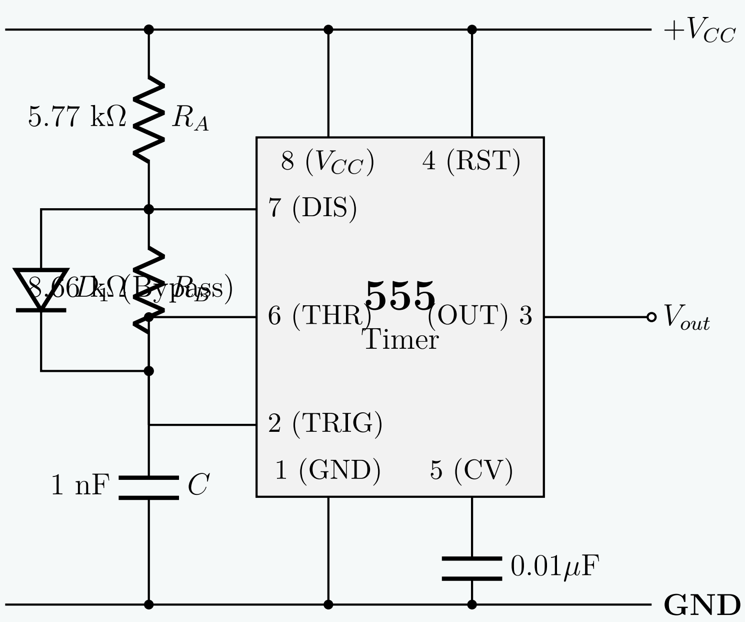
$$4 \times 10^{-6} = 0.693 \cdot R_A \cdot (1 \times 10^{-9})$$
$$R_A = \frac{4 \times 10^{-6}}{0.693 \times 10^{-9}} \approx 5772\ \Omega \approx \mathbf{5.77\ k\Omega}$$

Calculate R_B using the t_{low} equation:

$$6 \times 10^{-6} = 0.693 \cdot R_B \cdot (1 \times 10^{-9})$$
$$R_B = \frac{6 \times 10^{-6}}{0.693 \times 10^{-9}} \approx 8658\ \Omega \approx \mathbf{8.66\ k\Omega}$$

Note: In practice, 5.77 kΩ can be implemented using a 5.6 kΩ standard resistor in series with a 180 Ω resistor or a precision trimpot. 8.66 kΩ can be implemented using an 8.2 kΩ in series with 470 Ω.

3. Final Circuit Connection Diagram



Q5. Design a square wave generator with 10 ms period and 75% duty cycle using a 555 timer. **(15 marks)** **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

1. Theory and Principle of Operation

A 555 timer configured in **astable mode** operates as a free-running oscillator, producing a continuous square wave output. In the standard configuration, the external timing capacitor C charges through two resistors, R_A and R_B, and discharges through only one resistor, R_B.

- **Charging Phase (Output HIGH):** The internal discharge transistor (connected to pin 7) is OFF. The capacitor C charges from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$ through the series combination of R_A and R_B.
- **Discharging Phase (Output LOW):** The internal discharge transistor turns ON, shorting pin 7 to ground. The capacitor C discharges from $\frac{2}{3}V_{CC}$ down to $\frac{1}{3}V_{CC}$ through R_B alone.

Since the charging path $(R_A + R_B)$ always has a higher resistance than the discharging path (R_B), the HIGH time is always strictly greater than the LOW time. Therefore, the standard astable configuration is perfectly suited for generating square waves with a duty cycle strictly greater than 50%. The required 75% duty cycle can be natively achieved without any bypass diodes.

2. Mathematical Design & Calculations

Given Specifications:

- Time Period, $T = 10$ ms $= 10 \times 10^{-3}$ s
- Duty Cycle, $D = 75\% = 0.75$

Step 1: Calculate HIGH and LOW times The duty cycle D determines the HIGH time (t_{high}) and LOW time (t_{low}):

$$t_{high} = D \times T = 0.75 \times 10 \text{ ms} = 7.5 \text{ ms}$$
$$t_{low} = (1 - D) \times T = 0.25 \times 10 \text{ ms} = 2.5 \text{ ms}$$

Step 2: Establish Resistor Relationships The standard design equations for the 555 astable multivibrator are:

$$t_{high} = \ln(2) \cdot (R_A + R_B) \cdot C \approx 0.693(R_A + R_B)C \quad (1)$$
$$t_{low} = \ln(2) \cdot R_B \cdot C \approx 0.693 R_B C \quad (2)$$

Dividing Equation (1) by Equation (2):

$$\frac{t_{high}}{t_{low}} = \frac{R_A + R_B}{R_B}$$
$$\frac{7.5 \text{ ms}}{2.5 \text{ ms}} = \frac{R_A}{R_B} + 1$$
$$3 = \frac{R_A}{R_B} + 1 \implies \frac{R_A}{R_B} = 2 \implies \mathbf{R_A = 2R_B}$$

Step 3: Component Selection Let us select a standard timing capacitor value. For a 10 ms period (100 Hz frequency), a 1 μF capacitor is appropriate.

$$\mathbf{C = 1}\ \mu\text{F} = \mathbf{1 \times 10^{-6}}\ \text{F} \quad (3)$$

Substitute C and t_{low} into Equation (2) to find R_B:

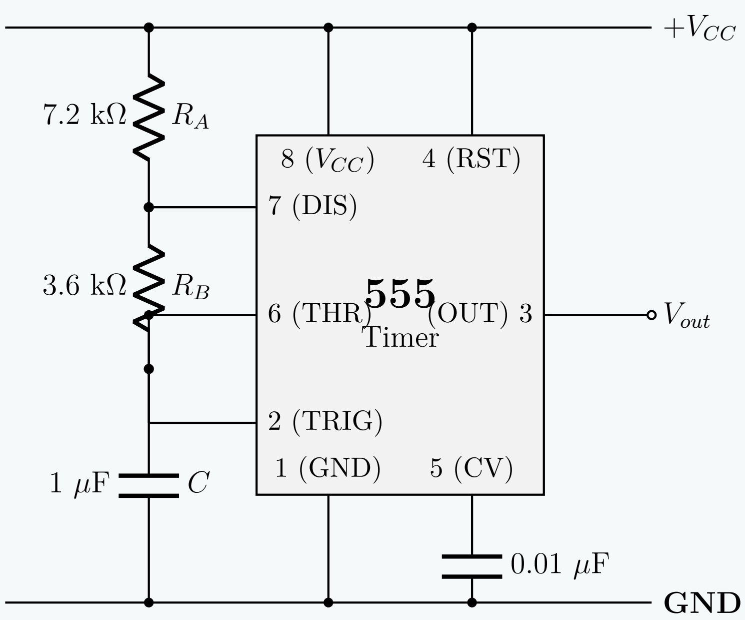
$$2.5 \times 10^{-3} = 0.693 \cdot R_B \cdot (1 \times 10^{-6})$$
$$R_B = \frac{2.5 \times 10^{-3}}{0.693 \times 10^{-6}} \approx 3607.5\ \Omega \approx \mathbf{3.6\ k\Omega}$$

Using the relationship derived in Equation (??), calculate R_A:

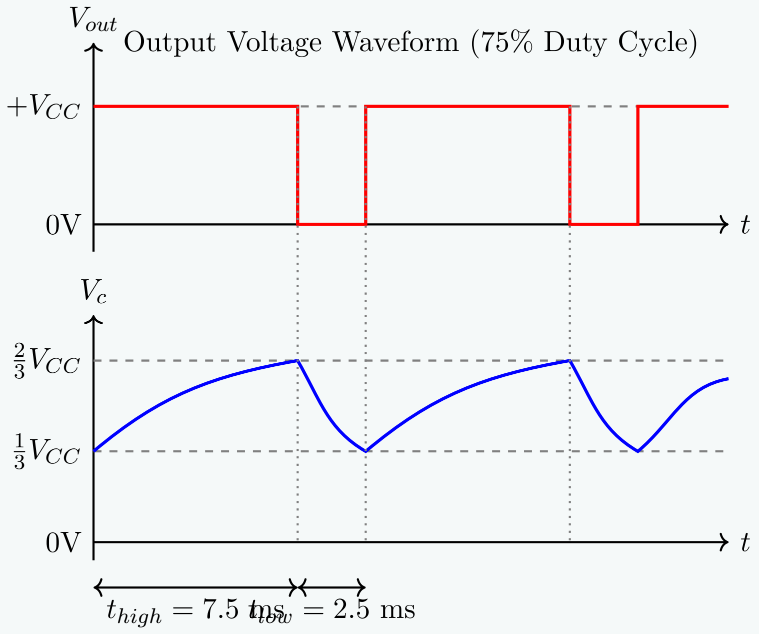
$$R_A = 2R_B = 2 \times 3.6\ \text{k}\Omega = \mathbf{7.2\ k\Omega} \quad (4)$$

Note: Both 3.6 kΩ and 7.2 kΩ can be implemented using standard precision resistors or variable trimpots to perfectly calibrate the 10 ms period.

3. Designed Circuit Schematic



4. Expected Waveforms



Q6. To get an output pulse of 10 ms width when a trigger is applied, an external resistor and a 1 μF capacitor are connected with a 555 timer. What should be the value of the resistor? **(10 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Theory and Principle of Operation

A 555 timer circuit that generates a single output pulse of a specific duration in response to an external trigger is operating in **Monostable Multivibrator** (or "one-shot") mode.
In this configuration:

- **Stable State:** The output is normally LOW. The internal discharge transistor is ON, keeping the external timing capacitor C discharged at 0V.
- **Quasi-Stable State:** When a negative-going trigger pulse (falling below $\frac{1}{3}V_{CC}$) is applied to pin 2, the internal flip-flop sets. The output goes HIGH, and the discharge transistor turns OFF. The capacitor C begins to charge exponentially towards V_{CC} through the external resistor R.
- **Recovery:** When the voltage across the capacitor reaches the threshold level of $\frac{2}{3}V_{CC}$, the upper comparator resets the flip-flop. The output immediately returns to LOW, and the discharge transistor turns ON, rapidly discharging C.

2. Mathematical Derivation

The voltage across the charging capacitor is governed by the step-response equation:

$$v_c(t) = V_{CC}\left(1 - e^{-t/RC}\right) \tag{1}$$

The output pulse ends at time $t = T$ when the capacitor voltage reaches $\frac{2}{3}V_{CC}$. Substituting these conditions:

$$\begin{aligned}
\frac{2}{3}V_{CC} &= V_{CC}\left(1 - e^{-T/RC}\right) \\
\frac{2}{3} &= 1 - e^{-T/RC} \\
e^{-T/RC} &= \frac{1}{3} \\
-\frac{T}{RC} &= \ln\left(\frac{1}{3}\right) = -\ln(3) \\
T &= \ln(3) \cdot R \cdot C \approx 1.1RC
\end{aligned}$$

3. Step-by-Step Solution

Given Parameters:

- Output pulse width, $T = 10$ ms $= 10 \times 10^{-3}$ s
- Timing capacitor, $C = 1\ \mu\text{F} = 1 \times 10^{-6}$ F

Using the standard derived formula for the 555 monostable multivibrator:

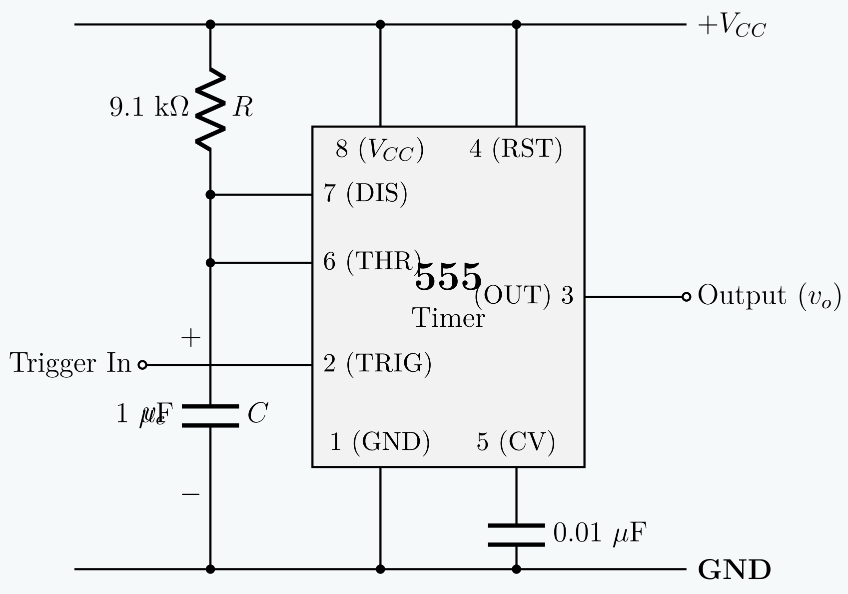
$$T = 1.1 \cdot R \cdot C \tag{2}$$

Rearranging the formula to solve for the unknown resistance R:

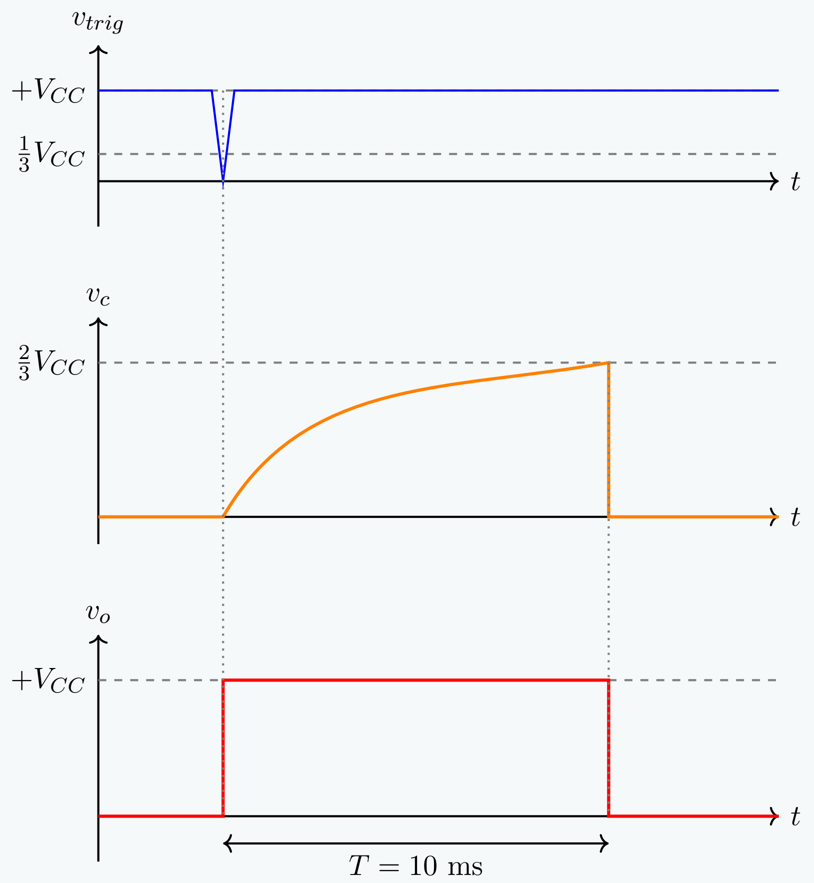
$$R = \frac{T}{1.1 \cdot C}$$
$$R = \frac{10 \times 10^{-3}}{1.1 \times 1 \times 10^{-6}}$$
$$R = \frac{10}{1.1} \times 10^{3}$$
$$R \approx 9090.9\ \Omega$$

Final Answer: The required value for the external resistor is approximately **9.1 kΩ**.

4. Monostable Circuit Diagram and Waveforms



Waveform Illustration:



Q7. Design an astable multivibrator using a 555 timer with a period of 10 ms and 75% duty cycle. **(10 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Theory and Principle of Operation

A 555 timer configured in **astable mode** operates as a free-running oscillator, producing a continuous square wave output. The timing of the output is controlled by an external RC network consisting of two resistors (R_A and R_B) and a capacitor (C).

- **Charging Phase (Output HIGH):** The internal discharge transistor (connected to pin 7) is OFF. The external capacitor C charges from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$ through the series combination of R_A and R_B.
- **Discharging Phase (Output LOW):** The internal discharge transistor turns ON, shorting pin 7 to ground. The capacitor

C discharges from $\frac{2}{3}V_{CC}$ down to $\frac{1}{3}V_{CC}$ through R_B alone.

Since the charging path ($R_A + R_B$) always has a higher resistance than the discharging path (R_B), the HIGH time is inherently strictly greater than the LOW time in this standard configuration. Therefore, it is naturally suited for generating square waves with a duty cycle strictly greater than 50%. The required 75% duty cycle can be achieved without any bypass diodes.

2. Mathematical Design & Calculations

Given Specifications:

- Time Period, $T = 10$ ms $= 10 \times 10^{-3}$ s
- Duty Cycle, $D = 75\% = 0.75$

Step 1: Calculate HIGH and LOW times The duty cycle D determines the HIGH time (t_{high}) and LOW time (t_{low}):

$$t_{high} = D \times T = 0.75 \times 10 \text{ ms} = 7.5 \text{ ms}$$
$$t_{low} = (1 - D) \times T = 0.25 \times 10 \text{ ms} = 2.5 \text{ ms}$$

Step 2: Establish Resistor Relationships The standard design equations for the 555 astable multivibrator are:

$$t_{high} = \ln(2) \cdot (R_A + R_B) \cdot C \approx 0.693(R_A + R_B)C \tag{1}$$
$$t_{low} = \ln(2) \cdot R_B \cdot C \approx 0.693 R_B C \tag{2}$$

Dividing Equation (1) by Equation (2):

$$\frac{t_{high}}{t_{low}} = \frac{R_A + R_B}{R_B}$$
$$\frac{7.5 \text{ ms}}{2.5 \text{ ms}} = \frac{R_A}{R_B} + 1$$
$$3 = \frac{R_A}{R_B} + 1 \implies \frac{R_A}{R_B} = 2 \implies \mathbf{R_A = 2R_B}$$

Step 3: Component Selection Let us select a standard, commercially available timing capacitor value. For a 10 ms period, a 1 μF capacitor is highly appropriate to keep resistor values in the optimal 1 kΩ to 100 kΩ range.

$$\mathbf{C = 1}\ \mu\text{F} = \mathbf{1 \times 10^{-6}} \text{ F} \tag{3}$$

Substitute C and t_{low} into Equation (2) to solve for R_B:

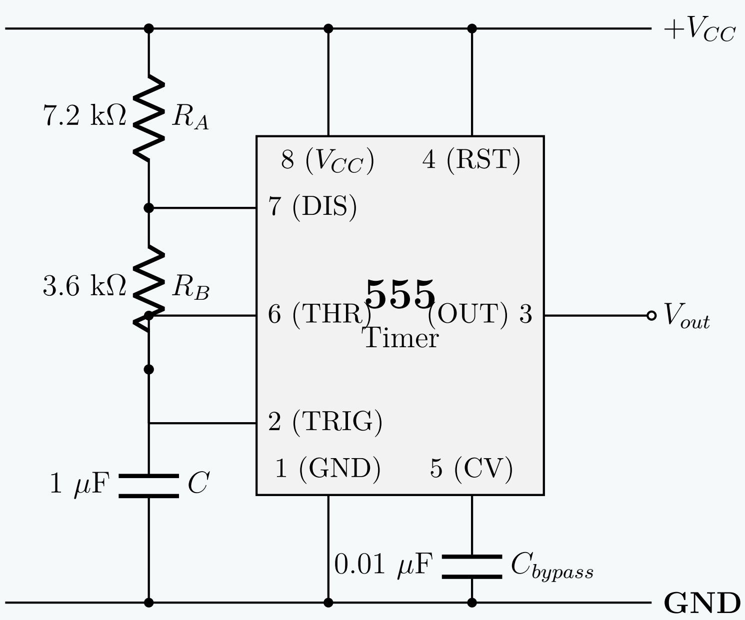
$$2.5 \times 10^{-3} = 0.693 \cdot R_B \cdot (1 \times 10^{-6})$$
$$R_B = \frac{2.5 \times 10^{-3}}{0.693 \times 10^{-6}} \approx 3607.5\ \Omega \approx \mathbf{3.6}\ \mathbf{k\Omega}$$

Using the relationship established in Equation (**??**), calculate R_A:

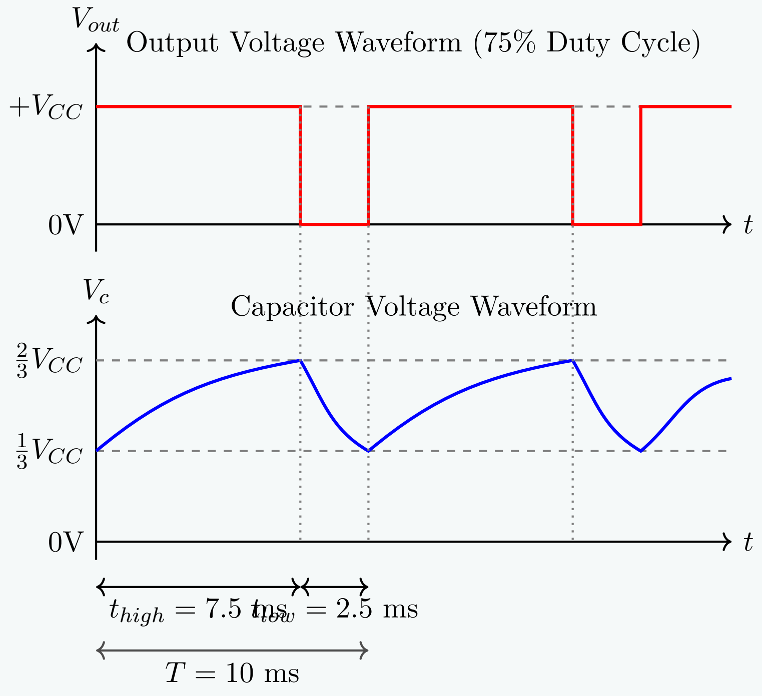
$$R_A = 2R_B = 2 \times 3607.5\ \Omega \approx 7215\ \Omega \approx \mathbf{7.2}\ \mathbf{k\Omega} \tag{4}$$

Note: Both 3.6 kΩ and 7.2 kΩ can be implemented using standard 1% precision resistors, or by utilizing standard E12 values (e.g., 3.3 kΩ + 300 Ω for R_B) in series for exact timing.

3. Designed Circuit Schematic



4. Expected Waveforms



Q8. Design a circuit using a 555 IC which results in a square wave with oscillation frequency of 100 kHz and 75% duty cycle. Draw your designed circuit and output wave shape. **(20 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Theory and Principle of Operation

A 555 timer integrated circuit can be operated as an **astable multivibrator**, acting as a free-running oscillator that generates a continuous square wave. The oscillation frequency and duty cycle are determined by two external resistors, R_A and R_B, and one timing capacitor, C.

- **Charging Phase (Output HIGH):** The internal discharge transistor (connected to pin 7) is OFF. The external capacitor C charges exponentially from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$ through the series combination of resistors R_A and R_B.
- **Discharging Phase (Output LOW):** The internal discharge transistor turns ON, providing a low-resistance path to ground at pin 7. The capacitor C discharges from $\frac{2}{3}V_{CC}$ to $\frac{1}{3}V_{CC}$ through resistor R_B only.

Since the charging time constant depends on $(R_A + R_B)C$ and the discharging time constant depends solely on $R_B C$, the charging time (t_{high}) is always strictly greater than the discharging time (t_{low}). This standard configuration naturally yields a duty cycle greater than 50%, making it perfect for generating the required 75% duty cycle square wave without any additional components (like bypass diodes).

2. Mathematical Design & Component Selection

Given Specifications:

- Oscillation Frequency, $f = 100$ kHz $= 100 \times 10^3$ Hz
- Duty Cycle, $D = 75\% = 0.75$

Step 1: Calculate the Time Period and Phase Durations The total time period (T) of the square wave is:

$$T = \frac{1}{f} = \frac{1}{100 \times 10^3} = 10\ \mu\text{s} \tag{1}$$

The duty cycle dictates the HIGH time (t_{high}) and LOW time (t_{low}):

$$t_{high} = D \times T = 0.75 \times 10\ \mu\text{s} = 7.5\ \mu\text{s}$$
$$t_{low} = (1 - D) \times T = 0.25 \times 10\ \mu\text{s} = 2.5\ \mu\text{s}$$

Step 2: Establish Resistor Relationships The design equations for the standard 555 astable multivibrator are:

$$t_{high} = \ln(2) \cdot (R_A + R_B) \cdot C \approx 0.693(R_A + R_B)C \tag{2}$$
$$t_{low} = \ln(2) \cdot R_B \cdot C \approx 0.693 R_B C \tag{3}$$

Taking the ratio of t_{high} to t_{low}:

$$\frac{t_{high}}{t_{low}} = \frac{R_A + R_B}{R_B}$$
$$\frac{7.5\ \mu\text{s}}{2.5\ \mu\text{s}} = \frac{R_A}{R_B} + 1$$
$$3 = \frac{R_A}{R_B} + 1 \implies \frac{R_A}{R_B} = 2 \implies \mathbf{R_A = 2R_B}$$

Step 3: Calculate Component Values For a relatively high frequency of 100 kHz, a small timing capacitor is suitable to keep resistor values in a practical range. Let us select:

$$\mathbf{C} = \mathbf{1}\text{ nF} = \mathbf{1} \times \mathbf{10^{-9}}\text{ F} \tag{4}$$

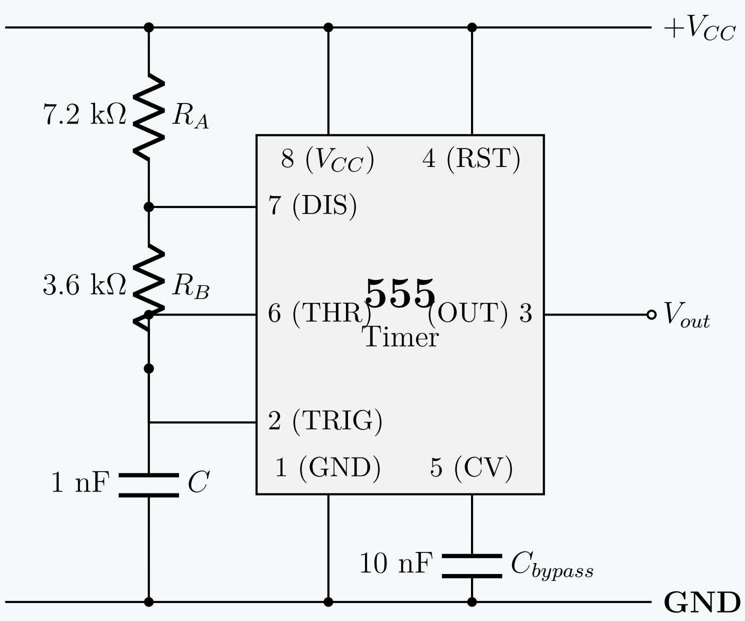
Using the discharging time equation (3) to find R_B:

$$2.5 \times 10^{-6} = 0.693 \cdot R_B \cdot (1 \times 10^{-9})$$
$$R_B = \frac{2.5 \times 10^{-6}}{0.693 \times 10^{-9}} \approx 3607.5\ \Omega \approx \mathbf{3.6}\text{ k}\boldsymbol{\Omega}$$

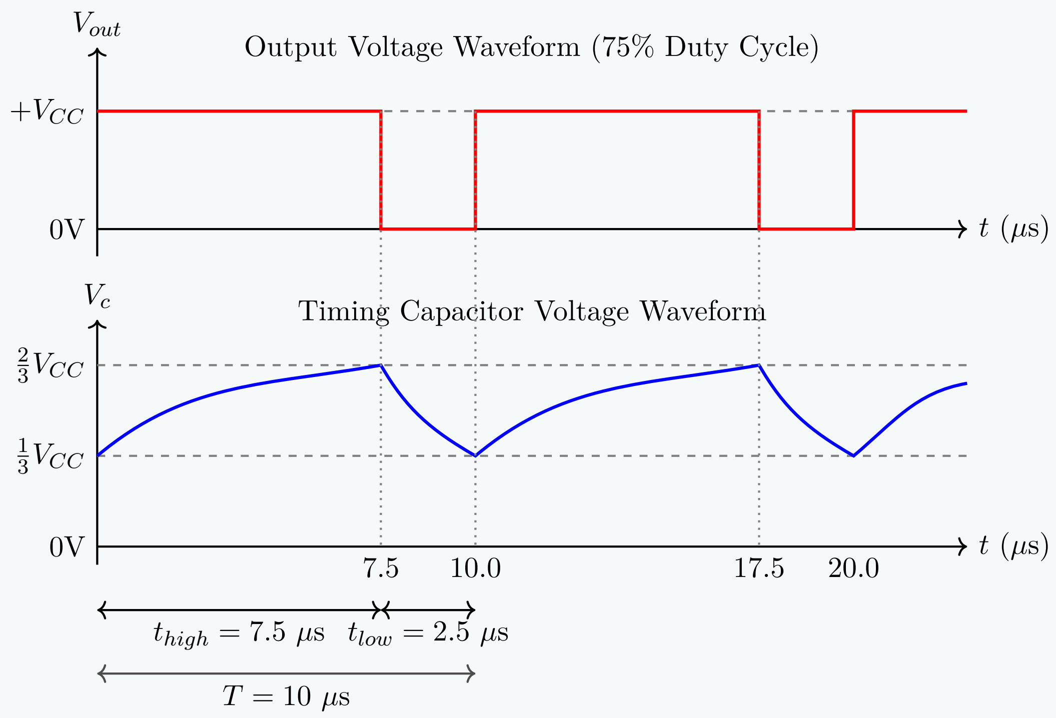
Using the relationship $R_A = 2R_B$:

$$R_A = 2 \times 3607.5\ \Omega = 7215\ \Omega \approx \mathbf{7.2}\text{ k}\boldsymbol{\Omega} \tag{5}$$

3. Designed Circuit Schematic



4. Expected Output Waveforms



Q9. For the 555 IC based astable multivibrator circuit shown with $R_1 = 7\,\text{k}\Omega$, $R_2 = 3\,\text{k}\Omega$, $C = 0.5\,\mu$F:

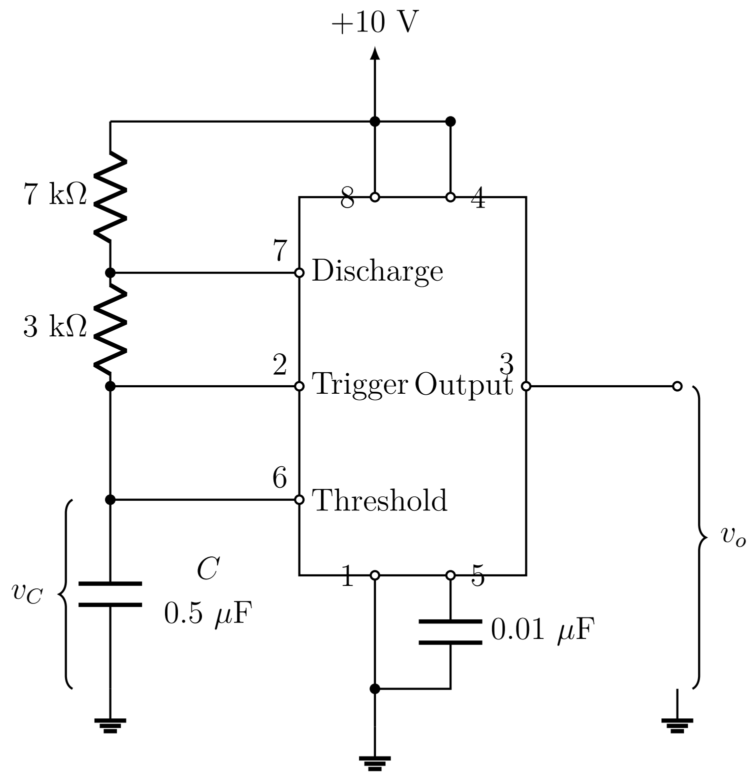
(a) draw waveshapes of V_C and V_O;

(b) determine t_{low} and t_{high};

(c) find duty cycle and free-running frequency;

(d) discuss how frequency depends on C with a graph;

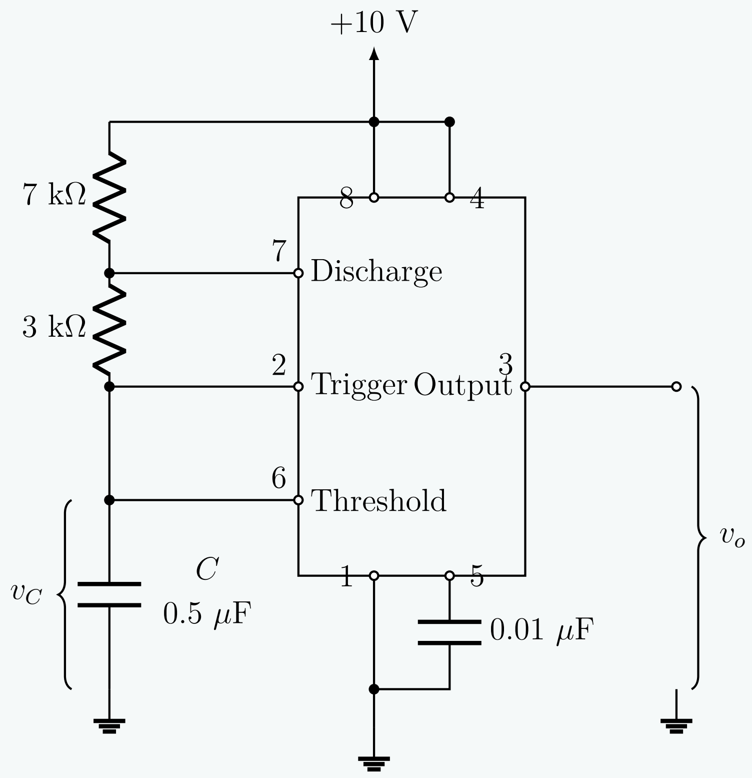
(e) design resistors and capacitors so that duty cycle becomes 0.40.



(29 marks) **[EEE 263 | L-2/T-1 | Jan–2021]**

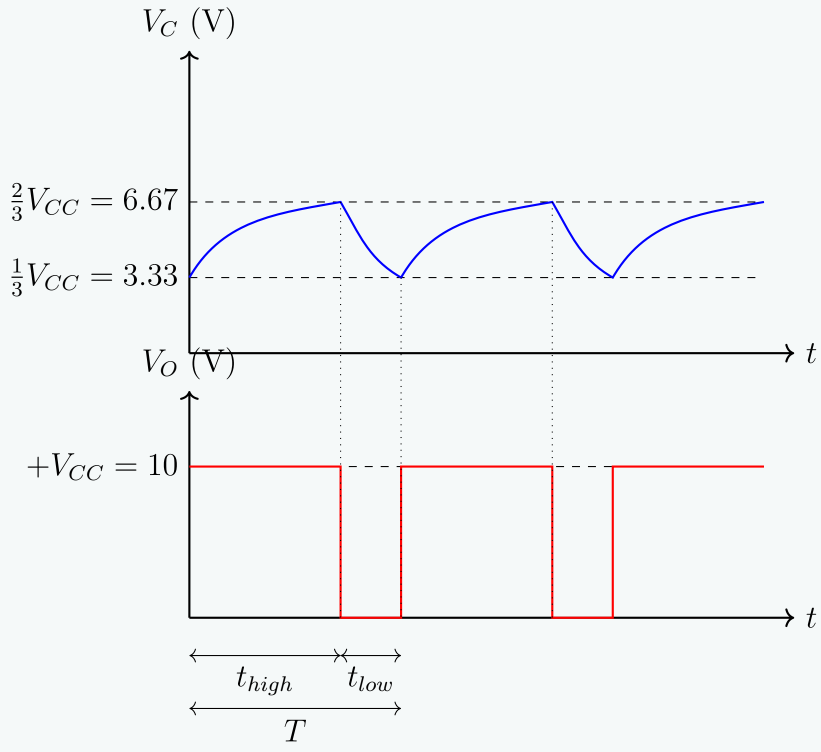
Answer

Circuit Diagram



(a) Waveshapes of V_C and V_O

In the 555 astable multivibrator, the capacitor C charges through $(R_1 + R_2)$ until it reaches $\frac{2}{3}V_{CC}$ (Upper Threshold Voltage, V_{TH}). Then, the internal discharge transistor turns ON, discharging C through R_2 until the voltage drops to $\frac{1}{3}V_{CC}$ (Lower Threshold Voltage, V_{TL}). For $V_{CC} = 10$ V, $V_{TH} = 6.67$ V and $V_{TL} = 3.33$ V.



(b) Determination of t_{low} and t_{high}

The capacitor charges through R_1 and R_2. The HIGH time (t_{high}) is given by:

$$\begin{aligned} t_{high} &= C(R_1 + R_2)\ln(2) \approx 0.693(R_1 + R_2)C \\ &= 0.693 \times (7 \text{ k}\Omega + 3 \text{ k}\Omega) \times 0.5\ \mu\text{F} \\ &= 0.693 \times 10 \times 10^3 \times 0.5 \times 10^{-6} \\ &= \mathbf{3.465} \text{ ms} \end{aligned}$$

The capacitor discharges through R_2 only. The LOW time (t_{low}) is given by:

$$\begin{aligned} t_{low} &= CR_2\ln(2) \approx 0.693R_2C \\ &= 0.693 \times 3 \text{ k}\Omega \times 0.5\ \mu\text{F} \\ &= 0.693 \times 3 \times 10^3 \times 0.5 \times 10^{-6} \\ &= \mathbf{1.0395} \text{ ms} \end{aligned}$$

(c) Duty Cycle and Free-Running Frequency

The total time period T of the oscillation is:

$$\begin{aligned} T &= t_{high} + t_{low} \\ &= 3.465 \text{ ms} + 1.0395 \text{ ms} = 4.5045 \text{ ms} \end{aligned}$$

The free-running frequency f is:

$$f = \frac{1}{T} = \frac{1}{4.5045 \times 10^{-3}} \approx \mathbf{222} \text{ Hz}$$

(Alternatively, using the exact formula: $f = \frac{1.44}{(R_1+2R_2)C} \approx 221.5$ Hz).
The duty cycle (D) is the percentage of the time the output is HIGH:

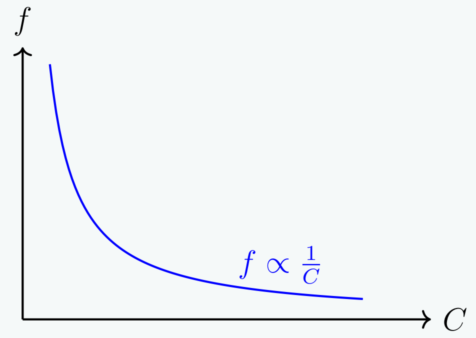
$$\begin{aligned} D &= \frac{t_{high}}{T} \times 100\% \\ &= \frac{R_1 + R_2}{R_1 + 2R_2} \times 100\% \\ &= \frac{10 \text{ k}\Omega}{13 \text{ k}\Omega} \times 100\% \approx \mathbf{76.92}\% \end{aligned}$$

(d) Dependence of Frequency on Capacitance C

The frequency of the standard astable multivibrator is inversely proportional to the timing capacitance C:

$$f = \frac{1.44}{(R_1 + 2R_2)C} = \frac{K}{C} \tag{1}$$

where K is a constant determined by the resistors. As C increases, the time required to charge and discharge it increases, lowering the oscillation frequency. This yields a hyperbolic curve in the first quadrant.



(e) Design for Duty Cycle = 0.40

Theory: In the standard 555 astable circuit, the duty cycle $D = \frac{R_1+R_2}{R_1+2R_2}$ is always strictly greater than 50% because the charging path uses $(R_1 + R_2)$ while the discharging path uses only R_2. To achieve a duty cycle $\leq 50\%$ (like 0.40), we must modify the circuit by adding a bypass diode (D_1) in parallel with R_2.
Modified Operation:

- **Charging:** Current flows through R_1 and D_1 (bypassing R_2). $t_{high} \approx 0.693R_1C$.
- **Discharging:** Current flows through R_2 as normal. $t_{low} = 0.693R_2C$.

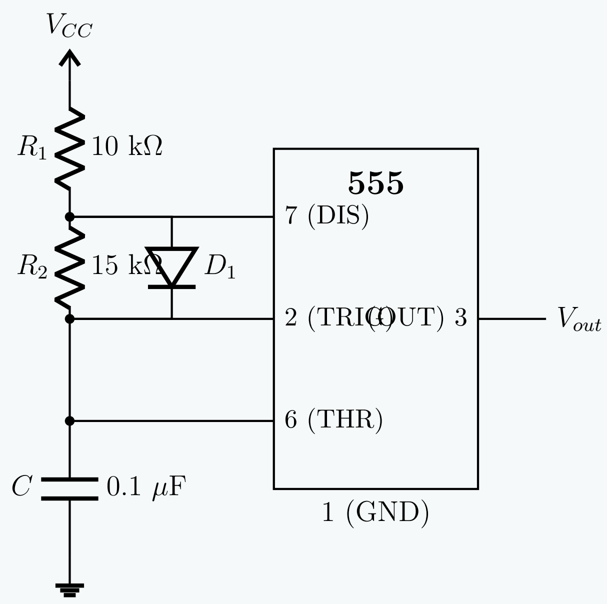
Mathematical Design: Given required Duty Cycle $D = 0.40$:

$$D = \frac{t_{high}}{t_{high} + t_{low}} = \frac{R_1}{R_1 + R_2}$$
$$0.40 = \frac{R_1}{R_1 + R_2}$$
$$0.40(R_1 + R_2) = R_1$$
$$0.40R_2 = 0.60R_1 \implies R_2 = 1.5R_1$$

Let us arbitrarily select a standard resistor value for R_1 and a standard capacitor C:

- Let $\mathbf{R_1} = \mathbf{10}$ k$\mathbf{\Omega}$.
- Then $R_2 = 1.5 \times 10$ kΩ = **15** k$\mathbf{\Omega}$.
- Let $\mathbf{C} = \mathbf{0.1}$ μF.

Modified Circuit Schematic:



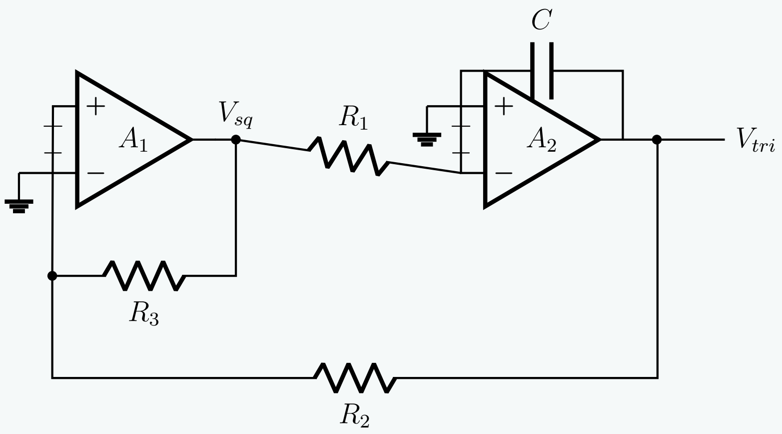
Triangular Wave Generator

Q1. Draw and explain the operation of a triangular wave generator circuit using op-amps and a Schmitt trigger. Derive the frequency of oscillation for the circuit. **(20 marks)** **[EEE 263 | L-2/T-1 | 2022–23]**

Answer

1. Circuit Diagram

A standard triangular wave generator consists of two stages: a **Non-inverting Schmitt Trigger** (comparator with hysteresis) followed by an **Ideal Integrator**.



2. Theory and Principle of Operation

This circuit operates as a **relaxation oscillator**. It does not depend on the Barkhausen criterion but rather on the charging and discharging of a capacitor between two defined threshold voltage levels.

- **Ideal Op-Amp Assumptions:** We assume no current flows into the op-amp input terminals ($I_{in} = 0$), and the integrator maintains a virtual ground at its inverting terminal. The Schmitt trigger operates in saturation, meaning its output V_{sq} is either $+V_{sat}$ or $-V_{sat}$.
- **Stage 1 (A_1) - Schmitt Trigger:** It compares the feedback voltage at its non-inverting terminal (V^+) with ground. Its output V_{sq} is a square wave that drives the integrator.
- **Stage 2 (A_2) - Integrator:** It integrates the constant voltage V_{sq} from A_1. The integral of a constant is a linear ramp. Therefore, the square wave input produces a triangular wave output V_{tri}.
- **Feedback Loop:** The output of the integrator V_{tri} is fed back to the non-inverting input of the Schmitt trigger through R_2, controlling when the Schmitt trigger changes states.

3. Mathematical Derivation of Frequency

Step 3.1: Determine Threshold Voltages (V_{TH} and V_{TL})
Apply Kirchhoff's Current Law (KCL) or Superposition at the non-inverting terminal (V^+) of A_1. Since the op-amp draws zero current:

$$\frac{V_{sq} - V^+}{R_3} + \frac{V_{tri} - V^+}{R_2} = 0$$

The state of the Schmitt trigger changes when V^+ crosses 0V. Setting $V^+ = 0$:

$$\frac{V_{sq}}{R_3} + \frac{V_{tri}}{R_2} = 0 \implies V_{tri} = -V_{sq}\frac{R_2}{R_3}$$

This gives the two peak threshold voltages for the triangular wave:

$$\text{Upper Threshold, } V_{TH} = +V_{sat}\frac{R_2}{R_3} \quad (\text{Occurs when } V_{sq} = -V_{sat})$$
$$\text{Lower Threshold, } V_{TL} = -V_{sat}\frac{R_2}{R_3} \quad (\text{Occurs when } V_{sq} = +V_{sat})$$

The peak-to-peak amplitude of the triangular wave is:

$$V_{PP} = V_{TH} - V_{TL} = 2V_{sat}\frac{R_2}{R_3} \tag{1}$$

Step 3.2: Integrator Sweep Time
The output of the ideal integrator is given by:

$$V_{tri}(t) = -\frac{1}{R_1C}\int V_{sq}dt \tag{2}$$

Assume $V_{sq} = -V_{sat}$. The integrator output ramps linearly upwards from V_{TL} to V_{TH}. The time taken for this upward ramp is exactly half the time period ($T/2$).

$$\Delta V_{tri} = -\frac{1}{R_1C}(-V_{sat})\left(\frac{T}{2}\right)$$
$$V_{TH} - V_{TL} = \frac{V_{sat}}{R_1C}\left(\frac{T}{2}\right)$$

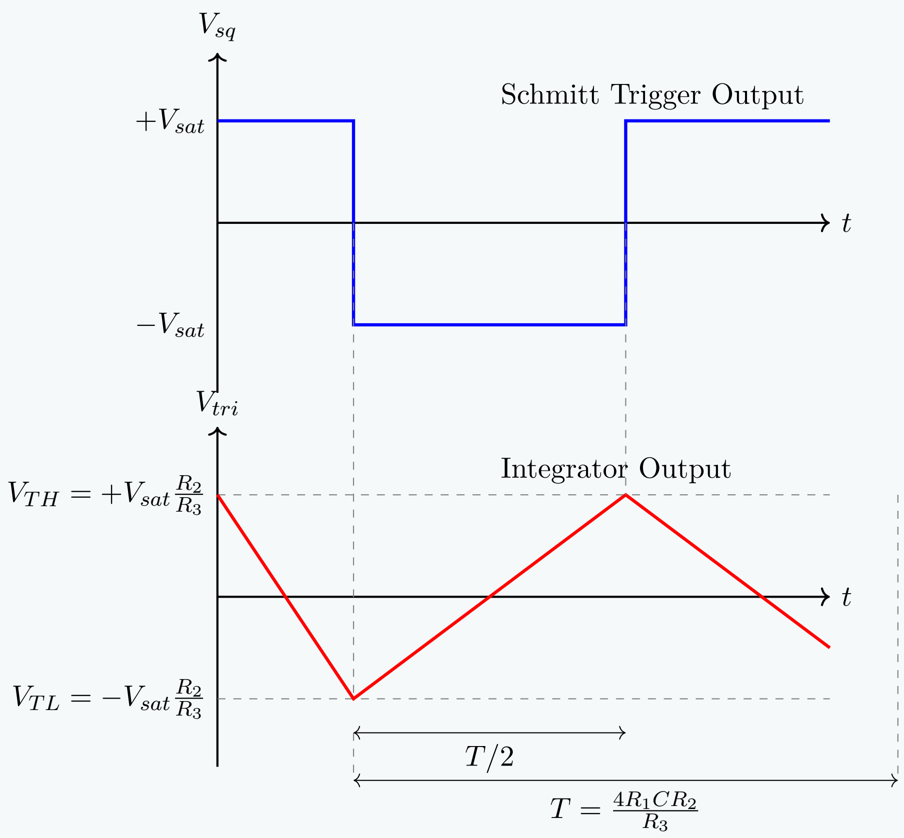
Step 3.3: Final Frequency Calculation
Equating (1) and (**??**):

$$2V_{sat}\frac{R_2}{R_3} = \frac{V_{sat}}{R_1C}\frac{T}{2}$$
$$T = \frac{4R_1CR_2}{R_3}$$

The free-running frequency of oscillation f_o is the reciprocal of the time period T:

$$f_o = \frac{1}{T} = \frac{R_3}{4R_1CR_2} \tag{3}$$

4. Expected Output Waveforms

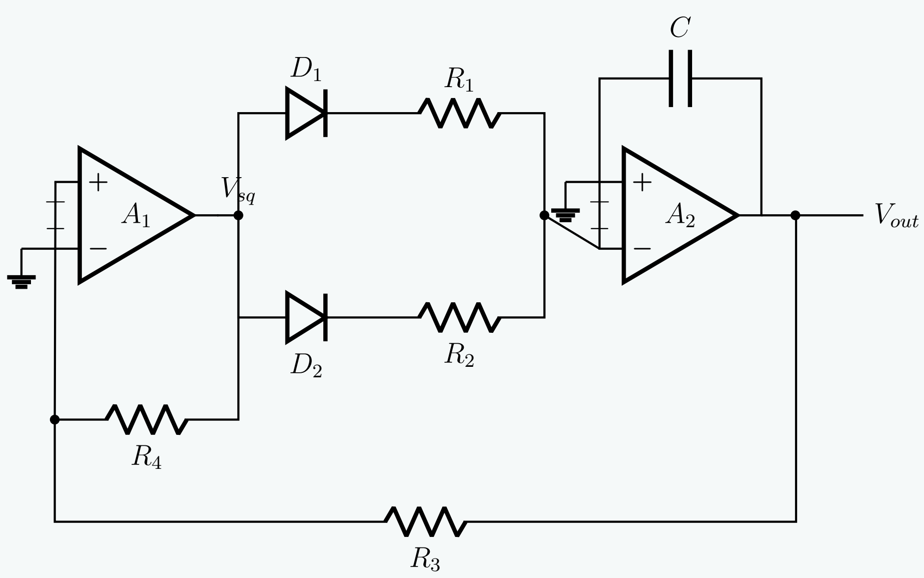


Q2. Draw a sawtooth wave generator circuit using an op-amp. Explain the operation of the circuit. **(23 marks)** [**EEE 263 | L-2/T-1 | 2021–22**]

Answer

1. Circuit Diagram

A standard Op-Amp Sawtooth Wave Generator is essentially an asymmetric triangular wave generator. It utilizes two op-amps: a **Non-inverting Schmitt Trigger** and an **Asymmetric Integrator**. The asymmetry is introduced by using two parallel feedback/input paths with diodes to create different charging and discharging time constants.



2. Theory and Principle of Operation

This circuit operates by generating an asymmetric triangular wave, which serves as a sawtooth wave. To achieve this, the integration rates (slopes) during the positive and negative half-cycles of the square wave must be different.

- **Stage 1 (A_1) - Schmitt Trigger:** This comparator has hysteresis, meaning its output V_{sq} switches between $+V_{sat}$ and $-V_{sat}$ based on the feedback from the integrator's output V_{out}. The resistors R_3 and R_4 establish the Upper Threshold (V_{UT}) and Lower Threshold (V_{LT}) voltages.
- **Stage 2 (A_2) - Asymmetric Integrator:**
  - **Fall Time (T_{fall}):** When $V_{sq} = +V_{sat}$, diode D_1 is forward-biased (ON) and D_2 is reverse-biased (OFF). The capacitor C charges through R_1. If R_1 is large, the integrator output ramps down slowly.

  - **Rise Time (T_{rise}):** When $V_{sq} = -V_{sat}$, diode D_2 is forward-biased (ON) and D_1 is reverse-biased (OFF). The capacitor C discharges through R_2. If R_2 is small ($R_2 \ll R_1$), the integrator output ramps up rapidly.

- By making R_1 significantly larger than R_2 (or vice versa), the slow ramp down and fast ramp up create a **sawtooth wave**.

3. Mathematical Derivation

Step 3.1: Schmitt Trigger Threshold Voltages
Assuming ideal op-amps, the non-inverting terminal voltage V^+ of A_1 is determined by the superposition of V_{out} and V_{sq}. Applying KCL at V^+:

$$\frac{V^+ - V_{out}}{R_3} + \frac{V^+ - V_{sq}}{R_4} = 0$$

Switching occurs when $V^+ = 0\text{V}$. Setting $V^+ = 0$:

$$\frac{V_{out}}{R_3} = -\frac{V_{sq}}{R_4} \implies V_{out} = -V_{sq}\frac{R_3}{R_4}$$

This yields the two threshold voltages defining the peak-to-peak amplitude of the sawtooth:

$$V_{UT} = +V_{sat}\frac{R_3}{R_4} \quad \text{(Occurs when } V_{sq} = -V_{sat}\text{)}$$
$$V_{LT} = -V_{sat}\frac{R_3}{R_4} \quad \text{(Occurs when } V_{sq} = +V_{sat}\text{)}$$

The peak-to-peak voltage is $\Delta V = V_{UT} - V_{LT} = 2V_{sat}\frac{R_3}{R_4}$.
Step 3.2: Fall Time (T_{fall}) and Rise Time (T_{rise})
Assuming ideal diodes ($V_D = 0$ for simplicity), the integration equation is:

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{sq}dt \tag{1}$$

For the **ramp-down (fall) period**, $V_{sq} = +V_{sat}$ and the active resistor is R_1:

$$\Delta V = \frac{V_{sat}}{R_1 C}T_{fall}$$
$$2V_{sat}\frac{R_3}{R_4} = \frac{V_{sat}}{R_1 C}T_{fall} \implies T_{fall} = \frac{2R_1 C R_3}{R_4} \tag{2}$$

For the **ramp-up (rise) period**, $V_{sq} = -V_{sat}$ and the active resistor is R_2:

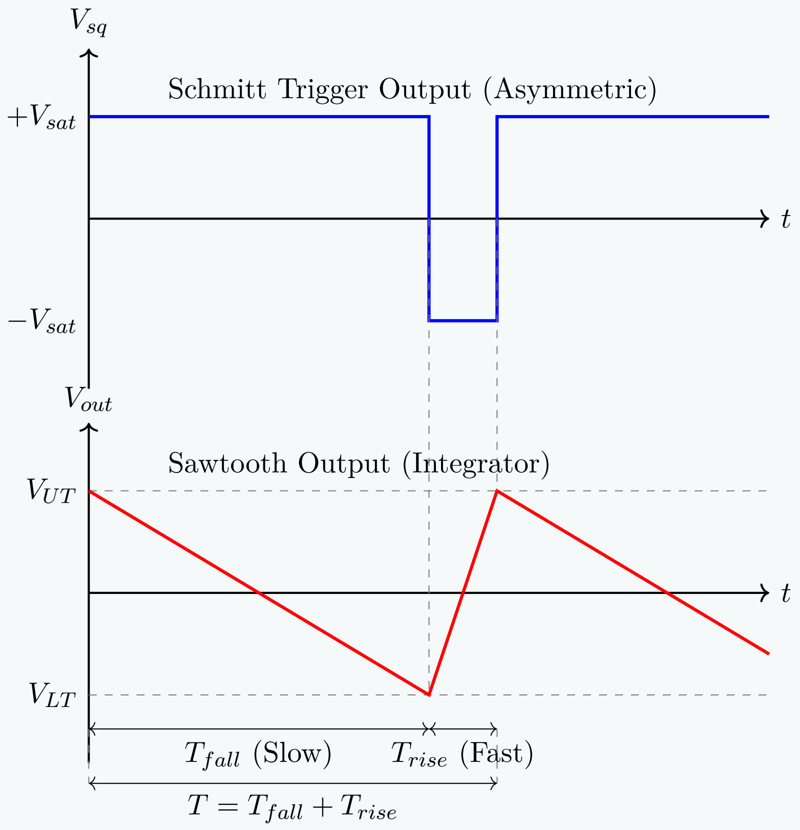
$$\Delta V = \frac{V_{sat}}{R_2 C}T_{rise}$$
$$2V_{sat}\frac{R_3}{R_4} = \frac{V_{sat}}{R_2 C}T_{rise} \implies T_{rise} = \frac{2R_2 C R_3}{R_4} \tag{3}$$

Step 3.3: Total Frequency
The total time period is $T = T_{fall} + T_{rise}$. The oscillation frequency is:

$$f = \frac{1}{T} = \frac{1}{\frac{2CR_3}{R_4}(R_1 + R_2)} = \frac{R_4}{2CR_3(R_1 + R_2)}$$

4. Expected Output Waveforms

Assuming $R_1 \gg R_2$, the fall time is much longer than the rise time, yielding a standard sawtooth profile.

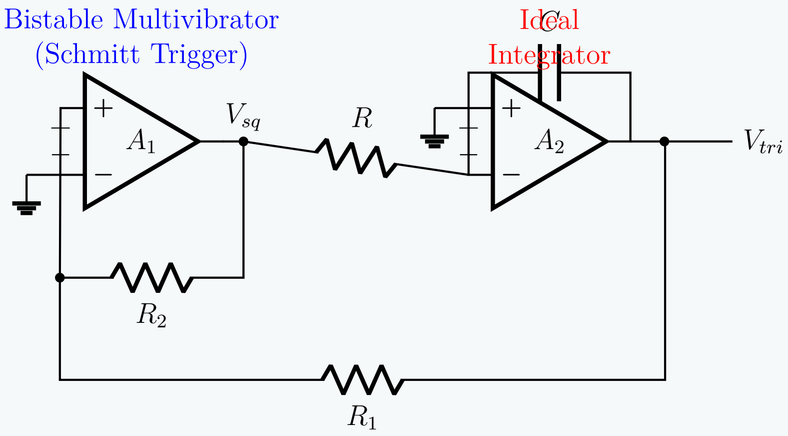


Q3. Drawing appropriate circuits, show how you can generate triangular waves using a bistable multivibrator and appropriate Op-amp circuits. **(26 marks)** [EEE 263 | L-2/T-1 | 2020–21]

Answer

1. Circuit Diagram

A triangular wave generator can be constructed by cascading a **Bistable Multivibrator** (Schmitt Trigger) and an **Ideal Integrator**. The output of the integrator is fed back to the non-inverting input of the Schmitt trigger to sustain oscillations.



2. Theory and Circuit Operation

This circuit is a relaxation oscillator that simultaneously generates a square wave and a triangular wave.

- **Bistable Multivibrator (A_1):** Due to the positive feedback provided by resistors R_1 and R_2, the op-amp A_1 operates in saturation. Its output V_{sq} will always be in one of two stable states: $+V_{sat}$ or $-V_{sat}$.
- **Integrator (A_2):** The square wave V_{sq} is fed into the inverting integrator. The integral of a constant voltage is a linear ramp.
  - When $V_{sq} = +V_{sat}$, a constant positive voltage is applied to the integrator, causing V_{tri} to ramp downwards linearly.
  - When $V_{sq} = -V_{sat}$, a constant negative voltage is applied, causing V_{tri} to ramp upwards linearly.
- **Feedback Mechanism:** The triangular wave V_{tri} is fed back to the non-inverting terminal of A_1. When V_{tri} reaches a specific threshold (V_{TH} or V_{TL}), it forces the non-inverting terminal voltage to cross zero, triggering A_1 to switch states. This continuous switching creates sustained oscillations.

3. Mathematical Derivation

Step 3.1: Determine Threshold Voltages
Assuming ideal op-amps, the non-inverting terminal voltage V^+ of A_1 is found using superposition (since no current enters the op-amp input):

$$V^+ = V_{sq}\left(\frac{R_1}{R_1+R_2}\right) + V_{tri}\left(\frac{R_2}{R_1+R_2}\right)$$

The Schmitt trigger switches states when V^+ crosses 0V:

$$0 = V_{sq}R_1 + V_{tri}R_2$$
$$V_{tri} = -V_{sq}\left(\frac{R_1}{R_2}\right)$$

Since V_{sq} can be either $+V_{sat}$ or $-V_{sat}$, we get two threshold voltages for the triangular wave:

$$\text{Upper Threshold, } V_{TH} = +V_{sat}\left(\frac{R_1}{R_2}\right) \quad (\text{when } V_{sq} = -V_{sat})$$
$$\text{Lower Threshold, } V_{TL} = -V_{sat}\left(\frac{R_1}{R_2}\right) \quad (\text{when } V_{sq} = +V_{sat})$$

The peak-to-peak amplitude of the triangular wave is:

$$V_{PP} = V_{TH} - V_{TL} = 2V_{sat}\left(\frac{R_1}{R_2}\right) \tag{1}$$

Step 3.2: Integrator Output Equation
The output of the inverting integrator is:

$$V_{tri}(t) = -\frac{1}{RC}\int_0^t V_{sq}dt + V_{tri}(0) \tag{2}$$

During the time interval $0 \le t \le T/2$, let $V_{sq} = -V_{sat}$. The output ramps linearly from V_{TL} to V_{TH}. The change in voltage is:

$$\Delta V_{tri} = -\frac{1}{RC}(-V_{sat})\left(\frac{T}{2}\right) = \frac{V_{sat}T}{2RC}$$

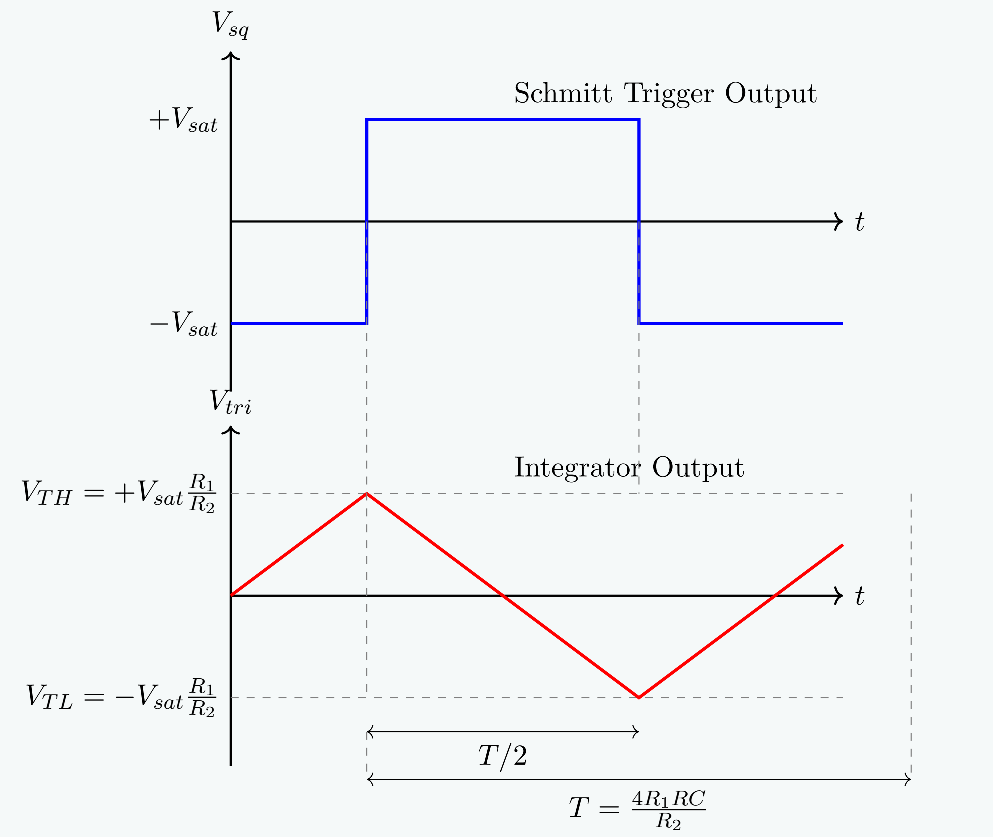
Step 3.3: Frequency of Oscillation
Equating the peak-to-peak voltage from (1) with the voltage sweep from (??):

$$2V_{sat}\left(\frac{R_1}{R_2}\right) = \frac{V_{sat}T}{2RC}$$
$$T = \frac{4R_1RC}{R_2}$$

The frequency of oscillation f_o is the reciprocal of the time period T:

$$f_o = \frac{1}{T} = \frac{R_2}{4R_1RC} \tag{3}$$

4. Waveform Diagrams



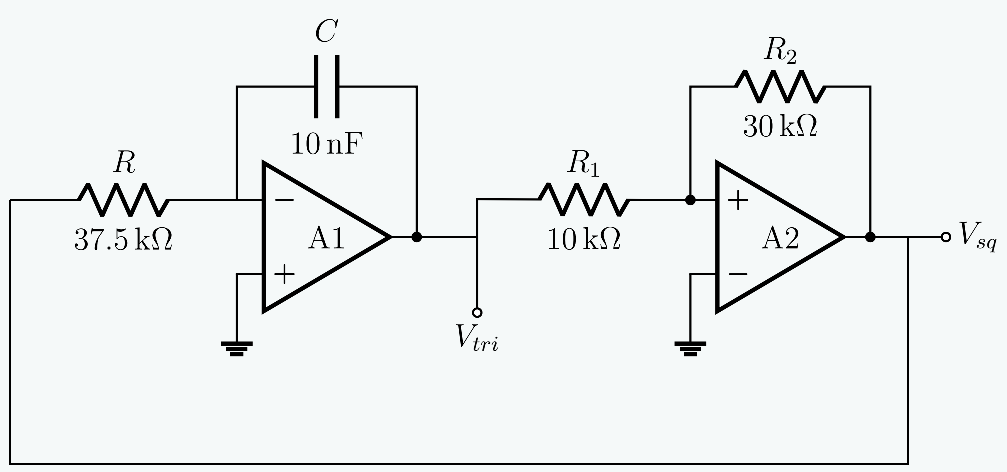
Q4. Design a circuit using a bistable multivibrator which generates a triangular waveform with oscillation frequency of 2 kHz and 10 V peak-to-peak amplitude. **(16 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

Theory & Principle
A triangular wave generator can be designed by cascading a bistable multivibrator (non-inverting Schmitt trigger) and an ideal integrator.

- **Integrator (Op-Amp A1):** Converts the square wave output of the Schmitt trigger into a linear triangular wave. When the input square wave is positive, the integrator's output ramps down linearly, and when the input is negative, it ramps up.
- **Schmitt Trigger (Op-Amp A2):** Compares the triangular wave to a reference level. It switches its output state (between $+V_{sat}$ and $-V_{sat}$) whenever the triangular wave reaches the upper threshold voltage ($+V_{UT}$) or the lower threshold voltage ($-V_{LT}$).

Circuit Diagram



Mathematical Derivation

1. Amplitude (V_{pp}):
Applying superposition at the non-inverting terminal (V_+) of A2:

$$V_+ = V_{tri}\left(\frac{R_2}{R_1+R_2}\right) + V_{sq}\left(\frac{R_1}{R_1+R_2}\right) \tag{1}$$

The Schmitt trigger changes state when $V_+ = 0$. Equating V_+ to 0, we find the threshold levels:

$$V_{tri} = -V_{sq}\frac{R_1}{R_2} \tag{2}$$

Assuming symmetric saturation voltages $V_{sq} = \pm V_{sat}$, the upper and lower thresholds are:

$$+V_{UT} = +V_{sat}\frac{R_1}{R_2}$$
$$-V_{LT} = -V_{sat}\frac{R_1}{R_2}$$

The peak-to-peak amplitude V_{pp} of the triangular wave is:

$$V_{pp} = V_{UT} - (-V_{LT}) = 2V_{sat}\frac{R_1}{R_2} \tag{3}$$

2. Frequency (f_0):
During one half-cycle ($t = T/2$), the input to the integrator is $-V_{sat}$. The output V_{tri} ramps linearly from $-V_{LT}$ to $+V_{UT}$. The integrator equation is:

$$\Delta V_{tri} = -\frac{1}{RC}\int_0^{T/2}(-V_{sat})\,dt = \frac{V_{sat}}{RC}\cdot\frac{T}{2} \tag{4}$$

Equating ΔV_{tri} to V_{pp}:

$$2V_{sat}\frac{R_1}{R_2} = \frac{V_{sat}\cdot T}{2RC}$$
$$T = 4RC\frac{R_1}{R_2}$$

Therefore, the oscillation frequency f_0 is:

$$f_0 = \frac{1}{T} = \frac{R_2}{4RCR_1} \tag{5}$$

Step-by-Step Design & Calculations
Assumption: We assume ideal op-amps powered by ± 15 V DC supplies, resulting in an output saturation voltage of approximately $V_{sat} = \pm 15$ V.
Step 1: Design for 10 V Peak-to-Peak Amplitude We require $V_{pp} = 10$ V.

$$10 = 2(15)\frac{R_1}{R_2}$$
$$\frac{R_1}{R_2} = \frac{10}{30} = \frac{1}{3}$$

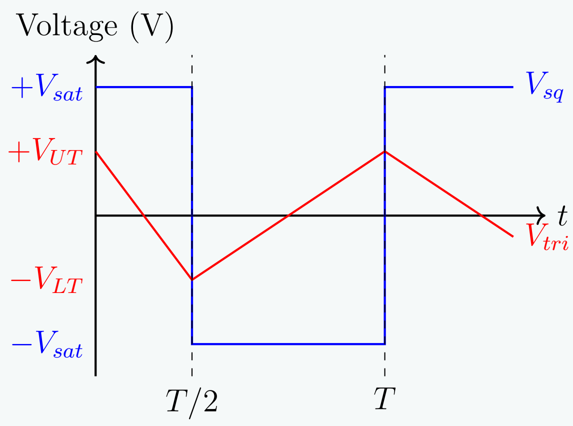
Choosing standard values: Let $\mathbf{R_1 = 10\,k\Omega}$, which implies $\mathbf{R_2 = 30\,k\Omega}$.
Step 2: Design for 2 kHz Oscillation Frequency We require $f_0 = 2000$ Hz.

$$2000 = \frac{30\,\text{k}\Omega}{4\cdot R\cdot C\cdot 10\,\text{k}\Omega}$$
$$2000 = \frac{3}{4RC}$$
$$RC = \frac{3}{8000} = 375\,\mu\text{s}$$

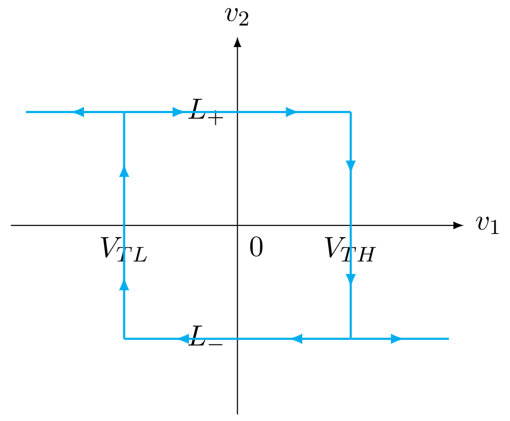
Choosing a standard capacitor value: Let $\mathbf{C = 10}$ nF (0.01 μF).

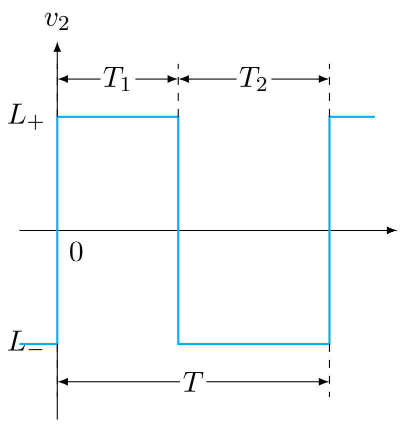
$$R = \frac{375\times 10^{-6}}{10\times 10^{-9}} = \mathbf{37.5\,k\Omega}$$

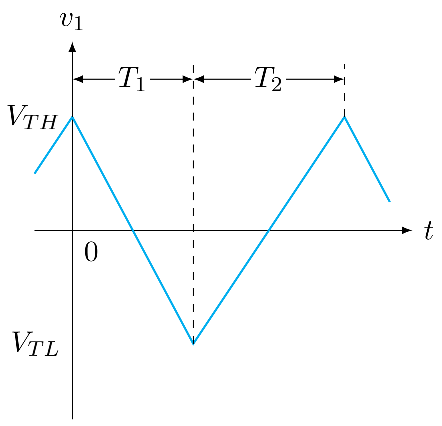
Expected Waveforms



Q5. A bistable multivibrator with a given hysteresis transfer characteristic is provided. Design a signal generator based on the bistable multivibrator that generates square and triangular waveforms. Choose values of your designed circuit components for which the frequency of oscillation is 1 MHz.







(18 marks) **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

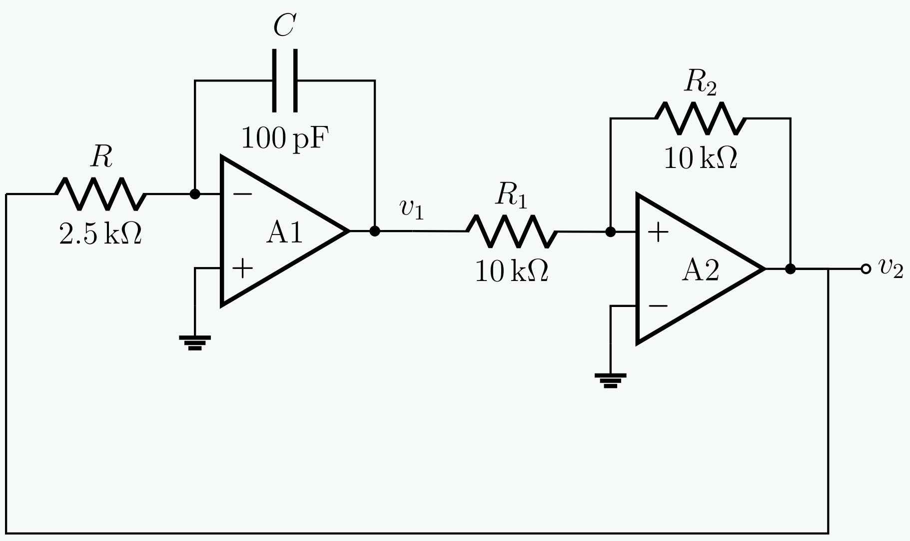
Theory & Principle

To design a signal generator that produces the given square (v_2) and triangular (v_1) waveforms, we must cascade an **inverting integrator** and a **non-inverting bistable multivibrator (Schmitt Trigger)**.

Analyzing the provided time-domain waveforms:

- During T_1, $v_2 = L_+$ (positive state), and v_1 ramps down from V_{TH} to V_{TL} with a negative slope. This confirms the use of an *inverting integrator*, where $v_1 = -\frac{1}{RC}\int v_2\, dt$.
- At the end of T_1, v_1 reaches V_{TL}, causing v_2 to switch from L_+ to L_-. Since v_2 switches to a lower state when v_1 decreases to a lower threshold, the bistable multivibrator must be *non-inverting*.

Circuit Diagram



Mathematical Derivation

1. Threshold Voltages (V_{TH}, V_{TL}):

Let A_2 be ideal. Applying superposition at the non-inverting terminal (v_+) of A2:

$$v_+ = v_1\left(\frac{R_2}{R_1+R_2}\right) + v_2\left(\frac{R_1}{R_1+R_2}\right) \tag{1}$$

The switching occurs when $v_+ = 0$:

$$0 = v_1 R_2 + v_2 R_1 \implies v_1 = -v_2\frac{R_1}{R_2} \tag{2}$$

Assuming symmetric output saturation levels $v_2 \in \{+L, -L\}$:

$$V_{TL} = -L\left(\frac{R_1}{R_2}\right) \quad \text{(when } v_2 = +L\text{)}$$
$$V_{TH} = -(-L)\left(\frac{R_1}{R_2}\right) = L\left(\frac{R_1}{R_2}\right) \quad \text{(when } v_2 = -L\text{)}$$

2. Integrator Operation & Period (T):
The output of the inverting integrator is:

$$v_1(t) = v_1(0) - \frac{1}{RC}\int_0^t v_2(\tau)\, d\tau \tag{3}$$

During interval T_1, $v_2 = +L$. The voltage v_1 ramps down linearly from V_{TH} to V_{TL}:

$$V_{TL} - V_{TH} = -\frac{L}{RC}T_1 \implies T_1 = \frac{RC(V_{TH} - V_{TL})}{L} \tag{4}$$

During interval T_2, $v_2 = -L$. The voltage v_1 ramps up linearly from V_{TL} to V_{TH}:

$$V_{TH} - V_{TL} = -\frac{-L}{RC}T_2 \implies T_2 = \frac{RC(V_{TH} - V_{TL})}{L} \tag{5}$$

The total period T is $T_1 + T_2$:

$$T = \frac{2RC(V_{TH} - V_{TL})}{L} = \frac{2RC\left[L\frac{R_1}{R_2} - \left(-L\frac{R_1}{R_2}\right)\right]}{L} = 4RC\frac{R_1}{R_2} \tag{6}$$

Thus, the oscillation frequency f is:

$$f = \frac{1}{T} = \frac{R_2}{4RCR_1} \tag{7}$$

Step-by-Step Solution & Component Selection
Target: Design for $f = 1\,\text{MHz}$ ($10^6\,\text{Hz}$).

(a) **Choose Resistor Ratio:** To simplify the design, we select $R_1 = R_2$.

$$\text{Let } R_1 = 10\,\text{k}\Omega \text{ and } R_2 = 10\,\text{k}\Omega. \implies \frac{R_1}{R_2} = 1.$$

(b) **Calculate RC Time Constant:**

$$f = \frac{1}{4RC} \implies 10^6 = \frac{1}{4RC} \implies RC = \frac{1}{4 \times 10^6} = 250\,\text{ns}$$

(c) **Select Capacitor and Resistor Values:** Choose a standard capacitor suitable for high frequencies.

$$\text{Let } C = 100\,\text{pF}\ (100 \times 10^{-12}\,\text{F})$$

$$R = \frac{250 \times 10^{-9}}{100 \times 10^{-12}} = 2500\,\Omega = \mathbf{2.5\,k\Omega}$$

Important Notes for 1 MHz Operation: While the nominal component values ($R = 2.5\,\text{k}\Omega$, $C = 100\,\text{pF}$, $R_1 = R_2 = 10\,\text{k}\Omega$) yield the correct mathematical frequency, a practical 1 MHz oscillator requires wide-bandwidth operational amplifiers. The op-amps must possess a high *Slew Rate* (SR $\gg 2L \cdot 10^6\,\text{V/s}$) and a high *Gain-Bandwidth Product* (GBW) to handle the rapid switching and prevent distortion in the triangular wave.

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

Part VI — Digital Logic and Sequential Circuits

CMOS logic design, flip-flops, counters, and digital circuit implementation

S9 Digital Logic & Sequential Circuits

CMOS Logic Design and Implementation

Q1. Discuss basic operational characteristics and parameters of CMOS and TTL logic families. **(23 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

Introduction to Logic Families
Digital integrated circuits are classified into various "logic families" based on the fundamental semiconductor technologies used to construct their basic logic gates. The two most prominent legacy and modern logic families are **TTL** (Transistor-Transistor Logic) and **CMOS** (Complementary Metal-Oxide-Semiconductor).

1. Transistor-Transistor Logic (TTL)

- **Basic Principle:** TTL utilizes Bipolar Junction Transistors (BJTs) and resistors. The logic functions are performed by multi-emitter NPN transistors at the input, while a "totem-pole" or open-collector configuration is typically used at the output stage.
- **Operation:** The BJTs in standard TTL circuits operate by switching between the *saturation* and *cutoff* regions.
- **Characteristics:** TTL is historically known for its relatively fast switching speeds. However, because BJTs are current-controlled devices and draw continuous base current when conducting, TTL exhibits substantial static power dissipation. It strictly operates on a nominal supply voltage of $V_{CC} = 5\,\text{V}$.

2. Complementary Metal-Oxide-Semiconductor (CMOS)

- **Basic Principle:** CMOS technology uses complementary and symmetrical pairs of p-type (PMOS) and n-type (NMOS) Enhancement-Mode MOSFETs.
- **Operation:** In any steady logic state, one transistor of the complementary pair is ON while the other is OFF. Consequently, there is no direct DC path from the power supply (V_{DD}) to ground (V_{SS}).
- **Characteristics:** CMOS has effectively zero static power dissipation, drawing significant power only during switching transients (dynamic power). It possesses a high input impedance, allows a wide range of supply voltages (typically $3\,\text{V}$ to $15\,\text{V}$), and offers excellent noise immunity.

3. Key Operational Parameters
To evaluate and compare logic families, the following fundamental parameters are defined:

- **Voltage Levels:**
  - $V_{IL(\max)}$: Maximum input voltage guaranteed to be recognized as a logic '0'.
  - $V_{IH(\min)}$: Minimum input voltage guaranteed to be recognized as a logic '1'.
  - $V_{OL(\max)}$: Maximum output voltage when driving a logic '0'.
  - $V_{OH(\min)}$: Minimum output voltage when driving a logic '1'.
- **Noise Margin (NM):** The maximum external noise voltage that can be superimposed on a standard logic signal without causing an erratic change in state.

$$\text{High-Level Noise Margin: } NM_H = V_{OH(\min)} - V_{IH(\min)}$$
$$\text{Low-Level Noise Margin: } NM_L = V_{IL(\max)} - V_{OL(\max)}$$

  CMOS generally has a wider noise margin (typically 45% of V_{DD}) compared to TTL.
- **Fan-out:** The maximum number of standard logic inputs that a single output can drive reliably without compromising its logic voltage levels. CMOS has a much higher static fan-out than TTL due to the near-infinite gate input impedance of MOSFETs.
- **Power Dissipation (P_D):** The electrical power consumed by the logic gate. For CMOS, power dissipation is heavily frequency-dependent (dynamic) and is given by:

$$P_{\text{dynamic}} = C_L V_{DD}^2 f \tag{1}$$

  where C_L is the load capacitance and f is the switching frequency. TTL power dissipation is largely static and continuous.
- **Propagation Delay (t_{pd}):** The time required for a signal to propagate from the input to the output. It is the average of the low-to-high (t_{pLH}) and high-to-low (t_{pHL}) transition times:

$$t_{pd} = \frac{t_{pHL} + t_{pLH}}{2} \tag{2}$$

- **Figure of Merit / Speed-Power Product (SPP):** A metric combining speed and power, ideally as small as possible.

$$\text{SPP} = t_{pd} \times P_D \quad \text{(measured in Joules)} \tag{3}$$

4. Comparative Summary

Parameter	Standard TTL	Standard CMOS
Active Device	Bipolar Junction Transistor (BJT)	MOSFET
Supply Voltage	Strictly $5\,\text{V} \pm 0.25\,\text{V}$	Flexible ($3\,\text{V}$ to $15\,\text{V}$)
Power Dissipation	High (typically $10\,\text{mW/gate}$)	Very Low (static: $\sim 10\,\text{nW/gate}$)
Propagation Delay	Fast ($10\,\text{ns/gate}$)	Slower at low V_{DD}, but modern sub-micron is faster
Noise Immunity	Moderate ($\sim 0.4\,\text{V}$)	Excellent ($\sim 0.45 \times V_{DD}$)
Fan-out	Low to Moderate (typically 10)	Very High (limited by capacitive loading/speed delays)
Input Impedance	Low	Very High ($10^{12}\,\Omega$)

Q2. Implement the function $f(A,B,C,D) = \sum m(0,4,8,9,10,11,12)$ using CMOS. **(13 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

Theory and Principle
Static CMOS logic gates are constructed using two complementary networks:

- **Pull-Up Network (PUN):** Consists of PMOS transistors connected between the positive supply voltage (V_{DD}) and the output node. PMOS transistors conduct when their gate input is LOW (Logic '0'). Series connections represent the logical `AND` operation, and parallel connections represent the logical `OR` operation for the active-low inputs.
- **Pull-Down Network (PDN):** Consists of NMOS transistors connected between the output node and ground (V_{SS}). NMOS transistors conduct when their gate input is HIGH (Logic '1'). Here, series connections represent `AND`, and parallel connections represent `OR`.

By design, the PUN and PDN are logical duals of each other. In any steady state, only one network conducts, ensuring zero static power dissipation (ideally) and a full voltage swing at the output.

Mathematical Derivation and Simplification
The given function is defined by its minterms:

$$f(A,B,C,D) = \sum m(0,4,8,9,10,11,12) \tag{1}$$

To simplify this boolean expression, we map the minterms onto a 4-variable Karnaugh Map (K-Map):

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	**1** (m_0)	0	0	0
$\bar{A}B$	**1** (m_4)	0	0	0
AB	**1** (m_{12})	0	0	0
$A\bar{B}$	**1** (m_8)	**1** (m_9)	**1** (m_{11})	**1** (m_{10})

Grouping the 1s to find the prime implicants:

(a) **Group 1 (Vertical column of four):** Cells m_0, m_4, m_8, m_{12} simplify to $\bar{C}\bar{D}$.

(b) **Group 2 (Horizontal row of four):** Cells m_8, m_9, m_{10}, m_{11} simplify to $A\bar{B}$.

The simplified Sum-of-Products (SOP) expression for the output f is:

$$f = \bar{C}\bar{D} + A\bar{B} \tag{2}$$

CMOS Implementation Strategy
To implement the gate directly in static CMOS without an output inverter, we define the PUN and PDN structures:

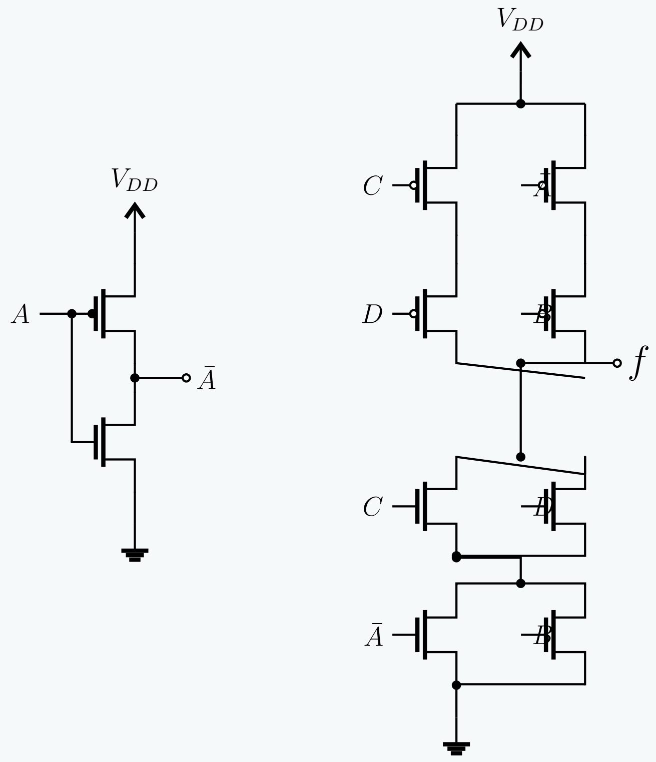
- **PUN (PMOS network):** Implements $f = 1$. The expression $f = \bar{C}\bar{D} + A\bar{B}$ indicates we need PMOS transistors that conduct on $C = 0, D = 0$ (meaning C and D directly drive the gates) and $A = 1, B = 0$ (meaning $\bar{A}$ and B drive the gates). Thus, the PUN consists of (C in series with D) in parallel with ($\bar{A}$ in series with B).
- **PDN (NMOS network):** Implements $f = 0$ (or $\bar{f} = 1$).

$$\bar{f} = \overline{\bar{C}\bar{D} + A\bar{B}} = \overline{(\bar{C}\bar{D})} \cdot \overline{(A\bar{B})} = (C + D)(\bar{A} + B) \tag{3}$$

The PDN consists of (C in parallel with D) in series with ($\bar{A}$ in parallel with B).

Notice that the input $\bar{A}$ is required for both networks, so an initial CMOS inverter stage is needed for input A.

Circuit Diagram



Final Answer Summary

- **Simplified Expression:** $f = \bar{C}\bar{D} + A\bar{B}$
- **Transistor Count:** The total implementation requires 10 transistors:
  - 2 transistors (1 PMOS, 1 NMOS) to invert input A to $\bar{A}$.
  - 4 PMOS transistors for the Pull-Up Network.
  - 4 NMOS transistors for the Pull-Down Network.

Q3. Draw a CMOS complex gate for the logic function $f(a, b, c, d) = \sum m(0, 1, 2, 4, 6, 8, 10, 12, 14)$. Use as few transistors as possible. **(15 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

Mathematical Derivation and Simplification
The given switching function is defined by the minterms:

$$f(a, b, c, d) = \sum m(0, 1, 2, 4, 6, 8, 10, 12, 14) \tag{1}$$

To design the complex CMOS gate with the fewest possible transistors, we must first simplify the boolean expression using a 4-variable Karnaugh Map (K-Map).

	$\bar{c}\bar{d}$	$\bar{c}d$	cd	$c\bar{d}$
$\bar{a}\bar{b}$	**1** (m_0)	**1** (m_1)	0	**1** (m_2)
$\bar{a}b$	**1** (m_4)	0	0	**1** (m_6)
ab	**1** (m_{12})	0	0	**1** (m_{14})
$a\bar{b}$	**1** (m_8)	0	0	**1** (m_{10})

We can form the following optimal groups from the K-Map:

(a) **Group 1 (Octet):** The entire first column ($\bar{c}\bar{d}$) and the entire fourth column ($c\bar{d}$) can be wrapped and grouped together into a block of 8 cells. This eliminates variables a, b, and c, leaving only $\bar{d}$.

(b) **Group 2 (Pair):** The minterm m_1 (0001) groups horizontally with m_0 (0000). This eliminates d, leaving $\bar{a}\bar{b}\bar{c}$.

Thus, the simplified Boolean expression in Sum-of-Products (SOP) form is:

$$f = \bar{d} + \bar{a}\bar{b}\bar{c} \tag{2}$$

CMOS Network Design
Static CMOS uses a Pull-Up Network (PUN) of PMOS transistors and a Pull-Down Network (PDN) of NMOS transistors.

- **Pull-Down Network (PDN):** Implements $\bar{f}$. NMOS transistors conduct when their gate is HIGH (1).

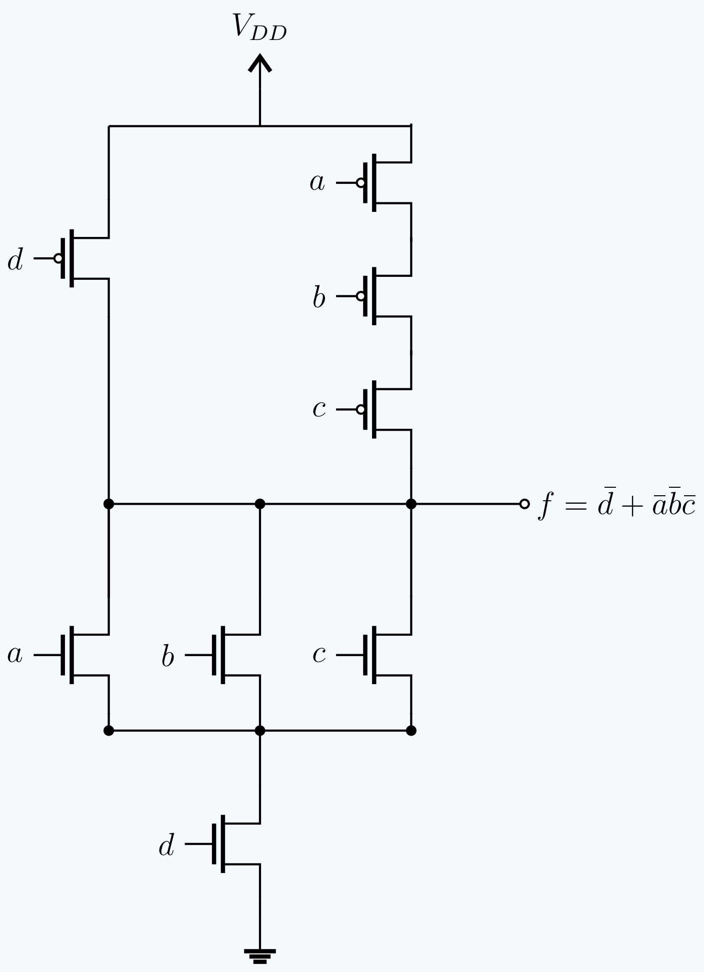
$$\bar{f} = \overline{\bar{d} + \bar{a}\bar{b}\bar{c}}$$
$$\bar{f} = d \cdot \overline{(\bar{a}\bar{b}\bar{c})}$$
$$\bar{f} = d \cdot (a + b + c) \tag{3}$$

  This requires an NMOS transistor d in **series** with a **parallel** combination of NMOS transistors a, b, and c.

- **Pull-Up Network (PUN):** Implements f. PMOS transistors conduct when their gate is LOW (0). Looking directly at $f = \bar{d} + \bar{a}\bar{b}\bar{c}$, the logic natively fits PMOS activation (active-low inputs). This requires a PMOS transistor d in **parallel** with a **series** combination of PMOS transistors a, b, and c.

Because both the PUN and PDN can directly use the non-inverted inputs (a, b, c, d), **no input inverters are required**, achieving an extremely low transistor count.

Circuit Diagram



Final Answer Summary

- **Simplified Function:** $f = \bar{d} + \bar{a}\bar{b}\bar{c}$
- **Transistor Count:** The logic function has been perfectly optimized, needing only **8 transistors** in total (4 PMOS and 4 NMOS). No inverters are needed.

Q4. Derive complex gates for the following logic functions using as few transistors as possible:

(a) $f(X_1, X_2, X_3, X_4) = \sum m(0, 4, 6, 8, 9, 15) + D(3, 7, 11, 13)$

(b) $\bar{f} = (A + B)(C + D)(\bar{A} + B + D)$

(21 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Answer

Theory and Principle of CMOS Complex Gates

A static CMOS complex gate consists of a Pull-Up Network (PUN) constructed from PMOS transistors and a Pull-Down Network (PDN) constructed from NMOS transistors.

- **PDN (NMOS):** Implements the complemented logic function $(\bar{f})$. Series connections perform the logical `AND`, and parallel connections perform the logical `OR`. NMOS transistors conduct when the gate input is HIGH ('1').
- **PUN (PMOS):** Implements the uncomplemented logic function (f). It is the structural dual of the PDN. Series branches in the PDN become parallel branches in the PUN, and vice versa. PMOS transistors conduct when the gate input is LOW ('0').

To use the minimum number of transistors, we must minimize the Boolean expressions for the networks.

Part 1: Minimization and Design for $f_1(X_1, X_2, X_3, X_4)$
1. Mathematical Derivation
The function is given by its minterms and don't care conditions:

$$f_1(X_1, X_2, X_3, X_4) = \sum m(0, 4, 6, 8, 9, 15) + D(3, 7, 11, 13) \tag{1}$$

To design the PDN optimally, we find the minimal Sum-of-Products (SOP) for $\bar{f}_1$ by grouping the **zeros** and don't cares (**X**) in the Karnaugh Map.
The zeros are located at minterms not specified in f_1: $m_1, m_2, m_5, m_{10}, m_{12}, m_{14}$.

	$\bar{X}_3\bar{X}_4$ (00)	$\bar{X}_3X_4$ (01)	X_3X_4 (11)	$X_3\bar{X}_4$ (10)
$\bar{X}_1\bar{X}_2$ (00)	1 (m_0)	**0** (m_1)	X (m_3)	**0** (m_2)
$\bar{X}_1X_2$ (01)	1 (m_4)	**0** (m_5)	X (m_7)	1 (m_6)
X_1X_2 (11)	**0** (m_{12})	X (m_{13})	1 (m_{15})	**0** (m_{14})
$X_1\bar{X}_2$ (10)	1 (m_8)	1 (m_9)	X (m_{11})	**0** (m_{10})

Grouping the **0s** (incorporating **X**s where beneficial):

- **Group 1 (Quad):** $m_1, m_5, X_3, X_7 \implies \bar{X}_1X_4$
- **Group 2 (Quad):** $m_2, m_{10}, X_3, X_{11} \implies \bar{X}_2X_3$
- **Group 3 (Pair):** $m_{12}, m_{14} \implies X_1X_2\bar{X}_4$

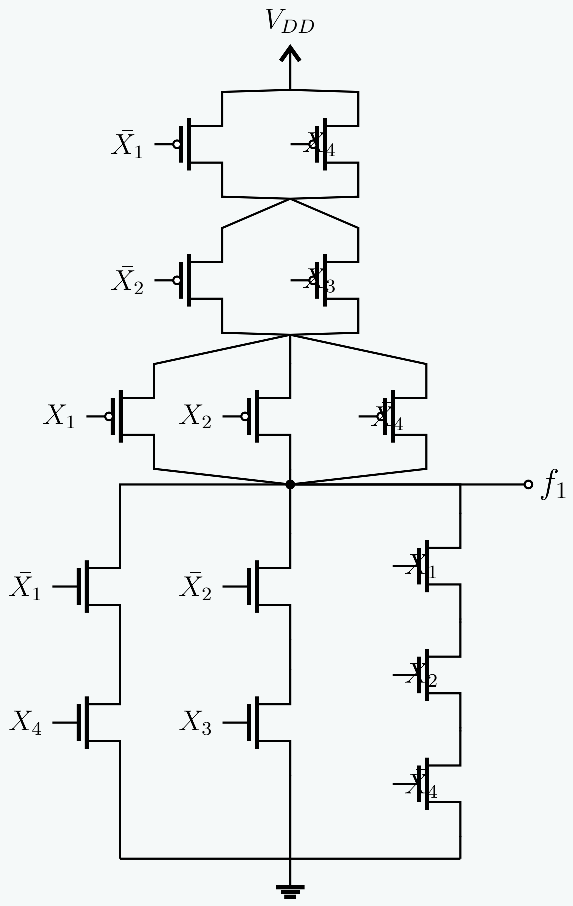
Thus, the minimized Boolean expression for the PDN is:

$$\bar{f}_1 = \bar{X}_1X_4 + \bar{X}_2X_3 + X_1X_2\bar{X}_4 \tag{2}$$

By applying De Morgan's laws, the PUN expression is the exact dual:

$$f_1 = (X_1 + \bar{X}_4) \cdot (X_2 + \bar{X}_3) \cdot (\bar{X}_1 + \bar{X}_2 + X_4) \tag{3}$$

2. Circuit Diagram for f_1
The PDN consists of 3 parallel branches connected to ground, utilizing 7 NMOS transistors in total. The PUN consists of 3 series blocks, utilizing 7 PMOS transistors. This requires just 14 transistors.



Part 2: Minimization and Design for $\bar{f}_2 = (A + B)(C + D)(\bar{A} + B + D)$
1. Mathematical Derivation
We simplify the provided Product-of-Sums (POS) expression for $\bar{f}_2$ using Boolean algebra rules to minimize the transistor count:

$$\begin{aligned}\bar{f}_2 &= (A + B)(C + D)(\bar{A} + B + D) \\ &= (A + B)(\bar{A} + B + D) \cdot (C + D) \\ &= (A\bar{A} + AB + AD + B\bar{A} + BB + BD) \cdot (C + D)\end{aligned} \tag{4}$$

Since $A\bar{A} = 0$ and $BB = B$:

$$\bar{f}_2 = (B + AB + \bar{A}B + BD + AD) \cdot (C + D) \tag{5}$$

Using the absorption law ($B + B \cdot X = B$), we extract B:

$$B(1 + A + \bar{A} + D) = B \tag{6}$$

Thus, the expression heavily simplifies to:

$$\begin{aligned}\bar{f}_2 &= (B + AD) \cdot (C + D) \\ &= BC + BD + ADC + ADD \\ &= BC + BD + ADC + AD\end{aligned} \tag{7}$$

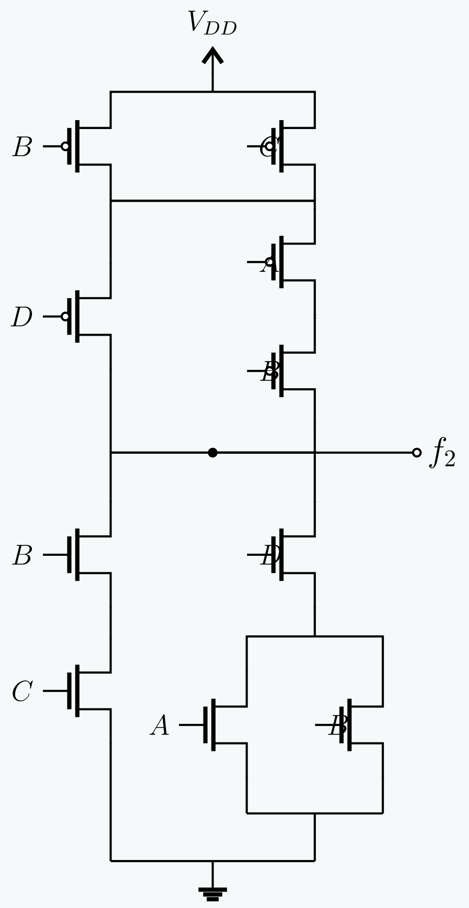
By factoring AD from the last two terms ($AD(C + 1) = AD$):

$$\begin{aligned}\bar{f}_2 &= BC + BD + AD \\ &= BC + D(A + B)\end{aligned} \tag{8}$$

This is the minimal SOP for the PDN. It requires a branch with B and C in series, placed in parallel with a branch containing D in series with the parallel combination of A and B. By duality, the PUN expression is $f_2 = (\bar{B} + \bar{C}) \cdot (\bar{D} + \bar{A}\bar{B})$.

2. Circuit Diagram for f_2

This implementation minimizes transistor count down to exactly 10 transistors (5 NMOS and 5 PMOS). No external inverters are required for the main inputs.



Q5. Implement the function $f(A, B, C, D) = \sum m(1, 2, 4, 7, 14, 15)$ using (i) CMOS circuit and (ii) pseudo NMOS circuit. Which circuit would you prefer between your two designed circuits and why? **(25 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

1. Mathematical Derivation and Logic Minimization

To implement the given logic function using standard static CMOS and pseudo-NMOS topologies, we first need to derive the minimized Boolean expressions. A static CMOS gate consists of a Pull-Down Network (PDN) utilizing NMOS transistors to implement the complement of the function ($\bar{f}$), and a Pull-Up Network (PUN) utilizing PMOS transistors to implement the true function (f) via the structural dual of the PDN.

The target function is given in sum-of-minterms form:

$$f(A, B, C, D) = \sum m(1, 2, 4, 7, 14, 15) \tag{1}$$

The complement of the function, which dictates the PDN, is represented by the remaining minterms:

$$\bar{f}(A, B, C, D) = \sum m(0, 3, 5, 6, 8, 9, 10, 11, 12, 13) \tag{2}$$

Implementing this directly from a standard Sum-of-Products (SOP) derived from a Karnaugh Map requires an excessive number of transistors. Instead, we perform algebraic factorization to find an optimized, shared-branch architecture.

By grouping minterms strategically, we can factor $\bar{f}$ as follows:

$$\bar{f} = A(\bar{B} + \bar{C}) + \bar{A} \cdot [\bar{B} \cdot (\bar{C}\bar{D} + CD) + B \cdot (\bar{C}D + C\bar{D})] \tag{3}$$

Notice the highly symmetric inner terms: $(\bar{C}\bar{D}+CD)$ represents the logical `XNOR` of C and D, and $(\bar{C}D+C\bar{D})$ represents the logical `XOR`. This factored form is extremely efficient for transistor-level layout.
The PUN implements the structural dual of the PDN. By applying De Morgan's Laws to our minimized $\bar{f}$, we obtain the topological equation for the PMOS network (where parallel branches become series, and series branches become parallel):

$$f_{PUN\_topology} = A \parallel (\bar{B}\cdot\bar{C}) \text{ in series with } \bar{A} \parallel \Big([\bar{B} \parallel ((\bar{C} \parallel \bar{D})\cdot(C \parallel D))]\cdot[B \parallel ((\bar{C} \parallel D)\cdot(C \parallel \bar{D}))]\Big) \tag{4}$$

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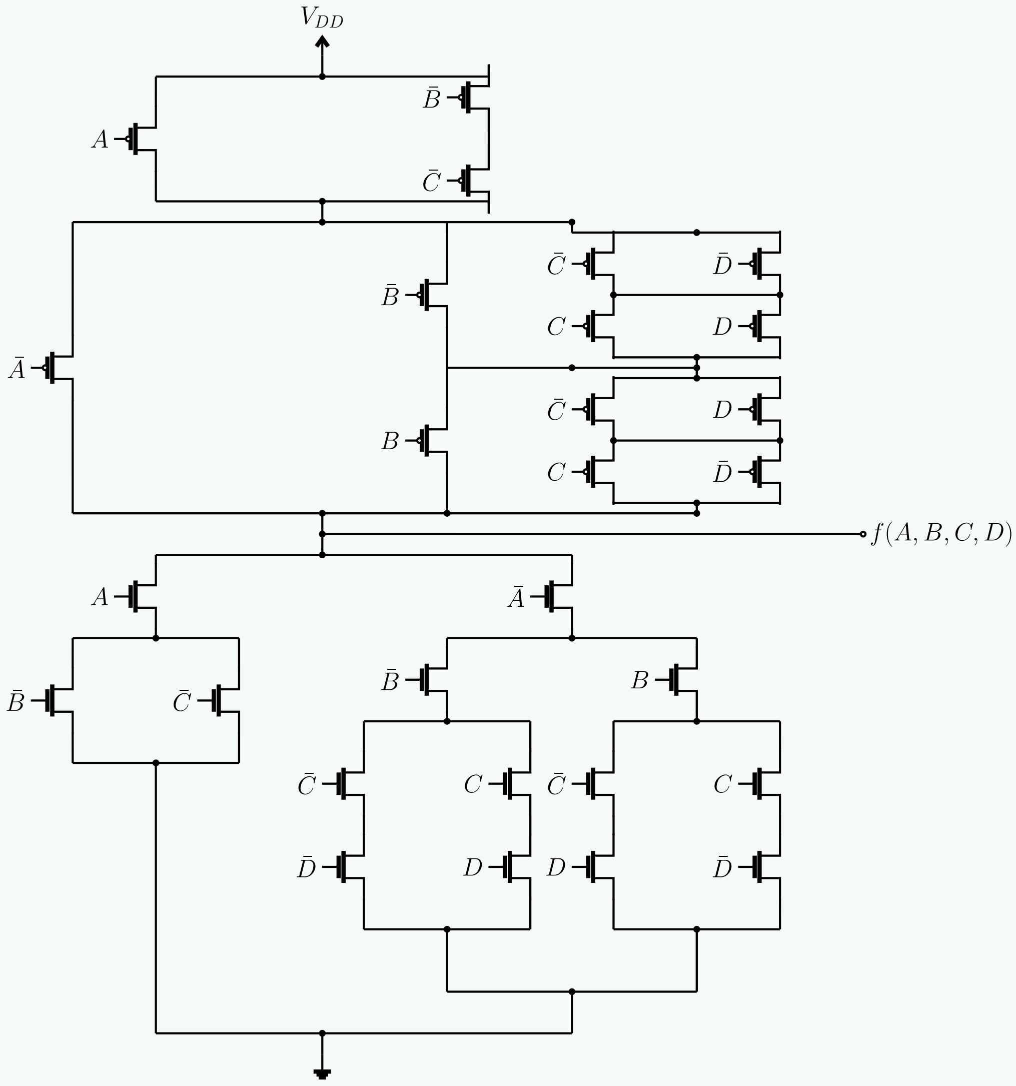
2. Circuit Implementation (i): Static CMOS Logic Gate
The static CMOS design pairs the 14-transistor NMOS PDN with its exact 14-transistor PMOS structural dual to achieve rail-to-rail switching with zero static power dissipation.



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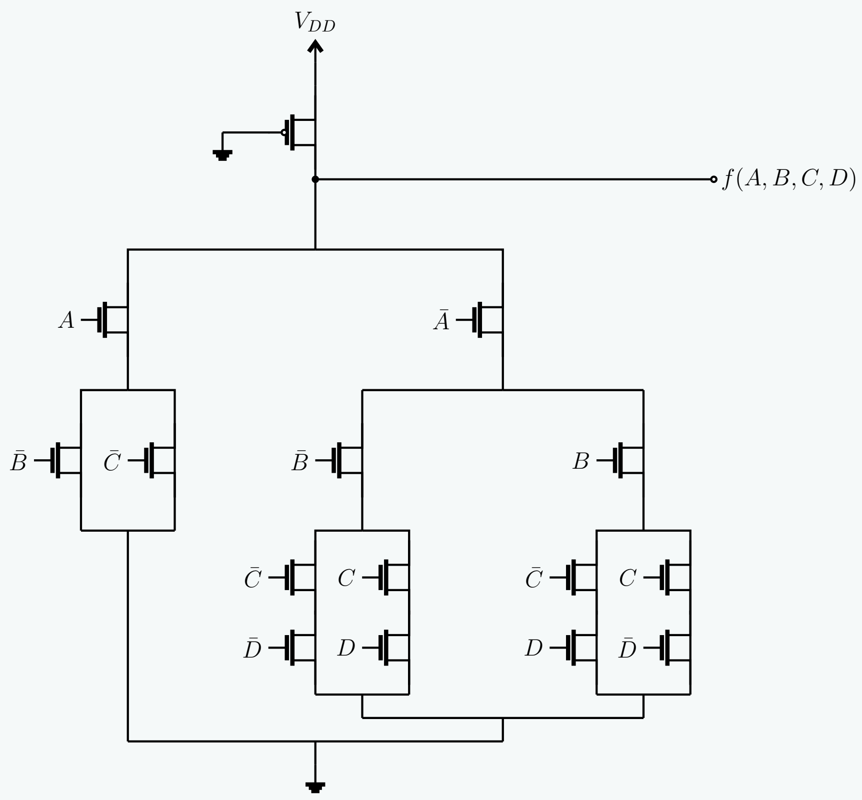
3. Circuit Implementation (ii): Pseudo-NMOS Logic Gate
Pseudo-NMOS topology simplifies the circuit by replacing the entire complex Pull-Up Network (14 PMOS transistors) with a single PMOS transistor whose gate is permanently grounded. This guarantees that the load device is strictly always ON, acting as a passive pull-up resistor. The Pull-Down Network remains exactly identical to the static CMOS design.



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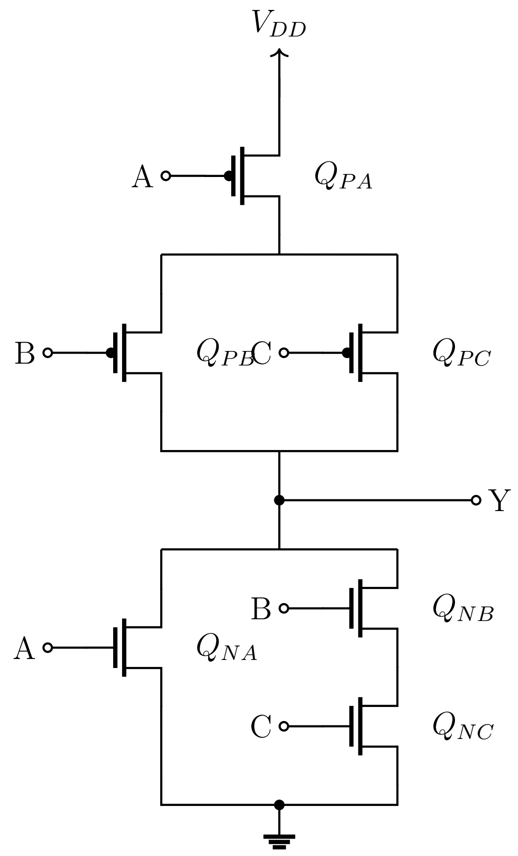
4. Performance Comparison and Design Preference

Parameter	Static CMOS Circuit	Pseudo-NMOS Circuit
Transistor Count	28 Transistors (14 PMOS + 14 NMOS)	15 Transistors (1 PMOS + 14 NMOS)
Static Power	**Zero** ($I_{DD} \approx 0$ under steady state).	**High** (Continuous DC current flows when $f = 0$).
Output Voltage Swing	Full rail-to-rail ($V_{OH} = V_{DD}$, $V_{OL} = 0$).	Reduced swing ($V_{OL} > 0$ due to ratioed voltage divider).
Noise Margins	Excellent and highly symmetric.	Reduced NM_L due to elevated V_{OL}.
Silicon Area	Large (requires substantial PMOS routing).	Highly compact (significantly less routing).

Final Preference: I would strongly prefer the **Static CMOS circuit** for this design.
Why? Despite the increased transistor count (28 vs. 15), static CMOS offers **zero static power dissipation** and full rail-to-rail voltage swings, maintaining robust noise margins. In modern digital logic and VLSI analog integration, power budgeting and signal integrity are practically always prioritized over raw geometric area minimization. Pseudo-NMOS introduces constant short-circuit current whenever the output is evaluated LOW, leading to unacceptable heating and power consumption in large-scale integrations, making it viable only for highly specialized, heavily area-constrained blocks where power isn't a concern.

CMOS Transistor Sizing

Q1. Calculate transistor W/L ratios for the logic circuit shown. Assume that for the basic inverter $n = 1.5$ and $P = 5$ and the channel length is $0.25\,\mu$m.



(16 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Theory and Principle of Transistor Sizing

To maintain equivalent performance (propagation delay and rise/fall times) to a basic reference inverter, the transistors in a complex CMOS logic gate must be sized such that the worst-case equivalent resistance of the Pull-Up Network (PUN) and Pull-Down Network (PDN) matches that of the reference inverter.

When transistors of the same type are connected in **series**, their equivalent resistance increases. Therefore, their W/L ratio must be scaled up by a factor equal to the number of transistors in the series path. When connected in **parallel**, the worst-case resistance is determined by the single longest branch (assuming only one transistor is ON).

Given parameters for the basic inverter:

- NMOS reference ratio: $n = (W/L)_n = 1.5$
- PMOS reference ratio: $p = (W/L)_p = 5$
- Channel length: $L = 0.25\,\mu\text{m}$

2. Circuit Logic Analysis

Analyzing the given schematic:

- **PDN (NMOS):** Q_{NA} is in parallel with the series combination of Q_{NB} and Q_{NC}. The logic evaluated is $f_{PDN} = A + BC$.
- **PUN (PMOS):** Q_{PA} is in series with the parallel combination of Q_{PB} and Q_{PC}. This is the exact dual of the PDN.
- The final logic function is $Y = \overline{A + BC}$.

3. Sizing the Pull-Down Network (NMOS)

The PDN must provide a worst-case equivalent discharge path equal to a single NMOS with $(W/L)_n = 1.5$.

- **Path 1 (through Q_{NA}):** Only one transistor is in this conduction path. Therefore, it requires no scaling.

$$(W/L)_{NA} = 1 \times 1.5 = 1.5 \tag{1}$$

- **Path 2 (through Q_{NB} and Q_{NC}):** There are two transistors in series. To maintain the same equivalent on-resistance, their widths must be doubled.

$$(W/L)_{NB} = (W/L)_{NC} = 2 \times 1.5 = 3.0 \tag{2}$$

4. Sizing the Pull-Up Network (PMOS)

The PUN must provide a worst-case equivalent charge path equal to a single PMOS with $(W/L)_p = 5$.

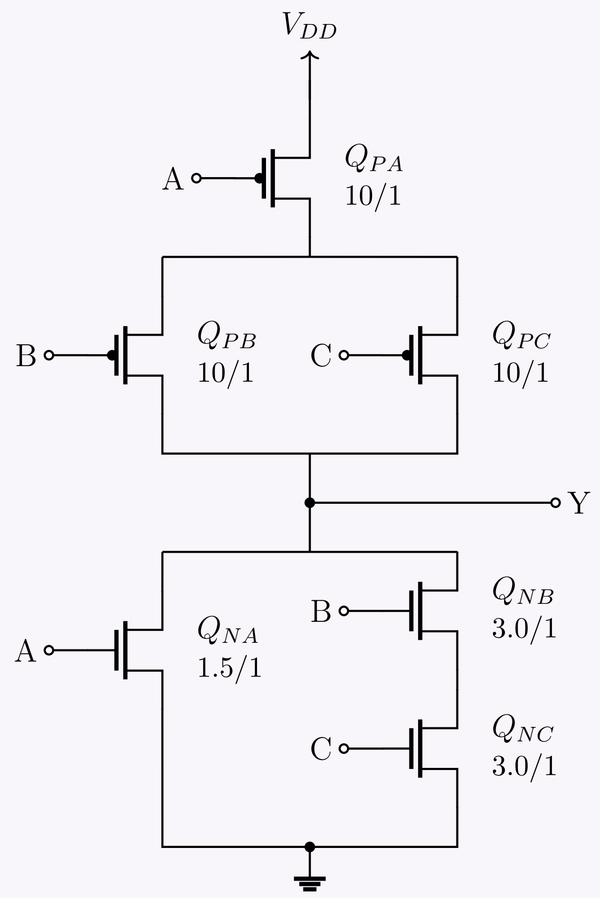
- The worst-case pull-up path occurs when either ($A = 0$ and $B = 0$) or ($A = 0$ and $C = 0$). In both cases, the current flows through exactly two PMOS transistors in series (Q_{PA} and Q_{PB}, or Q_{PA} and Q_{PC}).
- Since the worst-case path contains two transistors in series, all PMOS transistors in this network must be scaled by a factor of 2.

$$(W/L)_{PA} = 2 \times 5 = 10 \tag{3}$$
$$(W/L)_{PB} = 2 \times 5 = 10 \tag{4}$$
$$(W/L)_{PC} = 2 \times 5 = 10 \tag{5}$$

5. Summary of Transistor Geometries
Using the given constant channel length $L = 0.25\,\mu$m, the absolute width W for each transistor can be calculated as $W = (W/L) \times L$.

Transistor	**Type**	**Ratio** (W/L)	**Channel Width** W **(**μ**m)**
Q_{PA}	PMOS	10	$10 \times 0.25 = 2.50$
Q_{PB}, Q_{PC}	PMOS	10	$10 \times 0.25 = 2.50$
Q_{NA}	NMOS	1.5	$1.5 \times 0.25 = 0.375$
Q_{NB}, Q_{NC}	NMOS	3.0	$3.0 \times 0.25 = 0.75$

6. Final Redrawn Circuit with Sizing Details



Q2. Design transistor W/L ratios for the logic $Y = \overline{(AB + D)C}$ implemented using CMOS circuit. Assume that for the basic inverter $(W/L)_n = n = 2$ and $(W/L)_p = p = 7$ and the channel length is $0.15\,\mu$m. **(21 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Answer

1. Logic Function Analysis
The given Boolean function is $Y = \overline{(AB + D)C}$. This is an AND-OR-INVERT (AOI) complex logic gate. In static CMOS design, the function determines the topologies of the Pull-Down Network (PDN) and Pull-Up Network (PUN):

- **Pull-Down Network (NMOS):** The PDN evaluates the uncomplemented function $f_{PDN} = (AB + D)C$.
  - Transistors Q_{NA} and Q_{NB} (inputs A and B) are in **series**.
  - Transistor Q_{ND} (input D) is in **parallel** with the A-B series combination.
  - Transistor Q_{NC} (input C) is in **series** with the entire $(AB + D)$ block.
- **Pull-Up Network (PMOS):** The PUN evaluates the dual function $f_{PUN} = \overline{C} + \overline{D}(\overline{A} + \overline{B})$.
  - Transistors Q_{PA} and Q_{PB} are in **parallel**.
  - Transistor Q_{PD} is in **series** with the A-B parallel block.
  - Transistor Q_{PC} is in **parallel** with the entire $\overline{D}(\overline{A} + \overline{B})$ block.

2. Principle of Transistor Sizing
To ensure the complex logic gate has propagation delays and rise/fall times no worse than a basic reference inverter, we size the transistors such that the worst-case equivalent resistance of the PUN and PDN matches that of the reference inverter.

- NMOS reference equivalent ratio: $n = (W/L)_{n,eq} = 2$
- PMOS reference equivalent ratio: $p = (W/L)_{p,eq} = 7$
- Channel length: $L = 0.15\,\mu$m

When k identical transistors are in **series**, their equivalent resistance increases by k, so their individual (W/L) ratios must be multiplied by k. For transistors in **parallel**, the worst-case assumes only one branch is conducting.

3. Sizing the Pull-Down Network (NMOS)
The PDN must have a worst-case equivalent ratio of $n = 2$. The paths to ground are:

(a) **Path 1:** Through Q_{NC}, Q_{NA}, and Q_{NB} (3 series transistors).

(b) **Path 2:** Through Q_{NC} and Q_{ND} (2 series transistors).

To balance Path 1 (the longest path), we assign equal weights to Q_{NC}, Q_{NA}, and Q_{NB}:

$$\frac{1}{(W/L)_{NC}} + \frac{1}{(W/L)_{NA}} + \frac{1}{(W/L)_{NB}} = \frac{1}{2} \implies \frac{3}{(W/L)} = \frac{1}{2} \implies (W/L)_{NC,NA,NB} = 6 \tag{1}$$

Now, analyzing Path 2 to determine Q_{ND}:

$$\frac{1}{(W/L)_{NC}} + \frac{1}{(W/L)_{ND}} = \frac{1}{2} \implies \frac{1}{6} + \frac{1}{(W/L)_{ND}} = \frac{1}{2} \tag{2}$$

$$\frac{1}{(W/L)_{ND}} = \frac{1}{2} - \frac{1}{6} = \frac{3-1}{6} = \frac{2}{6} = \frac{1}{3} \implies (W/L)_{ND} = 3 \tag{3}$$

4. Sizing the Pull-Up Network (PMOS)
The PUN must have a worst-case equivalent ratio of $p = 7$. The paths to V_{DD} are:

(a) **Path 1:** Through Q_{PC} alone (1 transistor).

(b) **Path 2:** Through Q_{PD} and Q_{PA} (2 series transistors).

(c) **Path 3:** Through Q_{PD} and Q_{PB} (2 series transistors).

For Path 1, Q_{PC} alone must provide the full equivalent conductance:

$$(W/L)_{PC} = 7 \tag{4}$$

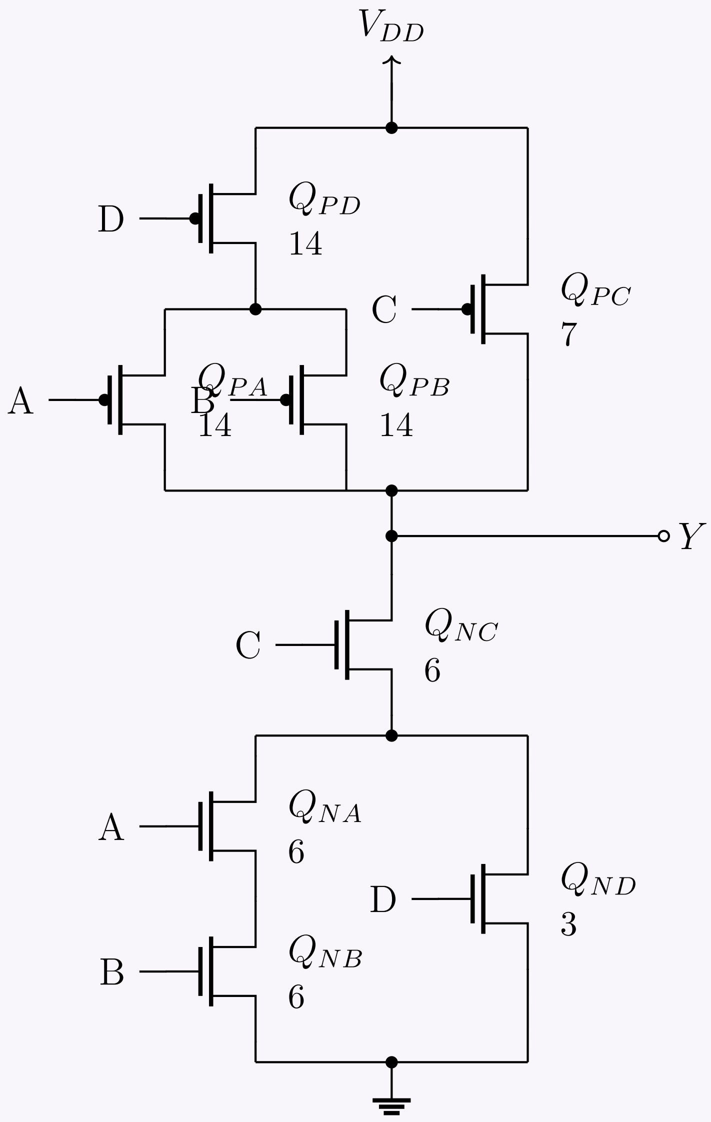
For Paths 2 and 3, there are 2 transistors in series. We scale their widths by a factor of 2:

$$(W/L)_{PD} = (W/L)_{PA} = (W/L)_{PB} = 2 \times 7 = 14 \tag{5}$$

5. Summary of Transistor Geometries
Using $W = (W/L) \times L$, where $L = 0.15\,\mu$m, we obtain the final physical channel widths:

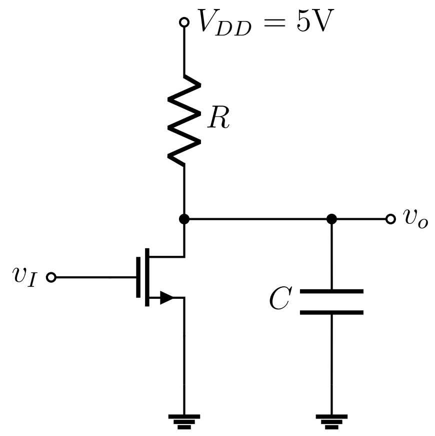
Transistor	**Type**	**Ratio** (W/L)	**Channel Width** W **(**μ**m)**
Q_{PC}	PMOS	7	$7 \times 0.15 = 1.05$
Q_{PA}, Q_{PB}, Q_{PD}	PMOS	14	$14 \times 0.15 = 2.10$
Q_{NA}, Q_{NB}, Q_{NC}	NMOS	6	$6 \times 0.15 = 0.90$
Q_{ND}	NMOS	3	$3 \times 0.15 = 0.45$

6. Final Sized CMOS Circuit Schematic



Inverter Propagation Delay

Q1. For the inverter shown with a capacitor C connected between the output node and ground: if at $t = 0$ the input goes low to high, find the equation of time for v_0 to reach $\frac{1}{2}(V_{OL} + V_{OH})$. Also calculate the high-to-low propagation time for $R = 25\,\text{k}\Omega$ and $C = 20\,\text{pF}$.



(10.67 marks) [EEE 263 | L-2/T-1 | 2018–19]

Answer

1. Circuit Principle and Operation

The given circuit is a Resistive-Load NMOS Inverter driving a load capacitor C.

- **Output HIGH (V_{OH}):** When the input v_I is LOW (less than the threshold voltage V_{tn}), the NMOS transistor M_1 is OFF. The capacitor C charges through the resistor R up to the supply voltage. Thus, $V_{OH} = V_{DD}$.
- **Output LOW (V_{OL}):** When the input v_I is HIGH (V_{DD}), M_1 turns ON. The steady-state output voltage drops to a low level V_{OL}, determined by the voltage divider action between R and the ON-resistance of the NMOS in the triode region.

2. Equation of Time for High-to-Low Transition (t_{PHL})

At $t = 0$, the input v_I transitions from LOW to HIGH ($0 \to V_{DD}$). The NMOS transistor turns ON, initiating the discharge of the load capacitor C from V_{OH} towards V_{OL}.

Applying Kirchhoff's Current Law (KCL) at the output node for $t \geq 0$:

$$i_D(v_o) + i_C(v_o) + i_R(v_o) = 0 \tag{1}$$

Substituting the expressions for the capacitor and resistor currents:

$$i_D(v_o) + C\frac{dv_o}{dt} + \frac{v_o - V_{DD}}{R} = 0 \tag{2}$$

Rearranging to solve for the differential time interval dt:

$$dt = \frac{-C}{i_D(v_o) - \frac{V_{DD}-v_o}{R}} dv_o \tag{3}$$

To find the equation of time t for v_o to reach the 50% midpoint $V_{50\%} = \frac{1}{2}(V_{OL} + V_{OH})$, we integrate from the initial voltage V_{OH} to $V_{50\%}$:

$$t_{PHL} = \int_{V_{OH}}^{\frac{1}{2}(V_{OL}+V_{OH})} \frac{-C}{i_D(v_o) - \frac{V_{DD}-v_o}{R}} dv_o \tag{4}$$

Important Note: The drain current $i_D(v_o)$ is non-linear. The NMOS starts in the **saturation region** (since $V_{OH} > V_{DD} - V_{tn}$) and transitions to the **triode region** as v_o falls below $V_{DD} - V_{tn}$. Evaluating this integral exactly requires the device parameters (e.g., $k'_n, W/L, V_{tn}$).

3. Numerical Calculation and Propagation Delay Analysis

The problem requests the calculation of the propagation time using only the passive components: $R = 25\,\text{k}\Omega$ and $C = 20\,\text{pF}$. Because the exact discharge time (t_{PHL}) depends heavily on the missing NMOS transistor parameters, standard textbook problems of this type rely on the charging time constant formed by R and C to evaluate the **low-to-high** propagation delay (t_{PLH}). If we analyze the complementary transition (input goes HIGH to LOW), M_1 turns OFF ($i_D = 0$), and the output node equation simplifies strictly to the RC network:

$$C\frac{dv_o}{dt} + \frac{v_o - V_{DD}}{R} = 0 \tag{5}$$

Solving this first-order differential equation with the initial condition $v_o(0) = V_{OL}$ yields:

$$v_o(t) = V_{DD} - (V_{DD} - V_{OL})e^{-t/RC} \tag{6}$$

Setting $v_o(t_{PLH}) = \frac{1}{2}(V_{OH} + V_{OL})$ and substituting $V_{OH} = V_{DD}$:

$$\frac{V_{DD} + V_{OL}}{2} = V_{DD} - (V_{DD} - V_{OL})e^{-t_{PLH}/RC} \tag{7}$$

$$\frac{V_{DD} - V_{OL}}{2} = (V_{DD} - V_{OL})e^{-t_{PLH}/RC} \tag{8}$$

$$\frac{1}{2} = e^{-t_{PLH}/RC} \tag{9}$$

Taking the natural logarithm of both sides gives the propagation time equation:

$$t_{PLH} = RC\ln(2) \approx 0.693RC \tag{10}$$

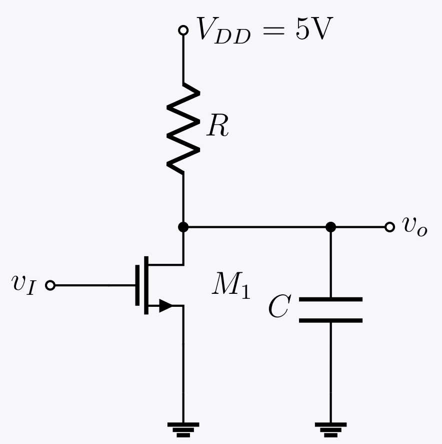
Assuming this RC-dominant transition is the intended calculation for the given passive values:

$$t_{prop} = \left(25 \times 10^3\,\Omega\right) \times \left(20 \times 10^{-12}\,\text{F}\right) \times 0.6931 \tag{11}$$

$$t_{prop} = \left(500 \times 10^{-9}\,\text{s}\right) \times 0.6931 \tag{12}$$

$$t_{prop} = 346.57\,\text{ns} \tag{13}$$

4. Reference Circuit Diagram



Transmission Gate Implementation

Q1. Implement the following function Y using CMOS transmission gate configuration:

$$Y = AB + \bar{A}C$$

(7 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Answer

1. Logical Analysis & Principle

The given Boolean function is:

$$Y = AB + \bar{A}C \tag{1}$$

This equation describes the behavior of a standard **2-to-1 Multiplexer (MUX)**, where:

- A acts as the **Select** signal.
- B and C act as the **Data Inputs**.

By evaluating the logic for the two possible states of the select line A:

$$\begin{aligned} \text{Case 1 } (A = 1): &\quad Y = (1)B + (0)C = B \\ \text{Case 2 } (A = 0): &\quad Y = (0)B + (1)C = C \end{aligned}$$

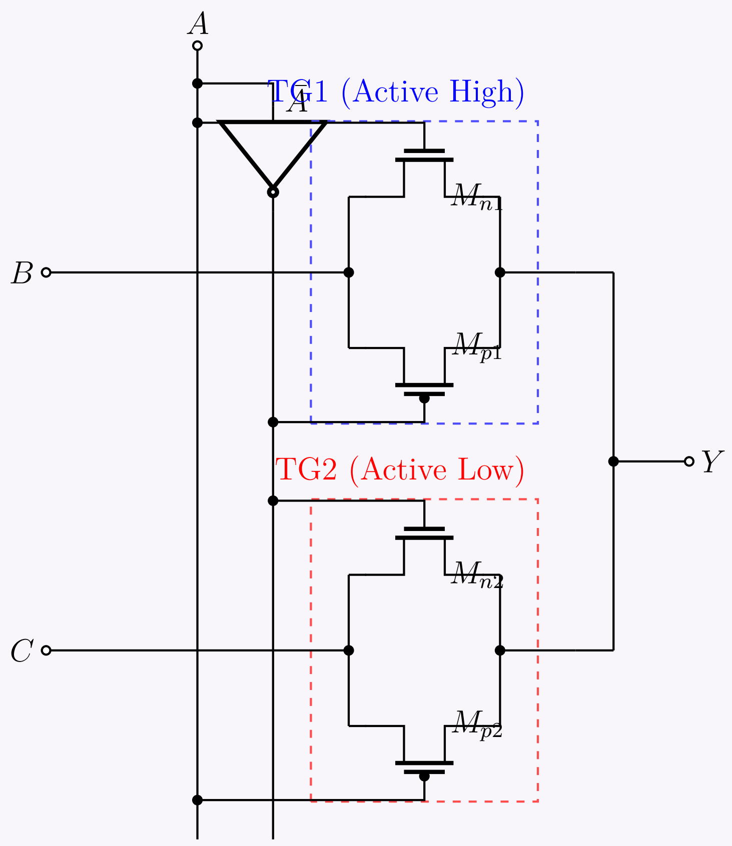
2. Circuit Operation

To implement this logic without voltage degradation (threshold drops), we use **CMOS Transmission Gates (TG)**. A transmission gate is formed by connecting an NMOS and a PMOS transistor in parallel.

- **TG1 (Top Branch):** Passes the input B to the output Y when $A = 1$. To turn ON both transistors simultaneously, the NMOS gate is driven by A and the PMOS gate is driven by $\bar{A}$.
- **TG2 (Bottom Branch):** Passes the input C to the output Y when $A = 0$. To turn ON both transistors, the NMOS gate is driven by $\bar{A}$ and the PMOS gate is driven by A.

An inverter is required to generate the complementary select signal $\bar{A}$ from the input A.

3. CMOS Transmission Gate Schematic



CMOS Flip-Flop Implementation

Q1. Show a CMOS implementation of edge-triggered D flip-flops with asynchronous reset. **(25 marks)** [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Theory and Principle of Operation

A Master-Slave D-Flip-Flop (D-FF) captures the input D precisely at the edge of a clock signal. The CMOS implementation leverages **Transmission Gates (TGs)** acting as ideal voltage-controlled switches.

- **Master Stage:** Acts as a level-sensitive latch that becomes transparent when $CLK = 0$.
- **Slave Stage:** Acts as a level-sensitive latch that becomes transparent when $CLK = 1$.
- **Edge-Triggering:** By cascading these two opposite-phase latches, the overall circuit securely captures and transfers data only on the rising edge ($0 \to 1$ transition) of the clock.

2. Asynchronous Reset Implementation

To incorporate an **active-low asynchronous reset** ($\overline{RST}$), the flip-flop must be forced to a known state ($Q = 0$) immediately, independent of the clock signal.

- We replace the standard forward-path inverters in both the master and slave latches with **NAND gates**.
- One input of the NAND gate receives the data stream, and the other is tied to the global $\overline{RST}$ line.
- To prevent logic inversion through the two NAND gates and to guarantee $Q = D$ under normal operation, we introduce an input inverter and an output inverter.

3. Mathematical / Logical Derivation

Let us define the logical states at each intermediate node during normal operation ($\overline{RST} = 1$). A NAND gate with one input tied to logic 1 acts as an inverter.

$$\text{Input node after INV:} \quad D_{int} = \overline{D} \tag{1}$$

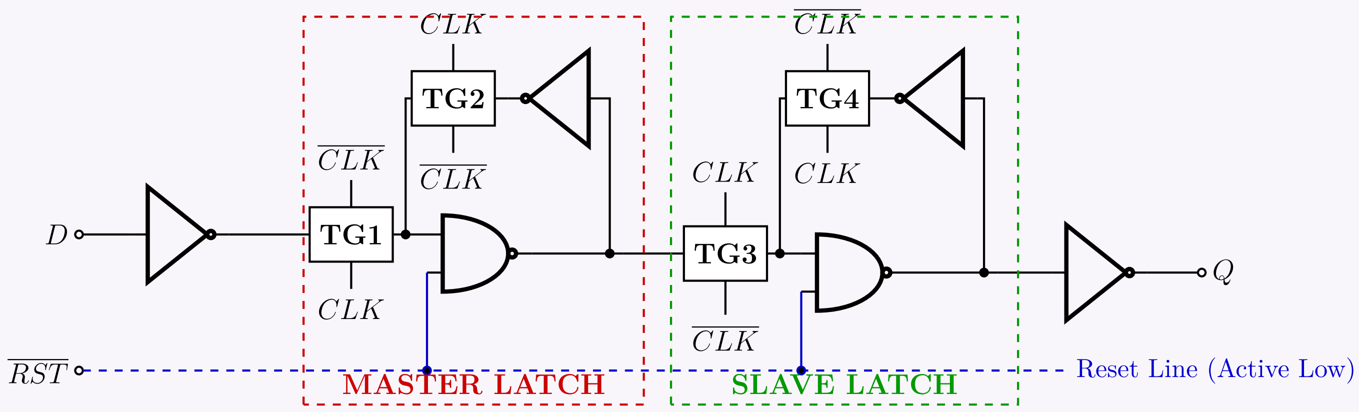
$$\text{Master output } (CLK = 0): \quad M_Q = \overline{D_{int} \cdot \overline{RST}} = \overline{\overline{D} \cdot 1} = D \tag{2}$$

$$\text{Slave output } (CLK = 1): \quad S_Q = \overline{M_Q \cdot \overline{RST}} = \overline{D \cdot 1} = \overline{D} \tag{3}$$

$$\text{Final output after INV:} \quad Q = \overline{S_Q} = \overline{\overline{D}} = D \tag{4}$$

Reset Condition: When $\overline{RST} = 0$, the NAND gates immediately output 1 regardless of the data or clock. Thus, $M_Q = 1$ and $S_Q = 1$. The final output inverter forces $Q = \overline{1} = 0$, achieving an instantaneous, asynchronous reset.

4. CMOS Logic & Transmission Gate Schematic



Flip-Flop Construction

Q1. Show how D flip-flop and T flip-flop can be constructed from J-K flip-flops. **(15 marks)** [EEE 263 | L-2/T-1 | 2016–17]

Answer

1. Theory and Principle of Flip-Flop Conversion

The J-K flip-flop is a universal flip-flop, meaning it can be configured to mimic the behavior of any other type of flip-flop (such as D or T types). To convert a J-K flip-flop into another type, we must determine the required inputs (J and K) as functions of the desired new input (e.g., D or T) and the current state (Q_n).

The fundamental behavior of a J-K flip-flop is governed by its characteristic equation:

$$Q_{n+1} = J\overline{Q}_n + \overline{K}Q_n \tag{1}$$

Additionally, we rely on the excitation table of the J-K flip-flop, which specifies the required J and K inputs for any desired state transition from Q_n to Q_{n+1}:

Current State (Q_n)	Next State (Q_{n+1})	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Note: 'X' denotes a "Don't Care" condition.

---

2. Construction of a D Flip-Flop from a J-K Flip-Flop

A Delay (D) flip-flop ensures that the next state always tracks the input, i.e., $Q_{n+1} = D$.

Step 2a: Conversion Table

We construct a table mapping the desired D input and current state Q_n to the required J and K inputs:

Input (D)	Current State (Q_n)	Next State (Q_{n+1})	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

Step 2b: Boolean Simplification
From the table, we can derive the logic expressions for J and K in terms of D:

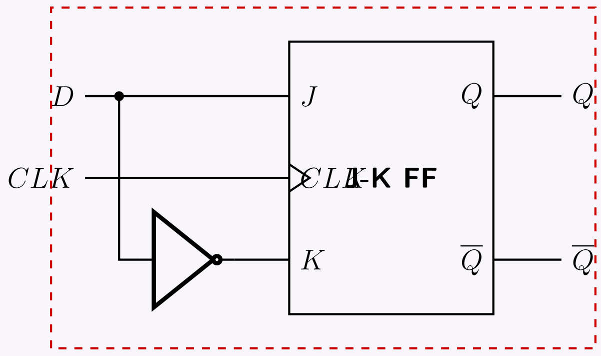
$$J = \sum m(2) + d(3) \implies J = D$$
$$K = \sum m(1) + d(0) \implies K = \overline{D}$$

Step 2c: Mathematical Verification
Substituting $J = D$ and $K = \overline{D}$ into the J-K characteristic equation:

$$\begin{aligned} Q_{n+1} &= D\overline{Q}_n + \overline{(\overline{D})}Q_n \\ &= D\overline{Q}_n + DQ_n \\ &= D(\overline{Q}_n + Q_n) \\ &= D \end{aligned}$$

This perfectly matches the characteristic equation of a D flip-flop.
Step 2d: Circuit Implementation
By connecting the D input directly to J and passing it through an inverter to K, we synthesize a D flip-flop.

Equivalent D Flip-Flop



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3. Construction of a T Flip-Flop from a J-K Flip-Flop
A Toggle (T) flip-flop reverses its state when $T = 1$ and holds its state when $T = 0$. Its characteristic equation is $Q_{n+1} = T \oplus Q_n$.
Step 3a: Conversion Table
We map the desired T input and current state Q_n to the required J and K inputs:

Input (T)	Current State (Q_n)	Next State (Q_{n+1})	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

Step 3b: Boolean Simplification
From the table, we extract the logic expressions for J and K:

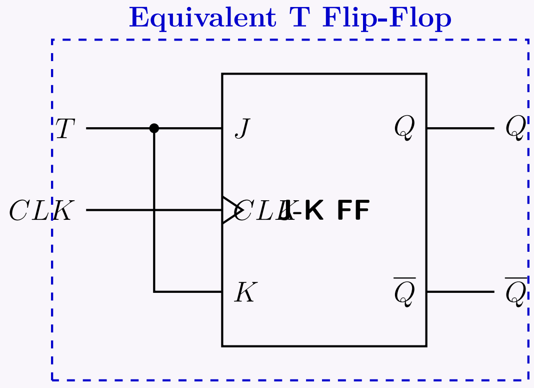
$$J = \sum m(2) + d(3) \implies J = T$$
$$K = \sum m(3) + d(2) \implies K = T$$

Step 3c: Mathematical Verification
Substituting $J = T$ and $K = T$ into the J-K characteristic equation yields:

$$\begin{aligned} Q_{n+1} &= T\overline{Q}_n + \overline{T}Q_n \\ &= T \oplus Q_n \end{aligned}$$

This is identically the characteristic equation of a T flip-flop.
Step 3d: Circuit Implementation
By tying both J and K inputs together and driving them with the common signal T, a T flip-flop is synthesized.

Equivalent T Flip-Flop



Counter Design

Q1. Design a MOD-3 counter using JK flip-flops and verify the design showing the truth table. **(15 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Answer

1. Theory and Design Principle

A **MOD-3 (Modulo-3) counter** is a sequential circuit that traverses three distinct states before resetting. Typically, the state sequence in binary is $00 \to 01 \to 10 \to 00$ (equivalent to decimal $0 \to 1 \to 2 \to 0$). The number of flip-flops (N) required is determined by the condition $2^N \geq 3$, which gives $N = 2$. We will denote the two flip-flops as **FF A** (Most Significant Bit, MSB) and **FF B** (Least Significant Bit, LSB).

2. State Transition and Excitation Table

To design the counter, we use the excitation table of a J-K flip-flop, which determines the required inputs (J and K) for a given state transition ($Q_n \to Q_{n+1}$):

$$0 \to 0 \implies J = 0, K = X \qquad 0 \to 1 \implies J = 1, K = X$$
$$1 \to 0 \implies J = X, K = 1 \qquad 1 \to 1 \implies J = X, K = 0$$

Note: 'X' denotes a "Don't Care" condition.

Applying this to our desired MOD-3 counting sequence, we construct the state table:

Present State		**Next State**		**Required J-K Inputs**			
A (MSB)	B (LSB)	A_{next}	B_{next}	J_A	K_A	J_B	K_B
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	0	0	X	1	0	X
1	1	X	X	X	X	X	X

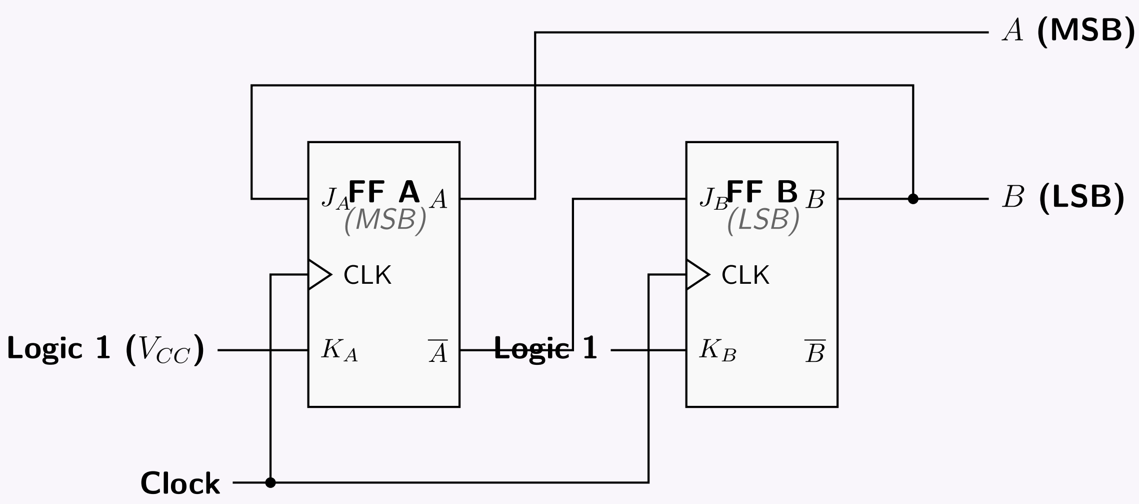
Note: State 11 is unused, so its next state and required inputs are treated as Don't Cares (X) to simplify logic.

3. Boolean Simplification

We derive the Boolean expressions for the J and K inputs using the tables above:

$$\textbf{For FF A:} \quad J_A = \sum m(1) + d(2,3) \implies \mathbf{J_A = B} \tag{1}$$
$$K_A = \sum m(2) + d(0,1,3) \implies \mathbf{K_A = 1} \tag{2}$$
$$\textbf{For FF B:} \quad J_B = \sum m(0) + d(1,3) \implies \mathbf{J_B = \overline{A}} \tag{3}$$
$$K_B = \sum m(1) + d(0,2,3) \implies \mathbf{K_B = 1} \tag{4}$$

4. Logic Circuit Implementation



5. Verification of the Design

To verify the design, we trace the logic states iteratively to ensure the counter operates identically to a MOD-3 sequence. We use the characteristic equation of a J-K flip-flop: $Q_{next} = J\overline{Q} + \overline{K}Q$. Given our logic equations: $J_A = B$, $K_A = 1$ and $J_B = \overline{A}$, $K_B = 1$.

Present State		**Applied Inputs**				**Next State**		**Decimal Output**
A	B	$J_A(B)$	$K_A(1)$	$J_B(\overline{A})$	$K_B(1)$	A_{next}	B_{next}	$(A\ B)_2$
0	0	0	1	1	1	0	1	1
0	1	1	1	1	1	1	0	2
1	0	0	1	0	1	0	0	0
1	*1*	*1*	*1*	*0*	*1*	*0*	*0*	*Unused state*

Conclusion: As demonstrated by the truth table, the states naturally progress $00 \to 01 \to 10 \to 00$, correctly satisfying the requirements of a MOD-3 counter. Additionally, if the circuit accidentally falls into the unused state (11), the inputs evaluate to $J_A = 1, K_A = 1, J_B = 0, K_B = 1$, which forces the next state to 00. This confirms the design is **self-starting** and immune to lock-out conditions.

Q2. Design a modulo-12 up counter with synchronous reset. **(15 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

1. Theory and Principle of Operation

A **Modulo-12 (MOD-12) counter** has 12 distinct states. In a standard binary sequence, it counts from 0 up to 11 (in decimal), which corresponds to 0000_2 to 1011_2 in binary. Because $2^3 < 12 \leq 2^4$, a minimum of 4 flip-flops (or a 4-bit binary counter) is required.

The design specifically calls for a **synchronous reset**. The fundamental characteristic of a synchronous reset is that the reset operation occurs *only* on the active edge of the clock. Therefore, to ensure the counter rolls over from state 11 to state 0 on the next clock edge, the reset logic must detect the *last state of the desired sequence* (state 11). This is in contrast to an asynchronous reset, which would require detecting state 12 to force an immediate reset.

2. Logic Derivation

Let the 4-bit outputs of the counter be Q_3 (MSB), Q_2, Q_1, and Q_0 (LSB). The counting sequence is:

$$0000 \to 0001 \to 0010 \to \cdots \to 1011 \to 0000$$

We must generate a low signal for the active-low synchronous reset pin ($\overline{SRST}$) when the counter reaches 11_{10}.

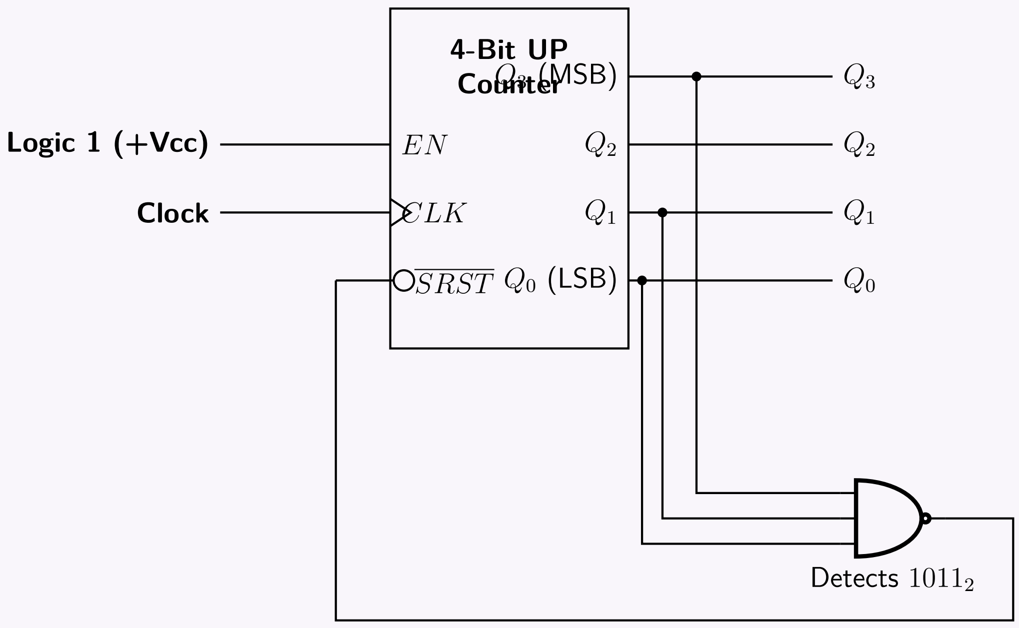
$$\text{Target State (11)}: \quad Q_3Q_2Q_1Q_0 = 1011_2 \tag{1}$$
$$\implies Q_3 = 1,\ Q_2 = 0,\ Q_1 = 1,\ Q_0 = 1 \tag{2}$$

To simplify the combinational logic, we look for the *first* state in the up-counting sequence where Q_3, Q_1, and Q_0 are simultaneously HIGH. Since state 11 (1011_2) is the first state to satisfy this condition, we do not need to explicitly include $\overline{Q_2}$ in our decoding logic. We can use a single 3-input NAND gate:

$$\overline{SRST} = \overline{Q_3 \cdot Q_1 \cdot Q_0} \tag{3}$$

When the counter is in state 11, the NAND gate outputs a 0. On the very next clock edge, this 0 is synchronously clocked in, clearing all flip-flops to 0000.

3. Circuit Diagram



4. State Sequence and Verification Table

The table below verifies that the counter correctly asserts the reset signal at state 11 and transitions to state 0.

Clock Cycle	Q_3	Q_2	Q_1	Q_0	Reset Signal ($\overline{SRST}$)	Next Action (on CLK edge)
0	0	0	0	0	1 (Inactive)	Count up to 0001_2
1	0	0	0	1	1 (Inactive)	Count up to 0010_2
⋮	⋮	⋮	⋮	⋮	⋮	⋮
10	1	0	1	0	1 (Inactive)	Count up to 1011_2
11	**1**	**0**	**1**	**1**	**0 (Active)**	**Synchronous Clear to** 0000_2
12	0	0	0	0	1 (Inactive)	Sequence repeats

Important Note: Because the counter features a *synchronous* reset, state 11 (1011_2) remains stable for a full clock period. No shortened or transient states (glitches) appear at the output, which is the primary advantage of synchronous design over asynchronous design.

Q3. Design a 4-bit up/down counter using T flip-flops. It should include a control pin $\overline{UP}/DOWN$. When $\overline{UP}/DOWN = 0$ the circuit behaves as an up counter; when $\overline{UP}/DOWN = 1$ it behaves as a down counter. The circuit should also have a synchronous active-low $\overline{RESET}$ pin. **(20 marks)** **[EEE 263 | L-2/T-1 | 2016–17]**

Answer

1. Theory and Principle of Operation
A synchronous up/down counter requires all flip-flops to be triggered by a common clock pulse simultaneously. The direction of counting is controlled by a mode selection line, which we will define as $M = \overline{UP}/DOWN$.

- When $M = 0$, the counter operates in **UP** mode ($0000 \to 0001 \to 0010 \to \dots$)
- When $M = 1$, the counter operates in **DOWN** mode ($0000 \to 1111 \to 1110 \to \dots$)

A Toggle (T) flip-flop changes state (toggles) if its input $T = 1$ on the active clock edge, and holds its state if $T = 0$. By analyzing the binary counting sequence, we observe:

- **UP Counting ($M = 0$):** A bit toggles when all less significant bits are 1.
- **DOWN Counting ($M = 1$):** A bit toggles when all less significant bits are 0.

2. Logic Derivation
Let the 4-bit outputs be Q_3 (MSB), Q_2, Q_1, and Q_0 (LSB). Based on the toggling principle, the excitation inputs for the T flip-flops are derived as follows:
For LSB (FF_0): The least significant bit must toggle on every clock cycle regardless of the counting direction.

$$T_0 = 1 \tag{1}$$

For FF_1: Toggles if $Q_0 = 1$ (UP mode) OR $\overline{Q}_0 = 1$ (DOWN mode).

$$T_1 = Q_0\overline{M} + \overline{Q}_0 M \tag{2}$$

For FF_2: Toggles if $Q_0Q_1 = 1$ (UP mode) OR $\overline{Q}_0\overline{Q}_1 = 1$ (DOWN mode).

$$\begin{aligned} T_2 &= (Q_0Q_1)\overline{M} + (\overline{Q}_0\overline{Q}_1)M \\ &= Q_1(Q_0\overline{M}) + \overline{Q}_1(\overline{Q}_0 M) \\ &= Q_1U_0 + \overline{Q}_1 D_0 \end{aligned} \tag{3}$$

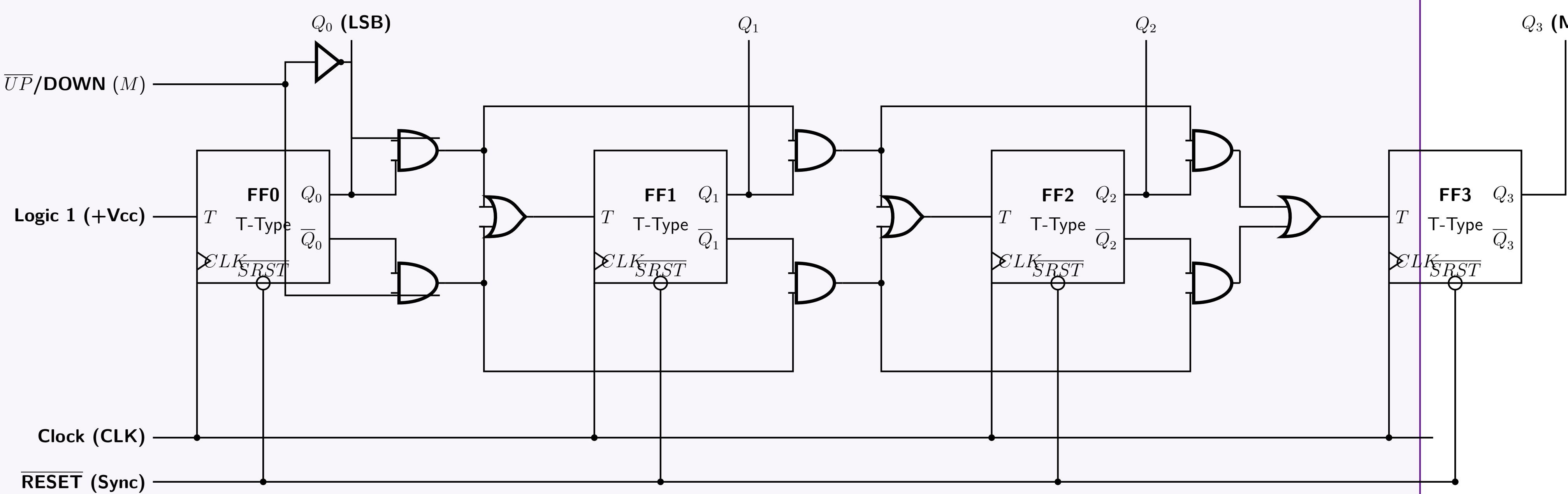
Where $U_0 = Q_0\overline{M}$ is the UP carry, and $D_0 = \overline{Q}_0 M$ is the DOWN carry.
For FF_3 (MSB): Toggles if $Q_0Q_1Q_2 = 1$ (UP) OR $\overline{Q}_0\overline{Q}_1\overline{Q}_2 = 1$ (DOWN).

$$\begin{aligned} T_3 &= (Q_0Q_1Q_2)\overline{M} + (\overline{Q}_0\overline{Q}_1\overline{Q}_2)M \\ &= Q_2(Q_1Q_0\overline{M}) + \overline{Q}_2(\overline{Q}_1\overline{Q}_0 M) \\ &= Q_2U_1 + \overline{Q}_2 D_1 \end{aligned} \tag{4}$$

Where $U_1 = U_0Q_1$ and $D_1 = D_0\overline{Q}_1$. This cascaded logic significantly reduces the fan-in requirements of the logic gates.

3. Synchronous Reset Operation
The circuit requires a synchronous active-low $\overline{RESET}$ pin. When $\overline{RESET} = 0$, all flip-flop states must transition to 0 on the next clock edge. Given the characteristic equation of a T flip-flop: $Q_{next} = T \oplus Q_{current}$. To force $Q_{next} = 0$, we require $T = Q_{current}$. Therefore, a built-in synchronous clear ($\overline{SRST}$) internally overrides the normal T input to feed the current state Q back into the toggle logic, ensuring a synchronous reset to 0000_2.

4. Circuit Implementation



Combinational Logic Design

Q1. Draw a PLA programmed to implement the following functions:

$$Y_1 = A\bar{B}C + \bar{A}C + B$$
$$Y_2 = \bar{B}AC + AB\bar{C} + \bar{B}$$
$$Y_3 = \bar{A}C + B\bar{C}A$$

(15 marks) **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

1. Theory and Principle of Operation

A **Programmable Logic Array (PLA)** is a programmable logic device utilized to implement complex combinational logic circuits. It consists of a fully programmable **AND plane** followed by a fully programmable **OR plane**.

A significant advantage of the PLA architecture is the ability to share *product terms* across multiple output functions. To exploit this, rather than minimizing each Boolean function independently (which might destroy common terms), we identify the unique unsimplified product terms from the given Sum-of-Products (SOP) expressions and program them into the AND array. These terms are then combined in the OR array as needed.

2. Logic Analysis and Product Term Identification

We are given the following three Boolean functions:

$$Y_1 = A\bar{B}C + \bar{A}C + B \tag{1}$$
$$Y_2 = \bar{B}AC + AB\bar{C} + \bar{B} \tag{2}$$
$$Y_3 = \bar{A}C + B\bar{C}A \tag{3}$$

First, let us rearrange the variables in alphabetical order for clarity:

$$Y_1 = A\bar{B}C + \bar{A}C + B \tag{4}$$
$$Y_2 = A\bar{B}C + AB\bar{C} + \bar{B} \tag{5}$$
$$Y_3 = \bar{A}C + AB\bar{C} \tag{6}$$

By inspecting the expressions, we extract the distinct product terms required to build the AND plane. There are exactly five unique product terms (P_1 to P_5):

- $P_1 = A\bar{B}C$ (Shared by Y_1 and Y_2)
- $P_2 = \bar{A}C$ (Shared by Y_1 and Y_3)
- $P_3 = B$ (Used only in Y_1)
- $P_4 = AB\bar{C}$ (Shared by Y_2 and Y_3)
- $P_5 = \bar{B}$ (Used only in Y_2)

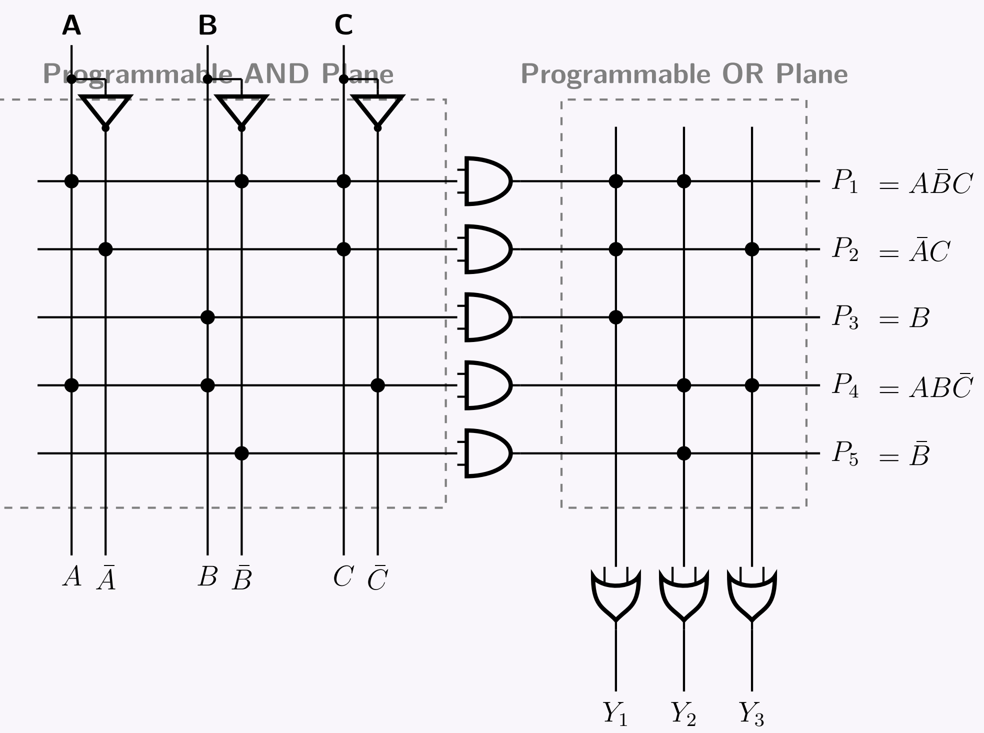
3. Output Expressions

Using the assigned product terms, we can define the outputs for the OR plane programming:

$$Y_1 = P_1 + P_2 + P_3 \tag{7}$$
$$Y_2 = P_1 + P_4 + P_5 \tag{8}$$
$$Y_3 = P_2 + P_4 \tag{9}$$

4. Programmable Logic Array (PLA) Schematic

The circuit below is the PLA implementation. The solid black dots (●) indicate the programmed interconnections in both the AND and OR arrays. Unmarked intersections are simply overlapping wires with no electrical connection.



Q2. Implement the function $f(x_1, x_2, x_3) = \sum m(2, 3, 4, 6, 7)$ using two-input LUTs (Look Up Tables). You can use as many LUTs as needed. **(16 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Answer

1. Logic Simplification
We are given a 3-variable Boolean function defined by its minterms:

$$f(x_1, x_2, x_3) = \sum m(2, 3, 4, 6, 7) \tag{1}$$

First, let us write the canonical Sum-of-Products (SOP) expression for this function:

$$\begin{aligned} f(x_1, x_2, x_3) &= m_2 + m_3 + m_4 + m_6 + m_7 \\ &= \bar{x}_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 \bar{x}_3 + x_1 x_2 x_3 \end{aligned} \tag{2}$$

To minimize this expression, we apply Boolean algebraic theorems (specifically, $A\bar{B} + AB = A$):

$$\begin{aligned} f(x_1, x_2, x_3) &= \bar{x}_1 x_2 (\bar{x}_3 + x_3) + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 (\bar{x}_3 + x_3) \\ &= \bar{x}_1 x_2 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 \end{aligned} \tag{3}$$

Now, group the terms containing x_2:

$$\begin{aligned} f(x_1, x_2, x_3) &= x_2(\bar{x}_1 + x_1) + x_1 \bar{x}_2 \bar{x}_3 \\ &= x_2 + x_1 \bar{x}_2 \bar{x}_3 \end{aligned} \tag{4}$$

Using the distributive law of Boolean algebra ($A + \bar{A}B = A + B$), we can further simplify:

$$f(x_1, x_2, x_3) = x_2 + x_1 \bar{x}_3 \tag{5}$$

2. Decomposition for 2-Input LUTs
A Look-Up Table (LUT) with k inputs can implement any Boolean function of up to k variables. Since we are restricted to using 2-input LUTs, and our simplified function has 3 variables, we must cascade multiple LUTs.
We can decompose the function into two separate 2-variable functions:

- **LUT 1:** Let us compute the intermediate function $y_1 = x_1 \bar{x}_3$. This function requires exactly two inputs (x_1 and x_3).
- **LUT 2:** The final output is then computed as $f = x_2 + y_1$. This function also requires exactly two inputs (x_2 and the intermediate output y_1).

Thus, exactly **two** 2-input LUTs are required to implement the given function.

3. LUT Programming (Truth Tables)
To program the LUTs, we simply write the truth tables for their respective functions. The configuration memory of the LUT stores the output column.

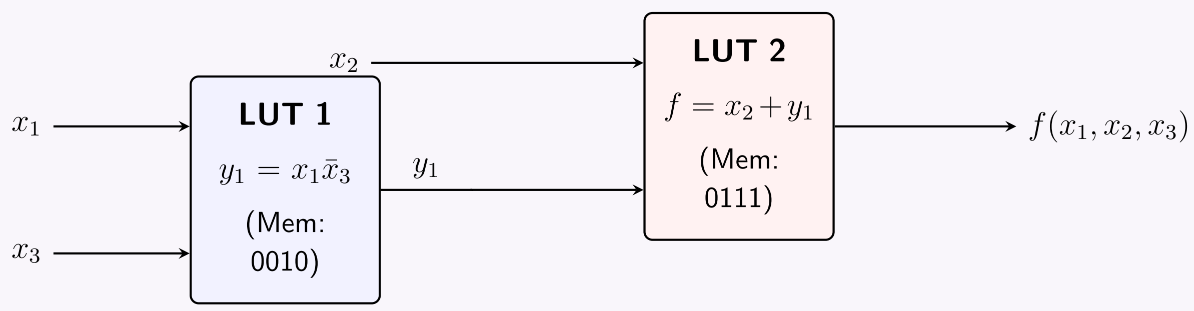
LUT 1 Configuration
Function: $y_1 = x_1 \bar{x}_3$

x_1	x_3	y_1
0	0	0
0	1	0
1	0	1
1	1	0

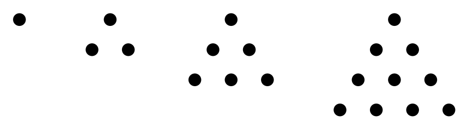
LUT 2 Configuration
Function: $f = x_2 + y_1$

x_2	y_1	f
0	0	0
0	1	1
1	0	1
1	1	1

4. Implementation Block Diagram
The circuit topology for the cascaded LUTs is shown below.



Q3. Triangular numbers are sequences formed by continuous summation of natural numbers (1, 3, 6, 10, ...). Design the simplest possible logic circuit that outputs 1 whenever it detects a 4-bit triangular number.



The triangular numbers
1, 3, 6, 10, ...

(11 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Answer

1. Theory and Principle

Triangular numbers are generated by the formula $T_n = \frac{n(n+1)}{2}$. For a 4-bit binary number system, the maximum decimal value is 15. Following the sequence provided in the problem statement $(1, 3, 6, 10, \dots)$, the triangular numbers that can be represented within 4 bits are:

- $T_1 = 1$
- $T_2 = 3$
- $T_3 = 6$
- $T_4 = 10$
- $T_5 = 15$

Therefore, our logic circuit must output $Y = 1$ when the 4-bit input corresponds to the decimal values $1, 3, 6, 10$, or 15. For all other combinations, the output must be 0.

2. Truth Table

Let the 4-bit input be represented by the variables A, B, C, D, where A is the Most Significant Bit (MSB) and D is the Least Significant Bit (LSB).

A	B	C	D	Y	A	B	C	D	Y
0	0	0	0	0	1	0	0	0	0
0	0	0	1	**1** (m_1)	1	0	0	1	0
0	0	1	0	0	1	0	1	0	**1** (m_{10})
0	0	1	1	**1** (m_3)	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	1	0	1	0
0	1	1	0	**1** (m_6)	1	1	1	0	0
0	1	1	1	0	1	1	1	1	**1** (m_{15})

The Boolean function in Sum-of-Minterms form is:

$$Y(A, B, C, D) = \sum m(1, 3, 6, 10, 15) \tag{1}$$

3. Karnaugh Map Simplification

We map the minterms onto a 4-variable K-map to find the optimal groupings.

$AB \backslash CD$	**00**	**01**	**11**	**10**
00	0	1 (m_1)	1 (m_3)	0
01	0	0	0	1 (m_6)
11	0	0	1 (m_{15})	0
10	0	0	0	1 (m_{10})

From the K-map, we derive the minimized Boolean expressions:

- Grouping m_1 and m_3: $\bar{A}\bar{B}D$
- m_6 is isolated: $\bar{A}BC\bar{D}$
- m_{10} is isolated: $A\bar{B}C\bar{D}$
- m_{15} is isolated: $ABCD$

The initial Sum-of-Products (SOP) expression is:

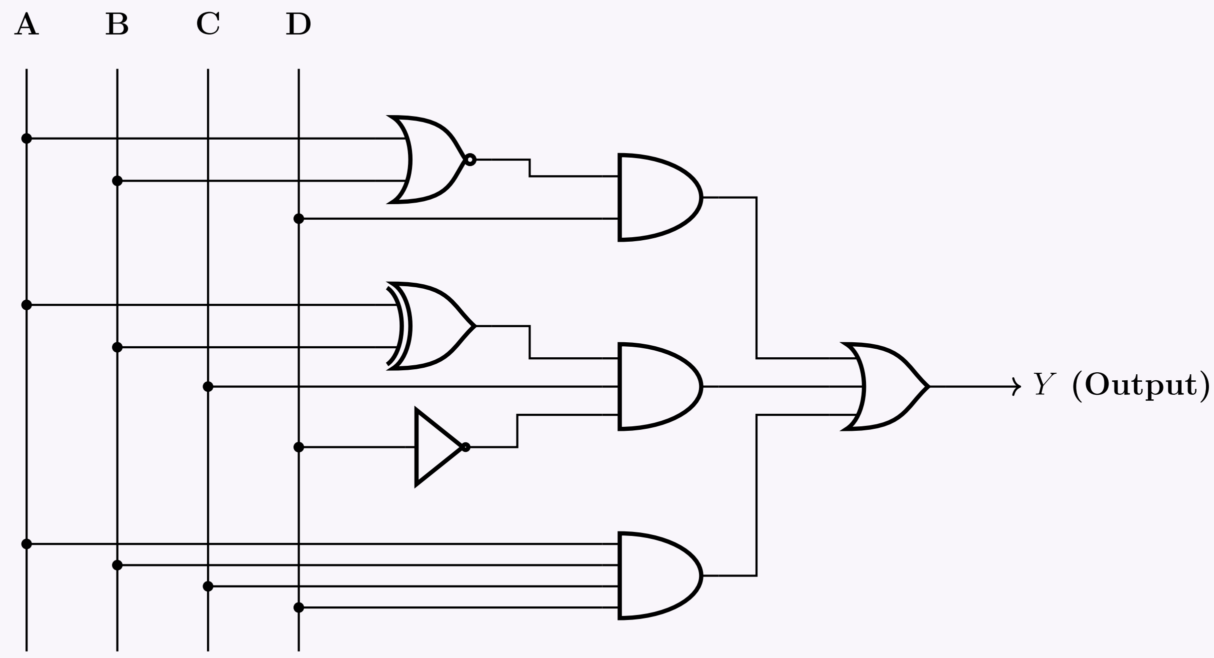
$$Y = \bar{A}\bar{B}D + \bar{A}BC\bar{D} + A\bar{B}C\bar{D} + ABCD \tag{2}$$

To design the *simplest possible* logic circuit (minimizing total gate count and inputs), we can factor the expression algebraically using XOR logic:

$$Y = \bar{A}\bar{B}D + C\bar{D}(\bar{A}B + A\bar{B}) + ABCD$$
$$Y = \overline{(A + B)}D + C\bar{D}(A \oplus B) + ABCD \tag{3}$$

4. Logic Circuit Implementation

The optimized expression $Y = \overline{(A + B)}D + C\bar{D}(A \oplus B) + ABCD$ requires fewer gates by utilizing a NOR gate for $\overline{(A + B)}$ and an XOR gate for $(A \oplus B)$.



Priority Encoder

Q1. Design a four-line to two-line priority encoder circuit with active high inputs and outputs, with priority assigned to the higher-order data input line ($I_4 > I_3 > I_1$). Use NOR gates only for the circuit implementation. **(16 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

1. Theory and Principle

A priority encoder is a combinational logic circuit that accommodates multiple active inputs and produces a binary output corresponding to the index of the highest-priority active input. In this problem, we are designing a 4-to-2 priority encoder with active-high inputs and outputs.

Based on the problem statement, the inputs are designated I_4, I_3, I_2, I_1 in descending order of priority (i.e., $I_4 > I_3 > I_2 > I_1$). The highest-order input (I_4) generates the maximum output state ($Y_1Y_0 = 11$), while the lowest-order input (I_1) generates the minimum output state ($Y_1Y_0 = 00$).

2. Truth Table

Let Y_1 (MSB) and Y_0 (LSB) be the outputs. We use "X" to denote a "don't care" condition, representing that lower-priority inputs have no effect when a higher-priority input is active.

Inputs				Outputs	
I_4	I_3	I_2	I_1	Y_1	Y_0
1	X	X	X	1	1
0	1	X	X	1	0
0	0	1	X	0	1
0	0	0	1	0	0
0	0	0	0	X	X

3. Boolean Derivation

From the truth table, we extract the Boolean expressions for Y_1 and Y_0 in Sum-of-Products (SOP) form. The condition for Y_1 being HIGH is when either I_4 is HIGH, or (I_4 is LOW and I_3 is HIGH):

$$Y_1 = I_4 + \overline{I_4}I_3 = I_4 + I_3 \tag{1}$$

The simplification $I_4 + \overline{I_4}I_3 = I_4 + I_3$ utilizes the distributive Boolean identity $A + \bar{A}B = A + B$.

The condition for Y_0 being HIGH is when I_4 is HIGH, or when (I_4 is LOW, I_3 is LOW, and I_2 is HIGH):

$$Y_0 = I_4 + \overline{I_4}\overline{I_3}I_2$$
$$Y_0 = I_4 + \overline{I_3}I_2 \tag{2}$$

Note that the input I_1 does not appear in the equations. It establishes the lowest priority level but does not dictate a logical HIGH on any output line.

4. Conversion to Universal NOR Logic
The problem strictly requires the use of NOR gates. A 2-input NOR gate performs the operation $\overline{A+B}$. We apply De Morgan's theorems and involution ($\overline{\overline{A}} = A$) to convert the derived SOP expressions into pure NOR logic.
For Y_1:

$$\begin{aligned} Y_1 &= I_4 + I_3 \\ Y_1 &= \overline{\overline{I_4 + I_3}} \quad \text{(Implementation: NOR followed by a NOT)} \end{aligned} \tag{3}$$

For Y_0:

$$\begin{aligned} Y_0 &= I_4 + \overline{I_3}I_2 \\ Y_0 &= I_4 + \overline{I_3 + \overline{I_2}} \\ Y_0 &= \overline{\overline{I_4 + \overline{I_3 + \overline{I_2}}}} \end{aligned} \tag{4}$$

This dictates the following cascaded NOR operations for Y_0:

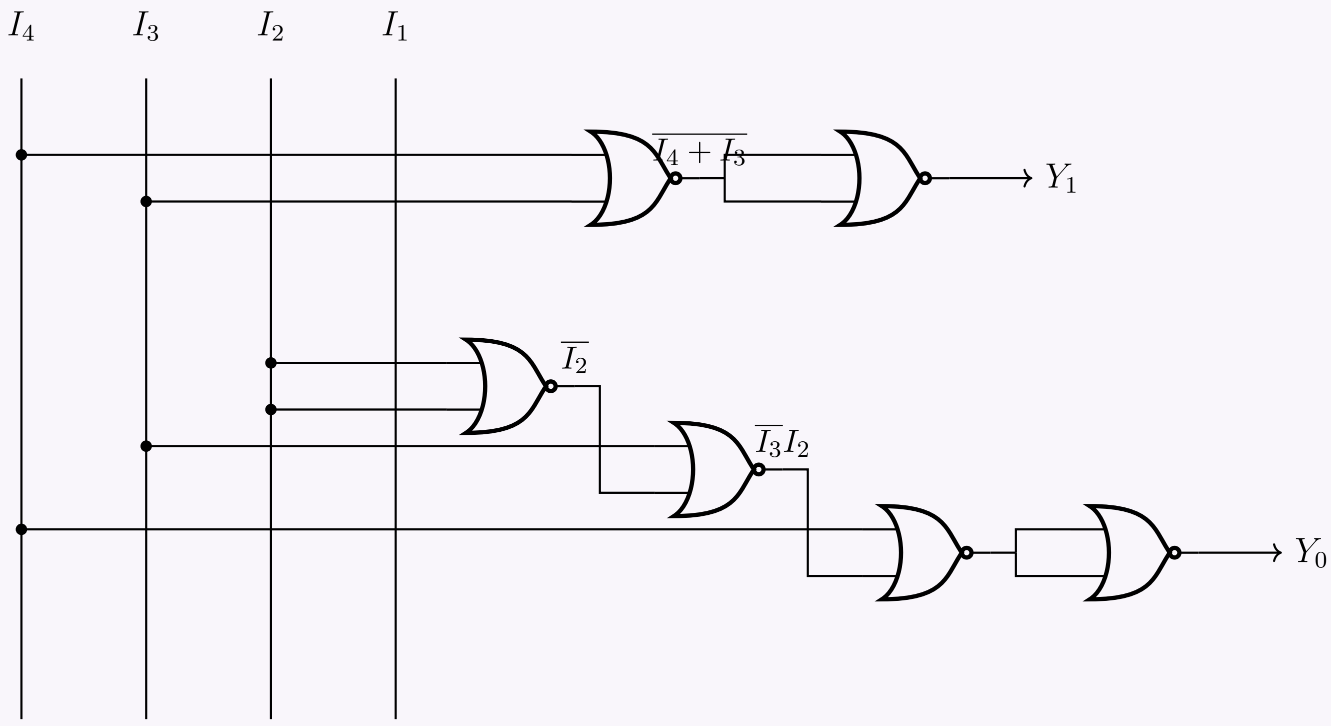
(a) Generate $\overline{I_2}$ using a NOR gate with tied inputs: $\overline{I_2} = I_2 \downarrow I_2$

(b) NOR the result with I_3: $\overline{I_3 + \overline{I_2}} = \overline{I_3}I_2$

(c) NOR the result with I_4: $\overline{I_4 + \overline{I_3}I_2}$

(d) Invert the output using a final NOR gate: $Y_0 = \overline{\overline{I_4 + \overline{I_3}I_2}}$

5. Logic Circuit Implementation
Below is the corresponding schematic containing solely NOR gates.



Shift Register Design

Q1. Design a 4-bit shift register using D flip-flops and a four-to-one multiplexer with mode selection inputs S_1 and S_0. The register operates according to the following function table:

S_1	S_0	**Register Operation**
0	0	No change
0	1	Complement the four outputs
1	0	Clear register to 0 (synchronous with clock)
1	1	Shift Right

(15 marks) **[EEE 263 | L-2/T-1 | 2017–18]**

Answer

1. Theory and Principle

A multi-mode shift register can be designed by using a generic register constructed from D-type flip-flops and routing the appropriate next-state signals to their D inputs. To accommodate four distinct operating modes, a 4-to-1 Multiplexer (MUX) is placed at the input of each D flip-flop.

The mode selection inputs, S_1 and S_0, serve as the select lines for the multiplexers. Because the operation must be synchronous, all flip-flops are triggered by a common Clock (CLK) signal. The MUX routes the correct signal to the D input based on the S_1S_0 state before the next active clock edge.

2. Derivation of Multiplexer Inputs

Let the four flip-flops be FF_3, FF_2, FF_1, and FF_0, where FF_3 holds the Most Significant Bit (MSB, Q_3) and FF_0 holds the Least Significant Bit (LSB, Q_0). For any given stage i (where $i \in \{0, 1, 2, 3\}$), the next state D_i is determined by the truth table:

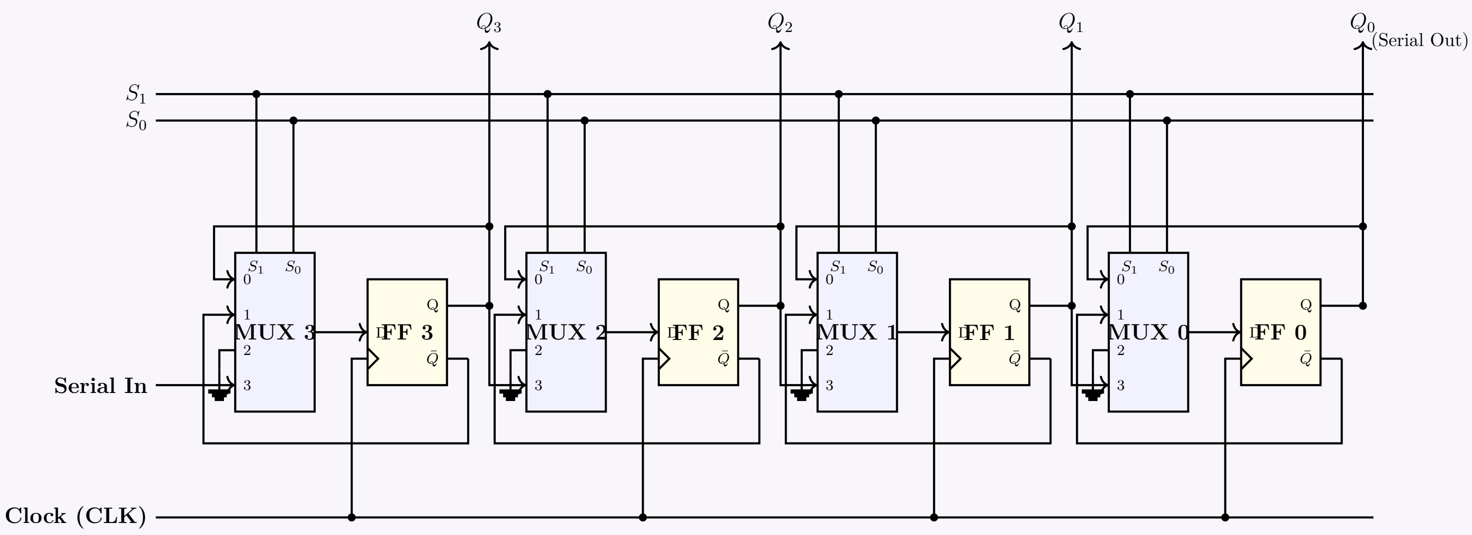
- **Mode 0 ($S_1S_0 = 00$): No Change.**
  The next state must equal the current state. Therefore, $D_i = Q_i$.
  Connection: Input 0 of the MUX is connected to the Q_i output of its own flip-flop.
- **Mode 1 ($S_1S_0 = 01$): Complement.**
  The next state must be the inverse of the current state. Therefore, $D_i = \overline{Q}_i$.
  Connection: Input 1 of the MUX is connected to the $\overline{Q}_i$ output of its own flip-flop.
- **Mode 2 ($S_1S_0 = 10$): Clear Register.**
  The register must be synchronously cleared to zero. Therefore, $D_i = 0$.
  Connection: Input 2 of the MUX is tied directly to Logic 0 (Ground).
- **Mode 3 ($S_1S_0 = 11$): Shift Right.**
  Data shifts from MSB to LSB. The next state of bit i takes the current state of bit $i+1$. Therefore, $D_i = Q_{i+1}$.
  Connection: Input 3 of the MUX is connected to the Q output of the preceding stage (Q_{i+1}). For the MSB stage (FF_3), input 3 is connected to an external **Serial Input**.

The complete Boolean expression for the D-input of stage i is:

$$D_i = \overline{S_1} \cdot \overline{S_0} \cdot Q_i + \overline{S_1} \cdot S_0 \cdot \overline{Q}_i + S_1 \cdot \overline{S_0} \cdot (0) + S_1 \cdot S_0 \cdot Q_{i+1} \tag{1}$$

3. Circuit Implementation

The detailed schematic below shows the 4-bit register using four D flip-flops and four 4-to-1 multiplexers. Common rails are utilized for the mode selection lines (S_1, S_0) and the synchronous Clock (CLK).



BUET · EEE 263 · Electronic Circuits

Electronic Circuits

Part VII — Data Converters, PLL & Memory

DAC, ADC, Phase-Locked Loop, DRAM, Flash Memory

S10 Data Converters, PLL & Memory

Digital-to-Analog Converter (DAC)

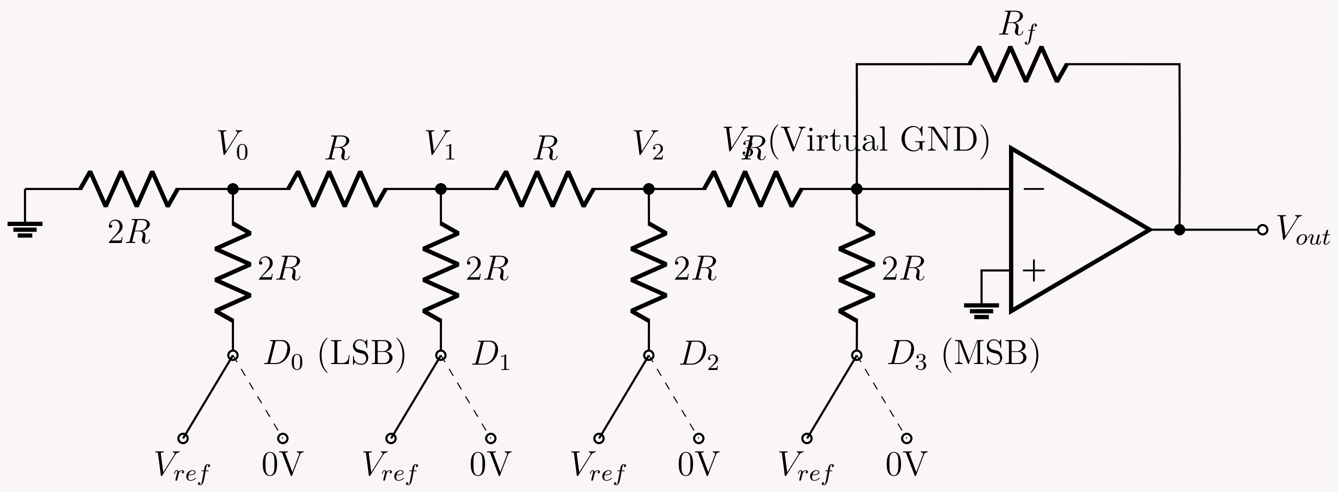
Q1. What are DAC circuits? How does an R-2R ladder DAC circuit operate? Explain using a circuit diagram and mathematical analysis. **(17 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Introduction to DAC Circuits
A **Digital-to-Analog Converter (DAC)** is an electronic circuit that transforms a discrete, digital binary signal (consisting of 0s and 1s) into a continuous analog equivalent (usually voltage or current). DACs are fundamental components in modern electronics, interfacing digital processing systems (like microcontrollers or DSPs) with the analog real world, enabling applications such as audio generation, motor control, and communication systems.

2. Principle of the R-2R Ladder DAC
The **R-2R Ladder DAC** overcomes the primary limitation of the binary-weighted resistor DAC (which requires a wide range of highly precise resistor values) by utilizing only two distinct resistor values: R and $2R$.
The core principle relies on the recurring parallel-series resistor geometry. Looking to the left or right of any node in the ladder, the Thevenin equivalent resistance exhibits a consistent property that splits the current by exactly half at each successive node. This creates the binary-weighted currents required for digital-to-analog conversion using a uniform cascaded structure.

3. Circuit Diagram
Below is the circuit diagram of a 4-bit R-2R ladder DAC coupled to an inverting operational amplifier configuration.



4. Mathematical Analysis and Derivation
Assumptions:

- The Operational Amplifier is ideal. Therefore, the inverting terminal (V_3) is at **virtual ground** (0V) because the non-inverting terminal is grounded.
- No current flows into the Op-Amp input terminals.
- The logic switches perfectly connect to V_{ref} (Logic 1) or 0V (Logic 0).

Thevenin Equivalent Resistance: If we analyze the ladder looking to the left from node V_1, the resistance is the terminating $2R$ in parallel with the D_0 branch $2R$ (assuming $D_0 = 0$). $R_{eq} = 2R \parallel 2R = R$. Adding the series resistor R between V_0 and V_1 yields $R + R = 2R$. This recursive property holds for every node. Looking left from any node k, the equivalent resistance is strictly $2R$.
Current Contribution per Bit (Superposition): We calculate the output by activating one binary input D_k at a time while setting the others to 0.
Case 1: Only MSB is high ($D_3 = 1$, all others 0)

- The MSB resistor $2R$ is connected to V_{ref}.
- Node V_3 is at virtual ground (0V).
- The current flowing from D_3 into the virtual ground is: $I_3 = \frac{V_{ref}}{2R}$.
- Because the Op-Amp has infinite input impedance, all of I_3 flows through the feedback resistor R_f.
- Output voltage: $V_{out,3} = -I_3 R_f = -V_{ref}\left(\frac{R_f}{2R}\right)$.

Case 2: Only D_2 is high ($D_2 = 1$, all others 0)

- Current from V_{ref} enters node V_2.
- At V_2, the resistance looking left is $2R$. The resistance looking right (towards virtual ground V_3) is the series resistor R.

- Total equivalent resistance seen by V_{ref} is: $R_{total} = 2R + (2R \parallel R) = 2R + \frac{2R}{3} = \frac{8R}{3}$.
- The total current leaving V_{ref} is $I_{D2} = \frac{V_{ref}}{8R/3} = \frac{3V_{ref}}{8R}$.
- Using the current divider rule, the fraction of I_{D2} flowing to the right (into V_3) is:

$$I_2 = I_{D2} \times \frac{2R}{2R + R} = \frac{3V_{ref}}{8R} \times \frac{2}{3} = \frac{V_{ref}}{4R} \tag{1}$$

- Output voltage: $V_{out,2} = -I_2 R_f = -V_{ref}\left(\frac{R_f}{4R}\right)$.

General Output Equation: Continuing this logic, each subsequent, less significant bit contributes exactly half the current of the preceding bit. The current into the virtual ground for any bit D_k is:

$$I_k = D_k \frac{V_{ref}}{2^{4-k}R} \quad \text{for } k = 0, 1, 2, 3 \tag{2}$$

By superposition, the total current I_{total} entering the virtual ground is the sum of the individual bit currents:

$$\begin{aligned} I_{total} &= I_3 + I_2 + I_1 + I_0 \\ I_{total} &= \frac{V_{ref}}{2R}D_3 + \frac{V_{ref}}{4R}D_2 + \frac{V_{ref}}{8R}D_1 + \frac{V_{ref}}{16R}D_0 \end{aligned} \tag{3}$$

The final output voltage $V_{out} = -I_{total}R_f$ evaluates to:

$$V_{out} = -V_{ref}\left(\frac{R_f}{R}\right)\left[\frac{D_3}{2^1} + \frac{D_2}{2^2} + \frac{D_1}{2^3} + \frac{D_0}{2^4}\right] \tag{4}$$

For an N-bit R-2R DAC, the generalized equation is:

$$V_{out} = -V_{ref}\left(\frac{R_f}{R}\right)\sum_{i=0}^{N-1}\frac{D_i}{2^{N-i}} \tag{5}$$

5. Advantages

- **Simplified Manufacturing:** Requires only two resistor values (R and $2R$), making it much easier to fabricate highly matched components in Monolithic ICs.
- **Scalability:** Can be easily extended to higher bit resolutions (e.g., 16-bit or 24-bit) without requiring impractically large or precise resistance values.
- **Constant Impedance:** The input impedance seen by the reference voltage source remains perfectly constant regardless of the digital input word.

Analog-to-Digital Converter (ADC)

Q1. A 6-bit A/D converter has reference voltage $V_{REF} = 8\,\text{V}$ and reference count $n_{REF} = 2^6$. Values of some consecutive analog sampled voltages $|V_A|$ are: $0\,\text{V}$, $3\,\text{V}$, $5.5\,\text{V}$, $7.125\,\text{V}$. If these voltages are given as inputs to the converter, find the corresponding digital outputs. **(12 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Theory and Principle
An Analog-to-Digital Converter (ADC) maps a continuous analog input voltage to a discrete digital binary output. The full-scale reference voltage, V_{REF}, is divided into a specific number of intervals (or quantization levels) defined by the reference count, n_{REF}. For an N-bit ADC, $n_{REF} = 2^N$.
The step size, also known as the **Least Significant Bit (LSB) voltage** or resolution (Δ), represents the smallest analog voltage change required to increment the digital output by one state. It is calculated as:

$$\Delta = \frac{V_{REF}}{n_{REF}} \tag{1}$$

The decimal equivalent of the digital output, D_{dec}, for a given sampled analog voltage, V_A, is determined by calculating how many step sizes fit into that voltage:

$$D_{dec} = \frac{V_A}{\Delta} = V_A\left(\frac{n_{REF}}{V_{REF}}\right) \tag{2}$$

Once D_{dec} is found, it is converted into its N-bit binary equivalent.

2. Given Parameters

- Resolution: $N = 6$ bits

- Reference Voltage: $V_{REF} = 8$ V
- Reference Count: $n_{REF} = 2^6 = 64$

3. Resolution (LSB) Calculation
First, we determine the LSB voltage (Δ) for this specific ADC:

$$\Delta = \frac{8\text{ V}}{64} = 0.125\text{ V/LSB} \tag{3}$$

This indicates that each increment in the digital output corresponds to an analog increase of 0.125 V.
The conversion formula thus simplifies to:

$$D_{dec} = \frac{V_A}{0.125} = 8 \cdot V_A \tag{4}$$

4. Step-by-Step Solution
We now apply the conversion formula to each given analog sampled voltage and convert the resulting decimal value into a 6-bit binary number.

Case 1: $V_A = 0\ V$

$$D_{dec} = 8 \times 0 = 0$$
$$\text{Digital Output (Binary)} = \mathbf{000000_2}$$

Case 2: $V_A = 3\ V$

$$D_{dec} = 8 \times 3 = 24$$
$$\text{Decimal to Binary}: 24 = 16 + 8 \implies \mathbf{011000_2}$$

Case 3: $V_A = 5.5\ V$

$$D_{dec} = 8 \times 5.5 = 44$$
$$\text{Decimal to Binary}: 44 = 32 + 8 + 4 \implies \mathbf{101100_2}$$

Case 4: $V_A = 7.125\ V$

$$D_{dec} = 8 \times 7.125 = 57$$
$$\text{Decimal to Binary}: 57 = 32 + 16 + 8 + 1 \implies \mathbf{111001_2}$$

5. Final Answer Summary

Analog Voltage (V_A)	**Decimal Value (D_{dec})**	**6-bit Digital Output**
0 V	0	**000000**
3 V	24	**011000**
5.5 V	44	**101100**
7.125 V	57	**111001**

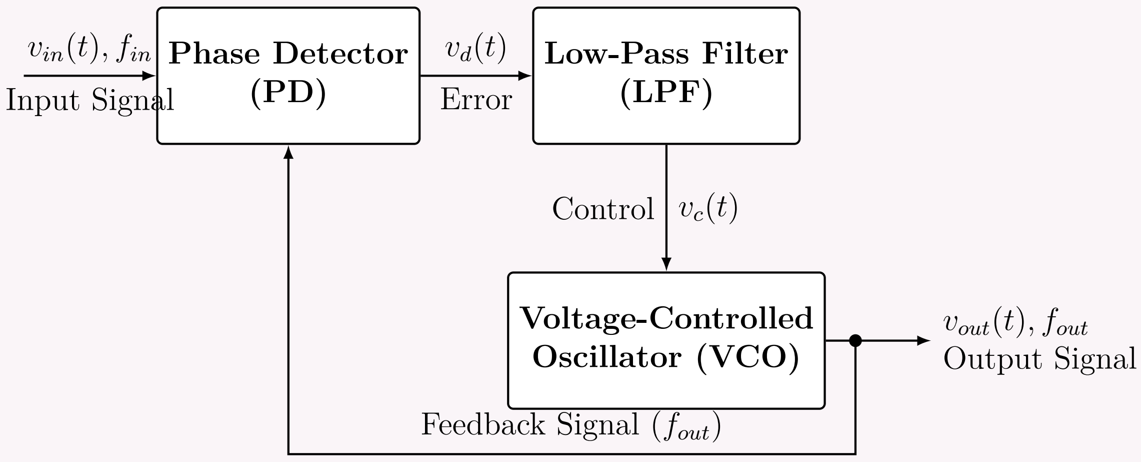
Phase-Locked Loop (PLL)

Q1. Draw a simplified block diagram of a PLL and briefly explain its operation. **(17 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Theory and Principle
A **Phase-Locked Loop (PLL)** is a negative feedback control system that automatically adjusts the frequency and phase of a locally generated signal to match those of an incoming reference signal. Once synchronized, the PLL is said to be "in lock," meaning the output frequency exactly matches the input frequency, and there is a constant (or zero) phase difference between them.

2. Simplified Block Diagram



3. Component Operation

A standard PLL consists of three primary functional blocks:

- **Phase Detector (PD):** This block acts as an error detector. It compares the phase and frequency of the incoming reference signal, $v_{in}(t)$, with the feedback signal originating from the VCO, $v_{out}(t)$. The output is an error voltage, $v_d(t)$, which is proportional to the phase difference ($\Delta\theta = \theta_{in} - \theta_{out}$) between the two signals. In practice, $v_d(t)$ contains a DC component and high-frequency AC components.
- **Low-Pass Filter (LPF):** The error signal $v_d(t)$ is fed into the LPF, which removes the high-frequency alternating components and noise, yielding a relatively clean, slow-varying DC voltage, $v_c(t)$. The LPF dictates the dynamic response of the PLL, including its capture range (the frequency range over which the PLL can acquire a lock) and lock range (the range over which it can maintain a lock).
- **Voltage-Controlled Oscillator (VCO):** The VCO generates the output signal, $v_{out}(t)$. Its oscillation frequency, f_{out}, is linearly determined by the filtered control voltage, $v_c(t)$, according to the relation:

$$f_{out} = f_0 + K_v v_c(t) \tag{1}$$

where f_0 is the free-running frequency of the VCO (when $v_c = 0$), and K_v is the voltage-to-frequency conversion gain of the VCO.

4. System Operation Stages

The operation of the PLL transitions through three distinct states:

(a) **Free-Running State:** If no input signal is applied ($v_{in} = 0$), the error voltage is zero, and the VCO oscillates at its natural free-running frequency, f_0.

(b) **Capture State:** When an input signal f_{in} is applied (and falls within the capture range), the PD generates a difference-frequency signal. The LPF filters this, modifying $v_c(t)$. The VCO frequency f_{out} begins to shift towards f_{in}.

(c) **Locked State:** The feedback loop successfully drives the control voltage $v_c(t)$ to a constant DC level such that the VCO frequency exactly equals the input frequency ($f_{out} = f_{in}$). At this point, the signals are perfectly synchronized, maintaining a steady, finite phase difference.

5. Important Applications

- **Frequency Synthesis:** Generating highly accurate multiple frequencies from a single stable crystal oscillator (used extensively in RF communications).
- **Demodulation:** Extracting the original modulating signal in Frequency Modulation (FM) or Phase Modulation (PM) receivers.
- **Clock Recovery:** Reconstructing the timing clock from high-speed digital data streams.
- **Motor Speed Control:** Ensuring precise rotational speeds in electromechanical systems.

Memory: DRAM and Flash

Q1. Explain the read and write operation in a DRAM cell. **(16 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

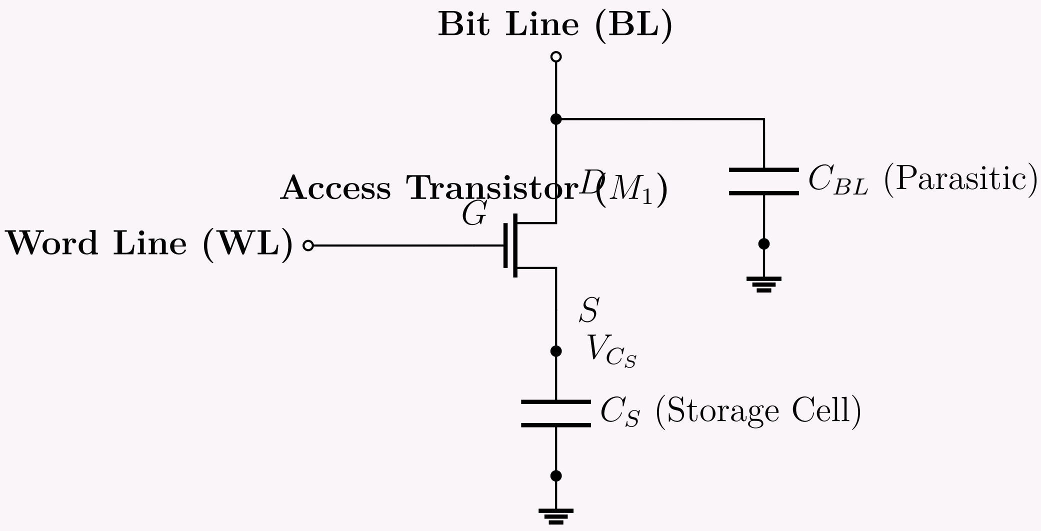
1. Theory and Principle of DRAM

Dynamic Random Access Memory (DRAM) is a type of volatile semiconductor memory that stores each bit of data in a distinct, microscopic capacitive structure within an integrated circuit. The fundamental building block of modern DRAM is the **1T1C (One-Transistor, One-Capacitor) cell**.

The logic state is defined by the presence or absence of charge on the storage capacitor (C_S):

- **Logic '1':** Capacitor is fully charged to V_{DD}.
- **Logic '0':** Capacitor is completely discharged to 0 V.

2. Equivalent Circuit Diagram



3. Write Operation
The write operation aims to store a distinct voltage level onto the storage capacitor C_S. The sequence is as follows:

(a) **Bit Line Conditioning:** The write driver circuitry forces the Bit Line (BL) to the desired logic voltage: V_{DD} for a Logic '1' and 0 V for a Logic '0'.

(b) **Word Line Activation:** The Word Line (WL) is asserted HIGH. To prevent a threshold voltage drop (V_{TH}) across the NMOS transistor and ensure a full V_{DD} is written to C_S, the WL voltage is typically boosted to a value $V_{WL} > V_{DD} + V_{TH}$.

(c) **Charge Transfer:** Transistor M_1 turns ON, providing a low-impedance conduction path. The storage capacitor C_S rapidly charges or discharges until its voltage V_{C_S} equals the BL voltage.

(d) **Data Trapping:** The WL is de-asserted (pulled LOW), turning M_1 OFF. The charge is now isolated and trapped in C_S.

4. Read Operation
The read operation in a DRAM cell is more complex and fundamentally **destructive**, meaning the act of reading temporarily corrupts the stored data. It relies on the principle of *charge sharing*.

(a) **Precharge:** Before a read, the Bit Line is disconnected from all active drivers and precharged to exactly half the supply voltage, $V_{BL} = \frac{V_{DD}}{2}$. The BL is then left floating.

(b) **Word Line Activation:** The WL is asserted HIGH, turning M_1 ON.

(c) **Charge Sharing:** The charge stored in the tiny cell capacitor C_S redistributes with the much larger parasitic capacitance of the Bit Line (C_{BL}). By applying the conservation of charge ($Q_{initial} = Q_{final}$):

$$C_S V_{C_S} + C_{BL}\left(\frac{V_{DD}}{2}\right) = (C_S + C_{BL})V_{BL,final} \tag{1}$$

The resulting small change in the Bit Line voltage (ΔV) is derived as:

$$\Delta V = V_{BL,final} - \frac{V_{DD}}{2} = \left(\frac{C_S}{C_S + C_{BL}}\right)\left(V_{C_S} - \frac{V_{DD}}{2}\right) \tag{2}$$

Because $C_{BL} \gg C_S$, ΔV is very small (typically in the tens of millivolts).

- If a '1' was stored ($V_{C_S} = V_{DD}$), ΔV is strictly positive.
- If a '0' was stored ($V_{C_S} = 0$), ΔV is strictly negative.

(d) **Sensing:** A highly sensitive, differential **Sense Amplifier** detects this minute ΔV. It quickly amplifies the slight deviation, pulling the Bit Line fully to V_{DD} (if $\Delta V > 0$) or fully to 0 V (if $\Delta V < 0$).

(e) **Restore (Write-Back):** Because the charge sharing equalized the voltages, the original data in C_S was destroyed. However, since M_1 is still ON while the Sense Amplifier drives the BL to the full logic level, the full voltage is automatically rewritten back into C_S, restoring the stored bit.

(f) **Completion:** The WL is finally de-asserted, isolating the restored charge in C_S.

5. Important Note: The Need for Refresh
Due to subthreshold leakage through the OFF-state transistor M_1 and dielectric leakage in C_S, the stored charge inevitably depletes over time. To prevent data loss, a **Refresh Cycle** must be periodically executed (typically every 64 ms). A refresh operation simply consists of a standard Read operation (which inherently restores the charge to its full logic level) across all rows in the memory array.

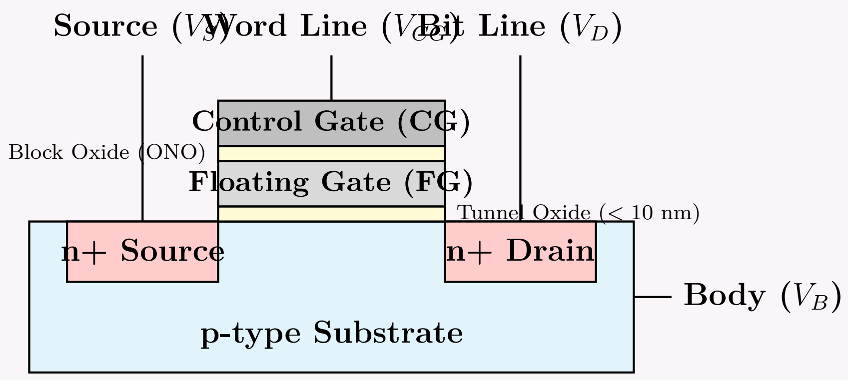
Q2. Explain briefly the programming and erase operation in a Flash Memory. **(15 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Answer

1. Theory and Principle

Flash memory is a non-volatile electrically erasable programmable read-only memory (EEPROM) that stores information using a specialized type of MOSFET known as a **Floating Gate MOSFET (FGMOS)**. The core principle relies on trapping electrons in a "floating gate," which is electrically isolated by an insulating oxide layer. The presence or absence of charge in this floating gate alters the threshold voltage (V_{TH}) of the transistor, defining the logic state ('0' or '1') of the memory cell.

2. FGMOS Cross-Sectional Structure



3. Programming Operation (Write)

Programming a flash memory cell involves injecting electrons into the isolated Floating Gate (FG). This is predominantly achieved using one of two mechanisms, depending on the flash architecture (NAND vs. NOR):

- **Fowler-Nordheim (FN) Tunneling (NAND Flash):** A very high positive voltage (e.g., +20 V) is applied to the Control Gate (CG), while the Source, Drain, and Substrate are grounded (0 V). This massive electric field across the extremely thin tunnel oxide effectively reduces the energy barrier, allowing electrons from the channel to quantum mechanically "tunnel" through the oxide into the floating gate.
- **Channel Hot Electron (CHE) Injection (NOR Flash):** A high voltage is applied to the Control Gate (e.g., +12 V) and a moderate voltage to the Drain (e.g., +5 V). Electrons flowing from the source to the drain gain massive kinetic energy ("hot" electrons). The strong vertical electric field from the CG attracts these highly energetic electrons, causing them to overcome the oxide barrier and jump into the floating gate.

Once the programming voltage is removed, the electrons are trapped in the floating gate, retaining the data indefinitely without power.

4. Erase Operation

The erase operation removes the trapped electrons from the floating gate, returning the cell to its default state. Unlike programming, which can be done on a per-byte or per-word basis, erasing in Flash memory is typically performed in large **blocks** simultaneously.

- **Erase Mechanism (FN Tunneling):** A large negative voltage (e.g., −20 V) is applied to the Control Gate, or alternatively, the CG is grounded (0 V) while a large positive voltage is applied to the bulk p-substrate (and Source/Drain).
- **Electron Evacuation:** The reverse electric field repels the trapped electrons in the floating gate. They undergo Fowler-Nordheim tunneling back through the tunnel oxide and into the substrate, clearing the floating gate of negative charge.

5. Effect on Threshold Voltage (V_{TH})

The memory logic state is differentiated by the threshold voltage, V_{TH}, of the FGMOS transistor. The shift in threshold voltage (ΔV_{TH}) is directly proportional to the trapped charge (Q_{FG}):

$$\Delta V_{TH} = -\frac{Q_{FG}}{C_{FC}} \tag{1}$$

where C_{FC} is the capacitance between the Control Gate and the Floating Gate.

- **Programmed State (Logic '0'):** Electrons are trapped in the FG (Q_{FG} is highly negative). This negative charge shields the channel from the Control Gate, meaning a much higher V_{CG} is required to form an inversion layer. Thus, V_{TH} **increases**. When a standard read voltage is applied, the transistor remains OFF.
- **Erased State (Logic '1'):** The FG has no trapped electrons (or is slightly positively depleted). The Control Gate has direct influence over the channel. Thus, V_{TH} **is low**. When a standard read voltage is applied, the transistor turns ON.